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Shouchuan Hu
Nikolaos S. Papageorgiou

Research Topics in Analysis, Volume II

Applications

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Shouchuan Hu • Nikolaos S. Papageorgiou

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To the memory of our mothers

Preface

This volume continues our presentation of certain research topics in modern analysis. In Volume I, we presented the basic background material necessary to understand the applications which we present in this volume. We start with "Degree Theory" which is a useful tool in the study of nonlinear equations. We present all major degree theories starting with the finite dimensional theory of Brouwer and continuing with its infinite dimensional analog (the Leray-Schauder theory), the degree theory for operators of monotone type and the degree theory for multifunctions. In this process, we address also some related topics of interest. In Chap. 2, we deal with Fixed Point Theory. We have a detailed presentation of Metric Fixed point Theory, Topological Fixed Point Theory and Order Fixes Point Theory. We also discuss fixed point theorems for multifunctions, needed in many applications such as mathematical economics and control theory. In Chap. 3, we present the Critical Pont Theory. We have a detailed presentation of theory starting with pseudogradients and the compactness conditions (PS and C conditions) and going all the way to critical groups, which with the use of algebraic topology provide a valuable tool in the study of boundary value problems. In Chap. 4, we present the main spectral properties of linear, nonlinear, anisotropic and double phase differential operators. Spectral Analysis of the various differential operators, is a valuable tool in the study of pineo and nonlinear boundary value problems. In Chap. 5, we study semilinear and nonlinear elliptic equations with different boundary conditions. In this chapter, we illustrate how the various analytical tools, developed thus far, are used to study elliptic equations (stationary problems). In Chap. 6, we do the same thing for dynamic systems monitored by ordinary and partial differential equations. In Chap. 7, we turn our attention to optimal control problems. We discuss the existence theory, the relaxation of problems, the relation between relaxability and value continuity (stability) and we develop some more

mathematical tools needed for an in-depth investigation of these issues. Finally in Chap. 8, we discuss some economic models (equilibrium models and dynamic ones) and we have a brief presentation of game theory, Nash equilibrium.

Springfield, IL, USA — Shouchuan Hu
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Note on Notations

In the text of the current book/volume, when we refer to a result or statement from Volume I, we adopt the following notations:

Proposition I-5.30
Definition I-5.29
Lemma I-6.118
Theorem I-6.132
etc.

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Chapter 1
Degree Theory



Introduction

In this chapter, we outline the main degree theories which we encounter in applications. We start with the Brouwer degree and then we consider its infinite dimensional extension, which is the Leray-Schauder degree. In Sect. 1.3, we examine the degrees for certain maps of monotone type, which appear in many boundary value problems. Finally, in Sect. 1.4, we see how the previous degrees can be extended to deal with multifunctions.

1.1 Brouwer Degree

Starting with an abstract equation $f(x) = b$, we want to know if this equation has a solution, whether this solution is unique or not and how these solutions are affected by changes in the function $f(\cdot)$ and the input b. An effective topological tool to perform these tasks, is the so called "Degree Theory".

In this section we deal with the finite dimensional part of this theory, which is represented by the "Brouwer Degree". We begin with some important definitions and facts concerning smooth functions.

Definition 1.1 Let $\Omega \subseteq \mathbb{R}^N$ be an open set and $f \in C^1(\Omega, \mathbb{R}^k)$. We say that $x \in \Omega$ is a "critical point of f", if rank $f'(x) < k$(recall that $f'(x) \in \mathcal{L}(\mathbb{R}^N, \mathbb{R}^k)$ and so it can be represented by a $k \times N$-matrix). The noncritical points, are called "regular points". A vector $b \in \mathbb{R}^k$ such that $b = f(x)$ for some critical point x, is said to be a "critical value". Otherwise, $b \in \mathbb{R}^k$ is said to be a "regular value".

Now let $k = N$. Then we can define $J_f(x) = \det f'(x)$(the "Jacobian" of $f(\cdot)$ at $x \in \Omega$). Then for $f \in C^1(\Omega, \mathbb{R}^N)$, the set of critical points of $f(\cdot)$ is

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$$S_f = \{x \in \Omega : J_f(x) = 0\}.$$

and $f(S_f)$ is the set of critical values of $f(\cdot)$. The next theorem, known in the literature as "Sard's Theorem", says that the set of critical values of $f(\cdot)$ is "small".

Theorem 1.2 *If $\Omega \subseteq \mathbb{R}^N$ is an open set and $f \in C^1(\Omega, \mathbb{R}^N)$, then $f(S_f)$ is Lebesgue-null.*

Proof Let $\mathcal{D}$ be a cube of side ϑ contained in Ω. We divide $\mathcal{D}$ into k^N sub-cubes each with side $\frac{\vartheta}{k}$. The uniform continuity of $f'(\cdot)$ implies that given $\varepsilon > 0$, we can find $\delta > 0$ such that

$$|x - u| < \delta \Rightarrow \|f'(x) - f'(u)\| < \varepsilon. \tag{1.1}$$

Choose $k \in \mathbb{N}$ big so that $\sqrt{N}\frac{\vartheta}{k} < \delta$.

By the mean value theorem, we have

$$|f(x) - f(u)| \leq \eta |x - u| \text{ for all } x, u \in \mathcal{D} \tag{1.2}$$

with $\eta = \sup_{x \in \mathcal{D}} \|f'(x)\|$.

Let $x \in \mathcal{D} \cap S_f$. We can find a sub-cube $\mathcal{D}' \subseteq \mathcal{D}$ such that $x \in \mathcal{D}'$. Then for every $u \in \mathcal{D}'$, we have

$$|f(x) - f(u)| \leq \eta\sqrt{N}\frac{\vartheta}{k} \text{ for all } u \in \mathcal{D}' \text{ (see (1.2))}.$$

Also, we have

$$\begin{aligned} f(u) - f(x) - f'(x)(u - x) &= \int_0^1 (f'(x + t(u - x)) - f'(x))(x - u)dt \\ &\leq \varepsilon |x - u| \text{ (see (1))} \\ &\leq \varepsilon\sqrt{N}\frac{\vartheta}{k}. \end{aligned} \tag{1.3}$$

Note that since $x \in \mathcal{D} \cap S_f$, we have $f'(x)(\mathbb{R}^N) = E$ a strict subspace of $\mathbb{R}^N$. So, by (1.3), we have

$$d(f(u), f(x) + E) \leq \varepsilon\sqrt{N}\frac{\vartheta}{k} \text{ for all } u \in \mathcal{D}'. \tag{1.4}$$

Hence, if $\lambda^N(\cdot)$ denotes the Lebesgue measure on $\mathbb{R}^N$, from (1.4), we infer that

$$\lambda^N(f(\mathcal{D}')) \leq 2c\sqrt{N}\frac{\vartheta}{k}(2\eta\sqrt{N}\frac{\vartheta}{k})^{N-1}$$

with $c = c(N) > 0$. It follows that

$$\begin{aligned}\lambda^N(f(\mathcal{D} \cap S_f)) &\leq \sum_{\mathcal{D}' \cap S_f \neq \Phi} \lambda^N(f(\mathcal{D}')) \\ &\leq 2^N c\eta^{N-1} N^{N/2} \vartheta^N \varepsilon \\ &= \gamma(N, \mathcal{D})\varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we have that

$$\lambda^N(f(\mathcal{D} \cap S_f)) = 0,$$

for any cube $\mathcal{D} \subseteq \Omega$. But Ω can be covered by a countable number of such cubes. Therefore,

$$\lambda^N(S_f) = 0.$$

□

Remark 1.3 We mention that there is a more general version of Sard's Theorem which says the following:

"If $\Omega \subseteq \mathbb{R}^N$ is open, $f \in C^1(\Omega, \mathbb{R}^N)$ and $A \subseteq \Omega$ is measurable, then $f(A) \subseteq \mathbb{R}^N$ is measurable and

$$\lambda^N(f(A)) \leq \int_A |J_f(x)| dx'' \tag{1.5}$$

(see Schwartz [437, p. 59]). Evidently, Theorem 1.2 follows from (1.5).

In what follows $\operatorname{sgn}(\cdot)$ is the function defined by

$$\operatorname{sgn}(u) = \begin{cases} 1 & \text{if } u > 0 \\ -1 & \text{if } u < 0 \end{cases}$$

and $\operatorname{sgn}(0)$ is not defined.

Now we are ready to make the first step in the direction of introducing a finite dimensional degree function.

Definition 1.4 Let $\Omega \subseteq \mathbb{R}^N$ be a bounded open set, $f \in C^1(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ and $b \notin f(S_f) \cup f(\partial\Omega)$. Then the (Brouwer) degree of f at b with respect to Ω, is defined by

$$d_B(f,\Omega,b)=\begin{cases}\sum\limits_{x\in f^{-1}(b)}\operatorname{sgn}(J_f(x)) & \text{if } f^{-1}(b)\neq\emptyset,\\ 0 & \text{otherwise}.\end{cases}$$

Remark 1.5 Since $b\notin f(S_f)$, by the inverse function theorem the set $f^{-1}(b)$ is finite and so the degree function $d_B(\cdot,\cdot,\cdot)$ is well-defined. According to the above definition $d_B(f,\Omega,b)=N_+-N_-$, with $N_+(resp.N_-)$ being the number of elements in $f^{-1}(b)$ with positive (resp. negative) Jacobian. This formulation explains why some people say that the degree function is an algebraic count of the solutions of $f(x)=b$(of zeros if $b=0$).

Example 1.6 If $id:\mathbb{R}^N\to\mathbb{R}^N$ is the identity map and $f=id|_{\Omega}$, then

$$d(f,\Omega,b)=\begin{cases}1 & \text{if } b\in\Omega\\ 0 & \text{if } b\notin\overline{\Omega}\end{cases}.$$

If $A:\mathbb{R}^N\to\mathbb{R}^N$ is linear, invertible and $f=A|_{\Omega}$, then we have

$$d(f,\Omega,b)=\begin{cases}\operatorname{sgn}(\det A) & \text{if } b\in A(\Omega)\\ 0 & \text{if } b\notin A(\overline{\Omega})\end{cases}.$$

Moreover, $sgn(\det A)=(-1)^{\vartheta}$ with $\vartheta=$the sum of the algebraic multiplicities of the negative eigenvalues of A.

Next we present an alternative definition of the degree in which the sum is replaced by a suitable integral. In what follows $\{\xi_\varepsilon\}_{\varepsilon>0}$ is a mollifier, that is, $\xi_\varepsilon\in C_c^\infty(\mathbb{R}^N)$,supp$\xi_\varepsilon=B_\varepsilon=\{z\in\mathbb{R}^N:|z|<\varepsilon\}$, $\xi_\varepsilon\geq 0$ and $\int_{\mathbb{R}^N}\xi_\varepsilon(z)dz=1$ (see Definition I-4.60).

Proposition 1.7 *If $f\in C^1(\Omega,\mathbb{R}^N)\cap C(\overline{\Omega},\mathbb{R}^N)$ and $b\notin f(S_f)\cup f(\partial\Omega)$, then there exists $\varepsilon_0>0$ such that for all $\varepsilon\in(0,\varepsilon_0)$, we have*

$$d_B(f,\Omega,b)=\int_{\Omega}\xi_\varepsilon(f(z)-b)J_f(z)dz.$$

Proof If $f^{-1}(b)=\emptyset$, then with $\varepsilon_0=d(b,f(\overline{\Omega}))$, we have

$$\xi_\varepsilon(f(z)-b)=0 \text{ for all } z\in\Omega,\ \text{all } 0<\varepsilon<\varepsilon_0$$

and so the assertion of the proposition is trivially true.

Next let $f^{-1}(b)=\{z_k\}_{k=1}^n$. Then for each $k\in\{1,\cdots,n\}$, we have $J_f(z_k)\neq 0$. Invoking the inverse function theorem, we can find $U_k\in\mathcal{N}(z_k)$ and $V_k\in\mathcal{N}(b)$ such that $U_k\cap U_m=\emptyset$ for all $k,m\in\{1,\cdots,n\}$ $k\neq m$ and

$$f|_{U_k}:U_k\to V_k \text{ is a homeomorphism onto.}$$

We can always assume that $J_f|_{U_k}$ has fixed sign (by shrinking the neighborhood U_k if necessary). Let $\varepsilon_0 > 0$ be such that

$$B_{\varepsilon_0}(b) \subseteq \bigcap_{k=1}^{n} V_k \quad (B_{\varepsilon_0}(b) = \{z \in \mathbb{R}^N : |z - b| < \varepsilon_0\}).$$

Let $A_k = f^{-1}(B_\varepsilon(b)) \cap U_k$, $k = 1, \cdots, n$. Then the sets A_k are open, mutually disjoint and $J_f|_{U_k}$ has fixed sign. We see that

$$0 < \varepsilon < \varepsilon_0 \Rightarrow \xi_\varepsilon(f(z) - b) = 0 \text{ on } \mathbb{R}^N \setminus \bigcup_{k=1}^{n} A_k). \tag{1.6}$$

Therefore, if $0 < \varepsilon < \varepsilon_0$, we have

$$\begin{aligned}
\int_\Omega \xi_\varepsilon(f(z) - b) J_f(z) dz &= \sum_{k=1}^{n} \int_{A_k} \xi_\varepsilon(f(z) - b) J_f(z) dz \text{ (see (1.6))} \\
&= \sum_{k=1}^{n} sgn(J_f(z_k)) \int_{A_k} \xi_\varepsilon(f(z) - b) |J_f(z)| dz \\
&= \sum_{k=1}^{n} sgn(J_f(z_k)) \int_{B_\varepsilon(0)} \xi_\varepsilon(v) dv \\
&\quad \text{(by a change of variables)} \\
&= d_B(f, \Omega, b) \quad (\text{since } \int_{B_\varepsilon(0)} \xi_\varepsilon(v) dv = 1).
\end{aligned}$$

□

Using this formula, we can show the stability of the degree function as we perform small changes in the function f and the reference point b. Consider the Jacobian matrix $\{\frac{\partial f_j}{\partial z_i}(z)\}_{i,j=1}^N$ and let $c_{ij}(z)$ be the cofactor of the entry $\frac{\partial f_j}{\partial z_i}(z)$, that is $(-1)^{i+j}$ times the determinant of the $(N-1) \times (N-1)$ matrix resulting from the Jacobian after deleting the $i \overset{\text{th}}{=}$ −column and the $j \overset{\text{th}}{=}$ −row. Also, for notational simplicity we write $\partial_i = \frac{\partial}{\partial z_i}$, $i \in \{1, \cdots, N\}$ and $\partial_{ij} = \frac{\partial}{\partial z_j \partial z_i}$, $i, j \in \{1, \cdots, N\}$.

Lemma 1.8 *If* $h \in C^2(\Omega, \mathbb{R}^{N-1})$ *and* $\vartheta_i = \det(\partial_1 h, \cdots, \partial_{i-1} h, \partial_{i+1} h, \cdots, \partial_N h)$, *then we have* $\sum_{i=1}^{N} (-1)^i \partial_i \vartheta_i = 0$.

Proof Let $1 \le i \le N$. We define d_{ij}, $i, j \in \{1, \cdots, N\}$ as follows:

- $d_{ii} = 0$;
- If $j < i$, then $d_{ij} = \det(\partial_1 h, \cdots, \partial_{j-1} h, \partial_{ij} h, \partial_{j+1} h, \cdots, \partial_{i-1} h, \partial_{i+1} h, \cdots, \partial_N h)$;
- If $i < j$, then $d_{ij} = \det(\partial_1 h, \cdots, \partial_{i-1} h, \partial_{i+1} h, \cdots, \partial_{j-1} h, \partial_{ij} h, \partial_{j+1} h, \cdots, \partial_N h)$.

We see that

$$\partial_i \vartheta_i = \sum_{j=1}^{N} d_{ij}.$$

It follows that

$$\sum_{i=1}^{N} (-1)^i \partial_i \vartheta_i = \sum_{i,j=1}^{N} (-1)^i d_{ij} = \eta. \tag{1.7}$$

But since $h \in C^2(\Omega, \mathbb{R}^{N-1})$, we have that $\partial_{ij} h = \partial_{ji} h$. Recall that transposing two columns in a matrix, we multiply the determinant by (-1). Hence $d_{ij} = (-1)^{i+j-1} d_{ji}$. Then from (7), we have

$$\eta = \sum_{i,j=1}^{N} (-1)^{j-1} d_{ji} = -\sum_{i,j=1}^{N} (-1)^j d_{ji} = -\eta,$$
$$\Rightarrow \eta = 0.$$

□

Lemma 1.9 *If $f \in C^2(\Omega, \mathbb{R}^N)$, then for all $j \in \{1, \cdots, N\}$, we have* $\sum_{i=1}^{N} \partial_i c_{ij} = 0$.

Proof We know that

$$c_{ij} = (-1)^{i+j} \det(\partial_m f_k)_{m \neq i, k \neq j}.$$

Fix $j \in \{1, \cdots, N\}$ and let $h = \{f_1, \cdots, f_{j-1}, f_{j+1}, \cdots, f_N\}$(recall that $f = (f_k)_{k=1}^N$). We apply Lemma 1.8 to this function and conclude that

$$\sum_{i=1}^{N} \partial_i c_{ij} = 0.$$

□

Using this lemma, we can have the stability of the degree function with respect to the reference point b and then extend the definition of degree to singular values.

Proposition 1.10 *If* $f \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$, $b \notin f(\partial\Omega)$, $r_0 = d(b, f(\partial\Omega)) > 0$ *and* $b_1, b_2 \in B_{r_0}(b)$, $b_1, b_2 \notin f(S_f)$, *then* $d_B(f, \Omega, b_1) = d_B(f, \Omega, b_2)$.

Proof Evidently $b_1, b_2 \notin f(\partial\Omega)$. So $b_1, b_2 \notin f(S_f) \cup f(\partial\Omega)$ and so $d(f, \Omega, b_1), d(f, \Omega, b_2)$ are well-defined. Let $0 < \mu < \min\{r_0 - \mid b - b_1 \mid, r_0 - \mid b - b_2 \mid\}$. By Proposition 1.7, we can find $\varepsilon \in (0, \mu)$ such that

$$d(f, \Omega, b_k) = \int_\Omega \xi_\varepsilon(f(z) - b_k) J_f(z) dz, \ k = 1, 2.$$

For all $y \in \mathbb{R}^N$, we have

$$\begin{aligned} \xi_\varepsilon(y - b_2) - \xi_\varepsilon(y - b_1) &= \int_0^1 \frac{d}{dt} \xi_\varepsilon(y - b_1 + t(b_2 - b_1)) dt \\ &= \int_0^1 D\xi_\varepsilon(y - b_1 + t(b_2 - b_1)) dt (b_1 - b_2) \\ &= \operatorname{div} w(y), \end{aligned} \tag{1.8}$$

with $w(y) = \int_0^1 \xi_\varepsilon(y - b_1 + t(b_2 - b_1)) dt (b_1 - b_2)$(recall that $\operatorname{div} w(y) = \sum_{k=1}^N \frac{\partial w_k(y)}{\partial y_k}$).

If $y \in f(\partial\Omega)$, then

$$\begin{aligned} \mid y - (1-t)b_1 - tb_2 \mid &= \mid y - b + (1-t)(b - b_1) + t(b - b_2) \mid \\ &\geq r_0 - (1-t)(r_0 - \mu) - t(r_0 - \mu) \\ &= \mu > \varepsilon. \end{aligned}$$

Recall that $\operatorname{supp}\xi_\varepsilon \subseteq B_\varepsilon(0)$. So, it follows that $w(y) = 0$ for all $y \in f(\partial\Omega)$. For $i \in \{1, \cdots, N\}$, we introduce the function

$$u_i(z) = \begin{cases} \sum_{j=1}^N w_j(f(z)) c_{ij}(z) & \text{if } z \in \overline{\Omega} \\ 0 & \text{if } z \in \mathbb{R}^N \setminus \overline{\Omega} \end{cases}.$$

Note that $u_i \mid_{\partial\Omega} = 0$ and

$$\frac{\partial u_i}{\partial z_i}(z) = \sum_{j,k=1}^{N} \partial_k w_j(f(z))(\sum_{i=1}^{N} \partial_i f_k(z) c_{ij}(z)) + \sum_{j=1}^{N} w_j(f(z))(\sum_{i=1}^{N} \partial_i c_{ij}(z))$$

$$= \sum_{j=1}^{N} \partial_j w_j(f(z)) J_f(z)$$

with the last inequality being a consequence of Lemma 1.9 and of the fact that $\sum_{i=1}^{N} \partial_i f_k(z) c_{ij}(z) = \delta_{jk} J_k(z)$. We conclude that

$$\operatorname{div} u(z) = \sum_{j=1}^{N} \partial_j w_j(f(z)) J_f(z) = \operatorname{div} w(f(z)) J_f(z). \tag{1.9}$$

Therefore from (8), (9) and Proposition 1.7, we infer that

$$d(f, \Omega, b_2) - d(f, \Omega, b_1) = \int_{\Omega} \operatorname{div}(w(f(z)) J_f(z) dz$$

$$= \int_{\Omega} \operatorname{div} u(z) dz = 0$$

using the divergence theorem and recalling that $u\,|_{\partial\Omega} = 0$. □

Now we are ready to remove from Definition 1.4 the condition that $b \notin f(S_f)$ at the expense of requiring that $f \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$. Indeed, on account of Theorem 1.2 (Sard's Theorem), the singular values of $f(\cdot)$ form a Lebesgue-null set in the ball $B_{r_0}(b)$. So, in the ball, we can find regular values of $f(\cdot)$ and by Proposition 1.10, the Brouwer degree at all the points in the ball is the same. So, the next definition makes perfect sense.

Definition 1.11 Let $\Omega \subseteq \mathbb{R}^N$ be a bounded open set, $f \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ and $b \notin f(\partial\Omega)$. Then

$$d_B(f, \Omega, b) = d_B(f, \Omega, b')$$

where b' is any regular value of $f(\cdot)$ in $B_{r_0}(b)$, where $r_0 = d(b, f(\partial\Omega)) > 0$.

Now we are ready to make the final step and remove the hypothesis that $f \in C^2(\Omega, \mathbb{R}^N)$. So, we will have the definition for all $f \in C(\overline{\Omega}, \mathbb{R}^N)$. This will be done with the help of the following proposition.

Proposition 1.12 *If $f, g \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ and $b \notin f(\partial\Omega)$, then there exists $\varepsilon = \varepsilon(f, g, \Omega) > 0$ such that*

$$d_B(f+tg,\Omega,b)=d_B(f,\Omega,b)\ \textit{for all}\ 0\le|t|<\varepsilon.$$

Proof First assume that $b\notin f(\overline{\Omega})$(that is, $f^{-1}(b)=\emptyset$). Let $\widehat{r}=d(b,f(\overline{\Omega}))>0$ and consider $\varepsilon=\dfrac{1}{2\|g\|_\infty}\widehat{r}$. Then for $|t|\le\varepsilon$, we have

$$\begin{aligned}&d(b,(f+tg)(\overline{\Omega}))\ge\frac{\widehat{r}}{2}>0\\ \Rightarrow&b\notin(f+tg)(\overline{\Omega}),\\ \Rightarrow&d_B(f,\Omega,b)=d_B(f+tg,\Omega,b)=0.\end{aligned}$$

Next we assume that $b\notin f(S_f)$ and let $f^{-1}(b)=\{z_k\}_{k=1}^m\subseteq\Omega$. Then $J_f(z_k)\neq 0$ for $1\le k\le m$. We set

$$h(t,z)=f(z)+tg(z)-b.$$

Then for all $k\in\{1,\cdots,m\}$, we have

$$h(0,z_k)=0,\ \frac{\partial}{\partial z}h(0,z_k)=f'(z_k).$$

Since $f'(z_k)$ is invertible, by the implicit function theorem, we can find mutually disjoint neighborhoods $U_k\in\mathcal{N}(z_k)$, open neighborhoods $(-\varepsilon_k,\varepsilon_k)$ of 0 in $\mathbb{R}$ and continuous functions $\xi_k:(-\varepsilon_k,\varepsilon_k)\to U_k$ such that $(t,\xi_k(t))$ is the only solution in $(-\varepsilon_k,\varepsilon_k)\times U_k$ of the equation $h(t,x)=0$. By shrinking the neighborhoods U_k if necessary, we can have that $\mathrm{sgn}J_f(z_k)=\mathrm{sgn}J_{f+tg}(z_k)$. Set $\varepsilon=\min\limits_{1\le k\le m}\varepsilon_k>0$. Then the equality of the degrees is a consequence of Definition 1.4.

Finally assume that $b\in f(S_f)$. Let $r_0=d(b,f(\partial\Omega))>0$. Let $b'\in B_{r_0/3}(b)$ such that b' is a regular value of $f(\cdot)$. So, by the previous part of the proof, there exists $\varepsilon'>0$ such that for $0\le|t|<\varepsilon'$, we have

$$d_B(f+tg,\Omega,b')=d_B(f,\Omega,b')=d_B(f,\Omega,b).\tag{1.10}$$

Let $\varepsilon<\min\{\dfrac{r_0}{3\|g\|_\infty},\varepsilon'\}$. Then for $|t|<\varepsilon$, we have $b\notin(f+tg)(\partial\Omega)$ and $|b-b'|<\dfrac{1}{3}r_0\le\dfrac{1}{2}d((f+tg)(\partial\Omega))$. We have

$$\begin{aligned}&d(f+tg,\Omega,b')=d(f+tg,\Omega,b),\\ \Rightarrow&d(f,\Omega,b)=d(f+tg,\Omega,b)\ \text{(see (10))}.\end{aligned}$$

□

This proposition says the degree map is constant for all C^2-functions sufficiently close to a continuous function. This allows us to extend the definition of the degree to all continuous functions on $\overline{\Omega}$. Let $f \in C(\overline{\Omega}, \mathbb{R}^N)$ and suppose $b \notin f(\partial\Omega)$. We set $r_0 = d(b, f(\partial\Omega)) > 0$. We can find $g, \widehat{g} \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ such that

$$\|f - g\|_\infty < r_0 \text{ and } \|f - \widehat{g}\|_\infty < r_0.$$

Let $h(t, z) = g(z) + t(\widehat{g}(z) - g(z))$, $\widehat{d}(t) = d_B(h(t, \cdot), \Omega, b)$, $t \in [0, 1]$. Note that $h(t, \cdot) = h(t_0, \cdot) + (t - t_0)(\widehat{g} - g)$. According to Proposition 1.12, we have that $\widehat{d}(\cdot)$ is constant in a neighborhood of t_0 and by Bolzano's theorem $\widehat{d}(\cdot)$ is constant on $[0, 1]$, hence

$$d_B(g, \Omega, b) = d_B(\widehat{g}, \Omega, b).$$

This implies that the following definition is not ambiguous.

Definition 1.13 Let $\Omega \subseteq \mathbb{R}^N$ be a bounded open set, $f \in C(\overline{\Omega}, \mathbb{R}^N)$, $b \notin f(\partial B)$. Then the Brouwer degree of $f(\cdot)$ in Ω, with respect to b is given by

$$d_B(f, \Omega, b) = d_B(g, \Omega, b),$$

where $g \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ such that $\|f - g\|_\infty < \dfrac{1}{2} d(b, f(\partial\Omega))$.

An immediate consequence of this definition, is the following result.

Proposition 1.14 *If $f \in C(\overline{\Omega}, \mathbb{R}^N)$ and $b \notin f(\partial\Omega)$, then $d_B(f, \Omega, b) = d_B(f - b, \Omega, 0)$.*

In the next theorem, we have gathered the main properties of the degree.

Theorem 1.15 *If $f \in C(\overline{\Omega}, \mathbb{R}^N)$, then*

(a) *if $b \notin f(\partial\Omega)$, then there exists $\varepsilon > 0$ such that*
$\|f - \widehat{f}\|_\infty < \varepsilon \Rightarrow b \notin \widehat{f}(\partial\Omega)$, $d_B(f, \Omega, b) = d_B(\widehat{f}, \Omega, b)$;
(b) *$d(f, \Omega, \cdot)$ is constant on each connected component of $\mathbb{R}^N \setminus f(\partial\Omega)$;*
(c) *If $h \in C([0, 1] \times \overline{\Omega}, \mathbb{R}^N)$ and $b \notin h([0, 1] \times \partial\Omega)$, then $d_B(h(t, \cdot), \Omega, b) = d_B(h(0, \cdot), \Omega, b)$ for all $t \in [0, 1]$;*
(d) *If Ω_1, Ω_2 are disjoint bounded open sets, $f \in C(\overline{\Omega}_1 \cup \overline{\Omega}_2, \mathbb{R}^N)$ and $b \notin f(\partial\Omega_1) \cup f(\partial\Omega_2)$, then $d_B(f, \Omega_1 \cup \Omega_2, b) = d_B(f, \Omega_1, b) + d_B(f, \Omega_2, b)$;*
(e) *If $K \subseteq \Omega$ is compact and $b \notin f(K) \cup f(\partial\Omega)$, then $d_B(f, \Omega, b) = d_B(f, \Omega \setminus K, b)$.*

Proof

(a) Let $r_0 = d(b, f(\partial\Omega)) > 0$ and set

$$B = \{\widehat{f} \in C(\overline{\Omega}, \mathbb{R}^N) : \|f - \widehat{f}\|_\infty < \frac{1}{4} r_0\}.$$

If $\widehat{f} \in B$, then $d(b, \widehat{f}(\partial\Omega)) \geq \frac{3}{4}r_0 > 0$ and so $b \notin \widehat{f}(\partial\Omega)$, hence the degree $d_B(\widehat{f}, \Omega, b)$ is well-defined. Let $g \in C(\overline{\Omega}, \mathbb{R}^N)$ such that $\|f - g\|_\infty < \frac{r_0}{8}$. Then we have

$$\|\widehat{f} - g\|_\infty < \frac{3}{8}r_0 < \frac{1}{2}d(b, \widehat{f}(\partial\Omega)).$$

So, by Definition 1.13, we have

$$d_B(\widehat{f}, \Omega, b) = d(g, \Omega, b) = d(f, \Omega, b).$$

(b) Let $\varepsilon \in (0, \frac{1}{4}r_0)$ with $r_0 = d(b, f(\partial\Omega))$ and $b' \in \mathbb{R}^N$. We introduce

$$f_1(z) = f(z) - b \text{ and } f_2(z) = f(z) - b' \text{ for all } z \in \overline{\Omega}.$$

If $|b - b'| < \varepsilon$, then $\|f_1 - f_2\|_\infty < \varepsilon$ and so by part (a) and Proposition 1.14, we have

$$d_B(f, \Omega, b) = d_B(f_1, \Omega, 0) = d_B(f_2, \Omega, 0) = d_B(f, \Omega, b).$$

(c) Let $\varepsilon = \frac{1}{4}r_0$, with $r_0 = d(b, h([0, 1] \times \partial\Omega) > 0$. The function $h(\cdot, \cdot)$ is uniformly continuous on $[0, 1] \times \overline{\Omega}$. So, we can find $\delta > 0$ such that

$$|t_1 - t_2| < \delta \Rightarrow |h(t_1, z) - h(t_2, z)| < \varepsilon \text{ for all } z \in \overline{\Omega},$$
$$\Rightarrow d_B(h(t_1, \cdot), \Omega, b) = d_B(h(t_2, \cdot), \Omega, b) \text{ (see } (a)).$$

Since $[0, 1]$ is compact, we can cover it by a finite number of intervals of length $\delta > 0$. Hence, we conclude that

$$d_B(h(t, \cdot), \Omega, b) = d_B(h(0, \cdot), \Omega, b) \text{ for all } t \in [0, 1].$$

(d) If $f \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$, then this property follows from Definition 1.11. For the general case, just approximate f by a C^2-function.

(e) Again choose $\widehat{f} \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ such that $d_B(f, \Omega, b) = d_B(\widehat{f}, \Omega, b)$ and $b \notin \widehat{f}(\partial\Omega)$. We choose $b' \in \mathbb{R}^N$ a regular value of $\widehat{f}(\cdot)$ very close to b such that $b' \notin f(K)$ and belonging to the same connected component as b of $\mathbb{R}^N \setminus \widehat{f}(\partial\Omega)$ and of $\mathbb{R}^N \setminus \widehat{f}(\partial(\Omega \setminus K))$. The result follows from Definition 1.11.

□

Properties (a) and (b) are known as "stability properties" of the degree map (stability with respect to the function f (part (a)) and stability with respect to the

reference point b (part (b)). Property (c) is known as "homotopy invariance" and arguably is the most important property of the degree. It allows one to transfer the computation of the degree to that of the degree of another function for which the situation is much simpler. Property (d) is known as "additivity property", while property (e) is known as "excision property".

Next we present some important consequences of the above properties. The first result essentially says that the degree depends uniquely on the boundary values.

Proposition 1.16 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded open set, $f, g \in C(\overline{\Omega}, \mathbb{R}^N)$, $f\mid_{\partial\Omega} = g\mid_{\partial\Omega}$ and $b \notin f(\partial\Omega)$, then $d_B(f, \Omega, b) = d_B(g, \Omega, b)$.*

Proof Consider the homotopy

$$h(t, z) = tf(z) + (1 - t)g(z).$$

Evidently $b \notin h([0, 1] \times \partial\Omega)$ and so by Theorem 1.15(c) (the homotopy invariance property), we have

$$\begin{aligned} &d_B(h(1, \cdot), \Omega, b) = d_B(h(0, \cdot), \Omega, b), \\ \Rightarrow &d_B(f, \Omega, b) = d_B(g, \Omega, b). \end{aligned}$$

□

Proposition 1.17 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded, open set, $f \in C(\overline{\Omega}, \mathbb{R}^N)$, $b \notin f(\partial\Omega)$ and $d_B(f, \Omega, b) \neq 0$, then there exists $u \in \Omega$ such that $f(u) = b$.*

Proof It suffices to show that if $f^{-1}(b) \neq \emptyset$, then $d_B(f, \Omega, b) = 0$. From Theorem 1.15(a), it suffices to show this for $f \in C^2(\Omega) \cap C(\overline{\Omega})$. Let $0 < \varepsilon < d(b, f(\overline{\Omega}))$ and consider a mollifier $\xi_\varepsilon(\cdot)$ such that $\text{supp}\xi_\varepsilon \subseteq [0, \varepsilon]$. According to Proposition 1.7, we have $d_B(f, \Omega, b) = 0$. □

Corollary 1.18 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded open set, $b \in \mathbb{R}^N$ and*

$$f(x) = \begin{cases} 1 \text{ if } b \in \Omega \\ 0 \text{ if } b \in \mathbb{R}^N \setminus \overline{\Omega} \end{cases}.$$

then

$$d_B(-f, \Omega, b) = \begin{cases} (-1)^N & \text{if } b \in \Omega \\ 0 & \text{if } b \in \mathbb{R}^N \setminus \overline{\Omega} \end{cases}.$$

Proposition 1.19 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded, open set, $f \in C(\overline{\Omega}, \mathbb{R}^N)$, $b \notin f(\partial\Omega)$ and $d_B(f, \Omega, b) \neq 0$, then $f(\Omega)$ is a neighborhood of b.*

Proof Let U_b be the connected component of $\mathbb{R}^N \setminus f(\partial\Omega)$, which contains b. Then for all $a \in U_b$, we have $d_B(f, \Omega, a) \neq 0$ and so $U_b \subseteq f(\Omega)$ (see Proposition 1.17). Since U_b is open, we conclude that $f(\Omega)$ is a neighborhood of b. □

Corollary 1.20 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded, open set, $f \in C(\overline{\Omega}, \mathbb{R}^N)$, $b \notin f(\partial\Omega)$ and $f(\Omega)$ is contained in a proper subspace of $\mathbb{R}^N$, then $d_B(f, \Omega, b) = 0$.*

Another consequence of the homotopy invariance property, similar to Proposition 1.16, is the following result.

Proposition 1.21 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded, open set, $f, g \in C(\overline{\Omega}, \mathbb{R}^N)$ and $h \in C([0, 1] \times \partial\Omega, \mathbb{R}^N)$ satisfies*

$$h(0, \cdot) = f\,|_{\partial\Omega},\ h(1, \cdot) = g\,|_{\partial\Omega}$$

and $b \notin h([0, 1], \partial\Omega)$, then $d_B(f, \Omega, b) = d_B(g, \Omega, b)$.

Proof By the Tietze extension theorem, we can find $\widehat{h} \in C([0, 1] \times \overline{\Omega}, \mathbb{R}^N)$ such that $\widehat{h}\,|_{[0,1]\times\partial\Omega} = h$. Then the homotopy invariance property of the degree implies that $d_B(f, \Omega, b) = d_B(g, \Omega, b)$. □

Proposition 1.22 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded open set, $f \in C(\overline{\Omega}, \mathbb{R}^N)$ and $b \notin f(\partial\Omega)$, then $d_B(f, \Omega, b) \in \mathbb{Z}$ (that is, the degree is integer-valued).*

Proof We know that we can find $f \in C^1(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ such that

$$d_B(f, \Omega, b) = d_B(\widehat{f}, \Omega, \mathbb{R}^N). \tag{1.11}$$

Recall $\widehat{f}(\partial\Omega) \subseteq \mathbb{R}^N$ is compact and $\widehat{f}(S_{\widehat{f}})$ is Lebesgue-null (see Theorem 1.2). Then $\mathbb{R}^N \setminus (\widehat{f}(\partial\Omega) \cup \widehat{f}(S_{\widehat{f}}))$ is dense in the open set $\mathbb{R}^N \setminus \widehat{f}(\partial\Omega)$. So, we can find $b' \in \mathbb{R}^N \setminus (\widehat{f}(\partial\Omega) \cup \widehat{f}(S_{\widehat{f}}))$ such that b, b' are in the same connected component of $\mathbb{R}^N \setminus f(\partial\Omega)$. From Theorem 1.15($b$), we know that the degree function is constant on each connected component of $\mathbb{R}^N \setminus \widehat{f}(\partial\Omega)$. Hence from (1.11), we infer that

$$d_B(f, \Omega, b) \in \mathbb{Z}.$$

□

Next we will present some important applications of the Brouwer degree. We start with the so-called "Hedgehog Theorem", also known as the "Hairy Ball Theorem".

Theorem 1.23 *If N is odd, $\Omega \subseteq \mathbb{R}^N$ is a bounded, open set, $0 \in \Omega$, $f \in C(\overline{\Omega}, \mathbb{R}^N)$ and $0 \notin f(\partial\Omega)$, then there exist $u \in \partial\Omega$, $\lambda \neq 0$ such that $f(u) = \lambda u$.*

Proof We consider the following two homotopies

$$h(t, u) = (1 - t) f(u) + tu,$$
$$\widehat{h}(t, u) = (1 - t) f(u) - tu, \text{ for all } t \in [0, 1], u \in \overline{\Omega}.$$

Arguing by contradiction, suppose that we can not find $u \in \partial\Omega$ and $\lambda \neq 0$ such that $f(u) = \lambda u$. Then for all $u \in \partial\Omega$, we have

$$h(t, u) \neq 0, \widehat{h}(t, u) \neq 0 \text{ for all } t \in (0, 1], \text{ all } u \in \partial\Omega.$$

Also, by hypothesis $h(0, u) \neq 0, \widehat{h}(0, u) \neq 0$ for all $u \in \partial\Omega$. Therefore we can use the homotopy invariance property and have

$$d_B(f, \Omega, 0) = d_B(id, \Omega, 0) = 1, \ d_B(f, \Omega, 0) = d_B(-id, \Omega, 0) = (-1)^N.$$

Hence, $1 = (-1)^N = -1$ (since N is odd), a contradiction. So, the assertion of the theorem holds. □

Remark 1.24 So, according to the above theorem, if N is odd and $S^{N-1} = \{u \in \mathbb{R}^N : |u| = 1\}$, then there is no continuous map $f : S^{N-1} \to \mathbb{R}^N$ such $f(u) \neq 0$ for all $u \in S^{N-1}$ and $(f(u), u)_{\mathbb{R}^N} = 0$. So, if $N = 3$ and you have a hairy ball, you can not comb it.

We recall the following basic notion from topology.

Definition 1.25 Let X be a Hausdorff topological space and $C \subseteq X$. We say that C is a "retract" of X, if there is a continuous map $r : X \to C$ such that $r\,|_C = id_C$ (that is, the identity map on C admits a continuous extension on all of X). The map $r(\cdot)$ is known as a "retraction" map for C.

Remark 1.26 Evidently a retract is always closed. From Theorem I-7.4, we know that every closed, convex set of a normed space is a retract. The notion of retract is a topological invariant, that is, it is preserved by homeomorphisms. A classical example of a retract is $S^{N-1} = \{u \in \mathbb{R}^N : |u| = 1\}$ in $\overline{B}_1^N \setminus \{0\}$. In this case the retraction is the map $r(u) = \dfrac{u}{|u|}$ for all $u \in \overline{B}_1^N \setminus \{0\}$. However, for $\overline{B}_1^N$ is a different story ($\overline{B}_1^N = \{u \in \mathbb{R}^N : |u| \leq 1\}$).

Proposition 1.27 *S^{N-1} is not a retract of $\overline{B}_1^N$.*

Proof Arguing by contradiction, suppose that exists a retraction $r(\cdot)$ of $\overline{B}_1^N$ onto S^{N-1}. Then by Proposition 1.16 we have

$$d_B(r, B_1^N, 0) = d_B(id, B_1^N, 0) = 1.$$

So, by Proposition 1.17 we can find $u \in B_1^N$ such that $r(u) = 0$, a contradiction since $r : \overline{B}_1^N \to S^{N-1}$. □

From elementary analysis we know that a continuous function $f : [-1, 1] \to [-1, 1]$ admits a fixed point, that is, there exists $u \in [-1, 1]$ such that $f(u) = u$. The proof is an easy application of Bolzano's theorem and can not be extended to

higher dimensions. In that setting the result is the so-called "Brouwer Fixed Point Theorem".

Theorem 1.28 *If $f : \overline{B}_1^N \to \overline{B}_1^N (N \geq 2)$ is continuous, then there exists $\widehat{u} \in \overline{B}_1^N$ such that $f(\widehat{u}) = \widehat{u}$.*

Proof We assume that $u - f(u) \neq 0$ for all $u \in S^{N-1}$ or otherwise we already have a fixed point on the boundary of the unit ball. Consider the homotopy

$$h(t, u) = u - tf(u) \text{ for all } t \in [0, 1], \text{ all } u \in \overline{B}_1^N.$$

Note that $0 \notin h(1, S^{N-1})$. If there exist $t \in [0, 1)$ and $u \in S^{N-1}$ such that $h(t, u) = 0$, then $t|f(u)| = 1$, a contradiction since $0 \leq t < 1$, $|f(u)| \leq 1$. Then by the homotopy invariance property of the degree, we have

$$d_B(h(1, \cdot), B_1^N, 0) = d_B(h(0, \cdot), B_1^N, 0) = d_B(id, B_1^N, 0) = 1.$$

So, there exist $\widehat{u} \in B_1^N$ such that $h(1, \widehat{u}) = 0$, hence $\widehat{u} = f(\widehat{u})$. □

We can extend this theorem as follows.

Theorem 1.29 *If $C \subseteq \mathbb{R}^N$ is compact, convex and $f : C \to C$ is continuous, then there exists $\widehat{u} \in C$ such that $f(\widehat{u}) = \widehat{u}$.*

Proof Let $\varrho > 0$ be big so that $C \subseteq \overline{B}_\varrho = \{u \in \mathbb{R}^N : |u| \leq \varrho\}$. Consider the metric projection map $p_C : \mathbb{R}^N \to C$ and let $\widehat{f} = f \circ p_C$. Then $\widehat{f} : \overline{B}_\varrho^N \to \overline{B}_\varrho^N$ and is continuous (see Proposition I-3.176). By Theorem 1.28, there exists $\widehat{u} \in \overline{B}_1^N$ such that $\widehat{f}(\widehat{u}) = \widehat{u} \Rightarrow f(p_C(\widehat{u})) = \widehat{u} \in C$ and so $f(\widehat{u}) = \widehat{u}$. □

As an application of the Brouwer fixed point theorem, let us prove the so-called "Perron-Frobenius Theorem".

Theorem 1.30 *If $A = (a_{ij})_{i,j=1}^N$ is an $N \times N$-matrix with $a_{ij} \geq 0$ for all $1 \leq i, j \leq N$, then A has a nonnegative eigenvalue with a corresponding eigenvector which belongs in $\mathbb{R}_+^N$.*

Proof Let $C = \{u = (u_k)_{k=1}^N \in \mathbb{R}^N : u_k \geq 0, k \in \{1, \cdots, N\}, \sum_{k=1}^N u_k = 1\}$. This is a compact, convex set in $\mathbb{R}^N$. If we can find $u_0 \in C$ such that $A(u_0) = 0$, then $\lambda = 0$ is an eigenvalue of A with $u_0 \in \mathbb{R}_+^N$ an eigenvector and so we are done. So, suppose that $Au \neq 0$ for all $u \in C$. Then $\sum_{k=1}^N (Ax)_k$ has a strictly positive minimum in C. Let $f : C \to C$ be defined by

$$f_k(u) = \frac{(Au)_k}{\sum_{i=1}^{N}(Au)_i} \text{ with } k = 1, \cdots, N.$$

Using Theorem 1.29 we see that $f(\cdot)$ has a fixed point $u_0 \in K$. We have $f(u_0) = u_0 \Rightarrow Au_0 = \lambda u_0$ with $\lambda = \sum_{k=1}^{N}(Au_0)_k$. □

When we want to find a solution in Ω of $f(u) = b$, according to Proposition 1.19, it suffices to show that $d_B(f, \Omega, b) \neq 0$. Borsuk's Theorem is very helpful in this respect. To state and prove Borsuk's Theorem we need some preparation.

Lemma 1.31 *If $K \subseteq \mathbb{R}^N$ is compact $f \in C(K, \mathbb{R}^n)$ with $n > N$, $0 \notin f(K)$ and $\mathcal{D}$ is a closed cube containing K, then there exists $\widehat{f} \in C(\mathcal{D}, \mathbb{R}^n)$ such that $\widehat{f} \mid_K = f$ and $0 \notin \widehat{f}(\mathcal{D})$.*

Proof Let $m = \min_{u \in K} |f(u)| > 0$ and $\varepsilon \in (0, m/2)$. Consider $g \in C^1(\mathcal{D}, \mathbb{R}^n)$ such that

$$\|f - g\|_{K,\infty} < \varepsilon/2.$$

If $0 \notin g(\mathcal{D})$, then we keep g. If $0 \in g(\mathcal{D})$, then let $k \in C^1(\mathcal{D} \times \mathbb{R}^{n-N}, \mathbb{R}^n)$ be defined by

$$k(u, y) = g(u) \text{ for all } u \in \mathcal{D}, y \in \mathbb{R}^{n-N}.$$

We see that $J_k(u, y) = 0$ for all $(u, y) \in \mathcal{D} \times \mathbb{R}^{n-N}$. So, by Theorem 1.2 (Sard's Theorem), $g(\mathcal{D})$ is Lebesgue-null. Hence, we can find $e \notin g(\mathcal{D})$ with $|e| < \varepsilon/2$. Then we consider $g - e$.

Therefore combining the two cases, we conclude that there exists $g \in C^1(\mathcal{D}, \mathbb{R}^n)$ such that $\|f - g\|_{K,\infty} < \varepsilon$ and $0 \notin g(\mathcal{D})$.

Let $\vartheta : \mathbb{R}_+ \to \mathbb{R}_+$ be defined by

$$\vartheta(t) = \begin{cases} \dfrac{2t}{m} & \text{if } t \leq \dfrac{m}{2} \\ 1 & \text{if } \dfrac{m}{2} < t \end{cases}.$$

We set $f_0(z) = \dfrac{g(z)}{\vartheta(|g(z)|)}$ for all $z \in \mathcal{D}$. We have

$$|f_0(z)| \geq \frac{m}{2} \text{ for all } z \in \mathcal{D}, 0 \notin f_0(\mathcal{D}).$$

If $z \in K$, then $|g(z)| \geq |f(z)| - |f(z) - g(z)| \geq m - \varepsilon \geq \dfrac{m}{2}$ and so $\vartheta(|g(z)|) = 1$ which means that $f_0 \mid_K = f \mid_K$.

Finally let $h = f_0 - f$ on K. Using the Tietze extension theorem, we can find $\widehat{h} \in C(\mathcal{D}, \mathbb{R}^n)$ such that $\widehat{h}\,|_K = h$ and $\|\widehat{h}\|_{\mathcal{D},\infty} < \varepsilon$ (since $\|h\|_{K,\infty} < \varepsilon$). We set $\widehat{f} = f_0 - \widehat{h}$ and we have a continuous extension of f on $\mathcal{D}$ and

$$|\widehat{f}(z)| \geq |f_0(z)| - |\widehat{h}(z)| \geq \frac{m}{2} - \varepsilon > 0 \text{ for all } z \in \mathcal{D},$$

$$\Rightarrow 0 \notin \widehat{f}(\mathcal{D}).$$

□

Lemma 1.32 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric, $0 \notin \Omega$, $f \in C(\partial\Omega, \mathbb{R}^n)$ with $n > N$, f is odd and $0 \notin f(\partial\Omega)$, then there exists $F \in C(\overline{\Omega}, \mathbb{R}^n)$ extension of f which is odd and $0 \notin F(\overline{\Omega})$.*

Proof We use induction on the dimension of $\mathbb{R}^N$.

So, first let $N = 1$. Then for $0 < a < c$, we have

$$\overline{\Omega} \subseteq [-c, -a] \cup [a, c].$$

Using Lemma 1.31 we can extend f to a continuous function f_1 on $[a, c]$ such that $0 \notin f([a, c])$. We set

$$F(z) = \begin{cases} -f_1(z) \text{ if } z \in [-c, -a] \\ f_1(z) \quad \text{if } z \in [a, c] \end{cases}.$$

Then this function has the desired properties.

Continuing the induction, suppose that the result holds for all dimensions $\leq N - 1$. We will show that the lemma is true when $\Omega \subseteq \mathbb{R}^N$. We split $\mathbb{R}^N$ into $\mathbb{R}^{N-1}$ (where for $z = (z_k)_{k=1}^N$, $z_N = 0$), $\mathbb{R}^N_+$ (where $z_N > 0$) and $\mathbb{R}^N_-$ (where $z_N < 0$). We set $\Omega_+ = \Omega \cap \mathbb{R}^N_+$, $\Omega_- = \Omega \cap \mathbb{R}^N_-$. We have $\partial(\Omega \cap \mathbb{R}^{N-1}) = \partial\Omega \cap \mathbb{R}^{N-1}$ and by the induction hypothesis, we can extend f to f^* on $\overline{\Omega} \cap \mathbb{R}^{N-1}$ with f^* odd and $0 \notin f(\overline{\Omega} \cap \mathbb{R}^{N-1})$ (nonvanishing). Let $\mathcal{D}$ be a cube in $\mathbb{R}^N_+$ containing $\overline{\Omega}_+$ and $\Omega \cap \mathbb{R}^{N-1}$. We introduce the continuous function

$$f_1(z) = \begin{cases} f(z) \quad \text{if } z \in \partial\Omega \cap \overline{\Omega}_+ \\ f^*(z) \text{ if } z \in \overline{\Omega} \cap \mathbb{R}^{N-1}. \end{cases}$$

This can be extend to a continuous function $\widehat{f}$ on D such that $0 \notin \widehat{f}(D)$. Finally let

$$F(z) = \begin{cases} \widehat{f}(z) \quad \text{if } z \in \Omega_+ \\ -\widehat{f}(-z) \text{ if } z \in \Omega_-. \end{cases}$$

This is the desired extension of $f(\cdot)$. □

Lemma 1.33 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric, $0 \notin \overline{\Omega}$, $f \in C(\partial\Omega, \mathbb{R}^N)$ is odd and $0 \notin f(\partial\Omega)$, then there exists $\widehat{f} \in C(\overline{\Omega}, \mathbb{R}^N)$ such that $\widehat{f}|_{\partial\Omega} = f$, $\widehat{f}$ is odd and $0 \notin \widehat{f}(\overline{\Omega} \cap \mathbb{R}^{N-1})$.*

Proof We consider $f_0 = f\,|_{\partial\Omega \cap \mathbb{R}^{N-1}}$. This is continuous, odd and nonvanishing. Applying Lemma 1.32 we can find $\widehat{F}_0 \in C(\overline{\Omega} \cap \mathbb{R}^{N-1}, \mathbb{R}^N)$ odd, nonvanishing and $\widehat{F}_0\,|_{\partial\Omega \cap \mathbb{R}^{N-1}} = f_0$. Let

$$f_1(z) = \begin{cases} f(z) & \text{if } z \in \partial\Omega \cap \overline{\Omega}_+ \\ \widehat{F}_0(z) & \text{if } z \in \overline{\Omega} \cap \mathbb{R}^{N-1}. \end{cases}$$

Using the Tietze extension theorem, we extend $f_1(\cdot)$ to a function on $\overline{\Omega}_+$ and finally, we consider its odd extension on all of $\overline{\Omega}$. □

Lemma 1.34 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric, $0 \notin \overline{\Omega}$, $f \in C(\partial\Omega, \mathbb{R}^N)$ is odd and $0 \notin f(\partial\Omega)$, then $d_B(f, \Omega, 0)$ is even.*

Proof Use Lemma 1.33 to produce $\widehat{f} \in C(\overline{\Omega}, \mathbb{R}^N)$ such that

$$\widehat{f}\,|_{\partial\Omega} = f,\ \widehat{f}(\cdot) \text{ is odd}, 0 \notin \widehat{f}(\overline{\Omega}) \text{ (nonvanishing)}.$$

Then by Proposition 1.16 we have

$$d_B(f, \Omega, 0) = d_B(\widehat{f}, \Omega, 0). \tag{1.12}$$

Choose $g \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$ such that $\|\widehat{f} - g\|_\infty < \varepsilon$ and set

$$\widehat{g}(z) = \frac{1}{2}[g(z) - g(-z)] \text{ for all } z \in \overline{\Omega}.$$

Evidently $\widehat{g} \in C^2(\Omega, \mathbb{R}^N) \cap C(\overline{\Omega}, \mathbb{R}^N)$, $\widehat{g}$ is odd and $\|\widehat{f} - \widehat{g}\|_\infty < \varepsilon$ choosing $\varepsilon > 0$ small we will have

$$d_B(\widehat{f}, \Omega, 0) \text{ and } 0 \notin \widehat{g}(\overline{\Omega} \cap \mathbb{R}^{N-1}). \tag{1.13}$$

Then the excision and additivity properties of the degree imply that

$$\begin{aligned} d_B(\widehat{g}, \Omega, 0) &= d_B(\widehat{g}, \Omega \setminus (\overline{\Omega} \cap \mathbb{R}^{N-1}), 0) \\ &= d_B(\widehat{g}, \Omega_+, 0) + d_B(\widehat{g}, \Omega_-, 0). \end{aligned} \tag{1.14}$$

Let $b \in \mathbb{R}^N$ be a regular value of $\widehat{g}$ such that

$$d_B(\widehat{g}, \Omega_+, 0) = d_B(\widehat{g}, \Omega_+, b) = \sum_{z \in \widehat{g}^{-1}(b)} \operatorname{sgn} J_{\widehat{g}}(z), \tag{1.15}$$

$$d_B(\widehat{g}, \Omega_-, 0) = d_B(\widehat{g}, \Omega_-, -b) = \sum_{z \in \widehat{g}^{-1}(-b)} \operatorname{sgn} J_{\widehat{g}}(z). \tag{1.16}$$

Since $\widehat{g}(\cdot)$ is odd $\widehat{g}^{-1}(-b) = -\widehat{g}^{-1}(b)$ and $J_{\widehat{g}}(\cdot)$ is even. So, from (1.15) and (1.16) it follows that

$$d_B(\widehat{g}, \Omega_+, 0) = d_B(\widehat{g}, \Omega_-, 0)$$

$$\Rightarrow d_B(\widehat{f}, \Omega, 0) \text{ is even (see (1.13) and (1.14))},$$

$$\Rightarrow d_B(f, \Omega, 0) \text{ is even (see (1.12))}.$$

□

Now we are ready to state and prove "Borsuk's Theorem".

Theorem 1.35 *If $\Omega \in \mathbb{R}^N$ is bounded, open, symmetric set, $0 \in \Omega$, $f \in C(\overline{\Omega}, \mathbb{R}^N)$ is odd and $0 \notin f(\partial\Omega)$, then $d_B(f, \Omega, 0)$ is an odd integer.*

Proof By hypothesis $0 \in \Omega$. So, we can find $\varrho > 0$ such that $\overline{B}_\varrho(0) \subseteq \Omega$. Consider the function

$$f_0(z) = \begin{cases} f(z) & \text{if } z \in \partial\Omega \\ z & \text{if } z \in \overline{B}_\varrho(0). \end{cases} \tag{1.17}$$

Using the Tietze extension theorem, we can find $\widehat{f_0} \in C(\overline{\Omega}, \mathbb{R}^N)$ extension of f_0. We set $\widehat{f_1}(z) = \frac{1}{2}[\widehat{f_0}(z) - \widehat{f_0}(-z)]$ and we have a continuous, odd function on $\overline{\Omega}$ with $\widehat{f_1}|_{\partial\Omega} = f$. Hence we have

$$\begin{aligned} d_B(f, \Omega, 0) &= d_B(\widehat{f_1}, \Omega, 0) \text{ (see Proposition 1.16)} \\ &= d_B(\widehat{f_1}, \Omega \setminus \overline{B}_\varrho(0), 0) + d_B(\widehat{f_1}, B_\varrho(0), 0) \text{ (additivity property)} \\ &= d_B(\widehat{f_1}, \Omega \cdot \overline{B}_\varrho(0), 0) + 1 \text{ (see (1.17))}, \\ &\Rightarrow d_B(f, \Omega, 0) \text{ is odd (see Lemma 1.34)}. \end{aligned}$$

□

Let us see some consequences of Borsuk's Theorem.

Corollary 1.36 *If $\Omega \in \mathbb{R}^N$ is a bounded, open, symmetric set, $0 \in \Omega$, $f \in C(\overline{\Omega}, \mathbb{R}^N)$, $0 \notin f(\partial\Omega)$ and for all $z \in \partial\Omega$, we have*

$$\frac{f(z)}{|f(z)|} \neq \frac{f(-z)}{|f(-z)|}, \tag{1.18}$$

then $d_B(f, \Omega, 0)$ is an odd integer.

Proof Let $h(t,z) = f(z) - tf(-z)$ for all $t \in [0,1]$, $z \in \overline{\Omega}$. Evidently $h \in C([0,1] \times \overline{\Omega}, \mathbb{R}^N)$ and $0 \notin f([0,1] \times \partial\Omega)$. The homotopy invariance property implies that

$$d_B(f, \Omega, 0) = d_B(h(1,\cdot), \Omega, 0) = \text{odd integer (by Theorem 1.35)}.$$

□

Remark 1.37 If $f \in C(\overline{\Omega}, \mathbb{R}^N)$ is odd and $0 \notin f(\partial\Omega)$, then (1.18) is clearly true. Condition (1.18) means that the vectors $f(z)$ and $f(-z)$ $z \in \partial\Omega$, do not point in the same direction.

Corollary 1.38 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric set, $0 \in \Omega$ and $f \in C(\overline{\Omega}, \mathbb{R}^N)$ is odd, then there exist $(u, v) \in \overline{\Omega} \times \overline{\Omega}$ such that*

$$f(u) = 0 \text{ and } f(v) = v.$$

Proof If $0 \in f(\partial\Omega)$, then there exists $u \in \partial\Omega$ such that $f(u) = 0$.

If $0 \notin f(\partial\Omega)$, then we can apply Borsuk's Theorem (Theorem 1.35) and find $u \in \Omega$ such that $f(u) = 0$.

Let $g(z) = z - f(z)$, $z \in \overline{\Omega}$. Then $g \in C(\overline{\Omega}, \mathbb{R}^N)$ and g is odd. So, we can apply the first part of the corollary and find $v \in \overline{\Omega}$ such that $g(v) = 0 \Rightarrow f(v) = v$. □

Corollary 1.39 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric set, $0 \in \Omega$, $f \in C(\overline{\Omega}, \mathbb{R}^N)$, $0 \notin f(\partial\Omega)$ and $\lambda f(z) \neq (1-\lambda) f(-z)$ for all $\lambda \in [\frac{1}{2}, 1]$, all $z \in \partial\Omega$, then $0 \in inf f(\Omega)$.*

Proof Observe that the assertion of the corollary follows immediately if we show that $d_B(f, \Omega, 0) \neq 0$. Consider the function $g(z) = \frac{1}{2}[f(z) - f(-z)]$. Evidently $g \in C(\overline{\Omega}, \mathbb{R}^N)$, $g(\cdot)$ is odd and $0 \notin g(\partial\Omega)$. So, by Borsuk's Theorem (Theorem 1.35), we have $d_B(g, \Omega, 0) \neq 0$. Consider the homotopy

$$h(t,z) = tf(z) - (1-t) f(-z)\ t \in [\frac{1}{2}, 1], z \in \overline{\Omega}.$$

Then $h \in C([\frac{1}{2}, 1] \times \overline{\Omega}, \mathbb{R}^N)$, $0 \notin h([\frac{1}{2}, 1], \partial\Omega)$, $h(\frac{1}{2}, \cdot) = g$, $h(1,\cdot) = f$. Hence the homotopy invariance property of the degree implies

$$d_B(f, \Omega, 0) = d_B(g, \Omega, 0) \neq 0.$$

□

Corollary 1.40 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric set, $0 \in \Omega$, $f \in C(\partial\Omega, \mathbb{R}^N)$ is odd and $f(\partial\Omega)$ is contained in a proper subspace of $\mathbb{R}^N$, then there exists $u \in \partial\Omega$ such that $f(u) = 0$.*

Proof Suppose $0 \notin f(\partial\Omega)$. Using the Tietze extension theorem, we can find $\widehat{f} \in C(\overline{\Omega}, \mathbb{R}^{N-1})$ such that $\widehat{f}|_{\partial\Omega \times \mathbb{R}^{N-1}} = f$. Since for all $z \in \partial\Omega$, $f(z)$, $f(-z)$ point in opposite directions, from Corollary 1.36, we have that $\operatorname{int}\widehat{f}(\Omega) \neq \emptyset$, a contradiction. So, there exist $u \in \partial\Omega$ such that $f(u) = 0$. □

This corollary leads to the so-called "Borsuk-Ulam Theorem".

Theorem 1.41 *If $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric set, $0 \in \Omega$, $f \in C(\partial\Omega, \mathbb{R}^N)$ and $f(\partial\Omega)$ is contained in a proper subspace of $\mathbb{R}^N$, then there exists $u \in \partial\Omega$ such that $f(u) = f(-u)$.*

Proof Consider the function $g(z) = f(z) - f(-z)$, $z \in \partial\Omega$ and apply Corollary 1.40. □

Remark 1.42 A nice interpretation of this theorem is the following. Suppose that the earth is a perfect sphere (that is, it is represented by S^2) and the temperature and pressure vary continuously on it. Then according to the Borsuk-Ulam Theorem there exist antipodal points on the surface of the earth which have the same temperature and pressure.

Another consequence of the Corollary 1.40 is the following result.

Corollary 1.43 *If $n < N$, then there is no odd map $f \in C(S^N, S^n)$.*

The next result concerns coverings of $\partial\Omega$ and is known in the literature as the "Ljusternik-Schnirelmann-Borsuk Theorem".

Theorem 1.44 *If $\Omega \in \mathbb{R}^N$ is bounded, open, symmetric set, $0 \in \Omega$ and $\{E_k\}_{k=1}^n$ is a covering of $\partial\Omega$ by closed sets $E_k \subseteq \partial\Omega$ $k = 1, \ldots, n$ with $E_k \cap (-E_k) = \emptyset$ for all $k = 1, \ldots, N$, then $n \geq N + 1$.*

Proof We claim $\bigcap_{k=1}^{n} E_k = \emptyset$. Arguing by contradiction, suppose $u_0 \in \bigcap_{k=1}^{n} E_k$. Because $\bigcup_{k=1}^{n} (-E_k) = \partial\Omega$, we must have that $u_0 \in E_k \cap (-E_k)$ for some $k = 1, \ldots, n$, a contradiction.

Again arguing indirectly, suppose $n \leq N$. Let $f_k(z) = d(z, E_k)$ for $k = 1, \ldots, n$ and $z \in \partial\Omega$. Let $f = (f_k)_{k=1}^{n-1} \in C(\partial\Omega, \mathbb{R}^{n-1})$. Then by Theorem 1.41 we can find $u \in \partial\Omega$ such that $f(u) = f(-u)$. Let $k_0 \in \{1, \ldots, n\}$ be the smallest index such that $u \in E_{k_0}$. Then we have two possibilities

(a) $k_0 \leq n - 1$. We have $f_{k_0}(u) = f_{k_0}(-u) = 0$ and so $u \in E_{k_0} \cap (-E_{k_0})$, a contradiction.

(b) $k_0 = n$. Then $u \in \bigcup_{k=1}^{n-1} E_k$ and so $f_k(u) = f_k(-u) > 0$ for $k = 1, \ldots, n-1$, and so $u \notin \bigcup_{k=1}^{n-1}(-E_k)$. Both $\{E_k\}_{k=1}^n$ and $\{-E_k\}_{k=1}^n$ are coverings of $\partial\Omega$. So, it follows that $u \in E_n \cap (-E_n)$, a contradiction.

□

Corollary 1.45 *If $S^N = \{u \in \mathbb{R}^{N+1} : |u| = 1\}$ admits a cover $\{E_k\}_{k=1}^{N+1}$ by closed sets, then some E_k must contain antipodal points (that is, for some $k = 1, \ldots, N+1$, $E_k \cap (-E_k) \neq \emptyset$).*

The notion of "genus" permits us to distinguish two closed, symmetric sets not containing the origin. The notion is useful in critical point theory (see Chap. 3).

So, let X be a Banach space and consider the following collection of subsets of X.

$\mathcal{S} = \{C \subseteq X$: is nonempty, closed, symmetric (i.e., $C = -C$) and $0 \notin C\}$.

Definition 1.46 Given $C \in \mathcal{S}$, the "genus" $\gamma(C)$ is defined by

$$\gamma(C) = \inf\{n \in \mathbb{N} : \textit{there exists } \xi : C \to \mathbb{R}^n \textit{ continuous, odd and } 0 \notin \xi(C)\}.$$

We set $\gamma(\emptyset) = 0$ and if no such $n \in \mathbb{N}$ exists, then $\gamma(C) = \infty$.

Example 1.47

(a) Let $\Omega \subseteq \mathbb{R}^N$ is bounded, open, symmetric set with $0 \in \Omega$. If $C = \partial\Omega \in \mathcal{S}$, then $\gamma(C) = N$. Indeed, consider the identity map $id : C \to \mathbb{R}^N$. Then from Definition 1.46 we infer that $\gamma(C) \leq N$. On the other hand, from Theorem 1.41 (the Borsuk-Ulam Theorem), we see that it can not happen that $\gamma(C) < N$. Therefore $\gamma(C) = N$. In particular $\gamma(S^{N-1}) = N$.

(b) Let X be a Banach space, $u \in X \setminus \{0\}$ and let $C = \overline{B}_r(u) \cup \overline{B}_r(-u)$ with $0 \leq r \leq ||u||_X$.

Evidently $C \in \mathcal{S}$ and $\gamma(C) = 1$. To see this, consider the map $\xi : C \to \mathbb{R}$ defined by

$$\xi(x) = \begin{cases} 1, & \text{if } x \in \overline{B}_r(u) \\ -1, & \text{if } x \in \overline{B}_r(-u) \end{cases}$$

This is continuous odd and so by Definition 1.46, $\gamma(C) = 1$.

Remark 1.48 From Example 1.47(b) we see that if $C \in \mathcal{S}$ is disconnected, then $\gamma(C) = 1$. So, if $\gamma(C) \geq 2$, then $card\ C = +\infty$.

The next theorem lists the main properties of the genus.

Theorem 1.49 *If X is a Banach space and $C, D \in \mathcal{S}$, then:*

(a) *If there is a continuous odd map $f : C \to D$, then $\gamma(C) \leq \gamma(D)$;*

(b) $C \subseteq D \Rightarrow \gamma(C) \leq \gamma(D)$ *(monotonicity property);*
(c) $\gamma(C \cup D) \leq \gamma(C) + \gamma(D)$ *(subadditivity property);*
(d) *If* $\gamma(D) < \infty$, *then* $\gamma(\overline{C \setminus D}) \geq \gamma(C) - \gamma(D)$;
(e) *If* $\xi : C \to D$ *is an odd homeomorphism, then* $\xi(C) = \xi(D)$;
(f) *If* C *is compact, then* $\gamma(C) < \infty$;
(g) *If* C *is compact, then for* $\varepsilon > 0$ *small we have*

$$\gamma(\overline{C}_\varepsilon) = \gamma(C),$$

where $\overline{C}_\varepsilon = \{x \in X : d(x, C) \leq \varepsilon\}$.

Proof

(a) Assume that $\gamma(D) = \eta < \infty$ or otherwise the statement it is obvious. If $g : D \to \mathbb{R}^N \setminus \{0\}$ is continuous, odd, then $g \circ f : A \to \mathbb{R}^N \setminus \{0\}$ is continuous and so the assertion follows from Definition 1.46.
(b) This part follows from (a) with $f = id$.
(c) We assume that both $\gamma(C), \gamma(D)$ are finite or otherwise the inequatity is obvious. Let $\gamma(C) = m$ and $\gamma(D) = n$. Then there exist continuous odd functions $f_C : C \to \mathbb{R}^m \setminus \{0\}$ and $f_D : D \to \mathbb{R}^n \setminus \{0\}$. Using the Tietze extension theorem, we can find continuous extensions $\widehat{f_C}$ and $\widehat{f_D}$ of f_C and f_D respectively. We can assume that these extensions are odd (if this is not true, we replace $\widehat{f_C}(z)$ by $\frac{1}{2}[\widehat{f_C}(z) - \widehat{f_C}(-z)]$ and $\widehat{f_D}(z)$ by $\frac{1}{2}[\widehat{f_D}(z) - \widehat{f_D}(-z)]$. Let $\xi = (\widehat{f_C}|_{C \cup D}, \widehat{f_D}|_{C \cup D}) : C \cup D \to \mathbb{R}^{m+n}$. Then clearly ξ is continuous, odd and nonvanishing.) Hence $\gamma(C \cup D) \leq m + n$.
(d) We see that $\overline{C \setminus D} = \overline{C \cap D^c} \in \mathcal{S}$ and $C \subseteq D \cup \overline{C \setminus D}$. Hence from (b) we have $\gamma(C) - \gamma(D) \leq \gamma(\overline{C \setminus D})$.
(e) Using (a) we have $\gamma(C) \leq \gamma(D) \leq \gamma(C)$, hence $\gamma(C) = \gamma(D)$.
(f) Let $u \in C$ and $0 < r < ||u||$ (recall that since $C \in \mathcal{S}$, $u \neq 0$). We know that $\gamma(\overline{B}_r(u) \cup \overline{B}_r(-u)) = 1$ (see Example 1.47(b)). Since C is compact we can cover C by a finite number of such balls and so from (b) we conclude that $\gamma(C) < \infty$.
(g) Let $\gamma(C) = n \in \mathbb{N}$ (see (f)) and let $f : C \to \mathbb{R}^n \setminus \{0\}$ be continuous, odd. Let $\widehat{f}$ be the continuous extension of f on all of X. Since $C \in \mathcal{S}$ is compact, we have $|f(z)| \geq m > 0$ for all $z \in C$. Then, for $\varepsilon > 0$ small we have $\widehat{f}(z) \neq 0$ for all $z \in \overline{C}_\varepsilon$. Hence $\gamma(\overline{C}_\varepsilon) \leq n$. On the other hand from (b) we have $n \leq \gamma(\overline{C}_\varepsilon)$. Therefore $\gamma(\overline{C}_\varepsilon) = n = \gamma(C)$.

□

Using the genus $\gamma(\cdot)$ we can have the following extension of Theorem 1.41 (the Borsuk-Ulam Theorem).

Proposition 1.50 *If* $\Omega \subseteq \mathbb{R}^N$ *is bounded, open, symmetric set,* $0 \in \Omega$, $n < N$, $f \in C(\partial\Omega, \mathbb{R}^n)$ *and* $C = \{u \in \partial\Omega : f(u) = 0\}$, *then* $N - n \leq \gamma(C)$.

Proof Let $\varepsilon > 0$ be small such that $\gamma(\overline{C}_\varepsilon) = \gamma(C)$ (see Proposition 1.50(g)). For $\delta > 0$, let $D_\delta = \{u \in \partial\Omega : |f(u)| \leq \delta\}$. We will show that for $\delta > 0$ small we

have $D_\delta \subseteq \overline{C}_\varepsilon$. Arguing indirectly, suppose we can find $\delta_n \downarrow 0$ and $u_n \in D_{\delta_n}$, $u_n \notin \overline{C}_\varepsilon$. Then $\varepsilon \leq d(u_n, C)$. The set $\partial\Omega$ is compact and so by passing to a subsequence if necessary, we may assume that $u_n \to u \in \partial\Omega$. Then $\varepsilon \leq d(u, C)$ and $f(u) = 0$, that is, $u \in C$, a contradiction. Hence for $\delta > 0$ small we have $C \subseteq D_\delta \subseteq \overline{C}_\varepsilon$ and so $\gamma(D_\delta) = \gamma(C)$ (see Proposition 1.49(g)). For $\vartheta > 0$, let $E_\vartheta = \{u \in \partial\Omega : |f(u)| \geq \vartheta\}$. Let $r_1(v) = \frac{v}{|v|}$ for all $v \in \mathbb{R}^n$(the radial retraction). We have that

$$r_1 \circ f : E_\vartheta \to \mathbb{R}^n \setminus \{0\}$$

is continuous, odd and so $\gamma(E_\vartheta) \leq n$. Then by Proposition 1.49(d) we have

$$N - n \leq \gamma(\partial\Omega) - \gamma(E_\vartheta) \leq \gamma(\overline{\partial\Omega \cdot E_\vartheta}).$$

Note that $\overline{\partial\Omega \setminus E_\vartheta} = D_\vartheta$ and so take $\vartheta = \delta$ to finish the proof. □

Corollary 1.51 *If $\Omega \in \mathbb{R}^N$ is a bounded, open, symmetric, $0 \in \Omega$, $n < N$, $f \in C(\partial\Omega, \mathbb{R}^n)$ and $C = \{u \in \partial\Omega : f(u) = f(-u)\}$, then $N - n \leq \gamma(C)$.*

Proof Just apply Proposition 1.50 to the function $g(z) = f(z) - f(-z)$. □

Lemma 1.52 *There exists a closed cover $\{C_k\}_{k=1}^N$ of $S^{N-1} = \{u \in \mathbb{R}^N : |u| = 1\}$ such that $C_k = A_k \cup (-A_k)$, $A_k \cap (-A_k) = \varnothing$ for $k = 1, \ldots, N$.*

Proof We proceed by induction on the dimension N. If $N = 1$, then $S^0 = \{-1, 1\}$ and so $C_1 = \{1\} \cup \{-1\}$. If $N = 2$, $S^1 = C_1 \cup C_2$, with $C_1 = \{(x, y) \in \mathbb{R}^2 : |x| \geq \frac{1}{2}\}$, $C_2 = \{(x, y) \in \mathbb{R}^2 : |y| \geq \frac{1}{2}\}$. Assume that the result holds for $N > 2$. We have $S^{N-1} = \bigcup_{k=1}^{N} C'_k$ with $C'_k = A'_k \cup (-A'_k)$, $A'_k \cap (-A'_k) = \varnothing$ $k = 1, \ldots, N$. Let $z = (z', z_{N+1}) \in \mathbb{R}^{N+1}$ with $z' \in \mathbb{R}^N$. We identify $\mathbb{R}^N$ with the hyperplane $H = \{z = (z_k)_{k=1}^N : z_{N+1} = 0\}$. Let

$$A_{N+1} = \{(z', z_{N+1}) \in S^N : z_{N+1} \geq \frac{1}{4}\}.$$

For $k = 1, \ldots, N$, define

$$A_k = \{(z', z_{N+1}) \in S^N : |z_{N+1}| \leq \frac{1}{2}, \frac{1}{\sqrt{1 - z_{N+1}^2}} z' \in A'_k\}.$$

The sets $\{A_k\}_{k=1}^{N+1}$ are closed, $A_k \cap (-A_k) = \varnothing$ and $C_k = A_k \cup (-A_k)$ cover S^N. □

This lemma permits the calculation of the genus of a set using sets of genus one.

Proposition 1.53 *If $D \in \mathcal{S}$, then $\gamma(D) = N$ if and only if N is the least integer such that there exist sets $\{A_k\}_{k=1}^N \subseteq \mathcal{S}$ such that $\gamma(A_k) = 1$ for all $k = 1, \ldots, N$, $D = \bigcup_{k=1}^N A_k$.*

Proof First we show that if $D \in \mathcal{S}$ and $\gamma(D) = m$, then we can find $\{A_k\}_{k=1}^m \subseteq \mathcal{S}$ with $\gamma(A_k) = 1$, $D \subseteq \bigcup_{k=1}^m A_k$. Since $\gamma(D) = m$, there exists $f : D \to \mathbb{R}^m \setminus \{0\}$ continuous, odd. Let $r_1(\cdot)$ be the radial retraction on S^{m-1}. Then $r_1 \circ f : D \to S^{m-1}$. Let C_k and A_k be as in Lemma 1.53. Then $\{(r_1 \circ f)^{-1}(C_k)\}_{k=1}^m$ is a closed cover of D. Set

$$E_k = (r_1 \circ f)^{-1}(C_k) \cup (r_1 \circ f)^{-1}(-A_k).$$

The right hand side sets are disjoint and $\gamma(E_k) = 1$

$\Rightarrow$ From the above argument we know that D can be covered by N sets of genus 1. If N is not minimal, then we can find integer $m < N$ such that D can be covered by m sets of genus 1 and so $\gamma(D) \leq m < N$, a contradiction to the hypothesis.

$\Leftarrow$ Let $\{A_k\}_{k=1}^N$ be as postulated by the hypothesis. Then $\gamma(D) \leq N$. If $\gamma(D) = m \leq N$, then the initial argument says that there exist m sets A_k' with the properties of A_k, contradicting the minimality of N. □

This proposition leads to the notion of "category", which is an extension of the genus to general topological spaces.

Definition 1.54 Let X be a Hausdorff topological space and $C \subseteq X$ a closed subset. We say that C is of "category 1" in X ($cat_X(C) = 1$), if there exists a deformation $h([0, 1] \times C, X)$ such that $h(0, u) = u$ for all $u \in C$, $h(1, u) = u_0$ for $u \in C$ (that is, C is continuously deformed to a singleton). We say that C is of "category n" in X ($cat_X(C) = n$), if n is the least integer such that there exist closed sets $\{A_k\}_{k=1}^n$ covering C such that $cat_X(A_k) = 1$. If no such integer exists, then $cat_X(C) = \infty$.

Remark 1.55 This is known in the literature as the Ljusternik-Schnirelmann Category. It is more general than the genus and has similar properties but it is in general more difficult to compute. Both notions provide information for the size of the solution set of nonlinear equations.

The classical example of the torus (donut) T^2 illustrates in a nice way the notion of category (as well as the deformation theorem which we will encounter later, see Theorem 3.20). So, consider a torus standing on the (x, y)-plane and let $\varphi(x, y, z) = z$ be the height functional. The sublevel sets $L_\lambda = \{u \in T^2 : \varphi(u) \leq \lambda\}$ are indicated in Figs. 1.1 and 1.2. If $\lambda \in [m_1, m_2)$, then L_λ is a spherical cup which is contractible. Hence

$$\text{cat}(L_\lambda, T^2) = 1.$$

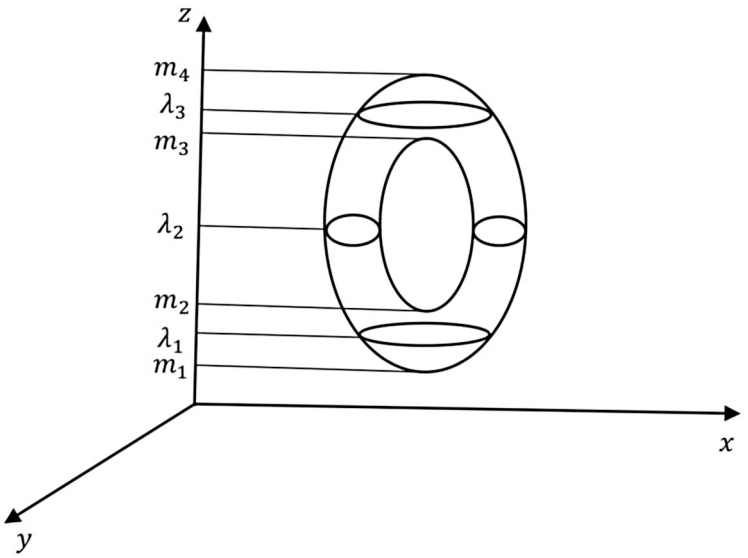


Fig. 1.1 Category

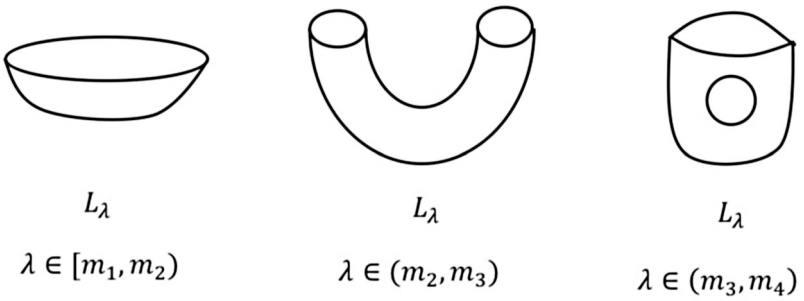


Fig. 1.2 Sublevel sets

If $\lambda \in [m_2, m_4)$, then L_λ is not contractible, but T^2 can be covered by two contractible sets. Hence

$$\text{cat}(L_\lambda, T^2) = 2.$$

Finally if $\lambda \geq m_4$, then $L_\lambda = T^2$ and so $\text{cat}(L_\lambda, T^2) = 3$.

Note also that the topological structure of the sublevel sets changes the moment we cross the critical levels m_1, m_2, m_3, m_4 (m_1, m_2 extremal levels, m_2, m_3 saddle point levels). This illustrates the deformation theorem (Theorem 3.20), since interesting things happen the moment we cross critical levels.

We conclude this section with a few observations that pave the way for the infinite dimensional extension of the degree.

Definition 1.56 Let X be an N-dimensional topological vector space, $\Omega \subseteq X$ is bounded open set, $f \in C(\overline{\Omega}, X)$ and $b \notin f(\partial\Omega)$. Then the degree of f at b with respect to Ω is defined by

$$d_B(f, \Omega, b) = d_B(\varphi \circ f \circ \varphi^{-1}, \varphi(\Omega), \varphi(b)),$$

where $\varphi : X \to \mathbb{R}^N$ is an isomorphism.

Remark 1.57 It is easy to check that this definition is independent of the isomorphism φ(that is, of the basis used to identify X and $\mathbb{R}^N$). So, the definition makes sense.

Proposition 1.58 *If X is a topological vector space with $\dim X = n$, $Y \subseteq X$ is a subspace with $\dim Y = m < n$, $\Omega \subseteq X$ is a bounded open set, $f \in C(\overline{\Omega}, Y)$, $\varphi = id - f$, $b \notin \varphi(\partial\Omega)$, then $d_B(\varphi, \Omega, b) = d_B(\varphi|_{\overline{\Omega \cap Y}}, \Omega \cap Y, b)$.*

Proof We may assume $X = \mathbb{R}^N$, $Y = \mathbb{R}^m$, $f \in C^1(\Omega, \mathbb{R}^m) \cap C(\overline{\Omega}, \mathbb{R}^m)$ and $b \notin \varphi(S_\varphi)$, we have

$$J_\varphi(z) = det[\begin{matrix} id_m - (\partial_j f_i(z)), & -(\partial_j f_i(z)) \\ 0, & id_{n-m} \end{matrix}], \ z \in \Omega \cap \mathbb{R}^m$$

$$\Rightarrow \quad J_\varphi(z) = J_{\varphi_m}(z) \text{ with } \varphi_m = \varphi|_{\overline{\Omega \cap \mathbb{R}^m}} .$$

□

1.2 Leray-Schauder Degree

In this section, we extend the notion of degree to infinite dimensional spaces. First we show that this can not be done to all continuous functions as was the case with the Brouwer degree. If this was possible, then we would have had the infinite dimensional version of the Brouwer fixed point theorem (see Theorem 1.28). However the next example (due to Kakutani [264]) shows that this is not possible.

Example 1.59 Let $X = l^2$, $\overline{B}_1 = \{u = (u_k)_{k\in\mathbb{N}} \in l^2 : ||u||^2 = \sum_{k=1}^{\infty} {u_k}^2 \leq 1\}$ and $f : \overline{B}_1 \to \overline{B}_1$ be defined by $f(u) = ((1 - ||u||^2)^{\frac{1}{2}}, u_1, \ldots, u_n, \ldots)$. Evidently $f \in C(\overline{B}_1, \overline{B}_1)$ and $f(\overline{B}_1) \subseteq \partial B_1 = \{u = (u_k)_{k\in\mathbb{N}} \in l^2 : ||u||^2 = \sum_{k=1}^{\infty} {u_k}^2 = 1\}$. Suppose that $f(\cdot)$ had a fixed point $\widehat{u} \in \overline{B}_1$. Then $f(\widehat{u}) = \widehat{u}$ and so $\widehat{u}_1 = (1 - ||\widehat{u}||^2)^{\frac{1}{2}})$, $\widehat{u}_{n+1} = \widehat{u}_n$. We have $||\widehat{u}|| = ||f(\widehat{u})|| = 1$ and so $\widehat{u}_1 = 0$, hence $\widehat{u} = 0$, which contradicts the fact $||\widehat{u}|| = 1$ (since $f(\overline{B}_1) \subseteq \partial B_1$).

For this reason, we limit ourselves to maps $\varphi(\cdot)$ which are compact perturbations of the identity, that is, $\varphi = id - f$ with $f \in K(\overline{\Omega}, X)$, We are going to exploit the

properties of compact maps and in particular that it can be approximated uniformly by finite rank maps (see Theorem I-3.151).

Let X be a Banach space, $\Omega \subseteq X$ a bounded, open set, $f \in K(\overline{\Omega}, X)$ (see Definition I-3.146) and $\varphi = id - f$. From Proposition I-7.22, we know that φ is closed (that is, maps closed sets to closed sets) and proper (that is, returns compact sets to compact sets). Let $b \notin \varphi(\partial\Omega)$. Then $r_0 = d(b, \varphi(\partial\Omega)) > 0$. According to Theorem I-3.151, given $\varepsilon > 0$ we can find $f_\varepsilon \in K_f(\overline{\Omega}, X)$ such that $||f(x) - f_\varepsilon(x)||_X < \varepsilon$ for all $x \in \overline{\Omega}$. We set

$$X_\varepsilon = \text{span} f_\varepsilon(\overline{\Omega}), \Omega_\varepsilon = \Omega \cap X_\varepsilon, \varphi_\varepsilon = id - f_\varepsilon.$$

Proposition 1.60 *If $\Omega \subseteq X$ is a bounded open set, $f \in K(\overline{\Omega}, X)$, $\varphi = id - f$, $b \notin \varphi(\partial\Omega)$ and $0 < \varepsilon < r_0 = d(b, \varphi(\partial\Omega))$, then $d_B(\varphi_\varepsilon, \Omega_\varepsilon, b)$ is well-defined and independent of ε.*

Proof Let $\partial_\varepsilon\Omega_\varepsilon$ be the boundary of Ω_ε in X_ε. Then $\partial_\varepsilon\Omega_\varepsilon \subseteq \partial\Omega$ and so $b \notin f_\varepsilon(\partial_\varepsilon\Omega_\varepsilon)$ for all $0 < \varepsilon < r_0$. Let $\varepsilon_1, \varepsilon_2 \in (0, r_0)$ and set $V = span\{X_{\varepsilon_1}, X_{\varepsilon_2}\}$ and $\Omega_V = \Omega \cap V$. From Proposition 1.58, we have

$$d_B(\varphi_{\varepsilon_1}, \Omega_V, b) = d_B(\varphi_{\varepsilon_1}, \Omega_{\varepsilon_1}, b) \text{ and } d_B(\varphi_{\varepsilon_2}, \Omega_V, b) = d_B(\varphi_{\varepsilon_2}, \Omega_{\varepsilon_2}, b). \tag{1.19}$$

Consider the continuous homotopy

$$h(t, u) = t\varphi_{\varepsilon_1} + (1 - t)\varphi_{\varepsilon_2}(u) \text{ for all } (t, u) \in [0, 1] \times \overline{\Omega}_V.$$

Note that $b \notin h(t, \partial\Omega_V)$. Hence by the homotopy invariance property we have

$$d_B(\varphi_{\varepsilon_1}, \Omega_V, b) = d_B(\varphi_{\varepsilon_2}, \Omega_V, b),$$

$$\Rightarrow \quad d_B(\varphi_{\varepsilon_1}, \Omega_{\varepsilon_1}, b) = d_B(\varphi_{\varepsilon_2}, \Omega_{\varepsilon_2}, b) \text{ (see (19))}. \qquad \square$$

This proposition leads to the following definition of an infinite dimensional degree.

Definition 1.61 Let X be a Banach space, $\Omega \subseteq X$ a bounded, open set $f \in K(\overline{\Omega}, X)$, $\varphi = id - f$, $b \notin \varphi(\partial\Omega)$. Then the "Leray-Schauder degree" of φ at b with respect to Ω, denoted by $d_{LS}(\varphi, \Omega, b)$ is defined by

$$d_{LS}(\varphi, \Omega, b) = d_B(\widehat{\varphi}, \Omega_V, b),$$

where $\widehat{\varphi} = id - \widehat{f}$, with $\widehat{f} \in K_f(\overline{\Omega}, X)$ such that $||f - \widehat{f}||_\infty < r_0 = d(b, \varphi(\partial\Omega))$, $V = span\{b, \widehat{\varphi}(\overline{\Omega})\}$, $\Omega_V = \Omega \cap V$.

Remark 1.62 A moment's reflection on this definition, reveals that it is not necessary for $\Omega \subseteq X$ to be bounded. It is enough to know that for every $V \subseteq X$ finite dimensional subspace, we have $\Omega \cap V$ is bounded.

Using this definition and properties of the finite dimensional degree from Sect. 1.1, we can easily deduce the main properties of the Leray-Schauder degree.

In what follows X is a Banach space, $\Omega \subseteq X$ is bounded, open set and $S(\overline{\Omega}, X) = \{\varphi = id - f : f \in K(\overline{\Omega}, X)\}$.

Theorem 1.63 *If $\varphi \in S(\overline{\Omega}, X)$ and $b \notin \varphi(\partial\Omega)$, then*

(a) *if Ω_1, Ω_2 are open subsets of Ω, $\Omega_1 \cup \Omega_2 = \Omega$, $\Omega_1 \cap \Omega_2 = \varnothing$, $b \notin \varphi(\partial\Omega_1 \cup \partial\Omega_2)$, then $d_{LS}(\varphi, \Omega, b) = d_{LS}(\varphi, \Omega_1, b) + d_{LS}(\varphi, \Omega_2, b)$;*
(b) *$d_{LS}(\varphi, \Omega, b)$ is constant on connected components of $X \setminus \varphi(\partial\Omega)$;*
(c) *if $b \notin \varphi(\overline{\Omega})$, then $d_{LS}(\varphi, \Omega, b) = 0$;*
(d) *if $d_{LS}(\varphi, \Omega, b) \neq 0$, then there exists $u \in \Omega$ such that $\varphi(u) = b$;*
(e) *there exists a neighborhood U of φ in $S(\overline{\Omega}, X)$ such that for all $\psi \in U$, we have $b \notin \psi(\partial\Omega)$ and $d_{LS}(\varphi, \Omega, b) = d_{LS}(\psi, \Omega, b)$;*
(f) *if $h \in K([0, 1] \times \overline{\Omega}, X)$ and $b \notin h([0, 1] \times \partial\Omega)$, then $d_{LS}(id - h(t, \cdot), \Omega, b)$ is independent of $t \in [0, 1]$ (this is the homotopy invariance property for the Leray-Schauder degree);*
(g) *if $C \subseteq \Omega$ is closed and $b \notin \varphi(C)$, then $d_{LS}(\varphi, \Omega, b) = d_{LS}(\psi, \Omega \setminus C, b)$;*
(h) *if $\psi \in S(\overline{\Omega}, X)$ and $\varphi|_{\partial\Omega} = \psi|_{\partial\Omega}$, then $d_{LS}(\varphi, \Omega, b) = d_{LS}(\psi, \Omega, b)$;*
(i) *if $V \subseteq X$ is a closed subspace such that $\varphi(\overline{\Omega}) \subseteq V$, then $d_{LS}(\varphi, \Omega, b) = d_{LS}(\varphi|_{\overline{\Omega} \cap V}, \Omega \cap V, b)$.*

Remark 1.64 Clearly we have $d_{LS}(id, \Omega, b) = 1$, $d_{LS}(id, \Omega, b) = 0$ for all $b \notin \overline{\Omega}$ and for all $c \in X$ $d_{LS}(\varphi, \Omega, b) = d_{LS}(\varphi - c, \Omega, b - c)$.

Next let us present the infinite dimensional version of the Brouwer fixed point theorem (see Theorem 1.28 and 1.29). The result is known as the "Schauder Fixed Point Theorem".

Theorem 1.65 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, bounded, convex and $f : C \to C$ is compact, then there exists $\widehat{u} \in C$ such that $f(\widehat{u}) = \widehat{u}$ (fixed point of f).*

Proof Recall that $C \subseteq X$ is a retract (see Theorem I-7.4) and let $R > 0$ be such that $C \subseteq B_R = \{u \in X : \|u\|_X < R\}$. Let $r : X \to C$ be the retraction map and consider $h(t, u) = t(for)(u)$ for all $t \in [0, 1]$, $u \in \overline{B}_R$. Evidently this is a compact homotopy and $0 \notin (id - h(t, \cdot))(\partial B_R)$. So, by the homotopy invariance property, we have

$$\begin{aligned} & d_{LS}(id - (for), B_R, 0) = 1, \\ \Rightarrow\ & f(r(\widehat{u})) = \widehat{u} \in C, \\ \Rightarrow\ & f(\widehat{u}) = \widehat{u}. \end{aligned}$$

□

Remark 1.66 Clearly it is sufficient that C is only homeomorphic to a closed, bounded, convex set.

Another such result which uses the compact homotopy $tf(\cdot)$ is the so-called "Leray-Schauder alternative principle".

Theorem 1.67 *If X is a Banach space and $f : X \to X$ is continuous and $\overline{f(X)}$ is compact, then one of the following statements holds:*

(a) *$f(\cdot)$ has a fixed point;*
(b) *the set $S = \{u \in X : u = tf(u)$ for some $0 < t < 1\}$ is unbounded.*

Proof Suppose that $f(\cdot)$ has no fixed point. We claim that for any $\varrho > 0$ there exist $u \in \partial B_\varrho = \{x \in X : \|x\|_X = \varrho\}$ and $t \in (0, 1)$ such that $u = tf(u)$. To see this, let $r : X \to \overline{B}_\varrho$ be the radial retraction map (that is, $r(x) = x$ if $\|x\|_X \leq \varrho$ and $r(x) = \frac{x}{\|x\|_X}$ if $\|x\|_X > \varrho$) and consider the map $rof : X \to \overline{B}_\varrho$. By Theorem 1.65, there exists $\widehat{u}$ such that $\widehat{u} = r(f(\widehat{u}))$ since $\widehat{u} \neq f(\widehat{u})$, we must have $\|f(\widehat{u})\|_X > \varrho$ and so $\widehat{u} = tf(\widehat{u})$ with $t \in (0, 1)$ (in fact $t = \frac{\varrho}{\|f(\widehat{u})\|_X}$). Since $\varrho > 0$ is arbitrary, we conclude that S is unbounded. □

Corollary 1.68 *If X is a Banach space, Ω is an open, bounded neighborhood of the origin and $f : \overline{\Omega} \to X$ is a compact map, then one of the following statements holds:*

(a) *$f(\cdot)$ has a fixed point in $\overline{\Omega}$;*
(b) *there exist $u \in \partial\Omega$ and $t \in (0, 1)$ such that $u = tf(u)$.*

We can also have an infinite dimensional version of Borsuk's theorem (see Theorem 1.35).

Theorem 1.69 *If X is a Banach space, $\Omega \subseteq X$ is a bounded, open, symmetric set, $0 \in \Omega$, $f \in K(\overline{\Omega}, X)$ and it is odd on $\partial\Omega$, $\varphi = id - f$, $0 \notin \varphi(\partial\Omega)$, then $d_{LS}(\varphi, \Omega, 0)$ is an odd integer.*

Remark 1.70 We can replace the oddness requirement on $f|_{\partial\Omega}$ by

$$\frac{\varphi(u)}{\|\varphi(u)\|_X} \neq \frac{\varphi(-u)}{\|\varphi(-u)\|_X} \quad \text{for all } u \in \partial\Omega.$$

Now we will introduce the notion of the "Leray-Schauder index". Let X be a Banach space and $f \in K(X, X)$ (that is, $f : X \to X$ is a compact map). Let $\varphi = id - f$. Given $b \in X$, we say that $u_0 \in X$ is a "b-point of φ, if it is an isolated solution of the equation $\varphi(u) = b$. So, we can find $\varepsilon > 0$ such that $\varphi^{-1}(b) \cap B_\varepsilon(u_0) = \{u_0\}$ ($B_\varepsilon(u_0) = \{u \in X : \|u - u_0\|_X < \varepsilon\}$). Then for every $\delta \in (0, \varepsilon)$, $d_{LS}(\varphi, B_\delta(u_0), b)$ is well-defined and by the excision property (see Theorem 1.63(g)), this degree is independent of $\delta \in (0, \varepsilon)$. This means that $d_{LS}(\varphi, B_\delta(u_0), b)$ stabilizes as $\delta \to 0^+$ and so the next definition makes sense.

Definition 1.71 The "Leray-Schauder index" of a b-point u_0 of φ, denoted by $i_{LS}(\varphi, u_0, b)$, is defined by

$$i_{LS}(\varphi, u_0, b) = \lim_{\delta \to 0^+} d_{LS}(\varphi, B_\delta(u_0), b).$$

Remark 1.72 If X is finite dimensional, then the above definition makes sense for every $\varphi \in C(X, X)$. Then we write $i_B(\varphi, u_0, b)$.

Proposition 1.73 *If X is a Banach space, $\Omega \subseteq X$ is a bounded, open set, $f \in K(\overline{\Omega}, X)$, $\varphi = id - f$, $b \in \varphi(\partial\Omega)$ and $\varphi^{-1}(b)$ is finite, then*

$$d_{LS}(\varphi, \Omega, b) = \sum_{u \in \varphi^{-1}(b)} i_{LS}(\varphi, u, b).$$

Proof Let $\varphi^{-1}(b) = \{u_k\}_{k=1}^m$ and let $\{U_k\}_{k=1}^m$ be mutually disjoint open subsets of Ω such that $u_k \in U_k$ for all $k = 1, \ldots, m$. From Definition 1.71, we have

$$i_{LS}(\varphi, u_k, b) = d_{LS}(\varphi, U_k, b), \ k = 1, \ldots, m.$$

Then using the excision property of the Leray-Schauder degree, we have

$$\begin{aligned}
\sum_{u \in \varphi^{-1}(b)} i_{LS}(\varphi, u, b) &= \sum_{k=1}^m i_{LS}(\varphi, u_k, b) \\
&= \sum_{k=1}^m d_{LS}(\varphi, U_k, b) \\
&= d_{LS}(\varphi, \bigcup_{k=1}^m U_k, b) \\
&= d_{LS}(\varphi, \Omega \setminus K, b) \quad \text{with } K = \overline{\Omega} \setminus \bigcup_{k=1}^m U_k) \\
&= d_{LS}(\varphi, \Omega, b).
\end{aligned}$$

□

Proposition 1.74 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded, open set, $\varphi \in C^1(\overline{\Omega}, \mathbb{R}^N)$, $u_0 \in \varphi^{-1}(b)$ and $J_\varphi(u_0) \neq 0$, then $i_B(\varphi, u_0, b) = (-1)^m$ with m is the number of real negative eigenvalues of $\nabla\varphi(u_0)$ counting algebraic multiplicities.*

Proof Let $u \subseteq \Omega$ be an open set such that $u_0 \in U$, $\varphi(u_0) = b$ and $\varphi(u) \neq b$ for every $u \in \overline{U} \setminus \{u_0\}$. From Definition 1.71, we have

$$i_B(\varphi, u_0, b) = d_B(\varphi, U, b) = sgn(J_\varphi(u_0)). \tag{1.20}$$

Let $\{\lambda_k\}_{k=1}^n$ be the eigenvalues of $\nabla\varphi(u_0)$. The complex eigenvalues occur in conjugate pairs $x, \overline{x}$ and so $x\overline{x} > 0$. Therefore

$$sgn(J_\varphi(u_0)) = (-1)^m,$$

$$\Rightarrow i_B(\varphi, u_0, b) = (-1)^m \ (see\ (20)).$$

□

Corollary 1.75 *If $A : \mathbb{R}^N \to \mathbb{R}^N$ is linear, 1 is not an eigenvalue of A and $\varphi = id - A$, then $i_B(\varphi, 0, 0) = (-1)^m$ with m being the sum of the algebraic multiplicities of the real eigenvalues of A in $(1, \infty)$.*

This corollary can also be stated for the Leray-Schauder index.

Proposition 1.76 *If X is a Banach space, $\Omega \subseteq X$ is a bounded, open set, $0 \in \Omega$, $A \in \mathcal{L}_c(X)$ and $\varphi = id - A$, then $i_{LS}(\varphi, 0, 0) = (-1)^m$ with m being the sum of the algebraic multiplicities of the real eigenvalues of A.*

Using this proposition, we can have the following infinite dimensional version of Proposition 1.74.

Proposition 1.77 *If X is a Banach space, $f \in C^1(X, X) \cap K(X, X)$, for some $u \in X$, $id - f'(u)$ is one-to-one, $\varphi = id - f$, then $i_{LS}(\varphi, u, h) = i_{LS}(id - f'(u), 0, 0) = (-1)^{m(u)}$ with $h = u - f(u)$ and $m(u)$ is the sum of the algebraic multiplicities of the real eigenvalues of $f'(u)$ in $(1, \infty)$.*

Proof We know that $f'(u) \in \mathcal{L}_c(X)$ (see Problem I-6.1). Let $U_u = \{v \in X : u + v \in \Omega\}$ and consider the map $\psi : U_u \to X$ defined by

$$\psi(v) = \varphi(u + v) - h - \varphi'(u)(v) \quad \text{for all } v \in U_u.$$

Then $\psi(\cdot)$ is compact and satisfies $\frac{\psi(v)}{\|v\|_X} \to 0$ as $v \to 0$. Since $\varphi'(u)$ is by hypothesis injective, we have $\mu = \min\{\|\varphi'(y)\|_{\mathcal{L}} : y \in \partial B_1(0)\} > 0$. Choose $\varepsilon > 0$ such that $B_\varepsilon(0) \subseteq U_u$ and $\|\psi(v)\|_X \leq \mu\|v\|_X$ for all $v \in B_\varepsilon(0)$. Then $\varphi'(u)(v) + t\psi(v) \neq 0$ for all $t \in [0, 1]$, all $v \in B_\varepsilon(0)$ and so we have

$$i_{LS}(\varphi, u, h) = i_{LS}(\varphi(u + \cdot) - h, 0, 0) = d_{LS}(id - f'(u), 0, 0) = (-1)^{m(u)}.$$

□

We conclude with a result about the index of critical points of potential operators. For details we refer to Amann [11] and Rabinowitz [425].

Proposition 1.78

(a) *If X is finite dimensional, $\varphi \in C^1(X, X)$ and $u_0 \in X$ is an isolated local minimizer of φ, then $i_B(\varphi', u_0, 0) = 1$;*
(b) *If H is a Hilbert space, $\varphi \in C^1(X, X)$, $\varphi = id - f$ with $f \in C^1(X, X) \cap K(X, X)$ and u_0 is an isolated local minimizer of φ, then $i_{LS}(\varphi, u_0, 0) = 1$.*

1.3 Degree for Maps of Monotone Type

In this section, we define a degree for certain maps of monotone type, which appear naturally in the study of boundary value problems. First we will define a degree for $(S)_+$-operators. Let us recall the definition (see Definition I-7.71). In this section, the standing hypotheses are the following: X is a reflexive Banach space. X^* is its topological dual and by $\langle \cdot, \cdot \rangle$ we denote the duality brackets for the pair (X, X^*).

Definition 1.79 Let $C \subseteq X$ and $f : C \to X^*$. We say that $f(\cdot)$ is an "$(S)_+$-map", if for every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq C$ such that $u_n \xrightarrow{w} u \in C$ in X and

$$\limsup_{n\to\infty} \langle f(u_n), u_n - u \rangle \leq 0,$$

we have $u_n \to u$ in X.

We say that $f : C \to X^*$ is "demicontinuous", if for every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq C$ with $u_n \to u \in C$ in X, we have

$$f(u_n) \xrightarrow{w} f(u).$$

Remark 1.80 The $(S)_+$ property is preserved if we perturb $f(\cdot)$ by a monotone or completely continuous map. A demicontinuous, $(S)_+$-map is quasimonotone, that is, $u_n \xrightarrow{w} u$ in X, implies $\limsup_{n\to\infty} \langle f(u_n), u_n - u \rangle \geq 0$.

From Theorem I-7.42 (Troyanskii's renorming theorem), we know that every reflexive Banach space can be equivalently renormed so that both X and X^* are locally uniformly convex So, without any loss of generality we may assume that both X and X^* are locally uniformly convex. Then the duality map $\mathscr{F} : X \to X^*$ defined by

$$\mathscr{F}(u) = \{u^* \in X^* : \langle u^*, u \rangle = \|u\|^2, \|u^*\|_* = \|u\|\} \quad \text{for all } u \in X$$

is single valued, monotone, demicontinuous (hence maximal monotone) and of type $(S)_+$. Moreover, $\mathscr{F}^{-1}$ is single valued, continuous and is the duality map of X^*.

The construction of the degree will use Galerkin approximations. We have already encountered them in the proof of Proposition I-7.40. Recall that the Galerkin approximation of $f : C \subseteq X \to X^*$ with respect to the finite dimensional subspace $V \subseteq X$, is the map $f_V : C \cap V \to V^*$ defined by

$$\langle f_V(u), h \rangle_V = \langle f(u), h \rangle \quad \text{for all } u \in \overline{C \cap V}, \text{ all } h \in V$$

with $\langle \cdot, \cdot \rangle_V$ denoting the duality brackets for the pair (V, V^*). Given $A \subseteq V$, by $\partial_V A$ we denote the boundary of A in V. We start with a lemma due Browder [73, (Proposition 11)], where the interested reader can find its proof. The lemma will be useful in the implementation of the Galerkin method.

Lemma 1.81 *If Y is a finite dimensional Banach space, $\Omega \subseteq Y$ is a bounded open set, $V \subseteq Y$ is a subspace of X, $\Omega \cap V \neq \emptyset$, $f \in C(\overline{\Omega}, Y^*)$, $f_V : \overline{\Omega \cap V} \to V^*$ is its Galerkin approximation and*

(a) $0 \in f(\partial\Omega) \cup f_V(\partial_V(\Omega \cap V))$ *or*
(b) $0 \notin f(\partial\Omega) \cup f_V(\partial_V(\Omega \cap V))$, $d_B(f, \Omega, 0) \neq d_B(f_V, \Omega \cap V, 0)$,

then there exists $\widehat{u} \in \partial\Omega$ such that

$$\langle f(\widehat{u}), \widehat{u}\rangle \leq 0 \quad \textit{and} \quad \langle f(\widehat{u}), h\rangle = 0 \quad \textit{for all } h \in V.$$

Now we consider $\Omega \subseteq X$ a bounded, open set and $f : \overline{\Omega} \to X^*$ a demicontinuous, $(S)_+$-map. We consider the family $\{V_\alpha\}_{\alpha\in J}$ of all finite dimensional subspaces of X such that $\Omega_\alpha = \Omega \cap V_\alpha \neq \emptyset$. Evidently V_α and V_α^* are both locally uniformly convex (since X and X^* are as we mentioned earlier). We introduce a partial order on J by defining for $\alpha,\ \beta \in J$,

$$\text{“}\alpha \leq \beta \quad \text{if and only if} \quad V_\alpha \subseteq V_\beta\text{”}.$$

We set $f_\alpha = f_{V_\alpha} : \overline{\Omega_\alpha} \to V_\alpha^*$ to be the Galerkin approximation of f with respect to V_α. With the next proposition, we show that the Brouwer degree of the Galerkin approximations, eventually stabilizes and this leads to the definition of the degree map for demicontinuous, $(S)_+$ maps.

Proposition 1.82 *If $\Omega \subseteq X$ is a bounded open set, $f : \overline{\Omega} \to X^*$ is demicontinuous and of type $(S)_+$ and $0 \notin f(\partial\Omega)$, then there exist $\alpha_0 \in J$ such that for all $\alpha \in J$ with $\alpha_0 \leq \alpha$ (that is, $V_{\alpha_0} \subseteq V_\alpha$), we have*

$$0 \notin f_\alpha(\partial_{V_\alpha}(\Omega_\alpha)) \quad \text{and} \quad d_B(f_\alpha, \Omega_\alpha, 0) = d_B(f_{\alpha_0}, \Omega_{\alpha_0}, 0).$$

Proof We argue by contradiction. So, we suppose that the assertion of the proposition is not true. Then given any $\alpha \in J$, we can find $\beta \in J$ with $\alpha \leq \beta$ such that either

$$0 \in f_\alpha(\partial_{V_\alpha}(\Omega_\alpha)) \cup f_\beta(\partial_{V_\beta}(\Omega_\beta))$$

$$\text{or } 0 \notin f_\alpha(\partial_{V_\alpha}(\Omega_\alpha)) \cup f_\beta(\partial_{V_\beta}(\Omega_\beta)) \text{ and } d_B(f_\alpha, \Omega_\alpha, 0) = d_B(f_\beta, \Omega_\beta, 0).$$

According to Lemma 1.81, we can find $\widehat{u} \in \partial_{V_\beta}(\Omega_\beta) \subseteq \partial\Omega$ such that

$$\langle f(\widehat{u}), \widehat{u}\rangle = \langle f_\beta(\widehat{u}), \widehat{u}\rangle_{V_\beta} \leq 0 \text{ and } \langle f(\widehat{u}), h\rangle = \langle f_\beta(\widehat{u}), h\rangle_{V_\beta} = 0 \text{ for all } h \in V_\beta.$$

So, if we introduce the set

$$E_\alpha = \{u \in \partial\Omega : \langle f(u), u\rangle \leq 0 \text{ and } \langle f(u), h\rangle = 0 \text{ for all } h \in V_\alpha\},$$

then for all $\alpha \in J$, $E_\alpha \neq \emptyset$. Note that, if $\alpha_1, \ldots \alpha_m \in J$, then $\bigcap_{k=1}^{m} E_{\alpha_k} \neq \emptyset$. So, $\{\overline{E}^w\}_{\alpha\in J}$ is a family of weakly closed subsets of $\partial\Omega$ with the finite intersection property. Since $\overline{\partial\Omega}^w \subseteq X$ is w-compact (due to the reflexivity of X), we infer that $\bigcap_{\alpha\in J} \overline{E_\alpha}^w \neq \emptyset$. Let $\widehat{u} \in \bigcap_{\alpha\in J} \overline{E_\alpha}^w$, $u \in X$. Choose $V_\alpha \subseteq X$, $\alpha \in J$ such that $\{u_0, u\} \subseteq V_\alpha$. Using Proposition I-3.136, we can find $\{v_n\}_{n\in\mathbb{N}} \subseteq E_\alpha$ such that $v_n \xrightarrow{w} \widehat{u}$ in X. Then we have

$$\langle f(v_n), v_n\rangle \leq 0 \text{ and } \langle f(v_n), h\rangle = 0 \text{ for all } h \in V_\alpha, \text{ all } n \in \mathbb{N}, \tag{1.21}$$

$$\Rightarrow \limsup_{n\to\infty} \langle f(v_n), v_n - \widehat{u}\rangle \leq 0,$$

$$\Rightarrow v_n \to \widehat{u} \in \partial\Omega \quad (\text{since } f(\cdot) \text{ is of type } (S)_+),$$

$$\Rightarrow f(v_n) \xrightarrow{w} f(\widehat{u}) \quad (\text{since } f(\cdot) \text{ is demicontinuous}).$$

Therefore we have

$$\langle f(\widehat{u}), u\rangle = 0 \quad (\text{see } (1.21)). \tag{1.22}$$

But $u \in X$ is arbitrary. Then from (1.22) we infer that $f(\widehat{u}) = 0$, which contradicts the hypothesis that $0 \notin f(\partial\Omega)$. □

According to the above proposition, the Brouwer degree of the Galerkin approximation at zero with respect to Ω, stabilizes and so the next definition makes sense.

Definition 1.83 Let $\Omega \subseteq X$ be a bounded open set and $f : \overline{\Omega} \to X^*$ be a demicontinuous, $(S)_+$-map with $0 \notin f(\partial\Omega)$. We define the degree map $d_{(S)_+}$ by

$$d_{(S)_+}(f, \Omega, 0) = d_B(f_\alpha, \Omega_\alpha, 0) \quad \text{for } \alpha \geq \alpha_0$$

with $\alpha_0 \in J$ as postulated by Proposition 1.82. Also, for any $u^* \in X^* \setminus \{0\}$, we define

$$d_{(S)_+}(f, \Omega, u^*) = d_{(S)_+}(f - w^*, \Omega, 0).$$

From this definition, we see that the properties of the new degree follow from the corresponding properties of the Brouwer degree. To do this, we need to specify the normalizing map and the admissible homotopies.

The normalizing map is the duality map $\mathfrak{F} : X \to X^*$ (recall that via Troyanski's renorming theorem, we assume that both X and X^* are locally uniformly convex).

For the admissible homotopies, we introduce the following definition.

Definition 1.84 Let $\Omega \subseteq X$ be a bounded open set and $h : [0, 1] \times \overline{\Omega} \to X^*$. We say that $h(\cdot, \cdot)$ is an "$(S)_+$-homotopy", if the following holds: "if $t_n \to t$ in $[0, 1]$,

$u_n \xrightarrow{w} u$ in X and $\limsup_{n\to\infty}\langle h(t_n, u_n), u_n - u\rangle \le 0$, then $u_n \to u$ in X and $h(t_n, u_n) \xrightarrow{w} h(t, u)$ in X^*."

The standard linear homotopy is such an $(S)_+$-homotopy.

Proposition 1.85 *If $f_1,\ f_2 : \overline{\Omega} \to X^*$ are both demicontinuous and $(S)_+$-maps and $h(t, u) = (1 - t) f_1(u) + t f_2(u)$ for all $t \in [0, 1]$ and all $u \in \overline{\Omega}$, then $h(\cdot, \cdot)$ is on $(S)_+$-homotopy.*

Proof Consider sequences $\{t_n\}_{n\in\mathbb{N}} \subseteq [0, 1]$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq \overline{\Omega}$ such that

$$t_n \to t,\ u_n \xrightarrow{w} u \text{ in } X \text{ and } \limsup_{n\to\infty}\langle h(t_n, u_n), u_n - u\rangle \le 0. \tag{1.23}$$

We will show that

$$\limsup_{n\to\infty}\langle (1-t_n) f_1(u_n), u_n - u\rangle \le 0 \text{ and } \limsup_{n\to\infty}\langle t_n f_2(u_n), u_n - u\rangle \le 0. \tag{1.24}$$

Arguing by contradiction, suppose that (1.24) is not true. Then we can say (by passing to a subsequence if necessary), that

$$\lim_{n\to\infty}\langle (1 - t_n) f_1(u_n), u_n - u\rangle > 0. \tag{1.25}$$

On account of (23) we have

$$\begin{aligned} &\lim_{n\to\infty}\langle t_n f_2(u_n), u_n - u\rangle < 0, \\ &\Rightarrow\ t_n \neq 0 \quad \text{for} \quad n \ge n_0. \end{aligned}$$

$$\begin{aligned} &\Rightarrow\ \lim_{n\to\infty}\frac{1}{t_n}\langle t_n f_2(u_n), u_n - u\rangle = \lim_{n\to\infty}\langle f_2(u_n), u_n - u\rangle < 0, \\ &\Rightarrow\ u_n \to u \text{ in } X \text{ (by the } (S)_+ - \text{property)}, \\ &\Rightarrow\ \langle f_2(u_n), u_n - u\rangle \to 0. \end{aligned} \tag{1.26}$$

From (1.26), (1.25) and (1.23), we have a contradiction. So (1.24) holds.

To infer that $u_n \to u$ in X we consider two distinct cases.

Case I: $0 < t \le 1$

If follows that $t_n \neq 0$ for all $n \ge n_1$, and so from the second inequality in (24), we have

$$\begin{aligned} &\limsup_{n\to\infty}\langle f_2(u_n), u_n - u\rangle \le 0, \\ &\Rightarrow\ u_n \to u \text{ in } X^* \text{ (by the } (S)_+ - \text{property of } f_2). \end{aligned}$$

Case II: $t = 0$

If follows that $t_n \neq 1$ for all $n \geq n_2$, and then the first inequality in (24) implies that

$$\limsup_{n\to\infty} \langle f_1(u_n), u_n - u\rangle \leq 0,$$

$$\Rightarrow\ u_n \to u \text{ in } X \text{ (by the } (S)_+ - \text{property of } f_1).$$

Therefore we have proved that $h(\cdot,\cdot)$ is an $(S)_+$-homotopy. □

Using Definition 1.83 and the corresponding properties of the Brouwer degree, we can have the following theorem which summarizes the main properties of the new degree map. We recall the setting of this degree. X is a reflexive Banach space, equivalently renormed so that both X and X^* are locally uniformly convex, $\Omega \subseteq X$ is a bounded open set and $f : \overline{\Omega} \to X^*$ is a demicontinuous, $(S)_+$-map and $u^* \notin f(\partial\Omega)$.

Theorem 1.86

(a) *Normalization.* $d_{(S)_+}(\mathfrak{F}, \Omega, u^*) = 1$ *if* $u^* \in \mathfrak{F}(\Omega)$.
(b) *Boundary Dependence: If* $\widehat{f} : \overline{\Omega} \to X^*$ *is another demicontinuous map of* $(S)_+$*-type and* $f|_{\partial\Omega} = \widehat{f}|_{\partial\Omega}$, *then*

$$d_{(S)_+}(f, \Omega, u^*) = d_{(S)_+}(\widehat{f}, \Omega, u^*).$$

(c) *Solution Property. If* $d_{(S)_+}(f, \Omega, u^*) \neq 0$, *then there exists* $u \in \Omega$ *such that* $f(u) = u^*$.
(d) *Homotopy invariance: If* $h(t,u)$ *is an* $(S)_+$*-homotopy (see Definition 1.84) and* $u^* \notin h(t, \partial\Omega)$ *for all* $t \in [0,1]$, *then* $d_{(S)_+}(h(t,\cdot), \Omega, u^*)$ *is independent of* $t \in [0,1]$.
(e) *Domain additivity: If* Ω_1, $\Omega_2 \subseteq \Omega$ *are open subsets,* $\Omega_1 \cap \Omega_2 = \emptyset$ *and* $u^* \notin f(\overline{\Omega} - (\Omega_1 \cup \Omega_2))$, *then*

$$d_{(S)_+}(f, \Omega_1 \cup \Omega_2, u^*) = d_{(S)_+}(f, \Omega_1, u^*) + d_{(S)_+}(f, \Omega_2, u^*).$$

(f) *Excision property: If* $C \subseteq \overline{\Omega}$ *is a closed subset and* $u^* \notin f(C)$ *(that is,* $u^* \notin f(C) \cup f(\partial\Omega)$*), then* $d_{(S)_+}(f, \Omega, u^*) = d_{(S)_+}(f, \Omega - C, u^*)$.
(g) $d_{(S)_+}(f, \Omega, u^*) = d_{(S)_+}(f - u^*, \Omega, 0)$.

An immediate consequence of the solution property, is the following result.

Corollary 1.87 *If* $f(u) \neq 0$ *for all* $u \in \overline{\Omega}$, *then* $d_{(S)_+}(f, \Omega, 0) = 0$.

Remark 1.88 On account of the boundary dependence property, for (b), (c), (d) in Theorem 1.86, we can assume that f is demicontinuous and $(S)_+$ on $\partial\Omega$. For (e), it suffices to assume that $f(\cdot)$ satisfies the $(S)_+$-condition on $\overline{\Omega} - (\Omega_1 \cup \Omega_2)$.

Next we will compute the degree for certain potential maps which are of $(S)_+$-type (see Definition I-7.7). We start with a finite dimensional result due to Amann [11].

Proposition 1.89 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded open set, $f \in C^1(\Omega, \mathbb{R})$ and there exist $\eta,\ \vartheta \in \mathbb{R}$, $\eta < \vartheta$, $r > 0$ and $u_0 \in \Omega$ such that*

(i) $U = \{f < \vartheta\}$ *is bounded and* $\overline{U} \subseteq \Omega$;
(ii) $\{f \leq \eta\} \subseteq \overline{B_r(u_0)} \subseteq U$;
(iii) $f'(u) \neq 0$ *for all* $u \in \{\eta \leq f \leq \vartheta\}$, *then* $d_B(f', U, 0) = 1$.

Remark 1.90 In fact a careful reading of the proof of Amann [11], reveals that hypothesis (ii) in the above proposition can be replaced by the following weaker one

$$(ii)' \quad u \in \{f \leq \eta\} \Rightarrow \{tu + (1-t)u_0 : 0 \leq t \leq 1\} \subseteq U.$$

Using Definition 1.83, we can extend Proposition 1.89 to the $(S)_+$-degree of demicontinuous, $(S)_+$-maps.

Theorem 1.91 *If X is a reflexive Banach space, $\Omega \subseteq X$ is nonempty, open, $f : \Omega \to \mathbb{R}$ is Gateaux differentiable, continuous and $f' : \Omega \to X^*$ is demicontinuous, $(S)_+$ and there exist $\eta,\ \vartheta \in \mathbb{R}$, $\eta < \vartheta$, $u_0 \in X$ such that*

(i) $U = \{f < \vartheta\}$ *is bounded and* $\overline{U} \subseteq \Omega$;
(ii) $u \in \{f < \eta\} \Rightarrow \{tu + (1-t)u_0 : 0 \leq t \leq 1\} \subseteq U$;
(iii) $f'(u) \neq 0$ *for all* $u \in \{\eta \leq f \leq \vartheta\}$, *then* $d_{(S)_+}(f', U, 0) = 1$.

Proof Note that $\partial U \subseteq f^{-1}(\{\vartheta\})$. Hence $f'(u) \neq 0$ for all $u \in \partial U$ (see (iii)). This means that $d_{(S)_+}(f', U, 0)$ is defined.

Consider the collection

$$\mathcal{S} = \{V \subseteq X :\ \text{finite dimensional subspace},\ V \cap U \neq \emptyset\}.$$

Given $V \in \mathcal{S}$, we set $f|_V = f|_{\Omega \cap V}$.

Claim We can find $V_0 \in \mathcal{S}$ such that

$$(f_V)'(u) \neq 0 \text{ for all } V \supseteq V_0,\ V \in \mathcal{S},\ \text{all } u \in \Omega \cap V \cap \{\eta \leq f \leq \vartheta\}.$$

If the Claim is not true, then for all $V \in \mathcal{S}$ we can find $\widehat{V} \in \mathcal{S}$ with $V \subseteq \widehat{V}$ and $u_V \in \Omega \cap \widehat{V} \cap \{\eta \leq f \leq \vartheta\}$ and $f'(u_V) = 0$. Then $\langle f'(u_V), h\rangle = 0$ for all $h \in \widehat{V}$. So, if we introduce the set

$$E_V = \{u \in \{\eta \leq f \leq \vartheta\} :\ \langle f'(u), u\rangle = 0 \text{ and } \langle f'(u), h\rangle = 0 \text{ for all } h \in V\},$$

then $u_V \in E_V$. Also $\bigcap_{k=1}^{n} E_{V_k} = E_{V_1+\ldots+V_n} \neq \emptyset$ for all $\{V_k\}_{k=1}^n \subseteq \mathcal{S}$ and so the family $\{\overline{E_V}^w\}_{V\in\mathcal{S}}$ has the finite intersection property. Note that $\{\eta \leq f \leq \mu\} \subseteq \overline{U}$. Indeed, if we can find $u \in \{\eta \leq f \leq \vartheta\} \setminus \overline{U}$, then u is going to be a minimizer of $f|_{\Omega\setminus\overline{U}}$ and so $f'(u) = 0$, contradicting hypothesis (iii). So, we have $\overline{E}_V^w \subseteq \overline{U}^w$ for all $V \in \mathcal{S}$. But $\overline{U}^w$ is w-compact (since X is reflexive). It follows that

$$E_0 = \bigcap_{V\in\mathcal{S}} \overline{E_V}^w \neq \emptyset.$$

Let $u \in E_0$, $h \in X$ and choose $V \in \mathcal{S}$ such that $\{u,\ h\} \subseteq V$. Then $u \in \overline{E_V}^w$. By Proposition I-3.136 (recall X_w is angelic), we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq E_V$ such that $u_n \xrightarrow{w} u$ in X. From the definition of E_V

$$\begin{aligned} &\langle f'(u_n), u_n - u\rangle = 0 \quad \text{for all } n \in \mathbb{N}, \\ \Rightarrow\ &u_n \to u \text{ in } X \text{ (since } f'(\cdot) \text{ is an } (S)_+\text{-map)}, \\ \Rightarrow\ &\langle f'(u), h\rangle = 0. \end{aligned}$$

Since $h \in X$ is arbitrary, we infer that $f'(u) = 0$, $u \in \{\eta \leq f \leq \vartheta\}$. But this contradicts hypothesis (iii). This proves the Claim.

The Claim implies that for all $V \in \mathcal{S}$, $V \supseteq V_0$, the Galerkin approximation satisfies the requirements of Proposition 1.89 (see also Remark 1.90). Hence we have

$$\begin{aligned} &d(f'_V, U \cap V, 0) = 1 \quad \text{for all } V \in \zeta,\ V \supseteq V_0, \\ \Rightarrow\ &d_{(S)_+}(f', U, 0) = 1 \text{ (see Definition 1.83)} \end{aligned}$$

(note that $f'_V = (f')_V$). □

A consequence of Theorem 1.91 is the following result.

Corollary 1.92 *If X is a reflexive Banach space, $f : X \to \mathbb{R}$ is Gateaux differentiable and continuous, $f' : X \to X^*$ is demicontinuous and $(S)_+$, $f(\cdot)$ is coercive (that is, $f(x) \to +\infty$ as $\|x\| \to \infty$) and we can find $\varrho_0 > 0$ such that $f'(x) \neq 0$ for all $\|x\| \geq \varrho_0$, then we can find $\varrho_1 \geq \varrho_0$ such that*

$$d_{(S)_+}(f', B_r(0), 0) = 1 \text{ for all } r \geq \varrho_1.$$

Proof Let $\eta > \sup[f(u) : u \in B_{\varrho_0}(0)]$ and $\varrho_1 = \sup[\|u\| : u \in \{f \leq \eta\}]$. For $r > \varrho_1$, we choose $\vartheta > \max\{\eta,\ \sup_{B_r(0)} f\}$. Apply Theorem 1.91, with $\Omega = X$, $U = \{f \leq \vartheta\}$ and use the excision property (see Theorem 1.86 (f)) with $C = \overline{U}\setminus B_r(0)$. We obtain $d_{(S)_+}(f', B_r(0), 0) = 1$ for all $r \geq \varrho_1$. □

Next we compute the degree at a local minimizer. To do this we will need two auxiliary results. The first one concerns C^1- maps with an $(S)_1$-derivative.

Lemma 1.93 *If X is a reflexive Banach space, $\Omega \subseteq X$ is nonempty, open convex and $f \in C^1(\Omega, X)$ with $f' : \Omega \to X^*$ on $(S)_+$-map, then $f(\cdot)$ is sequentially weakly lower semicontinuous.*

Proof We proceed indirectly. So, suppose that the claim of the lemma is not true. Then we can find $\{u_n, u\}_{n\in\mathbb{N}} \subseteq \Omega$, $u_n \xrightarrow{w} u$ in X and

$$\lim_{n\to\infty} f(u_n) < f(u). \tag{1.27}$$

Using the mean value theorem, we can find $t_n \in (0, 1)$ such that

$$\begin{aligned}
&f(u_n) - f(u) = \langle f'(u + t_n(u_n - u)), u_n - u\rangle \text{ for all } n \in \mathbb{N},\\
&\Rightarrow \limsup_{n\to\infty} \langle f'(u + t_n(u_n - u)), u_n - u\rangle \leq 0 \text{ (see (27))},\\
&\Rightarrow u + t_n(u_n - u) \to u \text{ (since } f'(\cdot) \text{ is } (S)_+),\\
&\Rightarrow f(u_n) \to f(u) \text{ (since } f'(\cdot) \text{ is continuous)},
\end{aligned}$$

which contradicts (1.27). □

The second auxiliary result concerns local minimizers.

Lemma 1.94 *If X is a reflexive Banach space, $\Omega \subseteq X$ is nonempty, open, $f : \Omega \to \mathbb{R}$ is sequentially weakly lower semicontinuous, Gateaux differentiable with $f' : \Omega \to X^*$ of $(S)_+$-type, $g : \Omega \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is sequentially weakly lower semicontinuous and $u_0 \in \Omega$ is such that*

$$(f + g)(u_0) < (f + g)(u) \text{ for all } u \in \overline{B}_{r_0}(u_0)\backslash\{u_n\}$$

with $\overline{B}_{r_0}(u_0) \subseteq \Omega$ (that is, u_0 is an isolated local minimizer of $(f + g)(\cdot)$), then for all $r \in (0, r_0)$ we have $\inf\{(f + g)(u) : u \in \overline{B}_{r_0}(u_0)\backslash B_r(u_0)\} > (f + g)(u_0)$.

Proof Again the proof is by contradiction. So, suppose that we can find $r \in (0, r_0)$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq \overline{B}_{r_0}(u_0)\backslash B_r(u_0)$ such that

$$\lim_{n\to\infty} (f + g)(u_n) = (f + g)(u_0) \tag{1.28}$$

The reflexivity of X implies that we may assume that

$$\begin{aligned}
&u_n \xrightarrow{w} u \text{ in } X,\\
&\Rightarrow (f + g)(u) \leq \liminf_{n\to\infty} (f + g)(u_n) = (f + g)(u_0),\\
&\Rightarrow u = u_0.
\end{aligned}$$

As before, via the mean value theorem, we can find $t_n \in (0, 1)$ such that

$$f(u_n) - f(\frac{u_n + u_0}{2}) = \langle f'(v_n), \frac{u_n - u_0}{2} \rangle \text{ for all } n \in \mathbb{N}, \tag{1.29}$$

with $v_n = t_n u_n + (1 - t_n)\frac{u_n+u_0}{2}$. We have $\frac{u_n+u_0}{2} \xrightarrow{w} u_0$. Note that

$$\begin{aligned}
& f(u_n) - f(\frac{u_n + u_0}{2}) \\
&= (f + g)(u_n) - (f(\frac{u_n + u_0}{2}) + g(u_n)), \\
&\Rightarrow \lim_{n\to\infty} (f + g)(u_n) - \liminf_{n\to\infty} (f(\frac{u_n + u_0}{2} + g(u_n))) \leq 0, \text{ (see (28))}, \\
&\Rightarrow \limsup_{n\to\infty} \langle f'(v_n), \frac{u_n - u_0}{2} \rangle \leq 0, \text{ (see (29))}, \\
&\Rightarrow \limsup_{n\to\infty} \langle f'(v_n), v_n - u_0 \rangle = \limsup_{n\to\infty} (1 + t_n)\langle f'(v_n), \frac{u_n - u_0}{2} \rangle \leq 0.
\end{aligned}$$

Since by hypothesis $f'(\cdot)$ is an $(S)_+$-map, we obtain

$$v_n \to u_0 \text{ in } X.$$

But note that

$$\| v_n - u_0 \| = \frac{1 + t_n}{2} \| u_n - u_0 \| \geq \frac{1}{2} r \text{ for all } n \in \mathbb{N},$$

a contradiction. □

Using the above two lemmata, we can compute the $(S)_1$-degree at a local minimizer.

Theorem 1.95 *If X is a reflexive Banach space, $\Omega \subseteq X$ is nonempty, open, $f \in C^1(\Omega, \mathbb{R})$ with $f' : \Omega \to X^*$ of $(S)_+$-type and u_0 is a local minimizer of $f(\cdot)$ which is an isolated critical point of $f(\cdot)$, then there exists $r^* > 0$ such that*

$$d_{(S)_+}(f', B_r(u_0), 0) = 1 \text{ for all } r \in (0, r^*].$$

Proof Consider the function $\widehat{f}(u) = f(u) - f(u_0), u \in \Omega$. Let $\Omega_0 \subseteq \Omega$ nonempty, open, convex with $u_0 \in \Omega_0$. Then $\widehat{f} \mid_{\Omega_0}$ is sequentially weakly lower semicontinuous (see Lemma 1.93). Note that $u_0 \in \Omega$ is a local minimizer and an isolated critical point of $\widehat{f}(\cdot)$. So, we can find $r_0 > 0$ such that $\overline{B}_{r_0}(u_0) \subseteq \Omega$ and

$$0 = \widehat{f}(u_0) < \widehat{f}(u), \widehat{f}'(u) \neq 0 \text{ for all } u \in \overline{B}_{r_0}(u_0) \backslash \{u_0\}. \tag{1.30}$$

Invoking Lemma 1.94 (with $f = \widehat{f}$ and $g = 0$), we have

$$\widehat{m}_r = \inf\{\widehat{f}(u) : u \in \overline{B}_{r_0}(u_0) \backslash B_r(u_0)\} > 0 \text{ for all } r \in (0, r_0].$$

Consider the nonempty open set

$$V = \{u \in B_{r_0}(u_0) : \widehat{f}(u) < \widehat{m}_{\frac{r}{2}}\}.$$

We have $V \subseteq B_{\frac{r_0}{2}}(u_0)$. Let $r_1 \in (0, \frac{r_0}{2})$ such that $\overline{B}_{r_1}(u_0) \subseteq V$ and $r \in (0, r_1)$. We choose $\vartheta \in (0, \widehat{m}_r)$ and we see that

$$\{u \in B_{r_0}(u_0) : \widehat{f}(u) \leq \vartheta\} \subseteq B_{r_1}(u_0) \subseteq \overline{B}_r(u_0) \subseteq V. \tag{1.31}$$

Then (1.30) and (1.31) permit the use of Theorem 1.91 for $\widehat{f}\,|_{\overline{B}_{r_0}(u_0)}$ and we have

$$d_{(S)_+}(f', V, 0) = 1.$$

If $C = \overline{V} \backslash B_r(u_0)$, then $0 \notin f'(C)$ and so the excision property of $d_{(S)_+}$ (see Theorem 1.86(f)), implies that

$$d_{(S)_+}(f', B_{r_0}(u_0), 0) = 1.$$

□

Next we will define a degree for maximal monotone perturbations of an $(S)_+$-map.

So, the setting is the following: X is a reflexive Banach space, X^* is its topological dual and we assume that both X and X^* are locally uniformly convex. Recall that by Troyanski's renorming theorem (see Theorem I-7.42), every reflexive Banach space can be equivalently renormed so that both X and X^* are locally uniformly convex. Let $\Omega \subseteq X$ be nonempty, bounded, open, $\varphi : \overline{\Omega} \to X^*$ is bounded, demicontinvous, $(S)_+$-map and $A : D(A) \subseteq X \to 2^{X^*}$ is a maximal monotone map such that $(0, 0) \in Gr\, A$. We assume that $u^* \notin (\varphi + A)(\partial\Omega)$. In what follows we will define a degree for the triple $(\varphi + A, \Omega, u^*)$.

For $\lambda > 0$, let $A_\lambda : X \to X^*$ be the "Yosida approximation" of $A(\cdot)$ (see Definition I-7.51). Then using Proposition I-7.51, we see that $u \to (\varphi + A_\lambda)(u)$ is bounded, demicontinuous and $(S)_+$. So, we can compute the $(S)_+$-degree for the triple $(\varphi + A_\lambda, \Omega, u^*)$. Using this computation, we can define a degree for the triple $(\varphi + A, \Omega, u^*)$. This is possible because of the next lemma, due to Browder [73, (Theorem 8)].

Lemma 1.96 *If $(\varphi + A, \Omega, u^*)$ is a triple as above, then we can find $\lambda_0 > 0$ such that*

(a) $u^* \notin (\varphi + A_\lambda)(\partial\Omega)$ *for all* $\lambda \in (0, \lambda_0]$.
(b) $d_{(S)_+}(\varphi + A_\lambda, \Omega, u^*)$ *is the same for all* $\lambda \in (0, \lambda_0]$.

Remark 1.97 Statement (a) implies that $d_{(S)_+}(\varphi + A_\lambda, \Omega, u^*)$ is well-defined. Statement (b) implies that $d_{(S)_+}(\varphi + A_\lambda, \Omega, u^*)$ stabilizes for small $\lambda > 0$ and so can be used as a degree for the triple $(\varphi + A, \Omega, u^*)$.

Definition 1.98 For a triple $(\varphi+A, \Omega, u^*)$ as above, we define $d_p(\varphi+A, \Omega, u^*) = d_{(S)_+}(\varphi + A_\lambda, \Omega, u^*)$ for $0 < \lambda \leq \lambda_0$ (see Lemma 1.96).

Remark 1.99 If $A \equiv 0$, then $d_p(\varphi, \Omega, u^*) = d_{(S)_+}(\varphi, \Omega, u^*)$.

We have to introduce the homotopies which preserve this degree. For the $(S)_+$-part of the map we will use $(S)_+$-homotopies introduced in Definition 1.85. For the maximal monotone part of the map we will use pseudomonotone homotopies, which we introduce next.

Definition 1.100 Consider a family $\{A_t\}_{t\in[0,1]}$ of maximal monotone operators from X into 2^{X^*} which satisfy $(0, 0) \in GrA_t$ for all $t \in [0, 1]$. We say that $\{A_t\}_{t\in[0,1]}$ is a "pseudomonotone homotopy", if it satisfies one of the following equivalent properties:

(a) If $t_n \to t$ in $[0, 1]$, $(u_n, u_n^*) \in GrA_{t_n}, n \in \mathbb{N}$, $u_n \xrightarrow{w} u$ in X, $u_n^* \xrightarrow{w} u^*$ in X^* and $\limsup_{n\to\infty} \langle u_n^*, u_n\rangle \leq \langle u^*, u\rangle$, then $(u, u^*) \in GrA_t$ and $\langle u_n^*, u_n\rangle \to \langle u^*, u\rangle$.

(b) Let $J_\lambda^{A_t} : X \to X$ be the resolvent operator for A_t (see Definition I-7.51). For every $\lambda > 0$ (equivalently for some $\lambda > 0$), the map $(t, u) \to J_\lambda^{A_t}(u)$ is continuous from $[0, 1] \times X$ into X.

(c) For every $\lambda > 0$ (equivalently for some $\lambda > 0$) and every $u \in X$, the map $t \to J_\lambda^{A_t}(u)$ is continuous from $[0, 1]$ into X.

(d) Suppose $t_n \to t$ in $[0, 1]$ and $(u, u^*) \in GrA_t$. We can find $(u_n, u_n^*) \in GrA_{t_n}, n \in \mathbb{N}$ such that

$$u_n \to u \text{ in } X, u_n^* \to u^* \text{ in } X^*.$$
$$\text{(that is, } GrA_t \subseteq s - \liminf_{n\to\infty} GrA_{t_n}, \text{ see Definition I-8.6).}$$

In contrast to $(S)_+$-homotopies, affine homotopies in general are not pseudomonotone. More precisely, we have:

Proposition 1.101 *If $A : D(A) \subseteq X \to 2^{X^*}$, $L : D(L) \subseteq X \to 2^{X^*}$ are two maximal montone maps with $(0, 0) \in GrA \cap GrL$ and $K : X \to X^*$ is continuous monotone (thus maximal monotone too, see Proposition I-7.36), then*

(a) *if $\overline{D(A)} \neq \overline{D(L)}$, then $S_t = (1 - t)A + tL, t \in [0, 1]$ is not a pseudomonotone homotopy;*

(b) *If $\overline{D(A)} = X$, then $S_t = (1 - t)A + tK, t \in [0, 1]$ is a pseudomonotone homotopy.*

Proof

(a) We may assume that $\overline{D(A)}\backslash\overline{D(L)} \neq \emptyset$. We can find $u_0 \in \overline{D(A)}\backslash\overline{D(L)}$. Also let $u_0^* \in A(u_0)$ and $t_n = \frac{1}{n+1}$. Suppose S_t is a pseudomontone homontopy. Then by Definition 1.100(d) we can $(u_n, u_n^*) \in S_{t_n}$, $n \in \mathbb{N}$ such that $u_n \to u$ in X and $u_n^* \to u^*$ in X^*. We have $u_n \in D(S_{t_n}) = D(A) \cap D(L)$ for all $n \in \mathbb{N}$, hence $u \in \overline{D(L)}$, a contradiction. This proves that $\{S_t\}_{t\in[0,1]}$ is not a pseudomonotone homotopy.

(b) Let $t_n \to t$ in $[0, 1]$ and $(u, u^*) \in GrS_t$. If $0 \leq t < 1$, then $u^* = (1-t)v^* + tK(u)$ with $v^* \in A(u)$. We set $u_n = u$ and $u_n^* = (1-t_n)v^* + t_nK(u_n)$, $n \in \mathbb{N}$. We have

$$(u_n, u_n^*) \in GrA_{t_n}, n \in \mathbb{N}, (u_n, u_n^*) \to (u, u^*) \text{ in } X \times X^*.$$

If $t = 1$, then for $n \in \mathbb{N}$ such that $t_n = 0$, we set $u_n = u$, $u_n^* = u^*$ and for $n \in \mathbb{N}$ such that $t_n > 0$ we set

$$u_n = J_{t_n}^A(u), u_n^* \in (1-t_n)A_{t_n}(u) + t_nK(u_n), n \in \mathbb{N}.$$

We have $(u_n, u_n^*) \in GrS_{t_n}$ for all $n \in \mathbb{N}$. Also

$$\begin{aligned} &u_n \to u \text{ in } X \text{ (since } \overline{D(A)} = X, \text{ see Proposition I-7.52)}, \\ \Rightarrow\ &K(u_n) \to K(u) \text{ (since } K(\cdot) \text{ is continuous)} \end{aligned}$$

We have

$$\begin{aligned} \| (1-t_n)A_{t_n}(u_n) \|_* &= \| \mathcal{F}(u - u_n) \|_* \text{ (see Volume I, } p.446) \\ &= \| u - u_n \| \to 0 \end{aligned}$$

So, finally $u_n^* \to K(u) = u^*$ in X^* and this proves that $\{S_t\}_{t\in[0,1]}$ is a pseudomonotone homotopy.

□

Now that we have the admissible homotopies, based on Definition 1.98 and Theorem 1.86, we can state the theorem which summarizes the main properties of the degree map $d_p(\cdot)$.

Theorem 1.102 *If $d_p(\varphi + A, \Omega, u^*)$ is the degree map from Definition 1.98, then*

(a) *Normalization: $d_p(\mathcal{F}, \Omega, u^*) = 1$ if $u^* \in \mathcal{F}(\Omega)$ (recall $\mathcal{F} : X \to X^*$ is the duality map);*

(b) *Solution property: If $d_p(\varphi + A, \Omega, u^*) \neq 0$, then there exist $u \in \Omega$ such that $u^* \in \varphi(u) + A(u)$;*

(c) *Homtopy invarivance: $d_p(\varphi_t + A_t, \Omega, u^*)$ is independent of $t \in [0, 1]$ when $\{\varphi_t\}_{t\in[0,1]}$ is an $(S)_+$-homotopy with each φ_t bounded, $\{A_t\}_{t\in[0,1]}$ is a pseudomonotone homotopy and $u^* \notin (\varphi_t + A_t)(\partial\Omega \cap D(A_t))$ for all $t \in [0, 1]$;*

(d) *Domain Additivity: If* $\Omega_1, \Omega_2 \subseteq \Omega$ *are nonempty open disjoint and* $u^* \notin (\varphi + A)(\overline{\Omega}\backslash(\Omega_1 \cup \Omega_2))$, *then* $d_p(\varphi + A, \Omega, u^*) = d_p(\varphi + A, \Omega_1, u^*) + d_p(\varphi + A, \Omega_2, u^*)$;

(e) *Excision property: If* $C \subseteq \overline{\Omega}$ *is nonempty closed and* $u^* \notin (\varphi + A)(C)$, *then* $d_p(\varphi + A, \Omega, u^*) = d_p(\varphi + A, \Omega\backslash C, u^*)$.

Remark 1.103 We can have an extended normalization properly which is the following: "If $\Omega \subseteq X$ is nonempty, bounded, open, $A : D(A) \subseteq X \to 2^{X^*}$ is maximal monotone with $(0, 0) \in Gr\, A$ and $u^* \in (\mathcal{F} + A)(\Omega)$, then $d_p(\mathcal{F} + A, \Omega, u^*) = 1$".

Finally, when $A = \partial g$ with $g \in \Gamma_0(X)$, then we can have an extension of Theorem 1.95 (see Kobayashi and Otani [273]).

Theorem 1.104 *If* $\Omega \subseteq X$ *is nonempty, open,* $f \in C^1(\Omega, \mathbb{R})$ *with* $f' : \Omega \to X^*$ *is bounded and* $(S)_+$, $g \in \Gamma_0(X)$ *with* $g(0) \leq g(u)$ *for all* $u \in X$ *and* $u_0 \in \Omega$ *is a local minimizer and an isolated critical point of* $(f + g)(\cdot)$ *(that is,* $0 \in f'(u_0) + \partial g(u_0)$*) and* $0 \notin f'(u) + \partial g(u)$ *for all* u *in a neighborhood of* u_0, *then we can find* $r^* > 0$ *such that* $d_p(f' + \partial g, B_r(u_0), 0) = 1$ *for all* $r \in (0, r^*]$.

If $\varphi = 0$, then we have a degree for maximal monotone operators $A : D(A) \subseteq X \to 2^{X^*}$ with $(0, 0) \in Gr\, A$. We denote this degree by $d_M(\cdot)$. This degree has all the basic properties, namely normalization, solution property, homotopy invariance (with admissible homotopies the pseudomonotone ones), domain additivity and excision property.

Proposition 1.105 *If* $\{A_t\}_{t\in[0,1]}$ *is a pseudomonotone homotopy and for some* $t_0 \in [0, 1]$, $d_M(A_{t_0}, \Omega, u^*) \neq 0$, *then for every* $t \in [0, 1]$, $\emptyset \neq A_t^{-1}(0) \subseteq \Omega$.

Proof Follows from the homotopy invariance and solution properties of the degree d_M. □

Proposition 1.106 *If* $A : D(A) \subseteq X \to 2^{X^*}$, $L : D(L) \subseteq X \to 2^{X^*}$ *are two maximal monotone maps,* $D(A) \cap int\, D(L) \neq \emptyset$, $u^* \notin A(\partial\Omega)$ *and* $A(u) = L(u)$ *for all* $u \in \partial\Omega$, *then* $d_M(A, \Omega, u^*) = d_M(L, \Omega, u^*)$ *(boundary dependence).*

Proof Consider the homotopy $S_t = (1 - t)A + tL$ for all $t \in [0, 1]$. It is easily seen that this is a pseudomonotone homotopy and $u^* \notin S_t(\partial\Omega)$ for all $t \in [0, 1]$. So, the homotopy invariance property implies that

$$d_M(S_0, \Omega, u^*) = d_M(S_1, \Omega, u^*),$$
$$\Rightarrow d_M(A, \Omega, u^*) = d_M(L, \Omega, u^*).$$

□

Proposition 1.107 *If* $A : D(A) \subseteq X \to 2^{X^*}$ *is a maximal monotone map with* $(0, 0) \in Gr\, A$ *and* $\vartheta(-\mathcal{F}(u) + u^*) \notin A(u)$ *for all* $u \in \partial\Omega$ *and all* $\vartheta > 1$ *(resp.all* $\vartheta \geq 0$*), then* $d_M(A + \mathcal{F}, \Omega, u^*) = 1$ *if* $u^* \in \mathcal{F}(\Omega)$ *and* $d_M(A + \mathcal{F}, \Omega, u^*) = 0$

if $u^* \notin \mathcal{F}(\Omega)$ *(resp.* $d_M(A, \Omega, u^*) = 1$ *if* $u^* \in \mathcal{F}(\Omega)$ *and* $d_M(A, \Omega, u^*) = 0$ *if* $u^* \notin \mathcal{F}(\Omega)$*).*

Proof We consider the homotopy

$$S_t = tA + \mathcal{F} + u^* \text{ for all } t \in [0, 1].$$

$$(\text{resp.} S_t = tA + (1-t)(\mathcal{F} + u^*)) \text{ for all } t \in [0, 1])$$

Evidently these are pseudomonotone homotopies. Also, we have $u^* \notin S_t(\partial\Omega)$ for all $t \in [0, 1]$. Then the homotopy invariance and normalization properties imply that

$$d_M(A + \mathcal{F}, \Omega, u^*) = d_M(\mathcal{F}, \Omega, u^*) = 1 \text{ if } u^* \in \mathcal{F}(\Omega),$$

$$d_M(A + \mathcal{F}, \Omega, u^*) = 0, \text{ if } u^* \notin \mathcal{F}(\Omega).$$

$$(resp. d_M(A, \Omega, u^*) = d_M(\mathcal{F}, \Omega, u^*) = 1) \text{ if } u^* \in \mathcal{F}(\Omega),$$

$$d_M(A, \Omega, u^*) = 0 \text{ if } u^* \notin \mathcal{F}(\Omega).$$

□

1.4 Degree for Multifunctions

We can extend the Leray-Schauder degree to multifunctions. So, our setting is the following: X is a Banach space and we consider maps of the form $id - F$ with $F(\cdot)$ being a multifunction with values in $P_{f_c}(X) = \{C \subseteq X :$ nonempty, closed, convex$\}$ and which is "compact" in the following sense.

Definition 1.108 We say that a multifunction $F : D \subseteq X \to P_{f_c}(X)$ is "compact" if it is usc and maps bounded sets to relatively compact sets.

From the theory of multifunctions, we know that such a multifunction admits an approximate continuous selection. More precisely, we have the following result (see Theorem I-5.91).

Theorem 1.109 *If* Y *is a metric space,* X *is a Banach space and* $F : Y \to P_{f_c}(X)$ *is an usc multifunction, then given* $\varepsilon > 0$*, we can find a locally Lipschitz function* $f_\varepsilon : Y \to X$ *such that*

$$f_\varepsilon(Y) \subseteq convF(Y), Grf_\varepsilon \subseteq GrF + \varepsilon B_1$$

$$(\text{recall } B_1 = \{x \in X :\| x \| < 1\}).$$

Using these approximate selections, we can define a Leray-Schauder degree for multifunctions of the form $id - F$ with $F(\cdot)$ compact in the sense of Definition 1.108. To do this, we need the following lemma.

Lemma 1.110 *If X is a Banach space, $\Omega \subseteq X$ is nonempty, bounded, open, $F : \overline{\Omega} \to P_{f_c}(X)$ is a compact multifunction and $y \notin (id - F)(\partial\Omega)$, then there exist $\varepsilon_0 > 0$ such that $y \notin (id - f_\varepsilon)(\partial\Omega)$ for all $\varepsilon \in (0, \varepsilon_0)$ (here $f_\varepsilon(\cdot)$ is the $\varepsilon-$approximate continuous selection of $F(\cdot)$), postulated by Theorem 1.109)*

Proof We argue by contradiction. So, suppose we can find $\varepsilon_n \to 0^+$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq \partial\Omega$ such that

$$y = u_n - f_{\varepsilon_n}(u_n) \text{ for all } n \in \mathbb{N} \tag{1.32}$$

Using Theorem 1.109, we can find $(v_n, z_n) \in GrF$ such that $\| u_n - v_n \| < \varepsilon_n$ and

$$\| f_{\varepsilon_n}(u_n) - z_n \| < \varepsilon_n \text{ for all } n \in \mathbb{N} \tag{1.33}$$

Note that $\{v_n\}_{n\in\mathbb{N}} \subseteq X$ is bounded and so the compactness of $F(\cdot)$ implies that $\{z_n\}_{n\in\mathbb{N}} \subseteq X$ is relatively compact. Therefore we may assume that $z_n \to z$ in X. From (1.33) it follows that

$$\begin{aligned} & f_{\varepsilon_n}(u_n) \to z \text{ in } X \text{ as } n \to \infty, \\ \Rightarrow\ & u_n \to y + z \in \partial\Omega \text{ (see (32))}, \\ \Rightarrow\ & v_n \to y + z \text{ (see (33))}. \end{aligned}$$

But since $GrF \subseteq X \times X$ is closed (see Proposition I-5.11) we have $(y + z, z) \in GrF$, which means that

$$\begin{aligned} & z \in F(y + z), \\ \Rightarrow\ & y \in (id - F)(\partial\Omega) \end{aligned}$$

a contradiction to our hypothesis. □

Form Theorem 1.109 we see that $f_\varepsilon(\cdot)$ is compact. Therefore we can define the Leray-Schauder degree of $id - f_\varepsilon$ for $\varepsilon > 0$ small. This leads to the following degree for multifunctions.

Definition 1.111 Let X be a Banach space, $\Omega \subseteq X$ is nonempty, bounded, open, $F : \overline{\Omega} \to P_{f_c}(X)$ is a compact multifunction and $y \notin (id - F)(\partial\Omega)$. Then the $Leray - Schauder$ degree on the triple $(id - F, \Omega, y)$ is defined by $d_0(id - F, \Omega, y) = \lim_{\varepsilon\to 0} d_{LS}(id - f_\varepsilon, \Omega, y)$.

Remark 1.112 We show that this is a well-defined notion. As we already observed, on account of Lemma 1.110 $d_{LS}(id - f_\varepsilon, \Omega, y)$ exists for $\varepsilon > 0$ small. In fact $d_{LS}(id - f_\varepsilon, \Omega, y)$ eventually stabilizes. To see this, we consider the homotopy

$$h_{\varepsilon,\delta}(t,u) = (1-t)f_\varepsilon(u) + tf_\delta(u) \text{ for all } (t,u) \in [0,1] \times \overline{\Omega}.$$

We show that we can find $\varepsilon_1 \in (0, \varepsilon_0]$ such that

$$y \notin (id - h_{\varepsilon,\delta}(t,\cdot))(\partial\Omega) \text{ for all } t \in [0,1], \textit{ all } \varepsilon, \delta \in (0, \varepsilon_0).$$

If this is not true, then there exist sequences $\{\varepsilon_n, \delta_n\}_{n\in\mathbb{N}} \subseteq (0,\infty)$ with $\varepsilon_n, \delta_n \to 0^+$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq \partial\Omega$ such that

$$y = u_n - h_{\varepsilon_n,d_n}(t_n, u_n) \text{ for all } n \in \mathbb{N}. \tag{1.34}$$

On account of the compactness of $F(\cdot)$ and using Theorem 1.109 we see that

$$\{f_{\varepsilon_n}(u_n)\}_{n\in\mathbb{N}}, \{f_{\delta_n}(u_n)\}_{n\in\mathbb{N}} \subseteq X$$

are relatively compact. So, we may assume that

$$t_n \to t \in [0,1],\ f_{\varepsilon_n}(u_n) \to v,\ f_{\delta_n}(u_n) \to z \text{ in } X.$$

Then from (1.34), it follows that

$$u_n \to y + (1-t)v + tz = w \in \partial\Omega$$

Note that $v, z \in (id - F)(\partial\Omega)$. Hence

$$\begin{aligned} &y = (1-t)(w-v) + t(w-z) \in (1-t)(id-F)(w) + t(id-F)(w) \\ &\Rightarrow y \in (id-F)(\partial\Omega) \text{ (since } F(\cdot) \text{ is convex-valued)}, \end{aligned}$$

a contradiction. So, $y \notin (id - h_{\varepsilon,\delta}(t,\cdot))(\partial\Omega)$ for all $t \in [0,1]$, all $\varepsilon, \delta > 0$ small. The homotopy invariance property of the Leray-Schauder degree implies that

$$d_{LS}(id - f_\varepsilon, \Omega, y) = d_{LS}(id - f_\delta, \Omega, y) \text{ for all } \varepsilon, \delta > 0 \textit{ small}.$$

So, $d_{LS}(id - f_\varepsilon, \Omega, y)$ eventually stabilizes and this implies that d_0 is well-defined (see Definition 1.111).

This degree has all the basic properties.

Theorem 1.113 *For every triple $(id - F, \Omega, y)$ as above, we have:*

(a) *Normalization: $d_0(id, \Omega, y) = 1$ provided $y \in \Omega$;*
(b) *Boundary dependence: if $G(\cdot)$ is another compact multifunction such that $G\,|_{\partial\Omega} = F\,|_{\partial\Omega}$, then $d_0(id - F, \Omega, y) = d_0(id - G, \Omega, y)$;*
(c) *Solution property: if $d_0(id - F, \Omega, y) \neq 0$, then there exist $u \in \Omega$ such that $y \in u - F(u)$;*

(d) *Homotopy invariance: if* $h : [0, 1] \times \overline{\Omega} \to P_{fc}(X)$ *is a compact multifunction and* $y \notin (id - h(t, \cdot))(\partial\Omega)$ *for all* $t \in [0, 1]$, *then* $d_0(id - h(t, \cdot), \Omega, y)$ *is independent of* $t \in [0, 1]$;
(e) *Domain additivity: if* Ω_1, Ω_2 *are disjoint open subsets of* Ω, $y \notin (id - F)(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$, *then* $d_0(id - F, \Omega, y) = d_0(id - F, \Omega_1, y) + d_0(id - F, \Omega_2, y)$;
(f) *Excision property: if* $C \subseteq \overline{\Omega}$ *is closed and* $y \notin (id - F)(C) \cup (id - F)(\partial\Omega)$, *then* $d_0(id - F, \Omega, y) = d_0(id - F, \Omega \setminus C, y)$.

The Borsuk theorem remains valid.

Theorem 1.114 *If* $(id - F, \Omega, 0)$ *is a triple as above, with* Ω *also symmetric and for every* $u \in \partial\Omega$ *we have*

$$F(-u) = -F(u), \tag{1.35}$$

then $d_0(id - F, \Omega, 0)$ *is odd.*

We can extend this result as follows.

Proposition 1.115 *If* $(id - F, \Omega, 0)$ *is a triple as above, with* Ω *also symmetric and for every* $u \in \partial\Omega$ *we have*

$$F(-u) \cap (-F(u)) \neq \varnothing,$$

then $d_0(id - F, \Omega, 0)$ *is odd.*

Proof Let $\widehat{F}(u) = F(-u) \cap (-F(u))$ for all $u \in \overline{\Omega}$. Then $\widehat{F}(\cdot)$ is compact and odd on $\partial\Omega$ (see (8.35)). 8.34 If $h(t, u) = (1-t)F(u) + t\widehat{F}(u)$ for all $(t, u) \in [0, 1] \times \overline{\Omega}$, then $h(\cdot, \cdot)$ is a multivalued compact homotopy and $0 \notin h(t, \partial\Omega)$ for all $t \in [0, 1]$. So, the homotopy invariance property of d_0 implies that

$$d_0(id - F, \Omega, 0) = d_0(id - \widehat{F}, \Omega, 0)$$

is odd (see Theorem 1.114). □

We present some more results concerning the computation of d_0. Such results are useful when we use topological methods in the study of boundary value problems.

Proposition 1.116 *If* $(id - F, \Omega, o)$ *is a triple as above, then*

(a) $d_0(id - F, \Omega, 0) = 0$ *if* $u \notin tF(u)$ *for all* $u \in \partial\Omega$ *and all* $t \in [0, 1]$;
(b) $d_0(id - F, \Omega, 0) = 0$ *if* $u \notin F(u) + ty_0$ *for every* $u \in \partial\Omega$, *every* $t \geq 0$ *and some* $y_0 \in X \setminus \{0\}$.

Proof

(a) Consider the multivalued homotopy $h(t, u) = tF(u)$ for all $(t, u) \in [0, 1] \times \overline{\Omega}$. Evidently this is a compact multivalued homotopy and by hypothesis $0 \notin (id -$

$h(t,\cdot))(\partial\Omega)$ for all $t \in [0,1]$. So, using the homotopy invariance property, we have $d_0(id - F, \Omega, 0) = d_0(id, \Omega, 0) = 1$;

(b) For this part, we consider the multivalued homotopy $h(t,u) = F(u) + ty_0$ for all $(t,u) \in [0,1] \times \overline{\Omega}$. This is a compact multivalued homotopy and by hypothesis $0 \notin (id - h(t,\cdot))(\partial\Omega)$ for all $t \in [0,1]$. So, the homotopy invariance property implies that $d_0(id - F, \Omega, 0) = d_0(id - F - ty_0, \Omega, 0)$. But for $t \geq 0$ large $ty_0 \notin (id - F)(\overline{\Omega})$ and so $d_0(id - F - ty_0, \Omega, 0) = d(id - F, \Omega, ty_0), \Rightarrow d_0(id - F, \Omega, 0) = 0$.

□

Corollary 1.117 *If $X = \mathbb{R}^N$, $\Omega \subseteq \mathbb{R}^N$ is nonempty, bounded, open, $G : \overline{\Omega} \to P_{k_c}(\mathbb{R}^m)$ is usc, $0 \notin G(\partial\Omega)$ and $m < N$, then $d_0(G, \Omega, 0) = 0$.*

Proof Let $G = id - F$ with $F = id - G$. Then for any $y_0 \in \mathbb{R}^N \setminus \mathbb{R}^m$, we use Proposition 1.116(b) and conclude that $d_0(G, \Omega, 0) = 0$. □

Next we consider multivalued perturbations of nonlinear maps of monotone type.

The mathematical setting is the following: X is a reflexive Banach space, equivalently renormed so that both X and X^* are locally uniformly convex (Troyanski's theorem), $\Omega \subseteq X$ is nonempty, bounded, open, $\varphi : \overline{\Omega} \to X^*$ is bounded, demicontinuous, $(S)_+$, $A : D(A) \subseteq X \to 2^{X^*}$ is maximal monotone with $(0,0) \in GrA$, $F : \overline{\Omega} \to P_{kc}(X^*)$ is compact (recall $P_{kc}(X^*) = \{C \subseteq X^* : \text{nonempty, compact, convex}\}$ and $u^* \notin (\varphi + A + F)(\partial\Omega)$). We are going to define a degree for such a triple $(\varphi + A + F, \Omega, u^*)$.

We will reduce the definition of the degree on the triple $(\varphi + A + F, \Omega, u^*)$ to the $d_p(\cdot)$ degree of $(\varphi + A + f_\varepsilon, \Omega, u^*)$ with f_ε being the ε-approximate continuous selection of $F(\cdot)$ postulated by Theorem 1.109. We can do that because $f_\varepsilon(\cdot)$ is compact and so the $(S)_+$-property is preserved by compact perturbations (that is, if $\varphi : \overline{\Omega} \to X^*$ is of type $(S)_+$ and $f_o : \overline{\Omega} \to X^*$ is compact, then $\varphi_0 + f_0$ is of type $(S)_+$).

In the next proposition, we execute this process.

Proposition 1.118 *If $(\varphi + A, \Omega, u^*)$ is a triple as above, then*

(a) *$u^* \notin (\varphi + A + f_\varepsilon)$ for all $\varepsilon > 0$ small;*
(b) *$d_p(\varphi + A, f_\varepsilon, \Omega, u^*)$ is defined and stabilizes for $\varepsilon > 0$ small.*

Proof

(a) We argue by contradiction. So, suppose we can find $\varepsilon_n \to 0^+$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq \partial\Omega$ such that

$$u^* \in (\varphi + A + f_{\varepsilon_n})(u_n) \tag{1.36}$$

for all $n \in \mathbb{N}$.

Since $\{u_n\}_{n\in\mathbb{N}} \subseteq \partial\Omega$ is bounded and X is reflexive, we may assume that $u_n \xrightarrow{w} u$ in X. From (1.36) we have

$$v_n^* = u^* - (\varphi(u_n) + f_{\varepsilon_n}) \in A(u_n) \tag{1.37}$$

for all $n \in \mathbb{N}$.

On account of Theorem 1.109 and the hypotheses on $\varphi(\cdot)$, we can say that

$$f_{\varepsilon_n}(u_n) \to y^* \text{ and } \varphi(u_n) \xrightarrow{w} z^* \text{ in } X^* \Rightarrow v_n^k \xrightarrow{w} v^* = u^* - (z^* + y^*)) \text{ in } X^*.$$

We have

$$\lim_{n\to\infty} \langle v_n^* + \varphi(u_n), u_n - u\rangle = \lim_{n\to\infty} \langle u^* - f_{\varepsilon_n}(u_n), u_n - u\rangle = 0$$

(see (1.37)).

By hypothesis φ is demicontinuous and of type $(S)_+$. Hence $\varphi(\cdot)$ is pseudomonotone (see proposition I-7.72) and so we have

$$0 \leq \liminf_{n\to\infty} \langle \varphi(u_n), u_n - u\rangle \text{ (see Definition I-7.67).}$$

Then from (8.37) we obtain

$$\lim_{n\to\infty} \sup \langle v_n^*, u_n - u\rangle \leq 0. \tag{1.38}$$

We have $v_n^* \in A(u_n)$ for all $n \in \mathbb{N}$ (see (1.37)) and by hypothesis $A(\cdot)$ is maximal monotone. So, it is generalized pseudomonotone (see Proposition I-7.73) and from (1.38) we have

$$(u, v^*) \in GrA, \langle v_n^*, u_n\rangle \to \langle v^*, u\rangle.$$

Therefore

$$\langle \varphi(u_n), u_n - u\rangle \to 0, \Rightarrow u_n \to u \in \partial\Omega \text{ in } X$$

(recall that $\varphi(\cdot)$ is of type $(S)_+$).

Also $\varphi(u) = z^*$. Moreover, since $F(\cdot)$ is a compact multifunction, we infer that $y^* \in F(u)$. Therefore finally we have $u^* \in (\varphi + A + F)(u)$ with $u \in \partial\Omega$, a contradiction.

(b) Since $f_\varepsilon(\cdot)$ is compact, $\varphi + f_\varepsilon$ remains bounded, demicontinuous and of type $(S)_+$. Hence on account of (a) $d_p(\varphi + A + f_\varepsilon, \Omega, u^*)$ is defined for $\varepsilon > 0$ small. We need to show that it stabilizes. Again we argue by contradiction. So, suppose we can find $0 < \varepsilon_n < \delta_n \to 0$ and $d_p(\varphi + A + f_{\varepsilon_n}, \Omega, u^*) \neq d_p(\varphi + A + f_{\delta_n}, \Omega, u^*)$. The homotopy invariance property of d_p implies that we can find $t_n \to t$ in $[0, 1]$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq \partial\Omega$ such that

$$u^* \in (\varphi + A + t_n f_{\varepsilon_n} + (1 - t_n) f_{\delta_n})(u_n)$$

for all $n \in \mathbb{N}$.

From Theorem 1.109 we have

$$t_n f_{\varepsilon_n} + (1 - t_n) f_{\delta_n}(x) \in conv[F(B_{\delta_n}(x) \cap \overline{\Omega}) + B^*_{\delta_n}]$$

for all $x \in \overline{\Omega}$.

Then arguing as in (a) we obtain

$$u^* \in (\varphi + A + F)(\partial\Omega),$$

a contradiction. So, $d_p(\varphi + A + f_\varepsilon, \Omega, u^*)$ stabilizes for $\varepsilon > 0$ small.

□

This proposition leads naturally to the following definition.

Definition 1.119 Let $\Omega \subseteq X$ be nonempty, bounded, open, $\varphi : \overline{\Omega} \to X^*$ is bounded, demicontinuous, of type $(S)_+$, $F : \Omega \to P_{kc}(X^*)$ is a compact multifunction. We define $\widehat{d}_0(\varphi + A + F, \Omega, u^*)$ to be the common value of $d_p(\varphi + A + f_\varepsilon, \Omega, u^*)$ for $\varepsilon > 0$ small.

Remark 1.120 If $F = \widehat{f}$ is single valued, then

$$\widehat{d}_0(\varphi + A + \widehat{f}, \Omega, u^*) = d_p(\varphi + A + \widehat{f}, \Omega, u^*).$$

This new degree has the basic properties.

Theorem 1.121

(a) *Normalization:* $\widehat{d}_0(\mathcal{F}, \Omega, u^*) = 1$ *if* $u^* \in \mathcal{F}(\Omega)$;

(b) *Solution property: If* $\widehat{d}_0(\varphi + A + F, \Omega, u^*) \neq 0$, *then we can find* $u \in \Omega$ *such that* $u^* \in (\varphi + A + F)(u)$;

(c) *Homotopy invariance property: Let* $\{\varphi_t\}_{t\in[0,1]}$ *be an* $(S)_+$*-homotopy,* $\{A_t\}_{t\in[0,1]}$ *a pseudomonotone homotopy with* $(0, 0) \in GrA_t$ *for all* $t \in [0, 1]$ *and* $\{F_t\}_{t\in[0,1]}$ *a compact multivalued homotopy and* $u^* \notin (\varphi_t + A_t + F_t)(\partial\Omega)$ *for all* $t \in [0, 1]$. *Then* $\widehat{d}_0(\varphi_t + A_t + F_t, \Omega, u^*)$ *is independent of* $t \in [0, 1]$;

(d) *Domain additivity: Let* Ω_1, Ω_2 *be nonempty disjoint open subsets of* Ω *and* $u^* \notin (\varphi + A + F)(\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2))$. *Then* $\widehat{d}_0(\varphi + A + F, \Omega, u^*) = \widehat{d}_0(\varphi + A + F, \Omega_1, u^*) + \widehat{d}_0(\varphi + A + F, \Omega_2, u^*)$;

(e) *Excision property: If* $C \subseteq \overline{\Omega}$ *is closed and* $u^* \notin (\varphi + A + F)(C) \cup (\varphi + A + F)(\partial\Omega)$, *then* $\widehat{d}_0(\varphi + A + F, \Omega, u^*) = \widehat{d}_0(\varphi + A + F, \Omega \setminus C, u^*)$.

Remark 1.122 We can extend this degree to maps of the form $\varphi + A + F$ with A and F as before, but now $\varphi(\cdot)$ is only bounded and generalized pseudomonotone (hence demicontinuous too). Note that in this case for $\varepsilon > 0$, $\varphi + \varepsilon \widetilde{f}$ is bounded, demicontinuous and of type $(S)_+$. Then using the same approach as above we can define $\widehat{d}^*_0(\varphi + A + F, \Omega, u^*) = \lim_{\varepsilon \downarrow 0} \widehat{d}_0(\varphi + A + F + \varepsilon \widetilde{f}, \Omega, u^*)$.

Finally we mention that the degree $\widehat{d}_0$ is the unique degree which can be defined on triples $(\varphi + A + F, \Omega, u^*)$ which satisfies the $\mathcal{F}$-normalization condition, the domain additivity and is invariant under all affine homotopies of the form $(1 - t)(\varphi + A + F) + th$ with $h : \overline{\Omega} \to X^*$ bounded, demicontinuous, of type $(S)_+$ and $\overline{D(A)} = X$.

Remarks

1.1: Degree theory started the work of Brouwer [70–72] and provides an algebraic count of the solutions of an equation $y_0 = f(x)$ with $f : \overline{\Omega} \to \mathbb{R}^N$ continuous, $\Omega \subseteq \mathbb{R}^N$ bounded, open and $y_0 \in \mathbb{R}^N$. The approach of Brouwer was based on combinatorial arguments and on elements from Algebraic Topology. This approach is also used in the books of Cronin [113], Granas and Dugundjc [215] and Hatcher [233]. Here we present an analytic construction of the Brouwer degree due to Heinz [238]. A similar analytic approach can be found in Nagumo [348]. The analytic construction is used in the books of Deimling [133], Denkowski et al. [136], Fonseca and Gangbo [167], Krawcewicz and Wu [280], Lloyd [317], Papageorgiou et al. [397], Schwartz [437].
If was proved independently by Führer [173] and Amann and Weiss [12] that the Brouwer degree is the only topological degree which can be defined in finite dimensional space. In fact Amann and Weiss [12] proved the uniqueness of the Leray-Schauder degree and as a consequence concluded also the uniqueness of the Brouwer degree. Theorem 1.35 was proved by Borsuk [56]. The Borsuk-Ulam Theorem (see Theorem 1.41) was conjectured by Ulam and proved by Borsuk [58]. Theorem 1.44 was first proved by Ljusternik and Schnirelmann [316] in their book on topological methods in analysis.

1.2: The limitation of the Brouwer degree is clear, since most problems we encounter in nonlinear functional analysis are infinite dimensional. So, we need an extension of the degree to infinite dimensional spaces. Example 1.59 shows that we can not do that for all continuous functions. The first major breakthrough in this direction, was made by Leray and Schauder [294], who were able to extend the degree to the class of maps which are compact perturbations of the identity.
In Proposition 1.27 we proved that S^{N-1} is not a retract of $\overline{B}_1^N$. In infinite dimensional spaces this changes and ∂B_1 is a retract of $\overline{B}_1$. This question was first asked by Ulam around 1935. It was answered positively for Hilbert spaces by Kakutani [264] and the general case was proved by Dugundji [147].

Let X be a Banach space and B is the family of all bounded subsets of X.

Definition 1.123

(a) The "Kuratowski measure of noncompactness" $\alpha : B \to \mathbb{R}_+$ is defined by $\alpha(C) = inf\{d > 0 : C \text{can be covered by a finite number of sets of diameter} \leq d\}$;

(b) The "Hausdorff (ball) measure of noncompactness" $\beta : B \to \mathbb{R}_+$ is defined by $\beta(C) = inf\{r > 0 : C \text{can be covered by a finite number of r-balls}\}$.

Remark 1.124 Evidently $\beta \leq \alpha \leq 2\beta$ and for Hilbert spaces we have $\sqrt{2}\beta \leq \alpha \leq 2\beta$. Moreover is $\omega = \alpha$ or β, then $\omega(C) = 0$ if and only if $\overline{C}$ is compact.

Definition 1.125 Let $K \subseteq X, Y$ is another Banach space and $f : K \to Y$ is a bounded, continuous map. In what follows by ω_X (resp. ω_Y) we denote either α or β on X (resp. on Y). We say that $f(\cdot)$ is "condensing", if for all $C \subseteq K$ with $\omega_X(C) > 0$ we have $\omega_Y(f(C)) < \omega_X(C)$.

Condensing maps were first considered by Sadovski [434] and Furi and Vignoli [174]. Nussbaum [352] extended the Leray-Schauder degree to maps of the form $id - f$, with $f(\cdot)$ considering (see also Deimling [133] and Hu and Papagcorgiou [247]).

1.3: The degree for $(S)_+$-maps was introduced by Skrypnik [441] using fundamental sets and by Browder [73] using Galerkin approximations. A third construction was proved by Berkovits and Mustonen [41], who used the Leray-Schauder degree and the Brower-Ton embedding theorem (see Browder and Ton [74]). The degree for the maps $\varphi + A$ with $\varphi(\cdot)$ bounded, demicontinuous, $(S)_+$ and A maximal monotone, is due to Browder [73]. The condition that $(0, 0) \in GrA$ can be removed as it was shown by Kien et al. [267] and Wang and Huang [468].

1.4: Degree theory for multifunctions, was initiated by Granas [214] using tools from Algebraic Toplogy. An analytic approach based on h-continuous approximations of $F(\cdot)$, can be found in Hukuhara [253], Cellina and Lasota [87], employed a more general approach based on continuous, ε-approximate selections which we present here. Extension to condensing multifunctions, can be found in Webb [469]. Finally the degree for multifunctions of the form $\varphi + A + F$, was defined by Hu and Papageorgiou [246] (see also Hu and Papageogiou [247], chapter IV).

Chapter 2
Fixed Point Theory



Introduction

In this chapter we deal with fixed point theory, which is one of the basic mathematical tools in proving the existence of solutions for different types of equations and inclusions. Fixed point theory uses tools from Analysis and Topology and this leads to an informal classification of the results to "metric fixed point theory"(here the metric structure of the ambient space and the metric properties of the maps play a central role), "topological fixed point theory"(here the topological properties of the spaces and/or of the maps are the main tools) and "order fixed point theory"(now the partial order induced by a cone, is the basic ingredient). The third class also leads to the study of the fixed point index. The first three sections of this chapter discuss the results of these three theories. Finally in Sect. 2.4, we present fixed point theorems for multifunctions.

2.1 Metric Fixed Point Theory

In this section we deal with fixed point theorems in which the metric structure on the ambient space and the corresponding geometric conditions on the functions, are the main ingredients. The prototype metric fixed point theorem, is the celebrated "Banach's Fixed Point Theorem".

Definition 2.1 Let (X, d) be a metric space. A map $\varphi : X \to X$ is said to be "Lipschitz", if there exists $\eta > 0$ such that

$$d(\varphi(u), \varphi(v)) \leq \eta d(u, v) \text{ for all } u, v \in X.$$

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The smallest $\eta > 0$ for which this is true, is known as the "Lipschitz constant of φ" and is denoted by $\mathrm{Lip}(\varphi)$. If $\mathrm{Lip}(\varphi) < 1$, then we say that $\varphi(\cdot)$ is a "contraction". If $\mathrm{Lip}(\varphi) = 1$, then we say that $\varphi(\cdot)$ is "nonexpansive". Finally we say that $\varphi(\cdot)$ is "contractive" if $d(\varphi(u), \varphi(v)) < d(u, v)$ for all $u, v \in X$, $u \neq v$.

Remark 2.2 If $\varphi, \psi : X \to X$ are both Lipschitz, then

$$\mathrm{Lip}(\varphi \circ \psi) \leq \mathrm{Lip}(\varphi)\,\mathrm{Lip}(\psi).$$

So, it follows that if $n \in \mathbb{N}_0$, then

$$\mathrm{Lip}(\varphi^{(n)}) \leq \mathrm{Lip}(\varphi)^n$$

with $\varphi^{(0)} = id$, $\varphi^{(n)} = \varphi \circ \varphi \circ \cdots \circ \varphi$(n-times). If $\vartheta > 0$, then $\mathrm{Lip}(\vartheta\varphi) = \vartheta\,\mathrm{Lip}(\varphi)$.

The next result is the "Banach contraction principle".

Theorem 2.3 *If (X, d) is a complete metric space and $\varphi : X \to X$ is a contraction, then $\varphi(\cdot)$ has a unique fixed point $u \in X$(that is, $\varphi(u) = u$) and for any $v \in X$ we have*

$$\lim_{n\to\infty} \varphi^{(n)}(v) = u,$$

$$d(u, \varphi^{(n)}(v)) \leq \frac{k^n}{1-k} d(v, \varphi(v)),$$

with $k = \mathrm{Lip}(\varphi) < 1$.

Proof The uniqueness of the fixed point is straightforward. Indeed if $\varphi(u) = u$ and $\varphi(v) = v$, then

$$\begin{aligned} d(u, v) = d(\varphi(u), \varphi(v)) &\leq k d(u, v), \\ \Rightarrow d(u, v) = 0, \text{ that is , } u &= v. \end{aligned}$$

Now we show the existence of the fixed point. For $n \in \mathbb{N}$ and $v \in X$, we have

$$\begin{aligned} d(\varphi^{(n)}(v), \varphi^{(n+1)}(v)) &= d(\varphi(\varphi^{(n-1)}(v)), \varphi(\varphi^{(n)}(v))) \\ &\leq k d(\varphi^{(n-1)}(v), \varphi^{(n)}(v)) \\ &\cdots \leq k^n d(v, \varphi(v)). \end{aligned} \tag{2.1}$$

If $n \in \mathbb{N}_0$, $m > n$ and $v \in X$, then

$$\begin{aligned} d(\varphi^{(n)}(v), \varphi^{(m)}(v)) &\leq d(\varphi^{(n)}(v), \varphi^{(n+1)}(v)) + \cdots + d(\varphi^{(m-1)}(v), \varphi^{(m)}(v)) \\ &\leq k^n d(v, \varphi(v)) + \cdots + k^{m-1} d(v, \varphi(v)) \\ &\leq k^n (\sum_{i \geq 0} k^i) d(v, \varphi(v)) \\ &= \frac{k^n}{1-k} d(v, \varphi(v)). \end{aligned} \tag{2.2}$$

Therefore $\{\varphi^{(n)}(v)\}_{n \in \mathbb{N}}$ is a Cauchy sequence in X. The completeness of X implies that $\varphi^{(n)}(v) \to u$ in X. We have

$$u = \lim_{n \to \infty} \varphi^{(n+1)}(v) = \lim_{n \to \infty} \varphi(\varphi^{(n)}(v)) = \varphi(u).$$

Passing to the limit as $m \to \infty$ in (2), we obtain

$$d(u, \varphi^{(n)}(v)) \leq \frac{k^n}{1-k} d(v, \varphi(v)).$$

□

Remark 2.4 If φ is nonexpansive (that is, $\text{Lip}(\varphi) = 1$), then the result fails. The following simple example illustrates this. Consider $\varphi : \mathbb{R} \to \mathbb{R}$ defined by $\varphi(u) = u + 1$ for all $u \in \mathbb{R}$. Evidently $\varphi(\cdot)$ is nonexpansive (in fact an isometry), but $\varphi(\cdot)$ does not have a fixed point.

We can have a local version of the above fixed point theorem.

Proposition 2.5 *If (X, d) is a complete metric space, $r > 0$, $B_r(u_0) = \{u \in X, d(u, u_0) < r\}$ and $\varphi : B_r(u_0) \to X$ is a contraction satisfying $d(u_0, \varphi(u_0)) < (1-k)r (k = \text{Lip}(\varphi))$, then $\varphi(\cdot)$ has a unique fixed point.*

Proof Let $\varrho \in (0, r)$ such that $d(u_0, \varphi(u_0)) \leq (1-k)\varrho < (1-k)r$. Let $\overline{B}_\varrho(u_0) = \{u \in X : d(u, u_0) \leq \varrho\}$. We claim that $\varphi : \overline{B}_\varrho(u_0) \to \overline{B}_\varrho(u_0)$. To this end let $v \in \overline{B}_\varrho(v_0)$. We have

$$\begin{aligned} d(u_0, \varphi(v)) &\leq d(u_0, \varphi(u_0)) + d(\varphi(u_0), \varphi(v)) \\ &\leq (1-k)\varrho + k\varrho = \varrho. \end{aligned}$$

So, indeed $\varphi : \overline{B}_\varrho(u_0) \to \overline{B}_\varrho(u_0)$ and we can apply Theorem 2.3 to produce a unique fixed point for $\varphi(\cdot)$. □

We can use this local version of the Banach fixed point theorem, to obtain an invariance of domain result.

Proposition 2.6 *If X is a Banach space, $\Omega \subseteq X$ is nonempty open, $\varphi : \Omega \to X$ is a contraction and $\vartheta = id - \varphi$, then*

(a) *$\vartheta : \Omega \to X$ is open;*
(b) *$\vartheta : \Omega \to \vartheta(\Omega)$ is a homeomorphism.*

Proof

(a) Let $u \in \Omega$ and choose $r > 0$ so that $B_r(u) \subseteq \Omega$. Choose $u_0 \in B_{(1-k)r}(\vartheta(u))$ and consider $\psi : B_r(u) \to X$ defined by $\psi(v) = u_0 + \varphi(v)$ for all $v \in B_r(u)$. Evidently $\psi(\cdot)$ is a contraction and $\mathrm{Lip}(\psi) = \mathrm{Lip}(\varphi) < 1$. We have

$$\|u - \psi(u)\| = \|u - u_0 - \varphi(u)\| = \|u_0 - \vartheta(u)\| < (1-k)r.$$

Apply Proposition 2.5 to obtain a unique fixed point $\widehat{u}$ of $\psi(\cdot)$. So, $\widehat{u} = u_0 + \varphi(\widehat{u})$, hence $\vartheta(\widehat{u}) = u_0$ and so

$$B_{(1-k)r}(\vartheta(u)) \subseteq \vartheta(B_r(u)),$$
$$\Rightarrow \vartheta(\cdot) \text{ is open.}$$

(b) Let $u, v \in \Omega$. Then

$$\|\vartheta(u) - \vartheta(v)\| \geq \|u - v\| - \|\varphi(u) - \varphi(v)\|$$
$$\geq (1-k)\|u - v\|,$$
$$\Rightarrow \vartheta(\cdot) \text{ is injective.}$$

So, $\vartheta : \Omega \to \vartheta(\Omega)$ is a continuous, open bijection, thus it is a homeomorphism. □

Corollary 2.7 *If X is a Banach space and $\varphi : X \to X$ is a contraction, then $\vartheta = id - \varphi : X \to X$ is a homeomorphism.*

Remark 2.8 If $X = H =$ *a* Hilbert space, then $\vartheta(\cdot)$ is maximal monotone, strictly monotone (see Example I-7.28).

A close examination of the proof of Theorem 2.3, leads to the following generalized version of the fixed point result.

Proposition 2.9 *If (X, d) is a complete metric space, $\varphi : X \to X$ and for some $n \in \mathbb{N}$, $\varphi^{(n)} : X \to X$ is a contraction, then $\varphi(\cdot)$ has a unique fixed point.*

Proof By Theorem 2.3, $\varphi^{(n)}(\cdot)$ has a unique fixed point $u \in X$. So,

$$\varphi^{(n)}(u) = u$$

$$\Rightarrow \varphi(u) = \varphi(\varphi^{(n)}(u)) = \varphi^{(n)}(\varphi(u)),$$

$$\Rightarrow \varphi(u) = u \text{ (uniqueness of the fixed point } \varphi^{(n)}(\cdot)).$$

So, u is a fixed point of $\varphi(\cdot)$ and clearly this is the unique fixed point of φ. □

Remark 2.10 Note that in the above result $\varphi(\cdot)$ need not be continuous.

We can also have a parametric version of the Banach fixed point theorem.

Proposition 2.11 *If (X, d) is a complete metric space, L is another metric space(the parameter space) and for every $\lambda \in L$, $\varphi_\lambda : X \to X$ is a contraction with $\text{Lip}(\varphi_\lambda) = \eta < 1$ (independent of $\lambda \in L$) and $\lambda \to \varphi_\lambda$ is continuous, then for every $\lambda \in L$, there exists a unique $u_\lambda \in X$ such that $\varphi_\lambda(u_\lambda) = u_\lambda$ and $\lambda \to u_\lambda$ is continuous.*

Proof The existence of a unique $u_\lambda \in X$ such that $\varphi_\lambda(u_\lambda) = u_\lambda, \lambda \in L$, follows from Theorem 2.3. Suppose $\lambda \to \lambda_0 \in L$. We have

$$\begin{aligned} d(u_\lambda, u_{\lambda_0}) &= d(\varphi_\lambda(u_\lambda), \varphi_{\lambda_0}(u_{\lambda_0})) \\ &\leq d(\varphi_\lambda(u_\lambda), \varphi_\lambda(u_{\lambda_0})) + d(\varphi_\lambda(u_{\lambda_0}), \varphi_{\lambda_0}(u_{\lambda_0})) \\ &\leq k d(u_\lambda, u_{\lambda_0}) + d(\varphi_\lambda(u_{\lambda_0}), \varphi_{\lambda_0}(u_{\lambda_0}))(k = \text{Lip}(\varphi)) \\ \Rightarrow &(1-k) d(u_\lambda, u_{\lambda_0}) \leq d(\varphi_\lambda(u_{\lambda_0}), \varphi_{\lambda_0}(u_{\lambda_0})). \end{aligned}$$

The continuity of $\lambda \to \varphi_\lambda(u_{\lambda_0})$, implies that

$$d(u_\lambda, u_{\lambda_0}) \to 0.$$

□

Using degree theory we can prove a fixed point result for contractions defined on a closed ball $\overline{B}_r = \{u \in X : \|u\| \leq r\}(r > 0)$ of a Banach space X.

Proposition 2.12 *If $\varphi : \overline{B}_r \to X$ is a contraction and $\varphi(\partial B_r) \subseteq \overline{B}_r$, then $\varphi(\cdot)$ has a unique fixed point in $\overline{B}_r$.*

Proof Since $\varphi(\cdot)$ is a contraction, a fixed point, if it exists, it is unique. Consider the map $\vartheta = id - \varphi$. Then if $0 \in \vartheta(\partial B_1)$, we can find $u \in \partial B_1$ such that $u = \varphi(u)$ and so we have produced a fixed point for $\varphi(\cdot)$. So, suppose $0 \notin \vartheta(\partial B_r)$, hence $\vartheta(\partial B_r) \subseteq B_r$. Consider the continuous homotopy $h(t, u) = (id - t\varphi)(u)$ for all $t \in [0, 1]$, all $u \in \overline{B}_r$. So $h(0, \cdot) = id$, $h(1, \cdot) = \vartheta$, hence $0 \notin h(t, \partial\Omega)$ for $t = 0, t = 1$. Suppose $0 \in h(t, \partial\Omega)$ for some $t \in (0, 1)$. Then

$$u = t\varphi(u),$$

$$\Rightarrow r \leq tr < r, \text{ a contradiction.}$$

Therefore, $0 \notin h(t, \partial\Omega)$ for all $t \in [0, 1]$. Note that a contraction is also a set-contraction for $\omega = \alpha$ or $\omega = \beta$. Hence the homotopy invariance of the Nussbaum degree denoted by d_N (see the Remarks of Chap. 1), we have

$$d_N(\vartheta, B_r, 0) = d_N(id, B_r, 0) = 1.$$

Finally the solution property produces $u \in B_r$ such that

$$0 = \vartheta(u),$$
$$\Rightarrow u = \varphi(u).$$

□

We can extend the above result to more general homotopies admissible for the Nussbaum degree.

Definition 2.13 Let X be a Banach space and $\Omega \subseteq X$ a nonempty open set. Consider the triple $(id - \varphi, \Omega, 0)$ with $\varphi : \overline{\Omega} \to \mathbb{R}$ a contraction and $0 \notin (id - \varphi)(\partial\Omega)$. A map $h : [0, 1] \times \overline{\Omega} \to X$ is an admissible homotopy for the Nussbaum degree d_N if the following hold:

(a) $u \neq h(t, u)$ for all $t \in [0, 1]$, all $u \in \partial\Omega$;
(b) there exist $\eta_1, \eta_2 > 0$ such that $d(h(t, u), h(t, v)) \leq \eta_1 d(u, v)$ for all $t \in [0, 1]$, all $u, v \in \overline{\Omega}$ and $d(h(t, u), h(s, u)) \leq \eta_2 |t - s|$ for all $t, s \in [0, 1]$, all $u \in \overline{\Omega}$;
(c) $h(0, \cdot) = \varphi(\cdot)$.

If $\psi : \overline{\Omega} \to X$ is another contraction such that

$$h(1, \cdot) = \psi(\cdot),$$

then we say that φ, ψ are homotopic.

The homotopy invariance property of the Nussbaum degree leads to the following extension of Proposition 2.12.

Proposition 2.14 *If X is a Banach space, $\Omega \subseteq X$ nonempty, bounded, open, $\varphi, \psi : \overline{\Omega} \to X$ are two homotopic contractions and ψ has a fixed point in Ω, then φ has a fixed point in Ω.*

Thus far we have considered contractions. Now we turn our attention to contractive and nonexpansive maps. First we consider contractive maps. In general they need not have a fixed point.

Example 2.15 Let $K = \{u \in C[0, 1] : 0 = u(0) \leq u(t) \leq u(1) = 1$ for all $t \in [0, 1]\}$. Consider the map $\varphi : K \to K$ defined by

$$\varphi(u)(t) = tu(t) \text{ for all } u \in K, \text{ all } t \in [0, 1].$$

We have

$$\|\varphi(u) - \varphi(v)\|_\infty = \sup_{t\in[0,1]} |t(u-v)(t)| < \|u - v\|_\infty$$

$$\Rightarrow \varphi(\cdot) \text{ is contractive.}$$

However, $\varphi(\cdot)$ can not have a fixed point. In this example, K is bounded, closed convex but not compact.

Under a compactness condition on the ambient space, we can have a fixed point.

Theorem 2.16 *If (X, d) is a compact metric space and $\varphi : X \to X$ is contractive, then $\varphi(\cdot)$ admits a unique fixed point.*

Proof The uniqueness follows as for contractions. For the existence, consider the map $\xi : X \to \mathbb{R}_+$ defined by $\xi(u) = d(u, \varphi(u))$. This is a continuous function and so since X is compact we can find $u_0 \in X$ such that $\xi(u_0) = \inf_{u\in X} \xi(u)$. If $\xi(u_0) > 0$, then

$$d(\varphi(\varphi(u_0)), \varphi(u_0)) < d(\varphi(u_0), u_0),$$

$$\Rightarrow \xi(\varphi(u_0)) < \xi(u_0) \text{ a contradiction.}$$

□

For nonexpansive maps, we have the following result which is a slight extension of Theorem 1.65. In Sect. 2.2, we will generalize further this result.

Proposition 2.17 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex, $\varphi : C \to C$ is nonexpansive and $\overline{\varphi(C)}$ is compact, then $\varphi(\cdot)$ has a fixed point.*

Proof Fix $u_0 \in C$ and for $n \in \mathbb{N}$, we define

$$\varphi_n(u) = (1 - \frac{1}{n})\varphi(u) + \frac{1}{n}u_0.$$

Then $\varphi_n(\cdot)$ is a contraction and so by Theorem 2.3 there exists unique $u_n \in C$ such that

$$\varphi_n(u_n) = (1 - \frac{1}{n})\varphi(u_n) + \frac{1}{n}u_0 = u_n.$$

By hypothesis $\{\varphi(u_n)\}_{n\in\mathbb{N}}$ is relatively compact and so we may assume that

$$\varphi(u_n) \to y \text{ in } X,$$

$$\Rightarrow \varphi_n(u_n) = (1 - \frac{1}{n})\varphi(y_n) + \frac{1}{n}u_0 = u_n \to y \text{ in } X.$$

So, the continuity of $\varphi(\cdot)$ implies that

$$\varphi(u_n) \to \varphi(y),$$
$$\Rightarrow \varphi(y) = y.$$

□

We continue with nonexpansive maps and we make the following simple but crucial observation.

Lemma 2.18 *If X is a Banach space, $C \subseteq X$ is nonempty, bounded, closed, convex and $\varphi : C \to C$ is nonexpansive, then* $\inf\{\|u - \varphi(u)\| : u \in C\} = 0$.

Proof As in the previous proof, we fix $u_0 \in C$ and for $t \in (0, 1)$ we introduce the map $\varphi_t : C \to C$ by

$$\varphi_t(u) = (1 - t)\varphi(u) + tu_0.$$

Then $\varphi_t(\cdot)$ is a contraction and so by Theorem 2.3, there exists unique $u_t \in C$ such that $\varphi_t(u_t) = u_t$. We have

$$\begin{aligned} \|u_t - \varphi(u_t)\| &= \|(1 - t)\varphi(u_t) + tu_0 - \varphi(u_t)\| \\ &= t\|u_0 - \varphi(u_t)\| \\ &\leq t \operatorname{diam} C \text{ (recall } C \subseteq X \text{ is bounded).} \end{aligned}$$

Let $t \to 0^+$ to conclude that

$$\inf\{\|u - \varphi(u)\| : u \in C\} = 0.$$

□

Using this observation, we can have another particular case of the Schauder fixed point theorem which will be proved in the next section.

Proposition 2.19 *If X is a Banach space, $C \subseteq X$ is nonempty, compact, convex and $\varphi : C \to C$ is nonexpansive, then $\varphi(\cdot)$ has a fixed point.*

Proof Using Lemma 2.18, let $\{u_n\}_{n\in\mathbb{N}} \subseteq C$ such that

$$\|u_n - \varphi(u_n)\| \downarrow 0 \text{ as } n \to \infty.$$

On account of the compactness of C we may assume that $u_n \to u \in C$. Then the continuity of $\varphi(\cdot)$ implies that $\varphi(u_n) \to \varphi(u)$. So, finally we have $\|u - \varphi(u)\| = 0$, hence $\varphi(u) = u$. □

Another consequence of Lemma 2.18, is the next fixed point result which goes beyond the framework of the Schauder fixed point theorem, since it does not involve a compactness hypothesis.

Theorem 2.20 *If X is a Banach space, $C \subseteq X$ is nonempty, bounded, closed, convex, $\varphi : C \to C$ is nonexpansive and $(id - \varphi)(C)$ is closed, then $\varphi(\cdot)$ has a fixed point.*

Proof Lemma 2.18 implies that $0 \in \overline{(id - \varphi)(C)} = (id - \varphi)(C)$ and so there exists $u \in C$ such that $\varphi(u) = u$. □

Let X be a Banach space and $C \subseteq X$ nonempty, bounded. For every $u \in X$, we define

$$r_u(C) = \sup\{\|v - u\| : v \in C\}.$$

Definition 2.21 A point $u \in C$ is said to be "diametral", if $r_u(C) = \operatorname{diam} C$. If this is not true, then we say that u is "nondiametral". A nonempty, bounded, convex set $C \subseteq X$ is said to have "normal structure", if every nonempty, convex subset of C which is not a singleton, contains a nondiametral point. A Banach space has a "normal structure" if every nonempty, bounded, convex subset has normal structure.

Proposition 2.22 *Every locally uniformly convex Banach space has normal structure.*

Proof Let $C \subseteq X$ be nonempty, bounded, convex, which is not a singleton. Let $u, v \in C$, $u \neq v$ and set $y = \dfrac{u+v}{2}$. We claim that y is nondiametral. Arguing by contradiction, suppose we can find $\{x_n\}_{n\in\mathbb{N}}$ such that $\|x_n - y\| \to \operatorname{diam} C$. We know that

$$\|x_n - u\| \leq \operatorname{diam} C, \ \|x_n - v\| \leq \operatorname{diam} C \text{ for all } n \in \mathbb{N}.$$

Hence we have

$$\begin{aligned}
\|x_n - y\| &= \|x_n - \frac{u+v}{2}\| \\
&= \frac{1}{2}\|(x_n - u) + (x_n - v)\| \\
&\leq \frac{1}{2}(\|x_n - u\| + \|x_n - v\|) \\
&\leq \operatorname{diam} C, \\
\Rightarrow \|x_n - u\| \to \operatorname{diam} C, &\ \|x_n - v\| \to \operatorname{diam} C.
\end{aligned}$$

Therefore

$$2\|x_n - u\|^2 + 2\|x_n - v\|^2 - \|(x_n - x) + (x_n - y)\|^2 \to 0.$$

But $(x_n - u) - (x_n - v) = v - u \neq 0$, for all $n \in \mathbb{N}$ and this contradicts the definition of local uniform convexity (see Definition I-3.240, p.170). □

We will use the notion of normal structure to prove the main fixed point theorem for nonexpansive maps. To do this, we proceed as follows. Let

$$\mathcal{S} = \{D \subseteq C : D \text{ is nonempty, closed, convex,}$$
$$\varphi(D) \subseteq D \text{ (that is, } D \text{ is } \varphi - \text{invariant)}\}.$$

We introduce a partial order $\preceq$ on $\mathcal{S}$ by

$$D_1 \preceq D_2 \Leftrightarrow D_2 \subseteq D_1 \text{ (inverse inclusion).}$$

Let $\widehat{D} = \bigcap_{D \in \mathcal{S}} D$. Evidently $\widehat{D}$ is closed, convex, φ-invariant, but it may be empty. However, if C is weakly compact, then $\widehat{D} \neq \emptyset$. Then $\widehat{D}$ is an upper bound for every chain in $\mathcal{S}$ and so by Zorn's lemma, we can find $D_0 \in \mathcal{S}$ a maximal element(note that D_0 ia nonempty, closed, convex, φ-invariant and minimal with respect to inclusion). We have $\overline{\text{conv}}\varphi(D_0) = D_0$ and so if D_0 is a singleton, then we have a fixed point for $\varphi(\cdot)$. But in general D_0 need not be a singleton (see Alspach [8]).

Now we can state and prove the main fixed point theorem for nonexpansive maps.

Theorem 2.23 *If X is a Banach space, $C \subseteq X$ is nonempty, weakly compact, convex and has normal structure and $\varphi : C \to C$ is nonexpansive, then $\varphi(\cdot)$ has a fixed point.*

Proof Let $D_0 \neq \emptyset$ be the minimal closed, convex φ-invariant subset of C. We introduce the map $\vartheta : D_0 \to \mathbb{R}$ defined by

$$\vartheta(u) = \sup\{\|v - u\| : v \in D_0\}.$$

We know that $\vartheta(\cdot)$ is lower semicontinuous and convex. We claim that $\vartheta(u) = \sup\{\|\varphi(v) - u\| : v \in C\}$. If for some $u \in D_0$ we have $m_\varphi = \sup\{\|\varphi(v) - u\| : v \in C\} < \sup\{\|v - u\| : v \in C\}$, then the set $D_0 \cap \overline{B}_{m_\varphi}(u)$ is nonempty, closed, convex(thus w-compact), φ-invariant and proper subset of D_0, a contradiction. Therefore

$$\begin{aligned} \vartheta(u) &= \sup\{\|\varphi(v) - u\| : v \in C\}, \\ \Rightarrow \vartheta(\varphi(u)) &= \sup\{\|\varphi(v) - \varphi(u)\| : v \in C\} \\ &\leq \sup\{\|v - u\| : v \in C\} = \vartheta(u). \end{aligned} \tag{2.3}$$

From (2.3) we can infer that $\vartheta(\cdot)$ is constant. To see this, let $\lambda \in \mathbb{R}$ and consider the set $L_\lambda = \{u \in D_0 : \vartheta(u) \leq \lambda\}$. Then L_λ is closed convex. If $\vartheta(\cdot)$ is not constant, then we can find $\lambda_0 \in \mathbb{R}$ such that $L_{\lambda_0} \neq D_0$. For $u \in L_{\lambda_0}$, we have $\vartheta(\varphi(u)) \leq \vartheta(u) \leq \lambda_0$ (see (2.3)). So, $\varphi(u) \in L_{\lambda_0}$ and so L_{λ_0} is φ-invariant, a contradiction to the minimality of D_0. Therefore $\vartheta(\cdot)$ is constant. But recall that C has normal structure and so D_0 must be a singleton $D_0 = \{u\}$. Evidently $\varphi(u) = u$. □

Remark 2.24 The fixed point is not in general unique.

Combining Proposition 2.22 and Theorem 2.23, we infer the following corollary.

Corollary 2.25 *If X is locally uniformly convex, reflexive, $C \subseteq X$ is nonempty, bounded, closed, convex and $\varphi : C \to C$ is nonexpansive, then $\varphi(\cdot)$ has a fixed point.*

In the case of Hilbert spaces, we can say more. First we introduce a nonexpansive map which is a valuable tool in fixed point theory. Let H be a Hilbert space and for $\varrho > 0$ let $\overline{B}_\varrho = \{u \in H : \|u\| \leq \varrho\}$. We introduce the map $r : H \to \overline{B}_\varrho$ defined by

$$r(u) = \begin{cases} u & \text{if } \|u\| \leq \varrho \\ \varrho \dfrac{u}{\|u\|} & \text{if } \varrho < \|u\| \end{cases}$$

This map is known as the ϱ-radial retraction. Evidently $r(u) = p_{\overline{B}_r}(u)$(=the metric projection on $\overline{B}_\varrho$). From Proposition I-3.176(b), we have the following result.

Proposition 2.26 *$r : H \to \overline{B}_\varrho$ is nonexpansive.*

Next we prove a nonlinear alternative theorem of the Leray-Schauder type, for nonexpansive maps.

Theorem 2.27 *If $\varphi : \overline{B}_\varrho \to H$ is nonexpansive, then at least one of the following two properties holds*

(a) *$\varphi(\cdot)$ has a fixed point in $\overline{B}_\varrho$;*
(b) *there exist $u \in \partial B_\varrho$ and $\lambda \in (0, 1)$ such that $u = \lambda\varphi(u)$.*

Proof Let $\vartheta : \overline{B}_\varrho \to \overline{B}_\varrho$ be the nonexpansive map defined by $\vartheta = r \circ \varphi$. Invoking Corollary 2.25, we can find $u \in \overline{B}_\varrho$ such that

$$r(\varphi(u)) = u.$$

If $\varphi(u) \in \overline{B}_\varrho$, then $r(\varphi(u)) = \varphi(u)$ and so we conclude that u is a fixed point of $\varphi(\cdot)$. If $\varphi(u) \notin \overline{B}_\varrho$, then

$$u = r(\varphi(u)) = \varrho \frac{\varphi(u)}{\|\varphi(u)\|} = \lambda\varphi(u) \text{ with } \lambda = \frac{\varrho}{\|\varphi(u)\|} \in (0, 1),$$

thus (b) holds. □

By imposing conditions on $\varphi(\cdot)$ which guarantee that part (b) of the above theorem does not hold, we have fixed point theorems.

Theorem 2.28 *If $\varphi : \overline{B}_\varrho \to H$ is nonexpansive and for all $u \in \partial B_\varrho$ one of the following conditions holds*

(i) $\|\varphi(u)\| \leq \|u\|$;
(ii) $\|\varphi(u)\| \leq \|u - \varphi(u)\|$;
(iii) $\|\varphi(u)\|^2 \leq \|u\|^2 + \|u - \varphi(u)\|^2$;
(iv) $(u, \varphi(u)) \leq \|u\|^2$;

then $\varphi(\cdot)$ has a fixed point.

Proof We prove the theorem when (ii) holds. The proofs are similar if one of the other three conditions holds.

So, suppose that (ii) holds and arguing by contradiction suppose that $\varphi(\cdot)$ is fixed point free. Then by the nonlinear alternative theorem (see Theorem 2.27), there exist $u \in \partial B_1$ and $\lambda \in (0, 1)$ such that $u = \lambda\varphi(u)$. So, $\varphi(u) \neq 0$ and we have

$$\begin{aligned} \|\varphi(u)\| &= \|\varphi(\lambda\varphi(u))\| \\ &\leq \|\lambda\varphi(u) - \varphi(\lambda\varphi(u))\| \\ &= (1 - \lambda)\|\varphi(u)\|, \end{aligned}$$

a contradiction since $\lambda \in (0, 1)$.

So, $\varphi(\cdot)$ has a fixed point. □

As an application of this theorem, we have the following surjectivity result.

Proposition 2.29 *If $\varphi : H \to H$ is nonexpansive and $(u, u - \varphi(u)) \geq \xi(\|u\|)\|u\|$ with $\xi(r) \to \infty$ as $r \to \infty$, then $\vartheta(u) = u - \varphi(u)$ is surjective.*

Proof Let $u_0 \in H$ and set $k(u) = u - (\varphi(u) + u_0)$, $u \in H$. We have

$$\begin{aligned} \frac{(k(u), u)}{\|u\|} &= \frac{(\vartheta(u), u)}{\|u\|} - \frac{(u_0, u)}{\|u\|} \\ &\geq \xi(\|u\|) - \|u_0\|. \end{aligned}$$

So, we can find $\varrho > 0$ large such that

$$(k(u), u) \geq 0 \text{ for all } u \in H \text{ with } \|u\| = \varrho. \tag{2.4}$$

Note that $\psi(u) = \varphi(u) + u_0$ is nonexpansive and from (2.4) we have

$$(\psi(u), u) \leq \|u\|^2 \text{ for all } u \in H, \|u\| = r.$$

Invoking Theorem 2.28(iv), we can find $\widehat{u} \in \overline{B}_\varrho$ such that $\psi(\widehat{u}) = \widehat{u}$. Hence

$$\widehat{u} - \varphi(\widehat{u}) = u_0,$$
$$\Rightarrow \vartheta(\cdot) \text{ is surjective.}$$

□

For completeness, we also recall "Caristi's fixed point theorem" proved in Theorem I-8.116.

Theorem 2.30 *If (X, d) is a complete metric space, $\varphi : X \to X$ and there exists a lower semicontinuous function $\vartheta : X \to \mathbb{R}_+$ such that*

$$d(u, \varphi(u)) \leq \vartheta(u) - \vartheta(\varphi(u)) \text{ for all } u \in X,$$

then $\varphi(\cdot)$ has a fixed point.

Remark 2.31 Note that $\varphi(\cdot)$ need not be continuous. In Proposition I-8.118 we proved that Theorem 2.30 is equivalent to the Ekeland variational principle (see Corollary I-8.101). If $\varphi(\cdot)$ is a k-contraction and $\vartheta(u) = \dfrac{1}{1-k} d(u, \varphi(u))$, then

$$d(u, \varphi(u)) \leq \vartheta(u) - \vartheta(\varphi(u)) \text{ for all } u \in X.$$

Using Theorem 2.30 we obtain $\widehat{u} \in X$ such that

$$\varphi(\widehat{u}) = \widehat{u},$$

that is, we have recovered the existence part of Theorem 2.3. Of course Theorem 2.3 contains more information.

We conclude this section with an application of fixed point theory to the existence and uniqueness of certain first order initial value systems.

So, let $T = [0, b]$ and $f : T \times \mathbb{R}^N \to \mathbb{R}^N$ a vector field. We consider the following system

$$\left\{ \begin{array}{l} u'(t) = f(t, u(t)) \text{ a.e on } T, \\ u(0) = u_0. \end{array} \right\}. \tag{2.5}$$

The hypotheses on the vector field $f(t, x)$ are the following.

H : $f : T \times \mathbb{R}^N \to \mathbb{R}^N$ is a function such that

(i) for all $x \in \mathbb{R}^N$ $t \to f(t, x)$ is measurable;
(ii) $|f(t, x) - f(t, z)| \leq k(t)|x - z|$ for a.a. $t \in T$, all $x, z \in \mathbb{R}^N$ with $k \in L^p(T)$, $1 \leq p < \infty$;
(iii) $f(\cdot, 0) \in L^p(T)$.

Remark 2.32 Clearly hypotheses $H(ii), (iii)$ imply that for every $\varrho > 0$, there exists $\eta_\varrho \in L^p(T)$ such that

$$|f(t,x)| \leq \eta_\varrho(t) \text{ for a.a. } t \in T, \text{ all } |x| \leq \varrho$$

For notational simplicity we assume that $N = 1$.

The solutions of (2.5) are in the Sobolev space $W^{1,p}(0,b)$ and from Theorem I-4.125 we know that every $u \in W^{1,p}(0,b)$ has an absolutely continuous representative and $u' \in L^p(0,b)$. We see that $u \in W^{1,p}(0,b)$ is a solution of (2.5) if and only if

$$u(t) = u_0 + \int_0^t f(s,u(s))ds \text{ for all } t \in T$$

So, if we consider the map $\varphi : C(T) \to C(T)$ defined by

$$\varphi(u)(t) = u_0 + \int_0^t f(s,u(s))ds \text{ for all } t \in T,$$

then we look for fixed points of $\varphi(\cdot)$.

Proposition 2.33 *If hypotheses H hold, then problem* (2.5) *has a unique solution* $\widehat{u} \in W^{1,p}_N(0,b)$.

Proof Let $\mu(t) = \int_0^t k(s)ds$ and consider the following norm on $C(T)$

$$|u| = \max_{t \in T} |e^{-\mu(t)}u(t)|$$

We have

$$e^{-||k||_1}||u||_\infty \leq |u| \leq ||u||_\infty,$$

$\Rightarrow ||\cdot||_\infty$ and $|\cdot|$ are equivalent norms on $C(T)$.

We work with this new norm. For $u, v \in C(T)$, we have

$$|\varphi(u)(t) - \varphi(v)(t)| \leq \int_0^t |f(s,u(s)) - f(s,v(s))|ds$$

$$\begin{aligned} \Rightarrow e^{-\mu(t)}|\varphi(u)(t) - \varphi(v)(t)| &\leq e^{-\mu(t)} \int_0^t |f(s,u(s)) - f(s,v(s))|ds \\ &\leq e^{-\mu(t)} \int_0^t k(s)e^{\mu(s)}e^{-\mu(s)}|u(s) - v(s)|ds \end{aligned}$$

$$\leq e^{-\mu(t)} \int_0^t k(s) e^{\mu(s)} ds |u - v|$$
$$= e^{-\mu(t)} [e^{\mu(t)} - 1] |u - v|$$
$$= [1 - e^{-\mu(t)}] |u - v|$$

$\Rightarrow \varphi(\cdot)$ is a contraction.

So, by Theorem 2.3, $\varphi(\cdot)$ has a unique fixed point $\widehat{u} \in C(T)$. Note that $\widehat{u} \in W^{1,p}(0, b)$ and is the unique solution of (2.5). □

Remark 2.34 The renorming technique used in the above proof, was introduced by Bielecki [46] and for this reason $(\cdot)$ is sometimes called the Bielecki norm on $C(T)$.

2.2 Topological Fixed Point Theory

In this section, we go outside the framework of metric spaces and rely primarily on topology in order to prove the existence of fixed points. In Sect. 1.1, using degree theoretic arguments, we already had two prototype such fixed point theorems, see Theorems 1.28, 1.29 (Brouwer's fixed point theorem, finite dimensional spaces) and Theorems 1.65, 1.67 (Schauder's fixed point theorem, infinite dimensional spaces).

We start with an extension of Theorem 1.29

Proposition 2.35 *If $C \subseteq \mathbb{R}^N$ is homeomorphic to a compact and convex subset of $\mathbb{R}^N$ and $\varphi : C \to C$ is continuous, then $\varphi(\cdot)$ has a fixed point.*

Proof Let $h : C \to \widehat{C}$ be the homeomorphism between C and a compact convex subset $\widehat{C}$ of $\mathbb{R}^N$. Then $\psi = h \circ \varphi \circ h^{-1} : \widehat{C} \to \widehat{C}$ is continuous and so by Theorem 1.29 we can find $\widehat{u} \in \widehat{C}$ such that $\psi(\widehat{u}) = \widehat{u}$. Let $\overline{u} = h^{-1}(\widehat{u}) \in C$. Then $h(\varphi(\overline{u})) = \widehat{u}$, hence $\varphi(\overline{u}) = h^{-1}(\widehat{u}) = \overline{u}$, that is $\overline{u}$ is a fixed point of $\varphi(\cdot)$. □

Remark 2.36 Every compact, convex set $C \subseteq \mathbb{R}^N$ is homeomorphic to the closed unit ball of $\mathbb{R}^m$ for some $m \leq N$.

Definition 2.37 A topological space X has the "fixed point property", if every continuous map $\varphi : X \to X$ has a fixed point. Using this notion, we can generalize Proposition2.35 as follows. The proof is identical.

Proposition 2.38 *If X, Y are homeomorphic topological spaces and X has the fixed point property, then so does Y.*

Remark 2.39 It was proved by Klee [269] that a convex set in a metrizable locally convex space (for example a Banach space) has the fixed point property if and only if it is compact. Later Lin and Sternfeld [299] proved that in a Banach space a convex set which has the fixed point property for Lipschitz maps, is necessarily compact.

Proposition 2.40 *If X has the fixed point property and $C \subseteq X$ is a retract of X (see Definition 1.25), then C has the fixed point property.*

Proof Let $r : X \to C$ be a retraction map for C and let $\varphi : C \to C$ be a continuous map. Then $\varphi \circ r : X \to C \subseteq X$ is continuous. Since X has the fixed point property, we can find $u \in X$ such that

$$\varphi(r(u)) = u$$

We have $u \in C$ and since $r|_C = id|_C$, we conclude that

$$\varphi(u) = u$$

□

Remark 2.41 Using the above Proposition, we can conclude that $S^{N-1} = \{u \in \mathbb{R}^N : |u| = 1\}$ is not a retract of $\overline{B}^N = \{u \in \mathbb{R}^N : |u| \leq 1\}$. To see this first note that $\overline{B}_1^N$ has the fixed point property (see Theorem 1.28). If S^{N-1} is a retract of $\overline{B}^N$, then according to Proposition 2.40. S^{N-1} has the fixed point property too. Consider the antipodal map $a : S^{N-1} \to S^{N-1}$ defined by $a(u) = -u$. This is a continuous map, so it must have a fixed point $\widehat{u} \in S^{N-1}$, that is

$$\widehat{u} = a(\widehat{u}) = -\widehat{u},$$

$$\Rightarrow \widehat{u} = 0, \text{ a contradiction.}$$

Therefore S^{N-1} can not be a retract of $\overline{B}^N$. In infinite dimensional Banach spaces the situation is different. We have the following result originally due to Nowak [351], extended by Lin and Sternfeld [299] who produced a Lipschitz retraction.

Proposition 2.42 *If X is an infinite dimensional Banach space, then there is a Lipschitz retraction of the closed unit ball $\overline{B}_1 = \{u \in X : ||u|| \leq 1\}$ on its boundary $\partial\overline{B}_1 = \{u \in X : ||u|| = 1\}$.*

Proposition 2.43 *If $\varphi : \mathbb{R}^N \to \mathbb{R}^N$ is a continuous map and there exists $\varrho > 0$ such that*

$$0 \leq (\varphi(u), u)_{\mathbb{R}^N} \text{ for all } u \in \mathbb{R}^N, |u| = \varrho,$$

then there exists $\widehat{u} \in \overline{B}_\varrho = \{u \in \mathbb{R}^N : |u| \leq \varrho\}$ such that $\varphi(\widehat{u}) = 0$.

Proof We argue by contradiction. So, suppose that

$$0 < |\varphi(u)| \text{ for all } u \in \overline{B}_\varrho.$$

We can define the map $\psi : \overline{B}_\varrho \to \partial B_\varrho$ defined by

$$\psi(u) = -\varrho \frac{\varphi(u)}{|\varphi(u)|} \text{ for all } u \in \overline{B}_\varrho.$$

Clearly $\psi(\cdot)$ is continuous and so Theorem 1.29 implies that there exists $\overline{u} \in \overline{B}_\varrho$ such that

$$\overline{u} = \psi(\overline{u}) = -\varrho \frac{\varphi(\overline{u})}{|\varphi(\overline{u})|}$$

$$\Rightarrow |\overline{u}| = \varrho.$$

Then we have

$$\varrho^2 = |\overline{u}|^2 = (\psi(\overline{u}), \overline{u})_{\mathbb{R}^N} = -\frac{\varrho}{|\varphi(\overline{u})|}(\varphi(\overline{u}), \overline{u})_{\mathbb{R}^N} \leq 0,$$

$$\Rightarrow \varrho = 0, \text{ a contradiction.}$$

So, there exists $\widehat{u} \in \overline{B}_\varrho$ such that $\varphi(\widehat{u}) = 0$. □

Remark 2.44 The condition $0 \leq (\varphi(u), u)_{\mathbb{R}^N}$ on ∂B_ϱ means that on ∂B_ϱ the vector field $\varphi(\cdot)$ points to the exterior of $\overline{B}_\varrho$. The above proposition leads to another proof of Proposition 1.27 (the lack of retraction of $\overline{B}_\varrho$ onto ∂B_ϱ). To see this, suppose that ∂B_ϱ is a retract of $\overline{B}_\varrho$. Let $r : \overline{B}_\varrho \to \partial B_\varrho$ be the retraction. Then

$$0 \leq |u|^2 = (r(u), u)_{\mathbb{R}^N} \text{ for all } u \in \mathbb{R}^N, |u| = \varrho.$$

So, we can use Proposition 2.43 and produce $\widehat{u} \in \overline{B}_\varrho$ such that $r(\widehat{u}) = 0$, a contradiction since the range of $r(\cdot)$ is ∂B_ϱ. Therefore ∂B_ϱ can not be a retract of $\overline{B}_\varrho$.

Next we will show that Brouwer's fixed point theorem is in fact equivalent to some of the basic results proved in Sect. 2.1. First we need to fix our terminology.

Definition 2.45 Let X, Y be two Hausdorff spaces.

(a) Two continuous maps $\varphi, \psi : X \to Y$ are said to be "homotopic", if there exists a continuous map $h : [0, 1] \times X \to Y$ such that

$$h(0, \cdot) = \varphi(\cdot) \; and \; h(1, \cdot) = \psi(\cdot).$$

The map $h(\cdot, \cdot)$ is the "homotopy" or "deformation" and if φ, ψ are homotopic, we write $\varphi \sim \psi$.

(b) If a continuous function $\varphi : X \to Y$ is homotopic to a constant function, then we say that $\varphi(\cdot)$ is "nullhomotopic" and we write $\varphi \sim 0$.

(c) We say that X is "contractible", if the identity map $id_X : X \to X$ is nullhomotopic (that is, $id_X \sim 0$).

Remark 2.46 The relation of homotopy is an equivalence relation. For reflexivity ($\varphi \sim \varphi$) consider the homotopy $h(t, \cdot) = \varphi(\cdot)$ for all $t \in [0, 1]$. For symmetry ($\varphi \sim \psi \Leftrightarrow \psi \sim \varphi$) consider the homotopy $h(t, x)$ producing $\varphi \sim \psi$ and then use $\widehat{h}(t.u) = h(1 - t, u)$ for all $t \in [0, 1]$, all $u \in X$, to conclude that $\psi \sim \varphi$. For transitivity ($\varphi \sim \psi, \psi \sim \vartheta \Rightarrow \varphi \sim \vartheta$), let $h_1(\cdot, \cdot)$ be the homotopy corresponding to $\varphi \sim \psi$ and $h_2(\cdot, \cdot)$ the homotopy corresponding to $\psi \sim \vartheta$. Let

$$h_0(t, u) = \begin{cases} h_1(2t, u) & \text{if } t \in [0, \frac{1}{2}] \\ h_2(2t - 1, u) & \text{if } t \in [\frac{1}{2}, 1] \end{cases}$$

Evidently $h_0(\cdot, \cdot)$ is continuous and $h_0(0, \cdot) = \varphi(\cdot)$, $h_0(1, \cdot) = \vartheta(\cdot)$. Hence $\varphi \sim \vartheta$. So, the relation $\sim$ gives a partition of $C(X, Y)$ into disjoint homotopy classes. A space X is contractible if and only if there exist a continuous map $h : [0, 1] \times X \to X$ and $u_0 \in X$ such that

$$h(0, u) = u \text{ and } h(1, u) = u_0 \text{ for all } u \in X.$$

Proposition 2.47 *A continuous function $\varphi : S^N \to Y$ is nullhomotopic*

if and only if

can be extended to a continuous function $\widehat{\varphi} : \overline{B}^{N+1} \to Y$ (that is, $\widehat{\varphi}|_{S^N} = \varphi$).

Proof $\Downarrow$ Since $\varphi \sim 0$, there exist a continuous map $h : [0, 1] \times X \to Y$ and $y_0 \in Y$ such that $h(0, u) = \varphi(u)$ and $h(1, u) = y_0$ for all $u \in X$. Let

$$\widehat{\varphi}(u) = \begin{cases} h(2|u|, \frac{u}{|u|}) & \text{if } 0 \le |u| \le \frac{1}{2} \\ h(2|u| - 1, \frac{u}{|u|}) & \text{if } \frac{1}{2} \le |u| \le 1 \end{cases}$$

Evidently $\widehat{\varphi}(\cdot)$ is continuous and $\widehat{\varphi}|_{S^N} = \varphi$

$\Uparrow$ Let $h(t, u) = \widehat{\varphi}((1 - t)u)$ for all $t \in [0, 1]$, all $u \in S^N$. Then $h(\cdot, \cdot)$ is continuous on $[0, 1] \times S^N$, $h(0, \cdot) = \widehat{\varphi}|_{S^N} = \varphi$, $h(1, \cdot) = \widehat{\varphi}(0)$ and so $\varphi \sim 0$. □

Theorem 2.48 *The following statements are equivalent:*

(a) *The Ljusternik-Schnirelmann-Borsuk Theorem (see Theorem 1.44).*
(b) *There is no continuous odd map $\varphi : S^N \to S^{N-1}$.*
(c) *If $\varphi : S^{N-1} \to S^{N-1}$ is continuous, odd, then $\varphi(\cdot)$ is not nullhomotopic.*
(d) *If $\varphi : S^N \to \mathbb{R}^N$ is continuous, then there exists $\widehat{u} \in S^N$ such that $\varphi(\widehat{u}) = \varphi(-\widehat{u})$.*

Proof $(a) \Rightarrow (b)$: We argue by contradiction. So, suppose we can find a continuous odd function $\varphi : S^N \to S^{N-1}$. Let D_N be the N-simplex centered at the origin. Then ∂D_N and S^{N-1} are homeomorphic. Let E_k be the images of the $(N - 1)$-face of D_N. Then each E_k is closed and does not contain antipodal points. Let $A_k = \varphi^{-1}(E_k)$ $k = 1, \ldots, N + 1$. We see that $\{A_k\}_{k=1}^{N+1}$ is a closed cover of S^N. So, by

Corollary 1.45, for some $k \in \{1, \ldots, N+1\}$ $A_k \cap (-A_k) \neq \emptyset$. Let $u \in A_k \cap (-A_k)$. Since $\varphi(\cdot)$ is odd $\varphi(-u) = -\varphi(u) \in E_k$, a contradiction.

$(b) \Rightarrow (c)$: Consider a continuous, odd map $\varphi : S^{N-1} \to S^{N-1}$. Arguing by contradiction, suppose that $\varphi(\cdot)$ is nullhomotopic. Then there exists a homotopy $h(t, u)$ which deforms φ to a constant map. Consider the map $\widehat{\varphi} : \overline{B}_1^N \to S^{N-1}$ defined by

$$\widehat{\varphi}(u) = \begin{cases} h(2|u|, \frac{u}{|u|}) \ if \ 0 \leq |u| \leq \frac{1}{2} \\ h(1, S^{N\perp 1}) \ if \ \frac{1}{2} \leq |u| \leq 1 \end{cases}$$

Clearly $\widehat{\varphi}(\cdot)$ is continuous and $\widehat{\varphi}|_{S^{N-1}} = \varphi$. Consider the two hemispheres of S^N, namely the sets

$$S_+^N = \{u = (u_k)_{k=1}^{N+1} \in S^N : u_{N+1} \geq 0\},$$

$$S_-^N = \{u = (u_k)_{k=1}^{N+1} \in S^N : u_{N+1} \leq 0\}.$$

Note that $\overline{B}_1^N$ is homeomorphic to each of these hemispheres (by the stereographic projection). So, we can define the map $\psi : S^N \to S^{N-1}$ by

$$\psi(u) = \begin{cases} \widehat{\varphi}(u) \quad if \ u \in S_+^N \\ -\widehat{\varphi}(u) \ if \ u \in S_-^N \end{cases}$$

which is continuous, odd, a contradiction to (b).

$(c) \Rightarrow (d)$: We argue indirectly. So, suppose we can find a continuous map $\varphi : S^N \to R^N$ such that$\varphi(u) \neq \varphi(-u)$ for all $u \in S^N$. Let $\psi : S^N \to S^{N-1}$ be defined by

$$\psi(u) = \frac{\varphi(u) - \varphi(-u)}{|\varphi(u) - \varphi(-u)|}$$

Clearly $\psi(\cdot)$ is continuous and $\varphi|_{S^{N-1}} : S^{N-1} \to S^{N-1}$ is continuous, odd. Note that $\psi|_{S_+^N}$ is an extension on $\overline{B}_1^N$, hence $\psi|_{S^{N-1}}$ is null homotopic (just use the homotopy $h(t, u) = \psi|_{S^{N-1}}(tu)$), a contradiction.

$(d) \Rightarrow (a)$: Consider $\{E_k\}_{k=1}^{N+1}$ a closed cover of S^N and suppose that $E_k \cap (-E_k) = \emptyset$ for all $k \in \{1, \ldots, N+1\}$. Using Urysohn's Lemma (see Theorem I-1.42), we can find a continuous function $\vartheta_k : S^N \to [0, 1]$ such that

$$\vartheta_k|_{E_k} = 0 \text{ and } \vartheta_k|_{-E_k} = 1, k = 1, \ldots, N$$

Let $\widehat{\vartheta} = (\vartheta_k)_{k=1}^N : S^N \to \mathbb{R}^N$. This map is continuous and since (d) holds, we can find $v \in S^N$ such that

$$\widehat{\vartheta}(v) = \widehat{\vartheta}(-v),$$

$$\Rightarrow \vartheta_k(v) = \vartheta_k(-v) \text{ for all } k = 1, \ldots, N,$$

$$\Rightarrow v \in S^N \setminus (\bigcup_{k=1}^{N} E_k \cup \bigcup_{k=1}^{N}(-E_k)).$$

Both $\{E_k\}_{k=1}^{N+1}$ and $\{-E_k\}_{k=1}^{N+1}$ are closed covers of S^N. Hence we must have $v \in E_{N+1} \cap (-E_{N+1})$, a contradiction. □

We know that S^{N-1} is not a retract of $\overline{B}_1^N$ (see Proposition 1.27). Next we show that this fact is equivalent to some fixed point results.

Theorem 2.49 *The following statements are equivalent*

(a) S^{N-1} *is not a retract of* $\overline{B}_1^N$.
(b) S^{N-1} *is not contractible in itself.*
(c) *If* $\varphi : \overline{B}_1^N \to \mathbb{R}^N$ *is continuous, then at least one of the following statements holds*

(1) $\varphi(\cdot)$ *has a fixed point;*
(2) there exists $\widehat{u} \in S^{N-1}$ *and* $\lambda \in (0, 1)$ *such that*

$$\widehat{u} = \lambda\varphi(\widehat{u}).$$

(d) *A continuous map:*$\varphi : \overline{B}_1^N \to \overline{B}_1^N$ *has a fixed point (Brouwer's fixed point theorem).*

Proof $(a) \Rightarrow (b)$: We argue by contradiction. So, suppose that S^{N-1} is contractible in itself and let $h(t, u)$ be the homotopy which deforms S^{N-1} to $v_0 \in S^{N-1}$ (that is, $id_{S^{N-1}} \sim 0$). Consider the map $r : \overline{B}_1^N \to S^{N-1}$ defined by

$$r(u) = \begin{cases} u_0 & if\ 0 \le |u| \le \frac{1}{2} \\ h(2 - 2|u|, \mathrm{u}|u|)\ if\ \frac{1}{2} \le |u| \le 1 \end{cases}$$

Evidently $r(\cdot)$ is continuous and $r|_{S^{N-1}} = id|_{S^{N-1}}$ and so, we conclude that S^{N-1} is a retract of $\overline{B}_1^N$, a contradiction.

$(b) \Rightarrow (c)$: Suppose $\varphi(u) \neq u$ for all $u \in \overline{B}_1^N$ and $v \neq t\varphi(v)$ for all $0 \le t \le 1$, all $v \in S^{N-1}$. Let $r : \mathbb{R}^N \setminus \{0\} \to S^{N-1}$ be defined by

$$r(u) = \frac{u}{|u|}.$$

(the radial retraction of $\mathbb{R}^N \setminus \{0\}$ onto S^{N-1}). We consider the homotopy $h(t, u)$ defined by

$$h(t,u)=\begin{cases} r(u-2t\varphi(u)) & if\ 0\le t\le \frac{1}{2} \\ r((2-2t)u-\varphi((2-2t)u)) & if\ \frac{1}{2}\le t\le 1 \end{cases}$$

Evidently $h(\cdot,\cdot)$ is continuous and

$$h(0,u)=r(u),\ h(1,u)=r(\varphi(0)),$$

Since $r|_{S^{N-1}}=id|_{S^{N-1}}$, we infer that $id|_{S^{N-1}}\sim 0$, that is S^{N-1} is contractible in itself, a contradiction.

$(c)\Rightarrow(d)$: Note that (c)(2) can not occur since $|\varphi(u)|\le 1$ for all $u\in S^{N-1}$. So, $\varphi(\cdot)$ must have a fixed point.

$(d)\Rightarrow(a)$: Arguing by contradiction suppose that S^{N-1} is a retract of $\overline{B}_1^N$ and $r(\cdot)$ is the corresponding retraction. Then $\varphi:\overline{B}_1^N\to\overline{B}_1^N$ defined by $\varphi(u)=-r(u)$ for all $u\in\overline{B}_1^N$, is continuous and fixed point free, a contradiction. □

Now we will state some consequences of the previous theorems. The first result relaxes in Theorem 2.48(c) the hypothesis that $\varphi(\cdot)$ is odd

Proposition 2.50 *If $\varphi:S^N\to S^N$ is continuous and $\varphi(u)\ne\varphi(-u)$ for all $u\in S^N$, then $\varphi(\cdot)$ is not nullhomotopic.*

Proof Consider the map $\psi:S^N\to S^N$ defined by

$$\psi(u)=\frac{\varphi(u)-\varphi(-u)}{|\varphi(u)-\varphi(-u)|}$$

This map is continuous, odd. We claim that

$$\varphi(u)\ne-\psi(u)\text{ for all }u\in S^N$$

Indeed if $\varphi(u)=-\psi(u)$ for some $u\in S^N$, then

$$[1+|\varphi(u)-\varphi(-u)|]\varphi(u)=\varphi(-u)$$

a contradiction since $|\varphi(u)|=|\varphi(-u)|=1$ and $|\varphi(u)-\varphi(-u)|\ne 0$. Consider the homotopy $h(t,u)$ defined by

$$h(t,u)=\frac{(1-t)\varphi(u)+t\psi(u)}{|(1-t)\varphi(u)+t\psi(u)|}$$

If is well-defined and $h(0,u)=\varphi(u)$, $h(1,u)=\psi(u)$ for all $u\in S^N$. If $\varphi(\cdot)$ is nullhomotopic and $\varphi\sim\psi$, by transitivity ψ is nullhomotopic and odd, contradicting Theorem 2.48(c). So, φ is not nullhomotopic. □

Another consequence of Theorem 2.48, is the so-called"Borsuk Fixed Point Theorem"

Proposition 2.51 *If $\Omega \subseteq \mathbb{R}^N$ is an open neighborhood of the origin which is bounded, symmetric, $\varphi : \overline{\Omega} \to \mathbb{R}^N$ is continuous and $\varphi|_{\partial\Omega}$ is odd, then $\varphi(\cdot)$ has a fixed point.*

Proof Let $\varrho_\Omega : \mathbb{R}^N \to \mathbb{R}_+$ be the Minkowski functional for the set Ω (that is, $\varrho_\Omega(u) = inf\{\lambda > 0 : u \in \lambda\Omega\}$, see Definition I-3.13, p. 107). Equip $\mathbb{R}^N$ with the norm $|\cdot|_0 = \varrho_\Omega(\cdot)$ and denote the space by V. Then the identity map $\widehat{i} : \mathbb{R}^N \to V$ is a homeomorphism which maps $\overline{\Omega}$ onto $\overline{B}_1^V$. Let $\psi = \widehat{i} \circ \varphi \circ \widehat{i}^{-1} : \overline{B}_1^V \to V$ is continuous and $\psi|_{\partial\overline{B}_1^V}$ is odd. We show that $\psi(\cdot)$ has a fixed point. Indeed, if $\psi(\cdot)$ is fixed point free, then $\psi(u) \neq u$ for all $u \in \overline{B}_1^V$ and so we can define the map

$$k(u) = \frac{\psi(u) - u}{|\psi(u) - u|_0} \text{ for all } u \in \overline{B}_1^V$$

Clearly $k : \overline{B}_1^V \to \partial B_1^V$ is continuous and $k|_{\partial B_1^V} : \partial B_1^V \to \partial B_1^V$ is nullhomotopic (see Proposition 2.47). On the other hand $k|_{\partial B_1^V}$ is odd and so by Theorem 2.48(c), we have a contradiction.

Therefore we can find $u \in \overline{B}_1^V$ such that

$$(\widehat{i} \circ \varphi \circ \widehat{i}^{-1})(u) = u,$$

$$\Rightarrow \varphi(\widehat{i}^{-1}(u)) = \widehat{i}^{-1}(u),$$

$$\Rightarrow \varphi(\cdot) \text{ has a fixed point.}$$

□

Also from Theorem 2.49, we have:

Proposition 2.52 *The identity map $id : S^N \to S^N$ is not nullhomotopic.*

We have a another surjectivity result (see Proposition 2.43)

Proposition 2.53 *If $\varphi : \mathbb{R}^N \to \mathbb{R}^N$ is continuous and strongly coercive (that is, $\frac{(\varphi(u),u)_{\mathbb{R}^N}}{|u|} \to +\infty$ as $|u| \to \infty$), then $\varphi(\cdot)$ is surjective.*

Proof Let $v \in \mathbb{R}^N$. The strong coercivity implies that we can find $\varrho > 0$ large such that

$$(\varphi(u) - v, u)_{\mathbb{R}^N} \geq (\varphi(u), u)_{\mathbb{R}^N} - |v||u| \geq 0 \text{ for all } u \in \mathbb{R}^N \text{ with } |u| \geq \varrho.$$

We apply Proposition 2.43 to obtain $\widehat{u} \in \mathbb{R}^N$ such that

$$\varphi(\widehat{u}) = v$$

Since $v \in \mathbb{R}^N$ is arbitrary we conclude that $\varphi(\cdot)$ is surjective. □

Now we pass to infinite dimensional Banach spaces.

Proposition 2.54 *If X is a Banach space, $D \subseteq X$ is nonempty, bounded, closed with $intD \neq \emptyset$, $\varphi \in K(D, X)$ and there exists $u_0 \in intD$ such that*

$$\varphi(u) - u_0 \neq \lambda(u - u_0) \text{ for all } u \in \partial D, \text{ all } \lambda > 1,$$

then $\varphi(\cdot)$ has a fixed point.

Proof We consider the compact homotopy $h(t, u)$ defined by

$$h(t, u) = t(\varphi(u) - u_0) + u_0 \text{ for all } t \in [0, 1], \text{ all } u \in \overline{\text{int}D}$$

We assume that $0 \notin (id - \varphi)(\partial D)$ (otherwise we already have a fixed point in ∂D). Suppose for some $t \in (0, 1)$ and some $u \in \partial(\text{int}D)$ we have

$$u - h(t, u) = 0$$

$$\Rightarrow \varphi(u) - u_0 = \frac{1}{t}(u - u_0)$$

which contradicts our hypothesis. Hence

$$0 \notin (id - h(t, \cdot))(\partial(\text{int}D)) \text{ for all } t \in [0, 1].$$

Then the homotopy invariance property of the Leray-Schauder degree implies

$$d_{LS}(id - u_0, \text{int}D, 0) = d_{LS}(id - \varphi, \text{int}D, 0),$$

$$\Rightarrow d_{LS}(id - \varphi, \text{int}D, 0) = 1.$$

So, by the solution property of d_{LS} we can find $\widehat{u} \in \text{int}D$ such that $\varphi(\widehat{u}) = \widehat{u}$. Therefore $\varphi(\cdot)$ has a fixed point. □

We mention that in the above fixed point principle, the set D need not be convex and only the values of $\varphi|_{\partial(\text{int}D)}$ matter. If D is convex, then $D = \overline{\text{int}D}$. In general $\partial(\text{int}D) \subseteq \partial D$.

Next we present a generalization of the Schauder fixed point theorem (see Theorem 1.65). First we state a more general version of the Dugundji Extension Theorem (see Theorem I-3.154) in which we remove the compactness condition of the map. For a proof of this result we refer to Granas and Dugundji [215, (Theorem 7.4, p. 163)].

Theorem 2.55 *If Y is a metric space, X is a locally convex space, $C \subseteq X$ is nonempty closed and $\varphi : C \to X$ is a continuous map, then there exists a continuous extension $\widehat{\varphi} : X \to Y$ of φ (that is, $\widehat{\varphi}|_C = \varphi$) such that $\widehat{\varphi}(Y) \subseteq \text{conv}\,\varphi(C)$.*

Next we prove a slightly more restrictive version of the extension of Schauder's fixed point theorem.

Proposition 2.56 *If X is a Banach space, $\Omega \subseteq X$ is nonempty, open, $u_0 \in \Omega$ and $\varphi : \overline{\Omega} \to \overline{\Omega}$ is continuous with the following property*

$$\begin{aligned} &\text{"}D \subseteq \overline{\Omega} \text{ countable and } D \subseteq \overline{conv}\,\{\{u_0\} \cup \varphi(D)\} \\ &\text{imply that } D \text{ is relatively compact,"} \end{aligned} \tag{2.6}$$

then $\varphi(\cdot)$ has a fixed point.

Proof Let

$$C_0 = \{u_0\} \text{ and } C_n = conv(\{u_0\} \cup \varphi(D_{n-1})) \text{ for all } n \in \mathbb{N}.$$

Using Theorem I-3.130 (Mazur's Theorem) we have that each C_n is relatively compact and the sequence of sets $\{C_n\}_{n\in\mathbb{N}_0}$ is increasing. Each C_n is separable and so we can find E_n countable such that $\overline{C_n} = \overline{E_n}$ for all $n \in \mathbb{N}_0$. We set

$$C = \bigcup_{n\in\mathbb{N}_0} C_n \quad \text{and} \quad E = \bigcup_{n\in\mathbb{N}_0} E_n.$$

We have

$$C = \bigcup_{n\in\mathbb{N}_0} C_n = \bigcup_{n\in\mathbb{N}_0} conv(\{v_0\} \cup \varphi(C_{n-1})) = conv(\{v_0\} \cup \varphi(C)) \tag{2.7}$$

(since $\{C_n\}_{n\in\mathbb{N}_0}$ is increasing),

$$\overline{\bigcup_{n\in\mathbb{N}_0} \overline{C_n}} = \overline{\bigcup_{n\in\mathbb{N}_0} C_n} = \overline{C}, \tag{2.8}$$

and

$$\overline{\bigcup_{n\in\mathbb{N}_0} \overline{C_n}} = \overline{\bigcup_{n\in\mathbb{N}_0} \overline{E_n}} = \overline{\bigcup_{n\in\mathbb{N}_0} E_n} = \overline{E}. \tag{2.9}$$

Then (2.7), (2.8), (2.9) and the continuity of $\varphi(\cdot)$ imply that

$$\begin{aligned} E \subseteq \overline{E} = \overline{C} = \overline{conv}(\{u_0\} \cup \varphi(C)) &= \overline{conv}(\{u_0\} \cup \varphi(\overline{C})) \\ &= \overline{conv}(\{u_0\} \cup \varphi(\overline{E})) \\ &= \overline{conv}(\{u_0\} \cup \varphi(E)) \end{aligned}$$

$$\begin{aligned}(\text{since } \{u_0\} \cup \varphi(C) \subseteq \{u_0\} \cup \varphi(\overline{C}) &\subseteq \overline{\{u_0\} \cup \varphi(C)} \\ &\text{(by the continuity of } \varphi) \\ &\subseteq \overline{conv}(\{u_0\} \cup \varphi(C))).\end{aligned}$$

Therefore we have

$$\overline{conv}(\{u_0\} \cup \varphi(C)) = \overline{conv}(\{u_0\} \cup \varphi(\overline{C})).$$

The set C is countable and so by (6) we infer that $\overline{C}$ is compact. Then (2.9) implies $\overline{C}$ is compact. Recall that

$$\begin{aligned}&\overline{C} = \overline{conv}(\{u_0\} \cup \varphi(\overline{C})) \\ \Rightarrow\ &\varphi(\overline{C}) \subseteq \overline{C}.\end{aligned}$$

Apply Schauder's fixed point theorem (see Theorem 1.65) to conclude that $\varphi(\cdot)$ has a fixed point. □

Now we can state the complete generalization of Schauder's fixed point theorem.

Theorem 2.57 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex, $u_0 \in C$ and $\varphi : C \to C$ is a continuous map with the following property:*

$$\begin{aligned}&\text{"}D \subseteq C \text{ countable and } D \subseteq \overline{conv}(\{u_0\} \cup \varphi(D)) \\ &\text{imply that } D \text{ is relatively compact, "}\end{aligned} \tag{2.10}$$

then $\varphi(\cdot)$ has a fixed point.

Proof Using Theorem 2.55, we can find $\widehat{\varphi} : X \to C$ a continuous extension of φ. Let $D \subseteq X$ countable and suppose

$$D \subseteq \overline{conv}(\{u_0\} \cup \widehat{\varphi}(C)). \tag{2.11}$$

We have $\widehat{\varphi}(D) \subseteq C$ and so (11) implies $D \subseteq C$, hence $\widehat{\varphi}(D) = \varphi(D)$. So, (10) implies that D is relatively compact. Let $\Omega = X$ and apply Proposition 2.56 to the map $\widehat{\varphi}$, to obtain $u \in X$ such that $\widehat{\varphi}(u) = u$. We have $\widehat{\varphi}(u) \in C$ and so $u \in C$, hence

$$\widehat{\varphi}(u) = \varphi(u) = u.$$

□

Using this theorem, we infer the "Sadovski Fixed Point Theorem".

Theorem 2.58 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex and $\varphi : C \to C$ is continuous, condensing (see Definition 1.125) with $\varphi(C) \subseteq C$ bounded, then $\varphi(\cdot)$ has a fixed point.*

The above fixed points theorems as well as Schauder's fixed point theorem, concern self maps, that is, maps which map a nonempty, closed, convex set into itself. In applications this is usually difficult to verify. For this reason we want to see what can be said about nonself maps.

Theorem 2.59 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex, $\Omega \subseteq C$ is nonempty open, $u_0 \in \Omega$ and $\varphi : \overline{\Omega} \to C$ is continuous with $\varphi(\overline{\Omega}) \subseteq C$ relatively compact, then one of the following statements holds:*

(a) *$\varphi(\cdot)$ has a fixed point in $\overline{\Omega}$;*
(b) *there exist $v \in \partial\Omega$ and $\lambda \in (0, 1)$ such that*

$$v = \lambda\varphi(v) + (1 - \lambda)u_0.$$

Proof Without any loss of generality, we may assume that $u_0 = 0 \in \Omega$ (otherwise replace Ω by $\Omega - u_0$ and $\varphi(\cdot)$ by $\varphi_0(u) = \varphi(u + u_0) - u_0$ for $u \in \Omega - u_0$).

Suppose that (b) does not hold and that $\varphi|_{\partial\Omega}$ is fixed point free. We consider the set:

$$D = \{u \in \overline{\Omega} : u = \lambda\varphi(u) \text{ for some } \lambda \in [0, 1]\}.$$

Since $0 \in \Omega$, $D \neq \emptyset$. Also D is closed since $\varphi(\cdot)$ is continuous. Moreover, we have $D \cap \partial\Omega = \emptyset$. Invoking Urysohn's Lemma (see Theorem I-1.42), we can find a continuous function $f : \overline{\Omega} \to [0, 1]$ such that

$$f|_D = 1 \quad \text{and} \quad f|_{\partial\Omega} = 0. \tag{2.12}$$

Consider the function $\psi : C \to C$ defined by

$$\psi(u) = \begin{cases} f(u)\varphi(u) & \text{if } u \in \overline{\Omega} \\ 0 & \text{if } u \in C \setminus \overline{\Omega}. \end{cases}$$

Evidently $\psi(\cdot)$ is continuous and $\psi(C) \subseteq C$ is relatively compact. So, Theorem 1.65 (Schauder's fixed point theorem) implies the existence of $u \in C$ such that $\psi(u) = u$. Since $0 \in \Omega$, we see that $u \in \Omega$. We have

$$u = f(u)\varphi(u) \text{ with } u \in \Omega,$$
$$\Rightarrow\ u \in D \text{ and so } f(u) = 1 \text{ (see (12))}.$$

We conclude that $\varphi(u) = u$. □

Remark 2.60 The negation of (b) can be written as

$$\varphi(u) - u_0 \neq \lambda(u - u_0) \quad \text{for all } u \in \partial\Omega, \text{ all } \lambda > 1.$$

In this form the negation of (b) is known as the Leray-Schauder boundary condition and is a consequence of the homotopy invariance of d_{LS} (consider the compact homotopy $h(t, u) = t(\varphi(u) - u_0) + u_0$). It contains most conditions existing in the literature:

(a) Ω is convex and $\varphi(\partial\Omega) \subseteq \overline{\Omega}$ (Rothe [430]);
(b) $\|\varphi(u)\|^2 - \|u\|^2 \leq \|\varphi(u) - u\|^2$ for all $u \in \partial\Omega$ (Altman [9]).

Note that (b) is equivalent to $(\varphi(u), u) \leq \|u\|^2$ if $X = H =$ a Hilbert space.

We extend Theorem 2.59 to maps which satisfy (6) (Mönch-type maps).

Theorem 2.61 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex, $\Omega \subseteq C$ is nonempty open, $u_0 \in \Omega$ and $\varphi : \overline{\Omega} \to C$ is continuous with $\varphi(\overline{\Omega}) \subseteq C$ satisfies (b) (Mönch's condition) and in addition*

$$u \neq \lambda\varphi(u) + (1 - \lambda)u_n \text{ for all } u \in \partial\Omega, \text{ all } \lambda \in [0, 1],$$

then $\varphi(\cdot)$ has a fixed point.

Proof The proof is similar to that of Theorem 2.59. In the present case the map $\psi(\cdot)$ satisfies Mönch's condition. So, we apply Theorem 2.57 and obtain $u \in C$ such that

$$\psi(u) = u.$$

Since $u_0 \in \Omega$, we have $u \in \Omega$. Therefore

$$\begin{aligned} u &= f(u)\varphi(u) + (1 - f(u))u_0 \\ \Rightarrow\ & u \in D \text{ and so } f(u) = 1. \end{aligned}$$

We conclude that $\varphi(u) = u$. □

Using Theorem 2.61, we have the following generalization of Theorem 2.56 (Sadovski's fixed point theorem).

Theorem 2.62 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex, $\Omega \subseteq C$ is nonempty open, $u_0 \in \Omega$ and $\varphi : \overline{\Omega} \to C$ is continuous, condensing with $\varphi(\overline{\Omega}) \subseteq C$ bounded and*

$$u \neq \lambda\varphi(u) + (1 - \lambda)u_0 \text{ for all } u \in \partial\Omega, \text{ all } \lambda \in [0, 1],$$

then $\varphi(\cdot)$ has a fixed point.

So, we can state the following Alternative Theorem.

Theorem 2.63 *If X is a Banach space, $C \subseteq X$ is nonempty, closed, convex, $\Omega \subseteq C$ is nonempty open, $u_0 \in \Omega$ and $\varphi : \overline{\Omega} \to C$ is continuous, condensing and $\varphi(\overline{\Omega}) \subseteq C$ is bounded, then one of the following two statements holds:*

(a) $\varphi(\cdot)$ *has a fixed point;*
(b) *there exist* $u \in \partial\Omega$ *and* $\lambda \in (0, 1)$ *such that* $u = \lambda\varphi(u) + (1 - \lambda)u_0$.

Remark 2.64 Another formulation of the Alternative Theorem is the following (see Corollary 1.68)

"If X is a Banach space, $C \subseteq X$ is nonempty, convex $0 \in C$, $\varphi : C \to C$ is compact and $E(\varphi) = \{u \in C : u = \lambda\varphi(u)$ for some $\lambda \in (0, 1)\}$,

then either $E(\varphi)$ is unbounded or $\varphi(\cdot)$ has a fixed point".

In this form the result is known as the "Leray-Schauder Alternative Principle".

Next we will prove a fixed point theorem which is formulated in purely topological terms and includes the Schauder fixed point theorem as a special case.

Definition 2.65 A metric space X is an "absolute retract" (AR for short), if for any metric space Y and every $C \subseteq Y$ closed, any continuous function $\varphi : C \to X$ admits a continuous extension.

Remark 2.66 So, an AR is a metric space which can replace $\mathbb{R}$ in the Tietze extension theorem (see Theorem I-1.47 for metric spaces). If X is an AR and V is homeomorphic to X, then V is an AR.

Using Theorem 2.55 (the Dugundji Extension Theorem), we see that every convex set of a Banach space, is an AR Also, we can show that if a metric space $X = C_1 \cup C_2$ with C_1, C_2 closed and $\widehat{C} = C_1 \cap C_2$, then we have

(a) if $\widehat{C}, C_1, C_2$ are AR, then X is an AR;
(b) if $X, \widehat{C}$, are AR, then C_1, C_2 is an AR.

Then it follows that:

Proposition 2.67 *If* Y, X *are metric spaces and* X *is an AR, then any two continuous maps* φ, $\psi : Y \to X$ *are homotopic(denoted by* $\varphi \sim \psi$*).*

Corollary 2.68 *If* Y, X *are metric spaces and* X *is an AR, then* $C(Y, X)/_\sim = \{Y, X\}$ *is a singleton.*

Definition 2.69 Let Y, X be metric spaces. We say that X and Y are "homotopy equivalent" (denoted by $X \sim Y$), if there exist two continuous maps $\varphi : X \to Y$ and $\psi : Y \to X$ such that

$$\psi \circ \varphi \sim id_X \quad \text{and} \quad \varphi \circ \psi \sim id_Y.$$

Remark 2.70 If X, Y are homeomorphic metric spaces, then $X \sim Y$ in the above sense. In Definition 2.45 (c) we introduced the notion of a contractible metric space X, by requiring that $id_X \sim 0$. Note that equivalently we can say that $X \sim \{u_0\}$. These definitions do not depend on the choice of the point u_0 belongs to X, since any two singletons are homeomorphic, thus homotopy equivalent.

From these definitions we infer that:

Proposition 2.71 *If a metric space X is an AR, then X is contractible.*

Remark 2.72 The converse of the above proposition is not true.

Definition 2.73 A nonempty, compact metric space X is an "R_δ-set" if it is equal to the intersection of a decreasing sequence of compact absolute retracts, that is,

$$X = \bigcap_{n\in\mathbb{N}} C_n$$

with $\{C_n\}_{n\in\mathbb{N}}$ a decreasing sequence of compact absolute retracts.

Remark 2.74 To topologist sine curve (see Example I-1.130) is an R_δ-set but it is not contractible. In fact an R_δ-set need not be an AR. However any intersection of a decreasing sequence of R_δ-sets is an R_δ-set.

Proposition 2.75 *If X is a metric space which is an AR and $D \subseteq X$ is a retract of X, then D is an AR.*

Proof Let Y be a metric space, $C \subseteq Y$ a nonempty closed set and $\varphi : C \to D$ a continuous map. Since $D \subseteq X$ and X is an AR, we can find a continuous extension $\widehat{\varphi} : Y \to X$ of φ (that is, $\widehat{\varphi}|_C = \varphi$). Let $r : X \to D$ be the retraction of X on D. Then $\widehat{\varphi}_0 = r \circ \varphi : Y \to D$ is continuous and $\widehat{\varphi}_0|_D = \varphi$, that is, $\widehat{\varphi}_0$ is the desired extension of φ and so D is an AR. □

Now we ready for the generalized version of the Schauder fixed point theorem.

Theorem 2.76 *If X is a metric space which ia an AR and $\varphi : X \to X$ is a continuous map with $\varphi(X)$ relatively compact, then $\varphi(\cdot)$ has a fixed point.*

Proof Let $K = \overline{\varphi(X)} \subseteq X$. This set is compact and so it is homeomorphic to a closed subset $\widehat{K}$ of the Hilbert cube $I^\infty = [0,1]^{\mathbb{N}}$. Let $h : K \to \widehat{K}$ be this homeomorphism. We have

$$X \overset{\varphi}{\to} K \overset{h}{\to} \widehat{K} \overset{h^{-1}}{\to} K \overset{id}{\to} X \quad \text{(id=identity map)}$$

Consider the map $id \circ h^{-1} : \widehat{K} \to X$. This map is continuous, $\widehat{K} \subseteq I^\infty$ is closed and by hypothesis X ia an AR. So, the map $id \circ h^{-1}$ admits a continuous extension $\psi : I^\infty \to X$. Let $\gamma = \widehat{i} \circ h \circ \varphi$ with $\widehat{i} : \widehat{k} \to I^\infty$ the inclusion map. Then $\gamma : X \to I^\infty$ is continuous. So $\gamma \circ \psi : I^\infty \to I^\infty$ is continuous. But the space I^∞ has the fixed point property since each factor $I = [0,1]$ in the Cartesian product has the fixed point property. Therefore we can find $\widehat{u} \in I^\infty$ such that

$$(\gamma \circ \psi)(\widehat{u}) = \widehat{u}.$$

Let $\widehat{v} = \psi(\widehat{u})$. We have

$$\psi(\widehat{v}) = (\gamma \circ \psi)(\widehat{u}) = \psi(\gamma(\psi(\widehat{u}))) = \psi(\widehat{u}) = \widehat{v},$$

$$\Rightarrow \varphi(\cdot) \text{ has a fixed point.}$$

□

2.3 Order Fixed Point Theory

In this section we prove fixed point theorems in which the order structure of the ambient space plays a central role.

In a Banach space (or more generally in a vector space), partial order is introduced by a cone.

Definition 2.77 Let X be a Banach space. A set $K \subseteq X$ is a "cone", if K is a closed convex set in X, $\lambda K \subseteq K$ for all $\lambda \geq 0$ and $K \cap (-K) = \{0\}$.

Remark 2.78 We will always consider nontrivial cones, that is, $K \neq \{0\}$.

Using the cone $K \subseteq X$, we can introduce a partial order $\leq$ in X by

$$u \leq v \quad \text{if and only if} \quad v - u \in K.$$

Evidently $\leq$ is a partial order on X since $\leq$ is reflexive, antisymmetric, transitive. A sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ is "monotone" if it is increasing (that is, $u_n \leq u_{n+1}$ for all $n \in \mathbb{N}$) or decreasing (that is, $u_{n+1} \leq u_n$ for all $n \in \mathbb{N}$). For every $u,\ v \in X$, by $[u,\ v]$ we denote the "order interval" between these two elements, that is,

$$[u, v] = \{x \in X :\ u \leq x \leq v\}.$$

Clearly $[u, v] \neq \emptyset$ if and only if $u \leq v$. A set $C \subseteq X$ is said to be "order bounded", if C is contained in some order interval. We say that $C \subseteq X$ is bounded above (resp. below), if there is $\overline{u} \in X$ such that $u \leq \overline{u}$ (resp. $\overline{u} \leq u$) for all $u \in C$.

In what follows a Banach space X together with a cone K which defines a partial order, is said to be an "Ordered Banach Space" (OBS for short) and K is the "order" (or "positive") cone. Next we introduce some important classes of order cones.

Definition 2.79 Let X be an OBS with order cone K. Then

(a) K is "reproducing", if $X = K - K$ and K is "total" if $X = \overline{K - K}$.
(b) K is "normal", if there exists $\delta > 0$ such that for all $u, v \in K \cap \partial B_1$, we have $\|u + v\| \geq \delta$.
(c) K is "regular", if every increasing sequence which is bounded above, converges.
(d) K is "fully regular", if every increasing sequence which norm bounded, converges.

(e) K is "minihedral", if $\sup\{u, v\}$ exists for all $u, v \in X$ and it is "strongly minihedral", if for any $C \subseteq X$ bounded above, $\sup C$ exists. Recall, $\sup C$ is an element $\widehat{u} \in X$ such that

$$u \leq \widehat{u} \quad \text{for all} \quad u \in C.$$

and if $u \leq \widehat{y}$ for all $u \in C$, then $\widehat{u} \leq \widehat{y}$. Similarly we define $\inf C$.

Remark 2.80 Evidently normality of K means that any two positive elements on the unit sphere, form an angle which is bounded away from π. So, a normal cone can not be too wide.

Proposition 2.81 *If X is an OBS with order cone K, then*

(a) *int $K \neq \emptyset \Rightarrow K$ is reproducing;*
(b) *K is reproducing $\Leftrightarrow \varrho\overline{B_1} \subseteq C - C$ for some $\varrho > 0$ with $C = K \cap \overline{B_1}$.*

Proof

(a) Let $u_0 \in \text{int}K$ and let $r > 0$ be such that $B_r(u_0) \subseteq K$. If $u \in X \setminus \{0\}$, then $u_0 + r\frac{u}{\|u\|} \in K$ and $u = \frac{\|u\|}{r}(u_0 + r\frac{u}{\|u\|}) - \frac{\|u\|}{r}u_0 \in K - K$ and so K is reproducing.
(b) $\Rightarrow$ We have $X = \bigcup_{n\in\mathbb{N}} n(C - C)$ and so by Baire's theorem we have $\text{int}D \neq \emptyset$ with $D = \overline{C - C}$. Since D is convex, symmetric we can find $\varrho_0 > 0$ small such that $\varrho_0 B_1 \subseteq D$. Then from Lemma I-3.81 it follows that for some $\varrho \in (0, \varrho_0)$, $\varrho B_1 \subseteq C - C$.

□

The converse of (a) is not true. Consider $X = L^p(0, 1)$ with $1 \leq p < \infty$ with order cone $K = L^p(0, 1)_+ = \{u \in L^p(0, 1) : u(z) \geq 0 \text{ for a.a. } z \in \Omega\}$. We know that for every $u \in X$, $u = u^+ - u^-$ with $u^+, u^- \in K$, but $int\, K = \emptyset$.

Definition 2.82 Let X be an OBS with positive cone K and norm $\|\cdot\|$. We say that $\|\cdot\|$ is "semimonotone" if there is a constant $c > 0$ such that for all $u, v \in K$ with $u \leq v$, we have $\|u\| \leq c\|v\|$.

The next proposition characterizes normal cones.

Proposition 2.83 *Let X be an OBS with positive cone K and norm $\|\cdot\|$.*

(a) *K is normal $\Leftrightarrow$ $\|\cdot\|$ is semimonotone.*
(b) *K is normal $\Leftrightarrow$ $(B_1 + K) \cap (B_1 - K)$ is bounded.*

Proof (a) $\Rightarrow$ Arguing by contradiction, suppose that $\|\cdot\|$ is not semimonotone. Then we can find $0 \leq u_n \leq v_n$ such that

$$\|u_n\| > n\|v_n\| \quad \text{for all } n \in \mathbb{N}. \tag{2.13}$$

Let $b_n = \frac{u_n}{\|u_n\|} + \frac{1}{n}\frac{v_n}{\|v_n\|}$, $c_n = \frac{1}{n}\frac{v_n}{\|v_n\|} - \frac{u_n}{\|u_n\|}$, $n \in \mathbb{N}$. Using (2.13) we see that $b_n,\ c_n \in K$ for all $n \in \mathbb{N}$. So, the normality of K implies that

$$\left\| \frac{b_n}{\|b_n\|} + \frac{c_n}{\|c_n\|} \right\| \geq \delta \quad \text{for all } n \in \mathbb{N}.$$

But we can calculate that

$$\frac{b_n}{\|b_n\|} + \frac{c_n}{\|c_n\|} = \frac{2v_n}{n\|v_n\|\|b_n\|} + \frac{\|b_n\| - \|c_n\|}{\|b_n\|\|c_n\|}c_n.$$

Note that

$$1 - \frac{1}{n} \leq \|b_n\|,\ \|c_n\| \leq 1 + \frac{1}{n},\ |\|b_n\| - \|c_n\|| \leq \frac{2}{n},\ n \in \mathbb{N}.$$

Therefore

$$\left\| \frac{b_n}{\|b_n\|} + \frac{c_n}{\|c_n\|} \right\| \leq \frac{4}{n-1} \to 0,$$

which contradicts the normality of K.
$\Leftarrow$ Let $u, v \in K \cap \partial B_1$. We have

$$\begin{aligned} & 1 = \|u\| \leq c\|u + v\| \text{ (since } \|\cdot\| \text{ is semimonotone)}, \\ \Rightarrow\ & \frac{1}{c} \leq \|u + v\| \text{ for all } u,\ v \in K \cap \partial B_1, \\ \Rightarrow\ & K \text{ is normal.} \end{aligned}$$

(b) $\Rightarrow$ Let $u \in (\overline{B_1} + K) \cap (\overline{B_1} - K)$. This means that

$$\begin{aligned} & y \leq u \leq h \quad \text{for some } y,\ h \in \overline{B_1} \\ \Rightarrow\ & 0 \leq u - y \leq h - y. \end{aligned} \tag{2.14}$$

Since K is normal by part (a) $\|\cdot\|$ is semimonotone. So, from (2.14) we have

$$\begin{aligned} & \|u - y\| \leq c\|h - y\|, \\ \Rightarrow\ & \|u\| \leq 1 + 2c \text{ (recall } y,\ h \in \overline{B_1}) \\ \Rightarrow\ & (\overline{B_1} + K) \cap (\overline{B_1} - K) \text{ is bounded.} \end{aligned}$$

$\Leftarrow$ By hypothesis we can find $r > 0$ s.t.

$$(\overline{B_1} + K) \cap (\overline{B_1} - K) \subseteq \overline{B_r}.$$

Suppose $0 \leq u \leq v$. We have

$$\frac{v-u}{\| v \|} \in (\overline{B_1} - K) \cap K,$$
$$\Rightarrow \| v - u \| \leq \eta \| v \| \text{ for some } \eta > 0,$$
$$\Rightarrow \| u \| \leq (1+\eta) \| v \|,$$
$$\Rightarrow \| \cdot \| \text{ is semimonotone and so } K \text{ is normal (see (a))}$$

□

In the next proposition, we relate the notions introduced in Definition 2.79.

Proposition 2.84 *Let X be an OBS with order cone K. Then we have $K =$ fully regular $\Rightarrow K =$ regular $\Rightarrow K$ is normal.*

Proof Suppose K is not normal. Then according to Proposition 2.83(a), we can find $\{u_n\}_{n\in\mathbb{N}}, \{v_n\}_{n\in\mathbb{N}} \subseteq K$ such that

$$0 \leq u_n \leq v_n, \| u_n \| > 2^n \| v_n \| \text{ for all } n \in \mathbb{N}$$

We set $b_n = \frac{u_n}{\| u_n \|}, c_n = \frac{v_n}{2^n \| u_n \|}, n \in \mathbb{N}$. We have

$$0 \leq b_n \leq \frac{u_n}{2^n \| u_n \|} \leq \frac{v_n}{2^n \| v_n \|} = c_n, b_n \neq 0, n \in \mathbb{N}$$
$$\text{and } \sum_{n\in\mathbb{N}} \| c_n \| = \sum_{n\in\mathbb{N}} \frac{1}{2^n} = 1.$$

So, $\sum\limits_{n\in\mathbb{N}} c_n$ is absolutely convergent and we can define

$$c = \sum_{n\in\mathbb{N}} c_n \in X$$

Let

$$z_n = \begin{cases} \sum\limits_{k=1}^{m} c_{2k}, & \text{if } n = 2m, m \in \mathbb{N} \\ \sum\limits_{k=1}^{m} c_{2k+1}, & \text{if } n = 2m+1, m \in \mathbb{N} \end{cases}$$

Evidently $\{z_n\}_{n\in\mathbb{N}} \subseteq K \setminus \{0\}$ is increasing and bounded above by c. Also $\sup\limits_{n\in\mathbb{N}} \| z_n \| \leq 2$. So, $\{z_n\}_{n\in\mathbb{N}}$ is norm bounded too. Note that

$$\begin{aligned} &\| z_{2m+1} - z_{2m} \| = \| c_{2m+1} \| = 1, \\ \Rightarrow\ &\{z_n\}_{n\in\mathbb{N}} \text{ is not convergent}, \\ \Rightarrow\ &K \text{ is not regular and fully regular.} \end{aligned}$$

It remains to show that full regularity implies regularity. So, let K be fully regular and consider an increasing sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq K$, which is bounded above by $y \in K$. As we have seen in the first part of the proof, full regularity implies normality, So, by Proposition 2.83(a), we have

$$\begin{aligned} &\| u_n \| \leq c \| y \| \text{ for all } n \in \mathbb{N}, \\ \Rightarrow\ &u_n \to u \text{ (by full regularity)}, \\ \Rightarrow\ &K \text{ is regular.} \end{aligned}$$

□

In reflexive Banach spaces the situation is better.

Proposition 2.85 *If X is a reflexive OBS with order cone K, then normal $\Leftrightarrow$ regular $\Leftrightarrow$ fully regular.*

Proof On account of Proposition 2.84, it suffices to show that normality implies full regularity.

So, consider $\{u_n\}_{n\in\mathbb{N}} \subseteq K$ increasing and norm bounded. By reflexivity, we can find a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}}$ such that

$$u_{n_k} \xrightarrow{w} \widehat{u} \text{ in } X \tag{2.15}$$

We have $u_{n_k} \leq \widehat{u}$ for all $k \in \mathbb{N}$. If this is not true, we can find $k_0 \in \mathbb{N}$ such that $\widehat{u} - u_{n_{k_0}} \notin K$. Invoking the strong separation theorem, we can find $u^* \in X^* \setminus \{0\}$ such that

$$\langle u^*, \widehat{u} - u_{n_{k_0}} \rangle < \eta \leq \langle u^*, y \rangle \text{ for all } y \in K, \tag{2.16}$$

$$\Rightarrow \langle u^*, \widehat{u} \rangle < \eta + \langle u^*, u_{n_{k_0}} \rangle, \tag{2.17}$$

Also since $u_{n_k} - u_{n_{k_0}} \in K \setminus \{0\}$ for $k > k_0$, we have

$$\langle u^*, u_{n_{k_0}} \rangle + \eta \leq \langle u^*, u_{n_k} \rangle,\ k > k_0 \text{ (see (16))}, \tag{2.18}$$

$$\Rightarrow \langle u^*, u_{n_{k_0}} \rangle + \eta \leq \langle u^*, \widehat{u} \rangle.$$

Comparing (2.17) and (2.18) we have a contradiction. Therefore we have $0 \leq u_{n_k} \leq \widehat{u}$ for all $k \in \mathbb{N}$.

We claim that $\{u_{n_k}\}_{k\in\mathbb{N}}$ contains a strongly convergent subsequence to $\widehat{u}$. If the claim is not true, then we can find $\varepsilon > 0$ and $m \in \mathbb{N}$ such that

$$\| v_{n_k} - \widehat{u} \| \geq \varepsilon \text{ for all } k \geq m \tag{2.19}$$

Consider the increasing sequence of convex sets

$$C_k = \{u \in X : u \leq u_{n_k}\}, k \in \mathbb{N}_m = \{m, m+1, \ldots\}$$

If we set $C = \bigcup_{k\in\mathbb{N}_m} C_k$, then C is convex and so $\overline{C}$ is convex too. We show that $\widehat{u} \notin \overline{C}$. Let $u \in C$, then $u \in C_k$ for some $k \in \mathbb{N}_m$ and so $u \leq u_{n_k}$. We have

$$\begin{aligned} &0 \leq \widehat{u} - u_{n_k} \leq \widehat{u} - u \\ &\Rightarrow \| \widehat{u} - u \| \geq \frac{\varepsilon}{c} \text{ (see (2.19) and use the normality of } K) \\ &\Rightarrow \widehat{u} \notin \overline{C}. \end{aligned}$$

So, by the strong separation theorem, we can find $v^* \in X^* \cdot \{0\}$ such that

$$\begin{aligned} &\langle v^*, \widehat{u} \rangle < \eta_0 \leq \langle v^*, h \rangle \text{ for all } h \in \overline{C}, \\ &\Rightarrow \eta_0 \leq \langle v^*, u_{n_k} \rangle \text{ for all } k \geq m \text{ (since } u_{n_k} \in C_k), \\ &\Rightarrow \langle v^*, \widehat{u} \rangle < \eta_0 \leq \langle v^*, \widehat{u} \rangle, \text{ a contradiction.} \end{aligned}$$

Therefore $\{u_{n_k}\}_{k\in\mathbb{N}}$ contains a subsequence which strongly converges to $\widehat{u}$.

Finally we show that the whole sequence which strongly converges to $\widehat{u}$ in X. We already know that for some subsequence $\{u_{n_i}\}_{i\in\mathbb{N}}$ of $\{u_n\}_{n\in\mathbb{N}}$ we have

$$u_{n_i} \to \widehat{u} \text{ in } X$$

Fix $n \in \mathbb{N}$. Then $u_n \leq u_{n_i}$ for $i \in \mathbb{N}$ large and so

$$\begin{aligned} &u_n \leq \widehat{u} \text{ for all } n \in \mathbb{N}, \\ &\Rightarrow 0 \leq \widehat{u} - u_n \leq \widehat{u} - u_{n_i} \text{ for } n \geq n_i, \\ &\Rightarrow \| \widehat{u} - u_n \| \leq c \| \widehat{u} - u_{n_i} \| \text{ for } n \geq n_i, \\ &\text{(by normality, see Proposition 2.83)} \\ &\Rightarrow u_n \to \widehat{u}. \end{aligned}$$

Therefore K is fully regular. □

For cones with nonempty interior (also known as solid cones), we have the following result which is a very useful tool in the study of variational problems.

Proposition 2.86 *If X is an OBS with order cone K, $int\,K \neq \emptyset$ and $u_0 \in int\,K$, then for every $u \in X$ we can find $t_u > 0$ such that $t_u u_0 - u \in K$.*

Proof Since $u_0 \in int\,K$, we can find $r > 0$ such that $B_r(u_0) \subseteq K$. Let $t_u > \frac{\|u\|}{r}$. We have

$$\begin{aligned} & u_0 - \frac{1}{t_u} u \in B_r(u_0) \subseteq K, \\ \Rightarrow\ & t_u u_0 - u \in K. \end{aligned}$$

□

Proposition 2.87 *If X is a separable OBS with order cone K and K is regular and minihedral, then K is strongly minihedral.*

Proof Let $E \subseteq X$ be a nonempty bounded above set. The set E is separable with $\{u_n\}_{n\in\mathbb{N}}$ dense. Let $v_n = \sup\{u_k\}_{k=1}^{n}$ (recall that K is minihedral). We have that $\{v_n\}_{n\in\mathbb{N}}$ is increasing and bounded above. We have $\{v_n - v_1\}_{n\in\mathbb{N}} \subseteq K$ is increasing and bounded above. So, by regularity we have

$$v_n \to \widehat{v} \text{ in } X \tag{2.20}$$

We claim that $\widehat{v} = \sup E$. We have

$$u_n \leq \widehat{v} \text{ for all } n \in \mathbb{N} \tag{2.21}$$

If $u \in E$, then we can find a subsequence of $\{u_n\}_{n\in\mathbb{N}}$ converging to u and so from (2.21) we infer that

$$\begin{aligned} & u \leq \widehat{v}, \\ \Rightarrow\ & \widehat{v} \text{ is an upper bound of } E. \end{aligned}$$

Suppose $h \in X$ is another upper bound of E. We have

$$\begin{aligned} & v_n \leq h \text{ for all } n \in \mathbb{N}, \\ \Rightarrow\ & \widehat{v} \leq h \text{ (see (2.20))}, \\ \Rightarrow\ & \widehat{v} = \sup E. \end{aligned}$$

□

From Propositions 2.85 and 2.87 we infer the following corollary.

Corollary 2.88 *If X is a separable, reflexive OBS with order cone K which is normal and minihedral, then K is strongly minihedral.*

We now introduce the important concept of dual cone (positive linear functionals). It is an important tool for the use of duality techniques in ordered Banach space.

Definition 2.89 Let X be an OBS with order cone K. The "dual cone" of K is the set

$$K^* = \{u^* \in X^* : \langle u^*, u\rangle \geq 0 \text{ for all } u \in K\}$$

Remark 2.90 In fact note that K^* need not be a cone, since it is not necessary that we have $K^* \cap (-K^*) = \{0\}$. However, if K is total, then K^* is indeed a cone.

The elements of K^* are called "positive linear functionals". If $u^* \in K^*$ and $\langle u^*, u\rangle > 0$ for all $u \in K \setminus \{0\}$, then we say that u^* is "strictly positive". Finally, if $u^* \in K^*$ and $\langle u^*, u\rangle \geq c \parallel u \parallel$ for some $c > 0$, all $u \in K$, then we say that u^* is "strongly positive".

In the next propositions, we prove some useful properties of the positive linear functionals.

Proposition 2.91 *If X is an OBS with order cone K, then*

(a) $u \in K \Leftrightarrow \langle u^*, u\rangle \geq 0$ *for all* $u^* \in K^*$;
(b) *If* $u \in K \setminus \{0\}$, *then there exists* $u^* \in K^*$ *such that* $\langle u^*, u\rangle > 0$;
(c) *If* $u \notin K$, *then we can find* $u^* \in K^*$ *such that* $\langle u^*, u\rangle < 0$.

Proof

(a) Let $v \notin K$. We can find $r > 0$ such that $B_r(v) \cap K = \emptyset$ and so by the weak separation theorem we can find $u^* \in X^* \setminus \{0\}$ such that

$$\langle u^*, b\rangle < \inf\{\langle u^*, u\rangle : u \in K\} = m \text{ for all } b \in B_r(v). \tag{2.22}$$

Since K is a cone, we know $\lambda K \subseteq K$ for all $\lambda > 0$ and so we infer that

$$m \leq \langle u^*, \lambda u\rangle \text{ for all } u \in K, \text{all}\lambda > 0,$$
$$\Rightarrow m = 0.$$

This proves (a).

(b) Note that $u \notin (-K)$ and by strong separation theorem we can find $y^* \in X^* \setminus \{0\}$ such that

$$\langle y^*, u\rangle < 0 = \inf\{\langle y^*, -v\rangle : v \in K\},$$
$$\Rightarrow \langle -y^*, u\rangle > 0 \text{ and } -y^* \in K^*.$$

(c) This too is an easy consequence of the strong separation theorem.

□

Proposition 2.92 *If X is an OBS with order cone K and $intK \neq \emptyset$, then $u \in intK \Leftrightarrow \langle u^*, u\rangle > 0$ for all $u^* \in K^* \setminus \{0\}$.*

Proof $\Rightarrow$ Let $u \in \text{int}K$. We can find $r > 0$ such that

$$\begin{aligned} & B_r(u) \subseteq K, \\ \Rightarrow\ & 0 \leq u \pm rh \text{ for all } h \in B_1, \\ \Rightarrow\ & 0 \leq \langle u^*, u \pm rh\rangle \text{ for all } u^* \in K^*, \text{all} h \in B_1, \\ \Rightarrow\ & 0 < r \parallel u^* \parallel_* \leq \langle u^*, u\rangle \text{ for all } u^* \in K^* \setminus \{0\}. \end{aligned}$$

$\Leftarrow$ If $v \in \partial K$, then since $intK$ is convex, by the weak separation theorem, we can find $u^* \in X^* \setminus \{0\}$ such that

$$\begin{aligned} & \langle u^*, v\rangle \leq \langle u^*, u\rangle \text{ for all } u \in intK, \\ \Rightarrow\ & \langle u^*, v\rangle \leq 0. \end{aligned}$$

If $\langle u^*, v\rangle < 0$, then we can find $r > 0$ such that

$$\langle u^*, y\rangle < 0 \text{ for all } y \in B_r(v). \tag{2.23}$$

But $B_r(v) \cap intK \neq \emptyset$ (since $v \in \partial K$). So, from (2.23) we have

$$\langle u^*, u\rangle < 0 \text{ for some } u \in intK,$$

Since $\lambda \text{int}K \subseteq \text{int}K$ for all $\lambda > 0$, we have a contradiction. Therefore $u^* \in K^* \setminus \{0\}$ and $\langle u^*, v\rangle = 0$. This proves this implication. □

Proposition 2.93 *If X is a separable OBS with order cone K, then there exists a strictly positive linear functional $u^* \in K^* \setminus \{0\}$.*

Proof Let $\{u_n\}_{n\in\mathbb{N}}$ be dense in $\overline{B}_1$ and consider the metric

$$d(u^*, v^*) = \sum_{n\in\mathbb{N}} \frac{1}{2^n} |\langle u^* - v^*, u_n\rangle| \text{ for all } u^*, v^* \in K^* \cap \overline{B}_1^*.$$

From the proof of Theorem I-3.117 we know that

$$u_n^* \xrightarrow{d} u^* \text{ if and only if } u_n^* \xrightarrow{w^*} u^*.$$

Theorem I-3.117 says that $(\overline{B}_1^*, d)$ is a compact metric space, so it is separable. Let $\{u_n^*\}_{n\in\mathbb{N}}$ be d-dense in $K^* \cap \overline{B}_1^*$. We set $u_0^* = \sum_{n\in\mathbb{N}} \frac{1}{n^2} u_n^* \in K^*$. Then we see that

$$\begin{aligned} &\langle u_0^*, u\rangle = 0, \text{ for some } u \in K, \\ &\Rightarrow \langle u^*, u\rangle = 0, \text{ for all } u^* \in K^*, \\ &\Rightarrow u = 0. \end{aligned}$$

Therefore u_0^* is the desired strictly positive functional. □

We already know (see Remarch 2.90), that if K is reproducing, then K^* is a cone and so (X^*, K^*) is an OBS. In fact more can be said about X^* in this case. To do this we will need the following lemma.

Lemma 2.94 *If X is an OBS with order cone K which is reproducing, then we can find $\varrho > 0$ such that $\varrho\overline{B}_1 \subseteq C - C$ with $C = K \cap \overline{B}_1$.*

Proof Since K is reproducing, we have that

$$X = \bigcup_{n\in\mathbb{N}} n(C - C).$$

Then Baire's theorem implies that $\text{int}(C - C) \neq \emptyset$ and so we can find $\varrho > 0$ such that

$$\varrho\overline{B}_1 \subseteq C - C.$$

□

Using this lemma, we can prove the following theorem.

Theorem 2.95 *If X is an OBS with order cone K which is reproducing, then K^* is normal.*

Proof From Lemma 2.94 we know that for some $\varrho > 0$ we have

$$\varrho\overline{B}_1 \subseteq C - C \text{ with } C = K \cap \overline{B}_1. \tag{2.24}$$

Let $u^*, v^* \in X^*$ with $0 \leq u^* \leq v^*$ (order induced by K^*) and let $u \in X$. From (2.24) we have

$$\varrho\frac{u}{\| u \|} = b - c \text{ with } b, c \in C.$$

It follows that

$$\begin{aligned} \langle u^*, u\rangle &= \frac{1}{\varrho} \| u \| \langle u^*, b - c\rangle, \\ &\leq \frac{1}{\varrho} \| u \| \langle v^*, b\rangle \end{aligned}$$

$$\leq \frac{1}{\varrho} \parallel u \parallel \parallel v^* \parallel_*,$$

$$\Rightarrow \parallel u^* \parallel_* \leq \tfrac{1}{\varrho} \parallel v^* \parallel_* \text{ (since } u \in X \text{is arbitrary)},$$
$$\Rightarrow K^* \text{ is normal (see Proposition 2.83(a))} \qquad \square$$

We have an extension theorem for positive linear functionals.

Theorem 2.96 *If X is an OBS with order cone K, $int\, K \neq \emptyset$, $V \subseteq X$ is a subspace such that $int\, K \cap V \neq \emptyset$ and u^* is a continuous linear functional on V such that*

$$0 \leq \langle u^*, u \rangle \text{ for all } u \in K \cap V,$$

then there exists a continuous, linear functional $\widehat{u}^*$ on X such that

$$\widehat{u}^*|_V = u^* \text{ and } \widehat{u}^* \in K^*.$$

Proof From Proposition 2.81(a) we know that K is reproducing and so (X^*, K^*) is an OBS.

Let $L = \{v \in V : \langle u^*, v \rangle = 0\}$. This is a closed, linear subspace of V. We claim that $L \cap int\, K = \emptyset$. Indeed, if $u \in L \cap int\, K$, then $\langle u^*, u \rangle = 0$ and we can find $u_1 \in L \cap K$ such that $\langle u^*, u_1 \rangle < 0$, which contradicts the positivity of u^*. Invoking the weak separation we can find $v^* \in K^* \setminus \{0\}$ separating L and $int\, K$. Let $M = \{v \in V : \langle v^*, v \rangle = 0\}$. Then $L \subseteq M$ and so by Lemma I-3.97 we have

$$u^* = \lambda v^* \text{ with } \lambda > 0 \text{ on } V.$$

Then $\widehat{u}^* = \lambda v^*$ is the desired positive extension of u^*. $\square$

In general we can say the following (the proofs are straightforward).

- K^* is w^*-closed.
- $(\bigcap_{\alpha \in I} K_\alpha)^* = \sum_{\alpha \in I} K_\alpha^*$ for any index set I.
- If $V \subseteq X$ is a subspace with $V \cap int\, K \neq \emptyset$, then $(K \cap V)^* = K^* + V^*$.

Let us introduce some concrete ordered Banach spaces with their order cones.

Example 2.97

(a) Let $\Omega \subseteq \mathbb{R}^{\mathbb{N}}$ be a bounded domain and consider the Lebesgue spaces $L^p(\Omega)$, $1 \leq p \leq \infty$. The order cone in these spaces is given by

$$K = \{u \in L^p(\Omega) : u(z) \geq 0 \text{ for a.a } z \in \Omega\}.$$

This cone is reproducing, $int\, K = \emptyset$ if $1 \leq p < \infty$ and $int\, K \neq \emptyset$ if $p = \infty$. The monotonicity of the L^p-norm ($1 \leq p \leq \infty$) implies that K is normal and in fact fully regular for all $1 \leq p \leq \infty$;

(b) Let D be a compact space and consider the Banach space $C(D)$. This is an OBS with order cone $K = \{u \in C(D) : u(z) \geq 0 \text{ for all } z \in D\}$. Then $intK \neq \emptyset$ (hence K is reproducing too) and K is normal (monotonicity of the sup-norm). On the other hand for $k \in \mathbb{N}$, $C^k(\overline{\Omega})$ is an OBS with order cone K. Then $intK \neq \emptyset$ but it is not normal.
(c) Let $\Omega \subseteq \mathbb{R}^{\mathbb{N}}$ be a bounded domain and $1 \leq p \leq \infty$. We consider the Sobolev space $W^{1,p}(\Omega)$. These are OBS with order cone

$$K = \{u \in W^{1,p}(\Omega) : u(z) \geq 0 \text{ for a.a } z \in \Omega\}.$$

Then $intK = \emptyset$ if $1 \leq p < \infty$ but it is reproducing. If $p = \infty$, then $intK \neq \emptyset$. Finally K is not normal for all $1 \leq p \leq \infty$.

Now we pass to fixed point theorems.

Definition 2.98 Let X be an OBS with order cone K and $\varphi : D \subseteq X \to X$ a map. We say

(a) $\varphi(\cdot)$ is "increasing", if $u \leq v \Rightarrow \varphi(u) \leq \varphi(v)$.
(b) $\varphi(\cdot)$ is "strictly increasing", if $v - u \in K \setminus \{0\} \Rightarrow \varphi(v) - \varphi(u) \in K \setminus \{0\}$.
(c) $\varphi(\cdot)$ is "strongly increasing", if $v - u \in intK \Rightarrow \varphi(v) - \varphi(u) \in intK$.

Theorem 2.99 *If X is an OBS with order cone K which is regular and $\varphi : [u_0, v_0] \to X$ is demicontinuous, increasing and*

$$u_0 \leq \varphi(u_0), \varphi(v_0) \leq v_0,$$

then $\varphi(\cdot)$ has a maximal and a minimal fixed point in $[u_0, v_0]$. (extremal fixed points)

Proof Let $u_n = \varphi(u_{n-1})$ for all $n \in \mathbb{N}$. Since $\varphi(\cdot)$ is increasing, we have

$$u_0 \leq u_1 \leq \ldots \leq u_n \leq \ldots \leq v_0 \text{ for all } n \in \mathbb{N}.$$

Then the regularity of K implies that

$$\begin{aligned} & u_n \to \widehat{u} \text{ in } X, \\ \Rightarrow\ & \varphi(u_n) \xrightarrow{w} \varphi(\widehat{u}) \text{ (since } \varphi(\cdot) \text{ is demicontinuous)} \\ \Rightarrow\ & u_{n+1} \xrightarrow{w} \varphi(\widehat{u}) \\ \Rightarrow\ & \varphi(\widehat{u}) = \widehat{u}. \end{aligned}$$

Similarly if $v_n = \varphi(v_{n-1})$ for all $n \in \mathbb{N}$, then

$$\begin{aligned} & u_0 \leq \ldots \leq v_n \leq v_{n-1} \leq \ldots \leq v_1 \leq v_0, \\ \Rightarrow\ & v_n \to \widehat{v} \text{ in } X \text{ by the regularity of } K, \end{aligned}$$

$$\Rightarrow v_{n+1} = \varphi(v_n) \xrightarrow{w} \varphi(\widehat{v}) \text{ (by the demicontinuity of } \varphi(\cdot)),$$
$$\Rightarrow \varphi(\widehat{v}) = \widehat{v}.$$

It remains to show that these are the extremal fixed points of $\varphi(\cdot)$. Suppose $y \in [u_0, v_0]$ is a fixed point of $\varphi(\cdot)$. Then since $\varphi(\cdot)$ is increasing, we have

$$\varphi(u_0) \leq \varphi(y) \leq \varphi(v_0),$$
$$\Rightarrow u_1 \leq \varphi(y) \leq v_1.$$

and continuing this way, we have

$$u_n \leq \varphi(y) \leq v_n \text{ for all } n \in \mathbb{N},$$
$$\Rightarrow \widehat{u} \leq \varphi(y) = y \leq \widehat{v}.$$

This proves the extremality of $\widehat{u}$ and $\widehat{v}$. □

Corollary 2.100 *If X is a reflexive OBS with order cone K which is normal and $\varphi : [u_0, v_0] \to X$ is demicontinuous, increasing and*

$$u_0 \leq \varphi(u_0), \varphi(v_0) \leq v_0,$$

then $\varphi(\cdot)$ has extremal fixed points in $[u_0, v_0]$.

Theorem 2.101 *If X is an OBS with order cone K which is strongly minihedral and $\varphi : [u_0, v_0] \to X$ is increasing and*

$$u_0 \leq \varphi(u_0), \varphi(v_0) \leq v_0,$$

then $\varphi(\cdot)$ has extremal fixed points in $[u_0, v_0]$.

Proof We introduce the set

$$S = \{u \in [u_0, v_0] : u \leq \varphi(u)\}$$

This set is nonempty since $u_0 \in S$ and S is bounded above by v_0. Since K is strongly minihedral, it follows that $\sup S = \widehat{v}$ exists. We show that $\widehat{v}$ is the maximal fixed point of $\varphi(\cdot)$ in $[u_0, v_0]$. We have $u_0 \leq u \leq \widehat{v} \leq v_0$ for all $u \in S$ and so since $\varphi(\cdot)$ is increasing. We have

$$u_0 \leq \varphi(u_0) \leq \varphi(u) \leq \varphi(\widehat{v}) \leq \varphi(v_0) \leq v_0 \text{ for all } u \in S,$$
$$\Rightarrow u \leq \varphi(\widehat{v}) \text{ for all } u \in S,$$
$$\Rightarrow \widehat{v} \leq \varphi(\widehat{v}) \text{ (since } \widehat{v} = \sup S).$$

Note that $\varphi(\widehat{v}) \leq \varphi(\varphi(\widehat{v}))$ and so $\varphi(\widehat{v}) \in S$. Hence

$$\widehat{v} = \varphi(\widehat{v}).$$

Finally we show that $\widehat{v}$ is the maximal fixed point of $\varphi(\cdot)$ in $[u_0, v_0]$. Suppose $y \in [u_0, v_0]$ is another fixed point of $\varphi(\cdot)$. Then $y \in S$ and so $y \leq \widehat{v}$. This proves the maximality of $\widehat{v}$.

For the minimal fixed point, we consider the set

$$T = \{u \in [u_0, v_0] : \varphi(u) \leq u\}.$$

Reasoning as above, we produce the minimal fixed point $\widehat{u} \in [u_0, v_0]$ of $\varphi(\cdot)$. □

From Proposition 2.83(b) we see that if K is normal, then the order intervals in X are bounded. So, from Theorem 1.65 (Schauder's fixed point theorem), we infer the following result.

Theorem 2.102 *If X is an OBS with order cone K which is normal and $\varphi : [u_0, v_0] \to X$ is compact, increasing and*

$$u_0 \leq \varphi(u_0), \varphi(v_0) \leq v_0,$$

then $\varphi(\cdot)$ has a fixed point in $[u_0, v_0]$.

By introducing one more condition on K in the above theorem, we can conclude the existence of extremal fixed points.

Theorem 2.103 *If X is an OBS with order cone K which is normal and minihedral and $\varphi : [u_o, v_0] \to X$ is compact, increasing and*

$$u_0 \leq \varphi(u_0), \varphi(v_0) \leq v_0,$$

then $\varphi(\cdot)$ has extremal fixed points.

Proof We introduce the set

$$S = \{u \in \varphi([u_0, v_0]) : u \leq \varphi(u)\}.$$

Note that $\varphi(u_0) \in S \neq \emptyset$. Let D be a chain in S. Since $\varphi(\cdot)$ is compact, we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq S$ increasing such that

$$\sup D = \sup_{n\in\mathbb{N}} u_n.$$

By compactness we may assume that $u_n \to \widetilde{v}$ in X. We have $u_n \leq \varphi(u_n)$ for all $n \in \mathbb{N}$ and so $\widetilde{v} \leq \varphi(\widetilde{v})$. Also $u \leq \widetilde{v}$ for all $u \in S$, hence $u \leq \varphi(u) \leq \varphi(\widetilde{v})$ (since $\varphi(\cdot)$ is increasing) and so $\widetilde{v} \leq \varphi(\widetilde{v})$, that is, $\widetilde{v} \in S$.

Therefore every chain in S has an upper bound in S. So, by Zorn's lemma S has a maximal element $\widehat{v} \in S$. We show that this is the maximal fixed point of $\varphi(\cdot)$ in $[u_0, v_0]$. Let v be another fixed point of $\varphi(\cdot)$. Since K is minihedral $y = \max\{\widehat{v}, v\} \in X$ exists. We have

$$\begin{aligned} &\varphi(\widehat{v}) = \widehat{v} \leq \varphi(y), \varphi(v) = v \leq \varphi(y), \\ &\Rightarrow y = \max\{\widehat{v}, v\} \leq \varphi(y), \\ &\Rightarrow \varphi(y) \in S. \end{aligned}$$

Hence $\varphi(y) \leq \widehat{v}$, thus $\varphi(y) = \widehat{v}$ and so $v \leq \widehat{v}$. Note that $\varphi(\widehat{v}) = \widehat{v}$. So, $\widehat{v}$ is the maximal fixed point of $\varphi(\cdot)$ in $[u_0, v_0]$.

For the minimal fixed point, we consider the set

$$T = \{u \in \varphi([u_0, v_0]) : \varphi(u) \leq u\}$$

and argue as above. We produce a minimal fixed point $\widehat{u} \in [u_0, v_0]$. □

We continue with X an OBS with order cone K and $\varphi : D \subseteq K \to X$ a map. However, we no longer require that $\varphi(\cdot)$ is increasing. Instead we will use degree theory. To deal with the case when int $K = \emptyset$, we need to modify the Leray-Schauder degree.

Definition 2.104 Let Y be a Banach space, $C \subseteq Y$ a nonempty retract, $\Omega \subseteq C$ a nonempty, bounded, relatively open set and $\varphi : \overline{\Omega} \to X$ a compact map such that $0 \notin (id - \varphi)(\partial\Omega)$ (that is, $\varphi(\cdot)$ is fixed point free on $\partial\Omega$). Then the "fixed point index" for the triple (φ, Ω, C) is defined by

$$i(\varphi, \Omega, C) = d_{LS}(id - \varphi \circ r, r^{-1}(\Omega), 0)$$

with $r : Y \to C$ a retraction.

Remark 2.105 The homotopy invariance and excision properties of the Leray-Schauder degree, imply that the fixed point index is independent of the retraction $r(\cdot)$ used.

From the properties of the Leray-Schauder degree we can infer the main properties of the fixed point index.

Theorem 2.106 *For every triple (φ, Ω, C) as above we have*

(a) *Normalization: $i(\varphi, \Omega, C) = 1$ if $\varphi(u) = u_0 \in \Omega$ for all $u \in \overline{\Omega}$.*
(b) *Solution property: $i(\varphi, \Omega, C) \neq 0$, then there exists $u \in \Omega$ such that $\varphi(u) = u$.*
(c) *Homotopy invariance: if $h : [0, 1] \times \overline{\Omega} \to C$ is a compact homotopy and $h(t, u) \neq u$ for all $(t, u) \in [0, 1] \times \partial\Omega$, then $i(h(t, \cdot), \Omega, C)$ is independent of $t \in [0, 1]$.*

(d) *Domain additivity: if Ω_1, Ω_2 are disjoint open subsets of Ω and $\varphi(\cdot)$ has no fixed points on $\overline{\Omega} \setminus (\Omega_1 \cup \Omega_2)$, then $i(\varphi, \Omega, C) = i(\varphi, \Omega_1, C) + i(\varphi, \Omega_2, C)$.*
(e) *Excision property: if Ω_0 is an open subset of Ω and $\varphi(\cdot)$ has no fixed points in $\overline{\Omega} \setminus \Omega_0$, then $i(\varphi, \Omega, C) = i(\varphi, \Omega_0, C)$.*
(f) *Permanence: if E is a retract of C and $\varphi(\overline{\Omega}) \subseteq E$, then $i(\varphi, \Omega, C) = i(\varphi, \Omega \cap E, E)$.*

We return to our original setting when X is an OBS with order cone K. Since K is closed convex (see Definition 2.77), it is a retract of X (see Theorem 2.55). Let $\Omega \subseteq X$ be nonempty, bounded open set with $K \cap \Omega \neq \emptyset$. Then $K \cap \Omega$ is bounded and open in K. Moreover, we have $\partial(K \cap \Omega) = K \cap \partial\Omega$, $\overline{K \cap \Omega} = K \cap \overline{\Omega}$.

Proposition 2.107 *If $0 \in \Omega$ and $\varphi : K \cap \overline{\Omega} \to K$ is compact such that $\varphi(u) \neq \mu u$ for all $u \in K \cap \partial\Omega$, all $\mu \geq 1$, then $i(\varphi, K \cap \Omega, K) = 1$.*

Proof Consider the compact homotopy

$$h(t, u) = t\varphi(u) \text{ for all } t \in [0, 1], \text{ all } u \in K \cap \overline{\Omega}.$$

The homotopy invariance property and the normalization property imply that $i(\varphi, K \cap \Omega, K) = 1$. □

Proposition 2.108 *If $\varphi : K \cap \overline{\Omega} \to K$ and $\psi : K \cap \partial\Omega \to K$ are both compact maps and*

(i) $\inf\{\|\psi(u)\| : u \in K \cap \partial\Omega\} > 0$;
(ii) $u - \varphi(u) \neq \lambda\psi(u)$ *for all* $u \in K \cap \partial\Omega$, *all* $\lambda \geq 0$,

then $i(\varphi, K \cap \Omega, K) = 0$.

Proof Using Theorem 2.55 we can extend $\psi(\cdot)$ to a compact map $\widehat{\psi} : K \cap \overline{\Omega} \to K$. We know $\widehat{\psi}(K \cap \overline{\Omega}) \subseteq conv\psi(K \cap \partial\Omega)$.

Let $E = \widehat{\psi}(K \cap \partial\Omega)$. We claim that

$$\inf\{\|v\| : v \in \overline{conv}E\} = \eta > 0. \tag{2.25}$$

Let $V = \overline{span}E$. The compactness of $\psi(\cdot)$ implies that E is relatively compact. Hence V is a separable Banach space, $V \subseteq X$. We define $K_0 = K \cap V$. We see that K_0 is a cone in V and $\overline{conv}E \subseteq K_0$. By Proposition 2.93, we can find $v_0^* \in K_0^* \setminus \{0\}$ such that

$$0 < \langle v_0^*, u\rangle \text{ for all } u \in K_0 \setminus \{0\}.$$

We show that

$$\inf\{\langle v_0^*, u\rangle : u \in E\} = \eta_0 > 0. \tag{2.26}$$

If (2.26) is not true, then $\eta_0 = 0$ and so we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq E$ such that $\langle v_0^*, u_n\rangle \to 0^+$. Recall that E is relatively compact. So, we may assume that $u_n \to \widehat{u}$ in X, $\widehat{u} \in K$. We have

$$\begin{aligned} &\langle v_0^*, u_n\rangle \to \langle v_0^*, \widehat{u}\rangle, \\ &\Rightarrow \langle v_0^*, \widehat{u}\rangle = 0 \text{ and so } \widehat{u} = 0. \end{aligned}$$

Therefore $u_n \to 0$ in X, a contradiction to hypothesis (i). This means that (2.26) is true.

Let $y \in conv E$. Then $y = \sum_{k=1}^{n} \lambda_k u_k$, $\lambda_k \geq 0$, $\sum_{k=1}^{n} \lambda_k = 1$, $u_k \in E$. We have

$$\begin{aligned} \langle v_0^*, y\rangle &= \sum_{k=1}^{n} \lambda_k \langle v_0^*, u_k\rangle \qquad (2.27) \\ &\geq \sum_{k=1}^{n} \lambda_k \eta_0 = \eta_0, \\ &\Rightarrow \langle v_0^*, y\rangle \geq \eta_0 > 0 \text{ for all } y \in \overline{conv} E. \end{aligned}$$

The set $\overline{conv} E \subseteq X$ is compact (see Theorem I-3.130). So, there exists $\widehat{v} \in \overline{conv} E$ such that

$$\begin{aligned} &\eta = \|\widehat{v}\| \ \text{(see (2.25))} \\ &\Rightarrow \widehat{v} \neq 0 \ \text{(see (2.27))} \text{ and so } \eta > 0. \end{aligned}$$

We have proved (7.25).

Arguing by contradiction, suppose that $i(\varphi, K \cap \Omega, K) \neq 0$. Hypothesis (ii) and the homotopy invariance property imply that

$$i(\varphi + t\widehat{\psi}, K \cap \Omega, K) \neq 0 \text{ for all } t > 0. \qquad (2.28)$$

Let $\vartheta_1 = \sup\{\|u\| : u \in K \cap \overline{\Omega}\}$, $\vartheta_2 = \sup\{\|\varphi(u)\| : u \in K \cap \overline{\Omega}\}$.Choose $t_0 > \frac{1}{\eta}[\vartheta_1 + \vartheta_2]$. From (2.28) we have

$$i(\varphi + t_0\widehat{\psi}, K \cap \Omega, K) \neq 0.$$

The solution property implies that there exists $\widehat{u} \in K \cap \Omega$ such that

$$\begin{aligned} &\varphi(\widehat{u}) + t_0\widehat{\psi}(\widehat{u}) = \widehat{u}, \\ &\Rightarrow t_0 = \frac{\|\widehat{u} - \varphi(\widehat{u})\|}{\|\widehat{\psi}(\widehat{u})\|} \leq \frac{1}{\eta}[\vartheta_1 + \vartheta_2], \end{aligned}$$

a contradiction. So $i(\varphi, K \cap \Omega, K) = 0$. □

If in the above proposition $\psi(\cdot)$ is a nonzero constant map, then we have the following corollary.

Corollary 2.109 *If $\varphi : K \cap \overline{\Omega} \to K$ is compact, $u_0 \in K \setminus \{0\}$ and $u - \varphi(u) \neq \lambda u_0$ for all $u \in K \cap \partial\Omega$, all $\lambda \geq 0$, then $i(\varphi, K \cap \Omega, K) = 0$.*

We will use the above computations of the fixed point index in order to prove fixed point theorems in conical shells, that is, sets of the form $K \cap (\Omega_2 \setminus \overline{\Omega}_1)$ with Ω_1, Ω_2 bounded open sets and $\overline{\Omega}_1 \subseteq \Omega_2$. When $\Omega_1 = \overline{B}_\varrho$, $\Omega_2 = \overline{B}_R$, $\varrho < R$, then we have the so-called fixed point theorems of cone expansion and compression.

Theorem 2.110 *If $\Omega_1, \Omega_2 \subseteq X$ are nonempty bounded open sets, $0 \in \Omega_1$, $\overline{\Omega}_1 \subseteq \Omega_2$, $\varphi : K \cap \Omega_2 \to K$ is compact and the following two conditions hold,*

(i) *$\varphi(u) \neq \lambda u$ for all $u \in \partial\Omega_2$, all $\lambda > 1$;*
(ii) *there exists $u_0 \in K \setminus \{0\}$ such that $u - \varphi(u) \neq \lambda u_0$ for all $u \in K \cap \partial\Omega_1$, all $\lambda > 0$,*

then $\varphi(\cdot)$ has a fixed point in $K \cap (\overline{\Omega}_2 \setminus \Omega_1)$.

Proof On account of hypothesis (i) we can apply Proposition 2.107 and have that

$$i(\varphi, K \cap \Omega_2, K) = 1. \tag{2.29}$$

On the other hand, hypothesis (ii) and Corollary 2.109 imply

$$i(\varphi, K \cap \Omega_1, K) = 0. \tag{2.30}$$

Suppose $\varphi(\cdot)$ is fixed point free on $K \cap \partial\Omega_2$ and on $K \cap \Omega_1$ (otherwise we already have a fixed point as claimed by the theorem). Then the domain additivity property implies that

$$i(\varphi, K \cap \Omega_2, K) = i(\varphi, K \cap \Omega_1, K) + i(\varphi, K \cap (\Omega_2 \setminus \overline{\Omega_1}), K)$$
$$\Rightarrow i(\varphi, K \cap (\Omega_2 \setminus \overline{\Omega_1}, K) = 1 (\text{ see } (8.29), (8.30)).$$

From the solution property, we conclude that $\varphi(\cdot)$ has a fixed point in $K \cap (\Omega_2 \setminus \overline{\Omega}_1)$. □

Corollary 2.111 *If $\Omega_1, \Omega_2 \subseteq X$ are nonempty, bounded, open sets, $0 \in \Omega_1$, $\overline{\Omega}_1 \subseteq \Omega_2$, $\varphi : K \cap \overline{\Omega}_2 \to K$ is compact and the following two conditions hold*

(i) *$\varphi(u) - u \notin K$ for all $u \in K \cap \partial\Omega_2$;*
(ii) *$u - \varphi(u) \notin K$ for all $u \in K \cap \partial\Omega_1$,*

then $\varphi(\cdot)$ has a fixed point in $K \cap (\Omega_2 \setminus \overline{\Omega}_1)$.

Remark 2.112 The above corollary remains true if (i) is replaced by

(i)' $\varphi(u) - u \notin K$ for all $u \in K \cap \partial\Omega_1$;
and hypothesis (ii) is replaced by
(ii)' $u - \varphi(u) \notin K$ for all $u \in K \cap \partial\Omega_2$,

Theorem 2.113 *If $\Omega_1, \Omega_2 \subseteq X$ are bounded open sets, $0 \in \Omega_1$, $\overline{\Omega}_1 \subseteq \Omega_2$, $\varphi : K \cap \overline{\Omega}_2 \to K$ is compact and one of the following hypotheses holds.*

(i) $\|\varphi(u)\| \geq \|u\|$ *for all* $u \in K \cap \partial\Omega_2$ *and* $\|\varphi(u)\| \leq \|u\|$ *for all* $u \in K \cap \partial\Omega_1$;
(ii) $\|\varphi(u)\| \leq \|u\|$ *for all* $u \in K \cap \partial\Omega_2$ *and* $\|\varphi(u)\| \geq \|u\|$ *for all* $u \in K \cap \partial\Omega_1$,

then $\varphi(\cdot)$ has a fixed point in $K \cap (\overline{\Omega}_2 \setminus \overline{\Omega}_1)$.

Proof We do the proof when (i) holds. The proof is similar if (ii) holds.

Evidently hypothesis (i) implies that

$$\varphi(u) \neq \lambda u \text{ for all } u \in K \cap \partial\Omega_2, \text{ all } \lambda \in (0, 1) \tag{2.31}$$

and also

$$\varphi(u) \neq \mu u \text{ for all } u \in K \cap \partial\Omega_1, \text{ all } \mu > 1. \tag{2.32}$$

Moreover, we have

$$inf\{\|\varphi(u)\| : u \in K \cap \partial\Omega_2\} \geq inf\{\|u\| : K \cap \partial\Omega_2\} > 0. \tag{2.33}$$

Now from (2.31), (2.33) and Proposition 2.108 (with $\psi = \varphi$), we have

$$i(\varphi, K \cap \Omega_2, K) = 0. \tag{2.34}$$

Also from (2.32) and Proposition 2.107 we have

$$i(\varphi, K \cap \Omega_1, K) = 1. \tag{2.35}$$

Then (2.34), (2.35) and the domain additivity property imply

$$i(\varphi, K \cap (\Omega_2 \setminus \overline{\Omega}_1), K) = 1$$

$\Rightarrow \varphi(\cdot)$ has a fixed point in $K \cap (\Omega_2 \setminus \overline{\Omega}_1)$ (by the solution property).

□

The next lemma is obvious.

Lemma 2.114 *If X is an OBS with order cone K, $C \subseteq K$ is compact and $0 \notin C$, then $0 \notin conv C$.*

Using this lemma we can have one more fixed point theorem for conical shells.

Theorem 2.115 *If $\Omega_1, \Omega_2 \subseteq X$ are nonempty, bounded open sets, $0 \in \Omega_1$, $\overline{\Omega}_1 \subseteq \Omega_2$ and $\varphi : K \cap \overline{\Omega}_2 \to K$ is a compact map which satisfies the following three conditions:*

(i) $\varphi(u) \neq \lambda u$ *for all* $u \in K \cap \partial\Omega_2$, *all* $\lambda > 1$;
(ii) $\varphi(u) \neq \mu u$ *for all* $u \in K \cap \partial\Omega_1$, *all* $\mu \in (0, 1)$;
(iii) $inf\{\|\varphi(u)\| : u \in K \cap \partial\Omega_1\} > 0$,

then $\varphi(\cdot)$ *has a fixed point in* $K \cap (\Omega_2 \setminus \overline{\Omega}_1)$.

Proof We assume that φ is fixed point free on $K \cap \partial\Omega_1$ and on $K \cap \partial\Omega_2$. From hypothesis (i) and Proposition 2.107, we have

$$i(\varphi, K \cap \Omega_2, K) = 1. \tag{2.36}$$

Then (2.36) and the domain additivity property imply that

$$i(\varphi, K \cap (\Omega_2 \setminus \overline{\Omega}_1), K) = 1 - i(\varphi, K \cap \Omega_1, K). \tag{2.37}$$

From Theorem 2.55 and Lemma 2.114, we know that there exists $\widehat{\varphi} : X \to conv\varphi(K \cap \partial\Omega_1)$ compact map such that

$$\widehat{\varphi}|_{K\cap\partial\Omega_1} = \varphi|_{K\cap\partial\Omega_1}$$

and $inf\{\|\widehat{\varphi}(u)\| : u \in X\} = \eta > 0$.

Consider the compact homotopy $h(t, u)$ defined by

$$h(t, u) = (1 - t)\varphi(u) + tk\widehat{\varphi}(u) \text{ for all } t \in [0, 1], u \in \overline{\Omega}_2, \text{ with } k > 1.$$

We see that $h(t, u) \neq u$ for all $t \in [0, 1]$, all $u \in K \cap \partial\Omega_1$. Hence, by the homotopy invariance property, we have

$$i(\varphi, K \cap \Omega_1, K) = i(k\widehat{\varphi}, K \cap \Omega_1, K). \tag{2.38}$$

For $k > 1$ large enough, we have $u \neq k\widehat{\varphi}(u)$ for all $u \in K \cap \Omega_1$.

Hence,

$$\begin{aligned} &i(k\widehat{\varphi}, K \cap \Omega_1, K) = 0, \\ &\Rightarrow i(\varphi, K \cap \Omega_1, K) = 0 \text{ (see (8.38))}, \\ &\Rightarrow i(\varphi, K \cap (\Omega_2 \setminus \overline{\Omega}_1), K) = 1. \end{aligned}$$

So, by the solution property we conclude that $\varphi(\cdot)$ has a fixed point in $K \cap (\Omega_2 \setminus \overline{\Omega}_1)$. □

2.4 Fixed Points for Multifunctions

In this section we present some basic fixed point theorems for set valued maps (multifunctions). We start with the multivalued generalization of the Banach fixed point theorem.

Let (X, d) be a metric space and consider the following hyperspaces of X

$$P_{(b)f}(X) = \{C \subseteq X : \text{nonempty, (bounded), closed}\}.$$

On $P_f(X)$ we consider the Hausdorff metric $h(\cdot, \cdot)$ defined by

$$h(A, C) = max\{h^*(A, C), h^*(C, A)\} \text{ for all } A, C \in P_f(X),$$

where $h^*(A, C) = sup\{d(a, C) : a \in A\} = inf\{\varepsilon > 0 : A \subseteq C_\varepsilon\}$ with $C_\varepsilon = \{u \in X : d(u, C) < \varepsilon\}$ (see Definition I-5.39). Then $(P_f(X), h)$ is a generalized metric space which is complete if X is complete. Also

$$h(A, C) = sup\{|d(u, A) - d(u, C)| : u \in X\}.$$

Note that $P_{bf}(X)$ is a closed subset of $(P_f(X), h)$.

Definition 2.116 A multifunction $F : \Omega \to P_f(X)$ is said to be an "h-contraction", if there exists $k \in (0, 1)$ such that $h(F(u), F(v)) \leq kd(u, v)$ for all $u, v \in X$.

Theorem 2.117 *If (X, d) is a complete metric space and $F : X \to P_f(X)$ is an "h-contraction", then $F(\cdot)$ has a fixed point, that is, there exists $u \in X$ such that $u \in F(u)$.*

Proof Let $l \in (k, 1)$ and $u_0 \in X$. Consider $u_1 \in F(u_0)$ such that $0 < d(u_1, u_0)$. If no such a u_1 exists, then $u_0 \in F(u_0)$ and we are done. We have

$$d(u_1, F(u_1)) \leq h(F(u_0), F(u_1)) \leq kd(u_0, u_1) < ld(u_0, u_1).$$

We can find $u_2 \in F(u_1)$ such that

$$d(u_1, u_2) < ld(u_0, u_1).$$

Inductively we generate a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that $u_{n+1} \in F(u_n)$ and

$$d(u_n, u_{n+1}) < l^n d(u_0, u_1) \text{ for all } n \in \mathbb{N}. \tag{2.39}$$

Since $l < 1$, we infer that $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ is Cauchy and since by hypothesis X is complete, we have $u_n \xrightarrow{d} u \in X$. From (2.39) in the limit as $n \to \infty$, we have $u \in F(u)$. □

Remark 2.118 We can not expect that this fixed point is unique. For example, if $F(\cdot)$ is the constant multifunction $F(u) = X$ for all $u \in X$, then every $u \in X$ is a fixed point.

For a multifunction $F : X \to 2^X \setminus \{\emptyset\}$, we set

$$Fix(F) = \{u \in X : u \in F(u)\}$$

(the set of fixed points of $F(\cdot)$). Evidently, $Fix(F)$ is closed, possibly empty.

We have a stability property for the set of fixed points of an h-contraction.

Theorem 2.119 *If (X, d) is a complete metric space and $F, G : X \to P_{bf}(X)$ are two h-contractions with the same contraction constant $k \in (0, 1)$, then $h(Fix(F), Fix(G)) \leq \frac{1}{1-k} \sup_{v \in X} h(F(v), G(v))$.*

Proof Choose $\vartheta > 0$ such that $\vartheta \sum_{n \in \mathbb{N}} nk^n < 1$ and let $\varepsilon > 0$. We set $\varepsilon_1 = \vartheta_\varepsilon \frac{1}{1-k}$, consider $u_0 \in F_{ix}(F)$ and choose $u_1 \in G(u_0)$ such that

$$d(u_0, u_1) \leq h(F(u_0), G(u_0)) + \varepsilon. \tag{2.40}$$

Next, let $u_2 \in G(u_1)$ such that

$$d(u_2, u_1) \leq h(G(u_1), G(u_0)) + k\varepsilon_1 \leq kd(u_0, u_1) + k\varepsilon_1.$$

Inductively we generate a sequence $\{u_n\}_{n \in \mathbb{N}} \subseteq X$ such that $u_{n+1} \in G(u_n)$ and

$$d(u_{n+1}, u_n) \leq k^n d(u_0, u_1) + nk^n \varepsilon_1, n \in \mathbb{N}. \tag{2.41}$$

We have

$$\sum_{n \geq m} d(u_{n+1}, u_n) \leq \frac{k^m}{1-k} d(u_0, u_1) + \varepsilon_1 \sum_{n \geq m} nk^n$$

$$\Rightarrow \{u_n\}_{n \in \mathbb{N}} \subseteq X \text{ is Cauchy.}$$

So, we have $u_n \xrightarrow{d} u$. Then if in (6.41) we pass to the limit as $n \to \infty$, we obtain $u \in G(u)$. Moreover, we have

$$\begin{aligned} d(u_0, u) &\leq \sum_{n \in \mathbb{N}_0} d(u_n, u_{n+1}) \\ &\leq \frac{1}{1-k} d(u_0, u_1) + \varepsilon_1 \sum_{n \geq m} nk^n \ (see\ (2.41)) \\ &\leq \frac{1}{1-k} [h(F(u_0), G(u_0)) + 2\varepsilon] \ (see\ (7.40)). \end{aligned}$$

Reversing the roles of F and G in the above argument and letting $\varepsilon \downarrow 0$, we obtain

$$h(Fix(F), Fix(G)) \leq \frac{1}{1-k} \sup_{v \in X} h(F(v), G(v)).$$

□

Corollary 2.120 *If (X, d) is a complete metric space, $F_n, F : X \to P_{bf}(X)$ are h-contractions with the same contraction constant $k \in (0, 1)$ and $\sup_{v \in X} h(F_n(v), F(v)) \to 0$, then $h(Fix(F_n), Fix(F)) \to 0$.*

Next we present a theorem which is the foundation for many topological fixed point theorems for multifunctions and is also a useful tool in the study of variational inequalities.

We start with a lemma which is actually equivalent to Brouwer's fixed point theorem (see Theorem 1.28). The result is known as the Knaster-Kuratowski-Mazurkiewicz lemma and its proof is based on a combinatorial result known as "Sperner's Lemma". For details we refer to Border [53].

Lemma 2.121 *If $\{u_k\}_{k=1}^n \subseteq \mathbb{R}^N$ and $\{C_k\}_{k=1}^n$ are nonempty closed subsets of $\mathbb{R}^N$ such that for every $I \subseteq \{1, \cdots, n\}$, we have*

$$conv\{u_i\}_{i \in I} \subseteq \bigcup_{i \in I} C_i,$$

then $(\bigcap_{k=1}^n C_k) \cap conv\{u_k\}_{k=1}^n \neq \emptyset$ and it is compact.

Motivated by this result, we make the following definition.

Definition 2.122 Let X be a vector space, $D \subseteq X$ and $F : D \to 2^X \setminus \{\emptyset\}$ a multifunction. We say that $F(\cdot)$ is a "KKM-multifunction", if for every finite set $\{u_k\}_{k=1}^n \subseteq D$ we have $conv\{u_k\}_{k=1}^n \subseteq \bigcup_{k=1}^n F(u_k)$.

Remark 2.123 Evidently for all $u \in D$ we have $u \in F(u)$.

The next lemma presents two useful properties of KKM.

Lemma 2.124 *Let $F : D \to 2^X \setminus \{\emptyset\}$ be a KKM-multifunction.*

(a) *For every $E \subseteq D$, $F|_E$ is still a KKM-multifunction.*
(b) *If $C \subseteq D$ is convex and $G(u) = F(u) \cap C$ for all $u \in C$, then $G : C \to 2^X \setminus \{\emptyset\}$ is a KKM-multifunction.*

Proof

(a) Obvious.

(b) Let $\{u_k\}_{k=1}^n \subseteq C$. Then $conv\{u_k\}_{k=1}^n \subseteq C$ and also since $F(\cdot)$ is a KKM-multifunction, we have $conv\{u_k\}_{k=1}^n \subseteq \bigcup_{k=1}^n F(u_k)$. Therefore we have

$$conv\{u_k\}_{k=1}^n \subseteq C \cap (\bigcup_{k=1}^n F(u_k))$$
$$= \bigcup_{k=1}^n C \cap F(u_k) = \bigcup_{k=1}^n G(u_k),$$

$\Rightarrow G(\cdot)$ is a KKM-multifunction.

□

Suppose V is another vector space and $F : X \to 2^V \setminus \{\emptyset\}$. The fibers of $F(\cdot)$ are the sets $F^{-1}(v) = \{u \in X : v \in F(u)\}$.

We introduce the "dual" multifunction $F^* : V \to 2^X$ defined by $F^*(v) = X \setminus F^{-1}(v)$ for all $v \in V$.

Lemma 2.125 *If $D \subseteq X$ is convex and $F : D \to 2^X \setminus \{\emptyset\}$ is a multifunction which has the following two properties,*

(i) *$u \in F(u)$ for all $u \in D$;*
(ii) *$F^*(y)$ is convex for all $y \in X$;*

then $F(\cdot)$ is a KKM-map.

Proof Let $\{u_k\}_{k=1}^n \subseteq D$ and let $y \in conv\{u_k\}_{k=1}^n$. By hypothesis (i) we have $y \in F(y)$ and so $y \notin F^*(y)$. Therefore $conv\{u_k\}_{k=1}^n$ is not a subset of $F^*(y)$. By hypothesis (ii) $F^*(y)$ is convex. Hence for some $k \in \{1, \cdots, n\}$ we have $u_k \notin F^*(y)$ and so $y \in F(u_k)$. We conclude that $conv\{u_k\}_{k=1}^n \subseteq \bigcup_{k=1}^n F(u_k)$ which means that $F(\cdot)$ is a KKM-multifunction. □

Let us see some characteristic examples of KKM-multifunctions.

Example 2.126

(a) Let $D \subseteq X$ be a convex set and $\varphi : D \to X$ a convex function. Consider the multifunction $F : D \to 2^D \setminus \{\emptyset\}$ defined by

$$F(u) = \{v \in D : \varphi(v) \leq \varphi(u)\}.$$

This is a KKM-multifunction. To show this we argue by contradiction. So, let $\{u_k\}_{k=1}^n \subseteq C$, $v = \sum_{k=1}^n \lambda_k u_k$ with $\lambda_k \geq 0$ for all $k \in \{1, \cdots, n\}$, $\sum_{k=1}^n \lambda_k = 1$ and suppose $v \notin \bigcup_{k=1}^n F(u_k)$. Then we have

$$\varphi(u_k) < \varphi(v) \text{ for all } k \in \{1, \cdots, n\},$$
$$\Rightarrow u_k \in \{u \in D : \varphi(u) < \varphi(v)\} = E \text{ for all } k \in \{1, \cdots, n\}.$$

Since $\varphi(\cdot)$ is convex, the set E is convex and so

$$v = \sum_{k=1}^{n} \lambda_k u_k \in E,$$
$$\Rightarrow \widehat{\varphi}(v) < \varphi(v), \text{ a contradiction.}$$

Therefore $F(\cdot)$ is a KKM-multifunction.

(b) Let X be a Banach space, $C \subseteq X$ nonempty convex and $\varphi : C \to X$ a continuous map. Consider the multifunction $F : C \to 2^C \setminus \{\emptyset\}$ defined by

$$F(u) = \{v \in C : \|\varphi(v) - v\| \leq \|\varphi(v) - u\|\}.$$

We show that $F(\cdot)$ is a KKM-multifunction. Again we argue by contradiction. So, let $\{u_k\}_{k=1}^n \subseteq C$, $v = \sum\limits_{k=1}^{n} \lambda_k u_k$ with $\lambda_k \geq 0$ for all $k \in \{1, \cdots, n\}$, $\sum\limits_{k=1}^{n} \lambda_k = 1$ and suppose that $v \notin \bigcup\limits_{k=1}^{n} F(u_k)$. Then

$$\|\varphi(v) - u_k\| < \|\varphi(v) - v\| \text{ for all } k \in \{1, \cdots, n\}$$
$$\Rightarrow u_k \in \{u \in D : \|\varphi(v) - u\| < \|\varphi(v) - v\|\} = E \text{ for all } k \in \{1, \cdots, n\}.$$

The set E is an open ball, thus convex. Hence

$$v = \sum_{k=1}^{n} \lambda_k u_k \in E,$$
$$\Rightarrow \|\varphi(v) - v\| < \|\varphi(v) - v\|, \text{ a contradiction.}$$

Therefore $F(\cdot)$ is a KKM-multifunction.

(c) Let $X = H = a$ Hilbert space, $C \subseteq H$ nonempty, convex and $\varphi : C \to H$ a continuous map. Then we can easily check as above that the multifunction $F : C \to 2^C \setminus \{\emptyset\}$ defined by $F(u) = \{v \in C : (\varphi(v), v - u) \leq 0\}$ is a KKM-multifunction.

The next proposition can be viewed as an infinite dimensional generalization of the KKM-Lemma (see Lemma 2.121).

Proposition 2.127 *If X is a topological vector space, $C \subseteq X$ is nonempty and $F : C \to P_f(X)$ is a KKM-multifunction with $F(u_0) \subseteq X$ compact for some $u_0 \in C$, then $\bigcap_{u \in C} F(u) \neq \emptyset$ and it is compact.*

Proof Since $F(u_0)$ is compact, according to Proposition I-1.81 it suffices to show that $\{F(u)\}_{u \in C}$ has the finite intersection property (see Remark I-1.80). To this end, let $\{u_k\}_{k=1}^n \subseteq C$ and let $Y = span\{u_k\}_{k=1}^n$. We define $C_k = F(u_k) \cap conv\{u_k\}_{k=1}^n$, $k \in \{1, \cdots, n\}$. These sets are closed and for every $I \subseteq \{1, \cdots, n\}$, we have $conv\{u_i\}_{i \in I} \subseteq \bigcup_{i \in I} C_i$. So, we can use Lemma 2.121 and obtain

$$\emptyset \neq \bigcap_{k=1}^{n} C_k \subseteq \bigcap_{k=1}^{n} F(u_k)$$

$$\Rightarrow \{F(u)\}_{u \in C} \text{ has the finite intersection property.}$$

□

Remark 2.128 A careful reading of the above proof reveals that it is enough to assume that $F(\cdot)$ has finitely closed values, that is, for each $u \in C$ and for each finite dimensional subspace V of X. $F(u) \cap V$ is closed.

The next theorem, is an existence result for variational inequalities and its proof can be found in Browder [72, (Theorem 6.1, p. 72)].

Theorem 2.129 *If X is a locally convex space, $C \subseteq X$ is nonempty, compact, convex and $\varphi : C \to X^*$ is continuous, then there exists $u_0 \in C$ such that $\langle \varphi(u_0), u_0 - u \rangle \leq 0$ for all $u \in C$.*

Remark 2.130 Using the above theorem, we can prove the fixed point theorem for nonexpansive maps when $X = H = a$ Hilbert space (see Theorem 2.23). So, suppose $C \subseteq H$ is nonempty, bounded, closed, convex (thus w-compact) and let $\varphi : C \to C$ be a nonexpansive map. We show that $\varphi(\cdot)$ has a fixed point. Let $\psi(u) = u - \varphi(u)$ for all $u \in C$. This map is continuous, monotone (see Example I-7.28(c)), hence maximal monotone (see Proposition I-7.36). Using Theorem 2.129, we can find $u_0 \in C$ such that

$$(\psi(u_0), u - u_0)_H \geq 0 \text{ for all } u \in C,$$

$$\Rightarrow (u_0 - \varphi(u_0), u - u_0)_H \geq 0 \text{ for all } u \in C.$$

Let $u = \varphi(u_0)$. Then $(u_0 - \varphi(u_0), \varphi(u_0) - u_0) \geq 0$ and by the monotonicity of $\psi(\cdot)$, we have $\varphi(u_0) = u_0$. Finally we mention that the proof of the above theorem is based on KKM-multifunctions.

Now we will use Theorem 2.129 to prove some basic topological fixed point theorems for multifunctions.

Definition 2.131 Let X be a vector space and $C \subseteq X$ nonempty. A multifunction $F : C \to 2^X \setminus \{\emptyset\}$ is said to be

(a) "inward", if for every $u \in C$, there exist $y \in F(u)$ and $\lambda > 0$ such that $u + \lambda(y - u) \in C$.
(b) "outward", if for every $u \in C$, there exist $y \in F(u)$ and $\lambda > 0$ such that $u - \lambda(y - u) \in C$.

Remark 2.132 If $F(\cdot)$ maps C into itself, then $F(\cdot)$ must be inward. Choose any $y \in F(u)$ and note that $u + 1.(y - u) = y \in C$. The identity map is both inward and outward since $u = u + 1.(u - u) = u - 1.(u - u)$.

Theorem 2.133 *If X is a locally convex space, $C \subseteq X$ is nonempty, compact, convex and $F : C \to P_{fc}(X)$ is usc and inward, then $F(\cdot)$ has a fixed point.*

Proof We argue by contradiction. So, suppose that $u \notin F(u)$ for all $u \in C$. By the strong separation theorem (see Theorem I-3.72), we can find $u_u^* \in X^* \setminus \{0\}$ and $\eta_u \in \mathbb{R}$ such that $u_u^*(v) < \eta_u < u_u^*(u)$ for all $v \in F(u)$.

Consider the set

$$U_u = F^+(\{c \in C : u_u^*(c) < \eta_u\}) \cap \{c \in C : u_u^*(c) > \eta_u\}.$$

The upper semicontinuity of $F(\cdot)$ implies that $U_u \subseteq C$ is open and $u \in U_u$. Consider the family $\{U_u\}_{u \in C}$. This is an open cover of C and the compactness of C implies that we can find a finite subcover $\{U_{u_k}\}_{k=1}^n$. Compactness implies paracompactness and so by Theorem I-1.191, we can find a partition of unity subordinate to $\{U_{u_k}\}_{k=1}^n$, that is, there exist continuous functions $\vartheta_k : C \to [0, 1]$, $k \in \{1, \cdots, n\}$ such that $\vartheta_k|_{C \setminus U_{u_k}} = 0$ and $\sum_{k=1}^n \vartheta_k(c) = 1$ for all $c \in C$.

If $\vartheta_k(c) > 0$, then $c \in U_{u_k}$, hence $u_{u_k}^*(v) < \eta_u$ for all $v \in F(t)$.

We introduce the map $g : K \to X^*$ defined by $g(c) = \sum_{k=1}^n g_k(c) u_{u_k}^*$.

This is a continuous map and so by Theorem 2.129, we can find $u_0 \in C$ such that

$$\langle g(u_0), u_0 - c\rangle \leq 0 \text{ for all } c \in C. \tag{2.42}$$

For $c \in K$ and $v \in F(c)$, we have

$$\begin{aligned} \langle g(c), c\rangle &= \sum_{k=1}^n \vartheta_k(c) u_{u_k}^*(c) \\ &> \sum_{k=1}^n \vartheta_k(c) \eta_{u_k} \\ &> \sum_{k=1}^n \vartheta_k(c) u_{u_k}^*(v) = \langle g(c), v\rangle. \end{aligned} \tag{2.43}$$

Since by hypothesis $F(\cdot)$ is inward, we can find $v_0 \in F(u_0)$ and $\lambda > 0$ such that $u_0 + \lambda(v_0 - u_0) \in C$. In (6.42) we set $c = u_0 + \lambda(y_0 - v_0)$ and obtain

$$\langle g(u_0), y_0 - v_0 \rangle \leq 0. \tag{2.44}$$

Comparing (2.43) and (2.44) we have a contradiction. □

Remark 2.134 With suitable modifications in the above proof, we can see that the result remains true if we assume that the multifunction $F(\cdot)$ is outward.

Corollary 2.135 *If X is a locally convex space, $C \subseteq X$ is nonempty, compact, convex and $F : C \to P_{fc}(C)$ is usc, then the set of fixed points of $F(\cdot)$ is nonempty and compact.*

When $F(\cdot)$ is single-valued, then the above theorem reduces to the so-called "Tychonov Fixed Point Theorem".

Corollary 2.136 *If X is a locally convex space, $C \subseteq X$ is nonempty, compact, convex and $\varphi : C \to C$ is continuous, then the set of fixed points of $\varphi(\cdot)$ is nonempty and compact.*

There is an alternative formulation of the above corollaries.

Corollary 2.137 *If X is a locally convex space, $C \subseteq X$ is nonempty, closed, convex and $F : C \to P_{fc}(C)$ is usc (resp. $\varphi : C \to C$ is continuous) and $\overline{F(C)}$ (resp. $\overline{\varphi(C)}$) is compact, then the set of fixed points of $F(\cdot)$ (resp. of $\varphi(\cdot)$) is nonempty and compact.*

We conclude this section with a multivalued version of the Leray-Schauder Alternative Theorem, due to Bader [23].

So, let X, V be Banach spaces and $C \subseteq X$, $D \subseteq V$ nonempty, closed and convex sets. By D_w we denote the set D equipped with the relative weak topology. Let $G : C \to 2^C \setminus \{\emptyset\}$ be a multifunction which admits the following decomposition.

$$G = K \circ N, \tag{2.45}$$

with $N : C \to 2^D \setminus \{\emptyset\}$ is usc into D_w and $K : D_w \to C$ is sequentially continuous.

Theorem 2.138 *If X is a Banach space, $G : C \to 2^C \setminus \{\emptyset\}$ is a compact multifunction (see Definition 1.108) which admits the decomposition (2.45) and $0 \in C$, then at least one of the following properties holds:*

(a) *$G(\cdot)$ has a fixed point;*
(b) *the set $S = \{u \in C : u \in \lambda G(u)$ for some $\lambda \in (0, 1)\}$ is unbounded.*

Remarks

2.1: Theorem 2.3 is due to Banach [34]. Essentially is a brilliant abstraction of the method of successive approximations developed earlier for differential equations by Liouville and Picard. There have been various generalizations of the result by weakening the contraction condition. We mention one due to Weissinger [458].

Theorem 2.139 *If (X, d) is a complete metric space and $\varphi : X \to X$ satisfies*

$$d(\varphi^{(n)}(u), \varphi^{(n)}(v)) \leq c_n d(u, v)$$

with $\sum_{n\in\mathbb{N}} c_n < \infty$, then $\varphi(\cdot)$ has a unique fixed point $u \in X$ and for every $v \in X$, $\varphi^{(n)}(v) \xrightarrow{d} u$.

The notion of normal structure (see Definition 2.21) is due to Brodskii and Milman [30]. They also proved that all compact, convex sets have normal structure. Theorem 2.23 is due to Kirk [268]. The fixed point theorem for nonexpansive maps in a noncompact setting was established independently by Browder [71], Göhde [212] and Kirk [268]. Browder [71] and Göhde [212] proved that in a uniformly convex Banach space, a nonexpansive self-map of a nonempty, bounded, closed convex set, has a fixed point, while Kirk [268] proved the slightly more general version which we have included here. More on metric fixed point theory can be found in the books of Goebel and Kirk [211], Smart [443], Zeidler [476].

Note that first DeBlasi-Myjak [a] showed using the Baire category approach that most nonexpansive mappings on a bounded, closed-convex subset of a Banach space have a unique fixed point to which all iterates converge (as in the Banach Fixed Point Theorem, see Theorem 2.3). The book of Reich-Zaslavski [b] contains additional results in this direction. Also, in this book the reader can find fixed point results for an important subclass of nonexpansive maps, known as "contractive mappings."

Definition 2.140 Let X be a Banach space, $C \subseteq X$ a nonempty, bounded closed convex set. Set $d(C)$ =diameter of C, namely,

$$d(C) = \sup\{\|u - v\|_X : u, v \in C\}.$$

Let $\beta : [0, d(C)] \to [0, 1]$ be a decreasing function such that $\beta(t) < 1$ for all $t \in (0, d(C)]$. We say that $\varphi : C \to C$ is contractive if $\|\varphi(u) - \varphi(v)\|_X \leq \beta(\|u - v\|_X)\|u - v\|_X$ for all $u, v \in C$.

As it is shown in Reich-Zaslavski [b] (Chap. 3), many nonexpansive maps are contractive and there are fixed points theorem involving contractive mappings.

2.2: For the topological fixed point theory, the main reference is the book of Granas·Dugundji [215]. The product of two compact spaces with the fixed point property ("fpp" for short), need not have the fpp (see Bredon [58]). In contrast, an

infinite product of compact spaces with the fpp, has the fpp provided every finite product of these spaces has the fpp. From the fixed point theorems of Schauder, Tychonov and Borsuk, we know that the following spaces have the fpp.

- compact, convex sets in a Banach space;
- compact, convex sets in a locally convex space;
- compact, absolute retracts.

Theorem 2.48(c) is known as "Borsuk's Antipodal Theorem" while Theorem 2.48(d) is known as the "Borsuk-Ulam Theorem" (see Borsuk [9, 54, 55]). Borsuk's antipodal theorem was extended to locally convex spaces by Altman [9]. Theorem 2.49(d) (Brouwer's fixed point theorem), was proved for $N = 3$ by Brouwer [70] and for any $N \geq 1$ by Hadamard [221]. In contrast to the Banach fixed point theorem, the proof of Brouwer is not constructive, an ironic fact given his later advocacy of intuitionism. The fact that Brouwer's fixed point theorem admits an equivalent formulation in terms of homotopy and retraction (see Theorem 2.49), was observed by Borsuk [54, 55]. Theorem 2.57 is due to Mönch [338]. Finally, we mention an interesting consequence of the Schauder-Tychonov fixed point theorem due to Markov [331].

Theorem 2.141 *If X is a locally convex space, $C \subseteq X$ is nonempty, compact, convex and Φ is a commuting family of continuous affine maps $L : C \to C$, then the family φ has a common fixed point, that is, there exists $u \in C$ such that $L(u) = u$ for all $L \in \Phi$.*

2.3: The main references for order fixed point theory are the books of Krasnoselskii [278], Deimling [133], Guo and Lakshmikantham [218] and the survey paper of Amann [10]. The different kinds of cones in Definition 2.79 are due to Krasnoselskii [278]. For the fixed point index we refer to Amann [10] and the books of Deimling [133] and Guo and Lakshmikantham [218]. We finally mention also the book of Carl and Heikkila [85] on fixed point theory on ordered sets.

2.4: Theorem 2.117 is due to Nadler [347]. The stability properties of the fixed point set presented in Theorem 2.119 and Corollary 2.120 can be found in Hu and Papageorgiou [247]. Lemma 2.121 is due to Knaster et al. [271]. Proposition 2.127 is due to Fan [155]. The notions of inward and outward multifunctions (see Definition 2.131) and Theorem 2.133 are due to Halpern [226] and Halpern and Bergman [227]. Corollary 2.135 is due to Kakutani [263], Fan [155] and Glicksberg [208], while Corollary 2.137 (the multivalued part) is due to Bohnenblust and Karlin [52]. For more on the topological fixed point theory of multifunctions, we refer to the book of Gorniewicz [213].

Chapter 3
Critical Point Theory



Introduction

In this chapter we present the main variational tool in the study of boundary value problems, which is the critical point theory. In Sect. 3.1 we introduce the compactness conditions on the functional which compensate for the fact that the ambient space is not locally compact (being infinite dimensional) and also a replacement of the gradient when the gradient flow can not be used. With these tools in Sect. 3.2 we prove the deformation theorems which in Sect. 3.3 produce the minimax theorems for the critical values of φ. Prominent among these results is the so-called "Mountain Pass Theorem". In Sect. 3.4 we present theorems which produce multiple critical points. This is more transparent when the functional exhibits some symmetry condition. In Sect. 3.5 we examine the critical points of constrained functionals. This leads to the Lagrange multipliers and the Nehari method. Finally in Sect. 3.6, we outline the theory of critical groups. Critical groups lead to multiplicity results and allow us to distinguish between critical points.

3.1 Compactness Conditions-Pseudogradients

Critical point theory is the natural development of the classical "Calculus of Variations". With the critical point theory, problems which appear in a differential equation form are reduced to problems of the calculus of variations. So we look for the critical points of the energy functional corresponding to the equation when the latter has variational structure. We look for maxima, minima and saddle points. A useful tool in the study of the critical points is the so-called "deformation technique", introduced by Ljusternik and Schnirelmann [316]. In the context of Hilbert spaces, this technique consists in deforming the energy functional $\varphi(\cdot)$, outside the critical set using the gradient flow $u' = -\nabla\varphi(u)$. For this technique

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to work we need some kind of compactness condition. Since the ambient space X is in general infinite dimensional (consequently is not locally compact), the burden of compactness is passed to the functional $\varphi(\cdot)$, as was done in the case of degree theory (Leray-Schauder degree). So, we introduce the following compactness-type conditions on $\varphi(\cdot)$. In what follows X is a Banach space, X^* is its topological dual and by $\langle \cdot, \cdot \rangle$ we denote the duality brackets for the pair (X^*, X).

Definition 3.1 Let $\varphi \in C^1(X)$.

(a) We say that $\varphi(\cdot)$ satisfies the "Palais-Smale condition at level $c \in \mathbb{R}$" (the "PS_c-condition" for short), if every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi(u_n) \to c \text{ and } \varphi'(u_n) \to 0 \text{ in } X^*,$$

admits a strongly convergent subsequence. If this property holds at every level $c \in \mathbb{R}$, then we say that $\varphi(\cdot)$ satisfies the "Palais-Smale condition" ("PS-condition" for short).

(b) We say that $\varphi(\cdot)$ satisfies the "C-condition at level $c \in \mathbb{R}$" (the "C_c-condition" for short), if every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi(u_n) \to c \text{ and } (1 + \|u_n\|_X)\varphi'(u_n) \to 0 \text{ in } X^*,$$

admits a strongly convergent subsequence. If this property holds at every level $c \in \mathbb{R}$, then we say that $\varphi(\cdot)$ satisfies the "C-condition".

Remark 3.2 Evidently $\varphi(\cdot)$ satisfies the "C-condition", if every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that $\{\varphi(u_n)\}_{n\in\mathbb{N}} \subseteq \mathbb{R}$ is bounded and $(1 + \|u_n\|_X)\varphi'(u_n) \to 0$ in X^*, admits a strongly convergent subsequence. The C-condition is weaker than the PS-condition. These compactness conditions are strong and there are functions which fail to satisfy them. For example constant functions do not satisfy the PS and C-conditions. Similarly $\varphi(u) = e^u$ satisfies the PS_c and C_c-conditions for $c \neq 0$ but not for $c = 0$.

For $\varphi \in C^1(X)$ and $c \in \mathbb{R}$, we introduce the following sets

$$\varphi^c = \{u \in X : \varphi(u) \leq c\},$$
$$K_\varphi = \{u \in X : \varphi'(u) = 0\} \text{(the critical set of } \varphi),$$
$$K_\varphi^c = \{u \in K_\varphi : \varphi(u) = c\}.$$

First we state a generalized version of the Ekeland variational principle (see Theorem I-8.99), due to Zhong [485]. This version of the Ekeland variational principle, is suitable to handle the C-condition.

Theorem 3.3 *If $h : \mathbb{R}_+ \to \mathbb{R}_+$ is continuous, nondecreasing, $\int_0^\infty \frac{dr}{1+h(r)} dr = \infty$, (V, d) is a complete metric space, $u_0 \in V$, $\varphi : V \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ lower*

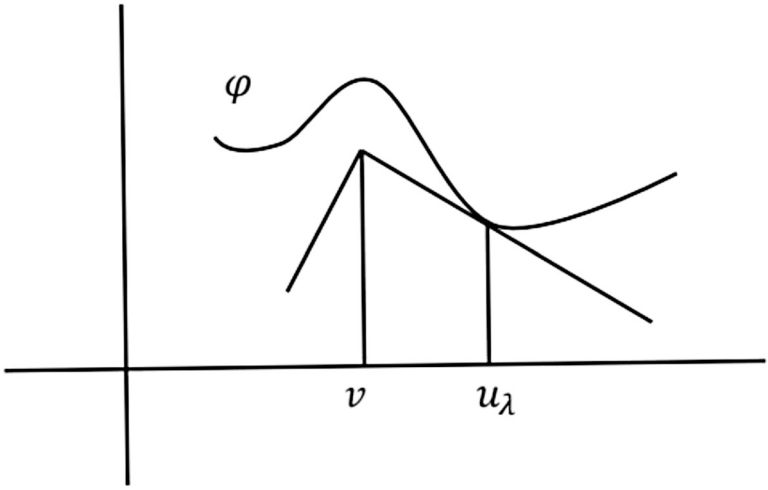


Fig. 3.1 Ekeland Variational Principle

semicontinuous ($\varphi \not\equiv +\infty$), bounded below, $\varepsilon > 0$, $\varphi(v) \leq \inf_X \varphi + \varepsilon$, $\lambda > 0$, then there exists $u_\lambda \in X$ such that

$$\varphi(u_\lambda) \leq \varphi(v), d(u_\lambda, u_0) \leq r_0 + \overline{r},$$

$$\varphi(u_\lambda) \leq \varphi(u) + \frac{\varepsilon}{\lambda} \frac{1}{1 + h(d(u_0, u_\lambda))} d(u, u_\lambda) \text{ for all } u \in X,$$

where $r_0 = d(u_0, v)$ and $\overline{r} > 0$ such that

$$\int_0^{r_0+\overline{r}} \frac{dr}{1 + h(r)} \geq \lambda.$$

Remark 3.4 Note that if $h \equiv 0$ and $u_0 = v$, then Theorem 3.3 reduces to Theorem I-8.99. The next figure reminds us of the geometric interpretation of the Ekeland variational principle (Fig. 3.1).

Using this theorem, we can investigate the consequences of the C-condition.

Proposition 3.5 *If $\varphi \in C^1(X)$ is bounded below, $m = \inf_X \varphi$ and φ satisfies the C_m-condition, then for some $\eta > 0$, $\varphi^{m+\eta}$ is bounded.*

Proof We argue by contradiction. So, suppose that for all $\eta > 0$, the set $\varphi^{m+\eta}$ is unbounded. So, we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$m \leq \varphi(u_n) \leq m + \frac{1}{n}, \|u_n\|_X \geq n \text{ for all } n \in \mathbb{N}.$$

We apply Theorem 3.3 with $h(r) = r$, $u_0 = u_n$, $\varepsilon = \dfrac{1}{n}$, $\lambda = \dfrac{1}{\sqrt{n}}$. We produce a sequence $\{v_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi(v_n) \downarrow m \text{ and } (1 + \|v_n\|_X)\varphi'(v_n) \to 0, \|v_n - u_n\|_X \leq \overline{r}_n \tag{3.1}$$

with $\overline{r}_n > 0$ such that $\ln(1 + \overline{r}_n) = \dfrac{1}{\sqrt{n}}$. Then $\overline{r}_n \to 0^+$ and $n \leq \|u_n\|_X \leq \overline{r}_n + \|v_n\|_X$. We have $\|v_n\|_X \to \infty$. But from (1) we see that this contradicts the hypothesis that $\varphi(\cdot)$ satisfies the C_m-condition. □

Proposition 3.6 *If $\varphi \in C^1(X)$, $c \in \mathbb{R}$, φ^λ is unbounded for all $\lambda > c$ and φ^λ is bounded for all $\lambda < c$, then we can find a sequence $\{y_n\}_{n\in\mathbb{N}} \subseteq X$ such that*

$$\varphi(y_n) \to c, (1 + \|y_n\|_X)\varphi'(y_n) \to 0 \text{ in } X^*,$$
$$\|y_n\| \to \infty \text{ as } n \to \infty.$$

Proof For every $n \in \mathbb{N}$, we can find $\varrho_n \geq n$ such that $\varphi^{c-\frac{1}{n}} \subseteq B_{\varrho_n} = \{u \in X : \|u\|_X < \varrho_n\}$. Let $C_n = B^c_{\varrho_n}$, $\widehat{\varphi}_n = \varphi|_{C_n}$ and note that

$$m_n = \inf_{C_n} \widehat{\varphi}_n \geq c - \frac{1}{n}. \tag{3.2}$$

By hypothesis $\varphi^{c+\frac{1}{n}}$ is unbounded and so we can find $v_n \in X$ such that

$$\varphi(v_n) \leq c + \frac{1}{n}, \|v_n\|_X \geq \varrho_n + 1 + \frac{1}{\sqrt{n}} \text{ for all } n \in \mathbb{N}. \tag{3.3}$$

We see that $v_n \in C_n$, $n \in \mathbb{N}$ and so from (3.2) and (3.3) we have

$$\varphi(v_n) \leq c + \frac{1}{n} \leq m_n + \frac{2}{n} \text{ for all } n \in \mathbb{N}. \tag{3.4}$$

We apply Theorem 3.3 with $h(r) = r$, $u_0 = v_n$, $\varepsilon = \dfrac{2}{n}$, $\lambda = \dfrac{1}{\sqrt{n}}$ and obtain a sequence $\{y_n\}_{n\in\mathbb{N}} \subseteq X$ with $y_n \in C_n$ and

$$c - \frac{1}{n} \leq m_n \leq \varphi(y_n) \leq \varphi(v_n) \leq c + \frac{1}{n} \leq m_n + \frac{2}{n} \text{ for all } n \in \mathbb{N} \text{ (see (3.4))},$$
$$(1 + \|y_n\|_X)\|\varphi'(y_n)\|_{X^*} \leq \frac{2}{\sqrt{n}} \text{ for all } n \in \mathbb{N},$$
$$\|y_n - v_n\|_X \leq \overline{r}_n \text{ for all } n \in \mathbb{N},$$

with $\bar{r}_n > 0$ such that $\ln(1+\bar{r}_n) = \dfrac{1}{\sqrt{n}}$. Then from the above we conclude that $\{y_n\}_{n\in\mathbb{N}} \subseteq X$ is the sequence claimed by the proposition. □

Corollary 3.7 *If $\varphi \in C^1(X)$ satisfies the C_c-condition for some $c \in \mathbb{R}$ and φ^λ is bounded for every $\lambda < c$, then there exists $\eta > 0$ such that $\varphi^{c+\eta}$ is bounded.*

Recall that $\varphi : X \to \mathbb{R}$ is said to be "coercive", if $\varphi(u) \to +\infty$ when $\|u\|_X \to \infty$ (see Definition I-6.120).

Proposition 3.8 *If $\varphi \in C^1(X)$ is bounded below but it is not coercive and*

$$\vartheta = \sup\{\eta \in \mathbb{R} : \varphi^\eta \text{ is bounded}\},$$

then φ does not satisfy the C_ϑ-condition.

Proof Let $D = \{\eta \in \mathbb{R} : \varphi^\eta \text{ is bounded}\}$. If $m = \inf_X \varphi > -\infty$, then we see that $(-\infty, m) \subseteq D \neq \emptyset$. The noncoercivity of $\varphi(\cdot)$ implies that $\vartheta < \infty$. For $\lambda > \vartheta$ we have that φ^λ is unbounded. Using Proposition 3.6, we conclude that $\varphi(\cdot)$ does not satisfy the C_ϑ-condition. □

An interesting consequence of this proposition is the following result relating the C-condition and coercivity.

Proposition 3.9 *If $\varphi \in C^1(X)$ is bounded below and satisfies the C-condition, then $\varphi(\cdot)$ is coercive.*

Using this proposition, we can have equivalence of the PS and C-conditions.

Theorem 3.10 *If $\varphi \in C^1(X)$ is bounded below, then the PS and C-conditions are equivalent.*

Proof We know that the C-condition is weaker. So, we have to show that the C-condition implies the PS-condition. We consider a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\{\varphi(u_n)\}_{n\in\mathbb{N}} \subseteq \mathbb{R} \text{ is bounded, } \varphi'(u_n) \to 0 \text{ in } X^* \text{ as } n \to \infty. \tag{3.5}$$

Since $\varphi(\cdot)$ satisfies the C-condition, from Proposition 3.9 we have that $\varphi(\cdot)$ is coercive. Then from (3.5) we infer that $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ is bounded. Therefore $(1 + \|u_n\|_X)\varphi'(u_n) \to 0$ in X^* and since $\varphi(\cdot)$ satisfies the C-condition, we conclude that $\{u_n\}_{n\in\mathbb{N}}$ admits a strongly convergent subsequence. This proves that $\varphi(\cdot)$ satisfies the PS-condition. □

As we already mentioned, the deformation approach examines the deformations of the energy functional outside the critical set. If the ambient space is a Hilbert space H and $\varphi \in C^1(H)$ with $\varphi'(\cdot)$ locally Lipschitz, then we can consider the negative gradient flow. However, in the more general setting of a Banach space or

of a not regular enough energy functional, this does not work. We need to introduce a suitable substitute of the gradient vector field.

Definition 3.11 Let $\varphi \in C^1(X)$. We say that $v \in X$ is a "pseudogradient vector" of $\varphi(\cdot)$ at u, if

$$\|v\|_X \leq 2\|\varphi'(u)\|_{X^*}, \|\varphi'(u)\|^2_{X^*} \leq \langle \varphi'(u), v\rangle.$$

We say that $\vartheta : X \setminus K_\varphi \to X$ is a "pseudogradient vector field", if $\vartheta(\cdot)$ is locally Lipschitz and for every $u \in X \setminus K_\varphi$, $\vartheta(u)$ is a pseudogradient vector of $\varphi(\cdot)$ at u.

Remark 3.12 More generally for $0 < \alpha < \beta$, we can require that

$$\alpha\|\varphi'(u)\|^2_{X^*} \leq \langle \varphi'(u), \vartheta(u)\rangle, \|\vartheta(u)\|_X \leq \beta\|\varphi'(u)\|_{X^*} \text{ for all } u \in X \setminus K_\varphi. \tag{3.6}$$

In most cases $\alpha = 1$, $\beta = 2$ as in Definition 3.11 above. If $X = H =$ a Hilbert space with $H = H^*$ and $\varphi \in C^2(H)$, then we can take

$$\vartheta(u) = \frac{\alpha + \beta}{2} \frac{1}{\|\varphi'(u)\|_{H^*}} \varphi'(u).$$

Finally from (3.6) (with $\alpha = 1$), we see that

$$\|\varphi'(u)\|_{X^*} \leq \|\vartheta(u)\|_X \text{ for all } u \in X \setminus K_\varphi.$$

We need to show that a pseudogradient vector field exists for any $\varphi \in C^1(X)$. To do this we will need the following lemma.

Lemma 3.13 *If (Y, d) is a metric space, E is a normed space, $F : Y \to 2^E \setminus \{\emptyset\}$ has convex values and for every $y \in Y$ we can find a neighborhood U of y such that*

$$\bigcap_{z \in U} F(z) \neq \emptyset,$$

then $F(\cdot)$ admits a locally Lipschitz selector, that is, there exists $\vartheta : Y \to E$ locally Lipschitz such that

$$\vartheta(y) \in F(y) \text{ for all } y \in Y.$$

Proof By hypothesis for every $y \in Y$, we can find U_y a neighborhood of y such that $\bigcap_{z \in U_y} F(z) \neq \emptyset$. Then $\{U_y\}_{y \in Y}$ is an open cover of Y and since Y is paracompact (being a metric space), we can find a locally finite refinement $\{V_i\}_{i \in I}$. First we assume that $V_i \neq Y$ for all $i \in I$. We set

$$\widehat{d}_i(y) = d(y, V_i^c) \text{ and } \mu(y) = \sum_{i \in I} \widehat{d}_i(y) \text{ for all } y \in Y.$$

Each $\widehat{d_i}(\cdot)$ is Lipschitz and on account of the local finiteness of the cover $\{V_i\}_{i\in I}$, we infer that $\mu(\cdot)$ is locally Lipschitz and $\mu(y) \neq 0$ for all $y \in Y$. So, we can define $k_i(y) = \dfrac{\widehat{d_i}(y)}{\mu(y)}$ for all $i \in I$, all $y \in Y$. Then $\{k_i\}_{i\in I}$ is a locally Lipschitz partition of unity subordinate to $\{V_i\}_{i\in I}$. Next we assume that for some $i_0 \in I$, $V_{i_0} = Y$. We set

$$k_{i_0} \equiv 1 \text{ and } k_i \equiv 0 \text{ for all } i \in I, i \neq i_0.$$

Again $\{k_i\}_{i\in I}$ is a locally Lipschitz partition of unity subordinate to $\{V_i\}_{i\in I}$. Since $\{V_i\}_{i\in I}$ is a refinement of $\{U_y\}_{y\in Y}$, we see that

$$\bigcap_{z\in V_i} F(z) \neq \emptyset.$$

Let $e_i \in \bigcap\limits_{z\in V_i} F(z_i)$ and consider the function $\vartheta : Y \to E$ defined by

$$\vartheta(y) = \sum_{i\in I} k_i(y)e_i.$$

Then $\vartheta(\cdot)$ is well-defined and locally Lipschitz. Since $F(\cdot)$ is convex valued and $\{k_i\}_{i\in I}$ is a partition of unity subordinate to $\{V_i\}_{i\in I}$, we conclude that $\vartheta(y) \in F(y)$ for all $y \in Y$. □

Using this lemma, we can show the existence of a pseudogradient vector field.

Theorem 3.14 *If $\varphi \in C^1(X)$, then $\varphi(\cdot)$ has a pseudogradient vector field $\vartheta(\cdot)$.*

Proof Let $Y = X \setminus K_\varphi$. For every $y \in Y$, let $F(y)$ =set of pseudogradient vectors for φ at $y \in Y$. Then $F(\cdot)$ is convex valued (see Definition 3.11). If $u \in Y$, we can find $v \in X$ such that

$$\|v\|_X \leq 1 \text{ and } \frac{4}{5}\|\varphi'(u)\|_{X^*} \leq \langle \varphi'(u), v\rangle. \tag{3.7}$$

We set

$$h = \frac{5}{3}\|\varphi'(u)\|_{X^*}v \in X.$$

We have

$$\|h\|_X \leq \frac{5}{3}\|\varphi'(u)\|_{X^*}, \frac{4}{3}\|\varphi'(u)\|^2_{X^*} \leq \langle \varphi'(u), v\rangle \text{(see (3.7))}.$$

Since $\varphi \in C^1(X)$, we can find U a neighborhood of u such that

$$\|h\|_X < 2\|\varphi'(z)\|_{X^*},\ \|\varphi'(z)\|^2_{X^*} < \langle \varphi'(z), h\rangle \text{ for all } z \in U$$
$$\Rightarrow h \in \bigcap_{z\in U} F(z) \neq \emptyset.$$

Apply Lemma 3.13 to produce a locally Lipschitz map $\vartheta : Y \to X$ such that $\vartheta(y) \in F(y)$ for all $y \in Y$. Then $\vartheta(\cdot)$ is a pseudogradient vector field for $\varphi(\cdot)$. □

Remark 3.15 The pseudogradient vector field is not in general unique. Also any convex combination of pseudogradient vector fields for φ is again a pseudogradient vector field.

The C-condition and the pseudogradient vector field will lead to the deformation theorems (see Sect. 3.2), which in turn will give the minimax theorems (see Sect. 3.3). For this reason we conclude this section with some additional remarks concerning the C-condition.

Proposition 3.16 *If $\varphi \in C^1(X)$ is bounded below, $m = \inf_X \varphi$ and φ satisfy the C_m-condition, then there exists $u_0 \in X$ such that $\varphi(u_0) = m$.*

Proof We apply Theorem 3.3 (with $h(r) = r$) and obtain $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi(u_n) \downarrow m \text{ and } (1 + \|u_n\|_X)\varphi'(u_n) \to 0 \text{ in } X^*.$$

Since φ satisfies the C_m-condition, we may assume that

$$u_n \to u_0 \text{ in } X,$$
$$\Rightarrow \varphi(u_n) \to \varphi(u_0) = m.$$

□

Remark 3.17 It is clear that every minimizing sequence of $\varphi(\cdot)$ admits a strongly convergent subsequence whose limit is a global minimizer of $\varphi(\cdot)$.

We have already encountered the next result (see Proposition 3.9). Here we provide an alternative proof.

Proposition 3.18 *If $\varphi \in C^1(X)$ is bounded below and satisfies the C-condition, then $\varphi(\cdot)$ is coercive.*

Proof We argue by contradiction. So, suppose that $\varphi(\cdot)$ is not coercive. Then we can find $c \in \mathbb{R}$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi(v_n) \leq c + \frac{1}{n},\ \|u_n\|_X \geq 2(e^n - 1) \text{ for all } n \in \mathbb{N}. \tag{3.8}$$

Apply Theorem 3.3 (with $h(r) = r$, $\varepsilon = c + \frac{1}{n} - \inf_X \varphi$, $\lambda = n$, $\bar{r} = e^n - 1$) to obtain a sequence $\{v_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\begin{aligned}\varphi(v_n) \leq \varphi(u_n),\ \|v_n - u_n\|_X \leq e^n - 1,\ \|\varphi'(v_n)\|_{X^*} \\ \leq \frac{c + \frac{1}{n} - \inf_X \varphi}{(1 + \|v_n\|_X)n} \text{ for all } n \in \mathbb{N}.\end{aligned} \tag{3.9}$$

From (3.8) and (3.9) it follows that

$$\begin{aligned}\|v_n\|_X \geq e^n - 1, \\ \Rightarrow \|v_n\|_X \to \infty.\end{aligned} \tag{3.10}$$

Also we have

$$\limsup_{n\to\infty} \varphi(v_n) \leq c \text{ and } (1 + \|v_n\|_X)\varphi'(v_n) \to 0 \text{ in } X^*. \tag{3.11}$$

But $\varphi(\cdot)$ is bounded below and satisfies the C-condition. So, from (3.11) we infer that we can find a strongly convergent subsequence of $\{v_n\}_{n\in\mathbb{N}}$, contradicting (3.10). Therefore $\varphi(\cdot)$ is coercive. □

Recall (see Definition I-7.71), that an operator $A : X \to X^*$ is of "type $(S)_+$", if it has the following property:

"If $u_n \xrightarrow{w} u$ in X and $\limsup_{n\to\infty} \langle A(u_n), u_n - u\rangle \leq 0$, then $u_n \to u$ in X." The next proposition is very convenient in applications on boundary value problems, because it shows that certain coercive energy functionals satisfy the PS-condition (equivalently the C-condition, see Theorem 3.10).

Proposition 3.19 *If X is a reflexive Banach space, $\varphi \in C^1(X)$ is coercive and $\varphi' = A + N$ with $A : X \to X^*$ of type $(S)_+$ and $N : X \to X^*$ completely continuous, then $\varphi(\cdot)$ satisfies the PS-condition.*

Proof Let $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ be a sequence such that

$$\{\varphi(u_n)\}_{n\in\mathbb{N}} \subseteq \mathbb{R} \text{ is bounded, } \varphi(u_n) \to 0 \text{ in } X^*. \tag{3.12}$$

The coercivity of $\varphi(\cdot)$ implies that $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ is bounded. Then the reflexivity of X and the Eberlein-Smulian theorem, imply that we may assume that

$$u_n \xrightarrow{w} u \text{ in } X. \tag{3.13}$$

We have

$$\langle \varphi'(u_n), u_n - u\rangle = \langle A(u_n), u_n - u\rangle + \langle N(u_n), u_n - u\rangle \text{ for all } n \in \mathbb{N}. \tag{3.14}$$

Note that the complete continuity of $N(\cdot)$ and (3.13), imply

$$N(u_n) \to N(u) \text{ in } X^*. \tag{3.15}$$

So, if in (3.14) we pass to the limit as $n \to \infty$ and use (3.12), (3.13) and (3.15), we obtain

$$\lim_{n\to\infty} \langle A(u_n), u_n - u\rangle = 0,$$

$$\Rightarrow u_n \to u \text{ in } X \text{ (by the } (S)_+ - \text{property)}.$$

We conclude that $\varphi(\cdot)$ satisfies the PS-condition. □

3.2 Deformation Theorem

In this section we develop the so-called "deformation approach" to critical point theory. We prove three deformation theorems which will be used in Sect. 3.3 to prove the main minimax theorems.

Theorem 3.20 *If $\varphi \in C^1(X)$ and satisfies the C_c-condition for some $c \in \mathbb{R}$, then for every $\varepsilon_0 > 0$, every neighborhood U of K_φ^c (if $K_\varphi^c = \emptyset$, then $U = \emptyset$) and every $\lambda > 0$, we can find $\varepsilon \in (0, \varepsilon_0)$ and a continuous map $h : [0,1] \times X \to X$ (a homotopy) such that for all $(t,u) \in [0,1] \times X$, we have*

(a) $\|h(t,u) - u\|_X \leq \lambda(1 + \|u\|_X)t$;
(b) $\varphi(h(t,u)) \leq \varphi(u)$;
(c) $h(t,u) \neq u \Rightarrow \varphi(h(t,u)) < \varphi(u)$;
(d) $|\varphi(u) - c| \geq \varepsilon_0 \Rightarrow h(t,u) = u$;
(e) $h(1, \varphi^{c+\varepsilon}) \subseteq \varphi^{c-\varepsilon} \cup U$.

Proof Since $\varphi(\cdot)$ satisfies the C_c-condition, the set $K_\varphi^c \subseteq X$ is compact (possibly empty). Therefore we can find $r > 0$ such that

$$B_{3r}(K_\varphi^c) = \{u \in X : d(u, K_\varphi^c) < 3r\} \subseteq U.$$

Next we show that there exist $\varepsilon_1 > 0$ and $\eta > 0$ such that

$$c - 2\varepsilon_1 \leq \varphi(u) \leq c + 2\varepsilon_1 \text{ and } u \notin B_r(K_\varphi^c) \Rightarrow (1 + \|u\|_X)\|\varphi'(u)\|_{X^*} \geq \eta. \tag{3.16}$$

Suppose that (3.16) is not true. This means that we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq X$, $u_n \notin B_r(K_\varphi^c)$ for all $n \in \mathbb{N}$ such that

$$\varphi(u_n) \to c \text{ and } (1 + \|u_n\|_X)\varphi'(u_n) \to 0 \text{ in } X^*. \tag{3.17}$$

Since $\varphi(\cdot)$ satisfies the C_c-condition from (3.17), it follows that we may assume that $u_n \to u \notin B_r(K_\varphi^c)$. We have $\varphi(u) = c$ and $\varphi'(u) = 0$, a contradiction. Therefore (3.16) is true.

We introduce the following sets

$$D_1 = \{u \in X : |\varphi(u) - c| \geq 2\varepsilon_1\} \cap \overline{B_r(K_\varphi^c)},$$
$$D_2 = \{u \in X : |\varphi(u) - c| \leq \varepsilon_1\} \cap B_{2r}(K_\varphi^c)^c.$$

These sets are closed and disjoint. So, we can find a locally Lipschitz function $\beta : X \to [0, 1]$ such that

$$\beta|_{D_1} = 0 \text{ and } \beta|_{D_2} = 1,$$

(for example consider $\beta(u) = \dfrac{d(u, D_1)}{d(u, D_1) + d(u, D_2)}$). We choose $\mu \in (0, 1)$ such that $e^\mu \leq \lambda$. Also let $\vartheta : X \setminus K_\varphi \to X$ be a pseudogradient vector field for φ (see Theorem 3.14). We define

$$S(u) = \begin{cases} -\eta\mu\beta(u)\dfrac{\vartheta(u)}{\|\vartheta(u)\|_X^2} & \text{if } |\varphi(u) - c| \leq 2\varepsilon_1, u \notin B_r(K_\varphi^c) \\ 0 & \text{otherwise} \end{cases}$$

This vector field is well-defined (see (3.16)) and locally Lipschitz. Moreover, if $|\varphi(u) - c| \leq 2\varepsilon_1$ and $u \notin B_r(K_\varphi^c)$, then

$$\|S(u)\|_X \leq \eta\mu\frac{1}{\|\vartheta(u)\|_X} \leq \eta\mu\frac{1}{\|\varphi'(u)\|_{X^*}} \leq \mu(1 + \|u\|) \text{ (see Remark 3.12)}, \tag{3.18}$$

and

$$\begin{aligned} \langle\varphi'(u), S(u)\rangle &= -\eta\mu\beta(u)\frac{\langle\varphi'(u), \vartheta(u)\rangle}{\|\vartheta(u)\|_X^2} \\ &\leq -\eta\mu\beta(u)\frac{\|\varphi'(u)\|_{X^*}^2}{\|\vartheta(u)\|_X^2} \text{ (see Definition 3.11)} \\ &\leq -\frac{1}{4}\eta\mu\beta(u) \text{ (again by Definition 3.11).} \end{aligned} \tag{3.19}$$

We consider the following abstract Cauchy problem

$$\frac{dk(t)}{dt} = S(k(t)) \text{ on } [0, 1], k(0) = u. \tag{3.20}$$

Since $S(\cdot)$ is locally Lipschitz with sublinear growth, we know that (3.20) admits a unique global flow $k(u) : [0, 1] \to X$. We set

$$h(t, u) = k(u)(t).$$

The continuous dependence of the flow on the initial condition implies that $h(\cdot,\cdot)$ is continuous. From the definition of $S(\cdot)$, we see that

$$|\varphi(u) - c| \geq 2\varepsilon_0 \Rightarrow h(t,u) = u.$$

This proves statement (d) of the theorem.
Also, we have

$$\begin{aligned}
\frac{\partial}{\partial t}\varphi(h(t,u)) &= \langle \varphi'(h(t,u)), \frac{\partial h(t,u)}{\partial t}\rangle \text{ (chain rule)} \\
&= \langle \varphi'(h(t,u)), S(h(t,u))\rangle \leq 0 \text{ (see (3.20), (3.19))}, \\
\Rightarrow \varphi(h(t,u)) &\leq \varphi(h(0,u)) = \varphi(u) \text{ for all } (t,u) \in [0,1]\times X.
\end{aligned}$$

This proves statement (b) of the theorem.
From the previous calculations, we know that

$$\frac{\partial}{\partial t}\varphi(h(t,u)) \leq -\frac{1}{4}\eta\mu\beta(h(t,u)).$$

So, if $h(t,u) \neq u$, then $\beta(h(t,u)) > 0$ and so $\varphi(h(t,u)) < \varphi(u)$ which proves statement (c) of the theorem.
We integrate (3.20) and obtain

$$\begin{aligned}
\|h(t,u) - u\|_X &\leq \int_0^t \|S(h(s,u))\|_X ds \\
&\leq \mu \int_0^t (1 + \|h(s,u)\|_X) ds \\
&\leq \mu \int_0^t \|h(s,u) - u\|_X ds + \mu(1 + \|u\|_X)t \\
\Rightarrow \|h(t,u) - u\|_X &\leq \lambda(1 + \|u\|_X)t \text{ (using Gronwall's inequality).}
\end{aligned}$$

So, we have proved statement (a) of the theorem.
It remains to show statement (e). For $0 \leq t_1 \leq t_2 \leq 1$, we have

$$\begin{aligned}
&h(t_2,u) = h(t_2 - t_1, h(t_1,u)) \text{ (problem (3.20) is autonomous)} \\
\Rightarrow &\|h(t_2,u) - h(t_1,u)\|_X \leq \lambda(1 + \|h(t_1,u)\|_X)(t_2 - t_1) \text{ (see statement (a)).}
\end{aligned} \tag{3.21}$$

Let $\varrho > 0$ and $\varepsilon \in (0, \varepsilon_1]$ be such that

$$\overline{B_{2r}(K_\varphi^c)} \subseteq B_\varrho, \quad 8\varepsilon \leq \eta\mu, \quad 8\lambda(1+\varrho)\varepsilon \leq \eta\mu r. \tag{3.22}$$

Suppose (e) is not true. We can find $u \in \varphi^{c+\varepsilon}$ such that

$$\varphi(h(1,u)) > c - \varepsilon, \quad h(1,u) \notin U.$$

We have seen that the homotopy h is φ-decreasing (see (b)). Hence

$$c - \varepsilon < \varphi(h(t,u)) \leq c + \varepsilon \text{ for all } t \in [0,1]. \tag{3.23}$$

We claim that

$$h([0,1],u) \cap B_{2r}(K_\varphi^c) \neq \emptyset.$$

Indeed if this is not true, then

$$\frac{1}{4}\eta\mu \leq \varphi(u) - \varphi(h(t,u)) \leq 2\varepsilon \text{ (see (3.23), (3.19))}. \tag{3.24}$$

Since $h(1,u) \notin U$ and $B_{2r}(K_\varphi^c) \subseteq U$, we can find $0 \leq t_1 \leq t_2 \leq 1$ such that

$$\left\{ \begin{array}{l} d(h(t_1,u), K_\varphi^c) = 2r, d(h(t_2,u), K_\varphi^c) = 3r, \\ 2r < d(h(t,u), K_\varphi^c) < 3r \text{ for all } t \in (t_1,t_2) \end{array} \right\}.$$

From the above relations and (3.24), we have

$$\begin{aligned} &\frac{1}{4}\eta\mu(t_2 - t_1) \leq \varphi(h(t_1,u)) - \varphi(h(t_2,u)) < 2\varepsilon \\ \Rightarrow\ & r \leq \|h(t_2,u) - h(t_1,u)\|_X \leq \lambda(1 + \|h(t_1,u)\|_X)(t_2 - t_1) \\ & < \lambda(1+\varrho)\frac{8\varepsilon}{\eta\mu} \leq r \text{ (see (3.21), (3.22))}, \end{aligned}$$

a contradiction. This proves statement (e) and completes the proof of the theorem. □

Corollary 3.21 *If H is a Hilbert space, $\varphi \in C^1(H)$, $\varphi'(\cdot)$ is locally Lipschitz, $\varphi'(u) \neq 0$ for all $u \in H$, then for $\varepsilon > 0$ small there exists $h \in C([0,1] \times H, H)$ such that for all $t \in [0,1]$, $u \in H$,*

(a) $h(0,u) = u$;
(b) $h(t,u) = u$, if $u \notin \varphi^{-1}[c - 2\varepsilon, c + 2\varepsilon]$;
(c) $h(1, \varphi^{c+\varepsilon}) \subseteq \varphi^{c-\varepsilon}$;
(d) $h(t,\cdot)$ is a homeomorphism on H.

Example 3.22 If $X = H =$ a Hilbert space and $\varphi \in C^2(H)$, then we use the gradient flow.

Remark 3.23 According to this theorem, if we start a little bit above a critical level c, either we will bypass the critical neighborhood U and reach a noncritical lower level, or we will end up in U, the neighborhood of K_φ^c, where topologically

interesting things can happen. On the other hand, when crossing a noncritical level, nothing is going to happen topologically. This is essentially the content of the so-called "Quantitative Deformation Theorem", which we present next.

Theorem 3.24 *If $\varphi \in C^1(X)$, $E \subseteq X$, $c \in \mathbb{R}$, $\varepsilon > 0$, $\delta \in (0, \frac{1}{2})$ are such that for all $y \in \varphi^{-1}([c-2\varepsilon, c+2\varepsilon] \cap E_{2\delta})$, we have*

$$\|\varphi'(y)\|_{X^*} \geq \frac{8\varepsilon}{\delta},$$

then there exists a homotopy $h \in C([0,1] \times X, X)$ such that

(a) $h(t,u) = u$, if $t = 0$ or if $u \notin \varphi^{-1}([c-2\varepsilon, c+2\varepsilon] \cap E_{2\delta})$;
(b) $h(1, \varphi^{c+\varepsilon} \cap E) \subseteq \varphi^{c-\varepsilon}$;
(c) for all $t \in [0,1]$, $h(t,\cdot)$ is a homeomorphism;
(d) $\|h(t,u) - u\|_X \leq \delta$ for all $t \in [0,1]$ and all $u \in X$;
(e) $\varphi(h(t,u)) \leq \varphi(h(s,u))$ for all $0 \leq s \leq t \leq 1$, all $u \in X$;
(f) $\varphi(h(t,u)) < c$ for all $t \in [0,1]$, all $u \in \varphi^c \cap E_\delta$.

Proof The proof is similar to that of Theorem 3.20.

We set

$$D_1 = \{u \in X : |\varphi(u) - c| \geq 2\varepsilon\} \cup E^c_{2S},$$
$$D_2 = \varphi^{-1}([c-\varepsilon, c+\varepsilon]) \cap E_\delta.$$

These are disjoint, closed sets and as before we consider the locally Lipschitz map $\beta : X \to [0,1]$ defined by

$$\beta(u) = \frac{d(u, D_1)}{d(u, D_1) + d(u, D_2)} \text{ for all } u \in X.$$

Evidently, $\beta|_{D_1} = 0$ and $\beta|_{D_2} = 1$.

We introduce the locally Lipschitz vector field $S : X \setminus K_\varphi \to X$ defined by

$$S(u) = \begin{cases} -\beta(u) \cdot \dfrac{\vartheta(u)}{\|\vartheta(u)\|^2_X} & \text{if } u \in D^c_1 \\ 0 & \text{if } u \in D_1 \end{cases}$$

Recall that $\vartheta(\cdot)$ is a pseudogradient vector field for $\varphi(\cdot)$. We have $\|S(u)\|_X \leq \frac{\delta}{8\varepsilon}$ for all $u \in X$ (see Definition 3.11). For each $u \in X$, we consider the abstract Cauchy problem

$$\frac{d\sigma(u)(t)}{dt} = S(\sigma(u)(t)) \text{ on } [0,1], \quad \sigma(u)(0) = u.$$

This has a unique global flow and $(t, u) \to \sigma(u)(t)$ is continuous. We define

$$h(t, u) = \sigma(u)(t) \text{ for all } (t, u) \in [0, 1] \times X.$$

For $t \geq 0$, we have

$$\begin{aligned} \|h(t, u) - u\|_X &= \|\sigma(u)(t) - u\|_X \\ &\leq \int_0^t \|S(\sigma(u)(s))\|_X ds \\ &\leq \frac{\delta t}{8\varepsilon}, \end{aligned}$$

and

$$\begin{aligned} \frac{\partial}{\partial t}\varphi(h(t, u)) &= \langle \varphi'(h(t, u)), \frac{\partial}{\partial t}h(t, u)\rangle \\ &= \langle \varphi'(h(t, u)), S(h(t, u))\rangle \\ &\leq -\frac{1}{4}\beta(h(t, u)). \end{aligned}$$

Then as before we verify (a), (c), (d), (e), (f).

Let $u \in \varphi^{c+\varepsilon} \cap E$. If we can find $t \in [0, 8\varepsilon]$ such that $\varphi(h(t, u)) < c - \varepsilon$, then $\varphi(h(8\varepsilon, u)) < c - \varepsilon$ and so (b) follows.

Finally if $h(t, u) \in \varphi^{-1}([c - \varepsilon, c + \varepsilon])$ for all $t \in [0, 8\varepsilon]$, then

$$\begin{aligned} \varphi(h(8\varepsilon, u)) &= \varphi(u) + \int_0^{8\varepsilon} \frac{\partial}{\partial t}\varphi(h(t, u))dt \\ &\leq \varphi(u) - \frac{1}{4}\int_0^{8\varepsilon} \beta(h(t, u))dt \\ &\leq c + \varepsilon - 2\varepsilon = c - \varepsilon. \end{aligned}$$

Therefore (b) is satisfied and this completes the proof of the theorem. □

Using this theorem, we can have the following variant of the Ekeland variational principle (see Corollary I-8.104).

Corollary 3.25 *If $\varphi \in C^1(X)$ is bounded below, $\varepsilon > 0$, $\delta \in (0, \frac{1}{2})$ and $v \in X$ satisfies $\varphi(v) \leq \inf_X \varphi + \varepsilon$, then there exists $u \in X$ such that*

(a) $\varphi(u) \leq \inf_X \varphi + 2\varepsilon$;

(b) $\|u - v\|_X \leq 2\delta$;

(c) $\|\varphi'(u)\|_{X^*} \leq \frac{8\varepsilon}{\delta}$.

Proof Apply Theorem 3.24 with $E = \{v\}$ and $c = \inf_X \varphi$. Consider $y \in \varphi^{-1}([c - 2\varepsilon, c + 2\varepsilon]) \cap E_{2\delta}$. Then if $\|\varphi'(y)\|_{X^*} \geq \frac{8\varepsilon}{\delta}$, from Theorem 3.24(b) we have $\varphi(h(1, v)) \leq c - \varepsilon$, contradicting the choice of c. □

Corollary 3.26 *If $\varphi \in C^1(X)$ is bounded below, then we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that*

$$\varphi(u_n) \downarrow \inf_X \varphi \text{ and } \varphi'(u_n) \to 0 \text{ in } X^*$$

(that is, $\{u_n\}_{n\in\mathbb{N}}$ is a PS-sequence for the level $\inf_X \varphi$).

Corollary 3.27 *If $\varphi \in C^1(X)$ satisfies $c = \liminf_{\|u\|_X\to\infty} \varphi(u) \in \mathbb{R}$, then for every $\varepsilon > 0$, $\delta \in (0, \frac{1}{2})$, $\varrho > 2\delta$, we can find $v \in X$ such that*

(a) $c - 2\varepsilon \leq \varphi(v) \leq c + 2\varepsilon$;
(b) $\|v\|_X \geq \varrho - 2\delta$;
(c) $\|\varphi'(v)\|_{X^*} \leq \frac{8\varepsilon}{\delta}$.

Proof We argue by contradiction. So, suppose that the claim of the corollary is not true. We set $E = B^c_\varrho$. From the definition of c, we see that $\varphi^{c+\varepsilon} \cap E$ is unbounded and $\varphi^{c-\varepsilon} \subseteq B_r$ for $r > 0$ large. From Theorem 3.24 we have $h(1, \varphi^{c+\varepsilon} \cap E) \subseteq \varphi^{c-\varepsilon}$ and $\varphi^{c+\varepsilon} \cap E \subseteq (\varphi^{c-\varepsilon})_\delta \subseteq B_{r+\delta}$, a contradiction. So, the corollary is true. □

Finally using the quantitative deformation theorem (Theorem 3.24), we can have the following generalization of Proposition 3.18.

Corollary 3.28 *If $\varphi \in C^1(X)$ is bounded below and for every $c \in \mathbb{R}$ a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that*

$$\varphi(u_n) \to c \text{ and } \varphi'(u_n) \to 0 \text{ in } X^*$$

is bounded, then $\varphi(\cdot)$ is coercive.

Proof Arguing by contradiction suppose that the claim of the corollary is not true. Then $c = \liminf_{\|u\|_X\to\infty} \varphi(u) \in \mathbb{R}$. Apply Corollary 3.27 with $\varepsilon = \frac{1}{n^2}$, $\delta = \frac{1}{n}$ and $\varrho = n$, $n \geq 3$. We can find a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi(u_n) \to c \text{ (see Corollary 3.27(a))},$$
$$\|u_n\|_X \to \infty \text{ (see Corollary 3.27(b))},$$
$$\varphi'(u_n) \to 0 \text{ (see Corollary 3.27(c))}.$$

This contradicts the hypothesis on $\varphi(\cdot)$. So, the corollary is true. □

Next we will present the so-called "Second Deformation Theorem". To do this, we need to introduce some topological notions which will be also important in the study of critical groups.

Definition 3.29 Let V be a Hausdorff topological space and $D \subseteq V$ nonempty.

(a) A "deformation" is a continuous map $h : [0, 1] \times V \to V$ such that $h(0, u) = u$ for all $u \in V$.
(b) We say that $h : [0, 1] \times V \to V$ is a "deformation of V into D", if $h(\cdot, \cdot)$ is a deformation and $h(1, V) \subseteq D$. We also say that V is "deformable" into D.
(c) We say that D is a "deformation retract"(resp. a "strong deformation retract") of V, if there exists a deformation $h : [0, 1] \times V \to V$ into D such that

$$h(1, \cdot)|_D = id|_D$$

(resp. $h(t, \cdot)|_D = \mathrm{id}|_D$ for all $t \in [0, 1]$).

Remark 3.30 According to the above definitions D is a deformation retract of V, if V is deformable into D and the points of D end up where they started. Similarly, D is a strong deformation retract of V, if V is deformable into D and during the deformation the elements of D remain fixed. If V is deformable into D and D is a retract of V(thus closed), then D is a deformation retract of V. If V is a metric space and an AR (see Definition 2.65) and $D \subseteq V$ is closed, then D is a strong deformation retract if and only if D is an AR. The set S^N is a retract of $\mathbb{R}^{N+1} \setminus \{\emptyset\}$ (via the radial retraction). In fact is a strong deformation retract. Just consider the deformation $h : [0, 1] \times \mathbb{R}^{N+1} \setminus \{0\} \to \mathbb{R}^{N+1} \setminus \{0\}$ defined by

$$h(t, u) = (1 - t)u + t\frac{u}{|u|} \text{ for all } (t, u) \in [0, 1] \times (\mathbb{R}^{N+1} \setminus \{0\}).$$

More generally, $D \subseteq V$ is a deformation retract if and only if D is a retract of V and V is deformable into D.

The next theorem is the "Second Deformation Theorem".

Theorem 3.31 *If $\varphi \in C^1(X)$, $a \in \mathbb{R}$, $b \in (a, +\infty]$, φ satisfies the C_c-condition for every $c \in [a, b)$ and $\varphi^{-1}(a)$ contains at most a finite number of critical points, then there is a deformation $h : [0, 1] \times (\varphi^b \setminus K_\varphi^b) \to \varphi^b \setminus K_\varphi^b$ of $\varphi^b \setminus K_\varphi^b$ into φ^a such that*

(a) φ^a is a strong deformation retract of $\varphi^b \setminus K_\varphi^b$;
(b) the deformation is φ-decreasing, that is, if $0 \le s \le t \le 1$, then

$$\varphi(h(t, u)) \le \varphi(h(s, u)) \text{ for all } u \in \varphi^b \setminus K_\varphi^b.$$

Proof Consider $\vartheta : X \setminus K_\varphi \to X$ a pseuologradient vector field for φ. Let $D = \varphi^b \setminus (\varphi^a \cup K_\varphi^b)$. Then $\varphi^b \setminus K_\varphi^b = D \cup \varphi^a$. We have $\varphi^a \subseteq X \setminus K_\varphi$. For $u \in D$ we consider the following abstract Cauchy problem

$$k'(t) = -\frac{\vartheta(k(t))}{(1+||k(t)||_X)||\vartheta(k(t))||_X^2} \text{ in } \mathbb{R}, k(0) = u \tag{3.25}$$

This problem has a unique solution $k_u(\cdot)$ defined on a maximal interval $[0, l_+(u)]$. We have

$$\begin{aligned} \frac{d}{dt}\varphi(k_u(t)) &= \langle \varphi'(k_u(t)), k_u'(t)\rangle \text{ (by the chain rule)} \\ &\leq -\frac{1}{4}\cdot\frac{1}{(1+||k_u(t)||_X)} \text{ (see Definition 3.11),} \end{aligned} \tag{3.26}$$

$\Rightarrow t \to \varphi(k_u(t))$ is decreasing on $[0, l_+(u)$.

We define

$$t(u) = \sup\{t \in [0, l_+(u)) : \varphi(k_u(t)) > a\}.$$

Note that $t : D \to (0, +\infty]$.

Claim 1 Given $\delta > 0$ and $c \in (a, b)$, we can find $\eta = \eta(\delta, c) > 0$ such that $u \in D$, $\varphi(u) \leq c$, $d(u, K_\varphi^a) > \delta \Rightarrow (1+||u||_X)||\varphi'(u)||_{X^*} \geq \eta$.

This is a consequence of the fact that φ satisfies the C_c condition and since $\varphi(K_\varphi) \cap (a, b) = \emptyset$

Claim 2 If $\{u_n\}_{n\in\mathbb{N}} \subseteq D$ and $\{\tau_n\}_{n\in\mathbb{N}}, \{t_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}_+$ satisfy

$$0 \leq \tau_n < t_n < t_n(u_n) \text{ for all } n \in \mathbb{N}$$

$$\text{and } \lim_{n\to+\infty} k_{u_n}(\tau_n) = \widehat{u} \text{ exists, } \widehat{u} \in \varphi^{-1}(a),$$

then $\lim_{n\to\infty} k_{u_n}(t_n) = \widehat{u}$.

Suppose that Claim 2 is not true. By passing to a subsequence if necessary, we can find $\varrho > 0$ such that

$$||k_{u_n} - \widehat{u}||_X \geq 2\varrho \text{ for all } n \in \mathbb{N}.$$

Since $\varphi(\widehat{u}) = a$ and K_φ^a is finite, we can find $\delta > 0$ and $c \in (a, b)$ such that

$$u \in D, \varrho \leq ||u - \widehat{u}||_X \leq 2\varrho \Rightarrow \varphi(u) \leq c, d(u, K_\varphi^a) > \delta.$$

We can find $n_0 \in \mathbb{N}$ such that

$$||k_{u_n}(\tau_n) - \widehat{u}||_X \leq \varrho \text{ for all } n \geq n_0$$

Therefore, we can find $\widehat{\tau}_n, \widehat{t}_n \geq 0$ such that

$$\tau_n \leq \widehat{\tau}_n < \widehat{t}_n \leq t_n, ||k_{u_n}(\widehat{\tau}_n) - \widehat{u}||_X = \varrho, ||k_{u_n}(\widehat{\tau}_n) - \widehat{u}||_X = 2\varrho,$$

$$\varrho \le ||k_{u_n}(t) - \widehat{u}||_X \le 2\varrho \text{ for all } t \in [\widehat{\tau}_n, \widehat{t}_n], n \in \mathbb{N}$$

By Claim 1 we can find $\eta = \eta(\delta, c) > 0$ such that

$$\begin{aligned} ||k'_{u_n}(t)||_X &\le \frac{1}{(1 + ||k_{u_n}(t)||_X|)||\vartheta(k_{u_n}(t))||_X} \\ &\le \frac{1}{(1 + ||k_{u_n}(t)||_X|)||\varphi'(k_{u_n}(t))||_X} \le \frac{1}{\eta} \text{ for all } t \in [\widehat{\tau}_n, \widehat{t}_n], \text{ all } n \ge n_0 \end{aligned}$$

$$\Rightarrow \varrho \le ||k_{u_n}(\widehat{t}_n) - k_{u_n}(\widehat{\tau}_n)||_X \le \frac{1}{\eta}(\widehat{t}_n - \widehat{\tau}_n). \tag{3.27}$$

Using (3.26), we have

$$\frac{d}{dt}\varphi(k_{u_n}(t)) \le -\frac{1}{4(1 + ||\widehat{u}||_X + 2\eta)} \text{ for all } t \in [\widehat{\tau}_n, \widehat{t}_n], \text{ all } n \ge n_0$$

$$\Rightarrow \varphi(k_{u_n}(\widehat{t}_n)) - \varphi(k_{u_n}(\widehat{\tau}_n)) \le -\frac{\eta\varrho}{4(1 + ||\widehat{u}||_X + 2\eta)} \text{ for all } n \ge n_0$$

Since $\widehat{t}_n < t(u_n)$, we have $a < \varphi(k_{u_n}(\widehat{t}_n))$ and $\varphi(k_{u_n}(\widehat{c}_n)) \le \varphi(k_{u_n}(\tau_n))$. Therefore

$$a + \frac{\eta\varrho}{4(1 + ||\widehat{u}||_X + 2\eta)} < \varphi(k_{u_n}(\tau_n)) \text{ for all } n \ge n_0$$

But $\varphi(k_{u_n}(\tau_n)) \to \varphi(0) = a$ as $n \to \infty$, a contradiction. This proves Claim 2.

Claim 3 Let $u \in D$. Then $t(u) < \infty$ and $\lim_{t \to t(u)} k_u(t) = k_u^* \in \varphi^{-1}(a)$

First assume that $\liminf_{t \to t(u)} d(k_u(t), K_\varphi^a) > 0$. Then we can find $\gamma \in (0, t(u))$ and $\delta > 0$ such that

$$d(k_u(t), K_\varphi^a) > \delta \text{ for all } t \in (\gamma, t(u)).$$

From (3.26) we have

$$\varphi(k_u(t)) \le \varphi(k_u(\gamma)) = c < b \text{ for all } t \in (\gamma, t(u)).$$

Let $\eta = \eta(\delta, c) > 0$ as postulated by Claim 1. We have

$$\begin{aligned} (1 + ||k_u(t)||_X)||\vartheta(k_u(t))||_X \ge (1 + ||k_u(t)||_X)||\varphi'(k_u(t))||_{X^*} \ge \eta \\ \text{for all } t \in [\gamma, t(u)), \\ \Rightarrow ||k'_u(t)||_X \le \frac{1}{\eta} \text{ (see (25))}, \end{aligned}$$

$$||k_u(t)||_X \leq ||k_u(\gamma)||_X + \frac{t-\gamma}{\eta} \text{ for all } t \in [\gamma, t(u)). \tag{3.28}$$

From (3.26) we have

$$\begin{aligned} a - \varphi(k_u(\gamma)) &< \varphi(k_u(t)) - \varphi(k_u(\gamma)) \\ &\leq -\int_\gamma^t \frac{\eta}{4(\beta+\tau-\gamma)} d\tau = -\frac{\eta}{4}(\ln\beta + t - \gamma) - \ln\beta \end{aligned}$$

for all $t \in [\gamma, t(u))$, with $\beta = \eta[1 + ||k_u(\gamma)||_X]$. Therefore $t(u) < +\infty$

Moreover, from (3.28) we have

$$\int_\gamma^t (u)||k_u'(t)||_X dt \leq \frac{t(u)-\gamma}{\eta} < \infty,$$

$$\Rightarrow \lim_{t \to t(u)} k_u(t) = k_u^* \text{ exists, } k_u^* \in \varphi^{-1}([a,c]) \setminus K_\varphi^a$$

So, $k_u^* \in X \setminus K_\varphi$. Since $k_u(\cdot)$ is the maximal solution, we infer $t(u) < l_+(u)$ and so $\varphi(k_u^*) = \varphi(k_u(t(u))) = a$

Next assume that $\liminf_{t \to t(u)} d(k_u(t), K_\varphi^a) = 0$. Then we can find $\{\tau_n\}_{n\in\mathbb{N}} \subseteq [0, t(u))$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq K_\varphi^a$ such that

$$\tau_n \to t(u), ||k_u(\tau_n) - u_n||_X \to 0 \tag{3.29}$$

Recall that K_φ^a is compact. So, we may say that $u_n \to \widehat{u} \in K_\varphi^a$ and from (3.29) we see that $k_u(\tau_n) \to \widehat{u}$. Then, using a contradiction argument and Claim 2, we see that

$$k_u(t) \to \widehat{u} \text{ as } t \to t(u). \tag{3.30}$$

Since $\widehat{u} \notin X \setminus K_\varphi$, $k_u(\cdot)$ is not defined at $t(u)$. (see (3.25)). Therefore $t(u) = l_+(u)$. If remains to show that $t(u) < \infty$. We can find $\mu > 0$ such that $||k_u(t)||_X \leq \mu$ for all $t \in [0, t(u))$ (see(3.30)). Then

$$a - \varphi(u) < \varphi(k_u(t)) - \varphi(k_u(0)) \leq -\frac{t}{4(1+\mu)} \text{ for all } t \in [0, t(u)).$$

$\Rightarrow t(u) < \infty$.

This proves Claim 3.

Claim 4 If $u \in D$, $k_u^* = \lim_{t \to t(u)} \in \varphi^{-1}(a)$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq D$, $t_n \in [0, t(u_n))$ for all $n \in \mathbb{N}$ satisfy $u_n \to u$, $\limsup_{n\to\infty} t_n \geq t(u)$, then $k_{u_n}(t_n) \to k_u^*$ as $n \to \infty$.

Let $\{\tau_n\}_{n\in\mathbb{N}} \subseteq [0, t(u))$ be increasing such that

$$\tau_n \to t(u) \text{ and } ||k_u(\tau_n) - k_u^*||_X < \frac{1}{n} \text{ for all } n \in \mathbb{N}.$$

The continuous dependence of the solution of (3.25) on the initial state, implies that for $n \in \mathbb{N}$ large, we have

$$||k_{u_n} - k_u^*||_X < \frac{2}{n}$$

Also by hypothesis, we may assume $t_n > \tau_n$. So, we can have an increasing map $k \to n_k$ such that

$$\tau_k < t_{n_k} \text{ for } k \in \mathbb{N},\ k_{u_{n_k}}(\tau_k) \to k_u^* \text{ as } k \to \infty$$

$$\Rightarrow k_{u_{n_k}}(t_{n_k}) \to k_u^* \text{ (see Claim 2).}$$

By Urysohn's criterion for convergent sequences, we conclude that Claim 4 holds.

Claim 5 $D \ni u \to t(u) \in (0, \infty)$ is continuous.

We argue by contradiction. So, suppose we can find a sequence $\{u_n\}_{n\in\mathbb{N}\subseteq D}$ such that $u_n \to u$ in X, but $t(u_n)$ does not converge to $t(u)$. Let $\varepsilon \in (0, t(u))$. The continuous dependence of the solution on the initial state, implies

$$l_+(u) - \varepsilon < l_+(u_n) \text{ and } a < \varphi(k_{u_n}(t(u) - \varepsilon)) \text{ for } n \in \mathbb{N} \text{ large,}$$

$\Rightarrow t(u) - \varepsilon < t(u_n)$ for $n \in \mathbb{N}$ large,
$\Rightarrow t(u) \le \liminf_{n\to\infty} t(u_n)$,
$\Rightarrow t(u) < \limsup_{n\to\infty} t(u_n)$.

Therefore we may assume that for some $\gamma > 0$, we have

$$t(u) + \gamma < t(u_n) \text{ for all } n \in \mathbb{N} \tag{3.31}$$

From Claim 4, we have that there exist $\varepsilon > 0$, $\beta > 0$ and $n_0 \in \mathbb{N}$ such that

$$||k_{u_n}(t)|| \le \beta \text{ for } t \in (t(u) - \varepsilon, t(u_n)),\ \text{all } n \ge n_0. \tag{3.32}$$

Let $\delta \in (0, \varepsilon)$. We have

$$\varphi(k_u(t(u) - \delta)) = \lim_{n\to\infty} \varphi(k_{u_n}(t(u) - \delta). \tag{3.33}$$

For $n \ge n_0$ and $\gamma \in (0, \mu)$, from (3.26) and (3.32),we have

$$\varphi(k_{u_n}(t(u_n) - \gamma)) - \varphi(k_{u_n}(t(u) - \delta))$$

$$\le -\frac{t(u_n) - \gamma - t(u) + \delta}{4(1 + \mu)}$$

Letting $\gamma \to 0^+$ and using (3.31) we obtain

$$\varphi(k_{u_n}(t(u) - \delta)) > a + \frac{\mu + \delta}{4(1+\beta)} \text{ for all } n \geq n_0,$$

$$\Rightarrow a \geq a + \frac{\mu + \delta}{4(1+\beta)} > a, \text{ a contradiction.}$$

This proves Claim 5.

Now we introduce the deformation $h : [0, 1] \times (\varphi^b \setminus K_\varphi^b) \to \varphi^b \setminus K_\varphi^b$ defined by

$$h(t, u) = \begin{cases} k_u(tt(u)) & \text{if } u \in D, t \in [0, 1), \\ k_u^* & \text{if } u \in D, t = 1, \\ u & \text{if } u \in \varphi^a. \end{cases} \tag{3.34}$$

for all $(t, u) \in [0, 1] \times (\varphi^b \setminus K_\varphi^b)$.

Claim 6 $h(\cdot, \cdot)$ is continuous.

From (3.34) it is clear that

$$h|_{[0,1]\times\varphi^a} \text{ is continuous.}$$

Also from Claim 5, we see that

$$h|_{[0,1]\times D} \text{ is continuous.}$$

Let $(t, u) \in (\{1\} \times D) \cup ([0, 1] \times \varphi^{-1}(a))$ and consider sequences $\{t_n\}_{n\in\mathbb{N}} \subseteq [0, 1]$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq D$ such that $(t_n, u_n) \to (t, u)$. For every $n \in \mathbb{N}$, $h(\cdot, u_n)$ is continuous on $[0, 1]$. So, we can find $\widehat{t_n} \in (0, 1)$ such that

$$|t_n - \widehat{t_n}| \leq \frac{1}{n}, \|h(t_n, u_n) - h(\widehat{t_n}, u_n)\| \leq \frac{1}{n} \text{ for all } n \in \mathbb{N}.$$

Evidently we may assume that $t_n \in (0, 1)$ for all $n \in \mathbb{N}$.

If $t = 1$ and $u \in A$, then $t_n t(u) \to t(u)$ and using Claim 4 we have

$$k_{u_n}(t_n t(u_n)) \to k_u^* = h(1, u) \text{ (see (3.34))}$$

If $u \in \varphi^{-1}(a)$, then for $\tau_n = 0$, we have $\tau_n = 0 < t_n t(u) < t(u_n)$ for all $n \in \mathbb{N}$ and $k_{u_n}(\tau_n) = u_n \to u$ (see (34)). Using Claim 2, we have

$$k_{u_n}(t_n t(u_n)) \to u = h(t, u) \text{ (see (3.34)).}$$

We conclude that $h(\cdot, \cdot)$ is continuous.

This completes the proof of the theorem. □

The next corollary, is sometimes called the "Noncritical Interval Theorem".

Corollary 3.32 *If $\varphi \in C^1(X)$, $a \in \mathbb{R}$, $a < b \leq \infty$, φ satisfies the C_c-condition for every $c \in [a, b)$ and has no critical values in $[a, b]$, then φ^a is a strong deformation retract of φ^b.*

Closing this section we sketch how these results can be extended to nonsmooth functionals.

The setting is the following. Let X be a Banach space, $\varphi \in C^1(X)$, $\psi \in \Gamma_0(X)$ (that is, $\psi : X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ not identically $+\infty$, it is convex and lower semicontinuous, see Definition I-6.102). Let

$$\xi = \varphi + \psi.$$

Definition 3.33

(a) The critical set K_ξ of $\xi(\cdot)$ is defined by

$$K_\xi = \{u \in X : 0 \in \varphi'(u) + \partial\psi(u)\}$$

with $\partial\psi(\cdot)$ being the convex subdifferential of $\psi(\cdot)$ (see Definition I-6.67).

(b) We say that $\xi(\cdot)$ satisfies the "Generalized Palais-Smale condition" (the "GPS-condition" for short), if it has the following property:
"Every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\xi(u_n) \to c \in \mathbb{R}$$

$$and \ -\varepsilon_n\|y - u_n\|_X \leq \langle\varphi'(u_n), y - u_n\rangle + \psi(y) - \psi(u_n)$$

for all $y \in X$, with $\varepsilon_n \to 0^+$, admits a strongly convergent subsequence.

Lemma 3.34 *If $f \in \Gamma_0(X)$, $f(0) = 0$ and $-\|u\|_X \leq f(u)$ for all $u \in X$, then there exists $\widehat{u}^* \in X^*$ with $\|\widehat{u}^*\|_{X^*} \leq 1$ such that*

$$\langle\widehat{u}^*, u\rangle \leq f(u) \ \textit{for all } u \in X.$$

Proof Let $\widehat{f}(u) = f(u) + \|u\|_X$ for all $u \in X$. Then $\widehat{f} \in \Gamma_0(X)$ and $u = 0$ is a minimizer of $\widehat{f}(\cdot)$. Hence $0 \in \partial\widehat{f}(0) = \partial f(0) + \overline{B}_1^*$, where $\overline{B}_1^* = \{u^* \in X^* : \|u^*\|_{X^*} \leq 1\}$. So, we can find $u^* \in \overline{B}_1^*$ such that $-u^* \in \partial f(0)$. Then for $\widehat{u}^* = -u^*$, we have

$$\langle\widehat{u}^*, u\rangle \leq f(u) \text{ for all } u \in X.$$

□

Using this lemma, we can have the following equivalent reformulation of the GPS-condition.

Proposition 3.35 *If $\xi = \varphi + \psi$ is as before, then $\xi(\cdot)$ satisfies the GPS-condition (see Definition 3.33(b)) if and only if every sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that*

$$\xi(u_n) \to c \in \mathbb{R}$$

and $\langle u_n^*, y - u_n\rangle \leq \langle \varphi'(u_n), y - u_n\rangle + \psi(y) - \psi(u_n)$ for all $y \in X$, with $u_n^* \to 0$ in X^* admits a strongly convergent subsequence.

Using this compactness-type condition on $\xi = \varphi + \psi$, we can have a deformation theorem (see Szulkin [450]).

Theorem 3.36 *If $\xi = \varphi + \psi$ is as before, satisfies the GPS-condition, U is an open neighborhood of K_ξ^c and $\xi_0 > 0$, then there exists $\varepsilon \in (0, \varepsilon_0)$ such that for every $K \subseteq X \setminus U$ compact with*

$$c \leq \sup_{u\in K} \xi(u) \leq c + \varepsilon,$$

we can find $C \subseteq X$ closed with $K \subseteq intC$, $t_0 > 0$ and a deformation $h : [0, t_0] \times C \to X$ such that

(a) $\|h(t, u) - u\|_X \leq t$ for all $u \in C$;
(b) $\xi(h(t, u)) - \xi(u) \leq 2t$ for all $u \in C$;
(c) $\xi(h(t, u)) - \xi(u) \leq -2\varepsilon t$ for all $u \in C$ with $\xi(u) \geq c - \varepsilon$ and

$$\sup_{u\in K} \xi(h(t, u)) - \sup_{u\in K} \xi(u) \leq -2\varepsilon t.$$

Moreover, if $C_0 \subseteq X$ is closed and $C_0 \cap K_\xi = \emptyset$, we construct C and the deformation $h(t, u)$ so that

$$\xi(h(t, u)) - \xi(u) \leq 0 \text{ for all } u \in C \cap C_0.$$

3.3 Minimax Theorems

In this section we use the deformation theorem to have minimax characterizations of the critical points of $\varphi \in C^1(X)$. The following notion is crucial in this approach.

Definition 3.37 Let V be a Hausdorff topological space and $E_0 \subseteq E$ and D nonempty subsets of V and $\gamma^* \in C(E_0, V)$. The pair (E_0, E) is "linking with D", if the following hold

(a) $E_0 \cap D = \emptyset$;
(b) for all $\gamma \in C(E, V)$ with $\gamma|_{E_0} = \gamma^*$, we have $\gamma(E) \cap D \neq \emptyset$.

Let us see some characteristic examples of linking sets.

Example 3.38

(a) Let X be a Banach space $u_0, u_1 \in X$, $u_0 \neq u_1$ and set $E_0 = \{u_0, u_1\}$, $E = conv\{u_0, u_1\} = \{(1-t)u_0 + tu_1\}_{t\in[0,1]}$, $D = \partial U$ with U an open neighborhood of u_0, with $u_1 \notin U$. Then the pair $\{E_0, E\}$ is linking with D. Clearly $E_0 \cap D = \emptyset$. Also let $\gamma \in C(E, X)$ with $\gamma|_{E_0} = id_{E_0}$. Then the set $\gamma(E)$ is connected and $\gamma(u_0) = u_0 \in U$, $\gamma(u_1) = u_1 \notin U$. So, by Bolzano's theorem, we have $\gamma(E) \cap D \neq \emptyset$.

(b) Let X be a Banach space such that

$$X = Y \oplus V \text{ with } \dim Y < \infty.$$

We set $E_0 = \partial B_\varrho \cap Y$, $E = \overline{B}_\varrho \cap Y$ and $D = V$. Recall that $\overline{B}_\varrho = \{u \in X : \|u\| \leq \varrho\}$. The pair $\{E_0, E\}$ is linking with D. To see this let p_Y be the Y-projection operator and let $\gamma \in C(E, X)$ such that $\gamma|_{E_0} = id|_{E_0}$. We need to show that $\gamma(E) \cap D \neq \emptyset$. For this it suffices to show that $0 \in p_Y(\gamma(E))$. Consider the continuous homotopy

$$h(t, y) = tp_Y(\gamma(y)) + (1 - t)y \text{ for all } t \in [0, 1], \text{ all } y \in Y.$$

Then, we have

$$h(1, \cdot) = p_Y \circ \gamma \text{ and } h(t, \cdot)|_{E_0} = id|_{E_0} \text{ for all } t \in [0, 1].$$

So, the homotopy invariance property of the Brouwer degree implies

$$\begin{aligned} &d_B(h(0, \cdot), B_\varrho \cap Y, 0) = d_B(h(1, \cdot), B_\varrho \cap Y, 0), \\ &\Rightarrow d_B(h(1, \cdot), B_\varrho \cap Y, 0) = 1, \\ &\Rightarrow 0 \in p_Y(\gamma(E)) (\text{ by the solution property}). \end{aligned}$$

Therefore $\gamma(E) \cap D \neq \emptyset$ and we have linking of the sets.

(c) Let X be a Banach space such that

$$X = Y \oplus V \text{ with } \dim Y < \infty,$$

$v_0 \in V$ with $\|v_0\|_X = 1$ and $0 < \varrho < r_1$, $0 < r_2$. We set
$E_0 = \{y + \lambda v_0 : y \in Y \text{ and } \|y\|_X = r_2, 0 < \lambda < r_2 \text{ or } \|y\|_X \leq r_2,\ \lambda \in \{0, r_1\}\}$,
$E = \{y + \lambda v_0 : y \in Y, \|y\|_X \leq r_2, 0 \leq \lambda \leq r_1\}$,
$D = \partial B_\varrho \cap V$.
We see that E is a cylinder with height r_1 and bottom basis $\partial B_{r_2} \cap Y$, while E_0 is the surface of the cylinder (the lateral surface and the bottom and top bases). Then the pair $\{E_0, E\}$ is linking with D. Let $\gamma \in C(E, X)$ such that $\gamma|_{E_0} = id_{E_0}$. We have to show that there exists $u \in E$ such that $\|\gamma(u)\|_X = \varrho$, $p_Y(\gamma(u)) = 0$. To this end consider the homotopy

$$h(t,u) = tp_Y(\gamma(u)) + (1-t)y + (t\|\gamma(u) - p_Y(\gamma(u))\|_X + (1-t)\lambda - r)v_0$$

for all $t \in [0,1]$, all $u = y + \lambda v_0 \in E$.
We see that $0 \notin h([0,1] \times E_0)$. Then the homotopy invariance property of the Brouwer degree implies that

$$d_B(h(0,\cdot), intE, 0) = d_B(h(1,\cdot), intE, 0).$$

Note that $h(0,u) = y + (\lambda - r)v_0$. So, we have

$$d_B(h(0,\cdot), intE, 0) = d(id, intE, rv_0) = 1$$

$$\Rightarrow d_B(h(1,\cdot), intE, 0) = 1$$

So, we can find $u = y + \lambda v_0$ such that

$$h(1,u) = 0,$$

$$\Rightarrow p_Y(\gamma(u)) = 0 \text{ and } \|\gamma(u)\|_X = \|\gamma(u) - p_Y(\gamma(u))\|_X = r.$$

(d) Let X be a Banach space such that

$$X = Y \oplus V \text{ with } \dim Y < \infty.$$

$v_0 \in V$, $\|v_0\| = 1$ and $0 < r < \varrho$. We set

$$E_0 = \{u = y + \lambda v_0 : y \in Y, (\lambda \geq 0, \|u\|_X = \varrho) \text{ or } (\lambda = 0, \|u\|_X \leq \varrho)\}$$
$$E = \{u = y + \lambda v_0 : y \in Y, \lambda \geq 0, \|u\|_X \leq \varrho\},$$
$$D = \partial B_r \cap V.$$

Evidently E is the upper half ϱ-ball, E_0 is the north hemisphere plus the equator disc and D is the r-sphere in V. Reasoning as in Example (c), we show that the pair $\{E, E_0\}$ links with D.

Using the notion of linking sets, we can prove the following general minimax principle.

To do this we will need the following variant of Theorem 3.20 which is proved in a similar way (see Gasinski and Papageorgiou [182], p.627)

Theorem 3.39 *If X is a Banach space, $\widehat{\varphi} \in C^1(X)$, $\widehat{c} \in \mathbb{R}$, $\widehat{\varphi}$ satisfies the $C_{\widehat{c}}$-condition, A, $C \subseteq X$ are disjoint closed sets with $A \cap K_{\widehat{\varphi}}^{\widehat{c}} = \emptyset$ and $\sup_A \widehat{\varphi} \leq c \leq \inf_C \widehat{\varphi}$, then there exist $\varepsilon > 0$ and a $\widehat{\varphi}$-decreasing, locally Lipschitz, one-parameter family of homeomorphisms $\{h(t,\cdot)\}_{t\in[0,1]}$ such that*

$$h(1, A) \subseteq \widehat{\varphi}^{\widehat{c}-\varepsilon}, h(t, u) = u$$

$$for\ all\ (t, u) \in [0, 1] \times (C \cup (X \setminus \widehat{\varphi}^{-1}([c - 2\varepsilon, c + 2\varepsilon])))$$

Now we can have the general minimax principle.

Theorem 3.40 *If X is a Banach space, $\{E, E_0, 0\}$ are closed linking sets $\gamma^*(E) \subseteq X$ is closed, $\Gamma = \{\gamma \in C(E, X) : \gamma|_{E_0} = \gamma^*\}$, $\varphi \in C^1(X)$,*

$$a = \sup_{\gamma^*(E_0)} \varphi \leq \inf_D \varphi = b \text{ and } c = \inf_{\gamma] \in \Gamma} \sup_{u \in E} \varphi(\gamma(u))$$

and $\varphi(\cdot)$ satisfies the C_c-condition, then $b \leq c$ and c is a critical value of φ (that is, there exists $u \in K_\varphi$ such that $c = \varphi(u)$); moreover, if $c = b$, then $D \cap K_\varphi^c \neq \emptyset$.

Proof If $\gamma \in \Gamma$, then $\gamma(E) \cap D \neq \emptyset$ and so we see that $b \leq c$.

First we assume that $b < c$ and let $\varepsilon_0 = c - b > 0$. Arguing by contradiction, we assume that $K_\varphi^c = \emptyset$. Using Theorem 3.20 with $\varepsilon_0 > 0$, $U = \emptyset$, we see that there exists a deformation $h : [0, 1] \times X \to X$ satisfying the statements of the theorem. Let $\gamma \in \Gamma$ such that $\varphi(\gamma(u)) \leq c + \varepsilon$ for all $u \in E$. We set $\vartheta = h(1, \cdot) \circ \gamma \in C(E, X)$. For $u \in E_0$, we have

$$\begin{aligned}
&\vartheta(u) = h(1, \gamma(u)) = h(1, \gamma^*(u)),\\
&\Rightarrow \varphi(\vartheta(u)) = \varphi(h(1, \gamma^*(u))) \leq \varphi(\gamma^*(u)) \leq a \leq c - \varepsilon_0,\\
&\Rightarrow \vartheta|_{E_0} = \gamma^*(u) \text{ and so } \vartheta \in \Gamma.
\end{aligned}$$

For $\varepsilon \in (0, \varepsilon_0)$ we have

$$\varphi(\vartheta(u)) \leq c - \varepsilon$$

which contradicts the definition of c. This proves that $K_\varphi^c \neq \emptyset$ and c is a critical value of φ.

Now assume that $c = b$. Again we argue indirectly. So, suppose that $D \cap K_\varphi^c = \emptyset$. We apply Theorem 3.39 with

$$\widehat{\varphi} = -\varphi, \widehat{c} = -c, A = D \text{ and } C = \gamma^*(E_0)$$

Then we can find a one parameter family of homeomorphisms $\{h(t, \cdot)\}_{t \in [0,1]}$ which satisfy the conclusions of Theorem 3.39. As before we choose $\gamma \in \Gamma$ such that $\varphi(\gamma(u)) < c + \varepsilon$ for all $u \in E$. Let $\vartheta_1 = h(1, \cdot)^{-1} \circ \gamma \in C(E, X)$. For $u \in E_0$, we have

$$\begin{aligned}
&\vartheta_1(u) = h(1, \cdot)^{-1}(\gamma(u)) = h(1, \cdot)^{-1}(\gamma^*(u)) = \gamma^*(u),\\
&\Rightarrow \vartheta_1 \in \Gamma.
\end{aligned}$$

Hence we have $\vartheta_1(E) \cap D \neq \emptyset$. We can find $u_0 \in E$ such that $\vartheta_1(u_0) \in D$ and $\varphi(\gamma(u_0)) = \varphi(h(1, \vartheta_1(u_0))) \geq c + \varepsilon$, which contradicts the choice of $\gamma \in \Gamma$. This proves that $D \cap K_\varphi^c \neq \emptyset$. □

We apply this general principle to the linking families of the Example 3.38. The family $\{E, E_0, D\}$ from Example 3.38 (a) gives the so-called "Mountain Pass Theorem"

Theorem 3.41 *If X is a Banach space, $\varphi \in C^1(X)$, $u_0, u_1 \in X$ with*

$$\|u_1 - u_0\| > \varrho > 0,$$

$$\max\{\varphi(u_0), \varphi(u_1)\} \leq \inf\{\varphi(u) : \|u - u_0\| = \varrho\} = m,$$

$$\Gamma = \{\gamma \in C([0, 1], X) : \gamma(0) = u_0, \gamma(1) = u_1\},$$

$$c = \inf_{\gamma \in \Gamma} \max_{0 \leq t \leq 1} \varphi(\gamma(t)),$$

and φ satisfies the $C_c - condition$, then $m \leq c$ and $K_\varphi^c \neq \emptyset$; moreover, if $c = m$, then we have

$$\partial B_\varphi(u_0) \cap K_\varphi^c \neq \emptyset.$$

Often in applications $u_0 = 0$ is a local minimizer of $\varphi(\cdot)$ and there exists $u_1 \neq 0$ satisfying $\varphi(u_1) \leq \varphi(0)$. Then the geometry of Theorem 3.41 is satisfied. The simple Fig. 3.2 illustrates this situation.

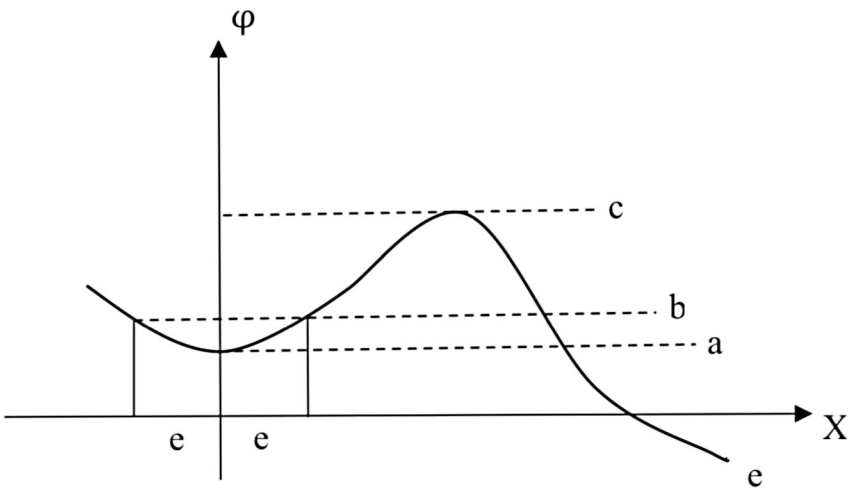


Fig. 3.2 Mountain Pass Theorem

If we use the linking family from Example 3.38 (b), we have the so-called "Saddle Point Theorem".

Theorem 3.42 *If X is a Banach space such that*

$$X = Y \oplus V \text{ with } \dim Y < \infty.$$

$\varphi \in C^1(X)$, there exists $\varrho > 0$ such that

$$\sup_{\partial B_\varrho \cap Y} \varphi \leq \inf_V \varphi = b,$$

$$\Gamma = \{\gamma \in C(\overline{B}_\varrho \cap Y, X) : \gamma|_{\partial B_\varrho \cap Y} = \mathrm{id}_{\partial B_\varrho \cap Y}\}$$

$$c = \inf_{\gamma \in \Gamma} \max_{u \in \overline{B}_\varrho \cap Y} \varphi(\gamma(u))$$

and φ satisfies the C_c-condition, then $b \leq c$ and $K_\varphi^c \neq \emptyset$; moreover, if $c = b$, then

$$V \cap K_\varphi^c \neq \emptyset.$$

If we use the linking family from Example 3.38 (c), we have the so-called "Generalized Mountain Pass Theorem".

Theorem 3.43 *If X is a Banach space such that*

$$X = Y \oplus V \text{ with } \dim Y < \infty,$$

$\{E, E_0, D\}$ is the linking family from Example 3.38 (c),

$$\sup_{E_0} \varphi \leq \inf_D \varphi = b,$$

$$\Gamma = \{\gamma \in C(E, X) : \gamma|_{E_0} = id_{E_0}\}, c = \inf_{\gamma \in \Gamma} \max_{u \in E} \varphi(\gamma(u))$$

and φ satisfies the C_c-condition, then $b \leq c$ and $K_\varphi^c \neq \emptyset$; moreover, if $c = b$, then $D \cap K_\varphi^c \neq \emptyset$.

Next we will present another general principle due to Ghoussoub and Preiss [202], which contains as a special case the mountain pass theorem (see Theorem 3.41). To do this, we change the metric on X and use the "geodesic metric".

Definition 3.44 Let X be a Banach space.

(a) If $\gamma \in C^1([0, 1], X)$, then the "geodesic length" $l(\gamma)$ of $\gamma(\cdot)$ is defined by

$$l(\gamma) = \int_0^1 \frac{\|\gamma'(t)\|_X}{1 + \|\gamma(t)\|_X} dt.$$

(b) The "geodesic distance" between $u_0, u_1 \in X$ is defined by

$$\vartheta(u_0, u_1) = \inf\{l(\gamma) : \gamma \in C([0, 1], X), \gamma(0) = u_0, \gamma(1) = u_1\}.$$

Remark 3.45 If $\gamma(t) = (1-t)u_0 + tu_1$, then we see that $\vartheta(u_0, u_1) \leq \|u_1 - u_0\|_X$. On the other hand, if $C \subseteq X$ is bounded, then we can find $\beta_C > 0$ such that $\vartheta(u_0, u_1) \geq \beta_C \|u_0 - u_1\|_X$. Hence on bounded sets, the two metrics are equivalent. If $u_0 = 0$, then the infimum in the definition of $\vartheta(\cdot, \cdot)$ is realized on the line segment from 0 to u_1 and so

$$\vartheta(0, u_1) = \int_0^1 \frac{\|u_1\|_X}{1 + t\|u_1\|_X} dt = \ln(1 + \|u_1\|_X).$$

Then ϑ-bounded and norm bounded sets in X coincide. If $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ is ϑ-Cauchy, then it is norm Cauchy too and so it converges in both the ϑ-metric and in the norm. We conclude that (X, ϑ) is a complete metric space.

Definition 3.46 Let X be a Banach space, $u_0, u_1 \in X$ and $C \subseteq X$ nonempty, closed. We say that C separates the two points u_0, u_1, if they belong in different connected components of $X \setminus C$.

The results of Ghoussoub and Preiss [202] asserts the following: (for a proof, we refer to Ekeland [149, p. 140]).

Theorem 3.47 *If X is a Banach space, $\varphi : X \to \mathbb{R}$ is continuous and Gateaux differentiable, $\varphi'_G : X \to X^*$ is norm-to-w*-continuous, $u_0, u_1 \in X$*

$$\Gamma = \{\gamma \in C([0, 1], X) : \gamma(0) = u_0, \gamma(1) = u_1\},$$

$$c = \inf_{\gamma\in\Gamma} \max_{0\leq t\leq 1} \varphi(\gamma(t))$$

and there exists $C \subseteq X$ nonempty closed such that the set $\{u \in C : \varphi(u) \geq c\}$ separates u_0, u_1, then we can find a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\vartheta(u_n, C) \to 0, \varphi(u_n) \to c,$$

$$(1 + \|u_n\|_X)\varphi'(u_n) \to 0 \text{ in } X^*$$

Remark 3.48 Consider the situation of the mountain pass theorem (see Theorem 3.41). Suppose that $c > \inf_{\partial B_\varrho(u_0)} \varphi$. Then taking $C = X$, we see that $\{u \in X : \varphi(u) \geq c\}$ separates u_0, u_1 and since φ satisfies the C_c-condition, from

Theorem 3.47 we can find $u \in K_\varphi^c$. If $c = \inf_{\partial B_\varrho(u_0)} \varphi$, then taking $C = \partial B_\varrho(v_0)$, we see that $\{u \in C : \varphi(u) \geq c\}$ separates u_0, u_1 and since φ satisfies the C_c-condition, from Theorem 3.47, we can find $u \in K_\varphi^c$.

Using Theorem 3.47, we can have the following generalization of the mountain pass theorem.

Corollary 3.49 *If X is a Banach space, $\varphi : X \to \mathbb{R}$ is continuous and Gateaux differentiable, $\varphi'_G : X \to X^*$ is norm-to-w^* continuous, $u_0, u_1 \in X$*

$$\Gamma = \{\gamma \in C([0,1], X) : \gamma(0) = u_0, \gamma(1) = u_1\}$$

$$c = \inf_{\gamma \in \Gamma} \max_{0 \leq t \leq 1} \varphi(\gamma(t))$$

$$\max\{\varphi(u_0), \varphi(u_1)\} < c$$

φ satisfies the C_c-condition and $C \subseteq X$ closed separates $u_0, u_1 \in X$, then $K_\varphi^c \neq \emptyset$; moreover, if

$$c = \max\{\varphi(u_0), \varphi(u_1)\},$$

then $C \cap K_\varphi^c \neq \emptyset$.

3.4 Multiple Critical Points

In this section we present results which guarantee the existence of multiple nontrivial critical points.

We start with those results that do not require symmetry conditions on the functional $\varphi(\cdot)$.

Definition 3.50 Let X be a Banach space, $X = Y \oplus V$ and $\varphi \in C^1(X)$. We say that $\varphi(\cdot)$ has "local linking at 0" (with respect to the (Y, V)-decomposition), if we can find $\varrho > 0$ such that

$$\varphi(u) \leq 0 \text{ if } u \in Y, \|u\|_X \leq \varrho$$
$$\varphi(u) \geq 0 \text{ if } u \in V, \|u\|_X \leq \varrho.$$

Remark 3.51 Note that $0 \in K_\varphi$. If $X = H =$ *a* Hilbert space, $\varphi \in C^2(H)$ and $\varphi''(0) \in \mathcal{L}(H)$ is an isomorphism, then $\varphi(\cdot)$ has a local linking at 0. We can easily see this by considering the negative spectrum of $\varphi''(0)$.

Lemma 3.52 *If X is a Banach space, $\varphi \in C^1(X)$, φ satisfies the PS-condition, $u_0 \in X$ is a unique global minimizer of φ, $v \in X$ satisfies $\varphi'(v) \neq 0$ and*

$$K_\varphi \cap \varphi^{-1}((\varphi(u_0), \varphi(v)))$$

then if $\vartheta : X \setminus K_\varphi \to X$ is a pseudogradient vector field of φ, the flow of the abstract Cauchy problem

$$u'(t) = -\frac{\vartheta(u(t))}{\|\vartheta(u(t))\|_X^2}, u(0) = v$$

exists on a maximal finite interval $[0, \tau_+(v))$, $t \to \varphi(u(t))$ is decreasing on this interval and $\lim_{t \to \tau_+(v)^-} u(t) = u_0$.

Proof By replacing φ by $\widehat{\varphi}(u) = \varphi(u_0 + u) - \varphi(u_0)$ if necessary, we see that we can assume that $u_0 = 0$ and $\varphi(u_0) = 0$. The Cauchy problem has a unique solution $u(\cdot)$ on a maximal interval $[0, \tau_+(v))$. Using the chain rule and the definition of a pseudogradient vector field we have

$$\begin{aligned}
\frac{d\varphi(u(t))}{dt} &= \langle \varphi'(u(t)), u'(t) \rangle \\
&= \langle \varphi'(u(t)), -\frac{\vartheta(u(t))}{\|\vartheta(u(t))\|_X^2} \rangle \\
&\leq -\frac{\|\varphi'(u(t))\|_{X^*}^2}{\|\vartheta(u(t))\|_X^2} \leq -\frac{1}{4} \text{ for all } t \in [0, \tau_+(v)).
\end{aligned}$$

$$\Rightarrow t \to \varphi(u(t)) \text{ is decreasing on } [0, \tau_+(v)).$$

Since by hypothesis $u_0 = 0$ is the unique global minimizer of $\varphi(\cdot)$, we have

$$0 = \varphi(0) < \varphi(u(t)) \leq -\frac{1}{4}t + \varphi(v) < \varphi(v) \text{ for all } t \in (0, \tau_+(v))$$

$$\Rightarrow \tau_+(v) \leq 4\varphi(v) < \infty$$

Hence $[0, \tau_+(v))$ is a finite interval.
First suppose that

$$\|\varphi'(u(t))\|_{X^*} \geq \beta_0 > 0 \text{ for all } t \in (0, \tau_+(v))$$

$$\Rightarrow \|\vartheta(u(t))\|_{X^8} \geq \beta_0 > 0 \text{ for all } t \in (0, \tau_+(v)) \text{ (see Remark 3.12)}$$

So, we have

$$\begin{aligned}
+\infty = \int_0^{\tau_+(v)} \|u'(t)\|_X dt &= \int_0^{\tau_+(v)} \frac{1}{\|\vartheta(u(t))\|_X} dt \\
&\leq \frac{\tau_+(v)}{\beta_0} < \infty
\end{aligned}$$

a contradiction.

This means that $t \to \varphi'(u(t))$ is not bounded away from zero and we can find $\{t_n\}_{n\in\mathbb{N}} \subseteq (0, \tau_+(v))$ such that

$$t_n \to \tau_+(v) \text{ and } \varphi'(u(t_n)) \to 0 \text{ in } X^*.$$

Also $\{\varphi(u(t_n))\}_{n\in\mathbb{N}} \subseteq \mathbb{R}$ is bounded. Since φ satisfies the PS-condition, we may assume that

$$u(t_n) \to y \in K_\varphi$$

Since by hypothesis $K_\varphi \cap (0, \varphi(v)) = \emptyset$, we infer that $y = 0$. Therefore recalling that $t \to \varphi(u(t))$ is decreasing, we have

$$\varphi(u(t)) \to 0 \text{ as } t \to \tau_+(v),$$

$$\Rightarrow u(t) \to 0 \text{ as } t \to \tau_+(v) \text{ (by the Ekeland variational principle).}$$

□

Using this lemma, we can have the following multiple critical point theorem, known in the literature as the "Local Linking Theorem".

Theorem 3.53 *If X is a Banach, $X = Y \oplus V$ with $dimY < \infty$, $\varphi \in C^1(X)$, $\varphi(0) = 0$, φ is bounded below, $\inf_X \varphi < 0$, it has local linking at 0 and satisfies the PS-condition, then K_φ has at least two nontrivial elements.*

Proof From Theorem 3.10 and Proposition 3.16 we know that there exists $u_0 \in X$ such that

$$\varphi(u_0) = \inf_X \varphi < 0 = \varphi(0),$$

$$\Rightarrow u_0 \neq 0, u_0 \in K_\varphi.$$

We assume that $K_\varphi = \{0, u_0\}$ or otherwise we already have a second nontrivial critical point and so we are done.

Case 1 $0 < \dim Y, 0 < \dim V$.

If $\varrho > 0$ is as postulated by Definition 3.50, then we can always assume that $\varrho = 1 < \|u_0\|_X$. Then for $y \in Y \cap \partial B_\varrho$, we have $\varphi'(y) \neq 0$ and so we can apply Lemma 3.52 and have that the flow $u(\cdot)$ of the Cauchy problem exists on a maximal interval $[0, \tau_+(y))$ and $\tau_+(y) \leq -4\varphi(u_0)$. We claim that we can find $\delta > 0$ such that

$$\overset{o}{\varphi}^{\varphi(u_0)+\delta} \subseteq B_{\frac{\|u_0\|_X}{2}}(u_0), \tag{3.35}$$

where for every $\lambda \in \mathbb{R}$, $\overset{o}{\varphi}{}^{\lambda} = \{u \in X : \varphi(u) < \lambda\}$. To show that (3.35) holds we argue by contradiction. So, suppose we can find a minimizing sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\|u_n - u_0\|_X \geq \frac{\|u_0\|_X}{2} \quad \text{for all } n \in \mathbb{N}.$$

From Corollary I-8.104, we have a minimizing sequence $\{v_n\}_{n\in\mathbb{N}} \subseteq X$ such that

$$\varphi'(v_n) \to 0 \text{ in } X^* \quad \text{and} \quad \|v_n - u_0\|_X \geq \frac{\|u_0\|_X}{4} \text{ for all } n \in \mathbb{N}.$$

The PS-condition implies that at least for a subsequence we have $v \to \widehat{v}$ in X. Then

$$\varphi'(\widehat{v}) = 0,\ \varphi(\widehat{v}) = \inf_X \varphi < 0 = \varphi(0),\ \|\widehat{v} - u_0\|_X \geq \frac{\|u_0\|_X}{4},$$

$$\Rightarrow \widehat{v} \in K_\varphi = \{0, u_0\} \text{ a contradiction. This proves (3.35).}$$

So, let $\delta > 0$ be small. Then we can find unique $t(y) \in (0, \tau_+(y))$ such that

$$\varphi(u(t(y))) = \varphi(u_0) + \delta \leq 0.$$

Moreover, on account of the uniqueness of $t(y)$, we see that $y \to t(y)$ is continuous. Let $v_0 \in V \cap \partial B_1$ and consider the set

$$E = \{u = y + \lambda v_0 : y \in Y,\ \lambda \geq 0,\ \|u\| \leq 1\}.$$

Let $\gamma_0 \in C(\partial E, X)$ be defined by

$$\gamma_0(y) = y \quad \text{for all } y \in Y,\ \|y\|_X \leq 1,\ \gamma_0(v_0) = u_0.$$

If $u \in \partial E,\ u \neq v_0$, then $\|u\|_X = 1$ and we can find unique $\mu \in (0, 1]$, $\lambda \in [0, 1]$ and $y \in Y$ such that $u = \mu y + \lambda v_0$. We set

$$\gamma_0(\mu y + \lambda v_0) = u(2\lambda t(y)),\ \lambda \in \left[0, \frac{1}{2}\right],$$

$$\Rightarrow \gamma_0(\mu y + \frac{1}{2} v_0) = u(t(y)),\ \|\mu y + \frac{1}{2} v_0 - v_0\|_X \leq \frac{\|u_0\|_X}{\delta}.$$

Finally if $\lambda \in \left(\frac{1}{2}, 1\right]$, then

$$\gamma_0(\mu y + \lambda v_0) = (2\lambda - 1)u_0 + (2 - 2\lambda)u(t(y)).$$

We see that as λ moves in $\left[\frac{1}{2}, 1\right]$, then $\gamma(\cdot)$ spans the segment from $u(t(y))$ to u_0 and so

$$\|\gamma_0(\mu y + \lambda v_0) - u_0\|_X \leq \frac{1}{2}\|u_0\|_X, \quad \lambda \in \left[\frac{1}{2}, 1\right].$$

So, the map γ_0 is continuous and $\varphi|_{\gamma_0(\partial E)} \leq 0$. Also $\|\gamma_0(\partial B_1)\|_X \geq \beta > 0$ for some $\beta \in [0, 1]$. Then the triple

$$E, \ E_0 = \partial E \quad \text{and} \quad D = V \cap \partial B_1$$

a linking family. We define

$$\Gamma = \{\gamma \in C(E, X) : \gamma|_{E_0} = \gamma_0\} \quad \text{and} \quad c = \inf_{\gamma \in \Gamma} \sup_{u \in E} \varphi(\gamma(u)).$$

We see that $c \geq 0$ and by Theorem 3.40 we have

$$K_\varphi^c \neq \emptyset.$$

If $c > 0$, then we have the second nontrivial critical point of $\varphi(\cdot)$ and so we are done. If $c = 0$, then by Theorem 3.40, we have $K_\varphi^c \cap D \neq \emptyset$ and so again we have a second nontrivial critical point.

Case 2 dim Y=0.

We know that there exists a global minimizer $u_0 \neq 0$ of $\varphi(\cdot)$. Since $\varphi(\cdot)$ has a local linking at 0 and $Y = \{0\}$, then from Definition 3.50, we see that $u = 0$ is a local minimizer of φ. Let $r \in (0, 1)$ small so that

$$\varphi(u_0) < \varphi(0) = 0 < m \leq \varphi(u) \quad \text{for all } \|u\|_X = r.$$

Since by hypothesis φ satisfies the PS-condition we can apply the mountain pass theorem (see Theorem 3.41) and have the second nontrivial critical point of φ.

Case 3 dim V=0 (in this case we can have dim Y=∞).

The local linking hypothesis implies that for $\varrho \in (0, 1)$ small $\varphi(u) \leq -\beta < 0$ for all $\|u\|_X = \varrho$. Apply the mountain pass theorem to $-\varphi$, to conclude that there is a second nontrivial critical point of φ. □

The next multiplicity result, will expand on the observation of Remark 3.51. First a definition which is also important in the study of critical groups (see Sect. 3.5).

Definition 3.54 Let H be a Hilbert space, $\varphi \in C^2(H)$ and $u \in K_\varphi$.

(a) We say that u is "nondegeneiate", if $\varphi''(u) \in \mathcal{L}(H)$ is invertible;
(b) The "Morse index" is the supremum of the dimensions of all subspaces of H on which $\varphi''(u)$ is negative definite;

(c) The "nullity" of u is the dim ker $\varphi''(u)$.

The next proposition is known in the literature as the "Morse Lemma" and its can be found Papageorgiou et al. [397, p. 413].

Proposition 3.55 *If H is a Hilbert space, $\varphi \in C^2(H)$, $u = 0$ is a nondegenerate critical point of φ, then there exists a Lipschitz homeomorphism $h(\cdot)$ of a neighborhood W of 0 into a neighborhood U of 0 such that*

$$h(0) = 0,\ \varphi(h(u)) = \varphi(0) + \frac{1}{2}(\varphi''(0)u, u)_H \quad \textit{for all } u \in W.$$

Remark 3.56 Let H_+ (resp. H_-) be the subspace of H for which we have $\varphi''(0)|_{H_+} > 0$ (resp. $\varphi''(0)|_{H_-} < 0$). We have

$$H = H_+ \oplus H_-.$$

Consider the orthogonal projection $\varrho_+ \in \mathcal{L}(H)$ onto H_+. Also on H consider the equivalent norm

$$|u|_H = |(\varphi''(0)u, u)_H|^{\frac{1}{2}} \quad \text{for all } u \in H.$$

Then we can equivalently rewrite the conclusion of Proposition 3.55 as follows

$$\varphi(h(u)) = \varphi(0) + \frac{1}{2}|\varrho_+(u)|_H^2 - \frac{1}{2}|(id - \varrho_+)(u)|_H^2.$$

This equivalent formulation of the conclusion of the Morse lemma, is more convenient, because we can see clearly that we have local linking at 0 and so we can apply Theorem 3.53 and have the following result.

Theorem 3.57 *If H is a Hilbert space, $\varphi \in C^2(H)$ is bounded below, if satisfies the PS-condition, $0 \in K_\varphi$ is nondegenerate with finite Morse index and*

$$\inf_X \varphi < 0 = \varphi(0),$$

then $\varphi(\cdot)$ has at least two nontrivial critical points.

In the local Linking theorem (see Theorem 3.53), we assumed that φ is bounded below. What can be said if $\varphi(\cdot)$ is unbounded below? This problem was investigated by Li and Willem [295].

Suppose $X = Y \oplus V$ and consider two increasing sequences of subspaces

$$\{Y_n\}_{n\in\mathbb{N}} \text{ of } Y \quad \text{and} \quad \{V_n\}_{n\in\mathbb{N}} \text{ of } V$$

such that

$$Y = \overline{\bigcup_{n\in\mathbb{N}} Y_n}, \quad V = \overline{\bigcup_{n\in\mathbb{N}} V_n}.$$

Let $\mu = (i, j) \in \mathbb{N} \times \mathbb{N} = \mathbb{N}^2$ and set

$$X_\mu = Y_i \times V_j.$$

We say that $\{\mu_n\}_{n\in\mathbb{N}} \subseteq \mathbb{N}^2$ is admissible, if

$$\mu_n = (i_n, j_n) \quad \text{with } i_n, j_n \to \infty, \quad \text{as } n \to \infty.$$

Definition 3.58 Let X be a Banach space and $\varphi \in C^1(X)$. We say that φ satisfies the (PS)*-condition, if every sequence $\{u_{\mu_n}\}_{n\in\mathbb{N}} \subseteq X$ with $\{\mu_n\}_{n\in\mathbb{N}} \subseteq \mathbb{N}^2$ admissible which satisfies

$$u_{\mu_n} \in X_{\mu_n},\ \{\varphi(u_{\mu_n})\}_{n\in\mathbb{N}} \subseteq \mathbb{R} \text{ is bounded},\ \varphi'(u_{\mu_n}) \to 0 \text{ in } X^*,$$

admits a strongly convergent subsequence.

Using the notion Li and Willem [295] (Theorem 2), proved the following existence theorem.

Theorem 3.59 *If X is a Banach space, $X = Y \oplus V$ with $V \neq 0$, $\varphi \in C^1(X)$ and*

(i) φ has a local linking at 0;
(ii) φ satisfies the (PS) condition;*
(iii) φ maps bounded sets to bounded sets;
(iv) for every $k \in \mathbb{N}$, $\varphi(u) \to -\infty$ as $\|u\|_X \to \infty$, $u \in Y \oplus V_k$,

then $\varphi(\cdot)$ has at least one nontrivial critical point.

More powerful multiplicity results can be proved if we impose symmetry conditions on the functional $\varphi(\cdot)$.

So, let X be a Banach and G a topological group.

Definition 3.60 A representation of G over X, is a family $\{L(g)\}_{g\in G} \subseteq \mathcal{L}(X)$ such that

(a) $L(e) = id$ (with e being the identity element of G).
(b) $L(g_1 g_2) = L(g_1)L(g_2)$ for all $g_1,\ g_2 \in G$.
(c) The map $(g, u) \to L(g)(u)$ from $G \times X$ into X is continuous.

The representation $\{L(g)\}_{g\in G}$ is "isometric" if

$$\|L(g)(u)\|_X = \|u\|_X \quad \text{for all } u \in X.$$

A set $C \subseteq X$ is "invariant" for the representation if

$$L(g)(C) \subseteq C \quad \text{for all } g \in G.$$

A functional $\varphi : X \to \mathbb{R}$ is "invariant" for the representation if

$$\varphi \circ L(g) = \varphi \quad \text{for all } g \in G.$$

A map $f : X \to X$ is "equivariant" for the representation if

$$L(g) \circ f = f \circ L(g) \quad \text{for all } g \in G.$$

The set of "fixed (or invariant) points" for the representation is the set

$$Fix(G) = \{u \in X : L(g)(u) = u \quad \text{for all } g \in G\}$$

Given $u \in X$, the "orbit" of u under the representation is

$$O(u) = \{L(g)(u) : g \in G\}.$$

Remark 3.61 In the sequel for notational simplicity we identify $L(g)$ with g and we speak about a "linear action" of G on X.

Proposition 3.62 *If G is a compact topological group with a linear action on $\mathbb{R}^N$, then we can find an invertible $N \times N$ matric S such that SgS^{-1} is an isometric representation of G on $\mathbb{R}^N$.*

Proof We know that a compact topological group admits a unique (up to scalar multiplication) G-invariant, finite regular Borel measure $\mu(\cdot)$, known as the "Haar measure". So, we can introduce the quadratic form

$$q(u) = \int_G |gu|^2 d\mu, \ u \in \mathbb{R}^N,$$

which is positive definite. Therefore we can find on $N \times N$ invertible matrix such that

$$q(u) = |Su|^2 \quad \text{for all } u \in \mathbb{R}^N.$$

Then SgS^{-1} is an isometric linear action of G on $\mathbb{R}^N$. □

Proposition 3.63 *If $\varphi \in C^1(X)$ is invariant, then*

(a) $\langle \varphi'(gu), y \rangle = \langle \varphi'(u), g^{-1}y \rangle$ for all $g \in G$, all $u, y \in X$;
(b) If the action is isometric, then

$$\|\varphi'(gu)\|_* = \|\varphi'(u)\|_*$$

for all $g \in G$, all $u \in X$.

Proof

(a) Using Definition 3.60, we can easily see that

$$L(g)^{-1} = L(g^{-1}) \quad \text{for all } g \in G.$$

We have

$$\begin{aligned}\langle \varphi'(gu), y\rangle &= \lim_{t\to 0} \frac{1}{l}\left[\varphi(gu + ty) - \varphi(gu)\right] \\ &= \lim_{t\to 0} \frac{1}{l}\left[\varphi(u + tg^{-1}y) - \varphi(u)\right] \\ &= \langle \varphi'(u), g^{-1}y\rangle \quad \text{for all } g \in G, \text{ all } u, y \in X.\end{aligned}$$

(b) We use (a) and take supremum of both sides over $\overline{B}_1$. Recalling that $\|y\|_X = \|g^{-1}y\|_X$ (since the action is isometric), we conclude that $\|\varphi'(gu)\|_* = \|\varphi'(u)\|_*$ for all $g \in G$, all $u \in X$.

□

Proposition 3.64 *If G is a compact topological group with a linear isometric action on the Banach space X and $\varphi \in C^1(X)$ is invariant, then we can find an equivariant pseudogradient vector field $\widehat{g} : X \setminus K_\varphi \to X$.*

Proof As before let $\mu(\cdot)$ be the Haar measure on G. We can always assume that $\mu(G) = 1$. From Theorem 3.14 we know that there is a pseudogradient vector field $\vartheta(\cdot)$ for φ. We set

$$\widehat{\vartheta}(u) = \int_G (g\vartheta g^{-1})(u)d\mu \quad \text{for all } u \in X \setminus K_\varphi.$$

For every $g' \in G$, we have

$$\begin{aligned}\widehat{\vartheta}(g'u) &= \int_G (g\vartheta g^{-1})(g'u)d\mu \\ &= \int_G g\vartheta(g^{-1}g'u)d\mu \\ &= \int_G g'(g')^{-1}g\vartheta(((g')^{-1}g)^{-1}u)d\mu \\ &= g'\int_G (g')^{-1}g\vartheta(((g')^{-1}g)^{-1}u)d\mu \\ &= g'\widehat{\vartheta}(u), \\ \Rightarrow \widehat{\vartheta}(\cdot) &\text{ is equivariant.}\end{aligned}$$

We verify that $\widehat{\vartheta}(\cdot)$ is a pseudogradient vector field (see Definition 3.11). We have

$$\begin{aligned}\|\widehat{\vartheta}(u)\|_X &\leq \int_G \|\vartheta(g^{-1}u)\|_X d\mu \text{ (recall the action is isometric)}\\ &\leq 2\int_G \|\varphi'(g^{-1}u)\|_* d\mu \text{ (see Definition 3.11)}\\ &= 2\int_G \|\varphi'(u)\|_* d\mu \text{ (see Proposition 3.63)}\\ &= 2\|\varphi'(u)\|_*.\end{aligned}$$

Also, we have

$$\begin{aligned}\langle \varphi'(u), \widehat{\vartheta}(u)\rangle &= \int_G \langle \varphi'(u), g\vartheta(g^{-1}u)\rangle d\mu\\ &= \int_G \langle \varphi'(g^{-1}u), \vartheta(g^{-1}u)\rangle d\mu \text{ (see Proposition 3.63)}\\ &\geq \int_G \|\varphi'(g^{-1}u)\|_*^2 d\mu \text{ (see Definition 3.11)}\\ &= \int_G \|\varphi'(u)\|_*^2 d\mu \text{ (see Proposition 3.63)}\\ &= \|\varphi'(u)\|_*^2.\end{aligned}$$

Finally we need to show that $\widehat{\vartheta}(\cdot)$ is locally Lipschitz. Let $u \in X \setminus K_\varphi$ and consider its orbit $O(u)$. On account of the compactness of G, $O(u) \subseteq X$ is compact. So, we can find $\delta > 0$ such that

$$\vartheta|_{O(u)_\delta} \text{ is k-Lipschitz } (k > 0),$$

where $O(u)_\delta = \{v \in X \setminus K_\varphi : d(v, O(u)) < \delta\}$. This set is invariant. So, for all $v,\ y \in \overline{B}_\delta(u) \cap (X \setminus K_\varphi)$ we have

$$\begin{aligned}\|\widehat{\vartheta}(v) - \widehat{\vartheta}(y)\|_X &\leq \int_G \|g(\vartheta(g^{-1}v) - \vartheta(g^{-1}y))\|_X d\mu\\ &= \int_G \|g(g^{-1}v) - \vartheta(g^{-1}y)\|_X d\mu\\ &\leq k\|v - y\|_X,\\ &\Rightarrow \widehat{\vartheta}(\cdot) \text{ is locally Lipschitz.}\end{aligned}$$

Therefore $\widehat{\vartheta}(\cdot)$ is an equivariant pseudogradient vector field for φ. □

This leads to the following version of the deformation theorem (see Theorem 3.20).

Theorem 3.65 *If G is a compact topological group with a linear isometric action on the Banach space X, $\varphi \in C^1(X)$ and φ satisfies the C_c-condition for some $c \in \mathbb{R}$, then for every $\varepsilon_0 > 0$, every U open invariant neighborhood of $K^c_{\varphi_-}$ (if $K^c_\varphi = \emptyset$, then $U = \emptyset$) and every $\lambda > 0$, we can find, we can find a deformation $h : [0,1] \times X \to X$ which satisfies (a)→(e) of Theorem 3.20 and in addition for all $t \in [0,1]$ $h(t,\cdot)$ is equivariant.*

Now we will develop the Ljusternik-Schnirelmann theory using the notion of index (or G-index). The idea behind that theory is to produce critical values of a given $\varphi \in C^1(X)$, as minimax values of φ over appropriate classes $\mathcal{S}_k$ of subsets of X. So, we consider

$$c_k = \inf_{C \in \mathcal{S}_k} \sup_{u \in C} \varphi(u), \quad k \in \mathbb{N},$$

where the classes $\mathcal{S}_k$ are defined using an index. Originally, Ljusternik-Schnirelmann used the notion of "LS-category". Here we will be using primarily the Krasnoselskii genus which corresponds to $G = \mathbb{Z}_2$. Alternatively, we could use $G = S^1$, in which case we have the "cohomological index" due to Fadell Rabinowitz [154].

So, let X be a Banach space G a compact topological group with a linear isometric action on X. We introduce the following family of subsets of X

$$\mathcal{S} = \{C \subseteq X : C \text{ is closed and G invariant}\}.$$

Definition 3.66 An "index" (or a "G-index") on X is a map $i : \mathcal{S}_k \to \mathbb{N} \cup \{+\infty\}$ such that

(a) $i(A) = 0 \Leftrightarrow A = \emptyset$.
(b) If $C,\ D \in \mathcal{S}$ and $h : C \to D$ is equivariant, then $i(C) \leq i(D)$ (monotonicity property).
(c) If $C,\ D \in \mathcal{S}$, then $i(C \cup D) \leq i(C) + i(D)$ (subadditivity property).
(d) If $C \in \mathcal{S}$ is compact, then there exists an invariant open neighborhood U of C such that $i(U) = i(C)$ (continuity property).

Example 3.67

(a) If $G = \mathbb{Z}_2 = \{id, -id\}$, then a set $C \subseteq X$ is G-invariant if and only if C is symmetric with respect to the origin. Then the index $i(\cdot)$ defined on $\mathcal{S} = \{C \subseteq X : \text{closed, symmetric}\}$, is the Krasnoselskii "genus" $\gamma(\cdot)$ introduced in Definition 1.46. Theorem 1.49 verifies that $\gamma(\cdot)$ is indeed an index.
(b) If $G = S^1$, then the index $i : \mathcal{S} \to \mathbb{N} \cup \{+\infty\}$ is defined by

$$i(C) = \inf\{n \in \mathbb{N} : \text{there exists } m \in \mathbb{N} \setminus \{0\} \text{ and } f \in C(C, \mathbb{R}^n \setminus \{0\})$$
$$\text{such that } f(gu) = e^{img} f(u) \text{ for all } u \in C,$$
$$\text{all } g \in S^1\}.$$

If no such $f(\cdot)$ exists, then we set $i(C) = \infty$. Also $i(\emptyset) = 0$. This index is known as the "cohomological index" and was introduced by Fadell and Rabinowitz [154] (see also Mawhin and Willem [335, p. 120]).

We will be using primarily the Krasnoselskii genus. Computation of the genus of some sets, can be found in Sect. 1.1. From those results we infer the following corollary.

Corollary 3.68 *If $C \subseteq X$ is a finite collection of symmetric vectors $\{\pm u_k\}_{k=1}^m$, then $\gamma(C) = 1$.*

Some more compulations of the genus of sets.

Proposition 3.69 *If $C \subseteq X$ is closed and $C \cap (-C) = \emptyset$, then $\gamma(C \cup (-C)) = 1$.*

Proof Consider the map $f : C \cup (-C) \to \mathbb{R} \setminus \{0\}$ defined by

$$f(u) = \begin{cases} -1, & \text{if } u \in -C \\ +1, & \text{if } u \in C. \end{cases}$$

This map is continuous, odd, hence $\gamma(C \cup (-C)) = 1$. □

Proposition 3.70 *If $C \subseteq \mathcal{S} = \{$closed, symmetric subsets of $X\}$ and there exists an odd homeomorphism $\xi : C \to S^{N-1} = \{u \in \mathbb{R}^N : |u| = 1\}$, then $\gamma(C) = N$.*

Proof Clearly $\gamma(C) \leq N$. Suppose $\gamma(C) = k < N$. Then by the definition of the genus, we can find $f : C \to \mathbb{R}^k \setminus \{0\}$ continuous, odd. We set

$$h = f \circ \xi^{-1} \in C(S^{N-1}, \mathbb{R}^k \setminus \{0\}).$$

This is an odd map. Since $k < N$, h is homotopic to a constant map $h_0 = u_0 \neq 0$, $u_0 \in \mathbb{R}^N \setminus \mathbb{R}^k$. We may assume $|u_0| > 1$. Then

$$d_B(h, B_1^N, 0) = 0 \quad \text{with} \quad B_1^N = \{u \in \mathbb{R}^N : |u| < 1\}.$$

which contradicts Borsuk's theorem (see Theorem 1.35). □

Proposition 3.71 *If $C \in \mathcal{S}$ with $0 \notin C$ and $2 \leq \gamma(C)$, then C is an infinite set.*

Proof If C is finite, then because $0 \notin C$, we have

$$C = D \cup (-D) \quad \text{with} \quad D \cap (-D) = \emptyset,$$
$$\Rightarrow \gamma(C) = 1,$$

by Corollary 3.68, a contradiction. So, C must be an infinite set. □

Now consider a G-index $i(\cdot)$ defined on

$$\mathcal{S} = \{C \subseteq X : \text{closed, G-invariant}\}.$$

For $k \in \mathbb{N}$ we define

$$\mathcal{S}_k = \{C \in \mathcal{S} : C \text{ is compact, } i(C) \geq k\}.$$

Suppose $\varphi \in C^1(X)$ and for $k \in \mathbb{N}$ define

$$c_k = \inf_{C \in \mathcal{S}_k} \sup_{u \in C} \varphi(u).$$

Clearly $-\infty \leq c_1 \leq c_2 \leq \ldots$. We have the following abstract multiplicity theorem.

Theorem 3.72 *if G is a compact topological group with a linear isometric action on the Banach space X, $\varphi \in C^1(X)$ is invariant, satisfies the C-condition and for some $k \in \mathbb{N}$, $c_k > -\infty$, then c_k is a critical value of φ and if $c_m = c_k = c > -\infty$ for some $m \geq k$, then $i(K_\varphi^c) \geq m - k + 1$.*

Proof Since φ satisfies the C-condition the set K_φ^c is compact. Also from Proposition 3.63 we have that K_φ^c is invariant. Then by Theorem 1.49 we can find U an open invariant neighborhood of K_φ^c such that

$$i(\overline{U}) = i(K_\varphi^c) \tag{3.36}$$

(note that Theorem 1.49 is stated for the genus, however it is true for any index). Then by Theorem 3.65 we can find a deformation $h : [0, 1] \times X \to X$ which satisfies conclusions (a)→(e) of that theorem. From (3.36) there exists $C \in \mathcal{S}_k$ such that

$$\max_C \varphi \leq c + \varepsilon.$$

Let $D = C \setminus U$. Then we have

$$\begin{aligned} m \leq i(C) &\leq i(D) + i(C \cap \overline{U}) \\ &\leq i(D) + i(\overline{U}) \\ &= i(D) + i(K_\varphi^c) \text{ (see Definition 3.66).} \end{aligned} \tag{3.37}$$

Note that $D \subseteq \varphi^{c+\varepsilon} \setminus U$ and so from Theorem 3.65, we have

$$E = h(i, D) \subseteq \varphi^{c-\varepsilon}.$$

Recall that the map $h(1,\cdot)$ is equivariant. So, E is compact and invariant and $\max_E \varphi \leq c - \varepsilon$. Therefore $i(E) \leq k-1$ and by Definition 3.66 (b) we have $i(D) \leq i(E) \leq k-1$. From (3.37) it follows that $m-k+1 \leq i(K_\varphi^c)$ and so $K_\varphi^c \neq \emptyset$. □

Now we pass to the special case of $G = \mathbb{Z}_2$ and so $\varphi \in C^1(X)$ which is invariant, must be even.

Theorem 3.73 *If $\varphi \in C^1(X)$ is even, bounded below, satisfies the C-condition and there exists a compact symmetric set C (hence $C \in \mathcal{S}$) such that $\gamma(C) = k$ and*

$$\sup_C \varphi < \varphi(0),$$

then $\varphi(\cdot)$ has at least k distinct pairs of critical points

$$\{\pm u_i\}_{i=1}^k \text{ with critical values } \varphi(-u_i) = \varphi(u_i) < \varphi(0).$$

Proof Let $\{c_i\}_{i\in\mathbb{N}}$ be as in (3.35) (in the present setting $\mathcal{S}_i = \{C \subseteq X : C \text{ is compact, symmetric, } \gamma(C) \geq i\}$). Since $\varphi(\cdot)$ is bounded below we have

$$-\infty < \inf_X \varphi \leq c_1 \leq c_2 \leq \ldots \leq c_k \leq \max_C \varphi < \varphi(0).$$

For each $i \in \{1,\ldots,k\}$, by Theorem 3.72 $K_\varphi^{c_i} \neq \emptyset$, it is symmetric and $0 \notin K_\varphi^{c_i}$. (thus $K_\varphi^{c_i}$ has at least two elements). If all the c_i's $(i = 1,\ldots,k)$ are distinct, then we have at least k distinct pairs of critical points with critical values less than $\varphi(0)$. If $c_i = c_j = c$ for some $1 \leq i < j \leq k$, then by Theorem 3.72 we have

$$\gamma(K_\varphi^c) \geq j - i + 1 \geq 2$$

and so Proposition 3.71 implies that K_φ^c is infinite. □

We can have a $\mathbb{Z}_2$-version (symmetric version) of the "Mountain Pass Theorem" (see Rabinowitz [424]).

Theorem 3.74 *If $\varphi \in C^1(X)$ is even and*

(i) there exists $V \subseteq X$ a subspace of finite codimension and $\beta, \varrho > 0$ such that

$$\beta \leq \inf\{\varphi(u) : u \in \partial B_\varrho \cap V\};$$

(ii) there exists $Y \subseteq X$ a finite dimensional subspace with codim $V <$ dim Y such that

$$\varphi|_Y \text{ is anticoercive}$$

$$(\text{that is } \varphi(y) \to -\infty \text{ as } \|y\| \to \infty,\ y \in Y),$$

then K_φ has at least dim Y − codim V distict pairs of critical points of φ.

There is also an "infinite" version of this result also due to Rabinowitz [424].

Theorem 3.75 *If X is an infinite dimensional Banach space $X = Y \oplus V$ with dim $Y < \infty$, $\varphi \in C^1(X)$ is even, satisfies the C-condition $\varphi(0) = 0$ and*

(i) there exist $\beta,\ \varrho > 0$ such that

$$\beta \leq \inf\{\varphi(u) : u \in \partial B_\varrho \cap V\};$$

(ii) for each finite dimensional subspace $E \subseteq X$ with $E \supseteq Y$, there exists $R = R(E) > 0$ such that

$$\varphi(u) \leq 0 \quad \text{for all } u \in E,\ \|u\| \geq R,$$

then $\varphi(\cdot)$ has an unbounded sequence of critical values.

In the same vein, we mention also a related result of Kajikiya [262].

Theorem 3.76 *If X is an infinite dimensional Banach space, $\varphi \in C^1(X)$ is even, bounded below, satisfies the C-condition, $\varphi(0) = 0$ and there exists $C \in \mathcal{S}_k$ such that*

$$\max_{u \in C} \varphi(u) < 0,$$

then one of the following two statements holds:

(a) there exists a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq K_\varphi$ such that

$$\varphi(u_n) < 0 \quad \text{for all } u \in \mathbb{N},\ u_n \to 0 \text{ in } X;$$

(b) there exists two sequences $\{u_n\}_{n\in\mathbb{N}},\ \{v_n\}_{n\in\mathbb{N}} \subseteq K_\varphi$ such that $\varphi(u_n) = 0$, $u_n \neq 0$ for all $n \in \mathbb{N}$, $u_n \to 0$ in X, $\varphi(v_n) < 0$, $\varphi(v_n) \to 0$ and $v_n \to v \neq 0$.

Remark 3.77 In any case the above theorem guarantees the existence of a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq K_\varphi$ with $\varphi(u_n) \leq 0$ for all $n \in \mathbb{N}$ and $u_n \to 0$ in X. If for every finite dimensional subspace E of X, we have $\inf_E \varphi < 0$, then the above theorem applies.

Another result that is useful when dealing with problems exhibiting symmetries, is the so-called "Symmetric Criticality Theorem".

Suppose H is a Hilbert space and G a topological group with a linear isometric action on H. Recall (see Definition 3.60) that $\mathrm{Fix}(G) = \{u \in H : gu = u \text{ for all } g \in G\}$.

Theorem 3.78 *If $\varphi \in C^1(H)$ is invariant and u_0 is a critical point of $\varphi|_{Fix(G)}$, then $u_0 \in K_\varphi$.*

Proof Let $(\cdot,\cdot)$ denote the inner product in H. Since the action of G isometric, it preserves inner products. So, we have

$$\begin{aligned}(\varphi'(gu),h) &= (\varphi'(u),g^{-1}h) \quad \text{for all } u,h\in H,\ g\in G\\ &= (g\varphi'(u),h),\\ \Rightarrow\ \varphi'(gu) &= g\varphi'(u),\ \text{that is, } \varphi'(\cdot) \text{ is equivariant.}\end{aligned}$$

Let $u_0\in \mathrm{Fix}(G)$. Then

$$\begin{aligned}&g\varphi'(u_0)=\varphi'(gu_0)=\varphi'(u_0),\\ \Rightarrow\ &\varphi'(u_0)\in \mathrm{Fix}(G).\end{aligned}$$

Suppose u_0 is a critical point of $\varphi|_{\mathrm{Fix}(G)}$. Then

$$\begin{aligned}&\varphi'(u_0)\in \mathrm{Fix}(G)^{\perp},\\ \Rightarrow\ &\varphi'(u_0)\in \mathrm{Fix}(G)\cap \mathrm{Fix}(G)^{\perp},\\ \Rightarrow\ &\varphi'(u_0)=0 \text{ and so } u_0\in K_\varphi.\end{aligned}$$

□

3.5 Critical Points Under Constraints

In applications often we have to find critical points of a functional $\varphi\in C^1(X)$ in the presence of constraints, that is, critical points of $\varphi|_M$ with $M\subseteq X$ representing a set of constraints. Usually

$$M=\{u\in X : g_k(u)=0,\ k=1,\dots,n\},\ g_k\in C^1(X).$$

To deal with such situations, we need to introduce some geometrical notions from the theory of differentiable manifolds.

Definition 3.79 Let X be a Banach space and $M\subseteq X$. For $m,n\in\mathbb{N}$, we say that M is a "C^m-submanifold of codimension n" if for every $u_0\in M$, we can find U an open neighborhood of u_0 and a function $\vartheta\in C^m(U,\mathbb{R}^m)$ such that

(a) $\vartheta'(u)$ is surjective for every $u\in U$;
(b) $M\cap U=\{u\in U : \vartheta(u)=0\}$.

Remark 3.80 Of special interest is the case when $n=1$ (submanifolds of codimension 1). In this case $M=\vartheta^{-1}(0)$ with $\vartheta\in C^1(X)$ and $\vartheta'(u)\neq 0$ for all $u\in M$. Such a situation arises in eigenvalue problems. Suppose X is a reflexive

Banach space with a locally uniformly convex dual (for example, a Hilbert space). Then the norm $\|\cdot\|$ of X is Frechet differentiable on $X\setminus\{0\}$ and if $\vartheta(u) = \|u\|^2 - \varrho^2$, then $\vartheta \in C^1(X)$ and

$$M = \vartheta^{-1}(0) = \partial B_\varrho = \{u \in X : \|u\| = \varrho\}$$

is a C^1-submonifold of codimension 1.

We need to define what we mean by a critical point of $\varphi|_M$. Again we appeal to Differential Geometry.

Definition 3.81 Let X be a Banach space, $M \subseteq X$ a C^m-manifold and $u_0 \in M$. We say that $h \in X$ is a tangent vector to M at u_0, if $h = \gamma'(0)$ for some C^1-path $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = u_0$ and $\gamma'(s) \neq 0$ for all $s \in (-\varepsilon, \varepsilon)$. The set of all tangent vectors, form the "tangent space to M at u_0", which we denote by $T_{u_0}M$.

Example 3.82 Let X be a Banach space, $\vartheta \in C^1(X)$ and define

$$M = \{u \in X : \vartheta(u) = 0\}.$$

We assume that $M \neq \emptyset$ and that $\vartheta'(u) \neq 0$ for all $u \in M$, we know that M is a C^m-submanifold of codimension 1. In this case for every $u \in M$ we have

$$T_u M = ker\vartheta'(u) = \{h \in X : \langle \vartheta'(u), h\rangle = 0\}.$$

The result is known as "Ljusternik's Theorem" (see Gasinski and Papageorgiou [182, p. 697] and Zeidler [475, p. 286]).

Definition 3.83 Let X be a Banach, $M \subseteq X$ a C^1-manifold and $\varphi \in C^1(X)$. We say that $u_0 \in M$ is a critical point of $\varphi|_M$ if and only if $\langle \varphi'(u_0), h\rangle = 0$ for all $h \in T_{u_0}M$.

Remark 3.84 According to this definition $u_0 \in M$ is a critical point of $\varphi|_M$ if and only if $\frac{d}{dt}\varphi(\gamma(t))|_{t=0} = 0$ for every C^1-path $\gamma(\cdot)$ as in Definition 3.79. Let $\vartheta \in C^1(X)$ and suppose

$$M = \{u \in X : \vartheta(u) = 0\},\ \vartheta'(u) \neq 0 \text{ for all } u \in M.$$

We know that this is a C^1-manifold of codimension 1. Suppose $\varphi \in C^1(X)$ and let $\varphi_M = \varphi|_M$. Then we have

$$\varphi'_M(u) = \varphi'(u) - \langle \varphi'(u), \xi(u)\rangle \vartheta'(u) \text{ for all } u \in M,$$

with $\xi : M \to X$ a continuous map such that

$$\langle \vartheta'(u), \xi(u)\rangle = 1 \text{ for all } u \in M.$$

This last example brings us to the doorsteps of the method of Lagrange multipliers. According to this method a critical point of a constrained functional is a critical point of a related unconstrained functional.

To prove the Lagrange multiplier rule, we will need to following factorization lemma.

Lemma 3.85 *If X, Y are Banach space, $A \in \mathcal{L}(X, Y)$ is surjective, $ker A = N(A)$ is complemented in X (see Definition I-3.76) and $u^* \in X^*$, then the following two statements are equivatent*

(a) $ker A \subseteq keru^*$;
(b) there exists $y^* \in Y^*$ such that $u^* = y^* \circ A$.

Proof (a)⇒(b): Let $y \in Y$. The surjectivity of A implies that $A^{-1}(y)$ is nonempty and closed. Consider $u_1, u_2 \in A^{-1}(y)$. Then $u_1 - u_2 \in ker A \subseteq keru^*$. Hence $\langle u^*, u_1\rangle = \langle u^*, u_2\rangle$. We denote by $y^*(y)$ the common value of u^* on $A^{-1}(y)$. Then $u^* = y^* \circ A$. We need to show that $y^* \in Y^*$.

First we check the linearity of y^*. If $y = A(u)$, then $y^*(y) = \langle u^*, u\rangle$. Hence for $\lambda \in \mathbb{R}$ we have

$$\begin{aligned} &\lambda y = \lambda A(u) = A(\lambda u), \\ &\Rightarrow y^*(\lambda y) = \langle u^*, \lambda u\rangle = \lambda\langle u^*, u\rangle = \lambda y^*(y). \end{aligned}$$

Next let $y_1 = A(u_1)$, $y_2 = A(u_2)$. We have

$$\begin{aligned} &y_1 + y_2 = A(u_1) + A(u_2) = A(u_1 + u_2), \\ &\Rightarrow y^*(y_1 + y_2) = \langle u^*, u_1 + u_2\rangle = \langle u^*, u_1\rangle + \langle u^*, u_2\rangle = y^*(y_1) + y^*(y_2). \end{aligned}$$

This proves the linearity of y^*.

Next we check the continuity of y^*. Let V be the topological complement of $ker A$. We have

$$V \subseteq Y \text{ is a closed subspace, } X = ker A \oplus V.$$

Then $A|_V$ is an isomorphism (see Corollary I-3.83). Also

$$\begin{aligned} &y^*(y) = 0 \Leftrightarrow y = A(u) \text{ with } u \in keru^*, \\ &\Rightarrow key^* = A(keru^*). \end{aligned}$$

Clearly

$$A|_V(V \cap keru^*) \subseteq A(keru^*). \tag{3.38}$$

Let $y \in A(keru^*)$. Then $y = A(u)$ with $u \in keru^*$. We can write $u = u_1 + u_2$ with $u_1 \in ker A$, $u_2 \in V$. By hypothesis $u_1 \in keru^*$ and so

$$\begin{aligned}
&\langle u^*, u\rangle = \langle u^*, u_2\rangle,\\
&\Rightarrow u_2 \in V \cap ker u^* \quad \text{and} \quad y = A(u) = A(u_2),\\
&\Rightarrow A(ker u^*) \subseteq A|_V(V \cap ker u^*),\\
&\Rightarrow A(ker u^*) = A|_V(V \cap ker u^*) \text{ (see (3.38))},\\
&\Rightarrow ker y^* \text{ is closed (recall } A|_V \text{ is isomorphism, } V \cap ker u^* \text{ is closed)},\\
&\Rightarrow y^* \in Y^*.
\end{aligned}$$

(b)$\Rightarrow$ (a): Obvious. □

Remark 3.86 The functional $y^* \in Y^*$ is unique. To see this suppose, $v^* \in Y^*$, $v^* \neq y^*$ such that $u^* = y^* \circ A = v^* \circ A$. We choose $y \in Y$ such that $\langle y^*, y\rangle \neq \langle v^*, y\rangle$. Let $u \in X$ such that $y = A(u)$ (recall A is surjective). Then

$$\langle u^*, u\rangle = \langle y^*, A(u)\rangle \neq \langle v^*, A(u)\rangle = \langle u^*, u\rangle,$$

a contradiction. Lemma 3.85 is the infinite dimensional counterpart of Lemma I-3.97, which was useful in the study of the weak topology.

Now we are ready for the Lagrange multiplier rule.

Theorem 3.87 *If X, Y are Banach spaces, $U \subseteq X$ nonempty open, $\varphi \in C^1(U)$, $g \in C^1(U, Y,)$, $M = g^{-1}(0)$, u_0 is a critical point of $\varphi|_M$, $g'(u_0) \in \mathcal{L}(X, Y)$ is surjective and $ker g'(u_0)$ is complemented in X, then there exists $y^* \in Y^*$ such that*

$$(\varphi - y^* \circ g)'(u_0) = 0$$

Proof If $ker g'(u_0) \subseteq ker \varphi'(u_0)$, then by Lemma 3.85 and Remark 3.86, we know that there exists unique $y^* \in Y^*$ such that

$$\varphi'(u_0) = y^* \circ g'(u_0) = (y^* \circ g)'(u_0) \text{ (chain rule)}.$$

Therefore we need to show that

$$ker g'(u_0) \subseteq ker \varphi'(u_0). \tag{3.39}$$

We know that $ker g'(u_0) \subseteq X$ is a closed subspace and by hypothesis it has a topological complement V. Let $\vartheta : ker g'(uo) \times V \to X$ be defined by

$$\vartheta(z, v) = z + v \text{ for all } z \in ker g'(u_0),\ v \in V.$$

Evidently $\vartheta \in \mathcal{L}(ker g'(u_0) \times V, X)$ and is a bijection. Therefore ϑ is an isomorphism (see Corollary I-3.83). Let $\widehat{U} = g^{-1}(U) \subseteq ker g'(u_0) \times V$ open and set $\widehat{g} = g \circ \vartheta : \widehat{U} \to Y$. We have

$$\widehat{g}'(z,v)(h,w) = g'(\vartheta(z,v)) \circ \vartheta'(z,v)(h,w) \text{ (by the chain rule)}, \tag{3.40}$$
$$= g'(z+v)(h+w).$$

Also note that, if by $\widehat{g}_i'(z,v)$ $i = 1,2$ we denote the partial derivatives with respect to the two variables of $\widehat{g}(\cdot,\cdot)$, then

$$\widehat{g}'(z,v)(h,w) = \widehat{g}_1'(z,v)(h) + \widehat{g}_2'(z,v)(w). \tag{3.41}$$

From (3.40) and (3.41) it follows that

$$\widehat{g}_1'(z,v) = \widehat{g}'(z+v)|_{kerg'(u_0)}, \quad \widehat{g}_2'(z,v) = \widehat{g}'(z+v)|_V$$

If $u_0 = \overline{u}_0 + \widehat{u}_0$ with $\overline{u}_0 \in kerg'(u_0), \widehat{u}_0 \in V$ then

$$\widehat{g}_1'(\overline{u}_0,\widehat{u}_0) = g'(u_0)|_{kerg'(u_0)}, \widehat{g}_2'(\overline{u}_0,\widehat{u}_0) = g'(u_0)|_V,$$
$$\Rightarrow \widehat{g}_1'(\overline{u}_0,\widehat{u}_0) = 0 \text{ and } \widehat{g}_2'(\overline{u}_0,\widehat{u}_0) \in \mathcal{L}(V,Y) \text{ is an isomorphism.}$$

We apply the Implicit Function Theorem. Then we can find open set $\widehat{U}' \subseteq \widehat{U}$ with $(\overline{u}_0,\widehat{u}_0) \in \widehat{U}'$, an open neighborhood W of u_0 and a C^1-map $\xi : W \to W$ such that

- $(z,v) \in \widehat{U}', \widehat{g}(z,v) = 0$;
- $z \in W$ and $v = \xi(z)$.

Let $\gamma : W \to \mathbb{R}$ be defined by $\gamma(z) = (\varphi \circ \vartheta)(z,\xi(z))$. Then $\overline{u}_0$ is a local minimizer of $\gamma(\cdot)$ and so

$$\gamma'(\overline{u}_0) = 0. \tag{3.42}$$

Let $h \in kerg'(u_0)$. Then

$$\langle \gamma'(\overline{u}_0), h\rangle = \varphi'(\vartheta(\overline{u}_0,\xi(\overline{u}_0))) \circ \vartheta'(\overline{u}_0,\xi(\overline{u}_0)) \circ (id|_{kerg'(u_0)}\xi'(\overline{u}_0))h,$$
$$= \langle \varphi'(u_0), \vartheta(h,\xi'(\overline{u}_0)h)\rangle,$$
$$= \varphi'(u_0)(h + \xi'(\overline{u}_0)h).$$

But for $z \in W$ we know that $\widehat{g}(z,\xi(z)) = 0$. Hence

$$\widehat{g}_1'(\overline{u}_0,\widehat{u}_0)h + \widehat{g}_2'(\overline{u}_0,\widehat{u}_0) \circ \xi'(\overline{u}_0)h = 0,$$
$$\Rightarrow \xi'(\overline{u}_0) = 0,$$
$$\Rightarrow \langle \gamma'(\overline{u}_0), h\rangle = \langle \varphi'(u_0), h\rangle \text{ for all } h \in kerg'(u_0) \text{ (see (43))},$$
$$\Rightarrow \langle \varphi'(u_0), h\rangle = 0 \text{ (see (42))}.$$

This shows that (3.39) is true and so the proof of the theorem is complete. □

Remark 3.88 If $X = H =$ a Hilbert space, then $ker g'(u_0)$ is always complemented in H. If $Y = \mathbb{R}^N$, then

$$M = \{u \in X : g_k(u) = 0, k = 1, \ldots, N\}$$

with $g_k \in C^1(X)$ and it is a C^1-manifold of codimension N. Then using Theorem 3.87 we can find $\{\lambda_k\}_{k=1}^N \subseteq \mathbb{R}$ such that

$$\varphi'(u_0) = \sum_{k=1}^{N} \lambda_k g_k'(u_0)$$

If $X = \mathbb{R}^m$, $Y = \mathbb{R}^N$, then

$$\nabla\varphi(u_0) = \sum_{k=1}^{N} \lambda_k \nabla g_k(u_0).$$

So, the gradient of φ at u_0, is a linear combination of the gradients of g_k's at u_0. The λ_k's are called the "Lagrange multipliers". Surjectivity of $g'(u_0) = \nabla g(u_0)$ is equivalent to the linear independence of the $\nabla g_k(u_0)$ $k = 1, \ldots, N$.

From Definition 3.83, we see that if $\varphi \in C^1(X)$, $M \subseteq X$ is a C^1-manifold and $u_0 \in K_\varphi$, then u_0 is also a critical point of $\varphi|_M$. The converse is not in general true. If this happens, we have a special kind of constraint.

Definition 3.89 Let X be a Banach space, $\varphi \in C^1(X)$, $M \subseteq X$ is a C^1-manifold. If $K_{\varphi|_M} = K_\varphi \cap M$, then we say that M is a "natural constraint for φ".

Remark 3.90 If $Y = \mathbb{R}^N$, then according to Remark 3.88 this means that all the Lagrange multipliers are equal to zero.

A useful natural constraint which we will encounter again in Chap. 5, is the so-called "Nehari manifold".

Definition 3.91 Let X be a Banach space and $\varphi \in C^1(X)$. The "Nehari manifold" of φ is the set

$$N = \{u \in X : \langle \varphi'(u), u\rangle = 0, u \neq 0\}.$$

Remark 3.92 Evidently $K_\varphi \setminus \{0\} \subseteq N$.

Theorem 3.93 *If X is a Banach space, $\varphi \in C^2(X)$, N is the Nehari manifold of φ, $N \neq \emptyset$ and it has the following properties*

(i) $N \cap B_\varrho = \emptyset$ *for some* $\varrho > 0$ $(B_\varrho = \{u \in X : \| u \|_X < \varrho\})$;
(ii) $\langle \varphi''(u)u, u\rangle \neq 0$ *for all* $u \in N$,

then M is a natural constraint for φ.

Proof Let $\vartheta(u) = \langle \varphi'(u), u \rangle$ for all $u \in X$. Then

$$N = \vartheta^{-1}(0) \setminus \{0\}.$$

Let $u \in N$. We have

$$\langle \vartheta'(u), u \rangle = \langle \varphi''(u)u, u \rangle + \langle \varphi'(u), u \rangle = \langle \varphi''(u)u, u \rangle \neq 0. \text{ (see hypothesis (ii))}$$
$$\Rightarrow \vartheta'(u) \neq 0 \text{ for all } u \in N.$$

From this and hypothesis (i), we have that N is a C^1-manifold of codimension 1. If $u_0 \in K_{\varphi|_M}$, then from Remark 3.88 we know that we can find $\lambda \in \mathbb{R}$ such that

$$\begin{aligned}
&\varphi'(u_0) = \lambda \vartheta'(u_0), \\
&\Rightarrow \langle \varphi'(u_0), u_0 \rangle = \lambda \langle \vartheta'(u_0), u_0 \rangle, \\
&\Rightarrow 0 = \vartheta(u_0) = \lambda \langle \vartheta'(u_0), u_0 \rangle \text{ and } \langle \vartheta'(u_0), u_0 \rangle \neq 0 \text{ (see (3.44))}, \\
&\Rightarrow \lambda = 0, \\
&\Rightarrow \varphi'(u_0) = 0 \text{ and so } u_0 \in K_\varphi.
\end{aligned}$$

Therefore we conclude that $K_\varphi \cap N = K_{\varphi|_N}$ and so N is a natural constraint for φ. □

Remark 3.94 The Nehari manifold is much smaller than the ambient space X. So, various properties of $\varphi(\cdot)$ which fail globally (especially if φ is indefinite), can be true for $\varphi|_N$.

3.6 Critical Groups

Critical groups are an efficient tool to distinguish between critical points of a functional $\varphi \in C^1(X)$ and to prove multiplicity theorems for boundary value problems (see Chap. 5). In this section we do not have a comprehensive presentation of the theory of critical groups. We will focus only on those results that we will need later in this volume. For the complete theory, we refer to Papageorgiou et al. [397] (Chap. 6).

Let X be a Banach space and (Y_1, Y_2) a topological pair such that $Y_2 \subseteq Y_1 \subseteq X$. For $k \in \mathbb{N}_0$, by $H_k(Y_1, Y_2)$ we denote the k-th relative singular homology group with integer coefficients. We could also choose for the coefficients a field of characteristic zero. Then we avoid torsion phenomena (that is, we may have $rank\, H_k(Y_1, Y_2) = 0$, but $H_k(Y_1, Y_2) \neq 0$) and the groups are in fact vector spaces. However, here we do not worry about such issues and use integer coefficients which correspond to the most standard singular homology.

Definition 3.95 Let $\varphi \in C^1(X)$ and $u \in K_\varphi$ is isolated. Then the "critical groups" of φ at u are defined

$$C_k(\varphi, u) = H_k(\varphi^c \cap U, \varphi^c \cap U \setminus \{u\}) \text{ for all } k \in \mathbb{N}_0,$$

with $c = \varphi(u)$ and U is a neighborhood of u such that

$$K_\varphi \cap U = \{u\}.$$

Remark 3.96 The excision property of singular homology implies that this notion is well defined (that is, it is independent of the choice of the isolating neighborhood U). Evidently critical groups are a local notion, that is, they depend only on the behavior of $\varphi(\cdot)$ near u. We mention that for $k \in -\mathbb{N}$ we have $C_k(\varphi, u) = 0$ and that critical groups are useful only at isolated critical points.

In what follows, we will use the so-called "Kronecker symbol" $\delta_{k,m}$ defined by

$$\delta_{k,m} = \begin{cases} 1, & \text{if } k = m \\ 0, & \text{if } k \neq m \end{cases} \quad k, m \in \mathbb{N}_0.$$

Before proceeding further with critical groups, let us mentions some elementary facts from Algebraic Topology. (for details, see Papageorgiou et al. [397], Sect. 6.1).

Proposition 3.97

(a) If the pairs (Y_1, Y_2) and (Y_1', Y_2') are homotopy equivalent, then $H_k(Y_1, Y_2) = H_k(Y_1', Y_2')$ for all $k \in \mathbb{N}_0$.
(b) If $Y_2 \subseteq Y_1$ is a deformation retract, then $H_k(Y_1, Y_2) = 0$ for all $k \in \mathbb{N}_0$; hence $H_k(X, X) = 0$.
(c) If $Y_3 \subseteq Y_2 \subseteq Y_1$ and Y_3 is a deformation retract of Y_2, then $H_k(Y_1, Y_2) = H_k(Y_1, Y_3)$ for all $k \in \mathbb{N}_0$.
(d) If $Y_3 \subseteq Y_2 \subseteq Y_1$ and Y_2 is a deformation retract of Y_1, then $H_k(Y_1, Y_3) = H_k(Y_2, Y_3)$ for all $k \in \mathbb{N}_0$.
(e) If Y is contractible and $* \in Y$, then $H_k(Y, *) = 0$ for all $k \in \mathbb{N}_0$.
(f) If $Y_2 \subseteq Y_1$ is contractible and $* \in Y_2$, then $H_k(Y_1, Y_2) = H_k(Y_1, *)$ for all $k \in \mathbb{N}_0$.

Next we compute the critical groups of some characteristic critical points of $\varphi \in C^1(X)$.

Proposition 3.98 *If $u_0 \in K_\varphi$ is isolated and it is a local minimizer of φ, then $C_k(\varphi, u_0) = \delta_{k,0}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.*

Proof Since u_0 is isolated in K_φ and also a local minimizer of $\varphi(\cdot)$, we can find U a neighborhood of u_0 such that

$$K_\varphi \cap U = \{u_0\} \text{ and } c = \varphi(u_0) < \varphi(u) \text{ for all } u \in U \setminus \{u_0\}.$$

Then according to Definition 3.95 we have

$$C_k(\varphi, u) = H_k(\{u\}, \emptyset) = H_k(\{u\}) = \delta_{k,0}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0.$$

□

Remark 3.99 If $X = \mathbb{R}$, then the critical groups of all critical points can be described completely.

- If $u_0 \in \mathbb{R}$ is a local minimizer, then $C_k(\varphi, u_0) = \delta_{k,0}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.
- If $u_0 \in \mathbb{R}$ is a local maximizer, then $C_k(\varphi, u_0) = \delta_{k,1}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.
- In all other cases $C_k(\varphi, u_0) = 0$ for all $k \in \mathbb{N}_0$.

Proposition 3.100 *If H is a Hilbert space, $\varphi \in C^2(H)$ and $u \in K_\varphi$ is isolated, nondegenerate and has Morse index m (see Definition 3.54), then $C_k(\varphi, u) = \delta_{k,m}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.*

Proof We may assume that $u = 0$ and $c = \varphi(u) = 0$. According to Proposition 3.55 (Morse lemma) we can find a homeomorphism $h : W \to U$ with W, U neighborhoods of the origin such that

$$h(0) = 0 \text{ and } \varphi(h(v)) = \frac{1}{2}(\varphi''(0)v, v) \text{ for all } v \in W.$$

We set $\psi = \varphi \circ h$ and $\overline{B}$ is a closed ball with center at 0 such that $\overline{B} \subseteq W$. Then

$$\begin{aligned} C_k(\varphi, 0) &= H_k(\varphi^0 \cap h(\overline{B}), \varphi^0 \cap h(\overline{B}) \setminus \{0\}), \\ &= H_k(\psi^0 \cap \overline{B}, \varphi^0 \cap \overline{B} \setminus \{0\}) \text{ for all } k \in \mathbb{N}_0. \end{aligned}$$

Recall that $u = 0$ is nondegenerate. Therefore

$$H = H_- \oplus H_+$$

where H_-(resp. H_+) is the subspace on which $\varphi''(0) \in \mathcal{L}(H)$ is negative (resp. positive). For $v \in H$ we have uniquely

$$v = v_- + v_+ \text{ with } v_- \in H_-, v_+ \in H_+.$$

Let $\vartheta : [0, 1] \times \overline{B} \to \overline{B}$ be the deformation defined by

$$\vartheta(t, v) = v_- + (1 - t)v_+ \text{ for all } t \in [0, 1], \text{ all } v \in \overline{B},$$

$$\Rightarrow \psi(\vartheta(t, v)) = \psi(v_-) + (1 - t)^2\psi(v_+) \leq \psi(v) \text{ for all } t \in [0, 1], \text{ all } v \in \overline{B}.$$

Then the pairs

$$(\psi^0 \cap \overline{B}, \psi^0 \cap \overline{B} \setminus \{0\}) \text{ and } (H_- \cap \overline{B}, H_- \cap \overline{B} \setminus \{0\})$$

are homotopy equivalent and so

$$C_k(\varphi, 0) = H_k(H_- \cap \overline{B}, H_- \cap \overline{B} \setminus \{0\}) = \delta_{k,m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0.$$

□

The degenerate case requires the generalized Morse lemma (see Chang [95], Mawhin and Willem [335, p. 184] and Papageorgiou et al. [397, p. 481]). Let $L_u \in \mathcal{L}(H)$ be the self-adjoint operator defined by $(L_u(v), y) = \varphi''(u)(v, y), v, y \in H$.

Proposition 3.101 *If H is a Hilbert space, $U \subseteq H$ nonempty open, $\varphi \in C^2(U)$ and $u \in K_\varphi$ is isolated with finite nullity (see Definition 3.54), then there exist a diffeomorphism $h : V \to H$ with V a neihborhood of the origin and a map $\widehat{\varphi} \in C^2(W)$ with W a neighborhood of the origin in $\ker L_u$ such that $h(0) = u, \widehat{\varphi}(0) = 0, \widehat{\varphi}'(0) = 0, \widehat{\varphi}''(0) = 0$ and*

$$\varphi(h(v)) = \varphi(u) + \frac{1}{2}\varphi''(u)(y, y) + \widehat{\varphi}(z)$$

for all $v \in V$, where $v = z + y$ with $z \in \ker L_u, y \in (\ker Lu)^\perp = im L_u$

Remark 3.102 If $w \in V \cap W \cap K_{\widehat{\varphi}}$, then $h(w) \in K_\varphi$. Since $h(0) = 0 \in K_\varphi$ is isolated, then $0 \in K_{\widehat{\varphi}}$ is isolated. Hence we can compute the critical groups $C_k(\widehat{\varphi}, 0)$ for all $k \in \mathbb{N}_0$.

Using Proposition 3.101, we have the "Shifting Theorem", which covers the degenerate case (see Chang [95, p. 50], Mawhin and Willem [335, p. 190], and Papageorgiou et al. [397, p. 485]).

Theorem 3.103 *If H is a Hibert space, $U \subseteq H$ is nonempty, open, $\varphi \in C^2(U)$ and $u \in K_\varphi$ is isolated, has finite nullity and finite Morse index m, then $C_k(\varphi, u) = C_{k-m}(\widehat{\varphi}, 0)$ for all $k \in \mathbb{N}_0$, with $\widehat{\varphi}$ as in Proposition 3.101.*

We set $m_u^* = m + \dim \ker L_u$ and call it the extended Morse index of $u \in K_\varphi$. Theorem 3.103 leads to the following corollary.

Corollary 3.104 *If H is a Hilbert space, $U \subseteq H$ is nonempty open, $\varphi \in C^2(U)$ and $u \in K_\varphi$ is isolated, has finite nullity and finite Morse index m, then if $C_k(\varphi, u) \neq 0$, we have $k \in [m, m_u^*] \cap \mathbb{N}_0$.*

We return to the Banach space setting. The next proposition reveals the importance of singular homology groups. Roughly speaking nontriviality of certain homology group, helps us detect the presence of a critical point.

Proposition 3.105 *If X is a Banach space, $\varphi \in C^1(X)$ and satisfies the C-condition and there exist $k_0 \in \mathbb{N}_0, a, b \in \mathbb{R}$ with $a < b$ such that $H_{k_0}(\varphi^b, \varphi^a) \neq 0$, then $K_\varphi \cap \varphi^{-1}([a, b]) \neq \emptyset$.*

Proof We proceed indirectly. So, suppose $K_\varphi \cap \varphi^{-1}([a,b]) = \emptyset$. Then by Corollary 3.32 (the "Noncritical Interval Theorem"), we have that

$$\varphi^a \text{ is a strong deformation retract of } \varphi^b,$$

$$\Rightarrow H_k(\varphi^b, \varphi^a) = 0 \text{ for all } k \in \mathbb{N}_0 \text{ (see Proposition 3.97).}$$

a contradiction. □

We make this result more precise, by computing $H_k(\varphi^b, \varphi^a)$ in terms of the critical groups of the elements in $K_\varphi \cap \varphi^{-1}([a,b])$.

Proposition 3.106 *If X is a Banach space, $\varphi \in C^1(X)$, $a, b \in \mathbb{R}$ with $a < b$, φ satisfies the C_c-condition for every $c \in [a,b)$ and the only critical value of φ in $[a,b]$ is $c_0 \notin \{a,b\}$ and $K_\varphi^{c_0} = \{u_i\}_{i=1}^m$, then $H_k(\varphi^b, \varphi^a) = \bigoplus_{i=1}^m C_k(\varphi, u_i)$ for all $k \in \mathbb{N}_0$.*

Proof Note that $\varphi^a \subseteq \varphi^{c_0} \setminus K_\varphi^{c_0} \subseteq \varphi^{c_0} \subseteq \varphi^b$. From Theorem 3.31 (the "Second Deformation Theorem"), we know that

$$\varphi^a \text{ is a strong deformation retract of } \varphi^{c_0} \setminus K_\varphi^{c_0},$$

while from Corollary 3.32 we have

$$\varphi^{c_0} \text{ is a strong deformation retract of } \varphi^b.$$

Therefore by Proposition 3.97, we have

$$H_k(\varphi^b, \varphi^a) = H_k(\varphi^{c_0}, \varphi^a) = H_k(\varphi^{c_0}, \varphi^{c_0} \setminus K_\varphi^{c_0}) \text{ for all } k \in \mathbb{N}_0.$$

Let $\{U_i\}_{i=1}^m$ be mutually disjoint open neighborhoods of $\{u_i\}_{i=1}^m$ such that

$$W = \bigcup_{i=1}^m U_i \subseteq \varphi^{-1}([a,b]).$$

We have $U_i \cap K_\varphi = \{u_i\}$ for every $i \in \{1, \ldots, m\}$. Hence

$$C_k(\varphi, u_i) = H_k(\varphi^{c_0} \cap U_i, \varphi^{c_0} \cap U_i \setminus \{u_i\}).$$

Using the properties of the singular homology groups (see Papageorgiou et al. [397, pp. 462, 465]), we have

$$\begin{aligned} H_k(\varphi^{c_0}, \varphi^{c_0} \setminus K_\varphi^{c_0}) &= H_k(\varphi^{c_0} \cap W, (\varphi^{c_0} \setminus K_\varphi^{c_0}) \cap W), \\ &= \bigoplus_{i=1}^m C_k(\varphi, u_i). \end{aligned}$$

□

If we cross multiple critical levels, then we have the following result.

Proposition 3.107 *If X is a Banach space, $\varphi \in C^1(X), a, b \in \mathbb{R}, a < b$ are regular values of φ, $K_\varphi \cap \varphi^{-1}([a,b])$ is finite and φ satisfies the C_c-condition for every $c \in [a,b]$, then $rank\, H_k(\varphi^b, \varphi^a) \leq \sum_{u \in K_\varphi^{[a,b]}} rank C_k(\varphi, u)$ with $K_\varphi^{[a,b]} = K_\varphi \cap \varphi^{-1}([a,b])$.*

Proof Let $\{c_i\}_{i=1}^m$ be the critical values of φ in (a,b) (recall that a, b are regular values). We may assume that the c_i's are in increasing order, that is, $a < c_1 < c_2 < \ldots < c_m < b$. Let $\{a_i\}_{i=1}^{m+1} \subseteq [a,b]$ be such that $a = a_1 < c_1 < a_2 < c_2 < \ldots < c_{m-1} < a_m < c_m < b = a_{m+1}$. Then from the properties of singular homotogy groups (see [397, Proposition 6.1.36, p. 472]), we have

$$\begin{aligned} rank\, H_k(\varphi^b, \varphi^a) &\leq \sum_{i=1}^m H_k(\varphi^{a_i+1}, \varphi^{a_i}), \\ &= \sum_{i=1}^m \sum_{u \in K_\varphi^{c_i}} C_k(\varphi, u) \text{ (see Proposition 3.106)}, \\ &= \sum_{u \in K_\varphi^{[a,b]}} C_k(\varphi, u). \end{aligned}$$

□

The results lead to the introduction of the following quantities which play an important role in the theory of critical groups.

Definition 3.108 Let X be a Banach space, $\varphi \in C^1(X), a, b \in \mathbb{R}, a < b$ are regular values of φ and $K_\varphi \cap \varphi^{-1}([a,b]) = \{u_i\}_{i=1}^m$. We introduce the following quantities:

(a) The "Morse-type number" of φ for (a,b) are defined by $M_k(a,b) = \sum_{i=1}^m C_k(\varphi, u_i)$ for all $k \in \mathbb{N}_0$. If $M_k(a,b) < \infty$ for all $k \in \mathbb{N}_0$ and $M_k(a,b) = 0$ for all $k \geq k_0$, then the "Morse polynomial" of φ for (a,b) is defined by $M(a,b)(t) = \sum_{k \in \mathbb{N}_0} M_k(a,b) t^k$ for all $t \in \mathbb{R}$.

(b) The "Betti-type number" of φ for (a,b) are defined by

$$\beta_k(a,b) = rank\, H_k(\varphi^b, \varphi^a) \text{ for all } k \in \mathbb{N}_0.$$

If $\beta_k(a,b) < \infty$ for all $k \in \mathbb{N}_0$ and $\beta_k(a,b) = 0$ for all $k \geq k_0$, then the "Poincare polynomial" of φ for (a,b) is defined by

$$P(a,b)(t) = \sum_{k \in \mathbb{N}_0} \beta_k(a,b) t^k \text{ for all } t \in \mathbb{R}.$$

These quantities are related via the so-called "Morse relation"(local version). To prove it, we need the following lemma about singular homology groups (see Papageorgious et al. [397, p. 489]).

Lemma 3.109 *If $\{A_k\}_{k=0}^m$ is an increasing family of Hausdorff topological spaces, then*

(a) $rank H_k(A_m, A_0) \leq \sum_{i=1}^{m} H_k(A_i, A_{i-1})$ *for all* $k \in \mathbb{N}_0$.

(b) If $rank H_k(A_i, A_{i-1}) < \infty$ *for all* $k \in \mathbb{N}_0$, *all* $i \in \{1, \ldots, m\}$ *and* $rank H_k(A_i, A_{i-1}) = 0$ *for all* $k \geq k_0$,

then $\sum_{i=1}^{m} (\sum_{k\in\mathbb{N}_0} rank(A_i, A_{i-1})t^k) = \sum_{k\in\mathbb{N}_0} rank H_k(A_m, A_0) + (1+t)Q(t)$ *with* $Q(t)$ *being a polynomial in* $t \in \mathbb{R}$ *with nonnegative integer coefficients.*

Having this lemma, we can prove the local "Morse relation".

Theorem 3.110 *If X is a Banach space, $\varphi \in C^1(X)$, $a, b \in \mathbb{R}$, $a < b$ are regular values, φ satisfies the C_c-condition for every $c \in [a, b]$ and $K_\varphi \cap \varphi^{-1}([a, b])$ is finite, then*

(a) $\beta_k(a, b) \leq M_k(a, b)$ *for all* $k \in \mathbb{N}_0$;

(b) if $M_k(a, b) < \infty$ *for all* $k \in \mathbb{N}_0$, $M_k(a, b) = 0$ *for all* $k \geq k_0$, *then the same hold for* $\beta_k(a, b)$ *and*

$$\sum_{k\in\mathbb{N}_0} M_k(a, b)t^k = \sum_{k\in\mathbb{N}_0} \beta_k(a, b)t^k + (1+t)Q(t) \tag{3.43}$$

with $Q(t)$ be a polynomial int with nonnegative integer coefficients.

Proof

(a) Let $\{c_i\}_{i=1}^m \subseteq \varphi(K_\varphi \cap [a, b])$ and let $\{a_i\}_{i=0}^m$ be numbers such that

$$a = a_0 < c_1 < a_1 < \ldots < a_{i-1} < c_i < a_i < \ldots < c_m = b = a_m$$

Using Proposition 3.107 and Lemma 3.109, we obtain

$$M_k(a, b) = \sum_{i=1}^{m} M_k(a_{i-1}, a_i) = \sum_{i=1}^{m} \beta_k(a_{i-1}, a_i) \geq \beta_k(a, b).$$

(b) Clearly from (a) we infer that

$$\beta_k(a, b) < \infty \text{ for all } k \in \mathbb{N}_0 \text{ and } \beta_k(a, b) = 0 \text{ for all } k \in \mathbb{N}_0$$

Then using Lemma 3.109, we have

$$\sum_{k\in\mathbb{N}_0}(\sum_{i=1}^{m} M_k(a_{i-1},a_i))t^k = \sum_{k\in\mathbb{N}_0}\beta_k(a,b)t^k + (1+t)Q(t), \tag{3.44}$$

with $Q(t)$ a polynomial in $t \in \mathbb{R}$ with nonnegative integer coefficients. Then

$$\sum_{k\in\mathbb{N}_0}(\sum_{i=1}^{m} M_k(a_{i-1},a_i))t^k = \sum_{k\in\mathbb{N}_0}\beta_k(a,b)t^k + (1+t)Q(t),$$

$$\Rightarrow \sum_{k\in\mathbb{N}_0} M_k(a,b)t^k = \sum_{k\in\mathbb{N}_0}\beta_k(a,b)t^k + (1+t)Q(t) \text{ for all } t \in \mathbb{R}.$$

□

Remark 3.111 If $t = -1$, then

$$\sum_{k\in\mathbb{N}_0}(-1)^k M_k(a,b) = \sum_{k\in\mathbb{N}_0}(-1)^k\beta_k(a,b),$$

which is known as the "Poincare-Hopf formula".

To have a global version of the Morse relation, we need the notion of critical groups of φ at infinity.

Definition 3.112 Let $\varphi \in C^1(X)$ which satisfies the C-condition and $\inf\varphi(K_\varphi) > -\infty$. The "critical groups of φ at infinity" are defined by

$$C_k(\varphi,\infty) = H_k(X,\varphi^c) \text{ for all } k \in \mathbb{N}_0,$$

with $c < \inf\varphi(K_\varphi)$.

Remark 3.113 On account of Corollary 3.32, the above definition is independent of the choice of the level $c < \inf\varphi(K_\varphi)$. To see this let $c_1 < \inf\varphi(K_\varphi)$ and suppose $c_1 < c$. Then φ^{c_1} is a strong deformation retract of φ^c (see Corollary 3.32). Hence by Proposition 3.97 (c), we have

$$H_k(X,\varphi^c) = H_k(X,\varphi^{c_1}) \text{ for all } k \in \mathbb{N}_0$$

The next proposition is an easy consequence of Theorem 3.31 and the properties of homology groups.

Proposition 3.114 *If X is a Banach space, $\varphi \in C^1(X)$, it satisfies the C-condition and $-\infty < \inf \varphi(K_\varphi)$, then*

(a) if $a < \inf \varphi(K_\varphi) \leq \sup \varphi(K_\varphi) < b$, then $C_k(\varphi, \infty) = H_k(\varphi^b, \varphi^a)$ for all $k \in \mathbb{N}_0$.
(b) if $K_\varphi = \emptyset$, then $C_k(\varphi, \infty) = 0$ for all $k \in \mathbb{N}_0$.
(c) if $K_\varphi = \{u\}$, then $C_k(\varphi, \infty) = C_k(\varphi, u)$.

Let us state some more useful facts about critical groups at infinity.

Proposition 3.115 *If X is a Banach space and $\varphi \in C^1(X)$ satisfies the C-condition, then*

(a) if $\varphi(\cdot)$ is bounded below, then $C_k(\varphi, \infty) = \delta_{k,0}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.
(b) if K_φ is finite, then $rank C_k(\varphi, \infty) \leq \sum_{u \in K_\varphi} rank C_k(\varphi, u)$ for all $k \in \mathbb{N}_0$.
(c) if K_φ is finite and $C_k(\varphi, \infty) \neq 0$ for some $k \in \mathbb{N}_0$,

then there exists $u \in K_\varphi$ such that $C_k(\varphi, u) \neq 0$.

Proof

(a) Let $c < \inf \varphi(X)$. We have $\varphi^c = \emptyset$ and so

$$C_k(\varphi, \infty) = H_k(X, \varphi^c) = H_k(X, \emptyset) = H_k(X) \text{ for all } k \in \mathbb{N}_0,$$
$$\Rightarrow C_k(\varphi, \infty) = \delta_{k,0}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0.$$

(b) Let $a < \inf \varphi(K_\varphi) \leq \sup \varphi(K_\varphi) < b$. If $K_\varphi = \{u_i\}_{i=1}^m$, then from Proposition 3.107

$$rank H_k(\varphi^b, \varphi^a) \leq \sum_{i=1}^m rank C_k(\varphi, u_i),$$
$$\Rightarrow rank H_k(X, \varphi^a) \leq \sum_{i=1}^m rank C_k(\varphi, u_i) \text{ (see Proposition 3.97 (d))}$$
$$\Rightarrow rank C_k(\varphi, \infty) \leq \sum_{i=1}^m rank C_k(\varphi, u_i).$$

(c) Follows from (b).

□

Proposition 3.116 *If X is a Banach space, $X = Y \oplus V$ with $\dim Y < \infty$, $\varphi \in C^1(X)$ satisfies the C-condition, $\inf \varphi(K_\varphi) > -\infty$, $\varphi|_V$ is bounded below and $\varphi|_Y$ is anticoercive, then $C_d(\varphi, \infty) \neq 0$ for $d = \dim Y$.*

Proof Let $c < \min\{\inf_V \varphi, \inf \varphi(K_\varphi)\}$. Since $\varphi|_Y$ is anticoercive. Let $\varrho > 0$ large so that

$$\partial B_\varrho^Y = \{y \in Y : \| y \| = \varrho\} \subseteq \varphi^c \subseteq X \setminus V \subseteq X. \tag{3.45}$$

Consider the deformation $h : [0, 1] \times (X \setminus V) \to X \setminus V$ defined by $h(t, u - v) = (1 - t)(u - v) + t\varrho \frac{u-v}{\|u-v\|}$ for all $t \in [0, 1]$, all $u \in X$, $v \in V$ (recall $X = Y \oplus V$).

It follows that ∂B_ϱ^Y is a strong deformation retract of $X \setminus V$. Therefore, by Proposition 3.97 we have

$$H_k(X \setminus V, \partial B_1^Y) = 0 \text{ for all } k \in \mathbb{N}_0. \tag{3.46}$$

We consider the following commutative diagram

$$\begin{array}{ccccc} H_k(\varphi^c, \partial B_1^Y) & \xrightarrow{i_*} & H_k(X, \partial B_1^Y) & \xrightarrow{j_*} & H_k(X, \varphi^c) = C_k(\varphi, \infty) \\ \downarrow \vartheta_* & \nearrow \eta_* & & & \\ H_k(X \setminus V, \partial B_1^Y) & & & & \end{array}$$

with $i_*, j_*, \vartheta_*, \eta_*$ being the group homomorphisms induced by the corresponding inclusion maps. In this diagram, the top row is exact (that is $imi_* = kerj_*$). Also $i_* = \eta_* \circ \vartheta_*$, hence $i_* = 0$ (see (3.46)), and so $kerj_* = 0$, that is, j_* is injective. We know that

$$\begin{aligned} & H_d(X, \partial B_1^Y) = H_{d-1}(\partial B_1^Y, x) = \mathbb{Z}(x \in \partial B_1^Y), \\ \Rightarrow\ & C_d(\varphi, \infty) \neq 0 \text{ (see (3.45)).} \end{aligned}$$

□

Combining Theorem 3.110 and Proposition 3.114, we can have the global Morse relation.

Theorem 3.117 *If X is a Banach space, $\varphi \in C^1(X)$ satisfies the C-condition, K_φ is finite and $rankC_k(\varphi, u) < \infty$ for all $k \in \mathbb{N}_0$, all $u \in K_\varphi$ and it is zero for all $k \geq k_0$, then*

$$\sum_{u \in K_\varphi} (\sum_{k \in \mathbb{N}_0} rankC_k(\varphi, u) t^k) = \sum_{k \in \mathbb{N}_0} rankC_k(\varphi, \infty) + (1 + t)Q(t)$$

with $Q(t)$ being a polynomial in t with nonnegative integer coefficients.

Next we present some rules for the computation of critical groups.

For the first computation rule, we will need the following lemma which is a consequence of the properties of long exact sequences in singular homology theory (see Papageorgiou et al. [397, p. 472]).

Lemma 3.118 *If $X_1 \subseteq X_2 \subseteq X_3 \subseteq X_4$ are Hausdorff topological spaces, then for all $n \in \mathbb{N}$*

$$rank H_k(X_3, X_1) - rank H_k(X_4, X_1) \leq rank H_{k-1}(X_2, X_1) + rank H_{k+1}(X_4, X_3).$$

With this lemma, we can prove our first computation rule.

Proposition 3.119 *If X is a Banach space, $\varphi \in C^1(X)$ satisfies the C-condition, K_φ is finite and $0 \in K_\varphi$, then*

(a) if for some $k \in \mathbb{N}_0$, $C_k(\varphi, 0) = 0$, $C_k(\varphi, \infty) \neq 0$, then we can find $u \in K_\varphi \setminus \{0\}$ such that $C_k(\varphi, u) \neq 0$;
(b) if for some $k \in \mathbb{N}_0$, $C_k(\varphi, 0) \neq 0$, $C_k(\varphi, \infty) = 0$, then we can find $u \in K_\varphi \setminus \{0\}$ such that $\varphi(u) < 0$ and $C_{k-1}(\varphi, u) \neq 0$ or $\varphi(u) > 0$ and $C_{k+1}(\varphi, u) \neq 0$.

Proof

(a) From Proposition 3.115(c), we know that there exists $u \in K_\varphi$ such that $C_k(\varphi, u) \neq 0$. Since $C_k(\varphi, 0) = 0$, we conclude that $u \in K_\varphi \setminus \{0\}$.
(b) We can always assume that $\varphi(0) = 0$. We can find $\varepsilon > 0$ such that $\varphi(K_\varphi) \cap [-\varepsilon, \varepsilon] = \{0\}$. Let $c < \min\{-\varepsilon, \inf \varphi(K_\varphi)\}$. By Proposition 3.114 we have

$$rank C_k(\varphi, 0) \leq rank H_k(\varphi^\varepsilon, \varphi^{-\varepsilon}).$$

Also by Definition 3.112, $C_k(\varphi, \infty) = H_k(X, \varphi^c)$. We consider the following sets

$$\varphi^c \subseteq \varphi^{-\varepsilon} \subseteq \varphi^\varepsilon \subseteq X.$$

$$\Rightarrow 0 < rank C_k(\varphi, 0) - rank C_k(\varphi, \infty) \leq rank H_{k-1}(\varphi^{-\varepsilon}, \varphi^c) + rank H_{k+1}(X, \varphi^\varepsilon) \text{ (see Lemma 3.118)}.$$

$$\Rightarrow rank H_{k-1}(\varphi^{-\varepsilon}, \varphi^c) \neq 0 \text{ or } rank H_{k+1}(X, \varphi^\varepsilon) \neq 0.$$

Therefore there exists $u \in K_\varphi$ such that
$\varphi(u) < 0$ and $C_{k-1}(\varphi, u) \neq 0$ (see Proposition 3.105)
or $\varphi(u) > 0$ and $C_{k+1}(\varphi, u) \neq 0$ (see Proposition 3.106).

□

Next we introduce a homological version of the notion of linking sets (see Definition 3.37).

Definition 3.120 Let X be a Banach space and E_0, E, D nonempty subsets such that

$$E_0 \subseteq E \text{ and } E_0 \cap D = \emptyset.$$

We say that the pair $\{E_0, E\}$ "homologically links" with D in dimension $d \in \mathbb{N}$, if the homomorphism

$$i_* : H_d(E, E_0) \to H_d(X, X \setminus D)$$

induced by the inclusion of (E, E_0) into $(X, X \setminus D)$, is nontrivial.

Remark 3.121 The linking notion introduced in Definition 3.37 is known as "homotopical linking". Evidently homological linking implies homotopical linking. In Example 3.38(a) the sets are homologically linking in dimension $d = 1$. In Example 3.38(b) the sets are homologically linking in dimension $d = dimY$. In Example 3.38(c) and (d) the sets are homologically linking in dimension $d \neq 1$.

Theorem 3.122 *If X is a Banach space, $\{E_0, E\}$ homologically links with D in dimension $d \geq 1$, $\varphi \in C^1(X)$ and $a < b \leq \infty$ are such that*

$$\varphi|_{E_0} \leq a < \varphi|_D \text{ and } \sup_E \varphi \leq b,$$

then

(a) $H_d(\varphi^b, \varphi^a) \neq 0$;
(b) if φ satisfies the C-condition, $a, b \notin \varphi(K_\varphi)$ and $K_\varphi \cap \varphi^{-1}([a, b])$ is finite, then there exists $u \in K_\varphi \cap \varphi^{-1}([a, b])$ such that $C_d(\varphi, u) \neq 0$.

Proof

(a) In addition to the inclusion $i : (E, E_0) \to (X, X \setminus D)$, we consider also the following inclusions

$$j : (E, E_0) \to (\varphi^b, \varphi^a), \quad \eta : (\varphi^b, \varphi^a) \to (X, X \setminus D).$$

We have $i = \eta \circ j$ and so $i_* = \eta_* \circ j_*$ for the corresponding group homomorphisms. By Definition 3.120 we have that

$$i_* = \eta_* \circ j_* : H_d(E, E_0) \to H_d(X, X \setminus D)$$

is nontrivial. Hence $j_* \neq 0$, $\eta_* \neq 0$ and so $H_d(\varphi^b, \varphi^a) \neq 0$.

(b) Use the local Morse relation (see Theorem 3.110) and (a).

□

An important consequence of this theorem is the following result concerning critical points of mountain pass-type.

Corollary 3.123 *If X is a Banach space, $\varphi \in C^1(X)$ satisfies the C-condition, K_φ is finite, $u_0, u_1 \in X$, $0 < \varrho < \|u_1 - u_0\|_X$ and $\max\{\varphi(u_0), \varphi(u_1)\} < \inf\{\varphi(u) : \|u - u_0\| = \varrho\} = m$, then there exists $u \in K_\varphi$ such that $m \leq \varphi(u)$ and $C_1(\varphi, u) \neq 0$.*

In Definition 3.50 we introduced the notion of local linking which resulted in a multiplicity result without any symmetry conditions. When we have local linking, then $0 \in K_\varphi$. Next we introduce a homological version of this notion.

Definition 3.124 Let X be a Banach space, $\varphi \in C^1(X)$, $\varphi(0) = 0$ and $0 \in K_\varphi$ is isolated. Let $m, n \in \mathbb{N}$. We say that φ has "local $(m, n)-$ linking" at the origin, if there exists a neighborhood U of the origin and nonempty sets $E_0 \subseteq E \subseteq U$, $D \subseteq X$ such that $E_0 \cap D = \emptyset$ and

(a) $\varphi^0 \cap U \cap K_\varphi = \{0\}$;
(b) $rank(imi_*^{m-1}) - rank(imj_*^{m-1}) \geq m$, with

$$i_*^{m-1} : H_{m-1}(E_0) \to H_{m-1}(X \setminus D),$$

$$j_*^{m-1} : H_{m-1}(E_0) \to H_{m-1}(E)$$

being the group homomorphisms induced by the inclusions

$$i : E_0 \to X \setminus D,\ j : E_0 \to E;$$

(c) $\varphi|_E \leq 0 \leq \varphi|_{U \cap D \setminus \{0\}}$.

Remark 3.125 If φ has a local linking at $u = 0$ (see Definition 3.50) and $u = 0$ is a strict local minimizer of $\varphi|_V$ or alternatively $X = H =$ a Hilbert space and φ' is Lipschitz near the origin, then $\varphi(\cdot)$ has local $(d, 1)-$ linking, with $d = dimY < \infty$.

The next theorem is due to Perera [421].

Theorem 3.126 *If X is a Banach, $\varphi \in C^1(X)$, $0 \in K_\varphi$ is isolated and $\varphi(\cdot)$ has a local $(m, n)-$ linking at the origin, then $n \leq rankC_m(\varphi, 0)$.*

In the case of the standard (homotopical) local linking (see Definition 3.50), we have the following result (see Remark 3.125).

Corollary 3.127 *If X is a Banach space, $X = Y \oplus V$ with $d = dimY < \infty$, $\varphi \in C^1(X)$, φ has a local linking at 0, $0 \in K_\varphi$ is isolated and $u = 0$ is a strict local minimizer of $\varphi|_V$ or $X = H$ is a Hilbert space and $\varphi'(\cdot)$ is Lipschitz near the origin, then $1 \leq rankC_d(\varphi, 0)$ (hence $C_1(\varphi, 0) \neq 0$).*

From Palais [357] we have.

Proposition 3.128 *If H is a Hilbert space, X is a Banach space, $X \to H$ continuously and densely, $\varphi \in C^2(H)$ and $u \in K_\varphi$ is isolated, then $C_k(\varphi, u) = C_k(\varphi|_X, u)$ for all $k \in \mathbb{N}_0$.*

We conclude this brief encounter with the theory of critical groups, with two basic properties of critical groups which highlight their usefulness in the study of boundary value problems (see Papageorgiou et al. [397, Section 6.3]).

The first theorem is about the C^1- continuity of critical groups.

Theorem 3.129 *If X is a Banach space, $\varphi, \psi \in C^1(X), u \in X$, there exists $\varrho > 0$ such that $\overline{B}_\varrho(u) \cap K_\varphi = \overline{B}_\varrho(u) \cap K_\psi = \{u\}$ and both satisfy the C-condition on $\overline{B}_r(u)$, then there exists $\delta > 0$ such that*

$$\|\varphi - \psi\|_{C^1(\overline{B}_r(u))} < \delta \Rightarrow rank C_k(\varphi, u) = rank C_k(\psi, u) \text{ for all } k \in \mathbb{N}_0.$$

Remark 3.130 If the singular homology theory is with coefficients in $\mathbb{R}$ (or more generally in a field $\mathbb{F}$ of characteristic zero), then the critical groups are vector spaces and in the conclusion of the above theorem we can say that $C_k(\varphi, u) = C_k(\psi, u)$ for all $k \in \mathbb{N}_0$.

The second theorem, establishes the homotopy invariance of critical groups.

Theorem 3.131 *If X is a Banach space, $\{h_t\}_{t\in[0,1]} \subseteq C^1(X)$, $u \in X$, there exists $\varrho > 0$ such that $\overline{B}_\varrho(u) \cap K_{h_t} = \{u\}$ for all $t \in [0, 1]$, $h_t(\cdot)(t \in [0, 1])$ satisfies the C-condition on $\overline{B}_\varrho(u)$ and $t \to h_t(\cdot)$ is continuous from $[0, 1]$ into $C^1(X)$, then $C_k(h_t, u)$ is independent of $t \in [0, 1]$.*

We have seen that critical groups make sense when the critical point is isolated. The final result of this chapter is related to this and describes the nature of local minimizers of a C^1- functional. The proof is a straightforward application of the generalized Ekeland variational principle (see Theorem 3.3).

Proposition 3.132 *If X is a Banach space, $\varphi \in C^1(X)$ satisfies the C-condition and u_0 is a local minimizer of $\varphi(\cdot)$, then one of the following statements holds:*

(a) there exists $\varrho > 0$ small such that

$$\inf\{\varphi(u) : \|u - u_0\| = \varrho\} > \varphi(u_0);$$

(b) for every $\varrho > 0$ small, φ has a local minimizer u_ϱ such that

$$\|u_\varrho - u_0\| = \varrho, \varphi(u_\varrho) = \varphi(u_0).$$

Remarks

3.1: The compactness conditions in Definition 3.1 are due to Palais and Smale [359] (the PS-condition) and to Cerami [88] (the C-condition). They are introduced in order to compensate for the fact that ambient space is not in general

locally compact (since it is usually infinite dimensional). So, as in degree theory, the burden of compactness is passed to the functional. Theorem 3.3 (the generalized Ekeland variational principle) is due to Zhong [485]. The relations between the compactness conditions, coercivity and the properties of the sublevel sets of the sublevel sets of the functional, were investigated by Calkovic et al. [83], Costa and Silva [111] and Kourogenis and Papageorgiou [275] (nonsmooth case), Struwe [446]. The pseudogradient vector field (Definition 3.11) was introduced by Palais [357], who also proved Theorem 3.14. This made possible the extension of the deformation approach to C^1- functionals on Banach space. The gradient flow required a Hilbert space (so that we may identify it with its dual) and a C^2- functional. Another result in the spirit of Proposition 3.19, is the following one by Aubin and Ekeland [20] (Proposition 3, p. 270).

Proposition 3.133 *If X is a reflexive Banach space and $\varphi : X \to \mathbb{R}$ is Gateaux differentiable, convex, lower semicontinuous and coercive, then every (PS)-sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq X$ has a cluster point in K_φ.*

Struwe [445] motivated by the Plateau problem, extended the $PS-$condition to functionals restricted on a nonempty closed, convex set and developed a critical point theory for such functionals. Kyritsi and Papageorgiou [282] obtained a nonsmooth extension of Struwe's theory.

3.2: In the literature, we can find two different approaches to critical point theory. The first (which we present here) is based on the deformation properties of the sublevel sets along the negative gradient flow or a suitable substitute of it, namely the pseudogradient flow. The second approach, uses the Ekeland variational principle and can be found in Cuesta [116], Ekeland [149], deFigueiredo [153]. A first version of Theorem 3.20 was proved by Clark [103], who developed a Ljusternik-Schnirelmann theory for even functionals defined on a Banach space using the Krasnoselskii genus (see Example 3.67(a)). Theorem 3.24 (the "Quantitative Deformation Theorem"), was proved by Willem [471]. Theorem 3.31 (the "Second Deformation Theorem") is due to Rothe [431], Marino and Prodi [329] and Chang [93], [96]. A nonsmooth version of it can be found in Corvellec [109]. The extension of the critical point theory to nonsmooth functionals started with the work of Chang [94] with subsequent contributions and extensions by Szulkin [450], Kourogenis and Papageorgiou [276]. The nonsmooth critical point theory with applications to boundary value problems, is presented in the books of Gasinski and Papageorgiou [181], Motreanu and Radulescu [342] and Radulescu [427]. For further details on the deformation approach to critical point theory, we refer to the books of Ambrosetti and Malchiodi [13], Gasinski and Papageorgiou [182], Ghoussoub [201], Motreanu et al. [344], Papageorgiou et al. [397], Struwe [446], Willem [471].

3.3: The notion of linking sets (see Definition 3.37) is due to Benci and Rabinowitz [39]. Slightly more restricted variants of Theorem 3.40 can be found in Ekeland [149], Mawhin and Willem [335], Struwe [446]. Theorem 3.41 (the "Mountain Pass Theorem"), was proved by Ambrosetti and Rabinowitz [14]. Theorems 3.42 and 3.43 (the "Saddle Point Theorem" and the "Generalized Mountain Pass Theorem") both are due to Rabinowitz [424]. Theorem 3.47 is due to Ghoussoub and Preiss [202] (see also Ekeland [149] and Ghoussoub [201]). We mention also the book of Jabri [258], which presents a detailed account of the variants, extensions and applications of the "Mountain Pass Theorem".

3.4: The notion of local linking (Definition 3.50), is originally due to Liu and Li [304] but under the stronger requirements

$$dim Y < \infty \text{ and } \varphi(v) \geq c > 0 \text{ for all } v \in V, \|v\|_X = \varrho.$$

The more general notion in Definition 3.50 is due to Brezis and Nirenberg [62]. Theorem 3.53 (the "Local Linking Theorem") is also due to Brezis and Nirenberg [62]. Proposition 3.55 (the "Morse Lemma"), was proved by Morse, for $H = \mathbb{R}^N$. Morse extended his theory to compact, smooth, finite dimensional manifolds. Extensions to Hilbert spaces, were obtained by Rothe [432], Palais [357], Palais and Smale [359], Smale [442]. Theorem 3.59 is due to Li and Willem [295]. We should also mention the "Fountain Theorem" due to Bartsch [37] (see also Willem [471, p. 58]). The setting is the following. Let X be a Banach and G a compact topological group with a linear isometric action on X. Also let Y be a finite dimensional Banach space and $Y^k = Y \times \cdots \times Y$ $k-$ times ($k \geq 2$). We assume that G acts on Y^k diagonally, that is $g(y_i)_{i=1}^k = (gy_i)_{i=1}^k$ and that is "admissible", that is, for each $k \geq 2$, each U bounded, open, invariant neighborhood of 0 in Y^b, we have that every continuous equivariant map $h : \partial u \to Y^k$ has a zero. If $G : Z_2 = \{id, -id\}$ and $Y = \mathbb{R}$, then by Theorem 1.41 (the "Borsuk-Ulam Theorem") the linear action of G on Y is admissible. Suppose that

$$X = \overline{\bigcup_{k \in \mathbb{N}} X_k}$$

with each X_k being invariant and there is an equivariant isomorphism between Y and X_k. We set

$$Z_m = \bigoplus_{k=1}^{m} X_k, \; V_m = \overline{\bigoplus_{k \geq m} X_k}.$$

The fountain theorem says:

Theorem 3.134 *If $\varphi \in C^1(X)$ is invariant, satisfies the PS_c- condition for every $c > 0$ and for every $m \in \mathbb{N}$ there are $\varrho_m > r_m > 0$ such that*

(i) $\max\{\varphi(z) : z \in Z_m, \|z\|_X = \varrho_m\} \leq 0$,
(ii) $\inf\{\varphi(v) : v \in V_m, \|v\|_X = r_m\} \to +\infty$ *as* $m \to \infty$,

then φ has an unbounded sequence of critical values.

The genus (see Example 3.67(a)) was introduced by Krasnoselskii [279] (see also Coffman [108]). The cohomological index (see Example 3.67(b)) was introduced by Fadell and Rabinowitz [154]. Theorem 3.72 goes back to the fundamental work of Ljusternik and Schnirelmann [316]. For additional results on the LS-theory, we refer to the books of Ambrosetti et al. [133], Gasinski and Papageorgiou [182], Struwe [446] and Zeidler [475]. Theorem 3.78 (the "Principle of Symmetric Criticality") is due to Palais [358] (for a nonsmooth extension, we refer to Kobayashi and Otani [274]).

3.5: The method of Lagrange multipliers, can be found in the books of Ioffe and Tichomirov [257], Gasinski and Papageorgiou [182], Papageorgiou and Kyritsi Yallourou [361], Zeidler [475]. For the Nehari manifold, we refer to Badiale-Serra [25] and Kuzin and Pohozaev [281] (fibering method).

3.6: For the necessary background on the algebraic topology, we refer to the books of Dold [143], Eilenberg and Steenrod [148], Hatcher [233], Maunder [333]. The first to make use of critical groups in the study of elliptic problems was Chang [93]. More on critical groups can be found in the books of Chang [95], [96], Motreanu et al. [344], Papageorgiou et al. [397], Perera et al. [421]. Extensions to nonsmooth functionals were obtained by Corvellec and Degiovanni [110].

We conclude with two more results on computation of critical groups.

Theorem 3.135 *If H is a Hilbert space, $\varphi \in C^2(H)$ and $u_0 \in K_\varphi$ is isolated, with finite nullity v_0 and finite Morse index m_0; moreover if $m_0 = 0$, assume that $v_0 \in \{0, 1\}$, then if $C_1(\varphi, u_0) \neq 0$, we have $C_k(\varphi, u_0) = \delta_{k,1}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.*

The second result is related to the so-called "Lyapunov-Schmidt reduction method". Let H be a Hilbert space with $H = Y \oplus V$, $dimY < \infty$ and $\varphi \in C^1(H)$, sequentially weakly lower semicontinuous and

$$c\|v - v'\|^2 \leq \langle \varphi'(y + v) - \varphi'(y + v'), v - v' \rangle$$

for all $y \in Y$, $v, v' \in H$, $c > 0$.

Then we can show that there exists $g : Y \to V$ continuous such that

$$\varphi(y + g(y)) = \inf\{\varphi(y + v) : v \in V\}.$$

We set $\varphi_0(y) = \varphi(y + g(y))$.

Theorem 3.136 *$\varphi_0 \in C^1(Y)$ and if $\widehat{y} \in K_{\varphi_0}$ is isolated, then $C_k(\varphi_0, \widehat{y}) = C_k(\varphi, \widehat{y} + g(\widehat{y}))$ for all $k \in \mathbb{N}_0$.*

Chapter 4
Spectra of Differential Operators



Introduction

In this chapter we prove the spectral properties of certain linear and nonlinear differential operators under various boundary conditions (Dirichlet, Neumann, Robin). In Sect. 4.1, we deal with scalar and vector differential operators (ordinary differential equations and systems). In Sect. 4.2, we consider linear operators with indefinite potential terms or indefinite weights. In Sect. 4.3, we study similar eigenvalue problems for the p-Laplacian. In the process we present also regularity results, a nonlinear Hopf's maximum principle, a nonlinear Picone's identity and a nonlinear Harnack inequality. Finally, in Sect. 4.4, we deal with eigenvalue problems related to double phase and anisotropic equations.

4.1 Scalar and Vector Operators

In this section, we present some basic spectral properties of the scalar and vector p-Laplace differential operators.

First, we deal with the scalar operator. The domain is $\Omega = (0, b)$ with $b > 0$. From Theorem I-4.130, we know that

$$W^{1,p}(0, b) \hookrightarrow C[0, b] \text{ compactly for all } 1 < p < \infty. \tag{4.1}$$

We will also use the following closed subspaces of $W^{1,p}(0, b)$

$$W_0^{1,p}(0, b) = \{u \in W^{1,p}(0, b) : u(0) = u(b) = 0\},$$
$$W_{per}^{1,p}(0, b) = \{u \in W^{1,p}(0, b) : u(0) = u(b)\}.$$

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On account of (4.1) the above definitions make sense and we have

$$W_0^{1,p}(0,b) \hookrightarrow W_{per}^{1,p}(0,b) \hookrightarrow W^{1,p}(0,b) \text{ continuously.}$$

The space $W_0^{1,p}(0,b)$ is for the Dirichlet problems and $W_{per}^{1,p}(0,b)$ is for the periodic problems. Finally $W^{1,p}(0,b)$ is for the Neumann problem. We formulate the corresponding eigenvalue problems. In what follows

$$\varphi_p(x) = \begin{cases} |x|^{p-2}x & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

This is a homeomorphism. We will study the following nonlinear eigenvalue problems

$$\left\{ \begin{array}{l} -\varphi_p(u'(t))' = \widehat{\lambda}\varphi_p(u(t)) \text{ in } (0,b), \\ u(0) = u(b) \text{ (Dirichlet eigenvalue problem),} \end{array} \right\} \tag{4.2}$$

$$\left\{ \begin{array}{l} -\varphi_p(u'(t))' = \widehat{\lambda}\varphi_p(u(t)) \text{ in } (0,b), \\ u(0) = u(b),\ u'(0) = u'(b) \text{ (periodic eigenvalue problem),} \end{array} \right\} \tag{4.3}$$

$$\left\{ \begin{array}{l} -\varphi_p(u'(t))' = \widehat{\lambda}\varphi_p(u(t)) \text{ in } (0,b), \\ u'(0) = u'(b) \text{ (Neumann eigenvalue problem).} \end{array} \right\} \tag{4.4}$$

We consider weak solutions of these problems. So, $\widehat{u} \in W_0^{1,p}(0,b)$(resp. $\widehat{u} \in W_{per}^{1,p}(0,b)$, $\widehat{u} \in W^{1,p}(0,b)$) is a "weak solution" of (2) (resp. (4.3), (4.4)), if

$$\int_\Omega \varphi_p(u'(t))h'(t)dt = \widehat{\lambda}\int_\Omega \varphi_p(u(t))h(t)dt \tag{4.5}$$

for all $h \in W_0^{1,p}(0,b)$(resp. all $h \in W_{per}^{1,p}(0,b)$, all $h \in W^{1,p}(0,b)$).

Definition 4.1 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" of (2) (resp. (4.3), (4.4)), if the problem has a nontrivial weak solution $\widehat{u}$. We call $\widehat{u}$ an "eigenfunction" corresponding to $\widehat{\lambda}$.

Remark 4.2 From (5) using integration by parts on the left hand side, we obtain

$$\begin{aligned} &-\varphi_p(\widehat{u}'(t))' = \widehat{\lambda}\varphi_p(\widehat{u}(t)) \text{ for a.a } t \in (0,b), \\ \Rightarrow &\varphi_p(\widehat{u}'(\cdot))' \in C[0,b], \\ \Rightarrow &\varphi_p(\widehat{u}'(\cdot)) \in W^{1,p'}(0,b) \hookrightarrow C[0,b] \ (\frac{1}{p} + \frac{1}{p'} = 1). \end{aligned}$$

Recall that $\varphi_p(\cdot)$ is a homeomorphism. Hence $\widehat{u}' \in C[0,b]$ and so we can say that $\widehat{u} \in C^1[0,b]$. Therefore in problems (4.3), (4.4) the boundary conditions make sense.

We introduce the following quantity

$$\pi_p = \frac{2\pi(p-1)^{1/p}}{p\sin\frac{\pi}{p}} = \int_0^{(p-1)^{1/p}} \frac{2}{(1-\frac{s^p}{p-1})^{1/p}} ds.$$

If $p=2$, then $\pi_2 = \pi$. Also, we introduce $\sin_p : \mathbb{R} \to \mathbb{R}$ a $2\pi_p$-periodic, odd function defined by

$$\int_0^{\sin_p t} \frac{ds}{(1-\frac{s^p}{p-1})^{1/p}} = t \text{ for } t \in [0, \frac{\pi_p}{2}].$$

We extend it to $[0,\pi_p]$, by oddness to $[-\pi_p,\pi_p]$ and by periodicity to all of $\mathbb{R}$. Evidently if $p=2$, then $\sin_2 = \sin$.

Using these quantities, we can have a complete description of the spectra for problems (4.2), (4.3), (4.4) (see Gasinski and Papageorgiou [182], Sect. 6.3).

Theorem 4.3 *For the Dirichlet eigenvalue problem (problem (4.2)), the eigenvalues are*

$$\widehat{\lambda}_k^D = (\frac{k\pi_p}{b})^p \text{ for all } k \in \mathbb{N}.$$

The corresponding eigenfunctions are

$$\widehat{u}_k^D(t) = \sin_p(\frac{k\pi_p}{b}t) \text{ for all } t \in [0,b].$$

All the eigenvalues are simple (this means that if $\widehat{u}^D$, $\widehat{v}^D$ are eigenfunctions corresponding to an eigenvalue $\widehat{\lambda}^D$, then $\widehat{u}^D = \vartheta\widehat{v}^D$ for some $\vartheta \in \mathbb{R}$).

Theorem 4.4 *For the periodic eigenvalue problem (problem (4.3)), the eigenvalues are*

$$\widehat{\lambda}_k^P = (\frac{2k\pi_p}{b})^p \text{ for all } k \in \mathbb{N}_0.$$

The corresponding eigenfunctions are

for $k=0$ ($\widehat{\lambda}_0^P = 0$), are the constant functions,

for $k \in \mathbb{N}$, $\widehat{u}_k^P(t) = \sin_p[\frac{2k\pi_p}{b}(t-\mu)]$ for all $t \in [0,b]$ with $\mu \in \mathbb{R}$.

Only the eigenvalue $\widehat{\lambda}_0^P$ is simple. The others are not.

Theorem 4.5 *For the Neumann eigenvalue problem (problem (4.4)), the eigenvalues are*

$$\widehat{\lambda}_k^N = (\frac{k\pi_p}{b})^p \text{ for all } k \in \mathbb{N}_0.$$

The corresponding eigenfunctions are

$$\widehat{u}_k^N(t) = \cos_p(\frac{k\pi_p}{b}t) \text{ for all } t \in [0, b],$$

with $\cos_p$ *being the* $2\pi_p$*-periodic even function defined by*

$$\int_0^{\cos_p t} \frac{ds}{(1 - \frac{s^p}{p-1})^{1/p}} = -t \text{ for } t \in [0, \frac{\pi_p}{2}],$$

we extend it to $[0, \pi_p]$*, by evenness to* $[-\pi_p, \pi_p]$ *and by periodicity to all of* $\mathbb{R}$*. Evidently* $\cos_2 = \cos$*. All eigenvalues are simple.*

Remark 4.6 It is worth describing the spectra for the case $p = 2$ (linear eigenvalue problem). We have

$$\widehat{\lambda}_k^D = (\frac{k\pi}{b})^2 \text{ and } \widehat{u}_k^D(t) = \sin(\frac{k\pi}{b}t) \text{ for all } t \in [0, b], \text{ all } k \in \mathbb{N},$$

$$\widehat{\lambda}_k^P = (\frac{2k\pi}{b})^2 \text{ and } \widehat{u}_k^P(t) = \sin(\frac{2k\pi}{b}(t - \mu)) \text{ for all } t \in [0, b], \text{ with } \mu \in \mathbb{R} \text{ and for all } k \in \mathbb{N}_0,$$

$$\widehat{\lambda}_k^N = (\frac{k\pi}{b})^2 \text{ and } \widehat{u}_k^N(t) = \cos(\frac{k\pi}{b}t) \text{ for all } t \in [0, b], \text{ all } k \in \mathbb{N}_0.$$

Next we will examine a weighted version of the Dirichlet eigenvalue problem. Consider the space $L^+(0, b) = \{u \in L^\infty(0, b) : u^+ \neq 0\}$. Then with $1 < p < \infty$ and $m \in L^+(0, b)$, we consider the following nonlinear eigenvalue problem

$$\left\{ \begin{array}{l} -\varphi_p(u'(t))' = \widehat{\lambda} m(t)\varphi_p(u(t)) \text{ in } (0, b), \\ u(0) = u(b) = 0. \end{array} \right\} \tag{4.6}$$

We will use the Ljusternik-Schnirelmann minimax scheme (see Sect. 3.4), based on the Krasnoselskii genus (see Example 3.67(a)).

So, we introduce the following families of sets:

$$\mathcal{F} = \{C \subseteq W_0^{1,p}(0, b) \text{ :compact, symmetric}\},$$

$$\mathcal{F}_k = \{C \in \mathcal{F} : \gamma(C) \geq k\} \text{ for all } k \in \mathbb{N}$$

($\gamma(\cdot)$ denotes the Krasnoselskii genus),

$$\mathcal{F}'_k = \mathcal{F}_k \cap \partial B_1 \text{ for all } k \in \mathbb{N}$$

$$(\text{with } \partial B_1 = \{u \in W_0^{1,p}(0,b) : \|u\| = 1\}).$$

We introduce the following sequence of eigenvalues of problem (4.6)

$$\widehat{\lambda}_k(m) = \inf_{C \in \mathcal{F}_k} \sup_{u \in C} \frac{\|u'\|_p^p}{\int_0^b m(t)|u|^p dt}, \ k \in \mathbb{N}.$$

Rewriting this minimax expressions as a maxmin one, we have

$$\begin{aligned} \widehat{\lambda}_k(m)^{-1} &= \sup_{C \in \mathcal{F}_k} \inf_{u \in C} \frac{\int_0^b m(t)|u|^p dt}{\|u'\|_p^p} \\ &= \sup_{C \in \mathcal{F}'_k} \inf_{u \in C} \int_0^b m(t)|u|^p dt \end{aligned} \tag{4.7}$$

(recall that by the Poincare inequality we have $\|u\| = \|u'\|_p$). Then we can show that $\widehat{\lambda}_1(m) > 0$, is simple, isolated and

$$0 < \widehat{\lambda}_1(m) < \widehat{\lambda}_2(m) \leq \widehat{\lambda}_3(m) \leq \cdots \leq \widehat{\lambda}_k(m) \to +\infty \text{ as } k \to \infty.$$

Moreover, the eigenfunctions belong in $C_0^1[0,b] = \{u \in C^1[0,b] : u(0) = u(b) = 0\}$ and those corresponding to $\widehat{\lambda}_1(m) > 0$ have constant sign while for all the other eigenvalues, the eigenfunctions are nodal (sign changing). Note that $C_0^1[0,b]$ is an ordered Banach space with positive cone $C_+ = \{u \in C_0^1[0,b] : u(t) \geq 0 \text{ for all } t \in [0,b]\}$. This cone has a nonempty interior

$$\text{int}C_+ = \{u \in C_+ : u(t) > 0 \text{ for all } t \in (0,b), u'_-(0) < 0 < u'_+(b)\}.$$

We will prove these facts in Sect. 4.3 for the more general setting of partial differential equations.

From (4.7), we see that if $m, \widehat{m} \in L^+(0,b)$, $m \leq \widehat{m}$, then $\widehat{\lambda}_k(\widehat{m}) \leq \widehat{\lambda}_k(m)$ for all $k \in \mathbb{N}_0$. In the sequel sometimes in order to stress the dependence on the underlying interval $T = (0,b)$, we will write $\widehat{\lambda}_k(m,T)$ for $k \in \mathbb{N}$. Given $u \in W_0^{1,p}(0,b) \subseteq C[0,b]$, we define

$$Z(u) = \{t \in (0,b) : u(t) = 0\},$$

the "zero set" of u. A "nodal domain" is a connected component of $(0,b) \setminus Z(u)$.

Proposition 4.7 *If $(\widehat{\lambda}(m), \widehat{u})$ is an eigenpair for problem (4.6) and $I \subseteq (0,b)$ is a nodal domain of $\widehat{u}$, then $m|_I \in L^+(I)$.*

Proof Let $\widehat{u}^* = \widehat{u}_{\chi_I} \in W_0^{1,p}(0,b)$ (see Proposition *I*-4.97). We use $\widehat{u}^*$ as a test function and have

$$0 < \int_I |(\widehat{u}^*)'|^p dt = \widehat{\lambda}(m) \int_I m(t)|\widehat{u}|^p dt,$$
$$\Rightarrow m \in L^+(I).$$

□

Proposition 4.8 *If $(\widehat{\lambda}(m), \widehat{u})$ is an eigenpair for problem (4.6) and $I \subseteq (0,b)$ is a nodal domain of $\widehat{u}$, then $\widehat{u}|_I$ is an eigenfunction corresponding to the eigenvalue $\widehat{\lambda}_1(m|_I, I) = \widehat{\lambda}(m)$.*

Proof Let $h \in W_0^{1,p}(I)$. We extend $h(\cdot)$ by zero on all of $(0,b)$ and we denote this extension by $\widetilde{h}$. We know that $\widetilde{h} \in W_0^{1,p}(0,b)$ and so we can use it as test function and obtain

$$\int_I |\widehat{u}'|^{p-2}\widehat{u}'h'dt = \widehat{\lambda}(m) \int_I m(t)|\widehat{u}|^{p-2}\widehat{u}hdt.$$

Since $h \in W_0^{1,p}(I)$ is arbitrary it follows that $\widehat{u}|_I$ is an eigenfunction of the eigenvalue problem on I. Since I is a nodal domain, $\widehat{u}|_I$ has a fixed sign. Recall that only the first eigenvalue has eigenfunctions of constant sign. Therefore $\widehat{\lambda}_1(m|_I, I) = \widehat{\lambda}(m)$. □

The next proposition will be helpful in establishing important properties of the spectrum.

Proposition 4.9 *If $(\widehat{\lambda}(m), \widehat{u})$ is an eigenpair for problem (6), then $Z(\widehat{u})$ is finite.*

Proof We claim that $\widehat{u}$ has a finite number of nodal domains. Arguing by contradiction, suppose we have a whole sequence $\{I_k\}_{k\in\mathbb{N}}$ of nodal domains. We have $I_i \cap I_j = \emptyset$ for all $i, j \in \mathbb{N}, i \neq j$. Using Proposition 4.8 for all $k \in \mathbb{N}$, we have

$$\begin{aligned}\widehat{\lambda}(m) = \widehat{\lambda}_1(m|_{I_k}, I_k) &\geq \widehat{\lambda}_1(\|m\|_\infty, I_k) \\ &= \frac{\widehat{\lambda}_1(1, I_k)}{\|m\|_\infty} \\ &= \frac{\widehat{\lambda}_1(1, (0,1))}{\|m\|_\infty |I_k|_1^p}\end{aligned} \tag{4.8}$$

with $|\cdot|_1$ denoting the Lebesgue measure on $\mathbb{R}$. From (4.8), we see that

$$|I_k|_1 \geq \left(\frac{\widehat{\lambda}_1(1,(0,1))}{\|m\|_\infty \widehat{\lambda}(m)}\right)^{1/p},$$

$$\Rightarrow |I|_1 = \sum_{k\geq 1} |I_k| = +\infty \text{ a contradiction.}$$

So, we have a finite number $\{I_k\}_{k=1}^n$ of nodal domains. From Proposition 4.8, we know that

$$\widehat{u}|_{I_k} \text{ is an eigenfunction corresponding to } \widehat{\lambda}_1(m|_{I_k}, I_k) = \widehat{\lambda}(m).$$

Therefore, $\widehat{u}|_{I_k}$ has constant sign (that is, $\widehat{u}|_{I_k} > 0$ or $\widehat{u}|_{I_k} < 0$). If $I_k = (a_k, b_k)$, then $Z(u) = \{a_k, b_k\}_{k=1}^n$, with $0 = a_1, b = b_n$. □

Next we prove the strict monotonicity of the maps $m \to \widehat{\lambda}_k(m, I)$ and of $I \to \widehat{\lambda}_k(m, I)$, $k \in \mathbb{N}$.

Proposition 4.10 *If $m, \widehat{m} \in L^+(0,b), m(t) \leq \widehat{m}(t)$ for a.a $t \in T$ with the inequality being strict on a set of positive measure and $I \subseteq (0,b)$ a strict subinterval such that $m|_I \in L^+(I)$, then $\widehat{\lambda}_1(\widehat{m}) < \widehat{\lambda}_1(m)$ and $\widehat{\lambda}_1(m) < \widehat{\lambda}_1(m|_I, I)$.*

Proof Let $\widehat{u}_1 \in \text{int}C_+$ be an eigenfunction corresponding to $\widehat{\lambda}_1(m)$ such that $\|\widehat{u}_1\| = 1$. Then

$$\frac{1}{\widehat{\lambda}_1(m)} = \int_0^b m(t)\widehat{u}_1^p dt < \int_0^b \widehat{m}(t)\widehat{u}_1^p dt \leq \frac{1}{\widehat{\lambda}_1(\widehat{m})} \text{ (see (4.7))}$$

$$\Rightarrow \widehat{\lambda}_1(\widehat{m}) < \widehat{\lambda}_1(m).$$

Next let $\widehat{u}_1(I) \in \text{int}C_+(I)$ be an eigenfunction corresponding to $\widehat{\lambda}_1(m|_I, I)$ with $\|\widehat{u}_1(I)\|_{W_0^{1,p}(I)} = 1$. We denote by $\widehat{u}_1(I)^*$ the extension by zero to $(0,b)$. We know that $\widehat{u}_1(I)^* \in W_0^{1,p}(0,b)$ and $\|\widehat{u}_1(I)^*\| = 1$. We have

$$\begin{aligned}\frac{1}{\widehat{\lambda}_1(m|_I, I)} &= \int_I m(t)\widehat{u}_1(I)^p dt \\ &= \int_0^b m(t)(\widehat{u}_1(I)^*)^p dt \\ &< \frac{1}{\widehat{\lambda}_1(m)} \text{ (see (4.7))}, \\ \Rightarrow \widehat{\lambda}_1(m) &< \widehat{\lambda}_1(m|_I, I).\end{aligned}$$

□

Now we look at the second eigenvalue $\widehat{\lambda}_2(m) > \widehat{\lambda}_1(m)$.

Proposition 4.11 *We can find* $\widehat{c} \in (0, b)$ *such that*

$$Z(\widehat{u}) = \{\widehat{c}\}$$

for every eigenfunction $\widehat{u}$ *corresponding to* $\widehat{\lambda}_2(m)$.

Proof We know that $\widehat{u}$ is nodal (sign changing). Let

$$I_1 = (0, c_1) \text{ and } I_2 = (c_2, b),$$

be two nodal domains for $\widehat{u}$. Suppose $c_1 < c_2$ and let $\widehat{c} \in (c_1, c_2)$. We set $E_1 = (0, \widehat{c})$, $E_2 = (\widehat{c}, b)$. Then

$$E_1 \cap E_2 = \emptyset,\ I_1 \subseteq E_1,\ I_2 \subseteq E_2 \text{ strictly.}$$

From Propositions 4.7 and 4.10, we have

$$\widehat{\lambda}_1(m|_{E_1}, E_1) < \widehat{\lambda}_1(m|_{I_1}, I_1) = \widehat{\lambda}_2(m), \tag{4.9}$$

$$\widehat{\lambda}_1(m|_{E_2}, E_2) < \widehat{\lambda}_1(m|_{I_2}, I_2) = \widehat{\lambda}_2(m), \tag{4.10}$$

Let $\widehat{u}_1(E_1) \in \text{int}C_+(E_1)$ and $\widehat{u}_1(E_2) \in \text{int}C_+(E_2)$ be eigenfunctions corresponding to the eigenvalues $\widehat{\lambda}_1(m|_{E_1}, E_1)$ and $\widehat{\lambda}_1(m|_{E_2}, E_2)$ respectively. Then

$$\frac{1}{\widehat{\lambda}_1(m|_{E_1}, E_1)} = \int_{E_1} m(t)\widehat{u}_1(E_1)^p dt,\ \frac{1}{\widehat{\lambda}_1(m|_{E_2}, E_2)} = \int_{E_2} m(t)\widehat{u}_1(E_2)^p dt.$$

We extend by zero both these eigenfunctions. We denote these extensions by $\widehat{u}_1(E_1)^*$ and $\widehat{u}_1(E_2)^*$ respectively. We know that

$$\widehat{u}_1(E_1)^*,\ \widehat{u}_1(E_2)^* \in W_0^{1,p}(0, b).$$

Let $V = \text{span}\{\widehat{u}_1(E_1)^*, \widehat{u}_1(E_2)^*\} \subseteq W_0^{1,p}(0, b)$ and $C = V \cap \partial B_1$. We have $\gamma(C) = 2$ (see Proposition 3.70). Let $\beta_1, \beta_2 \in \mathbb{R}$ and set

$$\begin{aligned} v &= \beta_1\widehat{u}_1(E_1)^* + \beta_2\widehat{u}_1(E_2)^*, \\ \Rightarrow &\|v\| = 1 \text{ if and only if } |\beta_1|^p + |\beta_2|^p = 1. \end{aligned} \tag{4.11}$$

Then we have

$$\begin{aligned} &\frac{1}{\widehat{\lambda}_2}(m) \\ &\geq \inf_{u \in C} \int_0^b m(t)|u|^p dt \end{aligned}$$

$$= \inf_{\beta_1,\beta_2 \in K} [|\beta_1|^p \int_{E_1} m(t)\widehat{u}_1(E_1)^p dt + |\beta_2|^p \int_{E_2} m(t)\widehat{u}_1(E_2)^p dt]$$

with $K \subseteq \mathbb{R}$ compact

$$= |\widehat{\beta}_1|^p \int_{E_1} m(t)\widehat{u}_1(E_1)^p dt + |\widehat{\beta}_2|^p \int_{E_2} m(t)\widehat{u}_1(E_2)^p dt \text{ for some } \widehat{\beta}_1, \widehat{\beta}_2 \in \mathbb{R}$$

$$= \frac{|\beta_1|^p}{\widehat{\lambda}_1(m|_{E_1}, E_1)} + \frac{|\beta_2|^p}{\widehat{\lambda}_1(m|_{E_2}, E_2)}$$

$$> \frac{|\beta_1|^p + |\beta_2|^p}{\widehat{\lambda}_2(m)} \text{ (see (4.9), (4.10))}$$

$$= \frac{1}{\widehat{\lambda}_2(m)} \text{ (see (4.11))},$$

a contradiction. This proves that $c_1 = c_2 = \widehat{c}$ and this means that $\widehat{u}$ has a unique zero $\widehat{c} \in (0, b)$.

Now let $\widehat{v}$ be another eigenfunction corresponding to $\widehat{\lambda}_2(m)$. From the previous part of the proof we know that

$$Z(\widehat{v}) = \{\widehat{c}_0\}.$$

Suppose $\widehat{c}_0 \neq \widehat{c}$ and without any loss of generality, we may assume that $\widehat{c}_0 < \widehat{c}$. Using Propositions 4.8 and 4.10, we have

$$\widehat{\lambda}_2(m) = \widehat{\lambda}_1(m|_{(0,\widehat{c})}, (0, \widehat{c})) < \widehat{\lambda}_1(m|_{(0,\widehat{c}_0)}, (0, \widehat{c}_0)) = \widehat{\lambda}_2(m),$$

a contradiction. Therefore $\widehat{c}_0 = \widehat{c}$. □

Recall that an eigenvalue $\widehat{\lambda}$ of (4.6) is simple, if the corresponding eigenspace is one-dimensional, that is, if $\widehat{u}, \widehat{v}$ are eigenfunctions corresponding to $\widehat{\lambda}$, then $\widehat{u} = \vartheta\widehat{v}$ for some $\vartheta \in \mathbb{R} \setminus \{0\}$.

Proposition 4.12 *$\widehat{\lambda}_2(m)$ is simple and $\widehat{\lambda}_2(m) < \widehat{\lambda}_3(m)$.*

Proof Consider $\widehat{u}$, $\widehat{v}$ two eigenfunctions corresponding to $\widehat{\lambda}_2(m)$. From Proposition 4.11, we know that

$$Z(\widehat{u}) = Z(\widehat{v}) = \{\widehat{c}\}.$$

From Proposition 4.8, we know that

$\widehat{u}|_{(0,\widehat{c})}$, $\widehat{v}|_{(0,\widehat{c})}$ are eigenfunctions of $\widehat{\lambda}_1(m|_{(0,\widehat{c})}, (0, \widehat{c}))$ and $\widehat{u}|_{(\widehat{c},b)}$, $\widehat{v}|_{(\widehat{c},b)}$ are eigenfunctions of $\widehat{\lambda}_1(m|_{(\widehat{c},b)}, (\widehat{c}, b))$.

But we know that

$$\widehat{\lambda}_1(m|_{(0,\widehat{c})}, (0, \widehat{c})), \widehat{\lambda}_1(m|_{(\widehat{c},b)}, (\widehat{c}, b)) \text{ are simple}$$

Therefore $\widehat{u} = \vartheta\widehat{v}$ for some $\vartheta \in \mathbb{R} \setminus \{0\}$. This implies that $\widehat{\lambda}_2(m)$ is simple. Finally the simplicity of $\widehat{\lambda}_2(m)$ implies via Theorem 3.72 that $\widehat{\lambda}_2(m) < \widehat{\lambda}_3(m)$. □

As we did for the first eigenvalue, we examine the monotonicity of the maps $m \to \widehat{\lambda}_2(m)$ and $I \to \widehat{\lambda}_2(m|_I, I)$.

Proposition 4.13

(a) If $m, \widehat{m} \in L^+(0,b)$, $m(t) \le \widehat{m}(t)$ for a.a $t \in (0,b)$ and the inequality is strict on a set of positive measure, then $\widehat{\lambda}_2(\widehat{m}) < \widehat{\lambda}_2(m)$.
(b) If $I \subseteq (0,b)$ is a strict subinterval and $m|_I \in L^+(I)$, then $\widehat{\lambda}_2(m) < \widehat{\lambda}_2(m|_I, I)$.

Proof

(a) From Proposition 4.11, we know that

$$Z(\widehat{u}) = \{\widehat{c}\} \text{ for every eigenfunction } \widehat{u} \text{ of } \widehat{\lambda}_2(m),$$
$$Z(\widehat{v}) = \{\widehat{c}_0\} \text{ for every eigenfunction } \widehat{v} \text{ of } \widehat{\lambda}_2(\widehat{m}).$$

First assume that $\widehat{c} = \widehat{c}_0 = c^*$. At least one of the following holds

$$|\{t \in (0,b) : m(t) < \widehat{m}(t)\} \cap (0, c^*)|_1 > 0,$$
$$|\{t \in (0,b) : m(t) < \widehat{m}(t)\} \cap (c^*, b)|_1 > 0.$$

Using Propositions 4.8 and 4.10, we have

$$\widehat{\lambda}_2(\widehat{m}) = \widehat{\lambda}_1(\widehat{m}|_{(0,c^*)}, (0,c^*)) < \widehat{\lambda}_1(m|_{(0,c^*)}, (0,c^*)) = \widehat{\lambda}_2(m),$$
$$\widehat{\lambda}_2(\widehat{m}) = \widehat{\lambda}_1(\widehat{m}|_{(c^*,b)}, (c^*,b)) < \widehat{\lambda}_1(m|_{(c^*,b)}, (c^*,b)) = \widehat{\lambda}_2(m).$$

Therefore $\widehat{\lambda}_2(\widehat{m}) < \widehat{\lambda}_2(m)$.
Next suppose $\widehat{c} < \widehat{c}_0$. Then using again Propositions 4.8 and 4.11, we have

$$\begin{aligned}\widehat{\lambda}_2(\widehat{m}) &= \widehat{\lambda}_1(m|_{(0,\widehat{c}_0)}, (0,\widehat{c}_0)) \\ &< \widehat{\lambda}_1(m|_{(0,\widehat{c})}, (0,\widehat{c})) = \widehat{\lambda}_2(m).\end{aligned}$$

Finally if $\widehat{c}_0 < \widehat{c}$, then as above we have

$$\begin{aligned}\widehat{\lambda}_2(\widehat{m}) &= \widehat{\lambda}_1(m|_{(\widehat{c}_0,b)}, (\widehat{c}_0,b)) \\ &< \widehat{\lambda}_1(m|_{(\widehat{c},b)}, (\widehat{c},b)) = \widehat{\lambda}_2(m).\end{aligned}$$

(b) We know that

$$Z(\widehat{u}_I) = \{\widehat{c}_I\} \text{ for every eigenfunction } \widehat{u}_I \text{ of } \widehat{\lambda}_2(m|_I, I),$$
$$Z(\widehat{u}) = \{\widehat{c}\} \text{ for every eigenfunction } \widehat{u} \text{ of } \widehat{\lambda}_2(m).$$

Reasoning as in part (a)(by examining the three different possibilities), we obtain $\widehat{\lambda}_2(m) < \widehat{\lambda}_2(m|_I, I)$.

□

Next we prove the converse of Proposition 4.11. So, we show that $\widehat{\lambda}_2(m)$ is the only eigenvalue with eigenfunctions that have a singleton (the same singleton) as zero set.

Proposition 4.14 *If $\widehat{\lambda}(m)$ is an eigenvalue of (4.6) such that $Z(\widehat{u}) = \{\widehat{c}\}$ for every eigenfunction $\widehat{u}$ of $\widehat{\lambda}(m)$, then $\widehat{\lambda}(m) = \widehat{\lambda}_2(m)$.*

Proof From Proposition 4.11, we know that

$$Z(\widehat{v}) = \{\widehat{c}_0\} \text{ for every eigenfunction } \widehat{v} \text{ of } \widehat{\lambda}_2(m).$$

First assume that $\widehat{c} < \widehat{c}_0$. We have

$$\begin{aligned}\widehat{\lambda}(m) &= \widehat{\lambda}_1(m|_{(\widehat{c},b)}, (\widehat{c}, b)) \\ &< \widehat{\lambda}_1(m|_{(\widehat{c}_0,b)}, (\widehat{c}_0, b)) = \widehat{\lambda}_2(m).\end{aligned}$$

Similarly, we have

$$\begin{aligned}\widehat{\lambda}_2(m) &= \widehat{\lambda}_1(m|_{(0,\widehat{c}_0)}, (0, \widehat{c}_0)) \\ &< \widehat{\lambda}_1(m|_{(0,\widehat{c})}, (0, \widehat{c})) = \widehat{\lambda}(m),\end{aligned}$$

a contradiction. Hence $\widehat{c}_0 \leq \widehat{c}$.

In a similar way we show that $\widehat{c} \leq \widehat{c}_0$ and we conclude that $\widehat{c} = \widehat{c}_0$. Then it follows that $\widehat{\lambda}(m) = \widehat{\lambda}_2(m)$. □

Now for the higher eigenvalues, we prove these properties, using an induction process. Recall that every eigenvalue has eigenfunctions with finite zero set (see Proposition 4.9). The hypotheses for the induction process, are the following:

H_1 : For $k \in \mathbb{N}$, $k \geq 2$, every eigenfunction $\widehat{u}$ of $\widehat{\lambda}_k(m)$ satisfies $Z(\widehat{u}) = \{\widehat{c}_k^i\}_{i=1}^{k-1}$.
H_2 : For $k \in \mathbb{N}$, $k \geq 2$, $\widehat{\lambda}_k(m)$ is simple.
H_3 : For $k \in \mathbb{N}$, $k \geq 2$, $\widehat{\lambda}_1(m) < \widehat{\lambda}_2(m) < \cdots < \widehat{\lambda}_{k+1}(m)$.
H_4 : If $(\widehat{\lambda}(m, I), \widehat{u})$ is an eigenpair of $(4.6)|_I$ and $Z(\widehat{u}) = \{\widehat{c}_k^i\}_{i=1}^{k-1}$, then $\widehat{\lambda}(m) = \widehat{\lambda}_k(m)$.
H_5 : For $k \in \mathbb{N}$, $k \geq 2$, the following maps are strictly monotone $m \to \widehat{\lambda}_k(m)$ and $I \to \widehat{\lambda}_k(m|_I, I)$ provided $m|_I \in L^+(I)$.

From the previous results we know that these five hypotheses are satisfied for $k = 2$. Next we assume that they hold for $k \geq 2$, $k \in \mathbb{N}$ and we will show that they are also true for $k + 1$. First the property of H_1.

Proposition 4.15 *We can find* $\{\widehat{c}_i\}_{i=1}^k \subseteq (0, b)$ *such that*

$$Z(\widehat{u}) = \{\widehat{c}_i\}_{i=1}^k$$

for every eigenfunction $\widehat{u}$ *of* $\widehat{\lambda}_{k+1}(m)$.

Proof From the induction hypotheses H_1 and H_4, we see that $Z(\widehat{u})$ has at least k-elements. So, we have at least $(k+1)$-nodal domains

$$I_1 = (0, \widehat{c}_1), I_2 = (\widehat{c}_1, \widehat{c}_2), \cdots, I_k = (\widehat{c}_{k-1}, \widehat{c}_k), I_{k+1} = (c, b).$$

We claim that $c = \widehat{c}_k$. If this is not true, then $\widehat{c}_k < c$ and we choose $c^* \in (\widehat{c}_k, c)$. We set

$$E_1 = (0, c^*), E_2 = (c^*, b).$$

Using the induction hypotheses H_4 and H_5, we obtain

$$\widehat{\lambda}_k(m|_{E_1}, E_1) < \widehat{\lambda}_k(m|_{(0,\widehat{c}_k)}, (0, \widehat{c}_k)) = \widehat{\lambda}_{k+1}(m|_{E_1}, E_1), \tag{4.12}$$

$$\widehat{\lambda}_1(m|_{E_2}, E_2) < \widehat{\lambda}_1(m|_{(c,b)}, (c, b)) = \widehat{\lambda}_{k+1}(m|_{E_2}, E_2). \tag{4.13}$$

Let $(\widehat{\lambda}_1(m|_{E_2}, E_2), \widehat{u}_{k+1})$ be an eigenpair of $(4.6)|_{E_2}$ and $(\widehat{\lambda}_n(m|_{E_1}, E_1), y)$ an eigenpair of $(4.6)|_{E_1}$. We set $u_i = y|_{I_i}$ for $i = 1, \cdots, k$. We extend by zero to all of $(0, b)$. We denote these extensions by u_i^*. Let

$$V_{k+1} = \text{span}\{u_i^*\}_{i=1}^{k+1}, C_{k+1} = V_{k+1} \cap \partial B_1.$$

We know that $\gamma(C_{k+1}) = k+1$. Then

$$\frac{1}{\widehat{\lambda}_{k+1}(m)} \geq \min_{y \in C_{k+1}} \int_0^b m(t)|y|^p dt > \frac{1}{\widehat{\lambda}_{k+1}(m)},$$

a contradiction. Therefore $c = c_k$. So,

$$Z(\widehat{u}) = \{c_i\}_{i=1}^k.$$

Let $\widehat{v}$ be another eigenfunction of $\widehat{\lambda}_{k+1}(m)$. We have

$$Z(\widehat{v}) = \{c_i^*\}_{i=1}^k.$$

Suppose $c_1 \neq c_1^*$. Then

$$\widehat{\lambda}_{k+1}(m) = \widehat{\lambda}_1(m|_{(0,c_1^*)}, (0, c_1^*)) \neq \widehat{\lambda}_1(m|_{(0,c_1)}, (0, c_1)) = \widehat{\lambda}_{k+1}(m)$$

a contradiction. So $c_1 = c_1^*$. Similarly we show that $c_i = c_i^*$ for all $i = 2, \cdots, k$.

□

This proves that every eigenvalue $\widehat{\lambda}_k(m)$ has a finite zero set $Z(\widehat{u}) = \{\widehat{c}_i\}_{i=1}^{k-1}$ the same for all eigenfunctions $\widehat{u}$ corresponding to $\widehat{\lambda}_k(m)$.

Proposition 4.16 *$\widehat{\lambda}_{k+1}(m)$ is simple and $\widehat{\lambda}_{k+1}(m) < \widehat{\lambda}_{k+2}(m)$.*

Proof Let $\widehat{u}$, $\widehat{v}$ be two eigenfunctions corresponding to the eigenvalue $\widehat{\lambda}_{k+1}(m)$. From Proposition 4.15, we know that

$$Z(\widehat{u}) = Z(\widehat{v}) = \{\widehat{c}_i\}_{i=1}^{k}. \tag{4.14}$$

Consider the functions $\widehat{u}|_{(0,\widehat{c}_1)}, \widehat{v}|_{(0,\widehat{c}_1)}$ and $\widehat{u}|_{(\widehat{c}_1,b)}, \widehat{v}|_{(\widehat{c}_1,b)}$. From Proposition 4.8, we know that

$$\widehat{u}|_{(0,\widehat{c}_1)}, \widehat{v}|_{(0,\widehat{c}_1)} \text{ are eigenfunctions of } \widehat{\lambda}_1(m|_{(0,\widehat{c}_1)}, (0, \widehat{c}_1)),$$
$$\widehat{u}|_{(\widehat{c}_1,b)}, \widehat{v}|_{(\widehat{c}_1,b)} \text{ are eigenfunctions of } \widehat{\lambda}_1(m|_{(\widehat{c}_1,b)}, (\widehat{c}_1, b)).$$

Then from H_2, we have that there exist $\vartheta_1, \vartheta_2 \in \mathbb{R} \setminus \{0\}$ such that

$$\widehat{u}|_{(0,\widehat{c}_1)} = \vartheta_1 \widehat{v}|_{(0,\widehat{c}_1)} \text{ and } \widehat{u}|_{(\widehat{c}_1,b)} = \vartheta_2 \widehat{v}|_{(\widehat{c}_1,b)}.$$

We know that $\widehat{u}, \widehat{v} \in C^1[0, b]$ and $\widehat{u}'(\widehat{c}_1) \neq 0$. Therefore $\vartheta_1 = \vartheta_2$.
Finally from (4.14) (see also Theorem 3.72), we have

$$\widehat{\lambda}_{k+1}(m) < \widehat{\lambda}_{k+2}(m).$$

□

Therefore, we have proved that every eigenvalue $\widehat{\lambda}_k(m)$ is simple.
Reasoning as in the proof of Proposition 4.13, we obtain.

Proposition 4.17

(a) If $m, \widehat{m} \in L^+(0, b)$, $m(t) \leq \widehat{m}(t)$ and the inequality is strict on a set of positive measure, then $\widehat{\lambda}_{k+1}(\widehat{m}) < \widehat{\lambda}_{k+1}(m)$.
(b) If $I \subseteq (0, b)$ is strict subinterval and $m|_I \in L^+(I)$, then $\widehat{\lambda}_{k+1}(m) < \widehat{\lambda}_{k+1}(m|_I, I)$.

So by induction we have proved the strict monotonicity of the maps $m \to \widehat{\lambda}_{k+1}(m)$, $I \to \widehat{\lambda}_{k+1}(m|_I, I)$.
It remains to complete the induction for H_4.

Proposition 4.18 *If $\widehat{\lambda}(m)$ is an eigenvalue for which the eigenfunctions $\widehat{u}$ satisfy $Z(\widehat{u}) = \{c_i^*\}_{i=1}^k$, then $\widehat{\lambda}(m) = \widehat{\lambda}_{k+1}(m)$.*

Proof We already know that every $\widehat{v}$ eigenfunction of $\widehat{\lambda}_{k+1}(m)$ satisfies

$$Z(\widehat{v}) = \{c_i^*\}_{i=1}^k.$$

We need to show that $c_i^* = \widehat{c}_i$ for $i \in \{1, \cdots, k\}$.
First suppose that $\widehat{c}_1 < c_1^*$. We have

$$\begin{aligned}
\widehat{\lambda}(m) &= \widehat{\lambda}_1(m|_{(0,c_1^*)}, (0, c_1^*)) \text{ (see Proposition 4.8)} \\
&< \widehat{\lambda}_1(m|_{(0,\widehat{c}_1)}, (0, \widehat{c}_1)) \text{ (see Proposition 4.10)} \\
&= \widehat{\lambda}_{k+1}(m) \text{ (again Proposition 4.8)} \\
&= \widehat{\lambda}_k(m|_{(\widehat{c}_1,b)}, (\widehat{c}_1, b)) \text{ (by the induction hypothesis } H_5) \\
&< \widehat{\lambda}_k(m|_{(c_1^*,b)}, (c_1^*, b)) \text{ (see Proposition 4.17)} \\
&= \widehat{\lambda}(m) \text{ (by the induction hypothesis } H_5),
\end{aligned}$$

a contradiction.

Similarly we treat the case $c_1^* < \widehat{c}_1$. Therefore $c_1^* = \widehat{c}_1$ and by the induction hypothesis we have $\widehat{\lambda}(m) = \widehat{\lambda}_{k+1}(m)$. □

We have completed the induction and we can state the following theorem which gives us a description of the spectrum of problem (4.6).

Theorem 4.19 *If $m \in L^+(0, b)$, then problem (4.6) has a sequence $\{\widehat{\lambda}_k(m)\}_{k\in\mathbb{N}}$ of eigenvalues which have the following properties*

(a) $Z(\widehat{u}) = \{\widehat{c}_i\}_{i=1}^{k-1}$ *for all eigenfunctions $\widehat{u}$ of $\widehat{\lambda}_k(m)$, $k \in \mathbb{N}$.*
(b) For every $k \in \mathbb{N}$, $\widehat{\lambda}_k(m)$ is simple and

$$0 < \widehat{\lambda}_1(m) < \widehat{\lambda}_2(m) < \cdots < \widehat{\lambda}_k(m) \to \infty \text{ as } k \to \infty.$$

(c) If $m, \widehat{m} \in L^+(0, b)$, $m(t) \le \widehat{m}(t)$ for a.a $t \in T$ and the inequality is strict on a set of positive measure, then $\widehat{\lambda}_k(\widehat{m}) < \widehat{\lambda}_k(m)$ for all $k \in \mathbb{N}$.
(d) If $I \subseteq (0, b)$ is a strict subinterval and $m|_I \in L^+(I)$, then $\widehat{\lambda}_k(m) < \widehat{\lambda}_k(m|_I, I)$.

Remark 4.20 If $m \in L^\infty(0, b) \setminus \{0\}$ and $m(t) \ge 0$ for a.a $t \in I$, then the eigenvalues of the above theorem are the whole spectrum of (4.6). If $m^- \ne 0$, then there is also a negative part of the spectrum given by

$$\widehat{\lambda}_{-k}(m) = -\widehat{\lambda}_k(-m) \text{ for all } k \in \mathbb{N}.$$

Hence the full spectrum of (4.6) is

$$\sigma(m) = \{\widehat{\lambda}_k(m)\}_{k\in\mathbb{N}} \cup \{\widehat{\lambda}_{-k}(m)\}_{k\in\mathbb{N}}.$$

From (4.7) we have the following variational characterization of the eigenvalues.

Proposition 4.21 *If* $\widehat{\mathcal{F}}_k = \{V \subseteq W_0^{1,p}(0,b) : k-dimensional\ subspace\}$, $k \in \mathbb{N}$, *then* $\widehat{\lambda}_k(m)^{-1} = \sup\limits_{v\in\widehat{\mathcal{F}}_k} \inf\limits_{v\in V\cap\partial B_1} \int_0^b m(t)|v|^p dt$.

A similar analysis can be conducted for the following nonlinear Neumann problem

$$\left\{\begin{array}{l} -\varphi_p(u')' + \xi(t)\varphi_p(u) = \widehat{\lambda}m(t)\varphi_p(u) \text{ in } (0,b), \\ u'(0) = u'(b) = 0. \end{array}\right\} \tag{4.15}$$

In this problem $\xi \in L^\infty(0,b)\setminus\{0\}$, $\xi(t) \geq 0$ for a.a $t \in (0,b)$ and $m \in L^+(0,b)$. The following theorem describes the positive part of the spectrum of (4.15) and is the counterpart of Theorem 4.19 for Neumann problems.

Theorem 4.22 *If the data of* (4.15) *are as above, then there is a sequence* $\{\widehat{\lambda}_k(m)\}_{k\in\mathbb{N}}$ *of eigenvalues such that*

(a) $Z(\widehat{u}) = \{\widehat{c}_i\}_{i=1}^{k-1}$ *for every eigenfunction* $\widehat{u}$ *of* $\widehat{\lambda}_k(m)$, $k \in \mathbb{N}$.
(b) For every $k \in \mathbb{N}$, $\widehat{\lambda}_k(m)$ *is simple and*

$$0 < \widehat{\lambda}_1(m) < \widehat{\lambda}_2(m) < \cdots < \widehat{\lambda}_k(m) \to +\infty \text{ as } k \to +\infty.$$

(c) If $m, \widehat{m} \in L^+(0,b)$, $m(t) \leq \widehat{m}(t)$ *for a.a* $t \in T$ *and the inequality is strict on a set of positive measure, then* $\widehat{\lambda}_k(\widehat{m}) < \widehat{\lambda}_k(m)$ *for all* $k \in \mathbb{N}$.
(d) If $I \subseteq (0,b)$ *is a strict subinterval and* $m|_I \in L^+(I)$, *then* $\widehat{\lambda}_k(m) < \widehat{\lambda}_k(m|_I, I)$ *for all* $k \in \mathbb{N}$.

Now we turn our attention to the vector p-Laplacian and will briefly describe some basic spectral properties of this operator. The situation is more complicated compared to the scalar case.

Now we deal with systems and so the ambient space is $\mathbb{R}^N (N \geq 2)$. As before for $p \in (1,\infty)$ we consider the homeomorphism $\varphi_p : \mathbb{R}^N \to \mathbb{R}^N$ defined by

$$\varphi_p(x) = \left\{\begin{array}{ll} |x|^{p-2}x & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{array}\right., \ 1 < p < \infty.$$

We consider the following nonlinear eigenvalue problems.

$$-\varphi_p(u'(t))' = \widehat{\lambda}\varphi_p(u(t)) \text{ in } (0,b),\ u(0) = u(b) = 0 \quad \text{(Dirichlet eigenvalue problem)}, \tag{4.16}$$

$$-\varphi_p(u'(t))' = \widehat{\lambda}\varphi_p(u(t)) \text{ in } (0,b),\ u(0) = u(b),\ u'(0) = u'(b) \quad \text{(periodic eigenvalue problem)}, \tag{4.17}$$

$$-\varphi_p(u'(t))' = \widehat{\lambda}\varphi_p(u(t)) \text{ in } (0,b),\ u'(0) = u'(b) = 0 \quad \text{(Neumann eigenvalue problem)}. \tag{4.18}$$

The relevant Sobolev spaces for these problems are:

$W_0^{1,p}((0,b),\mathbb{R}^N) = W_{0,N}^{1,p}$ for (4.16),

$W_{per}^{1,p}((0,b),\mathbb{R}^N) = W_{per,N}^{1,p} = \{u \in W^{1,p}((0,b),\mathbb{R}^N) : u(0) = u(b)\}$ for (4.17),

$W^{1,p}(0,b),\mathbb{R}^N) = W_N^{1,p}$ for (4.18).

Definition 4.23 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" for problem (4.16) (resp. (4.17), (4.18)), if the problem has a nontrivial weak solution $\widehat{u} \in W_{0,N}^{1,p}$ (resp. $\widehat{u} \in W_{per,N}^{1,p}$, $\widehat{u} \in W_N^{1,p}$) known as an "eigenfunction" corresponding to the eigenvalue $\widehat{\lambda}$. So, we have

$$\int_0^b |\widehat{u}'|^{p-2}(u', h')_{\mathbb{R}^N} dt = \widehat{\lambda} \int_0^b |\widehat{u}|^{p-2}(\widehat{u}, h)_{\mathbb{R}^N} dt$$

for all $h \in W_{0,N}^{1,p}$ (resp. $h \in W_{per,N}^{1,p}$, $h \in W_N^{1,p}$).

Remark 4.24 As before, we can check that every eigenfunction $\widehat{u}$ belongs in $C^1([0,b],\mathbb{R}^N) = C_N^1[0,b]$.

Let $\sigma_p^D(N)$ (resp. $\sigma_p^P(N)$, $\sigma_p^N(N)$) be the spectrum for problem (4.16) (resp. (4.17), (4.18)).

For the Dirichlet and Neumann problems, there is no difference from the scalar case (see Manasevich and Mawhin [325]).

Theorem 4.25 $\sigma_p^D(N) = \{\widehat{\lambda}_k^D = (\frac{k\pi_p}{b})^p\}_{k\in\mathbb{N}}$, *the eigenvalues are simple and the corresponding eigenfunctions are*

$$\widehat{u}_k^D(t) = \sin_p(\frac{k\pi_p}{b}t)y \text{ for all } t \in [0,b], \text{ with } y \in \mathbb{R}^N.$$

Theorem 4.26 $\sigma_p^N(N) = \{\widehat{\lambda}_k^N = (\frac{k\pi_p}{b})^p\}_{k\in\mathbb{N}_0}$, *the eigenvalues are simple and the corresponding eigenfunctions are*

$$\widehat{u}_k^N(t) = \cos_p(\frac{k\pi_p}{b}t)y \text{ for all } t \in [0,b], \text{ with } y \in \mathbb{R}^N.$$

(recall $\cos_p t = \sin_p(t + \frac{\pi}{2})$ *and so* $\cos_p(\cdot)$ *is even,* $2\pi_p$ *periodic).*

For the Dirichlet eigenvalue problem consider the energy functional $\varphi : W_{0,N}^{1,p} \to \mathbb{R}$ defined by

$$\varphi(u) = \|u'\|_p \text{ for all } u \in W_{0,N}^{1,p}$$

and the set $C_D = W_{0,N}^{1,p} \cap \partial B_1^{L_N^p}$ where

$$\partial B_1^{L_N^p} = \{u \in L^p((0,b), \mathbb{R}^N) = L_N^p : \|u\|_p = 1\}.$$

We consider the constrained minimization problem

$$\inf\{\varphi(u) : u \in C_D\} = m_D. \tag{4.19}$$

Proposition 4.27 *There exists $\widehat{u} \in C_D$ such that $m_D = \varphi(\widehat{u})$ and $m_D = \widehat{\lambda}_1^D$.*

Proof Exploiting the compact embedding of $W_{0,N}^{1,p}$ into

$$C([0,b], \mathbb{R}^N) = C_N[0,b]$$

(see Theorem *I*-4.130), we see that $C_D \subseteq W_{0,N}^{1,p}$ is sequentially weakly closed. The functional $\varphi_D(\cdot)$ is continuous, convex thus sequentially weakly lower semicontinuous. Moreover, using the Poincare inequality (see Theorem *I*-4.112), we see that $\varphi_D(\cdot)$ is coercive. Therefore by the Weierstrass-Tonelli theorem (see Proposition *I*-1.180) we know that we can find $\widehat{u} \in C_D$ such that

$$\varphi_D(\widehat{u}) = m_D.$$

Hence $m_D > 0$, or otherwise $\widehat{u}$ =constant, thus $\widehat{u} \equiv 0$, a contradiction. □

From the Lagrange multiplier rule (see Theorem 3.87 and Remark 3.88), we can find $\vartheta \in \mathbb{R}$ such that

$$\int_0^b |\widehat{u}'|^{p-2}(\widehat{u}', h')_{\mathbb{R}^N} dt + \vartheta \int_0^b |\widehat{u}|^{p-2}(\widehat{u}, h)_{\mathbb{R}^N} dt = 0 \text{ for all } h \in W_{0,N}^{1,p}. \tag{4.20}$$

$\Rightarrow -\varphi_p(\widehat{u}'(t))' = -\vartheta\varphi_p(\widehat{u}(t))$ for a. a $t \in (0,b)$, $\widehat{u}(0) = \widehat{u}(b) = 0$
$\Rightarrow \widehat{u}$ is an eigenfunction with $-\vartheta$ as the eigenvalue.

In (4.20) we choose $h = \widehat{u} \in W_{0,N}^{1,p}$. We obtain

$$\|\widehat{u}'\|_p^p = -\vartheta\|\widehat{u}\|_p^p = -\vartheta \text{ (since } \widehat{u} \in C_D)$$

$$\Rightarrow -\vartheta = m_D = \widehat{\lambda}_1^D = (\frac{\pi_p}{b})^p.$$

□

According to the above proposition

$$\widehat{\lambda}_1^D = \inf\{\|u'\|_p^p : u \in W_{0,N}^{1,p}, \|u\|_p^p = 1\} \tag{4.21}$$

From (4.21) follows the Poincare inequality for the Sobolev space $W_{0,N}^{1,p}$.

Corollary 4.28 $(\frac{\pi_p}{b})^p \|u\|_p^p \leq \|u'\|_p^p$ *for all* $u \in W_{0,N}^{1,p}$.

For the Neumann problem according to Theorem 4.26 the smallest eigenvalues is $\widehat{\lambda}_0^N = 0$. We consider the first nonzero eigenvalue $\widehat{\lambda}_1^N$ and we will produce a variational characterization for this eigenvalue.

Let $\widehat{\lambda} \in \sigma_p^N(N)$ and let $\widehat{u} \in C_N^1[0, b]$ be a corresponding eigenfunction. We have

$$-\varphi_p(\widehat{u}')' = \varphi_p(\widehat{u})$$

Integrating on $(0, b)$ and using the Neumann boundary condition, we obtain

$$\int_0^b |\widehat{u}|^{p-2}\widehat{u}dt = 0$$

We introduce the following subset of $W_N^{1,p}$.

$$C_N = \{u \in W_N^{1,p} : \|u\|_p = 1, \int_0^b |u|^{p-2}udt = 0\}$$

and consider the following minimization problem

$$\inf\{\widehat{\varphi}(u) = \|u'\|_p^p : u \in C_N\} = m_N. \tag{4.22}$$

Proposition 4.29 *There exists* $\widehat{u} \in \widehat{D}$ *such that*

$$\widehat{\varphi}(u) = m_N$$

and if $p \geq 2$, *then* $m_N = (\frac{\pi_p}{b})^p = \widehat{\lambda}_1^N$.

Proof Exploiting the compact embedding of $W_N^{1,p} \hookrightarrow C_N[0, b]$, we see that C_N is sequentially weakly closed. Also, $\widehat{\varphi}(\cdot)$ is sequentially weakly lower semicontinuous and on account of the Poincare-Wirtinger inequality (see Theorem I-4.119), it is coercive. So, by the Weierstrass-Tonelli theorem, we can find $\widehat{u} \in C_N$ such that

$$\widehat{\varphi}(u) = \widehat{m}_N.$$

Let $\widehat{k} : W_N^{1,p} \to \mathbb{R}$ and $\widehat{\xi} : W_N^{1,p} \to \mathbb{R}^N$ be defined by $\widehat{k}(u) = \|u\|_p^p - 1$ and $\widehat{\xi}(u) = \int_0^b |u|^{p-2}udt$ for all $u \in W_N^{1,p}$. We have

$$m_N = \inf\{\widehat{\varphi}(u) : \widehat{k}(u) = 0, \widehat{\xi}(u) = 0, u \in W_N^{1,p}\}.$$

We know that $\widehat{k} \in C^1(W_N^{1,p})$, $\widehat{\xi} \in C^1(W_N^{1,p}, \mathbb{R}^N)$ and $\widehat{k}'(u) = p\varphi_p(u(\cdot))$, $\widehat{\xi}'(u) = \varphi_p'(u(\cdot))$ for all $u \in W_N^{1,p}$. Recall that

$$\varphi_p'(x) = |x|^{p-2}[id + (p-2)\frac{x \otimes x}{|x|^2}] \text{ for all } x \in \mathbb{R}^N \setminus \{0\}.$$

Using the Lagrange multiplier rule, we can find $(\mu, \vartheta, \eta) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^N$, $(\mu, \vartheta, \eta) \neq (0, 0, 0)$ such that

$$\begin{aligned} &\mu \int_0^b p|\widehat{u}'|^{p-2}(\widehat{u}', h')_{\mathbb{R}^N} dt + \vartheta \int_0^b p|\widehat{u}|^{p-2}(\widehat{u}, h)_{\mathbb{R}^N} dt \\ &+ \int_0^b |\widehat{u}|^{p-2}((\eta, h)_{\mathbb{R}^N} + (p-2)\frac{(\widehat{u}, \eta)_{\mathbb{R}^N}(\widehat{u}, h)_{\mathbb{R}^N}}{|\widehat{u}|^2})dt \text{ for all } h \in W_N^{1,p}. \end{aligned} \tag{4.23}$$

If we choose $h = \eta \in \mathbb{R}^N$, then since $h' = 0$ and

$$\int_0^b |\widehat{u}|^{p-2}(\widehat{u}, \eta)_{\mathbb{R}^N} dt = 0,$$

We obtain

$$\begin{aligned} &\int_0^b |\widehat{u}|^{p-2}|\eta|^2 dt + \int_0^b |\widehat{u}|^{p-4}(\widehat{u}, \eta)_{\mathbb{R}^N}^2 dt = 0, \\ &\Rightarrow |\eta|^2 \int_0^b |\widehat{u}|^{p-2} dt = 0, \\ &\Rightarrow \eta = 0. \end{aligned}$$

Therefore (4.23) becomes

$$\mu p \int_0^b |\widehat{u}'|^{p-2}(\widehat{u}', h')_{\mathbb{R}^N} dt + \vartheta p \int_0^b |\widehat{u}|^{p-2}(\widehat{u}, h)_{\mathbb{R}^N} dt = 0 \quad \text{for all } h \in W_N^{1,p}. \tag{4.24}$$

Choosing $h = \widehat{u} \in W_N^{1,p}$, we obtain

$$\mu p \|\widehat{u}'\|_p^p + \vartheta p \|\widehat{u}\|_p^p = 0$$

If $\mu = 0$, then $\vartheta p \|\widehat{u}\|_p^p = \vartheta p = 0$, hence $\vartheta = 0$, a contradiction since $(\mu, \vartheta, \eta) \neq (0, 0, 0)$. So, $\mu \neq 0$ and so (4.24) becomes

$$\int_0^b |\widehat{u}'|^{p-2}(\widehat{u}', h')_{\mathbb{R}^N} dt + \frac{\vartheta}{\mu} \int_0^b |\widehat{u}|^{p-2}(\widehat{u}, h)_{\mathbb{R}^N} dt = 0$$

for all $h \in W_N^{1,p}$.

$$\begin{aligned} &\Rightarrow -\varphi_p(\widehat{u}')' = -\frac{\vartheta}{\mu}\varphi_p(\widehat{u}), \\ &\Rightarrow \widehat{u} \text{ is an eigenfunction with eigenvalue } -\frac{\vartheta}{\mu}. \end{aligned} \tag{4.25}$$

In (4.25) use $h = \widehat{u} \in W_N^{1,p}$ to obtain

$$\|\widehat{u}'\|_p^p + \frac{\vartheta}{\mu}\|\widehat{u}\|_p^p$$

$$\Rightarrow \|\widehat{u}'\|_p^p = -\frac{\vartheta}{\mu} = m_N$$

Therefore $m_N \in \sigma_p^N$ and so $m_N = (\frac{\pi_p}{b})^p = \widehat{\lambda}_1^N > 0$. □

Remark 4.30 The restriction $p \geq 2$ guarantees that the constraint function $\widehat{\xi}(\cdot)$ is C^1 and we can apply the Lagrange multiplier rule. It is an open question what can be said for the singular case $1 < p < 2$.

Proposition 4.29, provides the following variational characterization of $\widehat{\lambda}_1^N > 0$ (the first nonzero element of $\sigma_p^N(N)$)

$$\widehat{\lambda}_1^N = \min\{\widehat{\varphi}(u) : \widehat{k}(u) = 0, \widehat{\xi} = 0, u \in W_N^{1,p}\}, \; p \geq 2.$$

This leads to the following version of the Poincare-Wirtinger inequality.

Corollary 4.31 *If* $p \geq 2$, $u \in W_N^{1,p}$ *and* $\widehat{\xi}(u) = \int_0^b |u|^{p-2}u dt = 0$, *then* $(\frac{\pi_p}{b})^p \|u\|_p^p \leq \|u'\|_p^p$.

For the vector periodic eigenvalue problem (see (4.17)), the situation is more complicated and differs from the scalar case.

Let u be a solution of (4.17). Then we claim that

$$\frac{1}{p'}|u'(t)|^p + \frac{\lambda}{p}|u(t)|^p = \frac{1}{p}|u'(b)| + \frac{\lambda}{p}|u(b)|^p \text{ for all } t \in [0, b] \tag{4.26}$$

Recall $\frac{1}{p} + \frac{1}{p'} = 1$. We know that $\varphi_p(\cdot)$ is a homeomorphism and $\varphi_p^{-1} = \varphi_{p'}$. Let $v(t) = \varphi_p(u(t))$. Then $u'(t) = \varphi_{p'}(v(t))$. From (4.17). We have

$$(\varphi_p(u'(t))', \varphi_{p'}(v(t)))_{\mathbb{R}^N} + \lambda(\varphi_p(u(t)), \varphi_{p'}(v(t)))_{\mathbb{R}^N} = 0 \text{ for a.a } t \in (0, b),$$

$$\Rightarrow (v'(t), \varphi_{p'}(v(t)))_{\mathbb{R}^N} + \lambda(\varphi_p(u(t)), u'(t))_{\mathbb{R}^N} \text{ for a.a } t \in (0, b),$$

$$\Rightarrow \tfrac{1}{p'} \tfrac{d}{dt} |v(t)|^{p'} + \tfrac{\lambda}{p} \tfrac{d}{dt} |u(t)|^p = 0$$

Integrating this last equality over $(0, t)$ and using the periodic boundary condition, we obtain (4.26).

From (4.26) we infer the following lemma.

Lemma 4.32 *If $u \in W_N^{1,p}$ is an eigenfunction of the periodic eigenvalue problem (4.17), then $|u(\cdot)|$ is constant if and only if $|u'(\cdot)|$ is constant.*

Proof From (4.26) we have

$$\frac{1}{p'}|u'(t)| + \frac{\lambda}{p}|u(t)| = c \in \mathbb{R}_+ \text{ for all } t \in [0, b]$$

$$\Rightarrow |u(\cdot)| \text{ is constant if and only if } |u'(\cdot)| \text{ is constant.}$$

□

Lemma 4.33 *If $\mu > 0$ and we consider the linear system*

$$u''(t) + \mu^2 u(t) = 0 \text{ in } (0, b),$$

then the solutions of this system for which $|u(\cdot)|$ and $|u'(\cdot)|$ are constant are given by

$$u(t) = \cos(\mu t)\widehat{\xi}_1 + \sin(\mu t)\widehat{\xi}_2$$

with $\widehat{\xi}_1, \widehat{\xi}_2 \in \mathbb{R}^N$ such that

$$|\widehat{\xi}_1| = |\widehat{\xi}_2|, (\widehat{\xi}_1, \widehat{\xi}_2)_{\mathbb{R}^N} = 0$$

and we have

$$|u(t)|^2 = \frac{1}{2}[|\widehat{\xi}_1|^2 + |\widehat{\xi}_2|^2]$$

$$|u'(t)|^2 = \frac{\mu}{2}[|\widehat{\xi}_1|^2 + |\widehat{\xi}_2|^2] = \mu^2 |u(t)|^2 \text{ for all } t \in [0, b].$$

Proof We know that the general solution of the system is given by

$$u(t) = \cos(\mu t)\widehat{\xi}_1 + \sin(\mu t)\widehat{\xi}_2 \text{ for all } t \in [0, b],$$

with $\widehat{\xi}_1, \widehat{\xi}_2 \in \mathbb{R}^N$. We have

$$\begin{aligned} |u(t)|^2 =& (u(t), u(t))_{\mathbb{R}^N} \\ =& \frac{1}{2}[|\widehat{\xi}_1|^2 + |\widehat{\xi}_2|^2] + \frac{1}{2}[|\widehat{\xi}_1|^2 + |\widehat{\xi}_2|^2]\cos(2\mu t) \\ &+ (\widehat{\xi}_1, \widehat{\xi}_2)_{\mathbb{R}^N} \sin(2\mu t), \end{aligned} \tag{4.27}$$

$$|u'(t)|^2 = \mu^2 |u(t)| \text{ for all } t \in [0, b]. \tag{4.28}$$

From (4.27) we see that $|u(\cdot)|$ is constant if and only if $|\widehat{\xi}_1| = |\widehat{\xi}_2|$, $(\widehat{\xi}_1, \widehat{\xi}_2)_{\mathbb{R}^N} = 0$ Also from (4.28) it follows that $|u'(\cdot)|$ is constant too. □

We can show that $\sigma_p^P(1) \subseteq \sigma_p^P(N)$ (see Manasevich and Mawhin [325]). Moreover, as in the scalar case, the functions

$$\widehat{u}(t) = \sin_p[\frac{2k\pi_p}{b}(t - \mu)]y \text{ for all } t \in [0, b], \text{ with } \mu \in \mathbb{R}, \ y \in \mathbb{R}^N,$$

are eigenfunctions of the periodic eigenvalue problem. These eigenfunction satisfy

$$|u(t_0)| = 0 \text{ for some } t_0 \in [0, b].$$

One way to achieve this, is to produce eigenfunctions $u(\cdot)$ such that $u(\cdot)$, $|u'(\cdot)|$ are nonzero constant on $[0, b]$.

Proposition 4.34 *We have* $\{(\frac{2k\Pi}{b})^p\}_{k\in\mathbb{N}} \subseteq \overline{o}_p^p(N)$ *and the corresponding eigenfunctions are given by*

$$\widehat{u}(t) = \cos(\frac{2k\pi}{b})\widehat{\xi}_1 + \sin(\frac{2k\pi}{b})\widehat{\xi}_2$$

with $\widehat{\xi}_1, \widehat{\xi}_2 \in \mathbb{R}^N$ *such that*

$$|\widehat{\xi}_1| = |\widehat{\xi}_2|, \ (\widehat{\xi}_1, \widehat{\xi}_2)_{\mathbb{R}^N} = 0.$$

Proof As we already indicated, we will look for eigenfunctions $\widehat{u}(\cdot)$ satisfying

$$|\widehat{u}(t)| = \vartheta, \ |\widehat{u}'(t)| = \eta \text{ for all } t \in [0, b], \text{ some } \vartheta, \ \eta \in \mathbb{R}_+$$

Such functions satisfy the linear system

$$u'' + \lambda(\frac{\vartheta}{\eta})^{p-2}u = 0 \tag{4.29}$$

Then by Lemma 4.33 we must have

$$\eta^2 = \lambda(\frac{\vartheta}{\eta})^{p-2}\vartheta^2,$$

$$\Rightarrow \lambda^{-\frac{1}{p}} = \frac{\vartheta}{\eta}.$$

Therefore (4.29) becomes

$$u'' + \lambda^{\frac{2}{p}}u = 0$$

which has b-periodic solutions if and only if $\lambda^{\frac{1}{p}} = \frac{2k\pi}{b}$, $k \in \mathbb{N}$. □

Corollary 4.35 *For $p \neq 2$, we have*

$$\{\frac{2k\pi_p}{b}\}_{k\in\mathbb{N}_0} \cup \{\frac{2k\pi}{b}\}_{k\in\mathbb{N}} \subseteq \sigma_p^p(N)$$

Hence the inclusion $\sigma_p^P(1) \subseteq \sigma_p^P(N)$ is strict.
The general form of $\sigma_p^P(N)$ is not known.

Remark 4.36 Note that any eigenfunction for one of the three vector eigenvalue problems (see (4.16), (4.17), (4.18)), has components which are nodal functions.

4.2 Linear Operators

Now we start examining eigenvalue problems for partial differential equations. We start with the Laplacian. So, let $\Omega \subseteq \mathbb{R}^N (N \geq 2)$ be a bounded domain with a C^2-boundary $\partial\Omega$. We study the following linear Robin eigenvalue problem

$$\left\{\begin{array}{l} -\Delta u(z) + \xi(z)u(z) = \widehat{\lambda}u(z) \text{ in } \Omega, \\ \frac{\partial u}{\partial n} + \beta(z)u = 0 \text{ on } \partial\Omega \end{array}\right\} \tag{4.30}$$

So, we consider the Laplacian operator plus a potential term which is indefinite. More precisely, we impose the following conditions on the data of (4.30).

$\mathbf{H_0}$: $\xi \in L^s(\Omega)$, $s > N$ if $N \geq 3$, $s > 1$ if $N = 2$ and $\beta \in W^{1,\infty}(\partial\Omega)$, $\beta \geq 0$.

On $\partial\Omega$ we use the $(N-1)$-dimensional Hausdorff measure (surface measure), denoted by $\sigma(\cdot)$. Using this measure we can define the "boundary spaces $L^q(\partial\Omega)$, $1 \leq q \leq \infty$". From Theorem I-4.99, we know that there exists a unique $\gamma_0 \in \mathcal{L}(H^1(\Omega), L^2(\partial\Omega))$ known as the "trace operator" (of order zero) such that

$$\gamma_0(u) = u|_{\partial\Omega} \text{ for all } u \in H_0^1(\Omega) \cap C(\overline{\Omega}).$$

Operator $\gamma_0(\cdot)$ extends the notion of boundary values to all Sobolev function. We know that $\ker\gamma_0 = H_0^1(\Omega)$. From now on all restrictions of Sobolev functions on $\partial\Omega$ are understood in the sense of traces.

Definition 4.37 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" of (4.30), if the problem admits a nontrivial weak solution, that is, there exists $\widehat{u} \in H^1(\Omega)$, $\widehat{u} \neq 0$ such that

$$\int_\Omega (D\widehat{u}, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)\widehat{u}hdz + \int_{\partial\Omega} \beta(z)\widehat{u}hd\sigma$$
$$= \widehat{\lambda} \int_\Omega \widehat{u}hdz \text{ for all } h \in H^1(\Omega).$$

Remark 4.38 Using the linear Green's identity (see Corollary I-4.103) we infer that

$$-\Delta\widehat{u}(z) + \xi(z)\widehat{u}(z) = \widehat{\lambda}\widehat{u}(z) \text{ for a.a } z \in \Omega, \frac{\partial\widehat{u}}{\partial n} + \beta(z)\widehat{u} = 0 \text{ on } \partial\Omega.$$

From the representation theorem for $H_0^1(\Omega)^* = H^{-1}(\Omega)$ (see Theorem I-4.111), we have $\Delta u \in H^{-1}(\Omega)$. Also let $\langle\cdot,\cdot\rangle_0$ denote the duality brackets for the pair $(H_0^1(\Omega), H^{-1}(\Omega))$. From the definition of the weak derivative, we have

$$\int_\Omega (D\widehat{u}, Dh)_{\mathbb{R}^N} dz = -\langle\Delta\widehat{u}, h\rangle_0 \text{ for all } h \in H_0^1(\Omega).$$

From this and Definition 4.37, we have

$$-\langle\Delta\widehat{u}, h\rangle_0 = \int_\Omega [\widehat{\lambda}\widehat{u} - \xi(z)\widehat{u}]hdz$$
$$= \langle\widehat{\lambda}\widehat{u} - \xi(z)\widehat{u}, h\rangle_0 \text{ for all } h \in H_0^1(\Omega)$$
$$\Rightarrow -\Delta\widehat{u}(z) = \widehat{\lambda}\widehat{u}(z) - \xi(z)\widehat{u}(z) \text{ for a.a } z \in \Omega. \tag{4.31}$$

Standard regularity theory (see, for example, Gilbarg and Trudinger [206], p. 230), implies that $\widehat{u} \in L^\infty(\Omega)$. Hence $\Delta\widehat{u} \in L^2(\Omega)$ and so from the linear Green's identity, we have

$$\int_\Omega (\Delta\widehat{u})hdz + \int_\Omega (D\widehat{u}, Dh)_{\mathbb{R}^N} dz = \langle\frac{\partial\widehat{u}}{\partial n}, h\rangle_{\partial\Omega} \text{ for all } h \in H^1(\Omega)$$

with $\langle\cdot,\cdot\rangle_{\partial\Omega}$ being the duality brackets for the pair $(H^{\frac{1}{2},2}(\partial\Omega), H^{-\frac{1}{2},2}(\partial\Omega) = H^{\frac{1}{2},2}(\partial\Omega)^*)$. Using (4.31), we obtain

$$\int_\Omega [\xi(z)\widehat{u} - \widehat{\lambda u}]h dz + \int_\Omega (D\widehat{u}, Dh)_{\mathbb{R}^N} dz = \langle \frac{\partial \widehat{u}}{\partial n}, h\rangle_{\partial\Omega},$$

$$\Rightarrow \int_{\partial\Omega} \beta(z)\widehat{u}h d\sigma = \langle \frac{\partial \widehat{u}}{\partial n}, h\rangle_{\partial\Omega} \text{ for all } h \in H^1(\Omega) \tag{4.32}$$

(see Definition 4.37 and (4.31)).

But we know that $\gamma_0(H^1(\Omega)) = H^{\frac{1}{2},2}(\partial\Omega)$. Hence from (4.32) we infer that

$$\frac{\partial \widehat{u}}{\partial n} + \beta(z)\widehat{u} = 0 \text{ on } \partial\Omega.$$

We introduce the C^1-functional $\vartheta : H^1(\Omega) \to \mathbb{R}$ defined by

$$\vartheta(u) = \|Du\|_2^2 + \int_\Omega \xi(z)u^2 dz + \int_{\partial\Omega} \beta(z)u^2 d\sigma \text{ for all } u \in H^1(\Omega).$$

The next lemma is a kind of Young's inequality between this functional, the L^2-norm and the Sobolev norm. By $\|\cdot\|$ we denote the norm of $H^1(\Omega)$ ($\|u\| = [\|u\|_2^2 + \|Du\|_2^2]^{\frac{1}{2}}$ for all $u \in H^1(\Omega)$).

Lemma 4.39 *If hypotheses H_0 hold, then there exist c_0, $c_1 > 0$ such that*

$$c_0\|u\|^2 \le \vartheta(u) + c_1\|u\|_2^2 \text{ for all } u \in H^1(\Omega).$$

Proof Arguing by contradiction, suppose that the result of the lemma is not true. Then we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ such that

$$\vartheta(u_n) + n\|u_n\|_2^2 < \frac{1}{n}\|u_n\|^2 \text{ for all } n \in \mathbb{N}. \tag{4.33}$$

Let $y_n = \frac{u_n}{\|u_n\|}$, $n \in \mathbb{N}$. Then $\|y_n\| = 1$ for all $n \in \mathbb{N}$ and so we may assume that

$$y_n \xrightarrow{w} y \text{ in } H^1(\Omega),\ y_n \to y \text{ in } L^2(\Omega) \text{ and } L^2(\partial\Omega) \tag{4.34}$$

Here we have used the Sobolev embedding theorem and the compactness of the trace operator. Similarly, we see that $\vartheta(\cdot)$ is sequentially weakly lower semicontinuous. Hence we have

$$\vartheta(y) \le \liminf_{n\to\infty} \vartheta(y_n)$$

From (4.33) we have

$$\vartheta(y_n) + n\|y_n\|_2^2 \leq \frac{1}{n} \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow n\|y_n\|_2^2 \to 0$$

$$\Rightarrow y_n \to 0 \text{ in } L^2(\Omega) \text{ and so } y = 0 \text{ (see (4.34))}$$

We have $\int_\Omega \xi(z) y_n^2 dz \to 0$, $\int_{\partial\Omega} \beta(z) y_n^2 d\sigma \to 0$. So finally we have $\|Dy_n\|_2 \to 0$, hence $y_n \to 0$ in $H^1(\Omega)$, a contradiction, since $\|y_n\| = 1$ for all $n \in \mathbb{N}$. This proves the lemma. □

The above lemma, will be helpful in determining the spectrum of (4.30). Let $c_1 > 0$ be as postulated by Lemma 4.39. For all $u, h \in H^1(\Omega)$, we define

$$(u, h)_* = \int_\Omega (Du, Dh)_{\mathbb{R}^N} dz + \int_\Omega (\xi(z) + c_1) uh dz + \int_{\partial\Omega} \beta(z) uh d\sigma$$

On account of Lemma 4.39 $(\cdot, \cdot)_*$ is an inner produced on $H^1(\Omega)$ equivalent to the standard inner product $(\cdot, \cdot)$ on $H^1(\Omega)$ defined by

$$(u, h) = \int_\Omega (Du, Dh)_{\mathbb{R}^N} dz + \int_\Omega uh dz \text{ for all } u, h \in H^1(\Omega)$$

Given $f \in L^2(\Omega)$ by the Riesz-Frechet representation theorem there exists unique $\widehat{u} \in H^1(\Omega)$ such that

$$(\widehat{u}, h)_* = \int_\Omega fh dz \text{ for all } h \in H^1(\Omega)$$

We consider the solution map $S : L^2(\Omega) \to H^1(\Omega)$ defined by $S(f) = \widehat{u}$. Also, let $i : H^1(\Omega) \to L^2(\Omega)$ be the embedding map. By the Sobolev embedding theorem $i \in \mathcal{L}_c(H^1(\Omega), L^2(\Omega))$. We consider the operator $S \circ i \in \mathcal{L}_c(H^1(\Omega))$ which is self-adjoint. We have

$$((S \circ i)(u), h)_* = \int_\Omega uh dz \text{ for all } u, h \in H^1(\Omega),$$

$$\Rightarrow ((S \circ i)(u), u)_* = \|u\|_2^2 \text{ for all } u \in H^1(\Omega).$$

Then the Spectral Decomposition Theorem (see Theorem I-3.222) implies that $S \circ i$ has decreasing sequence $\{\mu_n\}_{n\in\mathbb{N}} \subseteq (0, \infty)$ of eigenvalues such that $\mu_n \to 0^+$. We set

$$\widehat{\lambda}_n = \frac{1}{\mu_n} - c_1 \text{ for all } n \in \mathbb{N}.$$

Then $\{\widehat{\lambda}_n\}_{n\in\mathbb{N}}$ is an increasing sequence of eigenvalues of (4.30) such that $\widehat{\lambda}_n \to +\infty$. Also, there exists a corresponding sequence of eigenfunctions $\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq H_0^1(\Omega)$ which form an orthogonal basis of $H_0^1(\Omega)$ and an orthonormal basis of $L^2(\Omega)$. Moreover, standard regularity theory implies that $\widehat{u} \in C^{1,\alpha}(\overline{\Omega})$ for all $n \in \mathbb{N}$, some $\alpha \in (0,1)$. By $E(\widehat{\lambda}_k)$ we denote the eigenspace corresponding to $\widehat{\lambda}_k$. We know that $E(\widehat{\lambda}_k) \subseteq C^1(\overline{\Omega})$ is finite dimensional.

Definition 4.40 A subspace $V \subseteq H^1(\Omega)$ is said to have the "Unique Continuation Property" (UCP for short), if it has the following property: "Every $u \in V$ which vanishes on a set of positive measure is identically 0".

The eigenspaces have this property. This was proved by de Figueiredo and Gossez [130].

Proposition 4.41 *If hypotheses H_0 hold, then for every $n \in \mathbb{N}$ $E(\widehat{\lambda}_n)$ has the UCP and*

$$H^1(\Omega) = \overline{\bigoplus_{n\in\mathbb{N}} E(\widehat{\lambda}_n)}.$$

Next we will produce variational characterizations for the eigenvalues $\widehat{\lambda}_n$, $n \in \mathbb{N}$. To this end we introduce the following subspaces of $H^1(\Omega)$

$$\overline{H}_n = \bigoplus_{k=1}^{n} E(\widehat{\lambda}_k) \text{ and } \widehat{H}_n = \overline{H}_{n-1}^{\perp} = \overline{\bigoplus_{k\geq n} E(\widehat{\lambda}_k)}, n \in \mathbb{N}.$$

Proposition 4.42 *If hypotheses H_0 hold, then*

(a) $\widehat{\lambda}_1 = \inf\{\frac{\vartheta(u)}{\|u\|_2^2} : u \in H^1(\Omega), u \neq 0\}$ *and the infimum is realized on $E(\widehat{\lambda}_1)$;*

(b) for every $n \in \mathbb{N}$, $n \geq 2$ we have

$$\widehat{\lambda}_n = \inf\{\frac{\vartheta(u)}{\|u\|_2^2} : u \in \widehat{H}_n, u \neq 0\}$$

$$= \sup\{\frac{\vartheta(u)}{\|u\|_2^2} : u \in \overline{H}_n, u \neq 0\}$$

and both the infimum and the supremum are realized on $E(\widehat{\lambda}_n)$;

(c) If $\mathcal{D}_n = \{V \subseteq H^1(\Omega) :$ subspace with $\dim V = n\}$, then

$$\widehat{\lambda}_n = \inf_{V\in\mathcal{D}} \sup_{u\in V, u\neq 0} \frac{\vartheta(u)}{\|u\|_2^2}, n \in \mathbb{N}, n \geq 2.$$

Proof We see that $\widehat{\lambda}_n = \vartheta(\widehat{u}_n)$ for all $n \in \mathbb{N}$ (since $\|\widehat{u}_n\|_2 = 1$, $n \in \mathbb{N}$). Recall that $\{\widehat{u}_n\}_{n\in\mathbb{N}}$ is an orthonormal basis for $L^2(\Omega)$. So, if $u \in \widehat{H}_n$, then we have

$$u = \sum_{k \geq n} \widehat{\beta}_k \widehat{u}_k \text{ with } \widehat{\beta}_k = \int_\Omega u \widehat{u}_k dz.$$

We set $u_m = \sum_{k=n}^{m} \widehat{\beta}_k \widehat{u}_k$, $m \geq n$. Then

$$\frac{\vartheta(u_m)}{\|u_m\|_2^2} \to \frac{\vartheta(u)}{\|u\|_2^2} \text{ as } m \to \infty.$$

We have

$$\frac{\vartheta(u_m)}{\|u_m\|_2^2} = \frac{\sum_{k=n}^{m} \widehat{\beta}_k^2 \widehat{\lambda}_m}{\sum_{k=n}^{m} \widehat{\beta}_k^2} \geq \widehat{\lambda}_m \text{ for } m \geq n,$$

$$\Rightarrow \frac{\vartheta(u)}{\|u\|_2^2} \geq \widehat{\lambda}_n.$$

As we mentioned in the beginning of the proof,

$$\widehat{\lambda}_n = \vartheta(\widehat{u}_n),\ \|\widehat{u}_n\|_2 = 1 \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow \widehat{\lambda}_n = \inf\{\frac{\vartheta(u)}{\|u\|_2^2} : u \in \widehat{H}_n, u \neq 0\}$$

If $n = 1$, then we have (a) of the proposition. Also we have proved the infimum part of (b).

Now let $u \in \overline{H}_n$. Then $u = \sum_{k=1}^{n} \widehat{\beta}_k \widehat{u}_k$ and so

$$\frac{\vartheta(u)}{\|u\|_2^2} = \frac{\sum_{k=1}^{n} \widehat{\beta}_k^2 \widehat{\lambda}_k}{\sum_{k=1}^{n} \widehat{\beta}_k^2} \leq \widehat{\lambda}_n.$$

We know that $\vartheta(\widehat{u}_n) = \widehat{\lambda}_n$. Therefore

$$\widehat{\lambda}_n = \sup\{\frac{\vartheta(u)}{\|u\|_2^2} : u \in \overline{H}_n, u \neq 0\}$$

This proves (b).

Finally we will prove part (c) of the proposition. So, let $V \in \mathcal{D}_n$ and let $u \in V$ such that $u \in \overline{H}_{n-1}^{\perp} = \widehat{H}_n$. From part (b), we have

$$\begin{aligned} &\frac{\vartheta(u)}{\|u\|_2^2} \geq \widehat{\lambda}_n, \\ \Rightarrow &\sup_{u \in V, u \neq 0} \frac{\vartheta(u)}{\|u\|_2^2} \geq \widehat{\lambda}_n, \\ \Rightarrow &\inf_{V \in \mathcal{D}_n} \sup_{u \in V, u \neq 0} \frac{\vartheta(u)}{\|u\|_2^2} \geq \widehat{\lambda}_n. \end{aligned}$$

But the supremum over $\overline{H}_n \in \mathcal{D}$ equals $\widehat{\lambda}_n$ (see (b)). Hence

$$\widehat{\lambda}_n = \inf_{V \in \mathcal{D}_n} \sup_{u \in V, u \neq 0} \frac{\vartheta(u)}{\|u\|_2^2}.$$

□

Remark 4.43 The quantity $R(u) = \frac{\vartheta(u)}{\|u\|_2^2}$ involved in all these characterizations of the eigenvalues, is known as the "Rayleigh quotient", for the eigenvalue problem (4.30). Note that the minimax characterization of (c) is intrinsic in the sense that it is independent of the choice of the eigenfunctions. That part is often called the "Courant-Fischer Theorem". Note that we can have a maxmin characterization

$$\widehat{\lambda}_n = \sup_{V \in \mathcal{D}_{n-1}} \inf_{u \in V^{\perp}, u \neq 0} R(u).$$

We have a closer look at the first eigenvalue $\widehat{\lambda}_1$.

Proposition 4.44 *If hypotheses H_0 hold and for some $u \in H_0^1(\Omega) \setminus \{0\}$*

$$R(u) = \frac{\vartheta(u)}{\|u\|_2^2} = \widehat{\lambda}_1$$

then u is an eigenfunction for $\widehat{\lambda}_1$.

Proof Let $u \in H^1(\Omega)$ and $t \in \mathbb{R}$. Then from Proposition 4.42, we have

$$\widehat{\lambda}_1 = R(u) \leq R(u + th) \tag{4.35}$$

We can always assume that $\|u\|_2 = 1$. Since $\|u + th\|_2 \geq \|u\|_2 - t\|h\|_2 = 1 - t\|h\|_2$, we can choose $|t|$ small so that $\|u + th\|_2 \neq 0$. Then we consider the function $l(t) = R(u + th)$, $t \in \mathbb{R}$. From (4.35) we see that $l(0) = \inf_{\mathbb{R}} l$ and so

$$
\begin{aligned}
& l'(0) = 0, \\
& \Rightarrow \langle \vartheta'(u), h \rangle = \vartheta(u) \int_\Omega uhdz = \widehat{\lambda}_1 \int_\Omega uhdz, \\
& \Rightarrow \int_\Omega (Du, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z) uhdz + \int_{\partial\Omega} \beta(z) uhd\sigma = \widehat{\lambda}_1 \int_\Omega uhdz \\
& \Rightarrow -\Delta u + \xi(z)u = \widehat{\lambda}_1 u \text{ in } \Omega, \frac{\partial u}{\partial n} + \beta(z)u = 0 \text{ on } \partial\Omega.
\end{aligned}
$$

□

Now we can summarize the main properties of the first eigenvalue. Note that the space $C^1(\overline{\Omega})$ is an ordered Banach space with positive cone $\widehat{C}_+ = \{u \in C^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\}$. This cone has a nonempty interior given by

$$\text{int}\, \widehat{C}_+ = \{u \in \widehat{C}_+ : u(z) > 0 \text{ for all } z \in \overline{\Omega}\}.$$

Theorem 4.45 *If hypotheses H_0 hold, then $\widehat{\lambda}_1(z)$ is simple (that is, $\dim E(\widehat{\lambda}_1) = 1$) and the elements of $E(\widehat{\lambda}_1)$ have a fixed sign. Let $\widehat{u}_1$ be the positive eigenfunction with $\|\widehat{u}_1\|_2 = 1$. Then $\widehat{u}_1(z) > 0$ for all $z \in \Omega$ and if $\xi \in L^\infty(\Omega)$, we have $\widehat{u}_1 \in \text{int}\widehat{C}_+$.*

Proof Let $\widehat{u}$ be an eigenfunction corresponding to $\widehat{\lambda}_1$. From Wang [467] we know that $\widehat{u} \in C^1(\overline{\Omega})$. We have

$$
\begin{aligned}
& \int_\Omega (D\widehat{u}, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)\widehat{u}hdz + \int_{\partial\Omega} \beta(z)\widehat{u}hd\sigma \\
& = \widehat{\lambda}_1 \int_\Omega \widehat{u}hdz \text{ for all } h \in H^1(\Omega).
\end{aligned}
$$

Let $h = \widehat{u}^+ \in H^1(\Omega)$. Then

$$\vartheta(\widehat{u}^+) = \widehat{\lambda}_1 \|\widehat{u}^+\|_2^2.$$

Similarly, using $h = -\widehat{u}^- \in H^1(\Omega)$, we obtain

$$\vartheta(\widehat{u}^-) = \widehat{\lambda}_1 \|\widehat{u}^-\|_2^2.$$

Suppose $\widehat{u}$ is nodal (sign-changing). This means $\widehat{u}^+ \neq 0, \widehat{u}^- \neq 0$. We have

$$R(\widehat{u}^+) = \widehat{\lambda}_1 = R(\widehat{u}^-)$$

$$\Rightarrow \widehat{u}^+, \widehat{u}^- \text{ are both eigenfunctions for } \widehat{\lambda}_1 \text{ (see Proposition 4.44).}$$

Therefore

$$-\Delta\widehat{u}^+ + \xi(z)\widehat{u}^+ = \widehat{\lambda}_1\widehat{u}^+ \text{ for a.a } z \in \Omega,$$

$$-\Delta\widehat{u}^- + \xi(z)\widehat{u}^- = \widehat{\lambda}_1\widehat{u}^- \text{ for a.a } z \in \Omega.$$

The Harnack inequality (see Gilbarg and Trudinger [206], p. 199) implies that

$$0 < \widehat{u}^+(z), 0 < \widehat{u}^-(z) \text{ for all } z \in \Omega,$$

a contradiction. So $\widehat{u}$ has fixed sign.

Let $\widehat{u}_1$ be the positive, L^2-normalized (that is, $\|\widehat{u}_1\|_2 = 1$) eigenfunction corresponding to $\widehat{\lambda}_1$. Then

$$0 < \widehat{u}_1(z) \text{ for all } z \in \Omega.$$

Suppose $\xi \in L^\infty(\Omega)$. Then

$$\Delta\widehat{u}_1 \leq [\|\xi\|_\infty + \widehat{\lambda}_1]\widehat{u}_1,$$

$$\Rightarrow \widehat{u}_1 \in \text{int}\widehat{C}_+ \text{ (by the Hopf maximum principle).}$$

Finally we show that $\widehat{\lambda}_1$ is simple. If this is not true, then we can find $\widehat{v}_1$ another eigenfunction for $\widehat{\lambda}_1$ such that $\widehat{v}_1 \perp \widehat{u}_1$. This means that $\int_\Omega \widehat{u}_1\widehat{v}_1 dz = 0$. But since $\widehat{u}_1(z) > 0$ for all $z \in \Omega$ and $\widehat{v}_1 \perp \widehat{u}_1$, then $\int_\Omega \widehat{u}_1\widehat{v}_1 dz = 0$ can not happen. □

Remark 4.46 In the literature, these facts about the first eigenvalue are proved using the Krein-Rutman theorem. We recall the statement of that theorem: "If X is an ordered Banach space with order cone K, $\text{int}K \neq \emptyset$, $X = \overline{K - K}$ (that is, K is total, see Definition 2.79(a)), $A \in \mathcal{L}_c(X)$, $A(K) \subseteq K$ (that is $A \geq 0$) and the spectral radius

$$r(A) = \lim_{n\to\infty} \|A^n\|_{\mathcal{L}}^{\frac{1}{n}} > 0,$$

then $r(A)$ is an eigenvalue of A with positive eigenvector".

This result is an infinite dimensional generalization of the Perron-Frobenius theorem. Our proof here avoids the use of the Krein-Rutman theorem and can be extended to more general elliptic differential operators.

We can extend Proposition 4.44 to higher eigenvalues.

Proposition 4.47 *If hypotheses H_0 hold and $n \in \mathbb{N}$, then*

$$\gamma(\widehat{u}) = \widehat{\lambda}_n\|\widehat{u}\|_2^2 \Leftrightarrow \widehat{u} \in E(\widehat{\lambda}_n)$$

.

Proof For $n = 1$, this follows from Proposition 4.44. So, suppose $n \geq 2$. Let $\|\cdot\|_*$ denote the norm on $H^1(\Omega)$ induced by the inner product $(\cdot,\cdot)_*$ introduced in the proof of Lemma 4.39. We have

$$\vartheta(u) = \|u\|_*^2 - c_1\|u\|_2^2 \text{ for all } u \in H^1(\Omega) \tag{4.36}$$

Suppose that $\vartheta(\widehat{u}) = \widehat{\lambda}_n\|\widehat{u}\|_2^2$. We can write that

$$\widehat{u} = v_0 + \widehat{v} \text{ with } v_0 \in E(\widehat{\lambda}_n), \widehat{v} \in \widehat{H}_{n+1}.$$

The orthogonality of the eigenspaces, implies that

$$\widehat{\lambda}_n\|\widehat{u}\|_2^2 = \widehat{\lambda}_n\|\widehat{v_0}\|_2^2 + \widehat{\lambda}_n\|\widehat{v}\|_2^2 \tag{4.37}$$

(Pythagorean theorem).

If $\widehat{v} \neq 0$, then

$$\begin{aligned}
\vartheta(\widehat{u}) = \widehat{\lambda}_n\|\widehat{u}\|_2^2 &< \widehat{\lambda}_n\|\widehat{v_0}\|_2^2 + \widehat{\lambda}_{n+1}\|\widehat{v}\|_2^2 \text{ (see (4.37))} \\
&\leq \vartheta(v_0) + \vartheta(\widehat{v}) \text{ (see Proposition 4.42)} \\
&= \|v_0\|_*^2 - c_1\|v_0\|_2^2 + \|\widehat{v}\|_*^2 - c_1\|\widehat{v}\|_2^2 \\
&= \|v_0 + \widehat{v}\|_2^2 - c_1\|v_0 + \widehat{v}\|_2^2 \text{ (Pythagorean theorem)} \\
&= \vartheta(\widehat{u}) \text{ (see (4.36),}
\end{aligned}$$

a contradiction. Therefore $\widehat{v} = 0$ and so

$$\widehat{u} = v_0 \in E(\widehat{\lambda}_n).$$

For the converse, we use Proposition 4.42. □

This proposition and the UCP of the eigenspaces (see Proposition 4.41), lead to the following useful inequalities.

Proposition 4.48 *If hypotheses H_0 hold and $n \in \mathbb{N}$, then*

(a) if $\eta \in L^\infty(\Omega)$, $\eta(z) \leq \widehat{\lambda}_n$ and the inequality is strict on a set of positive measure, then there exists $c_3 > 0$ such that

$$c_3\|u\|^2 \leq \vartheta(u) - \int_\Omega \eta(z)u^2 dz \text{ for all } u \in \widehat{H}_n;$$

(b) if $\eta \in L^\infty(\Omega)$, $\eta(z) \geq \widehat{\lambda}_n$ for a.a $z \in \Omega$ and the inequality is strict on a set of positive measure, then there exists $c_4 > 0$ such that

$$\vartheta(u) - \int_\Omega \eta(z)u^2 dz \leq -c_4\|u\|^2 \text{ for all } u \in \overline{H}_n.$$

Proof

(a) We argue by contradiction. So, we assume that the inequality is not true. Then the 2-homogeneity of $\vartheta(\cdot)$, implies that we can find $\{u_k\}_{k\in\mathbb{N}} \subseteq \widehat{H}_n$ such that

$$\|u_k\| = 1 \text{ and } 0 \leq \vartheta(u_k) - \int_\Omega \eta(z)u_k^2 dz < \frac{1}{k} \text{ for all } k \in \mathbb{N}. \tag{4.38}$$

We may assume that

$$u_k \xrightarrow{w} u \text{ in } H'(\Omega),\ u_k \to u \text{ in } L^2(\Omega) \text{ and in } L^2(\partial\Omega). \tag{4.39}$$

We have $u \in \widehat{H}_n$. In (4.38) we pass to the limit as $k \to \infty$ and use (4.39). We obtain

$$\begin{aligned}
&\vartheta(u) \leq \int_\Omega \eta(z)u^2 dz \leq \widehat{\lambda}_n \|u\|_2^2, \\
&\Rightarrow \vartheta(u) = \widehat{\lambda}_n \|u\|_2^2 \text{ since } u \in \widehat{H}_n \text{ (see Proposition 4.42)}, \\
&\Rightarrow u \in E(\widehat{\lambda}_n) \text{ (see Propositions 4.44 and 4.47)}
\end{aligned}$$

If $u \neq 0$, then on account of Proposition 4.41 we have $u(z) \neq 0$ for a.a $z \in \Omega$. Since $\eta \not\equiv \widehat{\lambda}_n$, we have

$$\vartheta(u) < \widehat{\lambda}_n \|u\|_2^2,$$

which contradicts Proposition 4.42, since $u \in \widehat{H}_m \setminus \{0\}$. Hence $u = 0$. From (4.38) we have

$$\begin{aligned}
&\int_\Omega \eta(z)u_k^2 dz \leq \vartheta(u_k) \leq \frac{1}{k} + \int_\Omega \eta(z)u_k^2 dz, \\
&\Rightarrow \vartheta(u_k) \to 0 \text{ (see (4.39) and recall } u = 0), \\
&\Rightarrow u_k \to 0 \text{ in } H^1(\Omega),
\end{aligned}$$

a contradiction since $\|u_k\| = 1$ for all $k \in \mathbb{N}$.
So the inequality of (a) is true.

(b) Again we argue by contradiction. We assume that the inequality is not true and so we can find $\{u_k\}_{k\in\mathbb{N}} \subseteq \overline{H}_n$ such that

$$\|u_k\| = 1 \text{ and } \vartheta(u_k) - \int_\Omega \eta(z)u_k^2 dz > -\frac{1}{k}, k \in \mathbb{N}. \tag{4.40}$$

The space $\overline{H}_n$ is finite dimensional. By the Heine-Borel theorem, at least for a subsequence, we have

$$u_k \to u \text{ in } H^1(\Omega), u \in \overline{H}_m, \|u\| = 1.$$

Therefore from (40) in the limit as $k \to \infty$, we obtain

$$\begin{aligned}
&\vartheta(u) \geq \int_\Omega \eta(z)u^2 dz \geq \widehat{\lambda}_n \|u\|_2^2, \\
&\Rightarrow \vartheta(u) = \widehat{\lambda}_n \|u\|_2^2, \\
&\Rightarrow u \in E(\widehat{\lambda}_n) \setminus \{0\}, \\
&\Rightarrow u(z) \neq 0 \text{ for a.a } z \in \Omega \text{ (by the UCP).}
\end{aligned}$$

It follows that

$$\widehat{\lambda}_n \|u\|_2^2 < \vartheta(u),$$

contradicting Proposition 4.42.

□

From Proposition 4.42 (c) we know that the eigenvalues follow from the Ljusternik-Schnirelmann minimax scheme. More precisely we have.

Proposition 4.49 *If hypotheses H_0 hold and $n \in \mathbb{N}$, then*

(a) $\widehat{\lambda}_n = \inf_{V \in \mathcal{D}_n} \sup_{u \in V, \|u\|_2 = 1} \vartheta(u).$

(b) $\widehat{\lambda}_n = \sup_{V \in \mathcal{D}_n} \inf_{u \in V^\perp, \|u\|_2 = 1} \vartheta(u).$

Remark 4.50 The unsatisfactory feature in the expression in (b), is that the minimization is done in an infinite dimensional space.

Proposition 4.51 *If hypotheses H_0 hold and $n \in \mathbb{N}$, then the elements of $E(\widehat{\lambda}_n)$ are nodal functions.*

Proof Let $u \in E(\widehat{\lambda}_n)$. We know $u \perp \widehat{u}_1$ in $L^2(\Omega)$ and so $\int_\Omega u\widehat{u}_1 dz = 0$. On account of Theorem 4.45 this means that u must be nodal. □

Proposition 4.52 *If hypotheses H_0 hold, $\xi(z) \geq 0$ for a.a $z \in \Omega$ and either $\xi(\cdot)$ or $\beta(\cdot)$ is nonzero, then $\widehat{\lambda}_1 > 0$.*

Proof First assume $\xi \not\equiv 0$. We claim that there exists $c_5 > 0$ such that

$$\|Du\|_2^2 + \int_\Omega \xi(z)u^2 dz \geq c_5 \|u\|^2 \text{ for all } u \in H^1(\Omega).$$

If not, we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ such that

$$\|Du_n\|_2^2 + \int_\Omega \xi(z)u_n^2 dz < \frac{1}{n}, \|u_n\| = 1 \text{ for all } n \in \mathbb{N}. \tag{4.41}$$

We may assume that

$$u_n \xrightarrow{w} u \text{ in } H^1(\Omega), u_n \to u \text{ in } L^2(\Omega).$$

Then

$$\|Du\|_2^2 + \int_\Omega \xi(z)u^2 dz \leq 0 \tag{4.42}$$

If $u = 0$, then $u_n \to 0$ in $H^1(\Omega)$, a contradiction (see (4.41)).
From (4.42), we see that $u = c \in \mathbb{N} \setminus \{0\}$. Then

$$0 < c^2 \int_\Omega \xi(z)dz \leq 0,$$

a contradiction. So, the claim is true and we have

$$\widehat{\lambda}_1 = \vartheta(\widehat{u}_1) = \|D\widehat{u}_1\|_2^2 + \int_\Omega \xi(z)\widehat{u}_1^2 dz + \int_{\partial\Omega} \beta(z)\widehat{u}_1^2 d\sigma \geq c_5\|\widehat{u}_1^2\| > 0.$$

Next assume that $\beta \not\equiv 0$. We claim that there exists $c_6 > 0$ such that

$$\|Du\|_2^2 + \int_{\partial\Omega} \beta(z)u^2 d\sigma \geq c_6\|u\| \text{ for all } u \in H^1(\Omega).$$

This can be proved as the previous case, exploiting this time the compactness of the trace operator. So, we will have

$$u_n \xrightarrow{w} u \text{ in } H^1(\Omega), u_n \to u \text{ in } L^2(\partial\Omega).$$

From the claim if follows that $\widehat{\lambda}_1 > 0$. □

Remark 4.53 The above proof reveals that $\xi \in L^\infty(\Omega) \setminus \{0\}$ and $\xi(z) \geq 0$ for a.a $z \in \Omega$, then $u \to |u|_1 = (\|Du\|_2^2 + \int_\Omega \xi(z)u^2 dz)^{\frac{1}{2}}$ is an equivalent norm on $H^1(\Omega)$. Similarly, if $\beta \in L^2(\partial\Omega) \setminus \{0\}$, $\beta(z) \geq 0$ σ-a.e. then $u \to |u|_2 = (\|Du\|_2^2 + \int_{\partial\Omega} \beta(z)u^2 d\sigma)^{\frac{1}{2}}$ is an equivalent norm of $H^1(\Omega)$.

We can also consider a weighted version of the linear eigenvalue problem (4.30). So, let $m \in L^\infty(\Omega) \setminus \{0\}$, $m(z) \geq 0$ for a.a $z \in \Omega$. We consider the following linear eigenvalue problem

$$\left\{\begin{array}{l} -\Delta u(z)+\xi(z)u(z)=\widehat{\lambda}m(z)u(z) \text{ in } \Omega, \\ \frac{\partial u}{\partial n}+\beta(z)u=0 \text{ on } \partial\Omega. \end{array}\right\} \tag{4.43}$$

A similar analysis can be conducted for this eigenvalue problem and produce the same properties for the eigenvalues and eigenvectors of (4.43). So, we have a sequence of distinct eigenvalues $\{\widehat{\lambda}_k(m)\}_{k\in\mathbb{N}}$ such that $\widehat{\lambda}_k(m)\to\infty$ as $k\to+\infty$ and a sequence $\{\widehat{u}_k(m)\}_{k\in\mathbb{N}}$ of corresponding eigenfunctions which form an orthogonal basis of $H^1(\Omega)$ and an orthonormal basis of $L^2(\Omega)$. As before by $E(\widehat{\lambda}_k(m))$ we denote the eigenspace corresponding to the eigenvalue $\widehat{\lambda}_k(m)$. We have

$$H^1(\Omega)=\overline{\bigoplus_{k\in\mathbb{N}} E(\widehat{\lambda}_k(m))}.$$

Moreover, we have the following properties (as in the unweighted case).

(a) For every $k\in\mathbb{N}$, $E(\widehat{\lambda}_k(m))\subseteq C^1(\overline{\Omega})$, it is finite dimensional and has the UCP (see Definition 4.40).

(b) $\widehat{\lambda}_1(m)\in\mathbb{R}$ is simple and has the following variational characterization

$$\widehat{\lambda}_1(m)=\inf\{\frac{\vartheta(u)}{\int_\Omega m(z)u^2dz}: u\in H^1(\Omega), u\neq 0\}.$$

In this expression the infimum is realized on the one-dimensional eigenspace $E(\widehat{\lambda}_1(m))$. The elements of $E(\widehat{\lambda}_1(m))$ have fixed sign. By $\widehat{u}_1(m)$ we denote the positive, normalized (that is, $\int_\Omega m(z)\widehat{u}_1(m)^2dz=1$) eigenfunction, then $\widehat{u}(m)(z)>0$ for all $z\in\Omega$. If $\xi\in L^\infty(\Omega)$, then $\widehat{u}(m)(z)>0$ for all $z\in\overline{\Omega}$ (that is, $\widehat{u}_1(m)\in \mathrm{int}\widehat{C}_1$).

(c) For every $n\in\mathbb{N}$, we have

$$\widehat{\lambda}_n(m)=\inf\{\frac{\vartheta(u)}{\int_\Omega m(z)u^2dz}: u\in\widehat{H}_n=\overline{\bigoplus_{k\geq n} E(\widehat{\lambda}_k(m))}, u\neq 0\}$$

$$=\sup\{\frac{\vartheta(u)}{\int_\Omega m(z)u^2dz}: u\in\overline{H}_n=\bigoplus_{k=1}^{n} E(\widehat{\lambda}_k(m)), u\neq 0\}.$$

In both expressions the infimum and the supremum are realized on $E(\widehat{\lambda}_n(m))$. The nonzero elements of this eigenspace are nodal functions.

(d) For every $k\in\mathbb{N}$

$$\vartheta(u)=\widehat{\lambda}_k(m)\int_\Omega m(z)u^2dz \Leftrightarrow u\in E(\widehat{\lambda}_k(m)).$$

Next we examine the monotonicity properties of the map $m \to \widehat{\lambda}_k(m)$, $k \in \mathbb{N}$. We introduce the following quantities:

$$n_+(m) = \min\{k \in \mathbb{N} : \widehat{\lambda}_k(m) > 0\},$$

$$n_-(m) = \max\{k \in \mathbb{N} : \widehat{\lambda}_k(m) < 0\}.$$

Proposition 4.54 *If hypotheses H_0 hold and $m_1, m_2 \in L^\infty(\Omega)$ satisfy*

$$0 \leq m_1(z) \leq m_2(z), m_1 \not\equiv 0, m_2 \not\equiv m_1,$$

then

(a) $\widehat{\lambda}_k(m_2) < \widehat{\lambda}_k(m_1)$ *for all* $k \geq n_+(m_1)$;
(b) $\widehat{\lambda}_k(m_1) < \widehat{\lambda}_k(m_2)$ *for all* $1 \leq k \leq n_-(m_2)$.

Proof Let $\widehat{u}$ be an eigenfunction corresponding to $\widehat{\lambda}_k(m_1)$ and $\widehat{v}$ an eigenfunction corresponding to the eigenvalue $\widehat{\lambda}_k(m_2)$. We know that

$$\widehat{u}, \widehat{v} \in C^1(\overline{\Omega}) \text{ and } \widehat{u}(z) \neq 0 \neq \widehat{v}(z) \text{ for a.a } z \in \Omega.$$

(a) Suppose $k \geq n_+(m_1)$. Then $\widehat{\lambda}_k(m_1) > 0$. We have

$$\widehat{\lambda}_k(m_1) = \frac{\vartheta(\widehat{u})}{\int_\Omega m_1(z)\widehat{u}^2 dz} > \frac{\vartheta(\widehat{u})}{\int_\Omega m_2(z)\widehat{u}^2 dz} \geq \widehat{\lambda}_k(m_2).$$

(b) Now let $k \leq n_-(m_2)$. Then $\widehat{\lambda}_k(m_2) < 0$. We have

$$\widehat{\lambda}_k(m_2) = \frac{\vartheta(\widehat{v})}{\int_\Omega m_2(z)\widehat{v}^2 dz} > \frac{\vartheta(\widehat{v})}{\int_\Omega m_1(z)\widehat{v}^2 dz} \geq \widehat{\lambda}_k(m_1).$$

□

Remark 4.55 We see that $\widehat{\lambda}_k(m_1) = 0$ if and only if $\widehat{\lambda}_k(m_2) = 0$ for all $k \in \mathbb{N}$.

Next we want to determine the dependence of the eigenvalues on the potential function $\xi(\cdot)$ when $\xi \in L^\infty(\Omega)$. So, for every $k \in \mathbb{N}$ we consider the map $\widehat{p}_k : L^\infty \to \mathbb{R}$ which to $\xi \in L^\infty(\Omega)$ assigns the kth-eigenvalue $\widehat{\lambda}_k(m)$ of (4.43).

Proposition 4.56 *For every $k \in \mathbb{N}$, the function $\widehat{p}_k(\cdot)$ is continuous.*

Proof Assume that $\xi_n \to \xi$ in $L^\infty(\Omega)$ and let $S_n, S : L^2(\Omega) \to H^1(\Omega)$ be the corresponding solution maps introduced earlier in this section (see the discussion after the proof of Lemma 4.39). Consider these solution maps, as maps with values in $L^2(\Omega)$. Since $H^1(\Omega) \hookrightarrow L^2(\Omega)$ compactly, we see that $S_n, S \in \mathcal{L}_c(L^2(\Omega))$.

Note that $S_n - S \in \mathcal{L}_c(L^2(\Omega))$ and so $(S_n - S)(\overline{B}_1^{L^2}) \subseteq L^2(\Omega)$ is relatively compact ($\overline{B}_1^{L^2} = \{f \in L^2(\Omega) : \|f\|_2 \leq 1\}$). Therefore, we can find $f_n \in \overline{B}_1^{L^2}$ such that

$$\|S_n - S\|_{\mathcal{L}} = \|(S_n - S)(f_n)\|_2, n \in \mathbb{N}. \tag{4.44}$$

We have

$$\int_\Omega (DS_n(f_n), Dh)_{\mathbb{R}^N} dz + \int_\Omega (\xi(z) + c_1) S_n(f_n) h dz + \int_{\partial\Omega} \beta(z) S_n(f_n) h d\sigma$$
$$= \int_\Omega f_n h dz \text{ for all } h \in H^1(\Omega), \text{ all } n \in \mathbb{N}.$$

As before, the regularity theory of Wang [467], implies that $\{S_n(f_n)\}_{n\in\mathbb{N}} \subseteq C^1(\overline{\Omega})$ is relatively compact. So, we may assume that

$$S_n(f_n) \to \widehat{u} \text{ in } C^1(\overline{\Omega}). \tag{4.45}$$

Passing to the limit as $n \to \infty$ in (4.44) and use (4.45) we infer that

$$\widehat{u} = S(f)$$

$$\Rightarrow S_n(f_n) \to S(f) \text{ in } C^1(\overline{\Omega}).$$

Similarly $S(f_n) \to S(f)$ in $C^1(\overline{\Omega})$. Finally

$$\begin{aligned} &\|(S_n - S)(f_n)\|_2 \leq \|S_n(f_n) - S(f)\|_2 + \|S(f) - S(f_n)\|_2 \to 0, \\ &\Rightarrow \|S_n - S\|_{\mathcal{L}} \to 0. \end{aligned} \tag{4.46}$$

Recall that $\mu_k^n = \mu_k(S_n)$, $\mu_k = \mu_k(S)$, $k \in \mathbb{N}$ denote the eigenvalues of S_n, $S \in \mathcal{L}_c(L^2(\Omega))$ respectively. From the Courant-Fischer minimax principle, we have

$$|\mu_k^n - \mu_k| \leq \|S_n - S\|_{\mathcal{L}} \to 0 \text{ for all } k \in \mathbb{N} \text{ (see (4.46))}.$$

But recall that

$$\widehat{p}_k(\xi_n) = \frac{1}{\mu_k^n} - c_1 \text{ for all } k \in \mathbb{N}$$

$$\Rightarrow \widehat{p}_k(\xi_n) \to \frac{1}{\mu_k} - c_1 = \widehat{p}_k(\xi) \text{ as } n \to \infty \text{ for all } k \in \mathbb{N}.$$

□

The exact same results are also valid for the Dirichlet eigenvalue problem

$$-\Delta u(z) + \xi(z)u(z) = \widehat{\lambda} u(z) \text{ in } \Omega, u|_{\partial\Omega} = 0. \tag{4.47}$$

Hypotheses H_0 remain the same. Recall that if $\Omega' \subseteq \Omega$, then $H_0^1(\Omega') \hookrightarrow H_0^1(\Omega)$ continuously (just extend $u \in H_0^1(\Omega')$ by zero on $\Omega \setminus \Omega'$). This fact leads to another monotonicity property of the eigenvalues. In what follows we write $\widehat{\lambda}_k^D(\Omega)$ to stress the domain on which the eigenvalue problem (4.47) is defined.

Proposition 4.57 *If hypotheses H_0 hold and $\Omega' \subseteq \Omega$, then $\widehat{\lambda}_k(\Omega) \leq \widehat{\lambda}_k(\Omega')$ for all $k \in \mathbb{N}$.*

Next we will determine the full spectrum of a general second order elliptic differential operator, in the presence of an indefinite weight. To do this, first we will state a few basic facts about the spectrum of compact self-adjoint operators on a Hilbert space. The mathematical setting is the following. Here H is an infinite dimensional separable Hilbert space. By $(\cdot, \cdot)$ we denote the inner product of H and by $\|\cdot\|$ the corresponding norm. Let $A \in \mathcal{L}_c(H)$ (=space of compact operators on H). By $\sigma_p(A)$ we denote the point spectrum of A. We set

$$\widehat{\lambda}_1 = \sup\{(A(u), u) : \|u\| = 1\}.$$

Proposition 4.58 *If $\widehat{\lambda}_1 > 0$, then $\widehat{\lambda}_1 \in \sigma_p(A)$.*

Proof Note that $0 < \widehat{\lambda}_1 \leq \|A\|_{\mathcal{L}}$. From Proposition I-3.215, we have that $\widehat{\lambda}_1 \in \sigma_p(A)$. □

Also, let

$$\widehat{\lambda}_{-1} = \inf\{(A(u), u) : \|u\| = 1\}.$$

Using the same results from Volume I, we have:

Proposition 4.59 *If $\widehat{\lambda}_{-1} < 0$, then $\widehat{\lambda}_{-1} \in \sigma_p(A)$.*

Remark 4.60 Evidently $\widehat{\lambda}_1 > 0$ is the biggest eigenvalue of A, while $\widehat{\lambda}_{-1} < 0$ is the smallest.

Consider $\widehat{u} \in H$, $\|\widehat{u}_1\| = 1$ such that $A(\widehat{u}_1) = \widehat{\lambda}_1 \widehat{u}_1$ (normalized eigenfunction corresponding to $\widehat{\lambda}_1$). By considering $A|_{(\mathbb{R}\widehat{u}_1)^\perp}$ we can produce new eigenvalues. The whole process is based on the following simple fact. (See also Proposition I-3.213.)

Lemma 4.61 *If $A \in \mathcal{L}_c(H)$ is self adjoint and $V \subseteq H$ is a subspace which is A-invariant (that is, $A(V) \subseteq V$), then $V^\perp$ is A-invariant too and $\widehat{A} = A|_{V^\perp} \in \mathcal{L}_c(V^\perp)$ and it is self adjoint.*

Proof Let $y \in V^{\perp}$. For every $v \in V$, we have $A(v) \in V$ and so

$$0 = (A(v), y) = (v, A(y)) \text{ (since } A \text{ is self adjoint)},$$

$$\Rightarrow A(y) \in V^{\perp} \text{ and so } V^{\perp} \text{ is } A\text{-invariant},$$

Since $A \in \mathcal{L}_c(H)$ and it is self adjoint, we have $\widehat{A} = A|_{V^{\perp}} \in \mathcal{L}_c(V^{\perp})$ and it is self adjoint. □

Let $\sigma(A) = \{\widehat{\lambda}_k\}_{k\in\mathbb{N}} \cup \{0\}$ with $\widehat{\lambda}_k$, $k \in \mathbb{N}$, being distinct nonzero eigenvalues. This sequence contains $\|A\|_{\mathcal{L}}$ and it is either a finite sequence or a countable sequence converging to zero. The space H has an orthonormal basis $\{\widehat{u}_n\}_{n\in\mathbb{N}}$ consisting of eigenvectors corresponding to the eigenvalues of A. Motivated by Lemma 4.61, inductively we define

$$\widehat{\lambda}_n = \sup\{(A(u), u) : \|u\| = 1, u \in [span\{\widehat{u}_k\}_{k=1}^{n-1}]^{\perp}\}, \tag{4.48}$$

$$\widehat{\lambda}_{-n} = \inf\{(A(u), u) : \|u\| = 1, u \in [span\{\widehat{u}_{-k}\}_{k=1}^{n-1}]^{\perp}\}. \tag{4.49}$$

The next proposition, extends Propositions 4.58 and 4.59.

Proposition 4.62

(a) If $\widehat{\lambda}_n > 0$, *then*

$$\widehat{\lambda}_n \in \sigma_p(A) \tag{4.50}$$

(b) If $\widehat{\lambda}_{-n} < 0$, *then*

$$\widehat{\lambda}_{-n} \in \sigma_p(A).(n \geq 2) \tag{4.51}$$

Therefore (4.48) provides variational expressions for the positive eigenvalues of A, while (4.49) provides analogous variational expressions for the negative eigenvalues of A. However, expressions (4.48) and (4.49) have an unsatisfactory feature. They require knowledge of the previous positive and negative respectively eigenvalues. The next proposition remedies this and provides a Courant-Fischer characterization of the positive and negative eigenvalues. So, let

$$\mathcal{F}_n = \{V \subseteq H : \text{subspace with } dim V = n\}. \tag{4.52}$$

Proposition 4.63

(a) $\widehat{\lambda}_n = \inf_{V\in\mathcal{F}_{n-1}} \sup\{(A(u), u) : \|u\| = 1, u \in V^{\perp}\}.$

(b) $\widehat{\lambda}_{-n} = \sup_{V\in\mathcal{F}_{n-1}} \inf\{(A(u), u) : \|u\| = 1, u \in V^{\perp}\}.$

Proof

(a) Let $\vartheta_n = \inf_{V\in\mathcal{F}_{n-1}} \sup\{(A(u),u) : \|u\| = 1, u \in V^\perp\}$ and $V = \text{span}\{u_k\}_{k=1}^{n-1}$. Eigenvectors corresponding to distinct eigenvalues, are linearly independent (see Proposition *I*-3.206). So, from Proposition 4.62(a), we have $\vartheta_n \le \widehat{\lambda}_n$. Next, let $\{e_k\}_{k=1}^{n-1}$ be an orthogonal set and $V = \text{span}\{e_k\}_{k=1}^{n-1}$. Let $\widehat{u} = \sum_{k=1}^{n} \beta_k \widehat{u}_k$ satisfying $\|\widehat{u}\| = 1$ (that is, $\sum_{k=1}^{n} \beta_k^2 = 1$) and $(\widehat{u}, e_k) = 0$ for all $k = 1, \cdots, n-1$. Then

$$
\begin{aligned}
&(A(\widehat{u}),\widehat{u}) = \sum_{k=1}^{n} \beta_k^2 \widehat{\lambda}_k \ge \widehat{\lambda}_n,\\
&\Rightarrow \sup\{(A(u),u) : \|u\| = 1, u \in V^\perp\} \ge \widehat{\lambda}_n \text{ for all } V \in \mathcal{F}_{n-1},\\
&\Rightarrow \vartheta_n \ge \widehat{\lambda}_n \text{ and so finally } \widehat{\lambda}_n = \vartheta_n, n \in \mathbb{N}.
\end{aligned}
$$

(b) The proof of this part is similar to that of (a).

□

Again in the above expressions there is an undesirable feature. In (a), the maximization is over an infinite dimensional space, while in (b), the minimization is over such a subspace of H. The next proposition fixes these handicaps.

Proposition 4.64

(a) $\widehat{\lambda}_n = \sup_{V\in\mathcal{F}_n} \inf\{(A(u),u) : \|u\| = 1, u \in V\}$.

(b) $\widehat{\lambda}_n = \inf_{V\in\mathcal{F}_n} \sup\{(A(u),u) : \|u\| = 1, u \in V\}$.

Proof

(a) Let $\widehat{\vartheta}_n = \sup_{V\in\mathcal{F}_n} \inf\{(A(u),u) : \|u\| = 1, u \in V\}$ and consider $V = \text{span}\{\widehat{u}_k\}_{k=1}^{n}$. If $u \in V \cap \partial B_1$, then $u = \sum_{k=1}^{n} \beta_k \widehat{u}_k$ and $\sum_{k=1}^{n} \beta_k^2 = 1$ (recall $\{\widehat{u}_k\}_{k=1}^{n}$ are linearly independent). We have

$$
\begin{aligned}
&(A(u),u) = \sum_{k=1}^{n} \beta_k^2 \widehat{\lambda}_k \ge \widehat{\lambda}_n,\\
&\Rightarrow \widehat{\lambda}_n \le \widehat{\vartheta}_n.
\end{aligned}
$$

Consider $V \in \mathcal{F}_n$ and choose $u \in V \cap [\text{span}\{\widehat{u}_k\}_{k=1}^{n-1}\}]^{\perp}$. From (4.48), we have

$$(A(u), u) \leq \widehat{\lambda}_n,$$
$$\Rightarrow \widehat{\vartheta}_n \leq \widehat{\lambda}_n \text{ and so finally } \widehat{\lambda}_n = \widehat{\vartheta}_n.$$

(b) The proof of this part is similar to that of (a).

□

After these abstract results, we are ready to deal with linear eigenvalue problems with indefinite weight. Let $\Omega \subseteq \mathbb{R}^N (N \geq 2)$ be a bounded domain with a C^2-boundary $\partial\Omega$. We consider the following second order differential operator in divergence from

$$L(u) = -\text{div}(A(z)Du) + a_0(z)u$$
$$= -\sum_{i,j=1}^{N} D_j(a_{ij}(z)D_i u) + a_0(z)u.$$

Here $A(z) = (a_{ij}(z))_{i,j=1}^N$(an $N \times N$-matrix), $D_k = \dfrac{\partial}{\partial z_k}$ for all $k = 1, \cdots, N$ and $Du = (D_i u)_{i=1}^N$ (the gradient of u). We assume the following:

$\mathbf{H_1}$ $a_{ij} \in L^{\infty}(\Omega)$, $a_{ij} = a_{ji}$, $a_0 \in L^{\frac{N}{2}}(\Omega)$, $a_0(z) \geq 0$ for a.a $z \in \Omega$ and the $N \times N$-matrix $A(z)$ satisfies the following strong ellipticity condition

$$(A(z)\xi, \xi)_{\mathbb{R}^N} \geq \widehat{c}|\xi|^2$$
$$\text{for a.a } z \in \Omega, \text{ all } \xi \in \mathbb{R}^N, \text{ some } \widehat{c} \geq 0.$$

Let $m \in L^s(\Omega)$ with $s > \dfrac{N}{2}$ and consider the following linear eigenvalue problem with weight $m(\cdot)$

$$Lu(z) = \widehat{\lambda} m(z)u(z) \text{ in } \Omega, u|_{\partial\Omega} = 0. \tag{4.53}$$

Definition 4.65 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" of (4.53), if we can find $\widehat{u} \in H_0^1(\Omega) \setminus \{0\}$ such that

$$\int_{\Omega} (A(z)D\widehat{u}, Dh)_{\mathbb{R}^N} + \int_{\Omega} a_0(z)\widehat{u}hdz = \widehat{\lambda} \int_{\Omega} m(z)\widehat{u}hdz \text{ for all } h \in H_0^1(\Omega)$$

(that is, $\widehat{u}$ is a nontrivial weak solution of (4.53)). The solution $\widehat{u}$ is an "eigenfunction" corresponding to the eigenvalue $\widehat{\lambda}$. We call the pair $(\widehat{\lambda}, \widehat{u})$ an "eigenpair" of (4.53).

By the Sobolev embedding theorem (see Theorem I-4.114) we know that $H_0^1(\Omega) \hookrightarrow L^{2^*}(\Omega)$ continuously with $2^* = \frac{2N}{N-2}$ if $N \geq 3$ and $2^* = r$ for any $1 \leq r < \infty$ if $N = 2$. Then we have

$$\frac{2}{N} + \frac{2}{2^*} = 1$$

and this by the generalized Hölder inequality (see Corollary I-2.121) implies that the integrals

$$\int_\Omega a_0(z)\widehat{u}hdz \text{ and } \int_\Omega m(z)\widehat{u}hdz$$

are finite.

We introduce a new inner product $(\cdot,\cdot)_0$ on $H_0^1(\Omega)$ defined by

$$(u,h)_0 = \int_\Omega (A(z)Du, Dh)_{\mathbb{R}^N} dz + \int_\Omega a_0(z)uhdz$$

for all $u, h \in H_0^1(\Omega)$ (see hypotheses H_1)

From hypotheses H_1, we infer that

$$|(u,h)_0| \leq c_5\|u\|\|h\| \text{ for some } c_5 > 0, \text{ all } u, h \in H_0^1(\Omega),$$

$$\widehat{c}\|u\|^2 \leq (u,u)_0 \text{ for all } u \in H_0^1(\Omega)$$

(see hypotheses H_1), with $\|\cdot\|$ being the norm of $H_0^1(\Omega)$ (on account of the Poincare inequality, $\|u\| = \|Du\|_2$ for all $u \in H_0^1(\Omega)$). From the above inequality it follows that $(\cdot,\cdot)_0$ us equivalent to the usual inner product of $H_0^1(\Omega)$ (that is, $(u.h) = \int_\Omega (Du, Dh)_{\mathbb{R}^N} dz$ for all $u, h \in H_0^1(\Omega)$). Let $|\cdot|_0$ be the norm corresponding to the inner product $(\cdot,\cdot)_0$. Evidently $\|\cdot\|$ and $|\cdot|_0$ are equivalent norms on $H_0^1(\Omega)$.

Let $u \in H_0^1(\Omega)$ and consider the map $\mu(u) : H_0^1(\Omega) \to \mathbb{R}$ defined by

$$\mu(u)(h) = \int_\Omega m(z)uhdz \text{ for all } h \in H_0^1(\Omega).$$

Clearly this is a continuous linear functional on $H_0^1(\Omega)$. Therefore by the Riesz-Frechet representation theorem, we can find unique $\widehat{A}(u) \in H_0^1(\Omega)$ such that

$$(\widehat{A}(u), h)_0 = \int_\Omega m(z)uhdz \text{ for all } h \in H_0^1(\Omega). \tag{4.54}$$

Proposition 4.66 *If hypotheses H_1 hold, then $\widehat{A} \in \mathcal{L}_c(H_0^1(\Omega))$ and it is self adjoint.*

Proof Clearly $\widehat{A} \in \mathcal{L}(H_0^1(\Omega))$ and it is self adjoint (see (4.54)). It remains to show that $\widehat{A}(\cdot)$ is compact. According to Corollary I-3.149 we have to show that $\widehat{A}(\cdot)$ is completely continuous. So, suppose that $u_n \xrightarrow{w} u$ in $H_0^1(\Omega)$. From the Sobolev embedding theorem we know that

$$H_0^1(\Omega) \hookrightarrow L^r(\Omega) \text{ compactly for all } 1 \leq r \leq 2^*.$$

Also note that

$$\frac{2}{N} + \frac{2}{r} \to 1 \text{ as } r \to 2^*.$$

Since by hypothesis $m \in L^s(\Omega)$ with $s > \frac{N}{2}$, for r near 2^*, we have $\frac{1}{s} + \frac{2}{r} < 1$ and so as before using the generalized Hölder inequality, we have

$$\begin{aligned} |(\widehat{A}(u_n - u), h)_0| &= |\int_\Omega m(z)(u_n - u)h dz| \\ &\leq \|m\|_s \|u_n - u\|_r \|h\|_r \\ &\leq c_6 \|m\|_s \|u_n - u\|_r |h|_0 \text{ for some } c_6 > 0, \end{aligned}$$

$$\Rightarrow \|\widehat{A}(u_n) - \widehat{A}(u)\|_0 \leq c_6 \|m\|_s \|u_n - u\|_r \to 0,$$

$$\Rightarrow \widehat{A}(\cdot) \text{ is completely continuous, thus compact.}$$

□

We rewrite problem (4.53) in the following equivalent way

$$(u, h)_0 = \widehat{\lambda}(\widehat{A}(u), h)_0 \text{ for all } h \in H_0^1(\Omega),$$

$$\Rightarrow \widehat{A}(u) = \frac{1}{\widehat{\lambda}} u.$$

Now we can use the earlier abstract results and have the following proposition concerning the spectrum (4.53).

Proposition 4.67 *If hypotheses H_1 hold, then the eigenvalue problem (4.53) has a double sequence of eigenvalues $\ldots \leq \widehat{\lambda}_{-n} \leq \ldots \leq \widehat{\lambda}_{-2} \leq \widehat{\lambda}_{-1} < 0 < \widehat{\lambda}_1 \leq \widehat{\lambda}_2 \leq \ldots \leq \widehat{\lambda}_n \leq \ldots$ and we have*

$$\frac{1}{\widehat{\lambda}_n} = \sup_{v \in \mathcal{F}_n} \inf \{\int_\Omega m(z)u^2 dz : |u|_0 = 1, u \in V\}$$

$$\frac{1}{\widehat{\lambda}_{-n}} = \inf_{v \in \mathcal{F}_n} \sup \{\int_\Omega m(z)u^2 dz : |u|_0 = 1, u \in V\}.$$

For the corresponding eigenfunctions $\widehat{u}_n$ we have

$$(\widehat{u}_n, h)_0 = \widehat{\lambda}_n \int_\Omega m(z)\widehat{u}_n h dz \text{ for all } h \in H_0^1(\Omega),$$

$$|\widehat{u}_n|_0 = 1, \frac{1}{\widehat{\lambda}_n} = \int_\Omega m(z)\widehat{u}_n^2 dz.$$

Remark 4.68 Note that $\widehat{\lambda}_n \to +\infty$ and $\widehat{\lambda}_{-n} \to -\infty$ (since $\widehat{\lambda}_{\pm n} = \frac{1}{\widehat{\mu}_{\pm n}}$ with $\mu_{\pm n}$ being in the spectrum of $\widehat{A} \in \mathcal{L}_c(H_0^1(\Omega))$ and so $\widehat{\mu}_{\pm n} \to 0$).

Recall that in general the weight $m(\cdot)$ is indefinite. We set

$$\Omega_+ = \{z \in \Omega : m(z) > 0\}, \Omega_- = \{z \in \Omega : m(z) < 0\}.$$

By $|\cdot|_N$ we denote the Lebesgue measure on $\mathbb{R}^N$.

Proposition 4.69 *If hypotheses H_1 hold, then*

(a) $|\Omega_+|_N = 0 \Rightarrow$ *there are no positive eigenvalues;*
(b) $|\Omega_-|_N = 0 \Rightarrow$ *there are no negative eigenvalues;*
(c) $|\Omega_+|_N > 0 \Rightarrow$ *there is a sequence* $\{\widehat{\lambda}_n\}_{n\in\mathbb{N}}$ *of positive eigenvalues such that* $\widehat{\lambda}_n \to \infty$*;*
(d) $|\Omega_-|_N > 0 \Rightarrow$ *there is a sequence* $\{\widehat{\lambda}_{-n}\}_{n\in\mathbb{N}}$ *of negative eigenvalues such that* $\widehat{\lambda}_{-n} \to -\infty$.

Proof

(a) and (b) follow from Proposition 4.64.
(c) For each $n \in \mathbb{N}$, we consider mutually disjoint open balls $\{B_k\}_{k=1}^n$ such that

$$|\Omega_+ \cap B_k|_N > 0 \text{ for all } k = 1, \dots, n.$$

We choose functions $\{u_k\}_{k=1}^n \subseteq C_c^\infty(\Omega)$ such that $\sup u_k \subseteq B_k$ and $\int_\Omega m(z)u_k^2 dz > 0$ for all $k = 1, \dots, n$. Normalizing, we also have

$$\int_\Omega m(z)u_k^2 dz = 1 \text{ for all } k = 1, \dots, n.$$

Evidently $\{u_k\}_{k=1}^n$ is an orthogonal family. We set

$$V = span\{u_k\}_{k=1}^n \in \mathcal{F}_n$$

If $v \in V$, then $v = \sum_{k=1}^n \beta_k u_k$ and we have

$$\int_\Omega m(z)v^2 dz = \sum_{k=1}^n \beta_k^2 \int_\Omega m(z)u_k^2 dz = \sum_{k=1}^n \beta_k^2, \tag{4.55}$$

$$|v|_0^2 = \sum_{k=1}^{n} \beta_k^2 |u_k|_0^2 \leq c_7 \sum_{k=1}^{n} \beta_k^2 \text{ for some } c_7 > 0 \tag{4.56}$$

Finally from (4.55), (4.56) we obtain

$$\frac{1}{c_7}|v|_0^2 \leq \int_{\Omega} m(z)v^2 dz \text{ for all } v \in V.$$

Then the result follows from Proposition 4.67.

(d) This is proved as part (c).

□

What about the UCP for the eigenspaces (see Definition 4.40)? In this direction, we have the following result due to Jerison and Kenig [260]

Proposition 4.70 *If $L(u) = -\Delta u$ and $m \in L^s(\Omega)$ with $s > \frac{N}{2}$, then each eigenspace $E(\widehat{\lambda}_k)$ $k \in \mathbb{N}$ has the UCP.*

Remark 4.71 What about the more general elliptic differential operator of problem (53)? For such operators the UCP holds if $a_0 \equiv 0$ and the matrix valued map $A : \overline{\Omega} \to \mathbb{R}^{N\times N}$ is Lipschitz continuous, while the weight satisfies the following property: For every $z_0 \in \Omega$, there exist $r_0 > 0$ and $\widehat{c}$, $\widehat{c}^* > 0$ such that

$$m^+(z) \leq \frac{\widehat{c}}{|z - z_0|^2} \quad \text{and} \quad m^-(z) \leq \frac{\widehat{c}^*}{|z - z_0|^2} \quad \text{for a.a } z \in \Omega \cap B_{r_0}(z_0).$$

These weight functions form a subset of $L^{\frac{N}{2}}_{loc}(\Omega)$ which contains $L^{\infty}_{loc}(\Omega)$. The result is due to Garofalo and Lin [178]. Another situation where the UCP holds, is when $a_0 \equiv 0$, $A \in C^{\infty}(\Omega, \mathbb{R}^{N\times N})$ and $m \in L^{\frac{N}{2}}(\Omega)$ (see Sogge [444]).

Now we focus on the properties of the map $m \to \widehat{\lambda}_k(m)$ for all $k \in \mathbb{Z}_0 = \mathbb{Z} \setminus \{0\}$.

Proposition 4.72 *If hypotheses H_1 hold, m, $\widehat{m} \in L^s(\Omega)$, $s > \frac{N}{2}$, $m(z) \leq \widehat{m}(z)$ for a.a $z \in \Omega$ and the inequality is strict on a set of positive measure, $k \in \mathbb{Z}_0$ and the eigenspace $E(\widehat{\lambda}_k(m))$ has the UCP, then $\widehat{\lambda}_k(\widehat{m}) < \widehat{\lambda}_k(m)$.*

Proof First we assume that $k \in \mathbb{N}$.

From Proposition 4.67 (see also Remark 4.68), we can find $V_k \subseteq H_0^1(\Omega)$ a k-dimensional subspace such that

$$\frac{1}{\widehat{\lambda}_k(m)} = \inf\left\{\int_{\Omega} m(z)u^2 dz : |u|_0 = 1,\ u \in V_k\right\}.$$

If $u \in V_k$, $|u|_0 = 1$ is an eigenfunction corresponding to $\widehat{\lambda}_k(m)$, then on account of the UCP we have $u(z) \neq 0$ for a.a $z \in \Omega$. Therefore

$$\frac{1}{\widehat{\lambda}_k(m)} = \int_{\Omega} m(z)u^2 dz < \int_{\Omega} \widehat{m}(z)u^2 dz.$$

If $u \in V_k$, $|u|_0 = 1$ is not an eigenfunction corresponding to $\widehat{\lambda}_k(m)$ (that is, $u \notin E(\widehat{\lambda}_k(m))$), then

$$\frac{1}{\widehat{\lambda}_k(m)} < \int_\Omega m(z)u^2 dz \le \int_\Omega \widehat{m}(z)u^2 dz.$$

So, we see that

$$\frac{1}{\widehat{\lambda}_k(m)} < \int_\Omega \widehat{m}(z)u^2 dz \quad \text{for all } u \in V_k,\ |u_0| = 1.$$

Let

$$l_k = \inf\left\{\int_\Omega \widehat{m}(z)u^2 dz : |u|_0 = 1,\ u \in V_k\right\}.$$

Consider $\{u_i\}_{i\in\mathbb{N}} \subseteq V_k$, $|u_i|_0 = 1$, $i \in \mathbb{N}$, a minimizing sequence, that is, $\int_\Omega \widehat{m}(z)u_i^2 dz \downarrow l_k$ as $i \to \infty$. Since V_k is finite dimensional, we may assume that

$$\begin{aligned}
&u_i \to \widetilde{u} \text{ in } H_0^1(\Omega),\ |\widetilde{u}|_0 = 1,\ \widetilde{u} \in V_k,\\
\Rightarrow &\int_\Omega \widehat{m}(z)u^2 dz = l_k,\ |\widetilde{u}|_0 = 1,\ \widetilde{u} \in V_k,\\
\Rightarrow &\frac{1}{\widehat{\lambda}_k(m)} < l_k \le \frac{1}{\widehat{\lambda}_k(\widehat{m})} \text{ (see Proposition 4.67)},\\
\Rightarrow &\widehat{\lambda}_k(\widehat{m}) < \widehat{\lambda}_k(m).
\end{aligned}$$

The proof is similar if $k \in -\mathbb{N}$. □

From Proposition 4.67, we easily deduce the following continuity property of $m \to \widehat{\lambda}_k(m)$.

Proposition 4.73 *If hypotheses H_1 hold, $\{m_n, m\}_{n\in\mathbb{N}} \subseteq L^s(\Omega)$ with $s > \frac{N}{2}$, $m_n \to m$ in $L^s(\Omega)$ and $k \in \mathbb{Z}_0$, then $\widehat{\lambda}_k(m_n) \to \widehat{\lambda}_k(m)$ as $n \to \infty$.*

Using this proposition, we can show that the monotonicity property of Proposition 4.72 fails if the UCP is not true for $E(\widehat{\lambda}_k(m))$.

Proposition 4.74 *If hypotheses H_1 hold, $m \in L^s(\Omega)$ with $s > \frac{N}{2}$, $k \in \mathbb{Z}_0$ and $E(\widehat{\lambda}_k(m))$ does not have the UCP, then we can find a weight $\widehat{m} \in L^s(\Omega)$, $s > \frac{N}{2}$ with $m(z) \le \widehat{m}(z)$ for a.a $z \in \Omega$ and the inequality is strict on a set of positive measure such that for some $i \in \mathbb{Z}_0$ with $\widehat{\lambda}_i(m) = \widehat{\lambda}_k(m)$ we have*

$$\widehat{\lambda}_i(m) = \widehat{\lambda}_i(\widehat{m}).$$

Proof First we assume that $k \in \mathbb{N}$.

Let u be an eigenfunction corresponding to $\widehat{\lambda}_k(m)$, which vanishes on a set of positive Lebesgue measure. Since we have assumed that $E(\widehat{\lambda}_k(m))$ does not have the UCP, such a function exists. Let $i \in \mathbb{N}$ such that

$$\widehat{\lambda}_k(m) = \widehat{\lambda}_i(m) < \widehat{\lambda}_{i+1}(m).$$

Consider the following function

$$\widehat{m}(z) = \begin{cases} m(z), & \text{if } u(z) \neq 0 \\ m(z) + \varepsilon, & \text{if } n(z) = 0. \end{cases} \tag{4.57}$$

Then $\widehat{m} \in L^s(\Omega)$, $m(z) \leq \widehat{m}(z)$ and the inequality is strict on a set of positive measure. Using Proposition 4.73, we see that if $\varepsilon > 0$ is small, then

$$\widehat{\lambda}_i(m) < \widehat{\lambda}_{i+1}(\widehat{m}).$$

We have

$$L(u) = \widehat{\lambda}_i(m)mu = \widehat{\lambda}_i(m)\widehat{m}u \text{ (see (4.57))},$$
$$\Rightarrow \widehat{\lambda}_i(m) \text{ is an eigenvalue for the weight } \widehat{m}.$$

Hence $\widehat{\lambda}_i(m) = \widehat{\lambda}_j(\widehat{m})$ for some $j \in \mathbb{N}$ and we can always choose the biggest such $j \in \mathbb{N}$. Evidently $j < i + 1$. Also $\widehat{\lambda}_i(\widehat{m}) \leq \widehat{\lambda}_i(m)$. Therefore $i \leq j$. We conclude $i = j$ and so $\widehat{\lambda}_i(m) = \widehat{\lambda}_i(\widehat{m})$.

The some reasoning applies to the case $k \in -\mathbb{N}$. □

For the first positive and negative eigenvalues, we have nice properties.

Proposition 4.75

(a) If hypotheses H_1 hold, $m \in L^s(\Omega)$ with $s > \frac{N}{2}$ and $m^+ \neq 0$, then $\widehat{\lambda}_1(m) > 0$, it is simple (that is, dim $E(\widehat{\lambda}_1(m)) = 1$) and if u is an eigenfunction corresponding to $\widehat{\lambda}_1(m)$, then

$$u(z) > 0 \text{ or } u(z) < 0 \text{ for } a.a\ z \in \Omega.$$

(b) If hypotheses H_1 hold, $m \in L^s(\Omega)$ with $s > \frac{N}{2}$ and $m^- \neq 0$, then $\widehat{\lambda}_{-1}(m) < 0$, it is simple (that is, dim $E(\widehat{\lambda}_{-1}(m)) = 1$) and if v is an eigenfunction corresponding to $\widehat{\lambda}_{-1}(m)$, then

$$v(z) > 0 \text{ or } v(z) < 0 \text{ for } a.a\ z \in \Omega.$$

Proof

(a) Let $u \in H_0^1(\Omega)$ be an eigenfunction corresponding to $\widehat{\lambda}_1(m)$. We have

$$(u, h)_0 = \widehat{\lambda}_1(m) \int_\Omega m(z) u h dz \text{ for all } h \in H_0^1(\Omega)$$

Suppose $u^+ \neq 0$, $u^- \neq 0$. We know that $u^\pm \in H_0^1(\Omega)$. So, if we choose as test function $h = u^+ \in H_0^1(\Omega)$ and $h = -u^- \in H_0^1(\Omega)$, we obtain

$$|u^+|_0^2 = \widehat{\lambda}_1(m) \int_\Omega m(z)(u^+)^2 dz,$$
$$|u^-|_0^2 = \widehat{\lambda}_1(m) \int_\Omega m(z)(u^-)^2 dz.$$

If follows that both $u^\pm$ are eigenfunctions corresponding to $\widehat{\lambda}_1(m)$ (see (4.49)). So, we have

$$Lu^+(z) = \widehat{\lambda}_1(m) m(z) u^+(z) \quad \text{for a.a } z \in \Omega,$$
$$Lu^-(z) = \widehat{\lambda}_1(m) m(z) u^-(z) \quad \text{for a.a } z \in \Omega.$$

Using the Harnack inequality (see Gilbarg and Trudinger [206], p. 199), we have

$$u^+(z) > 0,\ u^-(z) > 0 \quad \text{for a.a } z \in \Omega,$$

a contradiction. Therefore $u(\cdot)$ has fixed sign.
Suppose $\widehat{\lambda}_1(m) > 0$ is not simple. Then we can have $u_1,\ \widetilde{u}_1$ two orthogonal positive eigenfunctions. So,

$$u_1(z) > 0,\ \widetilde{u}_1(z) > 0 \quad \text{for a.a } z \in \Omega,\ \int_\Omega u_1 \widehat{u}_1 dz = 0,$$

a contradiction. Therefore $\widehat{\lambda}_1(m) > 0$ is simple.

(b) Similarly for $\widehat{\lambda}_{-1}(m) < 0$.

□

Next we will say a few things about the regularity of the eigenfunctions.

First using the Moser iteration technique (see Moser [340] and Gilbarg and Trudinger [206], p. 191 also Proposition 4.82), we have.

Proposition 4.76 *If hypotheses H_1 hold, $m \in L^s(\Omega)$ with $s > \frac{N}{2}$, then every eigenfunction u of (4.53) satisfies*

$$u \in H_0^1(\Omega) \cap L^\infty(\Omega).$$

Using this proposition, we can have the following global (that is, up to the boundary) regularity result for the eigenfunctions. Recall that $A(z) = (a_{ij}(z))_{i,j=1}^N$ (matrix of coefficients).

Proposition 4.77

(a) If $A \in C^k(\overline{\Omega}, \mathbb{R}^{N\times N})$, $a_0 \in L^\tau(\Omega)$ with $\tau > \frac{N}{2}$, $a_0(z) \geq 0$ for a.a $z \in \Omega$ and $m \in L^s(\Omega)$ with $s > \frac{N}{2}$, then every eigenfunction u satisfies

$$u \in H_0^1(\Omega) \cap C(\overline{\Omega}).$$

(b) If $A \in C^k(\overline{\Omega}, \mathbb{R}^{N\times N})$, $a_0 \in L^\tau(\Omega)$ with $\tau > N$, $a_0(z) \geq 0$ for a.a $z \in \Omega$ and $m \in L^s(\Omega)$ with $s > N$, then every eigenfunction u satisfies

$$u \in C_0^1(\overline{\Omega}) = \{u \in C^1(\overline{\Omega}) : u|_{\partial\Omega} = 0\}.$$

Proof

(a) On account of Proposition 4.76, we have

$$(mu)(\cdot) \in L^s(\Omega) \text{ with } s > \frac{N}{2}.$$

Using Theorem 9.15, p. 241, of Gilbarg and Trudinger [206], we have that

$$u \in W^{2,\eta}(\Omega) \text{ with } \eta = \min\{\tau, s\}.$$

Note that $\eta > \frac{N}{2}$ and so by the Sobolev embedding theorem (see Theorem I-4.114), we have

$$u \in H_0^1(\Omega) \cap C(\overline{\Omega}).$$

(b) The reasoning is similar, only now the Sobolev embedding theorem implies that

$$u \in C^{1,\alpha}(\overline{\Omega}),\ 0 < \alpha < 1 - \frac{N}{\eta}.$$

□

We can have similar results for the following linear Robin eigenvalue problem

$$\left\{\begin{array}{ll} Lu(z) = \widehat{\mu} m(z)u(z) & \text{in } \Omega, \\ \frac{\partial u}{\partial n_A} + \beta(z)u = 0 & \text{on } \partial\Omega. \end{array}\right\} \tag{4.58}$$

As before

$$L(u) = -div(A(z)Du) + a_0(z)u,$$

with $A(z) = (a_{ij}(z))_{i,j=1}^N$, $a_{i,j} = a_{j,i}$ (symmetric matrix), $A \in L^\infty(\Omega, \mathbb{R}^{N\times N})$, $a_0 \in L^s(\Omega)$ with $s > N$ and $a_0(z) \geq 0$ for a.a $z \in \Omega$. In addition, we assume the strong ellipticity condition

$$(A(z)\xi, \xi)_{\mathbb{R}^N} \geq \widehat{c}|\xi|^2 \quad \text{for a.a } z \in \Omega, \text{ all } \xi \in \mathbb{R}^N, \text{ some } \widehat{c} > 0.$$

For the weight function $m(\cdot)$ we assume that

$$m \in L^s(\Omega) \text{ with } s > \frac{N}{2} \ (m(\cdot) \text{ is in general sign-changing}).$$

In the boundary condition $\frac{\partial u}{\partial n_A}$ denotes the conormal derivative, which is interpreted using Green's identity and

$$\frac{\partial u}{\partial n_A} = (A(z)Du, n)_{\mathbb{R}^N} \text{ for all } u \in C^1(\overline{\Omega})$$

with $n(\cdot)$ being the outward unit normal on $\partial\Omega$.

For the boundary coefficient $\beta(\cdot)$ we assume

$$\beta \in L^\infty(\partial\Omega), \ \beta(z) \geq 0 \text{ for } \sigma\text{-a.a } z \in \partial\Omega.$$

We make the following assumption.

$\mathbf{H_2}$: $a_0 \neq 0$ or $\beta \neq 0$.

The two lemmata that follow justify this hypothesis. In what follows by $\|\cdot\|$ we denote the norm of the Sobolev space $H^1(\Omega)$ defined by

$$\|u\| = (\|u\|_2^2 + \|Du\|_2^2)^{\frac{1}{2}} \text{ for all } u \in H^1(\Omega).$$

Lemma 4.78 *If $A(\cdot)$ is as above and $a_0 \neq 0$, then there exists $c_7 > 0$ such that*

$$\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz + \int_\Omega a_0(z)u^2 dz \geq c_7\|u\|^2 \tag{4.59}$$

for all $u \in H^1(\Omega)$.

Proof The strong ellipticity condition and the hypothesis that $a_0 \geq 0$, imply that the left hand side of (4.59) is nonnegative. Suppose the result of the lemma is not true we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ such that

$$\|u_n\| = 1, \ n \in \mathbb{N} \text{ and } \int_\Omega (A(z)Du_n, Du_n)_{\mathbb{R}^N} dz + \int_\Omega a_0(z)u_n^2 dz \downarrow 0.$$

Then at least for a subsequence, we have

$$u_n \xrightarrow{w} u \text{ in } H^1(\Omega),\ u_n \to u \text{ in } L^{s'}(\Omega)\ \left(\frac{1}{s}+\frac{1}{s'}=1\right)$$

(note that $s' < (\frac{N}{2})' < 2^*$). In the limit as $n \to \infty$ we have

$$\begin{aligned}
&\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz + \int_\Omega a_0(z)u^2 dz \le 0,\\
&\Rightarrow\ \widehat{c}\,\|Du\|_2^2 + \int_\Omega a_0(z)u^2 dz = 0.\\
&\Rightarrow\ u = \vartheta \in \mathbb{R}.
\end{aligned}$$

If $\vartheta \neq 0$, then $0 < \vartheta^2 \int_\Omega a_0(z)dz = 0$, a contradiction. So, $\vartheta = 0$. Then we see that $u_n \to 0$ in $H^1(\Omega)$ and this contradicts the fact that $\|u_n\| = 1$ for all $n \in \mathbb{N}$. This proves the lemma. □

Lemma 4.79 *If $A(\cdot)$ is as above and $\beta \neq 0$, then there exists $c_8 > 0$ such that*

$$\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz + \int_{\partial\Omega} \beta(z)uhd\sigma \ge c_8\|u\|^2$$

for all $u \in H^1(\Omega)$.

Proof For $u, h \in H^1(\Omega)$, we define

$$(u,h)_* = \int_\Omega (A(z)Du, Dh)_{\mathbb{R}^N} dz + \int_{\partial\Omega} \beta(z)uhd\sigma$$

This is an inner product on $H^1(\Omega)$. Indeed we have

$$\begin{aligned}
&(u,h)_* = (h,u)_* \text{ (since } A(z) \text{ is a symmetric matrix)},\\
&(u,u)_* \ge 0,\\
&(u,u)_* = 0 \Rightarrow \widehat{c}\,\|Du\|_2^2 \le 0 \Rightarrow u = \vartheta \in \mathbb{R}.
\end{aligned}$$

Then

$$\vartheta^2 \int_{\partial\Omega} \beta(z)d\sigma = 0 \text{ and so } \vartheta = 0.$$

By $|\cdot|_*$ we denote the norm induced by this inner product. We have

$$\begin{aligned}
|u|_*^2 &\le \|A\|_\infty \|Du\|_2^2 + \|\beta\|_{L^\infty(\partial\Omega)} \|u\|_{L^2(\partial\Omega)}^2\\
&\le \|A\|_\infty \|Du\|_2^2 + \|\beta\|_{L^\infty(\partial\Omega)} \|\gamma_0\|_{\mathcal{L}} \|u\|^2
\end{aligned}$$

with $\gamma_0(\cdot)$ being the trace operator (recall that $\gamma_0 \in \mathcal{L}(H^1(\Omega), L^2(\partial\Omega))$). So, we have

$$|u|_*^2 \leq c_g \|u\| \text{ for some } c_g > 0, \text{ all } u \in H^1(\Omega). \tag{4.60}$$

Next we show that there exists $c_{10} > 0$ such that

$$\|u\|_2 \leq c_{10}|u|_* \quad \text{for all } u \in H^1(\Omega). \tag{4.61}$$

Arguing by contradiction, suppose that (4.61) is not true and we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ such that

$$n|u_n|_* < \|u_n\|_2 \quad \text{for all } n \in \mathbb{N}.$$

Normalizing in $L^2(\Omega)$, we have

$$\begin{aligned} &|u_n|_* < \frac{1}{n}, \ \|u_n\|_2 = 1 \quad \text{for all } n \in \mathbb{N}, \\ \Rightarrow\ &|u_n|_* \to 0, \\ \Rightarrow\ &\|Du_n\|_2 \to 0 \text{ (by the strong ellipticity hypothesis).} \end{aligned}$$

Therefore $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ is bounded. So, we may assume that

$$u_n \xrightarrow{w} u \text{ in } H^1(\Omega), \ u_n \to u \text{ in } L^2(\partial\Omega). \tag{4.62}$$

Then using (4.62) in the limit as $n \to \infty$, we obtain

$$\begin{aligned} &\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz + \int_{\partial\Omega} \beta(z)u^2 dz \leq 0, \\ \Rightarrow\ &u = \vartheta \in \mathbb{R}. \end{aligned}$$

Since $\beta \not\equiv 0$, if follows that $\vartheta = 0$ and so $u_n \to 0$ in $L^2(\Omega)$ (see (4.62)), a contradiction since $\|u_n\|_2 = 1$. This proves (4.61).

From (4.61) if follows that

$$\|u\|^2 \leq c_{11}|u|_*^2 \quad \text{for some } c_{11} > 0, \text{ all } u \in H^1(\Omega). \tag{4.63}$$

Then (4.60) and (4.63) imply that $\|\cdot\|$ and $|\cdot|_*$ are equivalent norms on $H^1(\Omega)$ and this proves the lemma. □

These two lemmata and hypothesis H_2 imply that

$$(u,h)_0 = \int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz + \int_\Omega a_0 uh dz + \int_{\partial\Omega} \beta(z)uh dz \text{ for all } u \in H^1(\Omega).$$

and

$$(u,h)=\int_{\Omega}(Du,Dh)_{\mathbb{R}^N}dz+\int_{\Omega}uhdz \text{ for all } u,h\in H^1(\Omega).$$

are equivalent inner products on $H^1(\Omega)$ and so we can proceed as in the Dirichlet case. Therefore we can have the same results for the spectrum of (4.58). For the regularity of the eigenfunctiona, we assume that $\beta\in W^{1,\infty}(\partial\Omega)$ (see Wang [467]).

4.3 Nonlinear Operators

In this section we study the spectral properties of the p-Laplace differential operator.

We start by presenting some general results about the regularity of weak solutions of nonlinear elliptic differential equations which we will need in the sequel.

Let $\Omega\subseteq\mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$. We consider the following two problem

$$\operatorname{div}a(Du)=f(z,u)\text{ in }\Omega,\ u|_{\partial\Omega}=0, \tag{4.64}$$

$$\operatorname{div}a(Du)=f(z,u)\text{ in }\Omega,\ \frac{\partial u}{\partial n_\alpha}+\beta(z)|u|^{p-1}=0\text{ on }\partial\Omega. \tag{4.65}$$

The hypotheses on the map $a(\cdot)$ involved in the differential operator of (4.64) and (4.65).

H_0 : $a:\mathbb{R}^N\to\mathbb{R}^N$ is continuous and

(i) $|a(y)|\le c_0(1+|y|^{p-1})$ for all $y\in\mathbb{R}^N$, some $c_0>0$;
(ii) $\widehat{c}|y|^p\le(a(y),y)_{\mathbb{R}^N}$ for all $y\in\mathbb{R}^N$, some $\widehat{c}>0$.

H_1 : $f:\Omega\times\mathbb{R}\to\mathbb{R}$ is a Caratheodary function such that

$$|f(z,x)|\le\widehat{a}(z)(1+|x|^{r-1})\text{ for a.a }z\in\Omega,\ \text{all }x\in\mathbb{R},$$

with $\widehat{a}\in L^\infty(\Omega)$ and $p\le r<p^*$ (recall p^* is the critical Sobolev exponent defined by $p^*=\begin{cases}\frac{Np}{N-p}, & \text{if } p<N\\ +\infty, & \text{if } N\le p\end{cases}$).

For the Robin problem (problem (4.65)), $\frac{\partial u}{\partial n_a}$ denotes the conormal derivative of u corresponding to the map $\alpha(\cdot)$ which is interpreted using the nonlinear Green's identity (see Theorem I-4.106). In particular we have

$$\frac{\partial u}{\partial n_a}=(\alpha(Du),n)_{\mathbb{R}^N}\text{ for all }u\in C^1(\overline{\Omega})$$

with $n(\cdot)$ being the outward unit normal on $\partial\Omega$. For the boundary coefficient $\beta(\cdot)$, the hypotheses are the following.

H_3 : $\beta \in L^\infty(\partial\Omega)$, $\beta(z) \geq 0$ for σ-a.a $z \in \partial\Omega$.

Remark 4.80 If $\beta \equiv 0$, then problem (4.65) is the Neumann problem.

Definition 4.81

(a) A function $u \in W_0^{1,p}(\Omega)$ is a "weak solution" of (4.64), if we have

$$\int_\Omega (a(Du), Dh)_{\mathbb{R}^N} dz = \int_\Omega f(z,u)hdz \text{ for all } h \in W_0^{1,p}(\Omega).$$

(b) A function $u \in W^{1,p}(\Omega)$ is a "weak solution" of (4.65), if we have $\int_\Omega (a(Du), Dh)_{\mathbb{R}^N} dz + \int_{\partial\Omega} \beta(z)|u|^{p-2}uhdz = \int_\Omega f(z,u)hdz$ for all $h \in W^{1,p}(\Omega)$.

Under the above hypotheses, we can prove the boundedness of the weak solutions of problems (4.64) and (4.65). We do this using the so called "Moser iteration technique". In what follows we set

$$\widehat{p}^* = p^* \text{ if } p < N \quad \text{and} \quad \widehat{p}^* > r \text{ if } N \leq p.$$

Proposition 4.82 *If hypotheses H_0, H_1 and for problem (4.65) also H_2 hold, and $u \in W_0^{1,p}(\Omega)$ (resp. $u \in W^{1,p}(\Omega)$) is a weak solution for problem (4.64) (resp. problem (4.65)), then $u \in L^\infty(\Omega)$ and we have $\|u\|_\infty \leq M$ with $M > 0$ depending on $\{c_0, \widehat{c}, \widehat{a}, N, p, |\Omega|_N, \|u\|_\beta\}$.*

Proof We restrict ourselves to problem (4.65).

Since $W^{1,p}(\Omega) \hookrightarrow W^{1,q}(\Omega)$ continuously and densely for $1 \leq q < p$, we see that we need to consider the case $p < N$.

Also by writing $u = u^+ - u^-$ with $u^\pm \in W^{1,p}(\Omega)$, $u^\pm \geq 0$, we see that without any loss of generality, we may assume that $u \geq 0$.

For every $\mu > 0$, let $u_\mu = \max\{u^+, \mu\} \in W^{1,q}(\Omega)$. Also let $\vartheta \geq 0$. Using hypotheses H_0, H_1, H_2 and Definition 4.81 with $h = u_\mu$ we see that

$$\frac{p^\vartheta + 1}{(\vartheta+1)^p} \|Du_\mu^{\vartheta-1}\|_p^p \leq c_2 \left(1 + \frac{p^\vartheta + 1}{(\vartheta+1)^p}\right)(1 + \|u\|_{p^\vartheta+r}^{p^\vartheta+r}) \text{ for some } c_2 > 0.$$

Recall that $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ continuously and let $\mu \to \infty$. We obtain

$$1 + \|u\|_{p^*(\vartheta+1)}^{p^*(\vartheta+1)} \leq c_3 \frac{(\vartheta+1)^{p^*}}{(p\vartheta+1)^{\frac{p^*}{p}}}(1 + \|u\|_{p^\vartheta+r}^{p^\vartheta+r})^{\frac{p^*}{p}} \text{ for some } c_3 > 0.$$

Let $s_0 = p^*$ and $s_{k+1} = (s_k - r + p)\frac{p^*}{p}$ for $k \geq 0$. Evidently $s_k \to +\infty$ as $k \to +\infty$. If we choose $\vartheta = \frac{s_k - r}{p}$ $(k \in \mathbb{N}_0)$, then

$$1 + \|u\|_{s_k+1}^{s_k+1} \leq c_3 \frac{(s_k - r + p)^{p^*}}{p^*(s_k - r + 1)^{\frac{p^*}{p}}} (1 + \|u\|_{s_k}^{s_k})^{\frac{p^*}{p}}.$$

Let $\gamma_k = ln(1 + \|u\|_{s_k}^{s_k})$. From the last inequality, we have

$$\gamma_{k+1} \leq p^* ln(c_4(s_k - r + p)) + \frac{p^*}{p}\gamma_k \quad \text{for some } c_4 > 0.$$

Note that $s_k - r + p < (\frac{p^*}{p})^k p^*$. If follows that

$$\gamma_{k+1} \leq c_5(1 + k) + \frac{p^*}{p}\gamma_k \quad \text{for some } c_5 > 0,$$

$$\Rightarrow \gamma_{k+1} \leq c_5 \left(\sum_{m=0}^{k} \left(\frac{p^*}{p} \right)^m + \sum_{m=0}^{k} (k - m) \left(\frac{p^*}{p} \right)^m \right) + \gamma_0 \left(\frac{p^*}{p} \right)^{k+1},$$

$$\Rightarrow \|u\|_{s_k} \leq M \text{ with } M = M(c_0, \widehat{c}, \widehat{a}, N, p, |\Omega|_N, \|u\|_{p^*}) > 0.$$

Let $k \to \infty$ to conclude that

$$\|u\|_\infty \leq M.$$

□

As a consequence of the Moser iteration technique, we can have the following result concerning the weak solution of the equation

$$-\operatorname{div} a(Du) = g \quad \text{in } \Omega, \tag{4.66}$$

with Dirichlet or Robin boundary condition.

Proposition 4.83 *If hypotheses H_0, H_1 and for the Robin problem also H_2 hold, $g \in L^s(\Omega)$ with $s > \frac{N}{p}$ and $u \in W_0^{1,p}(\Omega)$ (resp. $u \in W^{1,p}(\Omega)$) is a weak solution of the Dirichlet (resp. Robin) problem (4.66), then $\|u\|_\infty \leq c\|g\|_s^{\frac{1}{p-1}}$ for some $c > 0$.*

The boundedness of the weak solutions leads to global (that is, up to boundary) regularity. For this we need to strengthen the conditions on the map involved in the differential operator.

So, let $\vartheta \in C^1(0, \infty)$ and assume that it satisfies the following growth conditions

$$0 < c^* \leq \frac{t\vartheta'(t)}{\vartheta(t)} \leq \widehat{c}_0,$$

$$\widehat{c}_1 t^{p-1} \leq \vartheta(t) \leq \widehat{c}_2(t^{s-1} + t^{p-1}) \text{ for some } \widehat{c}_1, \widehat{c}_2 > 0, 1 < s < p.$$

Then the new stronger conditions on the map $\alpha(\cdot)$ are the following:

$\boldsymbol{H_0'}$: $a(y) = a_0(|y|)y$ for all $y \in \mathbb{R}^N$ with $a_0(t) > 0$ for all $t > 0$ and

(a) $a_0 \in C^1(0, \infty)$, $t \to a_0(t)t$ is strictly increasing on $(0, \infty)$ and $a_0(t)t \to 0^+$ as $t \to 0^+$;
(b) $|\nabla a(y)| \leq \widehat{c}_3 \frac{\vartheta(|y|)}{|y|}$ for some $\widehat{c}_3 > 0$, all $y \in \mathbb{R}^N \setminus \{0\}$;
(c) $\frac{\vartheta(|y|)}{|y|}|\xi|^2 \leq (\nabla a(y)\xi, \xi)_{\mathbb{R}^N}$ for all $y \in \mathbb{R}^N \setminus \{0\}$, all $\xi \in \mathbb{R}^N$.

The hypotheses on the reaction $f(z, x)$ are:

$\boldsymbol{H_1'}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is Caratheodary such that for every $\varrho > 0$ there exists $\alpha_\varrho \in L^\infty(\Omega)$ such that $|f(z, x)| \leq \alpha_\varrho(z)$ for a.a $z \in \Omega$, all $|x| \leq \varrho$ and there exists $\widehat{\xi}_p > 0$ such that $f(z, x) + \widehat{\xi}_p x^{p-1} \geq 0$ for *a.a* $z \in \Omega$, all $0 \leq x \leq \varrho$.

For the Robin problem (4.65) the hypotheses on the boundary coefficient $\beta(\cdot)$ are:

$\boldsymbol{H_2'}$: $\beta \in C^{1,\eta}(\partial\Omega)$ with $0 < \eta < 1$.

In Lieberman [298], we can find the following global regularity result. Recall $C_0^{1,\alpha}(\overline{\Omega}) = \{u \in C_0^{1,\alpha}(\overline{\Omega}) : u|_{\partial\Omega} = 0\}$ $(0 < \alpha \leq 1)$.

Theorem 4.84

(a) If hypotheses H_0', H_1' hold and $u \in W_0^{1,p}(\Omega) \cap L^\infty(\Omega)$ is a weak solution of problem (4.64), then $u \in C_0^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0, 1)$ and

$$\|u\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq M$$

with $M > 0$ depending on $\|u\|_\infty$,

(b) If hypotheses H_0', H_1', H_2' hold and $u \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$ is a weak solution of problem (4.65), then $u \in C^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0, 1)$ and

$$\|u\|_{C^{1,\alpha}(\overline{\Omega})} \leq M$$

with $M > 0$ depending on $\|u\|_\infty$.

Remark 4.85 So, if the weak solutions of (4.64) and (4.65) are uniformly bounded in $L^\infty(\Omega)$, then they are also uniformly bounded in $C_0^{1,\alpha}(\overline{\Omega})$ and $C^{1,\alpha}(\overline{\Omega})$ respectively. This is useful to know because $C_0^{1,\alpha}(\overline{\Omega}) \hookrightarrow C_0^1(\overline{\Omega})$ and $C^{1,\alpha}(\overline{\Omega}) \hookrightarrow C^1(\overline{\Omega})$ compactly.

Let us have a closer look at the new hypotheses H_0'.

Lemma 4.86 *If hypotheses H_0' hold, then*

(a) $y \to a(y)$ *is continuous are strictly monotone (thus maximal monotone too);*
(b) $|a(y)| \leq \widehat{c}_4(|y|^{s-1} + |y|^{p-1})$ *for some* $\widehat{c}_4 > 0$, *all* $y \in \mathbb{R}^N$;
(c) $\frac{\widehat{c}_1}{p-1}|y|^p \leq (a(y), y)_{\mathbb{R}^N}$ *for all* $y \in \mathbb{R}^N$.

Proof

(a) Clearly $a(\cdot)$ is continuous. We show that it is strictly monotone. So, let $y, y' \in \mathbb{R}^N$. We have

$$\begin{aligned}(a(y) - a(y'), y - y')_{\mathbb{R}^N} &= \int_0^1 (\frac{d}{dt}a(y' + t(y - y')), y - y')_{\mathbb{R}^N} dt \\ &= \int_0^1 (\nabla a(y' + t(y - y'))(y - y'), y - y')_{\mathbb{R}^N} dt \\ &\qquad \text{(by the chain rule)} \\ &\geq \widehat{c}_1|y - y'|^p \text{ (see hypothesis } H_0' \ (iii))\end{aligned}$$

$\Rightarrow$ $a(\cdot)$ is strictly monotone.

A continuous monotone map is maximal monotone (see Proposition I-7.36).

(b) For every $y \in \mathbb{R}^N$, we have

$$a(y) = \int_0^1 \frac{d}{dt}a(ty)dt = \int_0^1 \nabla a(ty)ydt \text{ (by the chain rule)}$$

$$\Rightarrow |a(y)| \leq \int_0^1 |\nabla a(ty)||y|dt \leq \widehat{c}_5(|y|^s + |y|^p)$$

for some $\widehat{c}_5 > 0$ (see hypothesis $H_0'(ii)$).

(c) For every $y \in \mathbb{R}^N$, we have

$$\begin{aligned}(a(y), y)_{\mathbb{R}^N} &= \int_0^1 (\frac{d}{dt}a(ty), y)_{\mathbb{R}^N} dt \\ &= \int_0^1 (\nabla a(ty), y)_{\mathbb{R}^N} dt \geq \frac{\widehat{c}_1}{p-1}|y|^p \\ &\qquad \text{(see hypothesis } H_0'(iii)).\end{aligned}$$

□

Remark 4.87 Let $G_0(t) = \int_0^t a_0(s)sds$ for all $t \geq 0$. Evidently $G_0(\cdot)$ is strictly convex and strictly increasing. We set $G(y) = G_0(|y|)$ for all $y \in \mathbb{R}^N$. We have

$$\nabla G(y) = G_0'(|y|)\frac{y}{|y|} = a_0(|y|)y = a(y) \text{ for all } y \in \mathbb{R}^N \setminus \{0\},$$

$$\Rightarrow G(\cdot) \text{ is the peimitive of } a(\cdot).$$

Since $G(\cdot)$ is convex, $\nabla G(y) = a(y)$ and $G(0) = 0$, we have

$$G(y) \leq (a(y), y)_{\mathbb{R}^N}. \tag{4.67}$$

From (4.67) and Lemma 4.86, we infer the following bilateral growth conditions for the primitive $G(\cdot)$.

Corollary 4.88 *If hypotheses H_0' hold, then $\frac{\widehat{c}_1}{p(p-1)}|y|^p \leq G(y) \leq \widehat{c}_6(1+|y|^p)$ for some $\widehat{c}_6 > 0$, all $y \in \mathbb{R}^N$.*

If we strengthen a little hypotheses H_0', we can have a nonlinear version of the Hopf maximum principle (see Pucci and Serrin [423], p. 120). The new hypotheses on $a(\cdot)$ are:

$\widehat{\boldsymbol{H}}_{\boldsymbol{0}}'$: $a(y) = a_0(|y|)y$ is as in hypotheses H_0' with the addition in H_0^1 (i) that $\lim_{l\to 0^+} \frac{a_0'(t)t}{a_0(t)} > -1$.

Theorem 4.89 *If hypotheses $\widehat{H}_0'$, H_1^1 hold, $u \in C^1(\overline{\Omega})$ satisfies*

$$\operatorname{div} a(Du) + f(z,u) \leq 0, \text{ in } \Omega$$
$$u(z) > 0 \text{ for all } z \in \Omega,\ u(z_0) = 0 \text{ with } z_0 \in \partial\Omega,$$

then $\frac{\partial u}{\partial n_\alpha}(z_0) < 0$.

Next we give some characteristic examples of maps $\alpha(\cdot)$ which satisfy hypotheses H_0' (and $\widehat{H}_0'$).

Example 4.90

(a) $a(y) = |y|^{p-2}y$ with $1 < p < \infty$.
This map corresponds to the p-Laplace differential operator defined by

$$\Delta_p u = \operatorname{div}(|Du|^{p-2}Du) \text{ for all } u \in W^{1,p}(\Omega).$$

(b) $a(y) = |y|^{p-2}y + c|y|^{q-2}y$ with $1 < q < p,\ c > 0$.
This map corresponds to the (p,q)-Laplace differential operator

$$\Delta_p u + c\Delta_q u \text{ for all } u \in W^{1,p}(\Omega).$$

(c) $a(y) = (1+|y|^2)^{\frac{p-2}{2}}y$ with $1 < p < \infty$.

This map corresponds to a capillary-type operator

$$\operatorname{div}\left((1+|Du|^2)^{\frac{p-2}{2}}Du\right) \text{ for all } u \in W^{1,p}(\Omega).$$

(d) $a(y) = \left(1 + \frac{|y|^p}{(1+|y|^{2p})^{\frac{1}{2}}}\right)|y|^{p-2}y$ with $1 < p < \infty$.

The map corresponds to the differential operator

$$\operatorname{div}\left(1 + \frac{|Du|^p}{(1+|Du|^{2p})^{\frac{1}{2}}}\right)|Du|^{p-2}Du \text{ for all } u \in W^{1,p}(\Omega).$$

Another tool which we will need is the so-called "Picone identity". In its original form for the Laplace differential operator this identity says that if $u \geq 0$ and $v > 0$ are differentiable functions, then

$$|Du|^2 + \frac{u^2}{v^2}|Dv|^2 - 2\frac{u}{v}(Du, Dv)_{\mathbb{R}^N} = |Du|^2 - (D(\frac{u^2}{v}), Dv)_{\mathbb{R}^N} \geq 0, \text{ in } \Omega.$$

If turned out to be a very useful tool in the study of second order differential equations and systems. Here we will state and prove a generalized version of this identity, which covers also the case of the p-Laplace differential operator.

Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain. Let $a : \Omega \times \mathbb{R}^N \to \mathbb{R}^N$ be a map which satisfies the following hypotheses:

$\boldsymbol{H_0''}$: $a : \Omega \times \mathbb{R}^N \to \mathbb{R}^N$ is a Caratheodory function such that

(i) for a.a $z \in \Omega$, $y \to a(z, y)$ is maximal cyclically monotone and strictly monotone;

(ii) for a.a $z \in \Omega$ and all $\lambda > 0$

$$a(z, \lambda y) = \lambda^{p-1}a(z, y) \text{ for all } y \in \mathbb{R}^N, \text{ with } 1 < p < \infty$$

$((p-1)$-homogeneity).

From Theorem I-7.65, we know that there exists $\varphi : \Omega \times \mathbb{R}^N \to \mathbb{R}$ a measurable function such that for a.a $z \in \Omega$.

- $\varphi(z, \cdot)$ is continuously differentiable and strictly convex.
- $\varphi(z, \cdot)$ is p-homogeneous, that is,

$$\varphi(z, \lambda y) = \lambda^p \varphi(z, y) \text{ for all } y \in \mathbb{R}^N, \text{ all } \lambda > 0.$$

- $\nabla_y \varphi(z, y) = a(z, y)$ for all $y \in \mathbb{R}^N$.

It follows that

$$p\varphi(z, y) = (a(z, y), y)_{\mathbb{R}^N} \text{ for all } y \in \mathbb{R}^N. \tag{4.68}$$

The typical example of such a map $a(\cdot)$ is

$$a(y) = |y|^{p-2}y \text{ for all } y \in \mathbb{R}^N, \text{ with } 1 < p < \infty,$$

which corresponds to the p-Laplace differential operator. The "Picone identity" asserts the following:

Theorem 4.91 *If hypotheses H_0'' hold, $u, v \in C(\Omega)$ are differentiable a.e in Ω, $u(z) \geq 0$, $v(z) > 0$ for all $z \in \Omega$ and*

$$L(u,v) = p\varphi(z, Du) + p(p-1)\left(\frac{u}{v}\right)^p \varphi(z, Dv) - p\left(\frac{u}{v}\right)^{p-1}(a(z, Dv), Du)_{\mathbb{R}^N}$$

$$R(u,v) = (a(z, Du), Du)_{\mathbb{R}^N} - \left(a(z, Dv), D\left(\frac{u^p}{v^{p-1}}\right)\right)_{\mathbb{R}^N},$$

then

(a) $L(u,v) = R(u,v)$ for a.a $z \in \Omega$;
(b) $L(u,v) \geq 0$ for a.a $z \in \Omega$;
(c) $L(u,v) = 0$ for a.a $z \in \Omega \Leftrightarrow u = \vartheta v$ for some $\vartheta \in \mathbb{R}$.

Proof

(a) We have

$$\begin{aligned} D\left(\frac{u^p}{v^{p-1}}\right) &= \frac{1}{(v^{p-1})^2}\left[pu^{p-1}v^{p-1}Du - (p-1)v^{p-2}u^p Dv\right] \\ &= p\left(\frac{u}{v}\right)^{p-1} Du - (p-1)\left(\frac{u}{v}\right)^p Dv. \end{aligned} \tag{4.69}$$

Then

$$\begin{aligned} &(a(z, Du), Du)_{\mathbb{R}^N} - \left(a(z, Dv), D\left(\frac{u^p}{v^{p-1}}\right)\right)_{\mathbb{R}^N} \\ &= (a(z, Du), Du)_{\mathbb{R}^N} - p\left(\frac{u}{v}\right)^{p-1}(a(z, Dv), Du)_{\mathbb{R}^N} \\ &\quad + (p-1)\left(\frac{u}{v}\right)^p (a(z, Dv), Dv)_{\mathbb{R}^N} \\ &= p\varphi(z, Du) + p(p-1)\left(\frac{u}{v}\right)^p \varphi(z, Dv) - p\left(\frac{u}{v}\right)^{p-1}(a(z, Dv), Du)_{\mathbb{R}^N} \\ &\Rightarrow L(u,v) = R(u,v). \end{aligned}$$

(b) We have

$$\begin{aligned}
&\varphi(z, Du) + (p-1)\left(\frac{u}{v}\right)^p \varphi(z, Dv) - \left(\frac{u}{v}\right)^{p-1} (a(z, Dv), Du)_{\mathbb{R}^N} \\
=&\varphi(z, Du) + (p-1)\varphi\left(z, \frac{u}{v}Dv\right) - \left(a\left(z, \frac{u}{v}Dv\right), Du\right)_{\mathbb{R}^N} \\
&\qquad\qquad \text{(from the homogeneity)} \\
=&\varphi(z, Du) - \varphi\left(z, \frac{u}{v}Dv\right) + \left(a\left(z, \frac{u}{v}Dv\right), \frac{u}{v}Dv\right)_{\mathbb{R}^N} \\
&- \left(a\left(z, \frac{u}{v}Dv\right), Du\right)_{\mathbb{R}^N} \\
=&\varphi(z, Du) - \varphi\left(z, \frac{u}{v}Dv\right) - \left(a\left(z, \frac{u}{v}Dv\right), Du - \frac{u}{v}Dv\right)_{\mathbb{R}^N} \\
\geq&0 \text{ (from the convexity of } \varphi(z, \cdot) \text{ and since } \nabla_y \varphi(z, y) = a(z, y)), \\
\Rightarrow& L(u, v) \geq 0 \text{ for a.a } z \in \Omega.
\end{aligned}$$

(c) We have

$$\begin{aligned}
&L(u, v) = 0, \\
\Rightarrow& \varphi(z, Du) + (p-1)\left(\frac{u}{v}\right)^p \varphi(z, Dv) - \left(\frac{u}{v}\right)^{p-1} (a(z, Dv), Du)_{\mathbb{R}^N} = 0, \\
\Rightarrow& \varphi(z, Du) + (p-1)\varphi\left(z, \frac{u}{v}Dv\right) - \left(a\left(z, \frac{u}{v}Dv\right), Du\right)_{\mathbb{R}^N} = 0, \\
\Rightarrow& \varphi(z, Du) - \varphi\left(z, \frac{u}{v}Dv\right) - \left(a\left(z, \frac{u}{v}Dv\right), Du - \frac{u}{v}Dv\right)_{\mathbb{R}^N} = 0.
\end{aligned}$$

The strict convexity of $\varphi(z, \cdot)$ implies that

$$\begin{aligned}
&Du = \frac{u}{v}Dv, \\
\Rightarrow& vDu = uDv, \\
\Rightarrow& D\left(\frac{u}{v}\right) = 0 \text{ and so } u = \vartheta v \text{ with } \vartheta \in \mathbb{R}.
\end{aligned}$$

□

Remark 4.92 For the p-Laplacian we have

$$L(u, v) = |Du|^p + (p-1)\left(\frac{u}{v}\right)^p |Dv|^p - p\left(\frac{u}{v}\right)^{p-1} |Dv|^{p-2}(Dv, Du)_{\mathbb{R}^N},$$

$$R(u, v) = |Du|^p - |Dv|^{p-2}\left(Dv, D\left(\frac{u^p}{v^{p-1}}\right)\right)_{\mathbb{R}^N}.$$

We present two simple applications of this identity, which are useful in the spectral analysis of the p-Laplacian.

We strengthen the conditions on the map $a(z, y)$.

$\boldsymbol{H_0'''}$: $a : \Omega \times \mathbb{R}^N \to \mathbb{R}^N$ is a Caratheodory function which satisfies hypotheses $H_0'''(i)$, (ii) which are the same as the corresponding hypotheses $H_0''(i)$, (iii) and

$$\text{(iii)} |a(z, y)| \leq c|y|^{p-1} \text{ for a.a } z \in \Omega, \text{ all } y \in \mathbb{R}^N, \text{ some } c > 0.$$

Proposition 4.93 *If hypotheses H_0''' hold, $m \in L^\infty(\Omega)$, $m(z) \geq 0$ for a.a $z \in \Omega$, $\lambda > 0$ and $u \in C^1(\Omega)$, $u(z) > 0$ for all $z \in \Omega$ satisfies the differential inequality*

$$\lambda m(z) u^{p-1} \leq -\operatorname{div} a(Du) \text{ in } \Omega,$$

then for all $h \in W_0^{1,p}(\Omega)$, $h \geq 0$, we have

$$\lambda \int_\Omega m(z) h^p dz \leq c \|Dh\|_p^p.$$

Proof Let $\vartheta \in C_c^\infty(\Omega)$ with $\vartheta(z) \geq 0$ for all $z \in \Omega$. From Theorem 4.91, we have

$$\begin{aligned}
0 \leq \int_\Omega L(\vartheta, u) dz &= \int_\Omega R(\vartheta, u) dz \\
&= \int_\Omega (a(z, D\vartheta), D\vartheta)_{\mathbb{R}^N} dz \\
&\quad - \int_\Omega \left(a(z, Du), D\left(\frac{\vartheta^p}{u^{p-1}} \right) \right)_{\mathbb{R}^N} dz \\
&\leq c \|D\vartheta\|_p^p + \int_\Omega \operatorname{div}(a(z, Du)) \frac{\vartheta^p}{u^{p-1}} dz \\
&\leq c \|D\vartheta\|_p^p - \lambda \int_\Omega m(z) \vartheta^p dz, \\
\Rightarrow \lambda \int_\Omega m(z) \vartheta^p dz &\leq c \|D\vartheta\|_p^p.
\end{aligned}$$

Exploiting the density of $\vartheta \in C_c^\infty(\Omega)$, $\vartheta \geq 0$ in $W_0^{1,p}(\Omega)_+$, we conclude that

$$\lambda \int_\Omega m(z) h^p dz \leq c \|Dh\|_p^p \quad \text{for all } h \in W_0^{1,p}(\Omega), \ h \geq 0.$$

□

Next consider the following two nonlinear Dirichlet problems

$$-\Delta_p u = m_1(z)|u|^{p-2}u \quad \text{in } \Omega,\ u|_{\partial\Omega} = 0, \tag{4.70}$$

$$-\Delta_p v = m_2(z)|v|^{p-2}v \quad \text{in } \Omega,\ v|_{\partial\Omega} = 0. \tag{4.71}$$

The Banach space $C_0^1(\overline{\Omega}) = \{u \in C^1(\overline{\Omega}) : u|_{\partial\Omega} = 0\}$ is ordered with positive (order) cone $C_+ = \{u \in C_0^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\}$. We know that this cone has a nonempty interior given by

$$int\, C_+ = \left\{u \in C_+ : u(z) > 0 \text{ for all } z \in \Omega,\ \frac{\partial u}{\partial n}|_{\partial\Omega} < 0\right\}$$

with $n(\cdot)$ being the outward unit normal on $\partial\Omega$.

Proposition 4.94 *If $m_1, m_2 \in L^\infty(\Omega)$, $0 \leq m_1(z) \leq m_2(z)$ for a.a $z \in \Omega$, $m_1 \neq 0$, $m_2 \neq m_1$, $u \in W_0^{1,p}(\Omega)$, $u \geq 0$ is a nontrivial solution of (4.70), then any nontrivial solution of (4.71) is nodal (sign changing).*

Proof On account of Theorem 4.84 (a), we have

$$u, v \in C_0^1(\overline{\Omega}).$$

Moreover, Theorem 4.89 implies that $u \in int\, C_+$.

Arguing by contradiction, suppose that v has constant sign and we can assume $v \geq 0$. Again we have $v \in int\, C_+$. Using Theorem 4.91, we have

$$\begin{aligned} 0 \leq \int_\Omega L(u, v)dz &= \int_\Omega R(u, v)dz \\ &= \|Du\|_p^p - \int_\Omega |Dv|^{p-2}\left(Dv, D\left(\frac{u^p}{v^{p-1}}\right)\right)_{\mathbb{R}^N} dz. \end{aligned} \tag{4.72}$$

From (4.70), we have

$$\|Du\|_p^p = \int_\Omega m_1(z)u^p dz. \tag{4.73}$$

Since $v \in int\, C_+$, using Proposition 2.86, we can find $\eta > 0$ such that $u \leq \eta v$. Hence

$$0 \leq \frac{u^p}{v^{p-1}} \leq \eta^p v.$$

Therefore $\frac{u^p}{v^{p-1}} \in W_0^{1,p}(\Omega)$ (see Proposition I-4.90) and using it as a test function in (4.71), we obtain

$$\int_\Omega |Dv|^{p-2}\left(Dv, D\left(\frac{u^p}{v^{p-1}}\right)\right)_{\mathbb{R}^N} dz = \int_\Omega m_2(z)u^p dz.$$

Using this in (4.72), we have

$$\begin{aligned}
&\int_\Omega m_2(z)u^p dz \le \|Du\|_p^p,\\
&\Rightarrow 0 \le \int_\Omega (m_1(z) - m_2(z))u^p dz \quad \text{(see (4.73))},\\
&\Rightarrow u \text{ must be nodal.}
\end{aligned}$$

□

As a last application, consider the following system

$$\left\{\begin{array}{ll} -\operatorname{div} a(z, Du) = v^{p-1} & \text{in } \Omega,\\ -\operatorname{div} a(z, Dv) = \frac{v^{2(p-1)}}{u^{p-1}} & \text{in } \Omega,\\ u|_{\partial\Omega} = 0 = v|_{\partial\Omega} = 0, & u, v > 0. \end{array}\right\} \tag{4.74}$$

Proposition 4.95 *If hypotheses H_0''' hold and $(u, v) \in int\, C_+ \times int\, C_+$ is a solution of (4.74), then $u = \vartheta v$ for some $\vartheta > 0$.*

Proof In the first equation of (4.74), we use the test function u and in the second we use the test function $\frac{u^p}{v^{p-1}}$ (we have seen in the proof of Proposition 4.93 that $\frac{u^p}{v^{p-1}} \in W_0^{1,p}(\Omega)$). We have

$$\int_\Omega (a(z, Du), Du)_{\mathbb{R}^N} dz = \int_\Omega v^{p-1} u dz,$$

$$\int_\Omega \left(a(z, Dv), D\left(\frac{u^p}{v^{p-1}}\right)\right)_{\mathbb{R}^N} dz = \int_\Omega v^{p-1} u dz.$$

We subtract the second equation from the first and obtain

$$\begin{aligned}
&\int_\Omega R(u, v)dz = 0,\\
&\Rightarrow \int_\Omega L(u, v)dz = 0,\\
&\Rightarrow L(u, v)(z) = 0 \text{ for a.a } z \in \Omega,\\
&\Rightarrow u = \vartheta v,\ \vartheta > 0 \text{ (see Theorem 4.91)}.
\end{aligned}$$

□

Finally we will state the Harnack inequality. So, consider the following equation in divergence form

$$-\operatorname{div} a(z, u, Du) + f(z, u, Du) = 0. \tag{4.75}$$

The hypotheses on the data of (4.75) are the following:

$\boldsymbol{H_0^{(iv)}}$: $a : \Omega \times \mathbb{R}^N \to \mathbb{R}^N$ is a Caratheodory function such that

(i) $|a(z, x, y)| \le \widehat{a}_1(z)^{p-1} + \widehat{a}_2(z)|x|^{p-1} + \eta_1|y|^{p-1}$ for a.a $z \in \Omega$, all $x \in \mathbb{R}$, all $y \in \mathbb{R}$ with $\widehat{a}_1 \in L^{p\alpha}(\Omega)$ where $a = \frac{1}{1-\varepsilon}\max\left\{\frac{N}{p}, 1\right\}$, $\varepsilon \in (0, 1)$, $\widehat{a}_2 \in L^{\alpha}(\Omega)$ and $\eta_1 > 0$;

(ii) $(a(z, x, y), y)_{\mathbb{R}^N} \ge \eta_2|y|^p - \widehat{a}_3(z)|x|^p - \widehat{a}_4(z)^p$ for a.a $z \in \Omega$, all $x \in \mathbb{R}$, all $y \in \mathbb{R}$ with $\eta_2 > 0, \widehat{a}_3 \in L^{\alpha}(\Omega), \widehat{a}_4 \in L^{p\alpha}(\Omega)$.

$\boldsymbol{H_1''}$: $f : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$ is a Caratheodory function such that

$$|f(z, x, y)| \le \widehat{a}_5(z)^{p-1} + \widehat{a}_6(z)|x|^{p-1} + \eta_3|y|^{p-1}$$

for a.a $z \in \Omega$, all $x \in \mathbb{R}$, all $y \in \mathbb{R}$ with $\widehat{a}_5 \in L^{p\alpha}(\Omega), \widehat{a}_6 \in L^{\alpha}(\Omega), \eta_3 > 0$.

The Harnack inequality says the following (see Pucci and Serrin [423], p. 163 and Trudinger [450]).

Theorem 4.96 *If hypotheses H_0^{iv}, H_1'' hold, $u \in W^{1,p}(\Omega) \cap L^{\infty}(\Omega)$, $u \ge 0$ is a weak solution of (4.75), $\overline{B}_{\varrho} \subseteq \Omega$ and $0 < \vartheta < 1$, then $\sup_{\overline{B}_{\vartheta\varrho}} u \le c\left[\inf_{\overline{B}_{\vartheta\varrho}} u + k(\varrho)\right]$ with $k(\varrho) = \varrho^{\varepsilon}\|\widehat{a}_4\| + \varrho^{\varrho'\varepsilon}\|\widehat{a}_5\| + \|\widehat{a}_1\|$, the norms taken in the respective spaces over B_{ϱ}. In the particular case of the p-Laplacian $k(\varrho) = 0$.*

Remark 4.97 So, if the differential operator is the p-Laplacian (that is, $a(z, x, y) = |y|^{p-2}y$), $u \in W^{1,p}(\Omega) \cap L^{\infty}(\Omega)$ is a nonnegative weak solution of (75) $B_{\varrho} \subseteq \Omega$ and $0 < \vartheta < 1$, then $\sup_{\overline{B}_{\vartheta\varrho}} u \le c \inf_{\overline{B}_{\vartheta\varrho}} u$.

Now we turn our attention to the following eigenvalue problem for the p-Laplacian differential operator

$$\left\{\begin{array}{c} -\Delta_p u(z) + \xi(z)|u(z)|^{p-2}u(z) = \widehat{\lambda}|u(z)|^{p-2}u(z) \text{ in } \Omega, \\ \frac{\partial u}{\partial n_p} + \beta(z)|u|^{p-2}u = 0 \text{ on } \partial\Omega, \ 1 < p < N. \end{array}\right\} \tag{4.76}$$

Here $\Omega \subseteq \mathbb{R}^N (N \ge 2)$ is a bounded domain with a C^2-boundary $\partial\Omega$. In the Robin boundary condition $\frac{\partial u}{\partial n_p}$ denotes the conormal derivative corresponding to the p-Laplace operator. If $u \in C^1(\overline{\Omega})$, then

$$\frac{\partial u}{\partial n_p} = |Du|^{p-2}(Du, n)_{\mathbb{R}^N} = |Du|^{p-2}\frac{\partial u}{\partial n}$$

with $n(\cdot)$ being the outward unit normal. For general $u \in W^{1,p}(\Omega)$, this conormal derivative is interpreted using the nonlinear Green's identity (see Theorem I-4.107).

Our hypotheses on the potential function $\xi(\cdot)$ and on the boundary coefficient $\beta(\cdot)$, are the following:

$\boldsymbol{H_3}$: $\xi \in L^s(\Omega)$ with $s > \frac{N}{p}$ and $\beta \in C^{0,\alpha}(\partial\Omega)$ with $\alpha \in (0,1)$, $\beta(z) \geq 0$ for σ-a.a $z \in \partial\Omega$.

Remark 4.98 Again when $\beta \equiv 0$, we have the Neumann problem. Note that the potential function is in general indefinite (sign-changing).

Let $\gamma_p : W^{1,p}(\Omega) \to \mathbb{R}$ be the C^1-functional defined by

$$\gamma_p(u) = \|Du\|_p^p + \int_\Omega \xi(z)|u|^p dz + \int_{\partial\Omega} \beta(z)|u|^p d\sigma \quad \text{for all } u \in W^{1,p}(\Omega).$$

Definition 4.99 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" of (4.76), if the equation has a nontrivial weak solution $\widehat{u} \in W^{1,p}(\Omega)$, that is,

$$\int_\Omega |D\widehat{u}|^{p-2}(D\widehat{u}, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)|\widehat{u}|^{p-2}\widehat{u}h dz + \int_{\partial\Omega} \beta(z)|\widehat{u}|^{p-2}\widehat{u}h d\sigma = \widehat{\lambda} \int_\Omega |\widehat{u}|^{p-2}\widehat{u}h dz \quad \text{for all } h \in W^{1,p}(\Omega).$$

We say that $\widehat{u}$ is an "eigenfunction" corresponding to $\widehat{\lambda}$ and $(\widehat{\lambda}, \widehat{u})$ is an "eigenpair".

Remark 4.100 Note that in terms of $\gamma_p(\cdot)$, we have

$$\langle \gamma_p'(\widehat{u}), h\rangle = \widehat{\lambda} \int_\Omega |\widehat{u}|^{p-2}\widehat{u}h dz \quad \text{for all } h \in W^{1,p}(\Omega).$$

Using the nonlinear Green's identity, we can see that $\widehat{u}$ is in fact a strong solution of (4.76), that is,

$$-\Delta_p \widehat{u}(z) + \xi(z)|\widehat{u}(z)|^{p-2}u(z) = \widehat{\lambda}|\widehat{u}(z)|^{p-2}\widehat{u}(z) \quad \text{for a.a } z \in \Omega,$$

while the boundary condition is understood as an equality in $W^{-\frac{1}{p'},p'}(\partial\Omega) = W^{\frac{1}{p'},p}(\partial\Omega)^*$ (recall $\frac{1}{p} + \frac{1}{p'} = 1$).

Theorem 4.101 *If hypotheses H_3 hold, then problem (4.76) has a smallest eigenvalue $\widehat{\lambda}_1 \in \mathbb{R}$ with the following properties.*

(a) $\widehat{\lambda}_1$ has a positive eigenfunction $\widehat{u}_1 \in C^1(\Omega)$ (local regularity);
(b) $\widehat{\lambda}_1$ is simple (this means that if $\widehat{u}$, $\widehat{v}$ are eigenfunctions corresponding to $\widehat{\lambda}_1$, then $\widehat{u} = \vartheta\widehat{v}$ for some $\vartheta \in \mathbb{R} \setminus \{0\}$).

Proof Let $M = W^{1,p}(\Omega) \cap \partial B_1^{L^p} = \{u \in W^{1,p}(\Omega) : \|u\|_p = 1\}$. By Ljusternik's theorem (see, for example, Gasinski and Papageorgiou [182], p. 697). M is a C^1-Banach manifold. Let

$$\widehat{\lambda}_1 = \inf\{\gamma_p(u) : u \in M\}. \tag{4.77}$$

Claim 1 $\widehat{\lambda}_1 > -\infty$.

Since $s > \frac{N}{p}$, we have $s' < \left(\frac{N}{p}\right)' = \frac{N}{N-p}$ (recall for any $\tau \in (1, \infty)$, $\tau' = \frac{\tau}{\tau-1}$ or equivalently $\frac{1}{\tau} + \frac{1}{\tau'} = 1$). Therefore $s'p < \frac{Np}{N-p} = p^*$ and so we have that $|u|^p \in L^{s'}(\Omega)$ (Sobolev embedding theorem). Then by Hölder's inequality, we have

$$\left| \int_\Omega \xi(z)|u|^p dz \right| \leq \|\xi\|_s \|u\|_{ps'}^p. \tag{4.78}$$

We consider the spaces $W^{1,p}(\Omega), L^{ps'}(\Omega), L^p(\Omega)$. We know that $p < ps' < p^*$ and so

$$W^{1,p}(\Omega) \hookrightarrow L^{ps'}(\Omega) \hookrightarrow L^p(\Omega),$$

with the first embedding being compact (Sobolev embedding theorem) and the second simply continuous (see Proposition I-2.124). Invoking Ehrling's inequality (see Lemma I-4.206), given $\varepsilon > 0$, we can find $c_\varepsilon > 0$ such that

$$\|u\|_{ps'}^p \leq \varepsilon \|u\|^p + c_\varepsilon \|u\|_p^p, \tag{4.79}$$

with $\|\cdot\|$ being the norm of $W^{1,p}(\Omega)$ (hence $\|u\| = (\|u\|_p^p + \|Du\|_p^p)^{\frac{1}{p}}$ for all $u \in W^{1,p}(\Omega)$). Using (4.78) and (4.79), we have

$$\begin{aligned}
&\|Du\|_p^p + \|u\|_p^p - \int_\Omega \xi(z)|u|^p dz \\
\leq &\|Du\|_p^p + \|u\|_p^p + \varepsilon\|\xi\|_s\|u\|^p + c_\varepsilon\|\xi\|_s\|u\|_p^p \\
\Rightarrow &(1 - \varepsilon\|\xi\|_s)\|u\|^p \leq \gamma_p(u) + (1 + c_\varepsilon\|\xi\|_s)\|u\|_p^p \\
&\qquad\qquad \text{(recall } \beta \geq 0, \text{ see } H_3).
\end{aligned}$$

We choose $\varepsilon \in (0, \|\xi\|_s^{-1})$ and obtain

$$\widehat{c}_\varepsilon \|u\|^p \leq \gamma_p(u) + (1 + c_\varepsilon\|\xi\|_s)\|u\|_p^p \quad \text{for some } \widehat{c}_\varepsilon > 0. \tag{4.80}$$

If $u \in M$, then

$$-(1 + c_\varepsilon \|\xi\|_s) \leq \gamma_p(u),$$
$$\Rightarrow \widehat{\lambda}_1 > -\infty.$$

This proves Claim 1.

Claim 2 There exists $\widehat{u}_1 \in M$ such that $\widehat{\lambda}_1 = \gamma_p(\widehat{u}_1)$ (see (4.77)).

Let $\{u_n\}_{n\in\mathbb{N}} \subseteq M$ such that $\gamma_p(u_n) \downarrow \widehat{\lambda}_1$. From (4.80) we see that $\{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega)$ is bounded. So, we may assume that

$$u_n \xrightarrow{w} \widehat{u}_1 \text{ in } W^{1,p}(\Omega) \text{ and } u_n \to \widehat{u}_1 \text{ in } L^{ps'}(\Omega) \text{ and in } L^p(\partial\Omega). \tag{4.81}$$

Using (4.81), we have

$$\|D\widehat{u}_1\|_p^p \leq \liminf_{n\to\infty} \|Du_n\|_p^p, \quad \int_\Omega \xi(z)|u_n|^p dz \to \int_\Omega \xi(z)|\widehat{u}_1|^p dz$$
$$\text{and} \quad \int_{\partial\Omega} \beta(z)|u_n|^p d\sigma \to \int_{\partial\Omega} \beta(z)|\widehat{u}_1|^p d\sigma.$$

If follows that

$$\gamma_p(\widehat{u}_1) \leq \widehat{\lambda}_1.$$

But $\|\widehat{u}_1\|_p = 1$ (see (4.81)) and so $\widehat{u}_1 \in M$, hence

$$\gamma_p(\widehat{u}_1) = \widehat{\lambda}_1.$$

This proves Claim 2.

Using the Lagrange multiplier rule (see Theorem 3.87), we infer that $\widehat{\lambda}_1 \in \mathbb{R}$ is an eigenvalue of (4.76), with $\widehat{u}_1 \in M$ a corresponding eigenfunction. Proposition 4.82 implies that $\widehat{u}_1 \in L^\infty(\Omega)$. Interior regularity (see DiBenedetto [141]), implies that $\widehat{u}_1 \in C^1(\Omega)$. Since $\gamma_p(\widehat{u}_1) = \gamma_p(|\widehat{u}_1|)$, we can always assume $\widehat{u}_1 \geq 0$. Then by the Harnack inequality (see Theorem 4.96 and Remark 4.97), we have $\widehat{u}_1(z) > 0$ for all $z \in \Omega$.

(b) Let $\widehat{v}_1$ be another eigenfunction corresponding to $\widehat{\lambda}_1$. Again $\widehat{v}_1 \in C^{1,\alpha}(\Omega)$ and $\widehat{v}_1(z) > 0$ for all $z \in \Omega$. For $n \in \mathbb{N}$, let $\widehat{v}_j^n = \widehat{v}_1 + \frac{1}{n}$ and use Theorem 4.91 (the Picone identity), We have

$$
\begin{aligned}
0 \le \int_\Omega L(\widehat{u}_1, \widehat{v}_1^n)dz &= \int_\Omega R(\widehat{u}_1, \widehat{v}_1^n)dz \\
&= \|D\widehat{u}_1\|_p^p - \int_\Omega |D\widehat{v}_1|^{p-2}\left(D\widehat{v}_1, D\left(\frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}}\right)\right)_{\mathbb{R}^N} dz \\
&= \|D\widehat{u}_1\|_p^p - \int_\Omega (-\Delta_p \widehat{v}_1)\frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}}dz - \left\langle \frac{\partial \widehat{v}_1}{\partial n_p}, \frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}}\right\rangle_{\partial\Omega} \\
&\qquad \text{(using Green's identity, since } \frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}} \in W^{1,p}(\Omega)) \\
&= \|D\widehat{u}_1\|_p^p + \int_\Omega \xi(z)\widehat{v}_1^{p-1}\frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}}dz - \widehat{\lambda}_1 \int_\Omega \widehat{v}_1^{p-1}\frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}}dz \\
&\qquad + \int_{\partial\Omega} \beta(z)\widehat{v}_1^{p-1}\frac{\widehat{u}_1^p}{(\widehat{v}_1^n)^{p-1}}d\sigma.
\end{aligned}
$$

Passing to the limit as $n \to \infty$, we obtain

$$
\begin{aligned}
&0 \le \gamma_p(\widehat{u}_1) - \widehat{\lambda}_1 = 0 \quad (\text{recall } \|\widehat{u}_1\|_p = 1), \\
\Rightarrow &\int_\Omega L(\widehat{u}_1, \widehat{v}_1)dz = 0, \\
\Rightarrow &\widehat{u}_1 = \vartheta\widehat{v}_1 \text{ with } \vartheta \in \mathbb{R} \setminus \{0\}.
\end{aligned}
$$

This proves the simplicity of $\widehat{\lambda}_1$. □

Corollary 4.102

(a) If $\xi \in L^s(\Omega)$ with $s > \frac{Np}{p-1} = Np'$, then $\widehat{u}_1 \in C^{1,\alpha}(\Omega)$ for some $\alpha \in (0,1)$ (see DiBenedetto [141]).

(b) If $\xi \in L^\infty(\Omega)$, then $\widehat{u}_1 \in C^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0,1)$ and $\widehat{u}_1(z) > 0$ for all $z \in \overline{\Omega}$ (see Theorems 4.83 and 4.88).

Corollary 4.103 *If hypotheses H_3 hold, then $\gamma_p(\widehat{u}) = \widehat{\lambda}_1$ if and only if $\widehat{u}$ is an eigenfunction corresponding to $\widehat{\lambda}_1$.*

Corollary 4.104 *If hypotheses H_3 hold, then all eigenfunctions corresponding to $\widehat{\lambda}_1$ have fixed sign.*

In the next proposition we show that $\widehat{\lambda}_1$ is the only eigenvalue with the above property.

Proposition 4.105 *If hypotheses H_3 hold and $\widehat{\lambda} \ne \widehat{\lambda}_1$ is an eigenvalue of (4.76), then every eigenfunction $\widehat{u} \in W^{1,p}(\Omega)$ corresponding to $\widehat{\lambda}$ is nodal (sign-changing).*

Proof Arguing by contradiction, suppose that $\widehat{u}$ has constant sign. We may assume that $\widehat{u} \geq 0$. Local regularity implies $\widehat{u} \in C^1(\Omega)$ and then from Harnack's inequality, we have $\widehat{u}(z) > 0$ for all $z \in \Omega$. For $\varepsilon \in (0,1)$, we set $\widehat{u}_\varepsilon = \widehat{u} + \varepsilon$. Using Theorem 4.91 (Picone's identity), we have

$$
\begin{aligned}
0 \leq & \int_\Omega L(\widehat{u}_1, \widehat{u}_\varepsilon) dz = \int_\Omega R(\widehat{u}_1, \widehat{u}_\varepsilon) dz \\
= & \widehat{\lambda}_1 - \int_\Omega \xi(z) \widehat{u}_1^p dz - \int_{\partial\Omega} \beta(z) \widehat{u}_1^p d\sigma - \int_\Omega |D\widehat{u}|^{p-2} \left(D\widehat{u}, D\left(\frac{\widehat{u}_1^p}{\widehat{u}_\varepsilon^{p-1}} \right) \right)_{\mathbb{R}^N} dz \\
= & \widehat{\lambda}_1 - \int_\Omega \xi(z) \widehat{u}_1^p dz - \int_{\partial\Omega} \beta(z) \widehat{u}_1^p d\sigma - \int_\Omega (-\Delta_p \widehat{u}) \frac{\widehat{u}_1^p}{\widehat{u}_\varepsilon^{p-1}} dz \\
& - \left\langle \frac{\partial \widehat{u}}{\partial n_p}, \frac{\widehat{u}^p}{\widehat{u}_\varepsilon^{p-1}} \right\rangle_{\partial\Omega} \\
= & \widehat{\lambda}_1 - \int_\Omega \xi(z) \widehat{u}_1^p dz - \int_{\partial\Omega} \beta(z) \widehat{u}_1^p d\sigma \\
& - \widehat{\lambda} \int_\Omega \left(\frac{\widehat{u}}{\widehat{u}_\varepsilon} \right)^{p-1} \widehat{u}_1^p dz + \int_\Omega \xi(z) \widehat{u}^p \frac{\widehat{u}^p}{\widehat{u}_\varepsilon^{p-1}} dz + \int_{\partial\Omega} \beta(z) \widehat{u}^{p-1} \frac{\widehat{u}^p}{\widehat{u}_\varepsilon^{p-1}} d\sigma.
\end{aligned}
$$

Let $\varepsilon \to 0^+$ to obtain

$$
0 \leq \widehat{\lambda}_1 - \widehat{\lambda} < 0,
$$

a contradiction. So, $\widehat{u}$ must be nodal. □

Let $\sigma_p(\Omega)$ denote the set of eigenvalues of (4.76).

Proposition 4.106 *If hypotheses H_3 hold, then $\sigma_p(\Omega) \subseteq \mathbb{R}$ is closed.*

Proof Let $\{\widehat{\lambda}_n\}_{n \in \mathbb{N}} \subseteq \sigma_p(\Omega)$ and assume that $\widehat{\lambda}_n \to \widehat{\lambda}$. For every $n \in \mathbb{N}$, let $\widehat{u}_n \in W^{1,p}(\Omega)$ be an eigenfunction corresponding to $\widehat{\lambda}_n$. We have

$$
\begin{aligned}
& \int_\Omega |D\widehat{u}_n|^{p-2} (D\widehat{u}_n, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z) |\widehat{u}_n|^{p-2} \widehat{u}_n h dz \\
& + \int_{\partial\Omega} \beta(z) |\widehat{u}_n|^{p-2} \widehat{u}_n h d\sigma \qquad (4.82) \\
= & \lambda_n \int_\Omega |\widehat{u}_n|^{p-2} \widehat{u}_n h dz \quad \text{for all } h \in W^{1,p}(\Omega).
\end{aligned}
$$

Let $y_n = \frac{\widehat{u}_n}{\|\widehat{u}_n\|}$, $n \in \mathbb{N}$. Then $\|y_n\| = 1$ for all $n \in \mathbb{N}$ and so we may assume that

$$
y_n \xrightarrow{w} y \text{ in } W^{1,p}(\Omega) \text{ and } y_n \to y \text{ in } L^{ps'}(\Omega) \text{ and in } L^p(\partial\Omega). \qquad (4.83)
$$

From (4.82) we have

$$\begin{aligned}&\int_\Omega |Dy_n|^{p-2}(Dy_n, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)|y_n|^{p-2}y_n h dz \\ &+ \int_{\partial\Omega} \beta(z)|y_n|^{p-2}y_n h d\sigma \\ =&\widehat{\lambda}_1 \int_\Omega |y_n|^{p-2}y_n h dz \quad \text{for all } h \in W^{1,p}(\Omega).\end{aligned} \tag{4.84}$$

We choose the test function $h = y_n - y \in W^{1,p}(\Omega)$, pass to the limit as $n \to \infty$ and use (4.83). We obtain

$$\begin{aligned}&\lim_{n\to\infty} \int_\Omega |Dy_n|^{p-2}(Dy_n, Dy_n - Dy)_{\mathbb{R}^N} dz = 0 \\ &\Rightarrow y_n \to y \text{ in } W^{1,p}(\Omega) \text{ (see Proposition I-7.77).}\end{aligned} \tag{4.85}$$

So, if in (4.84) we pass to the limit as $n \to \infty$ and use (4.85), we obtain

$$\begin{aligned}&\int_\Omega |Dy|^{p-2}(Dy, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)|y|^{p-2}yh dz + \int_{\partial\Omega} \beta(z)|y|^{p-2}yh d\sigma \\ =&\widehat{\lambda} \int_\Omega |\widehat{y}|^{p-2}\widehat{y}h dz \quad \text{for all } h \in W^{1,p}(\Omega),\ \|y\|_p = 1, \\ \Rightarrow&\ \widehat{\lambda} \in \sigma_p(\Omega) \subseteq \mathbb{R} \text{ and so } \sigma_p(\Omega) \subseteq \mathbb{R} \text{ is closed.}\end{aligned}$$

□

As in Proposition 4.52, we have:

Proposition 4.107 *If hypotheses H_3 hold, $\xi(z) \geq 0$ for a.a $z \in \Omega$ and either $\xi \not\equiv 0$ or $\beta \not\equiv 0$, then $\widehat{\lambda}_1 > 0$.*

Next we will show that $\widehat{\lambda}_1 \in \sigma_p(\Omega)$ is isolated (that is, we can find $\varepsilon > 0$ such that $(\widehat{\lambda}_1, \widehat{\lambda}_1 + \varepsilon) \cap \sigma_p(\Omega) = \emptyset$). To do this, we need some preparation.

Definition 4.108 Let $u \in C(\Omega)$. We set

$$Z(u) = \{z \in \Omega : u(z) = 0\} \text{ (the zero set of } u(\cdot)).$$

Every connected component of $\Omega \setminus Z(u)$ is a "nodal domain" of u.

By $|\cdot|_N$ we denote the Lebesgue measure on $\mathbb{R}^N$. With the lemma that follows, we estimate the Lebesgue measure of a nodal domain of an eigenfunction of (4.76).

Lemma 4.109 *If hypotheses H_3 hold, $\widehat{\lambda} \in \sigma_p(\Omega)$, $\widehat{u} \in W^{1,p}(\Omega) \cap C^1(\Omega)$ is an eigenfunction corresponding to $\widehat{\lambda}$ and $D \subseteq \Omega$ is a nodal domain of $\widehat{u}$, then $1 \leq c^*|D|_N^{1-\frac{p}{p^*}}$ for some $c^* > 0$.*

Proof From Proposition I-4.97 we know that

$$\widehat{u}\chi_D \in W^{1,p}(\Omega) \quad \text{and} \quad D(\widehat{u}\chi_D) = (Du)\chi_D. \tag{4.86}$$

(recall χ_D is the characteristic function of the nodal domain D). We have

$$\int_\Omega |D\widehat{u}|^{p-2}(D\widehat{u}, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)|\widehat{u}|^{p-2}\widehat{u}h dz + \int_{\partial\Omega} \beta(z)|\widehat{u}|^{p-2}\widehat{u}h d\sigma$$
$$= \int_\Omega \widehat{\lambda}|\widehat{u}|^{p-2}\widehat{u}h dz \quad \text{for all } h \in W^{1,p}(\Omega).$$

We can choose $h = \widehat{u}\chi_D$ and obtain

$$\begin{aligned} &\int_D |D\widehat{u}|^p dz + \int_D \xi(z)|\widehat{u}|^p dz + \int_{D\cap\partial\Omega} \beta(z)|u|^p d\sigma = \int_D \widehat{\lambda}|\widehat{u}|^p dz, \\ &\Rightarrow \gamma_p(\widehat{u}\chi_D) = \widehat{\lambda}\int_D |\widehat{u}|^p dz \text{ (see (4.86))}. \end{aligned} \tag{4.87}$$

From (4.80), we have

$$\begin{aligned} \|\widehat{u}\chi_D\|^p &\leq \widehat{c}_1(\gamma_p(\widehat{u}\chi_D) + \|\widehat{u}\chi_D\|_p^p) \quad \text{for some } \widehat{c}_1 > 0 \\ &\leq \widehat{c}_1(\widehat{\lambda}+1)\|\widehat{u}\chi_D\|_p^p \ (see\ (4.87)) \\ &\leq \widehat{c}_1(\widehat{\lambda}+1)|D|_N^{1-\frac{p}{p^*}}\|\widehat{u}\chi_D\|_{p^*}^p \\ &\qquad \text{(since } L^p(\Omega) \hookrightarrow L^{p^*}(\Omega) \text{ continuously)} \\ &\leq \widehat{c}_2(\widehat{\lambda}+1)|D|_N^{1-\frac{p}{p^*}}\|\widehat{u}\chi_D\|^p \quad \text{for some } \widehat{c}_2 > 0 \\ &\qquad \text{(since } W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega) \text{ continuously)} \\ \Rightarrow 1 &\leq c^*|D|_N^{1-\frac{p}{p^*}} \quad \text{with } c^* = \widehat{c}_2(\widehat{\lambda}+1) > 0. \end{aligned}$$

□

Using this lemma, we can prove the isolation of the first eigenvalue $\widehat{\lambda}_1$.

Proposition 4.110 *If hypotheses H_1 hold, then $\widehat{\lambda}_1$ is isolated in $\sigma_p(\Omega)$.*

Proof We argue by contradiction. So, suppose $\widehat{\lambda}_1$ is not isolated and suppose we can find $\{\lambda_n\}_{n\in\mathbb{N}} \subseteq \sigma_p(\Omega)$ such that $\lambda_n \downarrow \widehat{\lambda}_1$. Let $u_n \in W^{1,p}(\Omega) \cap C^1(\Omega)$ be an

eigenfunction corresponding to λ_n, $n \in \mathbb{N}$. We have

$$\int_\Omega |Du_n|^{p-2}(Du_n, Dh)_{\mathbb{R}^N} dz + \int_\Omega \xi(z)|u_n|^{p-2}u_n h dz$$
$$+ \int_{\partial\Omega} \beta(z)|u_n|^{p-2}u_n h d\sigma$$
$$= \int_\Omega \lambda_n |u_n|^{p-2}u_n h dz \quad \text{for all } h \in W^{1,p}(\Omega), \text{ all } n \in \mathbb{N}.$$

Let $y_n = \frac{u_n}{\|u_n\|}$, $n \in \mathbb{N}$ and reasoning as in the proof of Proposition 4.106, at least for a subsequence, we have

$$y_n \to \widehat{y} \text{ in } W^{1,p}(\Omega), \ \widehat{y} = \vartheta \widehat{u}, \text{ for some } \vartheta \in \mathbb{R} \setminus \{0\}.$$

We may always assume that $\vartheta > 0$. Then $y(z) > 0$ for all $z \in \Omega$ (see Corollary 4.104). We set

$$D_-^n = \{z \in \Omega : y_n(z) < 0\} \quad n \in \mathbb{N}.$$

Then $|D_-^n|_N \to 0$, $|D_-^n|_N \neq 0$, $n \in \mathbb{N}$ (see Proposition 4.105). We contradict Lemma 4.109. Therefore $\widehat{\lambda}_1$ is isolated in $\sigma_p(\Omega)$. □

This proposition leads to the definition of the second eigenvalue of (4.76), by

$$\widehat{\lambda}_2 = \inf\{\widehat{\lambda} \in \sigma_p(\Omega) : \widehat{\lambda} \neq \widehat{\lambda}_1\} \in \sigma_p(\Omega).$$

We can have a useful minimax characterization of $\widehat{\lambda}_2$.

We know that if X is a Banach space, $\varphi \in C^1(X)$ and $\eta \in \mathbb{R}$ is a regular value of $\varphi(\cdot)$ (that is $\eta \notin \varphi(K_\varphi)$, with $K_\varphi = \{u \in X : \varphi'(u) = 0\}$, the critical set of $\varphi(\cdot)$). Then $M = \varphi^{-1}(\eta)$ is a C^1-Banach manifold. Moreover, if $\widehat{\varphi} = \varphi|_M$, then u_0 is a critical point of $\widehat{\varphi}$ if and only if $\langle \varphi'(u_0), h\rangle = 0$ for all $h \in T_{u_0}(M)$ = the tangent space of M at u_0. (Ljusternik's Theorem, see Gasinski and Papageorgiou [182], p. 697).

The next theorem provides a minimax characterization of $\widehat{\lambda}_2$.

Theorem 4.111 *If hypotheses H_3 hold, then* $\widehat{\lambda}_2 = \inf_{\widehat{\gamma} \in \widehat{\Gamma}} \max_{-1 \le t \le 1} \gamma_p(\widehat{\gamma}(t))$ *where*

$$\widehat{\Gamma} = \{\widehat{\gamma} \in C([-1, 1], M) : \widehat{\gamma}(-1) = -\widehat{u}_1, \widehat{\gamma}(1) = \widehat{u}_1\}$$

(recall $M = W^{1,p}(\Omega) \cap \partial B_1^{L^p}$ and $\widehat{u}_1 > 0$, $\|\widehat{u}_1\|_p = 1$).

Proof Let $\widehat{\gamma}_p = \gamma_p|_M$. We claim that $\widehat{\gamma}_p(\cdot)$ satisfies the PS-condition. To this end, let $\{u_n\}_{n\in\mathbb{N}} \subseteq M$ be such that

$$\{\widehat{\gamma}_p(u_n) = \gamma_p(u_n)\}_{n\in\mathbb{N}} \subseteq \mathbb{R} \text{ is bounded,} \tag{4.88}$$

$$|\langle \gamma_p'(u_n), h\rangle| \leq \varepsilon_n \|h\| \quad \text{for all } h \in T_{u_n}M. \text{ with } \varepsilon_n \in 0^+. \tag{4.89}$$

We know that

$$M = \{u \in W^{1,p}(\Omega) : \psi(u) = 0\} \text{ with } \psi(u) = \|u\|_p^p - 1.$$

We have

$$\begin{aligned}\psi'(u) &= p\|u\|^{p-1}\frac{\mathscr{F}(u)}{\|u\|} \quad \text{for all } u \neq 0\\ &= p\|u\|^{p-2}\frac{|u(\cdot)|^{p-2}u(\cdot)}{\|u\|^{p-2}}\\ &= p|u(\cdot)|^{p-2}u(\cdot) \in L^{p'}(\Omega)\end{aligned}$$

(here $\mathscr{F}(\cdot)$ is the duality map of $L^p(\Omega)$, see Proposition I-6.94). Therefore

$$T_{u_n}M = \{h \in W^{1,p}(\Omega) : \int_\Omega |u_n|^{p-2}u_n h dz = 0\}.$$

Given any $v \in W^{1,p}(\Omega)$, we set

$$h_n = v - \left(\int_\Omega |u_n|^{p-2}u_n v dz\right)u_n, \ n \in \mathbb{N}.$$

We see that

$$\begin{aligned}&\int_\Omega |u_n|^{p-2}u_n h_n dz = 0 \quad \text{for all } n \in \mathbb{N}\\ &\qquad\qquad\qquad\qquad (\text{since } \|u_n\|_p = 1 \text{ for all } n \in \mathbb{N}),\\ &\Rightarrow h_n \in T_{u_n}M \quad \text{for all } n \in \mathbb{N}.\end{aligned}$$

In (4.89) we use $h_n \in T_{u_n}M$ and obtain

$$\begin{aligned}&\Big|\int_\Omega |Du_n|^{p-2}(Du_n, Dv)_{\mathbb{R}^N}dz + \int_\Omega \xi(z)|u_n|^{p-2}u_n v dz\\ &+\int_{\partial\Omega}\beta(z)|u_n|^{p-2}u_n v d\sigma - \left(\int_\Omega |u_n|^{p-2}u_n v dz\right)\gamma_p(u_n)\Big| \leq \varepsilon_n\|h_n\| \leq \widehat{c}_3\varepsilon_n\|v\|\\ &\qquad\qquad \text{for some } \widehat{c}_3 > 0, \text{ all } n \in \mathbb{N}.\end{aligned} \tag{4.90}$$

From (4.80) we have

$$\begin{aligned}&\|u_n\|^p \leq \widehat{c}_4(1 + \gamma_p(u_n)) \quad \text{for some } \widehat{c}_4 > 0, \text{ all } n \in \mathbb{N},\\ &\Rightarrow \{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega) \text{ is bounded (see (4.88))}.\end{aligned}$$

So, we may assume that

$$u_n \xrightarrow{w} u \text{ in } W^{1,p}(\Omega) \text{ and } u_n \to u \text{ in } L^{ps'}(\Omega) \text{ and in } L^p(\partial\Omega). \tag{4.91}$$

In (4.90) we choose $v = u_n - u \in W^{1,p}(\Omega)$, pass to the limit as $n \to \infty$ and use (4.91). We obtain

$$\lim_{n\to\infty} \int_\Omega |Du_n|^{p-2}(Du_n, Du_n - Du)_{\mathbb{R}^N} dz = 0,$$

$$\Rightarrow u_n \to u \text{ in } W^{1,p}(\Omega) \text{ (see Proposition I-7.77).}$$

We conclude that $\widehat{\gamma}_p = \gamma_p|_M$ satisfies the PS-condition.

From Theorem 4.101 and Corollary 4.103, we know that we can find $\varrho \in (0, 1)$ small such that

$$\gamma_p(\pm\widehat{\widehat{u}}_1) < \inf\{\gamma_p(u) : u \in M, \| u - (\pm\widehat{\widehat{u}}_1) \| = \varrho\}, \varrho < \| \widehat{\widehat{u}}_1 \|$$

(see Proposition 3.132).

Set $\widehat{\lambda} = \inf\limits_{\widehat{\gamma}\in\widehat{\Gamma}} \max\limits_{-1\le t\le 1} \gamma_p(\widehat{\gamma}(t))$. We know that $\widehat{\lambda}$ is a critical value of $\widehat{\gamma}_p(\cdot)$ (see Chang [96], Theorem 5.2.2, p. 348).

Claim $\widehat{\lambda} = \widehat{\lambda}_2$.

Evidently it suffices to show that

$$\sigma_p(\Omega) \cap (\widehat{\lambda}_1, \widehat{\lambda}) = \emptyset.$$

We argue indirectly. So suppose $\lambda \in \sigma_p(\Omega) \cap (\widehat{\lambda}_1, \widehat{\lambda})$ and let $u \in W^{1,p}(\Omega) \cap \partial B_1^{L^p}$ be an eigenfunction corresponding to λ. From Proposition 4.105 we know that u is nodal and so $u^\pm \neq 0$. We consider the following two paths in the manifold M:

$$\widehat{\gamma}_1(t) = \tfrac{u^+ - tu^-}{\|u^+ - tu^-\|_p}, \ t \in [0, 1],$$
$$\widehat{\gamma}_2(t) = \tfrac{-u^- + (1-t)u^+}{\|-u^- + (1-t)u^+\|_p}, \ t \in [0, 1].$$

We see that

$$\widehat{\gamma}_1(\cdot) \text{ connects } \tfrac{u^+}{\|u^+\|_p} \text{ with } u \text{ (recall } \| u \|_p = 1),$$
$$\widehat{\gamma}_2(\cdot) \text{ connects } u \text{ with } \tfrac{-u^-}{\|u^-\|_p}.$$

So, joining these two paths, we generate a new path $\widehat{\gamma} = \widehat{\gamma}_1 \cup \widehat{\gamma}_2$ in M which connects $\frac{u^+}{\|u^+\|_p}$ with $\frac{-u^-}{\|u^-\|_p}$. Note that

$$\gamma_p(u^+) = \lambda \parallel u^+ \parallel_p^p, \gamma_p(u^-) = \lambda \parallel u^- \parallel_p^p \tag{4.92}$$

$$\Rightarrow \gamma_p(\widehat{\gamma}_1(t)) = \gamma_p(\widehat{\gamma}_2(t)) = \lambda \text{ for all } t \in [0,1]$$

$$(u^+, u^- \text{ have disjoint interior supports})$$

Let $L_\lambda = \{u \in W^{1,p}(\Omega) : \gamma_p(u) < \lambda\}$. This set can not be connected (recall $\lambda < \widehat{\lambda}$). Using the Ekeland variational principal (see Theorem I-8.99) and the fact that $\widehat{\gamma}_p = \gamma_p|_M$ satisfies the PS-condition, we infer that every path component of L_λ contains critical points of $\widehat{\gamma}_p$. But the only critical points of $\widehat{\gamma}_p$ in L_λ are $\pm\widehat{u}_1$. Therefore L_λ has exactly two path components. One containing $\widehat{u}_1$ and the other $-\widehat{u}_1$.

Note that $\frac{u^+}{\|u^+\|_p}$ can not critical point of $\widehat{\gamma}_p$ because it has fixed sign and $\gamma_p(\frac{u^+}{\|u^+\|_p}) = \lambda$ (see (4.92) and Proposition 4.105). So, we can find a path $\tau : [-\varepsilon, \varepsilon] \to M$ such that

$$\tau(0) = \frac{u^+}{\parallel u^+ \parallel_p} \text{ and } \frac{d}{dt}\gamma_p(\tau(t)) \neq 0, t \in [-\varepsilon, \varepsilon].$$

Starting with this path $\tau(\cdot)$, we can move from $\frac{u^+}{\|u^+\|_p}$ to a point $v \in M$ along a path in M which is in L_λ except at the starting point $\frac{u^+}{\|u^+\|_p}$. So, $v \in L_\lambda$ and let U_1 be the path-component containing v. We may assume that $\widehat{u}_1 \in U_1$. Therefore, we can connect $\widehat{u}_1$ and v with a path $\tau_+ \subseteq L_\lambda$. Let $\widehat{\gamma}_+ = \tau_+ \cup \tau$. Then $\widehat{\gamma}_+ : [0,1] \to U_1$ satisfies

$$\widehat{\gamma}_+(0) = \widehat{u}_1, \widehat{\gamma}_+(1) = \frac{u^+}{\parallel u^+ \parallel_p}, \widehat{\gamma}_+(t) \in U_1 \subseteq L_\lambda \text{ for all } t \in [0,1).$$

Similarly, if U_2 is the other path-component of L_λ (the one containing $-\widehat{u}_1$), then we generate a path $\widehat{\gamma}_- : [0,1] \to U_2$ such that

$$\widehat{\gamma}_-(0) = \frac{-u}{\parallel u \parallel_p}, \widehat{\gamma}_-(1) = -\widehat{u}_1, \widehat{\gamma}_-(t) \in U_2 \subseteq L_\lambda \text{ for all } t \in (0,1].$$

Let $\widehat{\gamma}_* = \widehat{\gamma}_- \cup \widehat{\gamma} \cup \widehat{\gamma}_+ \in \widehat{\Gamma}$ and $\gamma_p(\widehat{\gamma}_*(t)) \leq \lambda$ for all $t \in [-1,1]$. Therefore $\widehat{\lambda} \leq \lambda$, a contradiction. So

$$\sigma_p(\Omega) \cap (\widehat{\lambda}_1, \widehat{\lambda}) = \emptyset,$$
$$\Rightarrow \widehat{\lambda} = \widehat{\lambda}_2.$$

□

Remark 4.112 A careful inspection of the above proof, reveals that we can also have the following characterization of $\widehat{\lambda}_2$

$$\widehat{\lambda}_2 = \inf_{\widehat{\gamma}\in\widehat{\Gamma}_0} \max_{-1\le t\le 1} \gamma_p(\widehat{\gamma}(t))$$

with $\widehat{\Gamma}_0 = \{\widehat{\gamma} \in C([-1, 1], M) : \widehat{\gamma}(-1) \le 0 \le \widehat{\gamma}(1)\}$.

Now consider the following subspace of $W^{1,p}(\Omega)$

$$V = \{u \in W^{1,p}(\Omega) : \int_\Omega \widehat{u}_1 u dz = 0\}.$$

In case $p = 2$, then $V = E(\widehat{\lambda}_1)^\perp = (\mathbb{R}\widehat{u}_1)^\perp$. We define

$$\widehat{\lambda}_V = \inf\{\gamma_p(u) : u \in V \cap M\}. \tag{4.93}$$

The next proposition relates $\widehat{\lambda}_V$ with $\widehat{\lambda}_1$ and $\widehat{\lambda}_2$.

Proposition 4.113 *If hypotheses H_3 hold, then $\widehat{\lambda}_1 < \widehat{\lambda}_V \le \widehat{\lambda}_2$.*

Proof Evidently $\widehat{\lambda}_1 \le \widehat{\lambda}_V$. Suppose $\widehat{\lambda}_V = \widehat{\lambda}_1$. Then we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq V \cap M$ such that

$$\gamma_p(u_n) \downarrow \widehat{\lambda}_V = \widehat{\lambda}_1.$$

From (4.80) we see that $\{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega)$ is bounded and so we may assume that

$$u_n \xrightarrow{w} u \text{ in } W^{1,p}(\Omega) \text{ and } u_n \to u \text{ in } L^{ps'}(\Omega) \text{ and in } L^p(\partial\Omega). \tag{4.94}$$

From (4.94) and the sequential weak lower semicontinuity of $\gamma_p(\cdot)$, we have

$$\begin{aligned} &\widehat{\lambda}_1 \le \gamma_p(u) \le \liminf_{n\to\infty} \gamma_p(u_n) = \widehat{\lambda}_V = \widehat{\lambda}_1, \qquad (4.95)\\ &\Rightarrow \gamma_p(u) = \widehat{\lambda}_1,\\ &\Rightarrow u = \pm\widehat{u}_1 \text{ (since } \| u \|_p = 1, \text{ see Corollary 4.103).} \end{aligned}$$

Note that $u \in V \cap M$ (see (4.94)) and this contradicts (4.95). Hence we have that $\widehat{\lambda}_1 < \widehat{\lambda}_V$.

Next suppose $\widehat{\lambda}_2 < \widehat{\lambda}_V$. Using Theorem 4.111, we can find $\widehat{\gamma} \in \widehat{\Gamma}$ such that

$$\gamma_p(\widehat{\gamma}(t)) < \widehat{\lambda}_V \text{ for all } t \in [-1, 1]. \tag{4.96}$$

Consider the function $\eta : [-1, 1] \to \mathbb{R}$ defined by

$$\eta(t) = \int_\Omega \widehat{u}_1 \widehat{\gamma}(t) dt \text{ for all } t \in [-1, 1].$$

Evidently $\eta(\cdot)$ is continuous and $\eta(-1) < 0 < \eta(1)$. By Bolzano's theorem, we can find $t_0 \in (-1, 1)$ such that

$$\begin{aligned} &\eta(t_0) = 0, \\ \Rightarrow &\int_\Omega \widehat{u}_1 \widehat{\gamma}(t_0) dz = 0, \\ \Rightarrow &\widehat{\gamma}(t_0) \in V. \end{aligned}$$

Then from (4.93) we have $\widehat{\lambda}_V \leq \gamma_p(\widehat{\gamma}(t_0))$ which contradicts (4.96). Therefore $\widehat{\lambda}_V \leq \widehat{\lambda}_2$. □

Remark 4.114 Let $\mathcal{D} = \{V \subseteq W^{1,p}(\Omega) : W^{1,p}(\Omega) = \mathbb{R}\widehat{u}_1 \oplus V\}$ (so $\mathcal{D}$ consists of all the topological complements of the principal eigenspace $E(\widehat{\lambda}_1) = \mathbb{R}\widehat{u}_1$). We set

$$\widehat{\lambda}_{\mathcal{D}} = \inf_{V \in \mathcal{D}} \sup_{u \in V} \gamma_p(u).$$

It is an open problem if $\widehat{\lambda}_{\mathcal{D}} = \widehat{\lambda}_2$.

We can produce additional eigenvalues using the Ljusternik-Schnirelmann minimax scheme (see Sect. 3.4), using as index the Krasnoselskii genus $\gamma(\cdot)$ (see Example 3.67 (a)). So, let

$$\mathcal{D} = \{A \subseteq W^{1,p}(\Omega) : A \text{ is a compact, symmetric}\}.$$

Using the Krasnoselskii genus $\gamma(\cdot)$, we introduce the following decreasing families of subsets of M

$$\mathcal{F}_k = \{A \in \mathcal{D} \cap M : \gamma(A) \geq k\}, k \in \mathbb{N}.$$

We set

$$\widehat{\lambda}_k = \inf_{A \in \mathcal{F}_k} \sup_{u \in A} \gamma_p(u), k \in \mathbb{N}.$$

From Theorem 3.72, we have:

Theorem 4.115 *If hypotheses H_3 hold, then $\{\widehat{\lambda}_k\}_{k \in \mathbb{N}}$ are eigenvalues of (4.76) and $\widehat{\lambda}_k \to +\infty$ as $k \to \infty$.*

Of course we can execute the Ljusternik-Schnirelmann minimax scheme, using a different index. If we use the "cohomological index" (see Example 3.67 (b)), we will generate another sequence $\{\widehat{\mu}_k\}_{k \in \mathbb{N}}$ of eigenvalues of (4.76). We can also use the "cogenus" $\gamma_*(\cdot)$ defined by

$$\gamma_*(A) = \sup\{k \in \mathbb{N} : \text{ there exists odd } \psi \in C(S^{k-1}, A)\}$$

with $S^{k-1} = \{u \in \mathbb{R}^k : |u| = 1\}$. If no such $\psi(\cdot)$ exists $\gamma_*(A) = \infty$. Using the cogenus in the Ljusternik-Schnirelmann minimax scheme, we produce a third sequence $\{\widehat{\vartheta}_k\}_{k\in\mathbb{N}}$ of eigenvalues of (4.76). We have

$$\widehat{\lambda}_k \leq \widehat{\mu}_k \leq \widehat{\vartheta}_k \text{ for all } k \in \mathbb{N}.$$

and for $k = 1$ and $k = 2$, we have equality. We do not know if the equality is also true for $k \geq 3$.

Proposition 4.116 *If hypotheses H_3 hold, then every eigenfunction $\widehat{u}$ of (4.76) has a finite number of nodal domains.*

Proof Let $\{D_i\}_{i\in I}$ be the nodal domains of $\widehat{u}$. Using Lemma 4.109 we have

$$|\Omega|_N \geq \sum_{i\in I} |D_i|_N \geq (\frac{1}{c^*})^{\frac{p^*-p}{p^*}} \sum_{i\in I} 1,$$

$$\Rightarrow I \text{ is finite.}$$

□

The same spectral analysis can be conducted for the Dirichlet eigenvalue problem

$$\begin{cases} -\Delta_p u(z) + \xi(z)|u(z)|^{p-2}u(z) = \widehat{\lambda}|u(z)|^{p-2}u(z) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, 1 < p < N. \end{cases} \tag{4.97}$$

The same results can be proved for problem (4.97). Moreover, in this case, we can have monotonicity of the principal eigenvalue with respect to the domain Ω.

Proposition 4.117 *If $\xi \in L^s(\Omega)$ with $s > \frac{N}{p}$ and $\Omega_1, \Omega_2 \subseteq \mathbb{R}^N$ are bounded domains with C^2-boundaries and*

$$\Omega_1 \subseteq \Omega_2, \Omega_1 \neq \Omega_2,$$

then $\widehat{\lambda}_1(\Omega_2) < \widehat{\lambda}_1(\Omega_1)$.

Proof Let $u \in W_0^{1,p}(\Omega)$ be an eigenfunction corresponding to the eigenvalue $\widehat{\lambda}_1(\Omega_1)$. We extend by zero $\widehat{u}_1$ to all of Ω_2 and denote the resulting function by $\overline{u}$. We know that $\overline{u} \in W_0^{1,p}(\Omega_2)$. Then

$$\widehat{\lambda}_1(\Omega_2) \leq \frac{\gamma_p(\overline{u})}{\| \overline{u} \|_{p,\Omega_2}^p} \text{ (now } \gamma_p(u) = \| Du \|_p^p + \int_{\Omega_2} \xi(z)|u|^p dz, u \in W_0^{1,p}(\Omega_2)),$$

$$= \frac{\| Du \|_{p,\Omega_1}^p + \int_{\Omega_1} \xi(z)|u|^p dz}{\| u \|_{p,\Omega_1}} = \widehat{\lambda}_1(\Omega_1).$$

Suppose that we have equality. Then

$$\widehat{\lambda}_1(\Omega_2) = \frac{\gamma_p(\overline{u})}{\| \overline{u} \|^p_{p,\Omega_2}}$$

$\Rightarrow \overline{u}$ is an eigenfunction for $\widehat{\lambda}_1(\Omega_2)$ (see Corollary 4.103)

$\Rightarrow \overline{u}(z) > 0$ for all $z \in \Omega_2$, a contradiction.

Therefore $\widehat{\lambda}_1(\Omega_2) < \widehat{\lambda}_1(\Omega_1)$. □

Remark 4.118 This monotonicity property is not true for the principal eigenvalue of (4.76).

Next we turn our attention to nonlinear eigenvalue problems driven by the p-Laplacian and with an indefinite weight. For simplicity, we will do the analysis for the Dirichlet eigenvalue problem. The same results can be proved also for the Robin eigenvalue problem.

Let $\Omega \subseteq \mathbb{R}^N (N \geq 2)$ be a bounded domain with a C^2-boundary $\partial\Omega$. We examine the following Dirichlet eigenvalue problem

$$\begin{cases} -\Delta_p u(z) = \widehat{\lambda} m(z)|u(z)|^{p-2}u(z) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, 1 < p < N. \end{cases} \tag{4.98}$$

In this problem $m(\cdot)$ is an indefinite weight function for which we assume the following:

$\boldsymbol{H_4}$: $m \in L^s(\Omega), s > \frac{N}{p}$ and $m^+ \neq 0$.

Problem (4.98) is a nonlinear generalization of problem (53). If $m \equiv 1$, then we recover problem (4.97) with no potential term (that is, $\xi \equiv 0$). We will focus on positive eigenvalues.

Definition 4.119 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" of problem (4.98), if it admits a nontrivial weak solution $\widehat{u} \in W^{1,p}_0(\Omega)$, that is

$$\int_\Omega |D\widehat{u}|^{p-2}(D\widehat{u}, Dh)_{\mathbb{R}^N} dz = \widehat{\lambda} \int_\Omega m(z)|\widehat{u}|^{p-2}\widehat{u}h dz \text{ for all } h \in W^{1,p}_0(\Omega).$$

We call $\widehat{u}$ an "eigenfunction" corresponding to $\widehat{\lambda}$ and $(\widehat{\lambda}, \widehat{u})$ is an "eigenpair" for (4.98).

Remark 4.120 Local regularity theory implies $\widehat{u} \in W^{1,p}(\Omega) \cap C^1(\Omega)$. If we strengthen the condition on the weight $m(\cdot)$ and require that $m \in L^\infty(\Omega)$, then on account of Theorem 4.84, we have that $\widehat{u} \in C^{1,\alpha}_0(\overline{\Omega})$ for some $\alpha \in (0, 1)$. In fact we can do better.

Proposition 4.121 *If $m \in L^s(\Omega)$ with $s > N$ and $\widehat{u} \in W_0^{1,p}(\Omega)$ is a weak solution of Dirichlet problem*

$$-\Delta_p u(z) = \widehat{\lambda} m(z)|u(z)|^{p-2}u(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0,$$

then $\widehat{u} \in C_0^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0,1)$.

Proof We have $\widehat{u} \in L^\infty(\Omega)$ (see Ladyzhenskaya and Uraltseva [285], Theorem 7.1, p. 286). So, if $k_{\widehat{\lambda}}(\cdot) = \widehat{\lambda} m(\cdot)|\widehat{u}(\cdot)|^{p-2}\widehat{u}(\cdot) \in L^s(\Omega)$ with $s > N$. We consider the following linear Dirichlet problem

$$-\Delta v(z) = k_{\widehat{\lambda}}(z) \text{ in } \Omega,\ v|_{\partial\Omega} = 0. \tag{4.99}$$

From standard linear theory (see Gilbarg and Trudinger [206], Theorem 9.15, p. 241), we know that problem (4.99) has a unique solution $\widehat{v} \in W^{2,s}(\Omega) \hookrightarrow C^{1,\eta}(\overline{\Omega})$ with $\eta = 1 - \frac{N}{s} > 0$. Using (4.99), we can rewrite our equation as

$$-\operatorname{div}(|D\widehat{u}|^{p-2}D\widehat{u} - D\widehat{v}) = 0 \text{ in } \Omega.$$

Applying Theorem 4.84 we conclude that

$$\widehat{u} \in C_0^{1,\alpha}(\overline{\Omega}) \text{ for some } \alpha \in (0,1).$$

□

We introduce the following C^1-Banach manifold

$$M = \{u \in W_0^{1,p}(\Omega) : \psi(u) = \int_\Omega m(z)|u|^p dz = 1\}.$$

Since by hypotheses H_4, $m^+ \neq 0$, we see that $M \neq \emptyset$. For $u \in M$ by the Ljusternik theorem, we have that

$$T_u M = \{h \in W_0^{1,p}(\Omega) : \langle \varphi'(u), h\rangle = 0\}.$$

Let $\varphi(u) = \| Du \|_p^p$ for all $u \in W_0^{1,p}(\Omega)$ and set $\widehat{\varphi} = \varphi|_M$. We recall that "$u \in W_0^{1,p}(\Omega)$ is a critical point of $\widehat{\varphi}$ if and only if $\langle \varphi'(u), h\rangle = 0$ for all $h \in T_u M$" (see Definition 3.83).

We employ the Ljusternik-Schnirelmann minimax scheme to generate a sequence of eigenvalues. As before as index, we use the Krasnoselskii genus (see Example 3.67(a)). We introduce the following families

$$\mathcal{F} = \{A \subseteq W_0^{1,p}(\Omega) : A \text{ is closed, symmetric}\},$$
$$\mathcal{F}_k = \{A \in \varphi : \gamma(A) \geq k\}\ k \in \mathbb{N}.$$

with $\gamma(\cdot)$ being the Krasnoselskii genus. We define

$$\lambda_k = \inf_{A \in \mathcal{F}_k} \sup_{u \in A} \widehat{\varphi}(u), k \in \mathbb{N}. \tag{4.100}$$

From Theorem 3.72, we have:

Proposition 4.122 *If hypotheses H_4 hold, then $\{\widehat{\lambda}_k\}_{k \in \mathbb{N}} \subseteq \mathbb{R}$ are eigenvalues of (4.98) and $\widehat{\lambda}_k \to \infty$ as $k \to \infty$.*

As we already mentioned earlier, if in the Ljusternik-Schnirelmann process, we use a different index, then we generate additional sequences of eigenvalues. All these sequences coincide in the first two elements. Beyond them we do not know if this is true. Of course if $N = 1$ (scalar case), then the sequences coincide.

From (4.100) we have

$$\widehat{\lambda}_1 = \inf\{\varphi(u) = \| Du \|_p^p : u \in W_0^{1,p}(\Omega), \psi(u) = \int_\Omega m(z)|u|^p dz = 1\}.$$

This is the smallest positive eigenvalue of problem (4.98).

Proposition 4.123 *If hypotheses H_4 hold, then there exists $\widehat{u} \in M$ such that*

$$\widehat{\lambda}_1 = \varphi(\widehat{u}) = \| D\widehat{u} \|_p^p > 0;$$

moreover $\widehat{\lambda}_1 = \varphi(\widehat{u})$ for some $\widehat{u} \in M$ if and only if $\widehat{u}$ is an eigenfunction corresponding to $\widehat{\lambda}_1$.

Proof Let $\{u_n\}_{n \in \mathbb{N}} \subseteq M$ such that $\varphi(u_n) \downarrow \widehat{\lambda}_1$. Evidently $\{u_n\}_{n \in \mathbb{N}} \subseteq W_0^{1,p}(\Omega)$ is bounded and so we may assume that

$$u_n \xrightarrow{w} \widehat{u} \text{ in } W_0^{1,p}(\Omega) \text{ and } u_n \to \widehat{u} \text{ in } L^{ps'}(\Omega).$$

(recall that hypotheses H_4 imply $ps' < p^* = \dfrac{N_p}{N-p}$ and $\dfrac{1}{s} + \dfrac{1}{s'} = 1$).

We have

$$\varphi(\widehat{u}) = \| D\widehat{u} \|_p^p \leq \liminf_{n \to \infty} \| Du_n \|_p^p = \liminf_{n \to \infty} \varphi(u_n) = \widehat{\lambda}_1$$

and $\int_\Omega m(z)|u_n|^p dz \to \int_\Omega m(z)|\widehat{u}|^p dz$.

Therefore $\widehat{u} \in M$ and so $\varphi(\widehat{u}) = \widehat{\lambda}_1 > 0$.

The last part of the proposition follows from the Lagrange multiplier rule. □

From the Harnack inequality (see Theorem 4.96 and Remark 4.97), we have:

Proposition 4.124 *If hypotheses H_4 hold and $u \in W_0^{1,p}(\Omega)$ is a weak solution of (4.98) which is nonnegative, then either $\widehat{u} \equiv 0$ or $\widehat{u}(z) > 0$ for all $z \in \Omega$.*

This observation leads to the following fundamental property of the eigenfunctions corresponding to $\widehat{\lambda}_1 > 0$.

Proposition 4.125 *If $\widehat{u} \in W_0^{1,p}(\Omega)$ is an eigenfunction corresponding to $\widehat{\lambda}_1 > 0$, then $\widehat{u}(z) > 0$ or $\widehat{u}(z) < 0$ for all $z \in \Omega$.*

Proof From Proposition 4.123 we know that

$$\widehat{\lambda}_1 = \varphi(\widehat{u}) \text{ with } \psi(\widehat{u}) = 1.$$

We can always replace $\widehat{u}$ by $|\widehat{u}|$ and the above equalities remain valid. Since $|\widehat{u}| \in W_0^{1,p}(\Omega)$, Proposition 4.124 implies that $|\widehat{u}(z)| > 0$ for all $z \in \Omega$. □

Remark 4.126 In the sequel by $\widehat{u}_1$ we denote the positive eigenfunction which satisfies $\psi(\widehat{u}_1) = \int_\Omega m(z)|\widehat{u}_1|^p = 1$.

Every other positive eigenvalue $\widehat{\lambda} > \widehat{\lambda}_1$, has nodal eigenfunctions.

Proposition 4.127 *If hypotheses H_4 hold and $\widehat{u} \in W_0^{1,p}(\Omega)$ is an eigenfunction corresponding to an eigenvalues $\widehat{\lambda} > \widehat{\lambda}_1$, then $\widehat{u}$ is nodal (sign changing).*

Proof We argue by contradiction and assume that $\widehat{u}$ has fixed sign. We can always assume that $\widehat{u} \geq 0$. As before via Harnack's inequality we infer that $\widehat{u}(z) > 0$ for all $z \in \Omega$. Let $\widehat{u}_1$ be as in Remark 4.126 and for $\varepsilon \in (0, 1)$, let $\widehat{u}_\varepsilon = \widehat{u} + \varepsilon$. Note that

$$0 \leq \frac{\widehat{u}_1^p}{\widehat{u}_\varepsilon^{p-1}} \leq \frac{\widehat{u}_1^p}{\varepsilon^{p-1}} \text{ and so } \frac{\widehat{u}_1^p}{\widehat{u}_\varepsilon^{p-1}} \in W_0^{1,p}(\Omega).$$

Using the nonlinear Picone identity (see Theorem 4.91), we have

$$\begin{aligned}
0 \leq \int_\Omega L(\widehat{u}_1, \widehat{u}_\varepsilon)dz &= \int_\Omega R(\widehat{u}_1, \widehat{u}_\varepsilon)dz \\
&= \widehat{\lambda}_1 - \int_\Omega |D\widehat{u}|^{p-2}(D\widehat{u}, D(\frac{\widehat{u}_1^p}{\widehat{u}_\varepsilon^{p-1}}))_{\mathbb{R}^N}dz \\
&= \widehat{\lambda}_1 - \int_\Omega (-\Delta_p \widehat{u})\frac{\widehat{u}_1^p}{\widehat{u}_\varepsilon^{p-1}}dz \\
&\text{We let } \varepsilon \to 0^+ \text{ and obtain} \\
0 &\leq \widehat{\lambda}_1 - \widehat{\lambda} \text{ (recall } \psi(\widehat{u}_1) = 1) \\
&< 0, \text{ a contradiction.}
\end{aligned}$$

We conclude that $\widehat{u}$ must be nodal. □

We can estimate the Lebesgue measure of the nodal domains of $\widehat{u}$.

Proposition 4.128 *If hypotheses H_4 hold, $\widehat{u} \in W_0^{1,p}(\Omega)$ is an eigenfunction corresponding to an eigenvalue $\widehat{\lambda} > \widehat{\lambda}_1$ and D is a nodal domain of $\widehat{u}$, then $(\widehat{\lambda c} \parallel m \parallel_s)^{-\eta} \leq |D|_N$ with $\eta = \frac{N_S}{sp-N} > 0, \widehat{c} = \widehat{c}(N, p) > 0$.*

Proof We assume that $\widehat{u}(z) > 0$ for all $z \in D$. We define

$$v(z) = \begin{cases} \widehat{u}(z) \text{ if } z \in D, \\ 0 \quad \text{if } z \in \Omega \backslash D. \end{cases}$$

We know that $v \in W_0^{1,p}(\Omega)$. Using $h = v \in W_0^{1,p}(\Omega)$ as test function (see Definition 4.119), we obtain

$$\begin{aligned} \int_{\mathcal{D}} |Dv|^p dz &= \widehat{\lambda} \int_{\mathcal{D}} m(z)|v|^p dz \\ &\leq \widehat{\lambda} \parallel m \parallel_s \parallel v \parallel_{p^*}^p |D|_N^{\frac{p^*-ps'}{p^*s'}} \text{ (by Hölder's inequality)} \\ &\leq \widehat{\lambda c} \parallel m \parallel_s \parallel Dv \parallel_p^p |D|_N^{\frac{p^*-ps'}{p^*s'}} \text{ with } \widehat{c} = \widehat{c}(N, p) > 0. \end{aligned}$$

$$\Rightarrow (\widehat{\lambda c} \parallel m \parallel_s)^{-\eta} \leq |D|_N \text{ with } \eta = \frac{p^*s'}{p^* - ps'} = \frac{N_S}{sp - N}.$$

□

From this estimate, we infer that there are finitely many nodal domains (since $\Omega \subseteq \mathbb{R}^N$ is bounded).

Corollary 4.129 *If hypotheses H_4 hold, then every eigenfunction of (4.98) has a finite number of nodal domains.*

We continue with the properties of the first positive eigenvalue $\widehat{\lambda}_1 > 0$.

Proposition 4.130 *If hypotheses H_4 hold, then $\widehat{\lambda}_1 > 0$ is simple (that is, if $\widehat{u}, \widehat{v} \in W_0^{1,p}(\Omega)$ are eigenfunctions corresponding to $\widehat{\lambda}_1 > 0$, then $\widehat{u} = \vartheta \widehat{v}$ for some $\vartheta \in \mathbb{R}\backslash\{0\}$).*

Proof Let $\widehat{u}, \widehat{v} \in W_0^{1,p}(\Omega) \cap C^1(\Omega)$ be two eigenfunctions corresponding to $\widehat{\lambda}_1 > 0$. On account of Proposition 4.125, we may assume that

$$\widehat{u}(z) > 0, \widehat{v}(z) > 0 \text{ for all } z \in \Omega.$$

For $\varepsilon \in (0, 1)$, let $\widehat{v}_\varepsilon = \widehat{v} + \varepsilon$. We know that $\frac{\widehat{u}^p}{\widehat{v}_\varepsilon^{p-1}} \in W_0^{1,p}(\Omega)$ (another way to see this, is to observe that $\widehat{v}_\varepsilon \in intL^\infty(\Omega)_+$ and use Proposition 2.86). Using the nonlinear Picone's identity (Theorem 4.91) we have

$$0 \leq \int_\Omega L(\widehat{u}, \widehat{v}_\varepsilon)dz = \int_\Omega R(\widehat{u}, \widehat{v}_\varepsilon)dz$$
$$= \widehat{\lambda}_1 \int_\Omega m(z)\widehat{u}^p(1 - \frac{\widehat{v}^{p-1}}{\widehat{v}_\varepsilon^{p-1}})dz.$$

We let $\varepsilon \to 0^+$ and obtain

$$0 = \int_\Omega L(\widehat{u}, \widehat{v})dz,$$
$$\Rightarrow L(\widehat{u}, \widehat{v})(z) = 0 \text{ for a.a } z \in \Omega,$$
$$\Rightarrow \widehat{u} = \vartheta\widehat{v} \text{ with } \vartheta \in \mathbb{R}\backslash\{0\}.$$

□

Let σ_p^+ denote the set of positive eigenvalues of (4.98).

Proposition 4.131 *If hypotheses H_4 hold, then $\sigma_p^+ \subseteq (0, \infty)$ is closed.*

Proof Let $\{\lambda_n\}_{n\in\mathbb{N}} \subseteq \sigma_p^+$ and suppose that $\lambda_n \to \lambda \geq \widehat{\lambda}_1 > 0$. We consider a corresponding sequence $\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega)$ of eigenfunctions. We can assume that $\| \widehat{u}_n \| = 1$ for all $n \in \mathbb{N}$. Then the sequence $\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega)$ is bounded and so we may assume that

$$\widehat{u}_n \xrightarrow{w} \widehat{u} \text{ in } W_0^{1,p}(\Omega), \widehat{u}_n \to \widehat{u} \text{ in } L^{ps'}(\Omega). \tag{4.101}$$

We have

$$\int_\Omega |D\widehat{u}_n|^{p-2}(D\widehat{u}_n, D\widehat{u}_n - D\widehat{u})_{\mathbb{R}^N}dz = \lambda_n \int_\Omega m(z)|\widehat{u}_n|^{p-2}\widehat{u}_n(\widehat{u}_n - \widehat{u})dz$$
$$\text{for all } n \in \mathbb{N}.$$

Passing to the limit as $n \to \infty$ and using (4.101) we obtain

$$\lim_{n\to\infty} \int_\Omega |D\widehat{u}_n|^{p-2}(D\widehat{u}_n, D\widehat{u}_n - D\widehat{u})_{\mathbb{R}^N}dz = 0,$$
$$\Rightarrow \widehat{u}_n \to \widehat{u} \text{ in } W_0^{1,p}(\Omega) \text{ (see Proposition I-7.77)},$$
$$\Rightarrow \| \widehat{u} \| = 1.$$

Also we have

$$\int_\Omega |D\widehat{u}|^{p-2}(D\widehat{u}, Dh)_{\mathbb{R}^N}dz = \lambda \int_\Omega m(z)|\widehat{u}|^{p-2}\widehat{u}hdz \text{ for all } h \in W_0^{1,p}(\Omega),$$
$$\Rightarrow \lambda \in \sigma_p^+ \text{ and so } \sigma_p^+ \text{ is closed.}$$

□

Proposition 4.132 *If hypotheses H_4 hold, then $\widehat{\lambda}_1 > 0$ is isolated in σ_p^+.*

Proof Arguing by contradiction, suppose we can find $\{\widehat{\lambda}_n\}_{n\in\mathbb{N}} \subseteq \sigma_p^+ \setminus \{\widehat{\lambda}_1\}$ such that $\widehat{\lambda}_n \downarrow \widehat{\lambda}_1$. Let $\widehat{u}_n \in W_0^{1,p}(\Omega)$ be corresponding eigenfunctions with $\| u_n \| = 1$ for all $n \in \mathbb{N}$. As in the previous proof, at least for a subsequence, we have

$$\begin{aligned}
&\widehat{u}_n \to \widehat{u} \text{ in } W_p^{1,p}(\Omega), \| \widehat{u} \| = 1,\\
&\Rightarrow \int_\Omega |D\widehat{u}|^{p-2}(D\widehat{u}, Dh)_{\mathbb{R}^N} dz,\\
&= \widehat{\lambda}_1 \int_\Omega m(z)|\widehat{u}|^{p-2}\widehat{u}h dz \text{ for all } h \in W_0^{1,p}(\Omega),\\
&\Rightarrow \widehat{u} \text{ is an eigenfunction for } \widehat{\lambda}_1.
\end{aligned}$$

On account of Proposition 4.125, we may assume that $\widehat{u}(z) > 0$ for all $z \in \Omega$. If $\Omega_n^- = \{z \in \Omega : u_n(z) < 0\}$, then $|\Omega_n^-|_N \to 0$ as $n \to \infty$, which contradicts Proposition 4.128. Therefore $\widehat{\lambda}_1 > 0$ is isolated in σ_p^+. □

When we want to emphasize the dependence of the eigenvalues on the weight m, we write $\widehat{\lambda}_k(m)$, $k \in \mathbb{N}$. Then we examine the map $m \to \widehat{\lambda}_k(m)$.

Proposition 4.133 *If $m, \widehat{m} \in L^s(\Omega)$ with $s > \frac{N}{p}$, $m^+ \neq 0$ and $m(z) \leq \widehat{m}(z)$ for a.a $z \in \Omega$, $m \neq \widehat{m}$, then $\widehat{\lambda}_1(\widehat{m}) < \widehat{\lambda}_1(m)$.*

Proof Let $\widehat{u}_1(m)$ be a positive eigenfunction for $\widehat{\lambda}_1(m)$ (then $\widehat{u}_1(m)(z) > 0$ for all $z \in \Omega$). We have

$$\widehat{\lambda}_1(\widehat{m}) \leq \frac{\| D\widehat{u}_1(m) \|_p^p}{\int_\Omega \widehat{m}(z)\widehat{u}_1(m)^p dz} \leq \frac{\| D\widehat{u}_1(m) \|_p^p}{\int_\Omega m(z)\widehat{u}_1(z)^p dz} = \widehat{\lambda}_1(m)$$

(see Proposition 4.123).

If $\widehat{\lambda}_1(\widehat{m}) = \widehat{\lambda}_1(m)$, then from (4.102) we have

$$\begin{aligned}
&\int_\Omega \widehat{m}(z)\widehat{u}_1(m)^p dz = \int_\Omega m(z)\widehat{u}_1(m)^p dz,\\
&\Rightarrow \widehat{m}(z) = m(z) \text{ for a.a } z \in \Omega \text{ (recall } \widehat{u}_1(m)(z) > 0, z \in \Omega).
\end{aligned}$$

a contradiction. We conclude that

$$\widehat{\lambda}_1(\widehat{m}) < \widehat{\lambda}_1(m).$$

□

Following the proof of Proposition 4.132, we obtain the continuity of $m \to \widehat{\lambda}_1(m)$.

Proposition 4.134 *If* $\{m_n, m\}_{n\in\mathbb{N}} \subseteq L^s(\Omega)$ *with* $s > \frac{N}{p}$, $m_n^+ \neq 0, m^+ \neq 0$ *and* $m_n \to m$ *in* $L^s(\Omega)$, *then* $\widehat{\lambda}_k(m_n) \to \widehat{\lambda}_k(m)$ *as* $n \to \infty$ *for all* $k \in \mathbb{N}$.

In Proposition 4.133 we established the strict monotonicity of $m \to \widehat{\lambda}_1(m)$. It is natural to ask if something similar can be said for higher eigenvalues. We will prove a monotonicity property for $\widehat{\lambda}_2(m)$. To do this, we need some preparation.

Definition 4.135 Let $\widehat{\lambda} \in \sigma_p^+$ and $\widehat{u} \in W_0^{1,p}(\Omega)$ a corresponding eigenfunction. By $N(\widehat{u})$ we denote the number of nodal domains of $\widehat{u}$. We define

$$N(\widehat{\lambda}) = \max\{N(\widehat{u}) : \widehat{u} \text{ is an eigenfunction for } \widehat{\lambda}\}.$$

Remark 4.136 From Proposition 4.125, we have $N(\widehat{\lambda}_1) = 1$.

Proposition 4.137 *If hypotheses* H_4 *hold and* $\widehat{\lambda} \in \sigma_p^+$, *then* $\widehat{\lambda}_{N(\widehat{\lambda})} \leq \widehat{\lambda}$.

Proof Let $\widehat{u}$ be an eigenfunction corresponding to $\widehat{\lambda}$ such that $N(\widehat{u}) = N(\widehat{\lambda})$. Let $\{D_k\}_{k=1}^{N(\widehat{\lambda})}$ be the nodal domains of $\widehat{u}$. Let $\widehat{u}_k = \widehat{u}_{\lambda_{D_k}} \in W_0^{1,p}(\Omega)$ for all $k = 1, \ldots, N(\widehat{\lambda})$ (see Proposition I-4.97). Using as test functions $h = \widehat{u}_k \in W_0^{1,p}(\Omega)$, we obtain

$$\int_\Omega |D\widehat{u}_k|^p dz = \widehat{\lambda} \int_\Omega m(z)|\widehat{u}_k|^p dz \text{ for all } k = 1, \ldots, N(\lambda). \tag{4.102}$$

Let $V_{\widehat{\lambda}} = span\{\widehat{u}_k\}_{k=1}^{N(\lambda)}$. The functions $\{\widehat{u}_k\}$ have distinct interior supports and so they are linearly independent. Hence $\dim V_{\widehat{\lambda}} = N(\widehat{\lambda})$. Let $E(\widehat{\lambda}) = V_\lambda \cap \partial B_1 (\partial B_1 = \{v \in W_0^{1,p}(\Omega) : \|v\| = \|Dv\|_p = 1\})$. If $\gamma(\cdot)$ denotes the Krasnoselskii genus, then

$$\begin{aligned} &\gamma(E(\widehat{\lambda})) = N(\widehat{\lambda}) \text{ (see Proposition 3.70)}, \\ &\Rightarrow E(\widehat{\lambda}) \in \mathcal{F}_{N(\widehat{\lambda})}. \end{aligned}$$

If $v \in E(\widehat{\lambda})$, then $v = \sum_{k=1}^{N(\widehat{\lambda})} \beta_k \widehat{u}_k$ with $\beta_k \in \mathbb{R}$. Then

$$\begin{aligned} \varphi(v) = \| Dv \|_p^p &= \sum_{k=1}^{N(\widehat{\lambda})} |\beta_k|^p \| D\widehat{u}_k \|_p^p, \\ &= \sum_{k=1}^{N(\widehat{\lambda})} |\beta_k|^p \widehat{\lambda} \textstyle\int_\Omega m(z)|\widehat{u}_k|^p dz \text{ (see (4.102))}, \\ &= \widehat{\lambda} \textstyle\int_\Omega m(z)|v|^p dz, \end{aligned}$$

$\Rightarrow \widehat{\lambda}_{N(\widehat{\lambda})} \leq \widehat{\lambda}$ (see (4.100)). □

Corollary 4.138 *If hypotheses* H_4 *hold,* $m, n \in \mathbb{N}$ *and* $\widehat{\lambda}_n < \widehat{\lambda}_m$, *then* $N(\widehat{\lambda}_n) < m$.

Proof If $N(\widehat{\lambda}_n) \geq m$, then $\widehat{\lambda}_{N(\widehat{\lambda}_n)} \geq \widehat{\lambda}_m$. But from Proposition 4.137 we have $\widehat{\lambda}_{N(\widehat{\lambda}_n)} \leq \widehat{\lambda}_n$. It follows that $\widehat{\lambda}_n < \widehat{\lambda}_n$, a contradiction. □

Propositions 4.131 and 4.132, we lead to the following definition

$$\widehat{\lambda}_2^* = \inf\{\lambda \in \sigma_p^+ : \lambda > \widehat{\lambda}_1\} \in \sigma_p^+.$$

Evidently this is the second element of σ_p^+ and it is natural to ask what is the relation between $\widehat{\lambda}_2^*$ and $\widehat{\lambda}_2$.

Proposition 4.139 *If hypotheses H_4 hold, then $\widehat{\lambda}_2^* = \widehat{\lambda}_2$.*

Proof We have $\widehat{\lambda}_2^* \leq \widehat{\lambda}_2$. Also from Proposition 4.137, we have

$$\widehat{\lambda}_{N(\widehat{\lambda}_2^*)} \leq \widehat{\lambda}_2^* \tag{4.103}$$

and since the eigenfunctions corresponding to $\widehat{\lambda}_2^*$ are nodal (see Proposition 4.127), we have

$$\begin{aligned} & 2 \leq N(\widehat{\lambda}_2^*), \\ \Rightarrow\ & \widehat{\lambda}_2 \leq \widehat{\lambda}_{N(\widehat{\lambda}_2^*)} \leq \widehat{\lambda}_2^* \text{ (see (4.103))}, \\ \Rightarrow\ & \widehat{\lambda}_2^* = \widehat{\lambda}_2. \end{aligned}$$

□

Remark 4.140 The eigenvalues defined by (4.100) (Ljusternik-Schnirelmann scheme), are usually called "variational eigenvalues". Proposition 4.139 says that the second eigenvalue $\widehat{\lambda}_2^*$ and the second variational eigenvalue coincide.

Now we can establish the monotonicity of the map $m \to \widehat{\lambda}_2(m)$.

Proposition 4.141 *If $m, \widehat{m} \in L^s(\Omega)$ with $s > \frac{N}{p}$, $m(z) \leq \widehat{m}(z)$ for a.a $z \in \Omega$, $m^+ \neq 0$ and $m(z) < \widehat{m}(z)$ for a.a $z \in D_m^+ = \{z' \in \Omega : m(z') > 0\}$, then $\widehat{\lambda}_2(\widehat{m}) < \widehat{\lambda}_2(m)$.*

Proof Let $\widehat{u}_2(m) \in W_0^{1,p}(\Omega)$ be an eigenfunction corresponding to $\widehat{\lambda}_2(m)$. We know that $\widehat{u}_2(m)$ is nodal (see Proposition 4.127). Let

$$\Omega_+ = \{z \in \Omega : \widehat{u}_2(m)(z) > 0\}, \Omega_- = \{z \in \Omega : \widehat{u}_2(m)(z) < 0\}$$

These are nonempty open sets. Let

$$\begin{aligned} v_1 &= \frac{\widehat{u}_2(m)^+}{(\int_{\Omega_+} m(z)(\widehat{u}_2(m)^+)^p dz)^{1/p}} \chi_{\Omega_\pm}, \\ v_2 &= \frac{\widehat{u}_2(m)^-}{(\int_{\Omega_-} m(z)(\widehat{u}_2(m)^-)^p dz)^{1/p}} \chi_{\Omega_\pm}. \end{aligned}$$

We know that $v_1, v_2 \in W_0^{1,p}(\Omega)$ and let $V = span\{v_1, v_2\} \subseteq W_0^{1,p}(\Omega)$. We introduce the set

$$E = \{v \in V : \| Dv \|_p^p = \int_\Omega \widehat{m}(z)|v|^p dz\}.$$

Then $\gamma(E) = 2$ ($\gamma(\cdot)$ being the Krasnoselskii genus). Note that

$$|D_m^+ \cap \Omega_+|_N > 0 \text{ and } |D_m^+ \cap \Omega_-|_N > 0.$$

Therefore

$$\begin{array}{l} \int_{D_m^+ \cap \Omega_+} m(z)|v_k|^p dz < \int_{D_m^+ \cap \Omega_+} \widehat{m}(z)|v_k|^p dz \; k = 1, 2, \\ \int_{D_m^+ \cap \Omega_-} m(z)|v_k|^p dz < \int_{D_m^+ \cap \Omega_-} \widehat{m}(z)|v_k|^p dz \; k = 1, 2. \end{array}$$

It follows that

$$\begin{array}{l} \sup\limits_{v \in E} \varphi(v) < \widehat{\lambda}_2(m), \\ \Rightarrow \widehat{\lambda}_2(\widehat{m}) < \widehat{\lambda}_2(m) (see\ (4.100)). \end{array}$$

□

Remark 4.142 In the case of the Laplacian (p=2, linear eigenvalue problem), we know that if $m, \widehat{m} \in L^s(\Omega)$ $s > \frac{N}{2}, m^+ \neq 0, m(z) \leq \widehat{m}(z)$ for a.a $z \in \Omega, m \neq \widehat{m}$, then for all $k \in \mathbb{N}, \widehat{\lambda}_k(\widehat{m}) < \widehat{\lambda}_k(m)$ (see Propositions 4.70 and 4.72). For the nonlinear case ($p \neq 2$), we were able to prove this monotonicity property only for $k = 1, 2$ and for $k = 2$ under stronger hypotheses on the weights. The reason for this difference is the UCP, which is true in the linear case, but we do not know if it holds for the p-Laplacian. Actually, some counterexamples due to Martio [332] suggest that the UCP is unlikely to hold for the eigenfunctions of the p-Laplacian.

Reasoning as in the proof of Theorem 4.111, we can have the following variational characterization of $\widehat{\lambda}_2$.

Theorem 4.143 *If hypotheses H_4 hold, then* $\widehat{\lambda}_2 = \inf\limits_{\widehat{\gamma} \in \widehat{\Gamma}} \max\limits_{-1 \leq t \leq 1} \| D\widehat{\gamma}(t) \|_p^p$ *where*

$$\widehat{\Gamma} = \{\widehat{\gamma} \in C([-1, 1], M) : \widehat{\gamma}(-1) = -\widehat{u}_1, \widehat{\gamma}(1) = \widehat{u}_1\},$$

(recall $M = W_0^{1,p}(\Omega) \cap \partial B_1^{L^p}$).

Remark 4.144 In addition to the positive eigenvalues, we also have a sequence of negative eigenvalues defined by

$$\widehat{\lambda}_{-k}(m) = -\widehat{\lambda}_k(-m) \text{ for all } k \in \mathbb{N}.$$

A similar spectral analysis can be conducted for the following nonlinear Robin eigenvalue problem

$$\left\{\begin{array}{l} -\triangle_p u(z) + \xi(z)|u(z)|^{p-2}u(z) = \widehat{\lambda} m(z)|u(z)|^{p-2}u(z) \text{ in } \Omega, \\ \frac{\partial u}{\partial n_p} + \beta(z)|u|^{p-2}u = 0 \text{ on } \partial\Omega. \end{array}\right\} \tag{4.104}$$

The assumptions on the data of (4.104) are the following:

$\boldsymbol{H_5}$: $\xi, m \in L^\infty(\Omega)$, $\beta \in C^{0,\alpha}(\partial\Omega)$ with $\alpha \in (0,1)$, $\xi(z) \geq 0$ for a.a $z \in \Omega$, $m^+ \neq 0$, $\beta(z) \geq 0$ for all $z \in \partial\Omega$ and $\xi \neq 0$ or $\beta \neq 0$.

As before let $\| \cdot \|$ denote the norm of the Sobolev space $W^{1,p}(\Omega)$ defined by

$$\| u \| = [\| u \|_p^p + \| Du \|_p^p]^{1/p} \text{ for all } u \in W^{1,p}(\Omega).$$

As in Proposition 4.52 (see also Remark 4.53), we have:

Lemma 4.145

(a) If $\xi \in L^\infty(\Omega)$, $\xi(z) \geq 0$ for a.a $z \in \Omega$ and $\xi \neq 0$, then there exists $\widehat{c} > 0$ such that

$$\widehat{c} \| u \|^p \leq \| Du \|_p^p + \int_\Omega \xi(z)|u|^p dz \text{ for all } u \in W^{1,p}(\Omega)$$

(b) If $\beta \in L^\infty(\partial\Omega)$, $\beta(z) \geq 0$ for a.a $z \in \partial\Omega$ and $\beta \neq 0$, then $u \to |u| = [\|Du\|_p^p + \int_{\partial\Omega} \beta(z)|u|^p d\sigma]^{1/p}$ is an equivalent norm on the Sobolev space $W^{1,p}(\Omega)$.

Consider the C^1-functional $\gamma_p : W^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\gamma_p(u) = \| Du \|_p^p + \int_\Omega \xi(z)|u|^p dz + \int_{\partial\Omega} \beta(z)|u|^p d\sigma \text{ for all } u \in W^{1,p}(\Omega).$$

On account of Lemma 4.145, we have

$$\beta_1 \| u \|^p \leq \gamma_p(u) \leq \beta_2 \| u \|^p \text{ for some } 0 < \beta_1, \beta_2, \text{ all } u \in W^{1,p}(\Omega). \tag{4.105}$$

Using (4.105), there is a similar spectral analysis for problem (4.104) and produce analogous results. More precisely we have:

- There is a smallest positive eigenvalue $\widehat{\lambda}_1 > 0$ defined by

$$\widehat{\lambda}_1 = \inf\{\gamma_p(u) : u \in W^{1,p}(\Omega), \int_\Omega m(z)|u|^p dz = 1\}. \tag{4.106}$$

- The spectrum of (4.104) is closed, $\widehat{\lambda}_1$ is simple and isolated and the infimum in (4.106) is realized on the corresponding one dimensional eigenspace, the elements of which have fixed sign.
- Let $\widehat{u}_1$ be the positive, L^p-normalized (that is, $\| \widehat{u}_1 \|_p = 1$) eigenfunction corresponding to $\widehat{\lambda}_1$. Then $\widehat{u}_1 \in intC_+$
- There is a whole sequence of positive variational eigenvalue $\{\widehat{\lambda}_k\}_{k\in\mathbb{N}}$ such that $\widehat{\lambda}_k \to \infty$; moreover,

$$\widehat{\lambda}_2 = \widehat{\lambda}_2^* = \inf\{\widehat{\lambda} : \widehat{\lambda} > \widehat{\lambda}_1, \widehat{\lambda} \text{ is an eigenvalue of (4.104)}\}.$$

- If $m_1, \widehat{m} \in L^\infty L^\infty(\Omega), m(z) \leq \widehat{m}(z)$ for a.a $z \in \Omega, m^+ \neq 0, m \neq \widehat{m}$, then $\widehat{\lambda}_1(\widehat{m}) < \widehat{\lambda}_1(m)$.

What can be said if $\xi = 0, \beta = 0$. In this case we have the following eigenvalue problem

$$\left\{ \begin{array}{l} -\triangle_p u(z) = \widehat{\lambda} m(z)|u(z)|^{p-2}u(z) \text{ in } \Omega, \\ \frac{\partial u}{\partial n_p} = 0 \text{ on } \partial\Omega. \end{array} \right\} \tag{4.107}$$

As before we define

$$\widehat{\lambda}^*(m) = \inf\{\| Du \|_p^p : u \in W^{1,p}(\Omega), \int_\Omega m(z)|u|^p dz = 1\}. \tag{4.108}$$

Proposition 4.146 *If $m \in L^\infty(\Omega)$ and $\int_\Omega m(z)dz > 0$, then $\lambda^*(m) = 0$.*

Proof Let $i(m) = \int_\Omega m(z)dz > 0$ and consider the function $\widehat{u}(z) = \frac{1}{i(m)^{1/p}}$ for all $z \in \Omega$ (constant function). From (4.108) we see that $\widehat{\lambda}^*(m) = 0$. □

Proposition 4.147 *If $m \in L^\infty(\Omega), m^+ \neq 0$ and $\int_\Omega m(z)dz < 0$, then $\widehat{\lambda}^*(m) > 0, \widehat{\lambda}^*(m)$ is the unique nonzero principal eigenvalue of (4.107) and if $\lambda \in (0, \widehat{\lambda}^*(m))$, then λ is not an eigenvalue of (4.107).*

Proof Let $C = \{u \in W^{1,p}(\Omega) : \int_\Omega m(z)|u|^p dz = 1\}$. We claim that there exists $\widehat{c} > 0$ such that

$$\widehat{c} \| u \|_p^p \leq \| Du \|_p^p \text{ for all } u \in C, \tag{4.109}$$

(so on C we have a Poincare-type inequality). If (4.109) is not true, we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq C$ such that

$$\| Du_n \|_p^p < \frac{1}{n} \| u_n \|_p^p \text{ for all } n \in \mathbb{N}. \tag{4.110}$$

First assume that $\{u_n\}_{n\in\mathbb{N}} \subseteq L^p(\Omega)$ is bounded. Then at least for a subsequence, we have

$$u_n \to \vartheta \in \mathbb{R} \text{ in } L^p(\Omega) \text{ (see (4.110))}$$

$$\Rightarrow 1 = \int_\Omega m(z)|u_n|^p dz \to |\vartheta|^p \int_\Omega m(z)dz \leq 0.$$

a contradiction.

Now suppose that $\{u_n\}_{n\in\mathbb{N}} \subseteq L^p(\Omega)$ is unbounded. We may assume that $\| u_n \|_p \to \infty$. We set $y_n = \frac{u_n}{\|u_n\|_p}$ $n \in \mathbb{N}$. Then $\|y_n\|_p = 1$ for all $n \in \mathbb{N}$ and

$$\|Dy_n\|_p^p < \frac{1}{n} \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow y_n \to \vartheta \in \mathbb{R} \setminus \{0\} \text{ in } L^p(\Omega).$$

Since $u_n \in C$ $n \in \mathbb{N}$, we have

$$\int_\Omega m(z)|y_n|^p dz = \frac{1}{\|u_n\|_p^p} \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow 0 = |\vartheta|^p \int_\Omega m(z)dz < 0,$$

again a contradiction. So, (4.109) is true.

Let $\{u_n\}_{n\in\mathbb{N}} \subseteq C$ be a minimizing sequence for (4.108), that is,

$$\|Dv_n\|_p^p \downarrow \widehat{\lambda}^*(m).$$

On account of (4.109), we see that $\{u_n\}_{n\in N} \subseteq W^{1,p}(\Omega)$ is bounded and so we may assume that

$$u_n \xrightarrow{w} u \text{ in } W^{1,p}(\Omega) \text{ and } u_n \to u \text{ in } L^p(\Omega).$$

Then $\|Du\|_p^p \leq \widehat{\lambda}^*(m)$, $\int_\Omega m(z)|u|^p dz = 1$ (that is, $u \in C$) and so $\|Du\|_p^p = \widehat{\lambda}^*(m) > 0$. We can always replace u by $|u|$ and so we may assume that $u \geq 0$, $u \neq 0$. Using the Lagrange multiplier rule (see Theorem 3.87), we can find $\eta \in \mathbb{R}$ such that

$$\int_\Omega |Du|^{p-2}(Du, Dh)_{\mathbb{R}^N} dz = \eta \int_\Omega m(z)|u|^{p-2}uhdz \text{ for all } h \in W^{1,p}(\Omega).$$

Let $h = u \in W^{1,p}(\Omega)$. We obtain

$$\|Du\|_p^p = \eta \int_\Omega m(z)|u|^p dz = \eta,$$
$$\Rightarrow \eta = \widehat{\lambda}^* > 0,$$
$$\Rightarrow \widehat{\lambda}^* \text{ in an eigenvalue of (4.107).}$$

Now let $\lambda \in (0, \widehat{\lambda}^*(m))$. Suppose λ ia an eigenvalue of (4.107). We can find $u \in W^{1,p}(\Omega)$, $u \neq 0$ such that

$$\int_\Omega |Du|^{p-2}(Du, Dh)_{\mathbb{R}^N} dz = \lambda \int_\Omega m(z)|u|^{p-2}uhdz \text{ for all } h \in W^{1,p}(\Omega)$$

Let $h = u \in W^{1,p}(\Omega)$. Then

$$\|Du\|_p^p = \lambda \int_\Omega m(z)|u|^p dz.$$

Consider $v = \dfrac{1}{\left(\int_\Omega m(z)|u|^p dz\right)^{1/p}}$, $u \in W^{1,p}(\Omega)$. Then

$$\|Dv\|_p^p = \lambda < \widehat{\lambda}^*(m) \text{ and } \int_\Omega m(z)|v|^p dz = 1,$$

a contradiction (see (4.108)).

Finally we need to show the uniqueness of this principal eigenvalue $\widehat{\lambda}^*(m) > 0$. So, suppose (4.107) has a nontrivial solution $u \in W^{1,p}(\Omega)$ for some $\lambda > 0$. Then $u(z) > 0$ for all $z \in \overline{\Omega}$ (see Theorem 4.89) and

$$\lambda \int_\Omega m(z)|y|^p dz \leq \|Dy\|_p^p \text{ for all } y \in W^{1,p}(\Omega) \cap L^\infty(\Omega),$$

(see Godoy et al. [210], Proposition 2.4),

$$\Rightarrow \lambda \leq \widehat{\lambda}^*(m) \text{ (see Theorem I-4.68).}$$

Also $-\lambda \leq \widehat{\lambda}^*(-m) = 0$ (see Proposition 4.146). Therefore

$$\lambda \in [0, \widehat{\lambda}^*(m)].$$

Since $\lambda \in (0, \widehat{\lambda}^*(m))$ can not be eigenvalue of (4.107), we infer that $\lambda = \widehat{\lambda}^*(m)$ and so we conclude that $\widehat{\lambda}^*(m) > 0$ is the unique principal eigenvalue of (4.107).

□

Proposition 4.148 *If $m \in L^\infty(\Omega)\backslash\{0\}$ and $\int_\Omega m(z)dz = 0$, then $\widehat{\lambda}^*(m) = 0$ and this is the unique principal eigenvalue of (4.107).*

Proof Let $h \in W^{1,p}(\Omega)$ be such that $\int_\Omega m(z)h^p dz > 0$. We introduce

$$u_\varepsilon = \frac{1 + \varepsilon h}{[\int_\Omega m(z)(1 + \varepsilon h)^p dz]^{1/p}}$$

with $\varepsilon > 0$ small so that the denominator does not vanish. Then $\| Du_\varepsilon \|_p \to 0$ as $\varepsilon \to 0^+$ and so $\widehat{\lambda}^*(m) = 0$. The infimum in the definition of $\widehat{\lambda}^*(m)$ (see (4.108)) is not attained or otherwise the minimizer will be a constant function which can not satisfy the constraint $\int_\Omega m(z)|u|^p dz = 1$, since by hypothesis $\int_\Omega m(z)dz = 0$. Finally as before (see the proof of Proposition 4.147), we show that $\widehat{\lambda}^*(m) = 0$ is the unique principal eigenvalue of (4.107). □

We can approximate the eigenvalues of (4.107) by those of a perturbed problem in which the operator is coercive. So, for $\varepsilon > 0$, we consider the following nonlinear problem

$$\left\{ \begin{array}{l} -\triangle_p u(z) + \varepsilon |u(z)|^{p-2}u(z) = \widehat{\lambda} m(z)|u(z)|^{p-2}u(z) \text{ in } \Omega, \\ \frac{\partial u}{\partial n_p} = 0 \text{ on } \partial\Omega. \end{array} \right\} \tag{4.111}$$

Let $\varphi_\varepsilon(u) = \| Du \|_p^p + \varepsilon \| u \|_p^p$, $\varphi(u) = \| Du \|_p^p$ and for $k \in \mathbb{N}$ we set

$$\widehat{\mathcal{F}}_k := \{A \subseteq W^{1,p}(\Omega) : A \text{ is compact, symmetric,}$$
$$\gamma(A) \geq k, \int_\Omega m(z)|u|^p dz = 1 \text{ for all } u \in A\}.$$

We define

$$\widehat{\lambda}_k^\varepsilon = \inf_{A \in \widehat{\mathcal{F}}_k} \sup_{u \in A} \varphi_\varepsilon(u),$$
$$\widehat{\lambda}_k = \inf_{A \in \widehat{\mathcal{F}}_k} \sup_{u \in A} \varphi(u), k \in \mathbb{N}.$$

Proposition 4.149 *If $m \in L^s(\Omega)$ with $s > \frac{N}{p}$ and $m^+ \neq 0$, then*

(a) for all $k \in \mathbb{N}$ we have $\widehat{\lambda}_k^\varepsilon \to \widehat{\lambda}_k$ as $\varepsilon \to 0^+$;
(b) $\widehat{\lambda}_k \to +\infty$ as $k \to +\infty$.

Proof

(a) Since $\varphi_\varepsilon \geq \varphi$, we see that $\widehat{\lambda}_k^\varepsilon \geq \widehat{\lambda}_k$ for all $k \in \mathbb{N}$. Let $\vartheta > 0$ such that $\widehat{\lambda}_k < \vartheta$. Then we can find $A \in \widehat{\mathcal{F}}_k$ such that

$$\widehat{\lambda}_k \leq \sup_{u \in A} \varphi(u) < \vartheta.$$

Set $\mu = \sup_{u \in A} \| u \|_p$ (recall $A \subseteq W^{1,p}(\Omega)$ is compact). Then

$$\widehat{\lambda}_k \leq \widehat{\lambda}_k^\varepsilon \leq \sup_{u \in A} \varphi_\varepsilon(u) \leq \sup_{u \in A} \varphi(u) + \frac{\varepsilon\mu}{p}.$$

We choose $\varepsilon < \frac{p}{\gamma}[\vartheta - \sup_{u\in A}\varphi(u)]$ and obtain

$$\widehat{\lambda}_k \le \widehat{\lambda}_k^{\varepsilon} < \vartheta.$$

Since $\vartheta > \widehat{\lambda}_k$ is arbitrary, we conclude that for all $k \in \mathbb{N}$

$$\widehat{\lambda}_k^{\varepsilon} \to \widehat{\lambda}_k \text{ as } \varepsilon \to 0^+.$$

(b) If $m, \widehat{m} \in L^s(\Omega)$, $m(z) \le \widehat{m}(z)$ for a.a $z \in \Omega$, $m^+ \ne 0$, then

$$\widehat{\lambda}_k^{\varepsilon}(\widehat{m}) \le \widehat{\lambda}_k^{\varepsilon}(m) \text{ for all } k \in \mathbb{N},$$
$$\widehat{\lambda}_k(\widehat{m}) \le \widehat{\lambda}_k(m) \text{ for all } k \in \mathbb{N} \text{ (see (a))}.$$

Let $\delta > 0$ and define

$$\widehat{m}(z) = \begin{cases} \delta, & \text{if } m(z) < \delta \\ m(z), & \text{if } \delta \le m(z) \end{cases}$$

Clearly $\widehat{m} \in L^s(\Omega)$, $m(z) \le \widehat{m}(z)$ for a.a $z \in \Omega$. Then

$$\widehat{\lambda}_k^{\varepsilon}(\widehat{m}) \le \widehat{\lambda}_k(m) + \frac{\varepsilon}{\delta}. \tag{4.112}$$

Since φ_ε is coercive, we have $\widehat{\lambda}_k^{\varepsilon}(m) \to +\infty$ as $k \to +\infty$. From (4.112) it follows that $\widehat{\lambda}_k(m) \to +\infty$. □

Remark 4.150 Similar results can be proved for the negative eigenvalues $\widehat{\lambda}_{-k}(m) = -\widehat{\lambda}_k(-m)$ for all $k \in \mathbb{N}$.

4.4 Double Phase and Anisotropic Operators

In this section we will prove results which will be used in the study of double phase problems and of problems with variable exponents (anisotropic problems). Both situations require that we abandon the convenient setting of standard Sobolev spaces and consider Orlicz-Sobolev spaces.

First let us deal with the double phase operator. So, let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a Lipschitz boundary $\partial\Omega$ and let $a \in L^\infty(\Omega)\backslash\{0\}$, $a(z) \ge 0$ for a.a $z \in \Omega$. For $r \in (1, \infty)$, let $\triangle_r^a$ be the nonautonomous (weighted) r-Laplace differential operator defined by

$$\triangle_r^a u = \operatorname{div}(a(z)|Du|^{r-2}Du).$$

For $1 < q < p < N$, we consider the so-called "double phase" differential operator defined by

$$\triangle_p^a u + \triangle_q u.$$

The energy functional of this operator has a density of the form

$$\eta(z,t) = a(z)t^p + t^q \text{ for all } z \in \Omega, \text{ all } t \geq 0. \tag{4.113}$$

Since we do not assume that $a(\cdot)$ is bounded away from zero $\big($that is, we do not require that $0 < essinf_\Omega a(\cdot)\big)$, $\eta(z,\cdot)$ exhibits unbalanced growth ((p,q)-growth), that is,

$$t^q \leq \eta(z,t) \leq c(1+t^p) \text{ for a.a } z \in \Omega, \text{ all } t \geq 0, \text{ some } c > 0.$$

Such operators are used to describe diffusion-type processes in a space where certain subdomains are distinguished from others. For example, consider a composite material, which on $\{a = 0\}$ has an energy density with q-growth, but on $\{a > 0\}$ the growth is of order p. Such a situation is modelled using the double phase operator and the mathematical analysis requires the use of spaces with modular functions (Orlicz spaces). In the next chapter we will use these spaces to study double phase boundary value problems. So, let us very briefly state some basic facts about them. For a comprehensive presentation of these spaces, we refer to Harjulehto and Hästo [230]. In what follows $\eta(z,t)$ denotes the double phase density given in (4.113). Also by $L^0(\Omega)$ we denote the space of all measurable functions $u : \Omega \to \mathbb{R}$. As usual we identify two such functions which differ only on a Lebesgue-null set.

Definition 4.151 The "generalized Lebesgue-Orlicz space" $L^\eta(\Omega)$ is defined by

$$L^\eta(\Omega) = \{u \in L^0(\Omega) : \varrho_\eta(u) = \int_\Omega \eta(z,|u|)dz < \infty\}.$$

This space is equipped with the so-called "Luxemburg norm" defined by

$$\| u \|_\eta = \inf\{\lambda > 0 : \varrho_\eta(\frac{u}{\lambda}) \leq 1\}.$$

Remark 4.152 The function $\varrho_\eta(\cdot)$ is called "modular function" and it is continuous, convex, thus weakly lower semicontinuous.

The space $L^\eta(\Omega)$ is a Banach space, which is separable and reflexive (in fact uniformly convex, since $\eta(z,\cdot)$ is uniformly convex).

Using the space $L^\eta(\Omega)$, we can define the corresponding generalized Sobolev-Orlicz space.

Definition 4.153 The generalized Sobolev-Orlicz space $W^{1,\eta}(\Omega)$ is defined by

$$W^{1,\eta}(\Omega) = \{u \in L^\eta(\Omega) : |Du| \in L^\eta(\Omega)\},$$

with Du being the weak gradient of u. We equip this space with the following norm

$$\| u \|_{1,\eta} = \| u \|_\eta + \| Du \|_\eta \text{ for all } u \in W^{1,\eta}(\Omega).$$

with $\| Du \|_\eta = \| |Du| \|_\eta$. Also, we define

$$W_0^{1,\eta}(\Omega) = \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{1,\eta}}.$$

Remark 4.154 The spaces $W^{1,\eta}(\Omega)$ and $W_0^{1,\eta}(\Omega)$ are Banach spaces which are separable and reflexive (in fact they are uniformly convex).

In the space $W_0^{1,\eta}(\Omega)$, the Poincare inequality holds. More precisely we have:

Proposition 4.155

(a) There exists a $c > 0$ depending only on Ω such that $\| u \|_\eta \leq c \| Du \|_\eta$ for all $u \in W_0^{1,\eta}(\Omega)$.
(b) There exists $c > 0$ depending only on Ω such that $\| u - u_\Omega \|_\eta \leq c \| Du \|_\eta$ for all $u \in W^{1,\eta}(\Omega)$ with $u_\Omega = \frac{1}{|\Omega|_N} \int_\Omega u dz$.

As a consequence of this proposition, we have the following result about the space $W_0^{1,\eta}(\Omega)$.

Corollary 4.156 *$\| u \| = \| Du \|_\eta$ for all $u \in W_0^{1,\eta}(\Omega)$ is an equivalent norm on $W_0^{1,\eta}(\Omega)$.*

In the sequel $\| \cdot \|$ is the norm that we will be using on $W_0^{1,\eta}(\Omega)$.

Recall that we have assumed that $1 < q < p < N$ and $\frac{p}{q} < 1 + \frac{1}{N}$. This last inequality implies that $p < q^* = \frac{Nq}{N-q}$. Then we have the following generalized version of the Sobolev embedding theorem.

Theorem 4.157 *If $X = W_0^{1,\eta}(\Omega)$ or $W^{1,\eta}(\Omega)$, then*

(a) If $X \hookrightarrow W_0^{1,\eta}(\Omega)$ or $W^{1,\eta}(\Omega)$ continuously for all $1 \leq r \leq q$.
(b) $X \hookrightarrow L^r(\Omega)$ compactly if $1 \leq r < q^*$ and continuously if $1 \leq r \leq q^*$.
(c) $W^{1,\eta}(\Omega) \hookrightarrow L^r(\partial\Omega)$ compactly if $1 \leq r < q_* = \frac{(N-1)q}{N-q}$ and continuously if $1 \leq r \leq q_*$.
(d) $L^p(\Omega) \hookrightarrow L^\eta(\Omega)$ continuously.

Remark 4.158 Let $L_a^p(\Omega) = \{u \in L^0(\Omega) : \int_\Omega a(z)|u|^p dz < \infty\}$. This is a locally convex space equipped with the seminorm

$$\| u \|_{p,a} = (\int_\Omega a(z)|u|^p dz)^{1/p}.$$

Then we have $L^\eta(\Omega) \hookrightarrow L_a^p(\Omega)$ continuously.

There is a close relation between the norm

$$\| u \| = \| Du \|_\eta \text{ for all } u \in W_0^{1,\eta}(\Omega)$$

and the modular function

$$\varrho_\eta(Du) = \int_\Omega [a(z)|Du|^p + |Du|^q]dz \text{ for all } W_0^{1,\eta}(\Omega).$$

Proposition 4.159

(a) $\| u \| = \lambda \Leftrightarrow \varrho_\eta(\frac{|Du|}{\lambda}) = 1.$
(b) $\| u \| < 1 (\text{resp.} = 1, > 1) \Leftrightarrow \varrho_\eta(Du) < 1 (\text{resp.} = 1, > 1).$
(c) $\| u \| < 1 \Rightarrow \| u \|^p \leq \varrho_\eta(Du) \leq \| u \|^q.$
(d) $\| u \| > 1 \Rightarrow \| u \|^q \leq \varrho_\eta(Du) \leq \| u \|^p .$
(e) $\| u \| \to 0 \Leftrightarrow \varrho_\eta(Du) \to 0.$
(f) $\| u \| \to \infty \Leftrightarrow \varrho_\eta(Du) \to \infty.$

Remark 4.160 The result is also true for the space $W^{1,\eta}(\Omega)$ with the norm

$$\| u \|_{1,\eta} = \| u \|_\eta + \| Du \|_\eta \text{ for all } u \in W^{1,\eta}(\Omega)$$

and the modular function

$$\widehat{\varrho}_\eta(u) = \int_\Omega [a(z)|Du|^p + |Du|^q]dz + \int_\Omega [a(z)|u|^p + |u|^q]dz \text{ for all } u \in W^{1,\eta}(\Omega).$$

Boundary value problems driven by the double phase differential operator $\triangle_p^a + \triangle_q$ (see (4.112)), will depend also on the spectral properties of the operator $\triangle_p^a$. So, we need to look at the generalized Orlicz spaces generated by the density function

$$\eta_0(z,t) = a(z)t^p \text{ for all } z \in \Omega, t \geq 0. \tag{4.114}$$

Now we assume that $a(z) > 0$ for all $z \in \Omega$. This implies that $\eta_0(z,\cdot)$ is uniformly convex function. Then $L^{\eta_0}(\Omega)$ is a Banach space which is separable and uniformly convex (thus reflexive, see Theorem I-3.241). The corresponding generalized Sobolev-Orlicz space $W_0^{1,\eta_0}(\Omega)$ has the same properties (see Harjulehto and Hästo [230], Theorem 6.1.4, p. 132).

Definition 4.161 By a weight we mean a function $\widehat{a} \in L^0(\Omega)$ such that $0 < \widehat{a}(z) < \infty$ for a.a $z \in \Omega$. If $1 < p < \infty$, then we say that the weight $\widehat{a}(\cdot)$ belongs in the "p-Muckhenhoupt class", denoted by $\widehat{a} \in \mathcal{A}_p$, if

$$[\widehat{a}]_{\mathcal{A}_p} = \sup_Q \Big(\frac{1}{|Q|_N} \int_Q a(z)dz\Big)\Big(\frac{1}{|Q|_N} \int_Q a(z)^{1-p'}dz\Big)^{p-1} < \infty.$$

where the supremum is taken over all cubes with sides parallel to the coordinate axes (recall that $p' = \frac{p}{p-1}$).

Remark 4.162 If $\widehat{a} \in \mathcal{A}_p$, then the averaging operator

$$\widehat{A}_Q u(z) = \frac{1}{|Q|_N} \int_Q u(z')dz' X_Q(z)$$

is uniformly bounded on $L^{\eta_0}(\Omega)$ (with $a \in \mathcal{A}_p$) for all Q, see Cruz Uribe and Fiorenza [114], Proposition 4.33, p. 152.

Proposition 4.163 *If $a \in C_0^{0,1}(\overline{\Omega}) \cap \mathcal{A}_p$, $a(z) > 0$ for all $z \in \Omega$, $1 < q < p < N$, $\frac{p}{q} < 1 + \frac{1}{N}$, then $W_0^{1,\eta}(\Omega) \hookrightarrow L^{\eta_0}(\Omega)$ compactly.*

Proof We will use convolutions (mollifications). By hypothesis $a|_{\partial\Omega} = 0$. We extend $a(\cdot)$ on all of $\mathbb{R}^N$, by $a|_{\mathbb{R}^N \setminus \overline{\Omega}} = 0$.

Suppose that $u_n \xrightarrow{w} u$ in $W_0^{1,\eta_0}(\Omega)$. Let $y_n = u_n - u \in W_0^{1,\eta_0}(\Omega)$, $n \in \mathbb{N}$. Then $y_n \xrightarrow{w} 0$ in $W_0^{1,\eta_0}(\Omega)$. So, we can find $c_1 > 0$ such that

$$\| y_n \|_{1,\eta_0} \leq c_1 \text{ for all } n \in \mathbb{N}.$$

Since $W_0^{1,\eta_0}(\Omega) = \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{1,\eta_0}}$, we can find $\beta_n \in C_c^\infty(\Omega)$ such that

$$\|y_n - \beta_n\|_{1,\eta_0} \leq \tfrac{1}{\eta} \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow \beta_n \to 0 \text{ in } W_0^{1,\eta_0}(\Omega) \text{ as } n \to \infty.$$

Consider the standard mollifier $\{\vartheta_\varepsilon\}_{\varepsilon \in (0,1]}$. We have

$$\|\beta_n\|_{\eta_0} = \|\beta_n - \vartheta_\varepsilon * \beta_n + \vartheta_\varepsilon * \beta_n\|_{\eta_0}$$

$$\leq \|\beta_n - \vartheta_\varepsilon * \beta_n\|_{\eta_0} + \|\vartheta_\varepsilon * \beta_n\|_{\eta_0}.$$

We have

$$\begin{aligned}
&(\vartheta_\varepsilon * \beta_n)(s) - \beta_n(s) \\
&= \int_{\mathbb{R}^N} \vartheta_\varepsilon(z)[\beta_n(s-z) - \beta_n(s)]dz \\
&= \int_{\mathbb{R}^N} \vartheta_\varepsilon(z) \int_0^1 \frac{d}{dt}\beta_n(s - tz)dtdz
\end{aligned}$$

$$= \int_{\mathbb{R}^N} \vartheta_\varepsilon(z) \int_0^1 (D\beta_n(s - tz), z)_{\mathbb{R}^N} dt dz \text{ (by the chain rule)}$$

$$= \int_0^1 \int_{\mathbb{R}^N} \vartheta_\varepsilon(z)(D\beta_n(s - tz), z)_{\mathbb{R}^N} dz dt \text{ (by the Fubini's theorem)}$$

$$= \int_0^1 \int_{\mathbb{R}^N} \vartheta_{\varepsilon t}(z)(D\beta_n(s - z), \frac{z}{t})_{\mathbb{R}^N} dz dt \text{ (by a change of variables).}$$

For the mollifier $\vartheta_{\varepsilon t}(\cdot)$ we have $\vartheta_{\varepsilon t}(z) \neq 0$. If $\frac{|z|}{t} < \varepsilon$. Hence

$$\begin{aligned} &|(\vartheta_\varepsilon * \beta_n)(s) - \beta_n(s)| \\ &\leq \varepsilon \int_0^1 \int_{\mathbb{R}^N} \vartheta_{\varepsilon t}(z) |D\beta_n(s - tz|_{\mathbb{R}^N} dz dt \\ &= \varepsilon \int_0^1 (\vartheta_{\varepsilon t} * |D\beta_n|)(s) dt. \end{aligned}$$

Therefore we have

$$\begin{aligned} \|\vartheta_\varepsilon * \beta_n - \beta_n\|_{\eta_0} &\leq c_2 \varepsilon \int_0^1 \| \vartheta \|_1 \|D\beta_n\|_{\eta_0} dt \text{ for some } c_2 > 0 \\ &= c_2 \varepsilon \|D\beta_n\|_{\eta_0} \text{ (since } \|\vartheta\|_1 = 1) \end{aligned}$$

(see Harjulehto and Hästo [230], Lemma 4.43, p. 91 and Remark 4.162).

Therefore we can say that

$$\| \beta_n \|_{\eta_0} \leq c_2 \varepsilon \| D\beta_n \|_{\eta_0} + \| \vartheta_\varepsilon * \beta_n \|_{\eta_0} \text{ for all } n \in \mathbb{N}.$$

We know that

$$(\vartheta_\varepsilon * \beta_n)(s) = \int_{\mathbb{R}^N} \vartheta_\varepsilon(s - z)\beta_n(z) dz \to 0 \text{ as } n \to \infty.$$

Set $\overline{\Omega}_{-\varepsilon} = \{z \in \mathbb{R}^N : d(z, \Omega) \leq \varepsilon\}$. Then

$$(\vartheta_\varepsilon * \beta_n)(s) = 0 \text{ for all } s \in \mathbb{R}^N \setminus \overline{\Omega}_{-\varepsilon}.$$

Hence we have

$$\begin{aligned} &|(\vartheta_\varepsilon * \beta_n)(s)| \\ &\leq \int_{\mathbb{R}^N} \vartheta_\varepsilon(s - z)|\beta_n(z)| dz \end{aligned}$$

$$\leq c_3 \parallel \vartheta_\varepsilon(s-\cdot) \parallel_{\eta_0^*} \parallel \beta_n \parallel_{\eta_0} \text{ for some } c_3 > 0$$

(using Hölder's inequality, see Harjulehto and Hästo [230], p. 54)

$$\leq c_4 \frac{1}{\varepsilon^N} \parallel \chi_{\Omega_{-\varepsilon}} \parallel_{\eta_0^*} \text{ for some } c_4 > 0$$

$$\leq \frac{c_5}{\varepsilon^N} \text{ for some } c_5 > 0, \text{ all } s \in \overline{\Omega}_{-\varepsilon}, \text{ all } n \in \mathbb{N}.$$

Therefore we have

$$|(\vartheta_\varepsilon * \beta_n)(s)| \leq \frac{c_5}{\varepsilon^N} \chi_{\Omega_{-\varepsilon}}(s) \text{ for all } s \in \mathbb{R}^N,$$

and $\chi_{\Omega_{-\varepsilon}} \in L^{\eta_0}(\Omega)$ (since Ω is bounded).

Since $(\vartheta_\varepsilon * \beta_n)(s) \to 0$ for a.a $s \in \mathbb{R}^N$, from the Lebesgue dominated convergence theorem, we have

$$\vartheta_\varepsilon * \beta_n \to 0 \text{ in } L^{\eta_0}(\Omega).$$

Therefore finally we can say that

$$\begin{aligned} &\limsup_{n\to\infty} \parallel \beta_n \parallel_{\eta_0} \leq c_6 \varepsilon \text{ for some } c_6 > 0, \\ &\Rightarrow \beta_n \to 0 \text{ in } L^{\eta_0}(\Omega), \\ &\Rightarrow y_n \to 0 \text{ in } L^{\eta_0}(\Omega). \end{aligned}$$

This proves that $W_0^{1,\eta_0}(\Omega) \hookrightarrow L^{\eta_0}(\Omega)$ compactly. □

Using this proposition, we can now develop the spectral properties of $(-\triangle_p^a, W_0^{1,\eta_0}(\Omega))$.

Let $m \in L^\infty(\Omega)$ with $m(z) > 0$ for a.a $z \in \Omega$. We consider the following nonlinear eigenvalue problem

$$\left\{ \begin{array}{l} -\triangle_p^a u(z) = \widehat{\lambda} m(z) a(z) |u(z)|^{p-2} u(z) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, 1 < p < N. \end{array} \right\} \tag{4.115}$$

Recall that $a \in C_0^{0,1}(\overline{\Omega}) \cap \mathcal{A}_p$, $a(z) > 0$ for all $z \in \Omega$ and $\eta_0(z,t)$ is the density function given by (4.114).

As before $\widehat{\lambda} \in \mathbb{R}$ is an eigenvalue of (4.115), if problem (4.115) has a nontrivial weak solution $\widehat{u} \in W_0^{1,\eta_0}(\Omega)$ (eigenfunction), that is,

$$-\int_\Omega a(z) |D\widehat{u}|^{p-2} (D\widehat{u}, Dh)_{\mathbb{R}^N} dz = \widehat{\lambda} \int_\Omega m(z) a(z) |\widehat{u}|^{p-2} \widehat{u} h dz$$

$$\text{for all } h \in W_0^{1,\eta_0}(\Omega).$$

Proposition 4.164 *Problem (4.115) has a smallest eigenvalue*

$$\widehat{\lambda}_1^a(m) > 0$$

and every corresponding eigenfunction $\widehat{u} \in W_0^{1,\eta_0}(\Omega)$ *satisfies*

$$\widehat{u} \in L^\infty, \widehat{u}(z) > 0 \text{ or } \widehat{u}(z) < 0 \text{ for a.a } z \in \Omega.$$

Proof In what follows let

$$\varrho_a(Du) = \int_\Omega a(z)|Du|^p dz \text{ for all } u \in W_0^{1,\eta_0}(\Omega).$$

We define

$$\lambda_1^a(m) = \inf\{\frac{\varrho_a(Du)}{\int_\Omega m(z)a(z)|u|^p dz} : u \in W_0^{1,\eta_0}(\Omega), u \neq 0\},$$

$$\inf\{\varrho_a(Du) : u \in W_0^{1,\eta_0}(\Omega), \int_\Omega m(z)a(z)|u|^p dz = 1\}(\text{by homogeneity})$$

Consider a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta_0}(\Omega)$ such that $\varrho_a(Du_n) \downarrow \widehat{\lambda}_1^a(m)$ and $\int_\Omega m(z)a(z)|u_n|^p dz = 1$ for all $n \in \mathbb{N}$.

Evidently $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta_0}(\Omega)$ is bounded. Therefore by Proposition 4.163 we may assume that

$$u_n \xrightarrow{w} \widehat{u} \text{ in } W_0^{1,\eta_0}(\Omega), u_n \to \widehat{u} \text{ in } L^{\eta_0}(\Omega). \tag{4.116}$$

The modular function $\varrho_a(\cdot)$ is continuous, convex, thus sequentially weakly lower semicontinuous. So, from (4.116) we have

$$\varrho_a(D\widehat{u}) \leq \liminf_{n\to\infty} \varrho_a(Du_n).$$

Moreover, again (4.116) implies that

$$\int_\Omega m(z)a(z)|u_n|^p dz \to \int_\Omega m(z)a(z)|\widehat{u}|^p dz.$$

Therefore finally we can say that

$$\varrho_a(D\widehat{u}) \leq \widehat{\lambda}_1^a(m), \int_\Omega m(z)a(z)|\widehat{u}|^p dz = 1,$$

$$\Rightarrow \varrho_a(D\widehat{u}) = \widehat{\lambda}_1^a(m), \int_\Omega m(z)a(z)|\widehat{u}|^p dz = 1,$$

$$\Rightarrow \widehat{\lambda}_1 = \widehat{\lambda}_1^a(m) > 0.$$

The Lagrange multiplier rule (see Theorem 3.87) implies that

$$-\Delta_p^a \widehat{u} = \widehat{\lambda}_1 m(z) a(z) |\widehat{u}|^{p-2} \widehat{u} \text{ in } \Omega, \widehat{u}|_{\partial\Omega} = 0. \tag{4.117}$$

Suppose that $u^+ \neq 0$. On (4.117) we act with $\widehat{u}^+ \in W_0^{1,\eta_0}(\Omega)$ and obtain

$$\varrho_a(D\widehat{u}^+) = \widehat{\lambda}_1 \int_\Omega m(z) a(z) (\widehat{u}^+)^p dz.$$

So, $\widehat{u}^+$ realizes the infimum in the definition of $\widehat{\lambda}_1$. Hence $\widehat{u}^+$ is an eigenfunction for $\widehat{\lambda}_1$. Using the Moser iteration process, we show that $\widehat{u}^+ \in W_0^{1,\eta}(\Omega) \cap L^\infty(\Omega)$. Then Proposition 2.4 of Papageorgiou et al. [409], implies that

$$\begin{aligned} &\widehat{u}^+(z) > 0 \text{ for } a.a.\ z \in \Omega, \\ &\Rightarrow \widehat{u} = \widehat{u}^+. \end{aligned}$$

If $\widehat{u}^+ \equiv 0$, then $\widehat{u} = -\widehat{u}-$ and so $\widehat{u}(z) < 0$ for $a.a.\ z \in \Omega$. □

Using this above proposition, we can have the monotonicity of the map $m \to \widehat{\lambda}_1^a(m)$.

Proposition 4.165 *If $m, \widehat{m} \in L^\infty(\Omega)$, $0 \leq m(z) \leq \widehat{m}(z)$ for $a.a.\ z \in \Omega$ and $m \neq 0$, $m \neq \widehat{m}$, then $\widehat{\lambda}_1^a(\widehat{m}) < \widehat{\lambda}_1^a(m)$.*

Proof Let $\widehat{u} \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega)$ be an eigenfunction for $\widehat{\lambda}_1^a(m)$ and $\widehat{v} \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega)$ an eigenfunction for $\widehat{\lambda}_1^a(\widehat{m})$. We may assume that

$$\widehat{u}(z) > 0 \text{ and } \widehat{v}(z) > 0 \text{ for } a.a.\ z \in \Omega \text{ (see Proposition 4.164)}.$$

We have

$$\begin{aligned} \widehat{\lambda}_1^a(\widehat{m}) &= \frac{\varrho_a(D\widehat{v})}{\int_\Omega \widehat{m}(z) a(z) |\widehat{v}|^p dz} \\ &\leq \frac{\varrho_a(D\widehat{u})}{\int_\Omega \widehat{m}(z) a(z) |\widehat{u}|^p dz} \\ &< \frac{\varrho_a(D\widehat{u})}{\int_\Omega m(z) a(z) |\widehat{u}|^p dz} = \widehat{\lambda}_1^a(m). \end{aligned}$$

□

Let $A_p^a : W_0^{1,\eta_0}(\Omega) \to W_0^{1,\eta_0}(\Omega)^*$ be the operator defined by

$$\langle A_p^a(u), h \rangle = \int_\Omega a(z) |Du|^{p-2} (Du, Dh)_{\mathbb{R}^N} dz \text{ for all } u, h \in W_0^{1,\eta_0}(\Omega).$$

Proposition 4.166 *If $\widehat{u} \in W_0^{1,\eta_0}(\Omega)$ is an eigenfunction corresponding to an eigenvalue $\widehat{\lambda} > \widehat{\lambda}_1^a(m)$, then $\widehat{u}$ is nodal (sign-changing).*

Proof We argue by contradiction. So, suppose that $\widehat{u}$ has fixed sign. We can always assume that $\widehat{u} \geq 0$ and as before we will have that $\widehat{u}(z) > 0$ for $a.a.\ z \in \Omega$. Let $\widehat{u}_1 \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega)$ be the positive eigenfunction corresponding to $\widehat{\lambda}_1^a(m) > 0$. Then $u \geq \widehat{u}_1$. For $\varepsilon > 0$ we set $\widehat{u}_\varepsilon = \widehat{u} + \varepsilon$ and $(\widehat{u}_1)_\varepsilon = \widehat{u}_1 + \varepsilon$.

We introduce the following functions

$$h_1 = \frac{(\widehat{u}_1)_\varepsilon^p - \widehat{u}_\varepsilon^p}{(\widehat{u}_1)_\varepsilon^{p-1}} \text{ and } h_2 = \frac{\widehat{u}_\varepsilon^p - (\widehat{u}_1)_\varepsilon^p}{\widehat{u}_\varepsilon^{p-1}}.$$

Note that $h_1, h_2 \in W_0^{1,\eta_0}(\Omega)$. We have

$$Dh_1 = [1 + (p-1)(\frac{\widehat{u}_\varepsilon}{(\widehat{u}_1)_\varepsilon})^p]D(\widehat{u}_1)_\varepsilon - p(\frac{\widehat{u}_\varepsilon}{(\widehat{u}_1)_\varepsilon})^{p-1}D\widehat{u}_\varepsilon, \tag{4.118}$$

$$Dh_2 = [1 + (p-1)(\frac{(\widehat{u}_1)_\varepsilon}{\widehat{u}_\varepsilon})^p]D\widehat{u}_\varepsilon - p(\frac{(\widehat{u}_1)_\varepsilon}{\widehat{u}_\varepsilon})^{p-1}D(\widehat{u}_1)_\varepsilon. \tag{4.119}$$

Using h_1, h_2 as test functions, we have

$$\langle A_p^a(\widehat{u}_1), h_1\rangle = \widehat{\lambda}_1 \int_\Omega m(z)a(z)\widehat{u}_1^{p-1}h_1 dz,$$

$$\langle A_p^a(\widehat{u}_2), h_2\rangle = \widehat{\lambda} \int_\Omega m(z)a(z)\widehat{u}_2^{p-1}h_2 dz.$$

Adding these two equations and using (4.118), (4.119), we obtain

$$\begin{aligned}
&\widehat{\lambda}_1 \int_\Omega m(z)a(z)\widehat{u}_1^{p-1}(\frac{(\widehat{u}_1)_\varepsilon^p - \widehat{u}_\varepsilon^p}{(\widehat{u}_1)_\varepsilon^{p-1}})dz + \widehat{\lambda} \int_\Omega m(z)a(z)\widehat{u}^{p-1}(\frac{\widehat{u}_\varepsilon^p - (\widehat{u}_1)_\varepsilon^p}{\widehat{u}_\varepsilon^{p-1}})dz \\
&= \int_\Omega ([1 + (p-1)(\frac{\widehat{u}_\varepsilon}{(\widehat{u}_1)_\varepsilon})^p]|D\widehat{u}_1|^p + [1 + (p-1)(\frac{(\widehat{u}_1)_\varepsilon}{\widehat{u}_\varepsilon})^p]|D\widehat{u}|^p)dz \\
&- p\int_\Omega ((\frac{\widehat{u}_\varepsilon}{(\widehat{u}_1)_\varepsilon})^{p-1}|D\widehat{u}_1|^{p-2} + (\frac{(\widehat{u}_1)_\varepsilon}{\widehat{u}_\varepsilon})^{p-1}|D\widehat{u}|^{p-2})(D\widehat{u}_1, D\widehat{u})_{\mathbb{R}^N}dz \\
&= \int_\Omega ((\widehat{u}_1)_\varepsilon^p - \widehat{u}_\varepsilon^p)(|Dln(\widehat{u}_1)_\varepsilon|^p - |Dln\widehat{u}_\varepsilon|^p)dz \\
&- p\int_\Omega \widehat{u}_\varepsilon^p|Dln(\widehat{u}_1)_\varepsilon|^{p-2}(Dln(\widehat{u}_1)_\varepsilon, Dln\widehat{u}_\varepsilon - Dln(\widehat{u}_1)_\varepsilon)_{\mathbb{R}^N}dz \\
&- p\int_\Omega (\widehat{u}_1)_\varepsilon^p|Dln\widehat{u}_\varepsilon|^{p-2}(Dln\widehat{u}_\varepsilon, Dln(\widehat{u}_1)_\varepsilon - Dln\widehat{u}_\varepsilon)_{\mathbb{R}^N}dz.
\end{aligned}$$

Let $\varepsilon \to 0^+$. We obtain

$$0 \le (\widehat{\lambda}_1 - \widehat{\lambda}) \int_\Omega m(z)a(z)(\widehat{u}^p - \widehat{u}_1^p)dz,$$
$$\Rightarrow 0 \le \widehat{\lambda}_1 - \widehat{\lambda} < 0 (\text{ recall } \widehat{u}_1 \le \widehat{u}),$$

a contradiction. This proves that $\widehat{u}$ is nodal. □

Let $\widehat{\Omega}_+ = \{z \in \Omega : \widehat{u}(z) > 0\}$ and $\widehat{\Omega}_- = \{z \in \Omega : \widehat{u}(z) < 0\}$.

Proposition 4.167 *There exists $\widehat{c} > 0$ such that $\widehat{c} \le |\widehat{\Omega}_\pm|_N$.*

Proof Clearly by multiplying things, we may assume that

$$\|\widehat{u}^+\| \ge 1 \text{ and } \int_\Omega m(z)a(z)(u^+)^p dz \ge 1. \tag{4.120}$$

Using as test function $h = \widehat{u}^+ \in W_0^{1,\eta_0}(\Omega)$, we have

$$\begin{aligned}
\varrho_a(D\widehat{u}^+) &= \widehat{\lambda} \int_\Omega m(z)a(z)(\widehat{u}^+)^p \chi_{\Omega_+} dz \\
&\le \widehat{\lambda} c_1 [\int_\Omega m(z)a(z)(\widehat{u}^+)^p dz]^{\frac{q}{p}} |\widehat{\Omega}_+|_N^{\frac{p-q}{p}} \\
\Rightarrow \varrho_a(D\widehat{u}^+) &\le \widehat{\lambda} c_1 \int_\Omega m(z)a(z)(\widehat{u}^+)^p dz |\widehat{\Omega}_+|_N^{\frac{p-q}{p}} \text{ (see (4.120) and recall } q < p) \\
&\le \frac{\widehat{\lambda}}{\widehat{\lambda}_1} c_1 \varrho_a(D\widehat{u}^+) |\widehat{\Omega}_+|_N^{\frac{p-q}{p}} \\
&\Rightarrow 1 \le \frac{\widehat{\lambda}}{\widehat{\lambda}_1} c_1 |\widehat{\Omega}_+|_N^{\frac{p-q}{p}} \\
&\Rightarrow 0 < \widehat{c} = \frac{\widehat{\lambda}_1}{\widehat{\lambda} c_1} \le |\widehat{\Omega}_+|_N.
\end{aligned}$$

Similarly for $\widehat{\Omega}_-$. □

This estimate leads to the conclusion that $\widehat{\lambda}_1 = \widehat{\lambda}_1^a(m) > 0$ is isolated in the spectrum $\sigma_p^a(m)$.

Proposition 4.168 *The first eigenvalue $\widehat{\lambda}_1 = \widehat{\lambda}_1^a(m) > 0$ is isolated in the spectrum $\sigma_p^a(m)$.*

Proof We argue by contradiction. So, suppose that we can find $\lambda_n \in \sigma_p^a(m) \setminus \{\widehat{\lambda}_1\}$ $n \in \mathbb{N}$ such that $\lambda_n \downarrow \widehat{\lambda}_1$. Let $\widehat{u}_n \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega)$ be corresponding eigenfunctions. We can normalize them in the sense that

$$\int_\Omega m(z)a(z)|u_n|^p dz = 1 \text{ for all } n \in \mathbb{N}.$$

Using $h = u_n \in W_0^{1,\eta_0}(\Omega)$ as a test function, we have

$$\varrho_a(Du_n) = \lambda_n \text{ for all } n \in \mathbb{N},$$
$$\Rightarrow \{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta_0}(\Omega) \text{ is bounded.}$$

On account of Proposition 4.163 we may assume that

$$u_n \xrightarrow{w} \widehat{u} \text{ in } W_0^{1,\eta_0}(\Omega),\ u_n \to \widehat{u} \text{ in } L^{\eta_0}(\Omega).$$

Then we have

$$\varrho_a(D\widehat{u}) \leq \liminf_{n\to\infty} \varrho_a(Du_n),\ \int_\Omega m(z)a(z)|\widehat{u}|^p dz = 1,$$
$$\Rightarrow \varrho_a(D\widehat{u}) \leq \widehat{\lambda}_1,\ \int_\Omega m(z)a(z)|\widehat{u}|^p dz = 1,$$
$$\Rightarrow \varrho_a(D\widehat{u}) = \widehat{\lambda}_1.$$

Therefore $\widehat{u}$ is an eigenfunction for $\widehat{\lambda}_1 > 0$. So, we have $\widehat{u} \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega)$, $\widehat{u} \geq 0$ and for every $K \subseteq \Omega$ compact

$$0 < c_K \leq \widehat{u}(z) \text{ for } a.a.\ z \in K. \tag{4.121}$$

(see Papageorgiou et al. [409], Proposition 2.4)

At least for a subsequence, we have

$$\widehat{u}_n(z) \to \widehat{u}(z) \text{ for } a.a.\ z \in \Omega.$$

By Egorov's theorem, given $\varepsilon \in (0, |\Omega|_N)$ we can find $K_\varepsilon \subseteq \Omega$ compact such that $|\Omega \setminus K_\varepsilon|_N \leq \varepsilon$ and $\widehat{u}_n \to \widehat{u}$ uniformly on K_ε.

We know that $c_{K_\varepsilon} \leq \widehat{u}(z)$ for $a.a.\ z \in K_\varepsilon$ and we can find $n_0 = n_0(\varepsilon) \in \mathbb{N}$ such that

$$|\widehat{u}_n(z) - \widehat{u}(z)| \leq \frac{c_{K_\varepsilon}}{2} \text{ for all } n \geq n_0,\ \text{all } z \in K_\varepsilon,$$
$$\Rightarrow \frac{c_{K_\varepsilon}}{2} \leq \widehat{u}_n(z) \text{ for all } n \geq n_0,\ \text{all } z \in K_\varepsilon.$$

Let $\Omega_-^n = \{z \in \Omega_- : u_n(z) < 0\}$ $n \in \mathbb{N}$. From Proposition 4.167 we know that

$$0 < \widehat{c} \leq |\Omega_-^n|_N \text{ for all } n \in \mathbb{N}.$$

If we choose $\varepsilon = \frac{\widehat{c}}{2}$, then for $n \geq n_0$ we have

$$\begin{aligned} |\Omega|_N &\geq |\Omega_+^n|_N + |\Omega_-^n|_N \\ &\geq |K_\varepsilon|_N + \widehat{c} \\ &\geq |\Omega|_N - \frac{\widehat{c}}{2} + \widehat{c} > |\Omega|_N, \end{aligned}$$

a contradiction. This proves that $\widehat{\lambda}_1 > 0$ is isolated in $\sigma_p^a(m)$. □

Proposition 4.169 *The first eigenvalue $\widehat{\lambda}_1 = \widehat{\lambda}_1^a(m) > 0$ is simple.*

Proof Let u, v be two eigenfunctions corresponding to $\widehat{\lambda}_1 > 0$. We know that $u, v \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega)$ and without any loss of generality we may assume that $u, v \geq 0$ (see Proposition 4.164). For $\varepsilon > 0$, we define

$$u_\varepsilon = u + \varepsilon \text{ and } v_\varepsilon = v + \varepsilon.$$

We have

$$\langle A_p^a(u), h\rangle = \int_\Omega m(z)a(z)|u|^{p-2}uhdz, \tag{4.122}$$

$$\langle A_p^a(v), h\rangle = \int_\Omega m(z)a(z)|v|^{p-2}vhdz, \tag{4.123}$$

for all $h \in W_0^{1,\eta_0}(\Omega)$.

In (4.122) we use the test function

$$h_1 = \frac{u_\varepsilon^p - v_\varepsilon^p}{u_\varepsilon^{p-1}} \in W_0^{1,\eta_0}(\Omega),$$

while in (4.123) we use the test function

$$h_2 = \frac{v_\varepsilon^p - u_\varepsilon^p}{v_\varepsilon^{p-1}} \in W_0^{1,\eta_0}(\Omega).$$

Then reasoning as in the proof of Proposition 4.167 we obtain:

If $2 \leq p$, then

$$\begin{aligned} 0 &\leq c_2 \int_\Omega a(z)[\frac{1}{v_\varepsilon^p} - \frac{1}{u_\varepsilon^p}]|v_\varepsilon Du - u_\varepsilon Dv|dz \\ &\leq -\widehat{\lambda}_1 \int_\Omega m(z)a(z)[(\frac{u}{u_\varepsilon})^{p-1} - (\frac{v}{v_\varepsilon})^{p-1}](u_\varepsilon^p - v_\varepsilon^p)dz, \end{aligned} \tag{4.124}$$

for some $c_2 > 0$.

If $1 < 2 < p$, then

$$\begin{aligned} 0 &\leq c_3 \int_\Omega a(z)[u_\varepsilon + v_\varepsilon]^p (u_\varepsilon v_\varepsilon)^p \frac{|u_\varepsilon Dv - v_\varepsilon Du|}{(v_\varepsilon |Du| - u_\varepsilon |Dv|)^{2-p}} dz \\ &\leq -\widehat{\lambda}_1 \int_\Omega m(z)a(z)[(\frac{u}{u_\varepsilon})^{p-1} - (\frac{v}{v_\varepsilon})^{p-1}](u_\varepsilon^p - v_\varepsilon^p)dz. \end{aligned} \tag{4.125}$$

In both (4.124) and (4.125) we let $\varepsilon \to 0^+$ and obtain

$$\begin{aligned} &uDv = vDu, \\ &\Rightarrow u = \vartheta v \text{ for some } \vartheta > 0. \end{aligned}$$

This proves that $\widehat{\lambda}_1 = \widehat{\lambda}_1^a(m) > 0$ is simple. □

Recall that $A_p^a : W_0^{1,\eta_0}(\Omega) \to W_0^{1,\eta_0}(\Omega)^*$ is the nonlinear operator defined by

$$\langle A_p^a(u), h\rangle = \int_\Omega a(z)|Du|^{p-2}(Du, Dh)_{\mathbb{R}^N} dz, \text{ for all } u, h \in W_0^{1,\eta_0}(\Omega).$$

The next proposition establishes some important properties of this operator.

Proposition 4.170 *The operator $A_p^a : W_0^{1,\eta_0}(\Omega) \to W_0^{1,\eta_0}(\Omega)$ is continuous, strictly monotone (thus maximal monotone too) and of type $(S)_+$ (see Definition I-7.71).*

Proof Continuity and strict monotonicity are straightforward. Then from Proposition I-7.46, we infer that $A_p^a(\cdot)$ is maximal monotone.

Next we show that $A_p^a(\cdot)$ is of type $(S)_+$. To this end, consider a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta_0}(\Omega)$ such that

$$u_n \xrightarrow{w} u \text{ in } W_0^{1,\eta_0}(\Omega), \limsup_{n\to\infty} \langle A_p^a(u_n), u_n - u\rangle \leq 0. \tag{4.126}$$

The modular function $\varrho_a(\cdot)$ is convex, differentiable and

$$\varrho_a'(Du) = A_p^a(u) \text{ for all } u \in W_0^{1,\eta_0}(\Omega).$$

So, we have

$$\begin{aligned} &\frac{1}{p}[\varrho_a(Du_n) - \varrho_a(Du)] \leq \langle A_p^a(u_n), u_n - u\rangle, \\ &\Rightarrow \limsup_{n\to\infty} \varrho_a(Du) \leq \varrho_a(Du) \text{ (see (4.126))}. \end{aligned} \tag{4.127}$$

On the other hand since $\varrho_a(\cdot)$ is continuous, convex, it is sequentially weakly lower semicontinuous. Therefore on account of (4.126) we have

$$\varrho_a(Du) \leq \liminf_{n\to\infty} \varrho_a(Du_n). \tag{4.128}$$

Then from (4.127) and (4.128) we see that

$$\varrho_a(Du_n) \to \varrho_a(Du). \tag{4.129}$$

Recall that $\eta_0(z,\cdot)$ is uniformly convex. So, from (4.129) and Lemma 3.6.5, p. 64, of Harjulehto and H*ä*sto [230], we have

$$\varrho_a(Du_n - Du) \to 0,$$

$$\Rightarrow u_n \to u \text{ in } W_0^{1,\eta_0}(\Omega).$$

This proves that $A_p^a(\cdot)$ is of type $(S)_+$. □

Now we assume that $m \equiv 1$ and consider the Banach manifold

$$M = \{u \in W_0^{1,\eta_0}(\Omega) : \|u\|_{\eta_0} = 1\}.$$

Let $\sigma_p^a = \sigma_p^a(1)$ (the spectrum of (4.115) with $m \equiv 1$). We know that $\sigma_p^a \subseteq [\widehat{\lambda}_1, \infty)$ and it is closed. Then the second element of σ_p^a (the second eigenvalue of (4.115)), is given by

$$\widehat{\lambda}_2 = \min\{\widehat{\lambda} \in \sigma_p^a : \widehat{\lambda} > \widehat{\lambda}_1\}.$$

We will produce a minimax characterization similar to the one obtained in Theorem 4.111. In what follows, by $\widehat{u}_1$ we denote the positive $L^{\eta_0}(\Omega)$-normalized (that is, $\|\widehat{u}_1\|_{\eta_0} = 1$) eigenfunction corresponding to $\widehat{\lambda}_1 > 0$. We know that

$$\widehat{u}_1 \in W_0^{1,\eta_0}(\Omega) \cap L^\infty(\Omega), \widehat{u}_1(z) > 0, \text{ for } a.a.\ z \in \Omega.$$

Theorem 4.171 $\widehat{\lambda}_2 = \inf\limits_{\widehat{\gamma}\in\widehat{\Gamma}} \max\limits_{-1\leq t\leq 1} \varrho_a(D\gamma(t))$ *where*

$$\widehat{\Gamma} = \{\widehat{\gamma} \in C([-1,1], M) : \widehat{\gamma}(-1) = -\widehat{u}_1, \widehat{\gamma}(1) = \widehat{u}_1\}.$$

Proof First we show that $\varrho_a|_M$ satisfies the Palais-Smale condition. So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq M$ be a sequence such that

$$|\varrho_a(Du_n)| \leq c_1 \text{ for some } c_1 > 0, \text{ all } n \in \mathbb{N}, \tag{4.130}$$

$$|\langle A_p^a(u_n), h\rangle| \leq \varepsilon_n \|h\| \tag{4.131}$$
$$\text{for all } h \in T_{u_m} M, \text{ with } \varepsilon_n \to 0^+.$$

From Ljusternik's Theorem, we know that

$$T_{u_n} M = \{h \in W_0^{1,\eta_0}(\Omega) : \int_\Omega a(z)|u_n|^{p-2}u_n h dz = 0\}, n \in \mathbb{N}.$$

Let $y \in W_0^{1,\eta_0}(\Omega)$ and define

$$h = y - (\int_\Omega a(z)|u_n|^{p-2}u_n y dz)u_n \in T_{u_n} M \text{ for all } n \in \mathbb{N}.$$

Using this as test function, we have

$$|\langle A_p^a(u_n), y\rangle - \int_\Omega a(z)|u_n|^{p-2}u_n y dz \varrho_a(Du_n)| \leq \varepsilon_n c_2 \|y\| \tag{4.132}$$
$$\text{for some } c_2 > 0, \text{ all } n \in \mathbb{N},$$
$$\Rightarrow \{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta_0}(\Omega) \text{ is bounded (see (4.130))}.$$

So, we may assume that

$$u_n \xrightarrow{w} u \text{ in } W_0^{1,\eta_0}(\Omega), u_n \to u \text{ in } L^{\eta_0}(\Omega). \tag{4.133}$$

Via Hölder's inequality, we have

$$\int_\Omega (u_n - u)a(z)|u_n|^{p-2}u_n dz \leq c_3 \varrho_a(u_n - u)$$
$$\text{for some } c_3 > 0, \text{ all } n \in \mathbb{N},$$
$$\Rightarrow \int_\Omega (u_n - u)a(z)|u_n|^{p-2}u_n dz \to 0 \text{ (see (4.133))}.$$

Hence, if in (4.132) we use $h = u_n - u$, then passing to the limit as $n \to \infty$, we obtain

$$\lim_{n\to\infty} \langle A_p^a(u_n), u_n - u\rangle = 0,$$
$$\Rightarrow u_n \to u \text{ in } W_0^{1,\eta_0}(\Omega) \text{ (see Proposition 4.170)},$$
$$\Rightarrow \varrho_a|_M \text{ satisfies the Palais-Smale condition}.$$

We know that $\pm\widehat{u}_1 \in M$ are minimizers of $\varrho_a|_M$ and $\varrho_a(\pm D\widehat{u}_1) = \widehat{\lambda}_1$. Using Proposition 3.132, we can find $\widehat{\varrho} \in (0, 1)$ small such that

$$\varrho_a(\pm D\widehat{u}_1) < \inf\{\varrho_a(Du) : \|u - (\pm\widehat{u}_1)\|_{1,\eta_0} = \widehat{\varrho}\} = m_{\widehat{\varrho}}.$$

Then we can apply the mountain pass theorem (see Theorem 3.41) and infer that

$$\widetilde{\lambda} = \inf_{\widehat{\gamma}\in\widehat{\Gamma}} \max_{-1\le t\le 1} \varrho_a(D_\gamma(t)) \ge m_{\widehat{\varrho}} > \varrho_a(\pm D\widehat{u}_1), \tag{4.134}$$

is a critical value of $\varrho_a|_M$. Therefore $\widetilde{\lambda}$ is an eigenvalue of $(-\Delta_p^a, W_0^{1,\eta_0}(\Omega))$. We claim that $\widetilde{\lambda} = \widehat{\lambda}_2$. To show this, it suffices to prove that $(\widehat{\lambda}_1, \widetilde{\lambda}) \cap \sigma_p^a = \emptyset$. To this end, we argue by contradiction. So, suppose there is $\lambda \in (\widehat{\lambda}_1, \widetilde{\lambda}) \cap \sigma_p^a$ and let $u \in M$ be a corresponding eigenfunction. From Proposition 4.166 we know that u is nodal and so $u^+ \neq 0$, $u^- \neq 0$. We introduce the following two paths in the manifold M

$$\gamma_1(t) = \frac{u^+ - tu^-}{\|u^+ - tu^-\|_{\eta_0}} \text{ for all } t \in [0, 1],$$

$$\gamma_2(t) = \frac{-u^- + (1-t)u^+}{\| - u^- + (1-t)u^+\|_{\eta_0}} \text{ for all } t \in [0, 1].$$

Evidently

$$\gamma_1(\cdot) \text{ connects } \frac{u^+}{\|u^+\|_{\eta_0}} \text{ with } u \in M,$$

$$\gamma_2(\cdot) \text{ connects } u \in M \text{ with } \frac{-u^-}{\|u^-\|_{\eta_0}}.$$

Therefore concatenating these two paths, we have a path in M connecting $\frac{u^+}{\|u^+\|_{\eta_0}}$ with $\frac{-u^-}{\|u^-\|_{\eta_0}}$. We have

$$\varrho_a(Du^+) = \lambda\varrho_a(u^+), \varrho_a(Du^-) = \lambda\varrho_a(u^-),$$
$$\Rightarrow \varrho_a(D\gamma_1(t)) = \varrho_a(D\gamma_2(t)) = \lambda \text{ for all } t \in [0, 1].$$

Consider the following sublevel set of the modular function

$$L = \{u \in W_0^{1,\eta_0}(\Omega) : \varrho_a(Du) < \lambda\}.$$

This set can not be path-connected or otherwise we violate (4.134). Using the Ekeland variational principle and the fact that $\varrho_a|_M$ satisfies the Palais-Smale condition, we deduce that every path-component of L has a critical value of $\varrho_a|_M$.

But we know that $\pm\widehat{u}_1$ are the only critical points of $\varrho_a|_M$. Therefore L has two path-components, one containing $\widehat{u}_1$ and the other containing $-\widehat{u}_1$. Note that $\frac{u^+}{\|u^+\|_{\eta_0}}$ can not be a critical point of $\varrho_a|_M$. So, we can find a continuous path $\tau : [-\varepsilon, \varepsilon] \to M$ such that

$$\tau(0) = \frac{u^+}{\|u^+\|_{\eta_0}} \text{ and } \frac{d}{dt}(\varrho_a|_M)(D\tau(t)) \neq 0 \text{ for all } t \in [-\varepsilon, \varepsilon].$$

Starting with this path, we can move from $\frac{u^+}{\|u^+\|_{\eta_0}}$ to $y \in M$, staying in L except at the starting point. So, $y \in L$ and let U_1 be the connected component of L containing y. We can assume $\widehat{u}_1 \in U_1$. Then we connect y and $\widehat{u}_1$ with a path staying in U_1. We concatenate this path with $\tau(\cdot)$ and generate a path $\gamma_+ : [0, 1] \to \overline{U}_1$ such that

$$\gamma_+(0) = \widehat{u}_1, \gamma_+(1) = \frac{u^+}{\|u^+\|_{\eta_0}}, \gamma_+(t) \in U_1 \subseteq L \text{ for all } t \in [0, 1).$$

Similarly, if U_2 is the other path component of L with $-\widehat{u}_1 \in U_2$, then we produce a path $\gamma_- : [0, 1] \to \overline{U}_2$ such that

$$\gamma_-(0) = \frac{-u^-}{\|u^-\|_{\eta_0}}, \gamma_-(1) = -\widehat{u}_1, \gamma_-(t) \in U_2 \text{ for all } t \in (0, 1].$$

Let $\widehat{\gamma} = \gamma_- \cup \gamma \cup \gamma_+$. Then $\widehat{\gamma} \in \widehat{\Gamma}$ and $\varrho_a(D\widehat{\gamma}(t)) \leq \lambda$ for all $t \in [-1, 1]$, $\Rightarrow \widetilde{\lambda} \leq \lambda$, a contradiction.

Therefore $\widetilde{\lambda} = \widehat{\lambda}_2$. □

Now we turn our attention to the anisotropic p-Laplace differential operator. For this operator, the situation is complicated, since the anisotropic p-Laplacian is not homogeneous.

In Section I-4.2, we introduced variable Lebesgue spaces. Recall, if $p \in L^0(\Omega)$, then

$$p_- = \operatorname*{essinf}_{\Omega} p, \; p_+ = \operatorname*{esssup}_{\Omega} p, \; E_1 = \{p \in L^0(\Omega) : 1 < p_- \leq p_+ < \infty\}.$$

Let $p \in L^\infty(\Omega) \cap E_1$. The variable Lebesgue space $L^{p(z)}(\Omega)$ is defined by

$$L^{p(z)}(\Omega) = \{u \in L^0(\Omega) : \varrho_p(u) = \int_\Omega |u|^{p(z)} dz < \infty\}$$

and $\varrho_p(\cdot)$ is known as the modular function. This space is equipped with the "Luxemburg norm", defined by

$$\|u\|_{p(z)} = \inf\{\lambda > 0 : \int_\Omega (\frac{|u(z)|}{\lambda})^{p(z)} dz \leq 1\}.$$

With this norm $L^{p(z)}(\Omega)$ becomes a Banach space which is separable and reflexive (in fact uniformly convex).

Using the variable Lebesgue spaces, we can define the corresponding variable Sobolev spaces.

Definition 4.172 Let $p \in L^\infty(\Omega) \cap E_1$. The variable Sobolev space $W^{1,p(z)}(\Omega)$ is defined by

$$W^{1,p(z)}(\Omega) = \{u \in L^{p(z)}(\Omega) : |Du| \in L^{p(z)}(\Omega)\}$$

with Du being the weak gradient of u. We equip this space with the norm

$$\|u\|_{1,p(z)} = \|u\|_{p(z)} + \|Du\|_{p(z)}$$

where $\|Du\|_{p(z)} = \||Du|\|_{p(z)}$. Also we define

$$W_0^{1,p(z)}(\Omega) = \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{1,p(z)}}.$$

For a comprehensive analysis of the variable Lebesgue and Sobolev spaces, we refer to the books of Cruz Uribe and Fiorenza [114] and of Diening-Harjulehto-Hästo and Ruzicka [142].

Theorem 4.173 *$W^{1,p(z)}(\Omega)$ and $W_0^{1,p(z)}(\Omega)$ are both Banach spaces which are separable and reflexive (in fact uniformly convex). Moreover, if $p \in C^{0,1}(\overline{\Omega})$, then there exists $\widehat{c} = \widehat{c}(\Omega) > 0$ such that*

$$\|u\|_{p(z)} \leq \widehat{c}\,\|Du\|_{p(z)} \text{ for all } u \in W_0^{1,p(z)}(\Omega)$$

(Poincare inequality). So, on $W_0^{1,p(z)}(\Omega)$ we can consider the equivalent norm

$$\|u\| = \|Du\|_{p(z)} \text{ for all } u \in W_0^{1,p(z)}(\Omega).$$

Remark 4.174 On $W^{1,p(z)}(\Omega)$ we can also consider the norm

$$|u|_{1,p(z)} = \inf\{\lambda > 0 : \int_\Omega ((\frac{|u|}{\lambda})^{p(z)} + (\frac{|Du|}{\lambda})^{p(z)})dz \leq 1\} \text{ for all } u \in W^{1,p(z)}(\Omega).$$

Then $\|\cdot\|_{1,p(z)}$ and $|\cdot|_{1,p(z)}$ are equivalent norms.

For $\Omega \subseteq \mathbb{R}^N$ a bounded domain with Lipschitz boundary $\partial\Omega$ and $p \in C^{0,1}(\overline{\Omega})$, we consider the following anisotropic eigenvalue problem

$$\{-\Delta_{p(z)}u(z) = \widehat{\lambda}|u(z)|^{p(z)-2}u(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0.\} \tag{4.135}$$

where $\Delta_{p(z)}$ denotes the anisotropic p-Laplacian defined by

$$\Delta_{p(z)}u = div(|Du|^{p(z)-2}Du) \text{ for all } u \in W_0^{1,p(z)}(\Omega).$$

In contrast to the usual p-Laplacian ($p(\cdot)$ constant), this operator is not homogeneous and this is a source of difficulties in the study of anisotropic problems.

Definition 4.175 We say that $\widehat{\lambda} \in \mathbb{R}$ is an "eigenvalue" of $(-\Delta_{p(z)}, W_0^{1,p(z)}(\Omega))$, if problem (4.135) has a nontrivial weak solution, that is, there exists $\widehat{u} \in W_0^{1,p(z)}(\Omega) \setminus \{0\}$ such that

$$\int_\Omega |D\widehat{u}|^{p(z)-2}(D\widehat{u}, Dh)_{\mathbb{R}^N} dz = \widehat{\lambda} \int_\Omega |\widehat{u}|^{p(z)-2}\widehat{u}h dz, \text{ for all } h \in W_0^{1,p(z)}(\Omega).$$

We say that $\widehat{u}$ is an "eigenfunction" corresponding to the eigenvalue $\widehat{\lambda}$ and we call $(\widehat{\lambda}, \widehat{u})$ an "eigenpair" for problem (4.135).

Let $\sigma_{p(z)} = \{\lambda \in \mathbb{R} : \lambda \text{ is an eigenvalue of (4.135)}\}$. Using the Ljusternik-Schnirelmann minimax scheme, we can show that

$$\sigma_{p(z)} \neq \emptyset, \text{ is infinite and } \sup \sigma_{p(z)} = +\infty.$$

In the constant exponent case, we know that $\inf \sigma_{p(z)} > 0$. Can we claim this also for the variable exponent case? Below we investigate this issue.

Let $\lambda_* = \inf \sigma_{p(z)}$. Also, we introduce the following quantities

$$\mu_* = \inf\{\frac{\int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz}{\int_\Omega \frac{1}{p(z)}|u|^{p(z)}dz} : u \in W_0^{1,p(z)}(\Omega), u \neq 0\}$$

$$\widetilde{\mu}_* = \inf\{\frac{\varrho_p(Du)}{\varrho_p(u)} : u \in W_0^{1,p(z)}(\Omega), u \neq 0\}.$$

Clearly we have

$$\frac{p_-}{p_+}\widetilde{\mu}_* \leq \mu_* \leq \frac{p_+}{p_-}\widetilde{\mu}_*.$$

In fact we can easily check that:

Lemma 4.176 $\lambda_* > 0 \Leftrightarrow \mu_* > 0 \Leftrightarrow \widetilde{\mu}_* > 0$.

So, in order to have $\lambda_* > 0$ we need to show that $\mu_* > 0$ or $\widetilde{\mu}_* > 0$. In this direction, Fan et al. [162], proved the following result.

Theorem 4.177

(a) If there exists $\Omega_0 \subseteq \Omega$ open and $u_0 \in \Omega_0$ such that $p(u_0) < p(u)$ or $p(u_0) > p(u)$ for all $u \in \partial\Omega_0$, then $\widetilde{\mu}_* = 0$ (hence $\lambda_* = 0$, $\mu_* = 0$).

(b) If $N > 1$ and there exists a vector $\vartheta \in \mathbb{R}^N \setminus \{0\}$ such that for any $z \in \Omega$ $t \to k(t) = p(z + t\vartheta)$ is monotone on $T_z = \{t : z + t\vartheta \in \Omega\}$, then $\lambda_* > 0$.

Remark 4.178 So under the monotonicity-type condition from (b), problem (4.135) has a smallest eigenvalue $\lambda_* = \widehat{\lambda}_1$.

Remarks

4.1: The spectral properties of the scalar p-Laplacian with Dirichlet, periodic and Neumann boundary conditions, are discussed in Anane et al. [17], del Pino et al. [134], Gasinski and Papageorgiou [182] and Otani [356]. We also mention the works of Binding and Rynne [48], [49] on periodic eigenvalue problems with potential terms. For the vector p-Laplacian, the main references are the works of Manasevich and Mawhin [325] and Mawhin [334].

4.2: In the spectral analysis of problem (4.30), we follow the work D'Agui et al. [120]. The spectrum of the Dirichlet Laplacian with no potential term (that is, $\xi \equiv 0$), is investigated in Brezis [60], Evans [153], Gasinski and Papageorgiou [182], Jost [261] and Motreanu et al. [344]. Many interesting facts about the Laplacian under different boundary conditions, can be found in the book of Henrot [239]. The analysis of the eigenvalue problem (4.53), is based on the work of de Figueiredo [129]. The monotonicity property of $m \to \widehat{\lambda}_k(m)$ proved in Proposition 4.72, is due to de Figueiredo and Gossez [130].

Consider a sequence of strongly elliptic differential operators

$$L_n(u) = -div(A_n(z)Du), n \in \mathbb{N},$$

and a limit operator

$$L(u) = -div(A(z)Du).$$

Here $A_n(z) = (a_{ij}^n(z))_{i,j=1}^N$ and $A(z) = (a_{ij}(z))_{i,j=1}^N$ are $N \times N-$matrices with entries $a_{ij}^n, a_{ij} \in L^\infty(\Omega)$ for all $n \in \mathbb{N}$, all $i, j \in \{1, \ldots, N\}$. Also we assume that the strong ellipticity constant $\widehat{c} > 0$, can be chosen independent of $n \in \mathbb{N}$, that is,

$$(A_n(z)\xi, \xi)_{\mathbb{R}^N} \geq c|\xi|^2 \text{ for } a.a\ z \in \Omega, \text{ all } \xi \in \mathbb{R}^N, \text{ all } n \in \mathbb{N}.$$

Let $\{\widehat{\lambda}_k^n\}_{k\in\mathbb{N}}$ be the positive eigenvalues of $L_n, n \in \mathbb{N}$ and $\{\widehat{\lambda}_k\}_{k\in\mathbb{N}}$ the positive eigenvalues of L (see problem (4.53)). The continuity result that we present next is essentially due to Henrot [239], where $m = 1$.

Proposition 4.179 *If $|A_n(z)| \leq M$ for a.a $z \in \Omega$, all $n \in \mathbb{N}$, some $M > 0$ and*

$$A_n(z) \to A(z) \text{ in } \mathbb{R}^{N \times N} \text{ for a.a } z \in \Omega,$$

(that is, $a_{ij}^n(z) \to a_{ij}(z)$ for a.a $z \in \Omega$ as $n \to \infty$ for all $i, j \in \{1, \ldots, N\}$) then for all $k \in \mathbb{N}$, we have $\widehat{\lambda}_k^n \to \widehat{\lambda}_k$ as $n \to \infty$.

4.3: The Moser iteration technique (see the proof of Proposition 4.82) can be found in the paper of Moser [340], who proposed an alternative approach to prove the regularity result of De Giorgi [132] (the answer to the 20th problem of Hilbert). Another way to prove the boundedness of the weak solutions, can be found in Ladyzhenskaya and Uraltseva [285], Theorem 7.1, p. 286. In this direction, we mention also the work of Winkert [472] on nonlinear Neumann problems. Theorem 4.84 (global regularity of nonlinear equations), is due to Lieberman [297], [298]. The crucial fact here is the balanced growth of the primitive $G(\cdot)$ (see Corollary 4.88). Earlier DiBenedetto [141] and Tolksdorf [454], proved local regularity results building upon the earlier fundamental work of Ladyzhenskaya and Uraltseva [285]. As a simple application of the global regularity theory, consider the following semilinear (that is, p=2) Dirichlet problem

$$\{-\Delta u(z) + \xi(z)u(z) = f(z, u(z)) \text{ in } \Omega, u|_{\partial\Omega} = 0.\} \tag{4.136}$$

We assume that $\Omega \subseteq \mathbb{R}^N$ is a bounded domain with a C^2-boundary $\partial\Omega, \xi \in L^p(\Omega)$ with $p > N$ and $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that

$$|f(z, x)| \leq a_0(z)(1 + |x|^{r-1}) \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R}$$

with $a_0 \in L^\infty(\Omega)$ and $1 < r \leq 2^*$.

Proposition 4.180 *If $u \in H_0^1(\Omega)$ is a weak solution of (4.136), then $u \in C_0^{1,\alpha}(\overline{\Omega})$ with $\alpha \in (0, 1)$ and $\|u\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq M_0$ for some $M_0 > 0$ depending on $\|u\|_\infty$.*

Similarly for the Robin problem

$$\{-\Delta u(z) + \xi(z)u(z) = f(z, u(z)) \text{ in } \Omega, \frac{\partial u}{\partial n} + \beta(z)u = 0 \text{ on } \partial\Omega.\} \tag{4.137}$$

The potential function $\xi(\cdot)$ and the reaction $f(z, x)$ are as above. Also we assume that $\beta \in W^{1,\infty}(\partial\Omega)$.

Proposition 4.181 *If $u \in H^1(\Omega)$ is a weak solution of (4.137), then $u \in C^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0, 1)$ and $\|u\|_{C^{1,\alpha}(\overline{\Omega})} \leq \widehat{M}_0$ for some $\widehat{M}_0 > 0$ depending on $\|u\|_\infty$. For both problems, if we have boundedness of the solution set in $H_0^1(\Omega)$*

(problem (4.136)) or in $H^1(\Omega)$ *(problem (4.137)), then we have also boundedness in* $C^{1,\alpha}(\overline{\Omega})$.

Theorem 4.89 can be found in Pucci and Serrin [423].

The nonlinear Picone identity (Theorem 4.91), was first proved by Allegretto and Huang [7] (case of the p-Laplacian) and by Jaros [259] (more general nonautonomous operators).

The Harnack inequality (Theorem 4.96), was first stated for harmonic functions by Harnack (1887). For weak solutions of linear equations of the form

$$\operatorname{div} \sum_{i=1}^{N} a_{ij}(z) D_i(u) = 0,$$

the Harnack inequality was proved by Moser [341].

The nonlinear generalization of the Harnack inequality is due to Trudinger [459]. More on this inequality can be found in the books of Han and Lin [228], Maly and Ziemer [323] and Pucci and Serrin [423].

The spectral analysis of problem (4.76), is based on the works of Mugnai and Papageorgiou [345] (Neumann equations, that is, $\beta \equiv 0$) and of Fragnelli et al. [169] (Robin equations). We mention that the first systematic study of the spectrum of the p-Laplacian with no potential term can be found in the thesis of Anane [15] (see also Anane [16]). The minimax characterization of $\widehat{\lambda}_2$ (see Theorem 4.111), was first obtained by Cuesta et al. [118] for the Dirichlet p-Laplacian with no potential term. For the Neumann p-Laplacian with no potential term, we refer to Aizicovici et al. [4], where the authors also prove the index formula for the Neumann p-Laplacian (see also Aizicovici et al. [3]). Finally we mention also the paper of Le [290].

Problem (4.98) was first studied by Cuesta [115]. The eigenvalue problem with Neumann boundary condition, was investigated by Godoy et al. [209], [210].

We state the regularity result for singular problem.

Theorem 4.182 *If hypotheses* H_0' *hold,* $f \in L^{\infty}_{loc}(\Omega)$, $f(z) \leq c\widehat{d}(z)^{-\eta}$ *for a.a* $z \in \Omega$, *some* $c > 0$, $\eta \in (0,1)$ *and* $u \in W^{1,p}_0(\Omega)$ *is a weak solution of* $-\operatorname{div} a(Du) = f, u|_{\partial\Omega} = 0$, *then* $u \in C^{1,\alpha}_0(\overline{\Omega})(0 < \alpha < 1)$ *and* $\|u\|_{C^{1,\alpha}_0(\overline{\Omega})} \leq \vartheta$ *with* $\vartheta \leq \vartheta(c, \eta, \Omega) > 0$.

Also if $-\operatorname{div} a(Du) + \widehat{\varepsilon}(z)u^{p-1} - \beta u^{-\eta} \geq 0$ *in* Ω *with* $\widehat{\varepsilon} \in L^{\infty}(\Omega)_+$, $\beta > 0$, *then* $u \in \operatorname{int} C_+$.

4.4: For the generalized Orlicz space, the main reference is the book of Harjulehto and Hästo [230] (see also Musielak [346], Proposition 4.163 is due to Papageorgiou et al. [410]. The analysis of the eigenvalue problem (4.115), follows Papageorgiou et al. [390].).

For the spaces with variable exponents, we refer to the books of Cruz Uribe and Fiorenza [114] and of Diening et al. [142]. The eigenvalue problem (4.135) was studied by Fan et al. [162].

Chapter 5
Elliptic Boundary Value Problems



Introduction

In this chapter we use the results obtained thus far(including those of Volume *I*), to study certain stationary(elliptic) problems. In Sect. 5.1, we prove maximum and antimaximum principles, we obtain certain strong comparison results and we determine the properties of the maps which correspond to the differential operators of the boundary value problems. In Sect. 5.2, we study certain nonlinear, nonhomogeneous boundary values problems. First we deal with $(p, 2)-$equations and prove multiplicity theorems for both the degenerate $(p \geq 2)$ and singular $(1 < p < 2)$ cases. The second is treated using the Lyapunov-Schmidt reduction method. Then we consider a Dirichlet problem driven by a general nonlinear nonhomogeneous differential and produce very general conditions guaranteeing existence and uniqueness of positive solutions. We also show, that in the particular case of the p-Laplacian, these conditions are also necessary. Finally we examine an eigenvalue problem for the $(p, q)-$Laplacian and prove a bifurcation-type result. In Sect. 5.3, we examine singular Dirichlet problems. First we consider weakly singular problems$(0 < \eta < 1)$ and prove an existence and multiplicity result which is global in the parameter $\lambda > 0$. Then we deal with a strongly singular problem $(\eta \geq 1)$ for which we prove a multiplicity theorem under a compatibility condition between the exponents. In Sect. 5.4, we consider problems with convection(gradient dependent reaction). Now the approach is topological and uses the fixed point theory. Finally in Sect. 5.5, we examine anisotropic and double phase(unbalanced growth problems). For the anisotropic problem, we prove a multiplicity theorem with sign information for the solutions when the reaction exhibits asymmetric behavior as $x \to \pm\infty$. For the double phase case, we consider an eigenvalue problem and prove the existence of an unbounded and continuous spectrum.

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5.1 Maximum, Antimaximum and Comparison Principles

We start with some regularity results for anisotropic equations since in this chapter we will also treat boundary value problems with variable exponents(anisotropic problems).

So, let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$. We consider the following Dirichlet problem

$$\left\{\begin{array}{l} -\Delta_{p(z)}u(z) - \Delta_{q(z)}u(z) = f(z, u(z)) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0. \end{array}\right\} \tag{5.1}$$

We make the following hypotheses on the data of this problem

H_0: $p, q \in C(\overline{\Omega})$ with $1 < q_- \leq q_+ < p_- \leq p_+$(recall that if $r \in C(\overline{\Omega})$, then $r_+ = \max\limits_{\overline{\Omega}} r$ and $r_- = \min\limits_{\overline{\Omega}} r$.

H_1: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that

$$|f(z, x)| \leq \widehat{a}(z)[1 + |x|^{r(z)-1}] \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R},$$

with $r \in C(\overline{\Omega})$, $1 < r(z) < p^*(z)$ for all $z \in \overline{\Omega}$ and for some $\widehat{a} \in L^\infty(\Omega)$.

Remark 5.1 Recall that $p^*(\cdot)$ is the variable critical Sobolev exponent defined by

$$p^*(z) = \left\{\begin{array}{ll} \dfrac{Np(z)}{N - p(z)} & \text{if } p(z) < N, \\ +\infty & \text{if } N \leq p(z), \end{array}\right. \quad z \in \overline{\Omega}.$$

In what follows, given $r \in C(\overline{\Omega}) \cap E_1$ (see Sect. 4.4), by $A_{r(z)} : W_0^{1,r(z)}(\Omega) \to W^{-1,r'(z)}(\Omega) = W_0^{1,r(z)}(\Omega)^*(r'(z) = \dfrac{r(z)}{r(z) - 1}$ for all $z \in \overline{\Omega}$), we denote the operator defined by

$$\langle A_{r(z)}(u), h\rangle = \int_\Omega |Du|^{r(z)-2}(Du, Dh)_{\mathbb{R}^N} dz \text{ for all } u, h \in W_0^{1,r(z)}(\Omega).$$

Definition 5.2 We say that $u \in W_0^{1,p(z)}(\Omega)$ is a "weak solution" of problem (1), if

$$\langle A_{p(z)}(u), h\rangle = \int_\Omega f(z, u)h dz \text{ for all } h \in W_0^{1,p(z)}(\Omega).$$

Theorem 5.3 *If hypotheses H_0, H_1 hold and $u \in W_0^{1,p(z)}(\Omega)$ is a weak solution of problem (1), then $u \in L^\infty(\Omega)$.*

Proof Since $u = u^+ - u^-$, we may assume that $u \geq 0$.

Let $\eta : \mathbb{R} \to [0, 1]$ be a C^1-cut off function satisfying

$$\text{supp}\,\eta \subseteq \mathbb{R}_+,\ \eta|_{[1,\infty)} \equiv 1 \text{ and } \eta'(t) \geq 0 \text{ for all } t \in [0,1]. \tag{5.2}$$

For $\varepsilon > 0$, we define $\eta_\varepsilon(t) = \eta(\frac{t-1}{\varepsilon})$. These are Lipschitz and so we can say that

$$\eta_\varepsilon(u) \in W_0^{1,p(z)}(\Omega) \text{ and } D\eta_\varepsilon(u) = \eta'_\varepsilon(u)Du. \tag{5.3}$$

Let $v \in W_0^{1,p(z)}(\Omega)$, $v \geq 0$ and use the test function $h = \eta_\varepsilon(u)v$. We have

$$\begin{aligned}
\langle A_{p(z)}(u), h\rangle &= \int_\Omega |Du|^{p(z)-2}(Du, D(\eta_\varepsilon(u)v))_{\mathbb{R}^N}dz \\
&= \int_\Omega |Du|^{p(z)-2}\eta'_\varepsilon(u)vdz + \int_\Omega \eta_\varepsilon(u)|Du|^{p(z)-2}(Du, Dv)_{\mathbb{R}^N}dz \\
&\quad \text{(see (5.3))} \\
&\geq \int_\Omega \eta_\varepsilon(u)|Du|^{p(z)-2}(Du, Dv)_{\mathbb{R}^N}dz \text{ (see (5.2))}.
\end{aligned} \tag{5.4}$$

Similarly, we obtain

$$\langle A_{q(z)}(u), h\rangle \geq \int_\Omega \eta_\varepsilon(u)|Du|^{q(z)-2}(Du, Dv)_{\mathbb{R}^N}dz. \tag{5.5}$$

Then using (5.4) and (5.5), we can say that

$$\begin{aligned}
&\int_\Omega \eta_\varepsilon(u)|Du|^{p(z)-2}(Du, Dv)_{\mathbb{R}^N}dz + \int_\Omega \eta_\varepsilon(u)|Du|^{q(z)-2}(Du, Dv)_{\mathbb{R}^N}dz \\
&\leq \langle A_{p(z)}(u), h\rangle + \langle A_{q(z)}(u), h\rangle \\
&= \int_\Omega f(z,u)hdz \\
&\leq \|\widehat{a}\|_\infty \int_\Omega [1 + u^{r(z)-1}]\eta_\varepsilon(u)vdz.
\end{aligned} \tag{5.6}$$

Let $\Omega_1 = \{z \in \Omega : u(z) > 1\}$. In (5.6), let $\varepsilon \to 0^+$. Then

$$\begin{aligned}
&\int_{\Omega_1} |Du|^{p(z)-2}(Du, Dv)_{\mathbb{R}^N}dz + \int_{\Omega_1} |Du|^{q(z)-2}(Du, Dv)_{\mathbb{R}^N}dz \\
&\leq \|\widehat{a}\|_\infty \int_\Omega [1 + u^{r(z)-1}]vdz.
\end{aligned}$$

For $\lambda \geq 1$, we define $u_\lambda = \min\{u, \lambda\} \in W_0^{1,p(z)}(\Omega)$ and then for $\mu > 0$, we use $v = uu_\lambda^{\mu p_+}$. Then we have

$$
\begin{aligned}
&\int_{\Omega_1} |Du|^{p(z)-2}(Du, D(uu_\lambda^{\mu p_+}))_{\mathbb{R}^N} dz \\
&= \int_{\Omega_1} |Du|^{p(z)} u_\lambda^{\mu p_+} dz + \mu p_+ \int_{\Omega_1 \cap \{u \leq \lambda\}} |Du|^{p(z)} u_\lambda^{\mu p_+} dz \\
&\geq \frac{1}{2} \int_{\Omega_1 \cap \{u > \lambda\}} |Du|^{p(z)} u_\lambda^{\mu p_+} dz + \mu p_+ \int_{\Omega_1 \cap \{u \leq \lambda\}} |Du|^{p(z)} u_\lambda^{\mu p_+} dz \\
&\geq \frac{\mu p_+ + 1}{2(\mu+1)^{p_+}} \int_{\Omega_1 \cap \{u > \lambda\}} |Du|^{p(z)} u_\lambda^{\mu p_+} dz + \frac{\mu p_+ + 1}{2} \int_{\Omega_1 \cap \{u \leq \lambda\}} |Du|^{p(z)} u_\lambda^{\mu p_+} dz \\
&\quad \text{(using Bernoulli's inequality)} \\
&\geq \frac{\mu p_+ + 1}{2(\mu+1)^{p_+}} \int_{\Omega_1} |Du|^{p(z)} u_\lambda^{\mu p_+} dz \\
&\geq \frac{\mu p_+ + 1}{2(\mu+1)^{p_+}} \int_{\Omega_1} |D(uu_\lambda^{\mu})|^{p(z)} dz \\
&\quad \text{(since } Du_\lambda(z) = 0 \text{ for a.a } z \in \{u > \lambda\} \text{ and } \lambda \geq 1).
\end{aligned}
$$

In a similar way, we show that

$$
\int_\Omega |Du|^{q(z)-2}(Du, D(uu_\lambda^{\mu p_+}))_{\mathbb{R}^N} dz \geq 0.
$$

Using the test function $h = uu_\lambda^{\mu p_+}$, we obtain

$$
\begin{aligned}
&\frac{\mu p_+ + 1}{2(\mu+1)^{p_+}} \int_{\Omega_1} |D(uu_\lambda^{\mu})|^{p(z)} dz \\
&\leq c_1 \Big[\int_{\Omega_1} uu_\lambda^{\mu p_+} dz + \int_{\Omega_1} u^{r(z)} u_\lambda^{\mu p_+} dz \Big] \text{ for some } c_1 > 0 \\
&\quad \text{(recall } \Omega_1 = \{z \in \Omega : u(z) > 1\}) \\
&\leq c_2 \int_{\Omega_1} u^{r_+} u_\lambda^{\mu p_+} dz \text{ for some } c_2 > 0.
\end{aligned} \tag{5.7}
$$

Without any loss of generality, we may assume that $\|uu_\lambda^{\mu}\| \geq 1$. Then

$$\frac{\mu p_+ + 1}{(\mu+1)^{p_+}} \int_{\Omega_1} |D(uu_\lambda^\mu)|^{p(z)} dz = \frac{\mu p_+ + 1}{(\mu+1)^{p_+}} \varrho_p^{\Omega_1}(D(uu_\lambda^\mu))$$

$$\geq \frac{\mu p_+ + 1}{(\mu+1)^{p_+}} \|D(uu_\lambda^\mu)\|_{p(z),\Omega_1}^{p_-} \text{ (see Corollary } I - 4.35)$$

$$\geq c_3 \frac{\mu p_+ + 1}{(\mu+1)^{p_+}} \|uu_\lambda^\mu\|_{L^{p_+}(\Omega_1)}^{p_-} \text{ for some } c_3 > 0$$

(since $W_0^{1,p(z)}(\Omega) \hookrightarrow L^{p_+}(\Omega)$ continuously).

Using this estimate in (5.7), we have

$$\|uu_\lambda^\mu\|_{L^{p_+}(\Omega_1)}^{p_-} \leq c_4 \int_{\Omega_1} u^{p_+^*} u_\lambda^{\mu p_+} dz \text{ for some } c_4 > 0 \text{ (since } r_+ < p_+^*).$$

We choose $\mu_1 > 0$ so that $(\mu_1 + 1)p_+ = p_+^*$. Then

$$\|uu_\lambda^{\mu_1}\|_{L^{(\mu_1+1)p_+}(\Omega_1)} \leq c_5[1 + \|u\|^{\frac{p_+^*}{p_+^*}}] < \infty \text{ for some } c_5 > 0.$$

We perform a standard bootstrap procedure choosing at each step $\mu_k > 0, k \in \mathbb{N}$, $k \geq 2$ such that

$$(\mu_k + 1)p_+ = (\mu_{k-1} + 1)p_+^*, \ k \geq 2.$$

This process shows that for every $\mu > 0$

$$\|u\|_{L^{(\mu+1)p_+}(\Omega_1)} \leq c_6 \text{ for some } c_6 = c_6(\mu, u) > 0.$$

From this bound, (5.7), Fatou's lemma as $\lambda \to +\infty$ and a new bootstrap argument with $\{\mu_k\}_{k\in\mathbb{N}}$ increasing, we have

$$\|u\|_{L^{(\mu_k+1)p_+^*}(\Omega_1)} \leq c_7 \text{ for some } c_7 > 0, \text{ all } k \in \mathbb{N}.$$

Let $\mu_k \to +\infty$ to conclude that

$$u \in L^\infty(\Omega_1), \|u\|_{L^\infty(\Omega_1)} \leq c_7,$$
$$\Rightarrow u \in L^\infty(\Omega).$$

□

Remark 5.4 The above proof also works for singular problems, that is, if the reaction $f(z, x)$ satisfies

$$|f(z, x)| \leq \widehat{a}(z)[x^{-\eta(z)} + x^{r(z)-1} + 1] \text{ for a.a } x \in \Omega, \text{ all } x \geq 0,$$

with $\eta, r \in C(\overline{\Omega})$, $0 < \eta_- \leq \eta_+ < 1 < r_- \leq r(z) < p^*(z)$ for all $z \in \overline{\Omega}$. We can prove the boundedness of the weak solution of (1) using a different approach. We outline this alternative proof. Since $\Omega \subseteq \mathbb{R}^N$ is relatively compact(being bounded), it is totally bounded. So, given $\varrho > 0$, we can find of finite number of $\varrho-$balls $\{B_\varrho^k\}_{k=1}^m$ with centers in $\overline{\Omega}$, which cover $\overline{\Omega}$. We define

$$p_-^k = \min_{\overline{\Omega}\cap\overline{B}_\varrho^k} p,\ p_+^k = \max_{\overline{\Omega}\cap\overline{B}_\varrho^k} p,\ r_+^k = \max_{\overline{\Omega}\cap\overline{B}_\varrho^k} r.$$

We consider $\{\varphi_k\}_{k=1}^m$ a smooth partition of unity subordinate to this cover. We have

$$\operatorname{supp}\varphi_k \subseteq B_\varrho^k, 0 \leq \varphi_k \leq 1, \sum_{k=1}^m \varphi_k(z) = 1 \text{ for all } z \in \overline{\Omega}. \tag{5.8}$$

As before, for $\lambda \geq 1$, let $u_\lambda = \min\{u, \lambda\} \in W_0^{1,p(z)}(\Omega)$. For $\mu > 0$ use the test function $h = u_\lambda^{\mu+1} \in W_0^{1,p(z)}(\Omega)$. Then

$$(\mu+1)[\int_\Omega |Du_\lambda|^{p(z)} u_\lambda^\mu dz + \int_\Omega |Du_\lambda|^{q(z)} u_\lambda^\mu dz] = \int_\Omega f(z,u) u_\lambda^{\mu+1} dz. \tag{5.9}$$

We have

$$\begin{aligned}
&(\mu+1)\int_\Omega |Du_\lambda|^{p(z)} u_\lambda^\mu dz \\
=&(\mu+1)\int_\Omega |Du_\lambda|^{p(z)} u_\lambda^\mu (\sum_{k=1}^m \varphi_k) dz \text{ (see (5.8))} \\
\geq&(\mu+1)\sum_{k=1}^m \int_\Omega |Du_\lambda|^{p_-^k} u_\lambda^\mu \varphi_k dz - (\mu+1)\sum_{k=1}^m \int_\Omega u_\lambda^\mu \varphi_k dz \\
\geq&\frac{\mu+1}{(\frac{\mu}{p_-}+1)^{p_+}} \sum_{k=1}^m \int_\Omega |Du_\lambda^{\frac{\mu}{p_-^k}+1}|^{p_-^k} \varphi_k dz \\
&-(\mu+1)c_1 \sum_{k=1}^m (\int_\Omega u^{(\frac{\mu}{p_-^k}+1)p_+^k} dz)^{\frac{p_-^k \mu}{(\mu+p_-^k)p_+^k}} \text{ for some } c_1 > 0 \\
\geq&\frac{\mu+1}{(\frac{\mu}{p_-}+1)^{p_+}} \sum_{k=1}^m \int_\Omega |Dv_\lambda^{\frac{\mu}{p_-^k}+1}|^{p_-^k} \varphi_k dz \\
&-(\mu+1)c_2 \sum_{k=1}^m [\|u^{\frac{\mu}{p_-^k}+1}\|_{r_+^k}^{p_-^k} + 1] \text{ for some } c_2 > 0.
\end{aligned} \tag{5.10}$$

Also we have

$$(\mu+1)\int_\Omega |Du_\lambda|^{q(z)}u_\lambda^\mu dz \geq 0.$$

Using the growth condition on the reaction, we obtain

$$\begin{aligned}
&\int_\Omega f(z,u)u_\lambda^{\mu+1}dz\\
\leq& c_3\int_\Omega [1+u^{r(z)-1}]u_\lambda^{\mu+1}dz \text{ for some } c_3>0\\
=& c_3\int_\Omega [1+u^{r(z)-1}]u^{\mu+1}(\sum_{k=1}^m \varphi_k)dz \text{ (see (5.8))}\\
\leq& c_4[\sum_{k=1}^m \int_\Omega u^{(\frac{\mu}{p_-^k}+1)p_-^k}u^{r_+^k-p_-^k}dz+1] \text{ for some } c_4>0\\
\leq& c_5[\sum_{k=1}^m \|u^{\frac{\mu}{p_-^k}+1}\|_{r_+^k}^{p_-^k}+1] \text{ for some } c_5>0.
\end{aligned}$$

So, from (5.9) and (5.10), it follows that

$$\begin{aligned}
&\frac{\mu+1}{(\frac{\mu}{p_-}+1)^{p_+}}\sum_{k=1}^m \int_\Omega |Du_\lambda^{\frac{\mu}{p_-^k}+1}|^{p_-^k}\varphi_k dz\\
\leq& c_6(\mu+1)[\sum_{k=1}^m \|u^{\frac{\mu}{p_-^k}+1}\|_{r_+^k}^{p_-^k}+1] \text{ for some } c_6>0.
\end{aligned}$$

We let $\lambda\to\infty$ and use Fatou's lemma. We obtain

$$\begin{aligned}
\|u\|_{L^{(\frac{\mu}{p_-^k}+1)(p_-^k)^*}(\Omega_k)} &\leq c_7[\|u\|^{(\frac{\mu}{p_-^k}+1)r_-^k}_{L^{(\frac{\mu}{p_-^k}+1)r_-^k}(\Omega_k)}+1]\\
&\text{for some } c_7>0, \text{ with } \Omega_k=\{\varphi_k=1\}.
\end{aligned}$$

From this estimate as in the proof of Theorem 5.3, we infer that $u\in L^\infty(\Omega)$.

Remark 5.5 In fact $\|u\|_\infty \leq M$ with $M>0$ depending on $\|\cdot\|_{p^*(z)}$.

As in the isotropic case (see Theorem 4.84), the boundedness of the weak solution, leads to global regularity. We need to strengthen the conditions on the exponents.

$\mathbf{H_0'}$: $p,q\in C^{0,1}(\overline{\Omega})$, $1<q_-\leq q(z)<p(z)$ for all $z\in\overline{\Omega}$.

The regularity result is due to Fan [157], who follows the steps of the corresponding isotropic regularity result of Lieberman [297], [298] (see Theorem 4.84).

Theorem 5.6 *If hypotheses H_0', H_1 hold and $u \in W_0^{1,p(z)}(\Omega)$ is a weak solution of* (1)*, then $u \in C_0^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0,1)$ and $\|u\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq M$ with $M > 0$ depending on $\|u\|_\infty$.*

Remark 5.7 This global regularity result was extended to singular problems by Giacomoni et al. [205] (isotropic problems) and Saoudi and Ghanmi[435] (anisotropic problems). So, consider the following anisotropic singular problem

$$\left\{ \begin{array}{l} -\Delta_{p(z)}u(z) - \Delta_{q(z)}u(z) = u(z)^{-\eta(z)} + f(z,u(z)) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, u > 0. \end{array} \right\} \tag{5.11}$$

We assume that hypotheses H_0', H_1 hold for a.a. $z \in \Omega$, $f(z,0) = 0$, $f(z,\cdot) \geq 0$ on $\mathbb{R}_+$ and that $\eta \in C(\overline{\Omega})$, $0 < \eta(z) < 1$ for all $z \in \overline{\Omega}$. By a "weak solution" of (5.11), we mean a function $u \in W_0^{1,p(z)}(\Omega)$ such that

$$u^{-\eta(z)}h \in L^1(\Omega) \text{ for all } h \in W_0^{1,p(z)}(\Omega),$$

$$\int_\Omega |Du|^{p(z)-2}(Du, Dh)_{\mathbb{R}^N}dz + \int_\Omega |Du|^{q(z)-2}(Du, Dh)_{\mathbb{R}^N}dz$$
$$= \int_\Omega f(z,u)hdz \text{ for all } h \in W_0^{1,p(z)}(\Omega).$$

As we already mentioned in Remark 5.4, every weak solution u belongs in $L^\infty(\Omega)$.

Theorem 5.8 *If hypotheses H_0', H_1 hold, $\eta \in C(\overline{\Omega})$, for a.a. $z \in \Omega$, $f(z,0) = 0$, $f(z,\cdot) \geq 0$ on $\mathbb{R}_+$, $0 < \eta(z) < 1$ for all $z \in \overline{\Omega}$ and $u \in W_0^{1,p(z)}(\Omega)$ is a weak solution of* (1)*, then $u \in C_0^{1,\alpha}(\overline{\Omega})$ for some $\alpha \in (0,1)$ and $\|u\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq M$ with $M > 0$ depending on $\|u\|_\infty$.*

Next we have the anisotropic version of the Hopf maximum principle. The regularity theory leads to the space

$$C_0^1(\overline{\Omega}) = \{u \in C^1(\overline{\Omega}) : u|_{\partial\Omega} = 0\}.$$

We know that this is an ordered Banach space with positive (order) cone

$$C_+ = \{u \in C_0^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\}.$$

This cone has a nonempty interior given by

$$\text{int}C_+ = \{u \in C_+ : u(z) > 0 \text{ for all } z \in \Omega, \frac{\partial u}{\partial n}|_{\partial\Omega} < 0\}.$$

with $n(\cdot)$ being the outward unit normal on $\partial\Omega$ and $\dfrac{\partial u}{\partial n} = (Du, n)_{\mathbb{R}^N}$.

Theorem 5.9 *If hypotheses H_0' hold, $\eta \in C(\overline{\Omega})$, $0 < \eta(z) < 1$ for all $z \in \overline{\Omega}$ and $u \in C_+ \setminus \{0\}$ satisfies*

$$-\Delta_{p(z)}u(z) - \Delta_{q(z)}u(z) + \widehat{\xi}u(z)^{p(z)-1} - \lambda u(z)^{-\eta(z)} \geq 0 \text{ in } \Omega,$$

with $\widehat{\xi}, \lambda > 0$, then $u \in \operatorname{int} C_+$.

Proof First we show that $u(z) > 0$ for all $z \in \Omega$. Arguing by contradiction suppose we can find $z_1, z_2 \in \Omega$ and an open ball $B_{2\varrho}(z_2)(\varrho > 0)$ such that $\overline{B}_{2\varrho}(z_2) \subseteq \Omega$ and $z_1 \in \partial B_{2\varrho}(z_2)$, $u(z_1) = 0$, $u|_{B_{2\varrho}(z_2)} > 0$. So, $2\varrho = |z_1 - z_2|$ and $\varrho > 0$ can be chosen small by fixing z_1 and letting z_2 vary.

Let $m = \min\{u(z) : |z - z_2| = \varrho\} > 0$. We have

$$m \to 0^+ \text{ and } \frac{m}{\varrho} \to 0^+ \text{ as } \varrho \to 0^+ \text{ (L'Hospital's rule).}$$

Since $u \in C_+$, we have $Du(z_1) = 0$. Consider the following ring (annulus)

$$R = \{z \in \Omega : \varrho < |z - z_2| < 2\varrho\}.$$

We know that $u|_R > 0$. By hypotheses H_0' the exponents p, q are Lipschitz functions and so by Rademacher's theorem (see Theorem I-6.135), they are differentiable almost everywhere. Therefore we can define

$$\vartheta = \sup\{\max\{|Dp(z)|, |Dq(z)|\} : z \in R\} < \infty.$$

We choose $\varrho \in (0, 1)$ small so that

$$\frac{m}{\varrho} < 1 \text{ and } \frac{p(z) - 1}{p(z_1) - 1} \geq \frac{1}{2} \text{ for all } z \in R.$$

In what follows we define

$$p_1 = p(z_1) \text{ and } \tau = -k\vartheta \ln(\frac{m}{\varrho}) + \frac{2(N-1)}{\varrho}, \quad k > 2.$$

We consider the function

$$v(t) = \frac{m(e^{\frac{\tau t}{p_1 - 1}} - 1)}{e^{\frac{\tau t}{p_1 - 1}} - 1} \text{ for all } t \in [0, \varrho].$$

Note that $v \in C^2[0, \varrho]$ and $v', v'' > 0$ on $[0, \varrho]$. So, v, v' are both strictly increasing and in fact we have

$$\frac{m}{\varrho} < v'(t) < 1 \text{ for all } t \in [0, \varrho]. \tag{5.12}$$

For notational simplicity and without any loss of generality, we may assume that $z_2 = 0$. Let

$$s = |z - z_2| = |z| \text{ and } t = 2\varrho - s.$$

We introduce the function

$$w(s) = v(2\varrho - s) = v(t) \text{ for all } s \in [\varrho, 2\varrho].$$

We see that $w \in C^2[\varrho, 2\varrho]$ and $w'(s) = -v'(t)$, $w''(s) = v''(t)$. For $z \in R$ with $|z| = s$, we write $w(z) = w(s)$. Then $w \in C^2(R)$ and we have

$$\begin{aligned}
&\operatorname{div}[|Du|^{p(z)-2}Du + |Du|^{q(z)-2}Du] \\
=&[p(z) - 1]v'(t)^{p(z)-1}v''(t) - \frac{N-1}{s}v'(t)^{p(z)-1} \\
&- v'(t)^{p(z)-1} \ln v'(t) \sum_{k=1}^{N} \frac{\partial p}{\partial z_k} \frac{z_k}{s} \\
&+ [q(z) - 1]v'(t)^{q(z)-1}v''(t) - \frac{N-1}{s}v'(t)^{q(z)-1} \\
&- v'(t)^{q(z)-1} \ln v'(t) \sum_{k=1}^{N} \frac{\partial q}{\partial z_k} \frac{z_k}{s}.
\end{aligned} \tag{5.13}$$

We have

$$(p_1 - 1)v''(t) = \tau v'(t). \tag{5.14}$$

Also note that

$$|\sum_{k=1}^{N} \frac{\partial p}{\partial z_k} \frac{z_k}{s}|, \ |\sum_{k=1}^{N} \frac{\partial q}{\partial z_k} \frac{z_k}{s}| \le \vartheta. \tag{5.15}$$

Using (5.12), (5.14), (5.15) in (5.13), we obtain

$$\begin{aligned}
&\operatorname{div}[|Dw|^{p(z)-2}Dw + |Dw|^{q(z)-2}Dw] \\
>&v'(t)^{p(z)-1}[\tau + 2\vartheta \ln(\frac{m}{\varrho})^2 - \frac{2(N-1)}{s}] \\
\ge&\widehat{\xi}v'(t)^{p(z)-1} \text{ in } R, \text{ for } \varrho \in (0, 1) \text{ small},
\end{aligned}$$

$$\begin{aligned} &\Rightarrow -\Delta_{p(z)} w - \Delta_{q(z)} w + \widehat{\xi} w^{p(z)-1} - \lambda w^{-\eta(z)} \\ &\leq -\Delta_{p(z)} u - \Delta_{q(z)} u + \widehat{\xi} u^{p(z)-1} - \lambda u^{-\eta(z)} \text{ in } R. \end{aligned} \tag{5.16}$$

Note that $v(0) = 0$, $v(\varrho) = m$. Hence $w|_{\partial R} \leq u|_{\partial R}$ and so from (5.16) it follows that

$$w(z) \leq u(z) \text{ for all } z \in \overline{R}.$$

So, we have

$$\frac{\partial u}{\partial n}(z_1) \leq \frac{\partial w}{\partial n}(z_1) = -v'(0) < 0 \text{ (since } u(z_1) = 0).$$

But recall that $Du(z_1) = 0$, a contradiction. This proves that

$$0 < u(z) \text{ for all } z \in \Omega.$$

Now suppose that $z_1 \in \partial\Omega$, $B_{2\varrho}(z_2) \subseteq \Omega$. From the previous part of the proof, we know that we can find a function $w \in C^2(R) \cap C^1(\overline{R})$ such that

$$w(z_1) = 0, \frac{\partial w}{\partial n}(z_1) < 0 \text{ and } w|_{\overline{R}} \leq u|_{\overline{R}}.$$

It follows that $\dfrac{\partial u}{\partial n}(z_1) < 0$ and so $u \in \text{int}C_+$. □

There is an analogous result for the double phase differential operator.

$$u \to \Delta_p^a u + \Delta_q u, \ u \in W_0^{1,\eta}(\Omega).$$

In this case the hypotheses on the data are the following:

$\mathbf{H_2}$: $a \in C^{0,1}(\overline{\Omega}) \setminus \{0\}$, $a(z) \geq 0$ for all $z \in \overline{\Omega}$, $1 < q < p < N$, $\dfrac{p}{q} < 1 + \dfrac{1}{N}$.

Unfortunately for double phase problems there is no global regularity theory. Only local regularity results.

Theorem 5.10 *If hypotheses H_2 hold, $\widehat{\xi} > 0$ and $u \in W_0^{1,\eta}(\Omega) \setminus \{0\}$, $u(z) \geq 0$ for a.a $z \in \Omega$ satisfies*

$$-\Delta_p^a u(z) - \Delta_q u(z) + \widehat{\xi} u(z)^{p-1} \geq 0 \text{ in } \Omega,$$

then $u \in W_0^{1,\eta}(\Omega) \cap L^\infty(\Omega)$ and for every $K \subseteq \Omega$ compact, we have

$$0 < c_K \leq u(z) \text{ for a.a } z \in K.$$

Proof First assume that $u \in C^1(\Omega)$. Reasoning as in the proof of Theorem 5.9, we show that

$$0 < u(z) \text{ for all } z \in \Omega.$$

Next consider the general case $u \in W_0^{1,\eta}(\Omega)$. Let

$$\Omega_0 = \{z \in \Omega : B_\varepsilon(z) \subseteq \Omega, u|_{B_\varepsilon(z)} = 0 \text{ a.e}, \varepsilon > 0\},$$
$$\Omega_+ = \Omega \setminus \Omega_0.$$

For $z_0 \in \Omega_+$, we can find $\varepsilon > 0$ such that

$$\overline{B}_{2\varepsilon}(z_0) \subseteq \Omega, \ u|_{\partial B_{2\varepsilon}(z_0)} \neq 0.$$

We consider the following

$$\left\{ \begin{array}{l} -\Delta_p^a y(z) - \Delta_q y(z) + \widehat{\xi} y(z)^{p-1} = 0 \text{ in } B_{2\varepsilon}(z_0), \\ y|_{\partial B_{2\varepsilon}(z_0)} = u|_{\partial B_{2\varepsilon}(z_0)}. \end{array} \right\} \tag{5.17}$$

Using Theorem *I*-7.47, we see that (5.17) has a unique weak solution $y \in W_0^{1,\eta}(\Omega)$ and moreover Theorem 1.1 of Baroni et al. [36], implies that $y \in C_{\text{loc}}^{1,\alpha}(B_{2\varepsilon}(z_0))$, $\alpha \in (0,1)$. Since $u|_{\partial B_{2\varepsilon}(z_0)} \geq 0$, from the Dirichlet boundary condition of (5.17), we infer that $y(z) \geq 0$ for a.a $z \in B_{2\varepsilon}(z_0)$. From the first part of the proof it follows that

$$\begin{aligned} & y(z) > 0 \text{ for all } z \in B_{2\varepsilon}(z_0), \\ \Rightarrow & y(z) \geq c > 0 \text{ for all } z \in \overline{B}_\varepsilon(z_0), \\ \Rightarrow & u(z) \geq c > 0 \text{ for a.a} z \in \overline{B}_\varepsilon(z_0). \end{aligned}$$

Now we show that $\Omega_0 = \emptyset$. If $\Omega_0 \neq \emptyset$, then Ω_0 is a nonempty open set and let $z_0 \in \Omega_+ \cap \partial\Omega_0$. Then from the previous part, we know that we can find $\varepsilon > 0$ such that

$$\begin{aligned} & u|_{B_\varepsilon(z_0)} > 0 \text{ a.e.}, \\ \Rightarrow & u|_{B_\varepsilon(z_0)\cap\Omega_0} > 0 \text{ a.e, a contradiction} \\ \Rightarrow & \Omega_0 = \emptyset. \end{aligned}$$

Finally let $K \subseteq \Omega$ compact. We can find balls $\{B_i\}_{i=1}^m$ such that $K \subseteq \bigcup_{i=1}^m B_i$ and $u|_{B_i} \geq c_i > 0$ a.e. Let $c_K = \min_{1 \leq i \leq m} c_i > 0$. Then

$$0 < c_K \leq u(z) \text{ for a.a } z \in K.$$

□

We can also have "antimaximum principles". Let $\Omega \subseteq \mathbb{R}^N (N \geq 2)$ be a bounded domain with a C^2-boundary $\partial\Omega$. Consider the follow linear parametric Dirichlet problem

$$-\Delta u(z) = \lambda u(z) + g(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0. \tag{5.18}$$

Suppose $g \in L^s(\Omega)$, $s > N$. Standard linear regularity theory implies that a weak solution of (5.18) belongs in $C_0^1(\overline{\Omega})$. Moreover, if $\lambda < \widehat{\lambda}_1 (\widehat{\lambda}_1 > 0$ being the principal eigenvalue of $(-\Delta, H_0^1(\Omega)))$ and $h \geq 0$, $h \neq 0$, then the solution belongs in intC_+(Hopf's maximum principle). We ask what happens if $\lambda > \widehat{\lambda}_1$. The "antimaximum principle" says that there exists $\lambda_g > \widehat{\lambda}_1$ such that for every $\lambda \in (\widehat{\lambda}_1, \lambda_g)$, the solution of (5.18) belongs in $-$intC_+. Furthermore λ_g cannot be chosen independent of g.

In the sequel, we will investigate this phenomenon in the context of linear and nonlinear Neumann and Dirichlet problems.

We start with the linear Neumann problem which in general is more difficult than the corresponding Dirichlet problem, due to the failure of the Poincare inequality.

We consider the following Neumann problem

$$Lu(z) = \lambda m(z)u(z) + g(z) \text{ in } \Omega,\ \frac{\partial u}{\partial n_L} = 0 \text{ on } \partial\Omega. \tag{5.19}$$

In this problem $L(\cdot)$ is a second order elliptic differential operator defined by

$$L(u) = -\operatorname{div}(A(z)Du) \text{ for all } u \in H^1(\Omega),$$

with $A(z) = (a_{ij}(z))_{i,j=1}^N \in \mathbb{R}^{N\times N}$, $a_{ij} \in L^\infty(\Omega)$, $a_{ij} = a_{ji}$ for all $i, j = 1, \cdots, N$ and

$$(A(z)\xi, \xi)_{\mathbb{R}^N} \geq \widehat{c}|\xi|^2 \text{ for a.a } z \in \Omega, \text{ all } \xi \in \mathbb{R}^N, \text{ some } \widehat{c} > 0.$$

(strong ellipticity condition)

The hypotheses on the weight $m(\cdot)$ and the perturbation $h(\cdot)$ are the following:

H$_3$: $m \in L^\infty(\Omega)$, $m^+ \neq 0$, $m^- \neq 0$ and $g \in L^s(\Omega)$, $s > N$.

Remark 5.11 If we assume also $m \geq 0$, $m \neq 0$, then from Sect. 4.4, we know that there are only positive eigenvalues.

In the boundary $\dfrac{\partial u}{\partial n_L}$ denotes the normal derivative of u associated with the operator L. It is defined using the Green's identity (see Corollary I-4.103). We know that if $u \in C^1(\overline{\Omega})$, then

$$\frac{\partial u}{\partial n_L} = (A(z)Du, n)_{\mathbb{R}^N},$$

with $n(\cdot)$ being the outward unit normal.

As always by a solution of (5.19), we mean a weak solution, that is a function $u \in H^1(\Omega)$ such that

$$\int_\Omega (A(z)Du, Dh)_{\mathbb{R}^N} dz = \lambda \int_\Omega m(z)uhdz + \int_\Omega ghdz \text{ for all } h \in H^1(\Omega).$$

Hypotheses H_3 and standard linear regularity theory imply that every weak solution u belongs in $C^1(\overline{\Omega})$. The space $C^1(\overline{\Omega})$ is an ordered Banach space with positive (order) cone $\widehat{C}_+ = \{u \in C^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\}$. This cone has a nonempty interior given by

$$\text{int}\widehat{C}_+ = \{u \in \widehat{C}_+ : u(z) > 0 \text{ for all } z \in \overline{\Omega}\}.$$

We start by considering the following linear Neumann problem

$$Lu(z) + a_0(z)u(z) = g(z) \text{ in } \Omega, \frac{\partial u}{\partial n_L} = 0 \text{ on } \partial\Omega. \tag{5.20}$$

Proposition 5.12 *If $a_0 \in L^\infty(\Omega)$, $a_0(z) \geq 0$ for a.a $z \in \Omega$ and $g \in L^s(\Omega) \setminus \{0\}$ with $s > N$, $g(z) \geq 0$ for a.a $z \in \Omega$, then any weak solution $u_0 \in H^1(\Omega)$ of (5.20) belongs in* $\text{int}\widehat{C}_+$.

Proof For the Schauder L^p-Theory, we have $u_0 \in W^{2,s}(\Omega)$ (see Gilbarg and Trudinger [206], Theorem 9.11, p. 235). From the Sobolev embedding theorem (see Theorem I-4.114), we know that $W^{2,s}(\Omega) \hookrightarrow C^1(\overline{\Omega})$ compactly(recall $s > N$). Using the test function $h = -u_0^- \in H^1(\Omega)$, we obtain

$$\widehat{c}\,\|Du_0^-\|_2^2 \leq 0,$$
$$\Rightarrow u_0^- = \eta \in \mathbb{R}_+.$$

If $\eta > 0$, then $0 \leq \eta^2 \int_\Omega a_0(z)dz = -\eta \int_\Omega g(z)dz < 0$, a contradiction. So, $\eta = 0$ and we have $u_0 \geq 0$, $u \neq 0$ (since $g \neq 0$). Then

$$Lu_0(z) + a_0u_0(z) = g(z) \geq 0 \text{ in } \Omega,$$
$$\Rightarrow u_0 \in \text{int}\widehat{C} \text{ (by the Hopf maximum principle).}$$

□

As we did in Sect. 4.2(in the context of Dirichlet problems), we define

$$\widehat{\lambda}^*(m) = \inf\{\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz : u \in H^1(\Omega), \int_\Omega m(z)u^2 dz = 1\}. \tag{5.21}$$

This quantity is related to the following eigenvalue problem

$$\left\{ \begin{array}{l} -\operatorname{div}(A(z)Du(z)) = \widehat{\lambda} m(z)u(z) \text{ in } \Omega, \\ \dfrac{\partial u}{\partial n_L} = 0 \text{ on } \partial\Omega. \end{array} \right\} \tag{5.22}$$

Proposition 5.13 *If $m \in L^\infty(\Omega)$, $m^+ \neq 0$ and $\int_\Omega m(z)dz > 0$, then $\widehat{\lambda}^*(m) = 0$.*

Proof Let $k(m) = \int_\Omega m(z)dz > 0$ and set $\widehat{u}(z) = \dfrac{1}{k(m)^{1/2}}$ for all $z \in \Omega$. Then from (5.21), we see that $\widehat{\lambda}^*(m) = 0$. □

Proposition 5.14 *If $m \in L^\infty(\Omega)$ with $m^+ \neq 0$ and $\int_\Omega m(z)dz < 0$, then $\widehat{\lambda}^*(m) > 0$ and this is the unique nonzero principal eigenvalue of (5.22) and any $\lambda \in (0, \widehat{\lambda}^*(m))$ is not an eigenvalue of (5.22).*

Proof First we show that there exists $c_1 > 0$ such that

$$c_1 \|u\|_2^2 \leq \int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz. \tag{5.23}$$

for all $u \in H^1(\Omega)$ satisfying $\int_\Omega m(z)u^2 dz = 1$.

Arguing by contradiction, suppose that (5.23) is not true. Then we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ such that

$$\int_\Omega (A(z)Du_n, Du_n)_{\mathbb{R}^N} dz < \frac{1}{n}\|u_n\|_2^2, \ \int_\Omega m(z)u_n^2 dz = 1, \ n \in \mathbb{N}. \tag{5.24}$$

If $\{u_n\}_{n\in\mathbb{N}} \subseteq L^2(\Omega)$ is bounded, then from (5.24) we infer that $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ is bounded and so for at least a subsequence, we have

$$u_n \to \vartheta \in \mathbb{R} \text{ in } L^2(\Omega) \text{ (see (5.24))}$$
$$\Rightarrow \vartheta^2 \int_\Omega m(z)dz = 1,$$

which contradicts our hypothesis that $\int_\Omega m(z)dz < 0$.

If $\{u_n\}_{n\in\mathbb{N}} \subseteq L^2(\Omega)$ is unbounded, then we can say that $\|u_n\|_2 \to \infty$. Let $y_n = \dfrac{u_n}{\|u_n\|}$, $n \in \mathbb{N}$. Then $\|y_n\|_2 = 1$ for all $n \in \mathbb{N}$ and from (5.24), it follows that

$$\widehat{c}\|Dy_n\|_2^2 < \frac{1}{n} \text{ for all } n \in \mathbb{N}$$

$$\Rightarrow y_n \to \vartheta \in \mathbb{R} \setminus \{0\} \text{ in } L^2(\Omega) \text{ (recall } \|y_n\|_2 = 1).$$

We have

$$\int_\Omega m(z) y_n^2 dz = \frac{1}{\|u_n\|_2^2} \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow 0 = \vartheta^2 \int_\Omega m(z) dz < 0, \text{ a contradiction.}$$

Therefore (5.23) holds.

Now consider $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ a minimizing sequence for (5.21). We have

$$\int_\Omega (A(z)Du_n, Du_n)_{\mathbb{R}^N} dz \downarrow \widehat{\lambda}^*(m), \quad \int_\Omega m(z) u_n^2 dz = 1 \text{ for all } n \in \mathbb{N}. \tag{5.25}$$

From (5.23) and (5.25), we see that $\{u_n\}_{n\in\mathbb{N}} \subseteq H^1(\Omega)$ is bounded. So, we may assume that

$$u_n \xrightarrow{w} u \text{ in } H^1(\Omega), \; u_n \to u \text{ in } L^2(\Omega). \tag{5.26}$$

Using (5.26) in (5.25), we obtain

$$\begin{aligned} &\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz \le \widehat{\lambda}^*(m), \quad \int_\Omega m(z) u^2 dz = 1, \\ \Rightarrow &\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz = \widehat{\lambda}^*(m) > 0, \quad \int_\Omega m(z) u^2 dz = 1. \end{aligned} \tag{5.27}$$

We can always replace u with $|u|$ and so we can assume that $u \ge 0$, $u \ne 0$. From (5.27) and using the Lagrange multiplier rule, we can find $\eta \in \mathbb{R}$ such that

$$\int_\Omega (A(z)Du, Dh)_{\mathbb{R}^N} dz = \eta \int_\Omega m(z) uh dz \text{ for all } h \in H^1(\Omega).$$

Using the test function $h = u \in H^1(\Omega)$ and (5.27), we obtain $\eta = \widehat{\lambda}^*(m)$. Therefore $\widehat{\lambda}^*(m) > 0$ is a principal eigenvalue of (5.22).

Now let $\lambda \in (0, \widehat{\lambda}^*(m))$. Suppose that λ is an eigenvalue of (5.22) and let $\widehat{u} \in H^1(\Omega) \setminus \{0\}$ be a corresponding eigenfunction. Then

$$\int_\Omega (A(z)D\widehat{u}, Dh)_{\mathbb{R}^N} dz = \widehat{\lambda} \int_\Omega m(z) \widehat{u} h dz \text{ for all } h \in H^1(\Omega).$$

Choosing $h = \widehat{u} \in H^1(\Omega)$, we obtain

$$\int_\Omega (A(z)D\widehat{u}, D\widehat{u})_{\mathbb{R}^N} dz = \lambda \int_\Omega m(z)\widehat{u}^2 dz.$$

If $y = \dfrac{\widehat{u}}{(\int_\Omega m(z)\widehat{u}^2 dz)^{1/2}}$, then

$$\int_\Omega (A(z)Dy, Dy)_{\mathbb{R}^N} dz = \lambda, \quad \int_\Omega m(z)y^2 dz = 1,$$

which contradicts (5.21)(recall $\lambda < \widehat{\lambda}^*(m)$).

Finally we show the uniqueness of this principal eigenvalue $\widehat{\lambda}^*(m) > 0$. Suppose that $(\widehat{\lambda}, \widehat{u})$ is a principal eigenpair for problem (5.22). Then regularity theory and the Hopf maximum principle imply that $\widehat{u} \in \operatorname{int} C_+$. Therefore $u = e^{-v}$ with $v \in C^1(\overline{\Omega})$ and for $y \in H^1(\Omega) \cap L^\infty(\Omega)$, we use the test function $h = e^v y^2 \in H^1(\Omega)$. Then

$$\widehat{\lambda} \int_\Omega m(z)y^2 dz \le \int_\Omega (A(z)Dy, Dy)_{\mathbb{R}^N} dz.$$

Since $y \in H^1(\Omega) \cap L^\infty(\Omega)$ is arbitrary, using Theorem I-4.68, we infer that $\widehat{\lambda} \le \lambda^*(m)$. Also $-\widehat{\lambda} \le \lambda^*(-m) = 0$ (see Proposition 5.13). Then $\lambda \in [0, \widehat{\lambda}^*(m)]$. So, if $\widehat{\lambda}$ is nonzero, then $\widehat{\lambda} = \widehat{\lambda}^*(m)$. □

Proposition 5.15 *If $m \in L^\infty(\Omega) \setminus \{0\}$ and $\int_\Omega m(z)dz = 0$, then $\lambda^*(m) = 0$ and this is the unique principal eigenvalue of (5.22).*

Proof Let $h \in H^1(\Omega)$ with $0 < \int_\Omega mh dz$ and define

$$u_\varepsilon = \frac{1 + \varepsilon h}{(\int_\Omega m(z)(1 + \varepsilon h)^2 dz)^{1/2}}$$

with $\varepsilon > 0$ small so that the denominator does not vanish. Then

$$\|Du_\varepsilon\|_2 \to 0 \text{ as } \varepsilon \to 0^+ \Rightarrow \widehat{\lambda}^*(m) = 0.$$

In this case the infimum in (5.21) is not attained. Because if it is attained, the minimizer is a constant function which violates the constraint $\int_\Omega m(z)u^2 dz = 1$. □

The next result will be useful for the general antimaximum principle.

Proposition 5.16 *If $m \in L^\infty(\Omega)$ with $m^+ \ne 0$, $m^- \ne 0$, $\int_\Omega m(z)dz \le 0$, $\lambda = 0$ or $\lambda = \widehat{\lambda}^*(m)$ and $g \in L^s(\Omega) \setminus \{0\}$ with $s > N$, $g \ge 0$, then problem (5.19) has no solution.*

Proof If $\lambda = 0$, then use as test function $h \equiv 1$, to see that (5.19) can not have a solution.

If $\lambda = \widehat{\lambda}^*(m) > 0$, then we argue by contradiction. So, suppose that problem (5.19) has a solution $u \in H^1(\Omega) \setminus \{0\}$. Regularity theory implies that $u \in C^1(\overline{\Omega})$. We use $h = -u^- \in H^1(\Omega)$ as a test function. Then

$$\int_\Omega (A(z)Du^-, Du^-)_{\mathbb{R}^N} dz = \widehat{\lambda}^*(m) \int_\Omega m(u^-)^2 dz - \int_\Omega gu^- dz,$$

$$\Rightarrow u^- \text{ is an eigenfunction for } \widehat{\lambda}^*(m) \text{ and } \int_\Omega gu^- dz = 0.$$

We know that $u^- \in \operatorname{int}\widehat{C}_+$ and so $\int_\Omega gu^- dz > 0$, a contradiction. Therefore $u \geq 0$ and so $u \in \operatorname{int}\widehat{C}_+$. Let $\widehat{u}_1 \in \operatorname{int}C_+$ be an eigenvalue for $\widehat{\lambda}^*(m)$. Using Picone's identity (see Theorem 4.91), we have

$$\begin{aligned}
0 &\leq \int_\Omega (A(z)D\widehat{u}_1, D\widehat{u}_1)_{\mathbb{R}^N} dz - \int_\Omega (Du, D(\frac{\widehat{u}_1^2}{u}))_{\mathbb{R}^N} dz \\
&= \int_\Omega (A(z)D\widehat{u}_1, D\widehat{u}_1)_{\mathbb{R}^N} dz - \int_\Omega Lu(\frac{\widehat{u}_1^2}{u}) dz \\
&= \int_\Omega (A(z)D\widehat{u}_1, D\widehat{u}_1)_{\mathbb{R}^N} dz - \int_\Omega \widehat{\lambda}^*(m)m(z)\widehat{u}_1^2 dz - \int_\Omega g\frac{\widehat{u}_1^2}{u} dz \\
&= -\int_\Omega g\frac{\widehat{u}_1^2}{u} dz < 0,
\end{aligned}$$

a contradiction.

So, problem (5.19) does not have a solution when $\lambda = \widehat{\lambda}^*(m)$. □

Also reasoning as in the last part in the proof of Proposition 5.14, we obtain the following result.

Proposition 5.17 *If $m \in L^\infty(\Omega)$, $m^+ \neq 0$, $m^- \neq 0$, $\int_\Omega m(z)dz \leq 0$ and $\lambda \notin [0, \widehat{\lambda}^*(m)]$, then problem (5.19) has no nontrivial, nonnegative solution.*

In addition to $\widehat{\lambda}^*(m)$ (see (5.21)), we introduce also the following quantity

$$\widetilde{\lambda}(m) = \inf\{\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz : u \in H^1(\Omega), \int_\Omega m(z)u^2 dz = 1 \quad u(\cdot) \text{ vanishes on some ball in } \Omega\}. \tag{5.28}$$

Both quantities $\widehat{\lambda}^*(m)$ and $\widetilde{\lambda}(m)$, play a central role in the antimaximum principle. It is clear that $\widehat{\lambda}^*(m) \leq \widetilde{\lambda}(m)$. In fact from Godoy et al. [209] (Lemma 3.1), we have

Proposition 5.18 $\widehat{\lambda}^*(m) = \widetilde{\lambda}(m)$.

Remark 5.19 If $N = 1$(recall we have assumed that $N \geq 2$), then $\widehat{\lambda}^*(m) < \widetilde{\lambda}(m)$ and every $\lambda \in (\widehat{\lambda}^*(m), \widetilde{\lambda}(m))$ is not an eigenvalue of (5.22).

First we show that the antimaximum principle does not hold far away to the right of $\widetilde{\lambda}(m) = \widehat{\lambda}^*(m)$ and to the left of $-\widehat{\lambda}^*(-m)$.

Proposition 5.20 *If $m \in L^\infty(\Omega)$ with $m^+ \neq 0$, $m^- \neq 0$ and $\int_\Omega m(z)dz \leq 0$, then*

(a) *for every $\varepsilon > 0$, we can find $g_\varepsilon \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g_\varepsilon \geq 0$ such that for all $\lambda \geq \widetilde{\lambda}(m) + \varepsilon$ problem (5.19) has no solution $u \in -\text{int}\widehat{C}_+$;*
(b) *for every $\varepsilon > 0$, we can find $\widehat{g}_\varepsilon \in L^s(\Omega) \setminus \{0\}$, $s > N$, $\widehat{g}_\varepsilon \geq 0$ such that for all $\lambda \leq -\widetilde{\lambda}(-m) - \varepsilon$ problem (5.19) has no solution $u \in -\text{int}\widehat{C}_+$.*

Proof

(a) We proceed indirectly. So, suppose we can find $\varepsilon > 0$ such that for all $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, there exists $\lambda \geq \widetilde{\lambda}(m) + \varepsilon$ so that problem (5.19) has a solution $u \in -\text{int}C_+$. Let $y \in H^1(\Omega) \cap L^\infty(\Omega)$ with $\int_\Omega m(z)y^2dz > 0$ and y vanishes on some ball in Ω. Also choose $g \geq 0$, $g \neq 0$ so that $\text{supp}\, g \cap \text{supp}\, y = \emptyset$(we can always choose $g \in C_c^\infty(\Omega)$ with support of arbitrarily small diameter in the ball where y vanishes). Let (λ, u) be the parameter-solution pair guaranteed by the contradiction argument. We have $u \in -\text{int}\widehat{C}_+$ and so $-u = e^v$ with $v \in C^1(\overline{\Omega})$. As before we use the test function $h = e^v y^2 \in H^1(\Omega)$. Then

$$\lambda \int_\Omega my^2dz = \int_\Omega (A(z)Dy, Dy)_{\mathbb{R}^N}dz + \int_\Omega ge^v y^2dz - \int_\Omega (A(z)(Dy + yDv), Dy + yDv)_{\mathbb{R}^N}dz.$$

Since $\text{supp}\, g \cap \text{supp}\, y = \emptyset$, we have

$$\int_\Omega ge^v y^2dz = 0.$$

Therefore

$$\widehat{\lambda}^*(m) + \varepsilon \leq \lambda \leq \frac{1}{\int_\Omega my^2dz} \int_\Omega (A(z)Dy, Dy)_{\mathbb{R}^N}dz$$
$$\Rightarrow \widehat{\lambda}^*(m) + \varepsilon \leq \widetilde{\lambda}(m) \text{ (see (5.28))},$$

which contradicts Proposition 5.18.

(b) This part is proved similarly.

□

Now we are ready for the antimaximum principle for problem (5.19).

Proposition 5.21 *If $m \in L^\infty(\Omega)$, $m^+ \neq 0$, $m^- \neq 0$, $\int_\Omega m(z)dz \leq 0$, then*

(a) *for every* $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$,*there exists* $\delta = \delta(g) > 0$ *such that for* $\lambda \in (\widehat{\lambda}^*(m), \widehat{\lambda}^*(m) + \delta)$ *or* $\lambda \in (-\delta, 0)$ *any solution* u *of* (5.19) *satisfies* $u \in -\text{int}C_+$*;*
(b) *we can not find* $\delta > 0$ *independent of* g.

Proof

(a) Arguing by contradiction, suppose we can find $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, a sequence $\{\lambda_k\}_{k\in\mathbb{N}}$ with $\lambda_k > \widehat{\lambda}^*(m)$, $\lambda_k \downarrow \widehat{\lambda}^*(m)$ and $u_k \in C^1(\overline{\Omega})$(regularity theory) such that

$$Lu_k(z) = \lambda_k m(z)u_k(z) + g(z) \text{ in } \Omega, \ \frac{\partial u_k}{\partial n_L} = 0 \text{ on } \partial\Omega, \tag{5.29}$$

$$u_k \notin -\text{int}C_+. \tag{5.30}$$

First assume that $\{u_k\}_{k\in\mathbb{N}} \subseteq L^2(\Omega)$ is bounded. Then from (5.29), using as test function $h = u_k$, we see that $\{u_k\}_{k\in\mathbb{N}} \subseteq H^1(\Omega)$ is bounded. The regularity theory implies that we can find $\alpha \in (0, 1)$ and $c > 0$ such that

$$u_k \in C^{1,\alpha}(\overline{\Omega}), \ \|u_k\|_{C^{1,\alpha}(\overline{\Omega})} \leq c \text{ for all } k \in \mathbb{N}.$$

Since $C^{1,\alpha}(\overline{\Omega}) \hookrightarrow C^1(\overline{\Omega})$ compactly, we see that at least for a subsequence, we have

$$u_k \to u \text{ in } C^1(\overline{\Omega}) \text{ as } k \to \infty.$$

From (5.29) in the limit as $k \to \infty$, we have

$$L(u) = \widehat{\lambda}^*(m)m(z)u(z) + g(z) \text{ in } \Omega, \ \frac{\partial u}{\partial n_L} = 0 \text{ on } \partial\Omega,$$

which contradicts Proposition 5.16.

Now we assume that $\{u_k\}_{k\in\mathbb{N}} \subseteq L^2(\Omega)$ is unbounded. We can say that $\|u_k\|_2 \to \infty$. Let $y_k = \dfrac{u_k}{\|u_k\|_2}$, $k \in \mathbb{N}$. We have

$$Ly_k(z) = \lambda_k m(z)y_k(z) + \frac{g(z)}{\|u_k\|_2} \text{ in } \Omega, \ \frac{\partial y_k}{\partial n_L} = 0 \text{ on } \partial\Omega. \tag{5.31}$$

As above, using the linear regularity theory, we can say that

$$y_k \to y \text{ in } C^1(\overline{\Omega}), \ \|y\|_2 = 1. \tag{5.32}$$

Passing to the limit as $k \to \infty$ in (5.31), we obtain

$$Ly(z) = \widehat{\lambda}^*(m)m(z)y(z) \text{ in } \Omega, \ \frac{\partial y}{\partial n_L} = 0 \text{ on } \partial\Omega,$$

$$\Rightarrow y(\cdot) \text{ is an eigenfunction for } \widehat{\lambda}^*(m) \text{ (see (5.32))},$$

$$\Rightarrow y \in \pm \text{int}C_+.$$

If $y \in \text{int}C_+$, then from (5.32) we see that $y_k \in \text{int}C_+$ for all $k \geq k_0$, which contradicts Proposition 5.17.

If $y \in -\text{int}C_+$, then $y_k \in -\text{int}C_+$ for all $k \geq \widehat{k}_0$, hence $u_k \in -\text{int}C_+$ for all $k \geq \widehat{k}_0$, contradicting (5.30).

This proves part (a).

(b) From Proposition 5.17, we know that $\widehat{\lambda}^*(m) = \widetilde{\lambda}(m)$ and $0 = \widehat{\lambda}^*(-m) = \widetilde{\lambda}(-m)$. Then the nonuniformity of the antimaximum principle follows from Proposition 5.20.

□

Remark 5.22 If $N = 1$(scalar case), then from Remark 5.19 we know that $\widehat{\lambda}^*(m) < \widetilde{\lambda}(m)$ and of course $-\widetilde{\lambda}(-m) < -\widehat{\lambda}^*(-m)$. In that case, for any $\lambda \in (\widehat{\lambda}^*(m), \widetilde{\lambda}(m)]$ and any $\lambda \in [-\widetilde{\lambda}(-m), -\widehat{\lambda}^*(-m))$ and if $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, any solution u of (5.19) satisfies $u \in -\text{int}\widehat{C}_+$. Moreover, $\widetilde{\lambda}(m)$ and $-\widetilde{\lambda}(-m)$ are the highest and smallest numbers for which the previous statement remains valid, but then the uniformity fails. Namely given $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, there exists $\delta = \delta(g) > 0$ such that for every $\lambda \in (\widetilde{\lambda}(m), \widetilde{\lambda}(m)+\delta)$ and every $\lambda \in (-\widetilde{\lambda}(-m)-\delta, -\widetilde{\lambda}(-m))$ any solution u of (5.19) satisfies $u \in -\text{int}\widehat{C}_+$. But now $\delta > 0$ can not be chosen independent of g. When $m \geq 0$, $m \neq 0$, the results on the antimaximum principle remain valid. In this case $\widehat{\lambda}^*(m) = 0$ (see Proposition 5.13) and this is the unique nonnegative principal eigenvalue of (5.22)($-\widehat{\lambda}^*(-m) < 0$ is the unique nonzero principal eigenvalue) and the antimaximum principle holds to the right of zero. The principle is nonuniform. If $N = 1$ is uniform on $(0, \widetilde{\lambda}(m))$ and it is still true beyond $\widetilde{\lambda}(m)$ but in a nonuniform fashion. We will illustrate this case below for the $p-$Laplacian.

Let us say a few things about the corresponding Dirichlet problem. The antimaximum principle remains true. We consider the following parametric Dirichlet problem

$$Lu(z) = \lambda m(z)u(z) + g(z) \text{ in } \Omega, \ u|_{\partial\Omega} = 0. \tag{5.33}$$

From the discussion in Sect. 4.2, we know that the corresponding linear Dirichlet eigenvalue problem with indefinite weight, has two principal eigenvalues

$$\widehat{\lambda}_1(m) = \inf\{\int_\Omega (A(z)Du, Du)_{\mathbb{R}^N} dz : u \in H_0^1(\Omega), \ \int_\Omega m(z)u^2 dz = 1\} > 0$$

and $-\widehat{\lambda}_1(-m) < 0$.

Now the relevant space is $C_0^1(\overline{\Omega}) = \{u \in C^1(\overline{\Omega}) : u|_{\partial\Omega} = 0\}$ which we know that it is an ordered Banach space with positive (order) cone $C_+ = \{u \in C_0^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\}$. Recall that this cone has a nonempty interior given by

$$\text{int}C_+ = \{u \in C_+ : u(z) > 0 \text{ for all } z \in \Omega, \ \frac{\partial u}{\partial n}|_{\partial\Omega} < 0\},$$

with $n(\cdot)$ being the outward unit normal on $\partial\Omega$.

Using the direct method of the calculus of variations, the linear regularity theory and the Hopf maximum principle, we obtain the following result.

Proposition 5.23 *If $\lambda \in (-\widehat{\lambda}_1(-m), \widehat{\lambda}_1(m))$ and $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, then problem (5.33) admits a unique solution $u \in \text{int}C_+$.*

Reasoning as in the proof of Proposition 5.21, via a contradiction argument, we show that the antimaximum principle is also true for the Dirichlet problem, to the right of $\widehat{\lambda}_1(m) > 0$ and to the left of $-\widehat{\lambda}_1(-m) < 0$ and it is nonuniform.

Theorem 5.24 *If $m \in L^\infty(\Omega)$ with $m^+ \neq 0$, $m^- \neq 0$, then*

(a) *for every $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, there exists $\delta = \delta(g) > 0$ such that for every $\lambda \in (\widehat{\lambda}_1(m), \widehat{\lambda}_1(m) + \delta)$ or $\lambda \in (-\widehat{\lambda}_1(-m) - \delta, -\widehat{\lambda}_1(-m))$, problem (5.33) has a solution $u \in -\text{int}C_+$;*
(b) *we can not find $\delta > 0$ independent of g.*

Proposition 5.25 *If $m \in L^\infty(\Omega)$ with $m^+ \neq 0$, $m^- \neq 0$, then for every $\varepsilon > 0$, we can find $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$ such that for every $\lambda \geq \widehat{\lambda}_1(m) + \varepsilon$ (resp. $\lambda \leq -\widehat{\lambda}_1(-m) - \varepsilon$) problem (5.33) has no solution $u \in -\text{int}C_+$.*

Proof We do the proof for $\widehat{\lambda}_1(m) > 0$, the proof for $-\widehat{\lambda}_1(-m) < 0$ being similar.

Arguing indirectly, suppose we can find $\varepsilon > 0$ such that for every $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, there exists $\lambda \geq \widehat{\lambda}_1(m) + \varepsilon$ for which problem (5.33) admits a solution $u \in -\text{int}C_+$. Let $y \in C_c^1(\Omega)$ such that $\int_\Omega m(z)y^2 dz > 0$. We choose $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$ such that $\text{supp}\, g \cap \text{supp}\, y = \emptyset$. Then by hypothesis we can find $\lambda \geq \widehat{\lambda}_1(m) + \varepsilon$ for which problem (5.33) has a solution $u \in -\text{int}C_+$. Then $-u(z) > 0$ for all $z \in \Omega$ and we can write $-u = e^v$ with $v \in C^1(\Omega)$. Using the test function $h = e^v y^2 \in C_c^1(\Omega)$, we obtain

$$\widehat{\lambda}_1(m) + \varepsilon \leq \lambda \leq \frac{\int_\Omega (A(z)Dy, Dy)_{\mathbb{R}^N} dz}{\int_\Omega m(z)y^2 dz},$$

$$\Rightarrow \widehat{\lambda}_1(m) + \varepsilon \leq \widehat{\lambda}_1(m) \text{ a contradiction.}$$

□

Remark 5.26 Similar results are true if $m(\cdot)$ has fixed sign.

Next we look at the case of the p-Laplacian. Now the distinction is not between $N \geq 2$ and $N = 1$; but between $p < N$ and $p \geq N$. Here we will restrict ourselves

to the more interesting case $p < N$. Also, for simplicity, we will assume that the weight function is nontrivial, nonnegative.

So, let $m \in L^\infty(\Omega) \setminus \{0\}$, $m(z) \geq 0$ for a.a $z \in \Omega$. We consider the nonlinear eigenvalue problem

$$-\Delta_p u(z) = \widehat{\lambda} m(z)|u(z)|^{p-2}u(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0. \tag{5.34}$$

By Proposition 4.123, we know that (5.34) has a smallest eigenvalue (principal eigenvalue)

$$\widehat{\lambda}_1(m) > 0.$$

Let $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$ and consider the following Dirichlet problem

$$\left\{ \begin{array}{l} -\Delta_p u(z) = \widehat{\lambda}_1(m) m(z)|u(z)|^{p-2}u(z) + g(z) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0. \end{array} \right\} \tag{5.35}$$

Proposition 5.27 *If everything is as above, then problem* (5.35) *has no solution.*

Proof Suppose we can find a weak solution $u \in W_0^{1,p}(\Omega)$ for problem (5.35). From Proposition 4.82, we know that $u \in L^\infty(\Omega)$. If we set

$$f(z) = \widehat{\lambda}_1(m) m(z)|u(z)|^{p-2}u(z) + g(z),$$

then $f \in L^s(\Omega)$, $s > N$. We consider the linear Dirichlet problem

$$-\Delta u(z) = f(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0.$$

We know (see Gilbarg and Trudinger [206], Theorem 9.15, p.241) that this problem has a unique solution

$$v \in W^{2,s}(\Omega) \cap H_0^1(\Omega).$$

By the Sobolev embedding theorem (see Theorem I-4.114), we have $v \in C_0^{1,\vartheta}(\overline{\Omega})$ with $\vartheta = \dfrac{N}{s} \in (0,1)$. Then $Dv \in C^{0,\vartheta}(\overline{\Omega})$. We consider the Dirichlet problem

$$-\operatorname{div}(|Du|^{p-2}Du - Dv) = 0 \text{ in } \Omega,\ u|_{\partial\Omega} = 0.$$

By Theorem 4.84, we have $u \in C_0^1(\overline{\Omega})$.

Claim $u \geq 0$.

We have

$$\int_\Omega |Du|^{p-2}(Du, Dh)_{\mathbb{R}^N} dz = \widehat{\lambda}_1(m) \int_\Omega m(z)|u|^{p-2}uhdz + \int_\Omega ghdz$$

$$\text{for all } h \in W_0^{1,p}(\Omega). \tag{5.36}$$

If the Claim is not true, then $u^- \neq 0$ and we can use $-u^- \in W_0^{1,p}(\Omega)$ as a test function in (5.36). We obtain

$$\|Du^-\|_p^p = \widehat{\lambda}_1(m) \int_\Omega m(z)(u^-)^p dz - \int_\Omega gu^- dz, \tag{5.37}$$

$$\Rightarrow \|Du^-\|_p^p \leq \widehat{\lambda}_1(m) \int_\Omega m(z)(u^-)^p dz,$$

$$\Rightarrow \|Du^-\|_p^p = \widehat{\lambda}_1(m) \int_\Omega m(z)(u^-)^p dz,$$

$$\Rightarrow u^- \text{ is an eigenfunction for } \widehat{\lambda}_1(m) > 0$$

$$\Rightarrow u^- \in \text{int}C_+.$$

But then $\int_\Omega gu^- dz > 0$ and so from (5.37), we infer that

$$\|Du^-\|_p^p < \widehat{\lambda}_1(m) \int_\Omega m(z)(u^-)^p dz,$$

a contradiction. This proves the Claim.

Hence we have $u \in C_+ \setminus \{0\}$ and using Theorem 4.89 we conclude that $u \in \text{int}C_+$.

Let $\widehat{u}_1 \in \text{int}C_+$ the positive $(m, p)-$normalized(that is, $\int_\Omega m(z)\widehat{u}_1^p dz = 1$) eigenfunction corresponding to $\widehat{\lambda}_1(m) > 0$. Using the nonlinear Picone's identity (see Theorem 4.91), we have

$$\begin{aligned}
0 &\leq \int_\Omega R(\widehat{u}_1, u) dz \\
&= \|D\widehat{u}_1\|_p^p - \int_\Omega |Du|^{p-2}(Du, D(\frac{\widehat{u}_1^p}{u^{p-1}}))_{\mathbb{R}^N} dz \\
&= \|D\widehat{u}_1\|_p^p - \int_\Omega (-\Delta_p u)\frac{\widehat{u}_1^p}{u^{p-1}} dz \\
&= \|D\widehat{u}_1\|_p^p - \widehat{\lambda}_1(m) \int_\Omega m(z)\widehat{u}_1^p dz - \int_\Omega g(z)\frac{\widehat{u}_1^p}{u^{p-1}} dz \\
&= -\int_\Omega g(z)\frac{\widehat{u}_1^p}{u^{p-1}} dz < 0,
\end{aligned}$$

a contradiction.

This proves that problem (5.35) can not have a solution. □

Now we can have the antimaximum principle for the Dirichlet $p-$Laplacian with a nonnegative weight. In this case, in contrast to the "linear" antimaximum principle, we allow also variation of the weight function $m(\cdot)$.

Theorem 5.28 *If $m \in L^\infty(\Omega) \setminus \{0\}$, $m \geq 0$ and $g \in L^s(\Omega) \setminus \{0\}$, $s > N$, $g \geq 0$, then there exists $\delta > 0$ such that if $\widehat{m} \in L^\infty(\Omega) \setminus \{0\}$, $\widehat{m} \geq 0$, $\|\widehat{m} - m\|_\infty < \delta$ and $\lambda \in (\widehat{\lambda}_1(\widehat{m}), \widehat{\lambda}_1(\widehat{m}) + \delta)$, then every weak solution of*

$$\left\{ \begin{array}{l} -\Delta_p u(z) = \lambda \widehat{m}(z)|u(z)|^{p-2}u(z) + g(z) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0. \end{array} \right\}$$

belongs to $-\text{int}C_+$.

Proof We argue by contradiction. So, suppose we can find $\{\widehat{m}_n\}_{n\in\mathbb{N}} \subseteq L^\infty(\Omega)$, $\{\lambda_n\}_{n\in\mathbb{N}} \subseteq (\widehat{\lambda}_1(\widehat{m}_n), \widehat{\lambda}_1(\widehat{m}_n) + \frac{1}{n})$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega)$ such that

$$\widehat{m}_n \in L^\infty(\Omega) \setminus \{0\},\ \widehat{m}_n \geq 0,\ \widehat{m}_n \to m \text{ in } L^\infty(\Omega), \tag{5.38}$$

$$\lambda_n \to \widehat{\lambda}_1(m), \tag{5.39}$$

$$-\Delta_p u_n = \lambda_n \widehat{m}_n(z)|u_n|^{p-2}u_n + g(z) \text{ in } \Omega,\ u_n|_{\partial\Omega} = 0,\ n \in \mathbb{N}, \tag{5.40}$$

$$u_n \notin -\text{int}C_+ \text{ for all } n \in \mathbb{N}. \tag{5.41}$$

From (5.40) and Proposition 4.82 we have $u_n \in L^\infty(\Omega)$ for all $n \in \mathbb{N}$.

If $\{u_n\}_{n\in\mathbb{N}} \subseteq L^\infty(\Omega)$ is bounded, then by Theorem 4.84 we can find $\alpha \in (0, 1)$ and $c > 0$ such that

$$u_n \in C_0^{1,\alpha}(\overline{\Omega}),\ \|u_n\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq c \text{ for all } n \in \mathbb{N}.$$

Since $C_0^{1,\alpha}(\overline{\Omega}) \hookrightarrow C_0^1(\overline{\Omega})$ compactly, we may assume that

$$u_n \to u \text{ in } C_0^1(\overline{\Omega}). \tag{5.42}$$

Using (5.38), (5.39), (5.40), (5.42), in the limit as $n \to \infty$, we have

$$-\Delta_p u = \widehat{\lambda}_1(m)m(z)|u|^{p-2}u + g(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0.$$

This contradicts Proposition 5.27.

Next assume that $\{u_n\}_{n\in\mathbb{N}} \subseteq L^\infty(\Omega)$ is not bounded. We assume that $\|u_n\|_\infty \to \infty$ and set $y_n = \dfrac{u_n}{\|u_n\|_\infty}$, $n \in \mathbb{N}$. Then $\|y_n\|_\infty = 1$ for all $n \in \mathbb{N}$. From (5.40), we have

$$-\Delta_p y_n = \lambda_n \widehat{m}_n(z)|y_n|^{p-2}y_n + \frac{g(z)}{\|u_n\|_\infty^{p-1}} \text{ in } \Omega,\ y_n|_{\partial\Omega} = 0 \quad \text{for all } n \in \mathbb{N}. \tag{5.43}$$

As before, using the nonlinear regularity theory (see Theorem 4.84), we obtain

$$y_n \to y \text{ in } C_0^1(\overline{\Omega}). \tag{5.44}$$

So, if in (5.43) we pass to the limit as $n \to \infty$ and use (5.44), we obtain

$$-\Delta_p y = \widehat{\lambda}_1(m)m(z)|y|^{p-2}y \text{ in } \Omega,\ y|_{\partial\Omega} = 0. \tag{5.45}$$

Note that $\|y\|_\infty = 1$ (see (5.44)) and so from (5.45) we infer that

$$y \in \text{int}C_+ \cup (-\text{int}C_+).$$

On account of (5.41) and (5.42), we see that we have

$$\begin{aligned} &y \in \text{int}C_+, \\ \Rightarrow &y_n \in \text{int}C_+ \text{ for all } n \geq n_0 \text{ (see (5.44))}. \end{aligned}$$

But then if $\widehat{f}(z) = [\lambda_n - \widehat{\lambda}_1(\widehat{m}_n)]\widehat{m}_n(z)y_n^{p-1} + \frac{g(z)}{\|u_n\|_\infty^{p-1}}$, then $\widehat{f} \in L^s(\Omega) \setminus \{0\}$, $s > N$, $\widehat{f} \geq 0$. So, we contradict Proposition 5.27. This proves the Theorem. □

Finally let us say a few things about the antimaximum principle for the Neumann problem. We consider, the following boundary value problem

$$\left\{ \begin{array}{l} -\Delta_p u(z) = \lambda m(z)|u(z)|^{p-2}u(z) + g(z) \text{ in } \Omega, \\ \dfrac{\partial u}{\partial n_p} = 0 \text{ on } \partial\Omega. \end{array} \right\} \tag{5.46}$$

Now we assume that $m, g \in L^\infty(\Omega) \setminus \{0\}$, $m \geq 0$, $g \geq 0$. We know that in this case the first eigenvalue $\widehat{\lambda}_1(m) = 0$. We will need the following lemma, which replaces Proposition 5.27.

Lemma 5.29 *If $g \in L^\infty(\Omega)$ and $\int_\Omega g(z)dz \neq 0$, then the Neumann problem*

$$-\Delta_p u(z) = g(z) \text{ in } \Omega,\ \frac{\partial u}{\partial n_p} = 0 \text{ on } \partial\Omega. \tag{5.47}$$

has no solution.

Proof If (5.47) had a weak solution $u \in W^{1,p}(\Omega)$, then using the test function $h \equiv 1 \in W^{1,p}(\Omega)$ we obtain

$$0 = \int_{\Omega} g(z)dz \neq 0,$$

a contradiction. Therefore (5.47) has no solution. □

Following the reasoning in the proof of Theorem 5.28, using this time Lemma 5.29 instead of Proposition 5.27, we obtain the following antimaximum principle for the Neumann problem (5.46).

Theorem 5.30 *If $m, g \in L^{\infty}(\Omega) \setminus \{0\}$, $m \geq 0$, $g \geq 0$, then there exists $\delta > 0$ such that if $\widehat{m} \in L^{\infty}(\Omega) \setminus \{0\}$, $\widehat{m} \geq 0$ and $\lambda > 0$ satisfy*

$$\|\widehat{m} - m\|_{\infty} < \delta, \ \lambda \in (0, \delta),$$

then every weak solution of

$$\left\{ \begin{array}{l} -\Delta_p u = \lambda \widehat{m}(z)|u|^{p-2}u + g(z) \text{ in } \Omega, \\ \dfrac{\partial u}{\partial n_p} = 0 \text{ on } \partial\Omega. \end{array} \right\}$$

belongs to $-\text{int}\widehat{C}_+$.

Remark 5.31 Similar results can be proved for other classes of problems, such as Robin problems or problems with indefinite potential terms.

Next we turn our attention to comparison results. We continue with $\Omega \subseteq \mathbb{R}^N$ a bounded domain with a C^2-boundary $\partial\Omega$.

Definition 5.32 Given $h, g \in L^{\infty}(\Omega)$ we write $h \prec g$ if and only if for every $K \subseteq \Omega$ compact, we can find $c_K > 0$ such that $c_K \leq g(z) - h(z)$ for a.a $z \in K$.

Remark 5.33 Evidently if $h, g \in C(\Omega)$ and $h(z) < g(z)$ for all $z \in \Omega$, then $h \prec g$.

We consider a differential operator $\text{div}a(Du)$, where the map $a : \mathbb{R}^N \to \mathbb{R}^N$ satisfies hypotheses H_0' from Sect. 4.3.

Theorem 5.34 *If $\widehat{\xi}, h, g \in L^{\infty}(\Omega)$, $\widehat{\xi}(z) \geq 0$ for a.a $z \in \Omega$, $0 < \widehat{c} \leq g(z) - h(z)$ for a.a $z \in \Omega$ and $u, v \in C^1(\overline{\Omega})$, $0 \leq u \leq v$ and satisfy the following equations*

$$\begin{aligned} &- div\, a(Du) + \widehat{\xi}(z)|u|^{p-2}u = h(z) \text{ in } \Omega, \\ &- div\, a(Dv) + \widehat{\xi}(z)v^{p-1} = g(z) \text{ in } \Omega, \end{aligned}$$

then $u(z) < v(z)$ and $\dfrac{\partial(v-u)}{\partial n}|_{\partial\Omega \cap \{u=v\}} < 0$.

Proof We have

$$-\text{div}[a(Dv) - a(Du)] = g(z) - h(z) - \widehat{\xi}(z)(v^{p-1} - |u|^{p-2}u) \text{ in } \Omega.$$

Let $\{a_k(\cdot)\}_{k=1}^N$ be the components of the map $a(\cdot)$. Then for $y = (y_k)_{k=1}^N$, $y' = (y'_k)_{k=1}^N \in \mathbb{R}^N$, we have

$$\begin{aligned} a(y) - a(y') &= \int_0^1 \frac{d}{dt} a(y' + t(y - y'))dt \\ &= \int_0^1 \nabla a(y' + t(y - y'))(y - y')dt \text{ (by the chaim rule),} \\ \Rightarrow a_k(y) - a_k(y') &= \int_0^1 \sum \frac{\partial a_k}{\partial y_i}(y' + t(y - y'))(y_i - y'_i)dt \text{ for all } k = 1, \cdots, N. \end{aligned}$$

With $\nabla_i = \dfrac{\partial}{\partial y_i}$, $i = \{1, \cdots, N\}$ we introduce the following functions

$$\widehat{c}_{k,i}(z) = \int_0^1 \nabla_i a_k(Du(z) + t(Dv(z) - Du(z))dt$$

for all $k, i = \{1, \cdots, N\}$.

We see that $\widehat{c}_{k,i} \in W^{1,\infty}(\Omega)$ and we introduce the following second order linear differential operator in divergence form

$$L(w) = -\text{div}(\sum_{i=1}^N \widehat{c}_{k,i}(z)\nabla_i w) = -\sum_{k,i=1}^N \nabla_k(\widehat{c}_{k,i}(z)\nabla_i w).$$

Note that $v \in C_0^{1,\alpha}(\overline{\Omega})$ (see Theorem 4.84) and let

$$\eta = v - u \in C^{1,\alpha}(\overline{\Omega}).$$

Since $h \le g$, we have $u \le v$ and so $\eta \ge 0$.
First suppose that for some $z_0 \in \Omega$, we have $\eta(z_0) = 0$. Then

$$v(z_0) = u(z_0).$$

The function $x \to \widehat{\xi}(z)|x|^{p-2}x$ is locally Lipschitz, thus uniformly continuous on $[-\mu, \mu]$ with $\mu = \max\{\|u\|_\infty, \|v\|_\infty\}$. Since $\widehat{\xi} \in L^\infty(\Omega)$, $v(z_0) > 0$ and using the fact that $0 < \widehat{c} \le g(z) - h(z)$ for a.a $z \in \Omega$, we can choose $\varrho > 0$ small such that

$$g(z) - h(z) - \widehat{\xi}(z)[v(z)^{p-1} - u(z)^{p-1}] \ge \frac{\widehat{c}}{2}$$

for a.a $z \in B_\varrho(z_0) = \{z' \in \mathbb{R}^N : |z' - z_0| < \varrho\} \subseteq \Omega$.

Then we have

$$L(\eta)(z) \geq c > 0 \text{ for a.a } z \in B_{\varrho}(z_0).$$

Using either Harnack's inequality (see Theorem 4.89) or the linear maximum principle, we have

$$0 < \eta(z) \text{ for all } z \in B_{\varrho}(z_0),$$

a contradiction.

Now suppose that $z_0 \in \partial\Omega$ and $\eta(z_0) = 0$. Since $\partial\Omega$ is a C^2-manifold, we can find $r > 0$ small and an open ball B_r such that

$$B_r \subseteq \Omega,\ z_0 \in \partial\Omega \cap \partial B_r.$$

Invoking the Hopf maximum principle, we obtain

$$\frac{\partial \eta}{\partial n}(z_0) < 0.$$

We conclude that

$$u(z) < v(z) \text{ for all } z \in \Omega,\ \frac{\partial(v-u)}{\partial n}|_{\partial\Omega\cap\{u=v\}} < 0.$$

□

Remark 5.35 We can introduce the open cone

$$D_+ = \{u \in C^1(\overline{\Omega}) : u(z) > 0 \text{ for all } z \in \Omega,\ \frac{\partial u}{\partial n}|_{\partial\Omega\cap\{u=0\}} < 0\}$$

and say that $v - u \in D_+$.

Theorem 5.36 *If $\widehat{\xi}, h, g \in L^{\infty}(\Omega)$, $\widehat{\xi} \geq 0$, $h \prec g$ and $u \in C^1(\overline{\Omega})$, $v \in intC_+$ solve the following problems*

$$-div\, a(Du) + \widehat{\xi}(z)|u|^{p-2}u = h(z) in\ \Omega,\ u|_{\partial\Omega} \leq 0,$$
$$-div\, a(Dv) + \widehat{\xi}(z)v^{p-1} = g(z)\ in\ \Omega,\ v|_{\partial\Omega} = 0.$$

then $v - u\ intC_+$.

Proof Clearly $u \leq v$. We introduce the following sets

$$E = \{z \in \Omega : u(z) = v(z)\},\ \widehat{E} = \{z \in \Omega : Du(z) = Dv(z) = 0\}.$$

Claim $E \subseteq \widehat{E}$.

Let $y = u - v$. Then $y \leq 0$. Suppose $z \in E$. Then

$$\begin{aligned} &Dy(z) = 0, \\ \Rightarrow &Du(z) = Dv(z). \end{aligned}$$

Arguing indirectly, suppose that $z \notin \widehat{E}$. Then $Dv(z) \neq 0$ and so we can find an open ball $B \subseteq \Omega$ centered at z such that

$$|Du(z')| > 0,\ |Dv(z')| > 0,\ (Du(z'), Dv(z'))_{\mathbb{R}^N} > 0 \text{ for all } z' \in B.$$

From the proof of Theorem 5.34, we know that

$$-\operatorname{div} L(y) = h(z) - g(z) - \widehat{\xi}(z)[|u|^{p-2}u - v^{p-1}] \text{ in } B. \tag{5.48}$$

Choosing B small enough we can have that the linear operator $L(\cdot)$ is strictly elliptic and the right hand side(the forcing term) of (5.48) is strictly negative. So, by the linear maximum principle, we have

$$y(z') < 0 \text{ for all } z' \in B,$$

a contradiction.

Therefore $z \in \widehat{E}$ and so $E \subseteq \widehat{E}$. This proves the Claim.

Recall that $v \in \operatorname{int} C_+$. Therefore $\widehat{E}$ is compact. We choose $U \subseteq \Omega$ open such that

$$\widehat{E} \subseteq U \subseteq \overline{U} \subseteq \Omega. \tag{5.49}$$

For $\varepsilon > 0$ small, we have

$$u(z) + \varepsilon < v(z) \text{ for all } z \in \partial U \text{ (see (5.49))}, \tag{5.50}$$

$$h(z) + \varepsilon < g(z) \text{ for a.a } z \in U \text{ (recall } h \prec g). \tag{5.51}$$

We choose $\delta > 0$ small so that

$$\begin{aligned} &|\widehat{\xi}(z)(|x|^{p-2}x - |w|^{p-2}w)| \\ \leq &\|\widehat{\xi}\|_\infty ||x|^{p-2}x - |w|^{p-2}w| < \varepsilon \text{ for a.a } z \in U,\ \text{all } |x - w| < \delta. \end{aligned}$$

Then for a.a $z \in U$, we have

$$
\begin{aligned}
&-\operatorname{div} a(D(u+\delta)+\widehat{\xi}(z)|u+\delta|^{p-2}(u+\delta)\\
=&-\operatorname{div} a(Du)+\widehat{\xi}(z)|u+\delta|^{p-2}(u+\delta)\\
=&h(z)+\widehat{\xi}(z)[|u+\delta|^{p-2}(u+\delta)-|u|^{p-2}u]\\
\leq&h(z)+\varepsilon\\
<&g(z) \text{ (see (5.51))}\\
=&-\operatorname{div} a(Dv)+\widehat{\xi}(z)v^{p-1},\\
\Rightarrow& u(z)+\delta\leq v(z) \text{ for all } z\in\overline{U},\\
\Rightarrow& u(z)<v(z) \text{ for all } z\in\overline{U},\\
\Rightarrow& E=\emptyset.
\end{aligned}
$$

Also by Theorem 4.89 we have

$$
\begin{aligned}
&\frac{\partial y}{\partial n}|_{\partial\Omega}>0,\\
\Rightarrow& v-u\in \operatorname{int} C_+.
\end{aligned}
$$

□

In fact we can extend Theorems 5.34 and 5.36 to singular problems. The proofs are similar with minor straightforward modifications.

So, let $\vartheta : (0,+\infty)\to(0,+\infty)$ be a locally Lipschitz function, which has the following two properties:

(a) there exists $\eta\in(0,1)$ such that

$$0<c_0\leq\liminf_{x\to 0^+}\vartheta(x)x^{\eta}\leq\limsup_{x\to 0^+}\vartheta(x)x^{\eta}\leq c_1.$$

(b) $\vartheta(\cdot)$ is nonincreasing.

Hypothesis (a) above implies that $\vartheta(\cdot)$ is singular at $x=0$. In the literature usually we encounter the function

$$\vartheta(x)=cx^{-\eta} \text{ for all } x>0, \text{ with } c>0.$$

Theorem 5.37

(a) *If $\vartheta(\cdot)$ is as above, $\widehat{\xi}, h, g\in L^{\infty}(\Omega)$, $\widehat{\xi}\geq 0$, $0<\widehat{c}\leq g(z)-h(z)$ for a.a $z\in\Omega$ and $u, v\in C^1(\overline{\Omega})$, $0\leq u\leq v$ and satisfy*

$$-\,div\, a(Du) + \widehat{\xi}(z)u^{p-1} - \vartheta(u) = h(z) \text{ in } \Omega,$$
$$-\,div\, a(Dv) + \widehat{\xi}(z)v^{p-1} - \vartheta(v) = g(z) \text{ in } \Omega,$$

then $v - u \in D_+$ *(see Remark 5.35).*

(b) *If* $\vartheta(\cdot)$ *is as above,* $\widehat{\xi}, h, g \in L^\infty(\Omega)$, $\widehat{\xi} \geq 0$, $h \prec g$ *and* $u \in C^1(\overline{\Omega})$, $v \in intC_+$ *satisfy*

$$-\,diva(Du) + \widehat{\xi}(z)u^{p-1} - \vartheta(u) = h \text{ in } \Omega,\ u|_{\partial\Omega} = 0,$$
$$-\,diva(Dv) + \widehat{\xi}(z)v^{p-1} - \vartheta(v) = g \text{ in } \Omega,\ v|_{\partial\Omega} = 0,$$

then $v - u \in intC_+$.

The same results are also true for anisotropic problems. Again the proofs are similar with minor straightforward changes. We formulate the results for the anisotropic $(p, q)-$Laplacian with the standard singularity.

Suppose $p, q \in C^{0,1}(\overline{\Omega})$ with $1 < q_- \leq q_+ < p_-$ and $\eta \in C(\overline{\Omega})$ with $0 < \eta(z) < 1$ for all $z \in \overline{\Omega}$.

Theorem 5.38

(a) *If* p, q, η *are as above,* $\widehat{\xi}, h, g \in L^\infty(\Omega)$, $\widehat{\xi} \geq 0$, $0 < \widehat{c} \leq g(z) - h(z)$ *for a.a* $z \in \Omega$, $u, v \in C^1(\overline{\Omega})$, $0 \leq u \leq v$ *satisfy*

$$-\Delta_{p(z)}u - \Delta_{q(z)}u + \widehat{\xi}(z)u^{p(z)-1} - cu^{-\eta(z)} = h(z) \text{ in } \Omega,$$
$$-\Delta_{p(z)}v - \Delta_{q(z)}v + \widehat{\xi}(z)v^{p(z)-1} - cv^{-\eta(z)} = g(z) \text{ in } \Omega,$$

then $v - u \in D_+$ *(see Remark 5.35).*

(b) *If* p, q, η *are as above,* $\widehat{\xi}, h, g \in L^\infty(\Omega)$, $\widehat{\xi} \geq 0$, $h \prec g$ *and* $u \in C_+$, $v \in intC_+$, $0 < u(z)$ *for all* $z \in \Omega$ *satisfy*

$$-\Delta_{p(z)}u - \Delta_{q(z)}u + \widehat{\xi}(z)u^{p(z)-1} - cu^{-\eta(z)} = h(z) \text{ in } \Omega,$$
$$-\Delta_{p(z)}v - \Delta_{q(z)}v + \widehat{\xi}(z)v^{p(z)-1} - cv^{-\eta(z)} = g(z) \text{ in } \Omega,$$

then $v - u \in intC_+$.

We continue with some more comparison results. The next one, is known in the literature as the "tangency principle".

Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with Lipschitz boundary $\partial\Omega$. We consider the following two differential inequalities in divergence form

$$\text{div } a(z, u, Du) + f(z, u, Du) \geq 0 \text{ in } \Omega, \tag{5.52}$$

$$\text{div } a(z, v, Dv) + f(z, v, Dv) \leq 0 \text{ in } \Omega. \tag{5.53}$$

Here $a : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}^N$ is a continuous function such that

- for all $z \in \Omega$, $a(z, \cdot, \cdot) \in C^1(\mathbb{R} \times \mathbb{R}^N, \mathbb{R}^N)$;
- for every $\varrho > 0$, there exists $c_\varrho > 0$ such that

$$|a(z, x, y)| \leq c_\varrho \text{ for all } z \in \Omega, \text{ all } |x|, |y| \leq \varrho.$$

Evidently these conditions incorporate the $p-$Laplacian$(a(y) = |y|^{p-2}y$, $1 < p < \infty)$ and the $(p, q)-$Laplacian$(a(y) = |y|^{p-2}y + |y|^{q-2}y$, $1 < q < p < \infty)$.

For the function $f : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$ we assume that it is a measurable and for a.a $z \in \Omega$, $f(z, \cdot, \cdot)$ is locally lower Lipschitz, that is,

$$f(z, x, y) - f(z, x', y') \geq -c_1(x - x') - c_2|y - y'|$$

for a.a $z \in \Omega$, all compact set of arguments with $x > x'$ and with $c_1, c_2 > 0$.

Theorem 5.39 *If $u, v \in C^1(\Omega)$ are weak solutions of the differential inequalities (5.52) and (5.53) respectively, $u \leq v$ and at least one of the following two $N \times N-$matrices*

$$\nabla_y a(z, u, Du) \text{ or } \nabla_y a(z, v, Dv)$$

is positive definite on Ω, then either $u = v$ or $u(z) < v(z)$ for all $z \in \Omega$.

Proof Let $\Omega_0 = \{z \in \Omega : u(z) = v(z)\}$. Evidently Ω_0 is relatively closed in Ω. If we show that Ω_0 is also open, then we can conclude that $\Omega_0 = \Omega$ or $\Omega_0 = \emptyset$.

Suppose $\Omega_0 \neq \emptyset$ and let $z_0 \in \Omega_0$. Then

$$u(z_0) = v(z_0) \text{ and } Du(z_0) = Dv(z_0) \text{ (recall } u \leq v).$$

Let $\varrho \in (0, 1)$ be small so that $\overline{B}_{3\varrho}(z_0) \subseteq \Omega$. We have

$$\text{div}[a(z, v, Dv) - a(z, u, Du)] + f(z, v, Dv) - f(z,u, Du) \leq 0 \text{ in } \Omega \quad \text{(see (5.52) and (5.53)).} \tag{5.54}$$

We set $y = v - u \geq 0$ and have

$$|a(z, v, Dv) - a(z, u, Du)| \leq c_1[y + |Dy|] \text{ in } B_{3\varrho}(z_0) \text{ for some } c_1 > 0, \tag{5.55}$$

$$-[f(z, v, Dv) - f(z, u, Du)] \leq c_2[y + |Dy|] \text{ in } B_{3\varrho}(z_0) \text{ for some } c_2 > 0.$$

To fix things, we assume that $\nabla_y a(z, u, Du) > 0$ in Ω. Let $\lambda > 0$ be the smallest eigenvalue of this matrix in $B_{3\varrho}(z_0)$. By the mean value theorem, for some ξ in the line segment determined by Du and Dv, we have

$$\begin{aligned}
&(a(z,u,Dv)-a(z,u,Du),Dy)_{\mathbb{R}^N}\\
=&(\nabla_y(a(z,u,\xi)Dy,Dy)_{\mathbb{R}^N}\\
=&(\nabla_y(a(z,u,Du)Dy,Dy)_{\mathbb{R}^N}+\circ(|Dy|^2)\\
\geq&\frac{1}{2}\lambda|Dy|^2.
\end{aligned}\tag{5.56}$$

Also we have

$$\begin{aligned}
&(a(z,v,Dv)-a(z,u,Du),Dy)_{\mathbb{R}^N}\\
=&(a(z,v,Dv)-a(z,u,Dv),Dy)_{\mathbb{R}^N}+(a(z,u,Dv)-a(z,u,Du),Dy)_{\mathbb{R}^N}\\
\geq&-c_3y|Dy|+\frac{1}{2}\lambda|Dy|^2 \text{ for some } c_3>0 \text{ (see (5.56))}\\
\geq&\frac{\lambda}{4}|Dy|^2-\frac{c_3^2}{\lambda}y^2 \text{ (by Young's inequality).}
\end{aligned}$$

From a weak form of the Harnack inequality (see Pucci and Serrin [423], p.34), we have

$$\begin{aligned}
&\int_{B_\varrho(z_0)} y dz\leq 0\\
\Rightarrow&y|_{B_\varrho(z_0)}=0,\\
\Rightarrow&B_\varrho(z_0)\subseteq\Omega_0 \text{ and so } \Omega_0 \text{ is open, thus } \Omega_0=\Omega.
\end{aligned}$$

This proves the theorem. □

The next comparison theorem is related to Theorem 5.36. Now we do not have the potential term $\widehat{\xi}(z)|u|^{p-2}u$, but we relax the hypothesis that $h\prec g$. So, let $a:\mathbb{R}^N\to\mathbb{R}^N$ be a map which satisfies hypotheses H_0^1 from Sect. 4.3 (the hypotheses used in the nonlinear regularity theory, see Theorem 4.84)

Theorem 5.40 *If $h,\ g\in L^\infty(\Omega)$, $h\leq g$, $h\not\equiv g$, $u,v\in C^1(\overline{\Omega})$, $u\leq v$ and they satisfy*

$$-diva(Du)=h(z)\ in\ \Omega,\ \frac{\partial u}{\partial n}|_{\partial\Omega}<0,$$

$$-diva(Dv)=g(z)\ in\ \Omega,\ \frac{\partial v}{\partial n}|_{\partial\Omega}<0,$$

then $v-u\in D_+$ (see Remark 5.35)

Proof We can find $\delta>0$ small such that if $\Omega_\delta=\{z\in\Omega: d(z,\partial\Omega)<\delta\}$, then

$$|D((1-t)u(z)+tv(z))| \geq \eta > 0 \text{ for all } z \in \overline{\Omega}_0, \text{ all } t \in [0,1].$$

By hypothesis we have

$$-\operatorname{div}[a(Dv)-a(Du)] = g(z)-h(z) \geq 0 \text{ in } \Omega.$$

As in the proof of Theorem 5.36, we obtain

$$L(v-u)(z) \geq 0 \text{ in } \Omega_0, \tag{5.57}$$

with $L(\cdot)$ being a strictly elliptic differential operator (see the proof of Theorem 5.36). We show that $v \neq u$ in Ω_δ. Suppose that $v(z) = u(z)$ for all $z \in \Omega_\delta$. Then $h_1 = h_2$ in Ω_δ. Since $h_1 \neq h_2$, we must have $h_1 \neq h_2$ in $\Omega \setminus \Omega_\delta$. We choose a function $\vartheta \in W^{1,p}(\Omega)$ such that

$$\vartheta > 0 \text{ in } \Omega, \ \vartheta|_{\Omega \setminus \Omega_\delta} = 1 \tag{5.58}$$

By $\langle \cdot, \cdot \rangle_{\partial\Omega}$ we denote the duality brackets for the pair $(W^{-\frac{1}{p'},p'}(\partial\Omega), W^{\frac{1}{p'},p}(\partial\Omega))(\frac{1}{p}+\frac{1}{p'}=1)$. Using the nonlinear Green's identity (see Theorem *I*-4.106), we obtain

$$\begin{aligned}
&\int_\Omega h\vartheta\, dz \\
&= \int_\Omega (a(Du), D\vartheta)_{\mathbb{R}^N} dz - \langle \frac{\partial u}{\partial n_a}, \vartheta \rangle_{\partial\Omega} \\
&= \int_{\Omega_\delta} (a(Du), D\vartheta)_{\mathbb{R}^N} dz - \langle \frac{\partial u}{\partial n_a}, \vartheta \rangle_{\partial\Omega} \text{ (see (5.58))} \\
&= \int_{\Omega_\delta} (a(Dv), D\vartheta)_{\mathbb{R}^N} - \langle \frac{\partial v}{\partial n_a}, \vartheta \rangle_{\partial\Omega} \\
&= \int_\Omega g\vartheta\, dz,
\end{aligned}$$

a contradiction since $\int_\Omega (g-h)dz > 0$ $(g \neq h)$. It follows that $v \neq u$ in Ω_δ. But then from (5.57) and the linear maximum principle, we infer that

$$u(z) < v(z) \text{ for all } z \in \Omega_\delta. \tag{5.59}$$

Let $E = \{z \in \Omega : u(z) = v(z)\}$. From (5.59) we see that $E \subseteq \Omega$ is compact. So, we can find $U \subseteq \mathbb{R}^N$ open such that

$$E \subseteq U \subseteq \overline{U} \subseteq \Omega,$$
$$\Rightarrow \partial U \cap E = \emptyset$$

It follows that $(v-u)|_{\partial U} > 0$. The set ∂U is compact, we can find $\varepsilon > 0$ such that

$$u \leq v - \varepsilon \text{ on } \partial U$$

We have $u, v - \varepsilon \in C^1(\widehat{U})$, $u \leq v - \varepsilon$ on ∂V. Using the test function $\widehat{h} = (u - (v-\varepsilon))^+ \in W^{1,p}(\Omega)$, we obtain

$$\int_{\{u > v-\varepsilon\} \cap U} (a(Du) - a(Dv), Du - Dv)_{\mathbb{R}^N} dz \leq 0,$$
$$\Rightarrow u \leq v - \varepsilon \text{ in } U \text{ (since } a(\cdot) \text{ is strictly monotone)}$$

a contradiction, unless $E = \emptyset$. Therefore we conclude that

$$u(z) < v(z) \text{ for all } z \in \Omega,$$
$$\Rightarrow v - u \in D_+.$$

□

Embedded in the above proof, we detect two useful results.
Consider the following differential inequality

$$-\operatorname{div} a(z, Du) + f(z,u) \leq -\operatorname{div} a(z, Dv) + f(z,v) \text{ in } \Omega. \tag{5.60}$$

Assume that

- $a : \Omega \times \mathbb{R}^N$ is a Caratheodory function such that for a.a $z \in \Omega$ $a(z, \cdot)$ is strictly monotone.
- $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that for a.a $z \in \Omega$ $f(z, \cdot)$ is nonincreasing.

Proposition 5.41 *If $u, v \in W^{1,\infty}(\Omega)$ satisfy (5.60) and $u \leq v$ on $\partial\Omega$, then $u(z) \leq v(z)$ for all $z \in \overline{\Omega}$.*

Proposition 5.42 *If $u, v \in C^1(\overline{\Omega})$, $u \leq v$, $-diva(z, Du) \leq diva(z, Dv)$ in Ω and $E = \{z \in \Omega : u(z) = v(z)\}$ is either compact or discrete, then $E = \emptyset$ and so $u(z) < v(z)$ for all $z \in \Omega$.*

Next we will present a result which is an outgrowth of the nonlinear regularity theory (see Theorem 4.84). The result says that a C_0^1-local minimizer of the energy functional, is also a $W_0^{1,p}$-local minimizer. In fact the result is also true for C^1, $W^{1,p}$ local minimizers, but for simplicity here we will present only the Dirichlet case.

Consider a map $a : \mathbb{R}^N \to \mathbb{R}^N$ which satisfies hypotheses H_0' from Sect. 4.3. Recall $G : \mathbb{R}^N \to \mathbb{R}$ is the primitive of $a(\cdot)$. Also let $f : \Omega \times \mathbb{R} \to \mathbb{R}$ be a Caratheodory function such that

$$|f(z,x)| \leq a(z)(1+|x|^{r-1}) \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R},$$

with $a \in L^\infty(\Omega)$ and $r \in (p, p^*)$. We set $F(z,x) = \int_0^x f(z,s)ds$ and consider the functional $\varphi : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\varphi(u) = \int_\Omega G(Du)dz - \int_\Omega F(z,u)dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

Then $\varphi \in C^1(W_0^{1,p}(\Omega))$ and $\langle \varphi'(u), h\rangle = \langle V(u), h\rangle - \int_\Omega f(z,u)hdz$ for all $u, h \in W_0^{1,p}(\Omega)$ and with $\langle V(u), h\rangle = \int_\Omega (a(Du), Dh)_{\mathbb{R}^N} dz$.

Definition 5.43

(a) We say that $u_0 \in W_0^{1,p}(\Omega)$ is a "$C_0^1(\overline{\Omega})$-local minimizer of φ", if there exists $\varepsilon > 0$ such that

$$\varphi(u_0) \leq \varphi(u_0+h) \text{ for all } h \in C_0^1(\overline{\Omega}), \|h\|_{C_0^1(\overline{\Omega})} \leq \varepsilon.$$

(b) We say that $u_0 \in W_0^{1,p}(\Omega)$ is a "$W_0^{1,p}(\overline{\Omega})$-local minimizer of φ", if there exists $\varepsilon > 0$ such that

$$\varphi(u_0) \leq \varphi(u_0+h) \text{ for all } h \in W_0^{1,p}(\Omega), \|h\| \leq \varepsilon.$$

(by $\|\cdot\|$ we denote the norm of $W_0^{1,p}(\Omega)$ which on account of the Poincare inequality is $\|u\| = \|Du\|_p$ for all $u \in W_0^{1,p}(\Omega)$).

Theorem 5.44 *If $u_0 \in W_0^{1,p}(\Omega)$ is a $C_0^1(\overline{\Omega})$-local minimizer of $\varphi(\cdot)$, then u_0 is also a $W_0^{1,p}(\Omega)$-local minimizer of $\varphi(\cdot)$.*

Proof The hypothesis implies that

$$\langle \varphi'(u_0), h\rangle = 0 \text{ for all } h \in C_0^1(\overline{\Omega})$$

The density of $C_0^1(\overline{\Omega})$ in $W_0^{1,p}(\Omega)$ implies that

$$\langle \varphi'(u_0), h\rangle = 0 \text{ for all } h \in W_0^{1,p}(\Omega) \text{ (that is, } u_0 \in K_\varphi),$$

$$\Rightarrow u_0 \in C_0^1(\overline{\Omega}) \text{ (see Theorem 4.84).}$$

Suppose that u_0 is not a $W_0^{1,p}(\overline{\Omega})$-local minimizer of $\varphi(\cdot)$. Since $W_0^{1,p}(\Omega) \hookrightarrow L^r(\Omega)$ continuously (recall $r \in (p, p^*)$), we can find $\delta > 0$ such that

$$-\infty < \widehat{m}_\delta = \inf\{\varphi(u_0 + h) : h \in W_0^{1,p}(\Omega), \|h\|_r \leq \delta\} < \varphi(u_0). \tag{5.61}$$

We can find $h_\delta \in W_0^{1,p}(\Omega)$ such that

$$\widehat{m}_\delta = \varphi(u_0 + h_\delta), \|h_\delta\|_r \leq \delta \tag{5.62}$$

From (5.61), (5.62) it follows

$$\{h_\delta\}_{\delta \in (0,1)} \subseteq W_0^{1,p}(\Omega) \setminus \{0\} \text{ is bounded} \tag{5.63}$$

If $\varrho_\delta = \|h_\delta\|_r^r > 0$, then

$$\varphi(u_0 + h_\delta) = \inf\{\varphi(u_0 + h) : h \in W_0^{1,p}(\Omega), \|h\|_r^r = \varrho_\delta\}.$$

By the Lagrange multiplier rule (see Theorem 3.87), we can find $\lambda_\delta \in \mathbb{R}$ such that

$$\varphi'(u_0 + h_\delta) = -\lambda_\delta |h_\delta|^{r-2} h_\delta \text{ in } W^{-1,p'}(\Omega), \tag{5.64}$$

$$\Rightarrow V(u_0 + h_\delta) = N_f(u_0 + h_\delta) - \lambda_\delta |h_\delta|^{r-2} h_\delta \tag{5.65}$$

with $N_f(\cdot)$ being the Nemytskii map corresponding to $f(\cdot, \cdot)$, that is, $N_f(u)(\cdot) = f(\cdot, u(\cdot))$ for all $u \in W_0^{1,p}(\Omega)$, see Definition I-6.14. On (5.64) we act with $h_\delta \in W_0^{1,p}(\Omega)$ and obtain

$$\begin{aligned} -\lambda_\delta \|h_\delta\|_r^r &= \langle \varphi'(u_0 + h_\delta), h_\delta \rangle \\ &= \lim_{\vartheta \downarrow 0} \frac{\varphi(u_0 + (1-\vartheta)h_\delta) - \varphi(u_0 + h_\delta)}{-\vartheta} \leq 0 \text{ (see (5.61))} \end{aligned}$$

$$\Rightarrow \lambda_\delta \geq 0$$

We have

$$V(u_0) = N_f(u_0) \text{ in } W^{-1,p'}(\Omega) \tag{5.66}$$

We subtract (5.66) from (5.65). Then

$$\widehat{V}(h_\delta) = N_{\widehat{f}}(h_\delta) - \lambda_\delta |h_\delta|^{r-2} h_\delta \text{ in } W^{-1,p'}(\Omega) \tag{5.67}$$

with $\widehat{V} : W_0^{1,p}(\Omega) \to W^{-1,p'}(\Omega)$ defined by

$$\langle \widehat{V}(u), h\rangle = \int_\Omega (\widehat{a}(z, Du), Dh)_{\mathbb{R}} dz \text{ for all } u, h \in W_0^{1,p}(\Omega),$$

where $\widehat{a}(z, y) = a(Du_0(z) + y) - a(Du_0(z))$ for all $y \in \mathbb{R}^N$ and

$$\widehat{f}(z, x) = f(z, u_0(z) + x) - f(z, u_0(z)) \text{ for all } x \in \mathbb{R}.$$

From (5.67) and the continuity of the Lagrange multiplier on $\delta > 0$, we infer that there exists $c_1 > 0$ such that

$$\|h_\delta\|_\infty \leq c_1 \text{ for all } \delta \in (0, 1) \text{ (see Proposition 4.82)},$$

$$\Rightarrow h_\delta \in C_0^{1,\alpha}(\overline{\Omega}), \|h_\delta\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq c_2 \text{ for all } \delta \in (0, 1), \text{ some } c_2 > 0.$$

The compact embedding of $C_0^{1,\alpha}(\overline{\Omega})$ into $C_0^1(\overline{\Omega})$ implies that

$$\|h_\delta\|_{C_0^1(\overline{\Omega})} \to 0 \text{ (see (5.62))},$$

which contradicts the hypothesis that u_0 is a $C_0^1(\overline{\Omega})$-local minimizer of $\varphi(\cdot)$. This proves that u_0 is a $W_0^{1,p}(\Omega)$-local minimizer of $\varphi(\cdot)$. □

The result remains true, if instead of the Dirichlet case, we consider the Neumann/Robin case, with the ambient space being $W^{1,p}(\Omega)$.

The result is also true for anisotropic functionals. So, let $p, q \in E_1 \cap C^{0,1}(\overline{\Omega})$, $q(z) < p(z)$ for all $z \in \overline{\Omega}$ and $l : \Omega \times \mathbb{R} \to \mathbb{R}$ Caratheodory such that

$$l(z, x) = f(z) + l_0(z, x)$$

with $f \in L^s(\Omega)$ $s > N$ and $l_0 : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function satisfying

$$|l_0(z, x)| \leq a(z)(1 + |x|^{r(z)-1}) \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R},$$

with $a \in L^\infty(\Omega)$, $r \in E_1 \cap C(\overline{\Omega})$, $r(z) < p^*(z)$ for all $z \in \overline{\Omega}$. We set $L(z, x) = \int_\Omega l(z, s)ds$ and consider the C^1-functional $\widehat{\varphi} : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{\varphi}(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega L(z, u)dz$$

for all $u \in W_0^{1,p(z)}(\Omega)$.

With a similar proof, using this time the anisotropic regularity theory, we obtain.

Theorem 5.45 *If $u_0 \in W_0^{1,p(z)}(\Omega)$ is a $C_0^1(\overline{\Omega})$-local minimizer of $\widehat{\varphi}(\cdot)$, then u_0 is also a $W_0^{1,p(z)}(\Omega)$-local minimizer of $\varphi(\cdot)$.*

We mention that this result is also true for singular problems that is, problems where the reaction is of the form

$$f(z, x) = \xi(z)x^{-\eta} + l_0(z, x)$$

with $\xi \in L^\infty(\Omega) \setminus \{0\}, \xi(z) \geq 0$ for $a.a\ z \in \Omega$ and $l_0(z, x)$ is a Caratheodory function as above. The proof is similar using this time the regularity theory for singular problems.

Finally we want to examine the properties of the differential operator $\mathrm{div} a(Du)$. Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with Lipschitz boundary $\partial\Omega$. Also, let $G : \Omega \times \mathbb{R}^N \to \mathbb{R}$ be a function which satisfies the following conditions.

H : $G : \Omega \times \mathbb{R}^N \to \mathbb{R}$ is a function such that $G(z, 0) = 0$ for a.a $z \in \Omega$ and

(i) for all $y \in \mathbb{R}^N$, $z \to G(z, y)$ is measurable;
(ii) for a.a $z \in \Omega$, $G(z, \cdot) \in C^1(\mathbb{R}^N, \mathbb{R})$ and it is strictly convex;
(iii) $|\nabla_y G(z, y)| \leq \widehat{a}(z)(1 + |y|^{p-1})$ for a.a $z \in \Omega$, all $y \in \mathbb{R}^N$, with $\widehat{a} \in L^\infty(\Omega)$ and $1 < p < \infty$;
(iv) $c_0|y|^p \leq pG(z, y)$ for a.a $z \in \Omega$, all $y \in \mathbb{R}^N$, for some $c_0 > 0$.

Remark 5.46 $G(z, y) = \dfrac{a_0(z)}{p}|y|^p$ and $G(z, y) = \dfrac{a_0(z)}{p}|y|^p + \dfrac{a_1(z)}{q}|y|^q$ satisfy the above hypotheses.

We set $a(z, y) = \nabla_y G(z, y)$ and consider the nonlinear map $V : W_0^{1,p}(\Omega) \to W^{1,p}(\Omega)^*$ is defined by

$$\langle V(u), h\rangle = \int_\Omega (a(z, Du), Dh)_{\mathbb{R}^N} dz \text{ for all } u, h \in W^{1,p}(\Omega).$$

In the next proposition we prove the main properties of this map.

Proposition 5.47 *If hypotheses H hold, then the map $V(\cdot)$ is bounded (that is, maps bounded sets to bounded sets), continuous monotone (therefore it is maximal monotone too) and of type $(S)_+$.*

Proof The boundedness and continuity of $V(\cdot)$ follow from hypothesis H(iii) and Theorem I-6.16. Since $G(z, \cdot)$ is strictly convex, $a(z, \cdot)$ is strictly monotone and so the map $V(\cdot)$ is monotone. Then Proposition I-7.36 implies that $V(\cdot)$ is maximal monotone. It remains to show that $V(\cdot)$ is of type $(S)_+$. To this end, let $\{v_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega)$ be such that

$$u_n \xrightarrow{w} u \text{ in } W^{1,p}(\Omega), \ \limsup_{n\to\infty}\langle V(u_n), u_n - u\rangle \leq 0. \tag{5.68}$$

(see Definition I-7.71). We have

$$\limsup_{n\to\infty}\langle V(u_n) - V(u), u_n - u\rangle \leq 0. \text{ (see (5.68))}$$

$$\Rightarrow \lim_{n\to\infty} \langle V(u_n) - V(n), u_n - u\rangle = 0 \text{ (sine } V(\cdot) \text{ is monotone).} \tag{5.69}$$

Let $\eta_n(z) = (a(z, Du_n) - a(z, Du), Du_n - Du)_{\mathbb{R}^N}$. Evidently

$$\eta_n(z) \geq 0 \text{ for a.a } z \in \Omega, \text{ all } n \in \mathbb{N}.$$

On account of (5.69) we can say that

$$\eta_n(z) \to 0 \text{ for a.a } z \in \Omega, 0 \leq \eta_n(z) \leq k(z) \text{ for a.a } z \in \Omega \tag{5.70}$$

with $k \in L^1(\Omega)$. The convexity of $G(z, \cdot)$ and hypothesis H(iv), imply that

$$\frac{c_0}{p}|y|^p \leq G(z, y) \leq (a(z, y), y)_{\mathbb{R}^N} \text{ for a.a } z \in \Omega, \text{ all } y \in \mathbb{R}^N. \tag{5.71}$$

Using (5.70), (5.71), we have

$$\begin{aligned} k(z) \geq \eta_n(z) \geq & \frac{c_0}{p}[|Du_n|^p + |Du|^p] \\ & -(\widehat{a}(z)(1 + |Du|^{p-1})|Du_n| \\ & -(\widehat{a}(z)(1 + |Du_n|^{p-1})|Du| \end{aligned} \tag{5.72}$$

for a.a $z \in \Omega$, all $n \in \mathbb{N}$.

From (5.72) we see that for all $z \in \Omega \setminus E$, $|E|_N = 0$,

$$\{Du_n(z)\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^N \text{ is bounded.}$$

Then for a suitable subsequence (in general depending on $z \in \Omega \setminus E$), we have

$$Du_n(z) \to \xi(z) \text{ for all } z \in \Omega \setminus E,$$

$$\Rightarrow a(z, Du_n(z)) \to a(z, \xi(z))$$

for all $z \in \Omega \setminus E$, $|E_1|_N = 0$, $E \subseteq E_1$.

In the limit as $n \to \infty$, we obtain

$$(a(z, \xi(z)) - a(z, Du(z)), \xi(z) - Du(z)) = 0 \text{ for all } z \in \Omega \setminus E_1,$$

$$\xi(z) = Du(z) \text{ for a.a } z \in \Omega$$

(since $a(z, \cdot)$ is strictly monotone).

Therefore, for the original sequence, we have

$$Du_n(z) \to Du(z) \text{ for a.a } z \in \Omega \tag{5.73}$$

From (5.72) we see that $\{|Du_n|^p\}_{n\in\mathbb{N}} \subseteq L^1(\Omega)$ is uniformly integrable. Then (5.73) and Vitali's Theorem (see Theorem I-2.147), imply that

$$\|Du_n\|_p \to \|Du\|_p \tag{5.74}$$

The space $L^p(\Omega, \mathbb{R}^N)$ is uniformly convex, thus it has the Kadee-Klee property (see Definition I-3.242 and Proposition I-3.244). So, from (5.68) and (5.74) it follows that

$$Du_n \to Du \text{ in } L^p(\Omega, \mathbb{R}^N),$$

$$\Rightarrow u_n \to u \text{ in } W^{1,p}(\Omega).$$

This proves that $V(\cdot)$ is of type $(S)_+$. □

Remark 5.48 If instead we consider the space $W_0^{1,p}(\Omega)$, then $V(\cdot)$ is strictly monotone.

If $\vartheta : W^{1,p}(\Omega) \to \mathbb{R}$ is defined by

$$\vartheta(u) = \int_\Omega G(z, Du)dz \text{ for all } u \in W^{1,p}(\Omega),$$

then $\vartheta \in C^1(W^{1,p}(\Omega))$ and

$$\langle \vartheta'(u), h\rangle = \langle V(u), h\rangle = \int_\Omega (a(z, Du), Dh)_{\mathbb{R}^N} dz$$

for all $u, h \in W^{1,p}(\Omega)$.

Proposition 5.47 will be useful in checking the C-condition for the energy function of the boundary value problem under consideration.

Consider a continuous function $\mu : \mathbb{R} \to \mathbb{R}_+ = [0, \infty)$ which satisfies the following property:

"For every $\varepsilon > 0$, there exist $c_\varepsilon > 0$ such that

$$|\mu(s + t) - \mu(s)| \le \varepsilon\mu(s) + c_\varepsilon\mu(t) \tag{5.75}$$

for all $s, t \in \mathbb{R}$".

Convex functions satisfy this property. The next lemma is the general form of the Brezis-Lieb Lemma, which we encountered in Problem I-2.15(there $\mu(s) = |s|^p$).

Lemma 5.49 *If $\mu : \mathbb{R} \to \mathbb{R}_+$ is continuous and satisfies (5.75), $\{u_n\}_{n\in\mathbb{N}}$ is a sequence of measurable functions on Ω such that*

$$u_n(z) \to u(z) \text{ for a.a } z \in \Omega,$$

$$\sup_{n\in\mathbb{N}} \int_\Omega \mu(u_n)dz < \infty, \int_\Omega \mu(u)dz < \infty,$$

then $\int_\Omega |\mu(u_n) - \mu(u) - \mu(u_n - u)|dz \to 0$ *as* $n \to \infty$.

Proof From (5.75) we have

$$|\mu(u_n) - \mu(u_n - u)| \leq \varepsilon\mu(u_n - u) + c_\varepsilon\mu(u)$$

$$\Rightarrow \mu(u_n - u) \leq 2[\mu(u_n) + c_{\frac{1}{2}}\mu(u)] \text{ use } \varepsilon = \frac{1}{2},$$

$$\Rightarrow M = \sup_{n\in\mathbb{N}} \int_\Omega \mu(u_n - u)dz < \infty.$$

Also, we have

$$k_{n,\varepsilon} = [|\mu(u_n) - \mu(u) - \mu(u_n - u)| - \varepsilon\mu(u_n - u)]^+ \leq (1 + c_\varepsilon)\mu(u),$$

$$\Rightarrow \int_\Omega k_{n,\varepsilon}dz \to 0 \text{ (by the dominated convergence theorem).}$$

But we have

$$|\mu(u_n) - \mu(u) - \mu(u_n - u)| \leq k_{n,\varepsilon} + \varepsilon\mu(u_n - u),$$

$$\Rightarrow \limsup_{n\to\infty} \int_\Omega |\mu(u_n) - \mu(u) - \mu(u_n - u)|dz \leq M\varepsilon.$$

Let $\varepsilon \to 0^+$ to conclude that

$$\int_\Omega |\mu(u_n) - \mu(u) - \mu(u_n - u)|dz \to 0 \text{ as } n \to \infty.$$

□

We consider the double phase differential operator defined by

$$u \to -div(a(z)|Du|^{p-2}Du + |Du|^{q-2}Du)$$

with $a \in L^\infty(\Omega) \setminus \{0\}$, $a(z) \geq 0$ for a.a $z \in \Omega$, $1 < q < p < N$, $\frac{p}{q} < 1 + \frac{1}{N}$ and $u \in W_0^{1,\eta}(\Omega)$, with $\eta(z, t)$ being the double phase density

$$\eta(z, t) = a(z)t^p + t^q \text{ for all } t \geq 0$$

We consider the operator $V : W_0^{1,\eta}(\Omega) \to W_0^{1,\eta}(\Omega)^*$ defined by

$$\langle V(u), h\rangle = \int_\Omega (a(z)|Du|^{p-2} + |Du|^{q-2})(Du, Dh)_{\mathbb{R}^N}dz$$

for all $u, h \in W_0^{1,\eta}(\Omega)$.

Proposition 5.50 *The operator $V : W_0^{1,\eta}(\Omega) \to W_0^{1,\eta}(\Omega)^*$ defined above is bounded, continuous, strictly monotone (thus maximal monitone too) and of type $(S)_+$.*

Proof First we show the boundedness of $V(\cdot)$. Let $\vartheta_1 = \|u\|$, $\vartheta_2 = \|h\|$. We have

$$
\begin{aligned}
&|\langle \frac{V(u)}{\vartheta_1}, \frac{h}{\vartheta_2}\rangle| = |\int_\Omega a(z)|\frac{Du}{\vartheta_1}|^{p-2}(\frac{Du}{\vartheta_1}, \frac{Dh}{\vartheta_2})_{\mathbb{R}^N}dz \\
&\quad + \int_\Omega |\frac{Du}{\vartheta_1}|^{q-2}(\frac{Du}{\vartheta_1}, \frac{Dh}{\vartheta_2})_{\mathbb{R}^N}dz| \\
&\leq (\int_\Omega a(z)|\frac{Du}{\vartheta_1}|^p dz)^{\frac{p-1}{p}}(\int_\Omega a(z)|\frac{Dh}{\vartheta_2}|^p dz)^{\frac{1}{p}} + (\int_\Omega |\frac{Du}{\vartheta_1}|^q dz)^{\frac{q-1}{q}}(\int_\Omega |\frac{Dh}{\vartheta_2}|^q dz)^{\frac{1}{q}} \\
&\quad \text{(using Hölder's inequality)} \\
&\leq \frac{p-1}{p}\int_\Omega a(z)|\frac{Du}{\vartheta_1}|^p dz + \frac{1}{p}\int_\Omega a(z)|\frac{Dh}{\vartheta_2}|^p dz \\
&\quad + \frac{q-1}{q}\int_\Omega |\frac{Du}{\vartheta_1}|^q dz + \frac{1}{q}\int_\Omega |\frac{Dh}{\vartheta_2}|^q dz \text{ (using Young's inequality)} \\
&\leq \frac{p-1}{p}[\int_\Omega a(z)|\frac{Du}{\vartheta_1}|^p dz + \int_\Omega |\frac{Du}{\vartheta_1}|^q dz] + \frac{1}{q}[\int_\Omega a(z)|\frac{Dh}{\vartheta_2}|^p dz \\
&\quad + \int_\Omega |\frac{Dh}{\vartheta_2}|^q dz] \text{ (since } x \to \frac{x-1}{x} \text{ is increasing on } (0,\infty)) \\
&\leq \frac{p-1}{p} + \frac{1}{q} \leq 2 \text{ (see Definition 4.151).} \\
&\Rightarrow \|V(u)\|_* = \sup[|\langle V(u), h\rangle| : \|h\| \leq 1] \leq 2\|u\|, \\
&\Rightarrow V(\cdot) \text{ is bounded.}
\end{aligned}
$$

The strict monotonicity of $V(\cdot)$ follows from the strict convexity of the double phase density $t \to \eta(z,t)$.

Next we show the continuity of $V(\cdot)$. So, let $u_n \to u$ in $W_0^{1,\eta}(\Omega)$. For every $h \in W_0^{1,\eta}(\Omega)$, we have

$$
\begin{aligned}
&|\langle V(u_n) - V(u), h\rangle| \\
&\leq \int_\Omega a(z)||Du_n|^{p-2}Du_n - |Du|^{p-2}Du||Dh|dz \\
&+ \int_\Omega ||Du_n|^{q-2}Du_n - |Du|^{q-2}Du||Dh|dz \\
&\leq [\int_\Omega a(z)||Du_n|^{p-2}Du_n - |Du|^{p-2}Du|^{p'}dz]^{\frac{1}{p'}}c_1\|h\| \\
&+ \||Du_n|^{q-2}Du_n - |Du|^{q-2}Du\|_q c_2\|h\| \text{ for some } c_1, c_2 > 0.
\end{aligned}
$$

Here we have used Hölder's inequality and the fact that $L^{\eta}(\Omega) \hookrightarrow L^{p}_{a}(\Omega)$ continuously (see Remark 4.158). Therefore

$$\begin{aligned} &\|V(u_n) - V(u)\|_* \\ \leq & c_3 \Big(\int_{\Omega} a(z) \big| |Du_n|^{p-2} Du_n - |Du|^{p-2} Du \big|^{p'} dz \Big)^{\frac{1}{p'}} \\ &+ \big\| |Du_n|^{q-2} Du_n - |Du|^{q-2} |Du| \big\|_{q'} \big] \text{ for some } c_3 > 0, \end{aligned} \tag{5.76}$$

Note that at least for a subsequence we have

$$Du_n(z) \to Du(z) \text{ for a.a } z \in \Omega.$$

Also the sequences.

$$\{a(\cdot) \big| |Du_n|^{p-2} Du_n - |Du|^{p-2} Du \big|^{p'}\}_{n \in \mathbb{N}} \subseteq L^1(\Omega)$$

$$\{ \big| |Du_n|^{q-2} Du_n - |Du|^{q-2} Du \big| \}_{n \in \mathbb{N}} \subseteq L^1(\Omega),$$

are uniformly integrable. So, from (5.76) and Vitali's theorem, we have

$$\begin{aligned} &\|V(u_n) - V(u)\|_* \to 0, \\ \Rightarrow & V(\cdot) \text{ is continuous.} \end{aligned}$$

Since $V(\cdot)$ is strictly monotone and continuous, it is maximal monotone (see Proposition I-7.36).

Finally we show the $(S)_+$-property. We know that $V(\cdot)$ is maximal monotone, hence it is generalized pseudomonotone (see Proposition I-7.73). Let $\{u_n\}_{n \in \mathbb{N}} \subseteq W^{1,\eta}_0(\Omega)$ be a sequence such that

$$u_n \xrightarrow{w} u \text{ in } W^{1,\eta}_0(\Omega), \ \limsup_{n \to \infty} \langle V(u_n), u_n - u \rangle \leq 0. \tag{5.77}$$

The boundedness of $V(\cdot)$, implies that $\{V(u_n)\}_{n \in \mathbb{N}} \subseteq W^{1,\eta}_0(\Omega)^*$ is bounded. Then the reflexivity of $W^{1,\eta}_0(\Omega)^*$ and the Eberlein-Smulian theorem, imply that we may assume that

$$V(u_n) \xrightarrow{w} u^* \text{ in } W^{1,\eta}_0(\Omega)^*. \tag{5.78}$$

Then (5.77), (5.78) and the fact that $V(\cdot)$ is generalized pseudomonotone, imply that

$$\begin{aligned} &u^* = V(u), \langle V(u_n), u_n \rangle \to \langle V(u), u \rangle \text{ (see Definition } I\text{-7.67).} \\ \Rightarrow & \varrho_{\eta}(Dv_n) \to \varrho_{\eta}(Du). \end{aligned}$$

Since $\eta(z,\cdot)$ is convex it satisfies (5.75) and so by Lemma 5.49, we have

$$\begin{aligned}
&\int_\Omega |\eta(z,|Du_n|) - \eta(z,|Du|) - \eta(z,|Du_n - Du|)|dz \to 0,\\
\Rightarrow &\int_\Omega (a(z)|Du_n - Du|^p + |Du_n - Du|^q)dz \to 0. \text{ (see (5.78))}\\
\Rightarrow &\varrho_\eta(Du_n - Du) \to 0,\\
\Rightarrow &\|u_n - u\| = \|D(u_n - u)\|_\eta \to 0,\\
\Rightarrow &V(\cdot) \text{ is an } (S)_+\text{-map.}
\end{aligned}$$

□

In the subsequent sections of this chapter, we will use the tools developed so far in order to study various type of boundary value problems.

5.2 Nonlinear Elliptic Equations

In this section, we examine boundary value problems driven by a nonhomogeneous differential operator.

First we deal with a $(p,2)$-equation (that is, a boundary value problem, where the differential operator is the sum of a p-Laplacian ($p \neq 2$) and a Laplacian). Such equations arise in many physical applications as, for example quantum physics (see Benci et al. [40], plasma physics (see Cherfils and llyasov [99]), nonlinear elasticity (see Zhikov [484]).

Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$. We will study the following Dirichlet boundary value problem

$$-\Delta_p u(z) - \Delta u(z) = f(z,u(z)) \text{ in } \Omega, u|_{\partial\Omega} = 0. \tag{5.79}$$

First we consider the case $2 < p < \infty$. The hypotheses on the reaction (forcing term) $f(z,x)$ are the following:

H_1 : $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such the $f(z,0) = 0$ for a.a $z \in \Omega$ and

(i) $|f(z,x)| \leq a(z)(1 + |x|^{r-1})$ for a.a $z \in \Omega$, all $x \in \mathbb{R}$, with $a \in L^\infty(\Omega)$, $p \leq r \leq p^*$;

(ii) if $F(z,x) = \int_0^x f(z,s)ds$ and $\widehat{\lambda}_1(p) > 0$ is the principal eigenvalue of $(-\Delta_p, W_0^{1,p}(\Omega))$, then $\limsup_{x\to\pm\infty} \frac{pF(z,x)}{|x|^p} \leq \widehat{\lambda}_1(p)$ uniformly for a.a $z \in \Omega$ and there exist $\xi, M > 0$ such that

$$-\xi \leq f(z,x)x - pF(z,x) \text{ for a.a } z \in \Omega, \text{ all } |x| \geq M;$$

(iii) if $\{\widehat{\lambda}_k(2)\}_{k\in\mathbb{N}} \subseteq (0,+\infty)$ are the eigenvalues of $(-\Delta, H_0^1(\Omega))$, then there exist $m \in \mathbb{N}$, $m \geq 2$, $\delta_0 > 0$ and $\eta \in L^\infty(\Omega)$ such that $\widehat{\lambda}_m(2) \leq \eta(z)$ for a.a $z \in \Omega$, $\eta \not\equiv \widehat{\lambda}_m(2)$ and

$$\eta(z)x^2 \leq f(z,x)x \leq \widehat{\lambda}_{m+1}(2)x^2 \text{ for a.a } z \in \Omega, \text{ all } |x| \leq \delta_0;$$

(iv) for every $\varrho > 0$, there exists $\widehat{\xi}_\varrho > 0$ such that for a.a $z \in \Omega$, the function $x \to f(z,x) + \widehat{\xi}_\varrho |x|^{p-2}x$ is nondecreasing on $[-\varrho, \varrho]$.

Remark 5.51 We see that these hypotheses allow for resonance to occur at $\pm\infty$ (hypothesis H_1(ii)) and at zero (hypothesis H_1(iii)). So, we are dealing with a doubly resonant problem.

The energy functional of this problem is $\varphi : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\varphi(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_\Omega F(z,u(z))dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

We know that $\varphi \in C^1(W_0^{1,p}(\Omega))$ and

$$\langle \varphi'(u), h\rangle = \langle A_p(u), h\rangle + \langle A(u), h\rangle - \int_\Omega f(z,u)hdz \text{ for all } u, h \in W_0^{1,p}(\Omega).$$

Recall that $A_p : W_0^{1,p}(\Omega) \to W^{-1,p'}(\Omega)$ and $A \in \mathcal{L}(H_0^1(\Omega), H^{-1}(\Omega))$ are defined by

$$\langle A_p(u), h\rangle = \int_\Omega |Du|^{p-2}(Du, Dh)_{\mathbb{R}^N}dz,$$

$$\langle A(u), h\rangle = \int_\Omega (Du, Dh)_{\mathbb{R}^N}dz.$$

To produce solutions of constant sign, we will also use the positive and negative truncations of the functional $\varphi(\cdot)$, namely the C^1-functionals $\varphi_\pm : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\varphi_\pm(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_\Omega F(z, \pm u^\pm(z))dz$$

for all $u \in W_0^{1,p}(\Omega)$.

We have

$$\langle \varphi'_\pm(u), h\rangle = \langle A_p(u), h\rangle + \langle A(u), h\rangle - \int_\Omega f(z, \pm u^\pm)hdz$$

for all $u, h \in W_0^{1,p}(\Omega)$.

Proposition 5.52 *If hypotheses H_1 hold, then the functions φ, $\varphi_\pm$ are coercive.*

Proof We do the proof for φ_+, the proofs for φ and φ_- being similar. For a.a $z \in \Omega$ and all $x \geq M$, we have

$$\frac{d}{dx}\frac{F(z,x)}{x^p} = \frac{f(z,x)x - pF(z,x)}{x^{p+1}} \geq -\frac{\xi}{x^{p+1}} \text{ (see } H_1\text{(iii))},$$

$$\Rightarrow \frac{F(z,x)}{x^p} - \frac{F(z,y)}{y^p} \geq \frac{\xi}{p}\left(\frac{1}{x^p} - \frac{1}{y^p}\right) \text{ for a.a } z \in \Omega, \text{ all } x \geq y \geq M.$$

We let $x \to +\infty$ and use hypothesis H_1(ii) to obtain

$$\frac{\widehat{\lambda}_1(p)}{p} - \frac{F(z,y)}{y^p} \geq -\frac{\xi}{py^p} \text{ for a.a } z \in \Omega, \text{ all } y \geq M,$$

$$\Rightarrow pF(z,y) - \widehat{\lambda}_1(p)y^p \leq \xi \text{ for a.a } z \in \Omega, \text{ all } y \geq M. \tag{5.80}$$

We will use (5.80) to show the coercivity of $\varphi_+(\cdot)$. Arguing by contradiction, suppose that $\varphi_+(\cdot)$ is not coercive. Then we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega)$ and $M_1 > 0$ such that

$$\|u_n\| \to \infty \text{ and } \varphi_+(u_n) \leq M_1 \text{ for all } n \in \mathbb{N} \tag{5.81}$$

Here $\|\cdot\|$ denotes the norm of $W_0^{1,p}(\Omega)$ ($\|u\| = \|Du\|_p$ for all $u \in W_0^{1,p}(\Omega)$ by the Poincare inequality).

From (5.80), (5.81) and the variational characterization of $\widehat{\lambda}_1(p) > 0$, we see that

$$\{u_n^-\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.} \tag{5.82}$$

Then (5.81) and (5.82) imply that $\|u_n^+\| \to \infty$. We set $y_n = \frac{u_n^+}{\|u_n^+\|}$, $n \in \mathbb{N}$. Then $\|y_n\| = 1$, $y_n \geq 0$ for all $n \in \mathbb{N}$ and so we may assume that

$$y_n \xrightarrow{w} y \text{ in } W_0^{1,p}(\Omega),\ y_n \to y \text{ in } L^r(\Omega). \tag{5.83}$$

From (5.81) we have

$$\frac{1}{p}\|Dy_n\|_p^p + \frac{1}{2\|u_n^+\|^{p-2}}\|Dy_n\|_2^2 - \int_\Omega \frac{F(z,u_n^+)}{\|u_n^+\|^p}dz \leq \frac{M_1}{\|u_n^+\|^p}$$

$$\Rightarrow \frac{1}{p}\|Dy_n\|_p^p + \frac{1}{2\|u_n^+\|^{p-2}}\|Dy_n\|_2^2 - \frac{\widehat{\lambda}_1(p)}{p}dz \leq \frac{M_2}{\|u_n^+\|^p}$$

for some $M_2 > 0$, all $n \in \mathbb{N}$,

$$\Rightarrow \frac{1}{p}\|Dy\|_p^p \leq \frac{\widehat{\lambda}_1(p)}{p}\|y\|_p^p \text{ (see (5.83))},$$
$$\Rightarrow \|Dy\|_p^p = \widehat{\lambda}_1(p)\|y\|_p^p,$$
$$\Rightarrow y = \mu \widehat{u}_1(p) \text{ for some } \mu \geq 0 \text{ (recall } y \geq 0 \text{ and see Theorem 4.101)}.$$

If $\mu = 0$, then $y = 0$ and so $y_n \to 0$ in $W_0^{1,p}(\Omega)$, which contradicts the fact that $\|y_n\| = 1$ for all $n \in \mathbb{N}$. Hence $\mu > 0$. Since $\widehat{u}_1(p) \in \text{int}C_+$, we have $u_n^+(z) \to +\infty$ for a.a $z \in \Omega$.

From (5.81) we have

$$\frac{1}{p}\|Du_n^+\|_p^p + \frac{1}{2}\|Du_n^+\|_2^2 - \int_\Omega F(z, u_n^+)dz \leq M_1 \text{ for all } n \in \mathbb{N},$$
$$\Rightarrow \int_\Omega \left[\frac{\widehat{\lambda}_1(p)}{p}(u_n^+)^p - F(z, u_n^+)\right]dz + \int_\Omega \frac{\widehat{\lambda}_1(2)}{2}(u_n^+)^2 dz \leq M_1 \text{ for all } n \in \mathbb{N},$$
$$\Rightarrow \int_\Omega \frac{\widehat{\lambda}_1(2)}{2}(u_n^+)^2 dz \leq M_3 \text{ for some } M_3 > 0, \text{ all } n \in \mathbb{N} \text{ (see (5.80))}.$$

From Fatou's lemma and since $u_n^+(z) \to +\infty$ for a.a $z \in \Omega$, we have a contradiction. This proves that $\varphi_+(\cdot)$ is coercive.

Similarly for the functionals φ, φ_-. □

This proposition and the direct method of the Calculus of Variations, will produce solutions of constant sign.

Proposition 5.53 *If hypothese H_1 hold, then problem (5.79) has at least two constant sign solutions $u_0 \in \text{int}C_+$, $v_0 \in -\text{int}C_+$.*

Proof From Proposition 5.52, we know that $\varphi_+(\cdot)$ is coercive. Also using the Sobolev embedding theorem (see Theorem I-4.114) and hypothesis H_1(i), we see that $\varphi_+(\cdot)$ is sequentially weakly lower semi-continuous. So, by the Weierstrass-Tonelli theorem (Proposition I-1.180) we can find $u_0 \in W_0^{1,p}(\Omega)$ such that

$$\varphi_+(u_0) = \inf\{\varphi_+(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.84}$$

Recall that $\widehat{u}_1(2) \in \text{int}C_+$. So, we can find $t \in (0, 1)$ small such that $t\widehat{u}_1(2)(z) \in [0, \delta_0]$ for all $z \in \overline{\Omega}$, with $\delta_0 > 0$ as postulated by hypothesis H_1(iii). We have

$$\begin{aligned}\varphi_+(t\widehat{u}_1(2)) &= \frac{t^p}{p}\|D\widehat{u}_1(2)\|_p^p + \frac{t^2}{2}\|D\widehat{u}_1(2)\|_2^2 - \int_\Omega F(t, t\widehat{u}_1(2))dz \\ &\leq \frac{t^p}{p}\|D\widehat{u}_1(2)\|_p^p + \frac{t^2}{2}[\widehat{\lambda}_1(2) - \widehat{\lambda}_m(2)] \text{ (see hypothesis } H_1\text{(iii))}\end{aligned}$$

$$\Rightarrow \varphi_+(t\widehat{u}_1(2)) \leq c_1 t^p - c_2 t^2 \text{ with } c_1, c_2 > 0 \text{ (recall } m \geq 2)$$

Choosing $t \in (0, 1)$ even smaller if necessary, we have

$$\begin{aligned} &\varphi_+(t\widehat{u}_1(2)) < 0 \\ \Rightarrow &\varphi_+(u_0) < 0 = \varphi_+(0) \text{ (see (5.84))}, \\ \Rightarrow &u_0 \neq 0. \end{aligned}$$

From (5.84), we have

$$\langle A_p(u_0), h\rangle + \langle A(u_0), h\rangle = \int_\Omega f(z, u_0^+)hdz \text{ for all } h \in W_0^{1,p}(\Omega). \tag{5.85}$$

Using the test function $h = -u_0^- \in W_0^{1,p}(\Omega)$, we obtain

$$u_0 \geq 0, u_0 \neq 0$$

From (5.85) we see that u_0 is a positive solution of (5.79). From Proposition 4.82 we have that $u_0 \in L^\infty(\Omega)$ and then Theorem 4.84 says that there exists $\alpha \in (0, 1)$ such that $u_0 \in C_0^{1,\alpha}(\overline{\Omega})$. Let $\varrho = \|u_0\|_\infty$ and let $\widehat{\xi}_\varrho > 0$ be as postulated by hypothesis H_1(iv). We have

$$-\Delta_p u_0 - \Delta u_0 + \widehat{\xi}_\varrho u_0^{p-1} \geq 0 \text{ in } \Omega,$$

$$\Rightarrow u_0 \in \text{int}C_+ \text{ (see Theorem 4.89)}.$$

Similarly working this time with the functional $\varphi_-(\cdot)$ we produce a negative solution $v_0 \in -\text{int}C_+$. □

Let S_+(resp.S_-) be the set of positive (resp.negative) solutions of (5.79). From Proposition 5.53 and its proof we know that

$$\emptyset \neq S_+ \subseteq \text{int}C_+ \text{ and } \emptyset \neq S_- \subseteq -\text{int}C_+.$$

We will show that there exist extremal constant sign solutions (barrier solutions), that is, there exist $u_* \in S_+$ and $v_* \in S_-$ such that

$$u_* \leq u \text{ for all } u \in S_+, v \leq v_* \text{ for all } x \in S_-.$$

These extremal constant sign solutions, will be used to generate a nodal (sign-changing) solution.

From hypotheses H(i),(iii), we see that

$$f(z, x)x \geq c_3 x^2 - c_4|x|^r \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R}, \tag{5.86}$$

with $c_3 > \widehat{\lambda}_1(2)$ and $c_4 > 0$. Motivated by (5.86), we consider the following auxiliary Dirichlet problem

$$\{-\Delta_p u(z) - \Delta u(z) = c_3 u(z) - c_4 |u(z)|^{r-2} u(z) \text{ in } \Omega, u|_{\partial\Omega} = 0.\} \tag{5.87}$$

Proposition 5.54 *Problem (5.87) has a unique positive solution $\overline{u} \in \text{int}C_+$. Since (5.87) is odd $\overline{v} = -\overline{u} \in -\text{int}C_+$ is the unique negative solution of (5.87).*

Proof First we show the existence of a positive solution for problem (5.87). To this end, we consider the C^1-functional $\psi_+ : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\psi_+(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \frac{c_3}{2}\|u^+\|_2^2 + \frac{c_4}{r}\|u^+\|_r^r$$

for all $u \in W_0^{1,p}(\Omega)$.

Since $2 < p \leq r < p^*$, we see that $\psi_+(\cdot)$ is coercive and sequentially weakly lower semicontinuous. So, we can find $\overline{u} \in W_0^{1,p}(\Omega)$ such that

$$\psi_+(\overline{u}) = \inf\{\psi_+(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.88}$$

Since $c_3 > \widehat{\lambda}_1(2)$ and $2 < p \leq r$, we see that for $t \in (0, 1)$ small we have

$$\begin{aligned} &\psi_+(t\widehat{u}_1(2)) < 0 \\ \Rightarrow &\psi_+(\overline{u}) < 0 = \psi_+(0) \text{ (see (5.88))}, \\ \Rightarrow &\overline{u} \neq 0. \end{aligned}$$

From (5.88) we have

$$\langle A_p(\overline{u}), h\rangle + \langle A(\overline{u}), h\rangle = \int_\Omega c_3(\overline{u}^+)h dz - \int_\Omega c_4(\overline{u}^+)^{r-1} h dz \tag{5.89}$$

for all $h \in W_0^{1,p}(\Omega)$.

Using the test function $h = -\overline{u}^- \in W_0^{1,p}(\Omega)$, we infer that

$$\overline{u} \geq 0, \overline{u} \neq 0.$$

Therefore $\overline{u}$ is a positive solution of (5.87) (see (5.88)) and as before using the nonlinear regularity theory (Theorem 4.84) and the nonlinear maximum principle (Theorem 4.89), we conclude that

$$\overline{u} \in \text{int}C_+.$$

Next we show the uniqueness of this positive solution. Let $G_0(t) = \frac{t^p}{p} + \frac{t^2}{2}$ for all $t \geq 0$ and $G(y) = G_0(|y|)$ for all $y \in \mathbb{R}^N$. We see that $G \in C^1(\mathbb{R}^N, \mathbb{R})$ and $\nabla G(y) = a(y) = |y|^{p-2}y + y$ for all $y \in \mathbb{R}^N$. The function $G_0(\cdot)$ is increasing an $\mathbb{R}^+$, $t \to G_0(t^{\frac{1}{2}})$ is convex and

$$\text{div}a(Du) = \Delta_p u + \Delta u \text{ for all } u \in W_0^{1,p}(\Omega).$$

We introduce the integral functional $j : L^1(\Omega) \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ defined by

$$j(u) = \begin{cases} \int_\Omega G(Du^{\frac{1}{2}})dz & \text{if } u \geq 0, u^{\frac{1}{2}} \in W_0^{1,p}(\Omega) \\ +\infty & \text{otherwise.} \end{cases}$$

Let $domj = \{u \in L^1(\Omega) : j(u) < \infty\}$ (the effective domain of $j(\cdot)$). Consider $u_1, u_2 \in \text{dom}j$ and let $v = (tu_1 + (1-t)u_2)^{\frac{1}{2}}$ for all $t \in [0, 1]$. We have

$$\begin{aligned} &|Dv| \leq (t|Du_1^{\frac{1}{2}}|^2 + (1-t)|Du_2^{\frac{1}{2}}|^2)^{\frac{1}{2}}, \\ \Rightarrow &G_0(|Dv|) \leq G_0((t|Du_1^{\frac{1}{2}}|^2 + (1-t)|Du_2^{\frac{1}{2}}|^2)\frac{1}{2}) \\ \leq &tG_0(|Du_1^{\frac{1}{2}}|) + (1-t)G_0(|Du_2^{\frac{1}{2}}|), \\ \Rightarrow &G(Dv) \leq tG(Du_1^{\frac{1}{2}}) + (1-t)G(Du_2^{\frac{1}{2}}), \\ \Rightarrow &j(\cdot) \text{ is convex.} \end{aligned}$$

Suppose $\overline{v} \in W_0^{1,p}(\Omega)$ is another positive solution of (5.87). Again we have $\overline{v} \in \text{int}C_+$. Using Proposition 2.86, we have

$$\frac{\overline{u}}{\overline{v}} \in L^\infty(\Omega), \frac{\overline{v}}{\overline{u}} \in L^\infty(\Omega).$$

Therefore if $h = \overline{u}^2 - \overline{v}^2 \in C_0^1(\overline{\Omega})$, then for $t \in (0, 1)$ small we have

$$\overline{u}^2 + th \in domj, \overline{v}^2 + th \in domj.$$

Hence we can compute the directional derivative of $j(\cdot)$ at $\overline{u}^2$ and at $\overline{v}^2$ in the direction h. From the convexity of $j(\cdot)$ and using Green's identity (see Theorem *I*-4.106), we have

$$j'(\overline{u}^2)(h) = \frac{1}{2}\int_\Omega \frac{-\Delta_p\overline{u} - \Delta\overline{u}}{\overline{u}}hdz$$

$$= \frac{1}{2}\int_\Omega (c_3 - c_4\overline{u}^{r-2})hdz$$

$$\text{and } j'(\overline{v}^2)(h) = \frac{1}{2}\int_\Omega \frac{-\Delta_p\overline{v} - \Delta\overline{v}}{\overline{v}}hdz$$

$$= \frac{1}{2}\int_\Omega (c_3 - c_4\overline{v}^{r-2})hdz.$$

The convexity of $j(\cdot)$, implies the monotonicity of $j'(\cdot)$. Therefore

$$0 \leq \int_\Omega c_4(\overline{v}^{r-2} - \overline{u}^{r-2})(\overline{u}^2 - \overline{v}^2)dz \leq 0,$$

$$\Rightarrow \overline{u} = \overline{v}.$$

This proves the uniqueness of the positive solution $\overline{u} \in \text{int}C_+$ of problem (5.87). The problem is odd, hence $\overline{v} = -\overline{u} \in -\text{int}C_+$ is the unique negative solution of (5.87). □

Proposition 5.55 *If hypotheses H_1 hold, then $\overline{u} \leq u$ for all $u \in S_+$ and $v \leq \overline{v}$ for all $v \in S_-$.*

Proof Let $u \in S_+$. Then $u \in \text{int}C_+$. Consider the Caratheodory function $\gamma_+(z, x)$ defined by

$$\gamma_+(z, x) = \begin{cases} c_3x^+ - c_4(x^+)^{r-1} & \text{if } x \leq u(z) \\ c_3u(z) - c_4u(z)^{r-1} & \text{if } u(z) < x. \end{cases} \tag{5.90}$$

We set $\Gamma_+(z, x) = \int_0^x \gamma_+(z, s)ds$ and consider the C^1-functional $\overline{o}_+ : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\sigma_+(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_\Omega \Gamma_+(z, u)dz$$

for all $u \in W_0^{1,p}(\Omega)$. It is clear from (5.90) that σ_+ is coercive. Also, it is sequentially weakly lower semicontinuous. Therefore, we can find $\widetilde{u} \in W_0^{1,p}(\Omega)$ such that

$$\sigma_+(\widetilde{u}) = \inf\{\sigma_+(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.91}$$

We know that $u \in \text{int}C_+$. So, using Proposition 2.86, we can find $t \in (0, 1)$ small such that

$$0 \leq t\widehat{u}_1(2)(z) \leq u(z) \text{ for all } z \in \overline{\Omega}. \tag{5.92}$$

From (5.90), (5.92) we have

$$\sigma_+(t\widehat{u}_1(2)) \leq c_5 t^p - c_6 t^2 \text{ for some } c_5, c_6 > 0$$
$$\text{(recall that } c_3 > \widehat{\lambda}_1(2)).$$

Since $p \geq 2$, choosing $t \in (0,1)$ even smaller, we have

$$\begin{aligned} &\sigma_+(t\widehat{u}_1(2)) < 0, \\ \Rightarrow &\sigma_+(\widetilde{u}) < 0 = \sigma_+(0) \text{ (see (5.91))}, \\ \Rightarrow &\widetilde{u} \neq 0. \end{aligned}$$

From (5.91) we have

$$\langle A_p(\widetilde{u}), h\rangle + \langle A(\widetilde{u}), h\rangle = \int_\Omega \gamma_+(z,\widetilde{u})h\,dz \text{ for all } h \in W_0^{1,p}(\Omega).$$

Choosing $h = -\widetilde{u}^- \in W_0^{1,p}(\Omega)$, we obtain that

$$\widetilde{u} \geq 0, \widetilde{u} \neq 0$$

Then we use the test function $h = (\widetilde{u} - u)^+ \in W_0^{1,p}(\Omega)$. We obtain

$$\begin{aligned} &\langle A_p(\widetilde{u}), (\widetilde{u}-u)^+\rangle + \langle A(\widetilde{u}), (\widetilde{u}-u)^+\rangle \\ =&\int_\Omega [c_3 u - c_4 u^{r-1}](\widetilde{u}-u)^+ dz \text{ (see (5.90))} \\ \leq &\int_\Omega f(z,u)(\widetilde{u}-u)^+ dz \text{ (see (5.86))} \\ =&\langle A_p(u), (\widetilde{u}-u)^+\rangle + \langle A(\widetilde{u}), (\widetilde{u}-u)^+\rangle \text{ (since } u \in S_+), \\ \Rightarrow &\widetilde{u} \leq u \text{ (since } V = A_p + A \text{ is monotone}). \end{aligned}$$

So, we have proved that

$$\widetilde{u} \in [0,u] = \{g \in W_0^{1,p}(\Omega) : 0 \leq g(z) \leq u(z) \text{ for a.a } z \in \Omega\},$$
$$\widetilde{u} \neq 0.$$

It follows (see (5.90)), that $\widetilde{u}$ is a positive solution of (5.87), hence $\widetilde{u} = \overline{u}$ (see Proposition 5.54). Therefore

$$\overline{u} \leq u \text{ for all } u \in S_+.$$

Similarly we show that

$$v \leq \overline{v} \text{ for all } v \in S_-.$$

□

Now we are ready to produce extremal constant sign solutions for problem (5.79).

Proposition 5.56 *If hypotheses H_1 hold, then there exist $u_* \in S_+$ and $v_* \in S_-$ such that*

$$u_* \leq u \text{ for all } u \in S_+,$$

$$v \leq v_* \text{ for all } v \in S_-$$

Proof Using the monotonicity of $V = A_p + A$, we can check that S_+ is downward directed, that is, if $u_1, u_2 \in S_+$, then there exists $u \in S_+$ such that $u \leq u_1$, $u \leq u_2$ (see Filippakis and Papageorgiou [165]). Then invoking Theorem I-5.109, we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq S_+$ decreasing such that

$$\inf S_+ = \inf_{n\in\mathbb{N}} u_n,$$

$$0 \leq \overline{u} \leq u_n \leq u_1 \text{ for all } n \in \mathbb{N}. \tag{5.93}$$

We have

$$\langle A_p(u_n), h\rangle + \langle A(u_n), h\rangle = \int_\Omega f(z, u_n)h\,dz \tag{5.94}$$

$$\text{for all } h \in W_0^{1,p}(\Omega), \text{ all } n \in \mathbb{N}.$$

Choosing $h = u_n \in W_0^{1,p}(\Omega)$ and using (5.93) and hypothesis H_1(i), we see that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded}$$

We may assume that

$$u_n \xrightarrow{w} u_* \text{ in } W_0^{1,p}(\Omega),\ u_n \to u_* \text{ in } L^r(\Omega). \tag{5.95}$$

In (5.94) we choose the test function $h = u_n - u_* \in W_0^{1,p}(\Omega)$, pass to the limit as $n \to \infty$ and use (5.95). We obtain

$$\lim_{n\to\infty} \langle V(u_n), u_n - u_*\rangle = 0,$$

$$\Rightarrow u_n \to u_* \text{ in } W_0^{1,p}(\Omega) \text{ (see Proposition 5.47).} \tag{5.96}$$

So, if in (5.94) we pass to the limit and use (5.96) and (5.93), we have

$$\langle A_p(u_*), h\rangle + \langle A(u_*), h\rangle = \int_\Omega f(z, u_*)hdz$$

for all $h \in W_0^{1,p}(\Omega)$,

$$\overline{u} \leq u_*.$$

It follows that $u_* \in S_+$, $u_* = \inf S_+$.

The set S_- is upward directed, that is, if $v_1, v_2 \in S_+$, then there exists $v \in S_-$ such that $v_1 \leq v$, $v_2 \leq v$. Working as above, we produce $v_* \in S_-$ such that $v \leq v_*$ for all $v \in S_-$. □

Using these extremal constant sign solutions, we can produce a nodal (sign-chunging) solution. The idea is simple. We look for nontrivial solutions distinct from u_* and v_* in the order interval

$$[v_*, u_*] = \{g \in W_0^{1,p}(\Omega) : v_*(z) \leq g(z) \leq u_*(z) \text{ for a.a } z \in \Omega\}.$$

The extremality of u_* and v_* guarantees that such a solution will be nodal. To generate such a solution, we will use truncations, comparisons and critical groups. For this reason we need to strengther the regularity of $f(z, \cdot)$. The new hypotheses on the reaction $f(z, x)$are the following:

$\mathbf{H_1'}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a measurable function such that for a.a $z \in \Omega$, $f(z, 0) = 0$, $f(z, \cdot) \in C^1(\mathbb{R})$ and (i)$|f_x'(z, x)| \leq o(z)(1 + |x|^{r-2})$ for a.a $z \in \Omega$, all $x \in \mathbb{R}$, with $a \in L^\infty(\Omega)$, $2 < p \leq r < p^*$;

hypotheses H_1'(ii),(iii),(iv) are the same as the corresponding hypotheses in H_1.

To restrict ourselves to $[v_*, u_*]$, we introduce the following truncation of $f(z, \cdot)$

$$\widehat{f}(z, x) = \begin{cases} f(z, v_*(z)) & \text{if } x < v_*(z) \\ f(z, x) & \text{if } v_*(z) \leq x \leq u_*(z) \\ f(z, u_*(z)) & \text{if } u_*(z) < x. \end{cases} \tag{5.97}$$

This is a Caratheodory function. We also introduce the positive and negative truncations of $\widehat{f}(z, \cdot)$, namely the Caratheodory functions

$$\widehat{f}_\pm(z, x) = \widehat{f}(z, \pm x^\pm). \tag{5.98}$$

We set $\widehat{F}(z, x) = \int_0^x \widehat{f}(z, s)ds$, $\widehat{F}_\pm(z, x) = \int_0^x \widehat{f}_\pm(z, s)ds$ and consider the C^1-functionals $\widehat{\varphi}, \widehat{\varphi}_\pm : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{\varphi}(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_\Omega \widehat{F}(z, u)dz,$$

$$\widehat{\varphi}_{\pm}(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_{\Omega} \widehat{F}_{\pm}(z,u)dz$$

for all $u \in W_0^{1,p}(\Omega)$.

For the next result we do not need the stronger hypotheses H_1' introduced above.

Proposition 5.57 *Let $u \in K_{\widehat{\varphi}}$. Then*

$$K_{\widehat{\varphi}} \subseteq [v_x, u_x] \cap C_0^1(\overline{\Omega}),\ K_{\widehat{\varphi}_+} = \{0, u_*\},\ K_{\widehat{\varphi}_-} = \{0, v_*\}.$$

Proof Let $u \in K_{\widehat{\varphi}}$. Then

$$\langle A_p(u), h\rangle + \langle A(u), h\rangle = \int_{\Omega} \widehat{f}(z,u)hdz \text{ for all } h \in W_0^{1,p}(\Omega).$$

First we choose the test function $(u - u_*)^+ \in W_0^{1,p}(\Omega)$. We have

$$\begin{aligned}
&\langle A_p(u), (u-u_*)^+\rangle + \langle A(u), (u-u_*)^+\rangle \\
=&\int_{\Omega} f(z,u_*)(u-u_*)^+dz \text{ (see (5.97))} \\
=&\langle A_p(u_*), (u-u_*)^+\rangle + \langle A(u_*), (u-u_*)^+\rangle \text{ (since } u_* \in S_+) \\
\Rightarrow&\langle V(u) - V(u_*), (u-u_*)^+\rangle = 0 \\
\Rightarrow&u \leq u_* \text{ (since } V(\cdot) \text{ is strictly monotone).}
\end{aligned}$$

Similarly using the test function $h = (v_* - u)^+ \in W_0^{1,p}(\Omega)$, we show that $v_* \leq u$. Therefore $u \in [v_*, u_*]$. Moreover, the nonlinear regularity theory implies $u \in C_0^1(\overline{\Omega})$. We conclude that

$$K_{\widehat{\varphi}} \subseteq [v_*, u_*] \cap C_0^1(\overline{\Omega}).$$

In a similar fashion, using (5.98) we show that

$$K_{\widehat{\varphi}_+} \subseteq [0, u_*] \cap C_+,$$

$$K_{\widehat{\varphi}_-} \subseteq [v_*, 0] \cap (-C_+).$$

The extremality of u_* and v_* implies that

$$K_{\widehat{\varphi}_+} = \{0, u_*\} \text{ and } K_{\widehat{\varphi}_-} = \{0, v_*\}.$$

□

In what follows by $\operatorname{int}_{C_0^1(\overline{\Omega})}[v_*, u_*]$ we denote the interior in $C_0^1(\overline{\Omega})$ of $[v_*, u_*] \cap C_0^1(\overline{\Omega})$. Also note that an account of hypotheses H_1', $\varphi \in C^2(W_0^{1,p}(\Omega))$ and for all $u, h, y \in W_0^{1,p}(\Omega)$, we have

$$\begin{aligned}\langle \varphi''(u)h, y\rangle = &\int_\Omega (1 + |Du|^{p-2})(Dh, Dy)_{\mathbb{R}^N} dz \\ &+ (p-2)\int_\Omega |Du|^{p-4}(Du, Dh)_{\mathbb{R}^N}(Du, Dy)_{\mathbb{R}^N} dz \\ &- \int_\Omega f_x'(z, u)hy dz.\end{aligned}$$

If $u_0 \in K_\varphi$, then $u_0 \in C_0^1(\overline{\Omega})$ (nonlinear regularity) and so

$$b = |Du_0|^{\frac{p-4}{2}} Du_0 \in L^\infty(\Omega, \mathbb{R}^N).$$

Consider the inner product

$$(h, y)_b = \int_\Omega [(1 + |b|^2)(Dh, Dy)_{\mathbb{R}^N} + (p-2)(b, Dh)_{\mathbb{R}^N}(b, Dy)_{\mathbb{R}^N}]dz$$

and let H_b the closure for this inner product of $C_c^\infty(\Omega)$. By $|\cdot|_b$ we denote the corresponding norm, which is clearly equivalent to the Sobolev norm $\|\cdot\|_{H_0^1(\Omega)}$. Therefore $W_0^{1,p}(\Omega) \hookrightarrow H_b$ continuously. Let $L_b \in \mathcal{L}(H_b, H_b^*)$ be defined by

$$\langle L_b(h), y\rangle = (h, y)_b - \int_\Omega f_x'(z, u_0)hy dz \text{ for all } h, y \in H_b.$$

This is a Fredholm operator of index zero (see Definition I-3.161) and it is the extension of $\varphi''(u_0)$ on H_b. We consider the orthogonal direct sum decomposition

$$H_b = H^- \oplus H^0 \oplus H^+$$

with H^-, H^0, H^+ respectively the negative, null and positive subspaces according to the spectral decomposition of L_b. We know that H^-, H_0 are finite dimensional. Since $u_0 \in C_0^1(\overline{\Omega})$, we have

$$H^- \oplus H^0 \subseteq W_0^{1,p}(\Omega) \cap L^\infty(\Omega).$$

Set $V = H^- \oplus H^0$, $W = H^+ \cap W_0^{1,p}(\Omega)$. Then

$$W_0^{1,p}(\Omega) = V \oplus W \text{ and } \langle \varphi''(u_0)h, h\rangle \geq c\|h\|_b^2$$

for all $h \in W$, some $c > 0$.

Proposition 5.58 *If hypotheses H_1' hold, then problem (5.79) has a nodal solution*

$$y_0 \in \operatorname{int}_{C_0^1(\overline{\Omega})}[v_*, u_*].$$

Proof The proof of this proposition, is divided into a number of steps

Step 1: $u \in \operatorname{int} C_+$ and $v_* \in -\operatorname{int} C_+$ are local minimizers of $\widehat{\varphi}(\cdot)$.

From (5.97) and (5.98), we see that $\widehat{\varphi}_+(\cdot)$ is coercive. Also it is sequentially weakly lower semicontinuous. So, we can find $\widehat{u}_* \in W_0^{1,p}(\Omega)$ such that

$$\widehat{\varphi}_+(\widehat{u}_*) = \inf\{\widehat{\varphi}_+(u) : u \in W_0^{1,p}(\Omega)\} \tag{5.99}$$

As before (see the proof of Proposition 5.55), for $t \in (0, 1)$ small we have

$$\begin{aligned} &\widehat{\varphi}_+(t\widehat{u}_1(2)) < 0, \\ \Rightarrow &\widehat{\varphi}_+(\widehat{u}_*) < 0 = \widehat{\varphi}_+(0) \\ \Rightarrow &\widehat{u}_* \neq 0. \end{aligned} \tag{5.100}$$

From (5.99) we see that $\widehat{u}_* \in K_{\widehat{\varphi}_+} = \{0, u_*\}$ (see Proposition 5.57). So, (5.100) implies that $\widehat{u}_* = u_* \in \operatorname{int} C_+$. From (5.97) and (5.96) it is clear that $\widehat{\varphi}|_{C_+} = \widehat{\varphi}_+|_{C_+}$. Hence we infer that

$$u_* \in \operatorname{int} C_+ \text{ is a } C_0^1(\overline{\Omega})\text{-local minimizer of } \widehat{\varphi}(\cdot),$$

$$\Rightarrow u_* \in \operatorname{int} C_+ \text{ is a } W_0^{1,p}(\Omega)\text{-local minimizer of } \widehat{\varphi}(\cdot).$$

Similarly for $v_* \in -\operatorname{int} C_+$ using this time $\widehat{\varphi}_-(\cdot)$.

This completes Step 1.

Step 2: There exists a solution $y \in \operatorname{int}_{C_0^1(\overline{\Omega})}[v_*, u_*]$ of problem (5.79)

Without any loss of generality, we assume that $\widehat{\varphi}(v_*) \leq \widehat{\varphi}(u_*)$ (the reasoning is similar if the opposite inequality holds). Also, we may assume that $K_{\widehat{\varphi}}$ is finite (otherwise on account of Proposition 5.57, we already have an infinity of smooth nodal solutions and so we are done). Then using Proposition 3.132, we can find $\varrho_0 \in (0, 1)$ small such that

$$\widehat{\varphi}(v_*) \leq \widehat{\varphi}(u_*) < \inf\{\widehat{\varphi}(u) : \|u - u_*\| = \varrho_0\} = \widehat{m}_0 \tag{5.101}$$

We know that $\widehat{\varphi}(\cdot)$ is coercive (see (5.97)). So, Proposition 3.19 implies that

$$\widehat{\varphi}(\cdot) \text{ satisfies the } C\text{-condition.} \tag{5.102}$$

Then (5.101), (5.102), permit the use of the mountain pass theorem (see Theorem 3.41) and so, we can find $y_0 \in W_0^{1,p}(\Omega)$ such that

$$y_0 \in K_{\widehat{\varphi}} \subseteq [v_*, u_*] \cap C_0^1(\overline{\Omega}) \tag{5.103}$$

(see Proposition 5.57)

$$\widehat{m}_0 \leq \widehat{\varphi}(y_0) \tag{5.104}$$

From (5.101) and (5.104) it follows that $y \notin \{u_*, v_*\}$. Recall that if $a(y) = |y|^{p-2}y + y$, $y \in \mathbb{R}^N$, we have

$$\text{div} a(Du) = \Delta_p u + \Delta u \text{ for all } u \in W_0^{1,p}(\Omega).$$

Since $2 < p$, we see that $a \in C^1(\mathbb{R}^N, \mathbb{R})$ and

$$\nabla a(y) = |y|^{p-2}[id + (p-2)\frac{y \otimes y}{|y|^2}] + id \text{ for } y \in \mathbb{R}^N$$

$$\Rightarrow (\nabla a(y)\vartheta, \vartheta)_{\mathbb{R}^N} \geq |\vartheta|^2 \text{ for all } y, \vartheta \in \mathbb{R}^N.$$

Then using Theorem 5.39, we have

$$v_*(z) < y(z) < u_*(z) \text{ for all } z \in \Omega. \tag{5.105}$$

Let $\varrho = \max\{\|u_*\|_\infty, \|v_*\|_\infty\}$ and let $\widehat{\xi}_\varrho > 0$ be as postulated by hypothesis H_1'(iv). We have

$$\begin{aligned} &- \Delta_p y_0 - \Delta y_0 + \widehat{\xi}_\varrho |y|^{p-2} y_0 \\ =& f(z, y_0) + \widehat{\xi}_\varrho |y_0|^{p-2} y_0 \\ \leq& f(z, u_*) + \widehat{\xi}_\varrho u_*^{p-1} \text{ (see (5.103) and hypothesis } H_1'\text{(iv))} \\ =& - \Delta_p u_* - \Delta u_* + \widehat{\xi}_\varrho u_*^{p-1} \text{ in } \Omega \text{ (since } u_* \in S_+). \end{aligned} \tag{5.106}$$

From (5.106), (5.105) and Theorem 5.34, we infer that

$$u_* - y_0 \in \text{int} C_+.$$

Similarly we show that

$$y_0 - v_* \in \text{int} C_+.$$

Therefore we can say that

$$y_0 \in \text{int}_{C_0^1(\overline{\Omega})}[v_*, u_*].$$

This completes Step 2.

Evidently y_0 will be desired nodal solution, if we can show that it is nontrivial. This will be done in the next two steps by computing the critical groups of $\widehat{\varphi}(\cdot)$ at y_0 and at the origin.

Step 3: $C_k(\widehat{\varphi}, y_0) = \delta_{k,1}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.

Using Step 2 and the local nature of critical groups, we have

$$C_k(\widehat{\varphi}|_{C_0^1(\overline{\Omega}}, y_0) = C_k(\varphi|_{C_0^1(\overline{\Omega})}, y_0) \text{ for all } k \in \mathbb{N}_0$$

$$\Rightarrow C_k(\widehat{\varphi}, y_0) = C_k(\varphi, y_0) \text{ for all } k \in \mathbb{N}_0 \tag{5.107}$$

From Step 2 we know that $y_0 \in K_{\widehat{\varphi}}$ is of mountain pass type, hence

$$C_1(\widehat{\varphi}, y_0) \neq 0 \text{ (see Corollary 3.123)},$$

$$\Rightarrow C_1(\varphi, y_0) \neq 0 \text{ (see (5.107))} \tag{5.108}$$

From Lemma 2.2 of Cingolani and Vannella [102], we know that we can find $\varrho > 0$ and $\eta : V \cap \overline{B}_\varrho \to \mathbb{R}$ (V as before and $\overline{B}_\varrho = \{u \in W_0^{1,p}(\Omega) : \|u\| \leq \varrho\}$), such that

$$\langle \eta''(0)h, y\rangle = \langle \varphi''(y_0)h, y\rangle \text{ for all } h, y \in V$$

The operator $\eta''(0)$ is Fredholm and $\ker\eta''(0) = H^0$. We have

$$C_k(\varphi, y_0) = C_k(\eta, 0) \text{ for all } k \in \mathbb{N}_0, \tag{5.109}$$

$$\Rightarrow C_1(\eta, 0) \neq 0 \text{ (see (5.108))}. \tag{5.110}$$

It follows that $d = \dim H^- \leq 1$. Let $d_0 = \dim H^0$

If $d_0 = 0$, then the origin is a nondegenerate critical point of η with Morse index d (see Definition 3.54). Hence

$$C_k(\eta, 0) = \delta_{k,d}\mathbb{Z} \text{ for all } k \in \mathbb{N} \text{ (see Proposition 3.100)},.$$

$$\Rightarrow d = 1 \text{ (see (5.110))},$$

$$\Rightarrow C_k(\varphi, y_0) = \delta_{k,1}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0 \text{ (see (5.109))},$$

$$\Rightarrow C_k(\widehat{\varphi}, y_0) = \delta_{k,1}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0 \text{ (see (5.107))}.$$

If $d_0 > 0$, then the origin is a degenerate critical point of $\eta(\cdot)$ and by the Shifting Theorem (see Theorem 3.103), we have

$$C_k(\eta, 0) = C_{k-d}(\widehat{\eta}, 0) \text{ for all } k \in \mathbb{N}_0,$$

with $\widehat{\eta} = \eta|_{H^0}$. Suppose $d = 1$. Then

$$C_0(\widehat{\eta}, 0) \neq 0,$$

$$\Rightarrow C_k(\eta, 0) = \delta_{k,1} Z \text{ for all } k \in \mathbb{N}_0 \text{ (see Corollary 3.104)}$$

Now suppose $d = 0$. Then

$$C_k(\eta, 0) = C_k(\widehat{\eta}, 0) \text{ for all } k \in \mathbb{N}_0,$$

$$\Rightarrow C_1(\widehat{\eta}, 0) \neq 0.$$

We show that "if $\sigma(\widehat{\eta}''(0)) \subseteq [0, \infty)$ (that is, the spectrum of $\widehat{\eta}''(0)$ is in $\mathbb{R}_0$), then $\dim\ker\eta''(0) \leq 1$". Hence for $\ker\eta''(0)$ to be nontrivial, it amounts to saying that 1 is the first eigenvalue of the weighted linear eigenvalue problem

$$-\operatorname{div}((1 + |b|^2)Du + (p-2)(b, Du)_{\mathbb{R}^N}) = \lambda f_x'(z, y_0)u,\ u|_{\partial\Omega=0}.$$

We know that the first eigenvalue of this problem is simple. So, we have

$$C_k(\varphi, y_0) = \delta_{k,1}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0$$

$$C_k(\widehat{\varphi}, y_0) = \delta_{k,1}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0 \text{ (see (5.107)).}$$

This completes Step 3.

Let $d_m = \dim \bigoplus_{k=1}^{m} E(\widehat{\lambda}_k(2)) \geq 2$ (since $m \geq 2$).

Step 4: $C_{d_m}(\widehat{\varphi}, 0) \neq 0$.

We set

$$\widehat{Y} = W_0^{1,p}(\Omega) \cap [\bigoplus_{k=1}^{m} E(\widehat{\lambda}_k(2))] \text{ and } \widehat{W} = W_0^{1,p}(\Omega) \cap \overline{[\bigoplus_{k\geq m+1} E(\widehat{\lambda}_k(2))]}.$$

We have $W_0^{1,p}(\Omega) = \widehat{Y} \oplus \widehat{W}$. Note that $\widehat{Y}$ is finite dimensional and $\widehat{Y} \subseteq C_0^1(\overline{\Omega})$. So, we can find $\varrho_0 \in (0, 1)$ such that

$$\|y\| \leq \varrho_0 \Rightarrow \|y\|_{C_0^1(\overline{\Omega})} \leq \delta_0 \text{ for all } y \in \widehat{Y}$$

(see Corollary I-3.41)

Here $\delta_0 > 0$ is as postulated by hypothesis H_1'(iii). If $y \in \widehat{Y}$, $\|y\| \leq \varrho_0$, we have

$$\varphi(y) \leq \frac{1}{p}\|Dy\|_p^p + \frac{1}{2}\|Dy\|_2^2 - \frac{1}{2}\int_\Omega \eta_0(z)y^2 dz$$

$$\leq \frac{1}{p}\|y\|^p - c\|y\|^2 \text{ for some } c > 0$$

(recall $\widehat{Y}$ is finite dimensional),

$$\Rightarrow \varphi(y) \leq 0 \text{ for all } y \in \widehat{Y} \text{ with } \|y\| \leq \widehat{\varrho}_0 \leq \varrho_0. \tag{5.111}$$

Now let $u \in \widehat{W}$. Using hypotheses H_1'(i),(iii) we have that

$$F(z,x) \leq \frac{\widehat{\lambda}_{m+1}(2)}{2}x^2 + \widehat{c}|x|^r$$

for a.a $z \in \Omega$, all $x \in \mathbb{R}$, some $\widehat{c} > 0$,

$$\Rightarrow \varphi(u) \geq \frac{1}{p}\|u\|^p - \widehat{c}_0\|u\|^r \text{ for some } \widehat{c}_0 > 0,$$

$$\varphi(u) > 0 \text{ for all } u \in \widehat{W}, 0 < \|u\| \leq \widetilde{\varrho}_0 \leq \varrho. \tag{5.112}$$

Then from (5.111), (5.112) and Theorem 3.126, we have

$$C_{d_m}(\varphi, 0) \neq 0,$$

$$\Rightarrow C_{d_m}(\widehat{\varphi}, 0) \neq 0 \text{ with } \widehat{d}_m \geq 2 \text{ (see (5.107))}.$$

This completes Step 4.
From Step 3 and 4 it follows that $y_0 \in \text{int}_{C_0^1}[v_*, u_*]$ is nontrivial and so it is a nodal solution of (5.79).

□

Concluding, we can state the following multiplicity theorem for problem (5.79).

Theorem 5.59 *If hypotheses H_1' hold, then problem (5.79) has at least three nontrivial solutions*

$$u_0 \in \text{int}C_+, v_0 \in -\text{int}C_+,$$

$$y_0 \in \text{int}_{C_0^1(\overline{\Omega})}[v_0, u_0] \text{ nodal.}$$

By strengthening hypothesis H_1'(iii) (the condition near the origin), we can improve the above multiplicity theorem and generate a second nodal solution, for a total of four nontrivial smooth solutions, all with sign information.

The new hypotheses on the reaction $f(z,x)$, are the following.

$\mathbf{H_1''}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a measurable function such that for a.a $z \in \Omega$, $f(z,0) = 0$, $f(z,\cdot) \in C^1(\mathbb{R})$, hypothese H_1''(i),(ii), are the same as the corresponding hypotheses H_1'(i),(ii) and (iii) there exists $m \in \mathbb{N}$, $m \geq 2$ such that

$$f'_x(z,0) \in [\widehat{\lambda}_m(2), \widehat{\lambda}_{m+1}(2)] \text{ for a.a } z \in \Omega,$$

$$f'_x(\cdot,0) \not\equiv \widehat{\lambda}_m(2),\ f'_x(\cdot,0) \not\equiv \widehat{\lambda}_{m+1}(2)$$

$$f'_x(z,0) = \lim_{x\to 0} \frac{f(z,x)}{x} \text{ uniformly for a.a } z \in \Omega.$$

Remark 5.60 With these hypotheses, we do not allow resonance at zero, only nonuniform nonresonance in the spectral interval $[\widehat{\lambda}_m(2), \widehat{\lambda}_{m+1}(2)]$. Also note that as in hypotheses H'_1, we drop condition (iv) since it is automatically satisfied on account of the differentiability of $f(z,\cdot)$ and of the growth condition on $f'_x(z,\cdot)$ (see H''_1(i)).

Theorem 5.61 *If hypotheses H''_1 hold, then problem (5.79) has at least four nontrivial solutions*

$$u_0 \in \operatorname{int} C_+,\ v_0 \in -\operatorname{int} C_+,$$

$$y_0, \widehat{y} \in \operatorname{int}_{C^1_0(\overline{\Omega})}[v_0, u_0] \text{ nodal.}$$

Proof From Theorem 5.59 we already have three nontrivial solutions $u_0 \in \operatorname{int} C_+$, $v_0 \in -\operatorname{int}_{C^1_0(\overline{\Omega})}[v_0, u_0]$ nodal.

$$u_0 \in \operatorname{int} C_+,\ v_0 \in -\operatorname{int} C_+,\ y_0 \in \operatorname{int}_{C^1_0(\overline{\Omega})}[v_0, u_0] \text{ nodal.}$$

From Step 1 in the proof of Proposition 5.58, we know that u_0 and v_0 are local minimizers of the functional $\widehat{\varphi}(\cdot)$. Therefore

$$C_k(\widehat{\varphi}, u_0) = C_k(\widehat{\varphi}, v_0) = \delta_{k,0}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.113}$$

Also from Step 3 in the proof of Proposition 5.58, we have

$$C_k(\widehat{\varphi}, y_0) = \delta_{k,1} Z \text{ for all } k \in \mathbb{N}_0. \tag{5.114}$$

Consider the C^2-functional $\varphi_0 : H'(\Omega) \to \mathbb{R}$ defined by

$$\varphi_0(u) = \frac{1}{2}\|Du\|_2^2 - \int_\Omega f'_x(z,0)u^2 dz \text{ for all } u \in H^1_0(\Omega).$$

Using hypothesis H''_1(iii) and Proposition 4.48, we see that $u = 0$ is a nondegenerate critical point of $\varphi_0(\cdot)$ with Morse index $d_m = \dim \bigoplus_{k=1}^{m} E(\widehat{\lambda}_k(2)) \geq 2$. Therefore from Proposition 3.100 we have

$$C_k(\varphi_0, 0) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.115}$$

Let $\widehat{\varphi}_0 = \varphi_0|_{W_0^{1,p}(\Omega)}$. From Proposition 3.128 and (5.115), we obtain

$$C_k(\widehat{\varphi}_0, 0) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.116}$$

Then (5.116) and the C^1-continuity of critical groups (see Theorem 3.129), imply that

$$C_k(\widehat{\varphi}, 0) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.117}$$

Recall that $\widehat{\varphi}(\cdot)$ is coercive. Hence Proposition 3.115 says that

$$C_k(\widehat{\varphi}, \infty) = \delta_{k,0}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.118}$$

Suppose $K_{\widehat{\varphi}} = \{0, u_0, v_0, y_0\}$. From (5.115), (5.116), (5.117), (5.118) and the global Morse relation (see Theorem 3.117) with $t = -1$, we have

$$(-1)^{d_m} + 2(-1)^0 + (-1)^1 = (-1)^0,$$

a contradiction. So, there exists $\widehat{y} \in K_{\widehat{\varphi}}$, $\widehat{y} \notin \{0, u_0, v_0, y_0\}$. From Proposition 5.57, we have

$$\widehat{y} \in [v_0, u_0] \cap C_0^1(\overline{\Omega}),$$

$$\Rightarrow \widehat{y} \text{ is a nodal solution of (5.79)}, \widehat{y} \neq y_0.$$

As in Step 2 in the proof of Proposition 5.58, we show that

$$\widehat{y} \in \text{int}_{C_0^1(\overline{\Omega})}[v_0, u_0].$$

□

So far we have assumed that $2 < p$ (degenerate case). Now we will consider the case $1 < p < 2$ (singular case). In this case the ambient space is the Hilbert space $H_0^1(\Omega)$ and this is more convenient for the use of Morse theoretic arguments. On the other hand the operator $-\Delta_p$ is singular.

The hypotheses on the reaction $f(z, x)$ are the following

H_2 : $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that for a.a $z \in \Omega$ $f(z, 0) = 0$, $|f(z, x)| \leq a(z)(1 + |x|)$ for all $x \in \mathbb{R}$ with $a \in L^\infty(\Omega)$ and

(i) there exist $m \in \mathbb{N}$, $m \geq 2$ and $\eta \in L^\infty(\Omega)$ such that

$$\eta(z) \leq \widehat{\lambda}_{m+1}(2) \text{ for a.a } z \in \Omega, \eta \not\equiv \widehat{\lambda}_{m+1}(2)$$

$$(f(z, x) - f(z, y))(x - y) \leq \eta(z)(x - y)^2 \text{ for a.a } z \in \Omega, \text{ all } x, y \in \mathbb{R};$$

(ii) $\widehat{\lambda}_m(2) \leq \liminf_{x\to\pm\infty} \dfrac{f(z,x)}{x}$ uniformly for a.a $z \in \Omega$.

(iii) for every $\varrho > 0$, there exists $\widehat{\xi}_\varrho > 0$ such that

$$f(z,x)x + \widehat{\xi}_\varrho x^2 \geq 0$$

for a.a $z \in \Omega$, all $x \in [-\varrho, \varrho]$.

Remark 5.62 Hypothesis H_2(ii) implies that as $x \to \pm\infty$, we can have resonance with respect to a nonprincipal eigenvalue.

To deal with the singular case, we will employ the so-called "Lyapunov-Schmidt reduction method". According to this method, the initial infinite dimensional problem, is reduced to finite dimensional one which is easier to solve.

Let $\varphi : H_0^1(\Omega) \to \mathbb{R}$ be the energy functional for problem (5.79), defined by

$$\varphi(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_\Omega F(z,u)dz \quad \text{for all } u \in H_0^1(\Omega).$$

We know that $\varphi \in C^1(H_0^1(\Omega))$. As before, to produce constant sign solutions, we consider the positive and negative truncations of $\varphi(\cdot)$, namely the C^1-functionals $\varphi_\pm : H_0^1(\Omega) \to \mathbb{R}$ defined by

$$\varphi_\pm(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{2}\|Du\|_2^2 - \int_\Omega F(z,\pm u^\pm)dz \quad \text{for all } u \in H_0^1(\Omega).$$

Proposition 5.63 *If $1 < p < 2$ and hypotheses H_2 hold, then the functionals φ, $\varphi_\pm$ satisfy the C-condition.*

Proof We do the proof for the function $\varphi_+(\cdot)$, the proofs for $\varphi_-(\cdot)$, $\varphi(\cdot)$ being similar.

So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq H_0^1(\Omega)$ be a sequence such that $\{\varphi_+(u_n)\}_{n\in\mathbb{N}} \subseteq \mathbb{R}$ is bounded and

$$(1 + \|u_n\|)\varphi'_+(u_n) \to 0 \text{ in } H^{-1}(\Omega) = H_0^1(\Omega)^* \text{ as } n \to \infty. \tag{5.119}$$

Now $\|\cdot\|$ denotes the norm of the Hilbert space $H_0^1(\Omega)$. On account of the Poincare inequality, we have $\|u\| = \|Du\|_2$ for all $u \in H_0^1(\Omega)$. From (5.119), we have

$$\left|\langle A_p(u_n), h\rangle + \langle A(u_n), h\rangle - \int_\Omega f(z,u_n^+)hdz\right| \leq \frac{\varepsilon_n\|h\|}{1+\|u_n\|} \tag{5.120}$$
$$\text{for all } h \in H_0^1(\Omega), \text{ with } \varepsilon_n \to 0^+.$$

In (5.120) we use the test function $h = -u_n^- \in H_0^1(\Omega)$. Then

$$\begin{aligned} &\|Du_n^-\|_p^p + \|Du_n^-\|_2^2 \le \varepsilon_n \quad \text{for all } n \in \mathbb{N}, \\ &\Rightarrow u_n^- \to 0 \text{ in } H_0^1(\Omega) \text{ as } n \to \infty. \end{aligned} \tag{5.121}$$

From (5.120) and (5.121), if follows that

$$\left|\langle A_p(u_n^+), h\rangle + \langle A(u_n^+), h\rangle - \int_\Omega f(z, u_n^+)h dz\right| \le \varepsilon_n' \|h\| \quad \text{for all } h \in H_0^1(\Omega), \text{ with } \varepsilon_n' \to 0. \tag{5.122}$$

Suppose that $\|u_n^+\| \to \infty$ and let $y_n = \frac{u_n^+}{\|u_n^+\|}$ $n \in \mathbb{N}$. Then $\|y_n\| = 1$, $y_n \ge 0$ for all $n \in \mathbb{N}$. We may assume that

$$y_n \xrightarrow{w} y \text{ in } H_0^1(\Omega), \ y_n \to y \text{ in } L^2(\Omega). \tag{5.123}$$

From (5.122), we have

$$\left|\frac{1}{\|u_n^+\|^{2-p}}\langle A_p(y_n), h\rangle + \langle A(y_n), h\rangle - \int_\Omega \frac{f(z, u_n^+)}{\|u_n^+\|}h dz\right| \le \varepsilon_n''\|h\| \quad \text{for all } h \in H_0^1(\Omega), \text{ with } \varepsilon_n'' \to 0. \tag{5.124}$$

Note that $\left\{\frac{f(\cdot, u_n^+(\cdot))}{\|u_n^+\|}\right\}_{n\in\mathbb{N}} \subseteq L^2(\Omega)$ is bounded. Hence we may assume that

$$\frac{f(\cdot, u_n^+(\cdot))}{\|u_n^+\|} \xrightarrow{w} g \text{ in } L^2(\Omega) \text{ as } n \to \infty \tag{5.125}$$

with $g(z) = \vartheta(z)y(z)$, $\widehat{\lambda}_m(2) \le \vartheta(z) \le \eta(z)$ for a.a $z \in \Omega$ (see hypotheses H_2(i),(ii)). In (5.124), we choose the test function $h = y_n - y \in H_0^1(\Omega)$, pass to the limit as $n \to \infty$ and use (5.123), (5.125). We obtain

$$\begin{aligned} &\lim_{n\to\infty}\langle A(y_n), y_n - y\rangle = 0, \\ &\Rightarrow \|Dy_n\|_2 \to \|Dy\|_2, \\ &\Rightarrow y_n \to y \text{ in } H_0^1(\Omega) \text{ (Kadec-Klee property)}, \ \|y\| = 1. \end{aligned} \tag{5.126}$$

So, in the limit as $n \to \infty$, we have

$$\begin{aligned} &\langle A(y), h\rangle = \int_\Omega \vartheta yh dz \text{ for all } h \in H_0^1(\Omega) \\ &\Rightarrow -\Delta y(z) = \vartheta(z)y(z) \text{ in } \Omega, \ y|_{\partial\Omega} = 0. \end{aligned} \tag{5.127}$$

Suppose that $\vartheta \not\equiv \widehat{\lambda}_m(2)$. Then from Proposition 4.72 we have

$$\widetilde{\lambda}_m(2,\vartheta) < \widetilde{\lambda}_m(2,\widehat{\lambda}_m(2)) = 1 \quad \text{and} \quad \widetilde{\lambda}_{m+1}(2,\widehat{\lambda}_{m+1}(2)) = 1 < \widetilde{\lambda}_{m+1}(2,\vartheta).$$

Therefore from (5.127) we infer that $y = 0$, a contradiction since $\|y\| = 1$ (see (5.127)).

So, suppose $\vartheta(z) = \widehat{\lambda}_m(2)$ for a.a $z \in \Omega$. Then from (5.127) we infer that $y(\cdot)$ must be nodal (see Proposition 4.51), a controdiction since $y \geq 0$. We conclude that $\{u_n^+\}_{n\in\mathbb{N}} \subseteq H_0^1(\Omega)$ is bounded and so

$$\{u_n\}_{n\in\mathbb{N}} \subseteq H_0^1(\Omega) \text{ is bounded (see (5.121))}.$$

We may assume that

$$u_n \xrightarrow{w} u \text{ in } H_0^1(\Omega),\ u_n \to u \text{ in } L^2(\Omega). \tag{5.128}$$

In (5.120), we use $h = u_n - u \in H_0^1(\Omega)$, pass to the limit as $n \to \infty$ and use (5.128). Then

$$\begin{aligned}
&\lim_{n\to\infty} \langle V(u_n), u_n - u\rangle = 0 \text{ (with } V = A_p + A),\\
\Rightarrow &u_n \to u \text{ in } H_0^1(\Omega),\\
\Rightarrow &\varphi_+(\cdot) \text{ satisfies the } C - \text{condition}.
\end{aligned}$$

Similarly for the functionals $\varphi_-(\cdot)$, $\varphi(\cdot)$. □

Remark 5.64 The above Proposition remains true even if $m = 1$, provided we introduce the extra condition that

$$\lim \frac{f(z,x)x - 2F(z.x)}{|x|^p} = -\infty \text{ uniformly for a.a } z \in \Omega$$

(recall $F(z,x) = \int_0^x f(z,s)ds$) or if H_2(ii) is replaced by the more restrictive condition

$$\eta_0(z) \leq \liminf_{x\to\pm\infty} \frac{f(z,x)}{x} \text{ uniformly for a.a } z \in \Omega,$$

with $\eta_0 \in L^\infty(\Omega)$, $\widehat{\lambda}_m(2) \leq \eta_0(z)$ for a.a $z \in \Omega$, $\eta_0 \not\equiv \widehat{\lambda}_m(2)$ (nonuniform nonresonance as $x \to \pm\infty$).

Proposition 5.65 *If $1 < p < 2$ and hupotheses H_2 hold, then $u = 0$ is a local minimizer of the functionals $\varphi_\pm(\cdot)$ and $\varphi(\cdot)$.*

Proof Clearly we have

$$\lim_{x\to 0}\frac{f(z,x)}{|x|^{p-2}x}=0 \text{ uniformly for a.a } z\in\Omega.$$

So, given $\varepsilon>0$, we can find $\delta=\delta(\varepsilon)>0$ such that

$$F(z,x)\le\frac{\varepsilon}{p}x^p \text{ for a.a } z\in\Omega,\ \text{all } 0\le x\le\delta.$$

Let $u\in C_0^1(\overline{\Omega})$ with $\|u\|_{C_0^1(\overline{\Omega})}\le\delta$. Then

$$\varphi_+(u)\ge\frac{1}{p}\left(1-\frac{\varepsilon}{\widehat{\lambda}_1(p)}\right)\|Du\|_p^p+\frac{1}{2}\|Du\|_2^2.$$

So choosing $\varepsilon\in(0,\widehat{\lambda}_1(p))$ we see that

$$\begin{aligned}&u=0 \text{ is a } C_0^1(\overline{\Omega})-\text{local minimizer of } \varphi_+(\cdot),\\ \Rightarrow\ &u=0 \text{ is a } H_0^1(\Omega)-\text{local minimizer of } \varphi_+(\cdot)\\ &\qquad\qquad\qquad

Proposition 5.67 *If $1 < p < 2$ and hupotheses H_2 hold, then problem (5.79) has at least two nontrivial constant sign solutions*

$$u_0 \in intC_+,\ v_0 \in -intC_+.$$

Proof We can easily check that $K_{\varphi_+} \subseteq C_+$. So, we assume that K_{φ_+} is finite or otherwise, we already have an infinity of positive solutions which by Theorem 4.89 belong in $intC_+$ and so we are done. From Proposition 5.65, we know that $u = 0$ is a local minimizer of $\varphi_+(\cdot)$. Using Proposition 3.132, we can find $\varrho \in (0, 1)$ small such that

$$\varphi_+(0) = 0 < \inf\{\varphi_+(u) : \|u\| = \varrho\} = m_+. \tag{5.129}$$

From Proposition 5.63 we know that

$$\varphi_+(\cdot) \text{ satisfies the } C - \text{condition.} \tag{5.130}$$

Moreover, if $u \in E(\widehat{\lambda}_m(2))$, $\|u\|_2 = 1$, then from Proposition 5.66, we have

$$\varphi_+(tu) \to -\infty \text{ as } t \to +\infty. \tag{5.131}$$

Then (5.129), (5.130), (5.131) permit the use of the mountain pass theorem. So, we can find $u_0 \in H_0^1(\Omega)$ such that

$$\begin{aligned} &u_0 \in K_{\varphi_+} \subseteq C_+,\ 0 < m_+ \leq \varphi_+(u_0), \\ \Rightarrow &u_0 \in C_+ \setminus \{0\}. \end{aligned}$$

Let $\varrho = \|u_0\|_\infty$ and let $\widehat{\xi}_\varrho > 0$ be as postulated by hypothesis H_2(iii). We have

$$\begin{aligned} &\Delta_p u_0 + \Delta u_0 \leq \widehat{\xi}_\varrho u_0, \\ \Rightarrow &u_0 \in intC_+ \text{ (see Theorem 4.89).} \end{aligned}$$

Similarly, working this time with the functional $\varphi_-(\cdot)$, we produce a negative solution $v_0 \in -intC_+$ of (5.79). □

As we already mentioned earlier, our approach uses the "Lyapunov-Schmidt reduction method". To implement it we consider the following orthogonal direct sum decomposition of the ambient space $H_0^1(\Omega)$

$$H_0^1(\Omega) = Y \oplus \widehat{H},$$

with $Y = \oplus_{k=1}^m E(\widehat{\lambda}_k(2))$ and $\widehat{H} = Y^\perp$. The reduction method with be based on this decomposition and the next proposition, is a major step in this direction.

Proposition 5.68 *If $1 < p < 2$ and hypotheses H_2 hold, then there exists a continuous map $\gamma_0 : \gamma \to \widehat{H}$ such that*

$$\varphi(y + \gamma_0(y)) = \inf\{\varphi(y + \widehat{u}) : \widehat{u} \in \widehat{H}\} \text{ for all } y \in Y.$$

Proof Fix $y \in Y$ and consider the C^1-functional $\varphi_y : H_0^1(\Omega) \to \mathbb{R}$ defined by

$$\varphi_y(u) = \varphi(y + u) \text{ for all } u \in H_0^1(\Omega).$$

Let $\widehat{i} : \widehat{H} \to H_0^1(\Omega)$ be the inclusion map and define

$$\widehat{\varphi}_y = \varphi_y \circ \widehat{i} : \widehat{H} \to \mathbb{R}.$$

Using the chain rule, we have

$$\widehat{\varphi}'_y(\widehat{u}) = p_{\widehat{H}^*}\varphi'_y(\widehat{u}) \text{ for all } \widehat{u} \in \widehat{H}$$

with $p_{\widehat{H}^*}$ being the orthogonal projection of $H^{-1}(\Omega) = H_0^1(\Omega)^*$ onto $\widehat{H}^*$. If $\widehat{u}_1$, $\widehat{u}_2 \in \widehat{H}$, then

$$\begin{aligned}
&\langle \widehat{\varphi}_y(\widehat{u}_1) - \widehat{\varphi}_y(\widehat{u}_2), \widehat{u}_1 - \widehat{u}_2\rangle_{\widehat{H}} \\
=&\langle \widehat{\varphi}'_y(\widehat{u}_1) - \widehat{\varphi}'_y(\widehat{u}_2), \widehat{u}_1 - \widehat{u}_2\rangle \\
=&\langle A_p(y + \widehat{u}_1) - A_p(y + \widehat{u}_2), \widehat{u}_1 - \widehat{u}_2\rangle + \langle A(y + \widehat{u}_1) - A(y + \widehat{u}_2), \widehat{u}_1 - \widehat{u}_2\rangle \\
&- \int_\Omega [f(z, y + \widehat{u}_1) - f(z, y + \widehat{u}_2)](\widehat{u}_1 - \widehat{u}_2)dz \\
\geq&\|D(\widehat{u}_1 - \widehat{u}_2)\|_2^2 - \int_\Omega \eta(z)(\widehat{u}_1 - \widehat{u}_2)^2 dz \\
&\qquad \text{(since } A_p(\cdot) \text{ is monotone and using hypothesis } H_2(\text{i}))\\
\geq&c\|\widehat{u}_1 - \widehat{u}_2\|^2 \text{ for some } c > 0 \\
&\qquad \text{(see Proposition 4.48 since } \widehat{u}_1,\ \widehat{u}_2 \in \widehat{H}),\\
\Rightarrow&\widehat{\varphi}'_y(\cdot) \text{ is strongly monotone,}\\
\Rightarrow&\widehat{\varphi}_y(\cdot) \text{ is strictly convex.}
\end{aligned}$$

Also, for every $\widehat{u} \in \widehat{H}$, we have

$$\begin{aligned}
\langle \widehat{\varphi}'_y(\widehat{u}), \widehat{u}\rangle &= \langle \widehat{\varphi}'_y(\widehat{u}) - \widehat{\varphi}'_y(0), \widehat{u}\rangle + \langle \varphi'_y(0), \widehat{u}\rangle \\
&\geq c\|\widehat{u}\|^2 - \widehat{c}\|\widehat{u}\| \text{ for some } \widehat{c} > 0, \qquad (5.132)\\
\Rightarrow&\widehat{\varphi}'_y(\cdot) \text{ is coercive.}
\end{aligned}$$

Therefore $\widehat{\varphi}'_y(\cdot)$ is continuous and strongly monotone, which means that $\widehat{\varphi}'_y(\cdot)$ is maximal monotone (see Proposition I-7.36). A coercive, maximal monotone map, is surjective (see Theorem I-7.47). So, we can find $\widehat{u}_0 \in \widehat{H}$ such that

$$\widehat{\varphi}'_y(\widehat{u}_0) = 0.$$

The strong monotonicity of $\widehat{\varphi}'_y(\cdot)$ implies that this solution $\widehat{u}_0 \in \widehat{H}$ is unique and it is a global minimizer of the strictly convex function $\widehat{\varphi}'_y(\cdot)$. Hence we can define a single valued map $\gamma_0 : Y \to \widehat{H}$ which to each $y \in Y$ assigns the unique global minimizer of $\widehat{\varphi}'_y(\cdot)$. We have

$$\left.\begin{array}{l} 0 = \widehat{\varphi}'_y(\gamma_0(y)) = p_{\widehat{H}^*}\varphi'(y + \gamma_0(y)), \\ \varphi(y + \gamma_0(y)) = \inf\{\varphi(y + \widehat{u}) : \widehat{u} \in \widehat{H}\}. \end{array}\right\} \tag{5.133}$$

Now we check the continuity of $\gamma_0 : Y \to \widehat{H}$. So, let $y_n \to y$ in Y. From (5.132), (5.133) if follows that

$$\{\gamma_0(u_n)\}_{n\in\mathbb{N}} \subseteq \widehat{H} \subseteq \widehat{H}^1_0(\Omega) \text{ is bounded.}$$

So, we may assume that

$$\gamma_0(u_n) \xrightarrow{w} \widehat{h} \text{ in } H^1_0(\Omega),\ \widehat{h} \in \widehat{H}.$$

Recall that $\varphi(\cdot)$ is sequentially weakly lower semicontinuous. Hence

$$\varphi(y + \widehat{h}) \leq \liminf_{n\to\infty} \varphi(y_n + \gamma_0(y_n)). \tag{5.134}$$

From (5.133) we know that

$$\begin{aligned} &\varphi(y_n + \gamma_0(u_n)) \leq \varphi(y_n + \widehat{u}) \text{ for all } n \in \mathbb{N},\ \text{all } \widehat{u} \in \widehat{H}, \\ \Rightarrow &\varphi(y + \widehat{h}) \leq \varphi(y + \widehat{u}) \text{ for all } \widehat{u} \in \widehat{H} \text{ (see (5.134))}, \\ \Rightarrow &\widehat{h} = \gamma_0(y), \\ \Rightarrow &\gamma_0(y_n) \xrightarrow{w} \gamma_0(y) \text{ in } H^1_0(\Omega). \end{aligned} \tag{5.135}$$

Once more, from (5.133) we have

$$\begin{aligned} &p_{\widehat{H}^*}\varphi'(y_n + \gamma_0(y_n)) = 0 \quad \text{for all } n \in \mathbb{N}, \\ \Rightarrow &p_{\widehat{H}^*}(A_p(y_n + \gamma_0(y_n)) + A(y_n + \gamma_0(y_n))) \\ = &p_{\widehat{H}^*}N_f(y_n + \gamma_0(y_n)) \quad \text{for all } n \in \mathbb{N}. \end{aligned} \tag{5.136}$$

Here $N_f(u)(\cdot) = f(\cdot, u(\cdot))$ is the Nemytskii map corresponding to $f(\cdot,\cdot)$ (see Definition I-6.14). On (5.136) we act with $\gamma_0(y_n) - \gamma_0(y)$ and pass to the limit as $n \to \infty$. We obtain

$$\begin{aligned} &\limsup \langle A(y_n + \gamma_0(y_n)) - A(y + \gamma_0(y)), \gamma_0(y_n) - \gamma_0(y)\rangle \leq 0, \\ &\Rightarrow \gamma_0(y_n) \to \gamma_0(y) \text{ (Kadec-Klee property and (5.135))}, \\ &\Rightarrow \gamma_0(\cdot) \text{ is continuous.} \end{aligned}$$

□

We consider the function $\psi : Y \to \mathbb{R}$ defined by

$$\psi(y) = \varphi(y + \gamma_0(y)) \quad \text{for all } y \in Y.$$

This function is defined on a finite dimensional space. Since $\gamma_0(\cdot)$ is only continuous, it is not immediately clear that $\psi(\cdot)$ is C^1.

Proposition 5.69 *If $1 < p < 2$ and hupotheses H_2 hold, then $\psi \in C^1(Y, \mathbb{R})$.*

Proof Let $y,\ v \in Y$ and $t > 0$ (the reasoning is similar for $t < 0$). Then

$$\begin{aligned} &\frac{\psi(y + tv) - \psi(y)}{t} \leq \frac{\varphi(y + tv + \gamma_0(y)) - \varphi(y + \gamma_0(y))}{t} \text{ (see (5.133))}, \\ &\Rightarrow \limsup_{t \to 0^+} \frac{\psi(y + tv) - \psi(y)}{t} \leq \langle \varphi'(y + \gamma_0(y)), v\rangle. \end{aligned} \tag{5.137}$$

In addition, we have

$$\begin{aligned} &\frac{\psi(y + tv) - \psi(y)}{t} \\ &\geq \frac{\varphi(y + tv + \gamma_0(y + tv)) - \varphi(y + \gamma_0(y + tv))}{t} \text{ (see (5.133))} \\ &\Rightarrow \limsup_{t \to 0^+} \frac{\psi(y + tv) - \psi(y)}{t} \geq \langle \varphi'(y + \gamma_0(y)), v\rangle. \end{aligned} \tag{5.138}$$

Then (5.137), (5.138) imply that $\psi(\cdot)$ is Gateaux differentiable and if $i : Y \to H_0^1(\Omega)$ is the inclusion map, then

$$\begin{aligned} &\langle \psi'_G(y), v\rangle_Y = \langle \varphi'(y + \gamma_0(y)), i(y)\rangle \quad \text{for all } y \in Y, \\ &\Rightarrow \psi'_G(y) = p_{Y^*}\varphi'(y + \gamma_0(y)), \end{aligned} \tag{5.139}$$

with p_{Y^*} being the orthogonal projection of $H^{-1}(\Omega) = H_0^1(\Omega)^*$ onto Y^*. From (5.139) and the continuity of $\gamma_0(\cdot)$ (see Proposition 5.58) we infer that $\varphi \in C^1(Y, \mathbb{R})$. □

For the next proposition, we need to strengthen our hypotheses on the reaction $f(z,x)$.

The new hypotheses on $f(z,x)$ are the following:

$\mathbf{H_2'}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that for a.a $z \in \Omega$ $f(z,0) = 0$, $|f(z,x)| \leq a(z)(1+|x|)$ for all $x \in \mathbb{R}$ with $a \in L^\infty(\Omega)$, hypotheses H_2'(i), (ii), (iii) are the same as the corresponding hypotheses H_2(i), (ii), (iii) and (iv) $\lim_{x \to \pm\infty} \dfrac{f(z,x)x - 2F(z,x)}{|x|^p} = -\infty$ uniformly for a.a $z \in \Omega$.

Proposition 5.70 *If $1 < p < 2$ and hupotheses H_2' hold, then $\psi(\cdot)$ is anticoercive (that is, if $\|y\| \to \infty$, $y \in Y$, then $\psi(y) \to -\infty$).*

Proof We argue by contradiction. So, suppose we can find $\{y_n\}_{n \in \mathbb{N}} \subseteq Y$ such that

$$\|y_n\| \to \infty \quad \text{and} \quad -c \leq \psi(y_n) \text{ for all } n \in \mathbb{N}, \text{ some } c > 0. \tag{5.140}$$

Then

$$-c \leq \frac{1}{p}\|Dy_n\|_p^p + \frac{1}{2}\|Dy_n\|_2^2 - \int_\Omega F(z,y_n)dz \text{ for all } n \in \mathbb{N}. \tag{5.141}$$

We set $v_n = \frac{y_n}{\|y_n\|}$, $n \in \mathbb{N}$. We see that $v_n \in Y$, $\|v_n\| = 1$ for all $n \in \mathbb{N}$. Therefore, we may assume that

$$v_n \to v \text{ in } H_0^1(\Omega), \ \|v\| = 1 \tag{5.142}$$

(recall Y is finite dimensional). From (5.141) we have

$$-\frac{c}{\|y_n\|^2} \leq \frac{1}{p}\frac{1}{\|y_n\|^{2-p}}\|Dv_n\|_p^p + \frac{1}{2}\|Dv_n\|_2^2 - \int_\Omega \frac{F(z,y_n)}{\|y_n\|^2}dz. \tag{5.143}$$

Note that $\left\{\frac{F(\cdot,y_n(\cdot))}{\|y_n\|^2}\right\}_{n \in \mathbb{N}} \subseteq L^1(\Omega)$ is uniformly integrable. So, by the Dunford-Pettis theorem (see Theorem I-4.26) and hypotheses H_2 we have

$$\frac{F(\cdot,y_n(\cdot))}{\|y_n\|^2} \xrightarrow{w} \frac{1}{2}\vartheta(\cdot)v^2 \text{ in } L^1(\Omega) \tag{5.144}$$

with $\vartheta \in L^\infty(\Omega)$ satisfying $\widehat{\lambda}_m(2) \leq \vartheta(z) \leq \eta(z)$ for a.a $z \in \Omega$. If in (5.143) we pass to the limit as $n \to \infty$ and use (5.142) and (5.144), we obtain

$$0 \leq \frac{1}{2}\left[\|Dv\|_2^2 - \int_\Omega \vartheta(z)v^2dz\right] \leq 0. \tag{5.145}$$

If $\vartheta \not\equiv \widehat{\lambda}_m(2)$, then the second inequality in (5.145) is strict and this leads to a contradiction.

If $\vartheta(z) = \widehat{\lambda}_m(2)$ for a.a $z \in \Omega$, then $v \in E(\widehat{\lambda}_m(2)) \setminus \{0\}$ (see (5.142)). Then $v(z) \neq 0$ for a.a $z \in \Omega$ (by the UCP), hence $|y_n(z)| \to +\infty$ for a.a $z \in \Omega$ and so by hypothesis H_2'(iv), we have

$$\frac{1}{\|y_n\|^p} \int_\Omega [f(z, y_n)y_n - 2F(z, y_n)]dz \to -\infty. \tag{5.146}$$

We know that

$$\begin{aligned}
&-2c \leq \frac{2}{p}\|Dy_n\|_p^p + \|Dy_n\|_2^2 - \int_\Omega 2F(z, y_n)dz, \\
&-\|Dy_n\|_p^p - \|Dy_n\|_2^2 + \int_\Omega f(z, y_n)y_n dz = 0 \quad \text{for all } n \in \mathbb{N}, \\
\Rightarrow &-(\frac{2}{p} - 1)\|Dy_n\|_p^p - 2c \leq \int_\Omega [f(z, y_n)y_n - 2F(z, y_n)]dz \\
\Rightarrow &-\widehat{c} \leq \frac{1}{\|y_n\|^p} \int_\Omega [f(z, y_n)y_n - 2F(z, y_n)]dz, \quad \text{for all } n \in \mathbb{N}, \text{ some } \widehat{c} > 0,
\end{aligned}$$

which contradicts (5.146).

Therefore we conclude that $\psi(\cdot)$ is anticoercive. □

Corollary 5.71 *If $1 < p < 2$ and hupotheses H_2' hold, then $\psi(\cdot)$ satisfies the C-condition.*

Now we can have a first multiplicity result for the singular case of problem (5.79).

Theorem 5.72 *If $1 < p < 2$ and hupotheses H_2' hold, then problem (5.79) has at least three nontrivial solutions $u_0 \in intC_+$, $v_0 \in -intC_+$ and $y_0 \in C_0^1(\overline{\Omega})$.*

Proof From Proposition 5.67 we already have two nontrivial constant sign solutions

$$u_0 \in intC_+ \quad \text{and} \quad v_0 \in -intC_+.$$

From the proof of Proposition 5.67 we know that u_0 is a critical point of φ_+ of mountain pass type and similarly v_0 is a critical point of φ_- of mountain pass type. Clearly

$$\varphi_+|_{C_+} = \varphi|_{C_+} \quad \text{and} \quad \varphi_-|_{-C_+} = \varphi|_{-C_+}.$$

Since $u_0 \in intC_+$, $v_0 \in -intC_+$, we have

$$
\begin{aligned}
&C_k(\varphi_+|_{C_0^1(\overline{\Omega})}, u_0) = C_k(\varphi|_{C_0^1(\overline{\Omega})}, u_0)\\
&C_k(\varphi_-|_{C_0^1(\overline{\Omega})}, v_0) = C_k(\varphi|_{C_0^1(\overline{\Omega})}, v_0) \text{ for all } k \in \mathbb{N}_0\\
\Rightarrow &C_k(\varphi_+, u_0) = C_k(\varphi, u_0),\ C_k(\varphi_-, v_0) = C_k(\varphi, v_0)\\
&\quad \text{for all } k \in \mathbb{N}_0 \text{ (since } C_0^1(\overline{\Omega}) \hookrightarrow H_0^1(\Omega) \text{ continuously and densely)},\\
\Rightarrow &C_1(\varphi, u_0) \neq 0,\ C_1(\varphi, v_0) \neq 0.
\end{aligned} \tag{5.147}
$$

From Proposition 5.70 we know that $\psi(\cdot)$ is anticoercive Therefore we can find $\overline{y}_0 \in Y$ such that

$$
\begin{aligned}
&\psi(\overline{y}_0) = \max\{\psi(y) : y \in Y\},\\
\Rightarrow &C_k(\psi, \overline{y}_0) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}\\
&\quad \text{with } d_m = dim Y = dim \oplus_{k=1}^{m} E(\widehat{\lambda}_k(2)) \geq 2\\
\Rightarrow &C_k(\varphi, \overline{y}_0 + \gamma_0(y_0)) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}\\
&\qquad\qquad \text{(see Theorem 3.136)}.
\end{aligned} \tag{5.148}
$$

Finally from Proposition 5.65, we have

$$
C_k(\varphi, 0) = \delta_{k,0}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.149}
$$

From (5.147), (5.148), (5.149) and (5.133) we conclude that $y_0 = \overline{y}_0 + \gamma_0(y_0)$ is the third nontrivial solution of (5.79) and the nonlinear regularity theory implies $y_0 \in C_0^1(\overline{\Omega})$. □

As in the degenerate case ($2 < p$), by strengthening the regularity of $f(z, \cdot)$, we can improve the above multiplicity theorem.

First we compute the critical groups of $\psi(\cdot)$ at infinity. To do this, we do not need the stronger set of hypotheses on the reaction $f(z, x)$.

Proposition 5.73 *If $1 < p < 2$, hupotheses H_2' hold and K_ψ is finite, then $C_k(\psi, \infty) = \delta_{k,d_m}\mathbb{Z}$ for all $k \in \mathbb{N}_0$.*

Proof Let $m_0 < \inf \psi(K_\psi)$. The anticoercivity of $\psi(\cdot)$ implies that we can find $\gamma < \beta < m_0$ and $0 < \varrho < R$ such that

$$
C_R \subseteq \psi^\gamma \subseteq C_\varrho \subseteq \psi^\beta,
$$

where for every $r > 0$, $C_r = \{y \in Y : \|y\| \geq r\}$.

For the triples (C_R, C_ϱ, Y) and $(\psi^\gamma, \psi^\beta, Y)$ we consider the corresponding long exact sequence of homology groups

$$\begin{array}{ccccccccc}
\ldots \to H_k(C_\varrho, C_R) & \xrightarrow{i_k} & H_k(Y, C_R) & \xrightarrow{j_k} & H_k(Y, C_\varrho) & \xrightarrow{\partial_*} & H_{k-1}(C_\varrho, C_R) \to \ldots \\
\downarrow h_*|_{C_\varrho} & & \downarrow h_* & & \downarrow h_* & & \\
\ldots \to H_k(\psi^\beta, \psi^\gamma) & \xrightarrow{\widehat{i}_k} & H_k(Y, \psi^\gamma) & \xrightarrow{\widehat{j}_k} & H_k(Y, \psi^\beta) & \xrightarrow{\widehat{\partial}_*} & H_{k-1}(\psi^\beta, \psi^\gamma) \to \ldots
\end{array} \tag{5.150}$$

In (5.150) all squares are commutative and the maps $i_*, \widehat{i}_*, j_*, \widehat{j}_*, h_*$ are the group homomorphisms induced by the corresponding inclusion maps. Also ∂_* and $\widehat{\partial}_*$ are the corresponding boundary homomorphisms. Since $\gamma < \beta < m_0$, the noncritical interval theorem (see Corollary 3.32) implies that ψ^γ is a strong deformation retract of ψ^β. Therefore

$$H_k(\psi^\beta, \psi^\gamma) = 0 \text{ for all } k \in \mathbb{N}_0 \text{ (see Proposition 3.97).} \tag{5.151}$$

Let $\tau : C_\varrho \to C_R$ be the map defined by

$$\tau(u) = \begin{cases} R\frac{u}{\|u\|} & \text{if } \varrho \le \|u\| \le R \\ u & \text{if } R < \|u\|. \end{cases}$$

Evidently $\tau(\cdot)$ is continuous and $\tau|_{C_R} = id|_{C_R}$. Hence C_R is a retract of C_ϱ. Consider the deformation $h : [0, 1] \times C_\varrho \to Y$ defined by

$$h(t, u) = (1 - t)u + tR\frac{u}{\|u\|} \text{ for all } t \in [0, 1], \text{ all } u \in C_\varrho.$$

We see that $h(0, \cdot) = id|_{C_\varrho}$ and $h(1, C_\varrho) \subseteq C_R$. Therefore C_ϱ is deformable into C_R. If follows that C_R is a deformation retract of C_ϱ and so

$$H_k(C_\varrho, C_R) = 0 \text{ for all } k \in \mathbb{N} \text{ (see Proposition 3.97).} \tag{5.152}$$

Then from the exactness of the long homology sequences in (5.150), we have

$$0 = im i_* = ker j_* \text{ (see (5.152))}, \ im j_* = ker \partial_* = H_k(Y, C_\varrho)$$

$$0 = im \widehat{i}_* = ker \widehat{j}_* \text{ (see (5.151))}, \ im \widehat{j}_* = ker \widehat{\partial}_* = H_k(Y, \psi^\beta).$$

Both j_* and $\widehat{j}_*$ are group isomorphisms. So, from (5.150) we infer that h_* is an isomorphisms. Hence

$$\begin{aligned} &H_k(Y, C_\varrho) = H_k(Y, \psi^\beta) \text{ for all } k \in \mathbb{N}_0, \\ \Rightarrow &H_k(Y, C_\varrho) = C_k(\psi, \infty) \text{ for all } k \in \mathbb{N}_0. \end{aligned} \tag{5.153}$$

Using the radial retraction, we see that

$$\begin{aligned}
&\partial B_\varrho = \{y \in Y : \|y\| = \varrho\} \text{ is a deformation retract of } C_\varrho,\\
\Rightarrow &H_k(Y, C_\varrho) = H_k(Y, \partial B_\varrho) \text{ for all } k \in \mathbb{N}_0 \text{ (see Proposition 3.97)},\\
\Rightarrow &H_k(Y, C_\varrho) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0\\
\Rightarrow &C_k(\psi, \infty) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0 \text{ (see (5.153))}.
\end{aligned}$$

□

Now we introduce the new stronger set of hypotheses on the reaction $f(z, x)$.

$\mathbf{H}_2''$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a measurable function such that for a.a $z \in \Omega$ $f(z, 0) = 0$, $f(z, \cdot) \in C^1(\mathbb{R})$ and

(i) there exist $m \in \mathbb{N}$, $m \geq 2$ and $\eta \in L^\infty(\Omega)$ such that

$$\eta(z) \leq \widehat{\lambda}_{m+1}(2) \text{ for a.a } z \in \Omega,\ \eta \not\equiv \widehat{\lambda}_{m+1}(2),$$

$$|f_x'(z, x)| \leq \eta(z) \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R};$$

(ii) $\widehat{\lambda}_m(2) \leq \liminf_{x\to\pm\infty} \dfrac{f(z, x)}{x}$ uniformly for a.a $z \in \Omega$;

(iii) $\lim_{x\to\pm\infty} \dfrac{f(z, x)x - 2F(z, x)}{|x|^p} = -\infty$ uniformly for a.a $z \in \Omega$.

Remark 5.74 The mean value theorem and H_2''(i) give H_2(i). Also, a simple differentiation reveals that for every $\varrho > 0$, we can find $\widehat{\xi}_\varrho > 0$ such that for a.a $z \in \Omega$ $x \to f(z, x) + \widehat{\xi}_\varrho x$ is nondecreasing on $[-\varrho, \varrho]$.

We are ready for the second multiplicity theorem for the singular case ($1 < p < 2$) of problem (5.79).

Theorem 5.75 *If $1 < p < 2$ and hupotheses H_2'' hold, then problem (5.79) has at least four nontrivial solutions*

$$u_0 \in intC_+,\ v_0 \in -intC_+,\ y_0,\ \widehat{y} \in C_0^1(\overline{\Omega}).$$

Proof From Theorem 5.72 we already have three nontrivial solutions

$$u_0 \in intC_+,\ v_0 \in -intC_+,\ y_0 \in C_0^1(\overline{\Omega}).$$

We know that $\varphi \in C^2(H_0^1(\Omega) \setminus \{0\})$ and

$$C_1(\varphi, u_0) \neq 0,\ C_1(\varphi, v_0) \neq 0 \text{ (see (5.147))}.$$

Since $u_0 \in intC_+,\ v_0 \in -intC_+$, if follows that

$$C_k(\varphi, u_0) = C_k(\varphi, v_0) = \delta_{k,1}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.154}$$

Also, we have

$$C_k(\varphi, y_0) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0 \text{ (see (5.148))}. \tag{5.155}$$

In addition from Proposition 5.73 we know that

$$\begin{aligned} &C_k(\psi, \infty) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0, \\ \Rightarrow &C_k(\varphi, \infty) = \delta_{k,d_m}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \end{aligned} \tag{5.156}$$

Finally Proposition 5.65 implies that

$$C_k(\varphi, 0) = \delta_{k,0}\mathbb{Z} \text{ for all } k \in \mathbb{N}_0. \tag{5.157}$$

Suppose $K_\varphi = \{0, u_0, v_0, y_0\}$. Then from (5.154), (5.155), (5.156), (5.157) and the global Moser relation with $t = -1$ (see Theorem 3.117) we have

$$(-1)^0 + 2(-1)^1 + (-1)^{dm} = (-1)^{dm},$$

a contradiction. So, we can find $\widehat{y} \in K_\varphi$, $\widehat{y} \notin \{0, u_0, v_0, y_0\}$. This is the fourth nontrivial solution of (5.79) and the nonlinear regularity theory implies $\widehat{y} \in C_0^1(\overline{\Omega})$. □

Next we consider the following nonlinear Robin problem

$$\left\{\begin{array}{ll} -div\, a(Du(z)) = f(z, u(z)) & \text{in } \Omega, \\ \frac{\partial u}{\partial n_\alpha} + \beta(z)u^{p-1} = 0 & \text{on } \partial\Omega,\ u \geq 0. \end{array}\right\} \tag{5.158}$$

We will produce general conditions on the reaction $f(z, x)$ which guarantee the existence of a unique positive solution for problem (5.158). In the particular case of the p-Laplacian, we show that these conditions are also necessary.

So, as before, $\Omega \subseteq \mathbb{R}^N$ is a bounded domain with a C^2-boundary $\partial\Omega$. In Sect. 4.3, we studied the following nonlinear Robin eigenvalue problem

$$\left\{\begin{array}{ll} -\Delta_p u(z) + \xi(z)|u(z)|^{p-2}u(z) = \widehat{\lambda}|u(z)|^{p-2}u(z) & \text{in } \Omega, \\ \frac{\partial u}{\partial n_p} + \beta(z)|u|^{p-2}u = 0 & \text{on } \partial\Omega. \end{array}\right\} \tag{5.159}$$

The next theorem summarizes the main properties of the first eigenpair of problem (5.159). Recall that

$$\widehat{C}_+ = \{u \in C^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\},$$

and this cone has a nonemply interior given by

$$int\, \widehat{C}_+ = \{u \in C_+ : u(z) > 0 \text{ for all } z \in \overline{\Omega}\}.$$

Theorem 5.76 *If $\xi \in L^\infty(\Omega)$, $\beta \in C^{0,\alpha}(\partial\Omega)$ with $0 < \alpha < 1$ and $\beta(z) \geq 0$ for all $z \in \partial\Omega$, then problem (5.159) admits a smallest eigenvalue $\widehat{\lambda}_1(p, \xi, \beta) \in \mathbb{R}$ which is simple, isolated and it has a corresponding L^p-normalized eigenfunxtion $\widehat{u}_1$ (that is, $\|\widehat{u}_1\|_p = 1$) such that*

$$\widehat{u}_1 \in C^{1,\alpha}(\overline{\Omega}) \cap int\, \widehat{C}_+,\ 0 < \alpha < 1.$$

The map $a(\cdot)$ involved in the definition of the differential operator of (5.159), satisfies conditions similar to those in $\widehat{H}_0'$ from Sect. 4.3. More precisely, let $\vartheta \in C^1(0, \infty)$ which satisfies

$$0 < c^* \leq \frac{t\vartheta'(t)}{\vartheta(t)} \leq \widehat{c}_0$$

$$\widehat{c}_1 t^{p-1} \leq \vartheta(t) \leq \widehat{c}_2(t^{s-1} + t^{p-1}) \text{ with } \widehat{c}_1,\ \widehat{c}_2 > 0 \text{ and } 1 < s < p.$$

Then the hypotheses on the map $a : \mathbb{R}^N \to \mathbb{R}^N$ are the following.

H_3: $a(y) = a_0(|y|)y$ for all $y \in \mathbb{R}^N$ with $a_0(t) > 0$ for all $t > 0$ and

(i) $a_0 \in C^1(0, \infty)$, $t \to a_0(t)t$ is strictly increasing on $(0, \infty)$, $a_0(t)t \to 0^+$ as $t \to 0^+$ and $\lim_{t\to 0^+} \frac{a_0'(t)t}{a_0(t)} > -1$;

(ii) $|\nabla a(y)| \leq \widehat{c}_3 \frac{\vartheta(|y|)}{|y|}$ for some $\widehat{c}_3 > 0$, all $y \in \mathbb{R}^N \setminus \{0\}$;

(iii) $\frac{\vartheta(|y|)}{|y|}|\xi|^2 \leq (\nabla a(y)\xi, \xi)_{\mathbb{R}^N}$ for all $y \in \mathbb{R}^N \setminus \{0\}$, all $\xi \in \mathbb{R}^N$;

(iv) if $G_0(t) = \int_0^t a_0(s)s\,ds$, then there exist $q \in (1, p]$ and $\delta > 0$ such that $t \to G_0(t^{\frac{1}{q}})$ is convex on $\mathbb{R}_+ = [0, \infty)$ and

$$G_0(t) \leq \widehat{c}_4 t^q \text{ for all } t \in [0, \delta],\ \text{with } \widehat{c}_4 > 0.$$

Remark 5.77 Hypotheses H_3(i), (ii), (iii) are dictated by the nonlinear regularity theory (see Theorem 4.84) and the nonlinear Hopf maximum principle (see Theorem 4.89). Hypothesis H_3(iv) serves the needs of our problem, it is mild and it is satisfied by all the particular cases presented in Example 4.90.

The hypotheses on the boundary coefficient $\beta(\cdot)$ are the following.

H_4: $\beta \in C^{0,\alpha}(\partial\Omega)$ with $0 < \alpha < 1$, $\beta(z) \geq 0$ for all $z \in \partial\Omega$.

Finally we present the hypotheses on the reaction $f(z, x)$ of problem (5.158).

H_5: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function which has the following properties

(i) $f(\cdot, x) \in L^\infty(\Omega)$ for all $x \geq 0$ and there exists $c > 0$ such that

$$f(z, x) \leq c(1 + x^{p-1}) \text{ for a.a } z \in \Omega,\ \text{all } x \geq 0;$$

(ii) for a.a $z \in \Omega$, $x \to \dfrac{f(z,x)}{x^{q-1}}$ is strictly decreasing on $(0,\infty)$ with $1 < q \leq p$ as in hypothesis H_3(iv);

(iii) If $\mu(z) = \lim_{x\to+\infty} \dfrac{f(z,x)}{x^{p-1}}$ for a.a $z \in \Omega$, $\widehat{\mu} = \frac{p-1}{\widehat{c}_1}\mu$, $\widehat{\beta} = \frac{p-1}{\widehat{c}_1}\beta$, then $\widehat{\lambda}_1(p, -\widehat{\mu}, \widehat{\beta}) > 0$;

(iv) If $\mu_0(z) = \lim_{x\to 0^+} \dfrac{f(z,x)}{x^{q-1}}$ for a.a $z \in \Omega$, $\widehat{\mu}_0 = \frac{1}{q\widehat{c}_4}\mu_0$, $\widehat{\beta} = \frac{1}{q\widehat{c}_4}\beta$, then $\widehat{\lambda}_1(q, -\widehat{\mu}_0, \widehat{\beta}_0) < 0$.

Remark 5.78

(a) All the above hypotheses concern the positive semiaxis $\mathbb{R}_+ = [0,\infty)$. Since we search for positive solutions, we may set $f(z,x) = f(z,0)$ for a.a $z \in \Omega$, all $x \leq 0$.

(b) Evidently the functions $\mu(\cdot)$ and $\mu_0(\cdot)$ are measurable. If $1 < q < p$ (case of the (p,q)-Laplacian), then $\mu \equiv 0$ and so for hypothesis H_5(iii) to be true, we must have $\beta \not\equiv 0$. Moreover, hypothesis H_5(iii) covers also the case $\mu(z) = -\infty$ on a set of positive measure.

(c) Combining hypotheses H_5(i), (ii), we see that

$$\frac{f(z,x)}{x^{p-1}} \leq f(z,1) \leq \|f(\cdot,1)\|_\infty = c_1 \text{ for a.a } z \in \Omega, \text{ all } x \geq 1,$$

$$\Rightarrow \mu(z) \leq c_1 \text{ for a.a } z \in \Omega,$$

$$\Rightarrow \widehat{\lambda}_1(p, -\widehat{\mu}, \widehat{\beta}) \in (-\infty, +\infty].$$

On the other hand, using once again hypotheses H_5(i), (ii), we see that

$$\frac{f(z,x)}{x^{q-1}} \geq f(z,1) \geq -\|f(\cdot,1)\|_\infty = -c_1 \text{ for a.a } z \in \Omega, \text{ all } 0 < x \leq 1,$$

$$\Rightarrow \mu_0(z) \geq -c_1 \text{ for a.a } z \in \Omega,$$

$$\Rightarrow \widehat{\lambda}_1(q, -\widehat{\mu}_0, \widehat{\beta}_0) \in [-\infty, \infty).$$

(d) If $f(z,x) = f(x)$ (autonomous reaction), then hypotheses H_5(iii), (iv) are equivalent to saying that

$$\mu < 0 < \mu_0.$$

(e) The function $f(x) = \lambda(x^{q-1} - x^{r-1})$, $x \geq 0$, $\lambda > 0$, with $1 < q \leq p < r$ is a reaction which corresponds to a subdiffusive logistic equation and satisfies hypotheses H_5. In this case $\mu = -\infty$ and $\mu_0 = \lambda > 0$.

We introduce the Caratheodory function $k(z,x)$ defined by

$$k(z,x) = \begin{cases} f(z,0) & \text{if } x \le 0 \\ f(z,x) + \frac{\widehat{c}_1}{p-1} x^{p-1} & \text{if } 0 < x. \end{cases} \tag{5.160}$$

Let $K(z,x) = \int_0^x k(z,s)ds$. Also recall that $G(y) = G_0(|y|)$ for all $y \in \mathbb{R}^N$. We consider the C^1-functional $\varphi : W^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\varphi(u) = \int_\Omega G(Du)dz + \frac{\widehat{c}_1}{p(p-1)}\|u\|_p^p + \frac{1}{p}\int_{\partial\Omega} \beta(z)|u|^p d\sigma - \int_\Omega K(z,u)dz$$

for all $u \in W^{1,p}(\Omega)$.

Proposition 5.79 *If hypotheses H_3(i), (ii), (iii), H_4, H_5 hold, then $\varphi(\cdot)$ is coercive.*

Proof We argue by contradiction. So, suppose that $\varphi(\cdot)$ is not coercive. This means that we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega)$ such that

$$\|u_n\| \to \infty \quad \varphi(u_n) \le c_2 \quad \text{for some } c_2 > 0, \text{ all } n \in \mathbb{N}. \tag{5.161}$$

Note that for some $c_3 > 0$, we have

$$K(z,x) \le c_3[1 + |x|^p] \text{ for a.a } z \in \Omega, \text{ all } x \in \mathbb{R} \quad \text{(see (5.160) and hypothesis } H_5\text{(i))}. \tag{5.162}$$

Using Corollary 4.88 and (5.161), (5.162), we obtain

$$\begin{aligned} &\frac{\widehat{c}_1}{p(p-1)}[\|Du_n\|_p^p + \|u_n\|_p^p] + \frac{1}{p}\int_{\partial\Omega} \beta(z)|u_n|^p d\sigma \le c_4(1 + \|u_n\|_p^p) \\ &\qquad \text{for some } c_4 > 0, \text{ all } n \in \mathbb{N}. \\ \Rightarrow &\frac{\widehat{c}_1}{p(p-1)}\|u_n\|^p \le c_4(1 + \|u_n\|_p^p) \\ &\qquad \text{(see hypotheses } H_4\text{)}, \\ \Rightarrow &\|u_n\|_p \to \infty \text{ (see (5.161))}. \end{aligned} \tag{5.163}$$

We set $y_n = \frac{u_n}{\|u_n\|_p}$, $n \in \mathbb{N}$. Then $\|y_n\|_p = 1$ for all $n \in \mathbb{N}$. We have

$$\begin{aligned} &\frac{\widehat{c}_1}{p(p-1)}\|y_n\|^p + \frac{1}{p}\int_{\partial\Omega} \beta(z)|y_n|^p d\sigma \le c_4\left(\frac{1}{\|u_n\|_p^p} + \|y_n\|_p^p\right) \\ &\qquad \text{for all } n \in \mathbb{N}, \\ \Rightarrow &\{y_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega) \text{ is bounded.} \end{aligned}$$

We may assume that

$$y_n \to y \text{ in } W^{1,p}(\Omega),\ y_n \to y \text{ in } L^p(\Omega) \text{ and in } L^p(\partial\Omega). \tag{5.164}$$

Evidently $\|y\|_p = 1$. Also using (5.160), (5.161), we have

$$\begin{aligned}&\frac{\widehat{c}_1}{p(p-1)}\|y_n\|^p + \frac{1}{p}\int_{\partial\Omega}\beta(z)|y_n|^p d\sigma\\ \leq &\frac{c_1}{\|u_n\|_p^p} + \int_{\{u_n>0\}}\left[\frac{F(z,u_n)}{\|u_n\|_p^p} + \frac{\widehat{c}_1}{p(p-1)}y_n^p\right]dz + \int_{\{u_n\leq 0\}}\frac{f(z,0)u_n}{\|u_n\|_p^p}dz\end{aligned} \tag{5.165}$$

(recall $F(z,x) = \int_0^x f(z,s)ds$). Hypothesis H_5(ii) implies that

$$\begin{aligned}&\frac{f(z,x)}{x^{p-1}} \geq f(z,1) \text{ for a.a } z \in \Omega,\ \text{all } x \in (0,1]\\ \Rightarrow &f(z,x) \geq -\|f(\cdot,1)\|_\infty x^{p-1} \text{ for a.a } z \in \Omega,\ \text{all } x \in (0,1],\\ \Rightarrow &f(z,0) \geq 0 \text{ for a.a } z \in \Omega.\end{aligned}$$

Therefore

$$\int_{\{u_n\leq 0\}}\frac{f(z,0)u_n}{\|u_n\|_p^p}dz \leq 0 \text{ for all } n \in \mathbb{N}. \tag{5.166}$$

We return to (5.165) and use (5.166). Then

$$\frac{\widehat{c}_1}{p(p-1)}\|y_n\|^p \leq \frac{c_1}{\|u_n\|_p^p} + \frac{c_1}{p(p-1)}\|y_n^+\|_p^p + \int_\Omega\frac{F(z,u_n^+)}{\|u_n\|_p^p}dz \quad \text{for all } n \in \mathbb{N}. \tag{5.167}$$

If $\{u_n^+\}_{n\in\mathbb{N}} \subseteq L^p(\Omega)$ is bounded, then $y_n^+ \to 0$ in $L^p(\Omega)$ (note that $y_n^+ = \frac{u_n^+}{\|u_n\|_p}$ for all $n \in \mathbb{N}$ and use (5.163)). Therefore $y \leq 0$. We know that for some $c_5 > 0$

$$\begin{aligned}&F(z,x) \leq c_5(1+x^p) \text{ for a.a } z \in \Omega,\ \text{all } x \geq 0,\\ \Rightarrow &\int_\Omega\frac{F(z,u_n^+)}{\|u_n\|_p^p}dz \leq c_5\left(\frac{1}{\|u_n\|_p^p} + \|y_n^+\|_p^p\right) \text{ for all } n \in \mathbb{N},\\ \Rightarrow &\limsup_{n\to\infty}\int_\Omega\frac{F(z,u_n^+)}{\|u_n\|_p^p}dz \leq 0.\end{aligned} \tag{5.168}$$

Now in (5.167) we pass to the limit as $n \to \infty$ and use (5.164), (5.168) and the fact that $y_n^+ \to 0$ in $L^p(\Omega)$ (we have assumed that $\{u_n^+\}_{n\in\mathbb{N}} \subseteq L^p(\Omega)$ is bounded). We obtain

$$\frac{\widehat{c}_1}{p(p-1)}\|y\|_p^p \le 0,$$

$$\Rightarrow y = 0, \text{ a contradiction since } \|y\|_p = 1.$$

Next suppose that $\{u_n^+\}_{n\in\mathbb{N}} \subseteq L^p(\Omega)$ is unbounded. We may assume that $\|u_n^+\|_p \to \infty$. From (5.161), we have

$$\frac{\widehat{c}_1}{p(p-1)}\|Dy_n^+\|_p^p + \frac{1}{p}\int_{\partial\Omega}\beta(z)(y_n^+)^p d\sigma \le \frac{c_1}{\|u_n^+\|_p^p} + \int_\Omega \frac{F(z,u_n^+)}{\|u_n^+\|_p^p}dz \quad \text{for all } n \in \mathbb{N}. \tag{5.169}$$

We analyze the integral in the right hand side of (5.169). We can write

$$\int_\Omega \frac{F(z,u_n^+)}{\|u_n^+\|_p^p}dz = \int_{\{y^+=0\}} \frac{F(z,u_n^+)}{\|u_n^+\|_p^p}dz + \int_{\{y>0\}\cap\{y_n>0\}} \frac{F(z,u_n^+)}{\|u_n^+\|_p^p}dz \quad \text{for all } n \in \mathbb{N}. \tag{5.170}$$

From (5.164) we have

$$\begin{aligned} & y_n^+ \to y^+ \text{ in } L^p(\Omega), \\ \Rightarrow & y_n^+(z) \to y^+(z) \text{ for a.a } z \in \Omega \\ & \qquad \text{(at least for a subsequence)}, \\ \Rightarrow & u_n^+(z) \to +\infty \text{ for a.a } z \in \{y^+ > 0\} = \{y > 0\} \end{aligned} \tag{5.171}$$

$$\text{and } \chi_{\{y>0\}\cap\{y_n>0\}}(z) \to \chi_{\{y>0\}}(z) \text{ for a.a } z \in \Omega. \tag{5.172}$$

Let $z \in \{\mu > -\infty\} \setminus E$ with $|E|_N = 0$ (as before, by $|\cdot|_N$ we denote the Lebesgue measure on $\mathbb{R}^N$), be a point for which we have

$$\frac{f(z,x)}{x^{p-1}} \to \mu(z) \text{ as } x \to +\infty.$$

Then given $\varepsilon > 0$, we can find $c_6 = c_6(\varepsilon) > 0$ such that

$$\begin{aligned} & f(z,x) \le [\mu(z)+\varepsilon]x^{p-1} \text{ for all } x \ge c_6, \\ \Rightarrow & F(z,x) \le \frac{1}{p}[\mu(z)+\varepsilon]x^p \text{ for all } x \ge c_6, \\ \Rightarrow & \limsup_{x\to+\infty} \frac{F(z,x)}{x^p} \le \frac{1}{p}\mu(z) \\ & \qquad \text{(since } \varepsilon > 0 \text{ is arbitrary)} \end{aligned}$$

Next suppose that $z \in \{\mu = -\infty\} \setminus E$ ($|E|_N = 0$). We have

$$\frac{f(z,x)}{x^{p-1}} \to -\infty \text{ as } x \to +\infty.$$

Given $M > 0$, we can find $c_7 = c_7(M) > 0$ such that

$$\begin{aligned} & f(z,x) \leq -Mx^{p-1} \text{ for all } x \geq c_7, \\ \Rightarrow & \frac{F(z,x)}{x^p} \leq -\frac{M}{p} \text{ for all } x \geq c_7, \\ \Rightarrow & \limsup_{x \to +\infty} \frac{F(z,x)}{x^p} \leq -\frac{M}{p} \end{aligned}$$

But $M > 0$ is arbitrary. So, we infer that

$$\lim_{x \to +\infty} \frac{F(z,x)}{x^p} = -\infty = \mu(z).$$

Therefore finally we can say that

$$\limsup_{x \to +\infty} \frac{F(z,x)}{x^p} \leq \frac{1}{p}\mu(z) \text{ for a.a } z \in \Omega. \tag{5.173}$$

Then (5.171), (5.172), (5.173) and Fatou's lemma, imply that

$$\limsup_{x \to +\infty} \int_{\{y>0\} \cap \{y_n > 0\}} \frac{F(z,u_n^+)}{(u_n^+)^p}(y_n^+)^p dz \leq \frac{1}{p} \int_{\{y>0\}} \mu(y^+)^p dz. \tag{5.174}$$

Also we have

$$\begin{aligned} & \left| \int_{\{y^+=0\}} \frac{F(z,u_n^+)}{\|u_n^+\|_p^p} dz \right| \leq c_5 \int_{\{y^+=0\}} \left(\frac{1}{\|u_n^+\|_p^p} + (y_n^+)^p \right) dz \text{ for all } n \in \mathbb{N}, \\ \Rightarrow & \lim_{n \to \infty} \int_{\{y^+=0\}} \frac{F(z,u_n^+)}{\|u_n^+\|_p^p} dz = 0. \end{aligned} \tag{5.175}$$

We return to (5.170), pass to the limit as $n \to \infty$ and use (5.174), (5.175). Then

$$\limsup_{x \to +\infty} \int_{\Omega} \frac{F(z,u_n^+)}{\|u_n^+\|_p^p} dz \leq \frac{1}{p} \int_{\{y^+ \neq 0\}} \mu(y^+)^p dz. \tag{5.176}$$

If in (5.169) we let $n \to \infty$ and use (5.164) and (5.176), we obtain

$$\frac{\widehat{c}_1}{p(p-1)}[\|Dy^+\|_p^p + \int_{\partial\Omega} \beta(z)(y^+)^p d\sigma] \leq \frac{1}{p}\int_{\{y^+\neq 0\}} \mu(y^+)^p dz. \tag{5.177}$$

If $y^+ = 0$, then from (5.167) we have

$$\frac{\widehat{c}_1}{p(p-1)}\|y\|^p \leq 0$$

$$\Rightarrow y = 0, \text{ a contradiction since } \|y\|_p = 1.$$

So, $y^+ \neq 0$. From (5.177) it folllows that $\widehat{\lambda}_1(p_1 - \widehat{\mu}, \widehat{\beta}) < 0$, a contradiction to hypothesis H_5(iii). This proves the coercivity of $\varphi(\cdot)$. □

Proposition 5.80 *If hypotheses H_3(i), (ii), (iii), H_4, H_5 hold, then the functional $\varphi(\cdot)$ is sequentially weakly lower semicontinuous.*

Proof Consider the C^1-functionals $\widehat{\gamma} : W^{1,p}(\Omega) \to \mathbb{R}$ and $\widehat{\psi} : W^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{\gamma}(u) = \int_\Omega G(Du)dz + \frac{\widehat{c}_1}{p(p-1)}\|u\|_p^p + \frac{1}{p}\int_{\partial\Omega} \beta(z)|u|^p d\sigma,$$

$$\widehat{\psi}(u) = -\int_\Omega K(z,u)dz \text{ for all } u \in W^{1,p}(\Omega).$$

The convexity of $G(\cdot)$, the Sobolev embedding theorem and the compactness of the trace map, imply that $\widehat{\gamma}(\cdot)$ is sequentially weakly lower semicontinuous.

Next we show the sequential weak lower emicontinuouty of $\widehat{\psi}(\cdot)$. Let $\vartheta \in \mathbb{R}$ and consider the sublevel set

$$L_\vartheta = \{u \in W^{1,p}(\Omega) : \widehat{\psi}(u) \leq \vartheta\}.$$

We need to show that $L_\vartheta \subseteq W^{1,p}(\Omega)$ is sequentially weakly closed. Let $\{u_n\}_{n\in\mathbb{N}} \subseteq L_\vartheta$ and assume that $u_n \xrightarrow{w} u$ in $W^{1,p}(\Omega)$. We have $u_n \to u$ in $L^p(\Omega)$ (by the Sobolev embedding theorem) it follows that

$$u_n^+ \to u^+ \quad \text{and} \quad u_n^- \to u^- \text{ in } L^p(\Omega),$$

$$\Rightarrow u_n^+(z) \to u^+(z), \text{ for a.a } z \in \Omega$$

(at least for a subsequence).

We have

$$\vartheta \geq -\int_\Omega K(z,u_n)dz = -\int_\Omega F(z,u_n^+)dz - \frac{\widehat{c}_1}{p(p-1)}\|u_n^+\|_p^p - \int_\Omega f(z,0)(-u_n^-)dz. \tag{5.178}$$

We see that

$$\frac{\widehat{c}_1}{p(p-1)}\|u_n^+\|_p^p \to \frac{\widehat{c}_1}{p(p-1)}\|u^+\|_p^p, \tag{5.179}$$

$$\int_\Omega f(z,0)(-u_n^-)dz \to \int_\Omega f(z,0)(-u^-)dz. \tag{5.180}$$

Also, using Fatou's lemma, we have

$$\liminf_{n\to\infty}(-\int_\Omega F(z,u_n^+)dz) = -\limsup_{n\to\infty}\int_\Omega F(z,u_n^+)dz \geq -\int_\Omega F(z,u^+)dz.$$

If in (5.178) we pass to the limit as $n \to \infty$ and use (5.179), (5.180), and the above inequality, we obtain

$$\begin{aligned}
&\widehat{\psi}(u) \leq \vartheta,\\
\Rightarrow &L_\vartheta \text{ is sequentially weakly closed,}\\
\Rightarrow &\widehat{\psi}(\cdot) \text{ is sequentially weakly lower semicontinuous.}
\end{aligned}$$

□

Now we can prove the existence and uniqueness of a positive solution for problem (5.158). At this point we use hypothesis H_3(iv) which has been idle so far.

Theorem 5.81 *If hypotheses H_3, H_4, H_5 hold, then problem (5.158) has a unique positive solution*

$$u_0 \in int\, \widehat{C}_+.$$

Proof The functional $\varphi(\cdot)$ is coercive (Proposition 5.79) and sequentially weakly lower semicontinuous (Proposition 5.80). Therefore, we can find $u_0 \in W^{1,p}(\Omega)$ such that

$$\varphi(u_0) = \inf\{\varphi(u) : u \in W^{1,p}(\Omega)\}. \tag{5.181}$$

Claim 1 $u_0 \geq 0$, $u_0 \neq 0$.

Arguing by contradiction suppose that $u_0^- \neq 0$. We have

$$
\begin{aligned}
&\varphi(u_0^+)\\
&= \int_\Omega G(Du_0^+)dz + \frac{\widehat{c}_1}{p(p-1)}\|u^-\|_p^p + \frac{1}{p}\int_{\partial\Omega}\beta(z)(u_0^+)^p d\sigma - \int_\Omega F(z,u_0^+)dz\\
&\qquad\qquad\qquad

$$\widehat{\mu}_0(z)\widehat{u}_n(z)^q \chi_{\{u_n \neq 0\}}(z) \geq -\|\widehat{f}(\cdot,1)\|_\infty u(z)^q \text{ for a.a } z \in \Omega.$$

Note that

$$\|\widehat{f}(\cdot,1)\|_\infty \widehat{u}^q \in L^1(\Omega)$$

$$\text{and } \widehat{\mu}_0(z)\widehat{u}_n(z)^q \chi_{\{u_n \neq 0\}}(z) \to \widehat{\mu}_0(z)\widehat{u}(z)^q \text{ for a.a } z \in \Omega.$$

Therefore we can use Fatou's lemma and obtain

$$\liminf_{n\to\infty} \int_{\{u_n \neq 0\}} \widehat{\mu}_0(z)\widehat{u}_n^q dz \geq \int_{\{u \neq 0\}} \widehat{\mu}_0(z)\widehat{u}^q dz. \tag{5.184}$$

Also, from (5.183) and the continuity of the trace map, we have

$$\|D\widehat{u}_n\|_q^q + \int_{\partial\Omega} \widehat{\beta}_0(z)\widehat{u}_n^q d\sigma \to \|Du\|_q^q + \int_{\partial\Omega} \beta(z)u^q d\sigma. \tag{5.185}$$

Then from (5.182), (5.184) and (5.185), we see that we can find $n_0 \in \mathbb{N}$ such that

$$\|D\widehat{u}_n\|_q^q + \int_{\partial\Omega} \widehat{\beta}_0(z)\widehat{u}_n^q d\sigma - \int_{\{u_n \neq 0\}} \widehat{\mu}_0(z)\widehat{u}_n^q dz < 0 \text{ for all } n \geq n_0.$$

Therefore there exists $u \in W^{1,q}(\Omega) \cap L^\infty(\Omega)$, $u \geq 0$ such that

$$\|Du\|_q^q + \int_{\partial\Omega} \widehat{\beta}_0(z)u^q d\sigma - \int_\Omega \widehat{\mu}_0(z)u^q dz < 0. \tag{5.186}$$

In (5.186) the left hand side is q-homogeneous and so, we can always assume that $\|u\|_q = 1$.

Let $\widehat{F}(z,x) = \int_0^x \widehat{f}(z,s)ds$. For every $(z,x) \in \Omega \times (0,\infty)$ we have

$$\begin{aligned}
&\widehat{F}(z,x) = \int_0^1 \frac{d}{dt}\widehat{F}(z,tx)dt = \int_0^1 \widehat{f}(z,tx)x dt \\
&\Rightarrow \frac{\widehat{F}(z,x)}{x^q} = \int_0^1 \frac{\widehat{f}(z,tx)}{x^{q-1}}dt \geq \int_0^1 \frac{\widehat{f}(z,x)}{x^{q-1}}t^{q-1}dt \\
&\qquad\qquad \text{(see hypothesis } H_5\text{(ii))} \qquad\qquad (5.187) \\
&\qquad\qquad = \frac{1}{q}\frac{\widehat{f}(z,x)}{x^{q-1}}, \\
&\Rightarrow \liminf_{x\to 0^+} \frac{\widehat{F}(z,x)}{x^q} \geq \frac{1}{q}\widehat{\mu}_0(z) \text{ for a.a } z \in \Omega.
\end{aligned}$$

Consider $u \in W^{1,q}(\Omega) \cap L^\infty(\Omega)$, $u \geq 0$ with $\|u\|_q = 1$ satisfying (5.186) and choose $\tau \in (0,1)$ small so that

$$\tau u(z) \in [0, \delta] \text{ for a.a } z \in \Omega,$$

with $\delta > 0$ as postulated by hypothesis H_3(iv). We can always take $\delta < 1$. Then

$$\begin{aligned} \frac{\widehat{F}(z, \tau u(z))}{\tau^q} &= \frac{1}{\tau^q} \int_0^{\tau u(z)} \widehat{f}(z, s) ds \\ &\geq -\frac{1}{\tau^q} \|\widehat{f}(\cdot, 1)\|_\infty \int_0^{\tau u(z)} s^{q-1} ds \\ &\geq -\frac{\|\widehat{f}(\cdot, 1)\|_\infty}{q} u(z)^q \\ &\geq -\frac{\|\widehat{f}(\cdot, 1)\|_\infty}{q} \|u\|_\infty^q. \end{aligned} \tag{5.188}$$

On account of (5.188), we can use Fatou's lemma and have

$$\liminf_{\tau \to 0^+} \int_{\{u \neq 0\}} \frac{\widehat{F}(z, \tau u)}{\tau^q} dz \geq \frac{1}{q} \int_{\{u \neq 0\}} \mu_0(z) u^q dz \text{ (see (5.187))}.$$

From (5.186) we see that for $\tau \in (0, 1)$ small we have

$$\widehat{c}_4 \|D(\tau u)\|_q^q + \frac{1}{q} \int_{\partial\Omega} \beta(z)(\tau u)^p d\sigma - \widehat{c}_4 \int_\Omega q \widehat{F}(z, \tau u) dz < 0.$$

(since $\widehat{\beta}_0 = \dfrac{1}{q\widehat{c}_4} \beta$). Using hypothesis H_3(iv) and since $\delta < 1$, we obtain

$$\begin{aligned} &\int_\Omega G(D(\tau u)) dz + \frac{1}{p} \int_{\partial\Omega} \beta(z)(\tau u)^p d\sigma - \int_\Omega F(z, \tau u) dz < 0, \\ &\Rightarrow \varphi(\tau u) < 0 \text{ (recall } u \geq 0), \\ &\Rightarrow \varphi(u_0) < 0 = \varphi(0) \text{ (see (5.181)) and so } u_0 \neq 0. \end{aligned}$$

This completes the proof of Claim 1.

Claim 2 $u_0 \in L^\infty(\Omega)$.

Let $k \in \mathbb{N}$ and consider the following truncation of $f(z, \cdot)$

$$f_k(z, x) = \begin{cases} f(z, 0) & \text{if } x \leq 0 \\ \max\{f(z, x), -kx^{p-1}\} & \text{if } 0 < x. \end{cases}$$

Recall that $f(z, 0) \geq 0$ for a.a $z \in \Omega$. Therefore $f_k(\cdot, \cdot)$ is a Caratheodory function and $f_k(\cdot, x) \in L^\infty(\Omega)$ for all $x \in \mathbb{R}$ (see hypothesis H_5(i)). Also, again by

hypothesis H_5(i), we have

$$\begin{aligned} &|f_k(z,x)| \le c_8(1+|x|^{p-1}) \\ &\qquad\qquad \text{for a.a } z \in \Omega,\ \text{all } x \in \mathbb{R},\ \text{some } c_8 > 0. \end{aligned} \tag{5.189}$$

We define

$$\mu^k(z) = \lim_{x\to+\infty} \frac{f_k(z,x)}{x^{p-1}} \quad \text{and} \quad \mu_0^k(z) = \lim_{x\to 0^+} \frac{f_k(z,x)}{x^{q-1}}.$$

We know that

$$\begin{aligned} &f(z,x) \le f_k(z,x) \text{ for a.a } z \in \Omega,\ \text{all } x \in \mathbb{R}, \\ &\Rightarrow \mu_0 \le \mu_0^k \quad \text{and so} \quad \widehat{\lambda}_1(q,-\widehat{\mu}_0^k,\widehat{\beta}_0) \le \widehat{\lambda}_1(q,-\widehat{\mu}_0,\widehat{\beta}_0) \\ &\qquad\qquad \text{for all } k \in \mathbb{N}. \end{aligned}$$

(hence $\widehat{\mu}_0^k = \frac{1}{q\widehat{c}_4}\mu_0^k$). Also note that $\mu_k \downarrow \mu$ as $k \to \infty$ and so

$$\begin{aligned} &\widehat{\lambda}_1(p,-\widehat{\mu}^k,\widehat{\beta}) \to \widehat{\lambda}_1(p,-\widehat{\mu},\widehat{\beta}) \ (\widehat{\mu}^k = \frac{p-1}{\widehat{c}_1}\mu^k, \widehat{\mu} = \frac{p-1}{\widehat{c}_1}\mu), \\ &\Rightarrow \widehat{\lambda}_1(p,-\widehat{\mu}^k,\widehat{\beta}) > 0 \text{ for } k \in \mathbb{N} \text{ large}. \end{aligned}$$

Let $\varphi_k : W^{1,p}(\Omega) \to \mathbb{R}$ be the C^1-energy functional corresponding to the reaction $f_k(z,s)$. This too is coercive and sequentially weakly lower semixontinuous. Reasoning as in Claim 1, we can find $u_0^k \ge 0$, $u_0^k \ne 0$ such that

$$\begin{aligned} &\varphi_k(u_0^k) = \inf\{\varphi_k(u) : u \in W^{1,p}(\Omega)\}, \\ &\Rightarrow \varphi_k'(u_0^k) = 0 \text{ in } W^{1,p}(\Omega)^*, \\ &\Rightarrow \langle V(u_0^k), h\rangle + \int_{\partial\Omega} \beta(z)(u_0^k)^{p-1} h\, d\sigma = \int_\Omega f_k(z,u_0^k) h\, dz \\ &\qquad\qquad \text{for all } h \in W^{1,p}(\Omega). \end{aligned} \tag{5.190}$$

Here $V : W^{1,p}(\Omega) \to W^{1,p}(\Omega)^*$ is the operator defined

$$\langle V(u), h\rangle = \int_\Omega (a(Du), Dh)_{\mathbb{R}^N} dz \text{ for all } u,\ h \in W^{1,p}(\Omega).$$

From (5.190) it follows that

$$\left\{ \begin{array}{ll} -div\, a(Du_0^k(z)) = f_k(z,u_0^k(z)) & \text{in } \Omega, \\ \frac{\partial u_0^k}{\partial n_a} + \beta(z)(u_0^k)^{p-1} = 0 & \text{on } \partial\Omega. \end{array} \right\}$$

Then Proposition 4.82 implies $u_0^k \in L^\infty(\Omega)$ and so by Theorem 4.84 we have

$$u_0^k \in C^1(\overline{\Omega}) \text{ for all } k \in \mathbb{N}.$$

We set

$$y_k = \min\{u_0, u_0^k\} \in W^{1,p}(\Omega) \cap L^\infty(\Omega),\ k \in \mathbb{N}.$$

Recall that u_0^k is a minimizer of $\varphi_k(\cdot)$. Hence

$$\varphi_k(u_0^k) \leq \varphi_k(v) \text{ for all } v \in W^{1,p}(\Omega).$$

Let $v = \max\{u_0, u_0^k\} \in W^{1,p}(\Omega)$. We have

$$\begin{aligned}
&\int_{\{u_0^k < u_0\}} G(Du_0^k)dz + \frac{1}{p}\int_{\{\widehat{\gamma}_0(u_0^k) < \widehat{\gamma}_0(u_0)\}} \beta(z)(u_0^k)^p d\sigma - \int_{\{u_0^k < u_0\}} F_k(z, u_0^k)dz \\
\leq &\int_{\{u_0^k < u_0\}} G(Du_0)dz + \frac{1}{p}\int_{\{\widehat{\gamma}_0(u_0^k) < \widehat{\gamma}_0(u_0)\}} \beta(z)u_0^p d\sigma - \int_{\{u_0^k < u_0\}} F_k(z, u_0)dz
\end{aligned}$$

with $\widehat{\gamma}_0(\cdot)$ being the trace map (see Definition I-4.100). Then

$$\begin{aligned}
&\int_{\{u_0^k < u_0\}} [G(Du_0^k) - G(Du_0)]dz + \frac{1}{p}\int_{\{\widehat{\gamma}_0(u_0^k) < \widehat{\gamma}_0(u_0)\}} \beta(z)((u_0^k)^p - u_0^p)d\sigma \\
\leq &\int_{\{u_0^k < u_0\}} [F_k(z, u_0^k) - F_k(z, u_0)]dz.
\end{aligned} \tag{5.191}$$

Therefore we have

$$\begin{aligned}
&\varphi(y_k) - \varphi(u_0) \\
= &\int_{\{u_0^k < u_0\}} [G(Du_0^k) - G(Du_0)]dz + \frac{1}{p}\int_{\{\widehat{\gamma}_0(u_0^k) < \widehat{\gamma}_0(u_0)\}} \beta(z)((u_0^k)^p - u_0^p)d\sigma \\
&- \int_{\{u_0^k < u_0\}} [F(z, u_0^k) - F(z, u_0)]dz \\
\leq &\int_{\{u_0^k < u_0\}} [F_k(z, u_0^k) - F_k(z, u_0) - F(z, u_0^k) + F(z, u_0)]dz \text{ (see (5.191))} \\
= &\int_{\{u_0^k < u_0\}} \int_{u_0^k}^{u} (-f_k(z, s) + f(z, s))ds dz \\
\leq &0 \text{ (recall that } f \leq f_k).
\end{aligned}$$

So, we have proved

$$\varphi(y_k) \leq \varphi(u_0) = \inf\{\varphi(u) : u \in W^{1,p}(\Omega)\},$$
$$\Rightarrow \varphi(y_k) = \varphi(u_0).$$

Since $y_k \in L^\infty(\Omega)$, we see that we can say that $u_0 \in L^\infty(\Omega)$. This proves Claim 2.

Claim 3 $v_0 \in int\, \widehat{C}_+$ is a solution of problem (5.158).

Let $h \in C^1(\overline{\Omega})$ and define

$$d(h) = \int_\Omega [K(z, u_0 + h) - K(z, u_0) - k(z, u_0)h]dz \text{ (see (5.160))}.$$

Using Fubini's theorem and Hölder's inequality, we have

$$|d(h)| \leq \int_\Omega |h| \int_0^1 |k(z, u_0 + th) - k(z, u_0)|dtdz$$
$$\leq \|h\|_p \int_0^1 \|N_k(u_0 + th) - N_k(u_0)\|_{p'}dt,$$

with $N_k(v)(\cdot) = k(\cdot, v(\cdot))$ for all $v \in W^{1,p}(\Omega)$ (the Nemytskii map corresponding to k) and $\frac{1}{p} + \frac{1}{p'} = 1$.

Note that $u_0 + th \in L^\infty(\Omega)$ (see Claim 2). So, using hypothesis H_5(ii), we see that for a.a $z \in \{u_0 + th > 0\}$

$$f(z, (u_0 + th)(z)) \geq \frac{f(z, \|u_0 + th\|_\infty)}{\|u_0 + th\|_\infty^{q-1}}(u_0 + th)(z)^{q-1}$$
$$\geq -\|f(\cdot, \|u_0 + th\|_\infty)\|_\infty \frac{(u_0 + th)(z)^{q-1}}{\|u_0 + th\|_\infty^{q-1}}$$
$$\geq -\|f(\cdot, \|u_0 + th\|_\infty)\|_\infty.$$

Also hypothesis H_5(i) implies that

$$f(z, (u_0 + th)(z)) \leq c_9$$
$$\text{for some } c_9 > 0, \text{ a.a } z \in \{u_0 + th > 0\}.$$

Therefore by the Lebesgue dominated convergence theorem we have

$$\int_0^1 \|N_k(u_0 + th) - N_k(u_0)\|_{p'}dt \to 0$$
$$\Rightarrow \frac{1}{\|h\|_p}|d(h)| \to 0.$$

Then from (5.160) it follows that $\varphi(\cdot)$ is Gateaux differentiable at u_0 in the direction $h \in C^1(\overline{\Omega})$. We have

$$\begin{aligned}
&\langle \varphi'_G(u_0), h\rangle = 0 \text{ for all } h \in W^{1,p}(\Omega)\\
\Rightarrow &\langle V(u_0), h\rangle + \int_{\partial\Omega} \beta(z) u_0^{p-1} h d\sigma = \int_\Omega f(z, u_0) h dz\\
&\qquad\qquad \text{for all } h \in W^{1,p}(\Omega),\\
\Rightarrow &- div\, a(Du_0(z)) = f(z, u_0(z)) \text{ in } \Omega,\\
&\frac{\partial u_0}{\partial n_a} + \beta(z) u_0^{p-1} = 0 \text{ on } \partial\Omega.\\
\Rightarrow &u_0 \in \widehat{C}_+ \setminus \{0\} \text{ (see Theorem 4.84).}
\end{aligned}$$

For a.a $z \in \{u_0 > 0\}$, we have

$$\begin{aligned}
&\frac{f(z, u_0(z))}{u_0(z)^{p-1}} \geq \frac{f(z, \|u_0\|_\infty)}{\|u_0\|_\infty^{p-1}} \text{ (see hypothesis } H_5\text{(ii))}\\
\Rightarrow &f(z, u_0(z)) \geq -\|f(\cdot, \|u_0\|_\infty)\|_\infty \frac{u_0(z)^{p-1}}{\|u_0\|_\infty^{p-1}}
\end{aligned}$$

Therefore we obtain

$$\begin{aligned}
&div\, a(Du_0(z)) \leq c_{10} u_0(z)^{p-1} \text{ in } \Omega \text{ for some } c_{10} > 0\\
\Rightarrow &u_0 \in int\, \widehat{C}_+ \text{ (see Theorem 4.84).}
\end{aligned}$$

This proves Claim 3.

Claim 4 $u_0 \in int\, \widehat{C}_+$ is the unique positive solution of problem (5.158).

We argue as in the proof of Proposition 5.54. So, we introduce the integral functional $j : L^1(\Omega) \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ defined by

$$j(u) = \begin{cases} \int_\Omega G(Du^{\frac{1}{q}}) dz + \frac{1}{p} \int_{\partial\Omega} \beta(z) u^{\frac{p}{q}} d\sigma & \text{if } u \geq 0,\ u^{\frac{1}{q}} \in W^{1,p}(\Omega)\\ +\infty & \text{otherwise.} \end{cases}$$

As in the proof of Proposition 5.54, using hypothesis H_3(iv), we check that $j(\cdot)$ is convex.

Let $dom j = \{u \in L^1(\Omega) : j(u) < \infty\}$ (the effective domain of $j(\cdot)$). If $v_0 \in W^{1,p}(\Omega)$ is another positive solution of (5.158), then again we have $v_0 \in int\, \widehat{C}_+$. Using Proposition 2.86, we see that

$$\frac{u_0}{v_0} \in L^\infty(\Omega) \quad \text{and} \quad \frac{v_0}{u_0} \in L^\infty(\Omega). \tag{5.192}$$

Let $h = u_0^q - v_0^q \in W^{1,p}(\Omega)$. Then for $t \in (0,1)$ small, using (5.192) we see that $u_0^q + th \in domj$, $v_0^q + th \in domj$. So, exploiting the convexity of $j(\cdot)$, we see that the directional derivatives of $j(\cdot)$ at u_0^q and at v_0^q in the direction h exist. A straightforward calculation gives

$$j'(u_0^q)(h) = \frac{1}{q}\int_\Omega \frac{-diva(Du_0)}{u_0^{q-1}}hdz,$$

$$j'(v_0^q)(h) = \frac{1}{q}\int_\Omega \frac{-diva(Dv_0)}{v_0^{q-1}}hdz.$$

The convexity of $j(\cdot)$, implies the monotonicity of $j'(\cdot)$. So, we have

$$\begin{aligned} 0 &\leq \langle j'(u_0^q) - j'(v_0^q), u_0^q - v_0^q\rangle \\ &= \frac{1}{q}\int_\Omega \left(\frac{f(z,u_0)}{u_0^{q-1}} - \frac{f(z,v_0)}{v_0^{q-1}}\right)(u_0^q - v_0^q)dz \\ \Rightarrow & u_0 = v_0 \text{ (see hypothesis } H_5\text{(ii))}. \end{aligned}$$

This proves Claim 4 and completes the proof of the theorem. □

Remark 5.82 When the differential operator is the p-Laplacion (that is, $a(y) = |y|^{p-2}y$ for all $y \in \mathbb{R}^N$, with $1 < p < \infty$), then the uniqueness of the solution $u_0 \in int\, \widehat{C}_+$ can be proved using Picone's identity (see Theorem 4.91).

Next we will have a closer look at the p-Laplacian problem. So, now we deal with the following nonlinear Robin problem

$$\left\{\begin{array}{ll} -\Delta_p u(z) = f(z,u(z)) & \text{in } \Omega, \\ \frac{\partial u}{\partial n_p} + \beta(z)u^{p-1} = 0 & \text{on } \partial\Omega,\ u \geq 0. \end{array}\right\} \tag{5.193}$$

This is a particular case of problem (5.158) when $a(y) = |y|^{p-2}y$ for all $y \in \mathbb{R}^N$ $(1 < p < \infty)$. Then

$$\widehat{c}_1 = p - 1 \quad \text{and so } \widehat{\mu} = \mu,\ \widehat{\beta} = \beta.$$

Also $q = p$ and $\widehat{c}_4 = \frac{1}{p}$. Therefore

$$\widehat{\mu}_0 = \mu_0,\ \widehat{\beta}_0 = \beta.$$

For problem (5.193) we will show that hypotheses H_5(iii), (iv) are also necessary for the existence of a unique positive solution for problem (5.193).

Theorem 5.83 *If hypotheses H_4 hold and $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function which satisfies hypotheses H_5(i), (ii), then problem (5.193) admits a unique positive solution*

$$u_0 \in int\, \widehat{C}_+ \text{ if and only if } \widehat{\lambda}_1(p, -\mu_0, \beta) < 0 < \widehat{\lambda}_1(p, -\mu, \beta).$$

Proof

Sufficiency: Follows from Theorem 5.81.
Necessity: We know that

$$\begin{aligned} \widehat{\lambda}_1(p, -\mu_0, \beta) &\leq \frac{\|Du_0\|_p^p + \int_{\partial\Omega} \beta(z) u_0^p d\sigma - \int_\Omega \mu_0(z) u_0^p dz}{\|u_0\|_p^p} \\ &= \frac{\int_\Omega f(z, u_0) u_0 dz - \int_\Omega \mu_0(z) u_0^p dz}{\|u_0\|_p^p}. \end{aligned} \tag{5.194}$$

Since $u_0 \in int\, C_+$, we have $m_0 = \min_{\overline{\Omega}} u_0 > 0$. Then hypothesis H_5(ii) implies that

$$\begin{aligned} &\frac{f(z, u_0(z))}{u_0(z)^{p-1}} \leq \frac{f(z, m_0)}{m_0^{p-1}} < \mu_0(z) \text{ for a.a } z \in \Omega, \\ &\Rightarrow f(z, u_0(z)) u_0(z) < \mu_0(z) u_0(z)^p \text{ for a.a } z \in \Omega. \end{aligned}$$

Using this strict inequality in (5.194), we obtain

$$\widehat{\lambda}_1(p, -\mu_0, \beta) < 0.$$

Next let

$$\xi(z) = \frac{f(z, \|u_0\|_\infty + 1)}{(\|u_0\|_\infty + 1)^{p-1}}.$$

Evidently $\xi \in L^\infty(\Omega)$. We consider the eigenvalue problem

$$\left\{ \begin{array}{ll} -\Delta_p u(z) + \xi(z)|u(z)|^{p-2}u(z) = \widehat{\lambda}|u(z)|^{p-2}u(z) & \text{in } \Omega, \\ \frac{\partial u}{\partial n_p} + \beta(z)|u|^{p-2}u = 0 & \text{on } \partial\Omega. \end{array} \right\}$$

This problem has a principal eigenfunction $\widehat{u}_1 \in int\, \widehat{C}_+$.
Choose $\tau > 0$ large so that

$$u_0 < \widetilde{u}_1 = \tau \widehat{u}_1 \in int\, C_+.$$

If $R(\widetilde{u}_1, u_0) = |D\widetilde{u}_1|^p - (|Du_0|^{p-2}Du_0, D(\frac{\widetilde{u}_1^p}{u_0^{p-1}}))_{\mathbb{R}^N}$ (see Theorem 4.91, Picone's identity), then

$$\begin{aligned}
&\int_\Omega \frac{f(z,u_0)}{u_0^{p-1}}(u_0^p - \widetilde{u}_1^p)dz \\
&= \int_\Omega (-\Delta_p u_0)(u_0 - \frac{\widetilde{u}_1^p}{u_0^{p-1}})dz \\
&= \int_\Omega |Du_0|^{p-2}(Du_0, D(u_0 - \frac{\widetilde{u}_1^p}{u_0^{p-1}}))_{\mathbb{R}^N}dz + \int_{\partial\Omega}\beta(z)u_0^{p-1}(u_0 - \frac{\widetilde{u}_1^p}{u_0^{p-1}})dz \\
&\qquad\qquad \text{(using Green's identity, see Theorem I-4.107)} \\
&= \|Du_0\|_p^p - \|D\widetilde{u}_1\|_p^p + \int_\Omega R(\widetilde{u}_1, u_0)dz + \int_{\partial\Omega}\beta(z)(u_0^p - \widetilde{u}_1^p)d\sigma.
\end{aligned} \tag{5.195}$$

Also, we have

$$\begin{aligned}
&\int_\Omega (\widehat{\lambda}_1(p,\xi,\beta) - \xi(z))(\widetilde{u}_1^p - u_0^p)dz \\
&= \int_\Omega R(u_0, \widetilde{u}_1)dz + \|D\widetilde{u}_1\|_p^p - \|Du_0\|_p^p + \int_{\partial\Omega}\beta(z)(\widetilde{u}_1^p - u_0^p)d\sigma.
\end{aligned} \tag{5.196}$$

Adding (5.195), (5.196), we obtain

$$\begin{aligned}
&\int_\Omega (\frac{f(z,u_0)}{u_0^{p-1}} + \xi(z) - \widehat{\lambda}_1(p,\xi,\beta))(u_0^p - \widetilde{u}_1^p)dz \\
&= \int_\Omega [R(\widetilde{u}_1, u_0) + R(u_0, \widetilde{u}_1)]dz \geq 0 \text{ (see Theorem 4.91)}.
\end{aligned} \tag{5.197}$$

Hypothesis H_5(ii) implies that

$$\begin{aligned}
&\frac{f(z,u_0)}{u_0^{p-1}} > \frac{f(z, \|u_0\|_\infty + 1)}{(\|u_0\|_\infty + 1)^{p-1}} = -\xi(z) \text{ for a.a } z \in \Omega, \\
&\Rightarrow \frac{f(z,u_0)}{u_0^{p-1}} + \xi(z) > 0 \text{ for a.a } z \in \Omega.
\end{aligned} \tag{5.198}$$

One the other hand since $u_0 < \tau\widehat{u}_1 = \widetilde{u}_1$, we see that

$$(u_0^p - \widetilde{u}_1^p)(z) < 0 \text{ for all } z \in \overline{\Omega}. \tag{5.199}$$

From (5.197) we have

$$\int_\Omega [\frac{f(z,u_0)}{u_0^{p-1}} + \xi(z)](u_0^p - \widetilde{u}_1^p)dz$$

$$\geq \widehat{\lambda}_1(p,\xi,\beta)) \int_\Omega (u_0^p - \widetilde{u}_1^p)dz,$$

$$\Rightarrow \widehat{\lambda}_1(p,\xi,\beta)) \int_\Omega (u_0^p - \widetilde{u}_1^p)dz < 0$$

(see (5.198), (5.199)),

$$\Rightarrow \widehat{\lambda}_1(p,\xi,\beta)) > 0 \text{ (see (5.199))}.$$

Note that

$$\xi(z) \geq -\mu(z) \text{ for a.a } z \in \Omega,$$

$$\Rightarrow \widehat{\lambda}_1(p,-\mu,\beta)) \geq \widehat{\lambda}_1(p,\xi,\beta)) > 0.$$

□

In the last part of this section, we study an eigenvalue problem for the (p,q)-Laplacian

$$\left\{ \begin{array}{c} -\Delta_p u(z) - \Delta_q u(z) = \lambda(u(z)^{\tau-1} - u(z)^{\mu-1}) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0,\ u > 0,\ \lambda > 0. \end{array} \right\} \qquad p_\lambda$$

As in the other problems of this section, $\Omega \subseteq \mathbb{R}^N$ is a bounded domain with a C^2-boundary $\partial\Omega$. Also $1 < \mu < \tau < q < p$. Problem (p_λ) formally looks like a logistic equation of superdiffusive type, but it is essentially different, since in the superdiffusive logistic equation, we require that $1 < q < p < \tau < \mu$. Our aim is to determine the existence and multiplicity of positive solutions as the parameter $\lambda > 0$ moves in $\mathring{\mathbb{R}}_+ = (0,\infty)$. We prove a result which is global in $\lambda > 0$ (a "bifurcation type" result).

We introduce the following two sets:

$$\mathcal{L} = \{\lambda > 0 : \text{problem } (p_\lambda) \text{ has a positive solution}\},$$

$$S_\lambda = \text{set of positive solutions of } (p_\lambda).$$

Also, let $\varphi_\lambda : W_0^{1,p}(\Omega) \to \mathbb{R}$ be the energy functional for problem (p_λ) defined by

$$\varphi_\lambda(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{q}\|Du\|_q^q + \frac{\lambda}{\mu}\|u\|_\mu^\mu - \frac{\lambda}{\tau}\|u\|_\tau^\tau \quad \text{for all } u \in W_0^{1,p}(\Omega).$$

Evidently $\varphi_\lambda \in C^1(W_0^{1,p}(\Omega))$. In what follows

$$C_0^1(\overline{\Omega}) = \{u \in C^1(\overline{\Omega}) : u|_{\partial\Omega} = 0\}.$$

We know that this is an ordered Banach space with positive (order) cone $C_+ = \{u \in C_0^1(\overline{\Omega}) : u(z) \geq 0 \text{ for all } z \in \overline{\Omega}\}$. This cone has a nonempty interior given by

$$int\, C_+ = \{u \in C_+ : u(z) > 0 \text{ for all } z \in \Omega,\ \frac{\partial u}{\partial n}|_{\partial\Omega} < 0\}.$$

Proposition 5.84 *$\mathcal{L} \neq \emptyset$ and for every $\lambda \in \mathcal{L}$, $S_\lambda \subseteq C_+ \setminus \{0\}$.*

Proof Since $1 < \mu < \tau < q < p$, we see that $\varphi_\lambda(\cdot)$ is coercive. Also, it is sequentially weakly lower semicontinuous. So, we can find $\widehat{u}_\lambda \in W_0^{1,p}(\Omega)$ such that

$$\varphi_\lambda(\widehat{u}_\lambda) = \inf\{\varphi_\lambda(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.200}$$

Consider $K \subseteq \Omega$ compact and $x_0 > 1$. Let $\widehat{v} \in C_0^1(\overline{\Omega})$ be a bounded function such that

$$\widehat{v}|_K = x_0 \quad \text{and} \quad 0 \leq \widehat{v}(z) \leq x_0 \text{ for all } z \in \overline{\Omega}. \tag{5.201}$$

Then

$$\begin{aligned}
\varphi_\lambda(\widehat{v}) &= \frac{1}{p}\|D\widehat{v}\|_p^p + \frac{1}{q}\|D\widehat{v}\|_q^q - \lambda\left[\frac{1}{\tau}x_0^\tau - \frac{1}{\mu}x_0^p\right]|K|_N \\
&\qquad - \lambda\int_{\Omega\setminus K}\left(\frac{1}{\tau}\widehat{v}^\tau - \frac{1}{\mu}\widehat{v}^\mu\right)dz \\
&\leq \frac{1}{p}\|D\widehat{v}\|_p^p + \frac{1}{q}\|D\widehat{v}\|_q^q - \lambda\left[\frac{1}{\tau}x_0^\tau - \frac{1}{\mu}x_0^\mu\right]|K|_N \\
&\qquad + \frac{\lambda}{\mu}x_0^\mu|\Omega\setminus K|_N \text{ (see (5.201))} \\
&= \frac{1}{p}\|D\widehat{v}\|_p^p + \frac{1}{q}\|D\widehat{v}\|_q^q - \frac{\lambda}{\tau}x_0^\tau|K|_N + \frac{\lambda}{\mu}x_0^\mu|\Omega|_N.
\end{aligned} \tag{5.202}$$

As before $|\cdot|_N$ denotes the Lebesgue measure on $\mathbb{R}^N$. Let

$$\vartheta_1 = \frac{1}{p}\|D\widehat{v}\|_p^p + \frac{1}{q}\|D\widehat{v}\|_q^q = \frac{1}{p}\int_{\Omega\setminus K}|D\widehat{v}|^p dz + \frac{1}{q}\int_{\Omega\setminus K}|D\widehat{v}|^q dz$$

$$\vartheta_2 = \frac{1}{\tau}x_0^\tau|K|_N - \frac{1}{\mu}x_0^\mu|\Omega|_N.$$

We know that $\mu < \tau$ and so if $x_0 > 1$ is large, then $\vartheta_2 > 0$. Therefore, there is $\lambda_0 > 0$ such that

$$\vartheta_1 - \lambda\vartheta_2 < 0 \text{ for all } \lambda > \lambda_0.$$

Hence from (5.202) we see that

$$\begin{aligned}
&\varphi_\lambda(\widehat{v}) < 0 \text{ for all } \lambda > \lambda_0,\\
\Rightarrow &\varphi_\lambda(\widehat{u}_\lambda) < 0 = \varphi_\lambda(0) \text{ (see (5.100))},\\
\Rightarrow &\widehat{u}_\lambda \neq 0.
\end{aligned}$$

Since we can always replace $\widehat{u}_\lambda$ by $|\widehat{u}_\lambda|$, we may assume that $\widehat{u}_\lambda \geq 0$. From (5.200), we infer that

$$\begin{aligned}
&\langle \varphi'_\lambda(\widehat{u}_\lambda), h\rangle = 0 \text{ for all } h \in W_0^{1,p}(\Omega)\},\\
\Rightarrow &\langle V(\widehat{u}_\lambda), h\rangle = \int_\Omega \lambda[\widehat{u}_\lambda^{\tau-1} - \widehat{u}_\lambda^{\mu-1}]hdz \text{ for all } h \in W_0^{1,p}(\Omega)\},\\
\Rightarrow &(\lambda_0, \infty) \subseteq \mathcal{L} \neq \emptyset.
\end{aligned}$$

If $u \in S_\lambda$, then Proposition 4.82 implies $u \in L^\infty(\Omega)$ and then from Theorem 4.84, we have $u \in C_+ \setminus \{0\}$. □

Remark 5.85 In this case since $\mu < \tau < q < p$, we can not use Theorem 4.89 and conclude that $S_\lambda \subseteq intC_+$ as is the case in superdiffusive logistic equations.

Let $\lambda_* = \inf \mathcal{L}$.

Proposition 5.86 $\lambda_* > 0$.

Proof We argue by contradiction. So, suppose $\lambda_* = 0$ and let $\{\lambda_n\}_{n\in\mathbb{N}} \subseteq \mathcal{L}$ such that $\lambda_n \downarrow 0$. For each $n \in \mathbb{N}$, let $u_n \in S_{\lambda_n} \subseteq C_+ \setminus \{0\}$. We have

$$\langle \varphi'_{\lambda_n}(u_n), h\rangle = 0 \text{ for all } h \in W_0^{1,p}(\Omega), n \in \mathbb{N}.$$

Using the test function $h = u_n \in W_0^{1,p}(\Omega)$, we obtain

$$\begin{aligned}
&\| Du_n \|_p^p + \| Du_n \|_q^q = \lambda_n \textstyle\int_\Omega (u_n^\tau - u_n^\mu)dz \leq \lambda_n \| u_n \|_\tau^\tau\\
\Rightarrow &\| u_n \|^p \leq \lambda_n c_1 \| u_n \|^\tau \text{ for some } c_1 > 0, \text{ all } n \in \mathbb{N}\\
&\quad \text{(recall } W_0^{1,p}(\Omega) \hookrightarrow L^\tau(\Omega) \text{ continuously)}, \qquad (5.203)\\
\Rightarrow &\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}
\end{aligned}$$

Then Proposition 4.82 and Theorem 4.84, imply that there exist $\alpha \in (0, 1)$ and $c_2 > 0$ such that

$$u_n \in C_0^{1,\alpha}(\overline{\Omega}) = C^{1,\alpha}(\overline{\Omega}) \cap C_0^1(\overline{\Omega}),\ \| u_n \|_{C_0^{1,\alpha}(\overline{\Omega})} \leq c_2 \text{ for all } n \in \mathbb{N}.$$

From the compact embedding of $C_0^{1,\alpha}(\overline{\Omega})$ into $C_0^1(\overline{\Omega})$ and (5.203), we can say that

$$u_n \to 0 \text{ in } C_0^1(\overline{\Omega}) \text{ (recall } \lambda_n \downarrow 0).$$

Therefore we can find $n_0 \in \mathbb{N}$ such that

$$u_n(z)^\tau - u_n(z)^\mu < 0 \text{ for all } z \in \overline{\Omega}, \text{ all } n \geq n_0, \text{ (since } \mu < \tau).$$

So,

$$0 \leq \| u_n \|^p \leq \lambda_n \int_\Omega [u_n^\tau - u_n^\mu] dz < 0 \text{ for all } n \geq n_0$$

a contradiction. This proves that $\lambda_* > 0$. □

Proposition 5.87 $\mathcal{L} = [\lambda_*, \infty)$.

Proof Let $\lambda \geq \lambda_*$. As before, we can find $u_\lambda \in W_0^{1,p}(\Omega)$ such that

$$\varphi_\lambda(u_\lambda) = \inf\{\varphi_\lambda(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.204}$$

Since $\mu < \tau$, given $\eta \in (0, |\Omega|_N)$, we can find $x_0 > 1$ large such that

$$\frac{1}{\tau} x_0^\tau (|\Omega|_N - \eta) - \frac{1}{\mu} x_0^\mu |\Omega|_N > \frac{p|\Omega|_N^{\frac{p-q}{p}} + q}{\lambda_* pq}. \tag{5.205}$$

Also, on account of the Radon property of the Lebesgue measure (see Theorem I-2.157), we can find $K \subseteq \Omega$ compact such that $|\Omega \setminus K|_N \leq \eta$.

We introduce a cut-off function $\overline{u} \in C_c^1(\Omega)$ such that

$$\overline{u}|_K = 1,\ 0 \leq \overline{u}(z) \leq x_0 \text{ for all } z \in \overline{\Omega},\ |Du(z)| \leq \frac{1}{\eta^{1/p}} \text{ for all } z \in \Omega. \tag{5.206}$$

Such a cut-off function can be generated using standard mollification techniques. Using (5.206), we have

$$\varphi_\lambda(\overline{u}) \leq \frac{1}{p} \int_{\Omega \setminus K} |D\overline{u}|^p dz + \frac{1}{q} \int_{\Omega \setminus K} |Du|^q dz - \lambda\left(\frac{1}{\tau} x_0^\tau - \frac{1}{\mu} x_0^\mu\right)(|\Omega|_N - \eta)$$
$$+ \frac{\lambda}{\mu} x_0^\mu |\Omega|_N$$

$$\leq \frac{1}{p}|\Omega \setminus K|_N \frac{1}{\eta} + \frac{1}{q}|\Omega \setminus K|_N \frac{1}{\eta^{q/p}} - \lambda_*\left(\frac{1}{\tau}x_0^\tau(|\Omega|_N - \eta) - \frac{1}{p}x_0^p|\Omega|_N\right)$$

(see (5.205), (5.206))

$$\leq \frac{p|\Omega|_N^{\frac{p-q}{p}} + q}{pq} - \lambda_*\left(\frac{1}{\tau}x_0^\tau(|\Omega|_N - \eta) - \frac{1}{\mu}x_0^\mu|\Omega|_N\right)$$

$$< 0$$

$$\Rightarrow \varphi_\lambda(u_\lambda) < 0 = \varphi_\lambda(0).$$

$$\Rightarrow u_\lambda \neq 0 \text{ and so } u_\lambda \in S_\lambda, \text{ hence } \lambda \in \mathcal{L}.$$

□

The above proposition implies that $\mathcal{L} = [\lambda_*, \infty)$.

Proposition 5.88 *If $\lambda \in \mathcal{L} = [\lambda_*, \infty)$, then problem (p_λ) has at least two positive solutions. $u_0, \widehat{u} \in S_\lambda \subseteq C_+ \cdot \{0\}$ such that*

$$0 < \widehat{u}(z) \leq u_0(z) \text{ for all } z \in \Omega,\ u_0 \neq \widehat{u}.$$

Proof We already have one solution $u_0 \in S_\lambda \subseteq C_+ \cdot \{0\}$ such that

$$\varphi_\lambda(u_0) < 0$$

(see the proof of Proposition 5.87). We introduce the Caratheodory function $k(z, x)$ defined by

$$k(z,x) = \begin{cases} (x^+)^{\tau-1} - (x^+)^{\mu-1}, & \text{if } x \leq u_0(z) \\ u_0(z)^{\tau-1} - u_0(z)^{\mu-1}, & \text{if } u_0(z) < x \end{cases} \tag{5.207}$$

We set $K(z,x) = \int_0^x k(z,s)ds$ and consider the C^1-functional $\psi_\lambda : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\psi_\lambda(u) = \frac{1}{p} \parallel Du \parallel_p^p + \frac{1}{q} \parallel Du \parallel_q^q - \lambda \int_\Omega K(z,u)dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

Claim 1 There exists $\varrho \in (0, \parallel u_0 \parallel)$ such that

$$\varphi_\lambda(u) \geq m_\lambda > 0 \text{ for all } u \in W_0^{1,p}(\Omega) \text{ with } \parallel u \parallel = \varrho.$$

We can find $\delta \in (0, 1)$ small such that

$$\frac{1}{\tau}x^\tau - \frac{1}{\mu}x^\mu \leq 0 \text{ for all } 0 \leq x \leq \delta. \tag{5.208}$$

(since $\mu < \tau$). Let $u \in W_0^{1,p}(\Omega)$ and set

$$\Omega_0 = \{z \in \Omega : u(z) > \delta\}.$$

Consider $z \in (\Omega \setminus \Omega_0) \cap \{u \leq u_0\}$. Then

$$K(z, u(z)) = \frac{1}{\tau} u^+(z)^\tau - \frac{1}{\mu} u^+(z)^\mu \leq 0 \text{ (see (5.207), (5.208))}. \tag{5.209}$$

Let $z \in (\Omega \setminus \Omega_0) \cap \{u_0 < u\}$. In this case, we have

$$\begin{aligned} K(z, u(z)) &= \int_0^{u(z)} k(z, s) ds \\ &= \int_0^{u_0(z)} k(z, s) ds + \int_{u_0(z)}^{u(z)} k(z, s) ds \\ &= \frac{1}{\tau} u_0(z)^\tau - \frac{1}{\mu} u_0(z)^\mu + (u_0(z)^{\tau-1} - u_0(z)^{\mu-1})(u - u_0)(z) \\ &\qquad \text{(see (5.207))} \\ &\leq (u_0(z)^{\tau-1} - u_0(z)^{\mu-1})(u - u_0)(z) \text{ (see (5.208))}. \end{aligned}$$

Since $u_0 < u$ and $z \in \Omega \setminus \Omega_0$, we have $u_0(z) < \delta$ and so choosing $\delta \in (0, 1)$ even smaller if necessary, we have

$$K(z, u(z)) \leq 0 \text{ for a.a } z \in (\Omega \setminus \Omega_0) \cap \{u_0 < u\}. \tag{5.210}$$

From (5.209) and (5.210) it follows that

$$K(z, u(z)) \leq 0 \text{ for a.a } z \in \Omega \setminus \Omega_0.$$

Then we have

$$\varphi_\lambda(u) \geq \frac{1}{p} \| u \|^p - \lambda \int_{\Omega_0} K(z, u) dz \tag{5.211}$$

We estimate the integral in (5.211).

$$\begin{aligned}\int_{\Omega_0} K(z,u)dz &= \int_{\Omega_0\cap\{u\le u_0\}} K(z,u)dz + \int_{\Omega_0\cap\{u_0<u\}} K(z,u)dz\\ &= \int_{\Omega_0\cap\{u\le u_0\}} [\frac{1}{\tau}u^\tau - \frac{1}{\mu}u^\mu]dz + \int_{\Omega_0\cap\{u_0<u\}} [\frac{1}{\tau}u_0^\tau - \frac{1}{\mu}u_0^\mu]dz\\ &\quad + \int_{\Omega_0\cap\{u_0<u\}} (u_0^{\tau-1} - u_0^{\mu-1})(u-u_0)dz\\ &\le \int_{\Omega_0\cap\{u\le u_0\}} \frac{1}{\tau}u^\tau dz + \int_{\Omega_0\cap\{u_0<u\}} 2u^\tau dz\\ &\le \int_{\Omega_0} 3u^\tau dz\\ &\le c\int_{\Omega_0} u^p dz \text{ for some } c>0 \text{ (recall } \tau<p).\end{aligned} \tag{5.212}$$

Let $r \in (p, p^*)$ and note that $u^p \in L^{r/p}(\Omega)$. So, if we use Holder's inequality and the Sobolev embedding theorem, we obtain

$$\begin{aligned}c\int_{\Omega_0} u^p dz &= c\int_\Omega \chi_{\Omega_0} u^p dz\\ &\le c(\int_\Omega u^r dz)^{\frac{p}{r}} |\Omega_0|_N^{\frac{r-p}{r}}\\ &\le c_1 \parallel u \parallel^p |\Omega_0|_N^{\frac{r-p}{r}} \text{ for some } c_1>0,\\ \Rightarrow \int_\Omega K(z,u)dz &\le c_1 \parallel u \parallel^p |\Omega_0|_N^{\frac{r-p}{r}} \text{ (see (5.212))}\\ \Rightarrow \psi_\lambda(u) &\ge \frac{1}{p}\parallel u \parallel^p - \lambda c_1 \parallel u \parallel^p |\Omega_0|_N^{\frac{r-p}{r}}\\ &= (\frac{1}{p} - \lambda c_1 |\Omega_0|_N^{\frac{r-p}{r}}) \parallel u \parallel^p .\end{aligned} \tag{5.213}$$

Note that if $\parallel u \parallel \to 0$, then $|\Omega_0|_N \to 0$ and so from (5.213) we see that we can find $\varrho \in (0, \min\{\parallel u_0 \parallel, 1\})$ small that

$$\psi_\lambda(u) \ge m_\lambda > 0 \text{ for all } u \in W_0^{1,p}(\Omega), \parallel u \parallel = \varrho.$$

This proves Claim 1.

Claim 2 $K_{\psi_\lambda} \subseteq [0, u_0] \cap C_+$.

Let $u \in K_{\psi_\lambda}$. Then

$$\langle V(u), h\rangle = \int_\Omega \lambda k(z,u)h\,dz \text{ for all } h \in W_0^{1,p}(\Omega). \tag{5.214}$$

with $V = A_p + A_q$. In (5.214) first we use $h = -u^- \in W_0^{1,p}(\Omega)$. Then

$$\| u^- \|_p^p \leq 0 \text{ (see (5.207))}, \text{ hence } u \geq 0.$$

Next in (5.214) we choose $h = (u - u_0)^+ \in W_0^{1,p}(\Omega)$. Using (5.207) we have

$$\begin{aligned}\langle V(u), (u-u_0)^+\rangle &= \int_\Omega [u_0^{\tau-1} - u_0^{\mu-1}](u-u_0)^+ dz \\ &= \langle V(u_0), (u-u_0)^+\rangle \\ \Rightarrow u &\leq u_0 \text{ (strict monotonicity of } V(\cdot)).\end{aligned}$$

So, we have $u \in [0, u_0]$ and by Theorem 4.84, we conclude that

$$K_{\psi_\lambda} \subseteq [0, u_0] \cap C_+.$$

This proves Claim 2.
We have

$$\psi_\lambda(u_0) = \varphi_\lambda(u_0) < 0 = \psi_\lambda(0) < m_\lambda \leq \inf\{\psi_\lambda(u) : \| u \| = \varrho\}. \tag{5.215}$$

Also it is clear from (5.207) that $\psi_\lambda(\cdot)$ is coercive. Therefore by Proposition 3.19

$$\psi_\lambda(\cdot) \text{ satisfies the C-condition.} \tag{5.216}$$

Then (5.215), (5.216) and the mountain pass theorem (see Theorem 3.41), give $\widehat{u} \in W_0^{1,p}(\Omega)$ such that

$$\begin{aligned}&\widehat{u} \in K_{\psi_\lambda} \subseteq [0, u_0] \cap C_+, m_\lambda \leq \psi_\lambda(\widehat{u}), \\ \Rightarrow &\widehat{u} \neq \{0, u_0\} \text{ and } \widehat{u} \in S_\lambda \text{ (see (5.207))}.\end{aligned}$$

Finally Harnack's inequality (see Theorem 4.96) implies

$$0 < \widehat{u}(z) \text{ for all } z \in \Omega.$$

□

Therefore we can state the following multiplicity theorem for problem (p_λ), which is global in $\lambda > 0$.

Theorem 5.89 *If $1 < \mu < \tau < q < p$, then there exists $\lambda_* > 0$ such that*

(a) for all $\lambda \geq \lambda_*$ problem (p_λ) has at least two positive solution $u_0, \widehat{u} \in C_+ \setminus \{0\}$ such that $u_0 \neq \widehat{u}$ and $0 < \widehat{u}(z) \leq u_0(z)$ for all $z \in \Omega$;
(b) for $\lambda \in (0, \lambda_*)$ problem (p_λ) has no solution.

5.3 Singular Problems

In this section, we study nonlinear elliptic equations with singular terms in the reaction. The difficulty that we encounter when dealing with such problems, is that on account of the presence of the singular term, the energy functional of such a problem is not C^1 and so we have difficulty in using the variational tools from the critical point theory (see Chap. 3).

As before, let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$. We consider the following Dirichlet problem driven by the (p, q)-Laplacian.

$$\begin{cases} -\triangle_p u(z) - \triangle_q u(z) = u(z)^{-\eta} + \lambda f(z, u(z)) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, u > 0, \lambda > 0, 0 < \eta < 1, 1 < q < p. \end{cases} \tag{p^1_λ}$$

We impose the following conditions on the perturbation $f(z, x)$ of the singular term.

$\mathbf{H_1}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that $f(z, 0) = 0$ for a.a $z \in \Omega$ and

(i) $f(z, x) \leq a(z)(1 + x^{r-1})$ for a.a $z \in \Omega$, all $x \geq 0$, with $a \in L^\infty(\Omega)$ $p < r < p^*$ (as always $p^* = \frac{Np}{N-p}$) if $p < N$ and $p^* = \infty$, if $p \leq N$;

(ii) if $F(z, x) = \int_0^x f(z, s)ds$, then $\lim\limits_{x \to \infty} \frac{F(z,x)}{x^p} = +\infty$ uniformly for a.a $z \in \Omega$;

(iii) there exists $\tau \in ((r - p) \max\{\frac{N}{p}, 1\}, p^*)$ with $\tau > q$ such that

$$0 < c_0 \leq \liminf_{x \to \infty} \frac{f(z, x)x - pF(z, x)}{x^\tau} \text{ uniformly for a.a } z \in \Omega;$$

(iv) $\lim\limits_{x \to 0^+} \frac{f(z,x)}{x^{q-1}} = 0$ uniformly for a.a $z \in \Omega$;

(v) there exists $\mu \in (q, p)$ such that

$$0 < c_1 \leq \liminf_{x \to 0^+} \frac{f(z, x)}{x^{\mu-1}} \text{ uniformly for a.a } z \in \Omega;$$

(vi) for every $s > 0$, we can find $m_s > 0$ such that

$$0 < m_s \leq f(z, s) \text{ for a.a } z \in \Omega, \text{ all } x \geq s$$

and for every $\varrho > 0$, there exists $\widehat{\xi}_\varrho > 0$ such that for a.a $z \in \Omega$ the function $x \to f(z, x) + \widehat{\xi}_\varrho x^{p-1}$ is nondecreasing on $[0, \varrho]$.

Remark 5.90 We observe that the above hypotheses concern the positive semiaxis $\mathbb{R}_+ = [0, \infty)$. Since we search for positive solutions, without any loss of generality, we may assume that $f(z, x) = 0$ for a.a $z \in \Omega$, all $x \leq 0$. Hypotheses H_1(ii), (iii) imply that

$$\lim_{x \to +\infty} \frac{f(z, x)}{x^{p-1}} = +\infty \text{ uniformly for a.a } z \in \Omega.$$

So, the perturbation $f(z, \cdot)$ is $(p-1)$-superlinear. Usually problems with superlinear reaction, are studied using the so-called "Ambrosetti-Rabinowitz condition", which says the following: "there exist $r > p$ and $M > 0$ such that

$$\begin{cases} 0 < rF(z, x) \leq f(z, x)x \text{ for a.a } z \in \Omega, \text{ all } x \geq M, \\ 0 < \operatorname{essinf}_{\Omega} F(\cdot, M). \end{cases} \tag{AR}$$

In fact this is a unilateral version of the condition, since in our case the negative semiaxis is out of the picture. Integrating (AR) we obtain

$$\begin{aligned} c_2 x^r \leq F(z, x) \text{ for a.a } z \in \Omega, \text{ all } x \geq M, \\ \Rightarrow c_2 x^{r-1} \leq f(z, x) \text{ for a.a } z \in \Omega, \text{ all } x \geq M. \end{aligned}$$

So, the (AR)-condition implies that near $+\infty$, $f(z, \cdot)$ has at least $(r-1)$-growth. Instead of (AR), here we employ the less restrictive hypothesis H_1(iii), which fits in our framework also $(p-1)$-superlinear functions with "slower" growth near $+\infty$, which fail to satisfy (AR). For example, consider the following function (for the sake of simplicity we drop the z-dependence)

$$f(x) = \begin{cases} (x^+)^{\mu-1} & \text{if } x \leq 1 \\ x^{p-1} ln x + x^{s-1} & \text{if } 1 < x \end{cases}$$

with $\mu \in (q, p)$ and $s < p$. This function satisfies hypotheses H_1, but fails to satisfy (AR).

In this case, by a "solution" of (p_λ^1), we mean a function $u \in W_0^{1,p}(\Omega) \setminus \{0\}$, $u \geq 0$ such that

$$u^{-\eta} h \in L^1(\Omega)$$

and

$$\langle A_p(u), h \rangle + \langle A_q(u), h \rangle = \int_\Omega u^{-\eta} h dz + \lambda \int_\Omega f(z, u) h dz \text{ for all } h \in W_0^{1,p}(\Omega).$$

The energy functional of problem (p_λ^1) is given by

$$\varphi_\lambda(u) = \frac{1}{p} \parallel Du \parallel_p^p + \frac{1}{q} \parallel Du \parallel_q^q - \frac{1}{1-\eta} \int_\Omega (u^+)^{1-\eta} dz$$
$$- \int_\Omega \lambda F(z, u^+) dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

As we already remarked earlier, $\varphi_\lambda \notin C^1(W_0^{1,p}(\Omega))$, so we can not use the minimax theorems of critical point theory directly on φ_λ. For this reason, we have to find ways to bypass the singularity and deal with relevant C^1-functionals. To this end, first we consider the following auxiliary, purely singular problem

$$\left\{ \begin{array}{l} -\Delta_p u(z) - \Delta_q u(z) = u(z)^{-\eta} \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, u > 0, 0 < \eta < 1, 1 < q < p. \end{array} \right\} \tag{5.217}$$

We will solve this problem and its solution will help us bypass the singularity and introduce C^1-functionals on which we can use the results of critical point theory. In the analysis of (5.217) we will need "Hardy's inequality", which we recall below (see Brezis [60], p.313).

Proposition 5.91 *If $1 < r < \infty$ and $\widehat{d}(\cdot) = d(\cdot, \partial\Omega)$, then there exists $\widehat{c} > 0$ such that*

$$\parallel \frac{u}{\widehat{d}} \parallel_r \leq \widehat{c} \parallel Du \parallel_r \text{ for all } u \in W_0^{1,r}(\Omega).$$

Conversely, if $u \in W_0^{1,r}(\Omega)$ satisfies $\frac{u}{\widehat{d}} \in L^r(\Omega)$, then $u \in W_0^{1,r}(\Omega)$.

We will use this inequality in the solution of problem (5.217).

Proposition 5.92 *If $1 < q < p, 0 < \eta < 1$, then problem (5.217) admits a unique positive solution $\overline{u} \in intC_+$.*

Proof Let $g \in L^p(\Omega)$ and $\varepsilon \in (0, 1]$. We consider the following Dirichlet problem

$$-\Delta_p u - \Delta_q u = [|g(z)| + \varepsilon]^{-\eta} \text{ in } \Omega, \; u|_{\partial\Omega} = 0.$$

Let $V = A_p + A_q : W_0^{1,p}(\Omega) \to W_0^{-1,p'}(\Omega) = W_0^{1,p}(\Omega)^*$. Proposition 5.47 says $V(\cdot)$ is maximal monotone and

$$\parallel u \parallel^p \leq \langle V(u), u \rangle \text{ for all } u \in W_0^{1,p}(\Omega),$$
$$\Rightarrow V(\cdot) \text{ is coercive.}$$

A maximal monotone, coercive operator is surjective (see Theorem I-7.47). Evidently $(|g| + \varepsilon)^{-\eta} \in L^\infty(\Omega)$ and so we can find $u_\varepsilon \in W_0^{1,p}(\Omega), u_\varepsilon \geq 0, u_\varepsilon \neq 0$ such that

$$V(u_\varepsilon) = (|g| + \varepsilon)^{-\eta}.$$

Since $V(\cdot)$ is also strictly monotone (see Remark 5.48), this solution u_ε is unique. So, we can define the solution map $\vartheta : L^p(\Omega) \to L^p(\Omega)$ by setting $\vartheta(g) = u_\varepsilon$ (we know that $W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ compactly, see Theorem I-4.115). We show that $\vartheta(\cdot)$ is continuous. To this end let $g_n \to g$ in $L^p(\Omega)$ and set $u_\varepsilon^n = \vartheta(g_n)$ $n \in \mathbb{N}$. We have

$$\langle V(u_\varepsilon^n), h\rangle = \int_\Omega \frac{h}{(|g_n| + \varepsilon)^\eta} dz \text{ for all } h \in W_0^{1,p}(\Omega), \text{ all } n \in \mathbb{N}. \tag{5.218}$$

Let $h = u_\varepsilon^n \in W_0^{1,p}(\Omega)$. Then

$$\| u_\varepsilon^n \|^p \leq \frac{1}{\varepsilon^\eta} \| u_\varepsilon^n \|_1 \leq \frac{1}{\varepsilon^\eta} c_3 \| u_\varepsilon^n \|_p \text{ for some } c_3 > 0, \text{ all } n \in \mathbb{N},$$

$$\Rightarrow \{u_\varepsilon^n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}$$

We may assume that

$$u_\varepsilon^n \xrightarrow{w} \overline{u}_\varepsilon \text{ in } W_0^{1,p}(\Omega), u_\varepsilon^n \to \overline{u}_\varepsilon \text{ in } L^p(\Omega) \tag{5.219}$$

In (5.218) we choose $h = u_\varepsilon^n - \overline{u}_\varepsilon \in W_0^{1,p}(\Omega)$, pass to the limit as $n \to \infty$ and use (5.219). We obtain

$$\begin{aligned} &\lim_{n\to\infty} \langle V(u_\varepsilon^n), u_\varepsilon^n - \overline{u}_\varepsilon\rangle = 0, \\ &\Rightarrow u_\varepsilon^n \to \overline{u}_\varepsilon \text{ in } W_0^{1,p}(\Omega) \text{ (see Proposition 5.47).} \end{aligned} \tag{5.220}$$

Passing to the limit as $n \to \infty$ in (5.218) and using (5.220) we obtain

$$\begin{aligned} &\langle V(\overline{u}_\varepsilon), h\rangle = \int_\Omega \frac{h}{(|g| + \varepsilon)^\eta} dz \text{ for all } h \in W_0^{1,p}(\Omega), \\ &\Rightarrow \vartheta(g) = \overline{u}_\varepsilon, \\ &\Rightarrow \vartheta(\cdot) \text{ is continuous.} \end{aligned}$$

From the above argument, we know that

$$\vartheta(L^p(\Omega)) \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}$$

Since $W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ compactly, it follows that $\overline{\vartheta(L^p(\Omega))} \subseteq L^p(\Omega)$ is compact. So, by Theorem 2.76, we can find $\widehat{u}_\varepsilon \in W_0^{1,p}(\Omega) \subseteq L^p(\Omega)$ such that $\vartheta(\widehat{u}_\varepsilon) = \widehat{u}_\varepsilon$. The nonlinear regularity theory and the nonlinear maximum principal, imply

$$\widehat{u}_\varepsilon \in intC_+.$$

Claim The map $\varepsilon \to \widehat{u}_\varepsilon$ is nonincreasing ($0 < \varepsilon' < \varepsilon \Rightarrow \widehat{u}_\varepsilon < \widehat{u}_{\varepsilon'}$).

Note that

$$-\Delta_p \widehat{u}_{\varepsilon'} - \Delta_q \widehat{u}_{\varepsilon'} = \frac{1}{(\widehat{u}_{\varepsilon'} + \varepsilon')^\eta} \geq \frac{1}{(\widehat{u}_{\varepsilon'} + \varepsilon)^\eta} \text{ in } \Omega \tag{5.221}$$

We introduce the following Caratheodory function

$$\ell_\varepsilon(z, x) = \begin{cases} (x^+ + \varepsilon)^{-\eta} & \text{if } x \leq \widehat{u}_{\varepsilon'}(z) \\ (\widehat{u}_{\varepsilon'}(z) + \varepsilon)^{-\eta} & \text{if } \widehat{u}_{\varepsilon'}(z) < x \end{cases} \tag{5.222}$$

We set $L_\varepsilon(z, x) = \int_0^x \ell_\varepsilon(z, x)ds$ and consider the C^1-functional $\sigma_\varepsilon : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\sigma_\varepsilon(u) = \frac{1}{p} \parallel Du \parallel_p^p + \frac{1}{q} \parallel Du \parallel_q^q - \int_\Omega L_\varepsilon(z, u)dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

From (5.222) it is clear that $\sigma_\varepsilon(\cdot)$ is coercive. Using the Sobolev embedding theorem, we see that $\sigma_\varepsilon(\cdot)$ is sequentially weakly lower semicontinuous. So, we can find $\widehat{u}_\varepsilon^* \in W_0^{1,p}(\Omega)$ such that

$$\sigma_\varepsilon(\widehat{u}_\varepsilon^*) = \inf\{\sigma_\varepsilon(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.223}$$

Let $u \in intC_+$ and $t > 0$. Then for $t \in (0, 1)$ small, we have $tu \leq \widehat{u}_{\varepsilon'}$ (recall $\widehat{u}_{\varepsilon'} \in intC_+$ and use Proposition 2.86). Using (5.222) we have

$$\begin{aligned} \sigma_\varepsilon(tu) &= \frac{t^p}{p} \parallel Du \parallel_p^p + \frac{t^q}{q} \parallel Du \parallel_q^q - \frac{t^{1-\eta}}{1-\eta} \int_\Omega u^{1-\eta}dz \\ &\leq c_4 t^q - c_5 t^{1-\eta} \text{ for some } c_4, c_5 > 0 \text{ (since } t \in (0, 1), q < p), \\ &\Rightarrow \sigma_\varepsilon(tu) < 0 \text{ for } t \in (0, 1) \text{ small}, \\ &\Rightarrow \sigma_\varepsilon(\widehat{u}_\varepsilon^*) < 0 = \sigma_\varepsilon(0) \text{ (see (5.223))}, \\ &\Rightarrow \widehat{u}_\varepsilon^* \neq 0. \end{aligned}$$

From (5.223) we have

$$\langle V(\widehat{u}_\varepsilon^*), h\rangle = \int_\Omega \ell_\varepsilon(z, \widehat{u}_\varepsilon^*)hdz \text{ for all } h \in W_0^{1,p}(\Omega). \tag{5.224}$$

Let $h = -(\widehat{u}_\varepsilon^*)^- \in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned} &\parallel D(\widehat{u}_\varepsilon^*)^- \parallel_p \leq 0 \text{ (see (5.233))}, \\ &\Rightarrow \widehat{u}_\varepsilon^* \geq 0, \widehat{u}_\varepsilon^* \neq 0. \end{aligned}$$

Also in (5.224) we use the test function $(\widehat{u}_\varepsilon^* - \widehat{u}_{\varepsilon'})^+ \in W_0^{1,p}(\Omega)$.
Then

$$\begin{aligned}\langle V(\widehat{u}_\varepsilon^*), (\widehat{u}_\varepsilon^* - \widehat{u}_{\varepsilon'})^+\rangle &= \int_\Omega \frac{(\widehat{u}_\varepsilon^* - \widehat{u}_{\varepsilon'})^+}{(\widehat{u}_{\varepsilon'} + \varepsilon)^\eta} dz \text{ (see (5.222))} \\ &\leq \int_\Omega \frac{(\widehat{u}_\varepsilon^* - \widehat{u}_{\varepsilon'})^+}{(\widehat{u}_{\varepsilon'} + \varepsilon')} dz \\ &= \langle V(\widehat{u}_{\varepsilon'}), (\widehat{u}_\varepsilon^* - \widehat{u}_{\varepsilon'})^+\rangle \text{ (see (5.221))}, \\ \Rightarrow \widehat{u}_\varepsilon^* &\leq \widehat{u}_{\varepsilon'}.\end{aligned}$$

So, we have proved that

$$\widehat{u}_\varepsilon^* \in [0, \widehat{u}_{\varepsilon'}] = \{h \in W_0^{1,p}(\Omega) : 0 \leq h(z) \leq \widehat{u}_{\varepsilon'}(z) \text{ for a.a } z \in \Omega\}.$$

Since $\widehat{u}_\varepsilon^* \neq 0$, from (5.222) it follows that

$$\widehat{u}_\varepsilon^* = \widehat{u}_\varepsilon \leq \widehat{u}_{\varepsilon'}.$$

This proves the Claim.

Now we will pass to the limit as $\varepsilon \to 0^+$ in order to produce a solution of (5.217). So, let $\varepsilon_n \to 0^+$ and set $\widehat{u}_n = \widehat{u}_{\varepsilon_n}$ $n \in \mathbb{N}$. Then

$$\langle V(\widehat{u}_n), h\rangle = \int_\Omega \frac{h}{(\widehat{u}_n + \varepsilon_n)^\eta} dz \text{ for all } h \in W_0^{1,p}(\Omega), \text{ all } n \in \mathbb{N}. \tag{5.225}$$

Let $h = \widehat{u}_n \in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned}\| \widehat{u}_n \|^p &\leq \int_\Omega \widehat{u}_n^{1-\eta} dz, \\ \Rightarrow &\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}\end{aligned}$$

We may assume that

$$\widehat{u}_n \xrightarrow{w} \overline{u} \text{ in } W_0^{1,p}(\Omega), \widehat{u}_n \to \overline{u} \text{ in } L^p(\Omega). \tag{5.226}$$

In (5.225) we choose $h = \widehat{u}_n - \overline{u} \in W_0^{1,p}(\Omega)$. Then

$$\langle V(\widehat{u}_n), \widehat{u}_n - \overline{u}\rangle = \int_\Omega \frac{\widehat{u}_n - \overline{u}}{(\widehat{u}_n + \varepsilon_n)^\eta} dz \tag{5.227}$$

$$\leq \int_\Omega \frac{\widehat{u}_n - \overline{u}}{\widehat{u}_1^\eta} dz \text{ (see the Claim).} \tag{5.228}$$

Since $\widehat{u}_1 \in intC_+$, we can find $c_4 > 0$ such that $\widehat{d} \leq c_4\widehat{u}_1$. Then using Proposition 5.91 (Hardy's inequality), we have

$$|\int_\Omega \frac{h}{\widehat{u}_1^\eta}dz| \leq c_5 \int_\Omega \frac{|h|}{\widehat{d}}dz \leq c_6 \parallel Dh \parallel_p \text{ for some } c_5, c_6 > 0, \text{ all } h \in W_0^{1,p}(\Omega).$$

Then (5.226) and (5.228), imply we can use Vitali's theorem (see Theorem I-2.147) and obtain

$$\int_\Omega \frac{\widehat{u}_n - \overline{u}}{\widehat{u}_1^\eta}dz \to 0 \text{ as } n \to \infty,$$

$$\Rightarrow \limsup_{n\to\infty} \langle V(\widehat{u}_n), \widehat{u}_n - \overline{u}\rangle \leq 0 \text{ (see (5.227))},$$

$$\Rightarrow \widehat{u}_n \to \overline{u} \text{ in } W_0^{1,p}(\Omega) \text{ (see Proposition 5.47)}.$$

Therefore if in (5.225) we pass to the limit as $n \to \infty$, we obtain

$$\langle V(\overline{u}), h\rangle = \int_\Omega \frac{h}{\overline{u}^\eta}dz \text{ for all } h \in W_0^{1,p}(\Omega).$$

Since $\widehat{u}_1 \leq \overline{u}$, we conclude that it is a positive solution of (5.217). Theorem 4.182 implies $\overline{u} \in intC_+$.

Finally, since $x \to x^{-\eta}$ is strictly decreasing on $\mathring{\mathbb{R}}_+ = (0, \infty)$, we see that the solution $\overline{u} \in intC_+$ is unique. □

Remark 5.93 Flattening $\partial\Omega$ and also using a partition of unity argument as in Lazer and McKenna [289] (proof of the Lemma, pp.726–727), we can show $\widehat{d}^{-\eta} \in L^s(\Omega)$ for all $1 \leq s < \frac{1}{\eta}$. Therefore we have $\overline{u}^{-\eta} \in L^s(\Omega)$ for all $1 \leq s < \frac{1}{\eta}$.

Next we consider the following auxiliary Dirichlet problem

$$\left\{ \begin{array}{l} -\Delta_p u(z) - \Delta_q u(z) = \overline{u}(z)^{-\eta} + 1 \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, 0 < \eta < 1, 1 < q < p. \end{array} \right\} \tag{5.229}$$

We show that this problem too has a unique solution.

Proposition 5.94 *If $1 < q < p$ and $0 < \eta < 1$, then problem (5.229) has a unique solution $\overline{u}_* \in intC_+$.*

Proof As we pointed out in Remark 5.93, we have

$$\overline{u}^{-\eta} + 1 \in L^s(\Omega) \text{ for all } 1 \leq s < \frac{1}{\eta}.$$

Since $\overline{u} \in intC_+$, we can find $c_5 > 0$ such that $\widehat{d} \leq c_5\overline{u}$. So, by Proposition 5.91, we have

$$\overline{u}^{-\eta} h \in L^p(\Omega), \| \frac{h}{\overline{u}^{\eta}} \|_p \leq \widehat{c} \| h \| \text{ for all } h \in W_0^{1,p}(\Omega).$$

Therefore $\overline{u}^{-\eta} \in W^{-1,p'}(\Omega) = W_0^{1,p}(\Omega)^*$. Since $V(\cdot)$ is surjective (see the proof of Proposition 5.92), we can find $\overline{u}_* \in W_0^{1,p}(\Omega)$ such that $V(\overline{u}_*) = \overline{u}^{-\eta} + 1$. On account of the strict monotonicity of $V(\cdot)$, this solution is unique. Finally the nonlinear regularity theory and the nonlinear maximum principal imply that $\overline{u}_* \in intC_+$. □

We have

$$A_p(\overline{u}) + A_q(\overline{u}) = \overline{u}^{-\eta} \leq \overline{u}^{-\eta} + 1 = A_p(\overline{u}_*) + A_q(\overline{u}_*) \text{ in } W^{-1,p'}(\Omega).$$

Acting with $(\overline{u} - \overline{u}_*)^+ \in W_0^{1,p}(\Omega)$ and using the monotonicity of $V = A_p + A_q$, we obtain $\overline{u} \leq \overline{u}_*$. Let $\xi \in L^\infty(\Omega) \setminus \{0\}, \xi(z) \geq 0$ for a.a $z \in \Omega$ and consider the Caratheodory function $\mu : \Omega \times \mathbb{R}_+ \to \mathbb{R}_+$ defined by

$$\mu(z,x) = \begin{cases} \xi(z)\overline{u}(z)^{-\eta} & \text{if } x < \overline{u}(z) \\ \xi(z)x^{-\eta} & \text{if } \overline{u}(z) \leq x \leq \overline{u}_*(z) \\ \xi(z)\overline{u}_*(z)^{-\eta} & \text{if } \overline{u}_*(z) < x. \end{cases} \tag{5.230}$$

We set $M(z,x) = \int_0^x \mu(z,s)ds$ and consider the functional $\beta : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\beta(u) = \int_\Omega M(z,u)dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

Lemma 5.95 *$\beta \in C^1(W_0^{1,p}(\Omega))$ and $\beta'(u) = N_\mu(u) = \mu(\cdot, u(\cdot))$ for all $u \in W_0^{1,p}(\Omega)$.*

Proof Let $u, h \in W_0^{1,p}(\Omega)$ and $t \in \mathbb{R} \setminus \{0\}$. From the integral form of the mean value theorem, we have

$$\frac{1}{t}[\beta(u+th) - \beta(u)] = \int_\Omega \left(\int_0^1 \mu(z, u+sth)ds\right)hdz. \tag{5.231}$$

We know that

$$\int_0^1 \mu(z, u+sth)ds \to \varrho(z,u) \text{ for a.a } z \in \Omega, \text{ as } t \to 0.$$

Also we have

$$\begin{aligned}
&\int_0^1 \mu(z, u + sth)ds \\
=&\xi(z)\overline{u}(z)^{-\eta}\chi_{\{u+sth<\overline{u}\}}(z) + \chi_{\{\overline{u}\leq u+sth\leq \overline{u}_*\}}\int_0^1 (u + sth)^{-\eta}ds \\
&+ \xi(z)\overline{u}_*(z)^{-\eta}\chi_{\{\overline{u}_*<u+sth\}}(z) \text{ (see (5.230))} \\
\leq&2\xi(z)\overline{u}(z)^{-\eta} + c(\max_{0\leq s\leq 1}|u + sth|^{-\eta})\chi_{\{\overline{u}\leq u+sth\leq \overline{u}_*\}}(z) \text{ for some } c > 0 \\
\leq&[2\xi(z) + c]\overline{u}(z)^{-\eta} \text{ for a.a } z \in \Omega.
\end{aligned} \tag{5.232}$$

Note that $[2\xi(\cdot) + c]\overline{u}(\cdot)^{-\eta} \in L^1(\Omega)$ (see Remark 5.93). On account of (5.231), (5.232), we can apply the Lebesgue dominated convergence theorem and obtain

$$\begin{aligned}
&\langle \beta'(u), h\rangle = \int_\Omega \mu(z, u)hdz \text{ for all } u, h \in W_0^{1,p}(\Omega), \\
\Rightarrow&\beta \in C^1(W_0^{1,p}(\Omega)), \beta'(u) = N_\mu(u).
\end{aligned}$$

□

Remark 5.96 This Lemma explains why the solutions $\overline{u}, \overline{u}^*$ of the two auxiliary problems, will help us bypass the singularity and have C^1-functions to work with.

We introduce the following two sets.

$$\mathcal{L} = \{\lambda > 0 : \text{problem } (p_\lambda^1) \text{ has a positive solution}\},$$

$$S_\lambda = \text{set of positive solutions.}$$

Proposition 5.97 *If hypotheses H_1 hold, then $\mathcal{L} \neq \emptyset$.*

Proof Let $\overline{u}_* \in intC_+$ be the unique solution of problem (5.229) (see Proposition 5.94). Since $f \geq 0$ (see hypothesis H_1(vi)), from hypothesis H_1(i), we have $f(\cdot, \overline{u}_*(\cdot)) \in L^\infty(\Omega)$. So, we can find $\lambda_0 > 0$ small such that

$$0 \leq \lambda_0 f(z, \overline{u}_*(z)) \leq 1 \text{ for a.a } z \in \Omega. \tag{5.233}$$

Recall that $\overline{u} \leq \overline{u}_*$. So, for $0 < \lambda \leq \lambda_0$, we can define the following truncation of the reaction of (p_λ^1)

$$g_\lambda(z,x)=\begin{cases}\overline{u}(z)^{-\eta}+\lambda f(z,\overline{u}(z)) & \text{if } x<\overline{u}(z)\\ x^{-\eta}+\lambda f(z,x) & \text{if } \overline{u}(z)\le x\le \overline{u}_*(z)\\ \overline{u}_*(z)^{-\eta}+\lambda f(z,\overline{u}_*(z)) & \text{if } \overline{u}_*(z)<x.\end{cases} \tag{5.234}$$

This is a Caratheodory function. We set $G_\lambda(z,x)=\int_0^x g_\lambda(z,s)ds$ and we introduce the functional $\psi_\lambda : W_0^{1,p}(\Omega)\to\mathbb{R}$ defined by

$$\psi_\lambda(u)=\frac{1}{p}\parallel Du\parallel_p^p+\frac{1}{q}\parallel Du\parallel_q^q-\int_\Omega G_\lambda(z,u)dz \text{ for all } u\in W_0^{1,p}(\Omega).$$

On account of Lemma 5.95, $\psi_\lambda\in C^1(W_0^{1,p}(\Omega))$. From (5.233) we see that $\psi_\lambda(\cdot)$ is coercive. Also, it is sequentially weakly lower semicontinuous. So, we can find $u_\lambda\in W_0^{1,p}(\Omega)$ such that

$$\begin{aligned}&\psi_\lambda(u_\lambda)=\min\{\psi_\lambda(u): u\in W_0^{1,p}(\Omega)\},\\ \Rightarrow&\langle\psi_\lambda'(u_\lambda),h\rangle=0 \text{ for all } h\in W_0^{1,p}(\Omega),\\ \Rightarrow&\langle V(u_\lambda),h\rangle=\int_\Omega g_\lambda(z,u_\lambda)hdz \text{ for all } h\in W_0^{1,p}(\Omega).\end{aligned}$$

First we use the test function $h=(\overline{u}-u_\lambda)^+\in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned}&\langle V(u_\lambda),(\overline{u}-u_\lambda)^+\rangle\\ =&\int_\Omega[\overline{u}^{-\eta}+\lambda f(z,\overline{u})](\overline{u}-u_\lambda)^+dz \text{ (see (5.234))}\\ \ge&\int_\Omega\overline{u}^{-\eta}(\overline{u}-u_\lambda)^+dz \text{ (since } f\ge 0)\\ =&\langle V(\overline{u}),(\overline{u}-u_\lambda)^+\rangle \text{ (see Proposition 5.92).}\\ \Rightarrow&\overline{u}\le u_\lambda.\end{aligned} \tag{5.235}$$

Next in (5.235) we choose the test function $h=(u_\lambda-\overline{u}_*)^+\in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned}&\langle V(u_\lambda),(\overline{u}-u_\lambda)^+\rangle\\ =&\int_\Omega[\overline{u}_*^{-\eta}+\lambda f(z,\overline{u}_*)](u_\lambda-\overline{u}_*)^+dz \text{ (see (5.234))}\\ \le&\int_\Omega[\overline{u}^{-\eta}+\lambda_0 f(z,\overline{u}_*)](u_\lambda-\overline{u}_*)^+dz \text{ (since } u\le\overline{u}_* \text{ and } 0<\lambda\le\lambda_0)\\ \le&\int_\Omega[\overline{u}^{-\eta}+1](u_\lambda-\overline{u}_*)^+dz \text{ (see (5.233))}\end{aligned}$$

$$= \langle V(\overline{u}_*), (u_\lambda - \overline{u}_*)^+ \rangle \text{ (see Proposition 5.94)},$$
$$\Rightarrow u_\lambda \leq \overline{u}_*.$$

So, we have proved that

$$u_\lambda \in [\overline{u}_1, \overline{u}_*] = \{h \in W_0^{1,p}(\Omega) : \overline{u}(z) \leq h(z) \leq \overline{u}_*(z) \text{ for a.a } z \in \Omega\}. \tag{5.236}$$

Then (5.236), (5.234) and (5.235) imply $u_\lambda \in S_\lambda$ and so

$$(0, \lambda_0] \subseteq \mathcal{L} \neq \emptyset.$$

□

Proposition 5.98 *If hypotheses H_1 hold and $\lambda \in \mathcal{L}$, then $\overline{u} \leq u$ for all $u \in S_\lambda$.*

Proof Let $u \in S_\lambda$ and introduce the Caratheodory function $k : \Omega \times (0, \infty) \to \mathbb{R}$ defined by

$$k(z, x) = \begin{cases} x^{-\eta} & \text{if } 0 < x < u(z) \\ u(z)^{-\eta} & \text{if } u(z) < x. \end{cases} \tag{5.237}$$

We consider the singular Dirichlet problem

$$-\Delta_p u - \Delta_q u = k(z, u) \text{ in } \Omega, u|_{\partial\Omega} = 0, u > 0. \tag{5.238}$$

Using regularizations of $k(z, \cdot)$ and a fixed point argument as in the proof of Proposition 5.92, we show that (5.238) has a unique positive solution $\widehat{u} \in intC_+$. We have

$$\langle V(\widehat{u}), h \rangle = \int_\Omega k(z, \widehat{u}) h dz \text{ for all } h \in W_0^{1,p}(\Omega). \tag{5.239}$$

Let $h = (\widehat{u} - u)^+ \in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned} \langle V(\widehat{u}), (\widehat{u} - u)^+ \rangle &= \int_\Omega u^{-\eta} (\widehat{u} - u)^+ dz \text{ (see (5.237))} \\ &\leq \int_\Omega [u^{-\eta} + \lambda f(z, u)](\widehat{u} - u)^+ dz \text{ (since } f \geq 0) \\ &= \langle V(u), (\widehat{u} - u)^+ \rangle \text{ (since } u \in S_\lambda) \\ \Rightarrow &= \widehat{u} \leq u. \end{aligned}$$

From (5.238) and (5.239) we infer that $\widehat{u}$ is a positive solution of problem (5.217). Therefore $\widehat{u} = \overline{u}$ (see Proposition 5.92). We conclude that $\overline{u} \leq u$ for all $u \in S_\lambda$ for all $u \in S_\lambda$. □

Then Proposition 5.98 and the regularity theory for singular problems (see Theorem 4.182), lead to the following property of the solution set S_λ.

Proposition 5.99 *If hypotheses H_1 hold and $\lambda \in \mathcal{L}$, then $S_\lambda \subseteq intC_+$.*

Let $\lambda^* = \sup \mathcal{L}$.

Proposition 5.100 *If hypotheses H_1 hold, then $\lambda^* < \infty$.*

Proof Hypotheses H_1(ii), (iii) imply that

$$\lim_{x \to +\infty} \frac{f(z,x)}{x^{p-1}} = +\infty \text{ uniformly for a.a } z \in \Omega.$$

So, we can find $M > 0$ such that

$$f(z,x) \geq x^{p-1} \text{ for a.a } z \in \Omega, \text{ all } x \geq M.$$

On the other hand, on account of hypotheses H_1(v), there exist $\delta \in (0,1)$ and $0 < \widehat{c}_1 < c_1$ such that

$$\begin{aligned} &f(z,x) \geq \widehat{c}_1 x^{\mu-1} \geq \widehat{c}_1 x^{p-1} \text{ for a.a } z \in \Omega, \text{ all } x \in [0,\delta] \\ \Rightarrow &\frac{1}{\widehat{c}_1} f(z,x) \geq x^{p-1} \text{ for a.a } z \in \Omega, \text{ all } x \in [0,\delta]. \end{aligned}$$

Finally hypotheses H_1(vi) implies that for $\widetilde{\lambda} > 0$ large, we have

$$\widetilde{\lambda} f(z,x) \geq M^{p-1} \text{ for a.a } z \in \Omega, \text{ all } x \in [\delta, M].$$

These observations, lead to the conclusion that for $\widetilde{\lambda} > 0$ large, we have

$$\widetilde{\lambda} f(z,x) \geq x^{p-1} \text{ for a.a } z \in \Omega, \text{ all } x \geq 0. \tag{5.240}$$

Take $\lambda > \widetilde{\lambda}$ and suppose $\lambda \in \mathcal{L}$. We can find $u_\lambda \in S_\lambda \subseteq intC_+$ (see Proposition 5.99). Consider $\Omega' \subseteq \Omega$ open with $\overline{\Omega}' \subseteq \Omega$ and C^2-boundary. Since $u_\lambda \in intC_+$, we have $m_\lambda = \min_{\overline{\Omega}'} u_\lambda > 0$. Let $\varrho = \| u_\lambda \|_\infty$ and let $\widehat{\xi}_\varrho > 0$ be as postulated by hypothesis H_1(vi). For $\vartheta > 0$, let $m_\lambda^\vartheta = m_\lambda + \vartheta$. On the set Ω' we have

$$
\begin{aligned}
&-\Delta_{\varrho} m_{\lambda}^{\vartheta} - \Delta_q m_{\lambda}^{\vartheta} + \lambda \widehat{\xi}_{\varrho} (m_{\lambda}^{\vartheta})^{p-1} - (m_{\lambda}^{\vartheta})^{-\eta} \\
&\leq \lambda \widehat{\xi}_{\varrho} m_{\lambda}^{p-1} + \chi(\vartheta) \text{ with } \chi(\vartheta) \to 0^+ \text{ as } \vartheta \to 0^+ \\
&\leq [\lambda \widehat{\xi}_{\varrho} + 1] m_{\lambda}^{p-1} + \chi(\vartheta) \\
&\leq \widetilde{\lambda} f(z, m_{\lambda}) + \lambda \widehat{\xi}_{\varrho} m_{\lambda}^{p-1} + \chi(\vartheta) \text{ (see (5.240))} \qquad (5.241) \\
&= \lambda [f(z, m_{\lambda}) + \widehat{\xi}_{\varrho} m_{\lambda}^{p-1}] - (\lambda - \lambda_0) f(z, m_{\lambda}) + \chi(\vartheta) \\
&\leq \lambda [f(z, m_{\lambda}) + \widehat{\xi}_{\varrho} m_{\lambda}^{p-1}] \text{ for } \widehat{u} > 0 \text{ small (see } H_1(\text{vi})) \\
&= -\Delta_{\varrho} u_{\lambda} - \Delta_q u_{\lambda} + \lambda \widehat{\xi}_{\varrho} u_{\lambda}^{p-1} - u_{\lambda}^{-\eta} \text{ in } \Omega' \text{ (since } u_{\lambda} \in S_{\lambda}).
\end{aligned}
$$

For $\vartheta > 0$ small we see that

$$
0 < c^* \leq (\lambda - \widetilde{\lambda}) f(z, m_{\lambda}) - \chi^{(\vartheta)} \text{ for a.a } z \in \Omega', \text{ some } c^* > 0.
$$

From (5.241) and Theorem 5.37(a) we infer that

$$
m_{\lambda}^{\vartheta} < u_{\lambda}(z) \text{ for all } z \in \Omega', \text{ all } \vartheta > 0 \text{ small},
$$

which contradicts the definition of m_{λ}. This means that $\lambda \notin \mathcal{L}$ and so $0 < \lambda^* \leq \widetilde{\lambda} < \infty$. □

The next proposition shows that $\mathcal{L}$ is connected, which in turn implies that

$$
(0, \lambda^*) \subseteq \mathcal{L} \subseteq (0, \lambda^*].
$$

Proposition 5.101 *If hypotheses H_1 hold, $\lambda \in \mathcal{L}$ and $0 < \mu < \lambda$, then $\mu \in \mathcal{L}$.*

Proof Since $\lambda \in \mathcal{L}$, we can find $u_{\lambda} \in S_{\lambda} \subseteq intC_+$. From Proposition 5.98 we know that $\overline{u} \leq u_{\lambda}$. So, we can define the following truncation of the reaction of problem (p_{μ}^1)

$$
e_{\mu}(z, x) = \begin{cases} \overline{u}(z)^{-\eta} + \mu f(z, \overline{u}(z)) & \text{if } x < \overline{u}(z) \\ x^{-\eta} + \mu f(z, x) & \text{if } \overline{u}(z) \leq x \leq u_{\lambda}(z). \\ u_{\lambda}(z)^{-\eta} + \mu f(z, u_{\lambda}(z)) & \text{if } u_{\lambda}(z) < x. \end{cases} \qquad (5.242)
$$

This is a Caratheodory function. We set $E_{\mu}(z, x) = \int_0^x e_{\mu}(z, s) ds$ and consider the function $\widehat{\varphi}_{\mu} : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$
\widehat{\varphi}_{\mu}(u) = \frac{1}{p} \| Du \|_p^p + \frac{1}{q} \| Du \|_q^q - \int_{\Omega} E_{\mu}(z, u) dz \text{ for all } u \in W_0^{1,p}(\Omega).
$$

Lemma 5.95 implies that $\widehat{\varphi}_\mu \in C^1(W_0^{1,p}(\Omega))$. The functional $\widehat{\varphi}_\mu(\cdot)$ is coercive (see (5.242)) and sequentially weakly lower semicontinuous. So, we can find $u_\mu \in W_0^{1,p}(\Omega)$ such that

$$\begin{aligned} &\widehat{\varphi}_\mu(u_\mu) = \inf\{\widehat{\varphi}_\mu(u) : u \in W_0^{1,p}(\Omega)\}, \\ &\Rightarrow \langle \widehat{\varphi}'_\mu(u_\mu), h\rangle = 0 \text{ for all } h \in W_0^{1,p}(\Omega), \\ &\Rightarrow \langle V(u_\mu), h\rangle = \int_\Omega e_\mu(z, u_\mu)h dz. \end{aligned} \tag{5.243}$$

As before using $h = (\overline{u} - u_\mu)^+ \in W_0^{1,p}(\Omega)$ and since $f \geq 0$, we obtain $\overline{u} \leq u_\mu$. Next, in (5.243) we choose the test function $(u_\mu - u_\lambda)^+ \in W_0^{1,p}(\Omega)$ and since $\mu < \lambda$, we obtain $u_\mu \leq u_\lambda$. Therefore

$$\begin{aligned} &u_\mu \in [\overline{u}, u_\lambda], \\ &\Rightarrow u_\mu \in S_\mu \subseteq intC_+ \text{ (see (5.242)) and so } \mu \in \mathcal{L}. \end{aligned}$$

□

Embedded in the above proof is the following corollary.

Corollary 5.102 *If hypotheses H_1 hold, $\lambda \in \mathcal{L}$, $u_\lambda \in S_\lambda \subseteq intC_+$ and $0 < \mu < \lambda$, then $\mu \in \mathcal{L}$ and there exists $u_\mu \in S_\mu \subseteq intC_+$ such that $u_\mu \leq u_\lambda$.*

Using Theorem 5.37 we can improve the conclusion of this corollary.

Proposition 5.103 *If hypotheses H_1 hold, $\lambda \in \mathcal{L}$, $u_\lambda \in S_\lambda \subseteq intC_+$ and $0 < \mu < \lambda$, then $\mu \in \mathcal{L}$ and there exists $u_\mu \in S_\mu \subseteq intC_+$ such that*

$$u_\lambda - u_\mu \in intC_+.$$

Proof From Corollary 5.102, we know that $\mu \in \mathcal{L}$ and there exists $u_\mu \in S_\mu \subseteq intC_+$ such that

$$u_\mu \leq u_\lambda. \tag{5.244}$$

Let $\varrho = \| u_\lambda \|_\infty$ and let $\widehat{\xi}_\varrho > 0$ be as postulated by hypothesis H_1(vi). Then we have

$$\begin{aligned} &- \Delta_\varrho u_\mu - \Delta_q u_\mu + \lambda\widehat{\xi}_\varrho u_\mu^{p-1} - u_\mu^{-\eta} \\ =&\mu f(z, u_\mu) + \lambda\widehat{\xi}_\varrho u_\mu^{p-1} \\ =&\lambda[f(z, u_\mu) + \widehat{\xi}_\varrho u_\mu^{p-1}] - (\lambda - \mu) f(z, u_\mu) \\ \leq&\lambda[f(z, u_\lambda) + \widehat{\xi}_\varrho u_\lambda^{p-1}] \text{ (see (5.244) and } H_1\text{(vi))} \\ =&- \Delta_p u_\lambda - \Delta_q u_\lambda + \lambda\widehat{\xi}_\varrho u_\lambda^{p-1} - u_\lambda^{-\eta} \text{ (since } u_\lambda \in S_\lambda). \end{aligned} \tag{5.245}$$

Since $u_\mu \in intC_+$, we have $0 \prec (\lambda - \mu) f(z, u_\mu)$ and so from (5.245) and Theorem 5.37(b), we conclude that $u_\lambda - u_\mu \in intC_+$. □

For $\lambda \in (0, \lambda^*)$ we have multiplicity of positive solutions.

Proposition 5.104 *If hypotheses H_1 hold and $0 < \lambda < \lambda^*$, then problem (p^1_λ) has at least two positive solutions*

$$u_0, \widehat{u} \in intC_+, u_0 \leq \widehat{u}, u_0 \neq \widehat{u}.$$

Proof Let $\lambda < \vartheta < \lambda^*$. From Proposition 5.101 we know that $\lambda_1, \vartheta \in \mathcal{L}$ and we can find $u_\vartheta \in S_\vartheta \subseteq intC_+$ and $u_0 \in S_\lambda \subseteq intC_+$ such that

$$u_\vartheta - u_0 \in intC_+. \tag{5.246}$$

We introduce the following truncation from below of the reaction of (p^1_λ)

$$\gamma_\lambda(z, x) = \begin{cases} u_0(z)^{-\eta} + \lambda f(z, u_0(z)) & \text{if } x < u_0(z) \\ x^{-\eta} + \lambda f(z, x) & \text{if } u_0(z) < x. \end{cases} \tag{5.247}$$

Also using (5.246), we introduce the following truncation of $\gamma_\lambda(z, \cdot)$

$$\widehat{\gamma}_\lambda(z, x) = \begin{cases} \gamma_\lambda(z, x) & \text{if } x < u_\vartheta(z) \\ \gamma_\lambda(z, u_\vartheta(z)) & \text{if } u_\vartheta(z) < x. \end{cases} \tag{5.248}$$

Both functions are Caratheodory functions. We set

$$\Gamma_\lambda(z, x) = \int_0^x \gamma_\lambda(z, s) ds \text{ and } \widehat{\Gamma}_\lambda(z, x) = \int_0^x \widehat{\gamma}_\lambda(z, s) ds.$$

and introduce the C^1-functionals $\sigma_\lambda, \widehat{\sigma}_\lambda : W^{1,p}_0(\Omega) \to \mathbb{R}$ defined by

$$\sigma_\lambda(u) = \frac{1}{p} \parallel Du \parallel^p_p + \frac{1}{q} \parallel Du \parallel^q_q - \int_\Omega \Gamma_\lambda(z, u) dz,$$

$$\widehat{\sigma}_\lambda(u) = \frac{1}{p} \parallel Du \parallel^p_p + \frac{1}{q} \parallel Du \parallel^q_q - \int_\Omega \widehat{\Gamma}_\lambda(z, u) dz \text{ for all } u \in W^{1,p}_0(\Omega).$$

From (5.247) and (5.248), we see that

$$\sigma_\lambda|_{[0,u_\vartheta]} = \widehat{\sigma}_\lambda|_{[0,u_\vartheta]}, \sigma'_\lambda|_{[0,u_\vartheta]} = \widehat{\sigma}'_\lambda|_{[0,u_\vartheta]}. \tag{5.249}$$

If $[u_0) = \{h \in W^{1,p}_0(\Omega) : u_0(z) \leq h(z) \text{ for a.a } z \in \Omega\}$, then using (5.247) and (5.248), we obtain

$$K_{\sigma_\lambda} \subseteq [u_0) \cap intC_+, \ K_{\widehat{\sigma}_\lambda} \subseteq [u_0, u_\vartheta] \cap intC_+. \tag{5.250}$$

From (5.250), we see that we may assume that

$$K_{\sigma_\lambda} \text{ is finite, } K_{\sigma_\lambda} \cap [u_0, u_\vartheta] = \{u_0\}. \tag{5.251}$$

or otherwise, we already have a second positive smooth solution bigger than u_0 and so we are done.

From (5.247) and (5.248), we see that $\widehat{\sigma}_\lambda(\cdot)$ is coercive. Also, it is sequentially weakly lower semicontinuous. So, we can find $\widetilde{u}_0 \in W_0^{1,p}(\Omega)$ such that

$$\widehat{\sigma}_\lambda(\widetilde{u}_0) = \inf\{\widehat{\sigma}_\lambda(u) : u \in W_0^{1,p}(\Omega)\}.$$

Since $\widetilde{u}_0 \in K_{\widehat{\sigma}_\lambda}$, from (5.251) we have $\widetilde{u}_0 = u_0 \in intC_+$. From (5.246) and (5.249), we have that

$$\begin{aligned} &u_0 \text{ is a } C_0^1(\overline{\Omega})\text{-local minimizer of } \sigma_\lambda, \\ \Rightarrow &u_0 \text{ is a } W_0^{1,p}(\Omega)\text{–local minimizer of } \sigma_\lambda. \end{aligned} \tag{5.252}$$

From (5.251) and Proposition 3.132, we know that we can find $\varrho \in (0, 1)$ small such that

$$\sigma_\lambda(u_0) < \inf\{\sigma_\lambda(u) : \| u - u_0 \| = \varrho\} = m_\lambda. \tag{5.253}$$

Note that hypothesis H_1(ii) (the superlinearity hypothesis), implies that for $u \in intC_+$, we have

$$\sigma_\lambda(tu) \to -\infty \text{ as } t \to +\infty. \tag{5.254}$$

Claim The functional $\sigma_\lambda(\cdot)$ satisfies the C-condition. Let $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega)$ be a sequence such that

$$|\sigma_\lambda(u_n)| \leq c_1 \text{ for some } c_1 > 0, \text{ all } n \in \mathbb{N}, \tag{5.255}$$

$$(1+ \| u_n \|)\sigma'_\lambda(u_n) \to 0 \text{ in } W^{-1,p'}(\Omega) \text{ as } n \to \infty. \tag{5.256}$$

From (5.256) we have

$$\begin{aligned} &|\langle V(u_n), h\rangle - \int_\Omega \gamma_\lambda(z, u_n)h dz| \leq \frac{\varepsilon_n \| h \|}{1+ \| u_n \|} \\ &\text{(for all } h \in W_0^{1,p}(\Omega) \text{ and with } \varepsilon_n \to 0^+). \end{aligned} \tag{5.257}$$

In (5.257) we choose $h = -u_n^- \in W_0^{1,p}(\Omega)$ and obtain that

$$\{u_n^-\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded (see (5.247)).} \tag{5.258}$$

Then from (5.255) and (5.258), we see that

$$\| Du_n^+ \|_p^p + \frac{p}{q} \| Du_n^+ \|_q^q - \int_\Omega \lambda_p F(z, u_n^+) dz \leq c_3[1 + \| u_n^+ \|_\tau] \tag{5.259}$$

(for some $c_3 > 0$, all $n \in \mathbb{N}$ (see (5.247))).

Moreover, if in (5.257) we use $h = u_n^+ \in W_0^{1,p}(\Omega)$, then

$$- \| Du_n^+ \|_p^p - \| Du_n^+ \|_q^q + \int_\Omega \lambda f(z, u_n^+) u_n^+ dz \leq c_4 \tag{5.260}$$

(for some $c_4 > 0$, all $n \in \mathbb{N}$ (see (5.247))).

We add (5.259) and (5.260) and obtain

$$\int_\Omega [f(z, u_n^+) u_n^+ - pF(z, u_n^+)] dz \leq c_5(1 + \| u_n^+ \|_\tau) \tag{5.261}$$

for some $c_5 = c_5(\lambda) > 0$, all $n \in \mathbb{N}$. Hypotheses H_1(i), (iii) imply that there exist $c_6, c_7 > 0$ such that

$$c_6 x^\tau - c_7 \leq f(z, x)x - pF(z, x) \text{ for a.a } z \in \Omega, \text{ all } x \geq 0. \tag{5.262}$$

Using (5.262) in (5.261), we infer that

$$\{u_n^+\}_{n \in \mathbb{N}} \subseteq L^\tau(\Omega) \text{ is bounded.} \tag{5.263}$$

First suppose $p \neq N$. From hypothesis H_1(iii) we see that we may assume that $\tau < r < p^*$. So, we can find $t \in (0, 1)$ such that

$$\frac{1}{r} = \frac{1-t}{\tau} + \frac{t}{p^*}. \tag{5.264}$$

Using the interpolation inequality (see Proposition I-2.123) we have

$$\| u_n^+ \|_r \leq \| u_n^+ \|_\tau^{1-t} \| u_n^+ \|_{p^*}^t,$$

hence

$$\| u_n^+ \|_r^r \leq c_8 \| u_n^+ \|^{tr}, \tag{5.265}$$

for some $c_8 > 0$ and all $n \in \mathbb{N}$

Here we have used (5.264) and the fact that $W_0^{1,p}(\Omega)$ is embedded continuously into $L^{p^*}(\Omega)$ (see Theorem I-4.115). In (5.257) we use the test function $h = u_n^+ \in W_0^{1,p}(\Omega)$ and have

$$\begin{aligned}
&\|u_n^+\|^p \leq \int_\Omega \gamma_\lambda(z, u_n^+)u_n^+ dz \\
\Rightarrow &\|u_n^+\|^p \leq c_9[1 + \|u_n^+\|_r^r] \text{ for some } c_9 > 0, \text{ all } n \in \mathbb{N} \\
&\qquad\qquad\qquad\qquad \text{(see hypothesis } H_1\text{(i))} \\
\Rightarrow &\|u_n^+\|^p \leq c_{10}[1 + \|u_n^+\|^{tr}] \text{ for some } c_{10} > 0, \text{ all } n \in \mathbb{N} \text{ (see (5.265))}.
\end{aligned} \tag{5.266}$$

If $N < p$, then $p^* = \infty$ and so from (5.264), we have $tr = r - \tau$, which by hypothesis H_1(iii) implies that $tr < p$. Hence

$$\{u_n^+\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded (see (5.266))}.$$

If $p < N$, then $p^* = \frac{Np}{N-p}$ and so from (5.264) we have

$$tr = \frac{(r-\tau)p^*}{p^* - \tau} = \frac{(r-\tau)Np}{N(p-\tau) + \tau p} < p \text{ (see hypothesis } H_1\text{(iii))}.$$

Therefore again we have

$$\{u_n^+\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded (see (5.266))}.$$

Finally we check the case $N = p$. In this case on account of Theorem I-4.115, we have that $W_0^{1,p}(\Omega)$ is embedded continuously into $L^s(\Omega)$ for all $1 \leq s < \infty$. Then in order to use above argument we replace p^* by $s > r > \tau$. We have

$$\begin{aligned}
&\frac{1}{r} = \frac{1-t}{\tau} + \frac{t}{s}, \\
\Rightarrow &tr = \frac{(r-\tau)s}{s-\tau} \to r - \tau < p \text{ as } s \to \infty \text{ (see hypothesis } H_1\text{(iii))}.
\end{aligned}$$

Hence we choose $s > r$ large so that $tr < p$ and then again from (5.266), we have

$$\{u_n^+\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.} \tag{5.267}$$

In all three cases, we have proved (5.267) and this combined with (5.258) gives

$$\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded}$$

We may assume that

$$u_n \xrightarrow{w} u \text{ in } W_0^{1,p}(\Omega),\ u_n \to u \text{ in } L^r(\Omega) \text{ (since } r < p^*). \tag{5.268}$$

In (5.257) we choose the test function $h = u_n - u \in W_0^{1,p}(\Omega)$, pass to the limit as $n \to \infty$ and use (5.268) . We obtain

$$\lim_{n\to\infty} \langle V(u_n), u_n - u\rangle = 0,$$

$$\Rightarrow u_n \to u \text{ in } W_0^{1,p}(\Omega) \text{ (see Proposition 5.50).}$$

This proves the Claim.

Then (5.253), (5.254) and the Claim, permit the use of the Mountain Pass Theorem (see Theorem 3.41) and we obtain $\widehat{u} \in W_0^{1,p}(\Omega)$ such that

$$\widehat{u} \in K_{\sigma_\lambda} \subseteq [u_0) \cap intC_+ \text{ (see (5.250)).} \tag{5.269}$$

$$\sigma_\lambda(u_0) < m_\lambda \leq \sigma_\lambda(\widehat{u}). \tag{5.270}$$

From (5.247) and (5.269), we see that $\widehat{u} \in S_\lambda \subseteq intC_+$ and $u_0 \leq \widehat{u}$. From (5.270) we have $\widehat{u} \neq u_0$. □

It remains to check the critical parameter λ^*.

Proposition 5.105 *If hypotheses H_1 hold, then $\lambda^* \in \mathcal{L}$.*

Proof Let $\{\lambda_n\}_{n\in\mathbb{N}} \subseteq (0, \lambda^*)$ such that $\lambda_n \uparrow \lambda^*$. Let $u_n \in S_{\lambda_n}$, $n \in \mathbb{N}$. We have

$$\overline{u} \leq u_n \text{ for all } n \in \mathbb{N} \text{ (see Proposition 5.98),} \tag{5.271}$$

$$\langle V(u_n), h\rangle = \int_\Omega [u_n^{-\eta} + \lambda_n f(z, u_n)]h dz \text{ for all } h \in W_0^{1,p}(\Omega), \text{ all } n \in \mathbb{N}. \tag{5.272}$$

Let $\varphi_\lambda : W_0^{1,p}(\Omega) \to \mathbb{R}$ be the energy functional for problem (p_λ^1) defined by

$$\varphi_\lambda(u) = \frac{1}{p} \parallel Du \parallel_p^p + \frac{1}{q} \parallel Du \parallel_q^q - \frac{1}{1-\eta} \int_\Omega (u^+)^{1-\eta} dz - \int_\Omega \lambda F(z, u^+) dz$$

for all $u \in W_0^{1,p}(\Omega)$.

For every $n \in \mathbb{N}$ let $\vartheta > \lambda_n$ and $u_\vartheta \in S_\vartheta \subseteq intC_+$. Truncating above the reaction at $u_\vartheta(z)$ and using the direct method of the calculus of variations, we can have

$$\varphi_\lambda(u_n) < 0 \text{ for all } n \in \mathbb{N}. \tag{5.273}$$

Using (5.272) and (5.273) and reasoning as in the "Claim" in the proof of Proposition 5.104, we obtain

$$u_n \to u_* \text{ in } W_0^{1,p}(\Omega). \tag{5.274}$$

From (5.271), (5.272), (5.274), we have

$$\overline{u} \leq u_*, \langle V(u_*), h\rangle = \int_\Omega [u_*^{-\eta} + \lambda^* f(z, u_*)]hdz \text{ for all } h \in W_0^{1,p}(\Omega),$$

$$\Rightarrow u_* \in S_{\lambda^*} \subseteq intC_+ \text{ and so } \lambda^* \in \mathcal{L}.$$

□

So, we have $\mathcal{L} = (0, \lambda^*]$ and we can state the following existence and multiplicity result for problem (p_λ^1) which is global in the parameter $\lambda > 0$.

Theorem 5.106 *If hypotheses H_1 hold, then there exist $\lambda^* > 0$ such that*

(a) for all $\lambda \in (0, \lambda^*)$ problem (p_λ^1) has at least two positive solutions $u_0, \widehat{u} \in intC_+$, $u_0 \leq \widehat{u}$, $u_0 \neq \widehat{u}$;
(b) for $\lambda = \lambda^*$ problem (p_λ^1) has at least one positive solution $u_* \in intC_+$;
(c) for all $\lambda > \lambda^*$ problem (p_λ^1) has no solution.

In the study of problem (p_λ^1), helpful was the hypothesis that $0 < \eta < 1$. Such problems are usually called "weakly singular". We have seen that for such problems if $u \in \text{int}C_+$ is a solution, we have $u^{-\eta} \in L^s(\Omega)$ for all $s \in [1, \frac{1}{\eta})$. Moreover, we can make effective use of Hardy's inequality (see Proposition 5.91). If $\eta \geq 1$, the situation is more complicated and then problem (p_λ^1) is said to be "strongly singular". We know (see Lazer and McKenna [289]) that

$$-\Delta u(z) = u(z)^{-\eta} \text{ in } \Omega, u|_{\partial\Omega} = 0,$$

has a solution in $H_0^1(\Omega)$ if and only if $\eta < 3$. Moreover, the solution is not in $C_0^1(\Omega)$ if $\eta > 1$.

So, it is interesting to study such singular problems. We consider the following Dirichlet problem

$$\left\{\begin{array}{l} -\Delta_p u(z) - \Delta_q u(z) = \lambda(u(z)^{-\eta} + f(z, u(z))) \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, 1 < q < p < N, 2 \leq p, \eta > 1, \lambda > 0, u > 0. \end{array}\right\} \tag{p_λ^2}$$

Our hypotheses on the data of (p_λ^2) are the following:

H_2 : $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that $f(z, 0) = 0$ for a.a $z \in \Omega$ and

(i) $0 \leq f(z, x) \leq a(z)[1 + x^{r-1}]$ for a.a $z \in \Omega$, all $x \geq 0$, with $a \in L^\infty(\Omega)$, $p < r < p^* = \frac{Np}{N-p}$;
(ii) if $F(z, x) = \int_0^x f(z, s)ds$, then $\lim\limits_{x \to +\infty} \frac{F(z,x)}{x^p} = +\infty$ uniformly for a.a $z \in \Omega$;
(iii) there exists $\tau \in ((r - p)\frac{N}{p}, p^*)$ such that

$$0 < \widehat{c}_0 \leq \liminf_{x \to +\infty} \frac{f(z,x)x - pF(z,x)}{x^\tau} \text{ uniformly for a.a } z \in \Omega;$$

(iv) $\lim\limits_{x \to 0^+} \frac{f(z,x)}{x^{q-1}} = 0$ uniformly for a.a $z \in \Omega$.

Remark 5.107 Again the hypotheses concern the positive semiaxis $\mathbb{R}_+ = [0, +\infty)$. So, since we search for positive solutions, we may assume that $f(z,x) = 0$ for a.a $z \in \Omega$, all $x \leq 0$. As for problem (p^1_λ), the perturbation of the singular term is $(p-1)$-superlinear without satisfying the (AR)-condition. For example the function

$$f(x) = \begin{cases} (x^+)^{\tau-1} & \text{if } x \leq 1 \\ x^{p-1} \ln x + x^{\vartheta-1} & \text{if } 1 < x. \end{cases}$$

with $q < \tau, \vartheta \leq p$, satisfies hypothesis H_2 without satisfying (AR).

To deal with the strong singularity, we need a compatibility condition between the exponents p and η.

$\mathbf{H_3}$: $1 < \eta < \dfrac{3p^* + 1}{2p^* + 1} = \dfrac{3N_p + N - p}{2N_p + N - p}$.

Remark 5.108 Note that $\frac{3N_p+N-p}{2N_p+N-p} < \frac{3}{2} < 2 - \frac{1}{p}$ (recall $p \geq 2$).

By a solution of (p^2_λ), we mean a function $u \in W^{1,p}_0(\Omega)$ such that $u \geq 0$,

$$u^{-\eta} h \in L^1(\Omega) \text{ for all } h \in W^{1,p}_0(\Omega)$$

$$\langle V(u), h \rangle = \int_\Omega \lambda [u^{-\eta} + f(z,u)] h dz \text{ for all } h \in W^{1,p}_0(\Omega) \text{ (recall } V = A_p + A_q)$$

As before, we start by considering the purely singular problem

$$\left\{ \begin{array}{l} -\Delta_p u(z) - \Delta_q u(z) = \lambda u(z)^{-\eta} \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, 1 < q < p < N, 2 \leq p, \eta > 0, \lambda > 0, u > 0. \end{array} \right\} \qquad (Q_\lambda)$$

We say that $u \in W^{1,p}_0(\Omega)$ is a (weakly) solution of (Q_λ), if $u \geq 0$, $u^{-\eta} h \in L^1(\Omega)$ for all $h \in W^{1,p}_0(\Omega)$, $\langle V(u), h \rangle = \int_\Omega \lambda u^{-\eta} h dz$ for all $h \in W^{1,p}_0(\Omega)$.

Proposition 5.109 *If $1 < \eta < 2 - \frac{1}{p}$ and $\lambda > 0$, then problem (Q_λ) has a unique solution $\overline{u} \in W^{1,p}_0(\Omega) \cap L^\infty(\Omega)$ and for every $K \subseteq \Omega$ compact we have*

$$0 < c_K \leq \overline{u}(z) \text{ for a.a } z \in K.$$

Proof First we consider the regularization of (Q_λ), namely, for every $\varepsilon > 0$, we consider the following Dirichlet problem

$$\{\begin{matrix} -\Delta_p u(z) - \Delta_q u(z) = \lambda[u+\varepsilon]^{-\eta} \text{ in } \Omega, \\ u|_{\partial\Omega} = 0, u > 0, \lambda > 0. \end{matrix}\} \qquad (Q_\lambda^\varepsilon)$$

As in the proof of Proposition 5.92, we show that Q_λ^ε has a unique positive solution $\overline{u}_\varepsilon \in intC_+$ and the map $\varepsilon \to \overline{u}_\varepsilon$ from $\overset{\circ}{\mathbb{R}}_+ = (0,\infty)$ into $C_0^1(\overline{\Omega})$ is nonincreasing.

Now we pass to the limit as $\varepsilon \to 0^+$ in order to produce a solution for (Q_λ). So, let $\varepsilon_n \to 0^+$ and let $\overline{u}_n = \overline{u}_{\varepsilon_n} \in \text{int}C_+$ be the unique solution of $(Q_\lambda^{\varepsilon_n})$. We know that

$$\{\overline{u}_n\}_{n\in\mathbb{N}} \subseteq intC_+ \text{ is nondecreasing.} \qquad (5.275)$$

For every $n \in \mathbb{N}$, we have

$$\langle V(\overline{u}_n), h\rangle = \int_\Omega \frac{\lambda h}{[\overline{u}_n + \varepsilon_n]^\eta} dz \text{ for all } h \in W_0^{1,p}(\Omega). \qquad (5.276)$$

We use the test function $h = \overline{u}_n \in W_0^{1,p}(\Omega)$ and obtain

$$\| \overline{u}_n \|^p \leq \int_\Omega \frac{\lambda}{\overline{u}_n^{\eta-1}} dz \leq \int_\Omega \frac{\lambda}{\overline{u}_1^{\eta-1}} dz \text{ (see (5.275))}. \qquad (5.277)$$

By hypothesis $\eta < 2$ and so $\eta - 1 < 1$, which implies that $\frac{1}{\overline{u}_1^{\eta-1}} \in L^1(\Omega)$. Therefore from (5.277) we have

$$\{\overline{u}_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}$$

So, we may assume that

$$\overline{u}_n \xrightarrow{w} \overline{u} \text{ in } W_0^{1,p}(\Omega), \overline{u}_n \to \overline{u} \text{ in } L^p(\Omega), \overline{u}_n(z) \to \overline{u}(z) \text{ for a.a } z \in \Omega. \qquad (5.278)$$

Note that

$$\begin{aligned} \left| \int_\Omega \frac{\lambda(\overline{u}_n - \overline{u})}{[\overline{u}_n + \varepsilon_n]} dz \right| &\leq \int_\Omega \frac{\lambda|\overline{u}_n - \overline{u}|}{\overline{u}_1^\eta} dz \text{ see (5.275)} \\ &\leq \int_\Omega \lambda \overline{u}_1^{1-\eta} \frac{|\overline{u}_n - \overline{u}|}{\overline{u}_1} dz. \end{aligned}$$

Since $\overline{u}_1 \in intC_+$ we can find $c_1 > 0$ such that $c_1 \widehat{d} \leq \overline{u}_1$. Then

$$\lambda \int_\Omega \overline{u}_1^{1-\eta} \frac{|\overline{u}_n - \overline{u}|}{\overline{u}_1} dz \leq \frac{\lambda}{c_1} \int_\Omega \overline{u}_1^{1-\eta} \frac{|\overline{u}_n - u|}{\widehat{d}} dz. \qquad (5.279)$$

From Proposition 5.91 (Hardy's inequality), we have

$$\frac{\overline{u}_n - u}{\widehat{d}} \in L^p(\Omega). \tag{5.280}$$

Since $\eta < 2 - \frac{1}{p}$, we have $(\eta - 1)p' < 1$ (recall $\frac{1}{p} + \frac{1}{p'} = 1$) and so

$$\overline{u}_1^{1-\eta} \in L^{p'}(\Omega). \tag{5.281}$$

We return to (5.279), using (5.280) and (5.281) and Hölder's inequality, we obtain

$$\lambda \int_\Omega \overline{u}_1^{1-\eta} \frac{|\overline{u}_n - \overline{u}|}{\overline{u}_1} dz \leq \frac{\lambda}{c_1} \| \overline{u}_1^{-1} \|_{(\eta-1)p'}^{\eta-1} \| \frac{\overline{u}_n - \overline{u}}{\widehat{d}} \|_p . \tag{5.282}$$

Note that

$$(\frac{\overline{u}_n - \overline{u}}{\widehat{d}})^p \leq (\frac{2\overline{u}}{\widehat{d}})^p \in L^1(\Omega) \text{ (see (5.275) and Proposition 5.91).} \tag{5.283}$$

Also (5.278) implies that

$$\frac{|(\overline{u}_n - \overline{u})(z)|}{\widehat{d}(z)} \to 0 \tag{5.284}$$

for a.a $z \in \Omega$, as $n \to \infty$. Therefore we can apply the Lebesgue dominated convergence theorem and infer that

$$\begin{aligned} &\| \frac{\overline{u}_n - \overline{u}}{\widehat{d}} \|_p \to 0 \text{ as } n \to \infty, \\ \Rightarrow &\lambda \int_\Omega \overline{u}_1^{1-\eta} \frac{\overline{u}_n - \overline{u}}{\overline{u}_1} dz \to 0 \text{ (see (5.282))}, \\ \Rightarrow &\int_\Omega \frac{\lambda(\overline{u}_n - \overline{u})}{[\overline{u}_n + \varepsilon_n]^\eta} dz \to 0 \text{ (see (5.283))}. \end{aligned}$$

In (5.276) we choose the test function $h = \overline{u}_n - \overline{u} \in W_0^{1,p}(\Omega)$, pass to the limit as $n \to \infty$ and use (5.284). We obtain $\lim_{n\to\infty} \langle V(\overline{u}_n), \overline{u}_n - \overline{u} \rangle = 0$, thus

$$\overline{u}_n \to \overline{u} \text{ in } W_0^{1,p}(\Omega), \ \overline{u}_1 \leq \overline{u}, \tag{5.285}$$

see Proposition 5.47).

Recall (see (5.276)), that

$$\langle V(\overline{u}_n), h\rangle = \int_\Omega \frac{\lambda h}{[\overline{u}_n + \varepsilon_n]} dz \text{ for all } h \in W_0^{1,p}(\Omega), \text{ all } n \in \mathbb{N}. \tag{5.286}$$

For every $n \in \mathbb{N}$, we have

$$\frac{|h|}{[\overline{u}_n + \varepsilon_n]^\eta} \leq \frac{|h|}{\overline{u}_1^\eta} \text{ for all } n \in \mathbb{N}$$

and as before

$$\frac{|h|}{\overline{u}_1^\eta} \in L^1(\Omega).$$

Since $\frac{h(z)}{(\overline{u}_n(z) + \varepsilon_n)^\eta} \to \frac{h(z)}{\overline{u}_1(z)^\eta}$ for a.a $z \in \Omega$, we can use the Lebesgue dominated convergence theorem, and obtain

$$\int_\Omega \frac{\lambda h}{(\overline{u}_n + \varepsilon_n)^\eta} dz \to \int_\Omega \frac{\lambda h}{\overline{u}^\eta} dz. \tag{5.287}$$

Passing to the limit as $n \to \infty$ in (5.288) and using (5.285) and (5.287), we have

$$\langle V(\overline{u}), h\rangle = \int_\Omega \frac{\lambda h}{\overline{u}^\eta} dz \text{ for all } h \in W_0^{1,p}(\Omega),$$

$$\overline{u}_1 \leq u \text{ and } \frac{h}{\overline{u}^\eta} \in L^1(\Omega) \text{ for all } h \in W_0^{1,p}(\Omega).$$

Therefore $u \in W_0^{1,p}(\Omega)$ is a positive solution of (Q_λ). The strict monotonicity of $V(\cdot)$ and since the map $x \to x^{-\eta}$ is strictly decreasing on $\overset{\circ}{\mathbb{R}}_+ = (0, \infty)$, we infer that $\overline{u}$ is unique.

Since $\overline{u}_1 \leq u$ and $\overline{u}_1 \in intC_+$, we see that for all $K \subseteq \Omega$ compact, we have

$$0 < c_K \leq \overline{u}(z) \text{ for a.a } z \in K.$$

Finally let $k > 1$ and set $\xi_k(t) = [t - k]^+$. This is a Lipschitz function. Hence $\xi_k(\overline{u}_n) \in W_0^{1,p}(\Omega)$ for all $n \in \mathbb{N}$ (see Theorem I-4.76). In (5.286) we use the test function $h = \xi_k(\overline{u}_n) \in W_0^{1,p}(\Omega)$. Then

$$\| D\xi_k(\overline{u}_n) \|_p^p \leq \int_\Omega \frac{\lambda \xi_k(\overline{u}_n)}{[\overline{u}_n + \varepsilon_n]^\eta} dz,$$

$$\Rightarrow \| D\xi_k(\overline{u}_n) \|_p^p \leq \lambda \int_\Omega \xi_k(\overline{u}_n) dz.$$

From this estimate, as in the Moser iteration technique (see the proof of Proposition 4.82), we conclude that $\overline{u} \in L^\infty(\Omega)$. □

Using $\overline{u} \in W_0^{1,p}(\Omega) \cap L^\infty(\Omega)$ from the above proposition, we introduce the Caratheodory function $\widehat{\mu}_\lambda(z,x)$ defined by

$$\widehat{\mu}_\lambda(z,x) = \begin{cases} \lambda \overline{u}(z)^{-\eta} & \text{if } x \le \overline{u}(z) \\ \lambda x^{-\eta} & \text{if } \overline{u}(z) \le x. \end{cases} \tag{5.288}$$

We set $\widehat{M}_\lambda(z,x) = \int_0^x \widehat{\mu}_\lambda(z,s)ds$ and consider the functional $\widehat{\beta}_\lambda : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{\beta}_\lambda(u) = \int_\Omega \widehat{M}_\lambda(z,u)dz \text{ for all } u \in W_0^{1,p}(\Omega). \tag{5.289}$$

Also we introduce the space $C_w^1(W_0^{1,p}(\Omega))$ which consists of all functions $\widehat{\beta} : W_0^{1,p}(\Omega) \to \mathbb{R}$ which are differentiable and the derivative $u \to \widehat{\beta}'(u)$ is continuous from $W_0^{1,p}(\Omega)$ with the norm topology into $W^{-1,p'}(\Omega) = W_0^{1,p}(\Omega)^*$ equipped with the weak topology.

Lemma 5.110 *If* $1 < \eta < 2 - \frac{1}{p}$ *and* $\lambda > 0$, *then* $\widehat{\beta}_\lambda \in C_w^1(W_0^{1,p}(\Omega))$ *and* $\widehat{\beta}_\lambda(u) = N_{\widehat{\mu}_\lambda}(u) = \widehat{\mu}_\lambda(\cdot, u(\cdot))$.

Proof Let $t \in \mathbb{R} \setminus \{0\}$ and $h \in C_c^\infty(\Omega)$. We have

$$\begin{aligned} &\frac{1}{t}[\widehat{\beta}_\lambda(u+th) - \widehat{\beta}_\lambda(u)] \\ =&\frac{1}{t}\int_\Omega [\widehat{M}_\lambda(z,u+th) - \widehat{M}_\lambda(z,u)]dz \\ =&\frac{1}{t}\int_\Omega [\int_0^1 \widehat{\mu}_\lambda(z,u+sth)ds]hdz. \end{aligned}$$

We have

$$\int_0^1 \widehat{\mu}_\lambda(z,u+sth)ds \to \widehat{\mu}_\lambda(z,u) \text{ as } t \to 0. \tag{5.290}$$

Also, we have

- on $\{u < \overline{u}\}$ for $|t| < 1$ small, we have

$$|\widehat{\mu}_\lambda(z,u+sth)| = \lambda \overline{u}^{-\eta} \le \lambda \overline{u}_1^{-\eta} \text{ (see (5.288) and recall } h \in C_c^\infty(\Omega));$$

- on $\{\overline{u} < u\}$ for $|t| < 1$ small we have

$$|\widehat{\mu}_\lambda(z,u+sth)| = \lambda(u+sth)^{-\eta} \le \lambda \overline{u}_1^{-\eta} \text{ (since } \overline{u}_1 \le u).$$

By continuity we have

$$|\widehat{\mu}_\lambda(z, u + sth)| \leq \lambda \overline{u}_1^{-\eta}.$$

Recall that $\lambda \overline{u}_1^{-\eta}|h| \in L^1(\Omega)$. Then using the Lebesgue dominated convergence theorem, we have

$$\begin{aligned}
&\lim_{t \to 0} \frac{1}{t}[\widehat{\beta}_\lambda(u + th) - \widehat{\beta}_\lambda(u)] \\
=&\lambda \int_\Omega (\max\{u, \overline{u}\}^{-\eta}) h dz \\
=&\int_\Omega \widehat{\mu}_\lambda(z, u) h dz \text{ for all } h \in C_c^\infty(\Omega) \\
\Rightarrow&\langle \widehat{\beta}'_\lambda(u), h\rangle = \int_\Omega \widehat{\mu}_\lambda(z, u) h dz \text{ for all } h \in C_c^\infty(\Omega).
\end{aligned}$$

The density of $C_c^\infty(\Omega)$ in $W_0^{1,p}(\Omega)$ implies that

$$\langle \widehat{\beta}'_\lambda(u), h\rangle = \int_\Omega \widehat{\mu}_\lambda(z, u) h dz \text{ for all } h \in W_0^{1,p}(\Omega).$$

Now let $u_n \to u$ in $W_0^{1,p}(\Omega)$. We have

$$|\langle \widehat{\beta}_\lambda(u_n) - \widehat{\beta}_\lambda(u), h\rangle| = |\int_\Omega [\widehat{\mu}_\lambda(z, u_n) - \widehat{\mu}_\lambda(z, u)] h dz|.$$

Observe that

$$|\widehat{\mu}_\lambda(z, u_n) - \widehat{\mu}_\lambda(z, u)| \leq \frac{2}{\overline{u}_1^\eta}$$

and at least for a subsequence, we have

$$\widehat{\mu}_\lambda(z, u_n(z)) \to \widehat{\mu}_\lambda(z, u(z)) \text{ for a.a } z \in \Omega.$$

Therefore by the Lebesgue dominated convergence theorem, we have

$$\begin{aligned}
&\langle \widehat{\beta}'_\lambda(u_n) - \widehat{\beta}'_\lambda(u), h\rangle \to 0 \text{ for all } h \in W_0^{1,p}(\Omega), \\
\Rightarrow&\widehat{\beta}_\lambda \in C_w^1(W_0^{1,p}(\Omega)).
\end{aligned}$$

□

Now we can have the multiplicity result for problem (p_λ^2).

Theorem 5.111 *If hypotheses H_2, H_3 hold, then for all $\lambda > 0$ small problem (p_λ^2) has at least two positive solutions $u_0, \widehat{u} \in W_0^{1,p}(\Omega) \cap L^\infty(\Omega)$, $u_0 \neq \widehat{u}$ and for all $K \subseteq \Omega$ compact*

$$0 < c_K \leq u_0(z), \widehat{u}(z) \text{ for a.a } z \in K.$$

Proof Let $\overline{u} \in W_0^{1,p}(\Omega)$ be the unique positive solution of problem (Q_λ) (see Proposition 5.109). We introduce the Caratheodory function $k(z, x)$ defined by

$$k(z,x) = \begin{cases} \overline{u}(z)^{-\eta} + f(z, \overline{u}(z)) & \text{if } x \leq \overline{u}(z) \\ x^{-\eta} + f(z,x) & \text{if } \overline{u}(z) < x. \end{cases} \tag{5.291}$$

We set $K(z,x) = \int_0^x k(z,s)ds$ and consider the functional $\vartheta_\lambda : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\vartheta_\lambda(u) = \frac{1}{p} \parallel Du \parallel_p^p + \frac{1}{q} \parallel Du \parallel_q^q - \int_\Omega \lambda K(z,u)dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

From Lemma 5.110, we have $\vartheta_\lambda \in C_w^1(W_0^{1,p}(\Omega))$. For every $u \in W_0^{1,p}(\Omega)$ we have

$$\begin{aligned} \int_\Omega K(z,u)dz &= \int_{\{u \leq \overline{u}\}} [\overline{u}^{-\eta} + f(z,\overline{u})]udz + \int_{\{\overline{u}<u\}} [\overline{u}^{-\eta} + f(z,\overline{u})]udz, \\ &+ \frac{1}{1-\eta} \int_{\{\overline{u}<u\}} [u^{1-\eta} - \overline{u}^{1-\eta}]dz + \int_{\{\overline{u}<u\}} [F(z,u) - F(z,\overline{u})]dz. \end{aligned} \tag{5.292}$$

We will estimate the terms in the right hand side of (5.292). So, we have

$$\begin{aligned} &| \int_{\{u \leq \overline{u}\}} \overline{u}^{-\eta} udz + \int_{\{\overline{u}<u\}} \overline{u}^{1-\eta} dz| \\ &\leq \int_\Omega \overline{u}^{-\eta} |u| dz \\ &\leq \int_\Omega \overline{u}^{1-\eta} \frac{|u|}{\overline{u}_1} dz \text{ (since } \overline{u}_1 \leq \overline{u}, \text{ see the proof of Proposition 5.109).} \end{aligned} \tag{5.293}$$

As before (see (5.283)) using hypothesis H_3, we have

$$| \int_\Omega \overline{u}^{1-\eta} \frac{u}{\overline{u}_1} dz| \leq \int_\Omega \overline{u}^{1-\eta} \frac{|u|}{\overline{u}_1} dz \leq c_1 \parallel Du \parallel_p \text{ for some } c_1 > 0. \tag{5.294}$$

Using (5.294) in (5.293), we obtain

$$| \int_{\{u \leq \overline{u}\}} \overline{u}^{-\eta} udz + \int_{\{\overline{u}<u\}} \overline{u}^{1-\eta} dz| \leq c_1 \parallel u \parallel \tag{5.295}$$

Hypothesis H_2(i) implies that

$$\begin{aligned} &|\int_{\{u\le\overline{u}\}} f(z,\overline{u})u dz + \int_{\{\overline{u}<u\}} f(z,\overline{u})\overline{u} dz| \\ &\le \int_\Omega f(z,\overline{u})|u| dz \le c_2 \parallel u \parallel \text{ for some } c_2 > 0. \end{aligned} \tag{5.296}$$

We estimate the third summand in the right hand side of (5.292). We have

$$\begin{aligned} |\frac{1}{1-\eta}\int_{\{\overline{u}<u\}} [u^{1-\eta} - \overline{u}^{1-\eta}] dz| &= \frac{1}{\eta-1}\int_{\{\overline{u}<u\}} [\frac{1}{\overline{u}^{\eta-1}} - \frac{1}{u^{\eta-1}}] \text{ (recall } \eta > 1) \\ &= \frac{1}{\eta-1}\int_{\{\overline{u}<u\}} \frac{u^{\eta-1} - \overline{u}^{\eta-1}}{(\overline{u}u)^{\eta-1}} dz \\ &\le \frac{1}{\eta-1}\int_{\{\overline{u}<u\}} (\frac{u}{\overline{u}_1^2})^{\eta-1} dz \text{ (recall } \overline{u}_1 \le \overline{u}). \end{aligned} \tag{5.297}$$

From hypothesis H_3 and Remark 5.108, we know that $\eta < \frac{3}{2} \Rightarrow 2(\eta-1) < 1$. Therefore, if $\tau \in (1, \frac{1}{2(\eta-1)})$, then since $\overline{u}_1 \in intC_1$, we have

$$\overline{u}_1^{-2(\eta-1)} \in L^\tau(\Omega). \tag{5.298}$$

We know that $\eta < \frac{3p^*+1}{2P^*+1}$ (see H_3). Hence $\frac{p^*}{p^*+1-\eta} < \frac{1}{2(\eta-1)}$ and so if we consider $\tau \in [\frac{p^*}{p^*+1-\eta}, \frac{1}{2(\eta-1)})$, we have

$$(\eta-1)\tau' = (\eta-1)\frac{\tau}{\tau-1} \le p^*.$$

So, by the Sobolev embedding theorem we have

$$u^{\eta-1} \in L^\tau(\Omega). \tag{5.299}$$

We return to (5.297) and use (5.298), (5.299) and Hölder's inequality. We obtain

$$\begin{aligned} &\frac{1}{1-\eta}\int_{\overline{u}<u} (\frac{u}{\overline{u}_1^2})^{\eta-1} dz \\ \le &\frac{1}{\eta-1}\int_\Omega (\frac{|u|}{u_1^2})^{\eta-1} dz \\ \le &[\int_\Omega |u|^{(\eta-1)\tau'} dz]^{\frac{1}{\tau'}} [\int_\Omega \overline{u}_1^{-2(\eta-1)\tau} dz]^{\frac{1}{\tau}} \end{aligned}$$

$$\leq c_3 \parallel u \parallel^{\eta-1}_{(\eta-1)\tau'} \text{ for some } c_3 > 0 \text{ (recall } \overline{u}_1 \in intC_+ \text{ and } 2(\eta-1)\tau < 1)$$

$$\leq c_4 \parallel u \parallel^{\eta-1} \text{ for some } c_4 > 0 \text{ (recall } (\eta-1)\tau' \leq p^*).$$

We have proved the following estimate

$$|\frac{1}{1-\eta}\int_{\{\overline{u}<u\}}[u^{1-\eta}-\overline{u}^{1-\eta}]dz| \leq c_4\|u\|^{\eta-1}. \tag{5.300}$$

Finally we will estimate the fourth summand in the right hand side of (5.292). Using hypotheses H_2(i),(iv), we see that given $\varepsilon > 0$, we can find $c_5 = c_5(\varepsilon) > 0$, such that

$$0 \leq F(z,x) \leq \frac{\varepsilon}{q}|x|^q + c_5|x|^r \text{ for } a.a\ z \in \Omega, \text{ all } x \in \mathbb{R}. \tag{5.301}$$

Then we have

$$\begin{aligned} 0 &\leq \int_{\{\overline{u}<u\}}[F(z,u)-F(z,\overline{u})]dz \\ &\leq \int_{\{\overline{u}<u\}}F(z,u)dz \text{ (since } F \geq 0) \\ &\leq \frac{\varepsilon}{q}\|u\|_q^q + c_6\|u\|^r \text{ for some } c_6 > 0 \text{ (see (5.301))} \\ &\leq \frac{\varepsilon}{\widehat{\lambda}_1(q)}\|Du\|_q^q + c_6\|u\|^r. \end{aligned} \tag{5.302}$$

Using (5.295), (5.296), (5.300) and (5.301), we have

$$\vartheta_\lambda(u) \geq \frac{1}{p}\|u\|^p + \frac{1}{q}[1-\frac{\varepsilon\lambda}{\widehat{\lambda}_1(q)}]\|Du\|_q^q - \lambda c_7[\|u\|^{\eta-1}+\|u\|+\|u\|^r]. \tag{5.303}$$

We know that $\eta - 1 < 1 < r$. So

$$\|u\| \leq \|u\|^{\eta-1} + \|u\|^r. \tag{5.304}$$

Choosing $\varepsilon \in (0, \frac{\widehat{\lambda}_1(q)}{\lambda})$ and using (5.304), from (5.303) we have

$$\vartheta_\lambda(u) \geq [\frac{1}{p} - 2\lambda c_7(\|u\|^{\eta-1-p} + \|u\|^{r-p})]\|u\|^p. \tag{5.305}$$

Consider the function $\mu(t) = t^{\eta-1-p} + t^{r-p}, t > 0$. Evidently $\mu \in C^1(0,\infty)$ and since $\eta - 1 < 1 < p < r$, we see that $\mu(t) \to +\infty$ as $t \to 0^+$ and as $t \to +\infty$. Therefore we can find $t_0 > 0$ such that

$$\begin{aligned}
&\mu(t_0) = \min\{\mu(t) : t > 0\},\\
&\Rightarrow \mu'(t_0) = 0,\\
&\Rightarrow t_0 = \left(\frac{p+1-\eta}{r-p}\right)^{\frac{1}{r+1-\eta}}.
\end{aligned}$$

If $u \in W_0^{1,p}(\Omega)$ with $\|u\| = t_0$, then from (5.303), we have

$$\vartheta_\lambda(u) \geq \left[\frac{1}{p} - 2\lambda c_7 \mu(t_0)\right] t_0^p.$$

We can find $\lambda^* > 0$ such that

$$\vartheta_\lambda(u) \geq c_\lambda > 0 \text{ for all } \|u\| = t_0, \lambda \in (0, \lambda^*). \tag{5.306}$$

Let $\overline{B}_0 = \{u \in W_0^{1,p}(\Omega) : \|u\| \leq t_0\}$ and consider the minimization problem

$$\inf\{\vartheta_\lambda(u) : u \in \overline{B}_0\} = m_\lambda.$$

By the Eberlein-Smulian theorem (see Theorem I-3.122), $\overline{B}_0$ is sequentially weakly compact. Also, using the Sobolev embedding theorem, we see that $\vartheta_\lambda(\cdot)$ is sequentially weakly lower semicontinuous. So, we can find $u_0 \in W_0^{1,p}(\Omega)$ such that

$$\vartheta_\lambda(u_0) = m_\lambda. \tag{5.307}$$

For $t > 0$, we have

$$\vartheta_\lambda(t\overline{u}) \leq \frac{t^p}{p}\|D\overline{u}\|_p^p + \frac{t^q}{q}\|Du\|_q^q - \lambda t \int_\Omega \overline{u}^{1-\eta} dz \text{ (since } f \geq 0). \tag{5.308}$$

We know that $\overline{u}^{1-\eta} \leq \overline{u}_1^{1-\eta}$ (recall that $\overline{u}_1 \leq u$ and $\eta > 1$). Since $\overline{u}_1 \in intC_+$, we have $\overline{u}^{1-\eta} \in L^1(\Omega)$. Therefore

$$\begin{aligned}
&\vartheta_\lambda(t\overline{u}) \leq c_8 t^q - \lambda c_9 t \text{ for some } c_8, c_9 > 0 \text{ (recall } t \in (0,1), q < p),\\
&\Rightarrow \vartheta_\lambda(t\overline{u}) < 0 \text{ for } t \in (0,1) \text{ small },\\
&\Rightarrow \vartheta_\lambda(u_0) = m_\lambda < 0 = \vartheta_\lambda(0) \text{ (see (5.307))},\\
&\Rightarrow u_0 \neq 0.
\end{aligned} \tag{5.309}$$

From (5.306) and (5.309), we see that

$$\begin{aligned} &0 < \|u_0\| < t_0, \\ &\Rightarrow \vartheta'_\lambda(u_0) = 0 \text{ in } W^{-1,p'}(\Omega) \text{ (see (5.307))}, \\ &\Rightarrow \langle V(u_0), h\rangle = \lambda \int_\Omega k(z,u_0)hdz \text{ for all } h \in W_0^{1,p}(\Omega). \end{aligned} \tag{5.310}$$

In (5.310) we choose the test function $h = [\overline{u} - u_0]^+ \in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned} \langle V(u_0), (\overline{u} - u_0)^+\rangle &= \int_\Omega \lambda[\overline{u}^{-\eta} + f(z,\overline{u})](\overline{u} - u_0)^+ dz \text{ (see (5.291))} \\ &\geq \int_\Omega \lambda \overline{u}^{-\eta}(\overline{u} - u_0)^+ dz \text{ (since } f \geq 0) \\ &= \langle V(\overline{u}), (\overline{u} - u_0)^+\rangle \text{ (see Proposition 5.109)}, \\ \Rightarrow \overline{u} &\leq u_0. \end{aligned}$$

Therefore $u_0 \in W_0^{1,p}(\Omega)$ is a positive solution for $(p_\lambda^2)(\lambda > 0$ small). We have $\overline{u}_1 \leq \overline{u} \leq u_0$ and so as before

$$u_0 \in W_0^{1,p}(\Omega) \cap L^\infty(\Omega),$$

$$0 < c_K \leq u_0(z) \text{ for } a.a\ z \in K,\ K \subseteq \Omega \text{ compact.}$$

On account of hypothesis H_2(ii), if $u \in intC_+$, then

$$\vartheta_\lambda(tu) \to -\infty \text{ as } t \to +\infty. \tag{5.311}$$

Reasoning as in the "Claim" in the proof of Proposition 5.104, we show that

$$\vartheta_\lambda(\cdot) \text{ satisfies the } C - \text{condition.} \tag{5.312}$$

Then (5.306), (5.311), (5.312) permit the use of the Mountain Pass Theorem (see Theorem 3.41). So, we can find $\widehat{u} \in W_0^{1,p}(\Omega)$,

$$\widehat{u} \in K_{\vartheta_\lambda} \subseteq [\overline{u}) \cap L^\infty(\Omega) \text{ (see (5.291))},\ 0 < c_\lambda \leq \vartheta_\lambda(\widehat{u}).$$

We infer that $\widehat{u} \in W_0^{1,p}(\Omega) \cap L^\infty(\Omega)$ is a positive solution of (p_λ^2) distinct from u_0. This completes the proof of the theorem. □

5.4 Problems with Convection

So far we have considered boundary value problems driven by nonlinear and in general nonhomogeneous differential operators, with a reaction that is independent of the gradient of the unknown function. Now we will consider problems with a gradient dependent reaction. This means that the problem under consideration is nonvariational. So, our approach is topological based on the fixed point theory.

Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$. We will study the following Dirichlet problem

$$\{-\Delta_p u(z) - \Delta_q u(z) = f(z, u(z), Du(z)) \text{ in } \Omega, u|_{\partial\Omega} = 0, u > 0, 1 < q < p.\} \tag{5.313}$$

We impose the following conditions on the reaction $f(z, x, y)$.

H: $f : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$ is a Caratheodory function such that $f(z, 0, y) = 0$ for $a.a\ z \in \Omega$, all $y \in \mathbb{R}^N$ and

(i) $|f(z, x, y)| \leq a(z)(1 + |x|^{p-1} + |y|^{p-1})$ for $a.a\ z \in \Omega$, all $x \in \mathbb{R}$, with $a \in L^\infty(\Omega), a(z) \geq 0$ for $a.a\ z \in \Omega$;

(ii) there exists $M > 0$ such that if $x_0 > M$, we can find $\delta > 0$ such that $f(z, x, y) \leq 0$ for $a.a\ z \in \Omega$, all $|x - x_0| + |y| \leq \delta$;

(iii) $\limsup\limits_{x \to +\infty} \frac{f(z,x,y)}{x^{p-1}} \leq 0$ uniformly for $a.a\ z \in \Omega$, all $y \in \mathbb{R}^N$ in a bounded set;

(iv) for every $\mu > 0$, there exists $\eta_\mu \in L^\infty(\Omega)$ such that

$$\widehat{\lambda}_1(q) \leq \eta_\mu(z) \text{ for } a.a\ z \in \Omega, \widehat{\lambda}_1(q) \not\equiv \eta_\mu,$$

$$\eta_\mu(z) \leq \liminf_{x \to 0^+} \frac{f(z, x, y)}{x^{q-1}} \text{ uniformly for } a.a\ z \in \Omega, \text{ all } |y| \leq \eta;$$

(v) for every $t \in (0, 1)$, we have

$$f(z, \frac{1}{t}x, y) \leq \frac{1}{t^{p-1}} f(z, x, y) \text{ for } a.a\ z \in \Omega, \text{ all } x \geq 0, \text{ all } y \in \mathbb{R}^N.$$

Remark 5.112 Hypothesis H(ii) is a Nagumo-Hartman type condition, used usually in the study of systems of ordinary differential equations (see Hartman [231], [232]). Hypothesis H(iv) is a nonuniform, nonresonance condition at zero with respect to the principal eigenvalue of $(-\Delta_q, W_0^{1,q}(\Omega))$ from the right. Hypothesis H(v) is satisfied, if for example, for $a.a\ z \in \Omega$, all $y \in \mathbb{R}^N$ $x \to \frac{f(z,x,y)}{x^{p-1}}$ is nonincreasing on $(1, \infty)$. The following function (for simplicity we drop the z-dependence) $f(x, y) = x^{\tau-1} - x|y|^{p-2}$ with $1 < \tau < 2 < q < p$ satisfies hypothesis H.

As we already mentioned, our approach is eventually topological and uses the Leray-Schauder Alternative Principle (see Theorem 1.67). To reach that point, we

employ the so-called "frozen variable method". More precisely, we fix $y \in \mathbb{R}^N$ in the reaction $f(z, x, y)$. This way, we have a variational problem, which can be treated using the critical point theory. Then we face the task of choosing in a canonical way a solution from the solution set of the "frozen problem". This is done by choosing the minimal solution, which we show that it exists. This way we have a well-defined map, which we show that it satisfies the requirements of Theorem 1.67. We generate a fixed point for the minimal solution map and this is the solution of problem (5.313).

First we introduce and study the "frozen problem". Let $v \in C_0^1(\overline{\Omega})$ and consider the Caratheodory function $g_v : \Omega \times \mathbb{R} \to \mathbb{R}$ defined by

$$g_v(z, x) = f(z, x, Dv(z)).$$

Let $\mu = \|v\|_{C_0^1(\overline{\Omega})}$. Then using hypothesis H(iv), we see that given $\varepsilon > 0$ and $r \in (p, p^*)$, we can find $c_\varepsilon = c_\varepsilon(r) > 0$ such that

$$g_v(z, x) \geq [\eta_\mu(z) - \varepsilon]x^{p-1} - c_\varepsilon x^{r-1} \text{ for } a.a\ z \in \Omega, \text{ all } x \geq 0. \tag{5.314}$$

The unilateral growth condition (5.314) leads to the following auxiliary Dirichlet problem

$$-\Delta_p u(z) - \Delta_q u(z) = (\eta_\mu(z) - \varepsilon)u(z)^{q-1} - c_\varepsilon u(z)^{r-1} \tag{5.315}$$

in Ω, $u|_{\partial\Omega} = 0$, $1 < q < p$, $u > 0$. Reasoning as in the proof of Proposition 5.54, we have the following result concerning problem (5.315).

Proposition 5.113 *For every $\varepsilon > 0$ small problem (5.315) has a unique solution $\overline{u}_v \in intC_+$.*

This solution will serve as a lower bound for the solutions of the frozen problem, which we introduce next.

$$-\Delta_p u(z) - \Delta_q u(z) = g_v(z, u(z)) \tag{5.316}$$

in Ω, $u|_{\partial\Omega} = 0$, $1 < q < p$, $u > 0$.

Let S_v denote the set of positive solutions of (5.316).

Proposition 5.114 *If hypotheses H hold, then $\emptyset \neq S_v \subseteq intC_+$ and $\overline{u}_v \leq u$ for all $u \in S_v$.*

Proof We set $G_v(z, x) = \int_0^x g_v(z, s)ds$ and consider the C^1-functional $\widehat{\varphi}_v : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{\varphi}_v(u) = \frac{1}{p}\|Du\|_p^p + \frac{1}{q}\|Du\|_q^q - \int_\Omega G_v(z, u^+)dz \text{ for all } u \in W_0^{1,p}(\Omega).$$

Hypotheses H(i),(iii) imply that given $\varepsilon > 0$, we can find $c_1 = c_1(\varepsilon) > 0$ such that

$$f(z, x, Dv(z)) \leq \varepsilon x^{p-1} + c_1 \text{ for } a.a\ z \in \Omega, \text{ all } x \geq 0,$$
$$\Rightarrow G_v(z, x) \leq \frac{\varepsilon}{p} x^p + c_1 x \text{ for } a.a\ z \in \Omega, \text{ all } x \geq 0.$$

Then for any $u \in W_0^{1,p}(\Omega)$, we have

$$\begin{aligned}\widehat{\varphi}_v(u) &\geq \frac{1}{p}[\|Du\|_p^p - \varepsilon\|u\|_p^p] - c_2\|u\| \text{ for some } c_2 > 0\\ &\geq \frac{1}{p}[1 - \frac{\varepsilon}{\widehat{\lambda}_1(p)}]\|u\|^p - c_2\|u\|.\end{aligned}$$

Choosing $\varepsilon \in (0, \widehat{\lambda}_1(p))$, we see that $\widehat{\varphi}_v(\cdot)$ is coercive (recall $p > 1$). Also by the Sobolev embedding theorem, $\widehat{\varphi}_v(\cdot)$ is sequentially weakly lower semicontinuous.Therefore we can find $\widehat{u} \in W_0^{1,p}(\Omega)$ such that

$$\widehat{\varphi}_v(\widehat{u}) = \inf\{\widehat{\varphi}_v(u) : u \in W_0^{1,p}(\Omega)\}. \tag{5.317}$$

From hypothesis H(iv) we see that given $\varepsilon > 0$, we can find $\delta = \delta(\varepsilon) > 0$ such that

$$G_v(z, x) \geq \frac{1}{q}[\eta_v(z) - \varepsilon]x^q \text{ for } a.a\ z \in \Omega, \text{ all } 0 \leq x \leq \delta. \tag{5.318}$$

Recall that $\widehat{u}_1(q)$ denotes the positive, L^q-normalized (that is, $\|\widehat{u}_1(q)\|_q = 1$) eigenfunction corresponding to $\widehat{\lambda}_1(q) > 0$. We know that $\widehat{u}_1(q) \in intC_+$. So, we can find $t \in (0, 1)$ small such that

$$t\widehat{u}_1(q)(z) \in [0, \delta] \text{ for all } z \in \overline{\Omega}. \tag{5.319}$$

Using (5.318), (5.319), we have

$$\widehat{\varphi}_v(t\widehat{u}_1(q))) \leq \frac{t^p}{p}\|D\widehat{u}_1(q)\|_p^p + \frac{t^q}{q}[\int_\Omega (\widehat{\lambda}_1(q) - \eta_v(z))\widehat{u}_1(q)^q dz + \varepsilon]$$
$$(\text{since } \|\widehat{u}_1(q)\|_q = 1).$$

Note that $\beta = \int_\Omega (\eta_v(z) - \widehat{\lambda}_1(q))\widehat{u}_1(q)^q dz > 0$. So, choosing $\varepsilon \in (0, \beta)$, we have

$$\varphi(t\widehat{u}_1(q)) \leq c_3 t^p - c_4 t^q \text{ for some } c_3, c_4 > 0,$$

Since $q < p$, choosing $t \in (0,1)$ even smaller if necessary, we obtain that

$$\begin{aligned}
&\widehat{\varphi}_v(t\widehat{u}_1(q)) < 0,\\
&\Rightarrow \widehat{\varphi}_v(\widehat{u}) < 0 \leq \widehat{\varphi}_v(0) \text{ (see (5.317))}\\
&\Rightarrow \widehat{u} \neq 0.
\end{aligned}$$

From (5.317) we have $\langle \widehat{\varphi}'_v(\widehat{u}), h\rangle = 0$ for all $h \in W_0^{1,p}(\Omega)$. Hence

$$\langle V(\widehat{u}), h\rangle = \int_\Omega g_v(z,\widehat{u})hdz \tag{5.320}$$

for all $h \in W_0^{1,p}(\Omega)$.

Choosing $h = -\widehat{u}^- \in W_0^{1,p}(\Omega)$ in (5.320), we obtain

$$\begin{aligned}
\|D\widehat{u}^-\|_p^p &= -\int_\Omega f(z,\widehat{u}^+, Dv)\widehat{u}^- dz\\
&= -\int_\Omega f(z,0,Dv)\widehat{u}^- dz \leq 0 \text{ (see hypotheses H)}\\
\Rightarrow \widehat{u} \geq 0, \widehat{u} &\neq 0.
\end{aligned}$$

From (5.320) it follows that $\widehat{u} \in S_v \neq \emptyset$.

If $u \in S_v$, then the nonlinear regularity theory and the nonlinear maximum principle, imply that

$$\begin{aligned}
&u \in intC_+,\\
&\Rightarrow \emptyset \neq S_v \subseteq intC_+.
\end{aligned}$$

Let $u \in S_v \subseteq intC_+$ and consider the Caratheodory function $k_u : \Omega \times \mathbb{R} \to \mathbb{R}$ defined by

$$k_u(z,x) = \begin{cases} [\eta_v(z) - \varepsilon](x^+)^{q-1} - c_\varepsilon(x^+)^{r-1} & if\ x \leq u(z)\\ [\eta_v(z) - \varepsilon]u(z)^{q-1} - c_\varepsilon u(z)^{r-1} & if\ u(z) < x \end{cases} \tag{5.321}$$

with $\varepsilon > 0$ small. We set $K_u(z,x) = \int_0^x k_u(z,s)ds$ and consider the C^1-functional $\psi_u : W_0^{1,p}(\Omega) \to \mathbb{R}$ defined by

$$\psi_u(y) = \frac{1}{p}\|Dy\|_p^p + \frac{1}{q}\|Dy\|_q^q - \int_\Omega K_u(z,y)dz \text{ for all } y \in W_0^{1,p}(\Omega).$$

This functional is coercive and sequentially weakly lower semicontinuous. So, we can find $\overline{y} \in W_0^{1,p}(\Omega)$ such that

$$\psi_u(\overline{y}) = \inf\{\psi_u(y) : y \in W_0^{1,p}(\Omega)\}. \tag{5.322}$$

As before since $\eta_v(z) \geq \widehat{\lambda}_1(q)$ for $a.a\ z \in \Omega$, $\eta_v \not\equiv \widehat{\lambda}_1(q)$ and $q < p < r$, we see that for $t \in (0, 1)$ small

$$\begin{aligned} &\psi_u(t\widehat{u}_1(q)) < 0, \\ &\Rightarrow \psi_u(\overline{y}) < 0 = \psi_u(0) \text{ and so } \overline{y} \neq 0. \end{aligned}$$

From (5.322), we have $\langle \psi_u'(\overline{y}), h\rangle = 0$ for all $h \in W_0^{1,p}(\Omega)$, thus

$$\langle V(\overline{y}), h\rangle = \int_\Omega k_u(z, \overline{y})h dz \tag{5.323}$$

for all $h \in W_0^{1,p}(\Omega)$, recall that $V = A_p + A_q$). In (5.323) first let $h = -\overline{y}^- \in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned} &\|D\overline{y}\|_p^p \leq \int_\Omega k_u(z, \overline{y})(-\overline{y}^-)dz = 0 \\ &\Rightarrow \overline{y} \geq 0, \overline{y} \neq 0. \end{aligned}$$

Next in (5.323) we choose $h = (\overline{y} - u)^+ \in W_0^{1,p}(\Omega)$. Then

$$\begin{aligned} \langle V(\overline{y}), (\overline{y} - u)^+\rangle &= \int_\Omega [(\eta_\mu(z) - \varepsilon)u^{q-1} - c_\varepsilon u^{r-1}](\overline{y} - u)^+ dz \text{ (see (5.321))} \\ &\leq \int_\Omega g_v(z, u)(\overline{y} - u)^+ dz \text{ (see (5.316))} \\ &= \langle V(u), (\overline{y} - u)^+\rangle \text{ (since } u \in S_v) \\ \Rightarrow \overline{y} &\leq u. \end{aligned}$$

We have proved that

$$\overline{y} \in [0, u], \overline{y} \neq 0. \tag{5.324}$$

From (5.324), (5.321), (5.323) and Proposition 5.113, we infer that

$$\begin{aligned} &\overline{y} = \overline{u}_v \in intC_+, \\ &\Rightarrow \overline{u}_v \leq u \text{ for all } u \in S_v. \end{aligned}$$

□

As we said earlier, the canonical way of choosing an element of S_v, will be achieved by choosing the smallest element of S_v (the "minimal solution"). Arguing as in the proof of Proposition 5.56, we show the existence of a minimal solution.

Proposition 5.115 *If hypotheses H hold, then there exists $u_v^* \in S_v$ such that $u_v^* \leq u$ for all $u \in S_v$.*

So, we can define the minimal solution map $\vartheta : C_0^1(\overline{\Omega}) \to C_0^1(\overline{\Omega})$ by

$$\vartheta(v) = u_v^* \in intC_+ \text{ for all } v \in C_0^1(\overline{\Omega}).$$

We want to apply Theorem 1.67 on this map. For this reason, we establish the properties of this map. Eventually we need to show that $\vartheta(\cdot)$ is compact. The next proposition will be helpful in this direction.

Proposition 5.116 *If hypotheses H hold, $u \in S_v$ and $v_n \to v$ in $C_0^1(\overline{\Omega})$, then we can find $u_n \in S_{v_n}$ for all $n \in \mathbb{N}$ such that*

$$u_n \to u \text{ in } S_v.$$

Proof For every $n \in \mathbb{N}$, we consider the following Dirichlet problem

$$\{-\Delta_p y(z) - \Delta_q y(z) = g_{v_n}(z, u(z)) \text{ in } \Omega,\ y|_{\partial\Omega} = 0.\} \tag{5.325}$$

Since $g_{v_n}(\cdot, u(\cdot)) \to g_v(\cdot, u(\cdot)) \neq 0$ in $L^{p'}(\Omega)$, we may assume that $g_{v_n}(\cdot, u(\cdot)) \not\equiv 0$ for all $n \in \mathbb{N}$. Then the maximal monotonicity, strict monotonicity, coercivity of $V = A_p + A_q$ and the fact that $g_{v_n}(\cdot, u(\cdot)) \in L^\infty(\Omega)$ (see hypothesis H(i)), imply that problem (5.325) has a unique solution $y_n^0 \in C_0^1(\overline{\Omega})$. Note that

$$\{g_{v_n}(\cdot, u(\cdot))\}_{n\in\mathbb{N}} \subseteq L^\infty(\Omega) \text{ is bounded },$$
$$\Rightarrow \{y_n^0\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded }.$$

Invoking Theorem 4.84, we can find $\alpha \in (0, 1)$ and $c > 0$ such that

$$y_n^0 \in C_0^{1,\alpha}(\overline{\Omega}),\ \|y_n^0\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq c \text{ for all } n \in \mathbb{N}.$$

Then the compact embedding of $C_0^{1,\alpha}(\overline{\Omega})$ into $C_0^1(\overline{\Omega})$ implies that at least for a subsequence, we have

$$y_n^0 \to \widetilde{u} \text{ in } C_0^1(\overline{\Omega}).$$

Evidently, $\widetilde{u}$ is the unique solution of

$$-\Delta_p y(z) - \Delta_q y(z) = g_v(z, u(z)) \text{ in } \Omega,\ y|_{\partial\Omega} = 0,$$
$$\Rightarrow \widetilde{u} = u.$$

So, for the original sequence we have

$$y_n^0 \to u \text{ in } C_0^1(\overline{\Omega}) \text{ as } n \to \infty.$$

Next we consider the following Dirichlet problem

$$-\Delta_p y(z) - \Delta_q y(z) = g_{v_n}(z, y_n^0(z)) \text{ in } \Omega,\ y|_{\partial\Omega} = 0.$$

This problem too has a unique solution $y_n^1 \in C_0^1(\overline{\Omega})$ and as above, we have

$$y_n^1 \to u \text{ in } C_0^1(\overline{\Omega}) \text{ as } n \to \infty.$$

This way we generate a sequence $\{y_n^k\}_{n,k\in\mathbb{N}} \subseteq C_0^1(\overline{\Omega})$ such that

$$-\Delta_p y_n^k(z) - \Delta_q y_n^k(z) = g_{v_n}(z, y_n^{k-1}(z)) \text{ in } \Omega,\ y_n^k|_{\partial\Omega} = 0, \tag{5.326}$$

for all $n, k \in \mathbb{N}$ and

$$y_n^k \to u \text{ in } C_0^1(\overline{\Omega}) \text{ as } n \to \infty, \text{ for all } k \in \mathbb{N}. \tag{5.327}$$

Fix $n \in \mathbb{N}$ and consider the sequence $\{y_n^k\}_{k\in\mathbb{N}} \subseteq C_0^1(\overline{\Omega})$. We will show that this sequence is bounded in $W_0^{1,p}(\Omega)$. Arguing by contradiction, suppose that

$$\|y_n^k\| \to \infty \text{ as } k \to \infty, \tag{5.328}$$

$\{\|y_n^k\|\}_{k\in\mathbb{N}} \subseteq \mathbb{R}_+^0 = (0, \infty)$ is increasing.

We set $w_k = \frac{y_n^k}{\|y_n^k\|}$ $k \in \mathbb{N}$ and have $\|w_k\| = 1$ for all $k \in \mathbb{N}$. So, we may assume that

$$w_k \xrightarrow{w} w \text{ in } W_0^{1,p}(\Omega),\ w_k \to w \text{ in } L^p(\Omega) \text{ as } k \to \infty. \tag{5.329}$$

From (5.326) we have

$$A_p(w_k) + \frac{1}{\|y_n^k\|^{p-q}} A_q(w_k) = \frac{N_{g_{v_n}}(y_n^{k-1})}{\|y_n^k\|^{p-1}} \tag{5.330}$$

with $N_{g_{v_n}}(y_n^{k-1})(\cdot) = g_{v_n}(\cdot, y_n^{k-1}(\cdot))$ (the Nemytskii map corresponding to $g_{v_n}(\cdot,\cdot)$). We have

$$\frac{|N_{g_{v_n}}(y_n^{k-1})|}{\|y_n^k\|^{p-1}} = \frac{|N_{g_{v_n}}(y_n^{k-1})|}{\|y_n^{k-1}\|^{p-1}} \frac{\|y_n^{k-1}\|^{p-1}}{\|y_n^k\|^{p-1}}$$

$$\leq \frac{|N_{g_{v_n}}(y_n^{k-1})|}{\|y_n^{k-1}\|^{p-1}} \text{ (see (5.328))}$$

$$\Rightarrow \{\frac{N_{g_{v_n}}(y_n^{k-1})}{\|y_n^k\|^{p-1}}\}_{n\in\mathbb{N}} \subseteq L^{p'}(\Omega) \text{ is bounded (see hypothesis H(i)).}$$

So, we may assume that

$$\frac{N_{g_{v_n}}(y_n^{k-1})}{\|y_n^k\|^{p-1}} \xrightarrow{w} \vartheta(z)|w|^{p-2}w \text{ in } L^{p'}(\Omega), \text{ as } k \to \infty. \tag{5.331}$$

with $\vartheta \in L^\infty(\Omega)$, $\vartheta(z) \leq 0$ for $a.a\ z \in \Omega$ (see hypothesis H(iii)). On (5.330), we act with $w_k - w \in W_0^{1,p}(\Omega)$, pass to the limit as $k \to \infty$ and use (5.328), (5.331) and the fact that $q < p$. We obtain $\lim_{k\to\infty}\langle A_p(w_k), w_k - w\rangle = 0$, thus

$$w_k \to w \text{ in } W_0^{1,p}(\Omega) \text{ and so } \|w\| = 1 \text{ (see Proposition 5.47)} \tag{5.332}$$

In (5.330), we pass to the limit and use (5.331), (5.332), (5.328) (recalling that $q < p$), to obtain

$$A_p(w) = \vartheta(\cdot)|w|^{p-2}w.$$

We act with w and have

$$\|Dw\|_p^p = \int_\Omega \vartheta(z)|w|^p dz \leq 0,$$

$$\Rightarrow w = 0,$$

which contradicts (5.332). Therefore

$$\{y_n^k\}_{k\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded .}$$

From this as before, using Theorem 4.84, we have that

$$\{y_n^k\}_{k\in\mathbb{N}} \subseteq C_0^1(\overline{\Omega}) \text{ is relatively compact for all } n \in \mathbb{N}.$$

We may assume that

$$y_n^k \to u_n \text{ in } C_0^1(\overline{\Omega}) \text{ as } k \to \infty, \text{ for all } n \in \mathbb{N}. \tag{5.333}$$

Now in (5.326) we pass to the limit as $k \to \infty$ and use (5.333). We obtain

$$-\Delta_p u_n(z) - \Delta_q u_n(z) = g_{v_n}(z, u_n(z)) \text{ in } \Omega,\ u_n|_{\partial\Omega} = 0. \tag{5.334}$$

As we did earlier, via a contradiction argument, we show that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p}(\Omega) \text{ is bounded .}$$

From this and the nonlinear regularity theory, we infer that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C_0^1(\overline{\Omega}) \text{ is relatively compact.}$$

Using the double limit lemma (see Lemma I-1.193), we have

$$u_n \to u \text{ in } C_0^1(\overline{\Omega}).$$

Since $u \in intC_+$, we see that $u_n \in intC_+$ for all $n \geq n_0$ with $n_0 \in \mathbb{N}$. Therefore we may assume that $u_n \in intC_+$ for all $n \in \mathbb{N}$ and so from (5.334) we can say that

$$u_n \in S_v \text{ for all } n \in \mathbb{N}, u_n \to u \text{ in } C_0^1(\overline{\Omega}).$$

□

Remark 5.117 This proposition says that the solution multifunction $v \to S_v$ from $C_0^1(\overline{\Omega})$ into $2^{C_0^1(\overline{\Omega})} \setminus \{\emptyset\}$ is lower semicontinuous (see Proposition I-5.6).

Using Proposition 5.116, we can show that the minimal solution map $\vartheta(\cdot)$ is compact. This way we make a crucial step in the application of Theorem 1.67, see also Remark 2.64 (the Leray-Schauder Alternative Principle).

Proposition 5.118 *If hypotheses H hold, then the minimal solution map* $\vartheta : C_0^1(\overline{\Omega}) \to C_0^1(\overline{\Omega})$ *is compact.*

Proof First we show that $\vartheta(\cdot)$ maps bounded sets in $C_0^1(\overline{\Omega})$ into relatively compact sets in $C_0^1(\overline{\Omega})$.

Consider $E \subseteq C_0^1(\overline{\Omega})$ bounded and let $\mu = |E| = \sup\{\|u\|_{C_0^1(\overline{\Omega})} : u \in E\}$. Hypotheses H(i), (iii) imply that given $\varepsilon > 0$, we can find $c_\varepsilon > 0$ such that

$$f(z, x, Dv(z)) \leq \varepsilon x^{p-1} + c_\varepsilon \text{ for } a.a\ z \in \Omega, \text{ all } x \geq 0. \tag{5.335}$$

Let $u_v = \vartheta(v) \in intC_+$. Then

$$\{-\Delta_p u_v(z) - \Delta_q u_v(z) = f(z, u_v(z), Dv(z)) \text{ in } \Omega, u_v|_{\partial\Omega} = 0.\} \tag{5.336}$$

Acting on (5.336) with u_v. we obtain

$$\begin{aligned} \|Du_v\|_p^p &\leq \int_\Omega f(z, u_v, Dv)u_v dz \\ &\leq \varepsilon\|u_v\|_p^p + c \text{ for some } c > 0 \text{ (see (5.335))} \end{aligned}$$

$$\Rightarrow [1 - \frac{\varepsilon}{\widehat{\lambda}_1(p)}]\|Du_v\|_p^p \leq c.$$

Choosing $\varepsilon \in (0, \widehat{\lambda}_1(p))$, we see that

$$\{u_v = \vartheta(v)\}_{v \in E} \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}$$

As before, the nonlinear regularity theory implies that

$$\vartheta(E) \subseteq C_0^1(\overline{\Omega}) \text{ is relatively compact.}$$

Now we show that $\vartheta(\cdot)$ is continuous. So, let $v_n \to v$ in $C_0^1(\overline{\Omega})$. From Proposition 5.116 we know that we can find $u_n \in S_{v_n}$ such that

$$u_n \to \vartheta(v) \text{ in } C_0^1(\overline{\Omega}). \tag{5.337}$$

We have

$$0 \leq \vartheta(v_n) \leq v_n \text{ for all } n \in \mathbb{N}. \tag{5.338}$$

and from the first part of the proof, we have that

$$\{\vartheta(v_n)\}_{n \in \mathbb{N}} \subseteq C_0^1(\overline{\Omega}) \text{ is relatively compact.}$$

So, we may assume that

$$\vartheta(v_n) \to \widehat{u} \in S_v \text{ in } C_0^1(\overline{\Omega}).$$

From (5.337), (5.338), we have $\widehat{u} \leq \vartheta(v)$, hence $\widehat{u} = \vartheta(v)$ and so $\vartheta(\cdot)$ is continuous. We conclude that $\vartheta(\cdot)$ is compact. □

We introduce the set

$$D(\vartheta) = \{u \in C_0^1(\overline{\Omega}) : u = t\vartheta(u), 0 < t < 1\}.$$

If we show that $D(\vartheta)$ is bounded, then according to Theorem 1.67, $\vartheta(\cdot)$ has a fixed point, which is a positive solution of (5.313).

Proposition 5.119 *If hypotheses H hold, then $D(\vartheta) \subseteq C_0^1(\overline{\Omega})$ is bounded.*

Proof Let $u \in D(\vartheta)$. Then

$$u = t\vartheta(u) \text{ with } 0 < t < 1,$$

$$\Rightarrow \frac{1}{t}u = \vartheta(u).$$

Claim $0 \leq u(z) \leq M$ for all $z \in \overline{\Omega}$ ($M > 0$ as in hypothesis H(ii)).

Arguing by contradiction, suppose that

$$u(z_0) = \max_{\overline{\Omega}} u > M, z_0 \in \Omega.$$

Then $Du(z_0) = 0$. On account of hypothesis H(ii), we can find $\Omega' \subseteq \Omega$ open, $\Omega' \subseteq \Omega$, $\partial\Omega'$ is Lipschitz, $z_0 \in \Omega'$ and

$$\frac{\partial u}{\partial n}|_{\partial\Omega'} < 0 \text{ and } f(z, u(z), Du(z)) \leq 0, \text{ for } a.a\ z \in \Omega',$$

(here $n(\cdot)$ is the outward unit normal on $\partial\Omega'$). We have

$$\begin{aligned}
&-\Delta_p(\frac{1}{t}u) - \Delta_q(\frac{1}{t}u) = f(z, \frac{1}{t}u, Du) \text{ in } \Omega \\
&\Rightarrow \int_{\Omega'} [-\Delta_p(\frac{1}{t}u) - \Delta_q(\frac{1}{t}u)]u dz \\
&= \int_{\Omega'} f(z, \frac{1}{t}u, Du)u dz \\
&\leq \frac{1}{t^{p-1}} \int_{\Omega'} f(z, u, Du)u dz \text{ (see hypothesis H(v))} \\
&\leq 0 \text{ (see (5.339)).}
\end{aligned}$$

On the other hand using Green's identity (see Theorem I-4.106) we have

$$\begin{aligned}
&\int_{\Omega'} [-\Delta_p(\frac{1}{t}u) - \Delta_q(\frac{1}{t}u)]u dz \\
&= \frac{1}{t^{p-1}} \|Du\|^p_{L^p(\Omega')} + \frac{1}{t^{q-1}} \|Du\|^q_{L^q(\Omega')} \\
&\quad - \int_{\partial\Omega'} [\frac{1}{t^{p-1}}|Du|^{p-2} + \frac{1}{t^{q-1}}|Du|^{q-2}]u \frac{\partial u}{\partial n} d\sigma \\
&\quad (\sigma(\cdot) \text{ being the Hausdorff measure on } \partial\Omega') \\
&\quad > 0 \text{ (see (5.339)).}
\end{aligned}$$

Comparing (5.340) and (5.341), we have a contradiction. This means that the Claim is true and $0 \leq u(z) = \xi(v)(z) \leq M$ for all $z \in \overline{\Omega}$.

We have

$$\langle A_p(\frac{1}{t}u), u\rangle + \langle A_q(\frac{1}{t}u), u\rangle = \int_{\Omega} f(z, \frac{1}{t}u, Du)u dz$$

$$\Rightarrow \frac{1}{t^{p-1}}\|Du\|_p^p + \frac{1}{t^{q-1}}\|Du\|_q^q$$

$$\leq \frac{1}{t^{p-1}}\int_\Omega f(z,u,Du)udz \text{ (see hypothesis H(v))},$$

$$\Rightarrow \|u\|^p \leq \widehat{c}[1+\|u\|^{p-1}] \text{ for some } \widehat{c}>0 \text{ (see the Claim and hypothesis H(i))}$$

$$\Rightarrow D(\vartheta) \subseteq W_0^{1,p}(\Omega) \text{ is bounded.}$$

From this and the nonlinear regularity theory we conclude that

$$D(\vartheta) \subseteq C_0^1(\overline{\Omega}) \text{ is bounded.}$$

□

Now we are ready to prove the existence of a positive solution for problem (5.313).

Theorem 5.120 *If hypotheses H hold, then problem (5.313) has a positive solution* $\widehat{u}_0 \in intC_+$.

Proof Propositions 5.118 and 5.119, together with Theorem 2.64 imply that $\vartheta(\cdot)$ has a fixed point $\widehat{u}_0 \in intC_+$. So, $\vartheta(\widehat{u}_0) = \widehat{u}_0$ which means that $\widehat{u}_0 \in intC_+$ is a positive solution of (5.313). □

5.5 Anisotropic and Double Phase Problems

In this section we study anisotropic and double phase problems. This requires that we abandon the classical Lebesgue and Sobolev spaces and use variable spaces (anisotropic problems) and generalized Orlicz spaces (double phase problems). In fact variable Lebesgue and Sobolev spaces, are a particular case of generalized Orlicz spaces.

We start with an anisotropic problem. Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$. We consider the following Dirichlet problem

$$\{-\Delta_{p(z)}u(z) - \Delta_{q(z)}u(z) = f(z,u(z)) \text{ in } \Omega, u|_{\partial\Omega} = 0\}. \tag{5.339}$$

Recall that if $r \in L^\infty(\Omega)$, $r_- = \operatorname*{essinf}_\Omega r$ and $r_+ = \operatorname*{esssup}_\Omega r$ and $C^{0,1}(\overline{\Omega})$ is the space of all Lipschitz function $\tau : \overline{\Omega} \to \mathbb{R}$.

Our hypotheses on the exponents $p(\cdot), q(\cdot)$ and on the reaction $f(z,x)$ are the following

$\mathbf{H_0}$: $p, q \in C^{0,1}(\overline{\Omega})$ with $1 < q_- \leq q_+ < p_-$ and there exists $\vartheta \in \mathbb{R}^N \setminus \{0\}$ such that $t \to k(t) = p(z+t\vartheta)$ is monotone on $T_z = \{t \in \mathbb{R} : z + t\vartheta \in \Omega\}$.

Remark 5.121 The extra hypothesis on the exponent $p(\cdot)$, implies that $(-\Delta_{p(z)}, W_0^{1,p(z)}(\Omega))$ has a smallest eigenvalue $\lambda_* > 0$ with eigenfunctions of constant sign (they belong in $intC_+$ or in $-intC_+$, see Theorem 4.177). Also since $p \in C^{0,1}(\overline{\Omega})$, we know that on $W_0^{1,p(z)}(\Omega)$ the Poincare inequality holds (see Theorem 4.173).

$\mathbf{H_1}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that $f(z, 0) = 0$ for $a.a\ z \in \Omega$ and

(i) $|f(z, x)| \leq a(z)[1 + |x|^{r(z)-1}]$ for $a.a\ z \in \Omega$, all $x \in \mathbb{R}$ with $a \in L^\infty(\Omega)$, $r \in C(\overline{\Omega})$ and $p_+ < r(z) < p^*(z)$ for all $z \in \overline{\Omega}$;

(ii) if $F(z, x) = \int_0^x f(z, s)ds$ and $e(z, x) = f(z, x)x - p_+F(z, x)$, then $\lim\limits_{x\to+\infty} \dfrac{F(z, x)}{x^{p_+}} = +\infty$ uniformly for $a.a\ z \in \Omega$ and there exists $\gamma \in L^1(\Omega)$ such that

$$e(z, x) \leq e(z, y) + \gamma(z) \text{ for } a.a\ z \in \Omega, \text{ all } 0 \leq x \leq y;$$

(iii) $\limsup\limits_{x\to-\infty} \dfrac{p(z)F(z, x)}{|x|^{p(z)}} < \mu_*$ uniformly for $a.a\ z \in \Omega$, where

$$\mu_* = \inf\{\frac{\int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz}{\int_\Omega \frac{1}{p(z)}|u|^{p(z)}dz} : u \in W_0^{1,p(z)}(\Omega), u \neq 0\};$$

(iv) there exists $\tau \in C(\overline{\Omega})$ such that $\tau(z) \leq q(z)$ for all $z \in \overline{\Omega}$, there exists $z_0 \in \Omega$ such that $\tau(z_0) < q(z_0)$ and there exist $\beta_0, \widehat{\delta} > 0$ such that

$$\beta_0 \leq \liminf_{x\to 0} \frac{f(z, x)}{|x|^{\tau(z)-2}x} \text{ uniformly for } a.a\ z \in \Omega;$$

$$f(z, x)x \leq \tau_- F(z, x) \text{ for } a.a\ z \in \Omega, \text{ all } |x| \leq \widehat{\delta};$$

(v) there exist $d, \widehat{\xi} > 0$ such that $f(z, x) \leq -\eta < 0$ for $a.a\ z \in \Omega$ and for $a.a\ z \in \Omega$, the function $x \to f(z, x) + \widehat{\xi}x^{p(z)-1}$ is nondecreasing on $[0, d]$.

Remark 5.122 From hypotheses H_1(ii), (iii) we see that $f(z, \cdot)$ exhibits asymmetric behavior as $x \to \pm\infty$. It is $(p_+ - 1)-$superlinear as $x \to +\infty$ and $(p_+ - 1)$ sublinear as $x \to -\infty$. Again the superlinearity is not formulated in terms of the AR-condition, but instead we use an alternative less restrictive condition involving the function $e(z, x)$. This condition is satisfied if there exists $M > 0$ such that for $a.a\ z \in \Omega$ $x \to \frac{f(z,x)}{x^{p_+-1}}$ is nondecreasing on $[M, \infty)$. From Lemma 4.176, we know that $\mu_* > 0$ if and only if $\lambda_* > 0$.

First we will produce positive solutions. To this end we introduce the C^1-functional $\varphi_+ : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ defined by

$$\varphi_+(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega F(z,u^+)dz$$
$$\text{for all } u \in W_0^{1,p(z)}(\Omega).$$

In addition we define the following truncation of $f(z,\cdot)$

$$g_+(z,x) = \begin{cases} f(z,x^+) \ if \ x \le d, \\ f(z,d) \quad if \ d < x. \end{cases} \tag{5.340}$$

This is a Caratheodory function. We set $G_+(z,x) = \int_0^x g_+(z,s)ds$ and consider the C^1-functional $\psi_+ : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ defined by

$$\psi_+(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega G_+(z,u)dz$$
$$\text{for all } u \in W_0^{1,p(z)}(\Omega).$$

Working with $\psi_+(\cdot)$, we will produce the first positive solution for problem (5.339). Recall that, if $h_1, h_2 \in L^0(\Omega)$, then

$$[h_1,h_2] = \{u \in W_0^{1,p(z)}(\Omega) : h_1(z) \le u(z) \le h_2(z) \text{ for } a.a \ z \in \Omega\},$$

$$int_{C_0^1(\overline{\Omega})}[h_1,h_2] = \text{ the interior in } C_0^1(\overline{\Omega}) \text{ of } [h_1,h_2] \cap C_0^1(\overline{\Omega}).$$

Proposition 5.123 *If hypotheses H_0, H_1 hold, then problem (5.339) has a positive solution*

$$u_0 \in int_{C_0^1(\overline{\Omega})}[0,d]$$

and u_0 is a local minimizer of $\varphi_+(\cdot)$.

Proof For all $u \in W_0^{1,p(z)}(\Omega)$, we have

$$\psi_+(u) \ge \frac{1}{p_+}\varrho_p(Du) - c_1 \text{ for some } c_1 > 0 \text{ (see (5.340))}$$
$$\Rightarrow \psi(\cdot) \text{ is coercive.}$$

Also from the compact embedding of $W_0^{1,p(z)}(\Omega)$ into $L^{p(z)}(\Omega)$ we infer that $\psi_+(\cdot)$ is sequentially weakly lower semicontinuous. So, we can find $u_0 \in W_0^{1,p(z)}(\Omega)$ such that

$$\psi_+(u_0) = \inf\{\psi_+(u) : u \in W_0^{1,p(z)}(\Omega)\}. \tag{5.341}$$

Hypothesis H_1(iv) implies that we can find $\beta_1 \in (0, \beta_0)$ and $\delta \in (0, d)$ such that

$$\frac{\beta_1}{\tau(z)} x^{\tau(z)} \leq F(z, x) \text{ for } a.a\ z \in \Omega, \text{ all } x \in [0, \delta]. \tag{5.342}$$

For $u \in intC_+$ with $|\{Du = 0\}|_N = 0$, let $t \in (0, 1)$ be small such that

$$0 \leq tu(z) \leq \delta \text{ for all } z \in \overline{\Omega}.$$

Then

$$\begin{aligned} \psi_+(tu) &\leq \int_\Omega \frac{t^{q(z)}}{q_-}(|Du|^{p(z)} + |Du|^{q(z)})dz - \int_\Omega \frac{\beta_1 t^{\tau(z)}}{\tau(z)}|u|^{\tau(z)}dz \\ &= \int_\Omega [t^{q(z)}\mu_1(z) - t^{\tau(z)}\mu_2(z)]dz \end{aligned} \tag{5.343}$$

with $\mu_1, \mu_2 \in C(\overline{\Omega})$, $\mu_1(z), \mu_2(z) > 0$ for $a.a\ z \in \Omega$. Let $\varepsilon > 0$ be such that $\tau(z) < q(z)$ for all $z \in B_\varepsilon(z_0) \subseteq \Omega$. Then from (5.343), we infer that

$$\begin{aligned} &\psi_+(tu) < 0, \\ &\Rightarrow \psi_+(u_0) < 0 = \psi_+(0) \text{ (see (5.341))} \\ &\Rightarrow u_0 \neq 0. \end{aligned}$$

From (5.341) we have $\langle \psi'_+(u_0), h\rangle = 0$ for all $h \in W_0^{1,p(z)}(\Omega)$, thus

$$\langle V(u_0), h\rangle = \int_\Omega g_+(z, u_0)hdz \tag{5.344}$$

for all $h \in W_0^{1,p(z)}(\Omega)$, with $V = A_{p(z)} + A_{q(z)}$. In (5.344) we choose $h = -u_0^- \in W_0^{1,p(z)}(\Omega)$ and obtain

$$\begin{aligned} &\varrho_p(Du_0^-) \leq 0, \\ &\Rightarrow u_0 \geq 0, u_0 \neq 0. \end{aligned}$$

Next, in (5.344) we use the test function $(u_0 - d)^+ \in W_0^{1,p(z)}(\Omega)$. Then

$$\begin{aligned} \langle V(u_0), (u_0 - d)^+\rangle &= \int_\Omega g_+(z, d)(u_0 - d)^+dz \text{ (see (5.341) and } H_1(\text{v})) \\ &\leq 0 = \langle V(d), (u_0 - d)^+\rangle, \\ &\Rightarrow u_0 \leq d. \end{aligned}$$

So, we have proved that

$$u_0 \in [0, d], u_0 \neq 0. \tag{5.345}$$

From (5.345), (5.341) and (5.344), we infer that u_0 is a positive solution of (5.339). From Theorem 5.3, we have that $u_0 \in L^\infty(\Omega)$ and then Theorem 5.6 implies that $u_0 \in C_+ \setminus \{0\}$. Let $\widehat{\xi} > 0$ be as postulated by hypothesis $H_1(\mathrm{v})$. Then

$$\begin{aligned} &- \Delta_{p(z)} u_0 - \Delta_{q(z)} u_0 + \widehat{\xi} u_0^{p(z)-1} \geq 0 \text{ in } \Omega, \\ &\Rightarrow u_0 \in intC_+ \text{ (see Theorem 5.9).} \end{aligned}$$

We have

$$\begin{aligned} &- \Delta_{p(z)} u_0 - \Delta_{q(z)} u_0 + \widehat{\xi} u_0^{p(z)-1} \\ &= f(z, u_0) + \widehat{\xi} u_0^{p(z)-1} \\ &\leq f(z, d) + \widehat{\xi} d^{p(z)-1} \text{ (see (5.345) and } H_1(\mathrm{v})) \\ &\leq -\Delta_{p(z)} d - \Delta_{q(z)} d + \widehat{\xi} d^{p(z)-1} \text{ in } \Omega \text{ (see } H_1(\mathrm{v})). \end{aligned} \tag{5.346}$$

Since $f(z, d) \leq -\eta < 0$ (see $H_1(\mathrm{v})$), from (5.346) we infer that $u_0(z) < d$ for all $z \in \overline{\Omega}$. Therefore $u_0 \in int_{C_0^1(\overline{\Omega})}[0, d]$. Note that

$$\begin{aligned} &\psi_+|_{[0,d]} = \varphi_+|_{[0,d]} \text{ (see (5.340)),} \\ &\Rightarrow u_0 \text{ is a } C_0^1(\overline{\Omega}) - \text{local minimizer of } \varphi_+(\cdot), \\ &\Rightarrow u_0 \text{ is a } W_0^{1,p(z)}(\Omega) - \text{local minimizer of } \varphi_+(\cdot) \text{ (see Theorem 5.45).} \end{aligned}$$

□

Proposition 5.124 *If hypotheses H_0, H_1 hold, then the functional $\varphi_+(\cdot)$ satisfies the $C-$condition.*

Proof Let $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p(z)}(\Omega)$ be a sequence such that

$$|\varphi_+(u_n)| \leq c_1 \text{ for some } c_1 > 0, \text{ all } n \in \mathbb{N}, \tag{5.347}$$

$$(1 + \|u_n\|)\varphi'_+(u_n) \to 0 \text{ in } W_0^{1,p(z)}(\Omega)^*. \tag{5.348}$$

From (5.348) we have

$$|\langle V(u_n), h\rangle - \int_\Omega f(z, u_n^+) h dz| \leq \frac{\varepsilon_n \|h\|}{1 + \|u_n\|} \tag{5.349}$$

$$\text{for all } h \in W_0^{1,p(z)}(\Omega), \text{ as } \varepsilon_n \to 0^+.$$

In (5.349) we choose $h = -u_n^- \in W_0^{1,p(z)}(\Omega)$. Then

$$\begin{aligned} &\varrho_p(Du_n^-) \leq \varepsilon_n \text{ for all } n \in \mathbb{N}, \\ &\Rightarrow u_n^- \to 0 \text{ in } W_0^{1,p(z)}(\Omega), \end{aligned} \tag{5.350}$$

Combining (5.350) and (5.347), we have

$$\begin{aligned} &\int_\Omega \frac{p_+}{p(z)}|Du_n^+|^{p(z)}dz + \int_\Omega \frac{p_+}{q(z)}|Du_n^+|^{q(z)}dz - \int_\Omega p_+F(z,u_n^+)dz \leq c_2 \\ &\text{for some } c_2 > 0, \text{ all } n \in \mathbb{N}, \\ &\Rightarrow \varrho_p(Du_n^+) + \varrho_q(Du_n^+) - \int_\Omega p_+F(z,u_n^+)dz \leq c_2 \text{ for all } n \in \mathbb{N}. \end{aligned} \tag{5.351}$$

In (5.349) use $h = u_n^+ \in W_0^{1,p(z)}(\Omega)$. Then

$$-\varrho_p(Du_n^+) - \varrho_q(Du_n^+) + \int_\Omega f(z,u_n^+)u_n^+dz \leq \varepsilon_n \text{ for all } n \in \mathbb{N}.$$

We add (5.351) and (5.352). We obtain

$$\begin{aligned} &\int_\Omega [f(z,u_n^+)u_n^+ - p_+F(z,u_n^+)]dz \leq c_3 \text{ for some } c_3 > 0, \text{ all } n \in \mathbb{N}, \\ &\Rightarrow \int_\Omega e(z,u_n^+)dz \leq c_3 \text{ for all } n \in \mathbb{N}. \end{aligned} \tag{5.352}$$

Claim $\{u_n^+\}_{n\in\mathbb{N}} \subseteq W_0^{1,p(z)}(\Omega)$ is bounded.

We argue by contradiction. Suppose that the Claim is not true. We may assume that

$$\|u_n\| \to \infty \text{ as } n \to \infty. \tag{5.353}$$

Let $y_n = \frac{u_n^+}{\|u_n^+\|}$ $n \in \mathbb{N}$. Then $y_n \geq 0$, $\|y_n\| = 1$ for all $n \in \mathbb{N}$. We may assume that

$$y_n \xrightarrow{w} y \text{ in } W_0^{1,p(z)}(\Omega),\ y_n \to y \text{ in } L^{r(z)}(\Omega),\ y \geq 0. \tag{5.354}$$

First we assume that $y \neq 0$. Let $\Omega_+ = \{z \in \Omega : y(z) > 0\}$. Since $y \geq 0$, $y \neq 0$, we see that $|\Omega_+|_N > 0$ ($|\cdot|_N$ being the Lebesgue measure on $\mathbb{R}^N$). Moreover, we

have

$$u_n^+(z) \to +\infty \text{ for all } z \in \Omega_+. \tag{5.355}$$

We may assume that $\|u_n^+\| \geq 1$ for all $n \in \mathbb{N}$ (see (5.353)). We have

$$\frac{F(z, u_n^+(z))}{\|u_n^+\|^{p_+}} = \frac{F(z, u_n^+(z))}{u_n^+(z)^{p_+}} y_n(z)^{p_+} \to +\infty \text{ for } a.a\ z \in \Omega_+ \tag{5.356}$$

(see (5.356) and H_1(ii)),

$$\Rightarrow \lim \int_{\Omega_+} \frac{F(z, u_n^+)}{\|u_n^+\|^{p_+}} dz = +\infty \text{(by Fatou's lemma)}.$$

From (5.347) and (5.350), we have

$$-\frac{1}{q_-}[\varrho_p(Du_n^+) + \varrho_q(Du_n^+)] + \int_\Omega F(z, u_n^+)dz \leq c_4 \text{ for some } c_4 > 0, \text{ all } n \in \mathbb{N}, \tag{5.357}$$

$$\Rightarrow \int_\Omega F(z, u_n^+)dz \leq \frac{1}{q_-}[\varrho_p(Du_n^+) + \varrho_q(Du_n^+)] + c_4$$

$$\leq \frac{2}{q_-}\varrho_p(Du_n^+) + c_5 \text{ for some } c_5 > 0, \text{ all } n \in \mathbb{N}$$

$$\Rightarrow \int_\Omega \frac{F(z, u_n^+)}{\|u_n^+\|^{p_+}} dz \leq \frac{2}{q_-} + \frac{c_5}{\|u_n^+\|^{p_+}}$$

(recall that $\|u_n^+\| \geq 1$ for all $n \in \mathbb{N}$).

We compare (5.356) and (5.357) and we have a contradiction.

Now assume that $y = 0$. Let $\sigma_+ : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ be the C^1-functional defined by

$$\sigma_+(u) = \frac{1}{p_+}\varrho_p(Du) - \int_\Omega F(z, u^+)dz \text{ for all } u \in W_0^{1,p(z)}(\Omega).$$

We see that

$$\sigma_+ \leq \varphi_+. \tag{5.358}$$

Let $t_n \in [0, 1]$ such that $\sigma_+(t_n u_n^+) = \max\limits_{0 \leq t \leq 1} \sigma_+(tu_n^+)$ for all $n \in \mathbb{N}$. We fix $\lambda > 1$ and define $v_n = (5.2\lambda)^{1/p_-} y_n \in W_0^{1,p(z)}(\Omega)$. We have

$$v_n \to 0 \text{ in } L^{r(z)}(\Omega) \text{ (see (5.354) and recall } y = 0), \tag{5.359}$$

$$\Rightarrow \int_\Omega F(z, v_n)dz \to 0.$$

From (5.353) we see that we can find $n_0 \in \mathbb{N}$ such that

$$\frac{(5.2\lambda)^{1/p_-}}{\|u_n^+\|} \in (0, 1) \text{ for all } n \geq n_0. \tag{5.360}$$

Therefore for $n \geq n_0$ we have

$$\begin{aligned}
\sigma_+(t_n u_n^+) &\geq \sigma_+\left(\frac{(5.2\lambda)^{1/p_-}}{\|u_n^+\|} u_n^+\right) \\
&= \frac{1}{p_+}\int_\Omega (5.2\lambda)^{\frac{p(z)}{p_-}} |Dy_n|^{p(z)}dz - \int_\Omega F(z, v_n)dz \\
&\geq \frac{2\lambda}{p_+}\varrho_p(Dy_n) - \int_\Omega F(z, v_n)dz \text{ (since } \lambda > 1) \\
&= \frac{2\lambda}{p_+} - \int_\Omega F(z, v_n)dz \text{ (since } \|y_n\| = 1) \\
&\geq \frac{\lambda}{p_+} \text{ for all } n \geq n_1 \text{ (see (5.359))}.
\end{aligned}$$

But $\lambda > 1$ is arbitrary. So, we infer that

$$\sigma_+(t_n u_n^+) \to +\infty \text{ as } n \to \infty. \tag{5.361}$$

We have $0 \leq t_n u_n^+ \leq u_n^+$. Therefore

$$\begin{aligned}
\int_\Omega e(z, t_n u_n^+)dz &\leq \int_\Omega e(z, u_n^+)dz + \|\gamma\|_1 \text{ (see hypothesis } H_1\text{(iii))} \qquad (5.362)\\
&\leq c_6 \text{ for some } c_6 > 0, \text{ all } n \in \mathbb{N} \qquad \text{(see (5.352))}.
\end{aligned}$$

We see that

$$\begin{aligned}
&\sigma_+(0) = 0, \sigma_+(u_n^+) \leq c_7 \text{ for some } c_7 > 0, \\
&\qquad \text{all } n \in \mathbb{N} \text{ (see (5.347), (5.350), (5.358))}.
\end{aligned}$$

From (5.361) it follows that there exists $n_2 \in \mathbb{N}$ such that

$$t_n \in (0, 1) \text{ for all } n \geq n_2. \tag{5.363}$$

Therefore for $n \geq n_2$ we have

$$\frac{d\sigma_+}{dt}(tu_n^+)|_{t=t_n} = 0, \tag{5.364}$$

$$\Rightarrow \langle A_{p(z)}(t_n u_n^+), u_n^+\rangle = \int_\Omega f(z, t_n u_n^+)u_n^+ dz,$$

$$\Rightarrow \langle A_{p(z)}(t_n u_n^+), t_n u_n^+\rangle = \int_\Omega f(z, t_n u_n^+)(t_n u_n^+)dz \text{ (see (5.363))},$$

$$\Rightarrow \varrho_p(D(t_n u_n^+)) = \int_\Omega f(z, t_n u_n^+)(t_n u_n^+)dz.$$

We return to (5.362) and use (5.364). We obtain

$$\varrho_p(D(t_n u_n^+)) - \int_\Omega p_+ F(z, t_n u_n^+)dz \leq c_6 \text{ for all } n \geq n_2, \tag{5.365}$$

$$\Rightarrow p_+\sigma_+(t_n u_n^+) \leq c_6 \text{ for all } n \geq n_2.$$

We compare (5.365) and (5.361) and have a contradiction. This proves the Claim. The Claim and (5.350), imply that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p(z)}(\Omega) \text{ is bounded.}$$

So, we may assume that

$$u_n \xrightarrow{w} u \text{ in } W_0^{1,p(z)}(\Omega),\ u_n \to u \text{ in } L^{r(z)}(\Omega). \tag{5.366}$$

In (5.349) we use the test function $h = u_n - u \in W_0^{1,p(z)}(\Omega)$, pass to the limit as $n \to \infty$ and use (5.366). Then

$$\lim_{n\to\infty} \langle V(u_n), u_n - u\rangle = 0,$$

$$\Rightarrow u_n \to u \text{ in } W_0^{1,p(z)}(\Omega),$$

$$\Rightarrow \varphi_+(\cdot) \text{ satisfies the } C - \text{condition.}$$

□

The anisotropic regularity theory (Theorems 5.3 and 5.6), implies that

$$K_{\varphi_+} \subseteq C_+.$$

We assume that K_{φ_+} is finite or otherwise we already have an infinity of positive smooth solutions for problem (5.339).

Next using variational tools and critical groups, we will generate a second positive solution for problem (5.339). For this purpose, we need to compute the critical groups of φ_+ at $u = 0 \in K_{\varphi_+}$.

In what follows, for notational convenience, we write

$$W_0^+ = \{u \in W_0^{1,p(z)}(\Omega) : u^+ \neq 0\} \cup \{0\}.$$

Lemma 5.125 *If hypotheses H_0, H_1 hold, then $C_k(\varphi_+, 0) = 0$ for all $k \in \mathbb{N}$.*

Proof Hypotheses H_1(i),(iv) imply that there exist $c_1, c_2 > 0$ such that

$$c_1 x^{\tau(z)} - c_2 x^{r(z)} \leq F(z, x) \text{ for } a.a\ z \in \Omega, \text{ all } x \geq 0. \tag{5.367}$$

Let $u \in W_0^+ \setminus \{0\}$ and $t \in [0, 1]$. Then

$$\varphi_+(tu) \geq \frac{t^{q_-}}{q_-}[\varrho_p(Du) + \varrho_q(Du) + c_3\varrho_r(u)] - c_1 t^{\tau_+}\varrho_\tau(u^+) \text{ for some } c_3 > 0.$$

Since $\tau_+ < q_-$, we see that there exists $t_* \in (0, 1)$ such that

$$\varphi_+(tu) < 0 \text{ for all } t \in (0, t^*). \tag{5.368}$$

Suppose $u \in W_0^+$ satisfies $0 < \|u\| \leq 1$, $\varphi_+(u) = 0$. Then

$$\begin{aligned}
\frac{d}{dt}\varphi_+(tu)|_{t=1} &= \langle \varphi'_+(u), u\rangle \text{ (using the chain rule)} \\
&= \varrho_p(Du) + \varrho_q(Du) - \int_\Omega f(z, u^+)u^+ dz \\
&= (1 - \frac{\tau_-}{p_-})\varrho_p(Du) + (1 - \frac{\tau_-}{q_-})\varrho_q(Du) \\
&\qquad + \int_\Omega [\tau_- F(z, u^+) - f(z, u^+)u^+]dz \text{ (since } \varphi_+(u) = 0) \\
&\geq (1 - \frac{\tau_-}{p_-})\varrho_p(Du) + (1 - \frac{\tau_-}{q_-})\varrho_q(Du) \\
&\qquad + \int_{\{u^+ > \delta\}} [\tau_- F(z, u^+) - f(z, u^+)u^+]dz \text{ (see } H_1\text{(iv))} \\
&\geq c_4\varrho_p(Du) - c_5\varrho_r(u) \text{ for some } c_4, c_5 > 0 \text{ (see hypothesis } H_1\text{(i))} \\
&\geq c_4\|u\|^{p_+} - c_5\|u\|^{r_-} \text{ for some } c_5 > 0 \text{ (since } \|u\| \leq 1).
\end{aligned}$$

Since $p_+ < r_-$, we see that for $\varrho \in (0, 1)$ small

$$\frac{d}{dt}\varphi_+(tu)|_{t=1} > 0 \text{ for all } u \in W_0^+, 0 < \|u\| \leq \varrho, \varphi_+(u) = 0. \tag{5.369}$$

Let $u \in W_0^+$ as in (5.369). We show that

$$\varphi_+(tu) \leq 0 \text{ for all } t \in [0, 1]. \tag{5.370}$$

If (5.370) is not true, then we can find $t_0 \in (0, 1)$ such that $\varphi_+(t_0 u) > 0$. Then we have

$$\begin{aligned} &\widehat{t} = \min\{t \in [t_0, 1] : \varphi_+(tu) = 0\} > t_0 > 0, \\ &\Rightarrow \varphi_+(tu) > 0 \text{ for all } t \in [t_0, \widehat{t}). \end{aligned} \tag{5.371}$$

We set $y = \widehat{t}u$ and have

$$0 < \|y\| = \widehat{t}\|u\| \leq \|u\| \leq \varrho, \varphi_+(y) = \varphi_+(\widehat{t}u) = 0.$$

Then from (5.369) it follows that

$$\frac{d}{dt}\varphi_+(ty)|_{t=1} > 0. \tag{5.372}$$

Then we have

$$\begin{aligned} &\varphi_+(y) = \varphi_+(\widehat{t}u) = 0 < \varphi_+(tu) \text{ for all } t \in [t_0, \widehat{t}) \text{ (see (5.371))}, \\ &\Rightarrow \frac{d}{dt}\varphi_+(ty)|_{t=1} = \widehat{t}\frac{d}{dt}\varphi_+(ty)|_{t=\widehat{t}} = \widehat{t}\lim_{t \to \widehat{t}^-} \frac{\varphi_+(tu)}{t - \widehat{t}} \leq 0. \end{aligned} \tag{5.373}$$

Comparing (5.372) and (5.373), we have a contradiction. This proves that (5.370) is true.

We define $\overline{B}_\varrho = \{u \in W_0^{1,p(z)}(\Omega) : \|u\| \leq \varrho\}$. Since we have assumed that K_{φ_+} is finite, by choosing $\varrho \in (0, 1)$ even smaller if necessary, we can have that $K_{\varphi_+} \cap \overline{B}_\varrho = \{0\}$. Consider the deformation $h_+ : [0, 1] \times (\varphi_+^0 \cap \overline{B}_\varrho) \to \varphi_+^0 \cap \overline{B}_\varrho$ defined by

$$h_+(t, u) = (1 - t)u \text{ for all } t \in [0, 1], \text{ all } u \in \varphi_+^0 \cap \overline{B}_\varrho.$$

From (5.370) we see that this is a well-defined deformation and shows that $\varphi_+^0 \cap \overline{B}_\varrho$ is contractible. Let $u \in \overline{B}_\varrho$ with $\varphi_+(u) > 0$. We show that there exists unique $t(u) \in (0, 1)$ such that

$$\varphi_+(t(u)u) = 0. \tag{5.374}$$

From (5.368) and Bolzano's theorem, we see that such a $t(u) \in (0,1)$ exists. We need to show its uniqueness. Suppose there exist $t_1^* = t_1(u) < t_2^* = t_2(u) < 1$ such that

$$\varphi_+(t_1^* u) = \varphi_+(t_2^* u) = 0.$$

From (5.370), we know that

$$\varphi_+(t t_2^* u) \leq 0 \text{ for all } t \in [0,1].$$

It follows that $t = \frac{t_1^*}{t_2^*} \in (0,1)$ is a maximizer of $t \to \varphi_+(t t_2^* u)$. Hence

$$\frac{d}{dt}\varphi_+(t t_1^* u)|_{t=1} = \frac{d}{dt}\varphi_+(t t_2^* u)|_{t=\frac{t_1^*}{t_2^*}} = 0.$$

Since $\varphi_+(t_1^* u) = 0$, the above equality contradicts (5.372). This proves the uniqueness of $t(u) \in (0,1)$ for which (5.374) holds. Then

$$\begin{aligned} &\varphi_+(tu) < 0 \text{ for } t \in (0, t(u)), \\ &\varphi_+(tu) > 0 \text{ for } t \in (t(u), 1]. \end{aligned} \tag{5.375}$$

We consider the map $l_+ : \overline{B}_\varrho \setminus \{0\} \to [0,1]$ defined by

$$l_+(u) = \begin{cases} 1 & if\ u \in \overline{B}_\varrho \setminus \{0\}, \varphi_+(u) \leq 0, \\ t(u) & if\ u \in \overline{B}_\varrho \setminus \{0\}, \varphi_+(u) > 0. \end{cases}$$

This map is well defined and we will show that it is continuous. From (5.375) we see that it suffices to show continuity at $u \in \overline{B}_\varrho \setminus \{0\}$ with $\varphi_+(u) = 0$. Let $u_n \to u$ in $W_0^{1,p(z)}(\Omega)$. From (5.375) we see that we may assume that $\varphi_+(u_n) > 0$ for all $n \in \mathbb{N}$. If we did not have continuity, then for at least a subsequence, we can say that

$$\begin{aligned} &t(u_n) \leq \widetilde{t} < 1 \text{ for all } n \in \mathbb{N} \\ &\Rightarrow \varphi_+(tu_n) > 0 \text{ for all } t \in (\widetilde{t}, 1], \text{ all } n \in \mathbb{N}, \\ &\Rightarrow \varphi_+(tu) \geq 0 \text{ for all } t \in (\widetilde{t}, 1], \\ &\Rightarrow \varphi_+(tu) = 0 \text{ for all } t \in (\widetilde{t}, 1] \text{ (see (5.370))}, \\ &\Rightarrow \frac{d}{dt}\varphi_+(tu)|_{t=1} = 0, \text{ a contradiction to (5.369)}. \end{aligned}$$

This proves the continuity of $l_+(\cdot)$. Then we define

$$\widehat{l}_+(u) = \begin{cases} u & if\ u \in \overline{B}_\varrho \setminus \{0\}, \varphi_+(u) \leq 0, \\ l_+(u)u & if\ u \in \overline{B}_\varrho \setminus \{0\}, \varphi_+(u) > 0. \end{cases}$$

Evidently $\widehat{l}_+(\cdot)$ is continuous (since $l_+(\cdot)$ is) and

$$\widehat{l}_+|_{(\varphi_+^0 \cap \overline{B}_\varrho)\setminus\{0\}} = id|_{(\varphi_+^0 \cap \overline{B}_\varrho)\setminus\{0\}},$$
$$\Rightarrow (\varphi^0 \cap \overline{B}_\varrho) \setminus \{0\} \text{ is a retract of } \overline{B}_\varrho \setminus \{0\}.$$

We know that $\overline{B}_\varrho \setminus \{0\}$ is contractible, hence $(\varphi_+^0 \cap \overline{B}_\varrho) \setminus \{0\}$ is contractible too. Therefore $H_k(\varphi_+^0 \cap \overline{B}_\varrho, (\varphi_+^0 \cap \overline{B}_\varrho) \setminus \{0\}) = 0$, hence $C_k(\varphi_+, 0) = 0$ for all $k \in \mathbb{N}_0$. □

Let $\varphi : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ be the energy functional for problem (5.339) defined by

$$\varphi(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega F(z,u)dz$$
$$\text{for all } u \in W_0^{1,p(z)}(\Omega).$$

Clearly $\varphi \in C^1(W_0^{1,p(z)}(\Omega))$. Following the steps of the previous proof, we compute the critical groups of $\varphi(\cdot)$ at $u = 0 \in K_\varphi$.

Lemma 5.126 *If hypotheses H_0, H_1 hold, then $C_k(\varphi, 0) = 0$ for all $k \in \mathbb{N}_0$.*

With the help of Lemma 5.125, we can have a second positive solution for problem (5.339).

Proposition 5.127 *If hypotheses H_0, H_1 hold, then problem (5.339) has a second positive solution*

$$\widehat{u} \in intC_+, \widehat{u} \neq u_0.$$

Proof From Proposition 5.123, we already have a positive solution

$$u_0 \in int_{C_0^1(\overline{\Omega})}[0, d],$$

which is a local minimizer of the functional $\varphi_+(\cdot)$. We know that K_{φ_+} is finite. So, by Proposition 3.132 we can find $\varrho \in (0, 1)$ small such that

$$\varphi_+(u_0) < \inf\{\varphi_+(u) : \|u - u_0\| = \varrho\} = m_+. \tag{5.376}$$

If $u \in intC_+$, then on account of hypothesis H_1(ii), we have

$$\varphi_+(tu) \to -\infty \text{ as } t \to +\infty. \tag{5.377}$$

From Proposition 5.124 we know that

$$\varphi_+(\cdot) \text{ satisfies the } C - \text{condition.} \tag{5.378}$$

Then (5.376), (5.377), (5.378) and the Mountain Pass Theorem (see Theorem 3.41), imply that there exists $\widehat{u} \in W_0^{1,p(z)}(\Omega)$ such that

$$\begin{aligned} &\widehat{u} \in K_{\varphi_+} \subseteq C_+ \text{ and } \varphi_+(u_0) < m_+ \leq \varphi_+(\widehat{u}), \\ &\Rightarrow \widehat{u} \neq u_0. \end{aligned}$$

If we show that $\widehat{u} \neq 0$, then $\widehat{u} \in intC_+$ (see Theorem 5.9) and is the desired second positive solution of (5.339). Since $\widehat{u}$ is a critical point of $\varphi_+(\cdot)$ of mountain pass type, then Corollary 3.123 implies that

$$C_1(\varphi_+, \widehat{u}) \neq 0. \tag{5.379}$$

On the other hand from Lemma 5.125 we have

$$C_k(\varphi_+, 0) = 0 \text{ for all } k \in \mathbb{N}_0. \tag{5.380}$$

Comparing (5.379) and (5.380), we conclude that $\widehat{u} \neq 0$ and so $\widehat{u} \in intC_+$ is the second positive solution of (5.339) distinct from u_0. □

Next we focus on the existence of negative solutions for problem (5.339). To this end, we introduce the C^1-functional $\varphi_- : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ defined by

$$\varphi_-(v) = \int_\Omega \frac{1}{p(z)}|Dv|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Dv|^{q(z)}dz - \int_\Omega F(z, -v^-)dz$$

$$\text{for all } v \in W_0^{1,p(z)}(\Omega).$$

Using $\varphi_-(\cdot)$ and the direct method of the calculus of variations, we can generate a negative solution for problem (5.339).

Proposition 5.128 *If hypotheses H_0, H_1 hold, then problem (5.339) admits a negative solution*

$$v_0 \in -intC_+.$$

Proof On account of hypotheses H_1(i),(iii) we can find $\mu \in (0, \mu_*)$ and $c_1 = c_1(\mu) > 0$ such that

$$F(z, x) \leq \frac{\mu}{p(z)}|x|^{p(z)} + c_1 \text{ for } a.a\ z \in \Omega, \text{ all } x \leq 0. \tag{5.381}$$

Then for $u \in W_0^{1,p(z)}(\Omega)$ we have

$$\varphi_-(u) \geq \int_\Omega \frac{1}{p(z)}(1 - \frac{\mu}{\mu_*})|Du|^{p(z)}dz - c_1|\Omega|_N$$
$$\geq c_2\varrho_5(Du) - c_1|\Omega|_N \text{ for some } c_2 > 0,$$

$\Rightarrow \varphi_-(\cdot)$ is coercive.

Also it is sequentially weakly lower semicontinuous. So, we can find $v_0 \in W_0^{1,p(z)}(\Omega)$ such that

$$\varphi_-(v_0) = \inf\{\varphi_-(v) : v \in W_0^{1,p(z)}(\Omega)\}.$$

On the other hand, using hypothesis H_1(iv) as in the proof of Proposition 5.123, we show that

$$\varphi_-(v_0) < 0 = \varphi_-(0),$$
$$\Rightarrow v_0 \neq 0.$$

From (5.381) we have

$$\langle \varphi'_-(v_0), h\rangle = 0 \text{ for all } h \in W_0^{1,p(z)}(\Omega),$$
$$\Rightarrow \langle V(v_0), h\rangle = \int_\Omega f(z, -v_0^-)hdz.$$

Choosing $h = v_0^+ \in W_0^{1,p(z)}(\Omega)$ we obtain $v_0 \leq 0, v_0 \neq 0$. As before the anisotropic regularity theory and the anisotropic maximum principle, imply $v_0 \in -intC_+$. □

We can have a negative solution by trading in some sense hypotheses between the exponents and the reaction. More precisely, now we impose the following conditions on the data of problem (5.339).

$\mathbf{H_0'}$: $p, q \in C^{0,1}(\overline{\Omega})$, $1 < q_- \leq q_+ < p_-$. $\mathbf{H_1'}$: $f : \Omega \times \mathbb{R} \to \mathbb{R}$ is a Caratheodory function such that $f(z, 0) = 0$ for $a.a\ z \in \Omega$, hypotheses H_1'(i),(ii),(iv),(v) are the same as the corresponding hypotheses H_1(i),(ii),(iv),(v) and

(iii) there exists $l \in C(\overline{\Omega})$ with $1 < l_- \leq l_+ < p_-$ such that $\lim\limits_{x\to-\infty} \frac{F(z,x)}{|x|^{l(z)}} = 0$ uniformly for $a.a\ z \in \Omega$.

Remark 5.129 We see that now the hypotheses on the exponent $p(\cdot)$ are weaker (we have dropped the monotonicity hypothesis), but we have strengthened the asymptotic condition on $F(z, \cdot)$ as $x \to -\infty$.

Proposition 5.130 *If hypotheses H_0', H_1' hold, then problem (5.339) admits a negative solution*

$$v_0 \in -intC_+.$$

Proof Hypotheses H_1'(i),(iii) imply that given $\varepsilon > 0$, we can find $c_1 = c_1(\varepsilon) > 0$ such that

$$F(z,x) \leq \frac{\varepsilon}{l_-}|x|^{l(z)} + c_1 \text{ for } a.a\ z \in \Omega,\ \text{all } x \leq 0. \tag{5.382}$$

Let $v \in W_0^{1,p(z)}(\Omega)$ with $\|v\|_{l(z)} \geq 1$ large. Then

$$\begin{aligned}
\varphi_-(v) &\geq \frac{1}{p_+}\varrho_p(Dv) - \frac{\varepsilon}{l_-}\varrho_l(v) - c_1|\Omega|_N \text{ (see (5.382))} \\
&\geq \frac{1}{p_+}\|v\|^{p_-} - \frac{\varepsilon}{l_-}c_2\|v\|^{l_+} - c_1|\Omega|_N \text{ for some } c_2 > 0 \\
&\qquad \text{(recall } W_0^{1,p(z)}(\Omega) \hookrightarrow L^{l(z)}(\Omega) \text{ continuously).} \\
&\Rightarrow \varphi_-(\cdot) \text{ coercive (since } l_+ < p_-).
\end{aligned}$$

Then continuing as in the proof of Proposition 5.128, we produce a negative solution $v_0 \in -intC_+$. □

We can show the existence of extremal constant sign solutions. We observe that hypotheses H_1(i),(iv)(= H_1'(i),(iv)), we can find $\beta \in (0, \beta_0)$ and $c = c(\beta) > 0$ such that

$$f(z,x)x \geq \beta|x|^{\tau(z)} - c|x|^{r(z)} \text{ for } a.a\ z \in \Omega,\ \text{all } x \in \mathbb{R}. \tag{5.383}$$

Motivated from (5.383), we consider the following auxiliary anisotropic Dirichlet problem

$$-\Delta_{p(z)}u(z) - \Delta_{q(z)}u(z) = \beta|u(z)|^{\tau(z)-2}u(z) - c|u(z)|^{r(z)}u(z) \text{ in } \Omega,\ u|_{\partial\Omega} = 0. \tag{5.384}$$

Reasoning as in the proof of Proposition 5.54, we obtain the following result concerning problem (5.384).

Proposition 5.131 *Problem (5.384) admits a unique positive solution*

$$\overline{u} \in intC_+.$$

Moreover, since problem (5.384) is odd $\overline{v} = -\overline{u} \in -intC_+$ is the unique negative solution of (5.384).

Let S_+ (resp.S_-) be the set of positive (resp.negative) solutions of problem (5.339). From Proposition 5.123, 5.128 and 5.130, we have that under hypotheses H_0, H_1 or H_0', H_1', we have

$$\emptyset \neq S_+ \subseteq intC_+ \text{ and } \emptyset \neq S_- \subseteq -intC_+.$$

Moreover, the set S_+ is downward directed, while S_- is upward directed.

Proposition 5.132 *If hypotheses H_0, H_1 or H_0', H_1' hold, then $\overline{u} \leq u$ for all $u \in S_+$ and $v \leq \overline{v}$ for all $v \in S_-$.*

Proof Let $u \in S_+ \subseteq intC_+$. We introduce the Caratheodory function $\widehat{k}_+(z,x)$ defined by

$$\widehat{k}_+(z,x) = \begin{cases} \beta(x^+)^{\tau(z)-1} - c(x^+)^{r(z)-1} & \text{if } x \leq u(z), \\ \beta u(z)^{\tau(z)-1} - cu(z)^{r(z)-1} & \text{if } u(z) < x. \end{cases} \tag{5.385}$$

We set $\widehat{K}_+(z,x) = \int_0^x \widehat{k}_+(z,s)ds$ and consider the C^1-functional $\widehat{\gamma}_+ : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{\gamma}_+(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega \widehat{K}_+(z,u)dz$$
$$\text{for all } u \in W_0^{1,p(z)}(\Omega).$$

From (5.385) we see that $\widehat{\gamma}_+(\cdot)$ is coercive. Also, it is sequentially weakly lower semicontinuous. Therefore, we can find $\widetilde{u} \in W_0^{1,p(z)}(\Omega)$ such that

$$\widehat{\gamma}_+(\widetilde{u}) = \inf\{\widehat{\gamma}_+(u) : u \in W_0^{1,p(z)}(\Omega)\}.$$

Since $\tau_+ < q_-$ and $u \in intC_+$, we have that $\widehat{\gamma}_+(\widetilde{u}) < 0 = \widehat{\gamma}_+(0)$. Hence $\widetilde{u} \neq 0$. Also since $\widetilde{u} \in K_{\widehat{\gamma}_+} \subseteq [0,u] \cap C_+$. Therefore $\widetilde{u}$ is a positive solution of (5.384). On account of Proposition 5.131, we have $\widetilde{u} = \overline{u} \in intC_+$ and so $\overline{u} \leq u$ for all $u \in S_+$.

Similarly, using $\overline{v} \in -intC_+$, we show that $v \leq \overline{v}$ for all $v \in S_-$. □

These bounds lead to extremal constant sign solutions for (5.339).

Proposition 5.133 *If hypotheses H_0, H_1 or H_0', H_1' hold, then there exist $u_* \in S_+$ and $v_* \in -S_-$ such that $u_* \leq u$ for all $u \in S_+$, $v \leq v_*$ for all $v \in S_-$.*

Proof Using Theorem I-5.109, we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq S_+$ decreasing such that

$$\inf S_+ = \inf_{n\in\mathbb{N}} u_n.$$

For all $n \in \mathbb{N}$, we have $\overline{u} \leq u_n \leq u_1$, see Proposition 5.132, thus

$$\langle V(u_n), h\rangle = \int_\Omega f(z,u_n)hdz \tag{5.386}$$

for all $h \in W_0^{1,p(z)}(\Omega)$.

Choosing $h = u_n \in W_0^{1,p(z)}(\Omega)$, from (5.386) it follows that $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p(z)}(\Omega)$ is bounded and so we may assume that

$$u_n \xrightarrow{w} u_* \text{ in } W_0^{1,p(z)}(\Omega),\ u_n \to u_* \text{ in } L^{p(z)}(\Omega). \tag{5.387}$$

In (5.386), we use the test function $h = u_n - u_* \in W_0^{1,p(z)}(\Omega)$, pass to the limit as $n \to \infty$ and use (5.387). Then

$$\lim_{n\to\infty} \langle V(u_n), u_n - u_* \rangle = 0,$$

$$\Rightarrow u_n \to u_* \text{ in } W_0^{1,p(z)}(\Omega).$$

If in (5.386) we pass to the limit as $n \to \infty$ and use (5.388), then

$$\langle V(u_*), h \rangle = \int_\Omega f(z, u_*) h dz \text{ for all } h \in W_0^{1,p(z)}(\Omega),\ \overline{u} \le u_*,$$

$$\Rightarrow u_* \in S_+,\ u_* = \inf S_+.$$

Similarly we produce $v_* = \sup S_-$. The set is upward directed and so we can find $\{v_n\}_{n\in\mathbb{N}} \subseteq S_-$ increasing such that $\sup S_- = \sup_{n\in\mathbb{N}} v_n$. □

Using the extremal constant sign solutions, we can generate a nodal solution for problem (5.339).

Proposition 5.134 *If hypotheses H_0, H_1 or H_0', H_1' hold, then problem (5.339) admits a nodal solution*

$$y_0 \in [v_*, u_*] \cap C_0^1(\overline{\Omega}).$$

Proof Let $u_* \in intC_+$ and $v_* \in -intC_+$, be the two extremal constant sign solutions of (5.339) produced in Proposition 5.133. We introduce the Caratheodory function $\widehat{\mu} : \Omega \times \mathbb{R} \to \mathbb{R}$ defined by

$$\widehat{\mu}(z,x) = \begin{cases} f(z, v_*(z)) & \text{if } x < v_*(z), \\ f(z,x) & \text{if } v_*(z) \le x \le u_*(z), \\ f(z, u_*(z)) & \text{if } u_*(z) < x. \end{cases} \tag{5.388}$$

Also, we introduce the positive and negative trunctions of $\widehat{\mu}(z,\cdot)$, namely the Caratheodory functions

$$\widehat{\mu}_\pm(z,x) = \widehat{\mu}(z, \pm x^\pm) \text{ for all } (z,x) \in \Omega \times \mathbb{R}. \tag{5.389}$$

We set $\widehat{M}(z,x) = \int_0^x \widehat{\mu}(z,s)ds$, $\widehat{M}_{\pm}(z,x) = \int_0^x \widehat{\mu}_{\pm}(z,s)ds$ and consider the C^1-functions $\widehat{w}, \widehat{w}_{\pm} : W_0^{1,p(z)}(\Omega) \to \mathbb{R}$ defined by

$$\widehat{w}(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega \widehat{M}(z,u)dz$$

$$\widehat{w}_{\pm}(u) = \int_\Omega \frac{1}{p(z)}|Du|^{p(z)}dz + \int_\Omega \frac{1}{q(z)}|Du|^{q(z)}dz - \int_\Omega \widehat{M}_{\pm}(z,u)dz$$

$$\text{for all } u \in W_0^{1,p(z)}(\Omega).$$

Using (5.388), (5.389) and the anisotropic regularity theory, we obtain

$$K_{\widehat{w}} \subseteq [v_*, u_*] \cap C_0^1(\overline{\Omega}),\ K_{\widehat{w}_+} \subseteq [0, u^*] \cap C_+,\ K_{\widehat{w}_-} \subseteq [v_*, 0] \cap (-C_+).$$

The extremality of u_* and v_*, implies that

$$K_{\widehat{w}} \subseteq [v_*, u_*] \cap C_0^1(\overline{\Omega}),\ K_{\widehat{w}_+} = \{0, u_*\},\ K_{\widehat{w}_-} = \{0, v_*\}. \tag{5.390}$$

Claim $u_* \in intC_+$ and $v_* \in -intC_+$ are local minimizers of $w(\cdot)$.

From (5.388), (5.389) we see that $w_+(\cdot)$ is coercive. Also, it is sequentially weakly lower semicontinuous. So, we can find $\widetilde{u}_* \in W_0^{1,p(z)}(\Omega)$ such that

$$\widehat{w}_+(\widetilde{u}_*) = \inf\{\widehat{w}_+(u) : u \in W_0^{1,p(z)}(\Omega)\}. \tag{5.391}$$

On account of hypothesis $H_1(\text{iv})(= H_1'(\text{iv}))$ and since $u_* \in intC_+$, we see that $\widehat{w}_+(\widetilde{u}_*) < 0 = \widehat{w}_+(0)$, hence $\widetilde{u}_* \neq 0$. From (5.391) we see that $\widetilde{u}_* \in K_{\widehat{w}_+} = \{0, u_*\}$ (see (5.390)). Therefore $\widetilde{u}_* = u_* \in intC_+$. Clearly

$$\widehat{w}|_{c_+} = \widehat{w}_+|_{c_+},$$

$$\Rightarrow u_* \text{ is a } C_0^1(\overline{\Omega}) - \text{local minimizer of } w(\cdot),$$

$$\Rightarrow u_* \text{ is a } W_0^{1,p(z)}(\Omega) - \text{local minimizer of } w(\cdot) \text{ (see Theorem 5.45)}.$$

Similarly for $v_* \in -intC_+$ using this time the functional $w_+(\cdot)$. This proves the Claim.

We may assume that $w(v_*) \leq w(u_*)$ and that K_w is finite (see (5.390)). Then using the "Claim", Proposition 3.132 and Theorem 3.41 (the Mountain Pass Theorem) (note that $w(\cdot)$ being coercive it satisfies the $C-$condition), we produce $y_0 \in W_0^{1,p(z)}(\Omega)$ such that

$$y_0 \in K_w \subseteq [v_*, u_*] \cap C_0^1(\overline{\Omega}),\ y_0 \notin \{u_*, v_*\},\ C_1(\widehat{w}, 0) \neq 0. \tag{5.392}$$

If we show that $y_0 \neq 0$, then we have that y_0 is the desired nodal solution (see (5.392) and take into account the extremality of u_* and v_*).

Consider the homotopy $\widehat{h}(t,u)$ defined by

$$\widehat{h}(t,u) = (1-t)\widehat{w}(u) + t\varphi(u) \text{ for all } t \in [0,1], \text{ all } u \in W_0^{1,p(z)}(\Omega).$$

Suppose that we can find $\{t_n\}_{n\in\mathbb{N}} \subseteq [0,1]$, $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p(z)}(\Omega)$ such that

$$t_n \to t \text{ in } [0,1], u_n \to 0 \text{ in } W_0^{1,p(z)}(\Omega), \widehat{h}'_u(t_n,u_n) = 0 \tag{5.393}$$

for all $n \in \mathbb{N}$.

From the equation in (5.393), we have that

$$\langle V(u_n), h\rangle = \int_\Omega [(1-t_n)\widehat{\mu}(z,u_n) + t_n f(z,u_n)]h\,dz \tag{5.394}$$

for all $h \in W_0^{1,p(z)}(\Omega)$, and all $n \in \mathbb{N}$.

If in (5.394) we choose the test function $h = u_n \in W_0^{1,p(z)}(\Omega)$ and use (5.393), we infer that $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,p(z)}(\Omega)$ is bounded. Then Theorem 5.3 implies that $\{u_n\}_{n\in\mathbb{N}} \subseteq L^\infty(\Omega)$ is bounded and so from Theorem 5.6, we can find $\alpha \in (0,1)$ and $c > 0$ such that

$$u_n \in C_0^{1,\alpha}(\overline{\Omega}), \|u_n\|_{C_0^{1,\alpha}(\overline{\Omega})} \leq c \text{ for all } n \in \mathbb{N}.$$

The compact embedding of $C_0^{1,\alpha}(\overline{\Omega})$ into $C_0^1(\overline{\Omega})$ and (5.393), imply that

$$\begin{aligned} &u_n \to 0 \text{ in } C_0^1(\overline{\Omega}), \\ &\Rightarrow u_n \in [v_*, u_*] \cap C_0^1(\overline{\Omega}) \text{ for all } n \geq n_0 \\ &\Rightarrow \{u_n\}_{n\geq n_0} \subseteq K_w \text{ (see (5.390))}, \end{aligned}$$

which contradicts our assumption that K_w is finite. Therefore (5.393) can not occur and then Theorem 3.131 (homotopy invariance of critical groups) implies that $C_k(w,0) = C_k(\varphi,0) = 0$ for all $k \in \mathbb{N}_0$ (see Lemma 5.126). From (5.392) we infer that $y_0 \neq 0$. So, $y_0 \in C_0^1(\overline{\Omega})$ is a nodal solution of (5.339) □

Concluding, we can state the following multiplicity theorem for problem (5.339).

Theorem 5.135 *If hypotheses H_0, H_1 or H'_0, H'_1 hold, then problem (5.339) has at least four nontrivial solutions $u_0, \widehat{u} \in intC_+$, $v_0 \in -intC_+$, $y_0 \in C_0^1(\overline{\Omega})$ nodal and $v_0 \leq y_0 \leq u_0, \widehat{u}$.*

Now we turn our attention to double phase problems (problems with unbalanced growth). So, let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with Lipschitz boundary $\partial\Omega$. We will study the following Dirichlet eigenvalue problem

$$\{-\Delta_p^a u(z) - \Delta_q u(z) = \lambda a(z)|u(z)|^{p-2}u(z) \text{ in } \Omega, u|_{\partial\Omega} = 0, 1 < q < p < N, \lambda > 0.\} \quad (E_\lambda)$$

Recall that Δ_p^a is the weighted p-Laplacian with weight $a \in L^\infty(\Omega)\setminus\{0\}$, $a(z) \geq 0$ for $a.a\ z \in \Omega$, defined by

$$\Delta_p^a u = div(a(z)|Du|^{p-2}Du).$$

If $a \equiv 1$, then we recover the usual p-Laplacian. However, here we do not assume that the weight $a(\cdot)$ is bounded away from zero. For this reason the right functional framework for problem (E_λ) is provided by the generalized Orlicz spaces. More, precisely, we will work with the space $W_0^{1,\eta}(\Omega)$, where $\eta(z,t)$ is the double phase density, given by

$$\eta(z,t) = a(z)t^p + t^q \text{ for all } z \in \Omega, \text{ all } t \geq 0.$$

By $\mathcal{A}_p$ we denote the p-Muckhenhoupt class (see Definition 4.161).

Our hypotheses on the data of problem (E_λ), are the following:

H_2: $1 < q < p < N$ with $\frac{p}{q} < 1 + \frac{1}{N}$ and $a \in C^{0,1}(\overline{\Omega}) \cap \mathcal{A}_p$ with $a(z) > 0$ for all $z \in \Omega$.

Remark 5.136 The condition $\frac{p}{q} < 1 + \frac{1}{N}$, says that the two exponents p, q can not be far apart and implies that $p < q^* = \frac{Nq}{N-q}$. This leads to compact embeddings of some relevant spaces (see Theorem 4.157). The hypothesis that $a \in C^{0,1}(\overline{\Omega})$ implies that the Poincare inequality is valid on $W_0^{1,\eta}(\Omega)$. So, on this Orlicz-Sobolev space, we use the norm $\|u\| = \|Du\|_\eta$ (see Proposition 4.155). The additional hypothesis that $a \in \mathcal{A}_p$, implies that the eigenvalue problem

$$-\Delta_p^a u(z) = \widehat{\lambda} a(z)|u(z)|^{p-2}u(z) \text{ in } \Omega, u|_{\partial\Omega} = 0,$$

has a first eigenvalue $\widehat{\lambda}_1^a > 0$ and the corresponding eigenfunctions $\widehat{u} \in W_0^{1,\eta}(\Omega) \cap L^\infty(\Omega)$ have constant sign and we have $\widehat{u}(z) > 0$ or $\widehat{u}(z) < 0$ for $a.a\ z \in \Omega$ (see Proposition 4.164).

As in Chap. 4, we set $\eta_0(z,t) = a(z)t^p$ for all $z \in \Omega$, all $t \geq 0$ and $\varrho_{\eta_0}(\cdot)$ is the corresponding modular function defined by

$$\varrho_{\eta_0}(y) = \int_\Omega a(z)|y|^p dz \text{ for all } y \in L^{\eta_0}(\Omega).$$

We introduce the following quantity

$$\lambda^* = \inf\{\frac{\varrho_{\eta_0}(Du) + \frac{p}{q}\|Du\|_q^q}{\varrho_{\eta_0}(u)} : u \in W_0^{1,\eta}(\Omega), u \neq 0\}. \quad (5.395)$$

Proposition 5.137 *If hypotheses H_2 hold, then $\lambda^* = \widehat{\lambda}_1^a > 0$.*

Proof From the proof of Proposition 4.164, we know that

$$\widehat{\lambda}_1^a = \inf\{\frac{\varrho_{\eta_0}(Du)}{\varrho_{\eta_0}(u)} : u \in W_0^{1,\eta}(\Omega), u \neq 0\}. \tag{5.396}$$

Then comparing (5.395) and (5.396), we see that $\widehat{\lambda}_1^a \leq \lambda^*$. Let $\widehat{u} \in W_0^{1,\eta}(\Omega) \cap L^\infty(\Omega)$ be an eigenfunction corresponding to $\widehat{\lambda}_1^a$. Then

$$\begin{aligned}\lambda^* \leq \frac{\varrho_{\eta_0}(D(t\widehat{u})) + \frac{p}{q}\|D(t\widehat{u})\|_q^q}{\varrho_{\eta_0}(t\widehat{u})} &= \frac{\widehat{\lambda}_1^a \varrho_{\eta_0}(t\widehat{u}) + \frac{p}{q}\|D(t\widehat{u})\|_q^q}{\varrho_{\eta_0}(tu)} \\ &= \widehat{\lambda}_1^a + \frac{p}{q}\frac{1}{t^{p-q}}\|D\widehat{u}\|_q^q.\end{aligned}$$

Let $t \to +\infty$. We obtain $\lambda^* \leq \widehat{\lambda}_1^a$ and so finally $\lambda^* = \widehat{\lambda}_1^a > 0$. □

For $\lambda > 0$, let $\varphi_\lambda : W_0^{1,\eta}(\Omega) \to \mathbb{R}$ be the energy functional for problem (E_λ) defined by

$$\varphi_\lambda(u) = \frac{1}{p}\varrho_{\eta_0}(Du) + \frac{1}{q}\|Du\|_q^q - \frac{\lambda}{p}\varrho_{\eta_0}(u) \text{ for all } u \in W_0^{1,\eta}(\Omega).$$

Evidently $\varphi_\lambda \in C^1(W_0^{1,\eta}(\Omega))$ and

$$\langle \varphi_\lambda'(u), h\rangle = \langle V(u), h\rangle - \lambda \int_\Omega a(z)|u|^{p-2}uhdz \text{ for all } u, h \in W_0^{1,\eta}(\Omega).$$

Recall that $V : W_0^{1,\eta}(\Omega) \to W_0^{1,\eta}(\Omega)^*$ is defined by

$$\langle V(u), h\rangle = \int_\Omega [a(z)|Du|^{p-2} + |Du|^{q-2}](Du, Dh)_{\mathbb{R}^N}dz \text{ for all } u, h \in W_0^{1,\eta}(\Omega).$$

We know (see Proposition 5.50), that $V(\cdot)$ is bounded, continuous, strictly monotone (thus maximal monotone too) and of type $(S)_+$. We introduce the Nehari manifold corresponding to $\varphi_\lambda(\cdot)$ which is

$$N_\lambda = \{u \in W_0^{1,\eta}(\Omega) : \langle \varphi_\lambda'(u), u\rangle = 0, u \neq 0\}.$$

We will determine the spectrum of (E_λ), that is, those parameters $\lambda > 0$ for which problem (E_λ) has a nontrivial solution. We denote the spectrum of (E_λ), by $\sigma(p, q) \subseteq (0, +\infty)$.

We start with a negative result.

Proposition 5.138 *If hypotheses H_2 hold and $0 < \lambda \leq \widehat{\lambda}_1^a$, then $\lambda \notin \sigma(p,q)$.*

Proof Arguing by contradiction, suppose $\lambda \in \sigma(p,q)$ and let u be a corresponding eigenfunction. Then

$$\begin{aligned}
&\varrho_{\eta_0}(Du) + \|Du\|_q^q = \lambda \varrho_{\eta_0}(u),\\
&\Rightarrow \widehat{\lambda}_1^a \varrho_{\eta_0}(u) < \widehat{\lambda}_1^a \varrho_{\eta_0}(u) + \|Du\|_q^q\\
&\qquad\qquad \leq \varrho_{\eta_0}(Du) + \|Du\|_q^q = \lambda \varrho_{\eta_0}(u) \text{ (see (5.396))},\\
&\Rightarrow (\widehat{\lambda}_1^a - \lambda)\varrho_{\eta_0}(u) < 0, \text{ a contradiction.}
\end{aligned}$$

□

Proposition 5.139 *If hypotheses H_2 hold and $\lambda > \widehat{\lambda}_1^a$, then $N_\lambda \neq \emptyset$.*

Proof From (5.396) we see that we can find $\widehat{u} \in W_0^{1,\eta_0}(\Omega) \setminus \{0\}$ such that

$$\varrho_{\eta_0}(Du) < \lambda \varrho_{\eta_0}(u).$$

Since $W_0^{1,\eta_0}(\Omega) \hookrightarrow W_0^{1,\eta}(\Omega)$ continuously and densely and the modular function $\varrho_{\eta_0}(\cdot)$ is continuous, we can have that $\widehat{u} \in W_0^{1,\eta}(\Omega) \setminus \{0\}$. Then we define

$$t_0 = \left[\frac{\|D\widehat{u}\|_q^q}{\lambda \varrho_{\eta_0}(\widehat{u}) - \varrho_{\eta_0}(D\widehat{u})}\right]^{\frac{1}{p-q}} > 0.$$

We have

$$\begin{aligned}
&t_0^p[\lambda \varrho_{\eta_0}(\widehat{u}) - \varrho_{\eta_0}(D\widehat{u})] = t_0^q \|D\widehat{u}\|_q^q,\\
&\Rightarrow \lambda \varrho_{\eta_0}(t_0\widehat{u}) - \varrho_{\eta_0}(D(t_0\widehat{u})) = \|D(t_0\widehat{u})\|_q^q\\
&\Rightarrow \langle \varphi_\lambda'(t_0\widehat{u}), t_0\widehat{u}\rangle = 0, \text{ hence } t_0\widehat{u} \in N_\lambda.
\end{aligned}$$

□

For $\lambda > \widehat{\lambda}_1^a$, we do not expect $\varphi_\lambda(\cdot)$ to be coercive, However, in the next proposition we show that $\varphi_\lambda|_{N_\lambda}$ is coercive. This result highlights the significance of the Nehari manifold.

Proposition 5.140 *If hypotheses H_2 hold and $\lambda > \widehat{\lambda}_1^a$, then $\varphi_\lambda|_{N_\lambda}$ is coercive.*

Proof We argue by contradiction. Suppose, we can find a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq N_\lambda$ such that

$$\|u_n\| \to \infty \text{ and } \varphi_\lambda(u_n) \leq c \text{ for some } c > 0, \text{ all } n \in \mathbb{N}. \tag{5.397}$$

Let $y_n = \dfrac{u_n}{\|u_n\|_{\eta_0}}$ $n \in \mathbb{N}$. Then $\|y_n\|_{\eta_0} = 1, n \in \mathbb{N}$. Since $u_n \in N_\lambda$, we have

$$\varrho_{\eta_0}(Du_n) + \|Du_n\|_q^q = \lambda \varrho_{\eta_0}(u_n), \tag{5.398}$$

$$\Rightarrow \frac{\varrho_{\eta_0}(Du_n)}{\|u_n\|_{\eta_0}^q} + \|Dy_n\|_q^q = \lambda \frac{\varrho_{\eta_0}(u_n)}{\|u_n\|_{\eta_0}^q}. \tag{5.399}$$

From (5.397) we have

$$\begin{aligned} &\frac{1}{q}\|Du_n\|_q^q \leq c + \frac{1}{p}[\lambda \varrho_{\eta_0}(u_n) - \varrho_{\eta_0}(Du_n)], \qquad (5.400)\\ &\Rightarrow \frac{1}{q}\|Dy_n\|_q^q \leq \frac{c}{\|u_n\|_{\eta_0}^q} + \frac{1}{p}[\frac{\lambda \varrho_{\eta_0}(u_n) - \varrho_{\eta_0}(Du_n)}{\|u_n\|_{\eta_0}^q}]\\ &\qquad = \frac{c}{\|u_n\|_{\eta_0}^q} + \frac{1}{p}\|Dy_n\|_q^q \text{ (see (5.398))},\\ &\Rightarrow [\frac{1}{q} - \frac{1}{p}]\|Dy_n\|_q^q \leq \frac{c}{\|u_n\|_{\eta_0}^q}. \end{aligned}$$

Note that on account of (5.397), we have $\varrho_\eta(Du_n) \to \infty$ (see Proposition 4.159). Then (5.398) implies that $\varrho_{\eta_0}(u_n) \to \infty$ and so $\|u_n\|_{\eta_0} \to \infty$. Then from (5.400) we see that

$$y_n \to 0 \text{ in } W_0^{1,q}(\Omega). \tag{5.401}$$

We have

$$\begin{aligned} \|Dy_n\|_q^q &= \|u_n\|_{\eta_0}^{p-q}[\lambda \varrho_{\eta_0}(y_n) - \varrho_{\eta_0}(Dy_n)]\\ &= \|u_n\|_{\eta_0}^{p-q}[\lambda - \varrho_{\eta_0}(Dy_n)] \text{ (since } \|y_n\|_{\eta_0} = 1, \text{ see Proposition 4.159)},\\ \Rightarrow \varrho_{\eta_0}(Dy_n) &= \lambda - \frac{\|Dy_n\|_q^q}{\|u_n\|_{\eta_0}^{p-q}} \text{ for all } n \in \mathbb{N},\\ \Rightarrow &\{y_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta_0}(\Omega) \text{ is bounded},\\ \Rightarrow &y_n \to 0 \text{ in } L^{\eta_0}(\Omega) \text{ (see (5.401) and Proposition 4.163)}. \end{aligned}$$

But this contradicts the fact that $\|y_n\|_{\eta_0} = 1$ for all $n \in \mathbb{N}$. This proves that $\varphi_\lambda|_{N_\lambda}$ is coercive. □

Let $m_\lambda = \inf_{N_\lambda} \varphi_\lambda$, $\lambda > \widehat{\lambda}_1^a$.

Proposition 5.141 *If hypotheses H_2 hold and $\lambda > \widehat{\lambda}_1^a$, then $m_\lambda > 0$.*

Proof Let $u \in N_\lambda$. We have

$$\varrho_{\eta_0}(Du) + \|Du\|_q^q = \lambda \varrho_{\eta_0}(u). \tag{5.402}$$

Since $p > q$, $u \neq 0$, we have

$$\begin{aligned} \varphi_\lambda(u) &= \frac{1}{p}\varrho_{\eta_0}(Du) + \frac{1}{q}\|Du\|_q^q - \frac{\lambda}{p}\varrho_{\eta_0}(u) \\ &= [\frac{1}{q} - \frac{1}{p}]\|Du\|_q^q \text{ (see (5.402))} \\ &> 0, \\ \Rightarrow m_\lambda &\geq 0. \end{aligned} \tag{5.403}$$

Suppose $m_\lambda = 0$. Then we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq N_\lambda$ such that

$$\begin{aligned} &\varphi_\lambda(u_n) \downarrow m_\lambda = 0 \text{ as } n \to \infty, \\ &\Rightarrow u_n \to 0 \text{ in } W_0^{1,q}(\Omega). \end{aligned} \tag{5.404}$$

From Proposition 5.140, we have that $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta}(\Omega)$ is bounded. The reflexivity of $W_0^{1,\eta}(\Omega)$ (see Remark 4.154) implies that we may assume that

$$\begin{aligned} &u_n \xrightarrow{w} \widehat{u} \text{ in } W_0^{1,\eta}(\Omega), \\ &\Rightarrow \widehat{u} = 0 \text{ (see (5.404))}. \end{aligned}$$

So, we have

$$u_n \xrightarrow{w} 0 \text{ in } W_0^{1,\eta}(\Omega),\ u_n \to 0 \text{ in } L^\eta(\Omega) \text{ (see Theorem 4.157)}. \tag{5.405}$$

Set $v_n = \frac{u_n}{\|u_n\|_\eta}$. Repeating the arguments in the proof of Proposition 5.141, we obtain

$$\begin{aligned} \|Dv_n\|_q^q &= \|u_n\|_\eta^{p-q}[\lambda \varrho_{\eta_0}(v_n) - \varrho_{\eta_0}(Dv_n)] \\ &\leq \|u_n\|_\eta^{p-q}[\lambda - \widehat{\lambda}_1^a]\varrho_{\eta_0}(v_n) \\ &\leq \|u_n\|_\eta^{p-q}[\lambda - \widehat{\lambda}_1^a]\varrho_\eta(v_n) \\ &= \|u_n\|_\eta^{p-q}(\lambda - \widehat{\lambda}_1^a) \\ \Rightarrow v_n &\to 0 \text{ in } W_0^{1,\eta}(\Omega),\ v_n \to 0 \text{ in } L^p(\Omega) \text{ (since } p < q^*). \end{aligned} \tag{5.406}$$

We have

$$0 \leq \varrho_{\eta_0}(v_n) = \int_\Omega a(z)|v_n|^p dz \leq \|a\|_\infty \|v_n\|_p^p \text{ for all } n \in \mathbb{N},$$
$$\Rightarrow \varrho_{\eta_0}(v_n) \to 0,$$
$$\Rightarrow \varrho_\eta(v_n) \to 0 \text{ (see (5.406))},$$
$$\Rightarrow v_n \to 0 \text{ in } L^\eta(\Omega), \text{ contradicting that } \|v_n\|_\eta = 1, n \in \mathbb{N}.$$

This proves that $m_\lambda > 0$. □

Proposition 5.142 *If hypotheses H_2 hold and $\lambda > \widehat{\lambda}_1^a$, then there exists $\widehat{u}_\lambda \in N_\lambda$ such that $\varphi_\lambda(\widehat{u}_\lambda) = m_\lambda$.*

Proof Consider a minimizing sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq N_\lambda$. On account of Proposition 5.140, $\{u_n\}_{n\in\mathbb{N}} \subseteq W_0^{1,\eta}(\Omega)$ is bounded. So, we may assume that

$$u_n \xrightarrow{w} \widehat{u}_\lambda \text{ in } W_0^{1,\eta}(\Omega), u_n \to \widehat{u}_\lambda \text{ in } L^\eta(\Omega). \tag{5.407}$$

Since $u_n \in N_\lambda$ $n \in \mathbb{N}$, we have

$$\varrho_\eta(Du_n) = \varrho_{\eta_0}(Du_n) + \|Du_n\|_q^q = \lambda \varrho_{\eta_0}(u_n) \leq \lambda \varrho_\eta(u_n), n \in \mathbb{N}. \tag{5.408}$$

Suppose $\widehat{u}_\lambda = 0$. Then from (5.407) and (5.408), we see that

$$u_n \to 0 \text{ in } W_0^{1,\eta}(\Omega),$$
$$\Rightarrow \varphi_\lambda(u_n) \to 0,$$
$$\Rightarrow m_\lambda = 0, \text{ contradicting Proposition 5.141}.$$

Therefore $\widehat{u}_\lambda \neq 0$.
From (5.407) and the sequential weak lower semicontinuity of $\varphi_\lambda(\cdot)$, we have

$$\varphi_\lambda(\widehat{u}_\lambda) \leq \liminf_{n\to\infty} \varphi_\lambda(u_n) = m_\lambda. \tag{5.409}$$

Assume that $\widehat{u}_\lambda \notin N_\lambda$ and consider the function $k_\lambda : [0, 1] \to \mathbb{R}$ defined by

$$k_\lambda(t) = \langle \varphi_\lambda'(t\widehat{u}_\lambda), t\widehat{u}_\lambda \rangle = \varrho_{\eta_0}(D(t\widehat{u}_\lambda)) + \|D(t\widehat{u}_\lambda)\|_q^q - \lambda \varrho_{\eta_0}(t\widehat{u}_\lambda).$$

Since $\widehat{u}_\lambda \notin N_\lambda$, we have that

$$\varrho_{\eta_0}(D\widehat{u}_\lambda) + \|D\widehat{u}_\lambda\|_q^q < \lambda \varrho_{\eta_0}(\widehat{u}_\lambda), \tag{5.410}$$
$$\Rightarrow k_\lambda(1) < 0.$$

On the other hand, using (5.396), we have

$$\begin{aligned} k_\lambda(t) &\geq t^q \|Du\|_q^q - t^p(\lambda - \widehat{\lambda}_1^q)\varrho_{\eta_0}(u_\lambda) \\ &= c_1 t^q - c_2 t^q \text{ for some } c_1, c_2 > 0. \end{aligned}$$

Since $q < p$, it follows that

$$k_\lambda(t) > 0 \text{ for } t \in (0,1) \text{ small.} \tag{5.411}$$

Then (5.410), (5.411) and Bolzano's theorem, imply that there exists $\widehat{t} \in (0,1)$ such that

$$\begin{aligned} &k(\widehat{t}) = \langle \varphi'_\lambda(\widehat{t}\widehat{u}_\lambda), \widehat{t}\widehat{u}_\lambda \rangle = 0 \\ &\Rightarrow \widehat{t}\widehat{u}_\lambda \in N_\lambda. \end{aligned}$$

We have

$$\begin{aligned} m_\lambda &\leq \varphi_\lambda(\widehat{t}\widehat{u}_\lambda) \\ &= [\frac{1}{q} - \frac{1}{p}]\widehat{t}^q \|D\widehat{u}_\lambda\|_q^q \\ &< [\frac{1}{q} - \frac{1}{p}]\|D\widehat{u}_\lambda\|_q^q \text{ (since } \widehat{t} \in (0,1)) \\ &\leq [\frac{1}{q} - \frac{1}{p}] \liminf_{n\to\infty} \|Du_n\|_q^q \text{ (see (5.407) and Theorem 4.157)} \\ &= \liminf_{n\to\infty} \varphi_\lambda(u_n) = m_\lambda, \end{aligned}$$

a contradiction. So, $\widehat{u}_\lambda \in N_\lambda$ and from (5.409) we have

$$\varphi_\lambda(\widehat{u}_\lambda) = m_\lambda.$$

□

Now we are ready to show that if $\lambda > \widehat{\lambda}_1^q$, then $\lambda \in \sigma(p,q)$.

Theorem 5.143 *If hypotheses H_2 hold, then $\sigma(p,q) = (\widehat{\lambda}_1^q, \infty)$ and the eigenfunctions $\widehat{u}_\lambda \in W_0^{1,\eta}(\Omega) \cap L^\infty(\Omega)$; moreover, we may assume that $\widehat{u}_\lambda$ is positive and we have $\widehat{u}_\lambda(z) > 0$ for a.a $z \in \Omega$.*

Proof Consider the function $\psi_\lambda : W_0^{1,\eta}(\Omega) \to \mathbb{R}$ defined by

$$\psi_\lambda(u) = \langle \varphi'_\lambda(u), u \rangle = \varrho_{\eta_0}(Du) + \|Du\|_q^q - \lambda \varrho_{\eta_0}(u) \text{ for all } u \in W_0^{1,\eta}(\Omega).$$

Evidently $\psi_\lambda \in C^1(W_0^{1,\eta}(\Omega))$. From Proposition 5.142, we know that there exists $\widehat{u}_\lambda \neq 0$ such that

$$\varphi_\lambda(\widehat{u}_\lambda) = \min\{\varphi_\lambda(u) : u \in W_0^{1,\eta}(\Omega), \psi_\lambda(u) = 0\}.$$

Invoking the Lagrange multiplier rule (see Theorem 3.87) we can find $\vartheta \in \mathbb{R}$ such that

$$\varphi'_\lambda|\widehat{u}_\lambda| = \vartheta\,\psi'_\lambda(\widehat{u}_\lambda).$$

Suppose $\vartheta \neq 0$. Since $\widehat{u}_\lambda \in N_\lambda$, we have

$$\begin{aligned} &\vartheta\,\langle \psi'_\lambda(\widehat{u}_\lambda), \widehat{u}_\lambda\rangle = 0, \\ &\Rightarrow 0 = \vartheta(q-p)\|D\widehat{u}_\lambda\|_q^q \neq 0. \end{aligned}$$

a contradiction. So, $\vartheta = 0$ and we have

$$\begin{aligned} &\varphi'_\lambda(\widehat{u}_\lambda) = 0, \\ &\Rightarrow \widehat{u}_\lambda \in K_{\varphi_\lambda}. \end{aligned}$$

So, $\widehat{u}_\lambda$ is a nontrivial solution of (E_λ) and $\widehat{u}_\lambda \in W_0^{1,\eta}(\Omega) \cap L^\infty(\Omega)$. Moreover, we can always replace $\widehat{u}_\lambda$ by $|\widehat{u}_\lambda|$ and so we may assume that $\widehat{u}_\lambda \geq 0$. Finally invoking Theorem 5.10 we conclude that $0 < \widehat{u}_\lambda(z)$ for $a.a\ z \in \Omega$. □

Remarks

5.1: The regularity theory for anisotropic elliptic equations, was first developed Fan and Zhao [161] (boundedness of the weak solutions) and Fan [157] (local and global continuity of weak solutions). The work of Fan and Zhao [161] on the boundedness of the weak solutions, was extended to more general anisotropic equations by Winkert and Zacher [473] and Papageorgiou et al. [408] (see also the isotropic work of Marino and Winkert [330]). The regularity of the weak solutions of singular anisotropic problems, was established by Saoudi and Ghanmi [435], following the arguments of Giacomoni et al. [204] for isotropic problems (see also Giacomoni and Saoudi [203] and Giacomoni et al. [205]). The anisotropic Hopf maximum principle (Theorem 5.9), was proved by Zhang [480] (nonsingular case) and Papageorgiou et al. [408] (singular case). The corresponding result for double phase problems (Theorem 5.10), can be found in Papageorgiou et al. [409]. The antimaximum principle (Proposition 5.21 (semilinear case) and Theorem 5.28 (nonlinear case)), is studied in the papers of Godoy et al. [209], [210]. (see also Motreanu et al. [344], Sect. 9.3). While

for semilinear equations, it is easy to have strong comparison results using the Hopf maximum principle, for nonlinear equations this is a considerably more complicated problem. We refer to the works of Arcoya and Ruiz [18], Cuesta and Takac [117], Damascelli [121], Damascell and Sciunzi [122], Gasinski and Papageorgiou [188], Roselli and Sciunzi [429] (isotropic equations), Papageorgiou et al. [398], Papageorgiou et al. [399] (anisotropic problems) and Giacomoni et al. [204], Papageorgiou et al. [401] (singular problems). We also refer to the book of Pucci and Serrin [423] where one can find the "tangency principle" (Theorem 5.39) as well as additional comparison results. The results on the comparison between C^1 and $W_0^{1,p}$ local minimizers, are an outgrowth of the nonlinear regularity theory. The first such result was proved for $H_0^1(\Omega)$ with $G(y) = \frac{1}{2}|y|^2$, $y \in \mathbb{R}^N$, by Brezis and Nirenberg [63]. It was extended to the space $W_0^{1,p}(\Omega)$ with $G(y) = \frac{1}{p}|y|^p$, $y \in \mathbb{R}^N$, by Garcia Azorero et al. [177] and Guo and Zhang [219]. Extensions to problems with more general differential operators and more general reactions, were obtained by Bai et al. [29], Gasinski and Papageorgiou [183], Giacomoni and Saoudi [203], Papageorgiou et al. [408]. The Brezis and Lieb lemma (see Lemma 5.49 and Brezis and Lieb[61]), is a refinement of Fatou's lemma and says that the difference $\|u_n\|_p - \|u\|_p$ behaves exactly like $\|u_n - u\|_p$. So, the Brezis-Lieb lemma identifies the missing term in the inequality of Fatou's lemma.

5.2: The analysis of problem (5.79) is based on the work of Papageorgiou and Radulescu [364]. We want to stress the result in Step 3 in the proof of Proposition 5.58. From Sect. 3.6, we know that in general, the precise computation of the critical groups requires a Hilbert space environment. In the case of $(p, 2)-$equations although the ambient space is $W_0^{1,p}(\Omega)(5.2 < p)$, using the results of Cingolani and Vanella [102], we were able locally to embed our setting to a Hilbert space and use the power of critical groups in such an ambient space. The "Lyapunov-Schmidt reduction method" used to treat the singular case of problem (5.79) ($1 < p < 2$, see Theorem 5.72) was first employed in the context of semilinear equations driven by the Laplacian, by Amann [11], Castro and Lazer [86] and Thews [453]. The work of Liu and Li [305] makes the connection between the reduction method and critical groups. Additional results on $(p, 2)-$equations can be found in the works of Baldelli et al. [31], Fu and Sun [172], Gasinski and Papageorgiou [191], [184], He et al. [236], Li et al. [296], Papageorgiou and Radulescu [365], Papageorgiou et al. [391], Papageorgiou et al. [406], Papageorgiou and Winkert [377], Papageorgiou and Zhang [386], Pei and Zhang [418], Sun et al. [448].

Brezis and Oswald [64] considered the following semilinear Dirichlet problem

$$-\Delta u(z) = f(z, u(z)) \text{ in } \Omega, u|_{\partial\Omega} = 0, u > 0. \tag{5.412}$$

Under very general conditions on the reaction $f(z, x)$, they produced necessary and sufficient conditions for the solvability of (5.412). The result of Brezis and Oswald [64], was extended to Dirichlet p-Laplacian equations, by Diaz and

Saa [138]. Here, we extend the result to Robin problems driven by a general nonhomogeneous differential operator, based on the work of Fragnelli et al. [170]. Equations driven by nonhomogeneous differential operators were studied in He et al. [235], He et al. [237], He et al. [234], Motreanu and Tanaka [343], Papageorgiou and Radulescu [366], [371], [367], [369], [368] Papageorgiou et al. [414], Tanaka [452].

The analysis of problem (p_λ) is based on the work of Gasinski and Papageorgiou [179]. Analogous results can be found in the works of Gasinski et al. [196], Liu and Papageorgiou [314].

5.3: The systematic study of singular problem started with the work of Crandall et al. [112], who proved the existence of a classical solution for an equation driven by a general linear elliptic differential operator. The solution was obtained by considering the regularized version of the problem. Since then there have been several papers on semilinear singular problems. We mention the works of Choi et al. [100], Coclite [106], Coclite and Palmieri [107], Ghergu and Radulescu [199], Gui and Lin [217], Haitao [233], Hernandez et al. [240], Lair and Shaker [286], Lazer and McKenna [289], Shi and Yao [439], Sun et al. [449]. A comprehensive presentation of the theory of semilinear singular problems, can be found in the book of Ghergu and Radulescu [200]. For nonlinear singular problems, Hardy's inequality (see Proposition 5.91), is a useful tool. Let us give two more versions of this inequality.

Proposition 5.144 *If* $1 < p < N$, $\eta \in [0, 1]$, $\frac{1}{r} = \frac{1}{p} - \frac{1-\eta}{N}$, *then there exists* $c = c(\Omega, N, p, \eta) > 0$ *such that* $\|\frac{u}{\widehat{d}^\eta}\|_r \leq c\|Du\|_r$ *for all* $u \in W_0^{1,p}(\Omega)$ *(recall that* $\widehat{d}(\cdot) = d(\cdot, \partial\Omega)$*).*

Proposition 5.145 *If* $u \in C_0^{1,\alpha}(\overline{\Omega})$ *with* $\alpha \in (0, 1]$, *then* $\frac{u}{\widehat{d}} \in C^{0,\beta}(\overline{\Omega})$ *with* $\beta = \frac{\alpha}{1+\alpha}$ *and there exists* $c = c(\Omega) > 0$ *such that*

$$\|\frac{u}{\widehat{d}}\|_{C^{0,\beta}(\overline{\Omega})} \leq c\|u\|_{C_0^{1,\alpha}(\overline{\Omega})}.$$

Remark 5.146 If $u \in C_0^1(\overline{\Omega})$, the $\frac{u}{\widehat{d}} \in C_0(\overline{\Omega})$ and there exists $c = c(\Omega) > 0$ such that $\|\frac{u}{\widehat{d}}\|_{C_0(\overline{\Omega})} \leq c\|u\|_{C_0^1(\overline{\Omega})}$.

Nonlinear singular problems, were studied Bai et al. [30], Canino et al. [84], Giacomoni et al. [204], Giacomoni et al. [205], Hai [222], Kyritsi and Papageorgiou [283], Leonardi and Papageorgiou [293], Papageorgiou and Scapellato [395], [372] Papageorgiou and Smyrlis [375], Papageorgiou et al. [412], [401], [402], Papageorgiou et al. [416], Papageorgiou and Winkert [380], [381], [382], Papageorgiou and Zhang [387], Perera and Zhang [420].

Strongly singular problems, are more difficult to deal with. We mention the works of Diaz et al. [140], Lazer and McKenna [289], Sun [447] (semilinear problems), Papageorgiou et al. [415] (equations driven by the $(p, q)-$Laplacian, see problem (p_λ^2)), Papageorgiou et al. [411] (double phase singular equations).

Finally we mention another interesting class of singular problems in which the singular term appears in the left hand side of the equation. These equations were studied by Diaz et al. [139], Davila and Monlenegro [123], Papageorgiou and Radulescu [363].

5.4: Problems with convection (gradient dependence), are nonvariational and so require either a topological approach usually based on fixed point theory or the use of the theory of nonlinear operators of monotone type. Various forms of these approaches can be found in the works of Bai [26], Bai et al. [28], de Figueiredo et al. [131], Faraci et al. [163], Gasinski et al. [195], Gasinski and Papageorgiou [185], Girardi and Matzeu [207], Guarnotta et al. [206], Hu and Papageorgiou [251], Liu and Papageorgiou [309], Motreanu and Tanaka [343], Papageorgiou et al. [402], Papageorgiou and Scapellato [373], Papageorgiou et al. [394], [405].

5.5: Anisotropic equations have been studied extensively and there are many existence and multiplicity results, primarily for equations driven by the $p(z)-$Laplacian. We can have an idea of the tools and techniques used in anisotropic problems in the works of Deng [135], Fan [159], [157], [158], Fan and Deng [160], Papageorgiou et al. [398], Papageorgiou et al. [412]. Concerning Lemma 5.125, the first computation of critical groups at zero, for functionals which have a concave term, was done by Moroz [339], but under more restrictive (global) conditions on $f(z,\cdot)$. For the isotropic case, a more general version of Lemma 5.129, can be found in Leonardi and Papageorgiou [291]. We mention that isotropic equations with asymmetric reaction, were investigated by Hu and Papageorgiou [250] and Papageorgiou and Winkert [377]. Anisotropic singular problems, are studied less extensively. In this direction, we mention the works of Byun and Ko [82], Gasinski and Papageorgiou [183], Papageorgiou et al. [408], Papageorgiou and Scapellato [374], Papageorgiou and Winkert [183], Saoudi and Ghanmi [435].

Problems with unbalanced growth (double phase problems) is an area of active research in the directions of regularity theory and on the existence and multiplicity of solutions. In the first direction, the papers of Marcellini [327], [328] and Mingione and Radulescu [337], provide a complete account of the progress made thus far. Unfortunately, a global (up to the boundary) regularity theory for double phase problems remains elusive and this makes the task in the second direction more difficult. The progress in this direction can be traced in the works of Gasinski and Papageorgiou [190], Gasinski and Winkert [193], [194], Leonardi and Papageorgiou [292], Liu and Dai [303], Liu and Winkert [315], Liu and Papageorgiou [309],[310], [308], [311], [312] Papageorgiou et al. [400], Papageorgiou et al. [409], Papageorgiou and Zhang [388], Zeng et al. [479], [477], Zeng et al. [478]. The first to make systematic use of the Nehari method in the study of elliptic problems, were Brown [75], Brown and Wu [76], Brown and Zhang [77], Wu [474]. Since then the Nehari manifold has been employed in many occasions, including double phase problems, see Papageorgiou et al. [413], Papageorgiou

and Zhang [386]. Results similar to Theorem 5.143 (continuous spectrum), were obtained by Barbu and Morosanu [35], Bhattacharaya et al. [45], Papageorgiou and Radulescu [371], Papageorgiou et al. [404]. Here we follow the very recent work of Gambera et al. [176].

Chapter 6
Evolution Equations



Introduction

In this chapter, we focus on dynamic (evolution) equations. In Sect. 6.1, we deal with Cauchy, periodic and antiperiodic systems defined in the framework of an evolution triple. We also study second order evolution inclusions and second order periodic systems involving maximal monotone operators which need not be defined on all $\mathbb{R}^N$ (differential variational inequalities). In addition we consider Lienard systems. In Sect. 6.2, we examine evolution inclusions involving nonlocal conditions. We show the nonemptiness (and compactness) of the solution set (the compactness for the convex problem). In Sect. 6.3, we show the topological triviality of the solution set. Finally in Sect. 6.4, using the notions of G and of PG convergences, we prove continuous dependence results for the solution multifunction. The main tools in this chapter, are the Lebesgue-Bochner spaces, multivalued analysis and the theory of operators of monotone-type.

6.1 Initial Value and Periodic Problems

In this section, we prove some existence results for initial value and periodic problems of evolution inclusions and of multivalued systems in $\mathbb{R}^N$. We consider multivalued systems, because we will use the results of this chapter to examine control systems in the next chapter (Chap. 7).

We start with an abstract subjectivity result, which will be a useful tool in our subsequent considerations. The setting is the following. Let (X, H, X^*) be an evolution triple (see Definition I-4.201). So, X is a separable, reflexive Banach space with dual X^*, H is a separable Hilbert space identified with its dual(by

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the Riesz-Frechet Representation Theorem, see Theorem I-3.181) and $X \hookrightarrow H$ continuously and densely. Then by Lemma I-4.200, we have $H \hookrightarrow X^*$ continuously and densely. Therefore

$$X \hookrightarrow H \hookrightarrow X^*$$

with all the embeddings being continuous and dense. We will also assume that the embedding $X \hookrightarrow H$ is in fact compact. Hence $H \hookrightarrow X^*$ is compact too (Schauder's Theorem, see Theorem I-3.156). In fact this is case in most applications. Consider $\Omega \subseteq \mathbb{R}^N$ a nonempty bounded domain with Lipschitz boundary $\partial\Omega$. We set

$$H = L^2(\Omega) \text{ and } X = W_0^{1,p}(\Omega) \text{ or } W^{1,p}(\Omega).$$

Then (X, H, X^*) is an evolution triple with the embeddings being compact. In what follows by $\|\cdot\|$(resp. $|\cdot|$, $\|\cdot\|_*$) we denote the norm of X(resp. of H, X^*). Also by $\langle\cdot,\cdot\rangle$ we denote the duality brackets for the pair (X, X^*) and by $(\cdot,\cdot)$ the inner product of H. The two are compatible in the sense that $\langle\cdot,\cdot\rangle|_{X\times H} = (\cdot,\cdot)$.

Also Y is a reflexive Banach space, both Y, Y^* are strictly convex and $L : D(L) \subseteq Y \to Y^*$ is linear, densely defined and maximal monotone.

Definition 6.1 Let $V : Y \to 2^{Y^*}$ be a multivalued map.

(a) We say that $V(\cdot)$ is "strongly coercive", if

$$\frac{\inf\{\langle y^*, y\rangle_Y : y^* \in V(y)\}}{\|y\|_Y} \to +\infty$$

as $\|y\|_Y \to \infty$(by $\|\cdot\|_Y$ we denote the norm of Y and by $\langle\cdot,\cdot\rangle_Y$ the duality brackets of the pair (Y, Y^*)).

(b) We say that $V(\cdot)$ is "L-pseudomonotone", if

[1] $V(y) \in P_{wkc}(Y^*) = \{C \subseteq Y^* : \text{nonempty, weakly compact, convex}\}$;

[2] the multifunction $V(\cdot)$ is upper semicontinuous from every $E \subseteq D(L)$ finite dimensional subspace into Y_w^*(=the space Y^* equipped with the w-topology);

[3] if $\{y_n, y\}_{n\in\mathbb{N}} \subseteq D(L)$, $y_n \xrightarrow{w} y$ in Y, $L(y_n) \xrightarrow{w} L(y)$ in Y^*, $y_n^* \in V(y_n)$ for all $n \in \mathbb{N}$, $y_n^* \xrightarrow{w} y^*$ in Y^* and

$$\limsup_{n\to\infty} \langle y_n^*, y_n\rangle_Y \le \langle y^*, y\rangle_Y,$$

then $y^* \in V(y)$ and $\langle y_n^*, y_n\rangle_Y \to \langle y^*, y\rangle_Y$.

Theorem 6.2 *If V is bounded, L-pseudomonotone, strongly coercive, then $L + V$ is surjective.*

Proof On $D(L)$ we introduce the graph norm

$$|y|_D = \|y\|_Y + \|L(y)\|_{Y^*}.$$

Then $D(L)$ becomes a reflexive Banach space. Consider the duality map $\mathcal{F} : Y \to Y^*$(see Example I-7.28(b)). The strict convexity of Y, Y^* implies that $\mathcal{F}(\cdot)$ is single-valued, bijective and $\mathcal{F}^{-1}$ is the duality map of Y^*. For every $m \in \mathbb{N}$ let $K_m : D(L) \to D(L)^*$ be defined by

$$\langle K_m(y), h\rangle_Y = \frac{1}{m}\langle \mathcal{F}^{-1}(L(y)), L(h)\rangle_Y + \langle L(y), h\rangle_Y \text{ for all } y, h \in D(L).$$

Clearly $K_m(\cdot)$ is bounded, maximal monotone, thus pseudomonotone (see Proposition I-7.73).

Let $S_m : D(L) \to 2^{D(L)^*}$ be defined by

$$S_m(y) = K_m(y) + V(y) \text{ for all } y \in D(L).$$

Note that $S_m(\cdot)$ is $P_{wkc}(D(L)^*)$-valued and usc from every finite dimensional subspace of $D(L)$ into $D(L)^*_w$. Let $\{(y_n, y_n^*)\}_{n\in\mathbb{N}} \subseteq \operatorname{Gr} S_m$ and assume that

$$y_n \xrightarrow{w} y \text{ in } D(L),\ y_n^* \xrightarrow{w} y^* \text{ in } D(L)^*,\ \limsup_{n\to\infty}\langle y_n^*, y_n - y\rangle \le 0. \tag{6.1}$$

Let $u_n^* \in V(y_n)$ such that

$$y_n^* = K_m(y_n) + u_n^*, n \in \mathbb{N}.$$

Then $\{u_n^*\}_{n\in\mathbb{N}} \subseteq D(L)^*$ is bounded and so we may assume that $u_n^* \xrightarrow{w} u^*$ in $D(L)^*$. Moreover, we have that

$$\limsup_{n\to\infty}[\langle K_m(y_n), y_n - y\rangle_Y + \langle u_n^*, y_n - y\rangle_Y] \le 0. \tag{6.2}$$

Suppose $\limsup_{n\to\infty}\langle u_n^*, y_n - y\rangle_Y = d > 0$. Then by passing to a subsequence, we may assume that $\lim_{n\to\infty}\langle u_n^*, y_n - y\rangle_Y = d > 0$. It follows from (6.2) that

$$\limsup_{n\to\infty}\langle K_m(y_n), y_n - y\rangle_Y \le -d < 0,$$

$$\Rightarrow \liminf_{n\to\infty}\langle K_m(y_n), y_n - y\rangle_Y \ge 0 \text{ (from the pseudomonotonicity of } K_m(\cdot))$$

a contradiction. Therefore, we have

$$\limsup_{n\to\infty}\langle u_n^*, y_n - y\rangle_Y \le 0,$$

$$\Rightarrow u^* \in V(y) \text{ and } \langle u_n^*, y_n\rangle_Y \to \langle u^*, y\rangle_Y \text{ (see Definition 6.1)},$$

$$\Rightarrow \limsup_{n\to\infty}\langle K_m(y_n), y_n - y\rangle_Y \le 0 \text{ (see (6.2))},$$

$$\Rightarrow K_m(y_n) \xrightarrow{w} K_m(y) \text{ in } D(L)^* \text{ (since } K_m(\cdot) \text{ is pseudomonotone)},$$

$$\Rightarrow y^* \in S_m(y).$$

We infer that $S_m(\cdot)$ is pseudomonotone.
Since $L(\cdot)$ is linear monotone, we have

$$\langle y^*, y\rangle \ge \frac{1}{m}\|L(y)\|_{Y^*}^2 + \langle u^*, y\rangle \text{ for all } y \in D(L), \text{ all } y^* \in S_m(y),$$

$$\Rightarrow \frac{\inf\{\langle y^*, y\rangle_Y : y^* \in S_m(y)\}}{|y|_D} \to +\infty \text{ as } |y|_D \to \infty \text{ (since } V(\cdot) \text{ is coercive)},$$

$$\Rightarrow S_m(\cdot) \text{ is strongly coercive.} \tag{6.3}$$

Invoking Theorem *I*-7.76, we have that $R(S_m) = D(L)^*$. Therefore given $y^* \in Y^*$, we can find $y_m \in D(L)$ such that $y^* \in S_m(y_m)$. Then

$$y^* = K_m(y_m) + u_m^* \text{ with } u_m^* \in V(y_m), m \in \mathbb{N}. \tag{6.4}$$

We have

$$\frac{1}{m}\langle \mathcal{F}^{-1}(L(y_m)), L(h)\rangle_Y = \langle y^* - u^* - L(y_m), h\rangle_Y \text{ for all } h \in D(L),$$

$$\text{all } m \in \mathbb{N},$$

$$\Rightarrow \mathcal{F}^{-1}(L(y_m)) \in D(L^*) \ \ (L^* \text{ being the dual of } L).$$

So, we can write that

$$\frac{1}{m}L^*(\mathcal{F}^{-1}(L(y_m))) + L(y_m) + u_m^* = y^* \text{ for all } m \in \mathbb{N}. \tag{6.5}$$

From (6.3) and (6.4) it follows $\{y_m\}_{m\in\mathbb{N}} \subseteq Y$ is bounded. Also from (6.5), we obtain

$$\frac{1}{m}\langle L^*(\mathcal{F}^{-1}(L(y_m))), \mathcal{F}^{-1}(L(y_m))\rangle_Y + \langle L(y_m), \mathcal{F}^{-1}(L(y_m))\rangle_Y$$

$$+\langle u_m^*, \mathcal{F}^{-1}(L(y_m))\rangle_Y$$

$$=\langle y^*, \mathcal{F}^{-1}(L(y_m))\rangle_Y \text{ for all } m \in \mathbb{N}.$$

Since L^* is monotone and $\mathcal{F}^{-1}$ is the duality map of Y^*, we have

$$\begin{aligned}
&\|L(y_m)\|^2_{Y^*} \leq \langle y^*, \mathcal{F}^{-1}(L(y_m))\rangle_Y - \langle u^*_m, \mathcal{F}^{-1}(L(y_m))\rangle_Y \\
\Rightarrow &\|L(y_m)\|^2_{Y^*} \leq \vartheta \|L(y_m)\|_{Y^*} \text{ for some } \vartheta > 0, \text{ all } m \in \mathbb{N} \\
&\quad \text{(since } V(\cdot) \text{ is bounded)}, \\
\Rightarrow &\{L(y_m)\}_{m\in\mathbb{N}} \subseteq Y^* \text{ is bounded.}
\end{aligned}$$

Therefore, we may assume that

$$y_m \xrightarrow{w} y \text{ in } Y,\ y \in D(L),\ L(y_m) \xrightarrow{w} L(y) \text{ in } Y^*,\ u^*_m \xrightarrow{w} u^* \text{ in } Y^*. \tag{6.6}$$

From (6.5) and (6.6), we obtain

$$\begin{aligned}
&\limsup_{m\to\infty} \langle u^*_m, y_m - y\rangle \leq 0, \\
\Rightarrow &u^* \in V(y) \text{ and } y^* \in L(y) + V(y), \\
\Rightarrow &R(L+V) = Y^*.
\end{aligned}$$

□

Let $T = [0, b]$, $1 < p < \infty$ and (X, H, X^*) an evolution triple as above. Recall (see Definition I-4.203), that

$$W_p(0, b) = \{u \in L^p(T, X) : u' \in L^{p'}(T, X^*)\} \text{ with } \frac{1}{p} + \frac{1}{p'} = 1.$$

Consider a map $A(t, x)$ which satisfies the following hypotheses.

$\mathbf{H_1}$: $A : T \times X \to X^*$ is a map such that

(i) for all $x \in X$, $t \to A(t, x)$ is measurable;
(ii) for a.a $t \in T$, $x \to A(t, x)$ is pseudomonotone;
(iii) $\|A(t, x)\|_* \leq a(t) + c\|x\|^{p-1}$ for a.a $t \in T$, with $a \in L^{p'}(T)$, $c > 0$ and $1 < p < \infty$;
(iv) $\langle A(t, x), x\rangle \geq \widehat{c}\|x\|^p - \vartheta(t)$ for a.a $t \in T$, all $x \in X$, with $\widehat{c} > 0$ and $\vartheta \in L^1(T)$.

Remark 6.3 On account of hypotheses H_1(ii), (iii) and Proposition I-7.70, we see that for a.a $t \in T$, $A(t, \cdot)$ is demicontinuous. So, for every $h \in X$, $(t, x) \to \langle A(t, x), h\rangle$ is a Caratheodory function thus jointly measurable (see Theorem I-2.47) and so for every $u : T \to X$ measurable, we have that $t \to A(t, u(t))$ is measurable (see Theorem I-4.144). Therefore, we can consider the lifting (Nemytskii map) of A on $L^p(T, X)$, $\widehat{A} : L^p(T, X) \to L^{p'}(T, X^*) = L^p(T, X)^*$(see Theorem I-4.173) defined by

$$\widehat{A}(u)(\cdot) = A(\cdot, u(\cdot)).$$

The next proposition reveals that the pseudomonotonicity property of $A(t,\cdot)$ is passed to $\widehat{A}(\cdot)$.

Proposition 6.4 *If hypotheses* H_1 *hold, then* $\widehat{A} : L^p(T,X) \to L^{p'}(T,X^*)$ *is bounded, pseudomonotone.*

Proof Boundedness follows from hypothesis H_1(iii).

Let $u_n \xrightarrow{w} u$ in $L^p(T,X)$ and assume that $\limsup_{n\to\infty} \langle \widehat{A}(u_n), u_n - u\rangle_p \leq 0$ with $\langle\cdot,\cdot\rangle_p$ denoting the duality brackets for the pair $(L^p(T,X), L^{p'}(T,X^*))$ (so $\langle u, h^*\rangle_p = \int_0^b \langle u(t), h^*(t)\rangle dt$ for all $u \in L^p(T,X)$, all $h^* \in L^{p'}(T,X^*)$). From Theorem *I*-4.204, we know that $W_p(0,b) \hookrightarrow C(T,H)$ continuously. Therefore we have $u_n(t) \xrightarrow{w} u(t)$ in H for all $t \in T$. We set

$$\eta_n(t) = \langle A(t,u_n(t)), u_n(t) - u(t)\rangle, \quad t \in T, n \in \mathbb{N}.$$

Let $N \subseteq T$ be the Lebesgue-null set outside of which H_1(ii),(iii),(iv) hold. For $t \in T \setminus N$, we have

$$\eta_n(t) \geq \widehat{c}\|u_n(t)\|^p - \vartheta(t) - (a(t) + c\|u_n\|^{p-1})\|u(t)\|. \tag{6.7}$$

Consider the set $C = \{t \in T : \liminf_{n\to\infty} \eta_n(t) < 0\}$. This set is measurable. We assume that $|C| > 0$($|\cdot|$ being the Lebesgue measure on $\mathbb{R}$). From (6.7) it is clear that for all $t \in C \cap (T \setminus N) \neq \emptyset$, the sequence $\{u_n(t)\}_{n\in\mathbb{N}} \subseteq X$ is bounded. For every $t \in T \setminus N$, we can find a subsequence(depending in general on t) such that

$$u_{n_k}(t) \xrightarrow{w} u(t) \text{ in } X.$$

The pseudomonotonicity of $A(t,\cdot)$, implies that

$$\langle A(t,u_{n_k}(t)), u_{n_k}(t) - u(t)\rangle \to 0,$$

a contradiction to the fact that $t \in C$. Hence $|C| = 0$ and so

$$0 \leq \liminf_{n\to\infty} \eta_n(t) \text{ for a.a } t \in T.$$

From the extended Fatou's lemma (see Proposition *I*-2.149), we have

$$\begin{aligned} 0 \leq \int_\Omega \liminf_{n\to\infty} \eta_n(t)dt &\leq \liminf_{n\to\infty} \int_\Omega \eta_n(t)dt \\ &\leq \limsup_{n\to\infty} \int_\Omega \eta_n(t)dt \leq 0, \end{aligned}$$

$$\Rightarrow \int_\Omega \eta_n(t)dt \to 0.$$

We know $|\eta_n(t)| = \eta_n^+(t) + \eta_n^-(t) = \eta_n(t) + 2\eta_n^-(t)$. We have seen above that $\eta_n^-(t) \to 0$ for a.a $t \in T$. From (6.7) we see that

$$\mu_n(t) \leq \eta_n(t) \text{ for a.a } t \in T,$$

with $\{\mu_n\}_{n\in\mathbb{N}} \subseteq L^1(T)$ uniformly integrable. We have

$$0 \leq \eta_n^- \leq \mu_n^- \text{ for all } n \in \mathbb{N} \text{ and } \{\mu_n^-\}_{n\in\mathbb{N}} \text{ is uniformly integrable.}$$

So, by Vitali's theorem (see Theorem I-2.147), we have

$$\int_0^b \eta_n^-(t)dt \to 0,$$

$$\Rightarrow \int_0^b |\eta_n(t)|dt \to 0.$$

Thus, we may assume that $\eta_n(t) \to 0$ for a.a $t \in T$. Then $\langle A(t, u_n(t)), u_n(t) - u(t)\rangle \to 0$ for a.a $t \in T$ and from the pseudomonotonicity of $A(t, \cdot)$, we have

$$A(t, u_n(t)) \xrightarrow{w} A(t, u(t)) \text{ in } X^*, \langle A(t, u_n(t)), u_n(t)\rangle \to \langle A(t, u(t)), u(t)\rangle$$
$$\text{for a.a } t \in T \text{ (see Proposition } I - 7.68),$$
$$\Rightarrow \widehat{A}(u_n) \xrightarrow{w} \widehat{A}(u) \text{ in } L^{p'}(T, X^*), \langle \widehat{A}(u_n), u_n\rangle_p \to \langle \widehat{A}(u), u\rangle_p.$$

So, we have proved that $\widehat{A}(\cdot)$ is bounded, generalized pseudomonotone. Proposition I-7.69 implies that $\widehat{A}(\cdot)$ is pseudomonotone. □

We can have a variant Proposition 6.4 by modifying the monotonicity property of $A(t, \cdot)$ (see hypothesis H_1(ii)). So, the new conditions on the map $A(t, x)$ are the following:

$\mathbf{H_1'}$: $A : T \times X \to X^*$ is a map such that hypotheses H_1'(i), (iii), (iv) are the same as hypotheses H_1(i), (iii), (iv) and

(ii) for a.a $t \in T$, $x \to A(t, x)$ is of type $(S)_+$(see Definition I-7.71).

Then with minor modifications in the proof of Proposition 6.4, we have the following result.

Proposition 6.5 *If hypotheses* H_1' *hold, then* $\widehat{A} : L^p(T, X) \to L^{p'}(T, X^*)$ *is bounded, demicontinuous and if* $u_n \xrightarrow{w} u$ *in* $W_p(0, b)$, $\limsup_{n\to\infty} \langle \widehat{A}(u_n), u_n - u\rangle_p \leq 0$, *then* $u_n \to u$ *in* $L^p(T, X)$.

We will use these results to study the following multivalued Cauchy problem

$$\left\{ \begin{array}{l} u'(t) + A(t, u(t)) \in F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = x_0 \in H. \end{array} \right\} \tag{6.8}$$

The hypotheses on the multivalued forcing term $F(t,x)$ are the following. By $P_{fc}(H)$ we denote the family of all nonempty, closed and convex subsets of H (see Section I-5.1).

$\mathbf{H_2}$: $F : T \times X \to P_{fc}(H)$ is a multifunction which satisfies

(i) for every $x \in H$, $t \to F(t,x)$ is graph measurable;
(ii) for a.a $t \in T$, Gr $F(t,\cdot)$ is sequentially closed in $H \times H_w$;
(iii) $|F(t,x)| = \sup\{|h| : h \in F(t,x)\} \leq a_0(t) + c_0|x|^{\tau-1}$ for a.a $t \in T$, all $x \in H$, with $a_0 \in L^{\tau'}(\Omega)(\frac{1}{\tau} + \frac{1}{\tau'} = 1)$, $c_0 > 0$, $1 < \tau < p$.

Definition 6.6 By a solution of (6.8), we mean a function $u \in W_p(0,b)$ such that there exists $f \in S^{\tau'}_{F(\cdot,u(\cdot))} = \{h \in L^{\tau'}(T,H) : h(t) \in F(t,u(t))$ for a.a $t \in T\}$ such that

$$u'(t) + A(t,u(t)) = f(t) \text{ for a.a } t \in T \text{ in } X^*,$$
$$u(0) = x_0.$$

Remark 6.7 Recall that $W_p(0,b) \hookrightarrow C(T,H)$(see Theorem I-4.204) and so the initial condition $u(0) = x_0$ makes sense.

By $S(x_0) \subseteq W_p(0,b)$ we denote the set of solutions of (6.8).

Theorem 6.8 *If hypotheses* H_1, H_2 *hold, then* $S(x_0) \neq \emptyset$ *and it is weakly compact in* $W_p(0,b)$ *and compact in* $C(T,H)$.

Proof First assume that $x_0 \in X$ and consider the operator $A_1(t,x)$ and the multifunction $F_1(t,x)$ defined by

$$A_1(t,x) = A(t,x+x_0) \text{ and } F_1(t,x) = F(t,x+x_0).$$

Evidently $A_1(t,x)$ still satisfies hypotheses H_1, while $F_1(t,x)$ satisfies hypotheses H_2. Then we consider the following Cauchy problem

$$\left\{\begin{array}{l} u'(t) + A_1(t,u(t)) \in F_1(t,u(t)) \text{ for a.a } t \in T, \\ u(0) = 0. \end{array}\right\} \tag{6.9}$$

Note that if $u \in W_p(0,b)$ is a solution of (6.9), then $\widehat{u} = u + x_0 \in W_p(0,b)$ is a solution of (6.8). So, we need to solve (6.9). Let $L : D(L) \subseteq L^p(T,X) \to L^{p'}(T,X^*)$ be the linear operator defined by

$$L(u) = u' \text{ for all } u \in D(L) = \{u \in W_p(0,b) : u(0) = 0\}.$$

From Proposition 9.3, p. 419, of Hu and Papageorgiou [247], we know that L is linear, densely defined and maximal monotone. Also, let $\widehat{A}_1 : L^p(T,X) \to L^{p'}(T,X^*)$ be defined by $\widehat{A}_1(u)(\cdot) = A_1(\cdot,u(\cdot))$ and $G_1 : L^p(T,X) \to$

$P_{wkc}(L^{\tau'}(T,H)) = \{C \subseteq L^{\tau'}(T,H) : \text{nonempty, weakly compact, convex}\}$ by defined by $G_1(u) = S^{\tau'}_{-F_1(\cdot,u(\cdot))}$. We set

$$V(u) = \widehat{A}_1(u) + G_1(u) \text{ for all } u \in L^p(T,X).$$

Claim 1 $V(\cdot)$ is bounded, L-pseudomonotone.

Hypotheses H_1(iii), H_2(iii), imply that $V(\cdot)$ is bounded. We show that the multivalued map $V(\cdot)$ is usc from $W_p(0,b)$ into $L^{p'}(T,X^*)_w$(recall that $L^{p'}(T,X^*) \hookrightarrow W_p(0,b)^*$ continuously and densely, see Lemma I-4.200). So, let $C \subseteq L^{p'}(T,X^*)$ be nonempty and weakly closed. We need to show that

$$V^-(C) = \{u \in W_p(0,b) : V(u) \cap C \neq 0\} \text{ (see Proposition } I\text{-5.5)}$$

is closed. So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq V^-(C)$ and assume that $u_n \to u$ in $W_p(0,b)$. We can find $v_n \in V(u_n) \cap C$, $n \in \mathbb{N}$ and

$$v_n = \widehat{A}_1(u_n) + g_n \text{ with } g_n \in G_1(u_n),\ n \in \mathbb{N}. \tag{6.10}$$

Evidently $\{g_n\}_{n\in\mathbb{N}} \subseteq L^{\tau'}(T,H)$ is bounded. So, we may assume that $g_n \xrightarrow{w} g$ in $L^{\tau'}(T,H)$. From Theorem I-8.38, we have

$$\begin{aligned} -g(t) &\in \overline{\text{conv}}\, w - \limsup_{n\to\infty}\{-g_n(t)\}, \\ &\subseteq \overline{\text{conv}}\, w - \limsup_{n\to\infty} F_1(t,u_n(t)) \\ &\subseteq F_1(t,u(t)) \text{ for a.a } t \in T \text{(see hypothesis } H_2\text{(ii))}. \end{aligned} \tag{6.11}$$

Then we have

$$\begin{aligned} &\int_0^b (g_n(t), u_n(t) - u(t))dt \to 0 \\ \Rightarrow &\lim_{n\to\infty} \langle \widehat{A}_1(u_n), u_n - u\rangle_p = 0, \\ \Rightarrow &\widehat{A}_1(u_n) \xrightarrow{w} \widehat{A}_1(u) \text{ in } L^{p'}(T,X^*) \text{ (see Proposition 6.4)}. \end{aligned}$$

So, finally we have

$$\begin{aligned} &v = \widehat{A}_1(u) + g \text{ with } g \in S^{\tau'}_{-F(\cdot,u(\cdot))} \text{ (see (6.11))}, \\ \Rightarrow &v \in V(u) \cap C, \end{aligned}$$

which implies that $V(\cdot)$ is usc from $W_p(0,b)$ into $L^{p'}(T,X^*)_w$.

We need to satisfy [4] in Definition 6.1. So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq D(L)$ and assume that $u_n \xrightarrow{w} u$ in $L^p(T,X)$, $L(u_n) = u_n' \xrightarrow{w} L(u) = u'$ in $L^{p'}(T,X^*)$(therefore $u_n \xrightarrow{w} u$ in $W_p(0,b)$), $v_n^* \in V(u_n)$, $n \in \mathbb{N}$, $v_n^* \xrightarrow{w} v^*$ in $L^{p'}(T,X^*)$ and $\limsup_{n\to\infty} \langle v_n^*, u_n - u\rangle_p \leq 0$. We have

$$v_n^* = \widehat{A}_1(u_n) + g_n \text{ with } g_n \in G_1(u_n), n \in \mathbb{N}. \tag{6.12}$$

Hypothesis H_2(iii) implies that $\{g_n\}_{n\in\mathbb{N}} \subseteq L^{\tau'}(T,H) \hookrightarrow L^{p'}(T,H)$ is bounded. So, we may assume that $g_n \xrightarrow{w} g$ in $L^{\tau'}(T,H)$. Also since $u_n \xrightarrow{w} u$ in $W_p(0,b)$, from Theorem *I*-4.207, we have that $u_n \to u$ in $L^p(T,H)$. Therefore,

$$\begin{aligned}
&\limsup_{n\to\infty} \langle \widehat{A}_1(u_n), u_n - u\rangle_p \\
=&\limsup_{n\to\infty} \langle v_n^* - g_n, u_n - u\rangle_p \text{ (see (6.12))} \\
\leq&\limsup_{n\to\infty} \langle v_n^*, u_n - u\rangle_p - \lim_{n\to\infty} \int_0^b (g_n(t), (u_n - u)(t))dt \\
=&\limsup_{n\to\infty} \langle v_n^*, u_n - u\rangle_p \\
\leq&0.
\end{aligned} \tag{6.13}$$

But from Proposition 6.4, we know that $\widehat{A}_1(\cdot)$ is pseudomonotone. So, from (6.13) it follows that

$$\widehat{A}_1(u_n) \xrightarrow{w} \widehat{A}_1(u) \text{ in } L^{p'}(T,X^*), \langle \widehat{A}_1(u_n), u_n\rangle_p \to \langle \widehat{A}_1(u), u\rangle_p.$$

Therefore, we have

$$v_n^* \xrightarrow{w} v^* = \widehat{A}_1(u) + g \text{ in } L^{p'}(T,X^*), g \in G_1(u).$$

Moreover, $\langle v_n^*, u_n\rangle_p \to \langle v^*, u\rangle_p$. We have proved Claim 1.

Claim 2 $V(\cdot)$ is strongly coercive.

Let $u \in L^p(T,X)$ and $v^* \in V(u)$. We have

$$v^* = \widehat{A}_1(u) + g \text{ with } g \in G_1(u).$$

Then

$$\langle v^*, u\rangle_p = \langle \widehat{A}_1(u), u\rangle_p + \int_0^b (g(t), u(t))dt. \tag{6.14}$$

From hypotheses $H_1(iii)$, (iv) and the definition of $A_1(t,\cdot)$, we have

$$\langle \widehat{A}_1(u), u\rangle \geq c_1\|u\|_p^p - c_2\|u\|_p^{p-1} - c_3 \text{ for some } c_1, c_2, c_3 > 0. \tag{6.15}$$

Also using Hölder's inequality, we have

$$\begin{aligned}\int_0^b (g(t), u(t))dt &\geq -c_4\|g\|_{\tau'}\|u\|_\tau \\ &\geq -c_5[\|u\|_p^\tau + 1] \text{ for some } c_4, c_5 > 0.\end{aligned} \tag{6.16}$$

We return to (6.14) and use (6.15), (6.16). Then

$$\begin{aligned}&\langle v^*, u\rangle_p \geq c_1\|u\|_p^p - c_2\|u\|_p^{p-1} - c_5\|u\|_p^\tau - c_6 \text{ for some } c_6 > 0\\ \Rightarrow &V(\cdot) \text{ is strongly coercive (recall } \tau < p).\end{aligned}$$

So, we have proved Claim 2.

On account of Claim 1 and Claim 2, we can apply Theorem 6.2 and infer that $S(x_0) \neq \emptyset$ if $x_0 \in X$.

Now we remove the restriction that $x_0 \in X$. So, let $x_0 \in H$. We can find $\{x_n\}_{n\in\mathbb{N}} \subseteq X$ such that $x_n \to x_0$ in H. For every initial condition x_n, we know that problem (6.8) has a solution $u_n \in W_p(0,b)$. Using the integration by parts formula (see Theorem I-4.205), the coercivity of $A(t,\cdot)$ and the fact that $\tau < p$, we see that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq W_p(0,b) \text{ is bounded.}$$

So, we may assume that

$$u_n \xrightarrow{w} u \text{ in } W_p(0,b). \tag{6.17}$$

We know that $u_n' + A(t,u_n) = f_n$ with $f_n \in S^{\tau'}_{F(\cdot,u_n(\cdot))}$, $n \in \mathbb{N}$. On account of hypothesis $H_2(iii)$ $\{f_n\}_{n\in\mathbb{N}} \subseteq L^{\tau'}(\Omega)$ is bounded. So, we may assume that

$$f_n \xrightarrow{w} f \text{ in } L^{\tau'}(T,H). \tag{6.18}$$

As before using (6.17) and Theorems I-4.207 and I-8.38, we infer that $f \in S^{\tau'}_{F(\cdot,u(\cdot))}$. We have

$$\langle \widehat{A}(u_n) - f_n, u_n - u\rangle_p = \langle u_n', u - u_n\rangle_p. \tag{6.19}$$

From the integration by parts formula (Theorem I-4.205), we have

$$\langle u_n', u_n - u\rangle_p = \frac{1}{2}|u_n(b) - u(b)|^2 - \frac{1}{2}|x_n - x_0|^2 + \langle u', u_n - u\rangle_p,$$

$$\Rightarrow \langle u_n', u - u_n\rangle_p \leq \frac{1}{2}|x_n - x_0|^2 + \langle u', u - u_n\rangle_p \to 0 \text{ (see (6.17))},$$

$$\Rightarrow \limsup_{n\to\infty} \langle \widehat{A}(u_n), u_n - u\rangle_p \leq 0 \text{ (see (6.18), (6.19))},$$

$$\Rightarrow \widehat{A}(u_n) \xrightarrow{w} \widehat{A}(u) \text{ in } L^{p'}(T, X^*) \text{ (see Proposition 6.4)}.$$

So, in the limit as $n \to \infty$, we obtain

$$u' + \widehat{A}(u) = f \text{ with } f \in S^{\tau'}_{F(\cdot,u(\cdot))}, u(0) = x_0,$$

$$\Rightarrow S(x_0) \neq \emptyset \text{ for all } x_0 \in H.$$

Now we prove the compactness of $S(x_0)$.

A straightforward a priori estimation, using hypotheses H_1(iv) and H_2(iii), reveals that there exists $M > 0$ such that

$$\|u\|_{W_p(0,b)} \leq M, \|u\|_{C(T,H)} \leq M \text{ for all } u \in S(x_0).$$

We set $\vartheta(t) = a_0(t) + c_0 M^{\tau-1}$. Then $\vartheta \in L^{\tau'}(T) \subseteq L^{p'}(T)$. So, we may assume that $|F(t,x)| \leq \vartheta(t)$ for a.a $t \in T$, all $x \in H$ (just replace $F(t,x)$ by $F(t, p_M(x))$ with $p_M(\cdot)$ being the M-radial retraction on H). We consider the set

$$K = \{h \in L^{p'}(T, H) : |h(t)| \leq \vartheta(t) \text{ for a.a } t \in T\}.$$

For each $h \in K$, we consider the following Cauchy problem

$$\left\{ \begin{array}{l} u'(t) + A(t, u(t)) = h(t) \text{ for a.a } t \in T, \\ u(0) = x_0. \end{array} \right\} \tag{6.20}$$

Problem (6.20) has a nonempty solution set (see Theorem 6.2). On K we consider the relative weak $L^{p'}(T, H)$-topology. Topologized this way, K becomes a compact metric space. Let $E : K \to 2^{C(T,H)} \setminus \{\emptyset\}$ be the multifunction which to each $h \in K$ assigns the nonempty solution set of (6.20).

Claim 3 $E(K) \subseteq C(T, H)$ is compact.

Let $\{u_n\}_{n\in\mathbb{N}} \subseteq E(K)$. We can find $\{h_n\}_{n\in\mathbb{N}} \subseteq K$ such that $u_n \in E(h_n)$ for all $n \in \mathbb{N}$. Then $\{u_n\}_{n\in\mathbb{N}} \subseteq W_p(0,b)$ is bounded and so we may assume that

$$u_n \xrightarrow{w} u \text{ in } W_p(0,b), u_n \to u \text{ in } L^p(T, H) \text{ (see Theorem } I\text{-4.207)},$$

$$u_n(t) \to u(t) \text{ in } H \text{ for all } t \in T \setminus N, |N| = 0, h_n \xrightarrow{w} h \text{ in } L^{p'}(T, H).$$

Note that $\{\langle u_n'(\cdot), (u_n - u)(\cdot)\rangle\}_{n\in\mathbb{N}} \subseteq L^1(T)$ is uniformly integrable. So, given $\varepsilon > 0$, we can find $t \in T \setminus N$ such that

$$\int_t^b |\langle u_n'(s), (u_n - u)(s)\rangle| ds < \varepsilon \text{ for all } n \in \mathbb{N}. \tag{6.21}$$

Let $\langle \cdot, \cdot \rangle_{p,t}$ denote the duality brackets for the pair

$$(L^p([0,t], X), L^{p'}([0,t], X^*)).$$

Using the integration by parts formula (see Theorem I-4.205), we have

$$\langle u_n', u_n - u\rangle_{p,t} = \frac{1}{2}|u_n(t) - u(t)|^2 + \langle u', u_n - u\rangle_{p,t}.$$

We observe that

$$\begin{aligned}
&\frac{1}{2}|u_n(t) - u(t)|^2 \to 0 \text{ (since } t \in T \setminus N),\\
&\langle u', u_n - u\rangle_{p,t} \to 0 \text{ (since } u_n \xrightarrow{w} u \text{ in } L^p([0,t], X)).
\end{aligned}$$

It follows that

$$\langle u_n', u_n - u\rangle_{p,t} \to 0 \text{ as } n \to \infty. \tag{6.22}$$

Note that

$$\langle u_n', u_n - u\rangle_p = \langle u_n', u_n - u\rangle_{p,t} + \int_t^b \langle u_n'(s), (u_n - u)(s)\rangle ds, \tag{6.23}$$

$$\begin{aligned}
&\Rightarrow \limsup_{n\to\infty} \langle u_n', u_n - u\rangle_p \le \varepsilon \text{ (see (6.21), (6.22))},\\
&\Rightarrow \limsup_{n\to\infty} \langle u_n', u_n - u\rangle_p \le 0.
\end{aligned} \tag{6.24}$$

On the other hand

$$\begin{aligned}
&\liminf_{n\to\infty} \langle u_n', u_n - u\rangle_p \ge -\varepsilon \text{ (see (6.21), (6.22))},\\
&\Rightarrow \liminf_{n\to\infty} \langle u_n', u_n - u\rangle_p \ge 0.
\end{aligned} \tag{6.25}$$

From (6.24), (6.25), it follows that

$$\langle u_n', u_n - u\rangle_p \to 0. \tag{6.26}$$

We know that

$$\langle u_n', u_n - u\rangle_p = \langle \widehat{A}(u_n), u_n - u\rangle_p + \int_0^b (h_n(t), (u_n - u)(t))dt$$
$$\Rightarrow \lim_{n\to\infty} \langle \widehat{A}(u_n), u_n - u\rangle_p = 0,$$
$$\Rightarrow \widehat{A}(u_n) \xrightarrow{w} \widehat{A}(u) \text{ (see Proposition 6.4)},$$
$$\Rightarrow u' + \widehat{A}(u) = h \in K \text{ and so } u \in E(K).$$

For every $t \in T$, we know that

$$\begin{aligned}
&\frac{1}{2}|u_n(t) - u(t)|^2 \\
&= \int_0^t (h_n(s) - h(s), u_n(s) - u(s))ds - \langle \widehat{A}(u_n) - \widehat{A}(u), u_n - u\rangle_{p,t} \\
&\le \int_0^b |(h_n(s) - h(s), u_n(s) - u(s))|ds \\
&+ \int_0^b |\langle A(s, u_n(s)), (u_n - u)(s)\rangle|ds + \langle \widehat{A}(u), u_n - u\rangle_{p,t}.
\end{aligned} \tag{6.27}$$

Observe that

$$\int_0^b |(h_n(s) - h(s), u_n(s) - u(s))|ds \to 0.$$

Also, from the proof of Proposition 6.4, we have

$$\int_0^b |\langle A(s, u_n(s)), (u_n - u)(s)\rangle|ds \to 0.$$

Finally let

$$\beta_n(t) = |\int_0^t \langle A(s, u(s)), (u_n - u)(s)\rangle ds|, t \in T.$$

We can find $t_n \in T$ such that $\beta_n(t_n) = \max_{t\in T} \beta_n(t)$. We may assume that $t_n \to t \in T$. We have

$$\beta_n(t_n) \to 0.$$

From (6.27), we infer that

$$\begin{aligned} &\|u_n - u\|_{C(T,H)} \to 0, u \in E(K), \\ \Rightarrow &E(K) \subseteq C(T, H) \text{ is compact.} \end{aligned} \tag{6.28}$$

Using the previous arguments, we can easily check that $S(x_0) \subseteq C(T, H)$ is closed. Since $S(x_0) \subseteq C(T, H)$, from (6.28) we conclude that $S(x_0)$ is compact in $C(T, H)$. Also, $S(x_0) \subseteq W_p(0, b)$ is bounded and weakly closed. Therefore $S(x_0) \subseteq W_p(0, b)$ is weakly compact. □

A careful reading of the above proof, leads to the following corollary.

Corollary 6.9 *If hypotheses* H_1, H_2 *hold, then the solution multifunction* $x_0 \to S(x_0)$ *is usc from* H *into* $P_k(C(T, H)) = \{D \subseteq C(T, H) : \text{nonempty, compact}\}$.

We can also have an existence theorem when the multivalued reaction $F(t, x)$ is nonconvex valued.

The hypotheses on $F(t, x)$ are the following. By $P_f(H)$ we denote the set of all nonempty, closed subsets of H.

H_3: $F : T \times H \to P_f(H)$ is a multifunction which satisfies

(i) $(t, x) \to F(t, x)$ is graph measurable;
(ii) for a.a $t \in T$, $x \to F(t, x)$ is lsc;
(iii) $|F(t, x)| \leq a(t) + c|x|^{\tau-1}$ for a.a $t \in T$, all $x \in H$, with $a \in L^{\tau'}(T)$, $c > 0$, $1 < \tau < p$.

Using Theorem I-5.137, we pass to a single-valued Cauchy problem and then we use Theorem 6.8 in order to deduce the following existence theorem.

Theorem 6.10 *If hypotheses* H_1, H_3 *hold, then* $S(x_0) \neq \emptyset$.

Remark 6.11 We point out a difference between the convex and the nonconvex cases. In hypothesis H_2(i) we required that for all $x \in H$, $t \to F(t, x)$ is graph measurable, while in hypothesis H_3(i) we required that $(t, x) \to F(t, x)$ is graph measurable. The issue here is to guarantee that for every $u : T \to H$ measurable, the set $S^{\tau'}_{F(\cdot,u(\cdot))}$ of measurable selections of $F(\cdot, u(\cdot))$ in $L^{\tau'}(T, H)$ is nonempty. Hypotheses H_2(i), (ii) do not imply joint measurability of $F(\cdot, \cdot)$ and so the nonemptiness of $S_{F(\cdot,u(\cdot))}$ can not follow from the classical measurable selection theorems(see Section I-5.3). To establish the nonemptiness of $S^{\tau'}_{F(\cdot,u(\cdot))}$, we argue as follows. Let $\{s_n\}_{n\in\mathbb{N}}$ be simple functions such that $s_n(t) \to u(t)$ for a.a $t \in T$. Then on account of hypothesis H_2(i) and using Theorem I-5.106, we can find $f_n \in S^{\tau'}_{F(\cdot,s_n(\cdot))}$, $n \in \mathbb{N}$. Then from hypothesis H_2(iii) we see that $\{f_n\}_{n\in\mathbb{N}} \subseteq L^{\tau'}(T, H)$ is bounded. Hence we may assume that $f_n \xrightarrow{w} f$ in $L^{\tau'}(T, H)$. Using Theorem I-8.38 and hypothesis H_2(ii), we infer that $f \in S^{\tau'}_{F(\cdot,u(\cdot))} \neq \emptyset$. Evidently this argument fails if instead $F(t, \cdot)$ is lsc. For this reason in hypothesis H_3(i) we require joint graph measurability.

Next we want to investigate the relation between the solution sets of the convex and nonconvex problems. More precisely, we want to show that the solution set of the nonconvex problem is $C(T, H)$-dense in the solution set of the convex problem. Such a result, known as "relaxation theorem", is important in the study of control systems, since it implies that we can approximate any state of the system, by economizing in the control functions. To have a relaxation theorem, we need to strengthen the hypotheses on the map $A(t, x)$. The new hypotheses on $A(\cdot, \cdot)$ are:

H_4: $A : T \times X \to X^*$ is a map such that

(i) for all $x \in X$, $t \to A(t, x)$ is measurable;
(ii) for a.a $t \in T$, $A(t, \cdot)$ is hemicontinuous,

$$\langle A(t, x) - A(t, y), x - y\rangle \geq c^* \|x - y\|^p$$

for some $c^* > 0$, all $x, y \in X$, with $1 < p < \infty$;
(iii) $\|A(t, x)\|_* \leq a(t) + c\|x\|^{p-1}$ for a.a $t \in T$, all $x \in X$, with $a \in L^{p'}(T)$, $c > 0$;
(iv) $\langle A(t, x), x\rangle \geq \widehat{c}\|x\|^p - \vartheta(t)$ for a.a $t \in T$, all $x \in X$, with $\vartheta \in L^1(T)$, $\widehat{c} > 0$.

Remark 6.12 Hypothesis H_4(ii) implies that for a.a $t \in T$, $A(t, \cdot)$ is maximal monotone (see Proposition I-7.36).

The hypotheses on the multivalued vector field $F(t, x)$ are the following.

H_5: $F : T \times X \to P_{wk}(H) = \{C \subseteq H : \text{nonempty, } w\text{-compact}\}$ is a multifunction such that

(i) for every $x \in H$, $t \to F(t, x)$ is graph measurable;
(ii) $h(\overline{\text{conv}}F(t, x), \overline{\text{conv}}F(t, y)) \leq k(t)|x - y|$ for a.a $t \in T$, all $x, y \in H$, with $k \in L^1(T)$($h(\cdot, \cdot)$ is the Hausdorff metric, see Definition I-5.39);
(iii) $|F(t, x)| \leq a_0(t) + c_0|x|^{\tau-1}$ for a.a $t \in T$, all $x \in H$, with $a_0 \in L^{\tau'}(T)$, $c_0 > 0$, $\tau < p$.

The "convex problem" is obtained by replacing F with $\overline{\text{conv}}F$. By $S_c(x_0)$ (resp. $S(x_0)$) we denote the solution set of the convex (resp. nonconvex) version of problem (6.8). As in Theorem 6.8, we have that $\emptyset \neq S_c(x_0) \subseteq W_p(0, b)$ is w-compact and $\emptyset \neq S_c(x_0) \subseteq C(T, H)$ is compact.

Theorem 6.13 *If hypotheses* H_4, H_5 *hold, then* $S_c(x_0) = \overline{S(x_0)}^{C(T,H)}$.

Proof Let $u \in S_c(x_0)$. Then

$$u'(t) + A(t, u(t)) = f(t) \text{ for a.a } t \in T,$$
$$u(0) = x_0$$

with $f \in S^{\tau'}_{\overline{\text{conv}}F(\cdot,u(\cdot))}$. From Proposition I-5.130, we know that we can find $f_n \in S^{\tau'}_{F(\cdot,u(\cdot))}$ such that

$$f_n \xrightarrow{w} f \text{ in } L^{\tau'}(T, H).$$

Let $\varepsilon_n \to 0^+$ and $y \in W_p(0, b)$. We introduce the multifunction $G_n : T \to 2^H$ defined by

$$G_n(t) = \{h \in H : |f_n(t) - h| < \varepsilon_n + d_H(f_n(t), F(t, y(t))), h \in F(t, y(t))\}.$$

From hypotheses H_5(i), (ii) we infer the graph measurability of $F(\cdot, \cdot)$ and so $\text{Gr}\, G_n \in \mathcal{L}_T \times B(H)$, with $\mathcal{L}_T$ being the Lebesgue σ-algebra of T and $B(H)$ is the Borel σ-algebra of H. We apply Theorem I-5.106(the Yankov-von Neumann-Aumann Selection Theorem) and produce a measurable selection of $G_n(\cdot)$, which clearly belongs in $L^{\tau'}(T, H)$. So, if we consider the multifunction $\Gamma_n : W_p(0, b) \to 2^{L^{\tau'}(T,H)}$ defined by

$$\Gamma_n(y) = \{h \in S^{\tau'}_{F(\cdot,y(\cdot))} : |f_n(t) - h(t)| < \varepsilon_n + d_H(f_n(t), F(t, y(t))) \text{ for a.a } t \in T\},$$

then $\Gamma_n(\cdot)$ has nonempty values which are w-compact in $L^{\tau'}(T, H)$ (see hypothesis H_5(iii)). Also the values of $\Gamma_n(\cdot)$ are decomposable and $\Gamma_n(\cdot)$ is easily seen to be lsc (see Proposition I-5.6). We apply Theorem I-5.137 and obtain $\gamma_n : W_p(0, b) \to L^{\tau'}(T, H)$ a continuous selection of $\overline{\Gamma_n(\cdot)}^{\|\cdot\|_{\tau'}}$, that is

$$\gamma_n(y) \in \overline{\Gamma_n(y)}^{\|\cdot\|_{\tau'}} \text{ for all } y \in W_p(0, b).$$

Then we consider the following Cauchy problem

$$\left\{ \begin{array}{l} u'(t) + A(t, u(t)) = \gamma_n(u)(t) \text{ for a.a } t \in T, \\ u(0) = x_0. \end{array} \right\}$$

As in Theorem 6.8, we show that this problem has a solution $u_n \in W_p(0, b)$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq C(T, H)$ is relatively compact. So, we may assume that

$$u_n \to \widehat{u} \text{ in } C(T, H). \tag{6.29}$$

We have

$$\int_0^t \langle u_n' - u', u_n - u\rangle ds + \int_0^t \langle A(s, u_n) - A(s, u), u_n - u\rangle ds$$

$$= \int_0^t (\gamma_n(u_n) - f, u_n - u)ds,$$

$$\Rightarrow \frac{1}{2}|u_n(t) - u(t)|^2 \leq \varepsilon_n b + \int_0^t k(s)|u_n(s) - u(s)|^2 ds \text{ (see Theorem } I\text{-4.205)},$$

$$\Rightarrow |\widehat{u}(t) - u(t)|^2 \leq \int_0^t 2k(s)|\widehat{u}(s) - u(s)|^2 ds \text{ (see (6.29))},$$

$$\Rightarrow \widehat{u} = u \text{ (by Gronwall's inequality).}$$

So, we have

$$u_n \to u \text{ in } C(T, H)$$

and since $u_n \in S(x_0)$ for all $n \in \mathbb{N}$, we conclude that

$$S_c(x_0) = \overline{S(x_0)}^{C(T,H)}.$$

□

Next we consider periodic problems. As before (X, H, X^*) is an evolution triple with $X \hookrightarrow H$ compactly. Also we assume that both X, X^* are strictly convex. We consider the following periodic problem

$$\left\{ \begin{array}{l} u'(t) + A(t, u(t)) \in F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b). \end{array} \right\} \tag{6.30}$$

The hypotheses on the data of (6.30) are the following

H_6: $A : T \times X \to X^*$ is a map such that

(i) for all $x \in X$, $t \to A(t, x)$ is measurable;
(ii) for a.a $t \in T$, $x \to A(t, x)$ is hemicontinuous, monotone;
(iii) $\|A(t, x)\|_* \leq a(t) + c\|x\|^{p-1}$ for a.a $t \in T$, all $x \in X$, with $a \in L^{p'}(T)$, $c > 0$;
(iv) $\langle A(t, x), x\rangle \geq \widehat{c}\|x\|^p$ for a.a $t \in T$, all $x \in X$.

H_7: $F : T \times X \to P_{fc}(H) = \{C \subseteq H : \text{nonempty, closed, convex}\}$ is a multifunction such that

(i) for every $x \in H$, $t \to F(t, x)$ is graph measurable;
(ii) for a.a $t \in T$, $\text{Gr } F(t, \cdot)$ is sequentially closed in $H \times H_w$;
(iii) there exist $M > 0$ and $a_M \in L^{p'}(T)$ such that

$$(h, x) \leq 0 \text{ for a.a } t \in T, \text{ all } |x| = M, \text{ all } h \in F(t, x),$$

$$|F(t, x)| \leq a_M(t) \text{ for a.a } t \in T, \text{ all } |x| \leq M.$$

Remark 6.14 Note that no growth condition is imposed on $F(t,\cdot)$. We have only the Hartman-type hypothesis H_7(iii).

Consider the duality map $\mathcal{F}_\varphi : X \to X^*$ for the weight $\varphi(t) = t^{p-1}$, $t \geq 0$. Since X and X^* are both strictly convex, $\mathcal{F}_\varphi$ is a bijection and $\mathcal{F}_\varphi^{-1} : X^* \to X$ is the duality map of X^* with weight φ^{-1}. We set $A_\varepsilon(t,x) = A(t,x) + \varepsilon\mathcal{F}_\varphi(x)$. Then $A_\varepsilon(t,x)$ satisfies the same conditions as $A(\cdot,\cdot)$ and in addition $A_\varepsilon(t,\cdot)$ is strictly monotone. Indeed, for all $x, y \in X$, we have

$$\begin{aligned}
&\langle A_\varepsilon(t,x) - A_\varepsilon(t,y), x-y\rangle \\
=&\langle A(t,x) - A(t,y), x-y\rangle + \varepsilon\langle \mathcal{F}_\varphi(x) - \mathcal{F}_\varphi(y), x-y\rangle \\
\geq&\varepsilon(\|x\|^{p-1} - \|y\|^{p-1})(\|x\| - \|y\|), \\
\Rightarrow\ & A_\varepsilon(t,\cdot) \text{ is strictly monotone.}
\end{aligned}$$

For $\varepsilon > 0$ and $h \in L^{p'}(T,H)$, we consider the periodic problem

$$\left.\begin{array}{l} u'(t) + A_\varepsilon(t,u(t)) = h(t) \text{ for a.a } t \in T, \\ u(0) = u(b). \end{array}\right\} \tag{6.31}$$

Let $\widehat{L} : D(L) \subseteq L^p(T,X) \to L^{p'}(T,X^*)$ be the linear operator defined by

$$\widehat{L}(u) = u' \text{ for all } u \in D(\widehat{L}) = \{u \in W_p(0,b) : u(0) = u(b)\}.$$

This is a linear, densely defined, maximal monotone operator. On the other hand, the map $\widehat{A}_\varepsilon : L^p(T,X) \to L^{p'}(T,X^*)$ defined by $\widehat{A}_\varepsilon(u)(\cdot) = A_\varepsilon(\cdot,u(\cdot))$ is bounded, maximal monotone(being demicontinuous, strictly monotone), hence it is pseudomonotone (see Proposition *I*-7.73). Moreover, the strict convexity of X and X^*, implies the strict convexity of the Lebesgue-Bochner spaces $L^p(T,X)$ and $L^{p'}(T,X^*)$. So, we can apply Theorem 6.2 and infer that problem (6.31) has a solution $\widehat{u}_\varepsilon(h) \in W_p(0,b)$, which is unique on account of the strict monotonicity of $\widehat{A}_\varepsilon(\cdot)$. So, we can define the solution map $k_\varepsilon : L^{p'}(T,H) \to C(T,H)$ by

$$k_\varepsilon(h) = \widehat{u}_\varepsilon(h).$$

(recall that $W_p(0,b) \hookrightarrow C(T,H)$). The next proposition establishes the continuity property of $k_\varepsilon(\cdot)$.

Proposition 6.15 *If hypotheses* H_6 *hold and* $\varepsilon > 0$*, then the solution map* $k_\varepsilon : L^{p'}(T,H) \to C(T,H)$ *is completely continuous.*

Proof Let $h_n \xrightarrow{w} h$ in $L^{p'}(T,H)$ and set $u_n = k_\varepsilon(h_n)$, $n \in \mathbb{N}$ and $u = k_\varepsilon(h)$. We have

$$\begin{aligned}
&\langle u_n'(t), u_n(t)\rangle + \varepsilon\|u_n(t)\|^p \leq (h_n(t), u_n(t)) \text{ for a.a } t \in T,\\
\Rightarrow &\varepsilon\|u_n\|^p_{L^p(T,X)} \leq c_1\|u_n\|_{L^p(T,X)} \text{ for some } c_1 > 0, \text{ all } n \in \mathbb{N},\\
\Rightarrow &\{u_n\}_{n\in\mathbb{N}} \subseteq L^p(T, X) \text{ is bounded.}
\end{aligned}$$

Then from (6.31) and hypothesis H_6(iii) we have that

$$\begin{aligned}
&\{u_n'\}_{n\in\mathbb{N}} \subseteq L^p(T, X^*) \text{ is bounded,}\\
\Rightarrow &\{u_n\}_{n\in\mathbb{N}} \subseteq W_p(0, b) \text{ is bounded.}
\end{aligned}$$

We may assume that

$$\left\{\begin{array}{l} u_n \xrightarrow{w} u \text{ in } W_p(0, b),\ u_n \to u \text{ in } L^p(T, H) \text{ (see Theorem I-4.207),}\\ \\ u_n(t) \to u(t) \text{ in } H \text{ for all } t \in T \setminus N,\ |N| = 0. \end{array}\right\} \tag{6.32}$$

Note that $\{\langle u_n'(\cdot), (u_n - u)(\cdot)\rangle\}_{n\in\mathbb{N}} \subseteq L^1(T)$ is uniformly integrable. So, given $\delta > 0$, we can find $t, s \in T \setminus N$, $s < t$, such that

$$\int_t^b \langle u_n'(\tau), (u_n - u)(\tau)\rangle d\tau < \frac{\delta}{2}, \int_0^s \langle u_n'(\tau), (u_n - u)(\tau)\rangle d\tau < \frac{\delta}{2}. \tag{6.33}$$

In what follows by $\langle\cdot,\cdot\rangle_{p,t,s}$ we denote the duality brackets for the pair $(L^p([s, t], X), L^{p'}([s, t], X^*))$. We have

$$\langle u_n', u_n - u\rangle_{p,t,s} = \frac{1}{2}|u_n(t) - u(t)|^2 - \frac{1}{2}|u_n(s) - u(s)|^2 + \langle u, u_n - u\rangle_{p,t,s}$$

(see Theorem I-4.205). (6.34)

Note that

$$\begin{aligned}
&|u_n(t) - u(t)|^2, |u_n(s) - u(s)|^2 \to 0 \text{ since } s, t \in T \setminus N,\\
&\langle u, u_n - u\rangle_{p,t,s} \to 0 \text{ (see (6.32)).}
\end{aligned}$$

Then from (6.34) we infer that

$$\langle u_n, u_n - u\rangle_{p,t,s} \to 0. \tag{6.35}$$

Using (6.33) we see that

$$\begin{aligned} &|\langle u_n', u_n - u\rangle_p| \le |\langle u_n', u_n - u\rangle_{p,t,s}| + \delta \\ &\Rightarrow \langle u_n', u_n - u\rangle_p \to 0 \text{ (since } \delta > 0 \text{ is arbitrary).} \end{aligned} \tag{6.36}$$

Then from the equation in the limit we have

$$u' + \widehat{A}_\varepsilon(u) = h, u(0) = u(b)$$

$$\text{and } \lim_{n\to\infty} \langle \widehat{A}_\varepsilon(u_n), u_n - u\rangle_p = 0.$$

From the maximal monotonicity of $\widehat{A}_\varepsilon(\cdot)$, we have

$$\left.\begin{aligned} &\widehat{A}_\varepsilon(u_n) \overset{w}{\to} \widehat{A}_\varepsilon(u) \text{ in } L^{p'}(T, X^*), \\ &\langle \widehat{A}_\varepsilon(u_n), u_n\rangle_p \to \langle \widehat{A}_\varepsilon(u), u\rangle_p \end{aligned}\right\} \tag{6.37}$$

Let $\{t_m\}_{m\in\mathbb{N}} \subseteq T \setminus N$ such that $t_m \to 0^+$. We have

$$u_n(t_m) \to u(t_m) \text{ in } H \text{ as } n \to \infty \text{ (see (6.32)))}$$

$$u(t_m) \to u(0) \text{ in } H \text{ as } m \to \infty$$

$$\text{(since } u \in W_p(0, b) \hookrightarrow C(T, H))$$

So, by Lemma I-1.193, we can find a sequence $\{t_{m_n}\} \subseteq \{t_m\}$ such that $u_n(t_{m_n}) \to u(0)$ in H. Then for every $t \in T$, we have

$$\begin{aligned} \frac{1}{2}|u_n(t) - u(t)|^2 \le &\frac{1}{2}|u_n(t_{m_n}) - u(t_{m_n})|^2 \\ &+ \int_0^b |\langle A_\varepsilon(s, u_n(s)) - A_\varepsilon(s, u(s)), u_n(s) - u(s)\rangle| ds \end{aligned}$$

Note that

$$|u_n(t_{m_n}) - u(t_{m_n})| \to 0$$

$$0 \le \int_0^b \langle A_\varepsilon(s, u_n(s)) - A_\varepsilon(s, u(s)), u_n(s) - u(s)\rangle ds \to 0$$

(see (6.32), (6.37)).

Therefore given $\delta > 0$, we can find $n_0 \in \mathbb{N}$ such that

$$|u_n(t) - u(t)| \le \delta \text{for all } n \ge n_0, \text{ all } t \in T,$$

$$\Rightarrow u_n \to u \text{ in } C(T, H).$$

This proves the complete continuity of the solution map $k_\varepsilon(\cdot)$. □

Using this proposition and the multivalued Leray-Schauder Alternative Theorem (see Theorem 2.138), we can prove that the solution set S_{per} of (6.30) is nonempty.

Theorem 6.16 *If hypotheses H_6 and H_7 hold, then $\emptyset \neq S_{per} \subseteq W_p(0,b)$ is w-compact and compact in $C(T,H)$.*

Proof Let $p_M : X \to \overline{B}_M = \{x \in H : |x| \leq M\}$ the M-radial retraction defined by

$$p_M(x) = \begin{cases} x & \text{if } |x| \leq M \\ M\frac{x}{|x|} & \text{if } M < |x| \end{cases}$$

Using $p_M(\cdot)$ we introduce the following truncation of the multivalued vector field $F(t,x)$

$$F_M(t,x) = \begin{cases} F(t,x) & \text{if } |x| \leq M \\ F(t,\, p_M(x)) & \text{if } M < |x| \end{cases} \tag{6.38}$$

Clearly $F_M(\cdot,\cdot)$ is graph measurable and for a.a $t \in T$ $\mathrm{Gr}F_M(t,\cdot)$ is sequentially closed in $H \times H_{w^+}$. Moreover, from (6.38) and hypothesis $H_7(iii)$, we see that

$$|F_M(t,x)| = \sup\{|h| : h \in F(t,x)\} \leq a_M(t) \tag{6.39}$$

for a.a $t \in T$, all $x \in H$.

Consider $N : C(T,H) \to 2^{L^{p'}(T,H)}$ the multivalued Nemytskii map corresponding to $F_M(t,x)$, defined by

$$N(u) = S^{p'}_{F_M(\cdot,u(\circ))} = \{h \in L^{p'}(T,H) : h(t) \in F_M(t,u(t)) \text{ for a.a } t \in T\}.$$

From Remark 6.11 and (6.39), we see that

$$N(u) \in P_{wkc}(L^{p'}(T,H)) = \{C \subseteq L^{p'}(T,H) : w\text{-compact, convex}\}$$

for all $u \in C(T,H)$

Claim 1 $N(\cdot)$ is usc from $C(T,H)$ into $L^{p'}(T,H)_w$.

On account of Proposition I-5.13, it suffices to show that $GrN \subseteq C(T,H) \times L^{p'}(T,H)_w$ is sequentially closed. So, we consider a sequence $\{(u_n,h_n)\}_{n\in\mathbb{N}} \subseteq GrN$ such that

$$u_n \to u \text{ in } C(T,H), h_n \xrightarrow{w} h \text{ in } L^{p'}(T,H).$$

Using Theorem I-8.38, we have

$$\begin{aligned} &h(t) \in \overline{\text{conv}} w \limsup_{n\to\infty}\{h_n(t)\} \text{ for a.a } t \in T, \\ \Rightarrow &h(t) \in F(t, u(t)) \text{ for a.a } t \in T \text{ (see hypothesis } H_7(ii)), \\ \Rightarrow &(u, h) \in GrN. \end{aligned}$$

This proves Claim 1.
We consider the following periodic evolution inclusion

$$\left\{ \begin{array}{l} u'(t) + A_\varepsilon(t, u(t)) \in F_M(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b). \end{array} \right\} \tag{6.40}$$

Existence of a solution of (6.40) is equivalent to finding a fixed point of the multivalued map $k_\varepsilon \circ N : C(T, H) \to 2^{C(T,H)} \setminus \{\emptyset\}$. To produce a fixed point of this multivalued map, we will use Theorem 2.138 (the multivalued "Leray-Schauder Alternative Theorem"). To this end, we consider the set

$$S = \{u \in C(T, H) : u \in \lambda(k_\varepsilon \circ N)(u), \lambda \in (0, 1]\}.$$

Claim 2 $|u(t)| \leq M$ for all $t \in T$, all $u \in S$.

We argue by contradiction. So, suppose that Claim 2 is not true. Then we have two possibilities

- $|u(t)| > M$ for all $t \in T$.
- There exist $\eta, \vartheta \in T$, $\eta < \vartheta$ such that

$$|u(\eta)| = M, |u(t)| > M \text{ for all } t \in (\eta, \vartheta].$$

If the first possibility is true, then acting on (6.40) with $u(t)$ and integrating over T, we obtain using Theorem I-4.205

$$|u(b)|^2 + 2\varepsilon \int_0^b \|u(t)\|^p dt \leq |u(0)|^2 + \int_0^b (f(t), u(t))dt \tag{6.41}$$

with $f \in S^{p'}_{F_M(\cdot,u(\cdot))}$. Note that

$$(f(t), u(t)) = (f(t), M\frac{u(t)}{|u(t)|})\frac{|u(t)|}{M} \leq 0$$

(see hypothesis $H_7(iii)$).

Using this in (6.41), we see that

$$|u(b)| < |u(0)|,$$

a contradiction.

If the second possibility happens, then repeating the above reasoning on the interval $[\eta, \vartheta]$, we obtain

$$|u(\vartheta)| < |u(\eta)|$$

again a contradiction.

Therefore Claim 2 is true.

Claims 1, 2 and Proposition 6.15, permit the use of Theorem 2.138 and so we can find $\widehat{u}_\varepsilon \in W_p(0, b)$ such that

$$\widehat{u}_\varepsilon \in (k_\varepsilon \circ N)(\widehat{u}_\varepsilon).$$

Since $|\widehat{u}_\varepsilon(t)| \leq M$ for all $t \in T$ (see Claim 2), we see that

$$F_M(t, \widehat{u}_\varepsilon(t)) = F(t, \widehat{u}_\varepsilon(t)) \text{ for all } t \in T \text{ (see (6.38))}.$$

Therefore $\widehat{u}_\varepsilon \in W_p(0, b)$ is a solution of the periodic problem

$$\left\{ \begin{array}{l} u'(t) + A_\varepsilon(t, u(t)) \in F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b). \end{array} \right\} \tag{6.42}$$

Now let $\varepsilon_n \to 0^+$ and let $\widehat{u}_n = \widehat{u}_{\varepsilon_n}$. Then from (6.42) and since $\|\widehat{u}\|_{C(T,H)} \leq M$, we see that $\{\widehat{u}_n\}_{n \in \mathbb{N}} \subseteq W_p(0, b)$ is bounded and so we may assume that

$$\widehat{u}_n \xrightarrow{w} \widehat{u} \text{ in } W_p(0, b), \widehat{u}_n \to \widehat{u} \text{ in } L^p(T, H).$$

As before, we obtain $\widehat{u} \in S_{per}$. Hence $S_{per} \neq \emptyset$ and from the above arguments, we see that $S_{per} \subseteq W_p(0, b)$ is w-compact. Finally, from Proposition 6.15, we conclude that $S_{per} \subseteq C(T, H)$ is compact. □

We can have a corresponding existence theorem for the "nonconvex problem". Now the hypotheses on the multivalued vector field $F(t, x)$ are the following

H_8: $F \times H \to P_f(H) = \{C \subseteq H$:nonempty, closed$\}$ is a multifunction such that

(i) $(t, x) \to F(t, x)$ is graph measurable;
(ii) for a.a $t \in T$, $x \to F(t, x)$ is lsc;
(iii) there exist $M > 0$ and $a_M \in L^{p'}(T)$ such that
$(h, x) \leq 0$ for a.a $t \in T$, all $|x| = M$, all $h \in F(t, x)$,
$|F(t, x)| \leq a_M(t)$ for a.a $t \in T$, all $|x| \leq M$.

Theorem 6.17 *If hypotheses H_6 and H_8 hold, then $S_{per} \neq \emptyset$.*

Proof Let $F_M(t,x)$ and $N(u)$ as in the proof of Theorem 6.16. Hypotheses H_8 imply that $N(\cdot)$ is well-defined and has decomposable values. Moreover, using Proposition I-5.6, we can easily check that $N(\cdot)$ is lsc. Invoking Theorem I-5.137, we can find a continuous map $\mu : C(T,H) \to L^{p'}(T,H)$ such that

$$\mu(u) \in N(u) \text{ for all } u \in C(T,H).$$

For $\varepsilon > 0$, we consider the following approximating periodic problem

$$\left\{ \begin{array}{l} u'(t) + A_\varepsilon(t,u(t)) = \mu(u)(t) \text{ for a.a } t \in T, \\ u(0) = u(b). \end{array} \right\} \tag{6.43}$$

The existence of a solution of problem (6.43) is equivalent to the existence of a fixed point for the map $k_\varepsilon \circ \mu$ on $C(T,H)$ (see the proof of Proposition 6.15). So, now we use the single-valued Leray-Schauder Alternative Theorem (see Theorem 1.67 and Remark 2.64) and obtain $\widehat{u}_\varepsilon \in W_p(0,b)$ such that $\widehat{u}_\varepsilon = (k_\varepsilon \circ \mu)(\widehat{u}_\varepsilon)$. Finally let $\varepsilon \to 0^+$ to produce a solution for problem (6.30). Therefore $S_{per} \neq \emptyset$. □

We can have similar results for the anti-periodic problem

$$\left\{ \begin{array}{l} u'(t) + A(t,u(t)) \in F(t,u(t)) \text{ for a.a } t \in T, \\ u(0) = -u(b). \end{array} \right\} \tag{6.44}$$

As before (X,H,X^*) is an evolution triple with the embedding X into H being compact. In this case we consider the linear operator $L : D(L) \subseteq L^p(T,X) \to L^{p'}(T,X^*)$ defined by

$$L(u) = u' \text{ for all } u \in D(L) = \{u \in W_p(0,b) : u(0) = -u(b)\}.$$

Proposition 6.18 *The operator $L(\cdot)$ is linear, densely defined and maximal monotone.*

Proof For all $u \in D(L)$, we have

$$\begin{aligned} \langle L(u), u \rangle_p &= \int_0^b \langle u'(t), u(t) \rangle dt \\ &= \frac{1}{2}[|u(b)|^2 - |u(0)|^2] \text{ (see Theorem } I\text{-4.205)} \\ &= 0, \end{aligned}$$

$\Rightarrow L(\cdot)$ is monotone.

We will show that $L(\cdot)$ is in fact maximal monotone. So, let $(y, y^*) \in L^p(T, X) \times L^{p'}(T, X^*)$ such that

$$0 \leq \langle y^* - L(u), y - u \rangle_p \text{ for all } u \in D(L). \tag{6.45}$$

We need to show $(y, y^*) \in GrL$. For this purpose, let $\vartheta \in C_c^\infty(0, b)$, $x \in X$ and define $u = \vartheta x$. Evidently $u \in D(L)$ and $u' = \vartheta' x$. From (6.45) we have

$$0 \leq \langle y^*, y \rangle_p - \int_0^b \langle \vartheta'(t) y(t) + \vartheta(t) y^*(t), x \rangle dt$$

(see Theorem I-4.205),

$$\Rightarrow \langle \int_0^b \vartheta(t)(-y'(t) + y^*(t))dt, x \rangle \leq \langle y^*, y \rangle_p$$

(see Proposition I-4.185).

Since $x \in X$ is arbitrary, we infer that

$$\int_0^b \vartheta(t)(-y'(t) + y^*(t))dt = 0 \text{ for all } \vartheta \in C_c^\infty(0, b),$$

$$\Rightarrow y^* = y', \text{ hence } y \in W_p(0, b).$$

We need to show that $y \in D(L)$. For all $u \in D(L)$ we have

$$\begin{aligned} 0 \leq & \langle y' - u', y - u \rangle_p \\ = & \frac{1}{2}[|y(b) - u(b)|^2 - |y(0) - u(0)|^2] \end{aligned}$$

(see Theorem I-4.205)

$$\Rightarrow 0 \leq |y(b)|^2 - |y(0)|^2 + 2(y((b) + y(0), u(0)) \tag{6.46}$$

Let $u(t) = (1 - \frac{2}{b}t)x$ for all $t \in T$ and with $x \in X$. Then $u(0) = -u(b)$ and so $u \in D(L)$. From (6.46) and since $x \in X$ is arbitrary we conclude that $y(0) = -y(b)$, that is, $y \in D(L)$. This completes the proof of the proposition. □

Let $S_{aper} \subseteq W_p(0, b)$ denote the solution set of the anti-periodic problem. On account of Proposition 6.18, we can use Theorem 6.2. Then reasoning as in the proof of Theorems 6.16 and 6.17, we obtain the following theorem.

Theorem 6.19 *If hypotheses H_6, H_7 or H_6, H_8 hold, then $\emptyset \neq S_{aper} \subseteq W_p(0, b)$; moreover, in the "convex case" S_{aper} is w-compact in $W_p(0, b)$ and compact in $C(T, H)$.*

So far we have considered first order evolution inclusions. Next we turn our attention to second order evolution inclusions. As before, (X, H, X^*) is an evolution triple with the embedding of X into H being compact (hence $H \hookrightarrow X^*$ compactly by Schauder's Theorem, see Theorem I-3.158).

The problem under consideration is the following one

$$\left\{\begin{array}{l} u''(t) + A(t, u'(t)) + Bu(t) \in F(t, u(t), u'(t)) \text{ for a.a } t \in T, \\ u(0) = x_0, u'(0) = x_1. \end{array}\right\} \tag{6.47}$$

In this problem $A : T \times X \to X^*$, $B \in \mathcal{L}(X, X^*)$, $F : T \times H \times H \to 2^H \setminus \{\emptyset\}$ and $x_0 \in X$, $x_1 \in H$. Below we will give precise hypotheses for these maps.

Definition 6.20 A function $u \in C(T, X)$ is said to be a "solution" of (6.47), if $u' \in W_p(0, b)$ and there exists $f \in S^{p'}_{F(\cdot,u(\cdot),u'(\cdot))} = \{h \in L^{p'}(T, X^*) : h(t) \in F(t, u(t), u'(t)) \text{ for a.a } t \in T\}$ such that

$$\left\{\begin{array}{l} u''(t) + A(t, u'(t)) + Bu(t) = f(t) \text{ for a.a } t \in T, \\ u(0) = x_0, u'(0) = x_1. \end{array}\right\}$$

By $S(x_0, x_1)$ we denote the set of solutions of (6.47).

Remark 6.21 From the above definition and since $W_p(0, b)$ is embedded continuously in $C(T, H)$ (see Theorem I-4.204), we infer that a solution $u(\cdot)$ of (6.47) belongs in $W^{1,p}((0, b), X)$ and in $C^1(T, H)$.

The map $A(t, x)$ satisfies hypotheses H_1, while for B and $F(t, x, y)$, we assume the following conditions.

H$_9$: $B \in \mathcal{L}(X, X^*)$, $\langle Bx, x\rangle \geq 0$ for all $x \in X$ (monotonicity) and $\langle Bx, y\rangle = \langle x, By\rangle$ (symmetry)

Remark 6.22 Evidently $B(\cdot)$ is maximal monotone.

H$_{10}$: $F : T \times H \times H \to P_{f_c}(H)$ is multifunction such that

(i) for every $x, y \in H$, $t \to F(t, x, y)$ is graph measurable;
(ii) for a.a $t \in T$, $GrF(t, \cdot, \cdot) \subseteq H \times H \times H_w$ is sequentially closed (recall that H_w denotes the Hilbert space H equipped with the weak topology);
(iii) $|F(t, x, y)| = \sup\{|h| : h \in F(t, x, y)\} \leq a(t) + c(|x|^{\frac{2}{p'}} + |y|^{\frac{2}{p'}})$ for a.a $t \in T$, all $x, y \in H$, with $a \in L^{p'}(T)$, $c > 0$.

Recall that $L : D(L) \subseteq L^p(T, X) \to L^{p'}(T, X^*)$ is the operator defined by

$$L(u) = u' \text{ for all } u \in D(L) = \{u \in W_p(0, b) : u(0) = x_1\}.$$

This operator is linear, densely defined and maximal monotone.

We start with an extension of the Gronwall inequality which is suitable for the study of integrodifferential equations.

Lemma 6.23 *If $u : T \to \mathbb{R}$ is continuous, $h, g \in L^1(T)_+$ and*

$$u(t) \leq x_0 + \int_0^1 h(s)u(s)ds + \int_0^t (\int_0^s g(\tau)u(\tau)d\tau)h(s)ds$$

for all $t \in T$, with $x_0 \geq 0$.

then $u(t) \leq (1 + \int_0^t h(s)exp(\int_0^s (h(\tau) + g(\tau))d\tau)ds)x_0$

for all $t \in T$.

Proof Let $v(t) = x_0 + \int_0^t h(s)u(s)ds + \int_0^t (\int_0^s g(\tau)u(\tau)d\tau)h(s)ds$ for all $t \in T$. Then $v(0) = x_0$ and $v'(t) = h(t)u(t) + h(t)\int_0^t g(s)u(s)ds$ for all $t \in T$. Therefore

$$v'(t) \leq h(t)[v(t) + \int_0^t g(s)v(s)ds] \text{ for a.a } t \in T$$

Let $y(t) = v(t) + \int_0^t g(s)v(s)ds$, $t \in T$. Then $v(t) \leq y(t)$ for all $t \in T$ and $y(0) = v(0) = x_0$. We obtain

$$y'(t) \leq [h(t) + g(t)]y(t) \text{ for a.a } t \in T$$

and this by Gronwall's inequality implies

$$y(t) \leq exp(\int_0^t (h(s) + g(s))ds)x_0$$

for a.a $t \in T$.

Integrating we obtain

$$v(t) \leq x_0 + \int_0^t h(s)exp(\int_0^s (h(\tau) + g(\tau))d\tau)dsx_0.$$

□

Remark 6.24 A careful reading of this proof reveals that the result remains true if $u(\cdot)$ is an X-valued function with X an ordered Banach space with order (positive) cone K, which has the RNP (see Definition I-4.168).

Now we are ready for the existence of solutions for the "convex problem".

Theorem 6.25 *If hypotheses H_1, H_9, H_{10} hold and $x_0 \in X$, $x_1 \in H$, then $\emptyset \neq S(x_0, x_1)$ and it is weakly compact in $W^{1,p}((0, b), X)$ and compact in $C^1(T, H)$.*

Proof We start by producing some a priori bounds for the elements of $S(x_0, x_1)$. Let $u \in S(x_0, x_1)$. By Definition 6.20, we have $u' \in W_p(0, b)$ and we can find $f \in S^{p'}_{F(\cdot,u(\cdot),u'(\cdot))}$ such that

$$u''(t) + A(t, u'(t)) + Bu(t) = f(t) \text{ for a.a } t \in T.$$

We act with $u'(t) \in X$ and obtain

$$\langle u''(t), u'(t)\rangle + \langle A(t, u'(t)), u'(t)\rangle + \langle Bu(t), u'(t)\rangle = (f(t), u'(t))$$

for a.a $t \in T$.

Note that

$$\langle u''(t), u'(t)\rangle = \frac{1}{2}\frac{d}{dt}|u'(t)|^2,$$

$$\langle Bu(t), u'(t)\rangle = \frac{1}{2}\frac{d}{dt}\langle Bu(t), u'(t)\rangle \text{ (see hypotheses } H_9),$$

$$\langle A(t, u'(t)), u'(t)\rangle \geq \widehat{c}\|u'(t)\| - \vartheta(t) \text{ for a.a } t \in T$$

(see hypothesis $H_1(iv)$)

Therefore we have

$$\frac{1}{2}\frac{d}{dt}|u'(t)|^2 + \widehat{c}\|u'(t)\|^p + \frac{1}{2}\frac{d}{dt}\langle Bu(t), u(t)\rangle$$

$$\leq (f(t), u'(t)) + \vartheta(t) \text{ for a.a } t \in T.$$

Integrating over $[0, t]$, we obtain

$$\begin{aligned} &\frac{1}{2}|u'(t)|^2 + \widehat{c}\int_0^t \|u'(s)\|^p ds + \frac{1}{2}\langle Bu(t), u(t)\rangle \\ &\leq \int_0^t (f(s), u'(s))ds + c_1 \text{ for some } c_1 > 0. \end{aligned} \tag{6.48}$$

Then hypotheses H_9 and H_{10}(iii) imply that

$$\begin{aligned} &\frac{1}{2}|u'(t)|^2 + \widehat{c}\int_0^t \|u'(s)\|^p ds \\ &\leq \int_0^t (a(s) + c(|u(s)|^{\frac{2}{p'}} + |u'(s)|^{\frac{2}{p'}}))|u'(s)|ds + c_1 \end{aligned}$$

Using Young's inequality with $\varepsilon > 0$, we have

$$\begin{aligned}
&\int_0^t (a(s) + c(|u(s)|^{\frac{2}{p'}} + |u'(s)|^{\frac{2}{p'}}))|u'(s)|ds \\
\leq &\frac{\varepsilon^p}{p}\int_0^t |u'(s)|^p ds + \frac{1}{\varepsilon^{p'}p'}\int_0^t (a(s) + c(|u(s)|^{\frac{2}{p'}} + |u'(s)|^{\frac{2}{p'}})^{p'})ds \\
\leq &\frac{\varepsilon^p}{p}\int_0^t |u'(s)|^p ds + \frac{c_2}{\varepsilon^{p'}p'}\int_0^t (a(s)^{p'} + |u(s)|^2 + |u'(s)|^2)ds \text{ for some } c_2 > 0.
\end{aligned} \tag{6.49}$$

From Definition 6.20, we know that $u' \in W_p(0,b)$ and so $u \in W^{1,p}((0,b),X) = AC^{1,p}(T,X)$ (see Theorem I-4.195). Hence

$$\begin{aligned}
&u(t) = x_0 + \int_0^t u'(s)ds \text{ in } X \text{ for all } t \in T, \\
\Rightarrow &|u(t)|^2 \leq 2|x_0|^2 + c_3\int_0^t |u'(s)|^2 ds \text{ for some } c_3 > 0, \text{ all } t \in T.
\end{aligned} \tag{6.50}$$

Using (6.50) we have

$$\begin{aligned}
&\frac{c_2}{\varepsilon^p p'}\int_0^t (a(s)^{p'} + |u(s)|^2 + |u'(s)|^2)ds \\
\leq &c_4(\varepsilon) + c_5(\varepsilon)\int_0^t |u'(s)|^2 ds + c_6(\varepsilon)\int_0^t\int_0^s |u'(\tau)|^2 d\tau ds \\
&\text{for some } c_4(\varepsilon), c_5(\varepsilon), c_6(\varepsilon) > 0.
\end{aligned} \tag{6.51}$$

We return to (6.48) and use (6.49), (6.51). Then

$$\begin{aligned}
&\frac{1}{2}|u'(t)|^2 + \widehat{c}\int \|u'(s)\|^p ds \\
\leq &c_7(\varepsilon) + c_5(\varepsilon)\int_0^t |u'(s)|^2 ds + c_6(\varepsilon)\int_0^t\int_0^s |u'(\tau)|^2 d\tau ds + \frac{\varepsilon^p\mu^p}{p}\int_0^t \|u(s)\|^p ds
\end{aligned} \tag{6.52}$$

with $\mu > 0$ such that $|\cdot| \leq \mu\|\cdot\|$ (recall $X \hookrightarrow H$ continuously) and for some $c_7(\varepsilon) > 0$. Choosing $\varepsilon > 0$ small (so that $\frac{\varepsilon^p\mu^p}{p} < \widehat{c}$), we obtain

$$\begin{aligned}
&\frac{1}{2}|u'(t)|^2 \leq c_7(\varepsilon) + c_5(\varepsilon)\int_0^t |u'(s)|^2 ds + c_6(\varepsilon)\int_0^t\int_0^s |u'(\tau)|^2 d\tau ds \\
\Rightarrow &|u'(t)| \leq M_1 \text{for some } M_1 > 0, \text{ all } t \in T, \text{ all } u \in S(x_0, x_1) \text{ (see Lemma 6.23).}
\end{aligned}$$

Then we can find $M_2 > 0$ such that

$$|u(t)| \leq M_2 \text{ for all } t \in T, \text{ all } u \in S(x_0, x_1). \tag{6.53}$$

From (6.51), using (6.52), (6.53), we see that there exists $M_3 > 0$ such that

$$\|u'\|_{L^p(T,X)} \leq M_3 \text{ for all } u \in S(x_0, x_1). \tag{6.54}$$

As we indicated earlier

$$u(t) = x_0 + \int_0^t u'(s)ds \text{ in } X \text{ for all } t \in T.$$

It follows that there exists $M_4 > 0$ such that

$$\|u(t)\| \leq M_4 \text{for all } t \in T, \text{ all } u \in S(x_0, x_1). \tag{6.55}$$

From the evolution inclusion, using (6.53), (6.54), (6.55), we infer that there exists $M_5 > 0$ such that

$$\|u''\|_{L^p(T,X^*)} \leq M_5 \text{for all } u \in S(x_0, x_1). \tag{6.56}$$

As before we replace $F(t, x, y)$ by

$$F_1(t, x, y) = F(t, p_{M_1}(x), p_{M_2}(y)).$$

Recall that for any $M > 0$, by $p_M(\cdot)$ we denote the M-radial retraction on H, defined by

$$p_M(x) = \begin{cases} x & \text{if } |x| \leq M \\ M\frac{x}{|x|} & \text{if } M < |x| \end{cases} \qquad \text{for all } x \in H$$

Then $F_1(t, x, y)$ satisfies hypotheses $H_{10}(i)$, (ii) and moreover, we have

$$|F_1(t, x, y)| = \sup\{|h| : h \in F_1(t, x, y)\} \leq \widehat{a}(t)$$

$$\text{for a.a } t \in T, \text{ all } x, y \in H, \text{ with } \widehat{a} \in L^{p'}(T)_+.$$

In the sequel we consider problem (6.47) with the multivalued vector field F replaced by F_1.

Let $K : L^p(T, X) \to C(T, X)$ be defined by

$$(Kv)(t) = x_0 + \int_0^t v(s)ds \text{ for all } t \in T, \text{ all } v \in L^p(T, X).$$

Using this map and if $\widehat{B}(y)(\cdot) = B(y(\cdot))$, (6.47) (with F replaced by F_1), can equivalently be rewritten as

$$\left\{ \begin{array}{l} v'(t) + A(t, v(t)) + \widehat{B}(Kv)(t) \in F_1(t, (Kv)(t), v(t)) \text{ for a.a } t \in T, \\ v(0) = x_1. \end{array} \right\} \tag{6.57}$$

As in the proof of Theorem 6.8, using Theorem 6.2 (with $L(\cdot)$ as indicated before the theorem), we obtain $v \in W_p(0,b)$ which is a solution of (6.57). Then $u \in K(v) \in S(x_0,x_1) \neq \emptyset$.

Now we will determine the topological properties of the solution set. To this end let

$$C = \{h \in L^{p'}(T,H) : |h(t)| \leq \widehat{a}(t) \text{ for a.a } t \in T\}.$$

This set equipped with the relative weak topology, becomes a compact, metrizable space. For each $h \in C$, we consider the Cauchy problem

$$\left\{ \begin{array}{l} v'(t) + A(t,v(t)) + \widehat{B}(Kv)(t) = h(t) \text{ for a.a } t \in T, \\ v(0) = x_0. \end{array} \right\}$$

This problem has a nonempty solution set

$$Q(h) \subseteq W_p(0,b) \hookrightarrow C(T,H).$$

As in the proof of Theorem 6.8, we have that

$$Q(C) \subseteq W_p(0,b) \text{ is } w\text{-compact},$$

$$Q(C) \subseteq C(T,H) \text{ is compact.}$$

Let $S'(x_0,x_1) = \{u' : u \in S(x_0,x_1)\}$. Then $S'(x_0,x_1) \subseteq C(T,H)$ is closed and $S'(x_0,x_1) \subseteq Q(C)$, so $S'(x_0,x_1) \subseteq C(T,H)$ is compact. But $S(x_0,x_1) = K(S'(x_0,x_1))$. Therefore $S(x_0,x_1) \subseteq C^1(T,H)$ is compact. Finally note that $S(x_0,x_1) \subseteq W^{1,p}((0,b),X)$ is bounded and w-closed, hence w-compact. □

So, we can define the solution multifunction

$$S : X \times H \to P_k(C^1(T,H)) = \{C \subseteq C^1(T,H) : \text{ nonempty, compact}\}.$$

We will determine the continuity properties of this solution multifunction.

Proposition 6.26 *If hypotheses H_1, H_9, H_{10} hold, then the solution multifunction $S : X \times H \to P_k(C^1(T,H))$ is usc.*

Proof According to Proposition *I*-5.5, we need to show that if $C \subseteq C^1(T,H)$ is closed, then $S^-(C) = \{(x_0,x_1) \in X \times H : S(x_0,x_1) \cap C \neq \emptyset\}$ is closed. To this end, let $\{(x_{0n},x_{1n})\}_{n\in\mathbb{N}} \subseteq S^-(C)$ and assume that $x_{0n} \to x_0$ in X and $x_{1n} \to x_1$ in H. From the a priori estimates in the first part of the proof of Theorem 6.25, we have that if $u_n \in S(x_{0n},x_{1n})$ $n \in \mathbb{N}$, then

$$\{u_n'\}_{n\in\mathbb{N}} \subseteq W_p(0,b) \text{ is bounded},$$

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C^1(T,H) \text{ is relatively compact.}$$

Therefore we may assume that

$$u_n' \xrightarrow{w} y \text{ in } W_p(0,b),\ u_n \to u \text{ in } C^1(T,H). \tag{6.58}$$

Clearly we have $u' = y$. We have

$$u_n'' + \widehat{A}(u_n') + \widehat{B}(u_n) = f_n \ n \in \mathbb{N}, \tag{6.59}$$

with $f_n \in S^{p'}_{F(\cdot,u_n(\cdot),u_n'(\cdot))}$. Evidently $\{f_n\}_{n\in\mathbb{N}} \subseteq L^{p'}(T,H)$ is bounded (see hypotheses H_{10}(iii)). So, we may assume that

$$f_n \xrightarrow{w} f \text{ in } L^{p'}(T,H) \tag{6.60}$$

Using (6.58) and Theorem I-8.38, we obtain that $f \in S^{p'}_{F(\cdot,u(\cdot),u'(\cdot))}$. On (6.59) we act with $u_n' - u'$. Then

$$\begin{aligned} &\langle u_n'', u_n' - u_n'\rangle_p + \langle \widehat{A}(u_n), u_n' - u'\rangle_p + \langle \widehat{B}(u_n), u_n' - u'\rangle_p \\ &= \int_0^b (f_n(t), u_n'(t) - u'(t))dt \end{aligned} \tag{6.61}$$

Note that

$$\int_0^b (f_n(t), u_n'(t) - u'(t))dt \to 0. \tag{6.62}$$

Also

$$\begin{aligned} \langle \widehat{B}(u_n), u_n' - u'\rangle_p &= \int_0^b \langle Bu_n(t), u_n'(t) - u'(t)\rangle dt \\ &\geq \int_0^b \langle Bu(t), u_n'(t) - u'(t)\rangle dt \\ &= \langle \widehat{B}(u), u_n' - u'\rangle_p \end{aligned}$$

$$0 = \lim_{n\to\infty} \langle \widehat{B}(u), u_n' - u'\rangle_p \leq \liminf_{n\to\infty} \langle \widehat{B}(u_n), u_n' - u'\rangle_p \tag{6.63}$$

Using Theorem I-4.205 (integration by parts formula) we have

$$\begin{aligned} \langle u_n'' - u'', u_n' - u'\rangle_p &= \int_0^b \langle u_n''(t) - u''(t), u_n'(t) - u'(t)\rangle dt \\ &= \int_0^b \frac{1}{2}\frac{d}{dt}|u_n'(t) - u'(t)|^2 dt \end{aligned}$$

$$= \frac{1}{2}|u_n'(b) - u'(b)|^2 - \frac{1}{2}|u_n'(0) - u'(0)|^2 \to 0 \text{ (see (6.58))},$$

$$\Rightarrow 0 = \lim \langle u', u_n' - u' \rangle_p \leq \liminf_{n\to\infty} \langle u_n', u_n' - u' \rangle_p. \tag{6.64}$$

Returning to (6.61), passing to the limit as $n \to \infty$ and using (6.62), (6.63) and (6.64), we obtain

$$\limsup_{n\to\infty} \langle \widehat{A}(u_n'), u_n' - u' \rangle_p \leq 0$$

Invoking Proposition 6.4, we have

$$\widehat{A}(u_n') \xrightarrow{w} \widehat{A}(u) \text{ in } L^{p'}(T, X^*). \tag{6.65}$$

If in (6.59), we pass to the limit as $n \to \infty$ and use (6.58), (6.60), (6.65), we have

$$u'' + \widehat{A}(u) + \widehat{B}(u) = f,\ f \in S^{p'}_{F(\cdot,u(\cdot),u'(\cdot))},$$

$$u(0) = x_0,\ u'(0) = x_1,$$

$$\Rightarrow u \in S(x_0, x_1) \cap C \text{ (see (6.58))}$$

$$\Rightarrow S(\cdot,\cdot) \text{ is usc.}$$

□

We can have a corresponding existence theorem for the "nonconvex problem". In this case we impose the following conditions on the multivalued vector field $F(t, x, y)$.

H_{11}: $F : T \times H \times H \to P_f(H) = \{C \subseteq H$:nonempty,closed $\}$ is a multifunction such that

(i) $(t, x, y) \to F(t, x.y)$ is graph measurable;
(ii) for a.a $t \in T$, $(x, y) \to F(t, x, y)$ is lsc;
(iii) $|F(t, x, y)| = \sup\{|h| : h \in F(t, x, y)\} \leq a(t) + c(|x|^{\frac{2}{p'}} + |y|^{\frac{2}{p'}})$ for a.a $t \in T$, all $x, y \in H$ with $a \in L^{p'}(T)$, $c > 0$.

Theorem 6.27 *If hypotheses H_1, H_9, H_{11} hold and $x_0 \in X$, $x_1 \in H$, then $S(x_0, x_1) \neq \emptyset$.*

Proof Consider the multifunction $G : L^p(T, H) \to P_f(L^{p'}(T, H))$ defined by $G(u) = S^{p'}_{F(\cdot,K(u)(\cdot),u(\cdot))} = \{h \in L^{p'}(T, H) : h(t) \in F(t, K(u)(t), u(t))$ for a.a $t \in T\}$.

We claim that this multifunction is lsc. According to Proposition I-5.6, we need to show that if $C \subseteq L^{p'}(T, H)$ is nonempty, closed, then $G^+(C) = \{u \in L^p(T, H) : G(u) \subseteq C\}$ is closed. So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq G^+(C)$ and assume that $u_n \to u$ in $L^p(T, H)$. We have to show that $u \in G^+(C)$. Let $h \in G(u)$ and consider the multifunction $\Gamma_n : T \to 2^H \setminus \{0\}$ defined by

$$\Gamma_n(t) = \left\{ \begin{array}{l} y \in F(t, (Ku_n)(t), u_n(t)) \text{ such that} \\ |h(t) - y| \leq d(h(t), F(t, (Ku_n)(t), u_n(t))) + \frac{1}{n} \end{array} \right\}$$

Let $\gamma_n(t, y) = |h(t) - y| - d(h(t), F(t, (Ku)(t), u(t)))$. On account of hypothesis $H_{11}(i)$, $\gamma_n(\cdot, \cdot)$ is a Caratheodory function, hence it is jointly measurable (see Theorem I-2.47). Therefore

$$Gr\Gamma_n = GrF(\cdot, (Ku_n)(\cdot), u_n(\cdot)) \cap \{(t, y) \in T \times H : \gamma_n(t, y) \leq \frac{1}{n}\}$$

$$\in \mathcal{L}_T \times B(H),$$

with $\mathcal{L}_T$ being the Lebesgue σ-algebra of T and $B(H)$ the Borel σ-algebra of H. So, we can use the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106) and produce a measurable map $h_n : T \to H$ such that $h_n(t) \in \Gamma_n(t)$ for a.a $t \in T$. We have

$$|h(t) - h_n(t)| \leq d(h(t), F(t, (Ku_n)(t), u_n(t))) + \frac{1}{n} \tag{6.66}$$

for a.a $t \in T$, all $n \in \mathbb{N}$.

For at least a subsequence, we have

$$u_n(t) \to u(t) \text{ and } (Ku_n)(t) \to (Ku)(t) \text{ in } H \text{ for a.a } t \in T.$$

Therefore by Proposition I-5.19, we have

$$\limsup_{n\to\infty} d(h(t), F(t, (Ku_n)(t), u_n(t))) \leq d(h(t), F(t, (Ku)(t), u(t))) = 0$$

for a.a $t \in T$

(recall $h \in G(u)$). From (66) it follows that

$$h_n(t) \to h(t) \text{ for a.a } t \in T,$$

$$\Rightarrow h_n \to h \text{ in } L^{p'}(T, H).$$

We have $h_n \in G(u_n) \subseteq C$, hence $h \in C$ and since $h \in G(u)$ is arbitrary, we infer that $G(u) \subseteq C$ and so $G^+(C) \subseteq L^p(T, H)$ is closed. This proves that $G(\cdot)$ is

lsc. Since the values of $G(\cdot)$ are decomposable sets, we can apply Theorem I-5.137 and find $g : L^p(T, H) \to L^{p'}(T, H)$ a continuous map such that $g(u) \in G(u)$ for all $u \in L^{p'}(T, H)$. We consider the following Cauchy problem

$$\left\{ \begin{array}{l} v'(t) + A(t, v(t)) + B((Kv)(t)) = g(v)(t) \text{ for a.a } t \in T, \\ v(0) = x_1. \end{array} \right\}$$

From Theorem 6.8 we know that this problem has a solution $v \in W_p(0, b)$. Then $u(t) = x_0 + \int_0^t v(s)ds$ for all $t \in T$ belongs in $S(x_0, x_1)$. □

We can compare the two solution sets (of the convex and of the nonconvex problems). As we already said, such a result is known as "Relaxation Theorem" and it is of practical interest in the study of control systems. By $S_c(x_0, x_1)$ (resp. $S(x_0, x_1)$), we denote the solution set of the convex (resp. nonconvex) problem.

Now we need to strengthen the hypotheses on both the map $A(t, x)$ and the multivalued vector field $F(t, x)$. The new stronger conditions are the following.

H_{12}: $A : T \times X \to X^*$ is a map such that

(i) for all $x \in X$, $t \to A(t, x)$ is measurable;
(ii) for a.a $t \in T$, $x \to A(t, x)$ is hemicontinuous, monotone;
(iii) $\|A(t, x)\|_* \leq a(t) + c\|x\|^{p-1}$ for a.a $t \in T$, all $x \in X$, with $a \in L^{p'}(T)$, $c > 0$ and $1 < p < \infty$;
(iv) $\langle A(t, x), x\rangle \geq \widehat{c}\|x\|^p - \vartheta(t)$ for a.a $t \in T$, all $x \in X$, with $\widehat{c} > 0$ and $\vartheta \in L'(T)$.

Remark 6.28 As before, in order to have relaxation results, we need monotonicity of $A(t, \cdot)$ (see Theorem 6.13). The reason being that the monotonicity condition guarantees uniqueness of the solution of the Cauchy problem $v'(t) + A(t, v(t)) + B(Kv)(t) = g(t)$ for a.a $t \in T$, $v(0) = x_1 \in H$. Hypothesis $H_{12}(ii)$ implies that for a.a $t \in T$, $A(t, \cdot)$ is demicontinuous and maximal monotone. It is an open question, whether the relaxation theorem remains valid if $A(t, \cdot)$ is only pseudomonotone.

H_{13}: $F : T \times H \to P_f(H)$ is a multifunction such that

(i) for all $x, y \in H$, $t \to F(t, x, y)$ is graph measurable;
(ii) $h(\overline{conv}F(t, x, y), \overline{conv}F(t, x', y')) \leq k(t)[|x - x'| + |y - y'|]$ for a.a $t \in T$, all $x, x', y, y' \in H$, with $k \in L^1(T)$;
(iii) $|F(t, x, y)| = \sup\{|h| : h \in F(t, x, y)\} \leq \widehat{a}(t) + \widehat{c}(|x|^{\frac{2}{p'}} + |y|^{\frac{2}{p'}})$ for a.a $t \in T$, all $x, y \in H$, with $\widehat{a} \in L^{p'}(T)$, $\widehat{c} > 0$.

Remark 6.29 In order to have relaxation results, we need h-Lipschitzness of the multivalued vector field. Simple h-continuity is not enough.

Theorem 6.30 *If hypotheses H_{12}, H_{13} hold and $x_0 \in X$, $x_1 \in H$, then $S_c(x_0, x_1) = \overline{S(x_0, x_1)}^{C(T,H)}$.*

Proof Let $u \in S_c(x_0, x_1)$. Then

$$\left\{ \begin{array}{l} u'(t) + A(t, u'(t)) + Bu(t) = f(t) \text{ for a.a } t \in T, \\ u(0) = x_0, u'(0) = x_1, \end{array} \right\}$$

with $f \in S^{p'}_{\overline{conv}F(\cdot,u(\cdot),u'(\cdot))}$. We rewrite this as an equivalent first order evolution equation using the map $Kv(t) = x_0 + \int_0^t v(s)ds$:

$$\left\{ \begin{array}{l} v'(t) + A(t, v(t)) + \widehat{B}(Kv)(t) = f(t) \text{ for a.a } t \in T, \\ v(0) = x_1. \end{array} \right\}$$

As in the proof of Theorem 6.13, we can find $v_n \in W_p(0, b)$ such that

$$\left\{ \begin{array}{l} v_n'(t) + A(t, v_n(t)) + B((Kv)(t)) = f_n(t) \text{ for a.a } t \in T, \\ v_n(0) = x_0. \end{array} \right\}$$

with $f_n \in S^{p'}_{F(\cdot,(Kv_n)(\cdot),v_n(\cdot))}$ $n \in \mathbb{N}$, such that

$$v_n \xrightarrow{w} v \text{ in } W_p(0, b),\ v_n \to v \text{ in } C(T, H).$$

Let $u_n(t) = x_0 + \int_0^t v_n(s)ds$ $n \in \mathbb{N}$. Then $v_n \in W^{1,p}((0, b), X)$ and so $u_n \in S(x_0, x_1)$ with $u_n \to u$ in $C^1(T, H)$. □

We turn our attention to multivalued periodic systems in $\mathbb{R}^N$. So, the problem under consideration is the following, second order periodic system

$$\left\{ \begin{array}{l} a(u'(t))' \in A(u(t)) + F(t, u(t), u'(t)) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b). \end{array} \right\} \tag{6.67}$$

In this problem $a : \mathbb{R}^N \to \mathbb{R}^N$ and $A : D(A) \subseteq \mathbb{R}^N \to 2^{\mathbb{R}^N}$ are two maximal monotone maps. The map $A(\cdot)$ is not necessarily defined on all of $\mathbb{R}^N$ (that is, $D(A)$ is not all of R^N) and so (6.67) incorporates systems with unilateral constraints. Also $F : T \times \mathbb{R}^N \times \mathbb{R}^N \to 2^{\mathbb{R}^N} \setminus \{\emptyset\}$ is a multivalued perturbation.

We introduce our hypotheses on the different items of problem (6.67).

H_{14}: $a : \mathbb{R}^N \to \mathbb{R}^N$ is of the form $a(y) = a_0(|y|)y$ for all $y \in \mathbb{R}^N$, with $a_0(t) > 0$ for all $t > 0$ and

(i) $a(\cdot)$ is continuous and strictly monotone;
(ii) $\widehat{c}|y|^p \leq (a(y), y)_{\mathbb{R}^N}$ for all $y \in \mathbb{R}^N$, with $\widehat{c} > 0$, $2 \leq p$;
(iii) $|a(y)| \leq c(1 + |y|^{p-1})$ for all $y \in \mathbb{R}^N$, some $c > 0$.

Remark 6.31 Note that $a(\cdot)$ is a maximal monotone map, a homeomorphism and we have $|a^{-1}(y)| \to +\infty$ as $|y| \to \infty$. We need the restriction $p \geq 2$, since we do not assume that $D(A) = \mathbb{R}^N$. If the map $A(\cdot)$ is defined on all of $\mathbb{R}^N$, then we can have $p > 1$. The hypotheses H_{14} incorporate the vector p-Laplacian ($a(y) = |y|^{p-2}y$, $2 \leq p$) and the vector (p,q)-Laplacian ($a(y) = |y|^{p-2}y + |y|^{q-2}y$, $1 < q < p$, $2 \leq p$).

$\mathbf{H_{15}}$: $A : \mathbb{R}^N \to 2^{\mathbb{R}^N}$ is a maximal monotone map and $0 \in A(0)$.

Remark 6.32 We do not require that $D(A) = \mathbb{R}^N$. This way our framework includes also problems with unilateral constraints. As we already mentioned in Remark 6.31, since $D(A)$ need not be all of $\mathbb{R}^N$, we need the restriction $p \geq 2$.

For the "convex case", the restrictions on the multivalued perturbation $F(t,x,y)$ are:

$\mathbf{H_{16}}$: $F : T \times \mathbb{R}^N \times \mathbb{R}^N \to P_{fc}(\mathbb{R}^N) = \{C \subseteq \mathbb{R}^N : \text{nonempty, closed, convex}\}$ is a multifunction such that

(i) for every $x, y \in \mathbb{R}^N$, $t \to F(t,x,y)$ is graph measurable;
(ii) for a.a $t \in T$, $(x,y) \to F(t,x,y)$ is closed;
(iii) if $k(t,x,y) = \inf\{(h,x)_{\mathbb{R}^N} : h \in F(t,x,y)\}$, then $\liminf_{|x|\to\infty} \frac{k(t,x,y)}{|x|^p} \geq \vartheta(t)$ uniformly for a.a $t \in T$, all $y \in \mathbb{R}^N$ with $\vartheta \in L^\infty(T)$, $\vartheta(t) \geq 0$ for a.a $t \in T$, $\vartheta \not\equiv 0$;
(iv) $|F(t,x,y)| = \sup\{|h| : h \in F(t,x,y)\} \leq \eta_1(t,|x|) + \eta_2(t,|x|)|y|^{q-1}$ for a.a $t \in T$, all $x, y \in \mathbb{R}^N$ with $1 < q < p$ and

$$\sup\{\eta_1(t,r) : 0 \leq r \leq M\} \leq \gamma_{1,M}(t) \text{ for a.a } t \in T, \text{ with } \gamma_{1,M} \in L^2(T),$$

$$\sup\{\eta_2(t,r) : 0 \leq r \leq M\} \leq \gamma_{2,M}(t) \text{ for a.a } t \in T, \text{ with } \gamma_{2,M} \in L^\infty(T).$$

As before for the "nonconvex case" we need to strengthen the measurability hypothesis (see Remark 6.11) and change hypothesis H_{16}(ii).

$\mathbf{H_{17}}$: $F : T \times \mathbb{R}^N \times \mathbb{R}^N \to P_f(\mathbb{R}^N) = \{C \subseteq \mathbb{R}^N : \text{nonempty, closed }\}$ is a multifunction such that

(i) $F(\cdot,\cdot,\cdot)$ is graph measurable (that is, $GrF = \{(t,x,y,h) \in T \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^N : h \in F(t,x,y)\} \in \mathcal{L}_T \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^N)$, with $\mathcal{L}_T$ being the Lebesgue σ-algebra of T and $B(\mathbb{R}^N)$ is the Borel σ-algebra of $\mathbb{R}^N$);
(ii) for a.a $t \in T$, $(x,y) \to F(t,x,y)$ is lsc;
(iii) the same as hypothesis $H_{16}(iii)$;
(iv) the same as hypothesis $H_{16}(iv)$.

We will need some auxiliary results, which are actually of independent interest Let $g \in L^{p'}(T, \mathbb{R}^N)$. We start by considering the following periodic problem

$$\left\{\begin{array}{l} -a(u'(t)) + |u(t)|^{p-2}u(t) = g(t) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b). \end{array}\right\} \tag{6.68}$$

Lemma 6.33 *If hypotheses H_{14} hold and $g \in L^{p'}(T, \mathbb{R}^N)$, then problem (6.68) has a unique solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$.*

Proof In what follows, for notational economy, we write

$$W_N^{1,p} = W_{per}^{1,p}((0,b), \mathbb{R}^N) = \{u \in W^{1,p}((0,b), \mathbb{R}^N) : u(0) = u(b)\}.$$

We introduce the nonlinear map $V : W_N^{1,p} \to (W_N^{1,p})^*$ defined by

$$\langle V(u), h\rangle = \int_0^b (a(u'), h')_{\mathbb{R}^N} dt \text{ for all } u, h \in W_N^{1,p}.$$

On account of hypotheses H_{14}, $V(\cdot)$ is continuous, monotone, hence maximal monotone. Also, let $\xi_p : L^p(T, \mathbb{R}^N) \to L^{p'}(T, \mathbb{R}^N)$ be defined by

$$\xi_p(u)(\cdot) = |u(\cdot)|^{p-2}u(\cdot) \text{ for all } u \in L^p(T, \mathbb{R}^N).$$

This map is continuous, strictly monotone, hence maximal monotone too. Let $\widehat{V} = V + \xi_p : W_N^{1,p} \to (W_N^{1,p})^*$. This map is continuous, strictly monotone (thus maximal monotone) and

$$\langle \widehat{V}(u), u\rangle \geq \widehat{c}\|u'\|_p^p + \|u\|_p^p,$$

$$\Rightarrow \widehat{V}(\cdot) \text{ is coercive.}$$

Therefore $V(\cdot)$ is maximal monotone, coercive. By Theorem I-7.47, $V(\cdot)$ is surjective. So,we can find $\widehat{u} \in W_N^{1,p} \subseteq C(T, \mathbb{R}^N)$ such that

$$\left\{\begin{array}{l} -a(\widehat{u}'(t))' + |\widehat{u}(t)|^{p-2}\widehat{u}(t) = g(t) \text{ for a.a } t \in T, \\ \widehat{u}(0) = \widehat{u}(b). \end{array}\right\} \tag{6.69}$$

The strict monotonicity of $\widehat{V}(\cdot)$ implies that $\widehat{v}(\cdot)$ is unique. From (6.69), we have that $a(\widehat{u}'') \in L^{p'}(T, \mathbb{R}^N)$. Since $\widehat{u}' \in L^p(T, \mathbb{R}^N)$, from hypothesis H_{14} (iii), we see that $a(\widehat{u}') \in L^{p'}(T, \mathbb{R}^N)$. Therefore, we have that $a(\widehat{u}') \in W_N^{1,p'} \subseteq C(T, \mathbb{R}^N)$. But $a(\cdot)$ is a homeomorphism. Hence $\widehat{u}' \in C(T, \mathbb{R}^N)$ and so we infer that $\widehat{u} \in C^1(T, \mathbb{R}^N)$. Moreover, we have $\widehat{u}'(0) = \widehat{u}'(b)$. □

Let $\widehat{a} : D(\widehat{a}) \subseteq L^p(T, \mathbb{R}^N) \to L^{p'}(T, \mathbb{R}^N)$ be defined by

$$\widehat{a}(u)(\cdot) = -a(u'(\cdot))',$$

for all $u \in D(\widehat{a}) = \{v \in C^1(T, \mathbb{R}^N) : a(v') \in W_N^{1,p'}, v(0) = v(b), v'(0) = v'(b)\}$. Using Lemma 6.33, we can prove the maximal monotonicity of this map.

Proposition 6.34 *If hypotheses H_{14} hold, then the map $\widehat{a}(\cdot)$ is maximal monotone.*

Proof From Lemma 6.33 we know that

$$R(\widehat{a} + \xi_p) = L^{p'}(T, \mathbb{R}^N). \tag{6.70}$$

Also note that $\widehat{a}(\cdot)$ is monotone. To see this let $u, v \in D(\widehat{a})$. We have

$$\begin{aligned}
&\langle \widehat{a}(u) - \widehat{a}(v), u - v \rangle_p \\
&= \int_0^b (-a(u')' + a(v')', u - v)_{\mathbb{R}^N} dt \\
&= \int_0^b (a(u') - a(v'), u' - v')_{\mathbb{R}^N} dt \text{ (by integration by parts)} \\
&\geq 0 \text{ (see hypotheses } H_{14}), \\
&\Rightarrow \widehat{a}(\cdot) \text{ is monotone.}
\end{aligned}$$

Let $v \in L^p(T, \mathbb{R}^N)$, $v^* \in L^{p'}(T, \mathbb{R}^N)$ and assume that

$$\langle \widehat{a}(u) - v^*, u - v \rangle_p \geq 0 \text{ for all } u \in D(\widehat{a}) \tag{6.71}$$

From (6.70), we know that we can find $u_1 \in D(\widehat{a})$ such that

$$\widehat{a}(u_1) + \xi_p(u_1) = v^* + \xi_p(v). \tag{6.72}$$

Using (6.72) in (6.71) with $u = u_1 \in D(\widehat{a})$, we obtain

$$\begin{aligned}
&0 \leq \langle \xi_p(v) - \xi_p(u_1), u_1 - v \rangle_p, \\
&\Rightarrow u_1 = v \text{ (since } \xi_p(\cdot) \text{ is strictly monotone)}, \\
&\Rightarrow \widehat{a}(u_1) = v^* \text{ (see (6.72))}.
\end{aligned}$$

Therefore $(v, v^*) \in Gr\widehat{a}$ and this proves the maximal monotonicity of $\widehat{a}(\cdot)$. □

Next we examine the "lifting" of the map $A(\cdot)$ on the dual pair $(L^p, L^{p'})$. Since the result is of independent interest, we prove it in a more general setting than the one needed for the analysis of problem (6.67).

Let X be a reflexive Banach and X^* its topological dual. We assume that both spaces are locally uniformly convex. Recall that by Troyanski's renorming theorem (see Theorem I-7.42), we can always equivalently renorm X, so that both X and X^* are locally uniformly convex. By $\|\cdot\|$ (resp. $\|\cdot\|_*$) we denote the norm of X (resp of X^*) and by $\langle \cdot, \cdot \rangle$ the duality brackets of the pair (X^*, X). Let $\mathcal{F} : X \to X^*$

be the duality map, which is single valued and a homeomorphism. We consider the "lifting" ("realization") of the map $A(\cdot)$ on the dual pair $(L^{p'}(T, X^*), L^p(T, X))$ $(1 < p < \infty$, see Theorem I-4.173), $\widehat{A} : L^p(T, X) \to 2^{L^{p'}(T,X^*)}$ defined by

$$\widehat{A}(u) = \{u^* \in L^{p'}(T, X^*) : u^*(t) \in A(u(t)) \text{ for a.a } t \in T\}.$$

for all $u \in D(\widehat{A}) = \{u \in L^p(T, X) : S^{p'}_{A(u(\cdot))} \neq \emptyset\}$.

As before by $\langle \cdot, \cdot \rangle_p$ we denote the duality brackets for the pair $(L^{p'}(T, X^*),$ $L^p(T, X))$. Therefore,

$$\langle u^*, u \rangle_p = \int_0^b \langle u^*(t), u(t) \rangle dt \quad \text{for all } u \in L^p(T, X), \text{ all } u^* \in L^{p'}(T, X^*).$$

Proposition 6.35 *If $A : X \to 2^{X^*}$ is a maximal monotone map with $(0, 0) \in GrA$, then $\widehat{A} : L^p(T, X) \to 2^{L^{p'}(T,X^*)}$ is maximal monotone too.*

Proof Let $\widehat{\eta} : L^p(T, X) \to L^{p'}(T, X^*)$ be the map defined by

$$\widehat{\eta}(u)(\cdot) = \|u(\cdot)\|^{p-2}\mathcal{F}(u(\cdot)) \quad \text{for all } u \in L^p(T, X).$$

Clearly $\widehat{\eta}(\cdot)$ is continuous and strictly monotone, thus maximal monotone too (see Proposition I-7.36).

Claim $R(\widehat{A} + \widehat{\eta}) = L^{p'}(T, X^*)$ (that is, $\widehat{A} + \widehat{\eta}$ is surjective).

Let $h^* \in L^{p'}(T, X^*)$ and consider the multifunction $L : T \to 2^X$ defined by

$$L(t) = \{x \in X : h^*(t) \in A(x) + \|x\|^{p-2}\mathcal{F}(x)\}.$$

We know that the map $x \to A(x) + \|x\|^{p-2}\mathcal{F}(x)$ is maximal monotone (see Theorem I-7.58) and coercive. Therefore it is surjective (see Theorem I-7.47). Hence $L(t) \neq \emptyset$ for all $t \in T \setminus N$, with N being Lebesgue null. We set $L(t) = \{0\}$ for all $t \in N$. Then

$$GrL = \{(t, x) \in T \times X : (x, h^*(t) - \|x\|^{p-2}\mathcal{F}(x)) \in GrA\}.$$

The maximal monotonicity of A, implies that $GrA \subseteq X \times X^*$ is closed (see Proposition I-7.35). Moreover, let $\vartheta : T \times X \to X \times X^*$ be the map defined by

$$\vartheta(t, x) = (x, h^*(t) - \|x\|^{p-2}\mathcal{F}(x)).$$

This is a Caratheodory map, hence jointly measurable (see Theorem I-2.47). Therefore,

$$\vartheta^{-1}(GrA) = GrL \in \mathcal{L}_T \otimes B(X),$$

with $\mathcal{L}_T$ being the Lebesgue σ-algebra of T and $B(X)$ the Borel σ-algebra of X. Then the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), produces a measurable map $u : T \to X$ such that $u(t) \in L(t)$ for all $t \in T$. So, we have

$$h^*(t) \in A(u(t)) + \|u(t)\|^{p-2}\mathcal{F}(u(t)) \quad \text{for all } t \in T. \tag{6.73}$$

We act on (6.73) with $u(t) \in X$ and using the hypothesis that $(0, 0) \in GrA$, we obtain

$$\begin{aligned} &\|u(t)\|^{p-1} \leq \|h^*(t)\|_* \quad \text{for a.a } t \in T, \\ \Rightarrow &u \in L^p(T, X) \text{ and } h^* \in \widehat{A}(u) + \widehat{\eta}(u). \end{aligned}$$

Since $h^* \in L^{p'}(T, X^*)$ is arbitrary, we conclude that

$$R(\widehat{A} + \widehat{\eta}) = L^{p'}(T, X^*).$$

This proves the Claim.

Evidently $\widehat{A}(\cdot)$ is monotone. Let $(v, g^*) \in L^p(T, X) \times L^{p'}(T, X^*)$ and suppose that

$$\langle u^* - g^*, u - v\rangle_p = \int_0^b \langle h^*(t) - g^*(t), u(t) - v(t)\rangle dt \quad \text{for all } (u, u^*) \in Gr\widehat{A}. \tag{6.74}$$

On account of the Claim, we can find $(\widehat{u}, \widehat{u}^*) \in Gr\widehat{A}$ such that

$$\widehat{u}^* + \widehat{\eta}(\widehat{u}) = g^* + \widehat{\eta}(v). \tag{6.75}$$

Then in (6.74) we choose $v = \widehat{u}$, $g^* = \widehat{u}^*$ and obtain

$$\begin{aligned} 0 &\leq \int_0^b \langle g^* + \widehat{\eta}(v) - \widehat{\eta}(\widehat{u}) - g^*, \widehat{u} - v\rangle dt \\ &= \int_0^b \langle \widehat{\eta}(v) - \widehat{\eta}(\widehat{u}), \widehat{u} - v\rangle dt \leq 0 \quad \text{(see (6.75))}, \\ \Rightarrow &\widehat{u} = v \text{ (since } \widehat{\eta}(\cdot) \text{ is strictly monotone)}. \end{aligned}$$

From (6.75) it follows $g^* = \widehat{\eta}(\widehat{u})$ and so $(v, g^*) \subseteq Gr\widehat{A}$ and this proves the maximality of $\widehat{A}(\cdot)$. □

For $\lambda > 0$, $A_\lambda(\cdot)$ denotes the Yosida approximation of $A(\cdot)$ (see Theorem I-7.51). We consider the following approximation of problem (67).

$$\left\{\begin{array}{l} a(u'(t))' \in A_\lambda(u(t)) + F(t, u(t), u'(t)) \quad \text{for a.a } t \in T, \\ u(0) = u(b),\ u'(0) = u'(b). \end{array}\right\}. \tag{6.76}$$

Proposition 6.36 *If hypotheses H_{14}, H_{15}, H_{16} hold and $\lambda > 0$, then problem (6.76) has a solution $u_\lambda \in C^1(T, \mathbb{R}^N)$.*

Proof Consider the Nemytskii map $\widehat{A}_\lambda : L^p(T, \mathbb{R}^N) \to L^{p'}(T, \mathbb{R}^N)$ corresponding to $A_\lambda(\cdot)$ and defined by

$$\widehat{A}_\lambda(u)(\cdot) = A_\lambda(u(\cdot)) \quad \text{for all } u \in L^p(T, \mathbb{R}^N).$$

Let $L_\lambda : D(\widehat{a}) \subseteq L^p(T, \mathbb{R}^N) \to L^{p'}(T, \mathbb{R}^N)$ be defined by

$$L_\lambda(u) = \widehat{a}(u) + \xi_p(u) + \widehat{A}_\lambda(u) \quad \text{for all } u \in D(\widehat{a}).$$

Using Lemma 6.34, Proposition I-7.52 and Theorem I-7.58, we see that

$$L_\lambda(\cdot) \text{ is maximal monotone.} \tag{6.77}$$

Also since $\widehat{A}_\lambda(0) = 0$, we infer that

$$L_\lambda(\cdot) \text{ is coercive.} \tag{6.78}$$

From (6.77), (6.78) and Theorem I-7.47, we infer that $L_\lambda(\cdot)$ is surjective. Also $L_\lambda(\cdot)$ is strictly monotone (since $\xi_p(\cdot)$ is) and so we can define the inverse map

$$L_\lambda^{-1} : L^{p'}(T, \mathbb{R}^N) \to D(\widehat{a}) \subseteq L^p(T, \mathbb{R}^N).$$

Claim 1 $L_\lambda^{-1} : L^p(T, \mathbb{R}^N) \to C^1(T, \mathbb{R}^N)$ is completely continuous (recall that $D(\widehat{a}) \subseteq C^1(T, \mathbb{R}^N)$ and see Definition I-3.146).

Let $g_n^* \xrightarrow{w} g^*$ in $L^{p'}(T, \mathbb{R}^N)$. We set

$$u_n = L_\lambda^{-1}(g_n^*)\ n \in \mathbb{N} \quad \text{and} \quad u = L_\lambda^{-1}(g^*).$$

Then $L_\lambda(u_n) = g_n^*$, $n \in \mathbb{N}$ and so

$$\begin{aligned} & g_n^* = \widehat{a}(u_n) + \xi_p(u_n) + \widehat{A}_\lambda(u_n),\ n \in \mathbb{N}, \\ \Rightarrow & \widehat{c}\|u_n'\|_p^p + \|u_n\|_p^p \leq \|g_n^*\|_{p'}\|u_n\|_p \\ & \quad (\text{since } (A_\lambda(x), x)_{\mathbb{R}^N} \geq 0 \text{ for all } x \in \mathbb{R}^N), \\ \Rightarrow & \{u_n\}_{n\in\mathbb{N}} \subseteq W_N^{1,p} \text{ is bounded.} \end{aligned} \tag{6.79}$$

For almost all $t \in T$ and all $n \in \mathbb{N}$, we have

$$\begin{aligned} &-a(u_n'(t))' + |u_n(t)|^{p-2}u_n(t) + A_\lambda(u_n(t)) = g_n^*(t) \\ &\Rightarrow \{\widehat{a}(u_n')'\}_{n\in\mathbb{N}} \subseteq L^p(T,\mathbb{R}^N) \text{ is bounded} \qquad (6.80) \\ &\qquad\qquad \text{(see (6.79) and Proposition I-7.56).} \end{aligned}$$

Then hypothesis $H_{14}(iii)$ and (6.79), imply that

$$\begin{aligned} &\{\widehat{a}(u_n')\}_{n\in\mathbb{N}} \subseteq L^{p'}(T,\mathbb{R}^N) \text{ is bounded,} \\ &\Rightarrow \{\widehat{a}(u_n')\}_{n\in\mathbb{N}} \subseteq W_N^{1,p'} \text{ is bounded.} \end{aligned}$$

Since $W_N^{1,p'} \hookrightarrow C(T,\mathbb{R}^N)$ compactly, it follows that

$$\{\widehat{a}(u_n')\}_{n\in\mathbb{N}} \subseteq C(T,\mathbb{R}^N) \text{ is relatively compact.} \qquad (6.81)$$

Recall that the map $a(\cdot)$ is a homeomorphism. So, if $\widehat{\gamma} : C(T,\mathbb{R}^N) \to C(T,\mathbb{R}^N)$ is defined by

$$\widehat{\gamma}(u)(\cdot) = a^{-1}(u(\cdot)) \quad \text{for all } u \in C(T,\mathbb{R}^N),$$

then $\widehat{\gamma}(\cdot)$ is continuous and bounded (in the sense that it maps bounded sets to bounded ones). So, from (6.81), we infer that

$$\{u_n'\}_{n\in\mathbb{N}} \subseteq C(T,\mathbb{R}^N) \text{ is relatively compact.} \qquad (6.82)$$

Moreover, from (6.79) and since $W_N^{1,p} \hookrightarrow C(T,\mathbb{R}^N)$ compactly, we have that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C(T,\mathbb{R}^N) \text{ is relatively compact.} \qquad (6.83)$$

From (6.82) and (6.83), we infer that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C^1(T,\mathbb{R}^N) \text{ is relatively compact.}$$

So, we may assume (at least for a subsequence), that we have

$$u_n \to \widehat{u} \text{ in } C^1(T,\mathbb{R}^N) \text{ as } n \to \infty. \qquad (6.84)$$

We know that $(u_n, g_n^*) \in GrL_\lambda$ for all $n \in \mathbb{N}$. The map $L_\lambda(\cdot)$ is maximal monotone and so GrL_λ is closed in $L^p(T,\mathbb{R}^N) \times L^{p'}(T,\mathbb{R}^N)_w$ (see Proposition I-7.35). Therefore $(\widehat{u}, g^*) \in GrL_\lambda$ and so for the original sequence, we have

$$u_n \to \widehat{u} = L_\lambda^{-1}(g^*) \text{ in } C^1(T, \mathbb{R}^N) \text{ as } n \to \infty,$$

$$\Rightarrow L_\lambda^{-1}(\cdot) \text{ is completely continuous.}$$

The proves Claim 1.

Next, let $N_F(\cdot)$ be the multivalued Nemytskii map corresponding to F and defined by

$$N_F(u) = S^{p'}_{F(\cdot,u(\cdot),u'(\cdot))}$$

Then we define $\widehat{N} : W^{1,p}_N \to 2^{L^{p'}(T,\mathbb{R}^N)}$ by

$$\widehat{N}(u) = -N_F(u) + \xi_p(u) \quad \text{for all } u \in W^{1,p}_N.$$

Evidently for all $u \in W^{1,p}_N$, we have

$$\widehat{N}(u) \in P_{wkc}(L^{p'}(T, \mathbb{R}^N))$$

$$= \{C \subseteq L^{p'}(T, \mathbb{R}^N) : \text{nonempty, } w\text{-compact, convex}\}.$$

Claim 2 $\widehat{N}(\cdot)$ is usc from $W^{1,p}_N$ into $L^{p'}(T, \mathbb{R}^N)_w$.

It is clear from hypothesis $H_{16}(iv)$, that $\widehat{N}(\cdot)$ is locally compact. Therefore by Proposition I-5.13, to show the upper semicontinuity of $\widehat{N}(\cdot)$, it suffices to show that $Gr\widehat{N} \subseteq W^{1,p}_N \times L^{p'}(T, \mathbb{R}^N)_w$ is closed. So, we consider a sequence $\{(u_n, h^*_n)\}_{n\in\mathbb{N}} \subseteq Gr\widehat{N}$ such that

$$u_n \to u \text{ in } W^{1,p}_N, \ h^*_n \xrightarrow{w} h^* \text{ in } L^{p'}(T, \mathbb{R}^N) \text{ as } n \to \infty. \tag{6.85}$$

We have $u_n \to u$ in $C(T, \mathbb{R}^N)$ and so there exists $M > 0$ such that

$$\|u_n\|_{C(T,\mathbb{R}^N)} \leq M \text{ for all } n \in \mathbb{N}. \tag{6.86}$$

We may assume that

$$u_n(t) \to u(t) \text{ for all } t \in T,$$

$$u'_n(t) \to u'(t) \text{ for a.a } t \in T,$$

$$|u'_n(t)| \leq \mu(t) \text{ for a.a } t \in T \text{ with } \mu \in L^p(T) \text{ (see (6.85))}.$$

It follows that

$$\limsup_{n\to\infty} F(t, u_n(t), u_n'(t)) \subseteq F(t, u(t), u'(t)) \text{ for a.a } t \in T. \tag{6.87}$$

Also we have

$$\xi_p(u_n) \to \xi_p(u) \text{ in } L^{p'}(T, \mathbb{R}^N). \tag{6.88}$$

Then from (6.85), (6.87), and (6.88), we infer that

$$\begin{aligned} &h^* \in \widehat{N}(u), \\ \Rightarrow &Gr\widehat{N} \text{ is closed in } W_N^{1,p} \times L^{p'}(T, \mathbb{R}^N)_w, \\ \Rightarrow &\widehat{N}(\cdot) \text{ is usc from } W_N^{1,p} \text{ into } L^{p'}(T, \mathbb{R}^N)_w. \end{aligned}$$

This proves Claim 2.
Let $S_\lambda = \{u \in W_N^{1,p} : u \in \widehat{\vartheta} L_\lambda^{-1} \widehat{N}(u), 0 < \widehat{\vartheta} < 1\}$.

Claim 3 $S_\lambda \subseteq W_N^{1,p}$ is bounded.

Let $u \in S_\lambda$. Then for some $\widehat{\vartheta} \in (0, 1)$, we have

$$\begin{aligned} &L_\lambda(\frac{1}{\widehat{\vartheta}} u) \in \widehat{N}(u), \\ \Rightarrow &\widehat{a}(\frac{1}{\widehat{\vartheta}} u) + \frac{1}{\widehat{\vartheta}^{p-1}} \xi_p(u) + \widehat{A}_\lambda(\frac{1}{\widehat{\vartheta}} u) \in -N_F(u) + \xi_p(u). \end{aligned} \tag{6.89}$$

On account of hypotheses $H_{16}(i)$,(iii), given $\varepsilon > 0$, we can find $c_1 = c_1(\varepsilon) > 0$ and $\mu \in L^{p'}(T)$ such that

$$(-h, x)_{\mathbb{R}^N} \leq (-\vartheta(t) + \varepsilon)|x|^p + c_1|y|^{q-1}|x| + \mu(t) \quad \text{for a.a } t \in T, \text{ all } x, y \in \mathbb{R}^N.$$

Acting on (6.89) with u, we obtain

$$\begin{aligned} &\frac{\widehat{c}}{\widehat{\vartheta}^{p-1}} \|u'\|_p^p + \frac{1}{\widehat{\vartheta}^{p-1}} \|u\|_p^p \\ &\leq \int_0^b [-\vartheta(t) + \varepsilon]|u|^p dt + c_1 \int_0^b |u'|^{q-1}|u| dt + \|\mu\|_1, \\ \Rightarrow &\widehat{c}\|u'\|_p^p + (1 - \varepsilon)\|u\|_p^p + \int_0^b \vartheta(t)|u|^p dt \leq c_1 \int_0^b |u'|^{q-1}|u| dt + \|\mu\|_1 \\ &\qquad \text{(since } 0 < \widehat{\vartheta} < 1). \end{aligned} \tag{6.90}$$

Using Young's inequality with $\varepsilon > 0$, we obtain

$$c_1 \int_0^b |u'|^{q-1}|u|dt \leq c_2\|u\|^\tau + \varepsilon\|u\|_p^p$$

$$\text{for some } c_2 = c_2(\varepsilon) > 0 \text{ (recall } q < p).$$

Using this estimate in (6.90), we have

$$\widehat{c}\|u'\|_p^p + (1-2\varepsilon)\|u\|_p^p + \int_0^b \vartheta(t)|u|^p dt \leq c_3(1+\|u\|^\tau)$$

$$\text{for some } c_3 > 0.$$

Choosing $\varepsilon \in (0, \frac{1}{2})$, we infer that

$$\|u\|^p \leq c_4(1+\|u\|^\tau) \text{ for some } c_4 > 0$$

$$\Rightarrow S_\lambda \subseteq W_N^{1,p} \text{ is bounded (since } \tau < p).$$

This proves Claim 3.

Note that Claims 1, 2 and 3 permit the use of Theorem 2.138 (the multivalued Leray-Schauder Alternative Principle) and we find $u_\lambda \in D(\widehat{a})$ such that

$$u_\lambda \in L_\lambda^{-1}\widehat{N}(u_\lambda),$$

$$\Rightarrow u_\lambda \in C^1(T, \mathbb{R}^N) \text{ is a solution of (6.76).}$$

□

Now we will let $\lambda \to 0^+$ in order to produce a solution of problem (67).

Theorem 6.37 *If hypotheses H_{14}, H_{15}, H_{16} hold, then problem (67) has a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$.*

Proof Let $\lambda_n \to 0^+$ and let $u_n = u_{\lambda_n} \in C^1(T, \mathbb{R}^N)$ be the solutions from Proposition 6.36. We have

$$\widehat{a}(u_n) + \widehat{A}_{\lambda_n}(u_n) + h_n^* = 0 \quad \text{with } h_n^* \in N_F(u_n),\ n \in \mathbb{N}. \tag{6.91}$$

Acting with u_n, we obtain

$$
\begin{aligned}
\widehat{c}\|u_n'\|_p^p &\leq \int_0^b (-h_n^*, u_n)_{\mathbb{R}^N} dt \\
&\leq \int_0^b (-\vartheta(t) + \varepsilon)|u_n|^p dt + c_1 \int_0^b |u_n'|^{q-1}|u_n| dt + \|\mu\|_1, \\
\Rightarrow \widehat{c}\|u_n'\|_p^p &+ \int_0^b \vartheta(t)|u_n|^p dt - \varepsilon\|u_n\|_p^p \leq c_1 \int_0^b |u_n'|^{q-1}|u_n| dt + \|\mu\|_1, \\
\Rightarrow c_5\|u_n\|^p &\leq c_1 \int_0^b |u_n'|^{q-1}|u_n| dt + \|\mu\|_1 \text{ for some } c_5 > 0 \\
&\qquad\qquad\qquad\qquad \text{(see hypothesis } H_{16}(iii)) \\
\Rightarrow \{u_n\}_{n\in\mathbb{N}} &\subseteq W_N^{1,p} \text{ is bounded (since } q < p).
\end{aligned}
\tag{6.92}
$$

On (6.91) we act with $\widehat{A}_{\lambda_n}(u_n) = A_{\lambda_n}(u_n(\cdot)) \in L^\infty(T, \mathbb{R}^N)$ since

$$
|A_{\lambda_n}(u_n(t))| \leq \frac{1}{\lambda_n}|u_n(t)| \quad \text{for all } t \in T, \text{ all } n \in \mathbb{N}.
$$

We have

$$
\begin{aligned}
&\int_0^b (-a(u_n')', A_{\lambda_n}(u_n))_{\mathbb{R}^N} dt + \|\widehat{A}_{\lambda_n}(u_n)\|_2^2 \\
&\leq \int_0^b |h_n^*||A_{\lambda_n}(u_n)| dt.
\end{aligned}
\tag{6.93}
$$

From Proposition I-7.56, we know that $x \to A_{\lambda_n}(x)$ is Lipschitz continuous. So, by Rademacher's theorem (see Theorem I-6.135), it is differentiable almost everywhere. Since $u_n \in C^1(T, \mathbb{R}^N)$, by the chain rule it follows that $t \to A_{\lambda_n}(u_n(t))$ is differentiable almost everywhere and

$$
\frac{d}{dt}A_{\lambda_n}(u_n(t)) = A_{\lambda_n}'(u_n(t))u_n'(t) \quad \text{for a.a } t \in T,
$$

with $A_{\lambda_n}'(u_n(t))$ being zero when $u_n'(t) = 0$ (even if $A_{\lambda_n}(\cdot)$ is not differentiable there). Exploiting the monotonicity of $A_{\lambda_n}(\cdot)$ at every point of differentiability $x \in \mathbb{R}^N$, we have

$$
0 \leq (y, A_{\lambda_n}'(x)y)_{\mathbb{R}^N} \quad \text{for all } y \in \mathbb{R}^N. \tag{6.94}
$$

From integration by parts, we obtain

$$\begin{aligned}
&\int_0^b (-a(u_n')', A_{\lambda_n}(u_n))_{\mathbb{R}^N} dt \\
&= \int_0^b (a(u_n'), \frac{d}{dt} A_{\lambda_n}(u_n))_{\mathbb{R}^N} dt \\
&= \int_0^b a_0(|u_n'|)(u_n', A_{\lambda_n}'(u_n)u_n')_{\mathbb{R}^N} dt \\
&\geq 0 \text{ (see (6.94))}.
\end{aligned} \tag{6.95}$$

We return to (6.93) and use (6.95). Then

$$\|\widehat{A}_{\lambda_n}(u_n)\|_2 \leq c_6 \text{ for some } c_6 > 0, \text{ all } n \in \mathbb{N} \quad (\text{since } 2 \leq p). \tag{6.96}$$

So, we may assume that

$$\widehat{A}_{\lambda_n}(u_n) \xrightarrow{w} \beta \text{ in } L^2(T, \mathbb{R}^N), \text{ as } n \to \infty \quad (\text{hence in } L^{p'}(T, \mathbb{R}^N) \text{ too since } 1 < p' \leq 2 \leq p). \tag{6.97}$$

As before (see the proof of Proposition 6.36), we show that

$$\{u_n\}_{n \in \mathbb{N}} \subseteq C^1(T, \mathbb{R}^N) \text{ is relatively compact.}$$

Hence we may assume that

$$u_n \to \widehat{u} \text{ in } C^1(T, \mathbb{R}^N) \text{ as } n \to \infty. \tag{6.98}$$

Also $\{h_n^*\}_{n \in \mathbb{N}} \subseteq L^{p'}(T, \mathbb{R}^N)$ is bounded and so

$$h_n^* \xrightarrow{w} h^* \text{ in } L^{p'}(T, \mathbb{R}^N)$$

with $h^*(t) \in F(t, u(t), u'(t))$ for a.a $t \in T$ (see Theorem I-8.38). In the limit as $n \to \infty$ in (6.91), and using (6.97), (6.98), we obtain

$$\widehat{a}(\widehat{u}) + \beta + h^* = 0.$$

If we can show that $\beta(t) \in A(\widehat{u}(t))$ for a.a $t \in T$, then we will complete the proof of the theorem. To this end note that

$$J_{\lambda_n}(u_n(t)) + \lambda_n A_{\lambda_n}(u_n(t)) = u_n(t) \quad \text{for all } t \in T, \text{ all } n \in \mathbb{N}$$

with $J_{\lambda_n}(\cdot)$ being the resolvent operator (see Definition I-7.51). Then if $\widehat{J}_\lambda(u)(\cdot) = J_\lambda(u(\cdot))$ for all $\lambda > 0$, all $u \in W_N^{1,p}$, we can write

$$\begin{aligned} &\widehat{J}_{\lambda_n}(u_n) + \lambda_n \widehat{A}_{\lambda_n}(u_n) = u_n \quad \text{for all } n \in \mathbb{N}, \\ \Rightarrow &\|\widehat{J}_{\lambda_n}(u_n) - u_n\|_2 = \lambda_n \|\widehat{A}_{\lambda_n}(u_n)\|_2 \leq \lambda_n c_6 \text{ (see (6.96))}, \\ \Rightarrow &\widehat{J}_{\lambda_n}(u_n) \to \widehat{u} \text{ in } L^2(T, \mathbb{R}^N). \end{aligned} \tag{6.99}$$

We know that

$$\begin{aligned} &A_{\lambda_n}(u_n(t)) \in A(J_{\lambda_n}(u_n(t))) \quad \text{for all } t \in T, \text{ all } n \in \mathbb{N}, \\ \Rightarrow &(\widehat{J}_{\lambda_n}(u_n), \widehat{A}_{\lambda_n}(u_n)) \in Gr\widehat{A} \quad \text{for all } n \in \mathbb{N}, \\ \Rightarrow &(\widehat{u}, \beta) \in Gr\widehat{A} \text{ (see (6.98) and Proposition 6.35)}, \\ \Rightarrow &\beta(t) \in A(\widehat{u}(t)) \quad \text{for a.a } t \in T. \end{aligned}$$

This completes the proof of the theorem. □

There is a corresponding existence result for the "nonconvex problem".

Theorem 6.38 *If hypotheses H_{14}, H_{15}, H_{17} hold, then problem (67) has a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$.*

Proof As before let $N_F : W_N^{1,p} \to 2^{L^{p'}(T,\mathbb{R}^N)}$ be the multivalued Nemytskii map corresponding to F, defined by

$$N_F(u) = S^{p'}_{F(\cdot,u(\cdot),u'(\cdot))}.$$

Hypothesis $H_{17}(i)$ and Theorem I-5.106, imply that for all $u \in W_N^{1,p} = \{u \in W^{1,p}((0,b), \mathbb{R}^N) : u(0) = u(b)\}$, we have

$$N_F(u) \in P_f(L^{p'}(T, \mathbb{R}^N)) = \{C \subseteq L^{p'}(T, \mathbb{R}^N) : \text{nonempty, closed}\}.$$

We claim that $N_F(\cdot)$ is lsc. According to Proposition I-5.19 it suffices to show that for every $g \in L^{p'}(T, \mathbb{R}^N)$, the function

$$u \to d(g, N_F(u)) = \inf\{\|g - h\|_{p'} : h \in N_F(u)\}$$

is an upper semicontinuous $\mathbb{R}_+$-valued function. To this end, we have

$$\begin{aligned} &d(g, N_F(u))^{p'} \\ &= \inf\{\|g - h\|^{p'}_{p'} : h \in N_F(u)\} \\ &= \int_0^b d(g(t), F(t, u(t), u'(u)))^{p'} dt \text{ (see Theorem I-5.126)}. \end{aligned} \tag{6.100}$$

To show the upper semicontinuity of $u \to d(g, N_F(u))$, we need to show that for every $\lambda \geq 0$, the superlevel set

$$U_\lambda = \{u \in W_N^{1,p} : d(g, N_F(u)) \geq \lambda\}$$

is closed. So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq U_\lambda$ and assume that $u_n \to u$ in $W_N^{1,p}$. Then at least for a subsequence we have

$$u_n(t) \to u(t) \text{ for all } t \in T,\ u_n'(t) \to u'(t) \text{ for a.a } t \in T. \tag{6.101}$$

Since $(x, y) \to d(g(t), F(t, x, y))$ is upper semicontinuous (see hypothesis $H_{17}(ii)$ and Proposition I-5.19), from (6.101), we infer that

$$\begin{aligned} \limsup_{n\to\infty} d(g(t), F(t, u_n(t), u_n'(t))) \leq & d(g(t), F(t, u(t), u'(t))) \\ & \text{for a.a } t \in T. \end{aligned} \tag{6.102}$$

Then using (6.102), Fatou's lemma and the fact that $\{u_n\}_{n\in\mathbb{N}} \subseteq U_\lambda$, we have

$$\begin{aligned} \lambda &\leq \liminf_{n\to\infty} d(g, N_F(u_n)) \\ &\leq \limsup_{n\to\infty} d(g, N_F(u_n)) \\ &= \limsup_{n\to\infty} \int_0^b d(g(t), F(t, u_n(t), u_n'(t)))dt \\ &\leq \int_0^b \limsup d(g(t), F(t, u_n(t), u_n'(t)))dt \\ &= d(g, N_F(u)) \\ \Rightarrow & u \in U_\lambda \text{ and so } u \to d(g, N_F(u)) \text{ is upper semicontinuous.} \end{aligned}$$

Since the values of $N_F(\cdot)$ are decomposable sets, we can apply Theorem I-5.137 and find a continuous map such $k : W_N^{1,p} \to L^{p'}(T, \mathbb{R}^N)$ such that

$$k(u) \in N_F(u) \text{ for all } u \in W_N^{1,p}.$$

We consider the following periodic system

$$\left\{ \begin{array}{ll} a(u'(t))' \in A(u(t)) + k(u)(t) & \text{for a.a } t \in T, \\ u(0) = u(b),\ u'(0) = u'(b). & \end{array} \right\}.$$

From Theorem 6.37, we know that this problem has a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$, which is a solution of the nonconvex problem. □

Remark 6.39 The two existence theorems (Theorems 6.37 and 6.38) are also valid if we assume Dirichlet boundary conditions (that is, $u(0) = u(b) = 0$), Neumann boundary conditions (that is $u'(0) = u'(b) = 0$) mixed boundary conditions (that is $u'(0) = u(b) = 0$). Note that for the Dirichlet and mixed problems, the principal eigenvalue $\widehat{\lambda}_1 > 0$ and so $\vartheta(t) \geq \widehat{\lambda}_1$ for a.a $t \in T$, $\vartheta \not\equiv \widehat{\lambda}_1$ (see hypotheses $H_{16}(iii)$ and $H_{17}(iii)$) while for the Neumann (and periodic) problem $\widehat{\lambda}_1 = 0$ and so $\vartheta(t) \geq 0$ for a.a $t \in T$, $\vartheta \not\equiv 0$.

When $domA = \mathbb{R}^N$, we can relax the hypotheses on the data of problem. The new conditions, are the following:

H_{18}: $a : \mathbb{R}^N \to \mathbb{R}^N$ is a map such that $a(0) = 0$ and

(i) $a(\cdot)$ is continuous and strictly monotone;
(ii) $\widehat{c}|y|^p \leq (a(y), y)_{\mathbb{R}^N}$ for all $y \in \mathbb{R}^N$, with $\widehat{c} > 0$, $1 < p < \infty$;
(iii) $|a(y)| \leq c(1 + |y|^{p-1})$ for all $y \in \mathbb{R}^N$, with $c > 0$.

Remark 6.40 We have removed the restriction $2 \leq p$ and so we incorporate in our setting also singular operators such as $y \to |y|^{p-2}y$ with $1 < p < 2$. Also the map $a(y) = |y|^{p-2}y + p_C(y)$ for all $y \in \mathbb{R}^N$, with $1 < p < \infty$ and $p_C(\cdot)$ the metric projection on the nonempty, closed, convex set $C \subseteq \mathbb{R}^N$ (see Proposition I-3.176).

H_{19}: $A : \mathbb{R}^N \to 2^{\mathbb{R}^N}$ is a maximal monotone map with $D(A) = \mathbb{R}^N$ and $0 \in A(0)$.

Remark 6.41 Now we require that $D(A) = \mathbb{R}^N$ which excludes from our setting systems with unilateral constraints. On the other hand, the above hypotheses incorporate gradient systems with nonsmooth potential, that is, $A = \partial\varphi$ with $\varphi(\cdot)$ continuous, convex.

For the multivalued vector field $F(t, x, y)$, we can improve the growth condition.

H_{20}: $F : T \times \mathbb{R}^N \times \mathbb{R}^N \to P_{kc}(\mathbb{R}^N)$ (resp. $F : T \times \mathbb{R}^N \times \mathbb{R}^N \to P_k(\mathbb{R}^N)$) is a multifunction satisfying hypotheses $H_{16}(i)$, (ii), (iii) (resp. $H_{17}(i)$, (ii), (iii)) and

(iv) for every $r > 0$, there exist $\eta_r \in L^{p'}(T)$ and $c_r > 0$ such that

$$|F(t, x, y)| = \sup\{|h| : h \in F(t, x, y)\} \leq \eta_r(t) + c_r|y|^{p-1}$$

for a.a $t \in T$, all $|x| \leq r$, all $y \in \mathbb{R}^N$.

As before $\widehat{A} : L^p(T, \mathbb{R}^N) \to 2^{L^{p'}(T,\mathbb{R}^N)}$ is the "lifting" of $A(\cdot)$ on the pair $(L^p(T, \mathbb{R}^N), L^{p'}(T, \mathbb{R}^N))$ and $D(\widehat{A}) = \{u \in L^p(T, \mathbb{R}^N) : S^{p'}_{A(u(\cdot))} \neq \emptyset\}$ (recall that $S^{p'}_{A(u(\cdot))} = \{h \in L^{p'}(T, \mathbb{R}^N) : h(t) \in A(u(t)) \text{ for a.a } t \in T\}$).

Lemma 6.42 *If hypotheses H_{19} hold, then $C(T, \mathbb{R}^N) \subseteq D(\widehat{A})$.*

Proof Hypotheses H_{19} imply that $A(\cdot)$ is $P_{kc}(\mathbb{R}^N)$-valued and usc. So $A(\cdot)$ maps compact sets to compact sets (see Proposition I-5.14). Therefore if $u \in C(T, \mathbb{R}^N)$, then

$$|A(u(t))| = \sup\{|g| : g \in A(u(t))\} \leq c \quad \text{for some } c > 0, \text{ all } t \in T. \tag{6.103}$$

The multifunction $t \in A(u(t))$ has closed graph. So, using Theorem I-5.106 (the Yankov-von Neumann-Aumann selection theorem), we can find a measurable map $g : T \to \mathbb{R}^N$ such that

$$\begin{aligned} &g(t) \in A(u(t)) \text{ for a.a } t \in T, \\ \Rightarrow &g \in L^\infty(T, \mathbb{R}^N) \subseteq L^{p'}(T, \mathbb{R}^N) \text{ (see (6.103))}, \\ \Rightarrow &S^{p'}_{A(u(\cdot))} \neq \emptyset. \end{aligned}$$

Therefore $C(T, \mathbb{R}^N) \subseteq D(\widehat{A})$. □

Also, we know that $\widehat{A}(\cdot)$ is maximal monotone (see Proposition 6.35). As before $N_F : W^{1,p}_N \to 2^{L^{p'}(T,\mathbb{R}^N)}$ is the multivalued Nemytskii map corresponding to F, defined by

$$N_F(u) = S^p_{F(\cdot,u(\cdot),u'(\cdot))} \quad \text{for all } u \in W^{1,p}_N.$$

We know that

$$N_F(u) \in P_{wkc}(L^{p'}(T, \mathbb{R}^N)) = \{C \subseteq L^{p'}(T, \mathbb{R}^N) : \text{ nonempty } w\text{-compact, convex}\} \quad \text{(convex problem)},$$

$$N_F(u) \in P_f(L^{p'}(T, \mathbb{R}^N)) = \{C \subseteq L^{p'}(T, \mathbb{R}^N) : \text{ nonempty, closed}\} \quad \text{(nonconvex problem)}.$$

Finally we introduce $V : W^{1,p}_N \to (W^{1,p}_N)^*$ defined by

$$\langle V(u), h\rangle = \int_0^b (a(u'), h')_{\mathbb{R}^N} dt \text{ for all } u,\ h \in W^{1,p}_N.$$

This map is continuous, monotone, thus maximal monotone.

Recall that $L^{p'}(T, \mathbb{R}^N) \hookrightarrow (W^{1,p}_N)^*$ continuously and densely (see Lemma I-4.200). Therefore, we can consider the multivalued map $L : W^{1,p}_N \to 2^{(W^{1,p}_N)^*} \setminus \{\emptyset\}$ defined by

$$L(u) = V(u) + \widehat{A}(u) + N_F(u) \text{ for all } u \in W_N^{1,p} \subseteq D(\widehat{A})$$

(see Lemma 6.42).

Proposition 6.43 *If hypotheses H_{18}, H_{19}, H_{20} (convex case) hold, then L is pseudomonotone.*

Proof It is clear that $L(\cdot)$ is bounded (that is, maps bounded sets to bounded sets). According to Proposition I-7.69, it is suffices to show that $L(\cdot)$ is generalized pseudomonotone. To this end, let $\{u_n\}_{n\in\mathbb{N}} \subseteq W_N^{1,p}$ and $\{u_n^*\}_{n\in\mathbb{N}} \subseteq (W_N^{1,p})^*$ such that

$$\left\{\begin{array}{l} u_n \xrightarrow{w} u \text{ in } W_N^{1,p},\ u_n^* \xrightarrow{w} u^* \text{ in } (W_N^{1,p})^*, \\ (u_n, u_n^*) \in GrL \text{ for all } n \in \mathbb{N}, \\ \limsup \langle u_n^*, u_n - u\rangle_W \leq 0. \end{array}\right\}, \tag{6.104}$$

with $\langle\cdot,\cdot\rangle_W$ being the duality brackets for the pair $((W_N^{1,p})^*, W_N^{1,p})$. For every $n \in \mathbb{N}$, we have

$$u_n^* = V(u_n) + g_n + h_n \text{ with } g_n \in \widehat{A}(u_n),\ h_n \in N_F(u_n).$$

Since $W_N^{1,p} \hookrightarrow C(T, \mathbb{R}^N)$ compactly, from (6.104) we have

$$\begin{aligned} &\|u_n\|_{C(T,\mathbb{R}^N)} \leq c_1 \text{ for some } c_1 > 0,\ \text{all } n \in \mathbb{N}, \\ \Rightarrow &|A(u_n(t))| : \sup\{|g| : g \in A(u_n(t))\} \leq c_2 \\ &\qquad\qquad \text{for some } c_2 > 0,\ \text{all } t \in T,\ \text{all } n \in \mathbb{N}, \\ \Rightarrow &\{g_n\}_{n\in\mathbb{N}} \subseteq L^{p'}(T, \mathbb{R}^N) \text{ is bounded.} \end{aligned}$$

So, we have

$$\langle g_n, u_n - u\rangle_W = \langle g_n, u_n - u\rangle_p \to 0, \tag{6.105}$$

$$\langle h_n, u_n - u\rangle_W = \langle h_n, u_n - u\rangle_p \to 0 \quad \text{(see (6.104))}. \tag{6.106}$$

From (6.104), (6.105), (6.106) it follows that

$$\begin{aligned} &\limsup_{n\to\infty} \langle V(u_n), u_n - u\rangle_W \leq 0 \\ \Rightarrow &\lim \langle V(u_n) - V(u), u_n - u\rangle = 0 \\ &\qquad\qquad \text{(from the monotonicity of } V(\cdot)). \end{aligned} \tag{6.107}$$

For every $n \in \mathbb{N}$, let

$$k_n(t) = (a(u_n'(t)) - a(u'(t)), u_n'(t) - u'(t))_{\mathbb{R}^N}.$$

Using hypothesis $H_{18}(iii)$ and Hölder's inequality, we see that $k_n \in L^1(T)$ for all $n \in \mathbb{N}$. Also the monotonicity of $a(\cdot)$, implies that $k_n \geq 0$ for all $n \in \mathbb{N}$. From (6.107), we have

$$\begin{aligned} &\int_0^b k_n(t)dt \to 0, \\ &\Rightarrow k_n \to 0 \text{ in } L^1(T). \end{aligned}$$

Therefore we may assume that

$$\begin{aligned} &k_n(t) \to 0 \text{ for a.a } t \in T,\ 0 \leq k_n(t) \leq \gamma(t) \text{ for a.a } t \in T, \text{ all } n \in \mathbb{N}, \\ &\qquad\qquad \text{with } \gamma \in L^1(T). \end{aligned} \tag{6.108}$$

We have

$$\begin{aligned} \gamma(t) &\geq k_n(t) \\ &\geq \widehat{c}\,[|u_n(t)|^p + |u(t)|^p] - c(1 + |u_n'(t)|^{p-1})|u'(t)| - c(1 + |u'(t)|^{p-1})|u_n'(t)| \\ &\qquad\qquad \text{(see hypotheses } H_{18}), \\ &\Rightarrow \{u_n'(t)\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^N \text{ is bounded.} \end{aligned}$$

So, at least for a subsequence (in general depending on $t \in T$), we have

$$u_n'(t) \to v(t) \text{ in } \mathbb{R}^N.$$

Using (6.108), in the limit as $n \to \infty$, we obtain

$$\begin{aligned} &\langle a(v(t)) - a(u'(t))), v(t) - u'(t)\rangle = 0, \\ &\Rightarrow v(t) = u'(t) \text{ for a.a } t \in T \\ &\qquad\qquad \text{(recall } a(\cdot) \text{ is strictly monotone)}. \end{aligned}$$

So, for the full sequence, we have

$$\begin{aligned} &u_n'(t) \to u'(t) \text{ for a.a } t \in T, \\ &\Rightarrow u_n' \to u' \text{ in } L^p(T, \mathbb{R}^N) \text{ (see (6.104))}, \\ &\Rightarrow u_n \to u \text{ in } W_N^{1,p}. \end{aligned} \tag{6.109}$$

The sequences $\{g_n\}_{n\in\mathbb{N}}$, $\{h_n\}_{n\in\mathbb{N}} \subseteq L^{p'}(T,\mathbb{R}^N)$ are bounded and so we may assume that

$$g_n \xrightarrow{w} g,\ h_n \xrightarrow{w} h \text{ in } L^{p'}(T,\mathbb{R}^N). \tag{6.110}$$

We have $(u_n, g_n) \in Gr\widehat{A}$ for all $n \in \mathbb{N}$. Hence

$$(u,g) \in Gr\widehat{A} \tag{6.111}$$

(from (6.109), (6.110) and since $\widehat{A}(\cdot)$ is maximal monotone).

Also, from (6.109), (6.110) and Theorem I-8.38, it follows that

$$(u,h) \in GrN_F. \tag{6.112}$$

Finally we have

$$\begin{aligned} &u^* = V(u) + g + h \text{ with } g \in \widehat{A}(u),\ h \in N_F(u) \\ &\qquad\qquad\qquad\qquad\qquad\qquad \text{(see (6.111), (6.112))}, \\ \Rightarrow &u^* \in L(u). \end{aligned}$$

This proves that $L(\cdot)$ is generalized pseudomonotone, thus pseudomonotone. □

Proposition 6.44 *If hypotheses H_{18}, H_{19}, H_{20} (convex case) hold, then $L(\cdot)$ is strongly coercive, that is,*

$$\frac{\inf\{\langle u^*, u\rangle_W : u^* \in L(u)\}}{\|u\|} \to +\infty \text{ as } \|u\| \to \infty.$$

Proof On account of hypotheses $H_{20}(iii)$, (iv), given $\varepsilon > 0$, we can find $\eta_\varepsilon \in L^{p'}(T)$ and $c_\varepsilon > 0$ such that

$$\begin{aligned} k(t,x,y) \geq [\vartheta(t) - \varepsilon]|x|^p - \eta_\varepsilon(t) - c_\varepsilon|y|^{p-1} \\ \text{for a.a } t \in T,\ \text{all } x, y \in \mathbb{R}^N. \end{aligned} \tag{6.113}$$

If $u^* \in L(u)$, we have

$$\begin{aligned} &u^* = V(u) + g + h \text{ with } g \in \widehat{A}(u),\ h \in N_F(u), \\ \Rightarrow\ &\langle u^*, u\rangle_W = \int_0^b (a(u'), u')_{\mathbb{R}^N} dt + \int_0^b (g(t), u(t))_{\mathbb{R}^N} dt + \int_0^b (h(t), u(t))_{\mathbb{R}^N} dt \\ &\geq \widehat{c}\,\|u'\|_p^p + \int_0^b \vartheta(t)|u|^p dt - \varepsilon\|u\|_p^p - c_\varepsilon\|u'\|_p^{p-1} - c_1 \end{aligned}$$

for some $\widehat{c}_1, c_1 > 0$ (see (6.113) and recall $(A(x), x)_{\mathbb{R}^N} \geq 0$ for all $x \in \mathbb{R}^N$)

$$\geq [\widehat{c} - \varepsilon c_2]\|u\|^p - c_3\|u\|^{p-1} - c_1$$

(for some $c_2 > 0,\ c_3 = c_3(\varepsilon) > 0$).

We choose $\varepsilon \in (0, \frac{\widehat{c}}{c_2})$ and conclude that $L(\cdot)$ is strongly coercive. □

Propositions 6.43 and 6.44 lead to the following existence theorem.

Theorem 6.45 *If hypotheses H_{18}, H_{19}, H_{20} (convex case) hold, then problem (67) has a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$.*

Proof Propositions 6.43 and 6.44, permit the use of Theorem I-7.76. So, we can find $\widehat{u} \in W^{1,p}_N$ such that

$$\begin{aligned}
&0 \in L(\widehat{u}),\\
\Rightarrow &V(\widehat{u}) + g + h = 0 \text{ with } g \in \widehat{A}(\widehat{u}),\ h \in N_F(\widehat{u}),\\
\Rightarrow &a(u'(t))' = g(t) + f(t) \text{ for a.a } t \in T,\ u(0) = u(b),\ u'(0) = u'(b),\\
\Rightarrow &a(\widehat{u}'(\cdot)) \in W^{1,p'}_N \text{ and so } \widehat{u}' \in C(T, \mathbb{R}^N)\\
&\qquad\qquad (a(\cdot) \text{ is a homeomorphism}),\\
\Rightarrow &\widehat{u} \in C^1(T, \mathbb{R}^N).
\end{aligned}$$

□

We can have a corresponding existence result for the nonconvex problem.

Theorem 6.46 *If hypotheses H_{18}, H_{19}, H_{20} (nonconvex case) hold, then problem (67) has a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$*

Proof Consider the multifunction $N_F : W^{1,p}_N \to 2^{L^{p'}(T,\mathbb{R}^N)}$ defined by

$$N_F(u) = S^p_{F(\cdot,u(\cdot),u'(\cdot))} \text{ for all } u \in W^{1,p}_N.$$

This multifunction has decomposable values in $P_f(L^{p'}(T, \mathbb{R}^N))$ and it is lsc (see the proof of Theorem 6.38). So, by Theorem I-5.137 we can find $k_0 : W^{1,p}_N \to L^{p'}(T, \mathbb{R}^N)$ a continuous map such that

$$k_0(u) \in N_F(u) \text{ for all } u \in W^{1,p}_N.$$

Then we consider the periodic system

$$\left\{\begin{array}{l} a(u'(t))' \in A(u(t)) + k_0(u)(t) \text{ for a.a } t \in T, \\ u(0) = u(b),\ u'(0) = u'(b). \end{array}\right\}.$$

As in the proof of Theorem 6.45, this system has a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$. □

The last problem that we will examine in this section, is the following multivalued periodic system

$$\left\{\begin{array}{l} a(u'(t))' + \frac{d}{dt}\nabla G(u(t)) \in A(u(t)) + F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b),\ u'(0) = u'(b). \end{array}\right\}. \tag{6.114}$$

As before $a : \mathbb{R}^N \to \mathbb{R}^N$ is a monotone homeomorphism which incorporates in our setting many differential operators of interest (the prototype being the vector p-Laplacian, $1 < p < \infty$). Also $A : D(A) \subseteq \mathbb{R}^N \to 2^{\mathbb{R}^N}$ is a maximal monotone multivalued map with $D(A)$ in general different from $\mathbb{R}^N$. This way, we can also consider systems with unilateral constraints (differential variational inequalities). Finally $G \in C^2(\mathbb{R}^N, \mathbb{R})$ and $F(t, x)$ is a multivalued perturbation. Note that if $\frac{d}{dt}\nabla G(u(t))$ is replaced by $r(t)|u(t)|^{p-2}u(t)$, with $r \in L^\infty(T)$, then we have a multivalued periodic Duffing system.

We impose the following conditions on the data of problem (6.114).

H_{21}: $a : \mathbb{R}^N \to \mathbb{R}^N$ is strictly monotone, $a(y) = a_0(|y|)y$ with $a_0 : (0, \infty) \to (0, \infty)$ continuous, $a_0(t)t \to 0$ as $t \to 0^+$ and

$$\widehat{c}\,|y|^p \leq (a(y), y)_{\mathbb{R}^N} \text{ for all } y \in \mathbb{R}^N,$$

with $\widehat{c} > 0$ and $1 < p < \infty$.

Remark 6.47 Clearly $a(\cdot)$ is continuous and so it is maximal monotone. Moreover, it is coercive, hence surjective. Then on account of the strict monotonicity of $a(\cdot)$, the map $a^{-1} : \mathbb{R}^N \to \mathbb{R}^N$ is well defined and continuous. Therefore $a(\cdot)$ is a homeomorphism and $|a^{-1}(y)| \to \infty$ as $|y| \to \infty$. We stress that in contrast to the previous periodic problems, we do not impose any growth condition on $a(\cdot)$, which can be a very general homeomorphism.

H_{22}: $G \in C^2(\mathbb{R}^N, \mathbb{R})$, $\nabla G(x) = \eta_0(|x|)x$ for all $x \in \mathbb{R}^N$, $\eta_0(r) > 0$ for all $r > 0$ and $\eta_0(\cdot)$ is continuous.

H_{23}: $A : D(A) \subseteq \mathbb{R}^N \to 2^{\mathbb{R}^N}$ is a maximal monotone map such that $(0, 0) \in Gr\,A$.

Let $h \in L^1(T, \mathbb{R}^N)$ and consider the following periodic system

$$\left\{\begin{array}{l} -a(u'(t))' + |u(t)|^{p-2}u(t) = h(t) \text{ for a.a } t \in T, \\ u(0) = u(b),\ u'(0) = u'(b). \end{array}\right\}. \tag{6.115}$$

For this system, we have the following existence and uniqueness result (see Manasevich and Mawhin [324], Corollary 3.1 and Kyritsi et al. [284], Proposition 2).

Proposition 6.48 *If hypotheses H_{21} hold, then problem (6.115) has a unique solution*

$$\widehat{u} \in C^1(T, \mathbb{R}^N).$$

We introduce the solution map $k : L^1(T, \mathbb{R}^N) \to C^1(T, \mathbb{R}^N)$ which to each input function $h \in L^1(T, \mathbb{R}^N)$ assigns the unique solution of problem (6.115). The next proposition establishes the continuity properties of this solution map.

Proposition 6.49 *If hypotheses H_{21} hold, then the solution map $k : L^1(T, \mathbb{R}^N) \to C^1(T, \mathbb{R}^N)$ is completely continuous, that is, if $h_n \xrightarrow{w} h$ in $L^1(T, \mathbb{R}^N)$, then $k(h_n) \to k(h)$ in $C^1(T, \mathbb{R}^N)$.*

Proof Consider a sequence $\{h_n\}_{n\in\mathbb{N}} \subseteq L^1(T, \mathbb{R}^N)$ such that $h_n \xrightarrow{w} h$ in $L^1(T, \mathbb{R}^N)$ and set $u_n = k(h_n)$ for all $n \in \mathbb{N}$. We have

$$\left\{ \begin{array}{l} -a(u_n'(t))' + |u_n(t)|^{p-2}u_n(t) = h_n(t) \text{ for a.a } t \in T, \\ u_n(0) = u_n(b),\ u_n'(0) = u_n'(b). \end{array} \right\}. \tag{6.116}$$

We act with $u_n(t)$, integrate over $T = [0, b]$ and perform integration by parts. The periodic boundary conditions, lead to

$$\int_0^b (a(u_n'), u_n')_{\mathbb{R}^N} dt + \|u_n\|_p^p \leq c_1 \|u_n\| \quad \text{for all } n \in \mathbb{N},\ \text{some } c_1 > 0$$

(recall that $\|\cdot\|$ denotes the norm of $W_N^{1,p}$),

$$\Rightarrow \widehat{c}\, \|u_n'\|_p^p + \|u_n\|_p^p \leq c_1 \|u_n\|,$$

$$\Rightarrow \{u_n\}_{n\in\mathbb{N}} \subseteq W_N^{1,p} \text{ is bounded.}$$

Recall that $W_N^{1,p} \hookrightarrow C(T, \mathbb{R}^N)$ compactly. Therefore

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N) \text{ is relatively compact.} \tag{6.117}$$

Integrating (6.116), we obtain

$$a(u_n'(t)) = a(u_n'(0)) + \int_0^t (h_n(s) - |u_n(s)|^{p-2}u_n(s))ds \quad \text{for all } t \in T,\ \text{all } n \in \mathbb{N}, \tag{6.118}$$

$$\Rightarrow u_n'(t) = a^{-1}(a(u_n'(0)) + \int_0^t (h_n(s) - |u_n(s)|^{p-2}u_n(s))ds$$

for all $t \in T$, all $n \in \mathbb{N}$.

Let $\beta_n(t) = \int_0^t (h_n(s) - |u_n(s)|^{p-2}u_n(s))ds$ for all $t \in T$, all $n \in \mathbb{N}$. We see that $\beta_n \in C(T, \mathbb{R}^N)$, $\{\beta_n\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N)$ is bounded and

$$\int_0^b u_n'(t)dt = 0.$$

So, from Proposition 2.2 (ii) of Manasevich and Mawhin [324], we have that

$$\{a(u_n'(0))\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^N \text{ is bounded.}$$

From (6.118), (6.117) and the Arzela-Ascoli theorem, it follows that

$$\begin{aligned} &\{a(u_n'(\cdot))\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N) \text{ is relatively compact,} \\ \Rightarrow &\{u_n'\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N) \text{ is relatively compact.} \end{aligned} \tag{6.119}$$

From (6.117) and (6.119), we infer that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C^1(T, \mathbb{R}^N) \text{ is relatively, compact.}$$

So, we may assume that

$$\begin{aligned} &u_n \to u \text{ in } C^1(T, \mathbb{R}^N) \\ \Rightarrow &u = k(h) \text{ (see (6.118)).} \end{aligned}$$

This proves the complete continuity of $k(\cdot)$. □

Let $\widehat{k} = k|_{L^2(T,\mathbb{R}^N)}$. Then Proposition 6.49 and Proposition I-3.148 imply the following continuity property of $\widehat{k}(\cdot)$.

Proposition 6.50 *If H_{21} hold, then the solution map $\widehat{k} : L^2(T, \mathbb{R}^N) \to C^1(T, \mathbb{R}^N)$ is compact.*

First we treat the case where the multivalued perturbation $F(t, x)$ is convex-valued ("convex problem"). So, the hypotheses on $F(t, x)$ are the following:

H_{24}: $F : T \times \mathbb{R}^N \to P_{fc}(\mathbb{R}^N) = \{C \subseteq \mathbb{R}^N : \text{nonempty, closed, convex}\}$ is a multifunction such that

(i) for all $x \in \mathbb{R}^N$, the map $t \to F(t, x)$ admits a measurable selection;
(ii) for a.a $t \in T$, $GrF(t, \cdot) \subseteq \mathbb{R}^N \times \mathbb{R}^N$ is closed;

(iii) for every $\varrho > 0$, there exists $a_\varrho \in L^2(T)$ such that

$$|F(t,x)| = \sup\{|h| : h \in F(t,x)\} \leq a_\varrho(t)$$

for a.a $t \in T$, all $|x| \leq \varrho$;

(iv) there exists $M > 0$ such that

$$0 \leq (h,x)_{\mathbb{R}^N} \text{ for a.a } t \in T, \text{ all } |x| = M, \text{ all } h \in F(t,x).$$

Remark 6.51 On account of the Yankov-von Neumann-Aumann Selection Theorem (see Theorem I-5.106), we see that if for $x \in \mathbb{R}^N$, $t \to F(t,x)$ is graph measurable, then hypothesis $H_{24}(i)$ is satisfied. Note that on $F(t,\cdot)$ we impose only a local growth condition (see $H_{24}(iii)$). Hypothesis $H_{24}(iv)$, is the Hartman condition already used in the context of evolution inclusions (see hypotheses H_7).

Motivated by the previous use of the Hartman condition (see the proof of Theorem 6.16), we introduce the M-radial retraction $P_M : \mathbb{R}^N \to \mathbb{R}^N$ defined by

$$p_M(x) = \begin{cases} x & \text{if } |x| \leq M \\ M\frac{x}{|x|} & \text{if } M < x. \end{cases}$$

Clearly $p_M(\cdot)$ is nonexpansive, that is,

$$|p_M(x) - p_M(x')| \leq |x - x'| \text{ for all } x, x' \in \mathbb{R}^N.$$

We modify F and the potential function $G(\cdot)$ as follows

$$\widehat{F}(t,x) = F(t, p_M(x)) - |p_M(x)|^{p-2}p_M(x) \text{ for all } t \in T, \text{ all } x \in \mathbb{R}^N,$$

$$B(x) = \nabla(Gop_M)(x) \text{ for all } x \in \mathbb{R}^N.$$

By Rademacher's Theorem (see Theorem I-6.135) and the chain rule, we see that $B(\cdot)$ well defined. In fact for every $u \in C^1(T,\mathbb{R}^N)$, we have

$$\left\{ \begin{array}{l} \frac{d}{dt}B(u(t)) \text{ exists for a.a } t \in T, \\ \frac{d}{dt}B(u(\cdot)) \in L^\infty(T). \end{array} \right\} \tag{6.120}$$

Finally for $\lambda > 0$, let $A_\lambda(\cdot)$ be the Yosida approximation of $A(\cdot)$ (see Definition I-7.51). Then we consider the following approximation of problem (6.114)

$$\left\{ \begin{array}{l} a(u'(t))' + \frac{d}{dt}B(u(t)) \in A_\lambda(u(t)) + \widehat{F}(t,u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b). \end{array} \right\} \tag{6.121}$$

We will solve (6.121) and then pass to the limit as $\lambda \to 0^+$ in order to produce a solution of (6.114). For this purpose, we introduce the following maps:

$$\widehat{A}_\lambda : C^1(T, \mathbb{R}^N) \to L^2(T, \mathbb{R}^N) \text{ defined by}$$

$$\widehat{A}_\lambda(u)(\cdot) = A_\lambda(u(\cdot)) \text{ for all } u \in C^1(T, \mathbb{R}^N)$$

and $N_{\widehat{F}} : C^1(T, \mathbb{R}^N) \to 2^{L^2(T,\mathbb{R}^N)}$ defined by

$$N_{\widehat{F}}(u) = S^2_{\widehat{F}(\cdot,u(\cdot))} \text{ for all } u \in C^1(T, \mathbb{R}^N).$$

Finally we consider the multifunction $N_\lambda : C^1(T, \mathbb{R}^N) \to 2^{L^2(T,\mathbb{R}^N)}$ defined by

$$N_\lambda(u) = \widehat{A}_\lambda(u) + N_{\widehat{F}}(u) - \frac{d}{dt}B(u) \text{ for all } u \in C^1(T, \mathbb{R}^N).$$

Proposition 6.52 *If hypotheses H_{21}, H_{22}, H_{23}, H_{24} hold, then the multifunction N_λ has values in*

$$P_{wkc}(L^2(T, \mathbb{R}^N)) = \{C \subseteq L^2(T, \mathbb{R}^N) : \textit{nonempty, w-compact, convex}\}$$

and it is usc from $C^1(T, \mathbb{R}^N)$ into $L^2(T, \mathbb{R}^N)_w$.

Proof If we show $N_{\widehat{F}}$ has values in $P_{wkc}(L^2(T, \mathbb{R}^N))$, then clearly we can say that N_λ has values in $P_{wkc}(L^2(T, \mathbb{R}^N))$. Evidently, the values of $N_{\widehat{F}}$ are closed and convex. However, on account of the weak measurability hypothesis on $F(\cdot, x)$ (see hypothesis $H_{24}(i)$), it is not immediately clear that for every $u \in C^1(T, \mathbb{R}^N)$, $N_{\widehat{F}}(u) \neq \emptyset$. Indeed, hypotheses $H_{24}(i), (ii)$ do not imply the graph measurability of $t \to \widehat{F}(t, u(t))$ and so we can not appeal to the Yankov-van Neumann-Aumann selection theorem (see Theorem I-5.106) to conclude that $N_{\widehat{F}}(u) \neq \emptyset$. To reach this conclusion, we argue as follows. Let $\{s_n\}_{n\in\mathbb{N}}$ be a sequence of step functions such that

$$\left\{ \begin{array}{l} s_n(t) \to u(t) \text{ for a.a } t \in T \text{ as } n \to \infty, \\ |s_n(t)| \leq |u(t)| \text{ for a.a } t \in T \text{ as } n \in \mathbb{N}. \end{array} \right\}$$

Then hypothesis $H_{24}(i)$ guarantees that we can find $h_n : T \to \mathbb{R}^N$ a measurable selection of $\widehat{F}(\cdot, s_n(\cdot))$, that is,

$$h_n(\cdot) \text{ is measurable and } h_n(t) \in \widehat{F}(t, s_n(t)) \text{ for a.a } t \in T.$$

From (6.121) and hypothesis $H_{24}(iii)$, we see that $\{h_n\}_{n\in\mathbb{N}} \subseteq L^2(T, \mathbb{R}^N)$ is bounded. So, we may assume that

$$h_n \xrightarrow{w} h \text{ in } L^2(T, \mathbb{R}^N).$$

Using Theorem I-8.38, we have

$$h(t) \in \overline{conv} \limsup_{n\to\infty} \widehat{F}(t, s_n(t)) \subseteq \widehat{F}(t, u(t)) \text{ for a.a } t \in T$$

(see hypothesis $H_{24}(ii)$),

$$\Rightarrow h \in S^2_{\widehat{F}(\cdot,u(\cdot))} = N_{\widehat{F}} \neq \emptyset.$$

Therefore, we have that $N_\lambda(u) \in P_{wkc}(L^2(T, \mathbb{R}^N))$ for all $u \in C^1(T, \mathbb{R}^N)$.

According to Proposition I-5.13 and the Eberlein-Smulian theorem, to show the upper semicontinuity of $N_\lambda(\cdot)$ from $C^1(T, \mathbb{R}^N)$ into $L^2(T, \mathbb{R}^N)_w$, it suffices to show that $GrN_\lambda \subseteq C^1(T, \mathbb{R}^N) \times L^2(T, \mathbb{R}^N)_w$ is sequentially closed. So, let $\{(u_n, g_n)\}_{n\in\mathbb{N}} \subseteq GrN_\lambda$ and assume that

$$u_n \to u \text{ in } C^1(T, \mathbb{R}^N),\ g_n \xrightarrow{w} g \text{ in } L^2(T, \mathbb{R}^N). \tag{6.122}$$

For some $h_n \in N_{\widehat{F}}(u_n)$, we have

$$g_n = \widehat{A}_\lambda(u_n) + h_n - \frac{d}{dt}B(u_n) \text{ for all } n \in \mathbb{N}. \tag{6.123}$$

We know that

$$\widehat{A}_\lambda(u_n) \to \widehat{A}_\lambda(u) \text{ in } L^2(T, \mathbb{R}^N) \text{ (see (6.122))}. \tag{6.124}$$

Also, the chain rule and (6.122), imply that

$$\frac{d}{dt}B(u_n) \to \frac{d}{dt}B(u) \text{ in } L^2(T, \mathbb{R}^N). \tag{6.125}$$

Then (6.122), (6.123), (6.124), and (6.125), imply that

$$h_n \xrightarrow{w} h = g - \widehat{A}_\lambda(u) - \frac{d}{dt}B(u) \text{ in } L^2(T, \mathbb{R}^N). \tag{6.126}$$

As above, via Theorem I-8.38, we have $h \in N_{\widehat{F}}(u)$ and so from (6.126) it follows that $(u, g) \in GrN_\lambda$. This proves the desired upper semicontinuity of $N_\lambda(\cdot)$. □

We introduce the set

$$S_\lambda = \{u \in C^1(T, \mathbb{R}^N) : u \in (k \circ (-N_\lambda))(u), 0 < \vartheta < 1\}.$$

Proposition 6.53 *If hypotheses H_{21}, H_{22}, H_{23}, H_{24} hold, then $S_\lambda \subseteq C^1(T, \mathbb{R}^N)$ is bounded.*

Proof Let $u \in S_\lambda$. Then

$$\frac{1}{\vartheta}u = k(-g) \text{ with } g \in N_\lambda(u),$$
$$\Rightarrow g = \widehat{A}_\lambda(\frac{u}{\vartheta}) + \widehat{h} - \frac{d}{dt}B(\frac{u}{\vartheta}) \text{ with } \widehat{h} \in N_{\widehat{F}}(\frac{u}{\vartheta}).$$

So, we have

$$\begin{cases} -a(\frac{1}{\vartheta}u'(t))' + \frac{1}{\vartheta^{p-1}}|u(t)|^{p-2}u(t), \\ = -A_\lambda(u(t)) - \widehat{h}(t) + \frac{d}{dt}B(u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b). \end{cases} \tag{6.127}$$

Claim $|u(t)| \leq M$ for all $t \in T$. (see hypothesis $H_{24}(iv)$).

Let $\gamma(t) = \frac{1}{2}|u(t)|^2, t \in T$. Consider $t_0 \in T$ such that

$$\gamma(t_0) = \max\{\gamma(t) : t \in T\}.$$

If the claim is not true, then $\gamma(t_0) > \frac{1}{2}M^2$.
Suppose $t_0 \in (0, b)$. Then we have

$$\gamma'(t_0) = (u'(t_0), u(t_0))_{\mathbb{R}^N} = 0.$$

Let $t_1 \in [0, t_0)$ such that

$$|u(t_1)| = M, \ |u(t)| > M \text{ for all } t \in (t_1, t_0].$$

In (6.127) we take inner product with $u(t) \in \mathbb{R}^N$. Then

$$\begin{aligned} &-\frac{d}{dt}(a(\frac{1}{\vartheta}u'(t)), u(t))_{\mathbb{R}^N} + (a(\frac{1}{\vartheta}u'(t)), u'(t))_{\mathbb{R}^N} + \frac{1}{\vartheta^{p-1}}|u(t)|^p \\ &= -(A_\lambda(u(t)), u'(t))_{\mathbb{R}^N} - (\widehat{h}(t), u(t))_{\mathbb{R}^N} \\ &\quad + (\frac{d}{dt}\nabla G(p_M(u(t))), u(t))_{\mathbb{R}^N} \text{ for a.a } t \in T. \end{aligned}$$

We know that $\widehat{h} \in N_{\widehat{F}}(u)$, hence

$$\widehat{h}(t) \in F(t, p_M(u(t))) - |p_M(u(t))|^{p-2}p_M(u(t)) \text{ for a.a } t \in T.$$

The Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), provides $h \in N_F(p_M(u))$ such that

$$\widehat{h}(t) = h(t) - |p_M(u(t))|^{p-2} p_M(u(t)) \text{ for a.a } t \in T.$$

Therefore on account of hypothesis $H_{24}(iv)$, we have

$$(\widehat{h}(t), u(t))_{\mathbb{R}^N} = (h(t), u(t))_{\mathbb{R}^N} - |u(t)|M^{p-1} \geq -|u(t)|M^{p-1} \text{ for a.a } t \in [t_1, t_0]. \tag{6.128}$$

Also, we have

$$\begin{aligned}
&(\frac{d}{dt}\nabla G(p_M(u(t))), u(t))_{\mathbb{R}^N} \\
&= \frac{|u(t)|}{M}(\frac{d}{dt}\nabla G(p_M(u(t))), p_M(u(t))) \\
&= \frac{|u(t)|}{M}[\frac{d}{dt}(\nabla G(p_M(u(t)), p_M(u(t))))_{\mathbb{R}^N} - (\nabla G(p_M(u(t))), \frac{d}{dt}p_M(u(t)))_{\mathbb{R}^N}] \\
&= \frac{|u(t)|}{M}[\frac{d}{dt}(\eta_0(p_M(u(t))))|p_M(u(t))| - \eta_0(|p_M(u(t))|)(p_M(u(t)), \\
&\quad \times \frac{d}{dt}p_M(u(t)))_{\mathbb{R}^N}] \\
&= \frac{|u(t)|}{M}[\frac{d}{dt}\eta_0(M)M - \eta_0(M)(M, \frac{d}{dt}M)_{\mathbb{R}^N}] \\
&= 0 \text{ for a.a } t \in [t_1, t_0] \text{ (see hypotheses } H_{22}).
\end{aligned} \tag{6.129}$$

Hypotheses H_{23} imply that $A_\lambda(0) = 0$. Hence

$$(A_\lambda(u(t)), u(t))_{\mathbb{R}^N} \geq 0 \text{ for all } t \in [t_1, t_0]. \tag{6.130}$$

We return to (6.127) and use (6.128), (6.129), (6.130) and hypotheses H_{21}. Then

$$|u(t)|[\frac{1}{\vartheta^{p-1}}|u(t)|^{p-1} - M^{p-1}] < \frac{d}{dt}(a(\frac{1}{\vartheta}u'(t)), u(t))_{\mathbb{R}^N} \text{ for a.a } t \in (t_1, t_0].$$

We know that $|u(t)| > M$ for all $t \in (t_1, t_0]$. Hence

$$\begin{aligned}
&0 < \frac{d}{dt}(a(\frac{1}{\vartheta}u'(t)))_{\mathbb{R}^N} \text{ for a.a } t \in [t_1, t_0], \\
&\Rightarrow (a(\frac{1}{\vartheta}u'(t)), u(t))_{\mathbb{R}^N} < (a(\frac{1}{\vartheta}u'(t_0)), u(t_0))_{\mathbb{R}^N} \text{ for a.a } t \in [t_1, t_0), \\
&\Rightarrow a_0(\frac{1}{\vartheta}|u'(t)|)(u'(t), u(t))_{\mathbb{R}^N} < a_0(\frac{1}{\vartheta}|u'(t_0)|)(u'(t_0), u(t_0))_{\mathbb{R}^N} = 0
\end{aligned}$$

Then we have

$$\gamma'(t) < 0 \text{ for all } t \in [t_1, t_0)$$
$$\Rightarrow \gamma(t_0) = \frac{1}{2}|u(t_0)|^2 < \gamma(t_1) = \frac{1}{2}|u(t_1)|^2 = \frac{1}{2}M^2$$

a contradiction.

Next suppose $t_0 = 0$ or $t_0 = b$. From the periodic boundary condition, we have

$$\gamma(0) = \gamma(b), \quad \text{and } \gamma'(0) \leq 0 \leq \gamma'(b).$$

Since $\gamma'(t) = (u'(t), u(t))_{\mathbb{R}^N}$, using the periodic boundary condition, we infer that

$$\gamma'(0) = 0 = \gamma'(b).$$

Then the previous argument remains valid.

Therefore $|u(t)| \leq M$ for all $t \in T$ and this proves the Claim.

We return to (6.127), take inner product with $u_n(t) \in \mathbb{R}^N$, integrate over $T = [0, b]$, perform integration by parts and use the periodic boundary condition, hypotheses H_{21} and the Claim. We obtain

$$\frac{1}{\vartheta^{p-1}}[\widehat{c}\,\|u\|_p^p + \|u'\|_p^p] \leq c \text{ for some } c > 0.$$
$$\Rightarrow S_\lambda \subseteq W_N^{1,p} \text{ is bounded.}$$

Reasoning as in the proof of Proposition 6.49, we have that $S_\lambda \subseteq C^1(T, \mathbb{R}^N)$ is bounded. □

We are ready to produce a solution for the approximate problem (6.121).

Proposition 6.54 *If hypotheses H_{21}, H_{22}, H_{23}, H_{24} hold and $\lambda > 0$, then problem (6.121) has a solution $\widehat{u}_\lambda \in C^1(T, \mathbb{R}^N)$.*

Proof Propositions 6.49, 6.52, and 6.53, permit the use of Theorem 2.138. So, we can find $\widehat{u}_\lambda \in C^1(T, \mathbb{R}^N)$ such that

$$\widehat{u}_\lambda \in (k \circ (-N_\lambda))(\widehat{u}_\lambda).$$

From the proof of Proposition 6.53, we have

$$|\widehat{u}_\lambda(t)| \leq M \text{ for all } t \in T.$$

Then $\widehat{F}(t, \widehat{u}_\lambda(t)) = F(t, \widehat{u}_\lambda(t)) - |u_\lambda(t)|^{p-2}u_\lambda(t)$ for all $t \in T$ and so from the definition of $k(\cdot)$ we conclude that $\widehat{u}_\lambda \in C^1(T, \mathbb{R}^N)$ is a solution of problem (6.121). □

Now we will pass to the limit as $\lambda \to 0^+$ in order to produce a solution for problem (6.114).

Theorem 6.55 *If hypotheses H_{21}, H_{22}, H_{23}, H_{24} hold, then problem (6.114) admits a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$.*

Proof Let $\lambda_n = \frac{1}{n}$ and $\widehat{u}_n = \widehat{u}_{\lambda_n} \in C^1(T, \mathbb{R}^N)$ be a solution of problem (6.121) (with $\lambda = \lambda_n$) obtained in Proposition 6.54. We know that

$$|\widehat{u}_n(t)| \leq M \text{ for all } t \in T, \text{ all } n \in \mathbb{N}. \tag{6.131}$$

Then from (6.131) as in the proof of Proposition 6.53, we infer that $\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq W_N^{1,p}$ is bounded and from this we obtain that $\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq C^1(T, \mathbb{R}^N)$ is relatively compact. So, we may assume that

$$\widehat{u}_n \to \widehat{u} \text{ in } C^1(T, \mathbb{R}^N). \tag{6.132}$$

For every $n \in \mathbb{N}$, we have

$$\left\{ \begin{array}{l} a(\widehat{u}_n'(t))' + \frac{d}{dt}\nabla G(\widehat{u}_n(t)) = A_{\lambda_n}(\widehat{u}_n(t)) + h_n(t) \text{ for a.a } t \in T, \\ \widehat{u}_n(0) = \widehat{u}_n(b), \widehat{u}_n'(0) = \widehat{u}_n'(b), \end{array} \right\} \tag{6.133}$$

with $h_n \in L^2(T, \mathbb{R}^N)$ such that

$$h_n(t) \in F(t, \widehat{u}_n(t)) \text{ for a.a } t \in T.$$

We take inner product of (6.133) with $A_{\lambda_n}(\widehat{u}_n(t))$ and then integrate over T. We obtain

$$\begin{aligned} &\int_0^b (a(\widehat{u}_n')', A_{\lambda_n}(\widehat{u}_n))_{\mathbb{R}^N} dt + \int_0^b (\frac{d}{dt}\nabla G(\widehat{u}_n), A_{\lambda_n}(\widehat{u}_n))_{\mathbb{R}^N} dt \\ &= \| A_{\lambda_n}(\widehat{u}_n) \|_2^2 + \int_0^b (h_n, A_{\lambda_n}(\widehat{u}_n))_{\mathbb{R}^N} dt. \end{aligned} \tag{6.134}$$

Performing integration by parts and using the periodic boundary conditions, we see that

$$\int_0^b (a(\widehat{u}_n')', A_\lambda(\widehat{u}_n))_{\mathbb{R}^N} dt = -\int_0^b (a(\widehat{u}_n'), \frac{d}{dt}A_\lambda(\widehat{u}_n))_{\mathbb{R}^N} dt. \tag{6.135}$$

From the chain rule (see Theorem I-4.76), we have

$$\frac{d}{dt}A_{\lambda_n}(\widehat{u}_n(t)) = A_{\lambda_n}'(\widehat{u}_n(t))\widehat{u}_n'(t) \text{ for a.a } t \in T.$$

We know that $A_{\lambda_n}(\cdot)$ is $\frac{1}{\lambda_n}$-Lipschitz (see Proposition I-7.56). So, by Rademacher's Theorem (see Theorem I-6.135), $A_{\lambda_n}(\cdot)$ is differentiable almost everywhere on $\mathbb{R}^N$. Moreover, the monotonicity of $A_{\lambda_n}(\cdot)$ implies that

$$(A'_{\lambda_n}(x)y, y)_{\mathbb{R}^N} \geq 0 \text{ for all } x, y \in \mathbb{R}^N. \tag{6.136}$$

Using hypotheses H_{21} and (6.136), we obtain

$$\int_0^b (a(u_n)', \frac{d}{dt} A_{\lambda_n}(\widehat{u}_n))_{\mathbb{R}^N} dt = \int_0^b a_0(|\widehat{u}'_n|)(\widehat{u}'_n, A'_\lambda(u'_n)u'_n)_{\mathbb{R}^N} dt \geq 0 \tag{6.137}$$

for all $n \in \mathbb{N}$.

Therefore returning to (6.134) and using (6.131), (6.135) and (6.137), we have

$$\begin{aligned} &\| A_{\lambda_n}(u_n) \|_2^2 \leq c_1 \| A_{\lambda_n}(u_n) \|_2 \text{ for some } c_1 > 0, \text{ all } n \in \mathbb{N} \\ &\Rightarrow \{A_{\lambda_n}(u_n(\cdot)) \|\}_{n\in\mathbb{N}} \subseteq L^2(T, \mathbb{R}^N) \text{ is bounded.} \end{aligned} \tag{6.138}$$

Let $\widehat{A}_\lambda(u_n)(\cdot) = A_{\lambda_n}(u_n(\cdot))$ (the Nemytskii map corresponding to $A_{\lambda_n}(\cdot)$). On account of (6.138), we may assume that

$$\widehat{A}_{\lambda_n}(\widehat{u}_n) \xrightarrow{w} \xi \text{ in } L^2(T, \mathbb{R}^N). \tag{6.139}$$

Let $\widehat{J}_{\lambda_n}(\cdot) = J_{\lambda_n}(u_n(\cdot))$ (the Nemytskii map corresponding to the resolvent operator). We know that

$$\begin{aligned} &\lambda_n \widehat{A}_{\lambda_n}(\widehat{u}_n) = \widehat{u}_n - \widehat{J}_{\lambda_n}(\widehat{u}_n) \text{ for all } n \in \mathbb{N}, \\ &\Rightarrow \widehat{u}_n - \widehat{J}_{\lambda_n}(\widehat{u}_n) \to 0 \text{ in } L^2(T, \mathbb{R}^N) \text{ (see (6.138))}, \\ &\Rightarrow \widehat{J}_{\lambda_n}(\widehat{u}_n) \to \widehat{u} \text{ in } L^2(T, \mathbb{R}^N) \text{ (see (6.132))}. \end{aligned} \tag{6.140}$$

Recall that

$$(\widehat{J}_{\lambda_n}(\widehat{u}_n), \widehat{A}_{\lambda_n}(\widehat{u}_n)) \in Gr\widehat{A} \text{ for all } n \in \mathbb{N}.$$

(recall $\widehat{A}(\cdot)$ is the lifting of $A(\cdot)$ on $L^2(T, \mathbb{R}^N)$). From Proposition 6.35, we know that $\widehat{A}(\cdot)$ is maximal monotone, hence $Gr\widehat{A} \subseteq L^2(T, \mathbb{R}) \times L^2(T, \mathbb{R}^N)_w$ is closed. Therefore from (6.139) and (6.140) we have

$$(\widehat{u}, \xi) \in Gr\widehat{A} \tag{6.141}$$

For all $\vartheta \in W_N^{1,2}$ we have

$$\int_0^b (a(\widehat{u}_n'), \vartheta)_{\mathbb{R}^N} dt = -\int_0^b (a(\widehat{u}_n'), \vartheta')_{\mathbb{R}^N} dt \text{ for all } n \in \mathbb{N},$$

$$\Rightarrow \lim_{n\to\infty} \int_0^b (a(\widehat{u}_n'), \vartheta)_{\mathbb{R}^N} dt = \int_0^b (a(\widehat{u}'), \vartheta)_{\mathbb{R}^N} dt \text{ (see (6.132))},$$

$$\Rightarrow a(\widehat{u}_n')' \xrightarrow{w} a(\widehat{u}')' \text{ in } L^2(T, \mathbb{R}^N) \text{ (since } W_N^{1,2} \text{is dense in } L^2(T, \mathbb{R}^N)).$$

Also we have

$$\frac{d}{dt}\nabla G(\widehat{u}_n) \to \frac{d}{dt}\nabla G(\widehat{u}) \text{ in } L^2(T, \mathbb{R}^N).$$

In addition hypothesis $H_{24}(iii)$ implies that $\{h_n\}_{n\in\mathbb{N}} \subseteq L^2(T, \mathbb{R}^N)$ is bounded and so we may assume that

$$h_n \xrightarrow{w} h \text{ in } L^2(T, \mathbb{R}^N).$$

As before using hypothesis $H_{24}(ii)$ and Theorem I-8.38, we have that $h \in S^2_{F(\cdot,\widehat{u}(\cdot))}$. Therefore in the limit as $n \to \infty$, we obtain

$$\left\{ \begin{array}{l} a(\widehat{u}'(t))' + \frac{d}{dt}\nabla G(\widehat{u}(t)) = \xi(t) + h(t) \text{ for a.a } t \in T, \\ \widehat{u}(0) = \widehat{u}(b), \widehat{u}'(0) = \widehat{u}'(b). \end{array} \right\}.$$

with $\xi \in \widehat{A}(\widehat{u})$, $h \in N_F(\widehat{u})$. Hence $\widehat{u} \in C^1(T, \mathbb{R}^N)$ is a solution of problem (6.114). □

We can have an existence theorem also when the orientor field $F(t, x)$ is nonconvex valued. In this case our hypotheses on the multifunction $F(t, x)$, are the following:

H_{25}: $F : T \times \mathbb{R}^N \to P_f(\mathbb{R}^N) = \{C \subseteq \mathbb{R}^N : \text{nonempty, closed}\}$ is a multifunction such that

(i) $(t, x) \to F(t, x)$ is graph measurable (that is, $GrF = \{(t, x) \in T \times \mathbb{R}^N \times \mathbb{R}^N : h \in F(t, x)\} \in \mathcal{L}_T \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^N)\}$, with $\mathcal{L}_T$ being the Lebesgue σ-algebra of T and $B(\mathbb{R}^N)$ the Borel σ-algebra of $\mathbb{R}^N$);

(ii) for a.a $t \in T$, $x \to F(t, x)$ is lsc;

(iii) for every $\varrho > 0$, there exists $a_\varrho \in L^2(T)$ such that

$$|F(t, x)| = \sup\{|h| : h \in F(t, x)\} \leq a_\varrho(t)$$

for a.a $t \in T$, all $|x| \leq \varrho$;

(iv) there exists $M > 0$ such that

$$c \leq (h, x)_{\mathbb{R}^N} \text{ for a.a } t \in T, \text{ all } |x| = M, h \in F(t, x).$$

Remark 6.56 Again we see that now we need a stronger measurability condition, since Theorem I-8.38 can not be used in this context.

Theorem 6.57 *If hypotheses H_{21}, H_{22}, H_{23}, H_{25} hold, then problem (6.114) admits a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$.*

Proof Consider the multifunction $N_F : C^1(T, \mathbb{R}^N) \to P_f(L^2(T, \mathbb{R}^N)) = \{C \subseteq L^2(T, \mathbb{R}^N) : \text{nonempty, closed}\}$ defined by

$$N_F(u) = S^2_{F(\cdot, u(\cdot))} \text{ for all } u \in C^1(T, \mathbb{R}^N).$$

As in the proof of Theorem 6.38, we show that $N_F(\cdot)$ is lsc and it has decomposable values. So, we can use Theorem I-5.137 and find a continuous map $k : C^1(T, \mathbb{R}^N) \to L^2(T, \mathbb{R}^N)$ such that

$$k(u) \in N_F(u) \text{ for all } u \in C^1(T, \mathbb{R}^N).$$

We consider the following periodic Lienard system

$$\left\{ \begin{array}{l} a(u'(t))' + \frac{d}{dt} \nabla G(u(t)) \in A(u(t)) + k(u)(t) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b), \end{array} \right\} \tag{6.142}$$

Invoking Theorem 6.55, we have a solution $\widehat{u} \in C^1(T, \mathbb{R}^N)$ for problem (6.142). Evidently $\widehat{u}$ solves (6.114). □

Finally we have a "relaxation theorem", namely we show that under stronger continuity hypotheses on the orientor field $F(t, \cdot)$, we show that the solutions of the convex problem can be obtained as limits in $C^1(T, \mathbb{R}^N)$ of the solutions of a certain initial value nonconvex problem. To do this we need to introduce a parameter $\mu > 0$ in the problem. The problems under consideration are the following:

$$\left\{ \begin{array}{l} a(u'(t))' + \mu \frac{d}{dt} \nabla G(u(t)) \in A(u(t)) + \overline{conv} F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b). \end{array} \right\} \tag{6.143}$$

and

$$\left\{ \begin{array}{l} a(u'(t))' + \mu \frac{d}{dt} \nabla G(u(t)) \in A(u(t)) + F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = y_0, u'(0) = y'_0. \end{array} \right\} \tag{6.144}$$

By $S_c^{\mu} \subseteq C^1(T, \mathbb{R}^N)$ we denote the solution of convex periodic problem (6.143) and by $S^{\mu}(y_0, y_0') \subseteq C^1(T, \mathbb{R}^N)$ the solution set of the nonconvex initial value system.

The hypotheses on the map $a(\cdot)$ defining the differential operator and on the orientor field, are now stronger:

H_{26}: $a : \mathbb{R}^N \to \mathbb{R}^N$ is a map such that $a(y) = a_0(|y|)y$ with $a_0 : (0, \infty) \to (0, \infty)$ continuous $a_0(t)t \to 0^+$ as $t \to 0^+$ and

$$\widehat{c}\,|y|^2 \leq (a(y), y)_{\mathbb{R}^N} \leq \widehat{c}_0(1 + |y|^2) \text{ for all } y \in \mathbb{R}^N,$$

with $\widehat{c}, \widehat{c}_0 > 0$ and for every $r > 0$, there exists $\widehat{c}_r > 0$ such that

$$\widehat{c}_r|y - \widehat{y}| \leq (a(y) - a(\widehat{y}), y - \widehat{y})_{\mathbb{R}^N}$$

for all $y, \widehat{y} \in \mathbb{R}^N$ with $|y|, |\widehat{y}| \leq r$.

Remark 6.58 Evidently $a(\cdot)$ is strictly monotone. Many maps corresponding to differential operators of interest, satisfy the above hypotheses. For example

$$a_1(y) = |y|^{p-2}y \text{ with } 1 < p \leq 2,$$

$$a_2(y) = |y|^{p-2}y + |y|^{q-2}y \text{ with } 1 < q < p, q \leq 2,$$

$$a_3(y) = (1 + |y|^2)^{\frac{p-2}{2}}y \text{ with } 1 < p < \infty.$$

The hypotheses on the multifunction $F(t, x)$ are the following:

H_{27}: $F : T \times \mathbb{R}^N \to P_f(\mathbb{R}^N) = \{C \subseteq \mathbb{R}^N : \text{nonempty, closed}\}$ is a multifunction such that

(i) for all $x \in \mathbb{R}^N$, $t \to F(t, x)$ admits a measurable selection;

(ii) for every $r > 0$, there exists $\vartheta_r \in L^{\infty}(T)$ such that

$$\|\vartheta_r\|_{\infty} b^2 < \widehat{c}_r \text{ (see hypotheses } H_{26}),$$

$$h(\overline{conv}F(t, x), \overline{conv}F(t, \widehat{x})) \leq \vartheta_r(t)|x - \widehat{x}|$$

for a.a $t \in T$, all $x, \widehat{x} \in \mathbb{R}^N$ with $|x|, |\widehat{x}| \leq r$;

(iii) for every $r > 0$, there exists $\widehat{a}_r \in L^2(T)$ such that

$$|F(t, x)| = \sup\{|h| : h \in F(t, x)\} \leq \widehat{a}_r(t) \text{ for a.a } t \in T, \text{ all } |x| \leq r;$$

(iv) there exists $M > 0$ such that

$$0 \leq (h, x)_{\mathbb{R}^N} \text{ for a.a } t \in T, \text{ all } |x| = M, \text{ all } h \in F(t, x).$$

Under the above hypotheses $S_c^\mu \neq \emptyset$ and in fact it is compact in $C^1(T, \mathbb{R}^N)$. Similarly for every $(y_0, y_0') \in \mathbb{R}^N \times \mathbb{R}^N$, $S^\mu(y_0, y_0') \neq \emptyset$.

Theorem 6.59 *If hypotheses $H_{22}, H_{23}, H_{26}, H_{27}$ hold and $\mu > 0$ is small, then given $u \in S_c^\mu$ we can find $\{u_n\}_{n\in\mathbb{N}} \subseteq S^\mu(u(0), u'(0))$ such that $u_n \to u$ in $C^1(T, \mathbb{R}^N)$.*

Proof Let $M > 0$ be as in hypothesis $H_{27}(iv)$ and the $\widehat{a}_M \in L^2(T)$ be as postulated by hypothesis $H_{27}(iii)$. We set

$$\overline{B}_M = \{h \in L^2(T, \mathbb{R}^N) : |h(t)| \leq \widehat{a}_M(t) \text{ for a.a } t \in T\}.$$

Let $C_0 = \widehat{k}(\overline{B}_M)$. This is compact in $C^1(T, \mathbb{R}^N)$ (see Proposition 6.50). Therefore $C = \overline{conv} C_0 \subseteq C^1(T, \mathbb{R}^N)$ is compact, convex (see Theorem I-3.130).

Since $u \in S_c^\mu$, we have

$$|u(t)| \leq M \text{ for all } t \in T, \tag{6.145}$$

$$\left\{ \begin{array}{l} a(u'(t))' + \mu \frac{d}{dt} \nabla G(u(t)) \in A(u(t)) + h(t) \text{ for a.a } t \in T, \\ u(0) = u(b), u'(0) = u'(b)'. \end{array} \right\} \tag{6.146}$$

with $h \in N_F(u) = S^2_{F(\cdot,u(\cdot))}$.

Given $\varepsilon > 0$, we introduce the multifunction $L_\varepsilon : C \to 2^{L^2(T,\mathbb{R}^N)}$ defined by

$$L_\varepsilon(v) = \{g \in S^2_{\overline{conv} F(\cdot,v(\cdot))} : |h(t) - g(t)| < \varepsilon + d(h(t), \overline{conv} F(t, v(t))) \text{ for a.a } t \in T\}.$$

Using Theorem I-5.106 (the Yankov-von Neumann-Aumann selection theorem), we see that $L_\varepsilon(v) \neq \emptyset$ for all $v \in C$. Also $v \to L_\varepsilon(v)$ is lsc (see Proposition I-5.6). Therefore $v \to \overline{L_\varepsilon(v)}$ is lsc too (see Proposition I-5.23). Then we can find a continuous map $\ell_\varepsilon : C \to L^2(T, \mathbb{R}^N)$ such that

$$\ell_\varepsilon(v) \in \overline{L_\varepsilon(v)} \text{ for all } v \in C.$$

Similarly, we have a continuous map $\gamma_\varepsilon : C \to L^2(T, \mathbb{R}^N)$ such that

$$\gamma_\varepsilon(v) \in N_F(v) = S^2_{F(\cdot,v(\cdot))}, \|\gamma_\varepsilon(v) - \ell_\varepsilon(v)\|_2 \leq \varepsilon \text{ for all } v \in C.$$

Let $\varepsilon_n \to 0^+$ and set $\ell_n = \ell_{\varepsilon_n}, \gamma_n = \gamma_{\varepsilon_n}$. We consider the following second order initial value system

$$\left\{\begin{array}{l} a(v'(t))' + \frac{d}{dt}\nabla G(v(t)) \in A(v(t)) + \gamma_n(v)(t) \text{ for a.a } t \in T, \\ v(0) = u(0), v'(0) = u'(0) \end{array}\right\}$$

This problem has a solution $u_n \in C^1(T, \mathbb{R}^N), n \in \mathbb{N}$ and $\{u_n\}_{n\in\mathbb{N}} \subseteq C^1(T, \mathbb{R}^N)$ is relatively compact (see (6.145)). So, we can say that

$$u_n \to \widehat{u} \text{ in } C^1(T, \mathbb{R}^N)$$

Using integration by parts, we have

$$\int_0^b (a(u_n') - a(u'), u_n' - u')_{\mathbb{R}^N} dt + \mu \int_0^b (\nabla G(u_n) - \nabla G(u), u_n' - u')_{\mathbb{R}^N} dt$$

$$\leq \int_0^b (\gamma_n(u_n) - \ell_n(u_n), u_n - u)_{\mathbb{R}^N} dt + \int_0^b (\ell_n(u_n) - h, u_n - u)_{\mathbb{R}^N} dt$$

(recall $A(\cdot)$ is monotone)

$$\Rightarrow \widehat{c}_r \|u_n' - u'\|_2^2 \leq \mu c_2 \|u_n' - u'\|_2 + \varepsilon_n \int_0^b |u_n - u| dt$$

$$+ \|\vartheta_r\|_\infty \|u_n - u\|_2^2 + \int_0^b (\gamma_n(u_n) - \ell_n(u_n), u - u_n)_{\mathbb{R}^N} dt$$

with $r = \max\{\sup_{n\geq 1} \|u_n\|_\infty, \|u\|_\infty\}, c_2 > 0.$

Passing to the limit as $n \to \infty$ and using Jensen's inequality we obtain

$$(\widehat{c}_r - \|\vartheta_r\|_\infty b^2)\|\widehat{u}' - u'\|_2^2 \leq \mu c_2 \|\widehat{u} - u\|_2 \|\widehat{u}' - u'\|_2,$$

$$\Rightarrow \|\widehat{u}' - u'\|_2 \leq \mu c_3 \|\widehat{u} - u\|_2 \text{ for some } c_3 > 0,$$

$$\Rightarrow \|\widehat{u} - u\|_2 \leq \mu c_4 \|\widehat{u} - u\|_2 \text{ for some } c_4 > 0.$$

If $\mu \in (0, \frac{1}{c_4})$, then we conclude

$$\widehat{u} = u,$$

$$\Rightarrow u_n \to u \text{ in } C^1(T, \mathbb{R}^N).$$

Since $u_n \in S(u(0), u'(0))$ for all $n \in \mathbb{N}$, we have the relaxation result. □

6.2 Evolution Inclusions with Nonlocal Conditions

In this section we study evolution inclusions with nonlocal conditions. Such problems have broader applications than traditional Cauchy problems.

The mathematical setting consists of a bounded time interval $T = [0, b]$ and an evolution triple (X, H, X^*) (see Definition I-4.201) with the embedding of X into H being compact Consequently $H \hookrightarrow X^*$ compactly (by Schauder's Theorem, see Theorem I-3.158). Then the evolution inclusion under consideration is the following:

$$\left\{ \begin{array}{l} -u'(t) \in A(t, u(t)) + F(t, u(t)) \text{ for a.a } t \in T, \\ u(0) = \beta(u). \end{array} \right\} \tag{6.147}$$

In this problem $A : T \times X \to 2^{X^*} \setminus \{\emptyset\}$ with $A(t, \cdot)$ maximal monotone, $F : T \times H \to 2^H \setminus \{\emptyset\}$ is a multifunction and $\beta : C(T, H) \to H$ represents the nonlocal condition. We will consider both the convex and the nonconvex problems (depending on the values of the multifunction $F(t, x)$).

We start with a compactness result which is of independent interest and can be a useful tool in the study of evolution equations. We consider the following Cauchy problem

$$\left\{ \begin{array}{l} u'(t) + A(t, u(t)) \ni h(t) \text{ for a.a } t \in T, \\ u(0) = x_0. \end{array} \right\} \tag{6.148}$$

We impose the following conditions on the map $A(t, x)$.

H_1: $A : T \times X \to 2^{X^*}$ is a multivalued map such that

(i) $(t, x) \to A(t, x)$ is graph measurable;
(ii) for a.a $t \in T$, $A(t, \cdot)$ is pseudomonotone;
(iii) $|A(t, x)|_* = \sup\{\|x^*\|_* : x^* \in A(t, x)\} \leq a(t) + c\|x\|^{p-1}$ for a.a $t \in T$, all $x \in X$, with $1 < p < \infty$, $a \in L^{p'}(T)$, $c > 0$;
(iv) for almost all $t \in T$, all $x \in X$ and all $x^* \in A(t, x)$, we have

$$\langle x^*, x \rangle \geq c_1 \|x\|^p - a_1(t)$$

with $c_1 > 0$, $a_1 \in L^1(T)$.

Remark 6.60 Since by hypothesis $H_1(ii)$, $A(t, \cdot)$ is pseudomonotone, it follows that for a.a $t \in T$, $A(t, \cdot)$ has values in $P_{wkc}(X^*) = \{C \subseteq X^* : \text{nonempty, weakly compact, convex}\}$ (see Definition I-7.67). Moreover, for a.a $t \in T$, $A(t, \cdot)$ is an usc multifunction from X into X^*_w. To see this, note that on account of hypothesis $H_1(iii)$, for a.a $t \in T$, $A(t, \cdot)$ maps bounded sets in X into bounded, hence relatively weakly compact subsets of X^*. So, according to Proposition I-5.13, to show the desired upper semicontinuity of $A(t, \cdot)$, it suffices to show that $GrA(t, \cdot) \subseteq X \times X^*_w$ is closed. So, let $\{(x_n, x_n^*)\} \subseteq GrA(t, \cdot)$ and

assume that $x_n \to x$ in X, $x_n^* \overset{w}{\to} x^*$ in X^*. Then $\lim_{n\to\infty} \langle x_n^*, x_n^* - x\rangle = 0$ and since $A(t,\cdot)$ is pseudomonotone, we infer that $x^* \in A(t,x)$ (see Proposition I-7.68). Hence $(x,x^*) \in GrA(t,\cdot)$ and so $A(t,\cdot)$ is usc from X into X_w^*. Finally note that hypothesis $H_1(i)$ implies that if $u : T \to X$ is measurable, then $t \to A(t,u(t))$ is a measurable multifunction (superpositional measurability).

We consider the multivalued map $\widehat{A} : L^p(T,X) \to 2^{L^{p'}(T,X^*)}$ defined by

$$\widehat{A}(u) = \{h \in L^{p'}(T,X^*) : h(t) \in A(t,u(t)) \text{ for a.a } t \in T\} \text{ for all } u \in L^p(T,X).$$

Recall that $L^p(T,X)^* = L^{p'}(T,X^*)$ (see Theorem I-4.173). Then the Remark 6.60, Theorem I-5.106 (the Yankov-von Neumann-Auman selection theorem) and hypothesis $H_1(iii)$ imply that for all $u \in L^p(T,X)$ we have $\widehat{A}(u) \in P_{wkc}(L^{p'}(T,X^*)) = \{C \subseteq L^{p'}(T,X^*) : \text{nonempty,w-compact, convex}\}$. Moreover, from Proposition 6.4, we know that $\widehat{A}(\cdot)$ is pseudomonotone (the fact that now $A(\cdot,\cdot)$ is multivalued does not change anything). Also, hypothesis $H_1(iv)$ implies that $\widehat{A}(\cdot)$ is strongly coercive. Then Theorem I-7.76 implies that the solution set $S(x_0,h) \subseteq W_p(0,b) \subseteq C(T,H)$ of (6.148) is nonempty and in fact from the proof of Theorem 6.8, we have that $S(x_0,h) \subseteq C(T,H)$ is compact. In the next proposition, we determine the continuity properties of the solution multifunction $S : H \times L^{p'}(T,H) \to P_k(C(T,H)) = \{C \subseteq C(T,H) : \text{nonempty, compact}\}$. First a lemma which gives us an extended version of the Gronwall inequality.

Lemma 6.61 *If $\vartheta \in L^1(T)$, $\vartheta(t) \geq 0$ for a.a $t \in T$, $\eta \geq 0$ and $\varphi \in C(T)$ satisfies $\frac{1}{2}\varphi^2(t) \leq \frac{1}{2}\eta^2 + \int_0^1 \vartheta(s)\varphi(s)ds$ for all $t \in T$, then $|\varphi(t)| \leq \eta + \int_0^t \vartheta(s)ds$ for all $t \in T$.*

Proof Let $\gamma_\varepsilon(t) = \frac{1}{2}[\eta+\varepsilon]^2 + \int_0^t \vartheta(s)\varphi(s)ds$ for all $t \in T$, with $\varepsilon > 0$. Then

$$\gamma_\varepsilon'(t) = \vartheta(t)\varphi(t),\ \frac{1}{2}\varphi^2(t) \leq \gamma_0(t) \leq \gamma_\varepsilon(t) \text{ for all } t \in T, \varepsilon > 0,$$
$$\Rightarrow \gamma_\varepsilon'(t) \leq \vartheta(t)(2\gamma_\varepsilon(t))^{1/2} \text{ for a.a } t \in T.$$

Note that $\frac{1}{2}\varepsilon^2 \leq \gamma_\varepsilon(t)$ for all $t \in T$. The function $t \to \gamma_\varepsilon(t)$ is absolutely continuous and $(\gamma_\varepsilon(t)^{1/2})' = \frac{1}{\gamma_\varepsilon(t)^{1/2}}\gamma_\varepsilon'(t)$ for a.a $t \in T$. Therefore

$$(\gamma_\varepsilon(t)^{1/2})' \leq \frac{1}{\sqrt{2}}\vartheta(t) \text{ for a.a } t \in T,$$
$$\Rightarrow \gamma_\varepsilon(t)^{1/2} \leq \gamma_\varepsilon(0)^{1/2} + \frac{1}{\sqrt{2}}\int_0^t \vartheta(s)ds \text{ for all } t \in T,$$
$$\Rightarrow |\varphi(t)| \leq (2\gamma_\varepsilon(t))^{1/2} \leq (2\gamma_\varepsilon(0))^{1/2} + \int_0^t \vartheta(t)ds$$
$$= \eta + \varepsilon + \int_0^t \vartheta(s)ds.$$

Let $\varepsilon \to 0^+$ to conclude that

$$|\varphi(t)| \leq \eta + \int_0^t \vartheta(s)ds \text{ for all } t \in T.$$

□

Proposition 6.62 *If hypotheses H_1 hold, then the solution multifunction $S(\cdot,\cdot)$ is sequentially usc from $H_w \times L^{p'}(T,H)_w$ into $C(T,H)$.*

Proof Let $C \subseteq C(T,H)$ be nonempty, closed. We need to show that

$$S^-(C) = \{(x_0,h) \in H \times L^{p'}(T,H) : S(x_0,h) \cap C \neq \emptyset\}$$

is sequentially closed in $H_w \times L^{p'}(T,H)_w$. So, let $\{(x_0^n,h_n)\}_{n\in\mathbb{N}} \subseteq S^-(C)$ and assume that $(x_0^n,h_n) \to (x_0,h)$ in $H_w \times L^{p'}(T,H)_w$. Let $u_n \in S(x_0^n,h_n), n \in \mathbb{N}$. We have

$$\left\{ \begin{array}{l} u_n'(t) + A(t,u_n(t)) \ni h_n(t) \text{ for a.a } t \in T, \\ u_n(0) = x_0^n, n \in \mathbb{N}. \end{array} \right\} \tag{6.149}$$

Acting with $u_n(t)$, integrating on $[0,t]$ and using hypothesis $H_1(iv)$, we obtain

$$|u_n(t)|^2 \leq c_2 + \int_0^t |h_n(s)||u_n(s)|ds \text{ for some } c_2 > 0, \text{ all } n \in \mathbb{N},$$

$$\Rightarrow |u_n(t)| \leq c_3 \text{ for some } c_3 > 0, \text{ all } t \in T, \text{ all } n \in \mathbb{N}, \text{ (see Lemma 6.61)}.$$

From (6.149) and hypothesis $H_1(iv)$ we have

$$\|u_n\|_p \leq c_4 \text{ for some } c_4 > 0, \text{ all } n \in \mathbb{N}.$$

Then hypothesis $H_1(iii)$ and Holder's inequality, imply

$$\|u_n'\|_{p'} \leq c_5 \text{ for some } c_5 > 0, \text{ all } n \in \mathbb{N},$$

$$\Rightarrow \{u_n\}_{n\in\mathbb{N}} \subseteq W_p(0,b) \text{ is bounded.}$$

Therefore we may assume that

$$u_n \xrightarrow{w} u \text{ in } W_p(0,b), u_n \to u \text{ in } L^p(T,H), \tag{6.150}$$

recall that $W_p(0,b) \hookrightarrow L^p(T,H)$ compactly, see Theorem I-4.207.

Then as in the proof of Theorem 6.8 (see Claim 3), we obtain

$$u_n \to u \text{ in } C(T,H), u \in C. \tag{6.151}$$

We have

$$u_n' + g_n = h_n, g_n \in \widehat{A}(u_n)\ n \in \mathbb{N}. \tag{6.152}$$

We may assume that $g_n \overset{w}{\to} g$ in $L^{p'}(T, X^*)$ (see hypothesis $H_1(iii)$). From (6.150) and (6.152), we have

$$\begin{aligned}
&\lim_{n\to\infty} \langle g_n, u_n - u\rangle_p = 0\\
&\Rightarrow g \in \widehat{A}(u) \text{ (since } \widehat{A}(\cdot) \text{ is pseudomonotone)},\\
&\Rightarrow u' + g = h, g \in \widehat{A}(u), u(0) = x_0,\\
&\Rightarrow u \in S(x_0, h) \cap C. \text{ (see (6.151))},\\
&\Rightarrow (x_0, h) \in S^-(C).
\end{aligned}$$

This proves the desired upper semicontinuity of the solution multifunction $S(\cdot, \cdot)$. □

Remark 6.63 If for a.a $t \in T$, $A(t, \cdot)$ is monotone and hemicontinuous (that is, for all $x, y \in X, \lambda \to A(t, x + \lambda y)$ is usc from $[0, 1]$ into X_w^*), then $S(x_0, h)$ is a singleton. Indeed, if $u_1, u_2 \in S(x_0, h)$, then

$$\begin{aligned}
&u_1' - u_2' + A(t, u_1) - A(t, u_2) \ni 0\\
&\Rightarrow \frac{1}{2}|u_1(t) - u_2(t)|^2 \le 0 \text{ for all } t \in T\\
&\Rightarrow u_1 = u_2.
\end{aligned}$$

We introduce the following conditions on the multivalued map $A(t, x)$.

$\mathbf{H_1'}$: $A : T \times X \to 2^{X^*}$ is a multivalued map such that hypotheses $H_1'(i)$, (iii), (iv) are the same as the corresponding hypotheses $H_1(i)$, (iii), (iv) are the same as the corresponding hypotheses $H_1(i)$, (iii), (iv) and

(ii) for a.a $t \in T$, $A(t, \cdot)$ is monotone, hemicontinuous.

Corollary 6.64 *If hypotheses H_1' hold, then the solution map $S : H \times L^{p'}(T, H) \to C(T, H)$ is single valued and completely continuous, thus compact.*

Remark 6.65 Note that hypothesis $H_1'(ii)$ implies that for a.a $t \in T$, $A(t, \cdot)$ is maximal monotone (see Proposition I-7.36). Also from the argument in Remark 6.63, it is clear that $(x_0, h) \to S(x_0, h)$ is Lipschitz continuous.

Now we can turn our attention to the nonlocal problem (6.147). The hypotheses on the multifunction $F(t,x)$ and the map $\beta(\cdot)$ are the following:

H_2: $F : T \times H \to P_{f_c}(H) = \{C \subseteq H : \text{nonempty, closed, convex}\}$ is a multifunction such that

(i) for all $x \in H$, $t \to F(t,x)$ is graph measurable;
(ii) for a.a $t \in T$, $GrF(t,\cdot) \subseteq H \times H_w$ is sequentially closed;
(iii) $|F(t,x)| = \sup\{|h| : h \in F(t,x)\} \leq \widehat{a}(t)(1+|x|)$ for a.a $t \in T$, all $x \in H$, with $\widehat{a} \in L^{p'}(T)$;

H_3: $\beta : C(T,H) \to H$ satisfies

$$|\beta(u) - \beta(v)| \leq k\|u - v\|_{C(T,H)} \text{ for all } u, v \in C(T,H), \text{ with } k \in (0,1)$$

and $\beta(0) = 0$.

Remark 6.66 Possible choices of the map $\beta(\cdot)$ are:

$$\beta(u) = \pm ku(b) \text{ with } 0 < k < 1,$$

$$\beta(u) = \frac{k}{b}\int_0^b u(s)ds \text{ with } 0 < k < 1,$$

$$\beta(u) = \sum_{k=1}^{m} \vartheta_k u(t_k) \text{ with } 0 < t_1 < \ldots < t_n \leq b, \sum_{k=1}^{m} \vartheta_k < 1.$$

$$\beta(u) = \|u\|_{C(T,H)} u(b).$$

Let $\widehat{S}(\beta) \subseteq W_p(0,b) \subseteq C(T,H)$ denote the solution set of problem (6.147).

Theorem 6.67 *If hypotheses H_1', H_2, H_3 hold, then $\widehat{S}(\beta) \subseteq C(T,H)$ is nonempty compact.*

Proof First we prove a priori bounds for the elements of $S(\beta)$. So, let $u \in \widehat{S}(\beta)$. We have

$$\left\{ \begin{array}{l} u'(t) + A(t,u(t)) + h(t) \ni 0 \text{ for a.a } t \in T, \\ u(0) = \beta(u), \end{array} \right\} \tag{6.153}$$

with $h \in N_F(u) = S^{\tau'}_{F(\cdot,u(\cdot))}$. On (6.153) we act with $u(t)$ and have

$$\frac{1}{2}\frac{d}{dt}|u(t)|^2 + c_1\|u(t)\|^p - a_1(t) + (h(t),u(t)) \leq 0 \text{ (see hypothesis } H_1(iv)).$$

Integrating, we obtain

$$\frac{1}{2}|u(t)|^2 + c_1\|u\|_p^p \leq \frac{1}{2}|\beta(u)|^2 + \|a_1\|_1 + \int_0^t |h(s)||u(s)|ds \text{ for all } t \in T,$$

$$\Rightarrow c_1\|u\|_p^p \leq \|a_1\|_1 + \int_0^b \widehat{a}(t)(1 + |u(t)|^{\tau-1})|u(t)|dt$$

(see hypotheses H_3 and $H_2(iii)$).

Using Holder's inequality and the fact that $u \in C(T, H)$, we have

$$\begin{aligned} & c_1\|u\|_p^p \leq \|a_1\|_1 + \|\widehat{a}\|_{\frac{p}{p-\tau}}[\|u\|_{L^{p/\tau}(T,H)} + \|u\|_{L^p(T,H)}^{\tau}] \\ & \Rightarrow c_1\|u\|_p^p \leq c_6(1 + \|u\|_p^{\tau}) \text{ for some } c_6 > 0 \\ & \text{(recall } L^p(T, X) \hookrightarrow L^p(T, X) \text{ continuously)}, \\ & \Rightarrow \|u\|_p \leq c_7 \text{ for some } c_7 > 0, \text{ all } u \in S(\beta) \end{aligned} \tag{6.154}$$

From (6.153), we have for a.a $t \in T$

$$\begin{aligned} \|u'(t)\|_* &\leq a(t) + c\|u(t)\|^{p-1} + h(t) \text{ (see hypothesis } H_2(iii)) \\ &\leq a(t) + c\|u(t)\|^{p-1} + \widehat{a}(t)(1 + |u(t)|^{\tau-1}) \text{ (see hypothesis } H_3(iii)) \\ &\leq a_0(t)(1 + \|u(t)\|^{p-1}) \text{ with } a_0 \in L^{\frac{p}{p-\tau}}(T). \end{aligned}$$

Using the generalized Holder's inequality (see Corollary I-2.121), we obtain

$$\|u'\|_{L^{p'}(T,X^*)} \leq c_8(1 + \|u\|_p) \leq c_9 \text{ for some } c_8, c_9 > 0 \text{ (see (6.154))}.$$

Therefore we infer that

$$\widehat{S}(\beta) \subseteq W_p(0, b) \text{ is bounded.}$$

Recall that $W_p(0, b) \hookrightarrow C(T, H)$ continuously (see Theorem I-4.204). Therefore we can find $M > 0$ such that

$$\|u\|_{C(T,H)} \leq M \text{ for all } u \in S(\beta) \tag{6.155}$$

So, we can replace $F(t, x)$ by $F(t, p_M(x))$ with $p_M(x)$ being the M-radial retraction defined by

$$p_M(x) = \begin{cases} x & \text{if } |x| \leq M \\ M\frac{x}{|x|} & \text{if } M < x. \end{cases}$$

So, we may assume that

$$|F(t,x)| = \sup\{|h| : h \in F(t,x)\} \leq \widehat{a}(t) \text{ for a.a } t \in T, \text{ all } x \in H,$$

with $\widehat{a} \in L^{\frac{p}{p-\tau}}(T)$. Note that $\frac{p}{\tau} < p$, hence $p' < (\frac{p}{\tau})' = \frac{p}{p-\tau}$. We introduce the set

$$V = \{h \in L^{\frac{p}{p-\tau}}(T,H) : |h(t)| \leq \widehat{a}(t) \text{ for a.a } t \in T\}.$$

For $h \in V$ and $\lambda \in (0,1)$, we consider the following problem

$$\left\{ \begin{array}{l} u'(t) + A(t,u(t)) \ni h(t) \text{ for a.a } t \in T, \\ u(0) = \lambda\beta(u). \end{array} \right\} \tag{6.156}$$

We will show that problem (6.156) has a unique solution in $W_p(0,b)$. To this end let $v \in C(T_1,H)$ and consider the following Cauchy problem

$$\left\{ \begin{array}{l} u'(t) + A(t,u(t)) \ni h(t) \text{ for a.a } t \in T, \\ u(0) = \lambda\beta(v). \end{array} \right\} \tag{6.157}$$

Problem (6.157) has a unique solution $k_\lambda(v,h) \in W_p(0,b)$. Exploiting the monotonicity of $A(t,\cdot)$ (see hypothesis $H_2(ii)$), we have

$$\begin{aligned} |k_\lambda(v_1,h) - k_\lambda(v_2,h)| &\leq \lambda|\beta(v_1) - \beta(v_2)| \\ &\leq \lambda\|v_1 - v_2\|_{C(T,H)} \text{ for all } v_1, v_2 \in C(T,H), \text{ all } \lambda \in (0,1). \end{aligned}$$

Invoking the Banach fixed point theorem, we can find $u \in C(T,H)$ such that

$$k_\lambda(u,h) = u.$$

Let $\widehat{k}_\lambda : L^{\frac{p}{p-\tau}}(T,H) \to C(T,H)$ being the solution map which to each $h \in L^{\frac{p}{p-\tau}}(T,H)$ assigns the unique solution of (6.156). From Corollary 6.64, we know that $\widehat{k}_\lambda(\cdot)$ is completely continuous. Also recall that $u \to N_{-F}(u)$ is usc from $C(T,H)$ into $L^{\frac{p}{p-\tau}}(T,H)$ (see the proof of Theorem 6.16).

Consider the set

$$D_\lambda = \{u \in C(T,H) : u \in \vartheta(\widehat{k}_\lambda \circ N_{-F})(u), 0 < \vartheta < 1\}.$$

If $u \in D_\lambda$, then

$$\begin{aligned} &\frac{1}{\vartheta}u = \widehat{k}_\lambda(h) \text{ for some } h \in N_{-F}(u), \\ &\Rightarrow \frac{1}{\vartheta}u' + A(t, \frac{1}{\vartheta}u) \ni h. \end{aligned}$$

Acting with $\frac{1}{\vartheta}u$ and reasoning as above, we infer that $D_\lambda \subseteq C(T, H)$ is bounded. Invoking Theorem 2.138 (the multivalued Leray-Schauder Alternative Principle), we can find $\widehat{u}_\lambda \in W_p(0, b) \subseteq C(T, H)$ such that

$$\widehat{u}_\lambda = \widehat{k}_\lambda(h) \text{ with } h \in N_{-F}(\widehat{u}).$$

Now let $\lambda_n \to 1^-$ and $\widehat{u}_n = \widehat{u}_{\lambda_n} \in W_p(0, b), n \in \mathbb{N}$ be the solution obtained above. From the first part of the proof (the a priori estimation), we have that

$$\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq W_p(0, b) \text{ is bounded.}$$

We may assume that

$$\widehat{u}_n \xrightarrow{w} \widehat{u} \text{ in } W_p(0, b) \tag{6.158}$$

As in the proof of Theorem 6.8, using (6.158), then $\widehat{u} \in S(\beta) \neq \emptyset$.

It is clear that $\widehat{S}(\beta) \subseteq C(T, H)$ is closed. Since by Corollary 6.64 the solution map $S : L^{\frac{p}{p-\tau}}(T, H) \to C(T, H)$ is compact and $\widehat{S}(\beta) \subseteq \overline{S(V)}$, we conclude that $\widehat{S}(\beta) \subseteq C(T, H)$ is compact. □

We can have a "nonconvex" version of the above existence theorem. The hypotheses on the multifunction $F(t, x)$ are the following:

H_4: $F : T \times H \to P_f(H) = \{C \subseteq H : \text{ nonempty, closed}\}$ is a multifunction such that

(i) $(t, x) \to F(t, x)$ is graph measurable;
(ii) for a.a $t \in T, x \to F(t, x)$ is lsc;
(iii) $|F(t, x)| = \sup\{|h| : h \in F(t, x)\} \leq \widehat{a}(t)(1 + |x|^{\tau-1})$ for a.a $t \in T$, all $x \in H$, with $1 < \tau < p$ and $\widehat{a} \in L^{\frac{p}{p-\tau}}(T)$.

We have the following existence theorem (see also Theorem 6.10).

Theorem 6.68 *If hypotheses H_1', H_3, H_4 hold, then $S(\beta) \neq \emptyset$.*

Strengthening the hypotheses on the multifunction $F(t, x)$ we can have a relaxation theorem, relating the solution sets of the convex and nonconvex problems. The new conditions on $F(t, x)$ are the following:

H_5: $F : T \times H \to P_f(H) = \{C \subseteq H : \text{ nonempty, closed}\}$ is a multifunction such that

(i) for all $x \in H, t \to F(t, x)$ is graph measurable;
(ii) $h(\overline{conv}F(t, x), \overline{conv}F(t, y)) \leq k_0(t)|x - y|$ for a.a $t \in T$, all $x, y \in H$, with $k_0 \in L^1(T)$;
(iii) $|F(t, x)| = \sup\{|h| \in F(t, x)\} \leq \widehat{a}(t)(1 + |x|^{\tau-1})$ for a.a $t < T$, all $x \in H$, with $1 < \tau < p$ and $\widehat{a} \in L^{\frac{p}{p-\tau}}(T)$.

By $S(\beta)$ we denote the solution set of the nonconvex problem and by $S_c(\beta)$ the solution set of the convex problem. We have

$$\emptyset \neq S(\beta) \subseteq S_c(\beta) \subseteq W_p(0,b) \subseteq C(T,H).$$

We know that $S_c(\beta) \subseteq C(T,H)$ is compact.

Reasoning as in the proof of Theorem 6.13, we obtain the following relaxation theorem.

Theorem 6.69 *If hypotheses H_1, H_3, H_5 hold, then $\overline{S(\beta)}^{C(T,H)} = S_c(\beta)$.*

We illustrate these results which an example of a parabolic system with a nonlocal initial condition.

Example 6.70 Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^1-boundary $\partial\Omega$. We consider the following parabolic system

$$\left\{ \begin{array}{l} \frac{\partial u}{\partial t} - \Delta_p^m u + f(t,z,u(t,z)) \ni 0 \text{ in } T \times \Omega, \\ u|_{T\times\partial\Omega} = 0, u(0,z) = \int_0^b \ell(t,z,u(t,z))dt \text{ for a.a } z \in \Omega. \end{array} \right\} \tag{6.159}$$

In this problem $u : \Omega \to \mathbb{R}^m$. Note that when $A \in \mathbb{R}^{m\times N}$, then $|A| = tr(AA^t)^{1/2}$. Then we define the vectorial p-Laplacian by

$$\Delta_p^m u = (div(|Du|^{p-2}Du_k))_{k=1}^m, 2 \leq p < \infty, \text{ for all } u \in W_0^{1,p}(\Omega,\mathbb{R}^m).$$

Also $f : T\times\Omega\times\mathbb{R}^m \to P_{f_c}(\mathbb{R}^m)$ is a multifunction which satisfies the following conditions

- for all $x \in \mathbb{R}^m$, $(t,z) \to f(t,z,x)$ is graph measurable;
- for all $(t,z) \in T \times \Omega$, $Grf(t,z,\cdot) \subseteq \mathbb{R}^m \times \mathbb{R}^m$ is closed;
- $|f(t,z,x)| = \sup\{|h| : h \in f(t,z,x) \leq \widehat{a}_0(t,x)\}(1+|x|^{\tau-1})$ for a.a $(t,z) \in T \times \Omega$, all $x \in \mathbb{R}^m$, with $\widehat{a}_0 \in L^{\frac{p}{p-\tau}}(T,\Omega)$, $1 < \tau < p$.

The function $\ell(t,z,x)$ involved in the nonlocal initial condition, is defined on $T \times \Omega \times \mathbb{R}^m$ with values in $\mathbb{R}^m$ and has the following properties

- for all $x \in \mathbb{R}^m$, $(t,z) \to \ell(t,z,x)$ is measurable;
- $|\ell(t,z,x) - \ell(t,z,y)| \leq k_0(t,z)|x-y|$ for a.a $(t,z) \in T \times \Omega$, all $x, y \in \mathbb{R}^m$, with $k_0 \in L^p(T, L^\infty(\Omega))$, $\int_0^b b\|k(t,\cdot)\|_\infty^2 dt \leq 1$;
- $\int_0^b \ell(t,z,0)dt = 0$ for a.a $z \in \Omega$.

The underlying evolution triple consists of the following spaces

$$X = W_0^{1,p}(\Omega,\mathbb{R}^m), H = L^2(\Omega,\mathbb{R}^m), X^* = W^{-1,p'}(\Omega,\mathbb{R}^m).$$

Since $2 \leq p$, we have that $X \hookrightarrow H$ compactly. Let $A : X \to X^*$ be defined by

$$\langle A(u), h\rangle = \sum_{k=1}^{m} \int_{\Omega} |Du|^{p-2}(Du_k, Dh_k)dz \text{ for all } u, h \in W_0^{1,p}(\Omega, \mathbb{R}^m).$$

It is easy to see that this map satisfies hypotheses H_1'. We set $F(t,u) = S^{\tau'}_{f(t,\cdot,u(\cdot))}$ for all $u \in L^2(\Omega, \mathbb{R}^m)$. The properties of $f(t,z,x)$ imply that hypotheses H_2 are satisfied by $F(t,u)$.

Also, let $\beta : C(T, L^2(\Omega, \mathbb{R}^m)) \to L^2(\Omega, \mathbb{R}^m)$ be defined by

$$\beta(u)(\cdot) = \int_0^b \ell(t, \cdot, u(t,\cdot))dt \text{ for all } u \in C(T, L^2(\Omega, \mathbb{R}^m))$$

Given $u, v \in C(T, L^2(\Omega, \mathbb{R}^m))$, we have

$$\begin{aligned}
&|\beta(u) - \beta(v)|^2 \\
&= \int_{\Omega} |\int_0^b \ell(t,z,u(t,z))|dt - \int_0^b \ell(t,z,v(t,z))dt|^2 dz \\
&\leq \int_{\Omega} |\int_0^b bk_0(t,z)^2 |u(t,z) - v(t,z)|^2 dt dz \text{ (by Jensen's inequality)} \\
&= \int_0^b \int_{\Omega} bk_0(t,z)^2 |u(t,z) - v(t,z)|^2 dz dt \text{ (by Fubini's theorem)} \\
&\leq \int_0^b b\|k_0(t,\cdot)\|_{\infty}^2 |u(t,\cdot) - v(t,\cdot)|^2 dt \\
&\leq \int_0^b b\widehat{k}_0(t)^2 dt \|u - v\|_{C(T,H)} \text{ with } \widehat{k}_0(t) = \|k_0(t,\cdot)\|_{\infty}.
\end{aligned}$$

Since $\beta(0) = 0$, we see that $\beta(\cdot)$ satisfies hypotheses H_3. Then problem (6.159) can be equivalently rewritten as the following evolution inclusion

$$u'(t) + A(u(t)) + F(t, u(t)) \ni 0 \text{ for } a.a\ t \in T, u(0) = \beta(u).$$

We can apply Theorem 6.67 and conclude that problem (6.159) has a nonempty solution set $\widehat{S}(\beta)$ which is compact in $C(T, L^2(\Omega, \mathbb{R}^m))$.

Finally, we consider the case when the nonlocal map $\beta(\cdot)$ is linear. In this case we can relax the conditions on the map $A(t,\cdot)$ and use the more general hypotheses H_1. For the map $\beta(\cdot)$ we assume the following:

H_6: $\beta \in \mathcal{L}(C(T,H), H)$.

Let $L : D(L) \subseteq L^p(T,X) \to L^{p'}(T, X^*)$ be defined by

$$L(u) = u' \text{ for all } u \in D(L) = \{u \in W_p(0,b) : u(0) = \beta(u)\}.$$

Then this is a linear, densely defined, maximal monotone operator. Also we set

$$V(u) = \widehat{A}(u) + N_F(u) \text{ for all } u \in L^p(T, X)$$

with $\widehat{A}(u)(\cdot) = A(\cdot, u(\cdot))$, $N_F(u) = S^{\tau'}_{F(\cdot,u(\cdot))}$. As in the proof of Theorem 6.8, we can show that $V(\cdot)$ is bounded, L-pseudomonotone and strongly coercive. Using Theorem 6.2, we obtain the following existence result.

Theorem 6.71 *If hypotheses H_1, H_2, H_6 hold, then $S(\beta) \subseteq C(T, H)$ is nonempty, compact.*

6.3 Structure of the Solution Set

Evolution inclusions usually have many solutions emanating from a given initial condition. This raises some interesting questions concerning the topological properties of the solution set. An important feature of the solution set is the R_δ-property. A subset of a metric space, is an R_δ-set if it can be written as the intersection of a decreasing sequence of compact, contractible sets. Such a set is acyclic and in particular compact and connected. From the point of view of algebraic topology, an R_δ-set is trivial, that is, it is equivalent to a singleton (it has the same homology group as a point). We recall the following notions (see Definitions 1.25 and 2.65).

Definition 6.72 Let X be a Hausdorff topological space

(a) A set $C \subseteq X$ is a "retract", if there is a continuous map $r : X \to C$ such that $r|_C = id_C$. The map $r(\cdot)$ is a "retraction" of X onto C (see also Definition 1.25).
(b) We say that X is an "absolute retract" ("AR" for short), if for each metrizable space Y and any closed set $C \subseteq Y$, each continuous map $\vartheta : C \to X$ is extendable on Y.
(c) We say that X is an "absolute neighborhood retract" ("ANR" for short), if for any metrizable space Y and any closed set $C \subseteq Y$, each continuous map $\vartheta : C \to X$ is extendable in some neighborhood U of C.

Remark 6.73 A retract is always closed. On account of Dugundji's extension theorem (see Theorem *I*-3.154), every closed and convex set in a locally convex space is a retract and an AR. If C is a retract of D and D is a retract of X, then C is a retract X. Often the definitions of AR and ANR are stated with the metrizable space Y being replaced by a normal space Y. Evidently an AR is an ANR. Every AR is path-connected and every ANR is locally path-connected. An alternative way to define an AR space X, is to say that every homeomorphic image of X in any metric space Y, is a retract of Y. More on absolute retracts can be found in Sect. 2.2.

We recall two more notions which we will use in this section (see Definition 2.45(c) and Definition 2.73). Here we formulate these notions in a different (but equivalent) way, more suitable for our needs in this section (see also Remark 2.46).

Definition 6.74

(a) Let X be a metric space and $C \subseteq X$ a nonempty subset. We say that C is "contractible", if there exist a deformation $h : [0,1] \times C \to C$ (that is, a continuous map $h(\cdot,\cdot)$ with $h(0,\cdot) = id$) and $x_0 \in C$ such that $h(1,x) = x_0$ for all $x \in C$.

(b) Let X be a metric space and $C \subseteq X$ a nonempty subset. We say that C is an "R_δ-set" if

$$C = \bigcap_{n\in\mathbb{N}} C_n$$

with $\{C_n\}_{n\in\mathbb{N}}$ a decreasing sequence of compact contractible subsets.

Remark 6.75 Any R_δ-set is compact and connected. As we already mentioned in Remark 2.74, an R_δ-set need not be an AR. In fact we have the following relations between the nonempty subsets of a metric space

$$\text{compact} + \text{connected} \subseteq \text{compact} + \text{AR}$$

$$\subseteq \text{compact} + \text{contractible} \subseteq R_\delta - \text{set}.$$

All these inclusions are proper. Recall that the intersection of a decreasing sequence of R_δ-set, is an R_δ-set.

To establish the topological regularity of the solution set of evolution inclusions, we will need to following approximation result for h-usc multifunctions (see Definition *I*-5.44).

Proposition 6.76 *If X is a Polish space, Y is a separable Banach space and* $F : T \times X \to P_{f_c}(Y) = \{C \subseteq Y : \text{nonempty, closed, convex}\}$ *is a multifunction such that*

(i) *for all* $x \in X$, $t \to F(t,x)$ *is graph measurable;*
(ii) *for a.a* $t \in T$, $x \to F(t,x)$ *is h-usc;*
(iii) $|F(t,x)| = \sup\{\|h\|_Y : h \in F(t,x)\} \leq \vartheta(t)$ *for a.a* $t \in T$, *all* $x \in X$ *with* $\vartheta \in L^1(T)$;

then there is a sequence of multifunctions

$$F_n : T \times X \to P_{f_c}(Y), n \in \mathbb{N},$$

such that for every $n \in \mathbb{N}$ *and* $x \in X$, *there exist* $k_n(x) > 0$ *and* $\varepsilon_n > 0$ *such that*

$$x_1, x_2 \in \overline{B}_{\varepsilon_n}(x) \Rightarrow h(F_n(t,x_1), F_n(t,x_2)) \leq k_n(x)\vartheta(t)d_X(x_1,x_2) \text{ for a.a } t \in T,$$

$$\{F_n\}_{n\in\mathbb{N}} \text{ is decreasing, } F(t,x) \subseteq F_n(t,x) \text{ for a.a } t \in T, \text{ all } x \in X,$$

$$|F_n(t,x)| = \sup\{\|h\|_Y : h \in F_n(t,x)\} \leq \vartheta(t) \text{ for a.a } t \in T, \text{ all } x \in X,$$

$$F_n(t,x) \xrightarrow{h} F(t,x) \text{ for a.a } t \in T, \text{ all } x \in X$$

and there exists $\eta_n : T \times X \to Y, n \in \mathbb{N}$, measurable in $t \in T$, locally Lipschitz in $x \in X$ such that $\eta_n(t,x) \in F_n(t,x)$ for all $(t,x) \in T \times X, n \in \mathbb{N}$.

Finally if $F(\cdot,\cdot)$ is measurable, then $t \to F_n(t,x)$ is measurable.

Proof Let $\mathcal{F} = \{B_{\frac{1}{3^n}}(x)\}_{x\in X}$. This is an open cover of X. We can find $P_n = \{\varphi_x^n\}_{x\in X}$ a locally Lipschitz partition of unity subordinate to the open cover $\mathcal{F}$. We set $G_x^n(t) = \overline{conv} F(t, B_{\frac{2}{3^n}}(x))$ and introduce the multifunction $F_n : T \times X \to P_{f_c}(Y)$ defined by

$$F_n(t,x) = \sum_{x\in X} \varphi_x^n(x) G_x^n(t).$$

This multifunction is well-defined since the partition of unity is locally finite (see Definition *I*-1.190). Let $f_x^n : T \to Y$ be a measurable selection of $G_x^n(\cdot)$ and define $\eta_n : T \times X \to Y$ by

$$\eta_n(t,v) = \sum_{x\in X} \varphi_x^n(v) f_x^n(t).$$

Then $\eta_n(\cdot,\cdot)$ is a selection of $F_n(\cdot,\cdot)$. Fix $t \in T$ and $x_0 \in X$. The partition of unity $\{\varphi_x^n\}_{x\in X}$ is locally finite, so we can find $r > 0$ such that $N(n,x_0) = \{x \in X : \operatorname{supp}\varphi_x^n \cap B_r(x_0) \neq \emptyset\}$ is nonempty, finite. Let $N(n,x_0) = \{x_i\}_{i=1}^m$. Then

$$F_n(t,v) = \sum_{i=1}^m \varphi_{x_i}^n(v) G_{x_i}^n(t).$$

Choose $\delta \in (0,r)$ so that $\varphi_{x_i}^n|_{B_\delta(x_0)}$ is k_i-Lipschitz for all $i \in \{1,\dots,m\}$. For $v_1, v_2 \in B_\delta(x_0)$ we have

$$\begin{aligned}
&h(F_n(t,v_1), F_n(t,v_2)) \\
&= h(\sum_{i=1}^m \varphi_{x_i}^n(v_1) G_{x_i}^n(t), \sum_{i=1}^m \varphi_{x_i}^n(v_2) G_{x_i}^n(t)) \\
&\le \sum_{i=1}^m |G_{x_i}^n(t)| \|\varphi_{x_i}^n(v_1) - \varphi_{x_i}^n(v_2)\|_Y \\
&\le k_n(x_0)\vartheta(t) d_X(v_1,v_2) \text{ with } k_n(x_0) > 0.
\end{aligned}$$

Next we show that $\{F_n\}_{n\in\mathbb{N}}$ is decreasing. Fix $v \in X$ and let $M(n,v)$ (resp. $M(n+1,v)$) be the finite set of $\varphi_x^n \in P_n$ whose support contains v (recall the partition of unity is locally finite). We claim that $G_{x_k^{n+1}}^{n+1}(t) \subseteq G_{x_i^n}^n(t)$ for all $t \in T$, all $1 \le i \le |M(n,v)|$ and all $1 \le k \le |M(n+1,v)|$. Clearly $v \in B_{\frac{1}{3^n}}(x_i^n)$, $v \in B_{\frac{1}{3^{n+1}}}(x_k^{n+1})$. Therefore, if $z \in B_{\frac{2}{3^{n+1}}}(x_k^{n+1})$, then

$$d_X(z, x_i^n) \le d_X(z, x_k^{n+1}) + d_X(x_k^{n+1}, x_i^n) < \frac{2}{3^{n+1}},$$
$$\Rightarrow B_{\frac{2}{3^{n+1}}}(x_k^{n+1}) \subseteq B_{\frac{2}{3^n}}(x_i^n),$$
$$\Rightarrow G^{n+1}_{x_k^{n+1}}(t) \subseteq G^n_{x_i^n}(t), t \in T,$$
$$\Rightarrow F_{n+1}(t, v) \subseteq F_n(t, v).$$

Similarly we show that

$$F(t, x) \subseteq F_n(t, x) \text{ for } (t, x) \in T \times X, n \in \mathbb{N}.$$

Next we show that $F_n(t, x) \xrightarrow{h} F(t, x)$ for all $(t, x) \in T \times X$. On account of the h-upper semicontinuity of $F(t, \cdot)$, given $\varepsilon > 0$ we can find $\delta = \delta(t, v, \varepsilon) > 0$ such that

$$F(t, z) \subseteq F(t, v) + \frac{\varepsilon}{2} B_1 \text{ for all } z \in X, d_X(z, v) \le \delta.$$

Choose $N_0 = N_0(t, v, \varepsilon) \ge 1$ such that $\frac{1}{3^n} \le \frac{\delta}{3}$ for all $n \ge N_0$. Let $x \in X$ be such that $\varphi_x^n \in M(n, v)$. For $z \in B_{\frac{2}{3^n}}(x)$, we have

$$F_n(t, z) \subseteq F_n(t, v) + \frac{\varepsilon}{3} B_1,$$
$$\Rightarrow G_x^n(t) \subseteq \overline{F(t, v) + \frac{\varepsilon}{3} B_1} \subseteq F(t, v) + \frac{2\varepsilon}{3} B_1$$
$$\text{for all } n \ge N_0, \text{ all } x \in X \text{ with } \varphi_x^n \in M(n, v),$$
$$\Rightarrow F_n(t, z) \subseteq F(t, v) + \varepsilon B_1 \text{ for all } n \ge n_0,$$
$$\Rightarrow F_n(t, v) \xrightarrow{h} F(t, v).$$

Finally if $F(\cdot, \cdot)$ is $\mathcal{L}_T \otimes B(X)$-measurable and $U_x^n = B_{\frac{2}{3^n}}(x)$ then for $y \in Y$ and $\lambda > 0$, let

$$D_\lambda = \{(t, v) \in T \times U_x^n : d(y, F(t, v)) < \lambda\} \in \mathcal{L}_T \otimes B(X)$$

(recall $\mathcal{L}_T$ is the Lebesgue σ-algebra of T and $B(X)$ the Borel σ-algebra of X). We have

$$\{t \in T : d(y, G_x^n(t)) < \lambda\}$$
$$= \{t \in T : \inf\{d(y, F(t, v)) : v \in U_x^n\} < \lambda\}$$
$$= \{t \in T : \text{ there exists } v \in U_x^n \text{ such that } (t, v) \in D_\lambda\}$$
$$= proj_T D_\lambda \in \mathcal{L}_T (\text{ see Corollary } I\text{-}2.55).$$

Therefore by Theorem *I*-5.78

$$t \to G_x^n(t) \text{ is measurable,}$$
$$\Rightarrow t \to F_n(t,x) \text{ is measurable.}$$

□

Using this approximation result, we can characterize the solution set of the nonlocal problem. So, the problem under consideration is

$$\{u'(t) + A(t,u(t)) + F(t,u(t)) \ni 0 \text{ for } a.a\ t \in T, u(0) = \beta(u).\} \tag{6.160}$$

Note that $\widehat{S}(\beta) \subseteq W_p(0,b) \subseteq C(T,H)$ denotes the solution set of (6.160).

Theorem 6.77 *If hypotheses H_1', H_2, H_3 hold, then $\widehat{S}(\beta) \subseteq C(T,H)$ is an R_δ-set.*

Proof From the a priori estimation in the first part of the proof of Theorem 6.67, we know that we may assume that

$$|F(t,x)| = \sup\{|h| : h \in F(t,x)\} \leq \widehat{a}(t)$$
$$\text{for } a.a\ t \in T, \text{ all } x \in H, \text{ with } \widehat{a} \in L^{\frac{p}{p-\tau}}(T).$$

We can apply Proposition 6.76 and produce $\{F_n\}_{n\in\mathbb{N}}$ a decreasing approximating sequence of multifunctions. We consider the following problem

$$\{u'(t) + A(t,u(t)) + F_n(t,u(t)) \ni 0 \text{ for } a.a\ t \in T, u(0) = \beta(u).\} \tag{6.161}$$

By $\widehat{S}_n(\beta) \subseteq W_p(0,b) \subseteq C(T,H)$ we denote the solution set of problem (6.161). From Theorem 6.67, we know that $\widehat{S}_n(\beta) \neq \emptyset$ and it is compact in $C(T,H)$.

We will show that $\widehat{S}_n(\beta)$ is contractible in $C(T,H)$. Let $\eta_n(t,x)$ be the measurable and locally Lipschitz in $x \in H$ selection of $F_n(\cdot,\cdot)$ postulated by Proposition 6.76. In what follows let $\widehat{k} : L^{\frac{p}{p-\tau}}(T,H) \to C(T,H)$ be the solution map which to each $h \in L^{\frac{p}{p-\tau}}(T,H)$ assigns the unique solution of the nonlocal problem

$$\{u'(t) + A(t,u(t)) \ni h(t) \text{ for } a.a\ t \in T, u(0) = \beta(u).\}$$

We know that this map is completely continuous (see the proof of Theorem 6.67). For $u \in \widehat{S}_n(\beta)$, let $\widehat{u} = \widehat{k}(\eta_n(\cdot,u(\cdot)))$ and then for $r \in [0,b)$, we consider the Cauchy problem

$$\{y'(t) + A(t,y(t)) = -\eta_n(t,y(t)) \text{ for } a.a\ t \in T_r = [r,b],$$
$$y(r) = \frac{r}{b}u(r) + (1 - \frac{r}{b})\widehat{u}(0).\}$$

This problem has a solution which due to the fact that $\eta_n(t,\cdot)$ is locally Lipschitz, it is unique. Denote by $y(\frac{r}{b},u) \in W_p(0,b) \subseteq C(T,H)$ this unique solution. For $r=b$, we have $y(1,u)(b)=u(b)$. We define $h_n(\cdot,\cdot)$ on $[0,1]\times \widehat{S}_n(\beta)$ as follows

$$h_n(\frac{r}{b},u)(t)=\begin{cases} \frac{r}{b}u(t)+(1-\frac{r}{b})\widehat{u}(0) \ if \ t\in[0,\frac{r}{b}], \\ y(\frac{r}{b},u)(t) \qquad\qquad if \ t\in(\frac{r}{b},b]. \end{cases} \tag{6.162}$$

We show that for all $r\in[0,b]$, $h_n(\frac{r}{b},u)\in\widehat{S}_n(\beta)$.

If $r=0$, then $h_n(0,u)=\widehat{u}(0)$ (see (6.162)).

If $r=b$, then $h_n(1,u)=y(1,u)=u$.

Finally let $r\in(0,b)$. Since $u\in\widehat{S}_n(\beta)$, we can find $h\in N_{F_n}(u)$ such that $u=\widehat{k}(-h)$. We set

$$g=h\chi_{[0,\frac{r}{b}]}+\eta_n(\cdot,u(\cdot))\chi_{(\frac{r}{b},b]}.$$

Note that $-g\in N_{F_n}(u)$ and we have

$$\widehat{k}(-g)=u \text{ on } [0,r], \widehat{k}(-g)=y(\frac{r}{b},u) \text{ on } (r,b],$$
$$\Rightarrow \widehat{k}(-g)=h_n(\frac{r}{b},u)\in\widehat{S}_n(\beta).$$

If we show that $h_n:[0,1]\times\widehat{S}_n(\beta)\to\widehat{S}_n(\beta)$ is continuous, then we can conclude that $\widehat{S}_n(\beta)$ is contractible in $C(T,H)$. To this end, let $r_m\to r$ in $[0,b]$, $u_m\to u$ in $C(T,H)$. We consider two distinct cases.

Case I $r_m\geq r$ for all $m\in\mathbb{N}$.

Let $z_m=h_n(\frac{r_m}{b},u_m)\in\widehat{S}_n(\beta)$ for all $m\in\mathbb{N}$. We know that $\widehat{S}_n(\beta)\subseteq C(T,H)$ is compact. So, we may assume that

$$z_m\to z \text{ in } C(T,H) \text{ as } m\to\infty.$$

We need to show that $z=h_n(\frac{r}{b},u)$. We have $[0,\frac{r}{b}]\subseteq[0,\frac{r_m}{b}]$ for all $m\in\mathbb{N}$ and

$$z_m=\frac{r_m}{b}u_m+(1-\frac{r_m}{b})\widehat{u}(0) \text{ on } [0,\frac{r_m}{b}] \text{ for all } m\in\mathbb{N},$$
$$\Rightarrow z=\frac{r}{b}u+(1-\frac{r}{b})\widehat{u}(0) \text{ on } [0,\frac{r}{b}].$$

Let $w\in W_p(0,b)\subseteq C(T,H)$ be the unique solution of the Cauchy problem

$$\{u'(t)+A(t,u(t))\ni-\eta_n(t,z(t)) \text{ for } a.a\ t\in[r,b], u(r)=z(r).\} \tag{6.163}$$

For $m \in \mathbb{N}$ large, say $m \geq N$, we have

$$z_m'(t) + A(t, z_m(t)) = -\eta_n(t, z_m(t)) \text{ for } a.a\ t \in [r_m, b].$$

Then exploiting the monotonicity of $A(t, \cdot)$, we obtain

$$\begin{aligned}
&\frac{1}{2}\frac{d}{dt}|w(t) - z_m(t)|^2 \leq (\eta_n(t, z_m(t)) - \eta_n(t, z(t)), w(t) - z_m(t)) \\
&\qquad\qquad \text{for } a.a\ t \in [r_N, b], \\
&\Rightarrow \frac{1}{2}|w(t) - z_m(t)|^2 \leq \frac{1}{2}|w(r_N) - z_m(r_N)|^2 \\
&\qquad + \int_{r_N}^t |\eta_n(s, z_m) - \eta_n(s, z)||w - z_m| ds \\
&\Rightarrow |w(t) - z_m(t)| \leq |w(r_N) - z_m(r_N)| + \int_{r_N}^t |\eta_n(s, z_m) - \eta_n(s, z)| ds \\
&\qquad\qquad \text{for all } t \in [r_N, b] (\text{see Lemma 6.61}).
\end{aligned}$$

We pass to the limit as $m \to \infty$ and obtain

$$|w(t) - z(t)| \leq |w(r_N) - z_m(r_N)| \text{ for all } t \in [r_N, b].$$

If we let $N \to \infty$, then $w(r_N), z_m(r_N) \to z(r)$ and so

$$w = z \text{ on } [r, b].$$

Therefore we infer that

$$\begin{aligned}
&z'(t) + A(t, z(t)) = -\eta_n(t, z(t)) \text{ for } a.a\ t \in [r, b], \\
&z(r) = \frac{r}{b}u(r) + (1 - \frac{r}{b})\widehat{u}(0), \\
&\Rightarrow z = h_n(\frac{r}{b}, u), \\
&\Rightarrow h_n(\frac{r_m}{b}, u_m) \to h_n(\frac{r}{b}, u) \text{ in } C(T, H) \text{ as } m \to \infty.
\end{aligned}$$

This takes care of Case I.

Case II $r_m \leq r$ for all $m \in \mathbb{N}$.

Now we have $[0, r_m] \subseteq [0, r]$ for all $m \in \mathbb{N}$ and

$$z_m = \frac{r_m}{b} u_m + (1 - \frac{r_m}{b})\widehat{u}(0) \text{ on } [0, r_m],$$
$$\Rightarrow z = \frac{r}{b} u + (1 - \frac{r}{b})\widehat{u}(0) \text{ on } [0, r].$$

Let $w \in W_p(0, b) \subseteq C(T, H)$ be the unique solution of (6.163).
For $m \in \mathbb{N}$ large, say $m \geq N$, we have

$$z'_m(t) + A(t, z_m(t)) = -\eta_n(t, z_m(t)) \text{ for } a.a\ t \in [r_N, b].$$

As before, we have

$$|w(t) - z_m(t)| \leq |w(r) - z_m(r)| + \int_r^t |\eta_n(s, z) - \eta_n(s, z_m)| ds$$
$$\text{for all } t \in [r, b].$$

Passing to the limit as $m \to \infty$, we obtain

$$|w(t) - z(t)| \leq |w(r) - z(r)| = 0 \text{ for all } t \in [r, b] \text{ (see (6.163))},$$
$$\Rightarrow w = z \text{ on } [r, b],$$
$$\Rightarrow z'(t) + A(t, z(t)) = -\eta_n(t, z(t)) \text{ for } a.a\ t \in [r, b],$$
$$z(r) = \frac{r}{b} u(r) + (1 - \frac{r}{b})\widehat{u}(0),$$
$$\Rightarrow z = h_n(\frac{r}{b}, u).$$

Therefore $h_n(\frac{r_m}{b}, u_m) \to h_n(\frac{r}{b}, u)$ in $C(T, H)$ as $m \to \infty$ and so we have proved also the continuity if Case II holds.

In general, we can always have a subsequence satisfying Case I or Case II and so we conclude that $h_n(\cdot, \cdot)$ is continuous and so it follows that $\widehat{S}_n(\beta) \subseteq C(T, H)$ is contractible for all $n \in \mathbb{N}$.

Finally we show that $\widehat{S}(\beta) = \bigcap_{n \in \mathbb{N}} \widehat{S}_n(\beta)$. Clearly we have that $\widehat{S}(\beta) \subseteq \bigcap_{n \in \mathbb{N}} \widehat{S}_n(\beta)$. Let $u \in \bigcap_{n \in \mathbb{N}} \widehat{S}_n(\beta)$. Then

$$u = \widehat{k}(h_n) \text{ with } -h_n \in N_{F_n}(u) \text{ for all } n \in \mathbb{N}.$$

We may assume that $h_n \xrightarrow{w} h$ in $L^{\frac{p}{p-\tau}}(T, H)$ and on account of Proposition 7.66, we have $-h \in N_F(u)$. Then

$$u = \widehat{k}(h) \text{ with } -h \in N_F(u),$$
$$\Rightarrow u \in \widehat{S}(\beta).$$

So, $\widehat{S}(\beta) = \bigcap_{n\in\mathbb{N}} \widehat{S}(\beta_n)$ and we conclude that $\widehat{S}(\beta) \subseteq C(T, H)$ is an R_δ-set (see Definition 6.74(b)). □

Corollary 6.78 *If hypotheses H_1', H_2, H_3 hold, then for every $t \in T$, the reachable set $R(t) = \widehat{S}(\beta)(t) = \{u(t) : u \in S(\widehat{\beta})\}$ is nonempty, compact and connected.*

We can have a structural result also for the nonconvex problem. We will need the following result due to Bressan et al. [59].

Theorem 6.79 *If (Ω, Σ, μ) is a finite nonatomic measure space, X is a separable Banach space, $Y = L^1(\Omega, X)$ and $F : Y \to 2^Y \setminus \{\emptyset\}$ is an h-continuous multifunction, then $FixF = \{y \in Y : y \in F(y)\}$ is an absolute retract.*

To be able to use this theorem, we will need to assume the following properties on the orientor field $F(t, x)$.

H_6: $F : T \times H \to P_f(H) = \{C \subseteq H : \text{nonempty, closed}\}$ is a multifunction such that

(i) for all $x \in H$, $t \to F(t, x)$ is graph measurable,
(ii) $h(F(t, x), F(t, y)) \leq l(t)|x - y|$ for $a.a\ t \in T$, all $x, y \in H$ with $l \in L^{p'}(T)$,
(iii) $|F(t, x)| = \sup\{|h| : h \in F(t, x)\} \leq \widehat{a}(t)(1 + |x|^{\tau-1})$ for $a.a\ t \in T$, all $x \in H$, with $1 \leq \tau < p$ and $\widehat{a} \in L^{\frac{p}{p-\tau}}(T)$.

Remark 6.80 Hypotheses $H_6(i)$, (ii) imply that $(t, x) \to F(t, x)$ is measurable.

Theorem 6.81 *If hypotheses H_1', H_6, H_3 hold, then $\widehat{S}(\beta) \subseteq C(T, H)$ is nonempty and path-connected.*

Proof Recall that on account of the a priori estimation we may assume that

$$|F(t, x)| \leq \widehat{a}_0(t) \text{ for } a.a\ t \in T, \text{ all } x \in H, \text{ with } \widehat{a}_0 \in L^{\frac{p}{p-\tau}}(T).$$

Let $r = \frac{p}{p-\tau}$ and on $L^r(T, H)$ consider the following equivalent norm (the Bielecki norm)

$$|h|_r = \int_0^b \exp(-\mu \int_0^t l(s)^q ds)|h(t)|^r dt \text{ for all } h \in L^r(T, H), \text{ with } \mu > 0 \text{ large.}$$

Let $h_r(\cdot,\cdot)$ denotes the Hausdorff metric corresponding to this equivalent norm. Consider the multifunction $N_F : L^r(T, H) \to P_f(L^r(T, H))$ defined by

$$N_F(g) = S^r_{F(\cdot,\widehat{k}(g)(\cdot))} \text{ for all } g \in L^r(T, H).$$

Note that for $g_1, g_2 \in L^r(T, H)$, we have

$$\begin{aligned}
&\frac{1}{2}|\widehat{k}(g_1)(t) - \widehat{k}(g_2)(t)|^2 \leq \int_0^t |\widehat{k}(g_1) - \widehat{k}(g_2)||g_1 - g_2|ds \\
&\Rightarrow |\widehat{k}(g_1)(t) - \widehat{k}(g_2)(t)| \leq \int_0^t |g_1 - g_2|ds \text{(see Lemma 6.61)} \qquad (6.164) \\
&\leq b^{\frac{1}{r'}}(\int_0^t |g_1 - g_2|^r ds)^{1/r}.
\end{aligned}$$

Using Theorem *I*-5.126, we have

$$\begin{aligned}
&h_r(N_F(g_1), N_F(g_2)) \\
&\leq (\int_0^b b^{r/r'} l(t)^r |\widehat{k}(g_1) - \widehat{k}(g_2)|^r \exp(-\mu \int_0^t l(s)^r ds)dt)^{1/r} \\
&\leq (\int_0^b b^{r/r'} \frac{1}{\mu} \exp(-\mu \int_0^t l(s)^q ds)|g_1 - g_2|^r dt)^{1/r} \text{(see (6.164))} \\
&\leq (\frac{1}{\mu})^{\frac{1}{r}} b^{1/r'} |g_1 - g_2|_r.
\end{aligned}$$

Choosing $\mu > 0$ large enough, we see that $N_F(\cdot)$ is an h-contraction. Invoking Theorem 2.117, we infer that

$$FixN_F \neq \emptyset.$$

Moreover, on account of Theorem 6.79 we have that $FixN_F$ is an absolute retract. But an absolute retract is path-connected (see Remark 6.73). Therefore

$$\begin{aligned}
&FixN_F \text{ is path-connected,} \\
&\Rightarrow \widehat{k}(FixN_F) \text{ is path-connected.}
\end{aligned}$$

But note that $\widehat{k}(FixN_F) = \widehat{S}(\beta)$. Therefore we conclude that $\widehat{S}(\beta) \subseteq C(T, H)$ is path-connected. □

6.4 Continuous Dependence Results

In this section we consider parametric evolution inclusions and examine how the set of solutions depends on the parameter.

So, let E be a compact metric space (the parameter space) and we consider the following parametric evolution inclusion with nonlocal conditions

$$\{u'(t) + A(t, u(t), \lambda) + F(t, u(t), \lambda) \ni 0 \text{ for } a.a\ t \in T, u(0) = \beta(u).\} \tag{6.165}$$

Let $\widehat{S}_\lambda(\beta) \subseteq W_p(0, b) \subseteq C(T, H)$ be the solution set of (6.165). Our aim in this section is to investigate the continuity properties of the solution multifunction.

The mathematical setting remains unchanged, with $T = [0, b]$ and (X, H, X^*) an evolution triple with $X \hookrightarrow H$ compactly (hence $H^* = H \hookrightarrow X^*$ compactly too). From Volume I, we recall the following definition (see Definition I-8.77).

Definition 6.82 Consider a sequence of operators $A_n : X \to 2^{X^*} n \in \mathbb{N}$. We say that the sequence $\{A_n\}_{n\in\mathbb{N}}$ "G-converges" to on operator $A : X \to 2^{X^*}$ if

$$Gr A_n \xrightarrow{K_\tau} Gr A$$

with K_τ denoting the Kuratowski convergence (see Definition I-8.4) in the topology $\tau = w \times s$ with w being the weak topology on X and s the norm (strong) topology on X^*.

Remark 6.83 Then according to Definition I-8.4, the G-convergence of the A_n's to A, is equivalent to the following two requirements:

(a) If $x_n \xrightarrow{w} x$ in X, $x_n^* \to x^*$ in X^* and $x_n^* \in A(x_n)$ for all $n \in \mathbb{N}$, then $x^* \in A(x)$ (that is, $\tau - \limsup_{n\to\infty} Gr A_n \subseteq Gr A$).

(b) If $x^* \in A(x)$, then we can find $\{x_n\}_{n\in\mathbb{N}} \subseteq X, \{x_n^*\}_{n\in\mathbb{N}} \subseteq X^*$ with $x_n^* \in A(x_n)$ $n \in \mathbb{N}$ such that

$$x_n \xrightarrow{w} x \text{ in } X, x_n^* \to x^* \text{ in } X^*$$

(that is, $Gr A \subseteq \tau - \liminf_{n\to\infty} Gr A_n$).

There is a "dynamic" version of this notion (parabolic G-convergence).

Definition 6.84 Let $P_n : W_p(0, b) \to L^{p'}(T, X^*) \times H$, $n \in \mathbb{N}$, and $P : W_p(0, b) \to L^{p'}(T, X^*) \times H$. We say that $\{P_n\}_{n\in\mathbb{N}}$ "PG-converges" to P, if $P_n^{-1}, P^{-1} : L^{p'}(T, X^*) \times H \to W_p(0, b)$ are defined and for every $(h^*, x_0) \in L^{p'}(T, X^*) \times H$ and every $t \in T$, we have

$$P_n^{-1}(h^*, x_0)(t) \to P^{-1}(h^*, x_0)(t) \text{ in } H.$$

Remark 6.85 In our case the operators P_n will be of the form

$$u \to (u'(\cdot) + A_n(\cdot, u(\cdot)), u(0))$$

with $A_n(t, \cdot)$ being monotone. We know then that P_n^{-1} exists.

Our hypotheses on the data of problem (6.165) are the following.

H_1: $A : T \times X \times E \to X^*$ is an operator such that

(i) $\|A(t+\tau, x, \lambda) - A(t, x, \lambda)\|_* \leq O(\tau)(1 + \|x\|^{p-1})$ for all $t, t+\tau \in T$, all $x \in X$, all $\lambda \in E$ and with $O(\tau)$ independent of λ and x;
(ii) for all $(t, \lambda) \in T \times E$, $x \to A(t, x, \lambda)$ is hemicontinuous;
(iii) $\|A(t, x, \lambda)\|_* \leq a(t) + c\|x\|^{p-1}$ for $a.a\ t \in T$, all $x \in X$, all $\lambda \in E$, with $a \in L^{p'}(T)$, $c > 0$;
(iv) $\langle A(t, x, \lambda) - A(t, y, \lambda)\rangle \geq c_0\|x - y\|^p$ for $a.a\ t \in T$, all $x, y \in X$, with $c_0 > 0$ (uniform monotonicity of $A(t, \cdot, \lambda)$);
(v) if $\lambda_n \to \lambda$ in E, then for all $t \in T$, we have

$$A(t, \cdot, \lambda_n) \xrightarrow{G} A(t, \cdot, \lambda).$$

Remark 6.86 The above hypotheses imply that

$$\frac{d}{dt} + A_{\lambda_n} \xrightarrow{PG} \frac{d}{dt} + A_\lambda \text{(see Definition 6.84)}. \tag{6.166}$$

This can be found in Hu and Papageorgiou [249], Theorem 6.9, p. 89. Also note that hypotheses $H_1(iii)$, (iv) imply that $\|A(t, x, \lambda)\|_* \geq \widehat{c}_0\|x\|^p - a_0(t)$ for $a.a\ t \in T$, all $x \in X$, all $\lambda \in E$ with $\widehat{c}_0 > 0$, $a_0 \in L^{p'}(T)$ (coercivity of $A(t, \cdot, \lambda)$).

First we consider the "convex" case and so our conditions on the multifunction $F(t, x)$ are the following:

H_2: $F : T \times H \times E \to P_{f_c}(H) = \{C \subseteq H : \text{nonempty, closed, convex}\}$ is a multifunction such that

(i) for all $(x, \lambda) \in H \times E$, $t \to F(t, x, \lambda)$ is graph measurable;
(ii) for $a.a\ t \in T$, $GrF(t, \cdot, \cdot) \subseteq H \times E \times H_w$ is sequentially closed;
(iii) for $a.a\ t \in T$ and all $(x, \lambda) \in H \times E$, we have

$$|F(t, x, \lambda)| = \sup\{|h| : h \in F(t, x, \lambda)\} \leq \widehat{a}(t)[1 + |x|^{\tau-1}],$$

with $1 < \tau < p$, $\widehat{a} \in L^{\frac{p}{p-\tau}}(T)$.

Finally for the nonlocal map $\beta(\cdot)$ we assume the following:

H_3: $\beta : C(T, H) \times E \to H$ is a map such that

$$|\beta(u, \lambda) - \beta(v, \lambda)| \leq \|u - v\|_{C(T,H)}$$

for all $(u, \lambda) \in C(T, H) \times E$, $\beta(0, \lambda) = 0$ for all $\lambda \in E$ and for all $u \in C(T, H)$ $\lambda \to \beta(u, \lambda)$ is continuous.

We know that the above conditions on the data of (6.165) imply that for all $\lambda \in E$, $\widehat{S}_\lambda(\beta) \in P_k(C(T, H)) = \{C \subseteq C(T, H) : \text{ nonempty, compact}\}$. Let $\widehat{\Gamma}_\beta : E \to P_k(C(T, H))$ be defined by $\widehat{\Gamma}_\beta(\lambda) = \widehat{S}_\lambda(\beta)$, $\lambda \in E$.

Theorem 6.87 *If hypotheses H_1, H_2, H_3 hold, then the solution multifunction $\widehat{\Gamma}_\beta : E \to P_k(C(T, H))$ is usc.*

Proof Since all the hypotheses are uniform with respect to the parameter $\lambda \in E$, from the previous proofs it follows that $\overline{\widehat{\Gamma}_\beta(E)} \subseteq C(T, H)$ is compact.

So, according to Proposition *I*-5.13, in order to prove the upper semicontinuity of $\widehat{\Gamma}_\beta(\cdot)$, it suffices to show that $Gr\widehat{\Gamma}_\beta \subseteq E \times C(T, H)$ is closed. To this end let $\{(\lambda_n, u_n)\}_{n\in\mathbb{N}} \subseteq Gr\widehat{\Gamma}_\beta$ and assume that

$$\lambda_n \to \lambda \text{ in } E, u_n \to u \text{ in } C(T, H).$$

We have

$$\{u_n'(t) + A(t, u_n(t), \lambda_n) = h_n(t) \text{ for } a.a\ t \in T, u_n(0) = \beta(u_n, \lambda_n)\}$$

with $-h_n \in N_{F_{\lambda_n}}(u_n) \subseteq L^{\frac{p}{p-\tau}}(T, H)$. On account of hypothesis $H_2(iii)$ we have that $\{h_n\}_{n\in\mathbb{N}} \subseteq L^{\frac{p}{p-\tau}}(T, H)$ is bounded and so we may assume that

$$h_n \xrightarrow{w} h \text{ in } L^{\frac{p}{p-\tau}}(T, H).$$

As before using hypothesis $H_2(ii)$ and Theorem *I*-8.38 we infer that $-h \in N_{F_\lambda}(u)$.

Consider the evolution inclusion

$$\{v'(t) + A(t, v(t), \lambda_n) = h(t) \text{ for } a.a\ t \in T, v(0) = \beta(v, \lambda).\} \tag{6.167}$$

From (6.166) it follows that

$$v_n(t) \to v(t) \text{ in } H \text{ for all } t \in T \tag{6.168}$$

with $v \in W_p(0, b) \subseteq C(T, H)$ being the unique solution of

$$\{v'(t) + A(t, v(t), \lambda) = h(t) \text{ for } a.a\ t \in T, v(0) = \beta(v, \lambda).\}$$

Acting on (6.167) with $v_n(t)$ and integrating over T, we obtain that $\{v_n\}_{n\in\mathbb{N}} \subseteq L^p(T, X)$ is bounded. It follows that $\{v_n'\}_{n\in\mathbb{N}} \subseteq L^{p'}(T, X^*)$ is bounded, hence

$\{v_n\}_{n\in\mathbb{N}} \subseteq W_p(0,b)$ is bounded. From the compact embedding of $W_p(0,b)$ into $L^p(T,H)$ (see Theorem I-4.207) and (6.168), we infer that

$$v_n \to v \text{ in } L^p(T,H). \tag{6.169}$$

We have

$$\begin{aligned}
&\langle u_n'(t) - v_n'(t), u_n(t) - v_n(t)\rangle + \langle A(t,u_n(t),\lambda_n) - A(t,v_n(t),\lambda_n), u_n(t) - v_n(t)\rangle \\
&= (h_n(t) - h(t), u_n(t) - v_n(t)) \text{ for } a.a\ t \in T, \\
&\Rightarrow \frac{1}{2}\frac{d}{dt}|u_n(t) - v_n(t)|^2 \le (h_n(t) - h(t), u_n(t) - v_n(t)) \text{ for } a.a\ t \in T, \\
&\Rightarrow \frac{1}{2}|u_n(t) - v_n(t)|^2 \le \frac{1}{2}|\beta(u_n,\lambda_n) - \beta(u,\lambda)|^2 \\
&\qquad + \int_0^t (h_n(s) - h(s), u_n(s) - v_n(s))ds.
\end{aligned} \tag{6.170}$$

Note that

$$\begin{aligned}
&\int_0^t (h_n(s) - h(s), u_n(s) - v_n(s))ds \\
&= \int_0^t (h_n(s) - h(s), u_n(s) - u(s))ds + \int_0^t (h_n(s) - h(s), u(s) - v(s))ds \\
&\qquad + \int_0^t (h_n(s) - h(s), v(s) - v_n(s))ds, \\
&\Rightarrow \int_0^t (h_n(s) - h(s), u_n(s) - v_n(s))ds \to 0.
\end{aligned} \tag{6.171}$$

Also, hypotheses H_3 imply

$$\begin{aligned}
&|\beta(u_n,\lambda_n) - \beta(u,\lambda)| \\
&\le |\beta(u_n,\lambda_n) - \beta(u,\lambda_n)| + |\beta(u,\lambda_n) - \beta(u,\lambda)| \\
&\le \|u_n - u\|_{C(T,H)} + |\beta(u,\lambda_n) - \beta(u,\lambda)| \to 0.
\end{aligned} \tag{6.172}$$

We return to (6.170), pass to the limit as $n \to \infty$ and use (6.171), (6.172). We obtain

$$|u(t) - v(t)| = 0 \text{ for all } t \in T \text{ (see (6.169))}.$$

So, we have

$$\{u'(t) + A(t, u(t), \lambda) = h(t) \text{ for } a.a\ t \in T, u(0) = \beta(u, \lambda)\}$$
$$\Rightarrow u \in \widehat{\Gamma}_\beta(\lambda)(\text{recall } -h \in N_{F_\lambda}(u)),$$
$$\Rightarrow Gr\,\widehat{\Gamma}_\beta \subseteq E \times C(T, H) \text{ is closed and so } \widehat{\Gamma}_\beta(\cdot) \text{ is usc.}$$

□

For the "nonconvex" case, we have lower semicontinuity of the solution multifunction. Now the hypotheses on the multifunction are the following:

H_4: $F : T \times H \times E \to P_f(H) = \{C \subseteq H : \text{nonempty, closed}\}$ is a multifunction such that

(i) for all $(x, \lambda) \in H \times E$, $t \to F(t, x, \lambda)$ is measurable;
(ii) $h(\overline{conv}F(t, x, \lambda), \overline{conv}F(t, y, \lambda)) \leq k_0(t)|x-y|$ for $a.a\ t \in T$, all $x, y \in H$, with $k_0 \in L^1(T)$;
(iii) for $a.a\ t \in T$ and all $(x, \lambda) \in H \times E$, we have

$$|F(t, x, \lambda)| = \sup\{|h| : h \in F(t, x, \lambda)\} \leq \widehat{a}(t)(1 + |x|^{\tau-1})$$

with $1 < \tau < p, \widehat{a} \in L^{\frac{p}{p-\tau}}(T)$;
(iv) for $a.a\ t \in T$, all $x \in H$, $\lambda \to F(t, x, \lambda)$ is lsc.

Theorem 6.88 *If hypotheses H_1, H_4, H_3 hold, then the solution multifunction $\widehat{\Gamma}_\beta : E \to 2^{L^p(T,H)} \setminus \{\emptyset\}$ is lsc.*

Proof To prove the lower semicontinuity of the solution multifunction, we need to show that if $\lambda_n \to \lambda$ in E, then $\widehat{\Gamma}_\beta(\lambda) \subseteq \liminf_{n\to\infty} \widehat{\Gamma}_\beta(\lambda_n)$ (see Proposition *I*-5.6).

Let $u \in W_p(0, b)$ be a solution of the convex problem, that is,

$$\{u'(t) + A(t, u(t), \lambda) = h(t) \text{ for } a.a\ t \in T, u(0) = \beta(u, \lambda), -h \in N_{\overline{conv}F}.\}$$

Let $\vartheta_n(t, x) = p_{\overline{conv}F(t,x,\lambda_n)}(h(t))$. This is a Caratheodory function. We consider the following nonlocal problem

$$\{u_n'(t) + A(t, u_n(t), \lambda_n) = \vartheta_n(t, u_n(t)) \tag{6.173}$$

for a.a. $t \in T$, and $u_n(0) = \beta(u_n, \lambda_n)$.

From Theorem 6.67, we know that problem (6.173) has a solution $u_n \in W_p(0, b)$. This is a solution of the convex problem (F replaced by $\overline{conv}F$). Let $v_n \in W_p(0, b)$ be the unique solution of the following Cauchy problem

$$\{v_n'(t) + A(t, v_n(t), \lambda_n) = h(t) \text{ for } a.a\ t \in T, v_n(0) = \beta(u, \lambda).\}$$

We know that

$$y_n \xrightarrow{w} u \text{ in } W_p(0,b),\ y_n \to u \text{ in } L^p(T,H) (\text{see hypothesis } H_1(v)). \tag{6.174}$$

We have

$$\begin{aligned}
&\langle u_n'(t) - v_n'(t), u_n(t) - v_n(t)\rangle + \langle A(t,u_n(t),\lambda_n) - A(t,v_n(t),\lambda_n), u_n(t) - v_n(t)\rangle \\
&= (\vartheta_n(t,u_n(t)) - h(t), u_n(t) - v_n(t)), \\
&\Rightarrow \frac{1}{2}\frac{d}{dt}|u_n(t) - v_n(t)|^2 \le (\vartheta_n(t,u_n(t)) - h(t), u_n(t) - v_n(t)), \\
&\Rightarrow \frac{1}{2}|u_n(t) - v_n(t)|^2 \le \frac{1}{2}|\beta(u_n,\lambda_n) - \beta(u,\lambda)|^2 \\
&\qquad + \int_0^t d(h(s), \overline{conv}F(s,u_n(s),\lambda_n))|u_n(s) - v_n(s)|ds \\
&\qquad \le \frac{1}{2}|\beta(u_n,\lambda_n) - \beta(u,\lambda)| \\
&\qquad + \int_0^t k_0(s)|u(s) - u_n(s)||u_n(s) - v_n(s)|ds \\
&\qquad (\text{see hypothesis } H_4(ii)).
\end{aligned} \tag{6.175}$$

Note that

$$\begin{aligned}
&\int_0^t k_0(s)|u(s) - u_n(s)||u_n(s) - v_n(s)|ds \\
&\le \int_0^t k_0(s)|u(s) - v_n(s)||u_n(s) - v_n(s)|ds + \int_0^t k_0(s)|v_n(s) - u_n(s)|^2 ds.
\end{aligned}$$

We know that

$$\int_0^t k_0(s)|u(s) - v_n(s)||u_n(s) - v_n(s)|ds \to 0 (\text{see } (6.174)).$$

Moreover, we have

$$\begin{aligned}
|\beta(u_n,\lambda_n) - \beta(u,\lambda)| &\le |\beta(u_n,\lambda_n) - \beta(u,\lambda_n)| + |\beta(u,\lambda_n) - \beta(u,\lambda)| \\
&\le \|u_n - u\|_{C(T,H)} + |\beta(u,\lambda_n) - \beta(u,\lambda)| \to 0 \\
&\qquad (\text{see hypotheses } H_3).
\end{aligned}$$

Then from (6.175) we see that given $\varepsilon > 0$, we can find $n_0(\varepsilon) \in \mathbb{N}$ such that

$$\frac{1}{2}|u_n(t) - v_n(t)|^2 \leq \varepsilon + \int_0^t k_0(s)|u_n(s) - v_n(s)|^2 ds,$$

$$\Rightarrow \|u_n - v_n\|_{C(T,H)} \leq 2\varepsilon \|k_0\|_1 \text{(by Gronwall's inequality)}.$$

From (6.174), it follows that

$$u_n \to u \text{ in } L^p(T, h). \tag{6.176}$$

So, if $\widehat{\Gamma}^c_\beta(\lambda)$ denotes the solution set of the convex problem, then we have proved that

$$\widehat{\Gamma}^c_\beta(\lambda) \subseteq \liminf_{n\to\infty} \widehat{\Gamma}^c_\beta(\lambda_n)$$

(see (6.176) and recall $u_n \in \widehat{\Gamma}^c_\beta(\lambda_n), n \in \mathbb{N}$).

If $u \in \widehat{\Gamma}_\beta(\lambda) \subseteq \widehat{\Gamma}^c_\beta(\lambda)$, then by the above argument there exists $u_n \in \widehat{\Gamma}^c_\beta(\lambda_n)$ $n \in \mathbb{N}$ such that $u_n \to u$ in $L^p(T, H)$. Theorem 6.69 implies for every $n \in \mathbb{N}$, we can find $u_n^m \in \widehat{\Gamma}_\beta(\lambda_n)$ $m \in \mathbb{N}$ such that

$$u_n^m \to u_n \text{ in } C(T, H) \text{ as } m \to \infty,$$

$$\Rightarrow u_n^m \to u_n \text{ in } L^p(T, H) \text{ as } m \to \infty.$$

Invoking Lemma *I*-1.193, we can find a sequence m_n such that $u_n^{m_n} \to u$ in $L^p(T, H)$ with $u_n^{m_n} \in \widehat{\Gamma}_\beta(\lambda_n)$ $n \in \mathbb{N}$. Therefore

$$\widehat{\Gamma}_\beta(\lambda) \subseteq \liminf_{n\to\infty} \widehat{\Gamma}_\beta(\lambda_n) \text{ in } L^p(T, H),$$

$$\Rightarrow \widehat{\Gamma}_\beta(\cdot) \text{ is lsc from } E \text{ into } L^p(T, H).$$

□

Strengthening the conditions on $F(t, x, \lambda)$, we can combine Theorems 6.87 and 6.88. The new hypotheses on the multifunction F are the following:

H_5: $F : T \times H \times E \to P_{f_c}(H) = \{C \subseteq H : \text{nonempty, closed, convex}\}$ is a multifunction such that

(i) for all $(x, \lambda) \in H \times E, t \to F(t, x, \lambda)$ is measurable;
(ii) $h(F(t, x, \lambda), F(t, y, \lambda)) \leq k_0(t)|x - y|$ for $a.a\ t \in T$, all $x, y \in H$, all $\lambda \in E$ with $k_0 \in L^1(T)$;
(iii) for $a.a\ t \in T$ and all $(x, \lambda) \in H \times E$, we have

$$|F(t, x, \lambda)| = \sup\{|h| : h \in F(t, x, \lambda)\} \leq \widehat{a}(t)(1 + |x|^{\tau-1})$$

with $1 < \tau < p, \widehat{a} \in L^{\frac{p}{p-\tau}}(T)$;
(iv) for $a.a\ t \in T$, all $x \in H, \lambda \to F(t, x, \lambda)$ is continuous.

Then from Theorems 6.87 and 6.88 combined with Proposition *I*-5.49, we have the next continuity result.

Theorem 6.89 *If hypotheses H_1, H_5, H_3 hold, then the solution multifunction $\widehat{\Gamma}_\beta : E \to P_k(L^p(T, H))$ is continuous and h-continuous.*

Remark 6.90 A function $\varphi : H \to \mathbb{R}$ is said to be

- "strongly convex", if for all $x, y \in H$, all $t \in [0, 1]$ we have

$$\frac{1}{2}(1-t)tc|x-y|^2 \leq (1-t)\varphi(x) + t\varphi(y) - \varphi((1-t)x + ty),$$

  with $c > 0$;
- "bounded convex", if for all $x, y \in H$, all $t \in [0, 1]$, we have

$$(1-t)\varphi(x) + t\varphi(y) - \varphi((1-t)x + ty) \leq \frac{1}{2}(1-t)t\widehat{c}\,|x-y|^2$$

  with $\widehat{c} > 0$.

If $\varphi : H \to \mathbb{R}$ is continuous, strongly convex and bounded convex, then $\varphi(\cdot)$ is Frechet differentiable and

$$c|x-y|^2 \leq (\varphi'(x) - \varphi'(y), x-y) \leq \widehat{c}\,|x-y|^2,$$

that is, the Frechet derivative is strongly monotone and Lipschitz continuous. Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with a C^2-boundary $\partial\Omega$ and consider an integrand $\widehat{\varphi} : \Omega \times \mathbb{R}^N \times E \to \mathbb{R}$ which satisfies

(i) for all $(x, \lambda) \in \mathbb{R}^N \times E$, $z \to \widehat{\varphi}(z, x, \lambda)$ is measurable;
(ii) for $a.a\ z \in \Omega$, all $\lambda \in E$, $x \to \widehat{\varphi}(z, x, \lambda)$ is continuous, strongly convex and bounded convex;
(iii) $c_1|x|^p - c_2 \leq \widehat{\varphi}(z, x, \lambda) \leq c_3(1 + |x|^p)$ for $a.a\ z \in \Omega$, all $x \in \mathbb{R}^N$, all $\lambda \in E$, with $c_1, c_2, c_3 > 0$, $2 \leq p < \infty$;
(iv) for $a.a\ z \in \Omega$, all $x \in \mathbb{R}^N$, $\lambda \to \widehat{\varphi}(z, x, \lambda)$ is continuous.

We set $a(z, x, \lambda) = \nabla_x \widehat{\varphi}(z, x, \lambda)$. We have the evolution triple

$$X = W_0^{1,p}(\Omega) \hookrightarrow H = L^2(\Omega) = H^* \hookrightarrow X^* = W^{-1,p'}(\Omega).$$

We introduce the operator $A : X \times E \to X^*$ defined by

$$\langle A(u, \lambda), h\rangle = \int_\Omega (a(z, Du, \lambda), Dh)_{\mathbb{R}^N} dz \text{ for all } u, h \in H_0^1(\Omega).$$

Let $\mathcal{B}$ be the family of open subsets of Ω and consider the integral functional $L : X \times \mathcal{B} \times E \to \mathbb{R}$ defined by

$$L(u, D, \lambda) = \int_D \widehat{\varphi}(z, Du, \lambda)dz.$$

Then we have

$$\Gamma_{seq}(W_0^{1,p}(\Omega)_w)L(u, D, \lambda_n) = L(u, D, \lambda).$$

This then implies that

$$A(\cdot, \lambda_n) \xrightarrow{G} A(\cdot, \lambda).$$

Remarks

6.1: Evolution triples were used by Gelfand and Shilov [198] and for this reason are also known as "Gelfand triples". They have been a basic tool in the study of evolution equations. We refer to the books of Gasinski and Papageorgiou [182], Hu and Papageorgiou [249], Lions [300], Showalter [440] and Zeidler [476], for further information on them. Theorem 6.2 can be found in Papageorgiou et al. [389] and it is a multivalued version of a result of Lions [300] (Theorem 1.2, p. 319). The Cauchy problem has been studied extensively for various types of evolution inclusions. We mention the works of Bin and Liu [47], Hu and Papageorgiou [249], Migorski et al. [336], Tolstonogov [457] and the references therein. Relaxation results (see Theorem 6.13), are of importance in control theory, since they suggest ways to economize in the use of control functions. More general relaxation theorems involving the extreme points of the orientor field $F(t, x)$ ("strong relaxation") can be found in Hu and Papageorgiou [249] (see Chaps. 1 and 2) and in Tolstonogov [456, 458]. Other works dealing with relaxation properties of evolution inclusions are those of Ahmed [1], DeBlasi and Pianigiani [125, 126], Fattorini [164], Hu and Papageorgiou [248, 249], Liu and Liu [307], Tolstonogov [455]. The periodic problem has been studied less extensively. We mention the works of Bader and Papageorgiou [24], Becker [38], Papageorgiou and Radulescu [370], Papageorgiou et al. [393], Papageorgiou and Yannakakis [383], Prüss [422], Vrabie [465]. Condition $H_7(iii)$ was first used by Hartman [231, 232] in the context of systems in $\mathbb{R}^N$. For antiperiodic solutions (see Theorem 6.19), we refer to the works of Aizicovici and Pavel [2], Chen [97], Gasinski and Papageorgiou [186], Haraux [229], Liu [302], Okochi [353]. For second order evolution inclusions (see problem (6.47)), we follow the works of Papageorgiou and Yannakakis [384, 385], see also Liu and Liu [306]. Systems like (6.67), with even more general nonlinear boundary conditions, were investigated by Frigon and Montoki [171], Gasinski and Papageorgiou [180], Halidias and Papageorgiou [224, 225], Kandilakis and Papageorgiou [266], Kyritsi et al. [284], Papageorgiou and Vetro [376], Papageorgiou et al. [395],

Zhang and Li [481, 482]. For multivalued Lienard and Duffing systems (see Problem 6.114), we refer to the papers of Gasinski and Papageorgiou [187], Kalita and Kowalski [265], Kowalski [277], Papageorgiou et al. [396].

6.2: Problems with nonlocal conditions (see problem (6.147)), provide a unified framework for many different kind of problems. Evolution inclusions with nonlocal conditions, were studied under more restrictive conditions on the map $A(t,\cdot)$ (uniform monotonicity) and on the nonlocal map $\beta(\cdot)$ (it is assumed to be linear). We mention the works of Cheng [98], Zhang et al. [483], Vrabie [466], Zhu and Li [486]. Here we remove these restrictions and we also prove a relaxation theorem (Theorem 6.69). Moreover, when the nonlocal map $\beta(\cdot)$ is linear, we show (Theorem 6.71) that we can consider nonmonotone maps $A(t,\cdot)$.

6.3: The approximation result (Proposition 6.76) is due to DeBlasi and Myjak [124]. In the proof of Theorem 6.77, we follow Hu and Papageorgiou [249] (Theorem I-3.5), who deal with a Cauchy problem. The investigation of the structure of the solution set of differential systems, started with the work of Kneser [272]. He proved that for systems in $\mathbb{R}^N$ with a continuous vector field, the reachable set (cross sections) at every time instant is a continuum of $\mathbb{R}^N$ (that is, it is a nonempty, compact and connected subset of $\mathbb{R}^N$). The result of Kneser, was extended by Hukuhara [252], who proved that the solution set itself is nonempty, compact and connected in $\mathbb{R}^N$. The R_δ-property of the solution set was proved by Aronszajn [19]. The extension to differential inclusions in $\mathbb{R}^N$ is due to Lasry and Robert [287] who proved acyclicity and compactness in $C(T,\mathbb{R}^N)$ of the solution set and to Himmelberg and Van Vleck [245] who proved the R_δ-property. For the Cauchy problem of evolution inclusions, we mention the works of DeBlasi et al. [127], Papageorgiou [360], Papageorgiou and Papalini [362].

6.4: Theorems 6.87, 6.88, and 6.89, extend and improve the results of Hu and Papageorgiou [249] (Section 1.6). Analogous results for subdifferential evolution inclusions, can be found in Hu and Papageorgiou [249] (Sect. 2.3).

Chapter 7
Calculus Of Variations



Introduction

In this chapter, we study variational problems. In Sect. 7.1, we deal with calculus of variations problems. We prove lower semicontinuity results for integral functionals. This also leads to a brief presentation of the theory of Young measures. We also develop the Euler-Lagrange equation and discuss the Lavrentiev gap phenomenon. In Sect. 7.2, we deal with optimal control problems involving maximal monotone operators in their dynamics(unilateral constraints). We present the existence theory, which via the reduction method relates optimal control and calculus of variations. For the existence theory to work, we need convexity properties. If these properties are not satisfied, we need to augment the problem appropriately (both the dynamics and the cost functional). This leads to the so-called "relaxed problem". We present two different relaxation methods, both leading to an admissible relaxed problem. Finally in Sect. 7.3, we study the stability of the optimal control problem when we perturb its data. We relate the stability properties with the relaxability of the system.

7.1 Calculus of Variations

In this section, we present some basic aspects from the theory of the "Calculus of Variations". The subject is as old as calculus itself and is closely connected with Functional Analysis and Optimal Control (see Sect. 7.2).

We will start with lower semicontinuity results for integral functionals. Such results are in the heart of the so-called "Direct Method". First we recall a notion from integration theory. Our definition here is more restrictive than the one given in Volume I(see Definition I-2.144). It is more convenient for our arguments in this section.

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Definition 7.1 Let (Ω, Σ, μ) be a finite measure space and $C \subseteq L^1(\Omega, \mathbb{R}^N)$. We say that C is "uniformly integrable" if

- C is L^1-bounded;
- for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $A \in \Sigma$, $\mu(A) \leq \delta \Rightarrow \sup_{u \in C} \int_A |u| d\mu \leq \varepsilon$.

Remark 7.2 In fact the above definition is equivalent to saying that

$$\lim_{\lambda \to \infty} \sup_{u \in C} \int_{\{|u| > \lambda\}} |u| d\mu = 0$$

or that there exists a function $k : \mathbb{R}_+ \to \mathbb{R}_+$(it can be taken to be convex and increasing) such that

$$\lim_{t \to +\infty} \frac{k(t)}{t} = +\infty \text{ and } \sup_{u \in C} \int_\Omega k(u) d\mu < \infty$$

("De la Vallee Poussin criterion"). It is easily seen that if $C \subseteq L^1(\Omega, \mathbb{R}^N)$ is compact, then C is uniformly integrable. The converse is not true. In this direction, we recall the "Dunford-Pettis Theorem" (see Theorem I-4.26).

Theorem 7.3 *If (Ω, Σ, μ) is a finite measure space and $C \subseteq L^1(\Omega, \mathbb{R}^N)$, then C is relatively w-compact if and only if it is uniformly integrable.*

Also we recall some other definitions from Volume I.

Definition 7.4 Let (Ω, Σ, μ) be a finite measure space and $\varphi : \Omega \times \mathbb{R}^N \to (-\infty, +\infty]$.

(a) We say that $\varphi(\omega, x)$ is a "Caratheodory integrand", if for all $x \in \mathbb{R}^N$, $\omega \to \varphi(\omega, x)$ is Σ-measurable and for $\mu-$a.a $\omega \in \Omega$, $x \to \varphi(\omega, x)$ is finite and continuous.

(b) We say that $\varphi(\omega, x)$ is a "normal integrand", if $(\omega, x) \to \varphi(\omega, x)$ is $\Sigma \otimes B(\mathbb{R}^N)$-measurable and for $\mu-$a.a $\omega \in \Omega$, $x \to \varphi(\omega, x)$ is lower semicontinuous.

(c) We say that $\varphi(\omega, x)$ is a "convex integrand", if $(\omega, x) \to \varphi(\omega, x)$ is $\Sigma \otimes B(\mathbb{R}^N)$-measurable and for $\mu-$a.a $\omega \in \Omega$, $x \to \varphi(\omega, x)$ is convex.

(d) We say that $\varphi(\omega, x)$ is a "convex normal integrand", if it is both a normal integrand and a convex integrand.

(e) If $\varphi : \Omega \times \mathbb{R}^N \times \mathbb{R}^m \to (-\infty, \infty]$, then we say that $\varphi(\omega, x, y)$ is a "convex normal integrand", if it is $\Sigma \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^m)$-measurable and there exists $N \subseteq \Omega$ μ-null set such that for all $\omega \in \Omega \setminus N$,

$$(x, y) \to \varphi(\omega, x, y) \text{ is lower semicontinuous,}$$

$$y \to \varphi(\omega, x, y) \text{ is convex for all } x \in \mathbb{R}^N.$$

Remark 7.5 We know that a Caratheodory integrand is jointly measurable (see Theorem I-2.47). This property fails if $\varphi(\omega, \cdot)$ is only semicontinuous (lower or upper). That is why in the definition of normal integrand, we require that $\varphi(\cdot, \cdot)$ is jointly measurable. Also we know that, if $\varphi(\cdot, \cdot)$ is a normal integrand which dominates a Caratheodory function ϑ(that is, $\vartheta(\omega, x) \leq \varphi(\omega, x)$ for all $\omega \in \Omega \setminus N$, with $\mu(N) = 0$, all $x \in \mathbb{R}^N$), then we can find an increasing sequence of Caratheodory integrands $\{\varphi_n\}_{n\in\mathbb{N}}$ such that $\varphi_n(\omega, x) \uparrow \varphi(\omega, x)$ for μ-a.a $\omega \in \Omega$, all $x \in \mathbb{R}^N$.

Proposition 7.6 *If (Ω, Σ, μ) is a finite, nonatomic measure space, $\varphi : \Omega \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a convex normal integrand, $1 \leq p < \infty$ and the integral functional $I_\varphi : L^p(\Omega, \mathbb{R}^N) \to \overline{\mathbb{R}}$ defined by*

$$I_\varphi(u) = \int_\Omega \varphi(\omega, u(\omega))d\mu \text{ for all } u \in L^p(\Omega, \mathbb{R}^N)$$

is continuous at a point, then $I_\varphi(\cdot)$ is continuous everywhere.

Proof Let $u_0 \in L^p(\Omega, \mathbb{R}^N)$ the point of continuity of $I_\varphi(\cdot)$. By considering if necessary the functional $u \to \widehat{I_\varphi}(u) = I_\varphi(u_0 + u) - I_\varphi(u)$, then without any loss of generality, we may assume that $u_0 = 0$ and $I_\varphi(0) = 0$. We can find $\delta > 0$ such that

$$\|u\|_p \leq \delta \Rightarrow I_\varphi(u) \leq 1.$$

Now let $v \in L^p(\Omega, \mathbb{R}^N)$. The nonatomicity of $\mu(\cdot)$ and the absolute continuity of the Lebesgue integral, imply that we can find $\delta_1 > 0$ and pairwise disjoint sets $\{A_k\}_{k=1}^m \subseteq \Sigma$ such that

$$\mu(A_k) \leq \delta_1 \text{ and } \|\chi_{A_k} v\|_p \leq \delta, k \in \{1, \cdots, m\}.$$

We have

$$\begin{aligned}\int_{A_k} \varphi(\omega, v(\omega))d\mu &= \int_\Omega \varphi(\omega, \chi_{A_k} v(\omega))d\mu - \int_{A_k^c} \varphi(\omega, 0)d\mu \\ &\leq 1 + \beta\end{aligned}$$

for some $\beta > 0$ independent of $k \in \{1, \cdots, m\}$. Then

$$\sum_{k=1}^m \int_{A_k} \varphi(\omega, v(\omega))d\mu = \int_\Omega \varphi(\omega, v(\omega))d\mu = I_\varphi(v) < \infty$$

$\Rightarrow I_\varphi(\cdot)$ is continuous everywhere (see Theorem $I - 6.72$).

□

For lower semicontinuous, convex functions, we have the following approximation result by affine functions. Recall that for $\overline{\mathbb{R}}$-valued functions we exclude the case that the function is identically $+\infty$.

Proposition 7.7 *If $\varphi : \mathbb{R}^m \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a lower semicontinuous function, for all $x \in \mathbb{R}^m$, $\varphi(x, \cdot)$ is convex and it has one of the following two properties*

(i) there exists $u_0 \in C(\mathbb{R}^m, \mathbb{R}^N)$ such that $x \to \varphi(x, u_0(x))$ is continuous and $\mathbb{R}$-valued;
(ii) there exists a function $\vartheta : \mathbb{R}_+ \to \mathbb{R}$ such that

$$\lim_{t \to +\infty} \frac{\vartheta(t)}{t} = +\infty, \varphi(x, y) \geq \vartheta(|y|) \text{ for all } (x, y) \in \mathbb{R}^m \times \mathbb{R}^N,$$

then there exist continuous functions

$$c_n : \mathbb{R}^m \to \mathbb{R}, \quad b_n : \mathbb{R}^m \to \mathbb{R}^N, \quad n \in \mathbb{N}$$

such that $\sup_{n \in \mathbb{N}} [c_n(x) + (b_n(x), y)_{\mathbb{R}^N}] = \varphi(x, y)$ for all $(x, y) \in \mathbb{R}^m \times \mathbb{R}^N$.

Proof First we assume that condition (i) is satisfied.

If we replace φ by $\widehat{\varphi}(x, y) = \varphi(x, y + u_0(x)) - \varphi(x, u_0(x))$, we see that without any loss of generality, we may assume that $\varphi(x, 0) = 0$ for all $x \in \mathbb{R}^m$. Consider the multifunction $L : \mathbb{R}^m \to 2^{\mathbb{R} \times \mathbb{R}^N}$ defined by

$$L(x) = \{(c, b) \in \mathbb{R} \times \mathbb{R}^N : \varphi(x, y) \geq c + (b, y)_{\mathbb{R}^N} \text{ for all } y \in \mathbb{R}^N\}.$$

Since $\varphi(x, \cdot)$ is lower semicontinuous, convex, by Proposition I-6.90, we see that for all $x \in \mathbb{R}^m$, $L(x) \neq \emptyset$ and clearly it is closed and convex.

Claim $L(\cdot)$ is lsc.

Let $U \subseteq \mathbb{R} \times \mathbb{R}^N$ open and consider

$$L^-(U) = \{x \in \mathbb{R}^m : L(x) \cap U \neq \emptyset\}.$$

Assume that $L^-(U) \neq \emptyset$ and let $x_0 \in L^-(U)$ and $(c_0, b_0) \in L(x_0) \cap U$. We can find $\varepsilon > 0$ such that

$$\{(c, b) \in \mathbb{R} \times \mathbb{R}^N : |c - c_0| + |b - b_0| \leq 2\varepsilon\} \subseteq U.$$

We will show that there is a neighborhood V of $x_0 \in \mathbb{R}^m$ such that

$$c_0 + (b_0, y)_{\mathbb{R}^N} - \varepsilon(1 + |y|) \leq \varphi(x, y) \text{ for all } (x, y) \in \mathbb{R}^m \times \mathbb{R}^N. \tag{7.1}$$

Arguing by contradiction, suppose that (7.1) is not true. Then we can find $x_n \to x_0$ in $\mathbb{R}^m$ and $\{y_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^N$ such that

$$\varphi(x_n, y_n) < c_0 + (b_0, y_n)_{\mathbb{R}^N} - \varepsilon(1 + |y_n|) \text{ for all } n \in \mathbb{N}. \tag{7.2}$$

First assume that $\{y_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^N$ is bounded. Then we can say that $y_n \to y_0$ in $\mathbb{R}^N$. From (7.2) and the lower semicontinuity of $\varphi(\cdot,\cdot)$, we have

$$\varphi(x_0, y_0) \leq c_0 + (b_0, y_0)_{\mathbb{R}^N} - \varepsilon(1 + |y_0|).$$

But this contradicts the fact that $(c_0, b_0) \in L(x_0)$.

Now assume that $\{y_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^N$ is not bounded. Then we can say that

$$|y_n| \to \infty.$$

Set $v_n = \dfrac{y_n}{|y_n|}$, $n \in \mathbb{N}$. Since $|v_n| = 1$, we may assume that

$$v_n \to v_0 \text{ in } \mathbb{R}^N, \quad |v| = 1.$$

Then recalling that we have $\varphi(x, 0) = 0$ for all $x \in \mathbb{R}^m$ and $\varphi(x,\cdot)$ is convex, for all $t > 0$ and $n \in \mathbb{N}$ large, we have

$$\begin{aligned}
\varphi(x_n, tv_n) &\leq \frac{t}{|y_n|}\varphi(x_n, y_n) \\
&\leq \frac{t}{|y_n|}[c_0 + (b_0, y_n)_{\mathbb{R}^N} - \varepsilon(1 + |y_n|)], \ n \in \mathbb{N} \\
\Rightarrow \varphi(x_0, tv_0) &\leq (b_0, tv_0)_{\mathbb{R}^N} - \varepsilon t.
\end{aligned}$$

If $t > 0$ is sufficiently large, then we contradict the fact that $(c_0, b_0) \in L(x_0)$. This proves (7.1).

Now let $x \in V$ and $(c, b) \in \mathbb{R} \times \mathbb{R}^N$ be such that $|c|, |b| \leq \varepsilon$ and

$$\begin{aligned}
&\varphi(x, y) - c_0 - (b_0, y)_{\mathbb{R}^N} \geq c + (b, y)_{\mathbb{R}^N} \geq -\varepsilon(1 + |y|) \text{ for all } y \in \mathbb{R}^N, \\
&\Rightarrow \varphi(x, y) \geq (c + c_0) + (b + b_0, y)_{\mathbb{R}^N} \text{ for all } y \in \mathbb{R}^N, \\
&\Rightarrow (c + c_0, b + b_0) \in L(x).
\end{aligned}$$

Also $|c| + |b| \leq 2\varepsilon$ and so $(c + c_0, b + b_0) \in U$. We conclude that $L^-(U) \subseteq \mathbb{R}^m$ is open and so $L(\cdot)$ is lsc. This proves the Claim.

On account of the Claim and of Theorem I-5.89 we can find $\{c_n\}_{n\in\mathbb{N}} \subseteq C(\mathbb{R}^m, \mathbb{R})$ and $\{b_n\}_{n\in\mathbb{N}} \subseteq C(\mathbb{R}^m, \mathbb{R}^N)$ such that

$$\varphi(x, y) = \sup_{n\in\mathbb{N}}(c_n(x) + (b_n(x), y)_{\mathbb{R}^N})$$

(since $\varphi(\cdot, \cdot)$ is by hypothesis lower semicontinuous). This proves the proposition when (i) holds.

Next suppose that (ii) holds. For every $n \in \mathbb{N}$, let

$$\varphi_n(x, y) = \inf\{\varphi(x', y) + n|x' - x| : x' \in \mathbb{R}^m\}. \tag{7.3}$$

We have that

(a) $\{\varphi_n\}_{n\in\mathbb{N}}$ is increasing and $\sup_{n\in\mathbb{N}} \varphi_n = \varphi$.
(b) $\varphi_n(x, y) \le \varphi_n(x', y) + n|x' - x|$ for all $n \in \mathbb{N}$, all $x, x' \in \mathbb{R}^m$, all $y \in \mathbb{R}^N$.
(c) For all $n \in \mathbb{N}$, all $x \in \mathbb{R}^m$, $\varphi_n(x, \cdot)$ is lower semicontinuous on $\mathbb{R}^N$.

For (a), (b), refer to the proof of Proposition I-5.154. For (c) we argue as follows. We fix $n \in \mathbb{N}$, $x \in \mathbb{R}^m$, $y \in \mathbb{R}^N$, $y_k \to y$ with $\lim_{k\to\infty} \varphi_n(x, y_k) < \infty$. From (7.3), we see that we can find $\{x_k\}_{k\in\mathbb{N}} \subseteq \mathbb{R}^m$ such that

$$\varphi(x_k, y_k) + n|x_k - x| \le \varphi_n(x, y_k) + \frac{1}{2^k} \text{ for all } k \in \mathbb{N}. \tag{7.4}$$

Evidently $\{x_k\}_{k\in\mathbb{N}} \subseteq \mathbb{R}^m$ is bounded and so we may assume that $x_k \to x'$ in $\mathbb{R}^m$. From (7.4) and the lower semicontinuity of $\varphi(\cdot, \cdot)$, we have

$$\begin{aligned} &\varphi(x, y) + n|x' - x| \le \liminf_{k\to\infty} \varphi_n(x, y_k), \\ \Rightarrow &\varphi_n(x, y) \le \liminf_{k\to+\infty} \varphi_n(x, y_k), \\ \Rightarrow &\varphi_n(x, \cdot) \text{ is lower semicontinuous.} \end{aligned}$$

Then for every $x \in \mathbb{R}^m$, we have

$$\varphi(x, \cdot) = \sup_{n\in\mathbb{N}}\{\varphi_n^{**}(x, \cdot)\} \text{ (see Theorem } I - 6.104). \tag{7.5}$$

From the previous case, for every $n \in \mathbb{N}$, we can find

$$\{c_{n,k}\}_{k\in\mathbb{N}} \subseteq C(\mathbb{R}^m, \mathbb{R}), \quad \{b_{n,k}\}_{k\in\mathbb{N}} \subseteq C(\mathbb{R}^m, \mathbb{R}^N)$$

such that

$$\varphi_n^{**}(x, y) = \sup_{k\in\mathbb{N}}\{c_{k,n}(x) + (b_{k,n}(x), y)_{\mathbb{R}^N}\}.$$

Using (7.5), we conclude the proof of the proposition. □

We can extend this approximation result to nonautonomous integrands. So, let (Ω, Σ, μ) be a finite, complete measure space and $\varphi : \Omega \times \mathbb{R}^m \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ an integrand which has the following properties.

(a) $\varphi(\cdot, \cdot, \cdot)$ is $\Sigma \otimes B(\mathbb{R}^m) \otimes B(\mathbb{R}^N)$-measurable;
(b) for all $\omega \in \Omega$, $\varphi(\omega, \cdot, \cdot)$ is lower semicontinuous;
(c) for all $\omega \in \Omega$ and all $x \in \mathbb{R}^m$, $\varphi(\omega, x, \cdot)$ is convex.

Proposition 7.8 *If $\varphi : \Omega \times \mathbb{R}^m \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is an integrand which satisfies (a), (b), (c) above and for all $\omega \in \Omega$, $\varphi(\omega, \cdot, \cdot)$ satisfies condition (i) or (ii) from Proposition 7.7, then we can find two sequences*

$$c_n : \Omega \times \mathbb{R}^m \to \mathbb{R} \text{ and } b_n : \Omega \times \mathbb{R}^m \to \mathbb{R}^N, \quad n \in \mathbb{N}$$

of Caratheodory integrands such that

$$\varphi(\omega, x, y) = \sup_{n \in \mathbb{N}} (c_n(\omega, x) + (b_n(\omega, x), y)_{\mathbb{R}^N})$$

for all $\omega \in \Omega$ and all $(x, y) \in \mathbb{R}^m \times \mathbb{R}^N$.

Proof Let $W = C(\mathbb{R}^m, \mathbb{R} \times \mathbb{R}^N)$ equipped with the topology of uniform convergence on compact subsets. Then W is a separable Frechet space (see Definition I-3.3). If $u \in W$, then $u = (u_1, u_2)$, with $u_1 \in C(\mathbb{R}^m, \mathbb{R})$, $u_2 \in C(\mathbb{R}^m, \mathbb{R}^N)$. We define

$$k_u(x, y) = u_1(x) + (u_2(x), y)_{\mathbb{R}^N} \text{ for all } (x, y) \in \mathbb{R}^m \times \mathbb{R}^N.$$

For every $V \subseteq \mathbb{R}^m \times \mathbb{R}^N$ open, we set

$$\widehat{k}_V(u) = \inf\{k_u(x, y) : (x, y) \in V\},$$
$$\widehat{\varphi}_V(\omega) = \inf\{\varphi(\omega, x, y) : (x, y) \in V\},$$
$$S(\omega) = \{u \in W : \varphi(\omega, x, y) \geq k_u(x, y) \text{ for all } (x, y) \in \mathbb{R}^m \times \mathbb{R}^N\}.$$

From Proposition I-1.182 we know that $\widehat{k}_V(\cdot)$ is upper semicontinuous on W. Let $\lambda \in \mathbb{R}$ and consider the strict λ-sublevel set of $\widehat{\varphi}_V(\cdot)$, namely the set

$$\begin{aligned}
\mathring{L}_\lambda &= \{\omega \in \Omega : \widehat{\varphi}_V(\omega) < \lambda\} \\
&= \{\omega \in \Omega : \varphi(\omega, x, y) < \lambda \text{ for some } (x, y) \in V\} \\
&= \text{proj}_\Omega\{(\omega, x, y) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^N : \varphi(\omega, x, y) < \lambda, (x, y) \in V\} \\
&\in \Sigma \text{ (see Theorem } I - 5.72), \\
&\Rightarrow \widehat{\varphi}_V(\cdot) \text{ is } \Sigma - \text{measurable.}
\end{aligned}$$

Consider $\mathcal{B}$ a countable base of open sets in $\mathbb{R}^m \times \mathbb{R}^N$. Then

$$\begin{aligned}\operatorname{Gr} S &= \{(\omega, u) \in \Omega \times W : u \in S(\omega)\} \\ &= \bigcap_{V \in \mathcal{B}} \{(\omega, u) \in \Omega \times W : \widehat{k}_V(u) \leq \widehat{\varphi}_V(\omega)\} \\ &\in \Sigma \otimes B(W), \\ \Rightarrow\ & S(\cdot) \text{ is graph measurable.}\end{aligned}$$

Then invoking Theorem I-5.107, we can find a sequence of functions $u_n : \Omega \to W$, $n \in \mathbb{N}$ which are Σ-measurable such that

$$S(\omega) = \overline{\{u_n(\omega)\}}_{n \in \mathbb{N}} \text{ for all } \omega \in \Omega. \tag{7.6}$$

We define

$$c_n(\omega, x) = u_n^1(\omega)(x), \quad b_n(\omega, x) = u_n^2(\omega)(x).$$

Evidently these are Caratheodory functions and by (7.6) and Proposition 7.7, we have

$$\varphi(\omega, x, y) = \sup_{n \in \mathbb{N}} (c_n(\omega, x) + (b_n(\omega, x), y)_{\mathbb{R}^N})$$

for all $\omega \in \Omega$ and all $(x, y) \in \mathbb{R}^m \times \mathbb{R}^N$. □

Remark 7.9 Propositions 7.7 and 7.8 remain valid if $\mathbb{R}^m$ is replaced by a general metric space X and $\mathbb{R}^N$ is replaced by a separable reflexive Banach space Y equipped with the weak topology. We know that Y topologized this way, is a Souslin space (see Remark I-1.163). Also if $\varphi \geq 0$, then $\varphi(\omega, x, y) = \sup_{n \in \mathbb{N}}((c_n(\omega, x) + (b_n(\omega, x), y)_{\mathbb{R}^N})^+)$.

For the first lower semicontinuity result of integral functionals, we will need the following lemma from integration theory.

Lemma 7.10 *If $f_n, f : \Omega \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$, $n \in \mathbb{N}$ are Σ-measurable functions, $f = \sup_{n \in \mathbb{N}} f_n$ and $\beta \leq f_n$ with $\beta \in L^1(\Omega)$, then $\int_\Omega f d\mu = \sup\{\sum_{i \in I} \int_{A_i} f_i d\mu : \{A_i\}_{i \in I} \in \mathcal{S}\}$ where $\mathcal{S} =$ finite partitions of Ω by pairwise disjoint sets; also for every $f \in L^1(\Omega)$*

$$\int_\Omega f^+ d\mu = \sup\{\int_A f d\mu : A \in \Sigma\}.$$

Proof Clearly we always have

$$\sup\{\sum_{i\in I}\int_{A_i} f_i d\mu : \{A_i\}_{i\in I} \in \mathcal{S}\} \le \int_{\Omega} f d\mu.$$

We set $h_n = \max\limits_{1\le i\le n} f_i$ and have $h_n \uparrow f$. Then by the monotone convergence theorem, we infer that

$$\int_{\Omega} f d\mu = \sup_{n\in\mathbb{N}} \int_{\Omega} h_n d\mu.$$

Let $A_i \in \Sigma$, $i \in \{1, \cdots, n\}$ mutually disjoint sets such that

$$\Omega = \bigcup_{i=1}^{n} A_i, \quad h_i = f_i \text{ on } A_i.$$

Therefore we have

$$\int_{\Omega} h_n d\mu = \sum_{i=1}^{n} \int_{A_i} f_i d\mu,$$

$$\Rightarrow \int_{\Omega} f d\mu = \sup\{\sum_{i\in I}\int_{A_i} f_i d\mu : \{A_i\}_{i\in I} \in \mathcal{S}\}.$$

If $f \in L^1(\Omega)$, then for $f^+ = \max\{f, 0\}$, we have

$$\int_{\Omega} f^+ d\mu = \sup\{\int_A f d\mu : A \in \Sigma\}.$$

□

Now we are ready for the first lower semicontinuity result for the integral functional

$$(u, v) \to I_{\varphi}(u, v) = \int_{\Omega} \varphi(\omega, u(\omega), v(\omega)) d\mu.$$

Theorem 7.11 *If $\varphi : \Omega \times \mathbb{R}^m \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is an integrand which satisfies properties (a), (b), (c) stated before Proposition 7.8, $1 \le p, \tau \le \infty$ and*

[D] "for every $\{u_n\}_{n\in\mathbb{N}} \subseteq L^p(\Omega, \mathbb{R}^m)$, $\{v_n\}_{n\in\mathbb{N}} \subseteq L^{\tau}(\Omega, \mathbb{R}^N)$, such that

$$u_n \to u \text{ in } L^p(\Omega, \mathbb{R}^m),$$

$$v_n \xrightarrow{w} v \text{ in } L^{\tau}(\Omega, \mathbb{R}^N) (w^* \text{ if } \tau = +\infty)$$

and

$$\int_\Omega \varphi^+(\omega, u_n, v_n)d\mu \leq \beta + \int_\Omega \varphi^-(\omega, u_n, v_n)d\mu$$

for all $n \in \mathbb{N}$, *some* $\beta > 0$, *we have that*

$$\{\varphi^-(\cdot, u_n(\cdot), v_n(\cdot))\}_{n\in\mathbb{N}} \subseteq L^1(\Omega) \text{ is } w - compact'',$$

then $I_\varphi(\cdot,\cdot)$ *is well defined on* $L^p(\Omega,\mathbb{R}^m) \times L^\tau(\Omega,\mathbb{R}^N)$ *with values in* $\overline{\mathbb{R}}$ *and it is sequentially lower semicontinuous on* $L^p(\Omega,\mathbb{R}^m) \times L^\tau(\Omega,\mathbb{R}^N)_w(w^*$ *if* $\tau = +\infty)$.

Proof That $I_\varphi(\cdot,\cdot)$ is well-defined with values in $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$, follows from condition [D]. We need to prove the lower semicontinuity.
Step 1 We prove the result under the following additional hypothesis on the integrand φ

$$\text{“}\vartheta(|y|) \leq \varphi(\omega, x, y) \text{ for all } \omega \in \Omega, \text{ all } (x, y) \in \mathbb{R}^m \times \mathbb{R}^N,$$

$$\text{with } \vartheta : \mathbb{R}_+ \to \mathbb{R}_+ \text{ such that } \lim_{t\to+\infty} \frac{\vartheta(t)}{t} = +\infty''.$$

In this case, since $\varphi \geq 0$, using Proposition 7.8 (see also Remark 7.9), we have

$$\varphi(\omega, x, y) = \sup_{n\in\mathbb{N}}((c_n(\omega, x) + (b_n(\omega, x), y)_{\mathbb{R}^N})^+)$$

with $c_n : \Omega \times \mathbb{R}^m \to \mathbb{R}$, $b_n : \Omega \times \mathbb{R}^m \to \mathbb{R}^N$, $n \in \mathbb{N}$, Caratheodory functions. Then using Lemma 7.10, it is enough to show that for every $A \in \Sigma$ and every $n \in \mathbb{N}$, the integral functional

$$(u, v) \to \int_A (c_n(\omega, u) + (b_n(\omega, u), v)_{\mathbb{R}^N})dz$$

is sequentially lower semicontinuous on $L^p(\Omega,\mathbb{R}^m) \times L^\tau(\Omega,\mathbb{R}^N)_w(w^*$ if $\tau = +\infty)$. This is immediate from the continuity of $c_n(\omega,\cdot)$ and of $b_n(\omega,\cdot)$.
Step 2 We prove the result under the additional hypothesis that φ is bounded below.

We may assume that $\varphi \geq 0$. Consider two sequences $\{u_n\}_{n\in\mathbb{N}} \subseteq L^p(\Omega,\mathbb{R}^m)$ and $\{v_n\}_{n\in\mathbb{N}} \subseteq L^\tau(\Omega,\mathbb{R}^N)$ such that

$$u_n \to u \text{ in } L^p(\Omega,\mathbb{R}^m), \quad v_n \xrightarrow{w} v \text{ in } L^\tau(\Omega,\mathbb{R}^N)(\omega^* \text{ if } \tau = +\infty).$$

We have that $\{v_n\}_{n\in\mathbb{N}}$ is uniformly integrable (see Theorem 7.3) and so by the De la Vallée Poussin criterion (see Remark 7.2), we can find a function $\vartheta : \mathbb{R}_+ \to \mathbb{R}_+$ such that

$$\lim_{t\to+\infty}\frac{\vartheta(t)}{t}=+\infty \text{ and } \int_\Omega \vartheta(|v_n|)d\mu\le 1 \text{ for all } n\in\mathbb{N}. \tag{7.7}$$

For every $\varepsilon>0$, let

$$\varphi_\varepsilon(\omega,x,y)=\varphi(\omega,x,y)+\varepsilon\vartheta(|y|).$$

Using Step 1, we have

$$\begin{aligned}\int_\Omega \varphi(\omega,u,v)d\mu &\le \int_\Omega \varphi_\varepsilon(\omega,u,v)d\mu\\ &\le \liminf_{n\to\infty}\int_\Omega \varphi_\varepsilon(\omega,u_n,v_n)d\mu\\ &\le \varepsilon+\liminf_{n\to\infty}\int_\Omega \varphi(\omega,u_n,v_n)d\mu \text{ (see (7.7))}.\end{aligned}$$

Recall that $\varepsilon>0$ is arbitrary. So, letting $\varepsilon\to 0^+$, we obtain

$$\int_\Omega \varphi(\omega,u,v)d\mu\le\liminf_{n\to\infty}\int_\Omega \varphi(\omega,u_n,v_n)d\mu.$$

Step 3 The general case.

Again we consider two sequences $\{u_n\}_{n\in\mathbb{N}}\subseteq L^p(\Omega,\mathbb{R}^m)$ and $\{v_n\}_{n\in\mathbb{N}}\subseteq L^\tau(\Omega,\mathbb{R}^N)$ such that

$$u_n\to u \text{ in } L^p(\Omega,\mathbb{R}^m),\quad v_n\xrightarrow{w} v \text{ in } L^\tau(\Omega,\mathbb{R}^N)\ (w^* \text{ if } \tau=\infty).$$

For $n,k\in\mathbb{N}$, we introduce the following items

$$\varphi_k=\max\{\varphi,-k\},\quad I_{\varphi_k}(u,v)=\int_\Omega \varphi_k(\omega,u,v)d\mu,$$
$$h_n(\cdot)=\varphi^-(\cdot,u_n(\cdot),v_n(\cdot)),\quad A_{n,k}=\{\omega\in\Omega: h_n(\omega)>k\}.$$

We may assume that $\lim_{n\to\infty} I_\varphi(u_n,v_n)$ is finite and then by condition [D], we have that

$$\{h_n\}_{n\in\mathbb{N}}\subseteq L^1(\Omega) \text{ is } w-\text{compact}.$$

Then by Step 2, for every $k\in\mathbb{N}$, we have

$$\begin{aligned} I_\varphi(u,v) &\le I_{\varphi_k}(u,v) \\ &\le \liminf_{n\to\infty} I_{\varphi_k}(u_n,v_n) \\ &= \liminf_{n\to\infty}[I_\varphi(u_n,v_n) + \int_{A_{n,k}} (h_n - k)d\mu] \\ &\le \liminf_{n\to\infty} I_\varphi(u_n,v_n) + \limsup_{n\to\infty} \int_{A_{n,k}} h_n d\mu. \end{aligned}$$

Let $k \to \infty$ and recall that $\{h_n\}_{n\in\mathbb{N}} \subseteq L^1(\Omega)$ are uniformly integrable. Hence

$$I_\varphi(u,v) \le \liminf_{n\to\infty} I_\varphi(u_n,v_n),$$

$\Rightarrow I_\varphi(\cdot,\cdot)$ is sequentially lower semicontinuous on $L^p(\Omega,\mathbb{R}^m) \times L^\tau(\Omega,\mathbb{R}^N)$.

□

Now we will study the necessary conditions for the lower semicontinuity of the integral functional I_φ. We start with a measure theoretic result.

Proposition 7.12 *If (Ω, Σ, μ) is a finite measure space with Σ countably generated (see Definition I-2.168), then μ is nonatomic if and only if for every $A \in \Sigma$ and $\eta \in [0,1]$, there exists a sequence $\{A_n\}_{n\in\mathbb{N}} \subseteq \Sigma$ with $A_n \subseteq A$ for all $n \in \mathbb{N}$ such that*

$$\chi_{A_n} \xrightarrow{w^*} \eta\chi_A \text{ in } L^\infty(\Omega).$$

Proof $\Leftarrow$ Consider $A \in \Sigma$ with $\mu(A) > 0$. By hypothesis we can find $\{A_n\}_{n\in\mathbb{N}} \subseteq \Sigma$ with $A_n \subseteq A$ for all $n \in \mathbb{N}$ such that

$$\begin{aligned} &\chi_{A_n} \xrightarrow{w} \frac{1}{2}\chi_A \text{ in } L^\infty(\Omega), \\ \Rightarrow &\mu(A_n) \to \frac{1}{2}\mu(A), \\ \Rightarrow &0 < \mu(A_n) < \mu(A) \text{ for all } n \ge n_0. \end{aligned}$$

This implies that $\mu(\cdot)$ is nonatomic (see Definition I-2.165).

$\Rightarrow$ Since Σ is countably generated, we can find $\{C_n\}_{n\in\mathbb{N}}$ such that $\Sigma = \sigma(\{C_n\})$. Let $\mathfrak{a}$ be the algebra generated by $\{C_n\}$. Since Σ is countably generated, we have that $\mathfrak{a} = \{D_n\}_{n\in\mathbb{N}}$. For $k \in \mathbb{N}$, let $\mathcal{S}^k$ be the sets of the form $F = \{i_j\}_{j=1}^n \subseteq \mathbb{N}^n$, $1 \le n \le k$ and $1 \le i_1 < \cdots < i_n \le k$. For every $F \in \mathcal{S}^k$, we define

$$D_F^k = (\bigcap_{i\in F} D_i) \setminus (\bigcup_{i\notin F} D_i).$$

These sets are in $\mathfrak{a}$ and are pairwise disjoint.
To simplify the notation, we write

$$\{D_I^k : I \in \mathcal{S}^k\} = \{E_i^k : i = 1, \cdots, l_k\}.$$

We can find measurable functions $u_i^k : E_i^k \to [0, 1)$ such that

$$\mu((u_i^k)^{-1}(B)) = |B|\mu(E_i^k),$$

with $|\cdot|$ denoting the Lebesgue measure on $\mathbb{R}$. We set

$$u_k = \sum_{i=1}^{l_k} \chi_{E_i^k} u_i^k.$$

We construct a sequence $\{B_n\}_{n\in\mathbb{N}} \in B([0, 1])$ such that

$$\chi_{B_n} \xrightarrow{w^*} \vartheta \text{ in } L^\infty_{\text{loc}}(\mathbb{R}).$$

We define $L_{n,k} = u_k^{-1}(B_n)$. For every $k \in \mathbb{N}$, $1 \le n \le k$, we have

$$\begin{aligned}\int_{E_i^k} \chi_{L_{n,k}} d\mu &= \mu(L_{n,k} \cap E_i^k) \\ &= \mu(u_k^{-1}(B_n) \cap E_i^k) \\ &= \mu((u_i^k)^{-1}(B_n) \cap E_i^k) \\ &= \mu((u_i^k)^{-1}(B_n)) \\ &= |B_n|\mu(E_i^k) \to \eta\mu(E_i^k) \text{ as } n \to \infty.\end{aligned}$$

Then for every $k \in \mathbb{N}$, $1 \le i \le k$, we have

$$\int_{D_i} \chi_{L_{n,k}} d\mu \to \eta\mu(D_i) \text{ as } n \to \infty.$$

Therefore for every $k \in \mathbb{N}$, we can find $n_k \in \mathbb{N}$ large such that

$$\left| \int_{D_i} \chi_{L_{n,k}} d\mu - \eta\mu(D_i) \right| \le \frac{1}{k} \text{ for all } 1 \le i \le k.$$

Then the set $A_k = L_{n_k,k}$ have the desired properties. □

Now we can have the necessary conditions for the lower semicontinuity of the integral functionals.

Theorem 7.13 *If (Ω, Σ, μ) is a finite measure space with Σ countably generated, $\varphi : \Omega \times \mathbb{R}^m \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a normal integrand such that*

$$-\vartheta(\omega, |x|, |y|) \leq \varphi(\omega, x, y)$$

for every $(\omega, x, y) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^N$ with ϑ defined on $\Omega \times \mathbb{R}_+ \times \mathbb{R}_+$ such that it is integrable in ω and increasing in $(t, s)(\vartheta(\omega, t, s))$ and

$$I_\varphi(u, v) = \int_\Omega \varphi(\omega, u(\omega), v(\omega)) d\mu$$

for all $(u, v) \in L^\infty(\Omega, \mathbb{R}^m) \times L^\infty(\Omega, \mathbb{R}^N)$, satisfies

(i) for every $u \in L^\infty(\Omega, \mathbb{R}^m)$, $I_\varphi(u, \cdot)$ is sequentially w^*-lower semicontinuous on $L^\infty(\Omega, \mathbb{R}^N)$;
(ii) for every $u \in L^\infty(\Omega, \mathbb{R}^m)$, there exists $v \in L^\infty(\Omega, \mathbb{R}^N)$ such that

$$I_\varphi(u, v) < \infty,$$

then $\varphi(\omega, x, y)$ is a convex normal integrand.

Proof Fix $u \in L^\infty(\Omega, \mathbb{R}^m)$. By hypothesis (ii) we can find $v_0 \in L^\infty(\Omega, \mathbb{R}^N)$ such that $\varphi(\cdot, u(\cdot), v_0(\cdot)) \in L^1(\Omega)$. Let $\eta \in [0, 1]$, $v, y \in L^\infty(\Omega, \mathbb{R}^N)$ and $D \in \Sigma$. We will show that

$$\begin{aligned} &\int_D \varphi(\omega, u, \eta v + (1-\eta) y) d\mu \\ &\leq \int_D [\eta \varphi(\omega, u, v) + (1-\eta)\varphi(\omega, u, y)] d\mu. \end{aligned} \tag{7.8}$$

From (7.8) we see that we may assume that

$$\varphi(\cdot, u(\cdot), v(\cdot)),\ \varphi(\cdot, u(\cdot), y(\cdot)) \in L^1(D).$$

Using Proposition 7.12, we can find $\{A_n\}_{n \in \mathbb{N}} \subseteq \Sigma$, $A_n \subseteq D$ for all $n \in \mathbb{N}$ such that

$$\chi_{A_n} \xrightarrow{w^*} \eta \text{ in } L^\infty(\Omega).$$

Then using hypothesis (i) we have

$$\begin{aligned}
&\int_D \varphi(\omega, u, \eta v + (1-\eta)y)d\mu \\
=&I_\varphi(u, \eta\chi_D v + (1-\eta)\chi_D y + \chi_{\Omega\setminus D} v_0) - \int_{\Omega\setminus D} \varphi(\omega, u, v_0)d\mu \\
\leq&\liminf_{n\to\infty} I_\varphi(u, \chi_{A_n} v + \chi_{D\setminus A_n} y + \chi_{\Omega\setminus D} v_0) - \int_{\Omega\setminus D} \varphi(\omega, u, v_0)d\mu \\
=&\int_D [\eta\varphi(\omega, u, v) + (1-\eta)\varphi(\omega, u, y)]d\mu.
\end{aligned}$$

So, we have proved (7.8). Since $D \in \Sigma$ is arbitrary, it follows that

$$\varphi(\omega, u(\omega), \eta y_1 + (1-\eta)y_2) \leq \eta\varphi(\omega, u(\omega), y_1) + (1-\eta)\varphi(\omega, u(\omega), y_2)$$

for all $\omega \in \Omega \setminus N$, with $\mu(N) = 0$, all $y_1, y_2 \in \mathbb{R}^N$ and all $\eta \in [0,1] \cap Q$. The normality of φ, implies that the inequality is true for all $\eta \in [0,1]$ and so we conclude that φ is a convex normal integrand. □

We can have a similar result for integrands of two variables.

Theorem 7.14 *If (Ω, Σ, μ) is a nonatomic, complete finite measure space, $\varphi : \Omega \times \mathbb{R}^m \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a normal integrand, there exists $u_0 \in L^1(\Omega, \mathbb{R}^m)$ such that*

$$I_\varphi(u_0) = \int_\Omega \varphi(\omega, u_0(\omega))d\mu < \infty$$

and $I_\varphi(\cdot)$ is weakly lower semicontinuous on $L^1(\Omega, \mathbb{R}^m)$, then for $\mu - a.a$ $\varphi(\omega, \cdot)$ is convex.

Proof By replacing $\varphi(\omega, x)$ by $\widehat{\varphi}(\omega, x) = \varphi(\omega, x) - \varphi(\omega, u_0(\omega))$ if necessary, we may assume that $\varphi(\omega, u_0(\omega)) = 0$ for all $\omega \in \Omega$. We introduce the multifunction $E : \Omega \to 2^{\mathbb{R}^m \times \mathbb{R}}$ defined by

$$E(\omega) = \operatorname{epi} \varphi(\omega, \cdot) = \{(x, \lambda) \in \mathbb{R}^m \times \mathbb{R} : \varphi(\omega, x) \leq \lambda\}$$

(the epigraphical multifunction of $\omega \to \varphi(\omega, \cdot)$). We have

$$(u_0(\omega), 0) \in E(\omega) \text{ for all } \omega \in \Omega$$

and so we see that $E(\cdot)$ has nonempty values. Moreover, since $\varphi(\omega, \cdot)$ is lower semicontinuous, we infer that these values of $E(\cdot)$ are closed. Let

$$S_E^1 = \{\widehat{u} \in L^1(\Omega, \mathbb{R}^m \times \mathbb{R}) : \widehat{u}(\omega) \in E(\omega) \ \ \mu - a.e\}.$$

Claim $S_E^1 \subseteq L^1(\Omega, \mathbb{R}^m \times \mathbb{R})$ is weakly closed.

Consider a net $\{\widehat{u}_\alpha\}_{\alpha\in J} \subseteq S_E^1$ such that

$$\widehat{u}_\alpha \xrightarrow{w} \widehat{u} \text{ in } L^1(\Omega, \mathbb{R}^m \times \mathbb{R}).$$

We have $\widehat{u}_\alpha = (u_\alpha, \lambda_\alpha)$ with $u_\alpha \in L^1(\Omega, \mathbb{R}^m)$, $\lambda_\alpha \in L^1(\Omega)$ for all $\alpha \in J$ and

$$u_\alpha \xrightarrow{w} u \text{ in } L^1(\Omega, \mathbb{R}^m), \quad \lambda_\alpha \xrightarrow{w} \lambda \text{ in } L^1(\Omega), \quad \widehat{u} = (u, \lambda).$$

For every $A \in \Sigma$, we have

$$\begin{aligned} &\chi_A u_\alpha \xrightarrow{w} \chi_A u \text{ in } L^1(\Omega, \mathbb{R}^m) \\ \Rightarrow &\chi_A u_\alpha + \chi_{\Omega\setminus A} u_0 \xrightarrow{w} \chi_A u + \chi_{\Omega\setminus A} u_0 \text{ in } L^1(\Omega, \mathbb{R}^N). \end{aligned}$$

Also, we have

$$\chi_A \lambda_\alpha \xrightarrow{w} \chi_A \lambda \text{ in } L^1(\Omega).$$

The weak lower semicontinuity of $I_\varphi(\cdot)$ on $L^1(\Omega, \mathbb{R}^m)$ implies that

$$\begin{aligned} &I_\varphi(\chi_A u + \chi_{\Omega\setminus A} u_0) \le \liminf_{\alpha\in J} I_\varphi(\chi_A u_\alpha + \chi_{\Omega\setminus A} u_0), \\ \Rightarrow &\int_A \varphi(\omega, u) d\mu \le \liminf_{\alpha\in J} \int_A \varphi(\omega, u_\alpha) d\mu \le \int_A \lambda d\mu, \\ \Rightarrow &\varphi(\omega, u(\omega)) \le \lambda(\omega) \text{ for } \mu - \text{a.a } \omega \in \Omega \text{ (since } A \in \Sigma \text{ is arbitrary)}, \\ \Rightarrow &(u, \lambda) \in S_E^1 \end{aligned}$$

and this proves the Claim.

Evidently since $\varphi(\cdot, \cdot)$ is a normal integrand, the multifunction $E(\cdot)$ is graph measurable. So, by Proposition I-5.130 and the Claim, we have

$$S_E^1 = S^1_{\overline{\text{conv}}\, E}. \tag{7.9}$$

We claim that (7.9) implies that $E(\omega) = \overline{\text{conv}} E(\omega)$ for all $\omega \in \Omega$. We argue by contradiction. So, for every $\omega \in \Omega$, let

$$L(\omega) = \overline{\text{conv}} E(\omega) \setminus E(\omega)$$

and assume that $L(\omega) \neq \emptyset$ for all $\omega \in A$, $\mu(A) > 0$. Note that

$$\text{Gr}\, L \in (\Sigma \cap A) \otimes B(\mathbb{R}^m).$$

By the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), we can find $(\Sigma \cap A)-$measurable functions

$$u : A \to \mathbb{R}^m, \quad \lambda : A \to \mathbb{R}$$

such that $\widehat{u}(\omega) = (u(\omega), \lambda(\omega)) \in L(\omega)$ for all $\omega \in A$. We set

$$\overline{u} = \chi_A \widehat{u} + \chi_{\Omega \setminus A} u_0, \quad \overline{\lambda} = \chi_A \lambda.$$

Then $(\overline{u}, \overline{\lambda}) \in S_E^1$ and $(\overline{u}(\omega), \overline{\lambda}(\omega)) \in L(\omega)$ for all $\omega \in A$ a contradiction. This proves that $E(\omega) = \overline{\text{conv}} E(\omega)$ for all $\omega \in \Omega$. This then implies the convexity of $\varphi(\omega, \cdot)$, $\omega \in \Omega$. □

Remark 7.15 The result remains true if $\mathbb{R}^m$ is replaced by a separable Banach space.

Next we will prove another lower semicontinuity result for $I_\varphi(u, v) = \int_\Omega \varphi(\omega, u, v) d\mu$, under a condition which is easier verifiable than condition $[D]$ of Theorem 7.11. To do this, we need to present a few basic things from the theory of Young measures. Young measures are useful in problems of Calculus of Variations and in Optimal Control Theory.

Let (Ω, Σ) be a measurable space. In Section I-4.5 we introduced the following spaces

$$\text{ca}(\Sigma) = \text{the space of all } \mathbb{R} - \text{valued, signed measures on } \Sigma,$$

$$\text{ca}_+(\Sigma) = \text{the space of all } \mathbb{R}_+ - \text{valued measures on } \Sigma.$$

We endow them with the "total variation norm"

$$\|\mu\| = |\mu|(\Omega) = \sup\{\sum_{k=1}^n |\mu(A_k)| : n \in \mathbb{N}, \{A_k\}_{k=1}^n \text{ is a } \Sigma - \text{partition of } \Omega\}.$$

This norm is equivalent to

$$\|\mu\|_\infty = \sup\{|\mu(A)| : A \in \Sigma\}.$$

We know that $(\text{ca}(\Omega), \|\cdot\| \text{ or } \|\cdot\|_\infty)$ are Banach spaces (see Propositions I-4.211 and I-4.212).

We also introduce some spaces of continuous functions. We limit ourselves to locally compact spaces, because we will use the notions and results valid in locally compact spaces.

Definition 7.16 Let X be a locally compact topological space. We define the following spaces of continuous functions.

(a) $C_b(x) = \{u : X \to \mathbb{R} : \text{continuous and bounded}\}$.

(b) $C_0(x) = \{u : X \to \mathbb{R} : \text{continuous and vanishes at infinity(that is, for every } \varepsilon > 0 \text{ we can find } K_\varepsilon \subseteq X \text{ compact such that } |u(x)| \leq \varepsilon \text{ for all } x \in X \setminus K_\varepsilon)\}$.
(c) $C_c(x) = \{u : X \to \mathbb{R} : \text{continuous and has compact support}\}$.

Remark 7.17 Evidently, we have

$$C_c(X) \subseteq C_0(X) \subseteq C_b(X)$$

and if X is compact, then all three spaces coincide.

The space $C_b(X)$ is equipped with the supremum norm

$$\|u\|_\infty = \sup\{|u(x)| : x \in X\}$$

and becomes a Banach space. Then $C_0(X)$ is a closed subspace and therefore a Banach space itself. Finally $\overline{C_c(X)}^{\|\cdot\|_\infty} = C_0(X)$.

Definition 7.18

(a) A (signed) measure defined on the Borel σalgebra $B(X)$, is said to be a "(signed) Borel measure".
(b) A Borel measure $\mu(\cdot)$ is said to be "regular", if for all $A \in B(X)$ we have

$$\mu(A) = \inf\{\mu(U) : U \text{ open }, A \subseteq U\} \text{ (outer regular)}$$

$$\mu(A) = \sup\{\mu(C) : C \text{ closed }, C \subseteq A\} \text{ (inner regular)}$$

(c) A Borel measure $\mu(\cdot)$ is said to be "tight" or "compact regular", if for all $A \in B(X)$ we have

$$\mu(A) = \sup\{\mu(K) : K \text{ compact }, K \subseteq A\}$$

(d) A Borel measure $\mu(\cdot)$ is said to be a "Radon measure" if it has the following properties

- $\mu(\cdot)$ is outer regular;
- $\mu(\cdot)$ is tight;
- $\mu(K) < \infty$ for all $K \subseteq X$ compact

For a signed Borel measure $\cdot$, we say that $\mu(\cdot)$ is "regular" (resp. "tight", "Radon"), if $|\mu|(\cdot)$ is regular (resp. tight, Radon). We recall that $|\mu|(\cdot)$ is the "total variation measure" defined by

$$|\mu|(A) = \sup\{\sum_{k=1}^{m} |\mu(A_k)| : \{A_k\}_{k=1}^n \text{ is a Borel partition of } A\}$$

for all $A \in B(X)$.

We know that $|\mu| = \mu^+ + \mu^-$ with

$$\mu^+ = \frac{1}{2}(|\mu| + \mu), \mu^- = \frac{1}{2}(|\mu| - \mu).$$

Remark 7.19 From the above definitions we see that

(a) A signed Borel measure is regular (resp. tight, Radon) if and only if $|\mu|(\cdot)$ is regular (resp. tight, Radon) Borel measure.
(b) A signed Borel measure $\mu(\cdot)$ is regular (resp. tight, Radon) if and only if $\mu^+(\cdot)$ and $\mu^-(\cdot)$ are regular (resp. tight, Radon) Borel measures.
So, we can limit ourselves to Borel measures and the same results also hold for signed Borel measures. Moreover, two regular Borel measures are equal if and only if they coincide on the open or closed subsets of X and two tight Borel measures are equal if they coincide on the compact subsets of X.

For the next five propositions, we can ignore the local compactness on X.

Proposition 7.20 *If X is a Hausdorff topological space and $\mu(\cdot)$ is a finite Borel measure $\gamma(\cdot)$ on X, then outer regularity $\Leftrightarrow$ inner regularity.*

Proof $\Rightarrow$ For all $C \in B(X)$, we have

$$\mu(C) = \inf\{\mu(U) : U \text{ open }, C \subseteq U\}. \tag{7.10}$$

Then for $A \in B(X)$ we have

$$\begin{aligned}\mu(X) - \mu(A) = \mu(X \setminus A) &= \inf\{\mu(U) : U \text{ open }, X \setminus A \subseteq U\} \text{ (see (7.10))} \\ &= \mu(X) - \sup\{\mu(C) : C \text{ closed }, C \subseteq A\},\end{aligned}$$

$\Rightarrow \mu(A) = \sup\{\mu(C) : C \text{ closed } C \subseteq A\}$ (since $\mu(\cdot)$ is finite).
$\Rightarrow \mu(\cdot)$ is inner regular.
$\Leftarrow$ The proof of this implication is similar. □

Proposition 7.21 *If X is a Hausdorff topological space and $\mu(\cdot)$ is a finite, tight Borel measure on X, then $\mu(\cdot)$ is a Radon measure.*

Proof Since $\mu(\cdot)$ is tight for all $A \in B(X)$ we have

$$\begin{aligned}\mu(A) &= \sup\{\mu(K) : K \text{ compact}, K \subseteq A\} \\ &\leq \sup\{\mu(C) : C \text{ closed }, C \subseteq A\} \leq \mu(A),\end{aligned}$$

$\Rightarrow \mu(\cdot)$ is regular (see Proposition 7.20)
Because $\mu(\cdot)$ is finite, we conclude that $\mu(\cdot)$ is Radon. □

Proposition 7.22 *If X is metrizable and $\mu(\cdot)$ is a finite Borel measurable on X, then $\mu(\cdot)$ is regular.*

Proof Let $\mathcal{F} = \{A \in B(X) : A \text{ is both outer and inner regular }\}$. We will show that $\mathcal{F}$ is a σ-algebra which contains all open sets, hence $\mathcal{F} = B(X)$. This is done in three steps.

Step 1 $A \in \mathcal{F} \Rightarrow X \setminus A \in \mathcal{F}$.

This is immediate from the definition of $\mathcal{F}$. (see the proof of Proposition 7.20).

Step 2 If $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{F}$, then $A = \bigcup_{n\in\mathbb{N}} A_n \in \mathcal{F}$.

For every $n \in \mathbb{N}$, we can find closed set C_n, open set U_n such that

$$C_n \subseteq A_n \subseteq U_n, \mu(U_n) \leq \mu(C_n) + \frac{\varepsilon}{2^n}. \tag{7.11}$$

Let $U = \bigcup_{n\in\mathbb{N}} U_n$. Then U is open and $A \subseteq U$. We have

$$U \setminus A \subseteq \bigcup\nolimits_{n\in\mathbb{N}} (U_n \setminus A_n)$$

$$\begin{aligned} \Rightarrow 0 \leq \mu(u) - \mu(A) = \mu(U \setminus A) &\leq \sum_{n\in\mathbb{N}} \mu(U_n \setminus A_n) \\ &= \sum_{n\in\mathbb{N}} (\mu(U_n) - \mu(A_n)) \\ &\leq \sum_{n\in\mathbb{N}} \frac{\varepsilon}{2^n} = \varepsilon, \end{aligned}$$

$\Rightarrow \mu(A) = \inf\{\mu(U) : U \text{ open }, A \subseteq U\}$ (outer regularity of A)

Next let $C = \bigcup\limits_{n\in\mathbb{N}} C_n$. Reasoning as above, we can show that

$$\mu(A) \leq \mu(C) + \varepsilon,$$

For every $m \in \mathbb{N}$, we set $\widehat{C}_m = \bigcup\limits_{n=1}^{m} C_n$. This is a closed subset of A and $\widehat{C}_m \uparrow C$. So, we can find $m_0 \in \mathbb{N}$ such that

$$\mu(A) \leq \mu(\widehat{C}_m) + 2\varepsilon \text{ for } m \geq m_0$$

$$\Rightarrow \mu(A) = \sup\{\mu(C) : C \text{ closed }, C \subseteq A\} \text{ (inner regularity of } A)$$

So, we conclude that $A \in \mathcal{F}$ and this completes Step.2.

Step 3 $\mathcal{F}$ contains all open sets of X.

Let $U \subseteq X$ open. From Proposition I-1.157, we know that U is an F_σ-set. So, we can find $\{C_n\}_{n\in\mathbb{N}}$ closed subsets of X such that $C_n \uparrow U$ and $\mu(C_n) \uparrow \mu(U)$. Hence

$$\mu(U) = \sup\{\mu(C) : C \text{ closed } C \subseteq U\},$$

$$\Rightarrow U \in \mathcal{F}.$$

This completes Step 3 and we conclude $\mathcal{F} = B(X)$. □

Proposition 7.23 *If X is metrizable and $\mu(\cdot)$ is a finite Borel measure on X, then $\mu(\cdot)$ is tight if and only if for every $\varepsilon > 0$, we can find $K_\varepsilon \subseteq X$ compact such that $\mu(X) - \varepsilon \leq \mu(K_\varepsilon)$.*

Proof $\Rightarrow$ This implication follows from Definition 7.18(c).

$\Leftarrow$ From Proposition 7.22 we know that $\mu(\cdot)$ is regular. So, it suffices to show that for every $C \subseteq X$ closed we have

$$\mu(C) = \sup\{\mu(K) : K \text{ compact }, K \subseteq C\}. \tag{7.12}$$

Arguing by contradiction, suppose that there exists $C \subseteq X$ closed set and $\varepsilon > 0$ such that

$$\sup\{\mu(K) : K \text{ compact }, K \subseteq C\} \leq \mu(C) - \frac{\varepsilon}{2}. \tag{7.13}$$

For $K \subseteq X$ compact, we have that $K \cap C$ is a compact subset of C. Hence using (7.13), we have

$$\begin{aligned}\mu(K) =& \mu(K \cap C) + \mu(K \cap (X \setminus C))\\ \leq& \mu(C) - \frac{\varepsilon}{2} + \mu(X \setminus C)\\ =& \mu(X) - \frac{\varepsilon}{2}.\end{aligned}$$

Since K is arbitrary compact subset, we have a contradiction to our hypothesis. This proves that $\mu(\cdot)$ is tight. □

On Polish spaces all finite Borel measures are Radon.

Proposition 7.24 *If X is a Polish space, then every finite Borel measure on X is Radon.*

Proof According to Proposition 7.21, it suffices to show that $\mu(\cdot)$ is tight. Let $D = \{x_k\}_{k\in\mathbb{N}} \subseteq X$ be dense. Consider the closed balls.

$$\overline{B}_{\frac{1}{n}}(x_k) = \{x \in X : d(x, x_k) \leq \frac{1}{n}\}, n, k \in \mathbb{N}.$$

with d a complete metric on X. We have

$$X = \bigcup_{k\in\mathbb{N}} \overline{B}_{\frac{1}{n}}(x_k) \text{ for all } n \in \mathbb{N}.$$

Then given $\varepsilon > 0$, for every $n \in \mathbb{N}$, we can find $m_\varepsilon \in \mathbb{N}$ such that

$$\mu(X \setminus \bigcup_{k=1}^{m_\varepsilon} \overline{B}_{\frac{1}{n}}(x_k)) \leq \frac{\varepsilon}{2^n}. \tag{7.14}$$

Let $K = \bigcap_{n\in\mathbb{N}} \bigcup_{k=1}^{m_\varepsilon} \overline{B}_{\frac{1}{n}}(x_k)$. Evidently K is closed and totally bounded, hence it is compact. We have

$$\begin{aligned}
\mu(X) - \mu(K) =& \mu(X \setminus K) \\
=& \mu(\bigcup_{n\in\mathbb{N}} (X \setminus \bigcup_{k=1}^{m_\varepsilon} \overline{B}_{\frac{1}{n}}(x_k))) \\
\leq& \sum_{n\in\mathbb{N}} \mu(X \setminus \bigcup_{k=1}^{m_\varepsilon} \overline{B}_{\frac{1}{n}}(x_k)) \\
\leq& \sum_{n\in\mathbb{N}} \frac{\varepsilon}{2^n} = \varepsilon,
\end{aligned}$$

$\Rightarrow \mu(\cdot)$ is tight and so by Proposition 7.21 it is Radon. □

We return to the hypothesis that X is locally compact. We define the following subspaces of $\text{ca}(B(X))$:

$$\text{ca}(B(X))_R = \{\mu \in \text{ca}(B(X)) : \mu \text{ is Radon}\},$$

$$\text{ca}(B(X))_r = \{\mu \in \text{ca}(B(X)) : \mu \text{ is regular}\}.$$

From Definition 7.18, we have

$$\text{ca}(B(X))_R \subseteq \text{ca}(B(X))_r \subseteq \text{ca}(B(X))$$

If X is metrizable, then

$$\text{ca}(B(X))_r = \text{ca}(B(X)) \text{ (see Proposition 7.22).}$$

If X is Polish, then

$$\text{ca}(B(X))_R = \text{ca}(B(X)).$$

Actually this equality is also true if X is Souslin.

From Theorem I-4.228, we have:

Theorem 7.25 *If X is a locally compact topological space, then $C_0(X)^* = \mathrm{ca}(B(X))_R$ in the sense that if $\xi \in C_0(X)^*$, then $\xi(u) = \int_X u(x)d\mu$ for some $\mu \in \mathrm{ca}(B(X))_R$, all $u \in C_0(X)$, and $\|\xi\|_* = \|\mu\|$.*

Remark 7.26 If X is compact, then we recover the well-known Riecz Representation Theorem which says that

$$C(X)^* = \mathrm{ca}(B(X))$$

(that is, if $\xi \in C(X)^*$, then $\xi(u) = \int_X u(x)d\mu$ for some $\mu \in \mathrm{ca}(B(X))$, all $u \in C(X)$ and $\|\xi\|_* = \|\mu\|$).

Using the spaces $C_b(X), C_0(X), C_c(X)$, we can introduce some topologies on $\mathrm{ca}(B(X))$. In what follows for u in one of these spaces, we consider the linear functional $k_u : \mathrm{ca}(B(X)) \to \mathbb{R}$ defined by

$$k_u(\mu) = \int_X u(x)d\mu.$$

Definition 7.27 Let X be a locally compact topological space.

(a) The weakest topology on $\mathrm{ca}(B(X))$ for which all the functional $\{k_u\}_{u\in C_b(X)}$ are continuous, is called the "narrow topology" on $\mathrm{ca}(B(X))$.
(b) The weakest topology on $\mathrm{ca}(B(X))$ for which all the functionals $\{k_u\}_{u\in C_0(X)}$ are continuous, is called the "weak topology" on $\mathrm{ca}(B(X))$.
(c) The weakest topology on $\mathrm{ca}(B(X))$ for which all the functionals $\{k_u\}_{u\in C_c(X)}$ are continuous, is called the "vague topology" on $\mathrm{ca}(B(X))$

By $\tau_n(X)$ (resp.$\tau_w(X)$, $\tau_v(X)$) we denote the narrow (resp. weak, vague) topology on $\mathrm{ca}(X)$.

Remark 7.28 On account of Theorem 7.25, the weak topology defined above is in fact the w^*-topology on the dual Banach space $\mathrm{ca}(B(X)) = C_0(X)^*$. The name "weak topology" comes from probability. Evidently, we have

$$\tau_v(X) \subseteq \tau_w(X) \subseteq \tau_n(X).$$

If X is compact, then these topologies coincide and are the w^*-topology on the dual Banach space $\mathrm{ca}(B(X)) = C(X)^*$. In fact $\tau_w(X) = \tau_n(X)$ if and only if X is compact. Finally if X is metrizable then using the fact that a lower semicontinuous function can be approximated by an increasing sequence of Lipschitz continuous functions (see Proposition I-1.183), we see that

- for every lower semicontinuous, bounded below function $g : X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$, the map

$$\mu \to k_g(\mu) = \int_X g(x)d\mu, \mu \in \mathrm{ca}_+(B(X))$$

is $\tau_n(X)$-lower semicontinuous on $\mathrm{ca}_+(B(X))$

- for upper every semicontinuous, bounded above function $g : X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty\}$, the map

$$\mu \to k_g(\mu) = \int_X g(x)d\mu, \mu \in \mathrm{ca}_+(B(X))$$

is $\tau_n(X)$-upper semicontinuous on $\mathrm{ca}_+(B(X))$

After this brief detour on the topic of spaces of measures, necessary to understand what follows, we are ready to introduce Young measures. Roughly speaking, Young measures are parametrized measures and can be viewed as the completion in some sense of the set of measurable functions.

So, let (Ω, Σ, μ) be a complete finite measure space and X a locally compact Polish space. As before by $B(X)$ denote the Borel σ-algebra of X. Also. by $P(X)$ we denote the set of probability measures on X equipped with the relative narrow topology $\tau_n(X)$. The next result is known as the "disintegration theorem" and can be found in Valadier [460].

Theorem 7.29 *If $p_\Omega : \Omega \times X \to \Omega$ is the projection map (that is, $p_\Omega(w, x) = w$), $\lambda \in \mathrm{ca}(\Sigma \otimes B(X))$ and $\mu = \lambda \circ p_\Omega^{-1}$, then we can find a Σ-measurable map $\widehat{\lambda} : \Omega \to P(X)$ such that $\lambda(A \times C) = \int_A \widehat{\lambda}(w)(C)d\mu$.*

Remark 7.30 The map $\widehat{\lambda}(\cdot)$ is unique. Indeed if $\widetilde{\lambda}(\cdot)$ is another such map, then $\widehat{\lambda}(w)(C) = \widetilde{\lambda}(w)(C)$ μ-a.e for all $C \in B$. But $B(X)$ is countably generated. So, $\widehat{\lambda}(w) = \widetilde{\lambda}(w)\mu$-a.e. in $P(X)$. The map $\lambda(\cdot)$ is known as the "disintegration" of $\lambda(\cdot, \cdot)$. with respect to μ. Using Proposition I-2.7, we obtain

$$\lambda(D) = \int_\Omega \widehat{\lambda}(w)(D_w)d\mu \tag{7.15}$$

with $D \in \Sigma \otimes B(X)$ and $D_w = \{x \in X : (w, x) \in D\}$. Using (7.15) and the fact that every $\Sigma \otimes B(X)$-measurable function $\varphi(w, x)$ can be approximated by simple ones, then

$$\int_{\Omega \times X} \varphi(w, x)d\lambda = \int_\Omega [\int_x \varphi(w, x)\widehat{\lambda}_w(dx)]d\mu. \tag{7.16}$$

Definition 7.31 A "Young measure" on $\Omega \times X$, is a $\lambda \in \mathrm{ca}_+(\Sigma \otimes B(X))$ such that $\mu = \lambda \circ p_\Omega^{-1}$, that is

$$\mu(A) = \lambda(A \times X) \text{ for all } A \in \Sigma.$$

We denote the set of Young measures on $\Omega \times X$ by $\mathrm{y}(\Omega \times X)$ (or simply y when the spaces Ω, X are clearly understood from the context).

Remark 7.32 If $\lambda \in \mathrm{y}(\Omega \times X)$, then using Theorem 7.29, we can identify λ with its disintegration $\widehat{\lambda}(\cdot)$. Therefore

$$\mathrm{y}(\Omega \times X) \subseteq L^{\circ}(\Omega, P(X)). \tag{7.17}$$

Also, if $\lambda \in \mathrm{y}(\Omega \times X)$, then for every $A \in \Sigma$, $\lambda(A, \cdot) \in \mathrm{ca}_{+}(B(X))$ and so it is a Radon measure (see Peoposition 7.24).

The next proposition, validates our earlier remark that Young measures are parametrized measures (see (7.17))

Proposition 7.33 *If $\widehat{\lambda} : \Omega \to P(X)$, then the following two statements are equivalent*

(a) $\widehat{\lambda}$ is Σ-measurable;
(b) for every $E \in B(X)$ $w \to \widehat{\xi}_E(w) = \widehat{\lambda}(w)(E)$ is Σ-measurable.

Proof $(a) \Rightarrow (b)$. Let $U \subseteq X$ be open and consider the map $\vartheta_U(\beta) = \beta(U)$ for all $\beta \in P(X)$. By Theorem I-4.237 (the Portmanteau Theorem), we know that ϑ_U is τ_n-lower semicontinuous. Then $\widehat{\xi}_U = \vartheta_U \circ \widehat{\lambda}$ and so $\widehat{\xi}_U$ is Σ-measurable. From Proposition 7.22, we know that the elements of $P(X)$ or regular, therefore $w \to \widehat{\xi}_E(w)$ is Σ-measurable for all $E \in B(X)$.

$(b) \Rightarrow (a)$. For $D \in \Sigma \otimes B(X)$, let $\sigma_D : \Omega \to [0, 1]$ be defined by

$$\sigma_D(w) = \widehat{\lambda}(w)(D_w)(D_w = \{x \in X : (w, x) \in D\}).$$

Let $\widehat{\mu} = \{D \in \Sigma \otimes B(X) : \sigma_D \text{ is } \Sigma\text{-measurable}\}$. We can easily check that $\widehat{\mu}$ is a monotone class (see Definition I-2.5) and it contains the algebra of measurable rectangles. Therefore by Proposition I-2.7, we have $\Sigma \otimes B(X) \subseteq \widehat{\mu}$. So, $\sigma_D(\cdot)$ is Σ-measurable for all $D \in \Sigma \otimes B(X)$. For $D = \Omega \times C$, $C \in B(X)$, we have

$$\sigma_D(w) = \widehat{\lambda}(w)(C) = \int_{\Omega} \chi_C(x) \widehat{\lambda}(w)(dx).$$

So, it follows that for every simple function $s(\cdot)$ on X, we have that

$$w \to \int_{\Omega} s(x) \widehat{\lambda}(w)(dx) \text{ is } \Sigma\text{-measurable.}$$

If $u \in C_b(X)$, then we can find a sequence $\{s_n\}_{n \in \mathbb{N}}$ of simple functions such that

$$|s_n(x)| \leq u(x) \text{ for all } x \in X, \text{ all } n \in \mathbb{N}, s_n(x) \to u(x) \text{ for all } x \in X$$

So, by the dominated convergence theorem we have

$$\int_{\Omega} s_n(x)\widehat{\lambda}(w)(dx) \to \int_{\Omega} u(x)\widehat{\lambda}(w)(dx),$$

$$\Rightarrow w \to \int_{\Omega} u(x)\widehat{\lambda}(w)(dx) \text{ is } \Sigma\text{-measurable},$$

$$\Rightarrow w \to \widehat{\lambda}(w) \text{ is } \Sigma\text{-measurable}$$

(see Definition 7.27(a)) □

Recall $L^{\circ}(\Omega, X) = \{u : \Omega \to X : \Sigma\text{-measurable}\}$. We identify two such functions which differ only on a μ-null set . Given $u \in L^{\circ}(\Omega, X)$, we define $\widehat{\lambda}^u : \Omega \to p(X)$ by setting $\widehat{\lambda}^u(w) = \delta_{u(w)}$ (=the Dirac measure of $u(w)$ defined by

$$\delta_{u(w)}(C) = \left\{ \begin{array}{l} 1 \text{ if } u(w) \in C \\ 0 \text{ otherwise} \end{array} \right).$$

We have

$$\widehat{\lambda}^u(w)(C) = \chi_{u^{-1}(C)}(w) \text{ for all } C \in B(X),$$

$$\Rightarrow w \to \widehat{\lambda}^u(w)(C) \text{ is } \Sigma - \text{measurable for all } C \in B(X)$$

$$\Rightarrow \widehat{\lambda}^u : \Omega \to P(X) \text{ is } \Sigma - \text{measurable}$$

(see Proposition 7.33)

Definition 7.34 If $u \in L^{\circ}(\Omega, X)$, then $\widehat{\lambda}^u$ defined above is the "Young measure associated with the measurable function $u(\cdot)$".

Remark 7.35 If $\lambda^u(A \times C) = \mu(A \cap u^{-1}(C))$ for all $A \in \Sigma$, $C \in B(X)$, then $\widehat{\lambda}^u$ is the disintegration of the Young measure λ^u. Then from (7.16) we have that if $\varphi : \Omega \times X \to \overline{R}_+ = \mathbb{R}_+ \cup \{+\infty\}$ is $\Sigma \otimes B(X)$-measurable then

$$\int_{\Omega \times X} \varphi(w, x) d\lambda^u = \int_{\Omega} \varphi(w, u(w)) d\mu$$

Note also that the map $u \to \lambda^u$ provides an embedding of $L^{\circ}(\Omega, X)$ into $\mathrm{y}(\Omega \times X)$ and so we can think of $L^{\circ}(\Omega, X)$ as a subspace of $\mathrm{y}(\Omega \times X)$. In particular, X is a subspace of $\mathrm{y}(\Omega \times X)$ (just identify $x \in X$ with the constant function $u(x') = x$ for all $x' \in X$).

Since Young measures λ^u play an important role in many applications, we would like to have a criterion that enables us to identify such Young measures.

Proposition 7.36 *If $\lambda \in y(\Omega \times X)$ and $u \in L^{\circ}(\Omega, X)$, then $\lambda = \lambda^u \Leftrightarrow \lambda((\Omega \times X) \setminus Gru) = 0$.*

Proof $\Rightarrow$. Let $\gamma_u : \Omega \to \Omega \times X$ be defined by $\gamma_u(w) = (w, u(w))$. For every $D \in \Sigma \otimes B(X)$, we have

$$\lambda(D) = \lambda^u(D) = \mu(\gamma_u^{-1}(D)) \tag{7.18}$$

We have $Gru \in \Sigma \otimes B(X)$ and so $D = (\Omega \times X) \setminus Gru \in \Sigma \otimes B(X)$. We have $\gamma_u^{-1}(D) = \emptyset$ and so from (7.18) it follows that

$$\lambda((\Omega \times X) \setminus Gru) = 0.$$

$\Leftarrow$ Let $D \in \Sigma \otimes B(X)$ and $A = \gamma_u^{-1}(D) \in \Sigma$. Then for $(Gru)^c = (\Omega \times X) \setminus Gru$, we have

$$\lambda(D) = \lambda(D \setminus (Gru)^c) \leq \lambda(A \times X) = \mu(A). \tag{7.19}$$

Also since $A \times X \setminus (Gru)^c \subseteq D$, we have

$$\mu(A) = \lambda(A \times X) = \lambda((A \times X) \setminus (Gru)^c) \leq \lambda(D). \tag{7.20}$$

From (7.19) and (7.20), we obtain

$$\lambda(D) = \mu(A) = \mu(\gamma_u^{-1}(D)) = \lambda^u(D),$$

$\Rightarrow \lambda = \lambda^u$. □

Next in analogy to Definition 7.27(a), we will introduce a topology on $y(\Omega \times X)$. If $\varphi : \Omega \times X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is $\Sigma \otimes B(X)$-measurable, then we say that $\varphi(\cdot, \cdot)$ is an "$L^1(\Omega)$-bounded integrand", if

$$|\varphi(w, x)| \leq \vartheta(w) \text{ for } \mu - a.a.w \in \Omega, \text{ all } x \in X,$$

with $\vartheta \in L^1(\Omega)$. Moreover, we will use the following notation:

$$\text{Car}(\Omega \times X) = \text{the Caratheodory integrands on } \Omega \times X,$$

$$\text{Car}_b(\Omega \times X) = \text{the } L^1(\Omega)\text{-bounded Caratheodory integrands on } \Omega \times X,$$

$$\text{Nor}(\Omega \times X) = \text{the normal integrands on } \Omega \times X$$

(see Definition 7.4, with $\mathbb{R}^N$ replaced by X).

Let $d(\cdot, \cdot)$ be the metric generating the topology on the Polish space X. Arguing as in the proof of Proposition I-1.183, we obtain the following approximation result.

Proposition 7.37 *If $\varphi \in \text{Nor}(\Omega \times X)$, then we can find a sequence $\{\varphi_n\}_{n \in \mathbb{N}} \subseteq \text{Car}(\Omega \times X)$, which are actually d-Lipschitz in $x \in X$ such that $\varphi_n \uparrow \varphi$ and if φ is $L^1(\Omega)$-bounded then so are the $\varphi_n's$ (that is, $\varphi_n \in \text{Car}_b(\Omega \times X)$, for all $n \in \mathbb{N}$.)*

Now using the family $\mathrm{Car}_b(\Omega \times X)$, we can define on $y(\Omega \times X)$ the counterpart of the narrow topology $\tau_n(X)$.

Definition 7.38 The "Young narrow topology" on $\mathrm{y}(\Omega \times X)$ is the weakest topology on $\mathrm{y}(\Omega \times X)$ for which the maps

$$\lambda \to I_\varphi(\lambda) = \int_{\Omega \times X} \varphi(w,x) d\lambda = \int_\Omega [\int_X \varphi(w,x) \widehat{\lambda}(w)(dx)] d\mu$$

are continuous for all $\varphi \in \mathrm{Car}_b(\Omega \times X)$ (recall $\widehat{\lambda}(\cdot)$ is the disintegration of λ, see Theorem 7.29). We denote this topology by $\tau_n^y(\Omega \times X)$ or simply by τ_n^y.

Remark 7.39 According to this definition we have

$$\lambda_n \xrightarrow{\tau_n^y} \lambda \Leftrightarrow \lambda_n(A \times \cdot) \xrightarrow{\tau_n(x)} \lambda(A \times \cdot) \text{ for all } A \in \Sigma. \tag{7.21}$$

In fact we can replace Σ by an algebra a_0 such that $\sigma(\mathrm{a}_0) = \Sigma$. The Dirac Young measures are τ_n^y-dense in $\mathrm{y}(\Omega \times \mathbb{R}^m)$.

The Definition 7.38, (7.21) and the previous discussion of the spaces of measures, leads to the following pormanteau-type theorem (see Theorem I-4.227).

Theorem 7.40 *If $\{\lambda_\alpha\}_{\alpha \in J} \subseteq y(\Omega \times X)$ is a net and $\lambda \in y(\Omega \times X)$, then the following statements are equivalent.*

(a) $\lambda_\alpha \xrightarrow{\tau_n^y} \lambda$;
(b) *if $\varphi = \chi_A \otimes \psi$ with $A \in \Sigma$ and $\psi : X \to \mathbb{R}$ is lower semicontinuous and bounded below, then $I_\varphi(\lambda) \le \liminf I_\varphi(\lambda_\alpha)$;*
(c) *if $\varphi = \chi_A \otimes \psi$ with $A \in \Sigma$ and $\psi \in U_d(X)$={bounded, d-uniformly continuous functions from X into $\mathbb{R}$}, then $I_\varphi(\lambda) = \lim_{\alpha \in J} I_\varphi(\lambda_\alpha)$;*
(d) $\lambda(A \times U) \le \liminf \lambda_\alpha(A \times U)$ *for all $A \in \Sigma$, all $U \subseteq X$ open;*
(e) $\limsup_{\alpha \in J} \lambda_\alpha(A \times C) \le \lambda(A \times C)$ *for all $A \in \Sigma$, all $C \subseteq X$ closed.*

Given $\vartheta \in P(X)$, it generates a Young measure by

$$\lambda^\vartheta = \mu \otimes \vartheta.$$

The disintegration of this Young measure is $\widehat{\lambda}^\vartheta(w) = \vartheta$ for all $w \in \Omega$.

Definition 7.41 We say that λ^ϑ is the "Young measure corresponding to $\vartheta \in P(X)$". The map $\vartheta \to \lambda^\vartheta$ is injective and so we can say that $P(X) \subseteq \mathrm{y}(\Omega \times X)$.

It is natural to ask how the narrow topologies (see Definitions 7.27 and 7.38) are related.

Proposition 7.42

(a) $\tau_n^y(\Omega \times X)|_{P(X)} = \tau_n(X)$

(b) $P(X)$ is τ_n^y-closed in $y(\Omega \times X)$.

Proof

(a) Consider $\{\vartheta_\alpha\}_{\alpha\in J} \subseteq P(X)$ a net and $\vartheta \in P(X)$. Then

$$\vartheta_\alpha \xrightarrow{\tau_n(x)} \vartheta \Leftrightarrow k_u(\vartheta_\alpha) \to k_u(\vartheta) \text{ for all } k \in C_b(X) \tag{7.22}$$

(see Definition 7.27).

For $A \in \Sigma$ and $u \in C_b(X)$, let $\varphi = \chi_A \otimes u$. From (7.22) it follows that

$$I_\varphi(\lambda^{\vartheta_\alpha}) = \mu(A)k_u(\vartheta_\alpha) \to \mu(A)k_u(\vartheta) = I_\varphi(\lambda^{\vartheta})$$

By Theorem 7.40, we infer that

$$\vartheta_\alpha \xrightarrow{\tau_n(x)} \vartheta \Leftrightarrow \lambda^{\vartheta_\alpha} \xrightarrow{\tau_n^y} \lambda^{\vartheta}.$$

(b) Consider a net in $P(X)$ and assume that $\lambda^{\vartheta_\alpha} \xrightarrow{\tau_n^y} \lambda$ in $y(\Omega \times X)$. Let $\widehat{\lambda}(\cdot)$ be the disintegration of $\lambda(\cdot)$ (see Theorem 7.29). We define

$$\vartheta(C) = \int_\Omega \widehat{\lambda}(w)(C)d\mu \text{ for all } C \in B(X), \lambda^{\vartheta} = \mu \otimes \vartheta.$$

For $\varphi = \chi_A \otimes u$ with $A \in \Sigma, u \in C_b(X)$, we have

$$\begin{aligned} I_\varphi(\lambda^{\vartheta}) = \mu(A)k_u(\vartheta) &= \mu(A) \int_X u(x)\widehat{\lambda}(w)(dx) \\ &= \mu(A) \lim_{\alpha\in J} k_u(\vartheta_\alpha) \text{ (see Definition 7.27)} \\ &= \lim_{\alpha\in J} I_\varphi(\lambda^{\vartheta_\alpha}), \end{aligned}$$

$\Rightarrow \lambda^{\vartheta_\alpha} \xrightarrow{\tau_n^y} \lambda^{\vartheta} = \lambda$ (see Theorem 7.40).

Therefore $P(X)$ is τ_n^y-closed in $y(\Omega \times X)$. □

In the above proof we used nets because we do not know yet the nature of τ_n^y. The next proposition provides some additional information on τ_n^y.

Proposition 7.43 *If Σ is countably generated, then $(y(\Omega \times X), \tau_n^y)$ is metrizable.*

Proof First assume that X is compact. Then $y(\Omega \times X)$ is bounded subset of $L^\infty(\Omega, \text{ca}(X)_w) = L^1(\Omega, C(X))^*$ (see Theorem I-4.177). Since Σ is countably generated and $C(X)$ is separable, it follows that $L^1(\Omega, C(X))$ is separable. Hence by Theorem I-3.117 we have $(y(\Omega \times X), \tau_n^y)$ is metrizable.

Now consider the general case where X is locally compact, Polish. Then X is homeomorphic to a subset of the Hilbert cube $H = [0,1]^{\mathbb{N}}$. Then $y(\Omega \times X)$ is homeomorphic to a subset of $y(\Omega \times H)$ and we conclude that $(y(\Omega \times H), \tau_n^y)$ is metrizable. □

The above proof leads to the following characterization of the τ_n^y topology.

Proposition 7.44 *If $\{\lambda_\alpha\}_{a\in J} \subseteq y(\Omega \times X)$ is a net and $\lambda \in y(\Omega \times X)$, then $\lambda_\alpha \xrightarrow{\tau_n^y} \lambda \Leftrightarrow I_\varphi(\lambda_\alpha) \to I_\varphi(\lambda)$ for all $\varphi \in \mathrm{Car}_b(\Omega \times X)$.*

Corollary 7.45 *If $\{\lambda_\alpha\}_{\alpha\in J} \subseteq y(\Omega \times X)$ is a net and $\lambda \in y(\Omega \times X)$, then $\lambda_\alpha \xrightarrow{\tau_n^y} \lambda \Leftrightarrow I_\varphi(\lambda) \le \liminf_{\alpha\in J} I_\varphi(\lambda_\alpha)$ for all $\varphi \in \mathrm{Nor}(\Omega \times X)$ which are bounded from below.*

We want to characterize the compact sets in $\tau_n(x)$ and in $\tau_n^y(\Omega \times X)$.

Definition 7.46

(a) Let $C \subseteq \mathrm{ca}(B(X))$. We say that C is "uniformly tight" if

(7.1) $\sup\{|\mu|(X) : \mu \in C\} < \infty$;
(7.2) for every $\varepsilon > 0$, there exists a compact set $K_\varepsilon \subseteq X$ such that $\sup\{|\mu|(X \setminus K_\varepsilon) : \mu \in C\} \le \varepsilon$

(b) Let $D \subseteq y(\Omega \times X)$. We say that D is "uniformly y-tight", if for every $\varepsilon > 0$, there exists a compact set $K_\varepsilon \subseteq X$ such that $\lambda(\Omega \times (X \setminus K_\varepsilon)) \le \varepsilon$ for all $\lambda \in D$.

Using these notions, we characterize the compact subsets of $\mathrm{ca}(B(X))$ (recall that since X is Polish, we have $\mathrm{ca}(B(X)) = \mathrm{ca}(B(X))_R$) and of $y(\Omega \times X)$.

For the compact subsets of $\mathrm{ca}(B(X))$, we have the following theorem, known in the literature as "Prokhorov's Theorem" (see Dudley [146, Theorem 11.5.4, p. 316]).

Theorem 7.47 *$C \subseteq \mathrm{ca}(B(X))$ is relatively $\tau_n(X)$-compact if and only if $C \subseteq \mathrm{ca}(B(X))$ is uniformly tight.*

For Young measures there is an analogous result (see Valadier [462]).

Theorem 7.48 *$D \subseteq y(\Omega \times X)$ is relatively sequentially τ_n^y-compact if and only if $D \subseteq y(\Omega \times X)$ is uniformly y-tight.*

Remark 7.49 If $C = \{\delta_x : x \in K\}$, then C is uniformly tight if and only if $K \subseteq X$ is relatively compact. If $B \subseteq L^1(\Omega, \mathbb{R}^N)$ is bounded and $D = \{\lambda^u : u \in B\}$, then D is uniformly $y-$tight. So, if $\{u_n\}_{n\in\mathbb{N}} \subseteq B$, then there exist a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}}$ of $\{u_n\}_{n\in\mathbb{N}}$ and $\lambda \in \mathrm{y}(\Omega \times N)$ such that $\lambda^{u_{n_k}} \xrightarrow{\tau_n^y} \lambda$. Moreover, if $\{u_n\}_{n\in\mathbb{N}} \subseteq B$ is uniformly integrable, then there exists a subsequence $\{u_{n_k}\}_{k\in\mathbb{N}}$ of $\{u_n\}_{n\in\mathbb{N}}$ and $u \in L^1(\Omega, \mathbb{R}^N)$, $\lambda \in \mathrm{y}(\Omega \times X)$ such that

$$\lambda^{u_{n_k}} \xrightarrow{\tau_n^y} \lambda \text{ and } u_{n_k} \xrightarrow{w} u \text{ in } L^1(\Omega, \mathbb{R}^N) \text{ with } u(w) = \int_{\mathbb{R}^N} x d\widehat{\lambda}(w)(dx)$$

where $\widehat{\lambda}(\cdot)$ is the disintegration of λ. We call the limit function u, the "barycenter" of $\widehat{\lambda}(\cdot)$. If $\mu(\cdot)$ is nonatomic, then $D = \{\lambda^u : u \in L^0(\Omega, X)\}$ is dense in $\mathrm{y}(\Omega \times X)$. Similarly, if

$$D = \{\lambda \in \mathrm{ca}(B(X)) : \lambda \text{ is a finite linear combination of Dirac measures}\},$$

then $\overline{D}^{\tau_n(x)} = \mathrm{ca}(B(X))$.

Now we can have another lower semicontinuity result for integral functionals.

Theorem 7.50 *If $\varphi : \Omega \times \mathbb{R}^m \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a measurable integrand such that*

(i) for μ-a.a. $w \in \Omega$, $(x, y) \to \varphi(w, x, y)$ is lower semicontinuous;
(ii) for μ-a.a. $w \in \Omega$, all $x \in \mathbb{R}$, $y \to \varphi(w, x, y)$ is convex;
(iii) $\eta(w) - c(|x| + |y|) \leq \varphi(w, x, y)$ for μ-a.a. $w \in \Omega$, all $(x, y) \in \mathbb{R}^m \times \mathbb{R}^N$ with $\eta \in L^1(\Omega)$ and $c > 0$ and if $1 \leq p, r \leq \infty$, then $(u, v) \to I_\varphi(u, v) = \int_\Omega \varphi(w, u, v) d\mu$ is sequentially lower semicontinuous on $L^p(\Omega, \mathbb{R}^m) \times L^r(\Omega, \mathbb{R}^N)_w$; If $r = \infty$, then we use the w^*-topology.

Proof For every $\vartheta \in \mathbb{R}$, we need to show that the sublevel set

$$L_\vartheta = \{(u, v) \in L^p(\Omega, \mathbb{R}^m) \times L^r(\Omega, \mathbb{R}^N) : J_\varphi(u, v) \leq \vartheta\}$$

is sequentially closed in $L^p(\Omega, \mathbb{R}^m) \times L^r(\Omega, \mathbb{R}^N)_w$ (if $r < \infty$) and in $L^p(\Omega, \mathbb{R}^m) \times L^\infty(\Omega, \mathbb{R}^N)_{w^*}$ (if $r = +\infty$).

So consider two sequences $\{u_n\}_{n\in\mathbb{N}}$, $\{v_n\}_{n\in\mathbb{N}}$ such that

$$u_n \to u \text{ in } L^p(\Omega, \mathbb{R}^m),\ v_n \xrightarrow{w} v \text{ in } L^r(\Omega, \mathbb{R}^N)\left(\xrightarrow{w^*} \text{ if } r = \infty\right). \tag{7.23}$$

By passing to a subsequence if necessary, we may also assume that

$$u_n(w) \to u(w) \text{ for } \mu - a.a\ w \in \Omega.$$

According to Remark 7.49 we also have

$$\lambda^{v_n} \xrightarrow{\tau_n^y} \lambda$$

For every $A \in \Sigma$ we have

$$\int_A v_n(w) d\mu = \int_A \int_{\mathbb{R}^N} z \delta_{v_n(w)}(dz) d\mu \to \int_A \int_{\mathbb{R}^N} z \widehat{\lambda}(w)(dz) d\mu$$

From (7.23) we have

$$\int_A v_n(w)d\mu \to \int_A v(w)d\mu.$$

Therefore we obtain

$$\int_A v(w)d\mu = \int_A \int_{\mathbb{R}^N} z\widehat{\lambda}(w)(dz)d\mu \text{ for all } A \in \Sigma,$$

$$\Rightarrow v(w) = \int_{\mathbb{R}^N} z\widehat{\lambda}(w)(dz) \text{ for } \mu - a.a\ w \in \Omega. \tag{7.24}$$

Let $\varphi_k : \Omega \times \mathbb{R}^m \times \mathbb{R}^N \to \mathbb{R}$, $k \in \mathbb{N}$ be Caratheodory integrands such that $\varphi_k(w, \cdot, \cdot)$ is k-Lipschitz, $\eta(w) - c(|x| + |y|) \leq \varphi_k(w, x, y)$ for μ-a.a $w \in \Omega$, all $(x, y) \in \mathbb{R}^m \times \mathbb{R}^N$ and $\varphi_k \uparrow \varphi$. Let $\{\vartheta_k\}_{k\in\mathbb{N}} \subseteq C_0(\mathbb{R}^m \times \mathbb{R}^n)$ be an increasing sequence converging pointwise to 1. We set

$$\widehat{\varphi}_k(w, x, y) = \min\{k\vartheta_k(x, y), \varphi_k(w, x, y)\}.$$

Clearly we have $\widehat{\varphi}_k(w, \cdot, \cdot) \uparrow \varphi(w, \cdot, \cdot)$ for μ-a.a $w \in \Omega$. Since $\text{ca}(B(\mathbb{R}^N)) = C_0(\mathbb{R}^N)^*$ (Theorem 7.25), from Theorem I-4.177 we have

$$L^\infty(\Omega, \text{ca}(B(\mathbb{R}^N))_{w^*}) = L^1(\Omega, C_0(\mathbb{R}^N))^*$$

Then viewing $y(\Omega \times \mathbb{R}^N)$ as a subset of $L^\infty(\Omega, \text{ca}(B(\mathbb{R}^N))_{w^*})$ we see that τ_n^y is the relative w^*-topology. In what follows by $((\cdot, \cdot))$, we denote the duality brackets for the pair

$$(L^\infty(\Omega, \text{ca}(B(\mathbb{R}^N))_{w^*}, L^1(\Omega, C_0(\mathbb{R}^N)))).$$

We have

$$\sup_{y\in\mathbb{R}^N} |\widehat{\varphi}_k(w, u_n(w), y) - \widehat{\varphi}_k(w, u(w), y)|$$

$$\leq k|u_n(w) - u(w)| \to 0 \text{ for } \mu - a.a\ w \in \Omega,$$

$$\Rightarrow \widehat{\varphi}_k(w, u_n(w), \cdot) \to \widehat{\varphi}_k(w, u(w), \cdot) \text{ in } C_0(\mathbb{R}^N)\mu - a.e.$$

We set

$$\widetilde{\varphi}_k(u_n)(w) = \widehat{\varphi}_k(w, u_n(w), \cdot), \widetilde{\varphi}_k(u)(w) = \widehat{\varphi}_k(w, u(w), \cdot)$$

By the dominated convergence theorem, we have

$$\begin{aligned}
&\widetilde{\varphi}_k(u_n) \to \widetilde{\varphi}_k(u) \text{ in } L^1(\Omega, C_0(\mathbb{R}^N)),\\
\Rightarrow &((\widetilde{\varphi}_k(u_n), \lambda^{v_n})) \to ((\widetilde{\varphi}_k(u), \lambda)),\\
\Rightarrow &\int_\Omega \int_{\mathbb{R}^N} \widehat{\varphi}_k(w, u_n(w), y)\lambda^{v_n(w)}(dy)d\mu\\
\to &\int_\Omega \int_{\mathbb{R}^N} \widehat{\varphi}_k(w, u(w), y)\widehat{\lambda}(w)(dy)d\mu
\end{aligned}$$

From the monotone convergence theorem, we have

$$\int_\Omega \int_{\mathbb{R}^N} \widehat{\varphi}_k(w, u(w), y)\widehat{\lambda}(w)(dy)d\mu \uparrow \int_\Omega \int_{\mathbb{R}^N} \varphi(w, u(w), y)\widehat{\lambda}(w)(dy)d\mu.$$

Invoking Lemma I-1.193, we can find a sequence $k(n) \to \infty$ such that

$$\begin{aligned}
&\int_\Omega \int_{\mathbb{R}^N} \widehat{\varphi}_{k(n)}(w, u_n(w), y)\lambda^{v_n(w)}(dy)d\mu\\
\to &\int_\Omega \int_{\mathbb{R}^N} \varphi(w, u(w), y)\widehat{\lambda}(w)(dy)d\mu.
\end{aligned}$$

Since $\widehat{\varphi}_k \le \varphi$, if follows that

$$\begin{aligned}
&\int_\Omega \int_{\mathbb{R}^N} \varphi(w, u(w), y)\widehat{\lambda}(w)(dy)d\mu \le \vartheta,\\
\Rightarrow &\int_\Omega \varphi(w, u(w), \int_{\mathbb{R}^N} z\widehat{\lambda}(w)(dz))d\mu \le \vartheta\\
&\qquad\qquad\qquad\qquad\qquad\qquad \text{(by Jensen's inequality)}\\
\Rightarrow &L_\vartheta \text{ is closed.}
\end{aligned}$$

This proves the lower semicontinuity of I_φ. □

Remark 7.51 In fact the result is true if $\mathbb{R}^m$ is replaced by a separable Banach space X and $\mathbb{R}^N$ by a separable, reflexive Banach space Y (see Balder [32]).

An interesting consequence of Theorem 7.50, is the following result.

Corollary 7.52 *If $\Omega \subseteq \mathbb{R}^N$ is a bounded domain with Lipschitz boundary $\partial\Omega$ and $\varphi : \Omega \times \mathbb{R}^m \times \mathbb{R}^{mN} \to \overline{\mathbb{R}} = \mathbb{R} \cup +\infty$ is a measurable integrand such that*

(i) for a.a $z \in \Omega$, $(x, y) \to \varphi(z, x, y)$ is lower semicontinuous;
(ii) for a.a $z \in \Omega$, all $x \in \mathbb{R}^m$, $y \to \varphi(w, x, y)$ is convex;
(iii) $\eta(z) - c(|x| + |y|) \le \varphi(z, x, y)$ for a.a $z \in \Omega$, all $(x, y) \in \mathbb{R}^m \times \mathbb{R}^{mN}$, with $\eta \in L^1(\Omega)$, $c > 0$, then for any $1 < p < \infty$, the integral functional

$$u \to J_\varphi(u) = \int_\Omega \varphi(z, u(z), Du(z))dz$$

is sequentially lower semicontinuous on $W^{1,p}(\Omega, \mathbb{R}^m)_w$.

Now we will discuss necessary conditions for optimality in a calculus of variations problem. The basic such condition, is the vanishing of the Gateaux variations, which in the smooth case leads to the "Euler equation". For simplicity in the presentation, we will limit ourselves to the one dimensional case.

Let $T = [0, b]$ and consider the integral functional

$$I_\varphi(u) = \int_0^b \varphi(t, u(t), u'(t))dt$$

with $u : T \to \mathbb{R}^N$ being a function belonging to some suitable function space. The integrand $\varphi(t, x, y)$ is usually called the "Lagrangian" of the integral functional $I_\varphi(\cdot)$ (sometimes known as the "variational integral"). A major role in the minimization of I_φ, is played by the direchonal derivatives (if they exist) of I_φ. Let V be the function space on which I_φ is defined. For $u, h \in V$ we set

$$\vartheta(\lambda) = I_\varphi(u + \lambda h) \quad \text{for } |\lambda| \text{ small.}$$

We recall the following definition (see Definition I-6.1).

Definition 7.53 The "Gateaux derivative" of $I_\varphi(\cdot)$ at $u \in V$ in the direction $h \in V$, is an element $I'_\varphi(u) \in V^*$ such that

$$\langle I'_\varphi(u), h\rangle = I'_\varphi(u)(h) = \lim_{\lambda \to 0} \frac{I_\varphi(u + \lambda h) - I_\varphi(u)}{\lambda}.$$

Remark 7.54 The existence of this directional derivative, presupposes that $I_\varphi(u)$ is defined and that for $|\lambda|$ small $I_\varphi(u + \lambda h)$ is defined too and then $I'_\varphi(u)(h) = \frac{d}{d\lambda} I_\varphi(u + \lambda h)|_{\lambda=0}$, if this ardinary derivative exists at $\lambda = 0$. Clearly this derivative depends only on the local behavior of $I_\varphi(\cdot)$ near u.

Example 7.55 Let $\vartheta \in C[0, b]$ and consider the integral functional

$$I_\varphi(u) = \int_0^b \vartheta(t)\sqrt{1 + u(t)^2}dt, \ u \in V = C^1[0, b].$$

Then for $h \in V = C^1[0, b]$

$$\begin{aligned} I'_\varphi(u)(h) &= \frac{d}{d\lambda} I_\varphi(u+\lambda h)|_{\lambda=0} \\ &= \int_0^b \frac{d}{d\lambda}[\vartheta(t)\sqrt{1+(u+\lambda h)^2}]|_{\lambda=0}dt \\ &= \int_0^b \frac{\vartheta(t)u'h'}{\sqrt{1+u^2}}dt. \end{aligned}$$

This functional is known as the one dimensional mean curvature functional.

Now we assume that $\varphi : T \times \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ is an integrand such that for all $(x, y) \in \mathbb{R}^N \times \mathbb{R}^N$, $t \to \varphi(t, x, y)$ is measurable and for a,a $t \in T$, $(x, y) \to \varphi(t, x, y)$ is C^1. Evidently $\varphi(\cdot, \cdot, \cdot)$ is jointly measurable and superpositionally measurable.

We consider the following simple minimization problem

$$\left\{ \begin{array}{l} I_\varphi(u) = \int_0^b \varphi(t, u(t), u'(t))dt \to \inf, \\ u(0) = x_0,\ u(b) = x_1, u \in C^1(T, \mathbb{R}^N) \end{array} \right\}. \tag{7.25}$$

Recall that

$$\begin{aligned} \vartheta(\lambda) &= I_\varphi(u+\lambda h) \\ &= \int_0^b \varphi(t, u+\lambda h, u'+\lambda h')dt, \end{aligned}$$

If $k(t, \lambda) = \varphi(t, u(t) + \lambda h(t), u'(t) + \lambda h'(t))$, then

$$\begin{aligned} \frac{\partial}{\partial\lambda}k(t, \lambda) =& \varphi_x(t, u(t) + \lambda h(t), u'(t) + \lambda h'(t))u(t) \\ &+ \varphi_y(t, u(t) + \lambda h(t), u'(t) + \lambda h'(t))u'(t). \end{aligned}$$

Differentiating under the integral sign, we obtain

$$\vartheta'(0) = I'_\varphi(u)(h) = \int_0^b [\varphi'_x(t, u, u')h + \varphi'_y(t, u, u')h']dt. \tag{7.26}$$

Note that if $C_0^1(T, \mathbb{R}^N) = \{h \in C^1(T, \mathbb{R}^N) : h(0) = h(b) = 0\}$, then $u + \lambda h$ satisfies the boundary conditions of (7.25) (that is, $u + \lambda h$ is admissible).

Definition 7.56 We say that $u \in C^1(T, \mathbb{R}^N)$ is a "weak extremal" of (7.25) if $\vartheta'(0) = I'_\varphi(u)(h) = 0$ for all $h \in C_0^1(T, \mathbb{R}^N)$ (see (7.26)). This means that u is a "critical point" of $I_\varphi(\cdot)$ in all directions $h \in C_0^1(T, \mathbb{R}^N)$.

The next lemma, is known as the "Lagrange lemma".

Lemma 7.57 *If $\psi \in C(T)$ and $\int_0^b \psi(t)h(t)dt = 0$ for all $h \in C_0^1(T)$, then $\psi(t) = 0$ for all $t \in T$.*

Proof The continuity of $\psi(\cdot)$ implies that it is enough to show that $\psi(t) = 0$ for all $t \in (0, b)$. Arguing by contradiction, suppose that $\psi(t) \neq 0$ for some $t \in (0, b)$. To fix things, we assume that $\psi(t) > 0$. Let $\varepsilon > 0$ small so that $[t - \varepsilon, t + \varepsilon] \subseteq (0, b)$ and $0 < \eta \leq \min\{\psi(s) : s \in [t - \varepsilon, t + \varepsilon]\}$. Consider $h \in C_c^1(T)$, $h \geq 0$, $h \neq 0$ with supp $h \subseteq [t - \varepsilon, t + \varepsilon]$. We have

$$\begin{aligned}\int_0^b \psi(s)h(s)ds &= \int_{t-\varepsilon}^{t+\varepsilon} \psi(s)h(s)ds \\ &= \psi(\tau) \int_{t-\varepsilon}^{t+\varepsilon} h(s)ds \quad \text{for some } \tau \in [t - \varepsilon, t + \varepsilon] \\ &\qquad\qquad \text{(mean value theorem for integrals)} \\ &\geq \eta \int_{t-\varepsilon}^{t+\varepsilon} h(s)ds \\ &> 0,\end{aligned} \tag{7.27}$$

a contradiction. Therefore $\psi(t) = 0$ for all $t \in T$. □

The above proof suggests that we can improve this lemma as follows.

Lemma 7.58 *If $\psi \in L^1(T)$ and $\int_0^b \psi(s)h(s)ds = 0$ for all $h \in C_0^1(T)$, then $\psi(s) = 0$ for a.a $s \in T$.*

Proof As in the previous proof, let $T_\varepsilon = [t - \varepsilon, t + \varepsilon]$ ($\varepsilon > 0$). Then

$$\int_{T_\varepsilon} \psi ds = \int_0^b \psi \chi_{T_\varepsilon} ds. \tag{7.28}$$

Since $\chi_{T_\varepsilon} \in L^1(T)$ and $C_0^1(T)$ is dense in $L^1(T)$, from (7.28) it follows that

$$\int_{T_\varepsilon} \psi ds = 0. \tag{7.29}$$

For $\delta > 0$ small, consider $h_\delta \in C_0^1(T)$ be defined

$$h_\delta(s) = \begin{cases} 1 & \text{if } s \in T_\varepsilon \\ 0 \qquad \text{if } s \in [0, b] \setminus [t - \varepsilon - \delta, t + \varepsilon + \delta] \\ \text{linear} & \text{in between.} \end{cases} \tag{7.30}$$

The density of $C_0^1(T)$ in $C_0(T)$ implies

$$\int_0^b \psi h_\delta ds = 0 \quad \text{for all } \delta \to 0^+,$$

$$\Rightarrow \int_{t-\varepsilon}^{t+\varepsilon} \psi ds = 0 \quad \text{as } \delta \to 0^+ \text{ (see (7.29), (7.30))},$$

$$\Rightarrow \psi(s) = 0 \quad \text{for a.a } s \in T.$$

□

Now we can have the Euler equation in the so-called "Lagrangian form".

Theorem 7.59 *If $\widehat{u} \in C^1(T, \mathbb{R}^N)$ is a weak extremal of (7.25), then*

$$\frac{d}{dt}\varphi'_y(t, \widehat{u}(t), \widehat{u}'(t)) - \varphi'_x(t, \widehat{u}(t), \widehat{u}(t)) = 0$$

for a.a $t \in T$.

Proof We know that $\widehat{u}$ satisfies (7.26). Then after integration by parts, we have

$$\int_0^b (\varphi'_x(t, \widehat{u}, \widehat{u}') - \frac{d}{dt}\varphi'_y(t, \widehat{u}, \widehat{u}'))h dt = 0$$

$$\text{for all } h \in C_0^1(T, \mathbb{R}^N),$$

$$\Rightarrow \varphi'_x(t, \widehat{u}(t), \widehat{u}'(t)) - \frac{d}{dt}\varphi'_y(t, \widehat{u}(t), \widehat{u}'(t)) = 0 \text{ for a.a } t \in T$$

(see Lemma 7.58).

□

Remark 7.60 So, we have produced a second order equation, which is necessary for the extremality of $\widehat{u}$. This equation is known as the "Euler equation" in Lagrangian form of problem (7.1) (or "Euler-Lagrange equation" of (7.25)).

We have the following particular case of the Euler equation.

Corollary 7.61 *If the Lagrangian φ is independent of y and $\widehat{u}$ is a weak extremal of (7.25), then $\varphi'_x(t, \widehat{u}(t)) = 0$ for a.a $t \in T$.*

The next lemma, analogous to Lemma 7.57, is known as the "DuBois-Reymond Lemma".

Lemma 7.62 *If $\psi \in L^1(T)$ and $\int_0^b \psi(t)h'(t)dt = 0$ for all $h \in C_0^1(T)$, then $\psi(t) = \eta \in \mathbb{R}$ for a.a $t \in T$.*

Proof Let $v_0, v_1 \in T$ be two Lebesgue points of ψ and let $\eta = \psi(v_0)$. We assume that $v_0 < v_1$ and let $\varepsilon > 0$ so that $[v_0 - \varepsilon, v_1 + \varepsilon] \subseteq (0, b)$. We consider the continuous function $\xi(\cdot)$ defined on T by

$$\xi(t) = \begin{cases} 1 & if\ t \in [v_0, v_1] \\ 0 & if\ t \notin [v_0 - \varepsilon, v_1 + \varepsilon] \\ \frac{t - v_0 + \varepsilon}{\varepsilon} & if\ t \in [v_0 - \varepsilon, v_0] \\ \frac{v_1 + \varepsilon - t}{\varepsilon} & if\ t \notin [v_1, v_1 + \varepsilon]. \end{cases}$$

Then we check that

$$\int_0^b \psi(t)h'(t)dt = 0$$
$$\Rightarrow \frac{1}{\varepsilon}\int_{v_0-\varepsilon}^{v_0} \psi(t)dt - \frac{1}{\varepsilon}\int_{v_1}^{v_1+\varepsilon} \psi(t)dt = 0.$$

Letting $\varepsilon \to 0^+$ and recalling that v_0, v_1 are Lebesgue points, we obtain

$$\psi(v_0) = \psi(v_1),$$
$$\Rightarrow \psi(v_1) = \eta \quad \text{for all } v_1 > v_0.$$

If $v_1 < v_0$, then by reversing the roles in the above argument we have the same result. Since almost all points are Lebesgue points, we conclude that $\psi(\cdot)$ is constant. □

Using this lemma, we can have the following result.

Proposition 7.63 *If the Lagrangian φ is independent of x and $\widehat{u}$ is a weak extremal of (7.25), then $\varphi'_y(t, u'(t)) = \eta \in \mathbb{R}$ for a.a $t \in T$.*

Proof From (7.26) we have

$$\int_0^b \varphi'_y(t, u')h'dt \quad \text{for all } h \in C_0^1(T),$$
$$\Rightarrow \varphi'_y(t, u'(t)) = \eta \in \mathbb{R} \quad \text{for a.a } t \in T$$

(see Lemma 7.62).

□

In fact using Lemma 7.62 and an integration by parts, we can have the following variant of the Euler equation.

Theorem 7.64 *If $\widehat{u} \in C^1(T, \mathbb{R}^N)$ is a weak extremal of (7.25), then there exists $\eta \in \mathbb{R}^N$ such that for all $t \in T$, we have*

$$\varphi'_y(t, \widehat{u}(t), \widehat{u}'(t)) = \eta + \int_0^t \varphi'_x(s, \widehat{u}(s), \widehat{u}'(s))ds.$$

Corollary 7.65 *If the Lagrangion φ is independent of $t \in T$ and $\widehat{u}$ is a solution of (7.25), then $\varphi_y(\widehat{u}(t), \widehat{u}'(t))\widehat{u}'(t) - \varphi(\widehat{u}(t), \widehat{u}'(t)) = \eta \in \mathbb{R}$ for a.a $t \in T$.*

Remark 7.66 In physical lerms, Proposition 7.63 says that the momentum $p(t) = \varphi_x(t, \widehat{u}(t))$ is constant along a weak extremal and Corollary 7.65 says that the energy is constant along an extremal (conservation of energy). It can happen that the solution of the Eular equation exists is unique and corresponds to a local but not to a global minimizer of the energy functional. For example consider the following problem

$$\left\{ \begin{array}{l} I(u) = \int_0^1 u'(t)^3 dt \to \inf, \\ u(0) = 0,\ u(1) = 1. \end{array} \right\}. \tag{7.31}$$

The Euler equation for this problem is

$$\frac{d}{dt}(3x^2) = 0.$$

Then the solution of this equation which satisfies the boundary conditions is

$$\widehat{u}(t) = t,\ t \in [0, 1].$$

If $u \in C_0^1[0, 1]$, then $\widehat{u} + u$ is admissible and

$$\begin{aligned} I(\widehat{u} + u) &= \int_0^1 (\frac{d}{dt}(t + u(t)))^3 dt \\ &= I(\widehat{u}) + \int_0^1 (3u'(t)^2 + u'(t)^3) dt. \end{aligned}$$

Suppose that $\|u\|_{C^1} \leq 3$, then $3(u')^2 + (u')^3 \geq 0$ and so

$$\begin{aligned} &I(\widehat{u} + u) \geq I(\widehat{u}), \\ \Rightarrow &\widehat{u} \text{ is a local minimum of (7.31).} \end{aligned}$$

On the other hand, if

$$u_n(t) = \left\{ \begin{array}{ll} -\sqrt{n} & if\ 0 \leq t \leq \frac{1}{n} \\ \sqrt{n}(n-1)^{-1} & if\ \frac{1}{n} < t \leq 1, \end{array} \right.$$

then $I(\widehat{u} + u_n) \to -\infty$ and so $\inf I = -\infty$, that is, there is no global minimum. Also, it can happen that the solution of the Euler equation exists, it is unique, corresponds to a global minimum of the energy functional but it is not of class C^1. To see this consider the problem

$$\left\{ \begin{array}{l} I(u) = \displaystyle\int_0^1 t^{\frac{2}{3}} u'(t)^2 dt \to \inf, \\ u(0) = 0,\ u(1) = 1. \end{array} \right\}. \tag{7.32}$$

For (7.32) the Euler equation is

$$\frac{d}{dt}(2t^{\frac{2}{3}} u') = 0.$$

Taking into account the boundary conditions, this equation has a unique solution $\widehat{u}(t) = t^{\frac{1}{3}}$, which corresponds to a global minimum of $I(\cdot)$, but it is not C^1. This example is due to Hibert.

Another interesting phenomenon that we would like to mention is the so-called "Lavrentiev gap phenomenon". It illustrates that some of the hypotheses used in order to have existence results, derive the Euler equation or to obtain regularity results for the minimizers, are optimal. It stresses that a careful choice of the space of admissible functions is neccessary. The Lavrentiev phenomenon says that it can happen that

$$\begin{aligned} &\inf\{I_\varphi(u) : u \in W^{1,\infty}(0,1) : u(0) = 0, u(1) = 1\} \\ &> \inf\{I_\varphi(u) : u \in W^{1,1}(0,1) : u(0) = 0, u(1) = 1\}. \end{aligned}$$

Recall that $W^{1,\infty}(0,1)$ is the space of Lipschitz functions while the elements of $W^{1,1}(0,1)$ are absolutely continuous functions with $u' \in L^1(0,1)$ (recall that absolutely continuous functions are differentiable almost everywhere). Therefore the boundary conditions are well-defined for $u \in W^{1,1}(0,b)$. We consider the integral functional

$$I_\varphi(u) = \int_0^1 (u^3 - t)^2 (u')^2 dt.$$

If $C = \{u \in W^{1,1}(0,1) : u(0) = 0, u(1) = 1\}$, then $\widehat{u}(t) = t^{\frac{1}{3}}$ is a minimizer of $I_\varphi(\cdot)$ on C. Indeed $I_\varphi(u) \geq 0$ for all $u \in C$ and $I_\varphi(\widehat{u}) = 0$. Moreover, we have

$$\begin{aligned} &\inf\{I_\varphi(u) : u \in W^{1,\infty}(0,1) : u(0) = 0, u(1) = 1\} \\ &\geq c_0 > 0 = \inf\{I_\varphi(u) : u \in W^{1,1}(0,1) : u(0) = 0, u(1) = 1\}. \end{aligned}$$

In fact, for any sequence of Lipschitz functions (that is, of functions in $W^{1,\infty}(0,1)$) $\{u_n\}_{n\in\mathbb{N}}$ with $u_n \to \widehat{u}$ uniformly, we have that $I_\varphi(u_n) \to \infty$.

A basic tool in the existence theory for calculus of variations problems, is the so-called "Weicrstrass-Tonelli Theorem", which says the following (see Proposition I-1.180).

Theorem 7.67 *If X is a Hausdorff topological space and $\varphi : X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous, then $\varphi(\cdot)$ attains its infimum on every compact subset of X.*

Remark 7.68 If $\varphi : X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous and one the sublevel sets $L_\lambda = \{u \in X : \varphi(u) \leq \lambda\}$, $\lambda \in \mathbb{R}$, is nonempty and compact, then $\varphi(\cdot)$ attains its infimum on X. A function $\varphi(\cdot)$ which has compact sublevel sets, is said to be "coercive" or "inf-compact".

The above remark, leads to the following corollary.

Corollary 7.69 *If X is a reflexive Banach space and $\varphi : X \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous, convex, bounded below and $\varphi(u) \to +\infty$ as $\|u\|_X \to +\infty$, then $\varphi(\cdot)$ attains its infimun on X.*

The use of Theorem 7.67 in problems of the calculus of variations, is usually known as the "Direct Method of the Calculus of Variations". A closer look at the requirements of Theorem 7.67, reveals that we have two compcting conditions, namely the lower semicontinuity of $\varphi(\cdot)$ and the compactness of the set. If the topdogy of X is rich, then it is easy to verify lower semicontinuity but it is more difficult to have compactness. So, we need to choose carefully the topology of X in order to achieve the right equilibrium between the two requirements. The lower semicontinuity results proved in the first part of this section, are helpful in this direction.

We consider problem (7.25) and for the Lagrangian $\varphi(t, x, y)$ we assume the following

(i) for all $(x, y) \in \mathbb{R}^N \times \mathbb{R}^N$, $t \to \varphi(t, x, y)$ is measurable;
(ii) for a.a $t \in T$, $(x, y) \to \varphi(t, x, y)$ is lower semicontinuous;
(iii) for a.a $t \in T$, all $x \in \mathbb{R}^N$, $y \to \varphi(t, x, y)$ is convex;
(iv) there exists a function $\vartheta : \mathbb{R}_1 \to \mathbb{R}_1$ such that

$$\lim_{t \to +\infty} \frac{\vartheta(t)}{t} = +\infty$$

and $\vartheta(|y|) \leq \varphi(t, x, y)$ for a.a. $t \in T$, all $(x, y) \in \mathbb{R}^N \times \mathbb{R}^N$;
(v) for every $M > 0$, there exists $\eta_M \in L^1(T)$ such that

$$|\varphi(t, x, y)| \leq \eta_M(t)$$

for a.a $t \in T$, all $|x|, |y| \leq M$.

Then we have the following existence theorem for problem (7.25).

Theorem 7.70 *If the Lagrangian $\varphi : T \times \mathbb{R}^N \times \mathbb{R}^N$ satisfies conditions $(i) \to (v)$ above, then problem (7.25) admits a solution $\widehat{u} \in W^{1,1}((0, b), \mathbb{R}^N)$*

Proof First note that we can find $M > 0$ such that

$$\varphi(t,x,y) \geq |y| \quad \text{for a.a } t \in T, \text{ all } x \in \mathbb{R}^N, \text{ all } |y| \geq M.$$

This combined with hypothesis (v), lead to

$$\varphi(t,x,y) \geq -\eta_M(t) \quad \text{for a.a } t \in T, \text{ all } x,\ y \in \mathbb{R}^N. \tag{7.33}$$

Then the infimum in (7.25) is finite and let $\{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,1}((0,b),\mathbb{R}^N)$ be a minimizing sequence. Then from (iv) it follows that

$$\begin{aligned} &\sup_{n\in\mathbb{N}} \int_0^b \vartheta(|u_n'|)dt < \infty, \\ &\Rightarrow \{u_n'\}_{n\in\mathbb{N}} \subseteq L^1(T,\mathbb{R}^N) \quad \text{is uniformly integrable} \\ &\qquad\qquad\qquad\qquad\qquad\qquad \text{(see Remark 7.2)}. \end{aligned} \tag{7.34}$$

We have

$$u_n(t) = x_0 + \int_0^t u_n'(s)ds \quad \text{for all } n \in \mathbb{N}, \text{ all } t \in T.$$

Hence by (7.34) and the Arzela-Ascoli theorem we have that

$$\{u_n\}_{n\in\mathbb{N}} \subseteq C(T,\mathbb{R}^N) \text{ is relatively compact.} \tag{7.35}$$

Then on account of (7.34), (7.35) and the Dunford-Pettis theorem, we may assume that

$$u_n \to \widehat{u} \text{ in } C(T,\mathbb{R}^N),\ u_n' \xrightarrow{w} v \text{ in } L^1(T,\mathbb{R}^N). \tag{7.36}$$

Clearly $v = \widehat{u}'$ and beacuse of (7.33) and (7.36), we can use Theorem 7.50 to obtain

$$I_\varphi(\widehat{u}) \leq \liminf_{n\to\infty} I_\varphi(u_n).$$

From (7.36), we see that $\widehat{u}(0) = x_0$, $\widehat{u}(1) = x_1$. Therefore $\widehat{u}$ is admissible and so a solution of (7.25). □

Remark 7.71 For $\mu \in \mathbb{R}$ and $1 < p$, consider the following generalization of problem (7.32) (with $N = 1$)

$$\left\{ \begin{aligned} &I_\varphi(u) = \int_0^1 t^\mu |u'|^p dt \to \inf, \\ &u(0) = x_0,\ u(1) = x_1. \end{aligned} \right\}. \tag{7.37}$$

The ambient space is $W^{1,1}(0,b)$. Weierstrass observed that if $\mu = p = 2$, then problem (7.37) has no solution when $x_0 \neq x_1$. To see this consider the following sequence of functions in $W^{1,1}(0,b)$:

$$u_n(t) = x_0 + (x_1 - x_0)\frac{\tan^{-1}(nt)}{\tan^{-1}n}.$$

Then we have

$$\begin{aligned} &I_\varphi(u_n) \to 0, \\ \Rightarrow &\inf\{I_\varphi(u) : u \in W^{1,1}(0,1) : u(0) = x_0,\ u(1) = x_1\} = 0. \end{aligned}$$

On the other hand, no function $u \in W^{1,1}(0,1)$ with $u(0) \neq u(1)$ satisfies $I_\varphi(u) = 0$, since this would imply $u' = 0$ (here $\varphi(t,x,y) = \varphi(t,y) = t^p|y|^p$) and so $u(0) = u(1)$.

We will close this section, by using the direct method of the calculus of to prove existence of solutions for a multidimensional problem.

So, let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with Lipschitz boundary $\partial\Omega$. The problem under consideration is the following

$$\inf\left\{\int_\Omega \varphi(z, Du)dz + \int_\Omega \psi(z,u)dz : u \in K\right\} = m \tag{7.38}$$

with $1 < p$ and $K \subseteq W^{1,p}(\Omega, \mathbb{R}^N)$ a nonemply, sequentially closed set.

The hypotheses on the two integrands φ and ψ are the following:

$\mathbf{H_\varphi}$: $\varphi : \Omega \times \mathbb{R}^N \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a convex normal integrand such that

$$\varphi(z,y) \geq c|y|^p - a(z)$$

for a.a $z \in \Omega$, all $y \in \mathbb{R}^N$, with $c \in \mathbb{R}$ and $a \in L^1(\Omega)$.

$\mathbf{H_\psi}$: $\psi : \Omega \times \mathbb{R} \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a normal integrand such that

$$\psi(z,x) \geq \widehat{c}|x|^p - \widehat{a}(z)$$

for a.a $z \in \Omega$, all $x \in \mathbb{R}^N$, with $\widehat{c} \in \mathbb{R}$ and $\widehat{a} \in L^1(\Omega)$.

We introduce the integral functionals $I_\varphi : W^{1,p}(\Omega) \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ and $J_\psi : W^{1,p}(\Omega) \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$, defined by

$$I_\varphi(u) = \int_\Omega \varphi(z, Du)dz \quad \text{and} \quad J_\psi(u) = \int_\Omega \psi(z,u)dz \quad \text{for all } u \in W^{1,p}(\Omega).$$

Proposition 7.72 *If hypotheses H_φ hold, then $I_\varphi(\cdot)$ is weakly lower semicontinuous on $W^{1,p}(\Omega)$.*

Proof We need to show that for every $\lambda \in \mathbb{R}$, the sublevel set

$$L_\lambda = \{u \in W^{1,p}(\Omega) : I_\varphi(u) \leq \lambda\}$$

is weakly closed. The integral functional $I_\varphi(\cdot)$ is convex and so by Mazur's theorem (see Theorem I-3.105), it suffices to check that L_λ is closed. So, let $\{u_n\}_{n\in\mathbb{N}} \subseteq L_\lambda$ and assume that $u_n \to u$ in $W^{1,p}(\Omega)$. Using Fatou's lemma we have

$$\begin{aligned}
&\int_\Omega (\varphi(z, Du) - c|Du|^p)dz \\
&\leq \liminf_{n\to\infty} \int_\Omega (\varphi(z, Du_n) - c|Du_n|^p)dz \leq \lambda - \int_\Omega c|Du|^p dz \\
&\Rightarrow u \in L_\lambda \quad \text{and so } L_\lambda \text{ is closed.}
\end{aligned}$$

We conclude that L_λ is w-closed and so $I_\varphi(\cdot)$ is weakly lower semicontinuous. □

Proposition 7.73 *If hypotheses H_ψ hold, then $J_\psi(\cdot)$ is sequentially weakly lower semicontinuous on $W^{1,p}(\Omega)$.*

Proof We need to show that for every $\lambda \in \mathbb{R}$, the sublevel set

$$\widehat{L}_\lambda = \{u \in W^{1,p}(\Omega) : J_\psi(u) \leq \lambda\}$$

is sequentially weakly closed in $W^{1,p}(\Omega)$.

Consider $\{u_n\}_{n\in\mathbb{N}} \subseteq L_\lambda$ such that $u_n \xrightarrow{w} u$ in $W^{1,p}(\Omega)$. From the Rellich-Kondrachov embedding theorem (see Theorem I-4.115), we have

$$u_n \to u \text{ in } L^p(\Omega).$$

Then as above, via Fatou's lemma, we have

$$\begin{aligned}
&J_\psi(u) \leq \lambda \\
&\Rightarrow L_\lambda \subseteq W^{1,p}(\Omega) \quad \text{is sequentially weakly closed.}
\end{aligned}$$

We conclude that $J_\psi(\cdot)$ is sequentially weakly lower semicontinuous on $W^{1,p}(\Omega)$. □

Using these two propositions, we can prove the following existence theorem for problem (7.38).

Theorem 7.74 *If hypotheses H_φ and H_ψ hold with c, $\widehat{c} > 0$ and $K \subseteq W^{1,p}(\Omega)$ is sequentioilly weakly closed, then*

(a) problem (7.38) has a solution;

(b) if $K \subseteq W^{1,p}(\Omega)$ is convex and for a.a $z \in \Omega$, $\psi(z,\cdot)$ is strictly convex, then the solution from part (a) is unique.

Proof

(a) Since c, $\widehat{c} > 0$, we have $m > -\infty$. Let $\{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega)$ be a minimizing sequence for problem (7.38). Then $\{u_n\}_{n\in\mathbb{N}} \subseteq W^{1,p}(\Omega)$ is bounded. So, we may assume that

$$\begin{aligned} & u_n \xrightarrow{w} \widehat{u} \text{ in } W^{1,p}(\Omega) \\ \Rightarrow & I_\varphi(\widehat{u}) + J_\psi(\widehat{u}) \leq \liminf_{n\to\infty} [I_\varphi(u_n) + J_\psi(u_n)] = m \end{aligned} \tag{7.39}$$

(see Propositions 7.72, 7.73).

Also $\widehat{u} \in K$ since $K \subseteq W^{1,p}(\Omega)$ is sequentially w-closed. Then from (7.39), we have

$$\begin{aligned} & I_\varphi(\widehat{u}) + J_\psi(\widehat{u}) = m \\ \Rightarrow & \widehat{u} \in W^{1,p}(\Omega) \text{ is a solution of (7.38).} \end{aligned}$$

(b) Let $\chi_K(\cdot)$ be the indicator function of the set K, that is,

$$\chi_K(u) = \begin{cases} 0 & \text{if } u \in K \\ +\infty \text{ if } u \notin K, \ u \in W^{1,p}(\Omega). \end{cases}$$

Then $m = \inf\{I_\varphi(u) + J_\psi(u) + \chi_K(u) : u \in W^{1,p}(\Omega)\}$. The functional $u \to I_\varphi(u) + J_\psi(u) + \chi_K(u)$ is strictly convex in $W^{1,p}(\Omega)$ and so any minimizer of it is unique.

□

7.2 Optimal Control-Relaxation

In this section we study the following optimal control problem

$$\left\{ \begin{array}{l} I_L(x,u) = \int_0^b L(t,x(t),u(t))dt \to \inf = \widehat{m}_L, \\ \text{subject to :} \\ -x'(t) \in A(x(t)) + f(t,x(t),u(t)) \text{ for a.a } t \in T = [0,b], \\ x(0) = x_0, \ u(t) \in U(t,x(t)) \text{ for a.a } t \in T. \end{array} \right\}. \tag{7.40}$$

In this problem in the dynamics, we have a multivalued maximal monotone map $A : \mathbb{R}^N \to 2^{\mathbb{R}^N}$ which is not defined on all of $\mathbb{R}^N$ (that is, $D(A) \neq \mathbb{R}^N$). This way we incorporate in our framework systems with unilateral constraints (differential

variational inequalities). The control constraint set $U(t,x)$ depends on the state of the system and this means that we have a priori feedback in the system, a setting of interest in engineering and economic problems. First we will deal with the problem of existence of an optimal "state-control" pair $(\widehat{x},\widehat{u})$, that is an admissible pair $(\widehat{x},\widehat{u})$ such that

$$I_L(\widehat{x},\widehat{u}) = \widehat{m}_L.$$

The existence theory, is based on the so-called "reduction method". According to this method, the original optimal control problem is reduced to an equivalent calculus of variations problem with multivalued dynamics. This is done by appropriate elimination of the control variable $u \in \mathbb{R}^m$. This method works and produces an optimal pair $(\widehat{x},\widehat{u})$, if we have "enough" convex structure in the system and the cost functional. In the absence of such a convex structure, a minimizing sequence of admissible "state-control" pairs need not converge to an admissible pair. To rectify this unpleasant situation, we need to pass to a "convexified" version of the problem, known as the "relaxed problem", which captures the asymptotic behavior of the minimizing pairs. The relaxation process is a delicate one since we have to balance between competing requirements. More precisely, we want the relaxed problem to have the following properties:

(a) Every original pair is also a relaxed pair (this means that the original problem, is embedded in the relaxed one).
(b) Every relaxed state can be approximated by original ones (with this requirement we want to make sure that the "convexification" did not augment the system too much).
(c) The values of the relaxed and original problems are equal and the relaxed problem admits an optimal pair.

The first two requirements involve the dynamics of the problem, while the third involves the cost functional. Any relaxation method which meets these three requirements is said to be "admissible".

In this section, first we develop the existence theory and then we present two admissible relaxation methods. The first method is an outgrowth of the existence theory, while the second method uses the theory of Young measures(parametrized measures).

For the existence theory, the hypotheses on the data of (7.40) are the following:

$\mathbf{H_0}$: $A : D(A) \subseteq \mathbb{R}^N \to 2^{\mathbb{R}^N}$ is a maximal monotone map, $0 \in A(0)$ and $x_0 \in \overline{D(A)}$.

Remark 7.75 We do not assume that $D(A) = \mathbb{R}^N$. This way we incorporate in our analysis also systems with unilateral constraints (differential variational inequalities).

$\mathbf{H_1}$: $f : T \times \mathbb{R}^N \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^N)$ is a map such that

(i) for all $(x, u) \in \mathbb{R}^N \times \mathbb{R}^m$, $t \to f(t, x)u$ is measurable;
(ii) for a.a $t \in T$ and all $u \in \mathbb{R}^m$, $x \to f(t, x)u$ is continuous;
(iii) $\|f(t, x)\|_{\mathcal{L}} \leq a(t)[1 + |x|]$ for a.a $t \in T$, all $x \in \mathbb{R}^N$, with $a \in L^1(T)_+$.

Remark 7.76 Using hypotheses $H_1(ii)$, (iii), we obtain that $(x, y) \to f(t, x)u$ is continuous from $\mathbb{R}^N \times \mathbb{R}^m$ into $\mathbb{R}^N$.

$\mathbf{H}_2$: $U : T \times \mathbb{R}^N \to P_k(\mathbb{R}^m) = \{C \subseteq \mathbb{R}^m : \text{nonempty, compact}\}$ is a multifunction such that

(i) for all $x \in \mathbb{R}^N$, $t \to U(t, x)$ is graph measurable;
(ii) for a.a $t \in T$, $x \to U(t, x)$ is continuous;
(iii) $|U(t, x)| = \sup\{|u| : u \in U(t, x)\} \leq c_1(1 + |x|)$ for a.a $t \in T$, all $x \in \mathbb{R}^N$, with $c_1 > 0$.

$\mathbf{H}_3$: $L : T \times \mathbb{R}^N \times \mathbb{R}^m \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ is a normal integrand (see Definition 7.4 (b)) such that

$$\eta(t) - c|x| \leq L(t, x, u)$$

for a.a $t \in T$, all $x \in \mathbb{R}^N$, all $u \in U(t, x)$ with $\eta \in L^1(T)$, $c > 0$.

As we already mentioned in the beginning of this section, in order to have existence of optimal pairs, we need convex structure in the problem. The next hypothesis, provides the needed structure for problem (7.40).

$\mathbf{H}_4$: For a.a $t \in T$ and all $x \in \mathbb{R}^N$, the set

$$Q(t, x) = \left\{ \begin{array}{l} (v, \vartheta) \in \mathbb{R}^N \times \mathbb{R} : \ -v \in A(x) + f(t, x)u \\ u \in U(t, x) \\ L(t, x, u) \leq \vartheta \end{array} \right\}.$$

is convex and $A^\circ(x) = p_{A(x)}(0)$ is bounded on compact subsets of $D(A)$.

Remark 7.77 Evidently, if the control constraint multifunction $U(\cdot, \cdot)$ is convex valued and $L(t, x, \cdot)$ is convex, then hypothesis H_4 is satisfied.

To establish existence of optimal state-control pairs, we will follow the so-called "reduction method". The idea of this method is simple and elegant. We will use hypothesis H_4 to pass to a control-free (deparametrized) variational problem (that is, a calculus of variations problem), which we solve using the direct method. Finally using a measurable selection theorem we associate to the optimal state an optimal control.

Theorem 7.78 *If hypotheses H_0, H_1, H_2, H_3, H_4 hold and $\widehat{m}_L < \infty$, then there exists $(\widehat{x}, \widehat{u})$ an admissible state-control pair such that*

$$I_L(\widehat{x}, \widehat{u}) = \widehat{m}_L.$$

(an optimal pair).

Proof Let $K : T \times \mathbb{R}^N \times \mathbb{R}^N \to 2^{\mathbb{R}^m}$ be defined by

$$K(t, x, v) = \{u \in U(t, x) : -v \in A(x) + f(t, x)u\}.$$

So, this multifunction gives all admissible controls which at time $t \in T$ and when the state of the system is x, generate the velocity v.

We have

$$\begin{aligned} GrK =&\{(t, x, v, u) \in T \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^m : \\ &(t, x, u) \in GrU,\ (x, -v - f(t, x)u) \in GrA\}. \end{aligned}$$

Note that since U is compact-valued, $U(t, \cdot)$ is also h-continuous (see Corollary I-5.50). Therefore $(t, u) \to U(t, u)$ is graph measurable. It follows that

$$GrK \in B(T) \times B(\mathbb{R}^N) \times B(\mathbb{R}^N) \times B(\mathbb{R}^m)$$

(recall that since A is maximal monotone, GrA is closed). We introduce the function

$$p(t, x, v) = \inf\{L(t, x, u) : u \in K(t, x, v)\}.$$

Then $p(t, x, v)$ represents the minimum instantaneous cost needed to achieve velocity $v \in \mathbb{R}^N$, when the state of the system is $x \in \mathbb{R}^N$. As always, we use the convention that $\inf \emptyset = +\infty$. Therefore p is $\overline{\mathbb{R}}$-valued.

Next we establish the properties of this function.

Claim A $p(\cdot, \cdot, \cdot)$ is Borel measurable.

Let $\eta \in \mathbb{R}$. We have

$$\begin{aligned} &\{(t, x, v) \in T \times \mathbb{R}^N \times \mathbb{R}^N : p(t, x, v) < \eta\}, \\ &= proj_{T \times \mathbb{R}^N \times \mathbb{R}^N}\{(t, x, v, u) \in GrK : L(t, x, u) < \eta\}, \\ &\in B(T) \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^N) \text{ (see Theorem I-5.76)} \\ &\Rightarrow p(\cdot, \cdot, \cdot) \text{ is Borel measurable.} \end{aligned}$$

This proves Claim A.

Claim B For a.a $t \in T$, $(x, v) \to p(t, x, v)$ is lower semicontinuous.

We need to show that for every $\eta \in \mathbb{R}$, the sublevel set

$$M_\eta = \{(x, v) \in \mathbb{R}^N \times \mathbb{R}^N : p(t, x, v) \leq \eta\}$$

is closed. To this end, let $\{(x_n, v_n)\}_{n \in \mathbb{N}} \subseteq M_\eta$ and assume that

$$(x_n, v_n) \to (x, v) \text{ in } \mathbb{R}^N \times \mathbb{R}^N.$$

We can find $u_n \in K(t, x_n, v_n)$ such that

$$L(t, x_n, x_n) \leq p(t, x_n, v_n) + \frac{1}{n} \text{ for all } n \in \mathbb{N}.$$

Then $u_n \in U(t, x_n)$ and $-v_n \in A(x_n) + f(t, x_n)u_n$ for all $n \in \mathbb{N}$.

On account of hypothesis $H_2(iii)$, we see that $\{u_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}^m$ is bounded. So, we may assume that $u_n \to u$ and hypothesis $H_2(ii)$ implies that $u \in U(t, x)$. Also, hypothesis $H_1(ii)$ implies that

$$f(t, x_n)u_n \to f(t, x)u \text{ in } \mathbb{R}^N.$$

Recall that $GrA \subseteq \mathbb{R}^N \times \mathbb{R}^N$ is closed (see Proposition I-7.35). Since $(x_n, -v_n - f(t, x_n)u_n) \in GrA$, it follows that

$$\begin{aligned} &(x, -v - f(t, x)u) \in GrA, \\ &\Rightarrow -v \in A(x) + f(t, x)u, \\ &\Rightarrow u \in K(t, x, v). \end{aligned}$$

Also hypothesis H_3 implies

$$\begin{aligned} &L(t, x, u) \leq \liminf_{n\to\infty} L(t, x_n, u_n) \leq \liminf_{n\to\infty} p(t, x_n, v_n) \leq \eta, \\ &\Rightarrow p(t, x, v) \leq \eta, \\ &\Rightarrow (x, v) \in M_\eta \text{ and so } p(t, \cdot, \cdot) \text{ is lower semicontinuous.} \end{aligned}$$

This proves Claim B.

Claim C For a.a $t \in T$, all $x \in \mathbb{R}^N$, $p(t, x, \cdot)$ is convex.

We have

$$\begin{aligned} epip(t, x, \cdot) &= \{(v, \eta) \in \mathbb{R}^N \times \mathbb{R} : p(t, x, v) \leq \eta\}, \\ &= \bigcap_{\varepsilon>0} \{(v, \eta) \in \mathbb{R}^N \times \mathbb{R} : -v \in A(x) + f(t, x)u, \\ &\qquad u \in U(t, x), L(t, x, u) \leq \eta + \varepsilon\}, \\ &\Rightarrow epip(t, x, \cdot) \text{ is convex (as intersection of convex sets, see } H_4), \\ &\Rightarrow p(t, x, \cdot) \text{ is convex.} \end{aligned}$$

This proves Claim C.

Now let $\{(x_n, u_n)\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N) \times L^1(T, \mathbb{R}^m)$ be a minimizing sequence of admissible state-control pairs for problem (7.40). Note that if $F(t, x) = f(t, x)U(t, x)$, then $\{x_n\}_{n\in\mathbb{N}}$ are solutions of the differential inclusion

$$-y'(t) \in A(y(t)) + F(t, y(t)) \text{ for a.a } t \in T,\ y(0) = x_0 \in \overline{D(A)}. \tag{7.41}$$

We know that the solution set of (7.41) is relatively compact in $C(T, \mathbb{R}^N)$. Using hypotheses H_4 we infer that $\{x_n'\}_{n\in\mathbb{N}} \subseteq L^1(T, \mathbb{R}^N)$ is uniformly integrable. So, we may assume that

$$x_n \to \widehat{x} \text{ in } C(T, \mathbb{R}^N),\ x_n' \xrightarrow{w} w \text{ in } L^1(T, \mathbb{R}^N).$$

Clearly $w = \widehat{x}'$. From Theorem 7.50, we have

$$\begin{aligned} -\infty < \int_0^b p(t, \widehat{x}, \widehat{x}')dt &\leq \liminf_{n\to\infty} \int_0^b p(t, x_n, x_n')dt \\ &\leq \liminf_{n\to\infty} \int_0^b L(t, x_n, u_n)dt = \widehat{m}_L < \infty, \\ \Rightarrow p(t, \widehat{x}(t), \widehat{x}'(t)) &\in \mathbb{R} \text{ for a.a } t \in T. \end{aligned}$$

Using the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), we can find $\widehat{u}_k : T \to \mathbb{R}^m$ measurable map such that

$$\widehat{u}_k(t) \in K(t, \widehat{x}(t), \widehat{x}'(t)) \text{ for a.a } t \in T, \tag{7.42}$$

$$L(t, \widehat{x}(t), \widehat{u}_k(t)) \leq p(t, \widehat{x}(t), \widehat{x}'(t)) + \frac{1}{k} \text{ for a.a } t \in T, \text{ all } k \in \mathbb{N}. \tag{7.43}$$

On account of hypotheses H_3 and (7.42), we see that

$$\{\widehat{L}_k(\cdot) = L(\cdot, \widehat{x}(\cdot), \widehat{u}_k(\cdot))\}_{k\in\mathbb{N}} \subseteq L^1(T) \text{ is uniformly integrable.}$$

Therefore, we may assume that

$$\widehat{L}_k \xrightarrow{w} \widehat{L} \text{ in } L^1(T).$$

From (7.42) we have

$$-\widehat{x}'(t) = e_k(t) + f(t, \widehat{x}(t))\widehat{u}_k(t) \text{ for a.a } t \in T,\ e_k(t) \in A(\widehat{x}(t)) \text{ for a.a } t \in T.$$

We may assume that

$$\begin{aligned} &e_k \xrightarrow{w} e \text{ in } L^1(T, \mathbb{R}^N), \\ &\Rightarrow e(t) \in A(\widehat{x}(t)) \text{ for a.a } t \in T \text{ (see Theorem I-8.38).} \end{aligned}$$

Also $\{h_k(\cdot) = f(\cdot, \widehat{x}(\cdot))\widehat{u}_k(\cdot)\}_{k\in\mathbb{N}} \subseteq L^1(T, \mathbb{R}^N)$ is uniformly integrable and so we may assume that

$$h_k \xrightarrow{w} h \text{ in } L^1(T, \mathbb{R}^N).$$

Then using hypotheses H_4 and Theorem I-8.38, we have

$$(-\widehat{x}'(t) - h(t), \widehat{L}(t)) \in Q(t, \widehat{x}(t)) \text{ for a.a } t \in T.$$

Via the Yankov-von Neumann-Aumann selection theorem, we obtain $\widehat{u} : T \to \mathbb{R}^m$ measurable such that

$$\begin{aligned}
&\widehat{u}(t) \in U(t, \widehat{x}(t)) \text{ for a.a } t \in T,\\
&-\widehat{x}'(t) \in A(\widehat{x}(t)) + f(t, \widehat{x}(t))\widehat{u}(t) \text{ for a.a } t \in T,\\
&L(t, \widehat{x}(u), \widehat{u}(t)) \leq \widehat{L}(t) \text{ for a.a } t \in T.
\end{aligned}$$

Hence $(\widehat{x}, \widehat{u})$ is an admissible state-control pair and

$$\begin{aligned}
&\int_0^b L(t, \widehat{x}(t), \widehat{u}(t))dt \leq \int_\Omega p(t, \widehat{x}(t), \widehat{x}'(t))dt \leq \widehat{m}_L.\\
&\Rightarrow I_L(\widehat{x}, \widehat{u}) = \widehat{m}_L.
\end{aligned}$$

□

Next we will investigate what can be said when the "convexity" hypothesis H_4 is no longer true. In this case as we already said, we need to augment the problem in order to capture the asymptotic behavior of the minimizing sequences. The augmented problem, is known as the "relaxed problem".

We need to stengthen the hypotheses on the data of (7.40).

$\mathbf{H}_1'$: $f : T \times \mathbb{R}^N \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^N)$ is a map such that

(i) for all $(x, u) \in \mathbb{R}^m \times \mathbb{R}^N$, $t \to f(t, x)u$ is measurable;
(ii) for every $\varrho > 0$, there exists $\vartheta_\varrho \in L^1(T)$ such that

$$\|f(t, x) - f(t, y)\|_{\mathcal{L}} \leq \vartheta_\varrho(t)|x - y|$$

for a.a $t \in T$, all $|x|, |y| \leq \varrho$;
(iii) $\|f(t, x)\|_{\mathcal{L}} \leq a(t)[1 + |x|]$ for a.a $t \in T$, all $x \in \mathbb{R}^N$ with $a \in L^1(T)$.

$\mathbf{H}_2'$: $U : T \times \mathbb{R}^N \to P_f(\mathbb{R}^m)$ is a multifunction such that

(i) for all $x \in \mathbb{R}^N$, $t \to U(t, x)$ is graph measurable;
(ii) for every $\varrho > 0$, there exists $k_\varrho \in L^\infty(T)$ such that

$$h(U(t, x), U(t, y)) \leq k_\varrho(t)|x - y|$$

for a.a $t \in T$, all $|x|, |y| \leq \varrho$;

(iii) there exists $\widehat{a} \in L^{\infty}(T)$ such that

$$|U(t,x)| = \sup\{|u| : u \in U(t,x)\} \leq \widehat{a}(t)$$

for a.a $t \in T$, all $x, y \in \mathbb{R}^N$.

$\mathbf{H}_3'$: $L : T \times \mathbb{R}^N \times \mathbb{R}^m \to \mathbb{R}$ is an integrand such that

(i) for all $(x,u) \in \mathbb{R}^N \times \mathbb{R}^m$, $t \to L(t,x,u)$ is measurable;
(ii) for every $\varrho > 0$, there exists $\gamma_r \in L^1(T)$ such that

$$|L(t,x,u) - L(t,y,v)| \leq \gamma_r(t)[|x-y| + |u-v|]$$

for a.a $t \in T$, all $|x|, |y|, |v| \leq \varrho$;
(iii) for every $\varrho > 0$, there exists $a_{\varrho}^* \in L^1(T)$ such that

$$|L(t,x,u)| \leq a_{\varrho}^*(t)$$

for a.a $t \in T$, all $|x|, |u| \leq \varrho$.

We consider the "convexified" differential inclusion governing the system, namely

$$\left\{ \begin{array}{l} -x'(t) \in A(x(t)) + f(t,y(t))u(t) \text{ for a.a } t \in T, \\ x(0) = x_0, u(t) \in convU(t,x(t)) \text{ for a.a } t \in T. \end{array} \right\} \tag{7.44}$$

Then we introduce the sets of admissible state-control pairs for both problems (7.40) and (7.44). So, we define the sets

$$P = \{(x,u) \in C(T,\mathbb{R}^N) \times L^1(T,\mathbb{R}^m) : \text{ the pair } (x,u) \text{ is admissible for (7.40)}\},$$

$$P_c = \{(x,u) \in C(T,\mathbb{R}^N) \times L^1(T,\mathbb{R}^m) : \text{ the pair } (x,u) \text{ is admissible for (7.44)}\}.$$

Also set $\tau = proj_{C(T,\mathbb{R}^N)}P$ and $\tau_c = proj_{C(T,\mathbb{R}^N)}P_c$.

This way we have the set of admissible trajectories of (7.40) for τ (resp. of (7.44) for τ_c).

Now we will describe the first relaxation method which is an outgrowth of the "reduction method" used in the proof of Theorem 7.78.

Proposition 7.79 *If hypotheses H_0, H_1', H_2', H_3' hold, then $P \neq \emptyset$ and there exists $\widehat{c} > 0$ such that*

$$\|x\|_{C(T,\mathbb{R}^N)} \leq \widehat{c} \text{ for all } x \in \tau_c.$$

Proof We will describe the dynamics of (7.44) using a differential inclusion which does not involve the control variable.

Let $F(t,x)$ be orientor field defined by

$$F(t,x) = f(t,x)U(t,x) = \bigcup_{u\in U(t,x)} f(t,x)u.$$

Evidently $F(t,x)$ is nonempty and compact subset of $\mathbb{R}^N$.

Claim 1 For every $x \in \mathbb{R}^N$, $t \to F(t,x)$ is measurable.

Using Theorem I-5.101, we can find $u_n : T \to \mathbb{R}^m n \in \mathbb{N}$, measurable functions such that

$$U(t,x) = \overline{\{u_n(t)\}}_{n\in\mathbb{N}}$$

Then we have

$$F(t,x) = f(t,x)\overline{\{u_n(t)\}}_{n\in\mathbb{N}} = \overline{\{f(t,x)u_n(t)\}}_{n\in\mathbb{N}} \text{ (see Proposition I-1.19),}$$

$$\Rightarrow t \to F(t,x) \text{ is measurable (see Theorem I-5.101).}$$

This proves Claim 1.

Claim 2 For a.a $t \in T$, $x \to F(t,x)$ is locally h-Lipschitz. Let $\varrho > 0$ and $|x|, |y| \le \varrho$. Suppose $h \in F(t,x)$. We have

$$h = f(t,x)u \text{ with } u \in U(t,x). \tag{7.45}$$

Given $\varepsilon > 0$, we choose $v \in U(t,y)$ such that

$$\begin{aligned} |u - v| &\le d(u, U(t,y)) + \varepsilon \\ &\le h(U(t,x), U(t,y)) + \varepsilon \\ &\le k_\varrho(t)|x-y| + \varepsilon. \end{aligned} \tag{7.46}$$

We have

$$\begin{aligned} d(h, F(t,y)) &= d(f(t,x)u, F(t,y)) \ (see\ (7.45)) \\ &\le |f(t,x)u - f(t,y)v| \\ &\le \|f(t,x)\|_{\mathcal{L}}|u-v| + \|f(t,x) - f(t,y)\|_{\mathcal{L}}|v| \\ &\le a_\varrho(t)|u-v| + \widehat{\vartheta}_\varrho|x-y| \end{aligned}$$

with $a_\varrho(t) = a(t)(1+\varrho)$ and $\widehat{\vartheta}_\varrho(t) = \vartheta_\varrho(t)\|\widehat{a}_\varrho\|_\infty$

$$\begin{aligned} &\le \widehat{k}_\varrho(t)[|x-y| + \varepsilon] \text{ with } \widehat{k}_\varrho \in L^1(T) \text{ (see (7.46)),} \\ &\Rightarrow h^*(F(t,x), F(t,y)) \le \widehat{k}_\varrho(t)[|x-y| + \varepsilon] \end{aligned}$$

Reversing the roles of $x, y \in \mathbb{R}^N$ in the above argument, we have

$$h^*(F(t,y), F(t,x)) \leq \widehat{k}_\varrho(t)[|y-x| + \varepsilon],$$
$$\Rightarrow h(F(t,x), F(t,y)) \leq \widehat{k}_\varrho(t)[|x-y| + \varepsilon].$$

Let $\varepsilon \to 0^+$ to conclude that $x \to F(t,x)$ is locally h-Lipschitz. This proves Claim 2.

Combining Claims 1 and 2, we have that

$$(t,x) \to F(t,x) \text{ is measurable (see Proposition I-5.156).}$$

In particular the $F(\cdot,\cdot)$ is superpositional measurable, that is, if $x : T \to \mathbb{R}^N$ is measurable, then the multifunction $t \to F(t, x(t))$ is measurable.

We consider the following multivalued, control-free Cauchy problem

$$\left\{ \begin{array}{l} -x'(t) \in A(x(t)) + F(t,x(t))u(t) \text{ for a.a } t \in T, \\ x(0) = x_0 \in \overline{D(A)} \end{array} \right\} \tag{7.47}$$

Using the tools and techniques of Chap. 6, we will solve this problem. So, let $h \in L^1(T, \mathbb{R}^N)$ and consider the problem

$$-x'(t) \in A(x(t)) + h(t) \text{ for a.a } t \in T, x(0) = x_0. \tag{7.48}$$

We know that (7.48) has a unique solution $x \in W^{1,1}((0,b), \mathbb{R}^N) = AC^1(T, \mathbb{R}^N)$ (see, for example, Vrabie [464, Theorem 1.91, p. 41]). Consider the solution map $K : L^1(T, \mathbb{R}^N) \to C(T, \mathbb{R}^N)$ which to each $h \in L^1(T, \mathbb{R}^N)$ assigns the unique solution of (7.48).

Claim 3 $K : L^1(T, \mathbb{R}^N) \to C(T, \mathbb{R}^N)$ is completely continuous.

Let $h_n \overset{w}{\to} h$ in $L^1(T, \mathbb{R}^N)$ and let $x_n = K(h_n)$ for all $n \in \mathbb{N}$. Then $\{x_n\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N)$ is relatively compact (see, for example, Vrabie [464, Theorem 2.3.2, p. 64]). So, we may assume that

$$x_n \to \widehat{x} \text{ is } C(T, \mathbb{R}^N). \tag{7.49}$$

Also let $x = K(h)$. On account of the monotonicity of $A(\cdot)$, we have

$$(x_n'(t) - x'(t), x_n(t) - x(t))_{\mathbb{R}^N} + (h_n(t) - h(t), x_n(t) - x(t))_{\mathbb{R}^N} \leq 0,$$
$$\text{for a.a } t \in T,$$
$$\Rightarrow \frac{1}{2}\frac{d}{dt}|x_n(t) - x(t)|^2 \leq (h_n(t) - h(t), x(t) - x_n(t))_{\mathbb{R}^N} \text{ for a.a } t \in T,$$
$$\Rightarrow \frac{1}{2}|x_n(t) - x(t)|^2 \leq \int_0^t (h_n(s) - h(s), x(s) - x_n(s))_{\mathbb{R}^N} ds \text{ for all } t \in T,$$

$$\Rightarrow \frac{1}{2}|\widehat{x}(t) - x(t)|^2 \leq 0 \text{ for a.a } t \in T$$

(due (7.49) and since $h_n \overset{w}{\to} h$ in $L^1(T, \mathbb{R}^N)$),

$$\Rightarrow \widehat{x} = x.$$

We infer that $K(\cdot)$ is completely continuous. This proves Claim 3.

Now let $N_F : C(T, \mathbb{R}^N) \to 2^{L^1(T,\mathbb{R}^N)}$ be the multivalued Nemytskii map corresponding to the orientor field $F(t, x)$, that is,

$$N_F(u) = \{h \in L^1(T, \mathbb{R}^N) : h(t) \in F(t, u(t)) \text{ for a.a } t \in T\}.$$

From Claims 1, 2 and hypotheses $H_1'(iii)$, $H_2'(iii)$ we see that for all $x \in C(T, \mathbb{R}^N)$, we have

$$N_F(x) \in P_{wk}(L^1(T, \mathbb{R}^N)) = \{C \subseteq L^1(T, \mathbb{R}^N) : \text{ nonempty, w-compact}\}.$$

Note also that $N_F(\cdot)$ has decomposable values. Then using Definition I-5.39 and Theorem I-5.126, we see that

$$\begin{aligned} h(N_F(x), N_F(y)) &= \int_0^b d(F(t, x(t)), F(t, y(t)))dt \\ &\leq \int_0^b \widehat{k}_\varrho(t)|x(t) - y(t)|dt \\ &\quad \text{with } \varrho = \max\{\|x\|_{C(T,X)'}, \|y\|_{C(T,X)}\}, \widehat{k}_\varrho \in L^1(T) \\ &\quad \text{(see Claim 2)} \\ \Rightarrow & N_F(\cdot) \text{ is locally h-Lipschitz.} \end{aligned}$$

In particular then $N_F(\cdot)$ is lsc (see Proposition I-5.47). So, we can apply Theorem I-5.137 and have a continuous map $\xi : C(T, \mathbb{R}^N) \to L^1(T, \mathbb{R}^N)$ such that

$$\xi(x) \in N_F(x) \text{ for all } x \in C(T, \mathbb{R}^N).$$

Then the map $x \to (k \circ \xi)(x)$ is compact from $C(T, \mathbb{R}^N)$ into itself.

Consider the set

$$S = \{x \in C(T, \mathbb{R}^N) : x = \lambda(K \circ \xi)(x), 0 < \lambda < 1\}.$$

Claim 4 The set $S \subseteq C(T, \mathbb{R}^N)$ is bounded.

Let $x \in S$. We have

$$\frac{1}{\lambda}x = K(\xi(x)) \text{ with } 0 < \lambda < 1,$$
$$\Rightarrow -x'(t) \in \lambda A(x(t)) + \lambda\xi(x)(t) \text{ for a.a } t \in T, x(0) = \lambda x_0.$$

We take inner product with $x(t) \in \mathbb{R}^N$ and use hypotheses H_0. We obtain

$$\frac{1}{2}\frac{d}{dt}|x(t)|^2 + \lambda(\xi(x)(t), x(t))_{\mathbb{R}^N} \leq 0 \text{ for a.a } t \in T,$$
$$\Rightarrow \frac{1}{2}|x(t)|^2 \leq \frac{1}{2}|x_0|^2 + \int_0^t |\xi(x)(s)||x(s)|ds \text{ for all } t \in T \text{ (recall } 0 < \lambda < 1),$$
$$\Rightarrow |x(t)| \leq |x_0| + \int_0^t |\xi(x)(s)|ds \text{ for all } t \in T \text{ (see Lemma 6.61)}$$
$$\Rightarrow |x(t)| \leq |x_0| + \int_0^t \beta(s)(1 + |x(s)|)ds$$

for all $t \in T$, with $\beta \in L^1(T)$ (see hypotheses $H_1'(iii)$, $H_2'(iii)$)

$$\Rightarrow |x(t)| \leq \widehat{c} \text{ for some } \widehat{c} > 0 \text{ independent of } t \in T, x \in S,$$
$$\Rightarrow S \subseteq C(T, \mathbb{R}^N) \text{ is bounded.}$$

This proves Claim 4.

Apply Theorem 1.67 (the Leray-Schauder Alternative Principle), to obtain $\widehat{x} \in C(T, \mathbb{R}^N)$ such that

$$\widehat{x} = (K \circ \xi)(\widehat{x}),$$
$$\Rightarrow -\widehat{x}'(t) \in A(\widehat{x}(t)) + \xi(\widehat{x})(t) \text{ for a.a } t \in T, \widehat{x}(0) = x_0.$$

We introduce the multifunction $E : T \to 2^{\mathbb{R}^m} \setminus \{\emptyset\}$ defined by

$$E(t) = \{u \in U(t, \widehat{x}(t)) : \xi(\widehat{x})(t) = f(t, \widehat{x}(t))u\},$$
$$\Rightarrow GrE \in \mathcal{L}_T \otimes B(\mathbb{R}^m) \ (\mathcal{L}_T = \text{ Lebesgue } \sigma\text{-algebra of } T).$$

Use the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), to obtain a measurable map $\widehat{u} : T \to \mathbb{R}^m$ such that

$$\widehat{u}(t) \in E(t) \text{ for a.a } t \in T,$$
$$\Rightarrow \xi(\widehat{x})(t) = f(t, \widehat{x}(t))\widehat{u}(t) \text{ for a.a } t \in T,$$
$$\Rightarrow (\widehat{x}, \widehat{u}) \in P.$$

Note that arguing as in the proof of Claim 4, we show that there exists $\widehat{c} > 0$ such that

$$\|x\|_{C(T,X)} \leq \widehat{c} \text{ for all } x \in \tau_c$$

□

Proposition 7.80 *If hypotheses H_0, H_1', H_2', H_3' hold, then $P_c \subseteq C(T, \mathbb{R}^N) \times L^\infty(T, \mathbb{R}^m)_{w^*}$ is sequentially compact.*

Proof Let $\{(x_n, u_n)\}_{n\in\mathbb{N}} \subseteq P_c$. On account of hypothesis $H_2'(iii)$, we see that at least for a subsequence, we have

$$u_n \overset{w^*}{\to} u \text{ in } L^\infty(T, \mathbb{R}^m). \tag{7.50}$$

From Proposition 7.79 we know that

$$\|x_n\|_{C(T,\mathbb{R}^N)} \leq \widehat{c} \text{ for all } n \in \mathbb{N}. \tag{7.51}$$

Then from (7.51), hypothesis $H_1'(iii)$ and the Dunford-Pettis theorem, we see that

$$\{h_n(\cdot) = f(\cdot, x_n(\cdot))u_n(\cdot)\}_{n\in\mathbb{N}} \subseteq L^1(T, \mathbb{R}^N)$$

is relatively w-compact. So, we may assume that

$$h_n \overset{w}{\to} h \text{ in } L^1(T, \mathbb{R}^N). \tag{7.52}$$

Recall that the solution map $K : L^1(T, \mathbb{R}^N) \to C(T, \mathbb{R}^N)$ from the proof of Proposition 7.79 is completely continuous (see Claim 3 in that proof). Therefore we have

$$x_n = K(h_n) \to K(h) = x \text{ in } C(T, \mathbb{R}^N). \tag{7.53}$$

For every $g \in L^\infty(T, \mathbb{R}^N) = L^1(T, \mathbb{R}^N)^*$, we have

$$\int_0^b (h_n, g)_{\mathbb{R}^N} dt = \int_0^b (u_n, f(t, x_n)^* g)_{\mathbb{R}^m} dt. \tag{7.54}$$

Note that

$$\begin{aligned} &\|f(t, x_n(t))^* - f(t, x(t))^*\|_{\mathcal{L}} \qquad (7.55)\\ &= \|f(t, x_n(t)) - f(t, x(t))\|_{\mathcal{L}} \\ &\leq \vartheta_{\widehat{c}}(t)|x_n(t) - x(t)| \text{(see (7.51) and hypothesis } H_1'(ii)), \\ &\Rightarrow \|(f(\cdot, x_n(\cdot))^* - f(\cdot, x(\cdot))^*)g(\cdot)\|_1 \to 0 \text{ (see (7.53))}. \end{aligned}$$

Therefore, if in (7.54) we pass to the limit as $n \to \infty$ and use (7.50), (7.52), (7.55), we obtain

$$\int_0^b (h, g)_{\mathbb{R}^N} dt = \int_0^b (u, f(t, x)^* g)_{\mathbb{R}^m} dt,$$

$$\Rightarrow \int_0^b (h, g)_{\mathbb{R}^N} dt = \int_0^b (f(t, x)u, g)_{\mathbb{R}^N} dt.$$

Since $g \in L^\infty(T, \mathbb{R}^N)$ is arbitrary, we infer that

$$h(t) = f(t, x(t))u(t) \text{ for a.a } t \in T.$$

Also, using Theorem I-8.38 and hypothesis $H_2'(ii)$ we see that

$$u(t) \in conv U(t, x(t)) \text{ for a.a } t \in T.$$

We know that for all $0 \leq s \leq t \leq b$, all $(y, y^*) \in Gr A$ and all $n \in N$, we have

$$\frac{1}{2}|x_n(t) - y|^2 \leq \frac{1}{2}|x_n(s) - y|^2 + \int_s^t (f(\tau, x_n(\tau))u_n(\tau) - y^*, x_n(\tau) - y)_{\mathbb{R}^N} d\tau.$$

Passing to the limit as $n \to \infty$, we obtain

$$\frac{1}{2}|x(t) - y|^2 \leq \frac{1}{2}|x(s) - y|^2 + \int_s^t (f(\tau, x(\tau))u(\tau) - y^*, x(\tau) - y)_{\mathbb{R}^N} d\tau$$

$$\Rightarrow -x'(t) \in A(x(t)) + f(t, x(t))u(t) \text{ for a.a } t \in \mathbb{N}, x(0) = x_0$$

$$\Rightarrow (x, u) \in P_c.$$

Then P_c is sequentially compact in $C(T, \mathbb{R}^N) \times L^\infty(T, \mathbb{R}^m)_{w^*}$. □

Let $Y = \mathbb{R}^m \times \mathbb{R}$ and let $\Gamma : T \times \mathbb{R}^N \to P_f(Y) = \{C \subseteq Y : \text{ nonempty, closed}\}$ be defined by

$$\Gamma(t, x) = \{(u, \eta) \in \mathbb{R}^m \times \mathbb{R} : u \in U(t, x), L(t, x, u) \leq \eta \leq a_{\widehat{c}}^*(t)\} \tag{7.56}$$

with $\widehat{c} > 0$ from (7.51) and $a_{\widehat{c}}^* \in L^1(T)$ from hypothesis $H_3'(iii)$.

The next proposition establishes the properties of this multifunction.

Proposition 7.81 *If hypotheses H_0, H_1, H_2', H_3' hold, then*

(a) $(t, x) \to \Gamma(t, x)$ is graph measurable;
(b) for a.a $t \in T$, $x \to \Gamma(t, x)$ is usc.

Proof

(a) Let $\overline{B}(t) = \{(x, u, \eta) \in \mathbb{R}^N \times \mathbb{R}^m \times \mathbb{R} : |u| \leq \widehat{a}(t), \eta \leq a_c^*(t)\}$. Evidently this is a graph measurable multifunction of T into $2^{\mathbb{R}^N \times \mathbb{R}^m \times \mathbb{R}}$. We have

$$\begin{aligned} &Gr\Gamma = (GrU \times \mathbb{R}) \cap epiL \cap Gr\overline{B}, \\ &\Rightarrow Gr\Gamma \in \mathcal{L}_T \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^m) \otimes B(\mathbb{R}), \\ &\Rightarrow \Gamma \text{ is graph measurable.} \end{aligned}$$

(b) Evidently $\Gamma(t, \cdot)$ is locally compact. So, by Proposition I-5.13, in order to show the upper semicontinuity of $x \to \Gamma(t, x)$, it suffices to show that the multifunction has closed graph.
So, let $\{(x_n, u_n, \eta_n)\}_{n\in\mathbb{N}} \subseteq Gr\Gamma(t, \cdot)$ and assume that

$$x_n \to x \text{ in } \mathbb{R}^N, u_n \to u \text{ in } \mathbb{R}^m, \eta_n \to \eta \text{ in } \mathbb{R}.$$

We have

$$\begin{aligned} &u_n \in U(t, x_n), L(t, x_n, u_n) \leq \eta_n \leq a_c^*(t) \text{ for all } n \in \mathbb{N}, \\ &\Rightarrow u \in U(t, x), L(t, x, u) \leq \eta \leq a_c^*(t) \text{ (see hypotheses } H_2'(ii), H_3'(ii)), \\ &\Rightarrow (x, u, \eta) \in Gr\Gamma(t, \cdot). \end{aligned}$$

This proves that the graph of $x \to \Gamma(t, x)$ is closed and so the multifunction is usc. □

Let $\widehat{L} : T \times \mathbb{R}^N \times \mathbb{R}^m \to \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty\}$ be defined by

$$\widehat{L}(t, x, u) = \begin{cases} L(t, x, u) & \text{if } u \in U(t, x) \\ +\infty & \text{otherwise.} \end{cases} \tag{7.57}$$

Let $\widehat{L}^{**}(t, x, u)$ denote the second convex conjugate of the function $u \to \widehat{L}(t, x, u)$. The first relaxation of problem (7.40), is the following optimal control problem

$$\overset{r}{\widehat{L}^{**}}(x, u) = \int_0^b \widehat{L}^{**}(t, x(t), u(t))dt \to \inf = \widehat{m}_{L^{**}}, \tag{7.58}$$

subject to $(x, u) \in P_c$.

Recall that $dom\widehat{L}^{**}(t, x, \cdot) = \{u \in \mathbb{R}^m : \widehat{L}^{**}(t, x, u) < \infty\}$ (the effective domain of $\widehat{L}^{**}(t, x, \cdot)$).

Proposition 7.82 *If hypotheses H_2', H_3' hold, then*

(a) $dom\widehat{L}^{**}(t,x,\cdot) = convU(t,x)$ *for all* $(t,x) \in T \times \mathbb{R}^N$;
(b) $\widehat{L}^{**}(t,x,\cdot) = \min\{\eta \in \mathbb{R} : (u,\eta) \in conv\Gamma(t,x)\}$ *for a.a* $t \in T$, *all* $(x,u) \in \mathbb{R}^N \times \mathbb{R}^m$.

Proof

(a) From the definition of the second conjugate, we have

$$epiL^{**}(t,x,\cdot) = \overline{conv}epiL(t,x,\cdot) \text{ for a.a } t \in T, \text{ all } x \in \mathbb{R}^N. \tag{7.59}$$

We have $domL(t,x,\cdot) = U(t,x)$ (see (7.57)). So, from (7.59) it follows that

$$dom\widehat{L}^{**}(t,x,\cdot) = convU(t,x) \text{ for a.a } t \in T, \text{ all } x \in \mathbb{R}^N.$$

(b) From (7.59) and (7.56), we see that

$$\widehat{L}^{**}(t,x,u) = \inf\{\eta \in \mathbb{R} : (u,\eta) \in conv\Gamma(t,x)\} \text{ a.a } t \in T, (x,y) \in \mathbb{R}^N \times \mathbb{R}^m.$$

□

To continue, we need to introduce some new notions and facts about the Lebesgue-Bochner space $L^1(T,X)$ (see Definition I-4.163).

Definition 7.83 Let X be a separable Banach space. The "weak norm" on $L^1(T,X)$ is defined by

$$\|h\|_w = \sup\{\|\int_s^t h(\tau)d\tau\|_X : 0 \le s \le t \le b\}.$$

Equivalently we can define the weak norm by

$$\|h\|_w = \sup\{\|\int_0^t h(\tau)d\tau\|_X : 0 \le t \le b\}.$$

Proposition 7.84 *If X^* has the RNP, $\{h_n, h\}_{n\in\mathbb{N}}$ are such that $1 \le p \le \infty$, $h_n \overset{\|\cdot\|_w}{\to} h$ and $\sup_{n\in\mathbb{N}} \|h_n\|_{L^p(T,X)} < \infty$, then $h_n \overset{w}{\to} h$ in $L^p(T,X)$.*

Proof Since X^* has the RNP, from Theorem I-4.173, we know that

$$L^p(T,X)^* = L^{p'}(T,X^*)(\frac{1}{p} + \frac{1}{p'} = 1).$$

Let $s(t) = \sum_{k=1}^{N} \chi_{(t_{k-1},t_k)}(t) v_k^*$ with $t_k \in T$, $v_k^* \in X^*$, $k \in \{1, \ldots, N\}$. By $\langle \cdot, \cdot \rangle_X$ we denote the duality brackets for the pair (X, X^*) and by $\langle \cdot, \cdot \rangle_p$ the duality brackets for pair $(L^p(T, X), L^{p'}(T, X^*))$. We have

$$|\langle s, h_n - h \rangle_p| \leq \sum_{k=1}^{N} \| \int_{t_{k-1}}^{t_k} (h_n - h)(\tau) d\tau \|_X \|v_k^*\|_{X^*}$$

$$\leq \|h_n - h\|_w \sum_{k=1}^{N} \|v_k^*\|_{X^*} \to 0 \text{ as } n \to \infty.$$

Since step functions are dense in $L^{p'}(T, X^*)$, we conclude that

$$h_n \xrightarrow{w} h \text{ in } L^p(T, X)$$

□

Remark 7.85 Recall that reflexive Banach spaces and separable dual Banach spaces, have the RNP (see Theorem I-4.171).

Clearly the weak norm $\|\cdot\|_w$ is strictly weaker than the usual norm $\|h\|_{L^1(T,X)} = \int_0^b \|h(t)\|_X dt$. We want to know how the $\|\cdot\|_w$-topology relates to the weak topology on $L^1(T, X)$. The next notion will help us in this direction.

Definition 7.86 We say that $W \subseteq L^1(T, X)$ has "property U" if

(a) W is uniformly integrable.
(b) For every $\varepsilon > 0$, we can find $K_\varepsilon \in P_k(X) = \{C \subseteq X : \text{ nonempty, compact}\}$ such that for every $h \in W$, there is $I_{h,\varepsilon} \subseteq T$ measurable such that

$$|T \setminus I_{h,\varepsilon}| < \varepsilon \text{ and } h(t) \in K_\varepsilon \text{ for all } t \in T_{h,\varepsilon},$$

(by $|\cdot|$ we denote the Lebesgue measure on $\mathbb{R}$).

Remark 7.87 From the above definition, it follows that a set $W \subseteq L^1(T, X)$ with the property U, is necessarily relatively w-compact. In fact we can say more (see Gutman [220] and Hu and Papageorgiou [249, Theorem 4.14, p. 195]).

Proposition 7.88 *If $W \subseteq L^1(T, X)$ has property U, then on W the weak topology and the $\|\cdot\|_w$-topology coincide.*

Using this proposition, we can have the following useful fact about the set of integrable selectors of a multifunction.

Proposition 7.89 *If $K : T \to P_k(\mathbb{R}^n) = \{C \subseteq \mathbb{R}^n$: nonempty, compact$\}(n \geq 1)$ is graph measurable and $S^1_K = \{h \in L^1(T, \mathbb{R}^n) : h(t) \in K(t)$ for a.a $t \in T\} \neq \emptyset$ and it is uniformly integrable, then $\overline{S^1_K}^{\|\cdot\|_w} = S^1_{convK}$.*

Proof From Proposition I-5.130, we know that

$$\overline{S^1_K}^{w} = S^1_{convK}. \tag{7.60}$$

Also using Proposition I-5.168 and Lusin's Theorem, given $\varepsilon > 0$ we can find $T_\varepsilon \subseteq T$ closed such that

$$|T \setminus I_\varepsilon| < \varepsilon \text{ and } K|_{I_\varepsilon} \text{ is h-continuous.}$$

Then $K(I_\varepsilon) \in P_k(\mathbb{R}^n)$ (see Proposition I-5.14). Moreover, since S^1_K is by hypothesis uniformly integrable, then so is $\overline{conv} S^1_K = S^1_{convK}$ (see Proposition I-5.127). Therefore S^1_{convK} has property U. So, by Proposition 7.88 the weak and $\|\cdot\|_w$-topologies coincide. Therefore from (7.60) we conclude that

$$\overline{S^1_K}^{\|\cdot\|_w} = S^1_{convK}.$$

□

This Proposition, leads to the following approximation result which is crucial in order to prove the admissibility of this relaxation method (see (7.58)).

Proposition 7.90 *If hypotheses H_0, H_1', H_2', H_3' hold and $(\widehat{x}, \widehat{u}) \in P_c$, then we can find $\widehat{u}_n \in S^1_{U(\cdot,\widehat{x}(\cdot))}$, $n \in \mathbb{N}$, such that*

$$\widehat{u}_n \overset{\|\cdot\|_w}{\to} \widehat{u} \text{ in } L^1(T, \mathbb{R}^m),$$

$$L(\cdot, \widehat{x}(\cdot), \widehat{u}_n(\cdot)) \overset{\|\cdot\|_w}{\to} \widehat{L}^{**}(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot)) \text{ in } L^1(T).$$

Proof From Proposition 7.82, we have

$$(\widehat{u}(t), \widehat{L}^{**}(t, \widehat{x}(t), \widehat{u}(t))) \in conv\Gamma(t, \widehat{x}(t)) \text{ for a.a } t \in T,$$

$$\Rightarrow \widehat{w}(\cdot) = (\widehat{u}(\cdot), \widehat{L}^{**}(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot))) \in S^1_{conv\Gamma(\cdot,\widehat{x}(\cdot))}.$$

Using Proposition 7.89, we can find $\widehat{w}_n \in S^1_{\Gamma(\cdot,\widehat{x}(\cdot))} n \in \mathbb{N}$ such that

$$\widehat{w}_n \overset{\|\cdot\|_w}{\to} \widehat{w} \text{ in } L^1(T, \mathbb{R}^m).$$

Note that $\widehat{w}_n(\cdot) = (\widehat{u}_n(\cdot), \eta_n(\cdot))$ with

$\widehat{u}_n \in S^1_{U(\cdot,\widehat{x}(\cdot))}$ and $L(t,\widehat{x}(t),\widehat{u}_n(t)) \leq \eta_n(t) \leq a^*_c(t)$ for a.a $t \in T$, all $n \in \mathbb{N}$.

Therefore

$$\widehat{u}_n \overset{\|\cdot\|_w}{\to} \widehat{u} \text{ in } L^1(T,\mathbb{R}^m) \text{ and } \sup_{t\in T}\int_0^t (\eta_n(s) - L^{**}(s,\widehat{x}(s),\widehat{u}_n(s)))ds \to 0.$$

It follows that

$$\sup_{t\in T}\int_0^t (L(s,\widehat{x}(s),\widehat{u}_n(s)) - \widehat{L}^{**}(s,\widehat{x}(s),\widehat{u}_n(s)))ds \to 0,$$

$$\Rightarrow L(\cdot,\widehat{x}(s),\widehat{u}_n(s)) \overset{\|\cdot\|_w}{\to} \widehat{L}^{**}(\cdot,\widehat{x}(\cdot),\widehat{u}(\cdot)) \text{ in } L^1(T) \text{(see hypothesis } H_3(ii)).$$

□

Now we are ready to show that the set of admissible pairs P is dense in the corresponding set P_c of the convexified problem in the product space $C(T,\mathbb{R}^N) \times L^1(T,\mathbb{R}^m)_w$.

Proposition 7.91 *If hypotheses H_0, H_1', H_2', H_3' hold and $(\widehat{x},\widehat{u}) \in P_c$, then we can find $\{(\widehat{x}_n,\widehat{u}_n)\}_{n\in\mathbb{N}} \subseteq P$ such that*

$$\widehat{x}_n \to \widehat{x} \text{ in } C(T,\mathbb{R}^N),$$

$$\widehat{u}_n \underset{\|\cdot\|_w}{\overset{w}{\to}} \widehat{u} \text{ in } C(T,\mathbb{R}^m).$$

$$L(\cdot,\widehat{x}_n(\cdot),\widehat{u}_n(\cdot)) \overset{\|\cdot\|_w}{\to} \widehat{L}^{**}(\cdot,\widehat{x}(\cdot),\widehat{u}(\cdot)) \text{ in } L^1(T).$$

Proof According to Proposition 7.90, we can find $\{\widehat{u}_n\}_{n\in\mathbb{N}} \subseteq S^1_{U(\cdot,\widehat{x}(\cdot))}$ such that

$$\widehat{u}_n \overset{\|\cdot\|_w}{\to} u \text{ in } L^1(T,\mathbb{R}^m) \text{ and } \widehat{L}(\cdot,\widehat{x}(\cdot),\widehat{u}_n(\cdot)) \overset{\|\cdot\|_w}{\to} \widehat{L}^{**}(\cdot,\widehat{x}(\cdot),\widehat{u}(\cdot)) \text{ in } L^1(T). \tag{7.61}$$

From hypothesis $H_2(iii)$ we see that

$$\sup_{n\in\mathbb{N}} \|\widehat{u}_n\|_1 < \infty.$$

Therefore from (7.61) and Proposition 7.84, we have

$$\widehat{u}_n \overset{w}{\to} \widehat{u} \text{ in } L^1(T,\mathbb{R}^m). \tag{7.62}$$

Consider the multifunction $G_n : T \times \mathbb{R}^N \to 2^{\mathbb{R}^m} \setminus \{\emptyset\}$ defined by

$$G_n(t,x) = \{v \in \mathbb{R}^m : |\widehat{u}_n(t) - v| \leq k_c(t)|\widehat{x}(t) - x| + \frac{1}{n}, \text{ with } v \in U(t,x).\} \tag{7.63}$$

(see hypothesis $H_2'(ii)$ and (7.51)).

We consider the following control system

$$-x'(t) \in A(x(t)) + f(t,x(t))v(t) \text{for a.a } t \in T, x(0) = 0, v \in S^1_{G_n(\cdot,x(\cdot))}. \tag{7.64}$$

As in the proof of Proposition 7.79, we show that for every $n \in \mathbb{N}$, problem (7.64) has admissible state-control pairs. Let $(\widehat{x}_n, \widehat{v}_n)$ be such a pair. Then

$$(\widehat{x}_n, \widehat{v}_n) \in P \text{ for all } n \in \mathbb{N}.$$

Hence we have that

$$\{\widehat{x}_n\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N) \text{ is relatively compact.}$$

We may assume that

$$\widehat{x}_n \to \widehat{x}^* \text{ in } C(T, \mathbb{R}^N). \tag{7.65}$$

Then we have

$$-\widehat{x}_n'(t) \in A(\widehat{x}_n(t)) + f(t,\widehat{x}_n(t))\widehat{v}_n(t) \text{ for a.a } t \in T, \widehat{x}_n(0) = x_0, n \in \mathbb{N}.$$

$$-\widehat{x}'(t) \in A(\widehat{x}(t)) + f(t,\widehat{x}(t))\widehat{u}(t) \text{ for a.a } t \in T, \widehat{x}(0) = x_0.$$

Using hypotheses H_0, we have

$$\begin{aligned}
0 &\geq \big(\widehat{x}_n'(t) - \widehat{x}'(t), \widehat{x}_n(t) - \widehat{x}(t)\big)_{\mathbb{R}^N} \\
&\quad + \big(f(t,\widehat{x}_n(t))\widehat{v}_n(t) - f(t,\widehat{x}(t))\widehat{u}(t), \widehat{x}_n(t) - \widehat{x}(t)\big)_{\mathbb{R}^N}, \\
&\Rightarrow \frac{1}{2}\frac{d}{dt}|\widehat{x}_n(t) - \widehat{x}(t)|^2 + \big(f(t,\widehat{x}_n(t))\widehat{v}_n(t) - f(t,\widehat{x}(t))\widehat{u}(t), \widehat{x}_n - \widehat{x}(t)\big)_{\mathbb{R}^N} \leq 0 \\
&\quad \text{for a.a } t \in T, \text{ all } n \in \mathbb{N}, \\
&\Rightarrow \frac{1}{2}|\widehat{x}_n(t) - \widehat{x}(t)|^2 + \int_0^t (f(s,\widehat{x}_n(s))\widehat{v}_n(s) \\
&- f(s,\widehat{x}(s))\widehat{u}(s), \widehat{x}_n(s) - \widehat{x}(s))_{\mathbb{R}^N} ds \leq 0 \text{ for all } n \in \mathbb{N}.
\end{aligned} \tag{7.66}$$

We pass to the limit as $n \to \infty$ and use (7.62), (7.63), (7.65) and hypotheses H_1', H_2' and H_3'. We obtain

$$|\widehat{x}^*(t) - \widehat{x}(t)| \leq 2 \int_0^t \mu(s)|\widehat{x}^*(s) - \widehat{x}(s)|^2 ds \text{ for some } \mu \in L^1(T)$$

$$\Rightarrow \widehat{x}^* = \widehat{x}.$$

Therefore we have

$$\widehat{x}_n \to \widehat{x} \text{ in } C(T, \mathbb{R}^N) \text{ (see (7.65))}. \tag{7.67}$$

From (7.63) we see that

$$\|\widehat{u}_n - \widehat{v}_n\|_1 \to 0,$$

$$\Rightarrow \widehat{v}_n \underset{\|\cdot\|_w}{\overset{w}{\to}} \widehat{u} \text{ in } L^1(T, \mathbb{R}^m) \text{ (see (7.62))}.$$

Finally from (7.61) we have

$$L(\cdot, \widehat{x}(\cdot), \widehat{u}_n(\cdot)) \overset{\|\cdot\|_w}{\to} \widehat{L}^{**}(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot)) \text{ in } L^1(T). \tag{7.68}$$

Also from hypothesis $H_3'(ii)$, (7.63) and (7.67), we see that

$$L(\cdot, \widehat{x}_n(\cdot), \widehat{u}_n(\cdot)) - L(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot)) \overset{\|\cdot\|_w}{\to} 0 \text{ in } L^1(T). \tag{7.69}$$

From (7.68) and (7.69) it follows that

$$L(\cdot, \widehat{x}_n(\cdot), v_n(\cdot)) \overset{\|\cdot\|_w}{\to} \widehat{L}^{**}(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot)) \text{ in } L^1(T).$$

□

Now we are ready to show that the first relaxation method which produced problem (7.58), is admissible.

Theorem 7.92 *If hypotheses H_0, H_1', H_2', H_3' hold, then (a) there exists $(\widehat{x}, \widehat{u}) \in P_c$ such that*

$$I^r_{\widehat{L}^{**}}(x, u) = \widehat{m}_{\widehat{L}^{**}}.$$

(b) $\widehat{m}_{\widehat{L}^{**}} = \widehat{m}_L$.
(c) there exists a sequence $\{(\widehat{x}_n, \widehat{u}_n)\}_{n\in\mathbb{N}} \subseteq P$ such that

$$I_L(\widehat{x}_n, \widehat{u}_n) \downarrow m_{L^{**}},$$

$$\widehat{x}_n \to \widehat{x} \text{ in } C(T, \mathbb{R}^N),$$

$$u_n \underset{\|\cdot\|_w}{\overset{w}{\to}} \widehat{u} \text{ in } L^1(T, \mathbb{R}^m).$$

Proof

(a) Let $\{(\widehat{y}_n, \widehat{v}_n)\}_{n\in\mathbb{N}} \subseteq P_c$ be a minimizing sequence for the relaxed problem (7.58). We have

$$I^r_{L^{**}}(\widehat{y}_n, \widehat{v}_n) \downarrow \widehat{m}_{L^{**}}.$$

From Proposition 7.80, we know that

$$\{(\widehat{y}_n, \widehat{v}_n)\}_{n\in\mathbb{N}} \subseteq C(T, \mathbb{R}^N) \times L^1(T, \mathbb{R}^m)_w \text{ is relatively compact.}$$

So, we may assume that

$$\widehat{y}_n \to \widehat{x} \text{ in } C(T, \mathbb{R}^N), \widehat{v}_n \xrightarrow{w} \widehat{u} \text{ in } L^1(T, \mathbb{R}^m). \tag{7.70}$$

Using Theorem 7.50, we have

$$I^r_{\widehat{L}^{**}}(\widehat{x}, \widehat{u}) \leq \liminf_{n\to\infty} I^r_{\widehat{L}^{**}}(\widehat{x}_n, \widehat{u}_n) = \widehat{m}_{\widehat{L}^{**}}.$$

Since $(\widehat{x}, \widehat{u}) \in P_c$ (see Proposition 7.80), we infer that

$$I^r_{\widehat{L}^{**}}(\widehat{x}, \widehat{u}) = \widehat{m}_{\widehat{L}^{**}}.$$

(b) $\oplus$(c): ***From Proposition 7.91, we know that we can find $\{(\widehat{x}_n, \widehat{u}_n)\}_{n\in\mathbb{N}} \subseteq P$ such that

$$\left\{\begin{array}{l} \widehat{x}_n \to \widehat{x} \text{ in } C(T, \mathbb{R}^N), \widehat{u}_n \xrightarrow[\|\cdot\|_w]{w} \widehat{u} \text{ in } L^1(T, \mathbb{R}^m), \\ L(\cdot, \widehat{x}_n(\cdot), \widehat{u}_n(\cdot)) \xrightarrow{\|\cdot\|_w} \widehat{L}^{**}(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot)) \text{ in } L^1(T). \end{array}\right\} \tag{7.71}$$

We have

$$\begin{aligned} &|I_L(\widehat{x}_n, \widehat{u}_n) - I^r_{\widehat{L}^{**}}(\widehat{x}, \widehat{u})| \\ &\leq \|L(\cdot, \widehat{x}_n(\cdot), \widehat{u}_n(\cdot)) - \widehat{L}^{**}(\cdot, \widehat{x}(\cdot), \widehat{u}(\cdot))\|_w \to 0 \text{ (see (7.71))}, \\ &\Rightarrow I_L(\widehat{x}_n, \widehat{u}_n) \to I^r_{\widehat{L}^{**}}(\widehat{x}, \widehat{u}) = \widehat{m}_{\widehat{L}^{**}}, \\ &\Rightarrow \widehat{m}_L \leq \widehat{m}_{\widehat{L}^{**}}, \\ &\Rightarrow \widehat{m}_L = \widehat{m}_{\widehat{L}^{**}} \text{ (since we always have } \widehat{m}_{\widehat{L}^{**}} \leq \widehat{m}_L). \end{aligned}$$

Therefore the relaxation method is admissible. □

The second relaxation method is based on Young measures. Let $M = \|\widehat{a}\|_\infty$ (see hypothesis $H'_2(iii)$) and set $\overline{B}_M = \{v \in \mathbb{R}^m : |v| \leq M\}$. Recall that using the disintegration theorem (see Theorem 7.29), we can view the Young measures $\mathcal{Y}(\Omega \times$

$\mathbb{R}^m$) as a subset of $L^0(\Omega, P(\mathbb{R}^m))$ (see (7.17)). The elements of $L^0(\Omega, P(\mathbb{R}^m))$ are also known as "transition probabilities".

For the new relaxed system, the control constraint set is

$$\Sigma(t,x) = \{\mu \in L^0(T, P(\overline{B}_M)) : \mu(U(t,x)) = 1\}.$$

Then the new relaxed optimal control problem is

$$\left\{\begin{array}{l} \widehat{I}_L^r(x,\lambda) = \int_0^b \int_{\overline{B}_M} L(t,x(t),u)\lambda(t)(du)dt \to \inf = \widehat{m}_L^* \\ \text{subject to:} \\ -x'(t) \in A(x(t)) + \int_{\overline{B}_M} f(t,x(t))u\lambda(t)(du) \text{ for } a.a\ t \in T, \\ x(0) = x_0, \lambda \in S_{\Sigma(\cdot,x(\cdot))} \end{array}\right\} \tag{7.72}$$

By $S_{\Sigma(\cdot,x(\cdot))}$ we denote the set of all measurable selectors of $t \to \Sigma(t,x(t))$ (recall that on $P(\overline{B}_M)$ we consider the narrow topology). Then the orientor field for problem (7.72) is

$$\widehat{F}(t,x) = \{\int_{\overline{B}_M} f(t,x)u\lambda(t)(du) : \lambda \in S_{\Sigma(\cdot,x(\cdot))}\}.$$

By $\widehat{P}_c$ we denote the set of admissible state-control pairs for problem (7.72). We have that $\widehat{P}_c$ is compact in $C(T,\mathbb{R}^N) \times \mathcal{Y}(T \times \overline{B}_M)_{\tau_n^y}$. We show that this relaxation method is admissible too.

Theorem 7.93 *If hypotheses H_0, H_1', H_2', H_3' hold, then there exists $(\widehat{x},\widehat{\lambda}) \in \widehat{P}_c$ such that*

$$\widehat{I}_L^r(\widehat{x},\widehat{\lambda}) = \widehat{m}_L^* = \widehat{m}_L = \widehat{m}_{L^{**}}$$

and we can find a sequence $\{(\widehat{x}_n,\widehat{u}_n)\}_{n\in\mathbb{N}} \subseteq P$ such that

$$\widehat{x}_n \to \widehat{x} \text{ in } C(T,\mathbb{R}^N) \text{ and } \delta_{\widehat{u}_n} \xrightarrow{\tau_n^y} \widehat{\lambda} \text{ in } y(\Omega \times \overline{B}_M).$$

Proof Let $\{(\widehat{x}_n,\widehat{\lambda}_n)\}_{n\in\mathbb{N}} \subseteq P_c$ be a minimizing sequence for problem (7.72). From Theorem 7.48, we know that $\{\widehat{\lambda}_n\}_{n\in\mathbb{N}} \subseteq \mathcal{Y}(\Omega \times \overline{B}_M)$ is relatively τ_n^y-compact and so we may assume that

$$\widehat{\lambda}_n \xrightarrow{\tau_n^y} \widehat{\lambda} \text{ in } \mathcal{Y}(\Omega \times \overline{B}_M) \text{ (see Proposition 7.43).} \tag{7.73}$$

Let $\widehat{x}_n \in W^{1,1}((0,b),\mathbb{R}^N) = AC^1(T,\mathbb{R}^N)$ be the relaxed state generated by the relaxed control $\widehat{\lambda}_n \in S_{\Sigma(\cdot,x_n(\cdot))}$. We have $\{\widehat{x}_n\}_{n\in\mathbb{N}} \subseteq C(T,\mathbb{R}^N)$ is relatively compact (see the proof of Proposition 7.79). So, we may assume that

$$\widehat{x}_n \to \widehat{x} \text{ in } C(T, \mathbb{R}^N). \tag{7.74}$$

Then form (7.73) and (7.74), we see easily that

$$(\widehat{x}, \widehat{\lambda}) \in P_c.$$

We have

$$\begin{aligned}\widehat{I}_L^r(\widehat{x}_n, \widehat{\lambda}_n) &= \int_0^b \int_{\overline{B}_M} L(t, \widehat{x}_n(t), u)\widehat{\lambda}_n(t)(du)dt \\ &= \int_0^b \int_{\mathbb{R}^N \times \overline{B}_M} L(t, x, u)(\delta_{\widehat{x}_n(t)} \otimes \widehat{\lambda}_n(t))(dx, du)dt.\end{aligned}$$

Note that

$$\delta_{x_n(\cdot)} \otimes \widehat{\lambda}_n \xrightarrow{\tau_n^y} \delta_{\widehat{x}(\cdot)} \otimes \widehat{\lambda} \text{ (see (7.73),(7.74)).}$$

Then hypotheses H_3' and the definition of the narrow topology τ_n^y (see Definition 7.38), imply

$$\begin{aligned}\widehat{m}_L^* &= \lim_{n\to\infty} \widehat{I}_L^r(\widehat{x}_n, \widehat{\lambda}_n) \\ &= \lim_{n\to\infty} \int_0^b \int_{\overline{B}_M} L(t, \widehat{x}_n(t), u)\widehat{\lambda}_n(t)(du)dt \\ &= \int_0^b \int_{\overline{B}_M} L(t, \widehat{x}(t), u)\widehat{\lambda}(t)(du)dt \\ &= \widehat{I}_L^r(\widehat{x}, \widehat{\lambda}) \geq \widehat{m}_L^*, \\ \Rightarrow \widehat{I}_L^r(\widehat{x}, \widehat{\lambda}) &= \widehat{m}_L^*.\end{aligned}$$

The Dirac Young measures are dense in $\mathcal{Y}(\Omega \times \overline{B}_M)$. So, we can find $\widehat{u}_n \in S_{u(\cdot,\widehat{x}(\cdot))}$ $n \in \mathbb{N}$ such that

$$\delta_{\widehat{u}_n(\cdot)} \xrightarrow{\tau_n^y} \widehat{\lambda}. \tag{7.75}$$

As before (see the proof of Proposition 7.91), we consider the multifunction

$$G_n(t, x) = \{v \in \mathbb{R}^m : |\widehat{u}_n(t) - v| \leq k_c(t)|\widehat{x}(t) - x| + \frac{1}{n}, v \in U(t, x)\}.$$

Arguing as in the proof of Proposition 7.91, we produce state-control pairs $(\widehat{x}_n, \widehat{v}_n)$ for the system with control constraint multifunction $G_n(t,x)$ such that

$$\widehat{x}_n \to \widehat{x} \text{ in } C(T,\mathbb{R}^N), \widehat{u}_n - \widehat{v}_n \xrightarrow[\|\cdot\|_w]{w} 0 \text{ in } L^1(T,\mathbb{R}^m). \tag{7.76}$$

Form (7.75) and (7.76), we have

$$\begin{aligned} & I_L(\widehat{x},\widehat{u}_n) \to \widehat{I}^r_L(\widehat{x},\widehat{\lambda}) = \widehat{m}^*_L, \\ & \Rightarrow \widehat{m}_L \leq \widehat{m}^*_L, \\ & \Rightarrow m_L = \widehat{m}^*_L = \widehat{m}_{L^{**}}. \end{aligned}$$

□

7.3 Sensitivity Analysis

In this section we examine the relation between relaxability and variational stability of finite dimensional systems driven by a maximal monotone map, which is not in general defined everywhere. This way we incorporate in our framework systems with unilateral constraints.

Relaxability means that convexification of the dynamics and of the cost functional, essentially does not alter the system. Variational stability means that small perturbations of the data, leads to small changes in the value of the problem. So,both notions correspond to some kind of "well-posedness" of the problem. Hence we can reasonably expect that the two notions are related. In this section under reasonable hypotheses on the data, we prove such a relation (equivalence).

Let Θ be a compact metric space, representing the parameter space and $0 \in \Theta$ is a point corresponding to the original unperturbed problem. We denote the parameters by $\vartheta \in \Theta$.

So, we are dealing with three problems. The original unperturbed (that is, $\vartheta = 0$) problem is

$$\left\{ \begin{array}{l} I_L(x,u) = \int_0^b L(t,x(t),u(t))dt \to \inf = m(0) \\ \text{subject to:} \\ -x'(t) \in A(x(t)) + f(t,x(t),u(t)) \text{ for } a.a\ t \in T, \\ x(0) = x_0, u(t) \in U(t) \text{ for } a.a\ t \in T. \end{array} \right\} \tag{7.77}$$

Assuming that the control constraint sets $U(t), t \in T$, are contained in a ball $\overline{B}_M = \{u \in \mathbb{R}^m : |u| \leq M\}$, then to problem (7.77) we associate a "relaxed" version using Young measures (see Section 7.2), which is defined by

$$\left\{\begin{array}{l} \widehat{I}_L^r(x,\lambda) = \int_0^b \int_{\overline{B}_M} L(t,x(t),u)\lambda(t)(du)dt \to \inf = m_r \\ \text{subject to:} \\ -x'(t) \in A(x(t)) + \int_{\overline{B}_M} f(t,x(t),u)\lambda(t)(du) \text{ for } a.a\ t \in T, \\ x(0) = x_0, \lambda \in S_\Sigma. \end{array}\right\} \tag{7.78}$$

Recall that

$$\Sigma(t) = \{\lambda \in P(\overline{B}_M) : \lambda(U(t)) = 1\} \tag{7.79}$$

and S_Σ is the set of measurable selectors of $\Sigma(\cdot)$.

Finally we introduce the "perturbed" problem. Note that we consider perturbations of both the dynamics and the cost functional

$$\left\{\begin{array}{l} I_L(x,u,\vartheta) = \int_0^b L(t,x(t),u(t),\vartheta)dt \to m(\vartheta), \\ \text{subject to:} \\ -x'(t) \in A(x(t),\vartheta) + f(t,x(t),u(t),\vartheta) \text{ for } a.a\ t \in T, \\ x(0) = x_0(\vartheta), u(t) \in U(t,\vartheta) \text{ for } a.a\ t \in T. \end{array}\right\} \tag{7.80}$$

We assume $A(x,0) = A(x)$, $f(t,x,u,0) = f(t,x,u)$, $x_0(0) = x_0$ and $U(t,0) = U(t)$. So problem (7.77) corresponds to the unperturbed version of (7.80) (that is, $\vartheta = 0$).

The hypotheses on the data of these three problems are the following:

$\mathbf{H}_4$: $A : \mathbb{R}^N \times \Theta \to 2^{\mathbb{R}^N}$ is a map such that for every $\vartheta \in \Theta$, $A(\cdot,\vartheta)$ is maximal monotone, $0 \in A(0,\vartheta)$ and if $\vartheta \to 0$ in Θ, then $A(\cdot,\vartheta) \xrightarrow{G} A$ (see Definition $I - 8.77$).

$\mathbf{H}_5$: $f : T \times \mathbb{R}^N \times \mathbb{R}^m \times \Theta \to \mathbb{R}^N$ is a function such that

(i) for all $(x,u,\vartheta) \in \mathbb{R}^N \times \mathbb{R}^m \times \Theta$, $t \to f(t,x,u,\vartheta)$ is measurable;
(ii) for $a.a\ t \in T$, $(x,u,\vartheta) \to f(t,x,u,\vartheta)$ is continuous;
(iii) for $a.a\ t \in T$, all $\vartheta \in \Theta$, all $u \in U(t,\vartheta)$ and all $x, y \in \mathbb{R}^N$ we have $|f(t,x,u,\vartheta) - f(t,y,u,\vartheta)| \le k(t)|x-y|$ with $k \in L^0(T)$;
(iv) $|f(t,x,u,\vartheta)| \le a(t)[1+|x|]$ for $a.a\ t \in T$, all $\vartheta \in \Theta$ and all $u \in U(t,\vartheta)$ with $a \in L^1(T)$.

$\mathbf{H}_6$: $U : T \times \Theta \to P_k(\mathbb{R}^m) = \{C \subseteq \mathbb{R}^m :$ nonempty, compact$\}$ is a multifunction such that

(i) for all $\vartheta \in \Theta$, $t \to U(t,\vartheta)$ is graph measurable;
(ii) for $a.a\ t \in T$, $\vartheta \to U(t,\vartheta)$ is h-continuous;
(iii) $|U(t,\vartheta)| = \sup\{|u| : u \in U(t,\vartheta)\} \le M$ for $a.a\ t \in T$, all $\vartheta \in \Theta$.

$\mathbf{H}_7$: $L : T \times \mathbb{R}^N \times \mathbb{R}^m \times \Theta \to \mathbb{R}$ is an integrand such that

(i) for all $(x,u,\vartheta) \in \mathbb{R}^N \times \mathbb{R}^m \times \Theta$, $t \to L(t,x,u,\vartheta)$ is measurable;
(ii) for $a.a\ t \in T$, $(x,u,\vartheta) \to L(t,x,u,\vartheta)$ is continuous;

(iii) for every $\varrho > 0$, there exists $\widehat{a}_\varrho \in L^1(T)$ such that $|L(t,x,u,\vartheta)| \leq \widehat{a}_\varrho(t)$ for $a.a\ t \in T$, all $|x| \leq \varrho$, all $|u| \leq M$, all $\vartheta \in \Theta$.

$\mathbf{H}_8 : x_0 : \Theta \to \overline{D(A)}$ is continuous.

Definition 7.94

(a) We say that problem (7.77) is "relaxable" if $m(0) = m_r$ (see problem (7.78));
(b) We say that problem (7.77) is "value continuous" (or "stable"), if

$$\vartheta \to 0 \text{ in } \Theta \Rightarrow m(\vartheta) \to m(0).$$

Consider the perturbed relaxed problem

$$\left\{\begin{array}{l} \widehat{I}_L^r(x,\lambda,\vartheta) = \int_0^b \int_{\overline{B}_M} L(t,x(t),u,\vartheta)\lambda(t)(du)dt \to \inf = \widehat{m}_r(\vartheta), \\ \text{subject to:} \\ -x'(t) \in A(x(t),\vartheta) + \int_{\overline{B}_M} f(t,x(t),u,\vartheta)\lambda(t)(du) \text{ for } a.a\ t \in T, \\ x(0) = x_0(\vartheta), \lambda \in S_{\Sigma(\cdot,\vartheta)}, \end{array}\right\} \tag{7.81}$$

where $\Sigma(t,\vartheta) = \{\mu \in P(\overline{B}_M) : \mu(U(t,\vartheta)) = 1\}$ and $S_{\Sigma(\cdot,\vartheta)}$ is the set of measurable selectors of $t \to \Sigma(t,\vartheta)$.

Let $P_c(0)$ be the set admissible state-control pairs for problem (7.77) and $P_c(\vartheta)$ the set of admissible state-control pairs for problem (7.81).

Proposition 7.95 *If hypotheses H_4, H_5, H_6, H_7, H_8 hold, then if $\vartheta_n \to 0$ in Θ, we have $P_c(\vartheta_n) \overset{K(s\times w^*)}{\longrightarrow} P_c(0)$, that is,*

$$(s \times w^*) - \limsup_{n\to\infty} P_c(\vartheta_n) \subseteq P_c(0), \tag{7.82}$$

$$P_c(0) \subseteq (s \times w^*) - \liminf_{n\to\infty} P_c(\vartheta_n), \tag{7.83}$$

in $C(T,\mathbb{R}^N) \times L^\infty(T, ca(\overline{B}_M)_{w^*})$.

Proof From Theorem $I - 4.177$, we know that

$$L^1(T, C(\overline{B}_M))^* = L^\infty(T, ca(\overline{B}_M)_{w^*})$$

(see also Theorem 7.25 and Remark 7.26).

First we show (7.82). So, let $(x,\lambda) \in (s \times w^*) - \limsup_{n\to\infty} P_c(\vartheta_n)$. Then by definition, we can find $(x_n,\lambda_n) \in P_c(\vartheta_n)\ n \in \mathbb{N}$ such that

$$x_n \to x \text{ in } C(T, \mathbb{R}^N), \lambda_n \xrightarrow{w^*} \lambda \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}). \tag{7.84}$$

For every $n \in \mathbb{N}$, we consider the following differential inclusion

$$\left\{ \begin{array}{l} -y_n'(t) \in A(y_n(t), \vartheta_n) + \int_{\overline{B}_M} f(t, x(t), u, \vartheta_n)\lambda_n(t)(du) \text{ for } a.a\ t \in T, \\ y_n(0) = x_0. \end{array} \right\} \tag{7.85}$$

Problem (7.85) has a unique solution $y_n \in W^{1,1}((0,b), \mathbb{R}^N) = AC^1(T, \mathbb{R}^N)$ and $\{y_n\}_{n\in\mathbb{N}} \subseteq W^{1,1}((0,b), \mathbb{R}^N)$ is relatively w-compact. So, we may assume that

$$\begin{aligned} & y_n \xrightarrow{w} y \text{ in } W^{1,1}((0,b), \mathbb{R}^N) \\ \Rightarrow & y_n \to y \text{ in } C(T, \mathbb{R}^N) \end{aligned} \tag{7.86}$$

(recall that $W^{1,1}((0,b), \mathbb{R}^N) \hookrightarrow C(T, \mathbb{R}^N)$ compactly). Then using hypotheses H_4 and Theorem $I - 8.38$, in the limit as $n \to \infty$, we obtain

$$\left\{ \begin{array}{l} -y'(t) \in A(y(t)) + \int_{\overline{B}_M} f(t, x(t), u)\lambda(t)(du) \text{ for } a.a\ t \in T, \\ y(0) = x_0. \end{array} \right\} \tag{7.87}$$

We set

$$\begin{aligned} h_n(t) &= \int_{\overline{B}_M} f(t, x_n(t), u, \vartheta_n)\lambda_n(t)(du), \\ h(t) &= \int_{\overline{B}_M} f(t, x(t), u)\lambda(t)(du). \end{aligned}$$

Then $\{h_n, h\}_{n\in\mathbb{N}} \subseteq L^1(T, \mathbb{R}^N)$ and from (7.85) we see that

$$h_n \xrightarrow{w} h \text{ in } L^1(T, \mathbb{R}^N). \tag{7.88}$$

Using hypotheses H_4, we have

$$\begin{aligned} & \frac{1}{2}\frac{d}{dt}|x_n(t) - y_n(t)|^2 \leq (h_n(t) - h(t), x_n(t) - y_n(t))_{\mathbb{R}^N} \text{ for } a.a\ t \in T, \\ & \Rightarrow \frac{1}{2}|x_n(t) - y_n(t)|^2 \leq \frac{1}{2}|x_0(\vartheta_n) - x_0|^2 + \int_0^t (h_n - h, x_n - y_n)_{\mathbb{R}^N} ds. \end{aligned}$$

Passing to the limit as $n \to \infty$ and using (7.85), (7.86), and (7.88), we obtain

$$\frac{1}{2}|x(t) - y(t)|^2 \leq 0 \text{ for all } t \in T,$$
$$\Rightarrow x = y.$$

Therefore $x \in W^{1,1}((0,b),\mathbb{R}^N)$ is a state of the relaxed system generated by the control measure $\lambda(\cdot)$ (see (7.87)). We need to show that λ is an admissible control, namely that $\lambda \in S_\Sigma$. To this end consider the integrands

$$k_n(t,u) = \chi_{U(t,\vartheta_n)}(u)\ n \in \mathbb{N} \text{ and } k(t,u) = \chi_{U(t)}(u).$$

(here $U(t) = U(t,0)$). These are measurable functions and since U is $P_k(\mathbb{R}^m)-$valued, it follows that $k_n(t,\cdot), k(t,\cdot)$ are both upper semicontinuous. Also using the Scorza-Dragoni theorem for multifunctions (see Hu and Papageorgiou [247, Proposition 7.17, p. 232]), for every $t \in (0,b]$, given $\delta > 0$, we can find $T_\delta^t \subseteq T^t = [0,t]$ closed such that

$$|T^t \setminus T_\delta^t| \leq \delta \text{ and } U|_{T_\delta^t \times \Theta} \text{ is lsc.}$$

Since U is $P_k(\mathbb{R}^m)-$valued from Proposition $I - 5.49$, we have

$$U|_{T_\delta^t \times \Theta} \text{ is h-lsc.}$$

Therefore given $\varepsilon > 0$, we can find $n_0 \in \mathbb{N}$ such that

$$U(s,\vartheta_n) \subseteq U(s) + \varepsilon \overline{B}_1 \text{ for all } n \geq n_0, \text{ all } s \in T^t.$$

($\overline{B}_1 = \{u \in \mathbb{R}^m : |u| \leq 1\}$). Then

$$k_n(s,u) \leq k(s,u) + \varepsilon \text{ for all } (s,u) \in T_\delta^t \times \mathbb{R}^N, \text{ all } n \in \mathbb{N}.$$

Then we have

$$\begin{aligned} &\int_0^t \int_{\overline{B}_M} k_n(t,u)\lambda_n(t)(du)dt \qquad (7.89)\\ &\leq \int_{T_\delta^t} \int_{\overline{B}_M} (k(t,u)+\varepsilon)\lambda_n(t)(du)dt + \int_{T^t \setminus T_\delta^t} \int_{\overline{B}_M} k_n(t,u)\lambda_n(t)(du)dt \\ &\leq \int_{T_\delta^t} \int_{\overline{B}_M} k(t,u)\widehat{\lambda}_n(t)(du)dt + \varepsilon + \delta. \end{aligned}$$

Using the approximation result from Proposition $I - 1.183$ we can see that for every measurable integrand $\varphi(t,u)$ which is upper semicontinuous in $u \in \mathbb{R}^m$

$$\mu \to \int_{T_\delta^t} \int_{\overline{B}_M} \varphi(s,u)\mu(s)(du)ds$$

is upper semicontinuous on $L^\infty(T^t, ca(\overline{B}_M)_{w^*}) = L^1(T^t, C(\overline{B}_M))^*$. So, if in (7.89) we pass to the limit as $n \to \infty$, we obtain

$$\begin{aligned}&\limsup_{n\to\infty} \int_0^t \int_{\overline{B}_M} k_n(s,u)\lambda_n(s)(du)ds\\ &\leq \int_{T_\delta^t} \int_{\overline{B}_M} k(s,u)\lambda(s)(du)ds + \varepsilon + \delta \text{ (see (7.84))}.\end{aligned}$$

Since $\varepsilon, \delta > 0$ are arbitrary, we let $\varepsilon, \delta \to 0^+$ and have

$$\begin{aligned}&\limsup_{n\to\infty} \int_0^t \int_{\overline{B}_M} k_n(s,u)\lambda_n(s)(du)ds\\ &\leq \int_0^b \int_{\overline{B}_M} k(s,u)\lambda(s)(du)ds\\ &\Rightarrow 1 \leq \lambda(t)(U(t)) \text{ for } a.a\ t \in T \text{(recall } t \in T \text{ was arbitrary)}.\end{aligned}$$

On the other hand, we have

$$\begin{aligned}&\|\lambda(t)\|_{ca(\overline{B}_M)} \leq 1 \text{ for } a.a\ t \in T,\\ \Rightarrow &\lambda(t)(U(t)) \leq 1 \text{ for } a.a\ t \in T,\\ \Rightarrow &\lambda(t)(U(t)) = 1 \text{ for } a.a\ t \in T \text{ and so } \lambda \in S_\Sigma.\end{aligned}$$

Therefore from (7.87) and since $y = x$ and $\lambda \in S_\Sigma$, we conclude that

$$\begin{aligned}&(x,\lambda) \in P_c(0),\\ \Rightarrow &(s \times w_*) - \limsup_{n\to\infty} P_c(\vartheta_n) \subseteq P_c(0).\end{aligned}$$

Next we will prove (7.83). So, let $(x,\lambda) \in P_c(0)$. We have

$$\left\{\begin{array}{l} -x'(t) \in A(x(t)) + \int_{\overline{B}_M} f(t,x(t),u)\lambda(t)(du)dt \text{ for } a.a\ t \in T,\\ x(0) = x_0, \lambda \in S_\Sigma.\end{array}\right\}$$

We can find $u^m \in S_U\ m \in \mathbb{N}$ such that

$$\delta_{u^m} \xrightarrow{w^*} \lambda \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}). \tag{7.90}$$

Through an easy application of the Yankov-von Neumann-Aumann selection (see Theorem $I-5.106$), we can find $u_n^m \in S_{U(\cdot,\vartheta_n)}$ such that

$$|v^m(t)-u_n^m(t)| = d(u^m(t), U(t,\vartheta_n)) \text{ for } a.a\ t \in T, \text{ all } n,m \in \mathbb{N}. \tag{7.91}$$

We have $\delta_{u_n^m} \in S_{\Sigma(\cdot,\vartheta_n)}$ for all $m,n \in \mathbb{N}$. From (7.91) and hypothesis $H_6(ii)$, we have

$$\delta_{u_n^m} \xrightarrow{w^*} \delta_{u^m} \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}) \text{ as } n \to \infty. \tag{7.92}$$

Invoking Lemma $I-1.193$, we can find $\{m(n)\}_{n\in\mathbb{N}}$ increasing to $+\infty$ such that

$$\lambda_n = \delta_{u_n^{m(n)}} \xrightarrow{w^*} \lambda \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}) (\text{see } (7.90), (7.92))).$$

Let $y_n \in W^{1,1}((0,b),\mathbb{R}^N)$ be the unique solution of

$$\left\{\begin{array}{l} -y'(t) \in A(y(t),\vartheta_n) + \int_{\overline{B}_M} f(t,x(t),u,\vartheta_n)\lambda_n(t)(du) \text{ for } a.a\ t \in T, \\ y(0) = x_0. \end{array}\right\}$$

The sequence $\{y_n\}_{n\in\mathbb{N}} \subseteq W^{1,1}((0,b),\mathbb{R}^N)$ is relatively weakly compact and so, we may assume that

$$y_n \xrightarrow{w} y \text{ in } W^{1,1}((0,b),\mathbb{R}^N),\ y_n \to y \text{ in } C(T,\mathbb{R}^N). \tag{7.93}$$

We introduce the following functions

$$g_n(t) = \int_{\overline{B}_M} f(t,x(t),u,\vartheta_n)\lambda_n(t)(du) \text{ for all } n \in \mathbb{N},$$

$$g(t) = \int_{\overline{B}_M} f(t,x(t),u)\lambda(t)(du).$$

Evidently $\{g_n, g\}_{n\in\mathbb{N}} \subseteq L^1(T,\mathbb{R}^N)$ and we have

$$g_n \xrightarrow{w} g \text{ in } L^1(T,\mathbb{R}^N) (\text{see } (7.85)).$$

Let $v_n \in W^{1,1}((0,b),\mathbb{R}^N)$ resp. $v \in W^{1,1}((0,b),\mathbb{R}^N)$ be the unique solutions of

$$\left\{\begin{array}{l} -v_n'(t) \in A(v_n(t),\vartheta_n) + g(t) \text{ for } a.a\ t \in T, \\ v_n(0) = x_0. \end{array}\right\}$$

$$\text{resp.} \left\{ \begin{array}{l} -v'(t) \in A(v(t)) + g(t) \text{ for } a.a\ t \in T, \\ v(0) = x_0. \end{array} \right\}$$

Evidently $v = x$ and $v_n \to x$ in $C(T, \mathbb{R}^N)$. We have

$$\begin{aligned} |y_n(t) - v_n(t)|^2 &\leq 2 \int_0^t (g_n - g, y_n - v_n)_{\mathbb{R}^N} ds \qquad (7.94) \\ &= 2 \int_0^t (g_n - g, y_n - y)_{\mathbb{R}^N} ds + 2 \int_0^t (g_n - g, y - v)_{\mathbb{R}^N} ds \\ &+ 2 \int_0^t (g_n - g, v - v_n)_{\mathbb{R}^N} ds, \\ &\Rightarrow y_n - v_n \to 0 \text{ in } C(T, \mathbb{R}^N) (\text{see } (7.94)), \\ &\Rightarrow y_n \to x \text{ in } C(T, \mathbb{R}^N). \end{aligned}$$

On account of hypothesis $H_5(iii)$, we see that for every $n \in \mathbb{N}$, the Cauchy problem

$$\left\{ \begin{array}{l} -x'(t) \in A(x(t), \vartheta_n) + \int_{\overline{B}_M} f(t, x_n(t), u, \vartheta_n) \lambda_n(t)(du) \text{ for } a.a\ t \in T, \\ x(0) = x_0(\vartheta_n). \end{array} \right\}$$

has a unique solution $x_n \in W^{1,1}((0, b), \mathbb{R}^N)$. Then as before, using hypotheses H_4, $H_5(iii)$ and H_8, given $\varepsilon > 0$, we can find $n_0 \in \mathbb{N}$ such that

$$\begin{aligned} &|x_n(t) - y_n(t)|^2 \leq \varepsilon + 2 \int_0^t k(s) |x_n(s) - y_n(s)|^2 ds \text{ for all } t \in T, \text{ all } n \geq n_0, \\ &\Rightarrow \|x_n - y_n\|_{C(T, \mathbb{R}^N)} \leq \varepsilon \|k\|_1 \text{ for all } n \geq n_0, \\ &\Rightarrow x_n - y_n \to 0 \text{ in } C(T, \mathbb{R}^N), \\ &\Rightarrow x_n \to x \text{ in } C(T, \mathbb{R}^N) (\text{see } (7.94)). \end{aligned}$$

Since $(x_n, \lambda_n) \in P_c(\vartheta_n)\ n \in \mathbb{N}$, we conclude that

$$P_c(0) \subseteq (s \times w^*) - \liminf_{n \to \infty} P_c(\vartheta_n).$$

Therefore finally we have

$$P_c(\vartheta_n) \overset{K_{s \times w^*}}{\longrightarrow} P_c(0) \text{ in } C(T, \mathbb{R}^N) \times L^\infty(T, ca(\overline{B}_M)_{w^*}).$$

□

Using this stability result, we can compare the notions of relaxability and value continuity introduced in Definition 7.94.

Theorem 7.96 *If hypotheses H_4, H_5, H_6, H_7, H_8 hold, then problem (7.77) is relaxable if and only if it is value continuous.*

Proof $\Rightarrow$ Let $\vartheta_n \to 0$ in Θ and for every $n \in \mathbb{N}$, let (x_n, u_n) be an admissible pair for (7.80) (with $\varepsilon = \varepsilon_n$) such that

$$I_L(x_n, u_n, \vartheta_n) \leq m(\vartheta_n) + \frac{1}{n}. \tag{7.95}$$

We may assume that

$$\begin{aligned} &x_n \xrightarrow{w} x \text{ in } W^{1,1}((0,b), \mathbb{R}^N), \delta_{u_n} \xrightarrow{w^*} \lambda \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}) \\ \Rightarrow &x_n \to x \text{ in } C(T, \mathbb{R}^N), \delta_{u_n} \xrightarrow{w^*} \lambda \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}). \end{aligned} \tag{7.96}$$

From (7.96) and Proposition 7.95, we have

$$(x, \lambda) \in P_c(0).$$

Also from hypotheses H_7 and (7.96), we have

$$\begin{aligned} \lim_{n\to\infty} I_L(x_n, u_n, \vartheta_n) &= \lim_{n\to\infty} \widehat{I}_L^r(x_n, \delta_{u_n}, \vartheta_n) \\ &= \widehat{I}_L^r(x, \lambda), \\ \Rightarrow m(0) &\leq \liminf_{n\to\infty} m(\vartheta_n). \end{aligned} \tag{7.97}$$

(see (7.95) and recall that we assume that the system is relaxable).
Let $(x, \lambda) \in P_c(0)$ be such that

$$\widehat{I}_L^r(x, \lambda) = \widehat{m}_r(0) = m(0). \tag{7.98}$$

We can find $(x_n, u_n) \in P(0)$ $n \in \mathbb{N}$ such that

$$x_n \to x \text{ in } C(T, \mathbb{R}^N), \delta_{u_n} \xrightarrow{w^*} \lambda \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}). \tag{7.99}$$

Also using Proposition 7.95 and hypotheses H_7, we see that given $\widehat{\delta} > 0$, we can (w_n, v_n) admissible pair for problem (7.80) such that

$$\begin{aligned} & I_L(w_n, v_n, \vartheta_n) \leq I_L(x_n, u_n, \vartheta_n) + \widehat{\delta}, \\ \Rightarrow & m(\vartheta_n) \leq \widehat{I}_L^r(x_n, \delta_{u_n}, \vartheta_n) + \widehat{\delta}, \\ \Rightarrow & \limsup_{n\to\infty} m(\vartheta_n) \leq \widehat{I}_L^r(x, \lambda) + \widehat{\delta} (\text{see (7.99)}), \\ \Rightarrow & \limsup_{n\to\infty} m(\vartheta_n) \leq m(0) (\text{since } \widehat{\delta} > 0 \text{ is arbitrary}). \end{aligned} \tag{7.100}$$

From (7.97) and (7.100), we infer that $m(\vartheta_n) \to m(0)$. So, we have proved value continuity.

$\Leftarrow$ Arguing by contradiction, suppose that the implication is not true. Then $0 < \delta = m(0) - \widehat{m}_L^*$. Let $(x, \lambda) \in P_c(0)$ such that

$$\widehat{J}_L^r(x, \lambda) \leq \widehat{m}_L^* + \frac{\delta}{8}. \tag{7.101}$$

We know that we can find a sequence $\{u_n\}_{n\in\mathbb{N}} \subseteq S_U$ such that $\delta_{u_n} \xrightarrow{w^*} \lambda$ in $L^\infty(T, ca(\overline{B}_M)_{w^*})$. Let $x_n \in W^{1,1}((0, b), \mathbb{R}^N)$ the unique state generated by the control u_n. We have that

$$x_n \xrightarrow{w} x \text{ in } W^{1,1}((0, b), \mathbb{R}^N), x_n \to x \text{ in } C(T, \mathbb{R}^N).$$

Let $\vartheta_m \to 0$ in Θ. According to Proposition 7.35, we can find $(y_n^m, \lambda_n^m) \in P_c(\vartheta_m)$ such that

$$y_n^m \to x_n \text{ in } C(T, \mathbb{R}^N), \lambda_n^m \xrightarrow{w^*} \delta_{u_n} \text{ in } L^\infty(\Omega, ca(\overline{B}_R)_{w^*}).$$

We can find (w_n^{mk}, u_n^{mk}) $k \in \mathbb{N}$ admissible state-control pairs of problem (7.80) such that

$$w_n^{mk} \to y_n^m \text{ in } C(T, \mathbb{R}^N), \delta_{u_n^{mk}} \xrightarrow{w^*} \lambda_n^m \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}) \text{ as } k \to \infty.$$

Using Lemma $I - 1.193$, we can find $\{m(k)\}_{k\in\mathbb{N}}$ increasing to $+\infty$ such that

$$w_n^{m(k)k} \to x_n \text{ in } C(T, \mathbb{R}^N), \delta_{u_n^{m(k)k}} \xrightarrow{w^*} \delta_{u_n} \text{ in } L^\infty(T, ca(\overline{B}_M)_{w^*}).$$

We have

$$\begin{aligned} & I_L(w_n^{m(k)k}, u_n^{m(k)k}, \vartheta_{m(k)}) = \widehat{I}_L^r(w_n^{m(k)k}, \delta_{u_n^{m(k)k}}, \vartheta_{m(k)}) \\ \to & \widehat{I}_L^r(x_n, \delta_{u_n}) = I_L(x_n, u_n) \text{ as } k \to \infty \end{aligned} \tag{7.102}$$

and

$$I_L(x_n, u_n, \vartheta_{m(k)}) = \widehat{I}_L^r(x_n, \delta_{u_n}, \vartheta_{m(k)}) \tag{7.103}$$
$$\to \widehat{I}_L^r(x_n, \delta_{u_n}) = I_L(x_n, u_n) \text{ as } k \to \infty.$$

From (7.102) and (7.103), we see that given $\delta > 0$, we can find (w_n, v_n) admissible state-control pair for (7.80) with $\vartheta \in \Theta$ in a neighborhood of 0 such that

$$I_L(w_n, u_n, \vartheta) \le I_L(x_n, u_n, \vartheta) + \frac{\delta}{4}.$$

If $\vartheta_n \to 0$ in Θ, for $n \in \mathbb{N}$ large enough, we have

$$I_L(w_n, v_n, \vartheta_n) \le I_L(x_n, u_n, \vartheta_n) + \frac{\delta}{4}. \tag{7.104}$$

Note that

$$I_L(x_n, u_n, \vartheta_n) = \widehat{I}_L^r(x_n, \delta_{u_n}, \vartheta_n) \tag{7.105}$$
$$\to \widehat{I}_L^r(x, \lambda) \text{ as } n \to \infty.$$

Therefore we may assume that

$$I_L(x_n, u_n, \vartheta_n) \le \widehat{I}_L^r(x, \lambda) + \frac{\delta}{8}. \tag{7.106}$$

Then from (7.101), (7.105), and (7.106), we have

$$m(\vartheta_n) \le I_L(w_n, v_n, \vartheta_n) \le \widehat{m}_L^* + \frac{\delta}{2} = m(0) - \frac{\delta}{2} (\text{ recall the definition of } \delta),$$
$$\Rightarrow \limsup m(\vartheta_n) \le m(0) - \frac{\delta}{2},$$

which contradicts our hypothesis that the system is value continuous. □

Remarks

7.1: For more details about uniform integrability, we refer to Fonseca and Leoni [168]. Normal integrands (see Definition 7.4(b)), were introduced by Rockafellar [428]. Theorem 7.11 (the first lower semicontinuity result) and the other proposition leading to it, are due to Buttazzo [79]. The same is true for Theorem 7.13. Theorem 7.14 can be found in Gasinski and Papageorgiou [182]. A detailed account of the theory of the spaces of measures on a locally compact space X (that is, the space $ca(B(X))$), can be found in Bourbaki [57] (see

also Section $I-4.5$ and the books of Aliprantis and Border [5], Bogachev [51], Choquet [101], Malliavin [322], Papageorgiou and Winkert [378], Schwartz [438]). For the theory of Young measures, we refer to the survey papers of Balder [33], Valadier [462], [461] and the books of Florescu and Godel Thobie [166], Pedregal [417], Roubiček [433]. Theorem 7.50 (the other lower semicontinuity result for integral functionals), is due to Balder [32]. Earlier similar results, were proved by Berkovitz [43], Cesari [90], Ioffe [256], Olech [355], [354]. We also refer to the books of Berkovitz [44], Buttazzo [79], Cesari [92], Dacorogna [119]. The Lavrentiev gap phenomenon, was first observed by Lavrentiev [288]. The example given here is due to Mania [326]. Conditions that exclude this phenomenon can be found in the works of Clarke and Vinter [105], Lowen [i], and Zaslavski [e] that discusses in detail this phenomenon (Chap. 6 on variational problems and Chap. 7–9 on optimal control problems) and proves the genericity of the nonoccurence of the Lavrentiev gap.

A detailed discussion of Calculus of Variations problems can be found in the books of Bliss [50], Buttazzo et al. [81], Clarke [104], Dacorogna [119], Gelfand and Fomin [197], Ioffe and Tichomirov [257].

7.2: The "reduction method" for the existence theory in optimal control problems, was developed by Cesari [89], [91] and Berkovitz [42] (see also the books of Berkovitz [44] and Cesari [92]). As we illustrate, this method consists of eliminating the control variable and examining a calculus of variations problem. This approach requires convex structure in the problem, expressed in terms of the "property Q" of Cesari [90]. In the absence of this, a minimizing sequence of the problem need not converge to an admissible pair. So, we need to augment the system in order to capture these limits. This leads to the so-called "relaxed problem". The two relaxed methods presented here, can be found in Papageorgiou et al. [403]. For infinite dimensional systems, we refer to Gasinski and Papageorgiou [189]. More results on the relaxation of control systems, can be found in the paper of Papageorgiou et al. [395] and in the books of Ekeland and Temam [119] and Roubiček [430]. A third relaxation method based on the formalism of $\Gamma-$limits can be found in the papers of Buttazzo [78] and Buttazzo and Dal Maso [80] and in the book of Buttazzo [79]. Finally for necessary conditions for optimality (the celebrated Pontryagin maximum principle), we refer to the books of Clarke [104] and Ioffe and Tichomirov [257] (see also Clarke and Vinter [105]). For nonconvex optimal control problems, we refer to the papers of Zaslavski [c], [d] and the book [e], which also deal with nonconvex variational problems in Chap. 4).

7.3: The results in this section are new. Related results for problems in game theory, can be found in the papers of Loridan and Morgan [318], Lucchetti and Patrone [319] and for optimal control problems in the work of Dontchev and Morduhovich [144]. We refer also to the book of Dontchev and Zolezzi [145]. Finally we mention that what we call here as "value continuity" of the problem, is also known as "well-posedness in the sense of Hadamard".

Chapter 8
Mathematical Economics and Game Theory



Introduction

In this section, we discuss some economic and game theoretic models. In Sect. 8.1, we deal with exchange economies and introduce the notions of core (Edgeworth) equilibrium and of competitive (Walras) equilibrium. In a setting of perfect competition (Aumann's model), we show the equivalence of these notions and also prove the existence of a competitive equilibrium. In Sect. 8.2, we discuss dynamic multisector models and prove the existence of optimal programs (paths) for both discounted and undiscounted models. For the latter, we use the optimality criterion of weak maximality. Finally in Sect. 8.3, we discuss noncooperative games, we introduce the notion of Nash equilibrium and prove an existence theorem for such equilibria.

8.1 Equilibrium Theory

In this section we deal with a static economic model and study the two main equilibrium notions. The first one is due to the French economist Leon Walras and it is about the non-cooperative allocation of resources by means of a price system. So, this is a competitive equilibrium. The second one is due to the British economist Francis Ysidro Edgeworth and it is about allocation of resources via a bargaining process. This is a cooperative equilibrium, since the core (or Edgeworth) optimal allocations, allow for the possibility of cooperation among the agents. If the size of the economy increases (that is, the number of agents increases), the core (Edgeworth optimal allocations) decreases. This is intuitively clear, since the bigger the number of agents, the bigger the number of coalitions and so the bigger the chance that a given allocation will be blocked (recontracted in the terminology of Edgeworth). It was conjectured by Edgeworth, that in the limit where the number of agents

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becomes infinite, the core shrinks to a set of competitive allocations which are a subset of the core. So, it is only in the limit, where the number of agents is infinite that the competitive allocations prevail. In that limit setting each individual agent is negligible compared to the size of the economy.

It was Aumann [21] in 1964 who first formulated and proved in a mathematically rigorous way the Edgeworth conjecture. His idea was to model perfect competition using a nonatomic measure space of agents. The result that he proved, is known as the "Core-Walras Equivalence Theorem".

The model considered by Aumann, is the following. The space of agents is a finite nonatomic measure space (Ω, Σ, μ). So Ω is the space of agents (consumers), the $\sigma-$algebra Σ is the set of all possible coalitions of agents (consumers) and $\mu(\cdot)$ measures the size of each coalition. Without any loss of generality, we may (and do) assume that $\mu(\Omega) = 1$(that is, $\mu(\cdot)$ is a nonatomic probability measure). This space of agents reflects a perfectly competitive economy, in which no individual agent can affect the social outcome of the economic activity by his/her individual decisions. The prices at which production and exchange take place, are totally independent of the actions of the agents as individuals. The agents take the prices as given. So, the supply and demand of each individual agent is negligible and does not influence the price. The commodity space is $\mathbb{R}^N$.

In more general models, the commodity space can be an infinite dimensional separable Banach space. However, we do not follow this path, since our goal in this chapter is not to have the most general model, but to illustrate how we can use the tools of mathematical analysis which we developed in these two volumes, in order to treat problems in mathematical economics. There is a multifunction $F : \Omega \to 2^{\mathbb{R}^N} \setminus \{\emptyset\}$, known as the "consumption multifunction"(or "consumption correspondence"). The set $F(\omega) \subseteq \mathbb{R}^N$ represents the consumption plans that are available to agent $\omega \in \Omega$. If $x \in F(\omega)$, then $x = (x_k)_{k=1}^N$ with $x_k \in \mathbb{R}$. If $x_k > 0$, then it is interpreted as an input for agent ω, while if $x_k < 0$ is interpreted as an output. So, we have consumption $(x_k > 0)$ versus supply of labor services $(x_k < 0)$. The consumption set $F(\omega)$ is equipped with an irreflexive binary relation $\prec_\omega$, which expresses the preferences of the agent ω. So, if $x \prec_\omega y$, then agent ω prefers quantity y to quantity x. Similarly $x \not\prec_\omega y$ means y is not preferred to x. Each agent starts his/her economic activity with an initial endowment $e(\omega) \in \mathbb{R}^N$. The function $\omega \to e(\omega)$ is $\Sigma-$measurable and it is known as the initial allocation function. This is the pure exchange economy model, that is, the quadruple

$$E = \{(\Omega, \Sigma, \mu), F, \prec, e\}. \tag{8.1}$$

In this model there is no production. We are only concerned with the economic exchange process, independently of the production sector.

As we already indicated agents(consumers) are given a price system $p \in \mathbb{R}^N_+$(that is, $p = (p_k)_{k=1}^N$ with $p_k \geq 0$). When the price system $p \in \mathbb{R}^N_+$ prevails in the market, then an agent has to make available the amount $\dfrac{p_k}{p_i}$ of commodity i in order to have one unit of commodity k. Then if $x \in \mathbb{R}^N$ is a commodity bundle, and

$p \in \mathbb{R}^N_+$ is a price system, the number $(p, x)_{\mathbb{R}^N}$ is the "value of x". Note that if $x, y \in \mathbb{R}^N$ are two commodity bundles and $p \in \mathbb{R}^N_+$ is the prevailing price system, then x and y can be exchanged in the market if $(p, x)_{\mathbb{R}^N} = (p, y)_{\mathbb{R}^N}$. We always assume that a price system $p \in \mathbb{R}^N_+ \setminus \{0\}$(it is nontrivial).

Definition 8.1

(a) The set $S^1_F = \{u \in L^1(\Omega, \mathbb{R}^N) : u(\omega) \in F(\omega)\ \mu - \text{a.a}\ \omega \in \Omega\}$ is the set of "allocations" of the economy E (see (8.1)).
(b) We say that an allocation $u \in S^1_F$ is "feasible"(or "attainable"), if

$$\int_\Omega u d\mu = \int_\Omega e d\mu.$$

Remark 8.2 So, a feasible allocation redistributes the initial endowment.

As we already mentioned in the beginning of this section, we want to achieve allocations which can not be blocked (improved) by any coalition of agents. This leads to the following fundamental notion.

Definition 8.3 We say that coalition $C \in \Sigma$ with $\mu(C) > 0$ can "improve upon" allocation $u \in S^1_F$, if there is an allocation $v \in S^1_F$ such that

(a) for $\mu-$a.a $\omega \in C$ $u(\omega) \prec_\omega v(\omega)$;
(b) $\int_C v d\mu = \int_C e d\mu$.

Remark 8.4 So, $v \in S^1_F$ is a feasible allocation, which can improve the well-being of each agent in coalition C. The requirement $\mu(C) > 0$ is understood as a critical mass condition. Then the "nontrivial" coalition $C \in \Sigma$ can block the proposed allocation u.

Definition 8.3 leads to the first basic equilibrium concept for a pure exchange economy. Note that no prices are involved in this equilibrium notion.

Definition 8.5 Given a pure exchange economy E, the set of all feasible allocations which can not be improved upon by any coalition with critical mass(nontrivial coalition), is the core of the economy E and it is denoted by $C(E)$.

Remark 8.6 The core of an economy, describes precisely the cooperative character of the economy.

The second equilibrium concept, is price-dependent.

Definition 8.7 A pair $(u, p) \in S^1_F \times \mathbb{R}^N_+$ is said to be a "competitive (Walras) equilibrium" for the economy E, if $u \in S^1_F$ is a feasible allocation and for $\mu-$a.a $\omega \in \Omega$ $u(\omega)$ is $\prec_\omega$ $-$maximal in the budget set

$$B_p(\omega) = B(\omega, p) = \{y \in F(\omega) : (p, y)_{\mathbb{R}^N} \leq (p, e(\omega))_{\mathbb{R}^N}\}.$$

We denote the set of competitive equilibria by $W(E)$.

Remark 8.8 The quantity $w(\omega) = (p, e(\omega))_{\mathbb{R}^N}$ is understood as the total wealth of agent $\omega \in \Omega$ and through the budget set $B(\omega, p)$ describes the constraint of the agent $\omega \in \Omega$, given the price system $p \in \mathbb{R}^N_+$. Let $D(\omega, p) = \{x \in B(\omega, p) :$ there is no $y \in B(\omega, p)$ such that $x \prec_\omega y\}$(the set of commodity bundles which are within the budget constraint of ω and which can not be improved in the $\prec_\omega$ $-$sense). Then $\omega \to D(\omega, p)$is the "demand correspondence" for the price system $p \in \mathbb{R}^N_+$. It describes the choices of the consumer $\omega \in \Omega$.

Our aim is to show that

$$C(E) = W(E)$$

(core-Walras equivalence theorem). To this end, we will need the following hypotheses on the data of the economy E.

H : (a) (Ω, Σ, μ) is a nonatomic probability measure space(perfect competition).
(b) For $\mu-$a.a $\omega \in \Omega$, $F(\omega) = \mathbb{R}^N_+$.
(c) $e \in L^1(\Omega, \mathbb{R}^N)$ and $\int_\Omega e d\mu \in \text{int}\mathbb{R}^N_+$(resource availability).
(d) For μ-a.a $w \in \Omega$, the preference relation $\prec_w$ is irreflexive and transitive and $\omega \to \prec_\omega$ is graph measurable (that is, $\{(\omega, x, y) \in \Omega \times \mathbb{R}^N_+ \times \mathbb{R}^N_+ : x \prec_\omega y\} \in \Sigma \otimes B(\mathbb{R}^N) \otimes B(\mathbb{R}^N)$).
(e) For μ-a.a $\omega \in \Omega$ and all $x \in \mathbb{R}^N_+$, the set $\{y \in \mathbb{R}^N_+ : y \prec_\omega x\}$ is open in $\mathbb{R}^N_+$.
(f) For μ-a.a $\omega \in \Omega$ and all $x, y \in \mathbb{R}^N_+$ with $x - y \in \mathbb{R}^N_+$ (that is, $y \leq x$), $x \neq y$, then $y \prec_\omega x$. (monotonicity of the preference relation).

Remark 8.9 In fact hypothesis $H(ii)$ is not necessary. We make it here for simplicity. The consumption multifunction can be more general. The graph measurability of $\omega \to \prec_\omega$ and hypothesis $H(e)$ are technical conditions and have no economic meaning.

Theorem 8.10 *If hypotheses H hold, then* $C(\varepsilon) = W(\varepsilon)$.

Proof First we show that

$$W(\varepsilon) \subseteq C(\varepsilon). \tag{8.2}$$

So let $u \in W(\varepsilon) \setminus C(\varepsilon)$. The we can find $C \in \Sigma$ with $\mu(C) > 0$ and $v \in S^1_F$ such that $u(\omega) \prec_\omega v(\omega)$ for μ-a.a $\omega \in C$ and $\int_C v d\mu = \int_C e d\mu$. We know that $u \in W(\varepsilon)$. Hence we have

$$\begin{aligned} &(p, e(\omega))_{\mathbb{R}^N} < (p, v(\omega)) \text{ for } \mu\text{-a.a } \omega \in C, \\ \Rightarrow &\int_C (p, e(\omega))_{\mathbb{R}^N} d\mu < \int_\Omega (p, v(\omega))_{\mathbb{R}^N} d\mu, \end{aligned}$$

$$\Rightarrow (p, \int_C e(\omega)d\mu)_{\mathbb{R}^N} < (p, \int_C v(\omega)d\mu)_{\mathbb{R}^N},$$

$$\Rightarrow \int_C e(\omega)d\mu \neq \int_C v(\omega)d\mu, \text{ a contradiction.}$$

This proves (8.2).

Next we will show the opposite inclusion, namely that

$$C(\varepsilon) \subseteq W(\varepsilon). \tag{8.3}$$

Let $u \in C(\varepsilon)$ and consider the multifunction $\Gamma : \Omega \to 2^{\mathbb{R}^N} \setminus \{\emptyset\}$ defined by

$$\Gamma(\omega) = \{y \in \mathbb{R}^N_+ : u(\omega) \prec_\omega y\} \cup \{e(\omega)\}$$

Claim 1 $\overline{(\int_\Omega \Gamma d\mu - \int_\Omega e d\mu)} \cap (-\text{int}\mathbb{R}^N_+) = \emptyset$

Evidently it suffices to show that

$$(\int_\Omega \Gamma d\mu - \int_\Omega e d\mu) \cap (-\text{int}\mathbb{R}^N_+) = \emptyset. \tag{8.4}$$

We proceed indirectly. So, suppose that (8.4) is not true. Then we can find $y \in \text{int}\mathbb{R}^N_+$ such that

$$\int_\Omega e d\mu - y \in \int_\Omega \Gamma d\mu,$$

$$\Rightarrow \int_\Omega e d\mu - y = \int_\Omega g d\mu \text{ with } g \in S^1_\Gamma \text{ (see Definition } I\text{-5.138)}.$$

We introduce the set

$$A = \{\omega \in \Omega : u(\omega) \prec_\omega g(\omega)\} \in \Sigma \tag{8.5}$$

(see hypothesis $H(d)$)

Note that

$$A^c = \{\omega \in \Omega : g(\omega) = e(\omega)\} (A^c = \Omega \setminus A). \tag{8.6}$$

Since $\int_\Omega e d\mu - \int_\Omega g d\mu \in \text{int}\mathbb{R}^N_+$, we have that $\mu(A) > 0$. Let $k(\omega) = g(\omega) + \frac{1}{\mu(A)} y$. Then $k(\omega) - g(\omega) \in \text{int}\mathbb{R}^N_+$ and so by hypothesis $H(f)$ (the monotonicity condition) we have

$$g(\omega) \prec_\omega k(\omega) \text{ for all } \omega \in \Omega,$$

$$\Rightarrow u(\omega) \prec_\omega k(\omega) \text{ for all } \omega \in A$$

(see (8.5) and use the transitivity of $\prec_w$).

Then

$$\int_A k d\mu = \int_A g d\mu + y = \int_\Omega g d\mu - \int_{A^c} e d\mu + y \text{ (see (8.6))}$$
$$= \int_\Omega e d\mu - \int_{A^c} e d\mu = \int_A e d\mu.$$

This means that the coalition $A \in \Sigma$ with $\mu(A) > 0$ blocks allocation $u \in S_F^1$. But $u \in C(E)$, a contradiction. This proves Claim 1.

From Proposition I-5.142, we know that

$$\int_\Omega \Gamma d\mu \subseteq \mathbb{R}^N \text{ is convex.}$$

Then we can find $p \in \mathbb{R}^N \setminus \{0\}$ such that

$$(p, \int_\Omega e d\mu)_{\mathbb{R}^N} \leq (p, h)_{\mathbb{R}^N} \text{ for all } h \in \int_\Omega \Gamma d\mu.$$

In fact $p \in \mathbb{R}^N_+$ (see hypothesis $H(f)$). Also using Theorem I-5.126, we have

$$\int_\Omega (p, e(\omega))_{\mathbb{R}^N} d\mu \leq \int_\Omega \inf\{(p, z)_{\mathbb{R}^N} : z \in \Gamma(\omega)\} d\mu \tag{8.7}$$

Claim 2 $(p, e(\omega))_{\mathbb{R}^N} \leq \inf\{(p, z)_{\mathbb{R}^N} : u(\omega) \prec_\omega z\}$. for μ-a.a $\omega \in \Omega$.

Again we argue by contradiction. So, suppose that Claim 2 is not true. Then we can find $C \in \Sigma$, with $\mu(C) > 0$ such that

$$\inf\{(p, z)_{\mathbb{R}^N} : u(\omega) \prec_\omega z\} < (p, e(\omega)) \text{ for all } \omega \in C.$$

Hypothesis $H(d)$ and the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), imply that we can find a find a $(\Sigma \cap C)$-measurable function $k_0 : c \to \mathbb{R}^N_+$ such that

$$u(\omega) \prec_\omega k_0(\omega) \text{ and } (p, k_0(\omega))_{\mathbb{R}^N} < (p, e(\omega))_{\mathbb{R}^N} \text{ for } \mu\text{-a.a } \omega \in C. \tag{8.8}$$

Set $\widehat{k_0} = \chi_C k_0 + \chi_{C^c} e \in S_F^1$ (by decomposability). Then

$$\int_\Omega (p, \widehat{k_0})_{\mathbb{R}^N} d\mu = \int_C (p, k_0)_{\mathbb{R}^N} d\mu + \int_{C^c} (p, e)_{\mathbb{R}^N} d\mu$$
$$< \int_\Omega (p, e)_{\mathbb{R}^N} d\mu \text{ (see (8))}$$

But this contradicts (8.7).

This proves Claim 2.

From Claim 2 and hypothesis $H(e)$ we have

$$(p, e(\omega))_{\mathbb{R}^N} = (p, u(\omega))_{\mathbb{R}^N} \text{ for } \mu\text{-a.a } \omega \in \Omega$$

We show that $u(\omega)$ is $\prec_\omega$ maximal on $B(\omega, p)$. Again we argue by contradiction. So, we assume that $u(\omega)$ is not $\prec_\omega$-maximal for all $\omega \in C \in \Sigma$, $\mu(C) > 0$. Again we can find $k_0 : C \to \mathbb{R}^N_+(\Sigma \cap C)$-measurable such that

$$u(\omega) \prec_\omega k_0(\omega) \text{ and } (p, k_0(\omega))_{\mathbb{R}^N} \leq (p, e(\omega))_{\mathbb{R}^N} \text{ for all } \omega \in C.$$

Then on account of hypotheses $H(d)$, (e), we can find $f \in L^1(C, \mathbb{R}^N)$ such that

$$f(\omega) \in \text{int}\mathbb{R}^N_+ \text{ and } u(\omega) \prec_\omega k_0(\omega) - f(\omega)$$

for μ-a.a $\omega \in C$

We have $(p, f(\omega))_{\mathbb{R}^N} > 0$ for μ-a.a $\omega \in C$ and so

$$(p, k_0(\omega) - f(\omega))_{\mathbb{R}^N} < (p, k_0(\omega))_{\mathbb{R}^N} \leq (p, e(\omega))_{\mathbb{R}^N}$$

for μ-a.a $\omega \in C$,

a contradiction to Claim 2

So, for μ-a.a $\omega \in \Omega$, $u(\omega)$ is $\prec_\omega$-maximal on $B(\omega, p)$ and this means that $u \in W(E)$. Therefore (8.3) holds. From (8.2) and (8.3) we have $C(E) = W(E)$. □

Remark 8.11 According to this theorem, an economy is perfectly competitive, if the sets of core and Walras allocations coincide In general we have $W(E) \subseteq C(E)$.

Theorem 8.10 provides conditions for the core to coincide with the Walras allocations, but does not say anything about the nonemptiness of these sets. Since we always have $W(E) \subseteq C(E)$ it is enough to show that $W(E) \neq \emptyset$. To do this we need to do some minor modifications in our model.

Now we equip each agent $\omega \in \Omega$ with a binary relation $\preceq_\omega$ which is a "preference-indifference" relation. More precisely $\preceq_\omega$ is reflexive (that is, $x \preceq_\omega x$ for all $x \in \mathbb{R}^N_+$), transitive (that is, for all $x, y, z \in \mathbb{R}^N$, $x \preceq_\omega y$ and $y \preceq_\omega z$, imply $x \preceq_\omega z$) and complete or total (that is, for every $x, y \in \mathbb{R}^N_+$, we have $x \preceq_\omega y$ or $y \preceq_\omega x$). We can break $\preceq_\omega$ into two component binary relations $\prec_\omega$ and $\sim_\omega$. These are defined as follows

$$x \prec_\omega y \Leftrightarrow (x \preceq_\omega y \text{ but not } y \preceq_\omega x),$$

$$x \sim_\omega y \Leftrightarrow (x \preceq_\omega y \text{ and } y \preceq_\omega x).$$

Evidently $\prec_\omega$ is irreflexive, transitive as in the initial model.

The following notion will be important in the existence theorem.

Definition 8.12 A commodity bundle $x \in \mathbb{R}^N_+$ "saturates" the desire of agent $\omega \in \Omega$, if for all $y \in \mathbb{R}^N_+$ we have $y \preceq_\omega x$.

Remark 8.13 In the previous model, saturation was impossible on account of hypothesis $H(f)$. In the new version of the model we will assume saturation, that is, we will assume that there is an allocation $v \in L^1(\Omega, \mathbb{R}^N)$, $v(\omega) \in \mathbb{R}^N_+$ for μ-a.a $\omega \in \Omega$ and $v(\omega)$ saturates the desire of μ-a.a $\omega \in \Omega$. So, if $\widehat{x} = (x_k)_{k=1}^N \in \mathbb{R}^N_+$ is a commodity bundle and $x - v(\omega) \in \mathbb{R}^N_+$ for μ-a.a $\omega \in \Omega$, then

$$\widehat{x} \sim_\omega (x_1, \cdots, x_{k-1}, v_k(w), x_{k+1}, \cdots, x_N)$$

In other words any change above the level $v_k(\omega)$ of the kth-component, does not add to the desirability of the agent $\omega \in \Omega$. From an economic viewpoint, the existence of a saturation allocation is reasonable. It means that there is an upper bound on the amount of commodity that can be profitably be used by an individual, no matter whether other commodities are or are not available.

The hypotheses for the new version of the model are:
$\widehat{\mathbf{H}}$:

(a) (Ω, Σ, μ) is a nonatomic probability measure space (perfect competition).
(b) For μ-a.a $\omega \in \Omega$, $F(\omega) = \mathbb{R}^N_+$.
(c) $e \in L^1(\Omega, \mathbb{R}^N)$ and $e(\omega) \in \text{int}\mathbb{R}^N_+$ for μ-a.a $\omega \in \Omega$ (resource availability).
(d) For μ-a.a $\omega \in \Omega$, the preference relation $\preceq_\omega$ is reflexive, transitive and complete (total); it decomposes to a strict preference relation $\prec_\omega$ and to an indifference relation $\sim_\omega$ and $\omega \to \prec_\omega$ is graph measurable;
(e) For μ-a.a $\omega \in \Omega$ and all $y \in \mathbb{R}^N_+$, the sets

$$U = \{x \in \mathbb{R}^N_+ : y \prec_\omega x\}$$

$$L = \{x \in \mathbb{R}^N_+ : x \prec_\omega y\}$$

are both open in $\mathbb{R}^N_+$.
(f) For μ-a.a $\omega \in \Omega$ and if $y \in \mathbb{R}^N_+$ does not saturate the desire of the agent, we have

$$\text{“}x - y \in \text{int}\mathbb{R}^N_+ \Rightarrow y \prec_\omega x''$$

("weak desirability")
(g) There exists an allocation $v \in L^1(\Omega, \mathbb{R}^N)$, $v(\omega) \in \mathbb{R}^N_+$ for μ-a.a $\omega \in \Omega$ such that for μ-a.a $\omega \in \Omega$, $v(\omega)$ is a saturation bundle (existence of a saturation allocation).
(h) For μ-a.a $\omega \in \Omega$, $x \in \mathbb{R}^N$ can not saturate the desire of agent ω, unless $x - e(\omega) \in \text{int}\mathbb{R}^N_+$ (saturation restriction).

Theorem 8.14 *If hypotheses $\widehat{H}$ hold, then $W(\varepsilon) \neq \emptyset$.*

Proof Given a price system $p \in \mathbb{R}^N_+ \backslash \{0\}$, the budget multifunction $B_p : \Omega \to 2^{\mathbb{R}^N} \setminus \{\emptyset\}$ is defined by

$$B_p(\omega) = \{x \in \mathbb{R}^N : (p, x)_{\mathbb{R}^N} \leq (p, e(\omega))_{\mathbb{R}^N}\}.$$

Then we define

$$C_p(\omega) = \{x \in \mathbb{R}^N : y \preceq_\omega x \text{ for all } y \in B_p(\omega)\}$$

So, $C_p(\omega)$ consists of all commodity bundles which are strictly preferred or indifferent to the budgetary commodity bundles. The set $\int_\Omega C_p(\omega) d\mu$ is the "aggregate preferred set". The commodity bundles of $C_p(\omega)$ are the ones that can be distributed among the agents, making them at least as satisfied as when he/she trades his/her initial endowment and buys the best that he/she can. Let

$$V(\omega) = \{x \in \mathbb{R}^N_+ : v(\omega) - x \in \mathbb{R}^N_+\},$$

$$D_p(\omega) = V(\omega) \cap C_p(\omega). \tag{8.9}$$

Then $\omega \to D_p(\omega)$ is graph measurable and

$$|D_p(\omega)| = \sup\{|x| : x \in D_p(\omega)\} \leq |v(\omega)|$$

for μ-a.a $\omega \in \Omega$.

Claim 1 For μ-a.a $\omega \in \Omega$, $p \to D_p(\omega)$ is continuous.

We have $V(\omega) \in P_k(\mathbb{R}^N)$ for all $\omega \in \Omega \setminus N$, $\mu(N) = 0$. Hence by Proposition *I*-5.13, to show the upper semicontinuity of $p \to D_p(\omega)$ it suffices to show that the multifunction has closed graph. So, let $p_n \to p$, $x_n \to x$ in $\mathbb{R}^N$ with $x \in D_{p_n}(\omega)$. Arguing by contradiction suppose that $x \notin D_p(\omega)$. Therefore we can find $y \in B_p(\omega)$ $x \prec_\omega y$. Then on account of hypothesis $\widehat{H}(e)$, we can find $\varepsilon > 0$ such that $z \in B_\varepsilon(y)$ and

$$x \prec_\omega z, (p, z)_{\mathbb{R}^N} < (p, e(\omega))_{\mathbb{R}^N} \text{ (see hypothesis } \widehat{H}(c)).$$

It follows that we can find $n_0 \in \mathbb{N}$ such that

$$x_n \prec_\omega y, (p_n, z)_{\mathbb{R}^N} < (p, e(\omega))_{\mathbb{R}^N} \text{ for all } n \geq n_0$$

$$\Rightarrow x_n \notin C_p(\omega) \text{ for all } n \geq n_0 \text{ a contradiction.}$$

So, $p \to D_p(\omega)$ is usc.

Next we show that for $\omega \in \Omega \setminus N$, $\mu(N) = 0$, $p \to D_p(\omega)$ is lsc. Suppose $p_n \to p$ in $\mathbb{R}^N_+$, $p \neq 0$ and let $x \in D_p(\omega)$. If x is a saturation bundle, then

$x \in D_{p_n}(\omega)$ for all $n \in \mathbb{N}$ and so we are done. Therefore we assume that x is not a saturation bundle. Since $D_{p_n}(\omega) \in P_k(\mathbb{R}^N_+)$ we can find $x_n \in D_{p_n(\omega)}$ such that

$$d(x, D_{p_n}(\omega)) = |x - x_n|$$

Let $v_\omega(x)$ be the bundle in $V(\omega)$ formed from $x \in \mathbb{R}^N_+$ by $v_\omega(x) = (\min\{x_k, v_k(\omega)\})_{k=1}^N$. For $\varepsilon > 0$ and $t = (\varepsilon_1, \cdots, \varepsilon) \in \text{int}\mathbb{R}^N_+$. Let $y_\varepsilon = v_\omega(x + t)$. Then

$$x \prec_\omega x + t \sim_\omega y_\varepsilon$$

Then either for all $\varepsilon > 0$ and $n \in \mathbb{N}$ large

$$y_\varepsilon \in D_{p_n}(\omega)$$

$$\text{or } y_\varepsilon \notin D_{p_{n_k}}(\omega)$$

for a subsequence $\{n_k\}_{k\in\mathbb{N}}$ and some $\varepsilon > 0$.
In the first case we have

$$|x_n - x| < |y_\varepsilon - x| \leq \varepsilon\sqrt{N} \text{ for all } n \in \mathbb{N} \text{ large},$$

$$\Rightarrow x_n \to x \text{ with } x_n \in D_{p_n}(\omega).$$

In the second case we can find $z_{n_k} \in D_{p_{n_k}}(\omega)$ such that $y_\varepsilon \prec_\omega z_{n_k}$. From (8.9) we see that we may assume that $z_{n_k} \to z$ and so in the limit we have

$$(p, z)_{\mathbb{R}^N} \leq (p, e(\omega))_{\mathbb{R}^N},$$

$$\Rightarrow z \in B_p(\omega)$$

Also from hypothesis $\widehat{H}(e)$ we have that $y_\varepsilon \preceq_\omega z$ and so $x \prec_\omega z$, a contradiction. Since $x \in D_p(\omega) \subseteq C_p(\omega)$. Therefore $p \to D_p(\omega)$ is lsc and so finally $p \to D_p(\omega)$ is continuous (and h-continuous, see Corollary I-5.50). This proves Claim 1.
Claim 2 For every $p \in \mathbb{R}^N_+ \setminus \{0\}$, the multifunctions

$$\omega \to C_p(\omega), \omega \to D_p(\omega)$$

are graph measurable.

The budget multifunction $\omega \to B_p(\omega)$ is measurable and $P_k(\mathbb{R}^N)$-valued. So, by Theorem I-5.101, we can find Σ-measurable functions $\beta_n : \Omega \to \mathbb{R}^N_+$ $n \in \mathbb{N}$ such that

$$B_p(\omega) = \overline{\{\beta_n(\omega)\}}_{n\in\mathbb{N}} \text{ for all } \omega \in \Omega \setminus N, \mu(N) = 0 \tag{8.10}$$

Since μ-null sets do not play any role, we may assume that (8.10) is true for all $\omega \in \Omega$. We have

$$C_p(\omega) = \bigcap_{n\in\mathbb{N}} \{x \in \mathbb{R}^N_+ : \beta_n(\omega) \preceq_\omega x\}.$$

$$\Rightarrow C_p(\cdot) \text{ is graph measurable}$$

Similarly $\omega \to D_p(\omega) = V(\omega) \cap C_p(\omega)$ is graph measurable. This proves Claim 2.

Claim 3 $\int_\Omega D_p(\omega)d\mu \in P_{kc}(\mathbb{R}^N_+)$ and is continuous in p.

Convexity follows from Theorem I-5.142. Also since $\omega \to D_p(\omega)$ is graph measurable, integrably bounded (by $|x(\cdot)|$) and $P_k(\mathbb{R}^N_+)$-valued, the compactness of the set-valued integral follows from Proposition I-5.144. Finally the continuity of $p \to \int_\Omega D_p(\omega)d\mu$ follows from Claim 1 and Theorem I-8.46. This proves Claim 3.

Then we can define

$$d(p) = p_{\int_\Omega D_p}(\int_\Omega e).$$

(recall for $C \subseteq \mathbb{R}^N$ nonempty closed, convex, by $p_C(\cdot)$ we denote the metric projection function).

Claim 4 $d : \mathbb{R}^N_+ \to \mathbb{R}^N_+$ is continuous.

The upper semicontinuity of $p \to d(p)$ follows from Proposition I-8.8. We need to show that $d \to d(p)$ is lower semicontinuous. Let $\eta > 0$ and consider the sublevel set

$$L_\eta = \{p \in \mathbb{R}^N_+ : d(p) \leq \eta\}$$

We have to show that L_η is closed. To this end let $\{p_n\}_{n\in\mathbb{N}} \subseteq L_\eta$ and assume that $p_n \to p$ in $\mathbb{R}^N_+$. Then

$$\begin{aligned} d(p_n) &= |\int_\Omega e d\mu - y_n| \text{ for some } y_n \in \int_\Omega D_p(\omega)d\mu \\ &= |\int_\Omega e d\mu - \int_\Omega g_n d\mu| \text{ with } g_n \in S^1_{D_p(\cdot)} \end{aligned}$$

We know that

$$|g_n(\omega)| \leq v(\omega) \text{ for } \mu\text{-a.a } \omega \in \Omega, \text{ all } n \in \mathbb{N},$$

$$\Rightarrow \{g_n\}_{n\in\mathbb{N}} \subseteq L^1(\Omega) \text{ is uniformly integrable}$$

By the Dunford-Pettis theorem (see Theorem I-4.26), we may assume that

$$g_n \xrightarrow{w} g \quad \text{in } L^1(\Omega, \mathbb{R}^N).$$

Then we have

$$\int_\Omega g_n d\mu \to \int_\Omega g d\mu \quad \text{and} \quad g \in \mathcal{S}^1_{D_p(\cdot)} \quad \text{(see Theorem I-8.46)},$$
$$\Rightarrow \left| \int_\Omega e d\mu - \int_\Omega g_n d\mu \right| \to \left| \int_\Omega e d\mu - \int_\Omega g d\mu \right|,$$
$$\Rightarrow d(p) \leq \eta.$$

Therefore L_η is closed. This proves the lower semicontinuity of $d(\cdot)$. We conclude that $d(\cdot)$ is continuous. This proves Claim 4.

Claim 5 For every $p \in \mathbb{R}^N_+$, $d(p) - \int_\Omega e d\mu \in \mathbb{R}^N_+$.

We argue indirectly. So, suppose that this claim is not true. This means that $d(p)$ has a coordinate strictly less than the corresponding coordinate of $\int_\Omega e d\mu$. Without any loss of generality we may assume that this is the first coordinate. So, we have

$$d_1(p) < \int_\Omega e_1 d\mu \tag{8.11}$$

(recall that $d(p) = (d_k(p))_{k=1}^N$ and $e(\cdot) = (e_k(\cdot))_{k=1}^N$). We know that

$$d(p) = p_{\int_\Omega D_p} \left(\int_\Omega e \right).$$

Since $\int_\Omega D_p d\mu \in P_{kc}(\mathbb{R}^N_+)$ (see Claim 3), we can find $h \in \mathcal{S}^1_{D_p}$ such that $d(p) = \int_\Omega h d\mu$. Let $g(\omega) = (v_1(\omega), h_2(\omega), \ldots, h_N(\omega))$. Then

$$h(\omega) \leq g(\omega) \quad \text{for } \mu - \text{a.a } \omega \in \Omega$$
(see hypothesis $\widehat{H}(h)$).

Also we have

$$g(\omega) \leq v(\omega) \quad \text{for } \mu - \text{a.a } \omega \in \Omega$$

(since $v(\cdot)$ is the saturation bundle). Therefore

$$g(\omega) \in D_p(\omega) \quad \text{for } \mu - \text{a.a } \omega \in \Omega$$
$$\Rightarrow \left(\int_\Omega v_1 d\mu,\ d_2(p), \ldots, d_N(p) \right) = \int_\Omega g d\mu \in \int_\Omega D_P d\mu.$$

From (8.11) and hypothesis $\widehat{H}(h)$, we have

$$d_1(p) < \int_\Omega e_1 d\mu < \int_\Omega v_1 d\mu.$$

Therefore we can find $t \in (0, 1)$ such that

$$t \int_\Omega v_1 d\mu + (1-t) d_1(p) = \int_\Omega e_1 d\mu.$$

We set

$$u = tg + (1-t)h \in \mathcal{S}^1_{D_p}.$$

The convexity of $\int_\Omega D_p d\mu$ implies that

$$\left(\int_\Omega e_1 d\mu,\ d_2(p), \ldots, d_N(p) \right) = \int_\Omega u d\mu \in \int_\Omega D_p d\mu.$$

We have

$$\begin{aligned} &\left| \int_\Omega u d\mu - \int_\Omega e d\mu \right|^2 \\ &= \sum_{k=2}^N \left(d_k(p) - \int_\Omega e_k d\mu \right)^2 \\ &< \sum_{k=1}^N \left(d_k(p) - \int_\Omega e_k d\mu \right)^2 \\ &= \left| d(p) - \int_\Omega e d\mu \right|^2, \end{aligned}$$

a contradiction to the definition of $d(p)$. This proves Claim 5.

We set $\vartheta(p) = d(p) - \int_\Omega e d\mu$. Claims 4 and 5 imply

$$\vartheta(p) \in \mathbb{R}^N_+ \quad \text{and} \quad \vartheta(\cdot) \text{ is continuous.}$$

Also we define

$$c(p) = p_{\int_\Omega C_p} \left(\int_\Omega e \right)$$

(the metric projection of $\int_\Omega C d\mu$ on $\int_\Omega C_p d\mu \in P_{k_c}(\mathbb{R}^N_+)$) and

$$\gamma(p) = c(p) - \int_\Omega e d\mu.$$

Claim 6 $\vartheta(p) = \gamma(p)$ for all $p \in \mathbb{R}^N_+$.

In the sequel for notational simplicity, for fixed $p \in \mathbb{R}^N_+$ we write

$$\vartheta = \vartheta(p),\ \gamma = \gamma(p),\ c = c(p),\ d = d(p).$$

If $\vartheta = 0$, then clearly $\gamma = 0$ (recall $D_p(\omega) \subseteq C_p(\omega)$) and so we are done. Otherwise, since ϑ is a support point of $\int_\Omega (D_p - e)d\mu$, we have

$$|\vartheta|^2 \leq (x, \vartheta)_{\mathbb{R}^N} \quad \text{for all } x \in \int_\Omega D_p d\mu - \int_\Omega e d\mu. \tag{8.12}$$

Suppose that we can find a vector in $\int_\Omega C_p d\mu$ which is closer to $\int_\Omega e d\mu$ that d. So, we can find $y \in \int_\Omega C_p d\mu - \int_\Omega e d\mu$ such that

$$|y|^2 < |\vartheta|^2. \tag{8.13}$$

Also, we have

$$0 < |y - \vartheta|^2 = |y|^2 - 2(y, \vartheta)_{\mathbb{R}^N} + |\vartheta|^2,$$
$$\Rightarrow (y, \vartheta)_{\mathbb{R}^N} + [(y, \vartheta)_{\mathbb{R}^N} - |\vartheta|^2] < |y|^2.$$

If $|\vartheta|^2 \leq (y, \vartheta)_{\mathbb{R}^N}$, then

$$|\vartheta|^2 < |y|^2$$

which contradicts (8.13). Therefore

$$(y, \vartheta)_{\mathbb{R}^N} < |\vartheta|^2. \tag{8.14}$$

We know that

$$y = \int_\Omega f d\mu - \int_\Omega e d\mu \quad \text{with } f \in \mathcal{S}^1_{C_p}.$$

Hypothesis $\widehat{H}(h)$ implies that

$$v_\omega(f(\omega)) \in D_p(\omega) \quad \text{for } \mu - \text{a.a } \omega \in \Omega.$$

Also, we have

$$v_\omega(f(\omega)) \leq f(\omega),\ v_\omega(f(\omega)) \leq v(\omega) \quad \text{for } \mu - \text{a.a } \omega \in \Omega.$$

We set $w(\omega) = v_\omega(f(\omega))$ and have

$$\int_\Omega w d\mu \in \int_\Omega D_p d\mu,\ \int_\Omega w d\mu - \int_\Omega e d\mu \leq y.$$

We know that $\vartheta \in \mathbb{R}^N_+$ (see Claim 5). Then

$$\left(\int_\Omega w d\mu - \int_\Omega e d\mu, \vartheta\right)_{\mathbb{R}^N} \leq (y, \vartheta)_{\mathbb{R}^N} < |\vartheta|^2 \text{ (see (8.14))}. \tag{8.15}$$

Note that

$$\int_\Omega w d\mu - \int_\Omega e d\mu \in \int_\Omega D_p d\mu - \int_\Omega e d\mu.$$

Then from (8.12) we have

$$|\vartheta|^2 \leq \left(\int_\Omega w d\mu - \int_\Omega e d\mu, \vartheta\right)_{\mathbb{R}^N}. \tag{8.16}$$

Comparing (8.15) and (8.16) we have a contradiction. This proves Claim 6.
Let P be the standard simplex in $\mathbb{R}^N_+$, that is,

$$P = \left\{ p \in (p_k)_{k=1}^N \in \mathbb{R}^N_+ : \sum_{k=1}^N p_k = 1 \right\}.$$

Consider the function $\varphi : P \to P$ defined by

$$\varphi(p) = \frac{1}{1 + \sum_{k=1}^N \gamma_k(p)}(p + \gamma(p)).$$

From Claims 4 and 6, we infer that $\varphi(\cdot)$ is continuous. Invoking Brouwer's fixed point theorem (see Theorem 1.29) we can find $q \in P$ such that $\varphi(q) = q$. Then

$$\begin{aligned} &(1 + \sum_{k=1}^N \gamma_k(q))q = q + \gamma(q) \\ \Rightarrow &\gamma(q) = \beta q \quad \text{with } \beta = \sum_{k=1}^N \gamma_k(q) \geq 0 \text{ (see Claims 5, 6)}. \end{aligned} \tag{8.17}$$

Claim 7 $\gamma(q) = 0$.
As before (see (8.12)), the definition of $\gamma(\cdot)$ and the convexity of $\int_\Omega c_q d\mu$, imply

$$|\gamma(q)|^2 \leq (y - \int_\Omega e d\mu, \gamma(q))_{\mathbb{R}^N} \quad \text{for all } y \in \int_\Omega C_q d\mu.$$

If $\gamma(q) \neq 0$, then $\beta > 0$ (see (8.17)) and so

$$0 < \beta|q|^2 \leq (y - \int_\Omega e d\mu, \gamma(q))_{\mathbb{R}^N} \quad \text{for all } y \in \int_\Omega C_q d\mu. \tag{8.18}$$

Consider the multifunction

$$\omega \to M(\omega) = \{x \in B_q(\omega) : x \text{ is } \prec_\omega \text{ maximal}\}.$$

The graph measurability of the budget multifunction $\omega \to B_q(\omega)$ together with hypotheses $\widehat{H}(d)$, (e), imply that

$$\omega \to M(\omega) \text{ is graph measurable.}$$

So, using the Yankov-von Neumann-Aumann selection theorem (see Theorem I-5.106), we can find $u : \Omega \to \mathbb{R}^N_+$ a $\Sigma-$measurable function such that

$$u(\omega) \in M(\omega) \text{ for } \mu - \text{a.a } \omega \in \Omega.$$

Then we have

$$\begin{aligned} &(q, u(\omega) - e(\omega))_{\mathbb{R}^N} \leq 0,\ u(\omega) \in C_q(\omega) \text{ for } \mu - \text{a.a } \omega \in \Omega \\ &\Rightarrow \int_\Omega (q, u - e)_{\mathbb{R}^N} d\mu = (q, \int_\Omega (u - e) d\mu)_{\mathbb{R}^N} \leq 0,\ \int_\Omega u d\mu \in \int_\Omega C_q d\mu, \end{aligned}$$

which contradicts (8.18). This proves Claim 7.

The equation $\gamma(q) = 0$ (see Claim 7), implies

$$\begin{aligned} &\int_\Omega e d\mu \in \int_\Omega C_p d\mu, \\ &\Rightarrow \int_\Omega e d\mu = \int_\Omega \xi d\mu \quad \text{with } \xi \in \mathcal{S}^1_{C_p}. \end{aligned} \tag{8.19}$$

From the definition of the multifunction $C_p(\cdot)$, we have

$$y \preceq_w \xi(\omega) \quad \text{for all } y \in B_q(\omega),\ \text{for } \mu - \text{a.a } \omega \in \Omega.$$

If we can show that $\xi(\omega) \in B_q(\omega)$ for $\mu-$a.a $\omega \in \Omega$, then (ξ, q) will be a Walras equilibrium pair.

First suppose that

$$(q, \xi(\omega))_{\mathbb{R}^N} < (q, e(\omega)) \quad \text{for all } \omega \in A \in \Sigma,\ \mu(A) > 0.$$

On account of hypothesis $\widehat{H}(h)$, $\xi(\omega)$ does not saturate the desire of agent ω. Therefore

$$\xi(\omega) \prec_w \xi(u) + \widehat{\varepsilon}, \widehat{\varepsilon} = (\varepsilon_k = \varepsilon)^N_{k=1},\ \varepsilon > 0,\ \omega \in A.$$

(see hypothesis $\widehat{H}(f)$).

If $\varepsilon > 0$ is small, then

$$\begin{aligned}&(q, \xi(\omega) + \widehat{\varepsilon})_{\mathbb{R}^N} < (q, e(\omega))_{\mathbb{R}^N} \quad \text{for all } \omega \in A,\\ \Rightarrow &\xi(\omega) + \widehat{\varepsilon} \in B_q(\omega) \quad \text{for all } \omega \in A.\end{aligned}$$

Since $\xi \in \mathcal{S}^1_{C_q}$, we have a contradiction. This implies that

$$(q, e(\omega))_{\mathbb{R}^N} \leq (q, \xi(\omega))_{\mathbb{R}^N} \quad \text{for } \mu - \text{a.a } \omega \in \Omega.$$

Suppose that this inequality is strict on a set of positive measure. Then

$$(q, \int_\Omega e d\mu)_{\mathbb{R}^N} < (q, \int_\Omega \xi d\mu)_{\mathbb{R}^N},$$

which contradicts (8.19). We conclude that

$$\begin{aligned}&(q, \xi(\omega))_{\mathbb{R}^N} = (q, e(\omega))_{\mathbb{R}^N} \quad \text{for } \mu - \text{a.a } \omega \in \Omega,\\ \Rightarrow &\xi(\omega) \in B_q(\omega) \quad \text{for } \mu - \text{a.a } \omega \in \Omega,\\ \Rightarrow &(\xi, q) \text{ is a Walras equilibrium pair.}\end{aligned}$$

□

8.2 Dynamic Models

In this section we consider a dynamic economic model in discrete time, with an infinite planning horizon. The infinite horizon is dictated by economic considerations. It has to do with the optimal ratio between "consumption" and "capital accumulation". More preeisely, we want to know what amount of resources should be optimally allocated to the production of capital equipment to be used for the production of consumption goods in the future. If the planning time horizon is finite, then in the last period all available reasources must be consumed an unpleasant feature since there are always concerns beyond the planning period. So, we encounter the fundamental economic fact that the consequences of investment are long lived and any proper evaluation of end-of-period capital stock should involve its value in use in the future.

First we will consider a discounted utility. In this case the intertemporal utility converges and then the question of existence of an optimal program can be established using the direct method of the calculus of variations. When the utility is not discounted, the intertemporal utility need not converge along a feasible program (path) and so we need to introduce an alternative (weaker) optimality criterion. This is the so-called "weak maximality criterion".

Our mathematical model of the economic process is described by a triple

$$(D, u, \delta),$$

with $D \subseteq \mathbb{R}^N_+ \times \mathbb{R}^N_+$ is the "technology constraint" $u : \mathbb{R}^N_+ \to \mathbb{R}$ is the utlilty function and $0 < \delta < 1$ is the discount factor. The commodity space is $\mathbb{R}^N$ and on it we consider the l^1-norm

$$|\widehat{x}| = \sum_{k=1}^{N} |x_k| \quad \text{for all } \widehat{x} = (x_k)_{k=1}^N \in \mathbb{R}^N.$$

The hypotheses on the items of this triple are following:

$\mathbf{H_1}$: The technology set (or production set) $D \subseteq \mathbb{R}^N_+ \times \mathbb{R}^N_+$ is nonempty closed, convex and

(a) $(0, 0) \in D$.
(b) $(0, y) \in D \Rightarrow y = 0$.
(c) there exists $M > 0$ such that

$$(x, y) \in D \quad \text{and} \quad |x| \geq M \Rightarrow |y| \leq M.$$

(d) If $(x, y) \in D$ and $x' - x \in \mathbb{R}^N_+$, $y - y' \in \mathbb{R}^N_+$, $y' \in \mathbb{R}^N_+$, then $(x', y') \in D$.

Remark 8.15 A pair $(x, y) \in D$ means that an input bundle x and an output bundle y, are technologically feasible in the economy. That the technology set D is convex, is common in economic models and it simply implies that it is possible to combine technological processes in arbitrary proportions. Hypothesis $H_1(a)$ says that inaction is an option, while hypothesis $H_1(b)$ says that you can not have production out of nothing. Hypothesis $H_1(c)$ says that if consumption exceeds certain level, then we have losses due to depreciation. Finally hypothesis $H_1(d)$, is a free disposability hypothesis.

Now we introduce the fundamental notion of feasible program (or feasible path) for the economy.

Definition 8.16 A "feasible program" (or "feasible path") for the economy starting from the initial endowment $\overline{y} \in \mathbb{R}^N_+$ is a sequence $\{x_n\}_{n \in \mathbb{N}_0}$ such that

$$x_0 = \overline{y} \quad \text{and} \quad (x_n, x_{n+1}) \in D \quad \text{for all } n \in \mathbb{N}_0.$$

We denote the set of all feasible programs starting from $\overline{y}$, by $A(\overline{y})$.

For the utility-function $u(\cdot)$ we assume the following:

$\mathbf{H_2}$: $u : D \to \mathbb{R}$ is a concave, upper semicontinuous function such that

- for all $y \in \operatorname{pro} j_2 D$, $x \to u(x, y)$ is nondecreasing, that is

$$x \leq x' (\text{i.e., } x' - x \in \mathbb{R}^N_+) \Rightarrow u(x, y) \leq u(x', y);$$

- for all $x \in \operatorname{pro} j_1 D$, $y \to u(x, y)$ is nonincreasing, that is

$$y \leq y' (\text{i.e., } y' - y \in \mathbb{R}^N_+) \Rightarrow u(x, y') \leq u(x, y)$$

As we already mentioned, first we will consider the "discounted" model. So, at each stage the utilility is discounted by a factor $\delta \in (0, 1)$. Therefore for a feasible program $\widehat{x} = \{x_n\}_{n\in\mathbb{N}_0} \in A(\overline{y})$ the intertemporal utility corresponding to it, is given by

$$\sum_{n\in\mathbb{N}_0} \delta^n u(x_n, x_{n+1}).$$

We are looking for feasible programs which maximize this intertemporal utility.

Definition 8.17

(*a*) A program $\widehat{x} = \{x_n\}_{n\in\mathbb{N}_0} \subseteq A(\overline{y})$ is said to be "optimal", if for any $\widehat{v} = \{v_n\}_{n\in\mathbb{N}_0} \subseteq A(\overline{y})$ we have

$$\sum_{n\in\mathbb{N}_0} \delta^n u(v_n, v_{n+1}) \leq \sum_{n\in\mathbb{N}_0} \delta^n u(x_n, x_{n+1}).$$

(*b*) An optimal program $\widehat{x} = \{x_n\}_{n\in\mathbb{N}_0}$ is said to be "optimal stationary", if $x_n = \overline{y}$ for all $n \in \mathbb{N}_0$.

(*c*) An "optimal stationary" program is nontrivial if

$$u(0, 0) < u(\overline{y}, \overline{y}).$$

(*d*) We say that $h \in \mathbb{R}^N_+$ is a "discounted golden rule stock" if $(h, h) \in D$, $u(x, y) \leq u(h, h)$ for all $(x, y) \in D$ such that $x \leq (1 - \delta)h + \delta y$ and $u(0, 0) < u(h, h)$.

We start by establishing an priori bound for the elements of $A(\overline{y})$.

Proposition 8.18 *If hypotheses H_1 hold and $\widehat{x} \in A(\overline{y})$, then $|x_n| \leq \max\{|\overline{y}|, M\}$ for all $n \in \mathbb{N}$.*

Proof Evidently it suffices to show that, see Definition 8.16,

$$(x, y) \in D \Rightarrow |y| \leq \max\{|\overline{x}|, M\}. \tag{8.20}$$

Arguing by contradiction, suppose that (8.20) is not true. Then on account of hypothesis $H_1(c)$, we can find a feasible pair $(x, y) \in D$ such that $|x| \leq M$ and $|y| > M$. Then hypothesis $H_1(b)$ implies that $x \neq 0$. Let $x' = M\frac{x}{|x|}$. We have

$$x' - x = \left(\frac{M}{|x|} - 1\right) x \geq 0.$$

Then hypothesis $H_1(d)$ (the free disposability hypothesis), implies

$$(x', y) \in D.$$

Note that $|x'| = M$. Hence by hypothesis $H_1(c)$ we have

$$|y| \leq M.$$

a contradiction. Therefore (8.20) is true and from this it follows that if $\widehat{x} = \{x_n\}_{n\in\mathbb{N}_0} \subseteq A(\overline{y})$, then $|x_n| \leq \max\{|\overline{y}|, M\}$. □

Using this a priori bound, we can prove the existence of optimal programs.

Theorem 8.19 *If hypotheses H_1, H_2 hold, then there exists an optimal program starting from the initial endowment $\overline{y} \in \mathbb{R}^N_+$.*

Proof On account of Proposition 8.18, we have $A(\overline{y}) \subseteq l^\infty(\mathbb{R}^N)$ is w^*-compact (Alaoglu's Theorem). Let $L : A(\overline{y}) \to \mathbb{R}$ be defined by

$$L(\widehat{v}) = \sum_{n\in\mathbb{N}_0} \delta^n u(v_n, v_{n+1}) \quad \text{for all } \widehat{v} = \{v_n\}_{n\in\mathbb{N}_0} \subseteq A(\overline{y}).$$

Using hypotheses H_2, we can see that $\widehat{L}(\cdot)$ is w^*-upper semicontinuous. So, we can find $\widehat{x} \in A(\overline{y})$ such that

$$L(\widehat{x}) = \sup\{L(\widehat{v}) : \widehat{v} \in A(\overline{y})\},$$
$$\Rightarrow \widehat{x} \in A(\overline{y}) \text{ is an optimal program.}$$

□

Remark 8.20 Let $\widehat{v} \in A(y)$ and define

$$V(\overline{y}) = \sum_{n\in\mathbb{N}_0} \delta^n u(v_n, v_{n+1}).$$

This is known as the "value function". This function satisfies the "principle of optimality", that is, if $\widehat{x} \in A(\overline{y})$ is an optimal program, then

$$V(\overline{y}) = \sum_{n=0}^{m} \delta^n u(x_n, x_{n+1}) + \delta^{m+1} V(x_{m+1}) \text{ for all } m \in \mathbb{N}_0$$

In Definition 8.17 we introduced the notions of optimal stationary program and of the discounted golden rule stock. In the next theorem, we relate the two notions without the use of prices.

Theorem 8.21 *If hypotheses H_1, H_2 hold, then every discounted golden rule stock is a nontrivial optimal stationary program.*

Proof From Definition 8.17(d) we know that a discounted golden rule stock $\overline{y} \in \mathbb{R}_+^N$ satisfies.

$$u(0, 0) < u(\overline{y}, \overline{y}).$$

Let $\widehat{x} \in A(\overline{y})$. We will show that $\widehat{x}$ does not generate greater utility than the stationary program $\widehat{\overline{y}} = \{\overline{y}_n = \overline{y}\}_{n \in \mathbb{N}_0}$.

Let $\vartheta(m) = \sum_{n=0}^{m+1} \frac{(1-\delta)\delta^n x_n}{1-\delta^m}$ and $\eta(m) = \sum_{n=0}^{m+1} \frac{(1-\delta)\delta^n x_{n+1}}{1-\delta^m}$. We have that $(\vartheta(m), \eta(m)) \in D$ for $m \in \mathbb{N}$. Then from Proposition 8.18 we have

$$|\vartheta(m)|, |\eta(m)| \leq \max\{|\overline{y}|, M\} \text{ for all } n \in \mathbb{N}.$$

Also the sequences

$$\{\vartheta(m)\}_{m \in \mathbb{N}}, \{\eta(m)\}_{m \in \mathbb{N}}$$

are increasing in $\mathbb{R}_+^N$. Therefore we can define

$$(\vartheta, \eta) = \lim_{m \to \infty} (\vartheta(m), \eta(m))$$

and $(\vartheta, \eta) \in D$ (since D is closed).

Using Jensen's inequality, we have

$$\sum_{n \in \mathbb{N}_0} (1 - \delta)\delta^m u(x_m, x_{n+1}) \leq u(\vartheta, \eta)$$

Note that

$$\vartheta - \delta\eta = (1 - \delta)\left(\sum_{n \in \mathbb{N}_0} \delta^n x_n - \sum_{n \in \mathbb{N}_0} \delta^{n+1} x_{n+1}\right) = (1 - \delta)\overline{y}$$

Since $\overline{y}$ is a discounted golden rule stock, we have

$$\begin{aligned}
&u(\vartheta, \eta) \leq u(\overline{y}, \overline{y}),\\
&\Rightarrow \sum_{n\in\mathbb{N}_0} \delta^n u(\vartheta, \eta) \leq \sum_{n\in\mathbb{N}_0} \delta^n u(\overline{y}, \overline{y})\\
&\Rightarrow \sum_{n\in\mathbb{N}_0} \delta^n u(x_n, x_{n+1}) \leq \frac{1}{1-\delta} u(\vartheta, \eta) \leq \frac{1}{1-\delta} u(\overline{y}, \overline{y}),\\
&\Rightarrow \widehat{\overline{y}} = \{\overline{y}_n = \overline{y}\}_{n\in\mathbb{N}_0} \text{ is optimal stationary.}
\end{aligned}$$

□

According to this theorem, in order to produce an optimal stationary program, it suffices to generate a discounted golden rule stock. To do this we will need the following notion.

Definition 8.22 We say that the economy is "δ-normal", if there exists $(x_0, y_0) \in D$ such that

$$x_0 \leq \delta y_0 \text{ and } u(0,0) < u(x_0, y_0).$$

We introduce the set $S \subseteq \mathbb{R}^N_+$ and the multifunctions $F, G : S \to 2^{\mathbb{R}^N_+ \times \mathbb{R}^N_+}$ defined by

$$S = \{x \in \mathbb{R}^N_+ : |x| \leq M\},$$

$$F(v) = \{(x, y) \in D : x \leq (1-\delta)v + \delta y\},$$

$$G(v) = \{(x, y) \in F(v) : u(x', y') \leq u(x, y) \text{ with } (x', y') \in F(v)\}.$$

Lemma 8.23 *The multifunction $G(\cdot)$ has nonempty, convex values and it is usc.*

Proof For any $v \in S$, we see that $(0,0) \in F(v)$. The convexity and closedness of D, imply that $F(\cdot)$ has nonempty, convex and closed values in $\mathbb{R}^N_+ \times \mathbb{R}^N_+$. Let $(x, y) \in F(v)$. If $|x| > M$, by hypothesis $H_1(c)$, we have

$$\begin{aligned}
&|x| \leq (1-\delta)|v| + \delta|y| \leq (1-\delta)|v| + \delta|x|,\\
&\Rightarrow |x| \leq |v| \leq M, \text{ a contradiction.}
\end{aligned}$$

Therefore we can say that

$$(x, y) \in F(v) \Rightarrow |x| \leq M.$$

Let $S_0 = \{(x, y) \in \mathbb{R}^N_+ \times \mathbb{R}^N_+ : |x| \le M, |y| \le M\}$. Then

$$F(S) \subseteq S_0,$$
$$\Rightarrow F(\cdot) \text{ is compact valued.}$$

Since $u(\cdot, \cdot)$ is upper semicontinuous, it follows that $G(v)$ is nonempty, compact. Moreover, the concavity of $u(\cdot, \cdot)$ implies that $G(v)$ is convex.

Now we will show that the multifunction $G(\cdot)$ is usc. According to Proposition I-5.13, it suffices to show that $G(\cdot)$ has a closed graph. So, consider a sequence $\{(v_n, x_n, y_n)\}_{n\in\mathbb{N}} \subseteq GrG$ and assume that

$$v_n \to v, x_n \to x, y_n \to y \text{ in } \mathrm{R}^N_+.$$

Since D is closed we have $(x, y) \in F(v)$. Suppose that $(x, y) \notin G(v)$. Then we can find $(x', y') \in F(v)$ and $\varepsilon > 0$ such that

$$u(t, y) + \varepsilon \le u(x', y') \tag{8.21}$$

The upper semicontinuity of $u(\cdot, \cdot)$ implies that we can find $n_1 \in \mathbb{N}$ such that

$$u(x_n, y_n) \le u(x, y) + \frac{\varepsilon}{3} \text{ for all } n \ge n_1. \tag{8.22}$$

From (8.21) and (8.22) we have

$$u(x_n, y_n) + \frac{2\varepsilon}{3} \le u(x', y') \text{ for all } n \ge n_1. \tag{8.23}$$

Let $t \in (0, 1)$ such that

$$-\frac{\varepsilon}{3} \le (1 - t)(u(0, 0) - u(x', y')).$$

We claim that we can find $n_2 \ge n_1$ such that

$$(tx', ty') \in F(v_2) \text{ for all } n \ge n_2. \tag{8.24}$$

Since $v_n \to v$ and $t \in (0, 1)$, we can find $n_2 \ge n_1$ such that

$$v_n \ge tv \text{ for all } n \ge n_2,$$
$$\Rightarrow (\delta - 1)v_n \le (\delta - 1)tv \le \delta tv - tx' \text{ for all } n \ge n_2.$$

So, (8.24) holds. Since $(x_n, y_n) \in G(v_n)$, we have

$$u(x_n, y_n) \ge tu(x', y') + (1 - t)u(0, 0) \ge u(x', y) - \frac{\varepsilon}{3}. \tag{8.25}$$

Using (8.25) in (8.23), we have

$$u(x', y') + \frac{\varepsilon}{3} \leq u(x_n, y_n) + \frac{2\varepsilon}{3} \leq u(x', y') \text{ for all } n \geq n_2,$$

a contradiction. So $(x, y) \in G(v)$ and this proves that $G(\cdot)$ has closed graph, therefore it is usc. □

This lemma leads to the existence of a discounted golden rule stock, which on account of Theorem 8.21 implies the existence of an optimal stationary program.

Proposition 8.24 *If hypotheses H_1, H_2 hold and the economy is δ-normal, then there exists a discounted golden rule stock.*

Proof We consider the multifunction $\Gamma : S \to 2^{\mathbb{R}^N_+} \setminus \{\emptyset\}$ defined by $\Gamma(v) = proj_1 G(v)$.

From Lemma 8.23 and its proof, we see that $\Gamma(\cdot)$ has nonempty, closed and convex values. We claim that $\Gamma(\cdot)$ is usc. Since $\Gamma(S) \subseteq S$ as before it suffices to show that $\Gamma(\cdot)$ has closed graph. We consider $\{(v_n, x_n)\}_{n\in\mathbb{N}} \subseteq Gr\Gamma$ and assume that $v_n \to v, x_n \to x$ in $\mathbb{R}^N_+$. We can find $y_n \in \mathbb{R}^N_+$ such that $(x_n, y_n) \in G(v_n)$. We know that $|y_n| \leq M$ for all $n \in \mathbb{N}$. So, we may assume that $y_n \to y$. From Lemma 8.23 we have

$$\begin{aligned} &(x, y) \in G(v), \\ &\Rightarrow x \in \Gamma(v) \text{ and so } \Gamma(\cdot) \text{ is usc.} \end{aligned}$$

By Corollary 2.135, we can find $x \in \mathbb{R}^N_+$ such that $x \in \Gamma(x)$. So, we can find $y \in \mathbb{R}^N_+$ such that $(x, y) \in G(x)$. This means that

$$(x, y) \in \Gamma(x) \text{ and } u(x', y') \leq u(x, y) \text{ for all } (x', y') \in \Gamma(x).$$

From the definition of $\Gamma(\cdot)$, we have

$$\begin{aligned} &x \leq y, \\ &\Rightarrow (x, x) \in D \text{ (see hypothesis } H_1(d)) \\ &\Rightarrow u(x', y') \leq u(x, y) \leq u(x, x) \end{aligned} \tag{8.26}$$

for all $(x', y') \in G$ with $\delta x - x \leq \delta y' - x'$ (see the definition of $\Gamma(\cdot)$). Since by hypothesis the economy is δ-normal (see Definition 8.22), we can find $(x_0, y_0) \in \Gamma(x)$ such that

$$\begin{aligned} &u(0, 0) < u(x_0, y_0), \\ &\Rightarrow x \in \mathbb{R}^N_+ \text{ is a discounted golden rule stock (see (8.26)).} \end{aligned}$$

□

Combining Theorem 8.21 and Proposition 8.24, we have the following fundamental result.

Theorem 8.25 *If hypotheses H_1, H_2 hold and the economy is δ-normal, then there exists nontrivial optimal stationary program.*

Next we will examine the undiscounted model, that is, $\delta = 1$. In this case the sum describing the intertemporal utility along a feasible program, need not converge. So, to evaluate the performance of a feasible program, we need to change the optimality criterion. The new criterion that we will employ is the so-called "weak maximality criterion".

Now the model involves two items. The technology multifunction $D(\cdot)$ and the utility function $u(\cdot, \cdot)$. The hypotheses on them are similar to those used in the discounted model.

H₃: $D : \mathbb{R}^N_+ \to P_{fc}(\mathbb{R}^N_+) = \{C \subseteq \mathbb{R}^N_+ : \text{nonempty, closed,convex}\}$ is a multifunction with closed and convex graph and

(a) $|D(x)| = \sup\{|y| : y \in D(x)\} \leq M$ for some $M > 0$.
(b) If $y \in D(x)$, $x \leq x'$ and $y' \leq y$, then $y' \in D(x')$.
(c) There exists $(x, y) \in GrD$ such that $y - x \in int\mathbb{R}^N_+$.

Remark 8.26 Note that $D(\cdot)$ is usc. Also the set of fixed points of $D(\cdot)$ is nonempty, compact, convex (see Corollary 2.137). Hypothesis $H_3(b)$ is the free disposability hypothesis. Essentially it says that the economy is ready to accept a supplement of goods, even if it does not have to do anything with them. Finally hypothesis $H_3(c)$ says that there is at least one expansible stock.

For the utility function $u(\cdot, \cdot)$ our hypotheses remain as before.

H₄: $u : GrD \to \mathbb{R}$ is concave, upper semicontinuous function such that $u(\cdot, y)$ is nondecreasing and $u(x, \cdot)$ is nonincreasing.

Definition 8.27 A "feasible program" (or "feasible path") starting from $y \in \mathbb{R}^N_+$, is a sequence $\widehat{x} = \{x_n\}_{n\in\mathbb{N}_0}$ such that $x_0 = \overline{y}$ and $x_{n+1} \in D(x_n)$ for all $n \in \mathbb{N}_0$. By $A(\overline{y})$ we denote the set of all feasible programs starting from $\overline{y} \in \mathbb{R}^N_+$.

Since we do not have a discount factor, we need to change the optimality criterion in order to take into account the possibility of divergence of the intertemporal utility.

Definition 8.28 A program $\widehat{x} = \{x_n\}_{n\in\mathbb{N}} \in A(\overline{y})$ is said to be "weakly maximal", if

$$\liminf_{m\to\infty} \sum_{n=0}^{m} [u(v_n, v_{n+1}) - u(x_n, x_{n+1})] \leq 0$$

for all $\widehat{v} = \{v_n\}_{n\in\mathbb{N}_0} \in A(\overline{y})$.

Remark 8.29 According to this definition $\widehat{x} = \{x_n\}_{n\in\mathbb{N}} \in A(\overline{y})$ is "weakly maximal", if for every $\varepsilon > 0$ and every $m_0 \geq 1$, there exists $m' \geq m_0$ such that

$$\sum_{n=0}^{m'} u(v_n, v_{n+1}) - \varepsilon \leq \sum_{n=0}^{m'} u(x_n, x_{n+1}).$$

So, we see that x is weakly maximal, if the finite horizon intertemporal utility of any other program $\widehat{v} \in A(\overline{y})$, can never "catchup" with the corresponding utility of $\widehat{x}$. That is why some authors call this optimality criterion "weakly overtaking". Note that if in the above definition we replace lim inf by lim sup, then we have the "strong maximality criterion". The drawback with the strong maximality is that it does not induce a total order on $A(\overline{y})$. More precisely, for two sequence $\{\vartheta_n\}_{n\in\mathbb{N}_0}, \{\eta_n\}_{n\in\mathbb{N}_0} \subseteq \mathbb{R}$, it may happen that

$$0 < \limsup_{m\to\infty} \sum_{n=0}^{m} (\vartheta_n - \eta_n) \text{ and } 0 < \limsup_{m\to\infty} \sum_{n=0}^{m} (\eta_n - \vartheta_n)$$

Consider the following sequence

$$\vartheta_n = (-1)^n \text{ and } \eta_n = \frac{1}{2^{n+2}}, \cdots, \ n \in pn_0.$$

Then both lim sup are $\frac{1}{2} > 0$. In contrast weak maximality induces a total order on $A(\overline{y})$ and this is why we use it.

As we did for the discounted model, we are interested on stationary programs, because they permit a balanced growth in the economy. First some easy observations resulting from hypotheses H_3 and H_4. We already pointed out that the set $\mathrm{Fix}(D)$ of fixed points of D is nonempty, compact and convex. Then the optimization problem

$$\widehat{u} = \sup\{u(x, x) : x \in \mathrm{Fix}(D)\}$$

has a solution. Moreover, hypotheses $H_3(ii)$ and H_4 imply that

$$\widehat{u} = \max\{u(x, y) : y \in G(x), x \leq y\} \tag{8.27}$$

Let S be the solution set of (8.27), that is

$$S = \{x \in \mathrm{Fix}(D) : \widehat{u} = u(x, x)\} \neq \emptyset.$$

Definition 8.30 A "weakly maximal stationary program" is a vector $x^* \in S$ such that the constant program

$$\widehat{x}^* = \{x_n^* = x^*\}_{n\in\mathbb{N}_0}$$

is weakly maximal among all programs in $A(x^*)$.

Our goal is to show that under hypotheses H_3, H_4 the economy has a weakly maximal program.

Proposition 8.31 *If hypotheses H_3, H_4 hold, then there exists $p \in \mathbb{R}^N_+$ such that*

$$u(x, y) + (p, y - x)_{\mathbb{R}^N_+} \leq \widehat{u}$$

for all $(x, y) \in GrD$.

Proof Let $k : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}^N$ be defined by

$$k(x, y) = y - x \text{ for all } x, y \in \mathbb{R}^N.$$

Using $k(\cdot, \cdot)$ we can equivalently rewrite problem (8.27) as follows

$$\widehat{u} = \max\{u(x, y) : (x, y) \in GrD, k(x, y) \geq 0\}. \tag{8.28}$$

Hypothesis $H_3(iii)$ says that there exists $(x, y) \in GrD$ such that $k(x, y) \in int\mathbb{R}^N_+$ (Slater condition). So, by the Kuhn-Tucker theorem we can find $p \in \mathbb{R}^N_+$ such that

$$u(x, y) + (p, y - x)_{\mathbb{R}^N} \leq \widehat{u} \text{ for all } (x, y) \in GrD.$$

□

Remark 8.32 We interpret $p \in \mathbb{R}^N_+$ as a price system, which supports the value of problem (8.27).

Using this proposition, we can show that there is no feasible program emanating from any initial stock $\overline{y} \in \overline{B}_M$, which is infinitely better than $\widehat{u}$ the value of (8.27).

Proposition 8.33 *If hypotheses H_3, H_4 hold, then there exists $\widehat{M} > 0$ such that for any $\widehat{x} = \{x_n\}_{n\in\mathbb{N}_0} \in A(\overline{y})$, $\overline{y} \in \overline{B}_M$ we have for any $m \in \mathbb{N}$,*

$$\sum_{n=0}^{m} (u(x_n, x_{n+1}) - \widehat{u}) \leq \widehat{M}.$$

Proof From Proposition 8.31, we know that there exists $p \in \mathbb{R}^N_+$ such that

$$u(x, y) + (p, y - x)_{\mathbb{R}^N} \leq \widehat{u} \text{ for all } (x, y) \in GrD.$$

Then we have

$$u(x_n, x_{n+1}) - \widehat{u} \leq (p, x_n - x_{n+1})_{\mathbb{R}^N} \text{ for all } \widehat{x} = \{x_n\}_{n\in\mathbb{N}_0} \in A(\overline{y}),$$

$$\Rightarrow \sum_{n=0}^{m} (u(x_n, x_{n+1}) - \widehat{u}) \leq (p, \overline{y} - x_{m+1}) \leq 2|p|M = \widehat{M}.$$

□

Then this proposition says that there is no feasible program which produces an intertemporal utility which is infinitely better than the intertemporal utility produced by the stationary program $x^* \in S$. On the other hand, it can happen that there is a feasible program which is infinitely worse than a stationary program $x^* \in S$. To exclude this possibility, we define the following particular class of feasible programs.

Definition 8.34 A feasible program $\widehat{x} = \{x_n\}_{n\in\mathbb{N}}$ is said to be "good" if there is $\eta \in \mathbb{R}$ such that for all $m \in \mathbb{N}$,

$$\eta \leq \sum_{n=0}^{m} [u(x_n, x_{n+1}) - \widehat{u}].$$

Evidently this is equivalent to saying that

$$-\infty < \liminf_{m\to\infty} \sum_{n=0}^{m} [u(x_n, x_{n+1}) - \widehat{u}].$$

Remark 8.35 Good programs appear in the literature also with the name "eligible". A feasible program is good if it can sustain a utility level close to $\widehat{u}$ (see (8.27)). Weakly maximal programs will be found among the good programs.

We show that good programs have an average asymptotic property, which leads to the solution set S of program (8.27).

Proposition 8.36 *If hypotheses H_3, H_4 hold, $\widehat{x} \in A(\overline{y})$ with $|\overline{y}| \leq M$ and $\widehat{x}$ is a good program, then $\mathcal{L} = \{$limit points of $\{\frac{1}{n+1} \sum_{k=0}^{n} x_h\}_{h\in\mathbb{N}_0}\}$ is nonempty and $\mathcal{L} \subseteq S$.*

Proof For every $n \in \mathbb{N}_0$, we have $\frac{1}{n+1} \sum_{k=0}^{n} x_k \in \overline{B}_M$. Therefore $\mathcal{L} \neq \emptyset$. If $x^* \in \mathcal{L}$, then we can find a subsequence $m \to n(m)$ such that

$$\frac{1}{n(m)+1} \sum_{k=0}^{n(m)} x_k \to x^* \text{ as } m \to \infty.$$

By hypothesis $GrD \subseteq \mathbb{R}^N_+ \times \mathbb{R}^N_+$ is convex. Therefore for every $m \in \mathbb{N}_0$, we have

$$\left(\frac{1}{n(m)+1} \sum_{k=0}^{n(m)} x_k, \frac{1}{n(m)+1} \sum_{k=1}^{n(m)+1} x_k \right) \in GrD. \tag{8.29}$$

We see that

$$\frac{1}{n(m)+1}\sum_{k=1}^{n(m)+1} x_k = \frac{1}{n(m)+1}\sum_{k=0}^{n(m)} x_k + \frac{1}{n(m)+1}[x_{n(m)+1} - \overline{y}].$$

Passing to the limit as $m \to \infty$, we obtain that

$$\frac{1}{n(m)+1}\sum_{k=1}^{n(m)+1} x_k \to x^*$$

(recall $|x_{n(m)+1}|, |\overline{y}| \leq M$). Since $GrD \subseteq \mathbb{R}^N_+ \times \mathbb{R}^N_+$ is closed, from (8.29) in the limit as $m \to \infty$, we obtain

$$(x^*, x^*) \in GrD, \Rightarrow x^* \in Fix(D). \tag{8.30}$$

With $\widehat{x} = \{x_n\}_{n\in\mathbb{N}} \in A(\overline{y})$ being the good program postulated by the hypothesis, we have

$$\begin{aligned}
\frac{\eta}{n(m)+1} &\leq \frac{1}{n(m)+1}\sum_{k=0}^{n(m)}[u(x_k, x_{k+1}) - \widehat{u}] \\
&\leq u\left(\frac{1}{n(m)+1}\sum_{k=0}^{n(m)} x_k, \frac{1}{n(m)+1}\sum_{k=1}^{n(m)+1} x_k\right) - \widehat{u} \\
&\quad \text{(by Jensen's inequality)} \\
\Rightarrow \widehat{u} &\leq \limsup_{m\to\infty} u\left(\frac{1}{n(m)+1}\sum_{k=0}^{n(m)} x_k, \frac{1}{n(m)+1}\sum_{k=1}^{n(m)+1} x_k\right) \\
&\leq u(x^*, x^*) \text{(see hypotheses } H_4), \\
\Rightarrow u(x^*, x^*) &= \widehat{u} \text{(see(8.30))}, \\
\Rightarrow \mathcal{L} &\subseteq S.
\end{aligned}$$

□

Now we can prove the existence of a weakly maximal stationary program. On account of Proposition 8.36, it is reasonable to expect to find such an optimal program among the elements of the set S (the solution set of problem (8.27)).

Theorem 8.37 *If hypotheses H_3, H_4 hold, then the economy admits a weakly maximal stationary program.*

Proof Let $p \in \mathbb{R}^N_+$ be the price system produced in Proposition 8.31 and let $x^* \in S$ be such that $(p, x^*)_{\mathbb{R}^N} \leq (p, y)_{\mathbb{R}^N}$ for all $y \in S$ (recall $S \subseteq \mathbb{R}^N_+$ is compact). Let $\widehat{x}^* = \{x^*_n = x^*\}_{n \in \mathbb{N}_n}$. We claim that $\widehat{x}^*$ is the desired weakly maximal stationary program in $A(x^*)$. Arguing by contradiction, suppose that is not weakly maximal in $A(x^*)$. Then we can find $\widehat{v} \in A(x^*)$, $\varepsilon > 0$ and $m_0 \in \mathbb{N}_0$ such that

$$U_m(\widehat{x}^*) + \varepsilon \leq U_m(\widehat{v}) \text{ for all } m \geq m_0,$$

where

$$U_m(\widehat{x}^*) = \sum_{n=0}^{m} u(x^*_n, x^*_{n+1}) = \sum_{n=0}^{m} u(x^*, x^*), \; U_m(\widehat{v}) = \sum_{n=0}^{m} u(v_n, v_{n+1}).$$

Then

$$\begin{aligned} \varepsilon \leq U_m(\widehat{v}) - U_m(\widehat{x}^*) &\text{ for all } m \geq m_0, \\ \Rightarrow \widehat{v} \in A(x^*) &\text{ is a good program.} \end{aligned} \tag{8.31}$$

Using Proposition 8.36, we may assume that

$$\frac{1}{m+1} \sum_{n=0}^{m} v_n \to y^* \in \mathcal{L} \subseteq S \text{ as } m \to \infty.$$

We have

$$\begin{aligned} &\lim_{m \to \infty} \frac{1}{m+1} \sum_{n=0}^{m} (p, v_n)_{\mathbb{R}^N} \\ &= \lim_{m \to \infty} (p, \frac{1}{m+1} \sum_{n=0}^{m} v_n)_{\mathbb{R}^N} \\ &\leq \limsup_{m \to \infty} (p, v_m)_{\mathbb{R}^N}, \\ &\Rightarrow (p, y^*) \leq \limsup_{m \to \infty} (p, v_m)_{\mathbb{R}^N}. \end{aligned} \tag{8.32}$$

From Proposition 8.31 and (8.31), we obtain

$$\begin{aligned} &0 < \varepsilon \leq (p, x^*)_{\mathbb{R}^N} - (p, v_{m+1})_{\mathbb{R}^N} \text{ for all } n \geq n_0, \\ &\Rightarrow 0 < \varepsilon \leq (p, x^*)_{\mathbb{R}^N} - \limsup_{m \to \infty} (p, v_{m+1})_{\mathbb{R}^N} \\ &\leq (p, x^*)_{\mathbb{R}^N} - (p, y^*)_{\mathbb{R}^N}, \\ &\Rightarrow (p, y^*) < (p, x^*)_{\mathbb{R}^N} \text{ with } y^* \in S, \end{aligned}$$

a contradiction to the choice of $x^* \in S$. This proves that $\widehat{x}^*$ is a weakly maximal stationary program. □

Remark 8.38 The above existence theorem fails if we look for a strongly maximal program. This can be illustrated, by a one-commodity, two processes von Neumann model (see Brock [65]).

8.3 Nash Equilibrium

In this section, we examine n-person games in normal form in which the players exhibit a noncooperative (individualistic) behavior. For such games, the main concept is the so-called "Nash equilibrium".

We have n-players $P = \{1, \ldots, n\}$ $n \geq 2$. Every player $k \in P$ has a strategy set X_k. H is the set of all strategies available to player $k \in P$. Let

$$X = \prod_{k=1}^{n} X_k \text{ (the multistrategy set).}$$

Also there is a set $S(P) \subseteq X$ which represents all the feasible multistrategies. Every player $k \in P$ has a utility function $u_k : S(P) \to \mathbb{R}$ which describes the preference of the player among the strategies available to him. For $x \in S(P)$, $u_k(x)$ measures the utility produced by x to the player k, who prefers multistrategies which generate maximum utility (an alternative interpretation is to think of u_k as a loss function, in which case $-u_k$ is a utility function). In economic applications u_k is interpreted as a utility function. Player $k \in P$, splits space X as follows

$$X = \widehat{X} \times X_k \text{ with } \widehat{X} = \prod_{i \neq k} X_i. \tag{8.33}$$

The set $k^c = P \setminus \{k\}$ is understood as a coalition adverse or complementary to $k \in P$. By p_k, resp.$\widehat{p}$, we denote the projection of X on X_k resp. on $\widehat{X}$. If $x \in X$, then $x = (\widehat{x}, x_k)$ with $\widehat{x} \in \widehat{X}$ and $x_k \in X_k$ (see (8.33)). This decomposition is important for player $k \in P$, since it distinguishes the component x_k which he/she controls, from the vector $\widehat{x} = \{x_i\}_{i \neq k}$ over which he/she has no control.

Definition 8.39 The collection $\{X_k, u_k\}_{k \in P}$ is a game in "normal form".

Suppose that no player has any information concerning the choices of any of the other players. In this case a cautious way to proceed, is for each player to maximize his/her minimum utility. So, if $k \in P$, then

$$v_k(x_k) = \inf\{u_k(x) : x \in S(P),\ p_k(x) = x_k\}. \tag{8.34}$$

This is the minimum utility for player $k \in P$ among all feasible multistrategies, when his/her choice of strategy is x_k. Then he/she will try to maximize $v_k(\cdot)$, that is,

$$\widehat{v}_k = \sup\{v_k(x_k) : x_k \in p_k(S(P))\}. \tag{8.35}$$

This is the outcome of his/her maxmin cautious approach.

Definition 8.40 The vector $\widehat{v} = \{\widehat{v}_k\}_{k=1}^n$ is the "conservative value" of the game and if $\widehat{x}^* = \{x_k^*\}_{k=1}^n \in S(P)$ is a multistrategy such that

$$\widehat{v}_k = v_k(x_k^*), k \in P$$

then it is a "conservative" or "maxmin" multistrategy.

Theorem 8.41 *If each strategy set X_k $k \in P$ is a Hausdorff topological space, $S(P) \subseteq X$ is compact in the product topology of X and for each $k \in P$ $u_k : S(P) \to \mathbb{R}$ is upper semicontinuous, then the game admits a conservative multistrategy $\widehat{x}^* \in S(P)$.*

Proof We have

$$\begin{aligned} v_k(x_k) &= \inf\{v_k(x) : x \in S(P),\ p_k(x) = x_k\} \\ &= \inf\{u_k(\widehat{x}, x_k) : (\widehat{x}, x_k) \in S(P)\}, \\ &\Rightarrow v_k : p_k(S(P)) \to \mathbb{R} \text{ is upper semicontinuous} \\ &\quad \text{(see Proposition } I - 5.51). \end{aligned}$$

Note that since the projection map is continuous, we have that $p_k(S(P)) \subseteq X_k$ is compact. So, according to Proposition $I - 1.180$ we can find $x_k^* \in p_k(S(P))$ such that

$$\widehat{v}_k = v_k(x_k^*)\ k \in P \text{ (see (8.35))}.$$

Then $\widehat{x}^* = \{x_k^*\}_{k=1}^n$ is a conservative multistrategy for the game. □

Remark 8.42 Evidently each player $k \in P$ will reject any multistrategy $x \in S(P)$ which generates utility $u_k(x)$ strictly less than $\widehat{v}_k$. So, $\widehat{v}_k$ is the minimum utility that player $k \in P$ can tolerate.

In the above setting each player chooses his/her strategy independently of the other players (of the elements in k^c). Now we will examine a different behavior. We will consider the case in which each player varies his/her strategy based on the choice of the complementary coalition k^c. The process is the following. As soon as the player $k \in P$ announces his intention to play, the complementary coalition

$k^c = P \setminus \{k\}$ moves first and then player k responds. In this setting, an equilibrium will be a multistrategy $x^* \in X$ in which no player k can increase his/her utility by abandoning strategy x_k^*, provided that the other players do not change their choice. So, we have

$$u_k(\widehat{x}^*, x_k^*) = \sup\{u_k(\widehat{x}^*, y) : y = p_k(x), x \in S(P)\}. \tag{8.36}$$

According to (8.36), if $\widehat{x}^*$ is the multistrategy of the complementary coalition $k^c = P\setminus\{k\}$, then player $k \in P$ responds by maximizing the utility

$$y \to u_k(\widehat{x}^*, y), y = p_k(x), x \in S(P).$$

This leads us to the fundamental notion of "Nash equilibrium".

Definition 8.43 If $\{X_k, u_k\}_{k\in P}$ is a game in normal form, then a "Nash equilibrium" is a feasible multistrategy $x^* \in S(P)$ such that for each $k \in P$, we have

$$u_k(\widehat{x}^*, y) \leq u_k(\widehat{x}^*, x_k^*) \text{ for all } y = p_k(x), x \in S(P).$$

Remark 8.44 So in the state of Nash equilibrium, there is no incentive for player $k \in P$ to change his/her strategy x_k^* by himself/herself. So, this is a noncooperative equilibrium notion.

To understand better the notion of Nash equilibrium, we introduce the function $\vartheta : S(P) \times S(P) \to \mathbb{R}$ defined by

$$\vartheta(x, v) = \sum_{k=1}^{n} [u_k(x) - u_k(\widehat{x}, v_k)]. \tag{8.37}$$

Using this function, we can have a sufficient condition for the existence of a Nash equilibrium.

Proposition 8.45 *If $\{X_k, u_k\}$ is a game in normal form and for some $x \in S(P)$ we have*

$$0 \leq \vartheta(x, v) \text{ for all } v \in S(P),$$

then $x \in S(P)$ is a Nash equilibrium for the game; the converse is also true provided $S(P)$ is a product space.

Proof Let $x \in S(P)$ be such that

$$0 \leq \vartheta(x, v) \text{ for all } v \in S(P).$$

Consider $v \in S(P)$ such that $\widehat{p}(v) = \widehat{x} = \{x_i\}_{i \neq k}$. We have

$$\begin{aligned} 0 \leq \vartheta(x, v) &= \sum_{i=1}^{n} |u_i(x) - u_i(\widehat{x}, v_i)| \\ &= \sum_{i \neq k} |u_i(x) - u_i(\widehat{x}, v_i)| + |u_k(x) - u_k(\widehat{x}, x_k)| \\ &= |u_k(x) - u_k(\widehat{x}, x_k)|, \end{aligned}$$

$\Rightarrow x \in S(P)$ is a Nash equilibrium for the game.

If $S(P) = \prod_{i=1}^{n} S_i \subseteq \prod_{i=1}^{n} X_i$ and $x^* \in S(P)$ is a Nash equilibrium for the game, then

$$0 \leq u_i(x^*) - u_i(\widehat{x}^*, v_i) \text{ for all } v_i \in S_i, i \in \{1, \ldots, n\} = P.$$

Adding these inequalities, we conclude that

$$0 \leq \vartheta(x^*, v) \text{ for all } v \in S(P).$$

□

So, according to this Proposition, in order to establish the existence of a Nash equilibrium x for a game in normal form, we need to show that

$$0 \leq \vartheta(x, v) \text{ for all } v \in S(P). \tag{8.38}$$

To do this we need a result known as the "Ky Fan inequality" (see Aubin and Ekeland [20, p. 327]).

Theorem 8.46 *If $K \subseteq \mathbb{R}^N$ is nonempty compact, convex and $\vartheta : K \times K \to \mathbb{R}$ is a function such that*

(i) for every $v \in K$, $x \to \vartheta(x, v)$ is upper semicontinuous;
(ii) for every $x \in K$, $v \to \vartheta(x, v)$ has convex sublevel sets (that is, $L_\lambda = \{v \in K : \vartheta(x, v) \leq \lambda\}$ is convex for all $\lambda \in \mathbb{R}$);
(iii) for all $v \in K$, $0 \leq \vartheta(v, v)$;
then we can find $x^* \in K$ such that

$$0 \leq \inf\{\vartheta(x^*, v) : v \in K\}.$$

Remark 8.47 A function with convex sublevel sets (see hypothesis (ii)), is known as a "quasiconvex" function. A convex function is quasiconvex, but the converse is not in general true. The function $x \to ln(|x| + 1)$ is quasiconvex but not convex. We say that a function $\varphi(\cdot)$ is quasiconcave, if $-\vartheta(\cdot)$ is quasiconvex.

Using Ky Fan's inequality, we can prove the existence of a Nash equilibrium.

Theorem 8.48 *If $\{X_k, u_k\}_{k\in P}$ is a game in normal form, $S(P) \subseteq \mathbb{R}^N$ is compact, convex, for every $k \in P$ $v_k(\cdot)$ is continuous and for every $\widehat{x} \in \widehat{p}(S(P))$ $v \to u_k(\widehat{x}, v)$ is quasiconvex, then the game admits a Nash equilibrium.*

Proof Let $\vartheta : S(P) \times S(P) \to \mathbb{R}$ be the function defined by

$$\vartheta(x, v) = \sum_{k=1}^{n} |u_k(x) - u_k(\widehat{x}, v_k)| \text{ for all } (x, v) \in S(P) \times S(P).$$

Evidently $\vartheta(\cdot, v)$ is continuous and $\vartheta(x, \cdot)$ is quasiconvex. Also

$$0 \leq \vartheta(x, v) \text{ for all } v \in S(P).$$

Therefore by Ky Fan's inequality (see Theorem 8.46), we can find $x^* \in S(P)$ such that

$$0 \leq \vartheta(x^*, v) \text{ for all } v \in S(P).$$

Then by Proposition 8.45 this means that $x^* \in S(P)$ is a Nash equilibrium of the game. □

Remarks

8.1: The study of equilibrium in exchange economies started with Edgeworth (1881), who introduced the so-called "contract curve". The core introduced in Definition 8.5, is a generalization of Edgeworth's contract curve. The notion of core is independent of perfect competition. In contrast the notion of competitive (Walras) equilibrium (see Definition 8.7), makes sense in an environment of perfect competition. The mathematical model of perfect competition through the use of a continuum of agents, is due to Aumann [21, 22]. Improvements in the initial model of Aumann, were made by Schmeidler [436] and Hildenbrand [241, 242]. Further generalizations with an infinite dimensional commodity space, can be found in the book of Aliprantis et al. [6]. We also mention the books of Hildenbrand [243], Hildenbrand and Kirman [244] and Klein and Thompson [270]. Finally we mention the paper of Vind [463] on the theory of core and core equivalence, where the primary notion is not the individual agent but coalitions of agents.

8.2: Dynamic models, started with the work of Malinvaud [321] and later was continued by Gale [175], Radner [426], Peleg and Ryder [463], Hurwicz [419], Brock and Majumdar [254]. Strong maximality was first proposed by Gale [66], who assumed that the utility function is strictly concave. Brock [65] objected to this hypothesis (since it excludes the von Neumann model) and instead suggested the weak maximality criterion. Additional material on such multisector growth

models, can be found in the books of Makarov and Rubinov [320] and Takayama [45], and Zaslavski [f],[g],[h].

8.3: Noncooperative games and the corresponding equilibrium notion were introduced by Nash [349, 350] who proved Theorem 8.48. Additional information on such games, can be found in the books of Aubin and Ekeland [20], Border [53], Ichishi [255].

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